ETSI TS 138 473 V19.3.0 (2026-07)

TECHNICAL SPECIFICATION

5G;
NG-RAN;
F1 Application Protocol (F1AP)
(3GPP TS 38.473 version 19.3.0 Release 19)

Reference

RTS/TSGR-0338473vj30

Keywords

5G

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI IPR online database.

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™**, **LTE™** and **5G™** logo are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Legal Notice

This Technical Specification (TS) has been produced by ETSI 3rd Generation Partnership Project (3GPP).

The present document may refer to technical specifications or reports using their 3GPP identities. These shall be interpreted as being references to the corresponding ETSI deliverables.

The cross reference between 3GPP and ETSI identities can be found at 3GPP to ETSI numbering cross-referencing.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Contents

Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

1 Scope

The present document specifies the 5G radio network layer signalling protocol for the F1 interface. The F1 interface provides means for interconnecting a gNB-CU and a gNB-DU of a gNB within an NG-RAN, or for interconnecting a gNB-CU and a gNB-DU of an en-gNB within an E-UTRAN. The F1 Application Protocol (F1AP) supports the functions of F1 interface by signalling procedures defined in the present document. F1AP is developed in accordance to the general principles stated in TS 38.401 [4] and TS 38.470 [2].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 38.470: "NG-RAN; F1 general aspects and principles".

[3] 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)".

[4] 3GPP TS 38.401: "NG-RAN; Architecture Description".

[5] ITU-T Recommendation X.691 (2002-07): "Information technology - ASN.1 encoding rules - Specification of Packed Encoding Rules (PER)".

[6] 3GPP TS 38.300: "NR; Overall description; Stage-2".

[7] 3GPP TS 37.340: "NR; Multi-connectivity; Overall description; Stage-2".

[8] 3GPP TS 38.331: "NR; Radio Resource Control (RRC); Protocol specification".

[9] 3GPP TS 36.423: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 Application Protocol (X2AP)".

[10] 3GPP TS 23.401: "General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access".

[11] 3GPP TS 23.203: "Policy and charging control architecture".

[12] ITU-T Recommendation X.680 (07/2002): "Information technology – Abstract Syntax Notation One (ASN.1): Specification of basic notation".

[13] ITU-T Recommendation X.681 (07/2002): "Information technology – Abstract Syntax Notation One (ASN.1): Information object specification".

[14] 3GPP TR 25.921: (version.7.0.0): "Guidelines and principles for protocol description and error".

[15] 3GPP TS 36.413: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Application Protocol (S1AP)".

[16] 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification".

[17] 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception".

[18] 3GPP TS 29.281: "General Packet Radio System (GPRS); Tunnelling Protocol User Plane (GTPv1-U) ".

[19] 3GPP TS 38.414: "NG-RAN; NG data transport".

[20] 3GPP TS 36.300: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Overall description; Stage 2".

[21] 3GPP TS 23.501: "System Architecture for the 5G System".

[22] 3GPP TS 38.472: "NG-RAN; F1 signalling transport".

[23] 3GPP TS 23.003: "Numbering, addressing and identification".

[24] 3GPP TS 38.304: "NR; User Equipment (UE) procedures in Idle mode and RRC Inactive state ".

[25] 3GPP TS 36.104: "Base Station (BS) radio transmission and reception".

[26] 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone".

[27] 3GPP TS 36.211: "Evolved Universal Terrestrial Radio Access (E-UTRA); Physical channels and modulation".

[28] 3GPP TS 38.423: "NG-RAN; Xn application protocol (XnAP)".

[29] 3GPP TS 32.422: "Trace control and configuration management".

[30] 3GPP TS 38.340: "NR; Backhaul Adaptation Protocol (BAP) specification".

[31] 3GPP TS 38.213: "NR; Physical layer procedures for control".

[32] 3GPP TS 38.314: " NR; Layer 2 measurements".

[33] 3GPP TS 38.211: "NR; Physical channels and modulation".

[34] 3GPP TS 38.214: "NR; Physical layer procedures for data".

[35] 3GPP TS 37.320: "Radio measurement collection for Minimization of Drive Tests (MDT)".

[36] 3GPP TS 23.032:"Technical Specification Group Services and System Aspects; Universal Geographical Area Description (GAD)".

[37] 3GPP TS 38.455: "NG-RAN; NR Positioning protocol A (NRPPa)".

[38] 3GPP TS 38.133: "NR; Requirements for support of radio resource management".

[39] 3GPP TS 37.355: "LTE Positioning Protocol (LPP)".

[40] 3GPP TS 23.287: "Architecture enhancements for 5G System (5GS) to support Vehicle-to-Everything (V2X) services".

[41] 3GPP TS 36.331: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification".

[42] 3GPP TS 38.305: "NG Radio Access Network (NG-RAN); Stage 2 functional specification of User Equipment (UE) positioning in NG-RAN".

[43] 3GPP TS 38.215: "NR; Physical layer (PHY); Measurements".

[44] 3GPP TS 23.304: "Proximity based Services (ProSe) in the 5G System (5GS)".

[45] Void

[46] 3GPP TS 37.213: "NR; Physical layer procedures for shared spectrum channel access".

[47] 3GPP TS 37.483: "E1 Application Protocol (E1AP)".

[48] IEEE Std 1588: "IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems", Edition 2019.

[49] 3GPP TS 23.273: "5G System (5GS) Location Services (LCS); Stage 2".

[50] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces; Stage 3".

[51] 3GPP TS 33.501: "Security architecture and procedures for 5G System".

3 Definitions and abbreviations

3.1 Definitions

elementary procedure: F1AP consists of Elementary Procedures (EPs). An Elementary Procedure is a unit of interaction between gNB-CU and gNB-DU. These Elementary Procedures are defined separately and are intended to be used to build up complete sequences in a flexible manner. If the independence between some EPs is restricted, it is described under the relevant EP description. Unless otherwise stated by the restrictions, the EPs may be invoked independently of each other as standalone procedures, which can be active in parallel. The usage of several F1AP EPs together is specified in stage 2 specifications (e.g., TS 38.470 [2]).

An EP consists of an initiating message and possibly a response message. Two kinds of EPs are used:

- **Class 1:** Elementary Procedures with response (success and/or failure).
- **Class 2:** Elementary Procedures without response.

For Class 1 EPs, the types of responses can be as follows:

Successful:

- A signalling message explicitly indicates that the elementary procedure successfully completed with the receipt of the response.

Unsuccessful:

- A signalling message explicitly indicates that the EP failed.
- On time supervision expiry (i.e., absence of expected response).

Successful and Unsuccessful:

- One signalling message reports both successful and unsuccessful outcome for the different included requests. The response message used is the one defined for successful outcome.

Class 2 EPs are considered always successful.

BH RLC channel: as defined in TS 38.300 [6].

CG-SDT-CS-RNTI: as defined in TS 38.300 [6].

Child UE: as defined in TS 38.300 [6].

Complete candidate configuration: one type of a candidate configuration as defined in TS 38.331 [8].

Conditional handover: as defined in TS 38.300 [6].

Conditional PSCell Addition: as defined in TS 37.340 [7].

Conditional PSCell Change: as defined in TS 37.340 [7].

DAPS Handover: as defined in TS 38.300 [6].

EN-DC operation: Used in this specification when the F1AP is applied for gNB-CU and gNB-DU in E-UTRAN.

F1-terminating IAB-donor: as defined in TS 38.401 [4].

First U2N Relay UE: as defined in TS 38.300 [6].

gNB: as defined in TS 38.300 [6].

gNB-CU: as defined in TS 38.401 [4].

gNB-CU UE F1AP ID: as defined in TS 38.401 [4].

gNB-DU: as defined in TS 38.401 [4].

gNB-DU UE F1AP ID: as defined in TS 38.401 [4].

en-gNB: as defined in TS 37.340 [7].

IAB-DU: as defined in TS 38.300 [6].

IAB-MT: as defined in TS 38.300 [6].

IAB-donor: as defined in TS 38.300 [6].

IAB-donor-CU: as defined in TS 38.401 [4].

IAB-donor-DU: as defined in TS 38.401 [4].

IAB-node: as defined in TS 38.300 [6].

Intermediate U2N Relay UE: as defined in TS 38.300 [6].

Last U2N Relay UE: as defined in TS 38.300 [6].

MBS-associated signalling: When F1AP messages associated to one MBS session uses the MBS-associated logical F1-connection for association of the message to the MBS session in gNB-DU and gNB-CU.

MBS-associated logical F1-connection: The MBS-associated logical F1-connection uses the identities *GNB-CU MBS F1AP ID* and *GNB-DU MBS F1AP ID* according to the definition in TS 38.401 [4]. For a received MBS-associated F1AP message the gNB-CU identifies the associated MBS session based on the *GNB-CU MBS F1AP ID* IE and the gNB-DU identifies the associated MBS session based on the *GNB-DU MBS F1AP ID* IE.

MBS Session context in a gNB-DU: as defined in TS 38.401 [4].

MBS session resource: as defined in TS 38.401 [4].

Mobile IAB-DU: as defined in TS 38.300 [6].

Mobile IAB-MT: as defined in TS 38.300 [6].

Mobile IAB-node: as defined in TS 38.300 [6].

MP Relay UE: as defined in TS 38.300 [6].

MP Remote UE: as defined in TS 38.300 [6].

Multi-path: as defined in TS 38.300 [6].

Multicast F1-U Context: as defined in TS 38.401 [4].

Other SI: as defined in TS 38.300 [6].

Parent UE: as defined in TS 38.300 [6].

PC5 Relay RLC channel: as defined in TS 38.300 [6].

Public network integrated NPN: as defined in TS 23.501 [21].

RRC-terminating IAB-donor: as defined in TS 38.401 [4].

SRAP: Sidelink relay adaptation protocol, as defined in TS 38.300 [6].

Stand-alone Non-Public Network: as defined in TS 23.501 [21].

UE-associated logical F1-connection: The UE-associated logical F1-connection uses the identities *GNB-CU UE F1AP ID* and *GNB-DU UE F1AP ID* according to the definition in TS 38.401 [4]. For a received UE associated F1AP message the gNB-CU identifies the associated UE based on the *GNB-CU UE F1AP ID* IE and the gNB-DU identifies the associated UE based on the *GNB-DU UE F1AP ID* IE. The UE-associated logical F1-connection may exist before the F1 UE context is setup in gNB-DU.

UE-associated signalling: When F1AP messages associated to one UE uses the UE-associated logical F1-connection for association of the message to the UE in gNB-DU and gNB-CU.

U2N Relay UE: as defined in TS 38.300 [6].

U2N Remote UE: as defined in TS 38.300 [6].

U2U Relay UE: as defined in TS 38.300 [6].

U2U Remote UE: as defined in TS 38.300 [6].

Uu Relay RLC channel: as defined in TS 38.300 [6].

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply.
An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

5GC	5G Core Network
5QI	5G QoS Identifier
A2X	Aircraft-to-Everything
AMF	Access and Mobility Management Function
ARP	Antenna Reference Point
ARPI	Additional RRM Policy Index
BH	Backhaul
CAG	Closed Access Group
CG	Cell Group
CG-SDT	Configured Grant-Small Data Transmission
CGI	Cell Global Identifier
CHO	Conditional Handover
CLI	Cross Link Interference
CN	Core Network
CP	Control Plane
CPA	Conditional PSCell Addition
CPAC	Conditional PSCell Addition or Change
CPC	Conditional PSCell Change
DAPS	Dual Active Protocol Stack
DL	Downlink
DL-PRS	Downlink Positioning Reference Signal
EN-DC	E-UTRA-NR Dual Connectivity
EPC	Evolved Packet Core
eRedCap	Enhanced Reduced Capability
FSA ID	MBS Frequency Selection Area (FSA) ID
GPSI	Generic Public Subscription Identifier
IAB	Integrated Access and Backhaul
IMEISV	International Mobile station Equipment Identity and Software Version number
LMF	Location Management Function
LP-SS	Low Power Synchronization Signal
LP-WUSPS	Low Power Wake UP Signal with Paging Subgrouping
LTM	L1/L2 Triggered Mobility
MBS	Multicast/Broadcast Service
MP	Multi-path
MT-SDT	Mobile Terminated Small Data Transmission

N3C	Non-3GPP Connection
NID	Network Identifier
NPN	Non-Public Network
NSAG	Network Slice AS Group
NSSAI	Network Slice Selection Assistance Information
OD-SIB1	On-demand SIB1
OOK	On-Off Keying
PDC	Propagation Delay Compensation
PEIPS	Paging Early Indication with Paging Subgrouping
PNI-NPN	Public Network Integrated NPN
posSIB	Positioning SIB
PSI	PDU Set Importance
PTM	Point to Multipoint
PTP	Point to Point
QMC	QoE Measurement Collection
QoE	Quality of Experience
RANAC	RAN Area Code
RedCap	Reduced Capability
RIM	Remote Interference Management
RIM-RS	RIM Reference Signal
RRC	Radio Resource Control
RSPP	Ranging/Sidelink Positioning Protocol
RSRP	Reference Signal Received Power
S-CPAC	Subsequent Conditional PSCell Addition or Change
S-NSSAI	Single Network Slice Selection Assistance Information
SDT	Small Data Transmission
SNPN	Stand-alone Non-Public Network
SUL	Supplementary Uplink
TAC	Tracking Area Code
TAG	Timing Advance Group
TAI	Tracking Area Identity
TEG	Timing Error Group
TRP	Transmission-Reception Point
TSS	Timing Synchronisation Status
U2N	UE-to-Network
U2U	UE-to-UE
UL-AoA	Uplink Angle of Arrival
UL-RSCP	UL Reference Signal Carrier Phase
UL-RTOA	Uplink Relative Time of Arrival
UL-SRS	Uplink Sounding Reference Signal
UL SRS-TDCT	UL SRS Time Domain Channel Timing
UL SRS-TDCP	UL SRS Time Domain Channel Power
V2X	Vehicle-to-Everything
Z-AoA	Zenith Angles of Arrival

4 General

4.1 Procedure specification principles

The principle for specifying the procedure logic is to specify the functional behaviour of the terminating node exactly and completely. Any rule that specifies the behaviour of the originating node shall be possible to be verified with information that is visible within the system.

The following specification principles have been applied for the procedure text in clause 8:

- The procedure text discriminates between:

  1) Functionality which "shall" be executed.

The procedure text indicates that the receiving node "shall" perform a certain function Y under a certain condition. If the receiving node supports procedure X but cannot perform functionality Y requested in the REQUEST message of a Class 1 EP, the receiving node shall respond with the message used to report unsuccessful outcome for this procedure, containing an appropriate cause value.

2) Functionality which "shall, if supported" be executed.

The procedure text indicates that the receiving node "shall, if supported," perform a certain function Y under a certain condition. If the receiving node supports procedure X, but does not support functionality Y, the receiving node shall proceed with the execution of the EP, possibly informing the requesting node about the not supported functionality.

- Any required inclusion of an optional IE in a response message is explicitly indicated in the procedure text. If the procedure text does not explicitly indicate that an optional IE shall be included in a response message, the optional IE shall not be included. For requirements on including *Criticality Diagnostics* IE, see clause 10.

4.2 Forwards and backwards compatibility

The forwards and backwards compatibility of the protocol is assured by mechanism where all current and future messages, and IEs or groups of related IEs, include ID and criticality fields that are coded in a standard format that will not be changed in the future. These parts can always be decoded regardless of the standard version.

4.3 Specification notations

For the purposes of the present document, the following notations apply:

Procedure When referring to an elementary procedure in the specification the Procedure Name is written with the first letters in each word in upper case characters followed by the word "procedure", e.g. Handover Preparation procedure.

Message When referring to a message in the specification the MESSAGE NAME is written with all letters in upper case characters followed by the word "message", e.g. HANDOVER REQUEST message.

IE When referring to an information element (IE) in the specification the *Information Element Name* is written with the first letters in each word in upper case characters and all letters in Italic font followed by the abbreviation "IE", e.g. *E-RAB ID* IE.

Value of an IE When referring to the value of an information element (IE) in the specification the "Value" is written as it is specified in the specification enclosed by quotation marks, e.g. "Value".

5 F1AP services

F1AP provides the signalling service between gNB-DU and the gNB-CU that is required to fulfil the F1AP functions described in clause 7. F1AP services are divided into the following groups:

Non UE-associated services: They are related to the whole F1 interface instance between the gNB-DU and gNB-CU utilising a non UE-associated signalling connection.

UE-associated services: They are related to one UE. F1AP functions that provide these services are associated with a UE-associated signalling connection that is maintained for the UE in question.

MBS-associated services: They are related to one MBS service. F1AP functions that provide these services are associated with a MBS-associated signalling connection that is maintained for the MBS service in question.

Unless explicitly indicated in the procedure specification, at any instance in time one protocol endpoint shall have a maximum of one ongoing F1AP procedure related to a certain UE.

Unless explicitly indicated in the procedure specification, at any instance in time one protocol endpoint shall have a maximum of one ongoing F1AP procedure related to a certain MBS session.

All considerations of gNB-DU in this specification also apply to the IAB-DU and IAB-donor-DU, unless stated otherwise. All considerations of gNB-CU in this specification apply to the IAB-donor-CU as well, unless stated otherwise.

6 Services expected from signalling transport

The signalling connection shall provide in sequence delivery of F1AP messages. F1AP shall be notified if the signalling connection breaks.

7 Functions of F1AP

The functions of F1AP are described in TS 38.470 [2].

8 F1AP procedures

8.1 List of F1AP Elementary procedures

In the following tables, all EPs are divided into Class 1 and Class 2 EPs (see subclause 3.1 for explanation of the different classes):

Table 1: Class 1 procedures

Elementary Procedure	Initiating Message	Successful Outcome	Unsuccessful Outcome
		Response message	Response message
Reset	RESET	RESET ACKNOWLEDGE	
F1 Setup	F1 SETUP REQUEST	F1 SETUP RESPONSE	F1 SETUP FAILURE
gNB-DU Configuration Update	GNB-DU CONFIGURATION UPDATE	GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE	GNB-DU CONFIGURATION UPDATE FAILURE
gNB-CU Configuration Update	GNB-CU CONFIGURATION UPDATE	GNB-CU CONFIGURATION UPDATE ACKNOWLEDGE	GNB-CU CONFIGURATION UPDATE FAILURE
UE Context Setup	UE CONTEXT SETUP REQUEST	UE CONTEXT SETUP RESPONSE	UE CONTEXT SETUP FAILURE
UE Context Release (gNB-CU initiated)	UE CONTEXT RELEASE COMMAND	UE CONTEXT RELEASE COMPLETE	
UE Context Modification (gNB-CU initiated)	UE CONTEXT MODIFICATION REQUEST	UE CONTEXT MODIFICATION RESPONSE	UE CONTEXT MODIFICATION FAILURE
UE Context Modification Required (gNB-DU initiated)	UE CONTEXT MODIFICATION REQUIRED	UE CONTEXT MODIFICATION CONFIRM	UE CONTEXT MODIFICATION REFUSE
Write-Replace Warning	WRITE-REPLACE WARNING REQUEST	WRITE-REPLACE WARNING RESPONSE	
PWS Cancel	PWS CANCEL REQUEST	PWS CANCEL RESPONSE	
gNB-DU Resource Coordination	GNB-DU RESOURCE COORDINATION REQUEST	GNB-DU RESOURCE COORDINATION RESPONSE	
F1 Removal	F1 REMOVAL REQUEST	F1 REMOVAL RESPONSE	F1 REMOVAL FAILURE
BAP Mapping Configuration	BAP MAPPING CONFIGURATION	BAP MAPPING CONFIGURATION	BAP MAPPING CONFIGURATION FAILURE

		ACKNOWLEDGE	
GNB-DU Resource Configuration	GNB-DU RESOURCE CONFIGURATION	GNB-DU RESOURCE CONFIGURATION ACKNOWLEDGE	GNB-DU RESOURCE CONFIGURATION FAILURE
IAB TNL Address Allocation	IAB TNL ADDRESS REQUEST	IAB TNL ADDRESS RESPONSE	IAB TNL ADDRESS FAILURE
IAB UP Configuration Update	IAB UP CONFIGURATION UPDATE REQUEST	IAB UP CONFIGURATION UPDATE RESPONSE	IAB UP CONFIGURATION UPDATE FAILURE
Resource Status Reporting Initiation	RESOURCE STATUS REQUEST	RESOURCE STATUS RESPONSE	RESOURCE STATUS FAILURE
Positioning Measurement	POSITIONING MEASUREMENT REQUEST	POSITIONING MEASUREMENT RESPONSE	POSITIONING MEASUREMENT FAILURE
Positioning Information Exchange	POSITIONING INFORMATION REQUEST	POSITIONING INFORMATION RESPONSE	POSITIONING INFORMATION FAILURE
TRP Information Exchange	TRP INFORMATION REQUEST	TRP INFORMATION RESPONSE	TRP INFORMATION FAILURE
Positioning Activation	POSITIONING ACTIVATION REQUEST	POSITIONING ACTIVATION RESPONSE	POSITIONING ACTIVATION FAILURE
E-CID Measurement Initiation	E-CID MEASUREMENT INITIATION REQUEST	E-CID MEASUREMENT INITIATION RESPONSE	E-CID MEASUREMENT INITIATION FAILURE
Broadcast Context Setup	BROADCAST CONTEXT SETUP REQUEST	BROADCAST CONTEXT SETUP RESPONSE	BROADCAST CONTEXT SETUP FAILURE
Broadcast Context Release	BROADCAST CONTEXT RELEASE COMMAND	BROADCAST CONTEXT RELEASE COMPLETE	
Broadcast Context Modification	BROADCAST CONTEXT MODIFICATION REQUEST	BROADCAST CONTEXT MODIFICATION RESPONSE	BROADCAST CONTEXT MODIFICATION FAILURE
Multicast Context Setup	MULTICAST CONTEXT SETUP REQUEST	MULTICAST CONTEXT SETUP RESPONSE	MULTICAST CONTEXT SETUP FAILURE
Multicast Context Release	MULTICAST CONTEXT RELEASE COMMAND	MULTICAST CONTEXT RELEASE COMPLETE	
Multicast Context Modification	MULTICAST CONTEXT MODIFICATION REQUEST	MULTICAST CONTEXT MODIFICATION RESPONSE	MULTICAST CONTEXT MODIFICATION FAILURE
Multicast Distribution Setup	MULTICAST DISTRIBUTION SETUP REQUEST	MULTICAST DISTRIBUTION SETUP RESPONSE	MULTICAST DISTRIBUTION SETUP FAILURE
Multicast Distribution Release	MULTICAST DISTRIBUTION RELEASE COMMAND	MULTICAST DISTRIBUTION RELEASE COMPLETE	
PDC Measurement Initiation	PDC MEASUREMENT INITIATION REQUEST	PDC MEASUREMENT INITIATION RESPONSE	PDC MEASUREMENT INITIATION FAILURE
PRS Configuration Exchange	PRS CONFIGURATION REQUEST	PRS CONFIGURATION RESPONSE	PRS CONFIGURATION FAILURE
Measurement Preconfiguration	MEASUREMENT PRECONFIGURATION REQUIRED	MEASUREMENT PRECONFIGURATION CONFIRM	MEASUREMENT PRECONFIGURATION REFUSE
Timing Synchronisation Status	TIMING SYNCHRONISATION STATUS REQUEST	TIMING SYNCHRONISATION STATUS RESPONSE	TIMING SYNCHRONISATION STATUS FAILURE
Multicast Context Notification	MULTICAST CONTEXT NOTIFICATION INDICATION	MULTICAST CONTEXT NOTIFICATION CONFIRM	MULTICAST CONTEXT NOTIFICATION REFUSE
Multicast Common Configuration	MULTICAST COMMON CONFIGURATION REQUEST	MULTICAST COMMON CONFIGURATION RESPONSE	MULTICAST COMMON CONFIGURATION REFUSE

DU-CU CSI-RS Coordination	DU-CU CSI-RS COORDINATION REQUEST	DU-CU CSI-RS COORDINATION RESPONSE	
CU-DU CSI-RS Coordination	CU-DU CSI-RS COORDINATION REQUEST	CU-DU CSI-RS COORDINATION RESPONSE	

Table 2: Class 2 procedures

Elementary Procedure	Message
Error Indication	ERROR INDICATION
UE Context Release Request (gNB-DU initiated)	UE CONTEXT RELEASE REQUEST
Initial UL RRC Message Transfer	INITIAL UL RRC MESSAGE TRANSFER
DL RRC Message Transfer	DL RRC MESSAGE TRANSFER
UL RRC Message Transfer	UL RRC MESSAGE TRANSFER
UE Inactivity Notification	UE INACTIVITY NOTIFICATION
System Information Delivery	SYSTEM INFORMATION DELIVERY COMMAND
Paging	PAGING
Notify	NOTIFY
PWS Restart Indication	PWS RESTART INDICATION
PWS Failure Indication	PWS FAILURE INDICATION
gNB-DU Status Indication	GNB-DU STATUS INDICATION
RRC Delivery Report	RRC DELIVERY REPORT
Network Access Rate Reduction	NETWORK ACCESS RATE REDUCTION
Trace Start	TRACE START
Deactivate Trace	DEACTIVATE TRACE
DU-CU Radio Information Transfer	DU-CU RADIO INFORMATION TRANSFER
CU-DU Radio Information Transfer	CU-DU RADIO INFORMATION TRANSFER
Resource Status Reporting	RESOURCE STATUS UPDATE
Access And Mobility Indication	ACCESS AND MOBILITY INDICATION
Reference Time Information Reporting Control	REFERENCE TIME INFORMATION REPORTING CONTROL
Reference Time Information Report	REFERENCE TIME INFORMATION REPORT
Access Success	ACCESS SUCCESS
Cell Traffic Trace	CELL TRAFFIC TRACE
Positioning Assistance Information Control	POSITIONING ASSISTANCE INFORMATION CONTROL
Positioning Assistance Information Feedback	POSITIONING ASSISTANCE INFORMATION FEEDBACK
Positioning Measurement Report	POSITIONING MEASUREMENT REPORT
Positioning Measurement Abort	POSITIONING MEASUREMENT ABORT
Positioning Measurement Failure Indication	POSITIONING MEASUREMENT FAILURE INDICATION
Positioning Measurement Update	POSITIONING MEASUREMENT UPDATE
Positioning Deactivation	POSITIONING DEACTIVATION
E-CID Measurement Failure Indication	E-CID MEASUREMENT FAILURE INDICATION
E-CID Measurement Report	E-CID MEASUREMENT REPORT
E-CID Measurement Termination	E-CID MEASUREMENT TERMINATION COMMAND
Positioning Information Update	POSITIONING INFORMATION UPDATE
Multicast Group Paging	MULTICAST GROUP PAGING

Elementary Procedure	Message
Broadcast Context Release Request	BROADCAST CONTEXT RELEASE REQUEST
Multicast Context Release Request	MULTICAST CONTEXT RELEASE REQUEST
PDC Measurement Report	PDC MEASUREMENT REPORT
PDC Measurement Failure Indication	PDC MEASUREMENT FAILURE INDICATION
PDC Measurement Termination	PDC MEASUREMENT TERMINATION COMMAND
Measurement Activation	MEASUREMENT ACTIVATION
QoE Information Transfer	QOE INFORMATION TRANSFER
Positioning System Information Delivery	POSITIONING SYSTEM INFORMATION DELIVERY COMMAND
DU-CU Cell Switch Notification	DU-CU CELL SWITCH NOTIFICATION
CU-DU Cell Switch Notification	CU-DU CELL SWITCH NOTIFICATION
DU-CU TA Information Transfer	DU-CU TA INFORMATION TRANSFER
CU-DU TA Information Transfer	CU-DU TA INFORMATION TRANSFER
QoE Information Transfer Control	QOE INFORMATION TRANSFER CONTROL
RACH Indication	RACH INDICATION
Timing Synchronisation Status Report	TIMING SYNCHRONISATION STATUS REPORT
Mobile IAB F1 Setup Triggering	MIAB F1 SETUP TRIGGERING
Mobile IAB F1 Setup Outcome Notification	MIAB F1 SETUP OUTCOME NOTIFICATION
Broadcast Transport Resource Request	BROADCAST TRANSPORT RESOURCE REQUEST
SRS Information Reservation Notification	SRS INFORMATION RESERVATION NOTIFICATION
DU-CU Access And Mobility Indication	DU-CU ACCESS AND MOBILITY INDICATION
CU-DU Mobility Initiation	CU-DU MOBILITY INITIATION REQUEST
CLI Indication	CLI INDICATION

8.2 Interface Management procedures

8.2.1 Reset

8.2.1.1 General

The purpose of the Reset procedure is to initialise or re-initialise the F1AP UE-related contexts, in the event of a failure in the gNB-CU or gNB-DU. This procedure does not affect the application level configuration data exchanged during, e.g., the F1 Setup procedure.

The procedure uses non-UE associated signalling.

8.2.1.2 Successful Operation

8.2.1.2.1 Reset Procedure Initiated from the gNB-CU

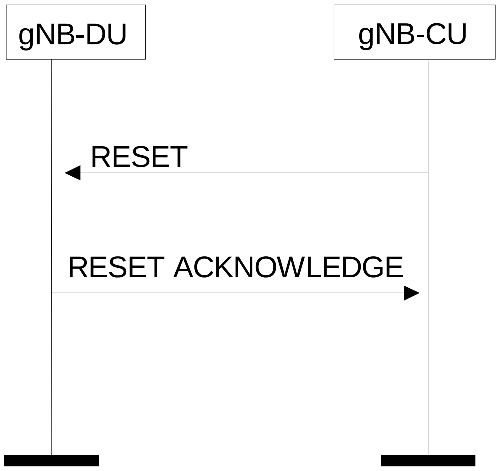

Figure 8.2.1.2.1-1: Reset procedure initiated from the gNB-CU. Successful operation

In the event of a failure at the gNB-CU, which has resulted in the loss of some or all transaction reference information, a RESET message shall be sent to the gNB-DU.

At reception of the RESET message the gNB-DU shall release all allocated resources on F1 and radio resources related to the UE association(s) indicated explicitly or implicitly in the RESET message and remove the indicated UE contexts including F1AP ID.

After the gNB-DU has released all assigned F1 resources and the UE F1AP IDs for all indicated UE associations which can be used for new UE-associated logical F1-connections over the F1 interface, the gNB-DU shall respond with the RESET ACKNOWLEDGE message. The gNB-DU does not need to wait for the release of radio resources to be completed before returning the RESET ACKNOWLEDGE message.

If the RESET message contains the *UE-associated logical F1-connection list* IE, then:

- The gNB-DU shall use the *gNB-CU UE F1AP ID* IE and/or the *gNB-DU UE F1AP ID* IE to explicitly identify the UE association(s) to be reset.
- The gNB-DU shall include in the RESET ACKNOWLEDGE message, for each UE association to be reset, the *UE-associated logical F1-connection Item* IE in the *UE-associated logical F1-connection list* IE. The *UE-associated logical F1-connection Item* IEs shall be in the same order as received in the RESET message and shall include also unknown UE-associated logical F1-connections. Empty *UE-associated logical F1-connection Item* IEs, received in the RESET message, may be omitted in the RESET ACKNOWLEDGE message.
- If the *gNB-CU UE F1AP ID* IE is included in the *UE-associated logical F1-connection Item* IE for a UE association, the gNB-DU shall include the *gNB-CU UE F1AP ID* IE in the corresponding *UE-associated logical F1-connection Item* IE in the RESET ACKNOWLEDGE message.
- If the *gNB-DU UE F1AP ID* IE is included in the *UE-associated logical F1-connection Item* IE for a UE association, the gNB-DU shall include the *gNB-DU UE F1AP ID* IE in the corresponding *UE-associated logical F1-connection Item* IE in the RESET ACKNOWLEDGE message.

Interactions with other procedures:

If the RESET message is received, any other ongoing procedure (except for another Reset procedure) on the same F1 interface related to a UE association, indicated explicitly or implicitly in the RESET message, shall be aborted.

8.2.1.2.2 Reset Procedure Initiated from the gNB-DU

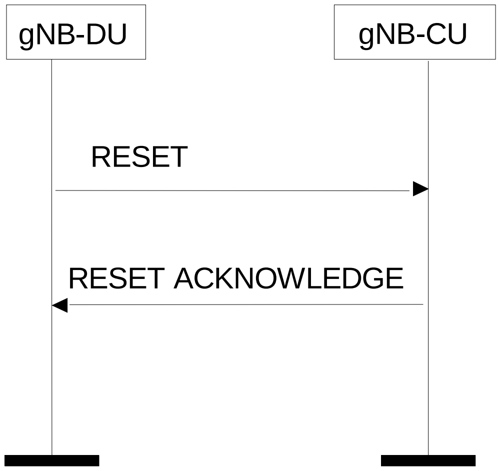

Figure 8.2.1.2.2-1: Reset procedure initiated from the gNB-DU. Successful operation

In the event of a failure at the gNB-DU, which has resulted in the loss of some or all transaction reference information, a RESET message shall be sent to the gNB-CU.

At reception of the RESET message the gNB-CU shall release all allocated resources on F1 related to the UE association(s) indicated explicitly or implicitly in the RESET message and remove the F1AP ID for the indicated UE associations.

After the gNB-CU has released all assigned F1 resources and the UE F1AP IDs for all indicated UE associations which can be used for new UE-associated logical F1-connections over the F1 interface, the gNB-CU shall respond with the RESET ACKNOWLEDGE message.

If the RESET message contains the *UE-associated logical F1-connection list* IE, then:

- The gNB-CU shall use the *gNB-CU UE F1AP ID* IE and/or the *gNB-DU UE F1AP ID* IE to explicitly identify the UE association(s) to be reset.
- The gNB-CU shall in the RESET ACKNOWLEDGE message include, for each UE association to be reset, the *UE-associated logical F1-connection Item* IE in the *UE-associated logical F1-connection list* IE. The *UE-associated logical F1-connection Item* IEs shall be in the same order as received in the RESET message and shall include also unknown UE-associated logical F1-connections. Empty *UE-associated logical F1-connection Item* IEs, received in the RESET message, may be omitted in the RESET ACKNOWLEDGE message.
- If the *gNB-CU UE F1AP ID* IE is included in the *UE-associated logical F1-connection Item* IE for a UE association, the gNB-CU shall include the *gNB-CU UE F1AP ID* IE in the corresponding *UE-associated logical F1-connection Item* IE in the RESET ACKNOWLEDGE message.
- If the *gNB-DU UE F1AP ID* IE is included in a *UE-associated logical F1-connection Item* IE for a UE association, the gNB-CU shall include the *gNB-DU UE F1AP ID* IE in the corresponding *UE-associated logical F1-connection Item* IE in the RESET ACKNOWLEDGE message.

Interactions with other procedures:

If the RESET message is received, any other ongoing procedure (except for another Reset procedure) on the same F1 interface related to a UE association, indicated explicitly or implicitly in the RESET message, shall be aborted.

8.2.1.3 Abnormal Conditions

Not applicable.

8.2.2 Error Indication

8.2.2.1 General

The Error Indication procedure is initiated by a node in order to report detected errors in one incoming message, provided they cannot be reported by an appropriate failure message.

If the error situation arises due to reception of a message utilising UE associated signalling, then the Error Indication procedure uses UE associated signalling. Otherwise the procedure uses non-UE associated signalling.

8.2.2.2 Successful Operation

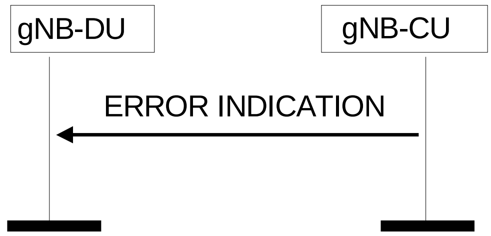

Figure 8.2.2.2-1: Error Indication procedure, gNB-CU originated. Successful operation

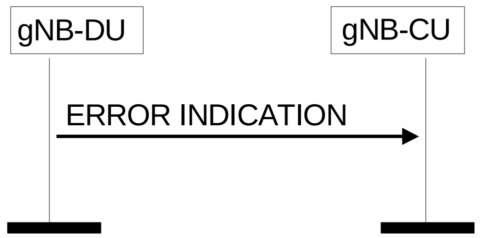

Figure 8.2.2.2-2: Error Indication procedure, gNB-DU originated. Successful operation

When the conditions defined in clause 10 are fulfilled, the Error Indication procedure is initiated by an ERROR INDICATION message sent from the receiving node.

The ERROR INDICATION message shall contain at least either the *Cause* IE or the *Criticality Diagnostics* IE. In case the Error Indication procedure is triggered by utilising UE associated signalling the *gNB-CU UE F1AP ID* IE and *gNB-DU UE F1AP ID* IE shall be included in the ERROR INDICATION message. If one or both of the *gNB-CU UE F1AP ID* IE and the *gNB-DU UE F1AP ID* IE are not correct, the cause shall be set to appropriate value, e.g., "Unknown or already allocated gNB-CU UE F1AP ID", "Unknown or already allocated gNB-DU UE F1AP ID" or "Unknown or inconsistent pair of UE F1AP ID".

8.2.2.3 Abnormal Conditions

Not applicable.

8.2.3 F1 Setup

8.2.3.1 General

The purpose of the F1 Setup procedure is to exchange application level data needed for the gNB-DU and the gNB-CU to correctly interoperate on the F1 interface. This procedure shall be the first F1AP procedure triggered for the F1-C interface instance after a TNL association has become operational.

NOTE: If F1-C signalling transport is shared among multiple F1-C interface instances, one F1 Setup procedure is issued per F1-C interface instance to be setup, i.e. several F1 Setup procedures may be issued via the same TNL association after that TNL association has become operational.

NOTE: Exchange of application level configuration data also applies between the gNB-DU and the gNB-CU in case the DU does not broadcast system information other than for radio frame timing and SFN, as specified in the TS 37.340 [7]. How to use this information when this option is used is not explicitly specified.

The procedure uses non-UE associated signalling.

This procedure erases any existing application level configuration data in the two nodes and replaces it by the one received. This procedure also re-initialises the F1AP UE-related contexts (if any) and erases all related signalling connections in the two nodes like a Reset procedure would do.

8.2.3.2 Successful Operation

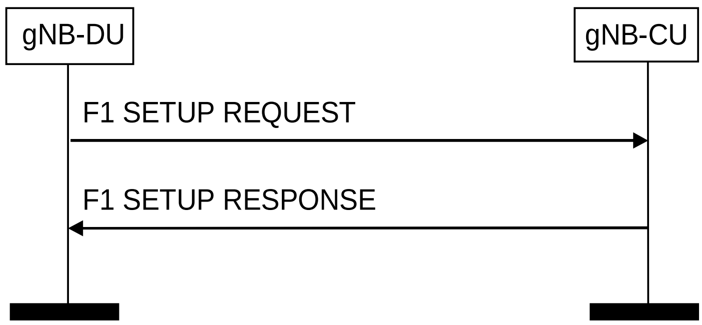

Figure 8.2.3.2-1: F1 Setup procedure: Successful Operation

The gNB-DU initiates the procedure by sending a F1 SETUP REQUEST message including the appropriate data to the gNB-CU. The gNB-CU responds with a F1 SETUP RESPONSE message including the appropriate data.

The exchanged data shall be stored in respective node and used as long as there is an operational TNL association. When this procedure is finished, the F1 interface is operational and other F1 messages may be exchanged.

If the F1 SETUP REQUEST message contains the *gNB-DU Name* IE, the gNB-CU may use this IE as a human readable name of the gNB-DU. If the F1 SETUP REQUEST message contains the *Extended gNB-DU Name* IE, the gNB-CU may use this IE as a human readable name of the gNB-DU and shall ignore the *gNB-DU Name* IE if included.

If the F1 SETUP RESPONSE message contains the *gNB-CU Name* IE, the gNB-DU may use this IE as a human readable name of the gNB-CU. If the F1 SETUP RESPONSE message contains the *Extended gNB-CU Name* IE, the gNB-DU may use this IE as a human readable name of the gNB-CU and shall ignore the *gNB-CU Name* IE if included.

If the F1 SETUP REQUEST message contains the *gNB-DU Served Cells List* IE, the gNB-CU shall take into account as specified in TS 38.401 [4].

For NG-RAN, the gNB-DU shall include the *gNB-DU System Information* IE and the *TAI Slice Support List* IE in the F1 SETUP REQUEST message.

The gNB-CU may include the *Cells to be Activated List* IE in the F1 SETUP RESPONSE message. The *Cells to be Activated List* IE includes a list of cells that the gNB-CU requests the gNB-DU to activate. The gNB-DU shall activate the cells included in the *Cells to be Activated List* IE and reconfigure the physical cell identity for cells for which the *NR PCI* IE is included.

If *Cells to be Activated List Item* IE is included in the F1 SETUP RESPONSE message, and the information for the cell indicated by the *NR CGI* IE includes the *IAB Info IAB-donor-CU* IE, the gNB-DU shall, if supported, apply the *IAB STC Info* IE therein to the indicated cell.

For NG-RAN, the gNB-CU shall include the *gNB-CU System Information* IE in the F1 SETUP RESPONSE message.

For NG-RAN, the gNB-DU may include the *RAN Area Code* IE in the F1 SETUP REQUEST message. The gNB-CU may use it according to TS 38.300 [6].

For NG-RAN, the gNB-DU may include *Supported MBS FSA ID List* IE in the *Served Cell Information* IE in the F1 SETUP REQUEST message. The gNB-CU may use it according to TS 38.300 [6].

For NG-RAN, the gNB-CU may include *Available PLMN List* IE, and optionally also *Extended Available PLMN List* IE in the F1 SETUP RESPONSE message, if the available PLMN(s) are different from what gNB-DU has provided in F1 SETUP REQUEST message, gNB-DU shall take this into account and only broadcast the PLMN(s) included in the received Available PLMN list(s).

For NG-RAN, the gNB-CU may include *Available SNPN ID List* IE in the F1 SETUP RESPONSE message. If the available SNPN(s) are different from what gNB-DU has provided in F1 SETUP REQUEST message, gNB-DU shall take this into account and only broadcast the SNPN(s) included in the received Available SNPN ID list.

The *Latest RRC Version Enhanced* IE shall be included in the F1 SETUP REQUEST message and in the F1 SETUP RESPONSE message.

If in F1 SETUP REQUEST message, the *Cell Direction* IE is present, the gNB-CU should use it to understand whether the cell is for UL or DL only. If in F1 SETUP REQUEST message, the *Cell Direction* IE is omitted in the *Served Cell Information* IE it shall be interpreted as that the Cell Direction is Bi-directional.

If the *Intended TDD DL-UL Configuration IE* is present in the F1 SETUP REQUEST message, the receiving gNB-CU shall use the received information for Cross Link Interference management and/or NR-DC power coordination. The gNB-CU may merge the Intended TDD DL-UL Configuration information received from two or more gNB-DUs. The gNB-CU shall consider the received *Intended TDD DL-UL Configuration* IE content valid until reception of an update of the IE for the same cell(s).

If the *Aggressor gNB Set ID* IE is included in the *Served Cell Information* IE in the F1 SETUP REQUEST message, the gNB-CU shall, if supported, take it into account.

If the *Victim gNB Set ID* IE is included in the *Served Cell Information* IE in the F1 SETUP REQUEST message, the gNB-CU shall, if supported, take it into account.

If the F1 SETUP REQUEST message contains the *Transport Layer Address Info* IE, the gNB-CU shall, if supported, take into account for IPSec tunnel establishment.

If the *SFN Offset* IE is contained in the *Served Cell Information* IE in the F1 SETUP REQUEST message, the gNB-CU shall, if supported, use this information to deduce the SFN0 offset of the reported cell.

If the F1 SETUP RESPONSE message contains the *Transport Layer Address Info* IE, the gNB-DU shall, if supported, take into account for IPSec tunnel establishment.

If the F1 SETUP RESPONSE message contains the *Uplink BH Non-UP Traffic Mapping* IE, the gNB-DU shall, if supported, consider the information therein for mapping of non-UP uplink traffic.

If the *BAP Address* IE is included in the F1 SETUP REQUEST message, without an accompanying *RRC Terminating IAB-Donor gNB-ID* IE, the receiving gNB-CU shall, if supported, consider the information therein for discovering the co-location of an IAB-DU and an IAB-MT.

If the F1 SETUP REQUEST message is received from an IAB-donor-DU, the gNB-CU shall, if supported, include the *BAP Address* IE in the F1 SETUP RESPONSE message.

NOTE: How to identify the IAB-donor-DU is up to gNB-CU implementation.

If the F1 SETUP RESPONSE message contains the *BAP Address* IE, the gNB-DU shall, if supported, store the received BAP address and use it as specified in TS 38.340 [30].

If the *NR PRACH Configuration List* IE is included in the *Served Cell Information* IE contained in the F1 SETUP REQUEST message, the gNB-CU may store the information, and forward it to other RAN nodes for RACH optimisation. If the *L139 Info* IE included in the *NR PRACH Configuration List* IE is present, it shall contain the *Root Sequence Index* IE.

If the *RedCap Broadcast Information* IE is included in the *Served Cell Information* IE in the F1 SETUP REQUEST message, the gNB-CU may store and use this information to determine a suitable target cell in case of mobility and paging involving RedCap UEs.

If the *eRedCap Broadcast Information* IE is included in the *Served Cell Information* IE in the F1 SETUP REQUEST message, the gNB-CU may store and use this information to determine a suitable target cell in case of mobility and paging involving eRedCap UEs.

If the *TAI NSAG Support List* IE is included in the *Served Cell Information* IE in the F1 SETUP REQUEST message, the gNB-CU shall, if supported, use this information as specified in TS 23.501 [21].

If both the *RRC Terminating IAB-Donor gNB-ID* IE and the *BAP Address* IE are included in the F1 SETUP REQUEST message, the gNB-CU shall, if supported, consider that the BAP address indicated by the *BAP Address* IE is assigned by the gNB-CU of the RRC-terminating IAB-donor indicated by the *RRC Terminating IAB-Donor gNB-ID* IE, and use this BAP address and gNB-ID for the subsequent IAB Transport Migration Management procedure towards the RRC-terminating IAB-donor of the mobile IAB-node, as specified in TS 38.423 [28].

If the F1 SETUP REQUEST message contains the *Mobile IAB-MT User Location Information* IE, the gNB-CU shall, if supported, take it into account when reporting UE location information to the AMF for a UE served by the mobile IAB-node.

If the *XR Broadcast Information* IE is included in the *Served Cell Information* IE in the F1 SETUP REQUEST message, the gNB-CU shall, if supported, consider the indicated cell does not allow 2Rx XR UEs access in case of mobility and paging involving XR UEs.

If the *NCGI to be Updated List* IE is included in the F1 SETUP RESPONSE message, the gNB-DU shall, if supported, change the NCGI of the cell indicated by the *Old NCGI* IE to the NCGI indicated by the *New NCGI* IE.

If the *Barring Exemption for Emergency Call Information* IE is included in the *Served Cell Information* IE in the F1 SETUP REQUEST message, the gNB-CU may store the information and consider the indicated cell allows emergency bearer services for UEs who would otherwise consider the cell as barred as specified in TS 38.304 [24].

If the *on-demand SIB1* IE is included and set to "Provision" in the *Served Cell Information* IE in the F1 SETUP REQUEST message, the gNB-CU shall, if supported, use the information indicated in the *On-demand SIB1 Config* IE for coordination of on-demand SIB1 transmission for network energy saving as specified in TS 38.300 [6].

8.2.3.3 Unsuccessful Operation

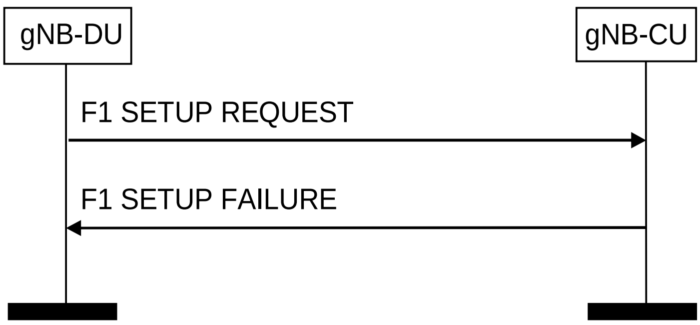

Figure 8.2.3.3-1: F1 Setup procedure: Unsuccessful Operation

If the gNB-CU cannot accept the setup, it should respond with a F1 SETUP FAILURE and appropriate cause value.

If the F1 SETUP FAILURE message includes the *Time To Wait* IE, the gNB-DU shall wait at least for the indicated time before reinitiating the F1 setup towards the same gNB-CU.

8.2.3.4 Abnormal Conditions

Not applicable.

8.2.4 gNB-DU Configuration Update

8.2.4.1 General

The purpose of the gNB-DU Configuration Update procedure is to update application level configuration data needed for the gNB-DU and the gNB-CU to interoperate correctly on the F1 interface. This procedure does not affect existing UE-related contexts, if any. The procedure uses non-UE associated signalling.

NOTE: Update of application level configuration data also applies between the gNB-DU and the gNB-CU in case the DU does not broadcast system information other than for radio frame timing and SFN, as specified in the TS 37.340 [7]. How to use this information when this option is used is not explicitly specified.

8.2.4.2 Successful Operation

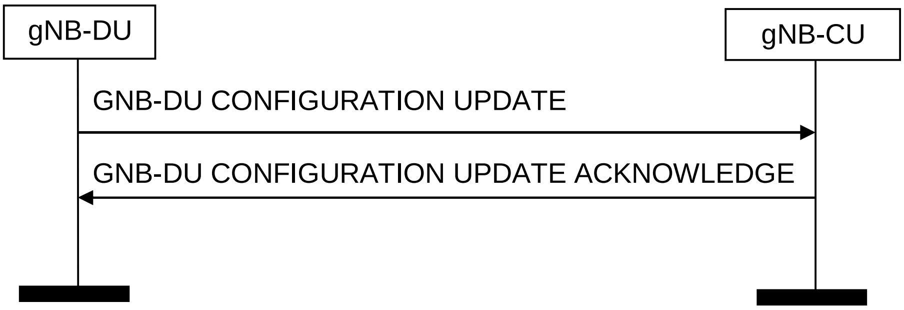

Figure 8.2.4.2-1: gNB-DU Configuration Update procedure: Successful Operation

The gNB-DU initiates the procedure by sending a GNB-DU CONFIGURATION UPDATE message to the gNB-CU including an appropriate set of updated configuration data that it has just taken into operational use. The gNB-CU responds with GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message to acknowledge that it successfully updated the configuration data. If an information element is not included in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall interpret that the corresponding configuration data is not changed and shall continue to operate the F1-C interface with the existing related configuration data.

The updated configuration data shall be stored in both nodes and used as long as there is an operational TNL association or until any further update is performed.

If g*NB-DU ID* IE is contained in the GNB-DU CONFIGURATION UPDATE message for a newly established SCTP association, the gNB-CU will associate this association with the related gNB-DU.

If *Served Cells To Add Item* IE is contained in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall add cell information according to the information in the *Served Cell Information IE*. For NG-RAN, the gNB-DU shall include the *gNB-DU System Information* IE.

If *Served Cells To Modify Item* IE is contained in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall modify information of cell indicated by *Old NR CGI* IE according to the information in the *Served Cell Informatio*n IE and overwrite the served cell information for the affected served cell. Further, if the *gNB-DU System Information* IE is present the gNB-CU shall store and replace any previous information received.

If *Served Cells To Delete Item* IE is contained in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall delete information of cell indicated by *Old NR CGI* IE.

If *Cells Status Item* IE is contained in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall update the information about the cells, as described in TS 38.401 [4]. If if the *Switching Off Ongoing* IE is present in the *Cells Status Item* IE, contained in the GNB-DU CONFIGURATION UPDATE message, and the corresponding *Service State IE* is set to "Out-of-Service", the gNB-CU shall ignore the *Switching Off Ongoing* IE.

If *Cells to be Activated List Item* IE is contained in the GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message, the gNB-DU shall activate the cell indicated by *NR CGI* IE and reconfigure the physical cell identity for cells for which the *NR PCI* IE is included.

If *Cells to be Activated List Item* IE is contained in the GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message and the indicated cells are already activated, the gNB-DU shall update the cell information received in *Cells to be Activated List Item* IE.

If *Cells to be Activated List Item* IE is included in the GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message, and the information for the cell indicated by the *NR CGI* IE includes the *IAB Info IAB-donor-CU* IE, the gNB-DU shall, if supported, apply the *IAB STC Info* IE therein to the indicated cell.

If *Cells to be Deactivated List Item* IE is contained in the GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message, the gNB-DU shall deactivate all the cells with NR CGI listed in the IE.

If *Dedicated SI Delivery Needed UE List* IE is contained in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU should take it into account when informing the UE of the updated system information via the dedicated RRC message.

For NG-RAN, the gNB-CU shall include the *gNB-CU System Information* IE in the GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message. The *SIB type to Be Updated List* IE shall contain the full list of SIBs to be broadcast.

For NG-RAN, the gNB-DU may include the *RAN Area Code* IE in the GNB-DU CONFIGURATION UPDATE message. The gNB-CU shall store and replace any previously provided *RAN Area Code* IE by the received *RAN Area Code* IE.

For NG-RAN, the gNB-DU may include the *Supported MBS FSA ID List* IE in the *Served Cell Information* IE in the GNB-DU CONFIGURATION UPDATE message. The gNB-CU shall store and replace any previously provided *MBS FSA ID list* IE by the received *MBS FSA ID list* IE.

If *Available PLMN List* IE, and optionally also *Extended Available PLMN List* IE, is contained in GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message, the gNB-DU shall overwrite the whole available PLMN list and update the corresponding system information.

If *Available SNPN ID List* IE is contained in GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message, the gNB-DU shall overwrite the whole available SNPN ID list and update the corresponding system information.

If in GNB-DU CONFIGURATION UPDATE message, the *Cell Direction* IE is present, the gNB-CU should use it to understand whether the cell is for UL or DL only. If in GNB-DU CONFIGURATION UPDATE message, the *Cell Direction* IE is omitted in the *Served Cell Information* IE it shall be interpreted as that the Cell Direction is Bi-directional.

If the GNB-DU CONFIGURATION UPDATE message includes *gNB-DU TNL Association To Remove List* IE, the gNB-CU shall, if supported, initiate removal of the TNL association(s) indicated by gNB-DU TNL endpoint(s) and gNB-CU TNL endpoint(s) if the *TNL Association Transport Layer Address gNB-CU* IE is present, or the TNL association(s) indicated by gNB-DU TNL endpoint(s) if the *TNL Association Transport Layer Address gNB-CU* IE is absent:

- if the received *TNL Association Transport Layer Address* IE includes the *Port Number* IE, the gNB-DU TNL endpoint is identified by the *Endpoint IP Address* IE and the *Port Number* IE. Otherwise, the gNB-DU TNL endpoints correspond to all gNB-DU TNL endpoints identified by the *Endpoint IP Address* IE and any port number(s).

- if the received *TNL Association Transport Layer Address gNB-CU* IE includes the *Port Number* IE, the *gNB-CU* TNL endpoint is identified by the *Endpoint IP Address* IE and the *Port Number* IE. Otherwise, the *gNB-CU* TNL endpoints correspond to all *gNB-CU* TNL endpoints identified by the *Endpoint IP Address* IE and any port number(s).

If the *Intended TDD DL-UL Configuration* IE is present in the GNB-DU CONFIGURATION UPDATE message, the receiving gNB-CU shall use the received information for Cross Link Interference management and/or NR-DC power coordination. The gNB-CU may merge the Intended TDD DL-UL Configuration information received from two or more gNB-DUs. The gNB-CU shall consider the received *Intended TDD DL-UL Configuration* IE content valid until reception of an update of the IE for the same cell(s).

If the *Aggressor gNB Set ID* IE is included in the *Served Cell Information* IE in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall, if supported, take it into account.

If the *Victim gNB Set ID* IE is included in the *Served Cell Information* IE in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall, if supported, take it into account.

If the GNB-DU CONFIGURATION UPDATE message includes *Transport Layer Address Info* IE, the gNB-CU shall, if supported, take into account for IPSec tunnel establishment.

If the GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message includes *Transport Layer Address Info* IE, the gNB-DU shall, if supported, take into account for IPSec tunnel establishment.

If the GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message contains the *Uplink BH Non-UP Traffic Mapping* IE, the gNB-DU shall, if supported, consider the information therein for mapping of non-UP uplink traffic.

If the *SFN Offset* IE is contained in the *Served Cell Information* IE in GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall, if supported, use this information to deduce the SFN0 offset of the reported cell.

If the *NR PRACH Configuration List* IE is included in the *Served Cell Information* IE contained in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU may store the information, and forward it to other RAN nodes for RACH optimisation. If the *L139 Info* IE included in the *NR PRACH Configuration List* IE is present, it shall contain the *Root Sequence Index* IE.

If the GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message contains the *BAP Address* IE, the gNB-DU shall, if supported, store the received BAP address and use it as specified in TS 38.340 [30].

If the *Coverage Modification Notification* IE is contained in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall, if supported, take it into account for Coverage and Capacity Optimization and network energy saving. If the *Coverage Modification Cause* IE is set to the "network energy saving", gNB-CU may consider those deactivated SSB beams are due to network energy saving.

If the *Cells for SON* IE is present in the GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE message, the gNB-DU may store or update this information and behaves as follows:

- For each served cell indicated by the *NR CGI* IE included within the *Cells for SON Item* IE, the gNB-DU may adjust the PRACH configuration of this served cell.

- If the *Neighbour NR Cells for SON List* IE is present in the *Cells for SON Item* IE, the gNB-DU may take the PRACH configuration of neighbour cells included in the *Neighbour NR Cells for SON List* IE into consideration when adjusting the PRACH configuration of the served cell.

If the *RedCap Broadcast Information* IE is contained in the *Served Cell Information* IE in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU may store and use this information to determine a suitable target cell in case of mobility and paging involving RedCap UEs.

If the *eRedCap Broadcast Information* IE is contained in the *Served Cell Information* IE in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU may store and use this information to determine a suitable target cell in case of mobility and paging involving eRedCap UEs.

If the *TAI NSAG Support List* IE is included in the *Served Cell Information* IE in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall, if supported, use this information as specified in TS 23.501 [21].

If the *gNB-DU Name* IE is included in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU may store it or update this IE value if already stored, and use it as a human readable name of the gNB-DU. If the *Extended gNB-DU Name* IE is included in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU may store it or update this IE value if already stored, and use it as a human readable name of the gNB-DU and shall ignore the *gNB-DU Name* IE if also included.

If the *RRC Terminating IAB-Donor Related Info* IE is included in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall, if supported, consider that the BAP address indicated by the *Mobile IAB-MT BAP Address* IE is assigned by the gNB-CU of the RRC-terminating IAB-donor indicated by the *RRC Terminating IAB-Donor gNB-ID* IE, and it shall use this BAP address and gNB ID for the subsequent IAB Transport Migration Management procedure towards the RRC-terminating IAB-donor of the mobile IAB-node as needed, as specified in TS 38.423 [28].

If the GNB-DU CONFIGURATION UPDATE message contains the *Mobile IAB-MT User Location Information* IE, the gNB-CU shall, if supported, take it into account when reporting UE location information to the AMF for a UE served by the mobile IAB-node.

If the *XR Broadcast Information* IE is included in the *Served Cell Information* IE in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall, if supported, consider the indicated cell does not allow 2Rx XR UEs access in case of mobility and paging involving XR UEs.

If the *Barring Exemption for Emergency Call Information* IE is included in the *Served Cell Information* IE in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU may store the information and consider the indicated cell allows emergency bearer services for UEs who would otherwise consider the cell as barred as specified in TS 38.304 [24].

If the *on-demand SIB1* IE is included and set to "Provision" in the *Served Cell Information* IE in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall, if supported, use this information indicated in the *On-*

demand SIB1 Config IE for coordination of on-demand SIB1 transmission for network energy saving as specified in TS 38.300 [6].

If the *on-demand SIB1* IE is included and set to "Stop provision" in the *Served Cell Information* IE in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall, if supported, stop the coordination of on-demand SIB1 transmission for network energy saving as specified in TS 38.300 [6].

If the *Future Coverage Modification Notification* IE is contained in the GNB-DU CONFIGURATION UPDATE message, the gNB-CU shall, if supported, take it into account for Coverage and Capacity Optimization.

If the *Future Coverage Modification Notification* IE is contained in the GNB-DU CONFIGURATION UPDATE message and if all the instances of the *Future Coverage Modification Cause* IE are set to “cancel”, the gNB-CU shall, if supported, consider it as a notification that the gNB-DU has cancelled the future coverage modifications indicated for the cell(s) and optionally beam(s) listed in the *Future Coverage Modification Notification* IE, as specified in TS 38.401[4].

8.2.4.3 Unsuccessful Operation

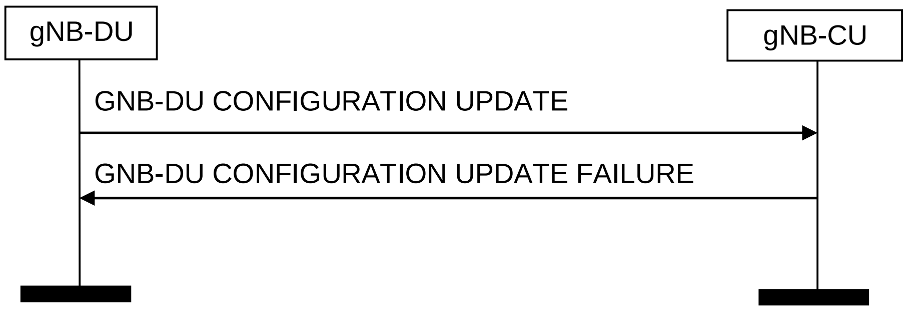

Figure 8.2.4.3-1: gNB-DU Configuration Update procedure: Unsuccessful Operation

If the gNB-CU cannot accept the update, it shall respond with a GNB-DU CONFIGURATION UPDATE FAILURE message and appropriate cause value.

If the GNB-DU CONFIGURATION UPDATE FAILURE message includes the *Time To Wait* IE, the gNB-DU shall wait at least for the indicated time before reinitiating the GNB-DU CONFIGURATION UPDATE message towards the same gNB-CU.

8.2.4.4 Abnormal Conditions

If the *Future Coverage Modification Notification* IE is contained in the GNB-DU CONFIGURATION UPDATE message and if some of the instances of the *Future Coverage Modification Cause* IE are set to “cancel”, while some other instances are set to values different from “cancel”, the gNB-CU shall ignore the *Future Coverage Modification Notification* IE.

If the *Future Coverage Modification Notification* IE is contained in the GNB-DU CONFIGURATION UPDATE message and if the *Future Coverage Modification Cause* IE is set to “cancel”, the *Time for Future Coverage Modification* IE is ignored.

8.2.5 gNB-CU Configuration Update

8.2.5.1 General

The purpose of the gNB-CU Configuration Update procedure is to update application level configuration data needed for the gNB-DU and gNB-CU to interoperate correctly on the F1 interface. This procedure does not affect existing UE-related contexts, if any. The procedure uses non-UE associated signalling.

8.2.5.2 Successful Operation

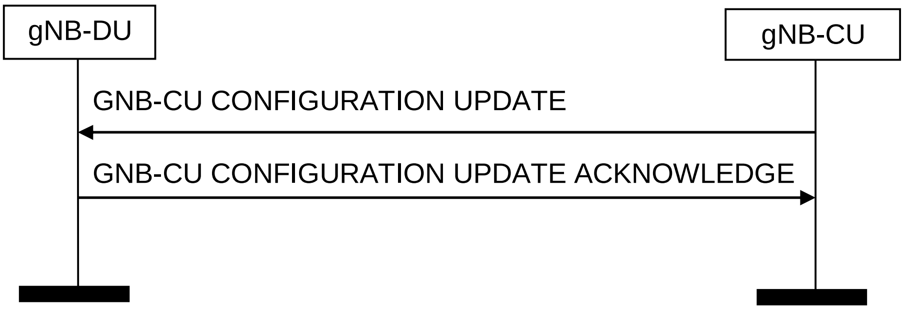

Figure 8.2.5.2-1: gNB-CU Configuration Update procedure: Successful Operation

The gNB-CU initiates the procedure by sending a GNB-CU CONFIGURATION UPDATE message including the appropriate updated configuration data to the gNB-DU. The gNB-DU responds with a GNB-CU CONFIGURATION UPDATE ACKNOWLEDGE message to acknowledge that it successfully updated the configuration data. If an information element is not included in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall interpret that the corresponding configuration data is not changed and shall continue to operate the F1-C interface with the existing related configuration data.

The updated configuration data shall be stored in the respective node and used as long as there is an operational TNL association or until any further update is performed.

If *Cells to be Activated List Item* IE is contained in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall activate the cell indicated by *NR CGI* IE and reconfigure the physical cell identity for which the *NR PCI* IE is included.

If the *SSBs within the cell to be Activated List* IE is included in the *Cells to be Activated List Item* IE within the gNB-CU CONFIGURATION UPDATE message, the gNB-DU shall, if supported, only activate those SSB beams indicated by the *SSB Index* IE.

If at least one requested SSB beam in the *SSBs within the cell to be Activated List* IE is activated, the gNB-DU includes the *Cells with SSBs Activated List* IE in the GNB-CU CONFIGURATION UPDATE ACKNOWLEDGE message. The gNB-CU shall consider that the SSB beams indicated by the *SSBs activated List* IE as activated.

If *Cells to be Deactivated List Item* IE is contained in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall deactivate the cell indicated by *NR CGI* IE.

If *Cells to be Activated List Item* IE is contained in the GNB-CU CONFIGURATION UPDATE message and the indicated cells are already activated, the gNB-DU shall update the cell information received in *Cells to be Activated List Item* IE.

If *Cells to be Activated List Item* IE is included in the GNB-CU CONFIGURATION UPDATE message, and the information for the cell indicated by the *NR CGI* IE includes the *IAB Info IAB-donor-CU* IE, the gNB-DU shall, if supported, apply the *IAB STC Info* IE therein to the indicated cell.

If the *Cells Allowed to be Deactivated List* IE is contained in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall, if supported, consider that it is allowed to deactivate the SSB beams within the indicated cells for network energy saving purpose.

If the *gNB-CU System Information* IE is contained in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall include the *Dedicated SI Delivery Needed UE List* IE in the GNB-CU CONFIGURATION UPDATE ACKNOWLEDGE message for UEs that are unable to receive system information from broadcast.

If *Dedicated SI Delivery Needed UE List* IE is contained in the GNB-CU CONFIGURATION UPDATE ACKNOWLEDGE message, the gNB-CU should take it into account when informing the UE of the updated system information via the dedicated RRC message.

If the *gNB-CU TNL Association To Add List* IE is contained in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall, if supported, use it to establish the TNL association(s) with the gNB-CU. If the *gNB-CU TNL Association To Add List* is included in the GNB-CU CONFIGURATION UPDATE message, and if the *TNL Association Transport Layer Information* IE does not include the *Port Number* IE, the gNB-DU shall assume that port number value 38472 is used for the endpoint. The gNB-DU shall report to the gNB-CU, in the gNB-CU CONFIGURATION UPDATE ACKNOWLEDGE message, the successful establishment of the TNL association(s) with the gNB-CU as follows:

- A list of TNL address(es) with which the gNB-DU successfully established the TNL association shall be included in the gNB-CU *TNL Association Setup List* IE;
- A list of TNL address(es) with which the gNB-DU failed to establish the TNL association shall be included in the *gNB-CU TNL Association Failed To Setup List* IE.

If the GNB-CU CONFIGURATION UPDATE message includes *gNB-CU TNL Association To Remove List* IE, the gNB-DU shall, if supported, initiate removal of the TNL association(s) indicated by gNB-CU TNL endpoint(s) and gNB-DU TNL endpoint(s) if the *TNL Association Transport Layer Address gNB-DU* IE is present, or the TNL association(s) indicated by gNB-CU TNL endpoint(s) if the *TNL Association Transport Layer Address gNB-DU IE* is absent:

- if the received *TNL Association Transport Layer Address* IE includes the *Port Number* IE, the gNB-CU TNL endpoint is identified by the *Endpoint IP Address* IE and the *Port Number* IE. Otherwise, the gNB-CU TNL endpoints correspond to all gNB-CU TNL endpoints identified by the *Endpoint IP Address* IE and any port number(s).
- if the received *TNL Association Transport Layer Address gNB-DU* IE includes the *Port Number* IE, the gNB-DU TNL endpoint is identified by the *Endpoint IP Address* IE and the *Port Number* IE. Otherwise, the gNB-DU TNL endpoints correspond to all gNB-DU node TNL endpoints identified by the *Endpoint IP Address* IE and any port number(s).

If the *gNB-CU TNL Association To Update List* IE is contained in the gNB-CU CONFIGURATION UPDATE message the gNB-DU shall, if supported, overwrite the previously stored information for the related TNL Association(s).

- if the received *TNL Association Transport Layer Address* IE includes the *Port Number* IE, the gNB-CU TNL endpoint is identified by the *Endpoint IP Address* IE and the *Port Number* IE. Otherwise, the gNB-CU TNL endpoints correspond to all gNB-CU TNL endpoints identified by the *Endpoint IP Address* IE and any port number(s).

If in the gNB-CU CONFIGURATION UPDATE message the *TNL Association usage* IE is included in the *gNB-CU TNL Association To Add List* IE or the *gNB-CU TNL Association To Update List* IE, the gNB-DU node shall, if supported, use it as described in TS 38.472 [22].

For NG-RAN, the gNB-CU shall include the *gNB-CU System Information* IE in the GNB-CU CONFIGURATION UPDATE message. The *SIB type to Be Updated List* IE shall contain the full list of SIBs to be broadcast.

If *Protected E-UTRA Resources List* IE is contained in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall protect the corresponding resource of the cells indicated by *E-UTRA Cells List* IE for spectrum sharing between E-UTRA and NR.

If the GNB-CU CONFIGURATION UPDATE message contains the *Protected E-UTRA Resource Indication* IE, the receiving gNB-DU should forward it to lower layers and use it for cell-level resource coordination. The gNB-DU shall consider the received *Protected E-UTRA Resource Indication* IE when expressing its desired resource allocation during gNB-DU Resource Coordination procedure. The gNB-DU shall consider the received *Protected E-UTRA Resource Indication* IE content valid until reception of a new update of the IE for the same gNB-DU.

If *Available PLMN List* IE, and optionally also *Extended Available PLMN List* IE, is contained in GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall overwrite the whole available PLMN list and update the corresponding system information.

If *Available SNPN ID List* IE is contained in GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall overwrite the whole available SNPN ID list and update the corresponding system information.

If *Cells Failed to be Activated Item* IE is contained in the GNB-CU CONFIGURATION UPDATE ACKNOWLEDGE message, the gNB-CU shall consider that the indicated cells are out-of-service as defined in TS 38.401 [4].

If the *Neighbour Cell Information List* IE is present in the GNB-CU CONFIGURATION UPDATE message, the receiving gNB-DU shall use the received information for Cross Link Interference management and/or NR-DC power coordination. The gNB-DU shall consider the received *Neighbour Cell Information List* IE content valid until reception of an update of the IE for the same cell(s). If the *Intended TDD DL-UL Configuration NR* IE is absent from the

Neighbour Cell Information List IE, whereas the corresponding *NR CGI* IE is present, the receiving gNB-DU shall remove the previously stored *Neighbour Cell Information* IE corresponding to the NR CGI.

If the GNB-CU CONFIGURATION UPDATE message includes *Transport Layer Address Info* IE, the gNB-DU shall, if supported, take into account for IPSec tunnel establishment.

If the GNB-CU CONFIGURATION UPDATE ACKNOWLEDGE message includes *Transport Layer Address Info* IE, the gNB-CU shall, if supported, take into account for IPSec tunnel establishment.

If the GNB-CU CONFIGURATION UPDATE message contains the *Uplink BH Non-UP Traffic Mapping* IE, the gNB-DU shall, if supported, consider the information therein for mapping of non-UP uplink traffic.

If the *IAB Barred* IE is included in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall, if supported, consider it as an indication of whether the cell allows IAB-node access or not.

If the *BAP Address* IE is included in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall, if supported, store the received BAP address and use it as specified in TS 38.340 [30].

If the *CCO Assistance Information* IE is contained in the GNB-CU CONFIGURATION UPDATE message, and the *NR CGI* IE contained in the *Affected Cells and Beams* IE is served by the gNB-DU, the gNB-DU may use it to determine a new cell and/or beam configuration.

If the *CCO Assistance Information* IE is contained in the GNB-CU CONFIGURATION UPDATE message and the *NR CGI* IE contained in the *Affected Cells and Beams* IE is not served by the gNB-DU, the gNB-DU may use it to adjust coverage of its cells.If the *CCO issue detection* IE set to "network energy saving" is included in the *CCO Assistance Information* IE, the gNB-DU may consider the indicated SSB beams by the *Affected Cells and Beam* IE are deactivated due to network energy saving.

If the *Cells for SON* IE is present in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU may store or update this information and it behaves as follows:

- For each served cell indicated by the *NR CGI* IE included within the *Cells for SON Item* IE, the gNB-DU may adjust the PRACH configuration of this served cell.
- If the *Neighbour NR Cells for SON List* IE is present in the *Cells for SON Item* IE, the gNB-DU may take the PRACH configuration of neighbour cells included in the *Neighbour NR Cells for SON List* IE into consideration when adjusting the PRACH configuration of the served cell.

If the *gNB-CU Name* IE is included in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU may store it or update this IE value if already stored, and use it as a human readable name of the gNB-CU. If the *Extended gNB-CU Name* IE is included in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU may store it or update this IE value if already stored, and use it as a human readable name of the gNB-CU and shall ignore the *gNB-CU Name* IE if also included.

If the *Mobile IAB Barred* IE is included in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall, if supported, consider it as an indication of whether the cell allows mobile IAB-node access.

If the *On-demand SIB1 Cell* IE is contained in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall, if supported, consider to start or stop the on-demand SIB1 operation as indicated by the *On-demand SIB1 indicator* IE for the cell indicated by the *NR CGI* IE.

If the *Predicted CCO Assistance Information* IE is contained in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU may use it to determine a future cell(s) and optionally beam(s) configuration.

If the *Predicted CCO Assistance Information* IE is contained in the GNB-CU CONFIGURATION UPDATE message and if the *Predicted CCO Issue* IE is set to “cancel”, the gNB-DU shall discard the *Predicted CCO Assistance Information* IE previously received together with the same list of cell(s) and optionally beam(s) included in the *Predicted Affected Cells and Beams* IE, and it should cancel the future coverage states associated to the *Predicted CCO Issue* IE for the same list of cell(s) and optionally beam(s) that have not been applied as specified in TS 38.401[4].

If the *Predicted CCO Assistance Information* IE is contained in the GNB-CU CONFIGURATION UPDATE message and if the *Time for Predicted CCO Issue* IE is included, the gNB-DU should take it into account to determine the timing of a future cell(s) and optionally beam(s) configuration, otherwise the gNB-DU assumes that the predicted CCO issue will occur shortly after the reception of the GNB-CU CONFIGURATION UPDATE message and it should apply a future cell(s) and optionally beam(s) configuration as soon as possible.

If the *Neighbour Future Coverage Modification Notification* IE is contained in the GNB-CU CONFIGURATION UPDATE message, the gNB-DU shall, if supported, take it into account for Coverage and Capacity Optimization. If the *Future Coverage Modification Cause* IE is included in the *Neighbour Future Coverage Modification Notification* IE and set to "cancel", the gNB-DU shall consider it as a notification of cancellation of the future coverage modifications included in the *Neighbour Future Coverage Modification Notification List* IE as specified in TS 38.401 [4].

8.2.5.3 Unsuccessful Operation

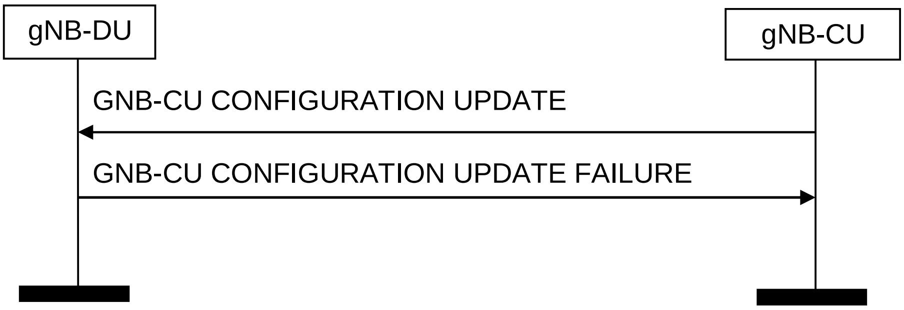

Figure 8.2.5.3-1: gNB-CU Configuration Update: Unsuccessful Operation

If the gNB-DU cannot accept the update, it shall respond with a GNB-CU CONFIGURATION UPDATE FAILURE message and appropriate cause value.

If the GNB-CU CONFIGURATION UPDATE FAILURE message includes the *Time To Wait* IE, the gNB-CU shall wait at least for the indicated time before reinitiating the GNB-CU CONFIGURATION UPDATE message towards the same gNB-DU.

8.2.5.4 Abnormal Conditions

If the *Predicted CCO Assistance Information* IE is contained in the GNB-CU CONFIGURATION UPDATE message, and if:

- the *Predicted CCO Issue* IE is set to "cancel", and
- the list of cells and beams contained in the *Predicted Affected Cells and Beams* IE is not the same as the list of cells and beams contained in a previously received *Predicted CCO Assistance Information* IE,

then the gNB-DU shall ignore the received *Predicted CCO Assistance Information* IE.

If the *CCO Assistance Information* IE is contained in the GNB-CU CONFIGURATION UPDATE message, and if:

- the *CCO Issue* IE is not present, and
- the *NR CGI* IE contained in the *Affected Cells and Beams* IE is served by the gNB-DU,

then the gNB-DU shall discard the received *CCO Assistance Information* IE.

If the *CCO Assistance Information* IE is contained in the GNB-CU CONFIGURATION UPDATE message, and if:

- the *CCO Issue* IE is present, and
- the *Affected Cells and Beams* IE is not present,

then the gNB-DU shall discard the received *CCO Assistance Information* IE.

8.2.6 gNB-DU Resource Coordination

8.2.6.1 General

The purpose of the gNB-DU Resource Coordination procedure is to enable coordination of radio resource allocation between a gNB-CU and a gNB-DU for the purpose of spectrum sharing between E-UTRA and NR. This procedure is to be used only for the purpose of spectrum sharing between E-UTRA and NR.

The procedure uses non-UE-associated signalling.

8.2.6.2 Successful Operation

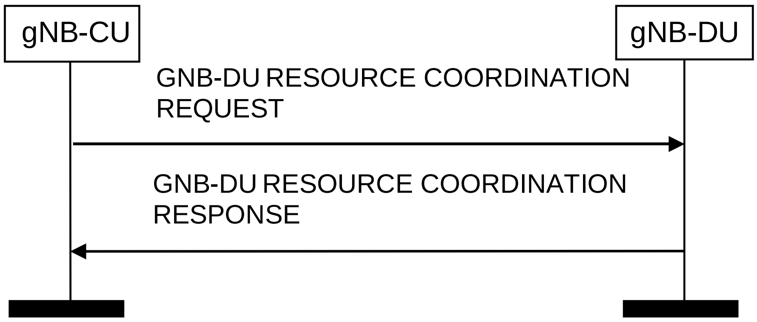

Figure 8.2.6.2-1: gNB-DU Resource Coordination, successful operation

A gNB-CU initiates the procedure by sending the GNB-DU RESOURCE COORDINATION REQUEST message to a gNB-DU over the F1 interface.

The gNB-DU extracts the *E-UTRA – NR Cell Resource Coordination Request Container* IE and it replies by sending the GNB-DU RESOURCE COORDINATION RESPONSE message.

In case of NR-initiated gNB-DU Resource Coordination procedure, the *Ignore Coordination Request Container* IE shall be present and set to "yes" and the *E-UTRA – NR Cell Resource Coordination Request Container* IE in the GNB-DU RESOURCE COORDINATION REQUEST message shall be ignored.

8.2.7 gNB-DU Status Indication

8.2.7.1 General

The purpose of the gNB-DU Status Indication procedure is informing the gNB-CU that the gNB-DU is overloaded so that overload reduction actions can be applied. This procedure is also used to inform the IAB-donor-CU about a downlink congestion at an IAB-DU or an IAB-donor-DU. The procedure uses non-UE associated signalling.

8.2.7.2 Successful Operation

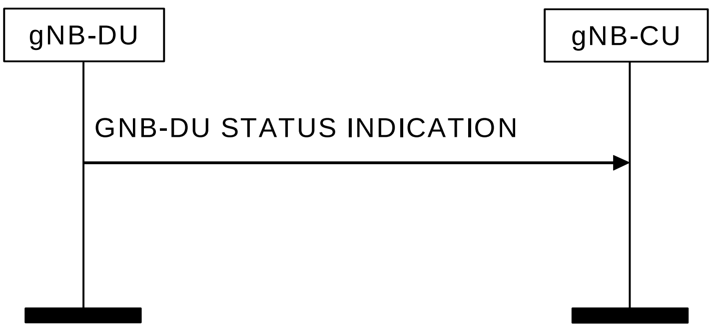

Figure 8.2.7.2-1: gNB-DU Status Indication procedure

If the *gNB-DU Overload Information* IE in the GNB-DU STATUS INDICATION message indicates that the gNB-DU is overloaded, the gNB-CU shall apply overload reduction actions until informed, with a new GNB-DU STATUS INDICATION message, that the overload situation has ceased.

The detailed overload reduction policy is up to gNB-CU implementation.

If the gNB-DU is an IAB-DU or an IAB-donor-DU, and if the *IAB Congestion Indication* IE is present in the GNB-DU STATUS INDICATION message and only includes the *Child Node Identifier* IE, the gNB-CU shall, if supported, consider that the backhaul link between the gNB-DU and the node identified by the *Child Node Identifier* IE is congested. If the *IAB Congestion Indication* IE is present in the GNB-DU STATUS INDICATION message and includes both the *Child Node Identifier* IE and the *BH RLC CH ID* IE, the gNB-CU shall, if supported, consider that congestion occurs on the corresponding BH RLC channel(s) over the link towards the node identified by the *Child Node Identifier* IE.

8.2.7.3 Abnormal Conditions

Void.

8.2.8 F1 Removal

8.2.8.1 General

The purpose of the F1 Removal procedure is to remove the interface instance and all related resources between the gNB-DU and the gNB-CU in a controlled manner. If successful, this procedure erases any existing application level configuration data in the two nodes.

NOTE: In case the signalling transport is shared among several F1-C interface instances, and the TNL association is still used by one or several F1-C interface instances, the initiating node should not initiate the removal of the TNL association.

The procedure uses non-UE-associated signaling.

8.2.8.2 Successful Operation

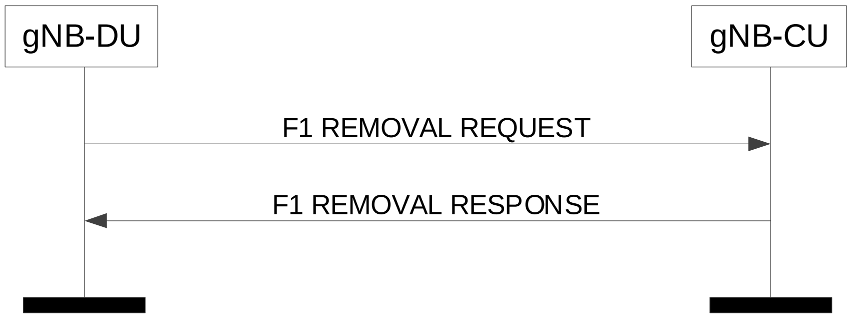

Figure 8.2.8-1: F1 Removal, gNB-DU initiated, successful operation

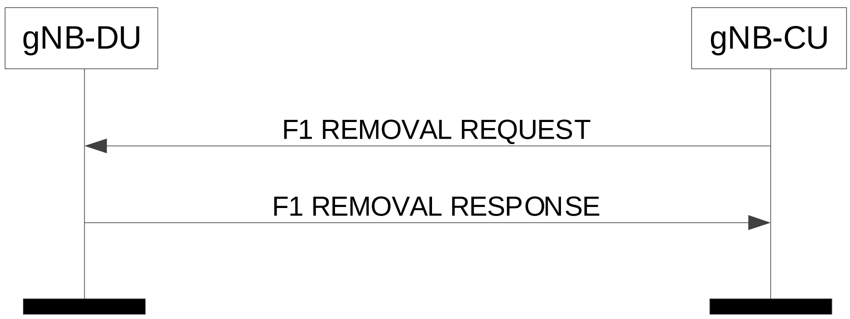

Figure 8.2.8.2-2: F1 Removal, gNB-CU initiated, successful operation

Successful F1 Removal, gNB-DU initiated

The gNB-DU initiates the procedure by sending the F1 REMOVAL REQUEST message to the gNB-CU. Upon reception of the F1 REMOVAL REQUEST message the gNB-CU shall reply with the F1 REMOVAL RESPONSE message. After receiving the F1 REMOVAL RESPONSE message, the gNB-DU may initiate removal of the TNL association towards the gNB-CU, if applicable, and may remove all resources associated with that interface instance. The gNB-CU may then remove all resources associated with that interface instance.

Successful F1 Removal, gNB-CU initiated

The gNB-CU initiates the procedure by sending the F1 REMOVAL REQUEST message to the gNB-DU. Upon reception of the F1 REMOVAL REQUEST message the gNB-DU shall reply with the F1 REMOVAL RESPONSE message. After receiving the F1 REMOVAL RESPONSE message, the gNB-CU may initiate removal of the TNL association towards the gNB-DU, if applicable, and may remove all resources associated with that interface instance. The gNB-DU may then remove all resources associated with that interface instance.

8.2.8.3 Unsuccessful Operation

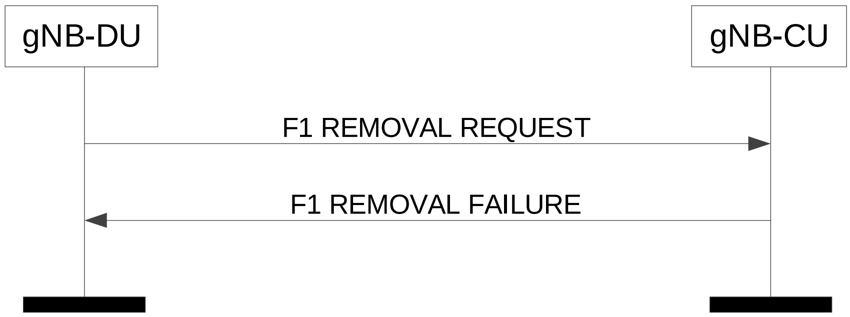

Figure 8.2.8.3-1: F1 Removal, gNB-DU initiated, unsuccessful operation

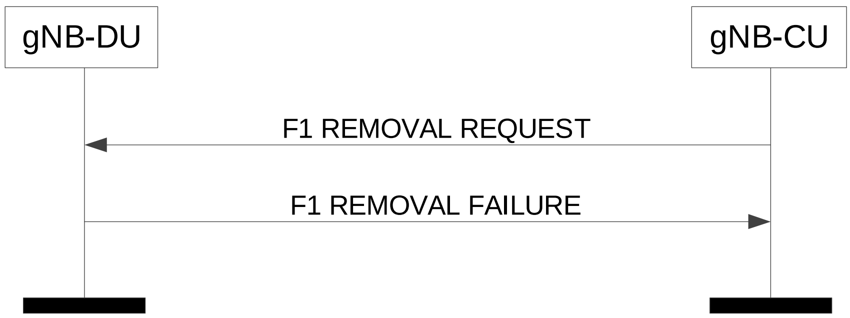

Figure 8.2.8.3-2: F1 Removal, gNB-CU initiated, unsuccessful operation

Unsuccessful F1 Removal, gNB-DU initiated

If the gNB-CU cannot accept to remove the interface instance with the gNB-DU it shall respond with an F1 REMOVAL FAILURE message with an appropriate cause value.

Unsuccessful F1 Removal, gNB-CU initiated

If the gNB-DU cannot accept to remove the interface instance with the gNB-CU it shall respond with an F1 REMOVAL FAILURE message with an appropriate cause value.

8.2.8.4 Abnormal Conditions

Not applicable.

8.2.9 Network Access Rate Reduction

8.2.9.1 General

The purpose of the Network Access Rate Reduction procedure is to indicate to the gNB-DU that the rate at which UEs are accessing the network need to be reduced from its current level.

The procedure uses non-UE associated signalling.

8.2.9.2 Successful operation

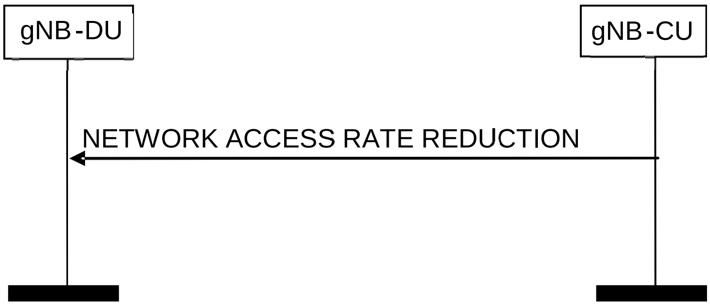

Figure 8.2.9.2-1: Network Access Rate Reduction, Successful operation

The gNB-CU initiates the procedure by sending a NETWORK ACCESS RATE REDUCTION message to the gNB-DU. When receiving the NETWORK ACCESS RATE REDUCTION message the gNB-DU should take into account the information contained in the *UAC Assistance Information* IE to set the parameters for Unified Access Barring.

If the *NID* IE is contained in the NETWORK ACCESS RATE REDUCTION message, the gNB-DU should take it into account and combine the *NID* IE with the *PLMN Identity* IE to identify the SNPN.

8.2.9.3 Abnormal Conditions

Not applicable

8.2.10 Resource Status Reporting Initiation

8.2.10.1 General

This procedure is used by an gNB-CU to request the reporting of load measurements to gNB-DU.

The procedure uses non UE-associated signalling.

8.2.10.2 Successful Operation

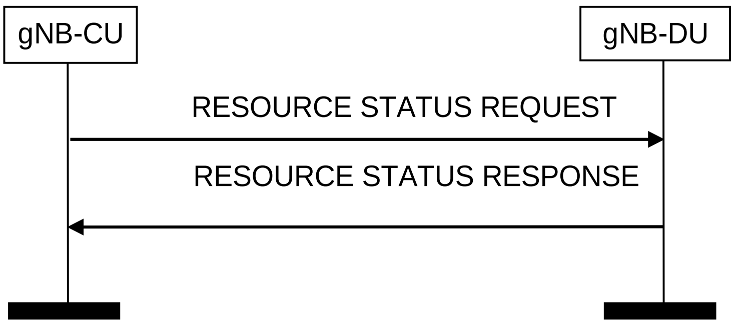

Figure 8.2.10.2-1: Resource Status Reporting Initiation, successful operation

gNB-CU initiates the procedure by sending the RESOURCE STATUS REQUEST message to gNB-DU to start a measurement, stop a measurement, or add cells to report for a measurement. Upon receipt, gNB-DU:

- shall initiate the requested measurement according to the parameters given in the request in case the *Registration Request* IE set to "start"; or
- shall stop all cells measurements and terminate the reporting in case the *Registration Request* IE is set to "stop"; or

- shall add cells indicated in the *Cell To Report List* IE to the measurements initiated before for the given measurement IDs, in case the *Registration Request* IE is set to "add". If measurements are already initiated for a cell indicated in the *Cell To Report List* IE, this information shall be ignored.

If the *Registration Request* IE is set to "start" in the RESOURCE STATUS REQUEST message and the *Report Characteristics* IE indicates cell specific measurements, the *Cell To Report List* IE shall be included.

If *Registration Request* IE is set to "add" in the RESOURCE STATUS REQUEST message, the *Cell To Report List* IE shall be included.

If gNB-DU is capable to provide all requested resource status information, it shall initiate the measurement as requested by gNB-CU, and respond with the RESOURCE STATUS RESPONSE message.

Interaction with other procedures

When starting a measurement, the *Report Characteristics* IE in the RESOURCE STATUS REQUEST indicates the type of objects gNB-DU shall perform measurements on. For each cell, gNB-DU shall include in the RESOURCE STATUS UPDATE message:

- the *Radio Resource Status* IE, if the first bit, "PRB Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1. If the cell for which *Radio Resource Status* IE is requested to be reported supports more than one SSB, the *Radio Resource Status* IE for such cell shall include the *SSB Area Radio Resource Status Item* IE for all SSB areas supported by the cell. If the *SSB To Report List* IE is included for a cell, the *Radio Resource* IE for such cell shall only include the *SSB Area Radio Resource Status List* IE; If the cell for which *Radio Resource Status* IE is requested to be reported supports more than one slice, and if the *Slice To Report List* IE is included for a cell, the *Radio Resource Status* IE for such cell shall, if supported, include the requested *Slice Radio Resource Status Item* IE; If the cell for which *Radio Resource Status* IE is requested to be reported supports MIMO the *Radio Resource Status* IE for such cell may include the *MIMO PRB usage Information* IE;

- the *TNL Capacity Indicator* IE, if the second bit, "TNL Capacity Ind Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1;

- the *Composite Available Capacity Group* IE, if the third bit, "Composite Available Capacity Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1. If *Cell Capacity Class Value* IE is included within the *Composite Available Capacity Group* IE, this IE is used to assign weights to the available capacity indicated in the *Capacity Value* IE. If the cell for which *Composite Available Capacity Group* IE is requested to be reported supports more than one SSB the *Composite Available Capacity Group* IE for such cell shall include the *SSB Area Capacity Value List* IE for all SSB areas supported by the cell, providing the SSB area capacity with respect to the *Cell Capacity Class Value* IE. If the *SSB To Report List* IE is included for a cell, the *Composite Available Capacity Group* IE for such cell shall include the requested *SSB Area Capacity Value List* IE providing the SSB area capacity with respect to the Cell Capacity Class Value. If the cell for which *Composite Available Capacity Group* IE is requested to be reported supports more than one slice, and if the *Slice To Report List* IE is included for a cell, the *Slice Available Capacity* IE for such cell shall include the requested *Slice Available Capacity Value Downlink* IE and *Slice Available Capacity Value Uplink* IE, providing the slice capacity with respect to the Cell Capacity Class Value.

- the *Hardware Load Indicator* IE, if the fourth bit, " HW LoadInd Periodic " of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1;

- the *Number of Active UEs* IE, if the fifth bit, "Number of Active UEs Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1;

- the *NR-U Channel List* IE, if the sixth bit, " NR-U Channel List Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to "1";

- the *Energy Cost* IE, if the seventh bit, "Energy Cost" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to "1".

If the *Reporting Periodicity* IE in the RESOURCE STATUS REQUEST is present, this indicates the periodicity for the reporting of periodic measurements. The gNB-DU shall report once, unless otherwise requested within the *Reporting Periodicity* IE.

8.2.10.3 Unsuccessful Operation

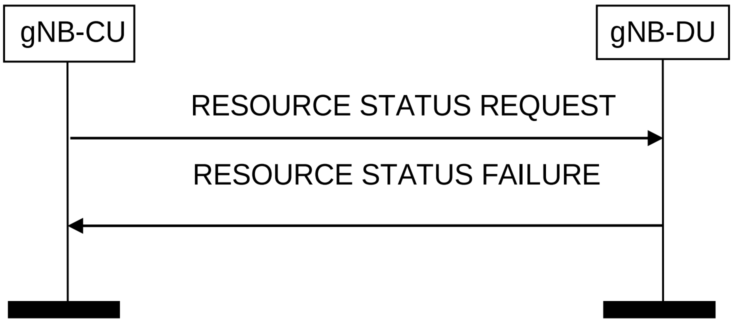

Figure 8.2.10.3-1: Resource Status Reporting Initiation, unsuccessful operation

If any of the requested measurements cannot be initiated, gNB-DU shall send the RESOURCE STATUS FAILURE message with an appropriate cause value.

8.2.10.4 Abnormal Conditions

If the initiating gNB-CU does not receive either RESOURCE STATUS RESPONSE message or RESOURCE STATUS FAILURE message, the gNB-CU may reinitiate the Resource Status Reporting Initiation procedure towards the same gNB-DU, provided that the content of the new RESOURCE STATUS REQUEST message is identical to the content of the previously unacknowledged RESOURCE STATUS REQUEST message with the same Transaction ID.

If the *Report Characteristics* IE bitmap is set to "0" (all bits are set to "0") in the RESOURCE STATUS REQUEST message then gNB-DU shall initiate a RESOURCE STATUS FAILURE message with an appropriate cause value.

If the gNB-DU receives a RESOURCE STATUS REQUEST message which includes the *Registration Request* IE set to "start" and the *gNB-CU Measurement ID* IE corresponding to an existing on-going load measurement reporting, for which a different Transaction ID is used, then gNB-DU shall initiate a RESOURCE STATUS FAILURE message with an appropriate cause value.

8.2.11 Resource Status Reporting

8.2.11.1 General

This procedure is initiated by gNB-DU to report the result of measurements admitted by gNB-DU following a successful Resource Status Reporting Initiation procedure.

The procedure uses non UE-associated signalling.

8.2.11.2 Successful Operation

Figure 8.2.11.2-1: Resource Status Reporting, successful operation

The gNB-DU shall report the results of the admitted measurements in RESOURCE STATUS UPDATE message. The admitted measurements are the measurements that were successfully initiated during the preceding Resource Status Reporting Initiation procedure.

If some results of the admitted measurements in RESOURCE STATUS UPDATE message are missing, the gNB-CU shall consider that these results were not available at the gNB-DU.

8.2.11.3 Unsuccessful Operation

Not applicable.

8.2.11.4 Abnormal Conditions

Void.

8.2.12 DU-CU TA Information Transfer

8.2.12.1 General

The purpose of the DU-CU TA Information Transfer procedure is to to enable the gNB-DU to send the TA related information to the gNB-CU. The procedure uses non-UE-associated signalling.

8.2.12.2 Successful Operation

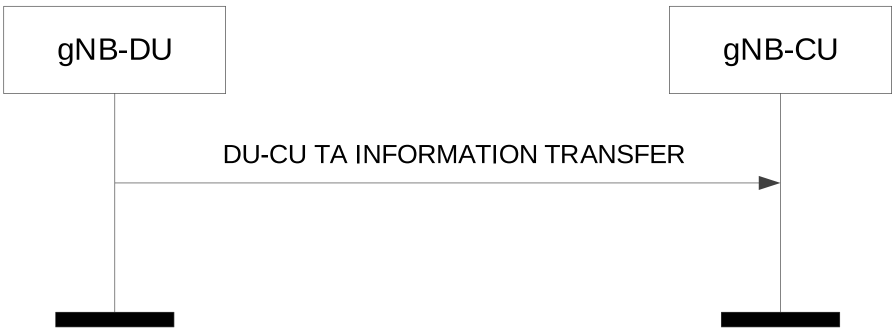

Figure 8.2.12.2-1: DU-CU TA Information Transfer procedure. Successful operation.

The gNB-DU initiates the procedure by sending a DU-CU TA Information Transfer message.

Upon reception of the DU-CU TA Information Transfer message, the gNB-CU shall, if supported, consider that the received information is the TA information from the candidate cell(s) that is indicated by the included *Candidate Cell ID* IE.

If the *Tag ID Pointer* IE is included, the gNB-CU shall, if supported, consider it to determine the TAG corresponding to the received TA value.

If the *LTM gNB ID* IE is included, the gNB-CU shall, if supported, consider that it is the gNB to which the associated TA information to be forwarded.

8.2.12.3 Unsuccessful Operation

Not applicable.

8.2.12.4 Abnormal Conditions

Not applicable.

8.2.13 CU-DU TA Information Transfer

8.2.13.1 General

The purpose of the CU-DU TA Information Transfer procedure is to to enable the gNB-CU to send the TA related information to the gNB-DU. The procedure uses non-UE-associated signalling.

8.2.13.2 Successful Operation

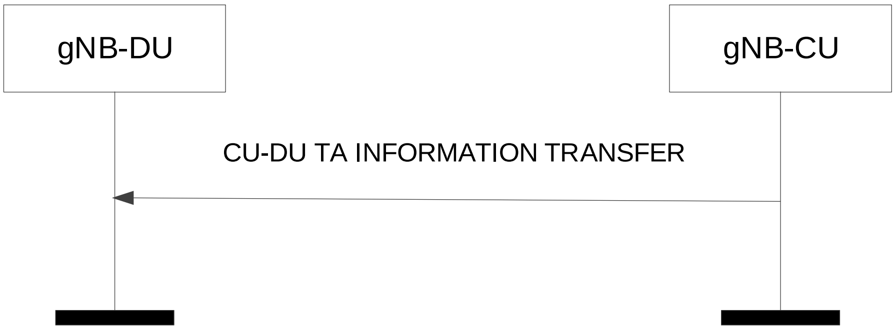

Figure 8.2.13.2-1: CU-DU TA Information Transfer procedure. Successful operation.

The gNB-CU initiates the procedure by sending a CU-DU TA Information Transfer message.

Upon reception of the CU-DU TA Information Transfer message, the gNB-DU shall, if supported, consider that the received information is the TA information from the candidate cell(s) that is indicated by the included *Candidate Cell ID* IE.

If the *Tag ID Pointer* IE is included, the gNB-DU shall, if supported, consider it to determine the TAG corresponding to the received TA value.

8.2.13.3 Unsuccessful Operation

Not applicable.

8.2.13.4 Abnormal Conditions

Not applicable.

8.2.14 RACH Indication

8.2.14.1 General

This procedure is initiated by the gNB-DU to inform the gNB-CU about the occurrences of successful random access procedures in the gNB-DU.

The procedure uses non-UE-associated signalling.

8.2.14.2 Successful Operation

Figure 8.2.14.2-1: RACH Indication procedure.

The gNB-DU initiates the procedure by sending the RACH INDICATION message to the gNB-CU.

The RACH INDICATION message contains information concerning one or more successful random access procedures occurring at the gNB-DU and not known to the gNB-CU as described in TS 38.401 [4].

Upon reception of the RACH INDICATION message, the gNB-CU may trigger retrieval of RACH Reports from the UE.

8.2.14.3 Abnormal Conditions

Not applicable.

8.2.15 CLI Indication

8.2.15.1 General

This procedure is initiated by the gNB-DU or the gNB-CU to report the result of gNB-to-gNB CLI measurements, the need for CLI mitigation and to indicate the need for SRS Resource Configuration information.

The procedure uses non UE-associated signalling.

8.2.15.2 Successful Operation

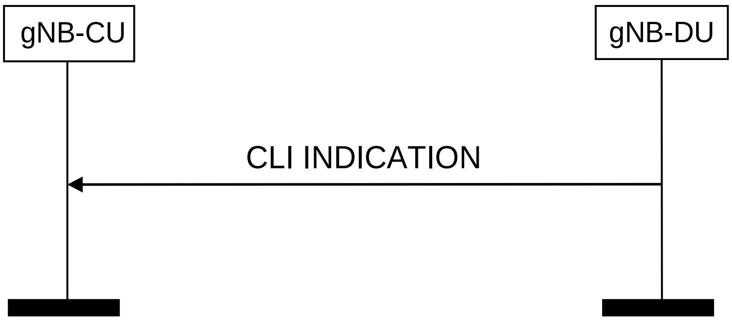

Figure 8.2.15.2-1: CLI Indication initiated from the gNB-DU, successful operation

The gNB-DU initiates the procedure by sending the CLIT INDICATION message to the gNB-CU. The gNB-DU reports the results of the gNB-to-gNB CLI measurements, possible need for gNB-to-gNB CLI mitigation and SRS Resource Indication in CLI INDICATION message to the gNB-CU.

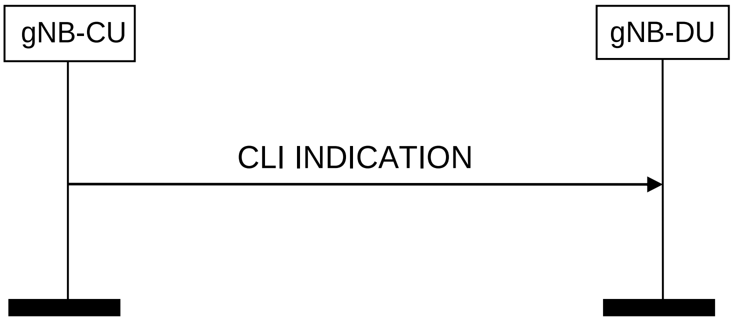

Figure 8.2.15.2-2: CLI Indication initiated from the gNB-CU, successful operation

The gNB-CU initiates the procedure by sending the CLI INDICATION message to the gNB-DU. The gNB-CU forwards the received results of the gNB-to-gNB CLI measurements, possible need for gNB-to-gNB CLI mitigation and SRS Resource Indication in the CLI INDICATION message to the gNB-DU.

8.3 UE Context Management procedures

8.3.1 UE Context Setup

8.3.1.1 General

The purpose of the UE Context Setup procedure is to establish the UE Context including, among others, SRB, DRB, BH RLC channel, Uu Relay RLC channel, PC5 Relay RLC channel, and SL DRB configuration. The procedure uses UE-associated signalling.

8.3.1.2 Successful Operation

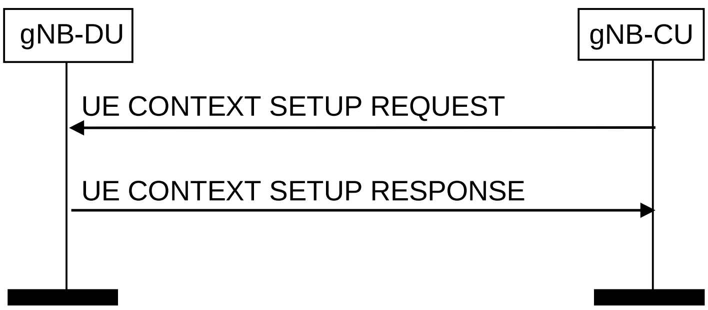

Figure 8.3.1.2-1: UE Context Setup Request procedure: Successful Operation

The gNB-CU initiates the procedure by sending UE CONTEXT SETUP REQUEST message to the gNB-DU. If the gNB-DU succeeds to establish the UE context, it replies to the gNB-CU with UE CONTEXT SETUP RESPONSE. If no UE-associated logical F1-connection exists, the UE-associated logical F1-connection shall be established as part of the procedure. Except for RACH based SDT and UE configured with BWP specific ServingCellMO, the gNB-CU shall perform RRC Reconfiguration or RRC connection resume to send UE to the RRC_CONNECTED state as described in TS 38.331 [8], and in this case, the *CellGroupConfig* IE shall transparently be signaled to the UE as specified in TS 38.331 [8]. In the cases of RACH based SDT procedure and UE configured with BWP specific ServingCellMO, the *CellGroupConfig* IE shall be ignored by the gNB-CU.

If the *UE-CapabilityRAT-ContainerList* IE is included in the UE CONTEXT SETUP REQUEST, the gNB-DU shall take this information into account for UE specific configurations.

If the *servingCellMO* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall configure servingCellMO for the indicated SpCell accordingly.

If the *servingCellMO List* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, select servingCellMO after determining the list of BWPs for the UE and include the list of servingCellMOs that have been encoded in *CellGroupConfig* IE as *ServingCellMO-encoded-in-CGC List* IE in the UE CONTEXT SETUP RESPONSE message.

If the *Configured BWP List* IE is included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, take it into account when requesting the gNB-DU for generating preconfigured measurement GAP for the indicated BWPs.

If the *SpCell UL Configured* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall configure UL for the indicated SpCell accordingly.

If the *SCell To Be Setup List* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall consider it as a list of candidate SCells to be set up. If the *SCell UL Configured* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall configure UL for the indicated SCell accordingly. If the *servingCellMO* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall configure servingCellMO for the indicated SCell accordingly. If the *servingCellMO-On-demand* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, configure the servingCellMO for on-demand SSB for the indicated SCell accordingly.

If the *DRX Cycle* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall use the provided value from the gNB-CU.

If the *Non-Integer DRX Cycle* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use the provided value from the gNB-CU.

If the *UL Configuration* IE in *DRB to Be Setup Item* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall take it into account for UL scheduling.

If the *SRB To Be Setup List* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall act as specified in TS 38.401 [4]. If *Duplication Indication* IE is contained in the *SRB To Be Setup List* IE, the gNB-DU shall, if supported, setup two RLC entities for the indicated SRB. If the *Additional Duplication Indication* IE is contained in the *SRB To Be Setup List* IE, the gNB-DU shall, if supported, setup the indicated RLC entities for the indicated SRB. If the *SDT RLC Bearer Configuration* IE is contained in the *SRB To Be Setup List* IE, the gNB-DU shall, if supported, use it for packet transmission belonging to the SDT SRB indicated by the *SRB ID* IE. If the *SRB Mapping Info* IE is contained in the *SRB To Be Setup List* IE, the gNB-DU shall, if supported, store the mapping information indicated in the S*RB Mapping Info* IE for the SRB identified by the *SRB ID* IE and the Uu Relay RLC channel identified by the *SRB Mapping Info* IE. The gNB-DU shall use the mapping information stored for the mapping of SRB data to Uu Relay RLC channel.

If the *DRB To Be Setup List* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall act as specified in TS 38.401 [4]. If the *QoS Flow Mapping Indication* IE is included in the *DRB To Be Setup List* IE for a QoS flow, the gNB-DU may take it into account that only the uplink or downlink QoS flow is mapped to the indicated DRB. If the *SDT RLC Bearer Configuration* IE is contained in the *DRB To Be Setup List* IE, the gNB-DU shall, if supported, use it for packet transmission belonging to the SDT DRB indicated by the *DRB ID* IE. If the *DRB Mapping Info* IE is contained in the *DRB To Be Setup List* IE, the gNB-DU shall, if supported, store the mapping information indicated in the *DRB Mapping Info* IE for the DRB identified by the *DRB ID* IE and the Uu Relay RLC channel identified by the *DRB Mapping Info* IE. The gNB-DU shall use the mapping information stored for the mapping of DRB data to Uu Relay RLC channel.

If the *PSI based SDU Discard UL* IE is included in the *DRB To Be Setup List* IE, the gNB-DU shall, if supported, take it into account to perform UL PSI based SDU discarding activation or deactivation for the indicated DRB as defined in TS 38.321 [16].

If the *PSI based SDU Discard DL* IE is included in the *DRB To Be Setup List* IE, the gNB-DU shall, if supported, take it into account to provide suggestion on whether to activate or deactivate DL PSI based SDU discarding for the indicated DRB.

For each GBR DRB, if the *Alternative QoS Parameters Sets* IE is included in the *GBR QoS Flow Information* IE in the UE CONTEXT SETUP REQUEST message, gNB-DU shall, if supported, behave the same as the NG-RAN node in the PDU Session Resource Setup procedure, specified in TS 38.413 [3].

If the *BH Information* IE is included in the *UL UP TNL Information to be setup List* IE or the *Additional PDCP Duplication TNL List* IE for a DRB, the gNB-DU shall, if supported, use the indicated BAP Routing ID and BH RLC channel for transmission of the corresponding GTP-U packets to the IAB-donor, as specified in TS 38.340 [30].

If the *BH RLC Channel To Be Setup List* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall act as specified in TS 38.401 [4]. If the *Traffic Mapping Information* IE is included in the *BH RLC Channel To Be Setup Item IEs* IE for a BH RLC Channel, the gNB-DU shall, if supported, process the *Traffic Mapping Information* IE as follows:

- if the *IP to layer2 Traffic Mapping Info* IE is included, the gNB-DU shall store the mapping information contained in the *IP to layer2 Traffic Mapping Info To Add* IE, if present, for the egress BH RLC channel identified by the *BH RLC CH ID* IE, and shall remove the previously stored mapping information as indicated by the *IP to layer2 Mapping Traffic Info To Remove* IE, if present. The gNB-DU shall use the mapping information stored for the mapping of IP traffic to layer 2, as specified in TS 38.340 [30].

- if the *BAP layer BH RLC channel Mapping Info* IE is included, the gNB-DU shall store the mapping information contained in the *BAP layer BH RLC channel Mapping Info To Add* IE, if present, for the egress or ingress BH RLC channel identified by the *BH RLC CH ID* IE, and shall remove the previously stored mapping information as indicated by the *BAP layer BH RLC channel Mapping Info To Remove* IE, if present. The gNB-DU shall use the mapping information stored when forwarding traffic on BAP sublayer, as specified in TS 38.340 [30].

If two *UL UP TNL Information* IEs are included in UE CONTEXT SETUP REQUEST message for a DRB, gNB-DU shall include two *DL UP TNL Information* IEs in UE CONTEXT SETUP RESPONSE message and setup two RLC entities for the indicated DRB. gNB-CU and gNB-DU use the *UL UP TNL Information* IEs and *DL UP TNL Information* IEs to support packet duplication for intra-gNB-DU CA as defined in TS 38.470 [2]. The first *UP TNL Information* IE of the two *UP TNL Information* IEs is for the primary path.

If one or two *Additional PDCP Duplication UP TNL Information* IEs are included in the UE CONTEXT SETUP REQUEST message for a DRB, the gNB-DU shall, if supported, include one or two *Additional PDCP Duplication UP TNL Information* IEs in the UE CONTEXT SETUP RESPONSE message and setup one or two additional RLC entities for the indicated DRB. The gNB-CU and the gNB-DU use the *Additional PDCP Duplication UP TNL Information* IEs to support packet duplication for intra-gNB-DU CA as defined in TS 38.470 [2].

If *Duplication Activation IE* is included in the UE CONTEXT SETUP REQUEST message for a DRB, gNB-DU should take it into account when activating/deactivating CA based PDCP duplication for the DRB. If the *RLC Duplication State List* IE is included in the *RLC Duplication Information* IE contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, take it into account when activating/deactivating CA based PDCP duplication for the DRB with more than two RLC entities.

If *DC Based Duplication Configured* IE is included in the UE CONTEXT SETUP REQUEST message for a DRB, gNB-DU shall regard that DC based PDCP duplication is configured for this DRB if the value is set to be "true" and it should take the responsibility of PDCP duplication activation/deactivation. If *DC Based Duplication Activation* IE is included in the UE CONTEXT SETUP REQUEST message for a DRB, gNB-DU should take it into account when activating/deactivating DC based PDCP duplication for this DRB. If the *RLC Duplication State List* IE is included in the *RLC Duplication Information* IE contained in the UE CONTEXT SETUP REQUEST message for a DRB, the gNB-DU shall, if supported, take it into account when activating/deactivating DC based PDCP duplication for the DRB with more than two RLC entities. If the *Primary Path Indication* IE is included in the *RLC Duplication Information* IE, the gNB-DU shall, if supported, take it into account when performing DC based PDCP duplication for the DRB with more than two RLC entities.

If *UL PDCP SN length* IE is included in the UE CONTEXT SETUP REQUEST message for a DRB, gNB-DU shall, if supported, store this information and use it for lower layer configuration.

For EN-DC operation, and if the *Subscriber Profile ID for RAT/Frequency priority* IE is received from an MeNB, the UE CONTEXT SETUP REQUEST message shall contain the *Subscriber Profile ID for RAT/Frequency priority* IE. If the *Additional RRM Policy Index* IE is received from an MeNB, the UE CONTEXT SETUP REQUEST message shall, if supported, contain the *Additional RRM Policy Index* IE. The gNB-DU shall store the received Subscriber Profile ID for RAT/Frequency priority in the UE context and use it as defined in TS 36.300 [20]. The gNB-DU shall, if supported, store the received Additional RRM Policy Index in the UE context and use it as defined in TS 36.300 [20].

If the *Index to RAT/Frequency Selection Priority* IE is available at the gNB-CU, the *Index to RAT/Frequency Selection Priority* IE shall be included in the UE CONTEXT SETUP REQUEST. The gNB-DU may use it for RRM purposes.

The gNB-DU shall report to the gNB-CU, in the UE CONTEXT SETUP RESPONSE message, the result for all the requested DRBs, SRBs, BH RLC channels, PC5 Relay RLC channels, and SL DRBs in the following way:

- A list of DRBs which are successfully established shall be included in the *DRB Setup List* IE;

- A list of DRBs which failed to be established shall be included in the *DRB Failed to Setup List* IE;
- A list of SRBs which failed to be established shall be included in the *SRB Failed to Setup List* IE.
- A list of successfully established SRBs with logical channel identities for primary path shall be included in the *SRB Setup List* IE only if CA based PDCP duplication is initiated for the concerned SRBs.
- A list of BH RLC channels which are successfully established shall be included in the *BH RLC Channel Setup List* IE;
- A list of BH RLC channels which failed to be established shall be included in the *BH RLC Channel Failed to be Setup List* IE;
- A list of SL DRBs which are successfully established shall be included in the *SL DRB Setup List* IE;
- A list of SL DRBs which failed to be established shall be included in the *SL DRB Failed to Setup List* IE.
- A list of Uu Relay RLC channels which are successfully established shall be included in the *Uu RLC Channel Setup List* IE;
- A list of Uu Relay RLC channels which failed to be established shall be included in the *Uu RLC Channel Failed to be Setup List* IE;
- A list of PC5 Relay RLC channels which are successfully established shall be included in the *PC5 RLC Channel Setup List* IE;
- A list of PC5 Relay RLC channels which failed to be established shall be included in the *PC5 RLC Channel Failed to be Setup List* IE;

If *Duplication Indication* IE in *SL DRB To Be Setup List* IE is contained in the UE CONTEXT SETUP REQUEST message,the gNB-DU shall, if supported, generate two PC5 RLC bearer configurations for the indicated SL DRB.

When the gNB-DU reports the unsuccessful establishment of a DRB or SRB or SL DRB or a BH RLC channel or a PC5 Relay RLC channel, the cause value should be precise enough to enable the gNB-CU to know the reason for the unsuccessful establishment.

For EN-DC operation, the gNB-CU shall include in the UE CONTEXT SETUP REQUEST the *E-UTRAN QoS* IE. The allocation of resources according to the values of the *Allocation and Retention Priority* IE included in the *E-UTRAN QoS* IE shall follow the principles described for the E-RAB Setup procedure in TS 36.413 [15].

For NG-RAN operation, the gNB-CU shall include in the UE CONTEXT SETUP REQUEST the *DRB Information* IE.

For DC operation, the *CG-ConfigInfo* IE shall be included in the *CU to DU RRC Information* IE at the gNB acting as secondary node. If the *CG-ConfigInfo* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall regard it as a reconfiguration with sync as defined in TS 38.331 [8].

For sidelink operation, the *CG-ConfigInfo* IE shall be included in the *CU to DU RRC Information* IE if the gNB-CU receives sidelink related UE information from UE. If the *CG-ConfigInfo* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall regard it as an indication of V2X sidelink information or NR sidelink information as defined in TS 38.331 [8].

If the *HandoverPreparationInformation* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU of the gNB acting as master node shall regard it as a reconfiguration with sync as defined in TS 38.331 [8]. The gNB-CU shall only initiate the UE Context Setup procedure for handover or secondary node addition when at least one DRB is setup for the UE, or at least one BH RLC channel is set up for IAB-MT. If the *HandoverPreparationInformation* IE containing the sidelink related UE information is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall regard it as an indication of V2X sidelink information or NR sidelink information as defined in TS 38.331 [8].

If the received *CU to DU RRC Information* IE does not include source cell group configuration, the gNB-DU shall generate the cell group configuration using full configuration. Otherwise, delta configuration is allowed.

If the gNB-CU includes the SMTC information of the measured frequency(ies) in the *MeasurementTimingConfiguration* IE of the *CU to DU RRC Information* IE that is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall generate the measurement gaps based on the received SMTC information. Then the gNB-DU shall send the measurement gaps information to the gNB-CU in the *MeasGapConfig* IE of the *DU to CU RRC Information* IE that is included in the UE CONTEXT SETUP RESPONSE message.

If the *MeasConfig* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall deduce that changes to the measurements configuration need to be applied. If the *measObjectToAddModList* IE is included in the *MeasConfig* IE, then the frequencies added in such IE are to be activated. Then the gNB-DU shall decide if measurement gaps are needed or not and, if needed, the gNB-DU shall send the measurement gaps information to the gNB-CU in the *MeasGapConfig* IE of the *DU to CU RRC Information* IE that is included in the UE CONTEXT SETUP RESPONSE message. If the *measObjectToRemoveList* IE is included in the *MeasConfig* IE, the gNB-DU shall ignore it.

If the *NeedForGapsInfoNR* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it as described in TS 38.331 [8]. If the *NeedForGapNCSG-InfoNR* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it as described in TS 38.331 [8].

If the *NeedForGapNCSG-InfoEUTRA* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it as described in TS 38.331 [8].

If the *NeedForInterruptionInfoNR* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it as described in TS 38.331 [8].

For EN-DC operation, if the gNB-CU includes the *Resource Coordination Transfer Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it for the purpose of resource coordination. If the *Ignore PRACH Configuration* IE is present and set to "true" the *E-UTRA PRACH Configuration* IE in the UE CONTEXT SETUP REQUEST message shall be ignored. If the gNB-CU received the MeNB Resource Coordination Information as defined in TS 36.423 [9], it shall transparently transfer it to the gNB-DU via the *Resource Coordination Transfer Container* IE in the UE CONTEXT SETUP REQUEST message. The gNB-DU shall use the information received in the *Resource Coordination Transfer Container* IE for reception of MeNB Resource Coordination Information at the gNB acting as secondary node as described in TS 36.423 [9]. If the *Resource Coordination E-UTRA Cell Information* IE is included in the *Resource Coordination Transfer Information* IE, the gNB-DU shall store the information replacing previously received information for the same E-UTRA cell, and use the stored information for the purpose of resource coordination.

For NGEN-DC or NE-DC operation, if the gNB-CU includes the *Resource Coordination Transfer Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it for the purpose of resource coordination. If the gNB-CU received the MR-DC Resource Coordination Information as defined in TS 38.423 [28], it shall transparently transfer it to the gNB-DU via the *Resource Coordination Transfer Container* IE in the UE CONTEXT SETUP REQUEST message. The gNB-DU shall use the information received in the *Resource Coordination Transfer Container* IE for reception of MR-DC Resource Coordination Information at the gNB as described in TS 38.423 [28].

The *UEAssistanceInformation* IE shall be included in *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message if the gNB-CU received this IE from the UE; if the *UEAssistanceInformation* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, take it into account when configuring resources for the UE.

The *UEAssistanceInformationEUTRA* IE shall be included in *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message if the gNB-CU received this IE from the UE; if the *UEAssistanceInformationEUTRA* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, take it into account when configuring LTE sidelink resources for the UE.

If the *Resource Coordination Transfer Container* IE is included in the UE CONTEXT SETUP RESPONSE, the gNB-CU shall transparently transfer this information for the purpose of resource coordination as described in TS 36.423 [9], TS 38.423 [28].

If the *Masked IMEISV* IE is contained in the UE CONTEXT SETUP REQUEST message the gNB-DU shall, if supported, use it to determine the characteristics of the UE for subsequent handling.

If the *SCell Failed To Setup List* IE is contained in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall regard the corresponding SCell(s) failed to be set up with an appropriate cause value for each SCell failed to setup.

If the *Inactivity Monitoring Request* IE is contained in the UE CONTEXT SETUP REQUEST message, gNB-DU may consider that the gNB-CU has requested the gNB-DU to perform UE inactivity monitoring. If the *Inactivity Monitoring*

Response IE is contained in the UE CONTEXT SETUP RESPONSE message and set to "Not-supported", the gNB-CU shall consider that the gNB-DU does not support UE inactivity monitoring for the UE.

If the *ServCellInfoList* IE is included in the *DU to CU RRC Information* IE contained in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall take it into account to generate the content of inter-node RRC message, i.e., *CG-Config* or *CG-ConfigInfo*, as described in TS 38.331 [8].

If the *Full Configuration* IE is contained in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall consider that the gNB-DU has generated the *CellGroupConfig* IE using full configuration.

If the *C-RNTI* IE is included in the UE CONTEXT SETUP RESPONSE, the gNB-CU shall consider that the C-RNTI has been allocated by the gNB-DU for this UE context.

The UE Context Setup Procedure is not used to configure SRB0.

If the UE CONTEXT SETUP REQUEST message contains the *RRC-Container* IE, the gNB-DU shall send the corresponding RRC message to the UE via SRB1.

If the *Notification Control* IE is included in the *DRB to Be Setup List* IE contained in the UE CONTEXT SETUP REQUEST message and it is set to active, the gNB-DU shall, if supported, monitor the QoS of the DRB and notify the gNB-CU if the QoS cannot be fulfilled any longer or if the QoS can be fulfilled again. The *Notification Control* IE can only be applied to GBR bearers.

If the *UL PDU Session Aggregate Maximum Bit Rate* IE is included in the *QoS Flow Level QoS Parameters* IE contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall store the received UL PDU Session Aggregate Maximum Bit Rate and use it when enforcing uplink traffic policing for non-GBR Bearers for the concerned UE as specified in TS 23.501 [21].

The gNB-DU shall store the received gNB-DU UE Aggregate Maximum Bit Rate Uplink and use it for non-GBR Bearers for the concerned UE.

If the UE CONTEXT SETUP REQUEST message contains the *QoS Flow Mapping Indication* IE, the gNB-DU may take it into account that only the uplink or downlink QoS flow is mapped to the DRB.

If the UE CONTEXT SETUP REQUEST message contains the *New gNB-CU UE F1AP ID* IE, the gNB-DU shall, if supported, replace the value received in the *gNB-CU UE F1AP ID* IE by the value of the *New gNB-CU UE F1AP ID* and use it for further signalling.

If the *RAN UE ID* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall store and replace any previous information received.

If the *Trace Activation* IE is included in the UE CONTEXT SETUP REQUEST message the gNB-DU shall, if supported, initiate the requested trace function as described in TS 32.422 [29].

In particular, the gNB-DU shall, if supported:

- if the *Trace Activation* IE includes the *MDT Activation* IE set to "Immediate MDT and Trace", initiate the requested trace session and MDT session as described in TS 32.422 [29];
- if the *Trace Activation* IE includes the *MDT Activation* IE set to "Immediate MDT Only", initiate the requested MDT session as described in TS 32.422 [29] and the gNB-DU shall ignore *Interfaces To Trace* IE, and *Trace Depth* IE. If the *Management Based MDT PLMN List* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, store the received information in the UE context, and use this information to allow subsequent selection of the UE for management based MDT defined in TS 32.422 [29].

For each QoS flow whose DRB has been successfully established and the *QoS Monitoring Request* IE was included in the *QoS Flow Level QoS Parameters* IE contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall store this information, and, if supported, perform delay measurement and QoS monitoring, as specified in TS 23.501 [21].

If the UE CONTEXT SETUP REQUEST message contains the *Configured BAP Address* IE, the gNB-DU shall, if supported, store this BAP address configured for the corresponding child IAB-node and use it as specified in TS 38.340 [30].

If the *BAP Control PDU Channel* IE is included in the *BH RLC Channel to be Setup List* IE, the gNB-DU shall, if supported, consider that the configured BH RLC channel can be used to transmit BAP Control PDUs, and use this BH RLC channel as specified in TS 38.340 [30].

If the *F1-C Transfer Path* IE is included in UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, take it into account.

If the *NR V2X Services Authorized* IE is contained in the UE CONTEXT SETUP REQUEST message and it contains one or more IEs set to "authorized", the gNB-DU node shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *LTE V2X Services Authorized* IE is contained in the UE CONTEXT SETUP REQUEST message and it contains one or more IEs set to "authorized", the gNB-DU node shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *NR UE Sidelink Aggregate Maximum Bit Rate* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it for the concerned UE's sidelink communication in network scheduled mode for NR V2X services.

If the *LTE UE Sidelink Aggregate Maximum Bit Rate* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it for the concerned UE's sidelink communication in network scheduled mode for LTE V2X services.

If the *PC5 Link Aggregate Bit Rate* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it for the concerned UE's sidelink communication in network scheduled mode for NR V2X services as defined in TS 23.287 [40].

If the *NR A2X Services Authorized* IE is contained in the UE CONTEXT SETUP REQUEST message and it contains one or more IEs set to "authorized", the gNB-DU node shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *LTE A2X Services Authorized* IE is contained in the UE CONTEXT SETUP REQUEST message and it contains one or more IEs set to "authorized", the gNB-DU node shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *NR UE Sidelink Aggregate Maximum Bit Rate for A2X* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it for the concerned UE's sidelink communication in network scheduled mode for NR A2X services.

If the *LTE UE Sidelink Aggregate Maximum Bit Rate for A2X* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it for the concerned UE's sidelink communication in network scheduled mode for LTE A2X services.

If the *TSC Traffic Characteristics* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, take into account the corresponding information received in the *TSC Traffic Characteristics* IE. If the *RAN Feedback Type* IE is included in the *TSC Assistance Information Uplink* IE of the *TSC Traffic Characteristics* IE, the gNB-DU shall, if supported, take this information into account when determining the feedback to provide in the *TSC Traffic Characteristics Feedback* IE in the UE CONTEXT SETUP RESPONSE message.

If the *Conditional Inter-DU Mobility Information* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall consider that the request concerns a conditional handover, conditional PSCell addition, conditional PSCell change, or subsequent CPAC for the included *SpCell ID* IE and shall include it as the *Requested Target Cell ID* IE in the UE CONTEXT SETUP RESPONSE message. The gNB-DU shall regard it as a reconfiguration with sync as defined in TS 38.331 [8].

If the *Target gNB-DU UE F1AP ID* IE is contained in the *Conditional Inter-DU Mobility Information* IE included in the UE CONTEXT SETUP REQUEST message, then the gNB-DU shall replace the existing prepared conditional handover, conditional PSCell addition or conditional PSCell change, or subsequent CPAC identified by the *Target gNB-DU UE F1AP ID* IE and the *SpCell ID* IE.

If the *Serving NID* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall combine the *Serving NID* IE with the *Serving PLMN* IE to identify the serving NPN, and may take it into account for UE context establishment.

If the *Estimated Arrival Probability* IE is contained in the *Conditional Inter-DU Mobility Information* IE included in the UE CONTEXT SETUP REQUEST message, then the gNB-DU may use the information to allocate necessary resources for the UE.

If the *S-CPAC Request* IE is included within the *Conditional Inter-DU Mobility Information* IE in the UE CONTEXT SETUP REQUEST message and is set to "initiation", the gNB-DU shall, if supported, consider that the procedure is triggered for S-CPAC preparation.

If for a given E-RAB for EN-DC operation the *ENB DL Transport Layer Address* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

If for a given Qos flow for NG-RAN operation the *PDCP Terminating Node DL Transport Layer Address* IE is included in the UE CONTEXT SETUP REQUEST message, then the gNB-DU shall, if supported, use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

If the *F1-C Transfer Path NRDC* IE is included in UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, take it into account.

If the *MDT Polluted Measurement Indicator* IE is included in the UE CONTEXT SETUP REQUEST, the gNB-DU shall take this information into account as specified in TS 38.401 [4].

If the *SCG Activation Request* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU may use it to configure SCG resources as specified in TS 37.340 [7], and if supported, shall include the *SCG Activation Status* IE in the UE CONTEXT SETUP RESPONSE message. If the *SCG Activation Request* IE in the UE CONTEXT SETUP REQUEST message is set to “Activate SCG”, the gNB-DU shall activate the SCG resources and set the *SCG Activation Status* IE in the UE CONTEXT SETUP RESPONSE message to “SCG Activated”.

If the *Old CG-SDT Session Info* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, retrieve the old CG-SDT resource configuration and old UE context based on the indicated gNB-CU F1AP UE ID and gNB-DU F1AP UE ID.

If the *5G ProSe Authorized* IE is contained in the UE CONTEXT SETUP REQUEST message and it contains one or more IEs set to "authorized", the gNB-DU node shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *5G ProSe UE PC5 Aggregate Maximum Bit Rate* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it for the concerned UE's sidelink communication in network scheduled mode for 5G ProSe services.

If the *5G ProSe PC5 Link Aggregate Bit Rate* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it for the concerned UE's sidelink communication in network scheduled mode for 5G ProSe services as defined in TS 23.304 [44].

If the *PC5 RLC Channel To Be Setup List* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, act as specified in TS 38.401 [4]. gNB-DU generates the PC5 Relay RLC channel configurations for a L2 U2N Remote UE, a L2 U2U Remote UE or a L2 U2U Relay UE.

If the *Path Switch Configuration* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it to configure the path switch from direct path to single-hop or multi-hop indirect path, from single-hop indirect path to single-hop or multi-hop indirect path, or from multi-hop indirect path to single-hop indirect path as specified in TS 38.401 [4] and TS 38.300 [6].

If the *MUSIM-GapConfig* IE is contained in the *CU to DU RRC Information* IE included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, decide to use this IE for MUSIM gap configuration or select another one based on the received *UEAssistanceInformation* IE. If gNB-DU selects a different MUSIM gap configuration from received *UEAssistanceInformation* IE, then it shall include the selected MUSIM gap information to the gNB-CU in the *MUSIM-GapConfig* IE of the *DU to CU RRC Information* IE that is included in the UE CONTEXT SETUP RESPONSE message.

If *MUSIM-GapConfig* IE is not contained in the *CU to DU RRC Information* IE, then gNB-DU shall, if supported, send the selected MUSIM gap configuration based on the received *UEAssistanceInformation* IE, to the gNB-CU in the *MUSIM-GapConfig* IE of the *DU to CU RRC Information* IE that is included in the UE CONTEXT SETUP RESPONSE message. When *MUSIM-GapConfig* IE is received, the gNB-CU should use this value.

If the *gNB-DU UE Slice Maximum Bit Rate List* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, store and use the information for the uplink traffic policing for each concerned slice as specified in TS 23.501 [21].

If the *Multicast MBS Session Setup List* IE is contained in the UE CONTEXT SETUP REQUEST message the gNB-DU shall, if supported, store and use the information for configuring MBS Session Resources, if applicable.

If the *UE Multicast MRB To Be Setup List* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, take it into account for configuring MBS Session Resources, if applicable. And if the *MBS PTP Retransmission Tunnel Required* IE is included in the *UE Multicast MRB to Be Setup Item IEs* IE, the gNB-DU shall, if supported trigger the establishment of the MBS PTP Retransmission F1-U tunnel. If the *MBS PTP Forwarding Tunnel Required Information* IE is included in the *UE Multicast MRB to Be Setup Item IEs* IE, the gNB-DU shall, if supported trigger the establishment of the MBS PTP Forwarding F1-U tunnel. If the *Source MRB ID* IE is included in the *UE Multicast MRB to Be Setup Item IEs* IE, the DU shall, if supported, use it to identify the MRB configuration as provided to the UE in the source cell and take it into account for configuring MBS Session Resources.

If the *Dedicated SI Delivery Indication* IE is included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, take it into account for the system information delivery to the UE as described in TS 38.331 [8].

If the *PDU Set QoS Parameters* IE or the *DL PDU Set Information Marking Support Indication* IE set to "true" is included in the *QoS Flow Level QoS Parameters* IE contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, store this information and use it as specified in TS 23.501 [21].

If the *ECN Marking or Congestion Information Reporting Request* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it accordingly for the specific DRB. If the *ECN Marking or Congestion Information Reporting Status* IE is included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, use it to deduce if ECN marking or congestion information reporting is active or not active.

If the *InterFrequencyConfig-NoGap* IE is included in the *DU to CU RRC Information* IE contained in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, use it as described in TS 38.331 [8].

If the *ul-GapFR2-Config* IE is contained in the *DU to CU RRC Information* IE that is included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, use it as described in TS 38.331 [8].

If the *TwoPHRModeMCG* IE or the *TwoPHRModeSCG* IE is contained in the *DU to CU RRC Information* IE that is included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, use this value as described in TS 38.331 [8].

If the *MBSInterestIndication* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, take it into account when configuring resources for the UE.

If the *ncd-SSB-RedCapInitialBWP-SDT* IE is contained in the *DU to CU RRC Information* IE that is included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, use it as described in TS 38.331 [8].

If the *Network Controlled Repeater Authorized* IE is contained in the UE CONTEXT SETUP REQUEST message and it is set to "authorized", the gNB-DU node shall, if supported, consider that the UE is authorized as Network Controlled Repeater.

If the *musim-CapabilityRestrictionIndication* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it as described in TS 38.331 [8].

If the *LTM Indicator* IE set to "true" is contained in the *LTM Information Setup* IE included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, consider that the request concerns LTM for the included *SpCell ID* IE and shall include it as the *Requested Target Cell ID* IE in the UE CONTEXT SETUP RESPONSE message. The gNB-DU shall regard it as a reconfiguration with sync as defined in TS 38.331 [8].

If the *LTM Indicator* IE set to "C-LTM" is contained in the *LTM Information Setup* IE included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, consider that the request concerns conditional LTM for the included *SpCell ID* IE and shall include it as the *Requested Target Cell ID* IE in the UE CONTEXT SETUP RESPONSE message.

If the *LTM Indicator* IE is set to "C-LTM" and the *Request for L1 Execution Condition* IE is present in the *LTM Information Setup* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall generate the conditional LTM L1 execution condition(s) for the candidate cell(s) included in the *Request for L1 Execution Condition* IE and provide the *L1 Execution Condition List* IE within the *LTM Configuration* IE in the UE CONTEXT SETUP RESPONSE message.

If the *Request for Lower Layer Configuration* IE set to "true" is contained within the *Reference Configuration* IE in the *LTM Information Setup* IE included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, provide the lower layer configuration in the *Reference Configuration Information* IE in the *LTM Configuration* IE in the UE CONTEXT SETUP RESPONSE message for the gNB-CU to generate the LTM reference configuration.

If the *Reference Configuration Information* IE is contained within the *Reference Configuration* IE in the *LTM Information Setup* IE included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, take it into account for generating the LTM lower layer configuration.

If the *CSI Resource Configuration* is contained in the *LTM Information Setup* IE included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it to generate the LTM CSI reporting configuration(s) in the *CellGroupConfig* IE for the requested LTM candidate cell.

If the *Request for CSI-RS Resource Configuration for L1 measurements* IE set to "true" is contained in the *LTM Information Setup* IE included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, include the *CSI-RS Resource Configuration for L1 measurements* IE in the *LTM Configuration* IE in the UE CONTEXT SETUP RESPONSE message.

If the *CSI-RS Resource Configuration for Early CSI Acquisition* IE is included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, act as specified in TS 38.401 [4].

If the *LTM Configuration ID Mapping List* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, consider this as the mapping information for the LTM candidate cell(s).

If the *Early Sync Information Request* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, include the *Early Sync Information* IE of the accepted candidate cell for early TA acquisition (early UL synchronisation), in the UE CONTEXT SETUP RESPONSE message. If the *Early UL Sync Configuration* IE is included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, consider it as the generated early UL sync information from the accepted candidate cell in the gNB-DU. If the *Early UL Sync Configuration for SUL* IE is included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, consider it as the generated early UL sync information for SUL from the accepted candidate cell in the gNB-DU.

If the *LTM Configuration* IE is included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, consider it as the generated configuration for LTM from the accepted candidate cell in the candidate gNB-DU.

If the *Complete Candidate Configuration Indicator* IE set to "complete" is contained in the *LTM Configuration* IE included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, consider that the LTM candidate configuration is a complete candidate configuration.

If the *LTM with SCG Indicator* IE set to “true” is contained in the *LTM Information SN Addition* IE included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, consider that the UE Context Setup procedure has been triggered as part of an MCG LTM. The gNB-DU shall consider that the request concerns a PSCell addition or change for the included *SpCell ID* IE and shall include it as the *Requested Target Cell ID* IE in the UE CONTEXT SETUP RESPONSE message. The gNB-DU shall regard it as a reconfiguration with sync as defined in TS 38.331 [8].

If the *Direct Path Addition* IE is contained in the *Path Addition Information* IE which is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, consider that the request concerns the direct path addition for the included *SpCell ID* IE as specified in TS 38.401 [4] and regard it as a reconfiguration with sync as defined in TS 38.331 [8].

If the *Indirect Path Addition* IE is contained in the *Path Addition Information* IE which is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, consider that the request concerns the indirect path addition for the MP Remote UE using PC5 link and use it as specified in TS 38.401 [4].

If the *N3C Indirect Path Addition* IE is contained in the *Path Addition Information* IE, the gNB-DU shall, if supported, consider that the request concerns the indirect path addition for the MP Remote UE using N3C and use it as specified in TS 38.401 [4].

If the *S-CPAC Lower Layer Reference Config Request* IE set to "true" is contained in the *Conditional Inter-DU Mobility Information* IE included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, provide the lower layer configuration in the *Reference Configuration Information* IE in the *S-CPAC Configuration* IE in the UE CONTEXT SETUP RESPONSE message for the gNB-CU to generate the S-CPAC reference configuration.

If the *Complete Candidate Configuration Indicator* IE set to "complete" is contained in the *S-CPAC Configuration* IE included in the UE CONTEXT SETUP RESPONSE message, the gNB-CU shall, if supported, consider that the S-CPAC candidate configuration is a complete candidate configuration.

If the *musim-CandidateBandList* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use it for temporary capability restriction.

If the *DL LBT Failure Information Request* IE is included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, consider that the gNB- CU requests collection of DL LBT failure information for the analysis of the MRO events of the UE specified in TS 38.300 [6], and act as specified in TS 38.401 [4].

If the *Ranging and Sidelink Positioning Service Information* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, take it into account for the UE's Ranging and Sidelink Positioning service.

For each GBR QoS flow whose DRB has been successfully established and the *Monitoring Request on Available Bitrate* IE was included in the *GBR QoS Flow Information* IE contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, store this information and perform available bitrate monitoring, as specified in TS 23.501 [21].

For each QoS flow whose DRB to be established, if the *MMSID* IE was included in the *QoS Flow Level QoS Parameters* IE contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, consider that the QoS flow is related to a multi-modal service, as described in TS 23.501 [21] and TS 38.300 [6].

If the *Indication of Bitrate Adaptation* IE set to "uplink" is included in the *QoS Flow Level QoS Parameters* IE contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, store the received indication and use it as defined in TS 38.300 [6].

For each DRB that has been successfully established and for which the *Performance Delay Monitoring* IE was included in the *DRB to Be Setup List* IE contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, store this information and use it to perform delay measurements on the successfully established DRBs.

Interaction with UE Inactivity Notification procedure

If the *SDT Volume Threshold* IE is contained in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, use the information during an SDT transaction to inform the gNB-CU via the UE INACTIVITY NOTIFICATION message as specified in TS 38.401 [4].

8.3.1.3 Unsuccessful Operation

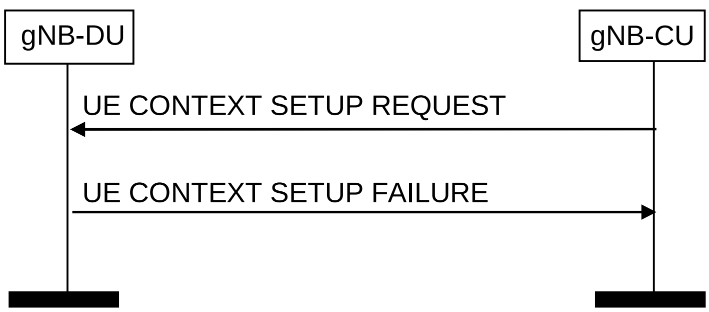

Figure 8.3.1.3-1: UE Context Setup Request procedure: unsuccessful Operation

If the gNB-DU is not able to establish an F1 UE context, or cannot even establish one bearer it shall consider the procedure as failed and reply with the UE CONTEXT SETUP FAILURE message. If the *Conditional Inter-DU Mobility Information* IE or the *LTM Indicator* IE was included in the UE CONTEXT SETUP REQUEST message, the gNB-DU shall include the received *SpCell ID* IE as the *Requested Target Cell ID* IE in the UE CONTEXT SETUP FAILURE message.

If the gNB-DU is not able to accept the *SpCell ID* IE in UE CONTEXT SETUP REQUEST message, it shall reply with the UE CONTEXT SETUP FAILURE message with an appropriate cause value. Further, if the *Candidate SpCell List* IE is included in the UE CONTEXT SETUP REQUEST message and the gNB-DU is not able to accept the *SpCell ID* IE, the gNB-DU shall, if supported, include the *Potential SpCell List* IE in the UE CONTEXT SETUP FAILURE message and the gNB-CU should take this into account for selection of an opportune SpCell. The gNB-DU shall include the cells in the *Potential SpCell List* IE in a priority order, where the first cell in the list is the one most desired and the last one is the one least desired (e.g., based on load conditions). If the *Potential SpCell List* IE is present but no *Potential SpCell Item* IE is present, the gNB-CU should assume that none of the cells in the *Candidate SpCell List* IE are acceptable for the gNB-DU.

8.3.1.4 Abnormal Conditions

If the gNB-DU receives a UE CONTEXT SETUP REQUEST message containing a *E-UTRAN QoS* IE for a GBR QoS DRB but where the *GBR QoS Information* IE is not present, the gNB-DU shall report the establishment of the corresponding DRB as failed in the *DRB Failed to Setup List* IE of the UE CONTEXT SETUP RESPONSE message with an appropriate cause value. If the gNB-DU receives a UE CONTEXT SETUP REQUEST message containing a *DRB QoS* IE for a GBR QoS DRB but where the *GBR QoS Flow Information* IE is not present, the gNB-DU shall report the establishment of the corresponding DRBs as failed in the *DRB Failed to Setup List* IE of the UE CONTEXT SETUP RESPONSE message with an appropriate cause value.

If the *Delay Critical* IE is included in the *Dynamic 5QI Descriptor* IE within the *DRB QoS* IE in the UE CONTEXT SETUP REQUEST message and is set to the value "delay critical" but the *Maximum Data Burst Volume* IE is not present, the gNB-DU shall report the establishment of the corresponding DRB as failed in the *DRB Failed to Setup List* IE of the of the UE CONTEXT SETUP RESPONSE message with an appropriate cause value.

In case of "CHO-replace" when the *Target gNB-DU UE F1AP ID* IE is included, if the candidate cell in the *SpCell ID* IE included in the UE CONTEXT SETUP REQUEST message was not prepared using the same UE-associated signaling connection, the gNB-DU shall ignore this candidate cell.

8.3.2 UE Context Release Request (gNB-DU initiated)

8.3.2.1 General

The purpose of the UE Context Release Request procedure is to enable the gNB-DU to request the gNB-CU to release the UE-associated logical F1-connection, candidate cells in conditional handover, conditional PSCell addition, conditional PSCell change, or LTM or subsequent CPAC. The procedure uses UE-associated signalling.

8.3.2.2 Successful Operation

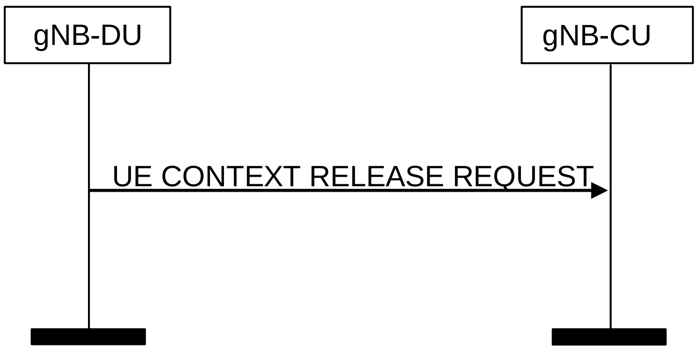

Figure 8.3.2.2-1: UE Context Release (gNB-DU initiated) procedure. Successful operation

The gNB-DU controlling a UE-associated logical F1-connection initiates the procedure by generating a UE CONTEXT RELEASE REQUEST message towards the affected gNB-CU node.

The UE CONTEXT RELEASE REQUEST message shall indicate the appropriate cause value.

If the *Candidate Cells To Be Cancelled List* IE is included in the UE CONTEXT RELEASE REQUEST message, the gNB-CU shall consider that the only the resources reserved for the candidate cells identified by the included NR CGIs and associated to the UE-associated signaling identified by the *gNB-CU UE F1AP ID* IE and the *gNB-DU UE F1AP ID* IE are about to be released by the gNB-DU.

If the *LTM Cells To Be Released List* IE is included in the UE CONTEXT RELEASE REQUEST message, the gNB-CU shall, if supported, consider that only the resources reserved for the LTM cells identified by the included NR CGIs and associated to the UE-associated signaling identified by the *gNB-CU UE F1AP ID* IE and the *gNB-DU UE F1AP ID* IE are about to be released by the gNB-DU.

Interactions with UE Context Release procedure:

The UE Context Release procedure may be initiated upon reception of a UE CONTEXT RELEASE REQUEST message.

Interactions with UE Context Setup procedure:

The UE Context Release Request procedure may be performed before the UE Context Setup procedure to request the release of an existing UE-associated logical F1-connection and related resources in the gNB-DU.

8.3.2.3 Abnormal Conditions

If one or more candidate cells in the *Candidate Cells To Be Cancelled List* IE included in the UE CONTEXT RELEASE REQUEST message were not prepared using the same UE-associated signaling connection, the gNB-CU shall ignore those non-associated candidate cells.

If one or more LTM cells in the *LTM Cells To Be Released List* IE included in the UE CONTEXT RELEASE REQUEST message were not prepared using the same UE-associated signaling connection, the gNB-CU shall ignore those non-associated LTM cells.

8.3.3 UE Context Release (gNB-CU initiated)

8.3.3.1 General

The purpose of the UE Context Release procedure is to enable the gNB-CU to order the release of the UE-associated logical connection, candidate cells in conditional handover, conditional PSCell addition, or conditional PSCell change or LTM or subsequent CPAC. The procedure uses UE-associated signalling.

8.3.3.2 Successful Operation

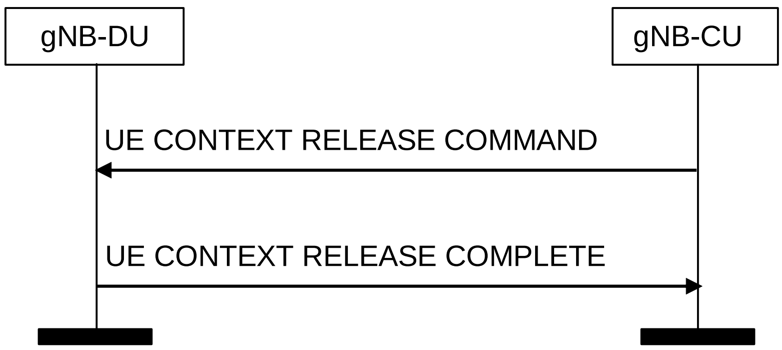

Figure 8.3.3.2-1: UE Context Release (gNB-CU initiated) procedure. Successful operation

The gNB-CU initiates the procedure by sending the UE CONTEXT RELEASE COMMAND message to the gNB-DU.

Upon reception of the UE CONTEXT RELEASE COMMAND message, the gNB-DU shall release all related signalling and user data transport resources and reply with the UE CONTEXT RELEASE COMPLETE message. If the *CG-SDT Kept Indicator* IE is contained in the UE CONTEXT RELEASE COMMAND message and set to "true", the gNB-DU shall, if supported, consider that the UE is sent to RRC_INACTIVE state with CG-SDT configuration and store the configured CG-SDT resources, C-RNTI, CG-SDT-CS-RNTI, the CG-SDT related RLC configurations and F1-U connections associated with the SDT bearers while releasing the UE context.

If the *old gNB-DU UE F1AP ID* IE is included in the UE CONTEXT RELEASE COMMAND message, the gNB-DU shall additionally release the UE context associated with the old gNB-DU UE F1AP ID.

If the UE CONTEXT RELEASE COMMAND message contains the *RRC-Container IE*, the gNB-DU shall send the RRC container to the UE via the SRB indicated by the *SRB ID* IE.

If the UE CONTEXT RELEASE COMMAND message includes the *Execute Duplication* IE, the gNB-DU shall perform CA based duplication or multi-path relay based duplication, if configured, for the SRB for the included *RRC-Container* IE.

If the *Candidate Cells To Be Cancelled List* IE is included in the UE CONTEXT RELEASE COMMAND message, the gNB-DU shall consider that the gNB-CU is cancelling only the conditional handover, conditional PSCell addition, conditional PSCell change, or subsequent CPAC associated to the cells identified by the included NR CGIs and associated to the UE-associated signaling identified by the *gNB-CU UE F1AP ID* IE and the *gNB-DU UE F1AP ID* IE.

If the *Positioning Context Reservation Indication* IE is included in the UE CONTEXT RELEASE COMMAND message, the gNB-DU shall not release the positioning context including the SRS configuration for the UE.

If the *LTM Cells To Be Released List* IE is included in the UE CONTEXT RELEASE COMMAND message, the gNB-DU shall, if supported, consider that the gNB-CU is cancelling only the LTM cells identified by the included NR CGIs and associated to the UE-associated signaling identified by the *gNB-CU UE F1AP ID* IE and the *gNB-DU UE F1AP ID* IE.

If the *Recommended SSBs for Paging List* IE is included in the UE CONTEXT RELEASE COMPLETE message, the gNB-CU shall, if supported, store it and may use it as assistance information for subsequent paging.

If the *DL LBT Failure Information Request* IE is included in the UE CONTEXT RELEASE COMMAND message, the gNB-DU shall, if supported, consider that the gNB-CU requests the reporting of DL LBT Failure information for the analysis of the MRO events of the UE specified in TS 38.300 [6], and act as specified in TS 38.401 [4].

Interactions with UE Context Setup procedure:

The UE Context Release procedure may be performed before the UE Context Setup procedure to release an existing UE-associated logical F1-connection and related resources in the gNB-DU, e.g. when gNB-CU rejects UE access it shall trigger UE Context Release procedure with the cause value of UE rejection.

8.3.3.3 Void

8.3.3.4 Abnormal Conditions

If one or more candidate cells in the *Candidate Cells To Be Cancelled List* IE included in the UE CONTEXT RELEASE COMMAND message were not prepared using the same UE-associated signalling connection, the gNB-DU shall ignore those non-associated candidate cells.

If one or more LTM cells in the *LTM Cells To Be Released List* IE included in the UE CONTEXT RELEASE COMMAND message were not prepared using the same UE-associated signalling connection, the gNB-DU shall ignore those non-associated LTM cells.

8.3.4 UE Context Modification (gNB-CU initiated)

8.3.4.1 General

The purpose of the UE Context Modification procedure is to modify the established UE Context, e.g., establishing, modifying and releasing radio resources or sidelink resources. This procedure is also used to command the gNB-DU to stop data transmission for the UE for mobility (see TS 38.401 [4]). The procedure uses UE-associated signalling.

8.3.4.2 Successful Operation

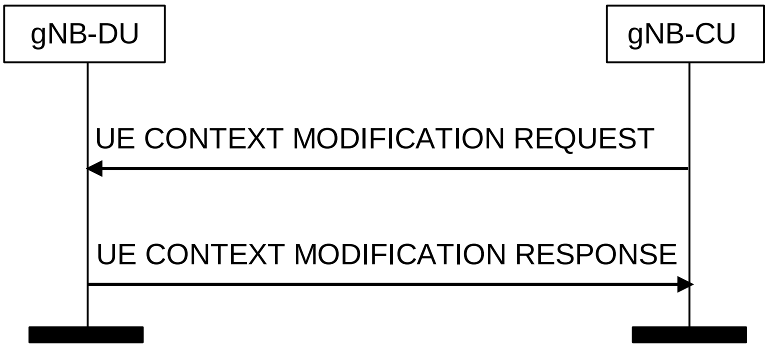

Figure 8.3.4.2-1: UE Context Modification procedure. Successful operation

The UE CONTEXT MODIFICATION REQUEST message is initiated by the gNB-CU.

Upon reception of the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall perform the modifications, and if successful reports the update in the UE CONTEXT MODIFICATION RESPONSE message.

If the *SpCell ID* IE is included in the UE CONTEXT MODIFICATION REQUEST message and neither the *LTM Information Modify* IE nor the *Conditional Intra-DU Mobility Information* IE is present, the gNB-DU shall replace any previously received value and regard it as a reconfiguration with sync as defined in TS 38.331 [8]. If the *ServCellIndex* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall take this into account for the indicated SpCell. If the *SpCell UL Configured* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall configure UL for the indicated SpCell accordingly. If the *servingCellMO* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall configure servingCellMO for the indicated SpCell accordingly. If the *servingCellMO List* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, configure servingCellMO after determining the list of BWPs for the UE and include the list of servingCellMOs that have been encoded in *CellGroupConfig* IE as *ServingCellMO-encoded-in-CGC List* IE in the UE CONTEXT MODIFICATION RESPONSE message.

If the *Configured BWP List* IE is included in the UE CONTEXT MODIFICATION RESPONSE message the gNB-CU shall, if supported, take it into account when requesting the gNB-DU for generating preconfigured measurement GAP for the indicated BWPs.

If the *Preconfigured Measurement GAP Request* IE is present in the *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATON REQUEST message, the gNB-DU shall, if supported, consider that the content of the previous *CellGroupConfig* IE was not sent to the UE and generate the pre-configured measurement GAP for the indicated BWPs in the *MeasConfig* IE. If the gNB-DU successfully generates pre-configured measurement GAP for the indicated BWPs, the gNB-DU shall update the *CellGroupConfig* IE with the content of the previous *CellGroupConfig* IE and the preconfigured measurement GAP configuration in the UE CONTEXT MODIFICATION RESPONSE message.

If the *SCell To Be Setup List* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall consider it as a list of candidate SCells to be set up. If the *SCell To Be Setup List* IE is included in the UE CONTEXT MODIFICATION REQUEST message and the indicated SCell(s) are already setup, the gNB-DU shall replace any previously received value. If the *SCell UL Configured* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall configure UL for the indicated SCell accordingly. If the *servingCellMO* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall configure servingCellMO for the indicated SCell accordingly. If the *servingCellMO-On-demand* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, configure the servingCellMO for on-demand SSB for the indicated SCell accordingly

If the *SCell To Be Removed List* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall consider it as a list of SCells to be removed.

If the *DRX Cycle* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall use the provided value from the gNB-CU. If the *DRX configuration indicator* IE is contained in the UE CONTEXT MODIFICATION REQUEST message and set to "release", the gNB-DU shall release DRX configuration.

If the *Non-Integer DRX Cycle* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use the provided value from the gNB-CU.

If the *SL DRX Cycle list* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use the provided value from the gNB-CU for the indicated RX UE of this UE. If the *SL DRX configuration indicator* IE is contained in the UE CONTEXT MODIFICATION REQUEST message and set to "release", the gNB-DU shall, if supported, release SL DRX configuration for the indicated RX UE of this UE.

If the *SRB To Be Setup List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall act as specified in the TS 38.401 [4], and replace any previously received value. If *Duplication Indication* IE is contained in the *SRB To Be Setup List* IE, the gNB-DU shall, if supported, setup two RLC entities for the indicated SRB if the value is set to be "true", or delete the RLC entity of secondary path if the value is set to be "false". If the *Additional Duplication Indication* IE is contained in the *SRB To Be Setup List* IE, the gNB-DU shall, if supported, setup the indicated RLC entities for the indicated SRB. If the *SRB Mapping Info* IE is contained in the *SRB To Be Setup List* IE, the gNB-DU shall, if supported, store the mapping information indicated in the *SRB Mapping Info* IE for the SRB identified by the *SRB ID* IE and the Uu Relay RLC channel identified by the *SRB Mapping Info*. The gNB-DU shall use the mapping information stored for the mapping of SRB data to Uu Relay RLC channel. If the *Path Addition Information* IE and the *SRB Mapping Info* IE are both contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, setup one RLC entity if necessary for the direct path and map the indicated SRB to the Uu Relay RLC channel based on the *SRB Mapping Info* IE. If the *Duplication Indication* IE and *SRB Mapping Info* IE are both contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, setup one RLC entity for the direct path if the value is set to be "true", and map the indicated SRB to the Uu Relay RLC channel based on the *SRB Mapping Info* IE. If the *Additional Duplication Indication* IE and *SRB Mapping Info* IE are both contained in the *SRB To Be Setup List* IE, the gNB-DU shall, if supported, setup the indicated RLC entities for the indicated SRB, and map the indicated SRB to the Uu Relay RLC channel based on the *SRB Mapping Info* IE. The number of RLC entities to be set up is the indicated value of *Additional Duplication Indication* IE minus 1.

If the *DRB To Be Setup List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall act as specified in the TS 38.401 [4]. If the *DRB Mapping Info* IE is contained in the *DRB To Be Setup List* IE, the gNB-DU shall, if supported, store the mapping information indicated in the *DRB Mapping Info* IE, if present, for the DRB identified by the *DRB ID* IE and the Uu Relay RLC channel identified by the *DRB Mapping Info*. The gNB-DU shall use the mapping information stored for the mapping of DRB data to Uu Relay RLC channel.

If the *PSI based SDU Discard UL* IE is included in the *DRB To Be Setup List* IE or the *DRB To Be Modified List* IE, the gNB-DU shall, if supported, take it into account to perform UL PSI based SDU discarding activation or deactivation for the indicated DRB as defined in TS 38.321 [16].

If the *PSI based SDU Discard DL* IE is included in the *DRB To Be Setup List* IE or the *DRB To Be Modified List* IE, the gNB-DU shall, if supported, take it into account to provide suggestion on whether to activate or deactivate DL PSI based SDU for the indicated DRB.

If the *BH Information* IE is included in the *UL UP TNL Information to be setup List* IE or the *Additional PDCP Duplication TNL List* IE for a DRB, the gNB-DU shall, if supported, use the indicated BAP Routing ID and BH RLC channel for transmission of the corresponding GTP-U packets to the IAB-donor, as specified in TS 38.340 [30].

If the *BH RLC Channel To Be Setup List* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall act as specified in TS 38.401 [4]. If the *Traffic Mapping Information* IE is included in the *BH RLC Channel To Be Setup Item IEs* IE for a BH RLC Channel, the gNB-DU shall, if supported, process the *Traffic Mapping Information* IE following the behaviour described for the UE Context Setup procedure.

If the *BH RLC Channel To Be Modified List* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall act as specified in TS 38.401 [4]. If the *Traffic Mapping Information* IE is included in the *BH RLC Channel To Be Modified Item IEs* IE for a BH RLC Channel, the gNB-DU shall, if supported, process the *Traffic Mapping Information* IE following the behaviour described for the UE Context Setup procedure.

If the *BH RLC Channel To Be Released List* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall release the BH RLC channels in the list.

If two *UL UP TNL Information* IEs are included and the *DRB Mapping Info* IE is not contained in UE CONTEXT MODIFICATION REQUEST message for a DRB, the gNB-DU shall include two *DL UP TNL Information* IEs in UE CONTEXT MODIFICATION RESPONSE message and setup two RLC entities for the indicated DRB. If the *UL UP TNL Information* IE with the *DRB Mapping Info* IE and the *UL UP TNL Information* IE without the *DRB Mapping Info* IE are both contained in the UE CONTEXT MODIFICATION REQUEST message for a DRB, the gNB-DU shall, if supported, include two *DL UP TNL Information* IEs in UE CONTEXT MODIFICATION RESPONSE message, setup one RLC entity for the *UL UP TNL Information* IE without the *DRB Mapping Info* IE, and map the indicated DRB to the Uu Relay RLC channel based on the *DRB Mapping Info* IE. gNB-CU and gNB-DU use the *UL UP TNL Information* IEs and *DL UP TNL Information* IEs to support packet duplication for intra-gNB-DU CA and multi-path relay as defined in TS 38.470 [2]. The first *UP TNL Information* IE of the two *UP TNL Information* IEs is for the primary path.

If one or two *Additional PDCP Duplication UP TNL Information* IEs are included in the UE CONTEXT MODIFICATION REQUEST message for a DRB, the gNB-DU shall, if supported, include one or two *Additional PDCP Duplication UP TNL Information* IEs in the UE CONTEXT MODIFICATION RESPONSE message and setup one or two additional RLC entities for the indicated DRB. The gNB-CU and the gNB-DU use the *Additional PDCP Duplication UP TNL Information* IEs to support packet duplication for intra-gNB-DU CA and multi-path relay as defined in TS 38.470 [2].

If *Duplication Activation* IE is included in the UE CONTEXT MODIFICATION REQUEST message for a DRB, the gNB-DU should take it into account when activating/deactivating CA based PDCP duplication or multi-path relay based PDCP duplication for the DRB. If the *RLC Duplication State List* IE is included in the *RLC Duplication Information* IE contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, take it into account for the DRB with more than two RLC entities.

If *DC Based Duplication Configured* IE is included in the UE CONTEXT MODIFICATION REQUEST message for a DRB, the gNB-DU shall regard that DC based PDCP duplication is configured for this DRB if the value is set to be "true" and it should take the responsibility of PDCP duplication activation/deactivation. Otherwise, the gNB-DU shall regard that DC based PDCP duplication is de-configured for this DRB id the value is set to be "false", and it should stop PDCP duplication activation/deactivation by MAC CE. If *DC Based Duplication Activation* IE is included in the UE CONTEXT MODIFICATION REQUEST message for a DRB, the gNB-DU should take it into account when activating/deactivating DC based PDCP duplication for this DRB. If the *RLC Duplication State List* IE is included in the *RLC Duplication Information* IE contained in the UE CONTEXT MODIFICATION REQUEST message for a DRB, the gNB-DU shall, if supported, take it into account when activating/deactivating DC based PDCP duplication for the DRB with more than two RLC entities. If the *Primary Path Indication* IE is included in the *RLC Duplication Information* IE, the gNB-DU shall, if supported, take it into account when performing DC based PDCP duplication for the DRB with more than two RLC entities.

For a certain DRB which was allocated with two GTP-U tunnels, if such DRB is modified and given one GTP-U tunnel via the UE Context Modification procedure, the gNB-DU shall consider that the CA based PDCP duplication or multi-path relay based PDCP duplication for the concerned DRB is de-configured. If such UE Context Modification procedure occurs, the *Duplication Activation* IE shall not be included for the concerned DRB.

If the *UL Configuration* IE in *DRB to Be Setup Item* IE or *DRB to Be Modified Item* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall take it into account for UL scheduling.

If the *RRC Reconfiguration Complete Indicator* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall consider the ongoing reconfiguration procedure involving changes of the L1/L2 configuration at the gNB-DU signalled to the gNB-CU via the *CellGroupConfig* IE for MR-DC operation or standalone operation has been successfully performed when such IE is set to 'true'; otherwise (when such IE is set to 'failure'), the gNB-DU shall consider the ongoing reconfiguration procedure has been failed and it shall continue to use the old L1/L2 configuration.

If *DL PDCP SN length* IE is included in the UE CONTEXT MODIFICATION REQUEST message for a DRB, gNB-DU shall, if supported, store this information and use it for lower layer configuration.

If *UL PDCP SN length* IE is included in the UE CONTEXT MODIFICATION REQUEST message for a DRB, gNB-DU shall, if supported, store this information and use it for lower layer configuration.

If the *RLC Failure Indication* IE is included in UE CONTEXT MODIFICATION REQUEST message, the gNB-DU should consider that the RLC entity indicated by such IE needs to be re-established when the CA-based packet duplication is active, and the gNB-DU may include the *Associated SCell List* IE in UE CONTEXT MODIFICATION RESPONSE by containing a list of SCell(s) associated with the RLC entity indicated by the *RLC Failure Indication* IE.

If the UE CONTEXT MODIFICATION REQUEST message contains the *RRC-Container* IE, the gNB-DU shall send the corresponding RRC message to the UE. If the UE CONTEXT MODIFICATION REQUEST message includes the *Execute Duplication* IE, the gNB-DU shall perform CA based duplication or multi-path relay based duplication, if configured, for the SRB for the included *RRC-Container* IE.

If the UE CONTEXT MODIFICATION REQUEST message contains the *Transmission Action Indicator* IE, the gNB-DU shall stop or restart (if already stopped) data transmission for the UE, according to the value of this IE. It is up to gNB-DU implementation when to stop or restart the UE scheduling.

For EN-DC operation, if the *DRB to Be Setup List* IE is present in the UE CONTEXT MODIFICATION REQUEST message the gNB-CU shall include the *E-UTRAN QoS* IE. The allocation of resources according to the values of the *Allocation and Retention Priority* IE included in the *E-UTRAN QoS* IE shall follow the principles described for the E-RAB Setup procedure in TS 36.413 [15]. For NG-RAN operation, the gNB-CU shall include the *DRB Information* IE in the UE CONTEXT MODIFICATION REQUEST message.

If the gNB-CU includes the SMTC information of the measured frequency(ies) in the *MeasurementTimingConfiguration* IE of the *CU to DU RRC Information* IE that is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall generate the measurement gaps based on the received SMTC information. Then the gNB-DU shall send the measurement gaps information to the gNB-CU in the *MeasGapConfig* IE of the *DU to CU RRC Information* IE that is included in the UE CONTEXT MODIFICATION RESPONSE message.

If the *MeasConfig* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall deduce that changes to the measurements' configuration need to be applied. The gNB-DU shall take the received info, e.g. the *measObjectToAddModList* IE, and/or the *measObjectToRemoveList* IE into account, when generating measurement gap and when deciding if a measurement gap is needed or not.

If the *NeedForGapsInfoNR* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it as described in TS 38.331 [8]. If the *NeedForGapNCSG-InfoNR* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it as described in TS 38.331 [8]. If the *NeedForGapNCSG-InfoEUTRA* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it as described in TS 38.331 [8]. If the *NeedForInterruptionInfoNR* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it as described in TS 38.331 [8].

For DC operation, if the gNB-CU includes the *CG-Config* IE in the *CU to DU RRC Information* IE that is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU may initiate low layer parameters coordination taking this information into account.

For sidelink operation, the *CG-ConfigInfo* IE shall be included in the *CU to DU RRC Information* IE if the gNB-CU receives sidelink related UE information from UE. If the *CG-ConfigInfo* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall regard it as an indication of V2X sidelink information or NR sidelink information as defined in TS 38.331 [8].

For EN-DC operation, if the gNB-CU includes the *Resource Coordination Transfer Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it for the purpose of resource coordination. If the gNB-CU received the MeNB Resource Coordination Information as defined in TS 36.423 [9], after completion of UE Context Setup procedures, the gNB-CU shall transparently transfer it to the gNB-DU via the *Resource Coordination Transfer Container* IE in the UE CONTEXT MODIFICATION REQUEST message. The gNB-DU shall use the information received in the *Resource Coordination Transfer Container* IE for reception of MeNB Resource Coordination Information at the gNB acting as secondary node as described in TS 36.423 [9]. If the *Resource Coordination E-UTRA Cell Information* IE is included in the *Resource Coordination Transfer Information* IE, the gNB-DU shall store the information replacing previously received information for the same E-UTRA cell, and use the stored information for the purpose of resource coordination. If the *Ignore PRACH Configuration* IE is present and set to "true" the *E-UTRA PRACH Configuration* IE in the UE CONTEXT MODIFICATION REQUEST message shall be ignored.

For NGEN-DC or NE-DC operation, if the gNB-CU includes the *Resource Coordination Transfer Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it for the purpose of resource coordination. If the gNB-CU received the MR-DC Resource Coordination Information as defined in TS 38.423 [28], after completion of UE Context Setup procedures, the gNB-CU shall transparently transfer it to the gNB-DU via the *Resource Coordination Transfer Container* IE in the UE CONTEXT MODIFICATION REQUEST message. The gNB-DU shall use the information received in the *Resource Coordination Transfer Container* IE for reception of MR-DC Resource Coordination Information at the gNB as described in TS 38.423 [28].

For EN-DC operation, and if the *Subscriber Profile ID for RAT/Frequency priority* IE is received from an MeNB, the UE CONTEXT MODIFICTION REQUEST message shall contain the *Subscriber Profile ID for RAT/Frequency priority* IE. If the *Additional RRM Policy Index* IE is received from an MeNB, the UE CONTEXT MODIFICATION REQUEST message shall, if supported, contain the *Additional RRM Policy Index* IE. The gNB-DU shall store the received Subscriber Profile ID for RAT/Frequency priority in the UE context and use it as defined in TS 36.300 [20].

The gNB-DU shall, if supported, store the received Additional RRM Policy Index in the UE context and use it as defined in TS 36.300 [20].

If the *Index to RAT/Frequency Selection Priority* IE is modified at the gNB-CU, the *Index to RAT/Frequency Selection Priority* IE shall be included in the UE CONTEXT MODIFICATION REQUEST. The gNB-DU may use it for RRM purposes.

Only one of the following IEs shall be contained in the UE CONTEXT MODIFICATION REQUEST message: the *Uplink TxDirectCurrentList Information* IE or the *Uplink TxDirectCurrentTwoCarrierList Information* IE or the *Uplink TxDirectCurrentMoreCarrierList Information* IE. If the UE CONTEXT MODIFICATION REQUEST message contains one of the *Uplink TxDirectCurrentList Information* IE or the *Uplink TxDirectCurrentTwoCarrierList Information* IE or the *Uplink TxDirectCurrentMoreCarrierList Information* IE, the gNB-DU may take that into account when selecting L1 configuration.

The *UEAssistanceInformation* IE shall be included in *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message if the gNB-CU received this IE from the UE; if the *UEAssistanceInformation* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, take it into account when configuring resources for the UE.

The *UEAssistanceInformationEUTRA* IE shall be included in *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message if the gNB-CU received this IE from the UE; if the *UEAssistanceInformationEUTRA* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, take it into account when configuring LTE sidelink resources for the UE.

The gNB-DU shall report to the gNB-CU, in the UE CONTEXT MODIFICATION RESPONSE message, the result for all the requested or modified DRBs, SRBs, BH RLC Channels, Uu Relay RLC channels, PC5 Relay RLC channels, and SL DRBs in the following way:

- A list of DRBs which are successfully established shall be included in the *DRB Setup List* IE;
- A list of DRBs which failed to be established shall be included in the *DRB Failed to be Setup List* IE;
- A list of DRBs which are successfully modified shall be included in the *DRB Modified List* IE;
- A list of DRBs which failed to be modified shall be included in the *DRB Failed to be Modified List* IE;
- A list of SRBs which failed to be established shall be included in the *SRB Failed to be Setup List* IE.
- A list of successfully established SRBs with logical channel identities for primary path shall be included in the *SRB Setup List* IE only if CA based PDCP duplication or multi-path relay based PDCP duplication is initiated for the concerned SRBs.
- A list of successfully modified SRBs with logical channel identities for primary path shall be included in the *SRB Modified List* IE only if CA based PDCP duplication or multi-path relay based PDCP duplication is initiated for the concerned SRBs.
- A list of BH RLC channels which are successfully established shall be included in the *BH RLC Channel Setup List* IE;
- A list of BH RLC channels which failed to be established shall be included in the *BH RLC Channel Failed to be Setup List* IE;
- A list of BH RLC channels which are successfully modified shall be included in the *BH RLC Channel Modified List* IE;
- A list of BH RLC channels which failed to be modified shall be included in the *BH RLC Channel Failed to be Modified List* IE;
- A list of Uu Relay RLC channels which are successfully established shall be included in the *Uu RLC Channel Setup List* IE;
- A list of Uu Relay RLC channels which failed to be established shall be included in the *Uu RLC Channel Failed to be Setup List* IE;

- A list of Uu Relay RLC channels which are successfully modified shall be included in the *Uu RLC Channel Modified List* IE;
- A list of Uu Relay RLC channels which are failed to be modified shall be included in the *Uu RLC Channel Failed to be Modified List* IE;
- A list of PC5 Relay RLC channels which are successfully established shall be included in the *PC5 RLC Channel Setup List* IE;
- A list of PC5 Relay RLC channels which failed to be established shall be included in the *PC5 RLC Channel Failed to be Setup List* IE;
- A list of PC5 Relay RLC channels which are successfully modified shall be included in the *PC5 RLC Channel Modified List* IE;
- A list of PC5 Relay RLC channels which failed to be modified shall be included in the *PC5 RLC Channel Failed to be Modified List* IE;
- A list of SL DRBs which are successfully established shall be included in the *SL DRB Setup List* IE;
- A list of SL DRBs which failed to be established shall be included in the *SL DRB Failed to be Setup List* IE;
- A list of SL DRBs which are successfully modified shall be included in the *SL DRB Modified List* IE;
- A list of SL DRBs which failed to be modified shall be included in the *SL DRB Failed to be Modified List* IE.

If *Duplication Indication* IE in *SL DRB To Be Setup List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, generate two PC5 RLC bearer configurations for the indicated SL DRB.

If *Duplication Indication* IE is contained in the *SL DRB To Be Modified List* IE, the gNB-DU shall, if supported, generate two PC5 RLC bearer configurations for the indicated SL DRB, if the value is set to be"true" and duplication is not already configured for the indicated SL DRB.

If *Duplication Indication* IE is contained in the *SL DRB To Be Modified List* IE, the gNB-DU shall, if supported, release the additional PC5 RLC configuration for the indicated SL DRB, if the value is set to be "false".

For each GBR DRB, if the *Alternative QoS Parameters Sets* IE is included in the *GBR QoS Flow Information* IE in the UE CONTEXT MODIFICATION REQUEST message, gNB-DU shall, if supported, behave the same as the NG-RAN node in the PDU Session Resource Setup procedure, specified in TS 38.413 [3].

If the *BAP Control PDU Channel* IE is included in the *BH RLC Channel to be Setup List* IE, the gNB-DU shall, if supported, consider that the configured BH RLC channel can be used to transmit BAP Control PDUs, and use this BH RLC channel as specified in TS 38.340 [30].

If the *BAP Control PDU Channel* IE is included in the *BH RLC Channel to be Modified List* IE, the gNB-DU shall, if supported, consider that the configured BH RLC channel can be used to transmit BAP Control PDUs, and use this BH RLC channel as specified in TS 38.340 [30]. Otherwise, if the *BAP Control PDU Channel* IE is not present for any BH RLC channel, any available BH RLC channel can be used to transmit BAP Control PDUs as specified in TS 38.340 [30].

If the *F1-C Transfer Path* IE is included in UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, take it into account.

When the gNB-DU reports the unsuccessful establishment of a DRB or SRB or SL DRB or a BH RLC channel or a Uu Relay RLC channel or a PC5 Relay RLC channel, the cause value should be precise enough to enable the gNB-CU to know the reason for the unsuccessful establishment.

If the *Resource Coordination Transfer Container* IE is included in the UE CONTEXT MODIFICATION RESPONSE, the gNB-CU shall transparently transfer this information for the purpose of resource coordination as described in TS 36.423 [9], TS 38.423 [28].

If the *DU to CU RRC Information* IE is included in the UE CONTEXT MODIFICATION RESPONSE message, except for the CG-SDT procedure and UE configured with BWP specific ServingCellMO, the gNB-CU shall perform RRC Reconfiguration as described in TS 38.331 [8]. The *CellGroupConfig* IE shall transparently be signaled to the UE as specified in TS 38.331 [8]. In the cases of CG-SDT, and UE configured with BWP specific ServingCellMO, the *CellGroupConfig* IE shall be ignored by the gNB-CU.

If the *ServCellInfoList* IE is included in the *DU to CU RRC Information* IE contained in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall take it into account to generate the content of inter-node message, i.e., *CG-Config* or *CG-ConfigInfo*, as described in TS 38.331 [8].

If the *UE-CapabilityRAT-ContainerList* IE is included in the UE CONTEXT MODIFICATION REQUEST, the gNB-DU shall take this information into account for UE specific configurations.

If the *SCell Failed To Setup List* IE is contained in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall regard the corresponding SCell(s) failed to be set up with an appropriate cause value for each SCell failed to setup.

If the *C-RNTI* IE is included in the UE CONTEXT MODIFICATION RESPONSE, the gNB-CU shall consider that the C-RNTI has been allocated by the gNB-DU for this UE context.

If the *Inactivity Monitoring Request* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, gNB-DU may consider that the gNB-CU has requested the gNB-DU to perform UE inactivity monitoring. If the *Inactivity Monitoring Response* IE is contained in the UE CONTEXT MODIFICATION RESPONSE message and set to "Not-supported", the gNB-CU shall consider that the gNB-DU does not support UE inactivity monitoring for the UE.

The UE Context Modify Procedure is not used to configure SRB0.

If in the UE CONTEXT MODIFICATION REQUEST, the *Notification Control* IE is included in the *DRB to Be Setup List* IE or the *DRB to Be Modified List* IE and it is set to active, the gNB-DU shall, if supported, monitor the QoS of the DRB and notify the gNB-CU if the QoS cannot be fulfilled any longer or if the QoS can be fulfilled again. The *Notification Control* IE can only be applied to GBR bearers.

If the *UL PDU Session Aggregate Maximum Bit Rate* IE is included in the *QoS Flow Level QoS Parameters* IE containded in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall replace the received UL PDU Session Aggregate Maximum Bit Rate and use it as specified in TS 23.501 [21].

If the *gNB-DU UE Aggregate Maximum Bit Rate Uplink* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall:

- replace the previously provided gNB-DU UE Aggregate Maximum Bit Rate Uplink with the new received gNB-DU UE Aggregate Maximum Bit Rate Uplink;

- use the received gNB-DU UE Aggregate Maximum Bit Rate Uplink for non-GBR Bearers for the concerned UE.

The *gNB-DU UE Aggregate Maximum Bit Rate Uplink* IE shall be sent in the UE CONTEXT MODIFICATION REQUEST if *DRB to Be Setup List* IE is included and the gNB-CU has not previously sent it. The gNB-DU shall store and use the received *gNB-DU UE Aggregate Maximum Bit Rate Uplink* IE.

If the *RLC Status IE* is included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall assume that RLC has been reestablished at the gNB-DU and may trigger PDCP data recovery.

If the GNB-*DU Configuration Query* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, gNB-DU shall include the *DU To CU RRC Information* IE in the UE CONTEXT MODIFICATION RESPONSE message.

If the *Bearer Type Change* IE is included in *DRB to Be Modified List* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall either reset the lower layers or generate a new LCID for the affected bearer as specified in TS 37.340 [7].

For NE-DC operation, if *NeedforGap* IE is included in the UE CONTEXT MODIFICATION REQUEST message,the gNB-DU shall generate measurement gap for the SeNB.

If the *QoS Flow Mapping Indication* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, replace any previously received value and take it into account that only the uplink or downlink QoS flow is mapped to the DRB.

If the *Lower Layer presence status change* IE set to "suspend lower layers" is included in the UE CONTEXT MODIFICATION REQUEST, the gNB-DU shall keep all lower layer configuration for UEs, and not transmit or receive data from UE.

If the *Lower Layer presence status change* IE set to "resume lower layers" is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall use the previously stored lower layer configuration for the UE.

If the *Full Configuration* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall generate a *CellGroupConfig* IE using full configuration and include it in the UE CONTEXT MODIFICATION RESPONSE.

If the *Full Configuration* IE is contained in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall consider that the gNB-DU has generated the *CellGroupConfig* IE using full configuration.

For each QoS flow whose DRB has been successfully established or modified and the *QoS Monitoring Request* IE was included in the *QoS Flow Level QoS Parameters* IE contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall store this information, and, if supported, perform delay measurement and QoS monitoring, as specified in TS 23.501 [21].

If the *NR V2X Services Authorized* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, update its V2X services authorization information for the UE accordingly. If the *NR V2X Services Authorized* IE includes one or more IEs set to "not authorized", the gNB-DU shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *LTE V2X Services Authorized* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, update its V2X services authorization information for the UE accordingly. If the *LTE V2X Services Authorized* IE includes one or more IEs set to "not authorized", the gNB-DU shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *LTE UE Sidelink Aggregate Maximum Bit Rate* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported:

- replace the previously provided UE LTE Sidelink Aggregate Maximum Bit Rate, if available in the UE context, with the received value;

- use the received value for the concerned UE’s sidelink communication in network scheduled mode for LTE V2X services.

If the *NR UE Sidelink Aggregate Maximum Bit Rate* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported:

- replace the previously provided UE NR Sidelink Aggregate Maximum Bit Rate, if available in the UE context, with the received value;

- use the received value for the concerned UE’s sidelink communication in network scheduled mode for NR V2X services.

If the *NR A2X Services Authorized* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, update its A2X services authorization information for the UE accordingly. If the *NR A2X Services Authorized* IE includes one or more IEs set to "not authorized", the gNB-DU shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *LTE A2X Services Authorized* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, update its A2X services authorization information for the UE accordingly. If the *LTE A2X Services Authorized* IE includes one or more IEs set to "not authorized", the gNB-DU shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *LTE UE Sidelink Aggregate Maximum Bit Rate for A2X* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported:

- replace the previously provided UE LTE Sidelink Aggregate Maximum Bit Rate for A2X, if available in the UE context, with the received value;

- use the received value for the concerned UE’s sidelink communication in network scheduled mode for LTE A2X services.

If the *NR UE Sidelink Aggregate Maximum Bit Rate for A2X* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported:

- replace the previously provided UE NR Sidelink Aggregate Maximum Bit Rate for A2X, if available in the UE context, with the received value;

- use the received value for the concerned UE's sidelink communication in network scheduled mode for NR A2X services.

If the *PC5 Link Aggregate Maximum Bit Rate* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported:

- replace the previously provided UE PC5 Link Aggregate Bit Rate, if available in the UE context, with the received value;

- use the received value for the concerned UE's sidelink communication in network scheduled mode for NR V2X services as defined in TS 23.287 [40].

If the *TSC Traffic Characteristics* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, take into account the corresponding information received in the *TSC Traffic Characteristics* IE. If the *RAN Feedback Type* IE is included in the *TSC Assistance Information Uplink* IE of the *TSC Traffic Characteristics* IE, the gNB-DU shall, if supported, take this information into account when determining the feedback to provide in the *TSC Traffic Characteristics Feedback* IE in the UE CONTEXT MODIFICATION RESPONSE message.

If the *CPAC MCG Information* IE is included in the UE CONTEXT MODIFICATION REQUEST message and the CPAC Trigger is set to:

- "CPAC-preparation": the gNB-DU shall, if supported, consider that the request concerns a conditional PSCell addition or conditional PSCell change or subsequent CPAC. The gNB-DU takes the included *CG-Config* and/or *CG-ConfigInfo* IE into account, and may provide a corresponding *CellGroupConfig* IE for MCG configuration preparation in the UE CONTEXT MODIFICATION RESPONSE message. The UE CONTEXT MODIFICATION RESPONSE message also includes a *Requested Target Cell ID* IE corresponding to the *PSCell ID* IE in the UE CONTEXT MODIFICATION REQUEST message.

- "CPAC-executed": the gNB-DU shall, if supported, consider that, for the included *PSCell ID* IE corresponding to the selected PSCell, the UE has successfully executed the CPAC preparation. The gNB-DU shall apply the corresponding *CellGroupConfig* IE for MCG configuration.

- "CPAC-cancel": the gNB-DU shall, if supported, consider that the gNB-CU is about to release the prepared MCG configuration(s) corresponding to the PSCell(s) identified by the included NR CGI(s) in the *Candidate PSCells To Be Cancelled List* IE.

If the *Conditional Intra-DU Mobility Information* IE is included in the UE CONTEXT MODIFICATION REQUEST message and the CHO Trigger is set to "CHO-initiation", the gNB-DU shall consider that the request concerns a conditional handover, conditional PSCell addition, conditional PSCell change, or subsequent CPAC for the included *SpCell ID* IE and shall include it as the *Requested Target Cell ID* IE in the UE CONTEXT MODIFICATION RESPONSE message. The gNB-DU shall regard it as a reconfiguration with sync as defined in TS 38.331 [8].

If the *Conditional Intra-DU Mobility Information* IE is included in the UE CONTEXT MODIFICATION REQUEST message and the CHO Trigger is set to "CHO-replace", the gNB-DU shall replace the existing prepared conditional mobility identified by the *gNB-DU UE F1AP ID* IE and the *SpCell ID* IE.

If the *Conditional Intra-DU Mobility Information* IE is included in the UE CONTEXT MODIFICATION REQUEST message and the CHO Trigger is set to "CHO-cancel", the gNB-DU shall consider that the gNB-CU is about to remove any reference to, and release any resources previously reserved for the cells identified by the included NR CGIs in the *Candidate Cells To Be Cancelled List* IE.

If the *S-CPAC Request* IE is included within the *Conditional Intra-DU Mobility Information* IE in the UE CONTEXT MODIFICATION REQUEST message and is set to "initiation", the gNB-DU shall, if supported, consider that the procedure is triggered for S-CPAC preparation.

If the *Transmission Stop Indicator* IE is included within the *DRB to Be Modified Item* IE in the UE CONTEXT MODIFICATION REQUEST message and set to “true”, the gNB-DU shall, if supported, stop the data transmission for the DRB. It is up to gNB-DU implementation when to stop the UE scheduling for that DRB.

If the *SCG Indicator* IE is contained in the UE CONTEXT MODIFICATION REQUEST message and it is set to “released”, the gNB-DU shall, if supported, deduce that an SCG is removed.

If the *Estimated Arrival Probability* IE is contained in the *Conditional Intra-DU Mobility Information* IE included in the UE CONTEXT MODIFICATION REQUEST message, then the gNB-DU may use the information to allocate necessary resources for the UE.

If the *Location Measurement Information* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, take it into account when configuring measurement gaps for the UE.

If the *F1-C Transfer Path NRDC* IE is included in UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, take it into account.

If for a given E-RAB for EN-DC operation the *ENB DL Transport Layer Address* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

If for a given Qos flow for NG-RAN operation the *PDCP Terminating Node DL Transport Layer Address* IE is included in the UE CONTEXT MODIFICATION REQUEST message, then the gNB-DU shall, if supported, use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

If the gNB-DU is an IAB-DU, and if the *IAB Conditional RRC Message Delivery Indication* IE is included in the UE CONTEXT MODIFICATION REQUEST message together with the *RRC-Container* IE, and if its value is set to “true”, and if the *RRC-Container* IE is for a child IAB-MT of the gNB-DU, the gNB-DU shall, if supported, withhold the RRC message until one of the following conditions is met:

> If the gNB-DU belongs to a migrating IAB-node, whose co-located IAB-MT has successfully performed the random-access procedure to the target parent node, and if the migrating IAB-node has one or more routing entries for the target path.

> The gNB-DU receives a subsequent F1AP message including an *RRC-Container IE* for the same child node.

> If the gNB-DU belongs to a descendant node of the migrating IAB-node, whose co-located IAB-MT has received an *RRCReconfiguration* message including the intra-donor migration configurations, e.g., new TNL address(es) and the new default UL BAP routing ID.

> If the gNB-DU belongs to a migrating IAB-node, whose co-located IAB-MT has successfully performed RLF recovery after handover failure, and if the migrating IAB-node has one or more routing entries for the target path.

If the *MDT Polluted Measurement Indicator* IE is included in the UE CONTEXT MODIFICATION REQUEST, the gNB-DU shall take this information into account as specified in TS 38.401 [4].

If the *SCG Activation Request* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU may use it to configure SCG resources as specified in TS 37.340 [7], and if supported, shall include the *SCG Activation Status* IE in the UE CONTEXT MODIFICATION RESPONSE message.

If the *CG-SDT Query Indication* IE is included in the UE CONTEXT MODIFICATION REQUEST message and set to ‘true’, the gNB-DU shall, if supported, provide the CG-SDT related resource configuration for the bearers indicated as SDT bearers in the *SDT-MAC-PHY-CG-Config* IE within the *DU to CU RRC Information* IE contained in the UE CONTEXT MODIFICATION RESPONSE message to the gNB-CU. If the *SDT-MAC-PHY-CG-Config* IE is also included in the UE CONTEXT MODIFICATION REQUEST message within the *CU to DU RRC Information* IE, the gNB-DU may provide the delta signalling version of the *SDT-MAC-PHY-CG-Config* IE within the *DU to CU RRC Information* IE contained in the UE CONTEXT MODIFICATION RESPONSE message to the gNB-CU.

If the *5G ProSe Authorized* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, update its 5G ProSe services authorization information for the UE accordingly. If the *5G ProSe Authorized* IE includes one or more IEs set to "not authorized", the gNB-DU shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *SDT Bearer Configuration Query Indication* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, provide the RLC bearer configuration in the *SDT Bearer Configuration Info* IE in the UE CONTEXT MODIFICATION RESPONSE message for each bearer indicated as SDT bearer.

If the *5G ProSe UE PC5 Aggregate Maximum Bit Rate* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported:

- replace the previously provided 5G ProSe UE PC5 Aggregate Maximum Bit Rate, if available in the UE context, with the received value;

- use the received value for the concerned UE's sidelink communication in network scheduled mode for 5G ProSe services.

If the *5G ProSe PC5 Link Aggregate Bit Rate* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported:

- replace the previously provided 5G ProSe PC5 Link Aggregate Bit Rate, if available in the UE context, with the received value;

- use the received value for the concerned UE's sidelink communication in network scheduled mode for 5G ProSe services as defined in TS 23.304 [44].

If the *Updated Remote UE Local ID* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, replace the previously provided Remote UE Local ID, if available in the UE context, with the received value.

If the *Uu RLC Channel To Be Setup List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, act as specified in TS 38.401 [4].

If the *Uu RLC Channel To Be Modified List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, act as specified in TS 38.401 [4].

If the *Uu RLC Channel To Be Release List* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, release the Uu Relay RLC channels in the list.

If the *PC5 RLC Channel To Be Setup List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, act as specified in TS 38.401 [4]. gNB-DU generates the PC5 Relay RLC channel configurations for a L2 U2N Remote UE, U2N Relay UE, a L2 U2U Remote UE or a L2 U2U Relay UE. If the F1AP-IDs are associated with a U2N Relay UE, the *PC5 RLC Channel to be Setup Item IEs* IE shall include the *Remote UE Local ID* in case of single-hop relay or for the PC5 RLC channel between the U2N Remote UE and the First U2N Relay UE in multi-hop relay and correspondingly, the *PC5 RLC Channel Setup Item IEs* IE and the *PC5 RLC Channel Failed to be Setup Item* IE in the UE CONTEXT MODIFICATION RESPONSE message shall include the *Remote UE Local ID* IE.

If the *PC5 RLC Channel To Be Modified List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, act as specified in TS 38.401 [4]. gNB-DU generates the PC5 Relay RLC channel configurations for a L2 U2N Remote UE , U2N Relay UE, a L2 U2U Remote UE or a L2 U2U Relay UE. If the F1AP-IDs are associated with a U2N Relay UE, the *PC5 RLC Channel to be Modified Item IEs* IE shall include the *Remote UE Local ID* IE in case of single-hop relay or for the PC5 RLC channel between the U2N Remote UE and the First U2N Relay UE in multi-hop relay and correspondingly, the *PC5 RLC Channel Modified Item IEs* IE and the *PC5 RLC Channel Failed to be Modified Item IEs* IE in the UE CONTEXT MODIFICATION RESPONSE message shall include the *Remote UE Local ID* IE.

If the *PC5 RLC Channel To Be Release List* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, release the PC5 Relay RLC channels in the list. If the F1AP-IDs are associated with a U2N Relay UE, the *PC5 RLC Channel to be Released Item IEs* IE shall include the *Remote UE Local ID* IE in case of single-hop relay or for the PC5 RLC channel between the U2N Remote UE and the First U2N Relay UE in multi-hop relay.

If the *Path Switch Configuration* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it to configure the path switch from direct path to single-hop or multi-hop indirect path, from single-hop indirect path to single-hop or multi-hop indirect path, or from multi-hop indirect path to single-hop indirect path, or to release the direct path during the MP as specified in in TS 38.401 [4] and TS 38.300 [6].

If the *MUSIM-GapConfig* IE is contained in the *CU to DU RRC Information* IE included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, decide to use this IE for MUSIM gap configuration or select another one based on the received *UEAssistanceInformation* IE. If gNB-DU selects a different MUSIM gap configuration from received *UEAssistanceInformation* IE, then it shall include the selected MUSIM gap information to the gNB-CU in the *MUSIM-GapConfig* IE of the *DU to CU RRC Information* IE that is included in the UE CONTEXT MODIFICATION RESPONSE message.

If *MUSIM-GapConfig* IE is not contained in the *CU to DU RRC Information* IE, then gNB-DU shall, if supported, send the selected MUSIM gap configuration based on the received *UEAssistanceInformation* IE, to the gNB-CU in the *MUSIM-GapConfig* IE of the *DU to CU RRC Information* IE that is included in the UE CONTEXT MODIFICATION RESPONSE message. When *MUSIM-GapConfig* IE is received, the gNB-CU should use this value.

If the *gNB-DU UE Slice Maximum Bit Rate List* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported,

- store and replace the previously provided gNB-DU UE Slice Maximum Bit Rate List, if any, with the new received *gNB-DU UE Slice Maximum Bit Rate List*;
- use the received *gNB-DU UE Slice Maximum Bit Rate List* for the uplink traffic policing for each concerned slice as specified in TS 23.501 [21].

If the *Multicast MBS Session Setup List* IE or the *Multicast MBS Session Remove List* IE or both IEs are contained in the UE CONTEXT MODIFICATION REQUEST message the gNB-DU shall, if supported, store and use the information for configuring MBS Session Resources, if applicable.

If the *UE Multicast MRB To Be Setup at Modify List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, take it into account for configuring MBS Session Resources, if applicable, and shall include the *Multicast F1-U Context Reference CU* IE, if available, in the UE CONTEXT MODIFICATION RESPONSE message. And if the *MBS PTP Retransmission Tunnel Required* IE is included in the *UE Multicast MRB to Be Setup at Modify Item IEs* IE, the gNB-DU shall, if supported trigger the establishment of the MBS PTP Retransmission F1-U tunnel.

If the *MBS PTP Forwarding Tunnel Required Information* IE is included in the *UE Multicast MRB to Be Setup at Modify Item IEs* IE, the gNB-DU shall, if supported trigger the establishment of the MBS PTP Forwarding F1-U tunnel.

If the *Management Based MDT PLMN Modification List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, overwrite any previously stored Management Based MDT PLMN List information in the UE context and use the received information to determine subsequent selection of the UE for management based MDT defined in TS 32.422 [29].

If the *Dedicated SI Delivery Indication* IE is included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, take it into account for the system information delivery to the UE as described in TS 38.331 [8].

If the *PDU Set QoS Parameters* IE or the *DL PDU Set Information Marking Support Indication* IE set to "true" is included in the *QoS Flow Level QoS Parameters* IE contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, store this information and use it as specified in TS 23.501 [21].

If the *ECN Marking or Congestion Information Reporting Request* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it accordingly for the specific DRB. If the *ECN Marking or Congestion Information Reporting Status* IE is included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, use it to deduce if ECN marking or congestion information reporting is active or not active.

If the *InterFrequencyConfig-NoGap* IE is included in the *DU to CU RRC Information* IE contained in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, use it as described in TS 38.331 [8].

If the *ul-GapFR2-Config* IE is contained in the *DU to CU RRC Information* IE that is included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, use it as described in TS 38.331 [8].

If the *TwoPHRModeMCG* IE or the *TwoPHRModeSCG* IE is contained in the *DU to CU RRC Information* IE that is included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, use this value as described in TS 38.331 [8].

If the *MBSInterestIndication* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, take it into account when configuring resources for the UE.

If the *ncd-SSB-RedCapInitialBWP-SDT* IE is contained in the *DU to CU RRC Information* IE that is included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, use it as described in TS 38.331 [8].

If the *Network Controlled Repeater Authorized* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, update its authorization information for the UE accordingly. If the *Network Controlled Repeater Authorized* IE is set to "not authorized", the gNB-DU shall, if supported, initiate actions to ensure that the UE is no longer accessing as a Network Controlled Repeater.

If the *LTM Indicator* IE set to "true" is contained in the *LTM Information Modify* IE included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, consider that the request concerns LTM for the included *SpCell ID* IE and shall include it as the *Requested Target Cell ID* IE in the UE CONTEXT MODIFICATION RESPONSE message. The gNB-DU shall regard it as a reconfiguration with sync as defined in TS 38.331 [8]. If the gNB-DU accepts the request for LTM for that *SpCell*, the gNB-DU shall generate and include the *CellGroupConfig* IE for the accepted LTM candidate cell in the UE CONTEXT MODIFICATION RESPONSE message.

If the *LTM Indicator* IE set to "C-LTM" is contained in the *LTM Information Modify* IE included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, consider that the request concerns conditional LTM for the included *SpCell ID* IE and shall include it as the *Requested Target Cell ID* IE in the UE CONTEXT MODIFICATION RESPONSE message. If the gNB-DU accepts the request for conditional LTM for that *SpCell*, the gNB-DU shall generate and include the *CellGroupConfig* IE for the accepted LTM candidate cell in the UE CONTEXT MODIFICATION RESPONSE message.

If the *LTM Indicator* IE is set to "C-LTM" and the *Request for L1 Execution Condition* IE is present in the *LTM Information Modify* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall generate the conditional LTM L1 execution condition(s) for the candidate cell(s) as included in the *Request for L1 Execution Condition* IE and provide the *L1 Execution Condition List* IE within the *LTM Configuration* IE in the UE CONTEXT MODIFICATION RESPONSE message.

If the *Request for Lower Layer Configuration* IE set to "true" is contained within the *Reference Configuration* IE in the *LTM Information Modify* IE included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, include the *Reference Configuration Information* IE in the *LTM Configuration* IE in the UE CONTEXT MODIFICATION RESPONSE message to provide lower layer configuration for the gNB-CU to generate the LTM reference configuration.

If the *Reference Configuration Information* IE is contained within the *Reference Configuration* IE in the *LTM Information Modify* IE included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, take it into account for generating the LTM lower layer configuration.

If the *CSI Resource Configuration* IE is contained in the *LTM Information Modify* IE included in the UE CONTEXT MODIFICATION REQUEST message and the *SpCell ID* IE is also included, the gNB-DU shall, if supported, use it to generate the LTM CSI reporting configuration for L1 measurement in the *CellGroupConfig* IE, and/or to generate the LTM CSI reporting configuration in the *CSI Report Configuration for CSI Acquisition* IE for CSI acquisition and/or CS-IM measurement for the requested LTM candidate cell identified by the *SpCell ID* IE.

If the *CSI Resource Configuration* IE received in the UE CONTEXT MODIFICATION REQUEST message and the *SpCell ID* IE is also included, and if the *CSI Resource Configuration* IE does not contain the CSI-RS resources for L1 measurement, the gNB-DU shall not use it to generate the LTM CSI reporting configuration for L1 measurement in the *CellGroupConfig* IE.

If the *CSI Resource Configuration* IE is contained in the *LTM Information Modify* IE included in the UE CONTEXT MODIFICATION REQUEST message while the *SpCell ID* IE is absent, the gNB-DU shall, if supported, use it to generate the LTM CSI reporting configuration in the *CellGroupConfig* IE for the serving cell.

If the *Request for CSI-RS Resource Configuration for L1 measurements* IE set to "true" is contained in the *LTM Information Modify* IE included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, include the *CSI-RS Resource Configuration for L1 measurements* IE in the *LTM Configuration* IE in the UE CONTEXT MODIFICATION RESPONSE message.

If the *CSI-RS Resource Configuration for Early CSI Acquisition* IE is included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, act as specified in TS 38.401 [4].

If the *LTM Configuration ID Mapping List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, consider this as the mapping information for the LTM candidate cell(s).

If the *Early Sync Information Request* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, include *Early Sync Information* IE of the accepted candidate cell for early TA acquisition (early UL synchronisation) in the UE CONTEXT MODIFICATION RESPONSE message. If the *Early UL Sync*

Configuration IE is included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, consider it as the generated early UL sync information from the accepted candidate cell in the gNB-DU. If the *Early UL Sync Configuration for SUL* IE is included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, consider it as the generated early UL sync information for SUL from the accepted candidate cell in the gNB-DU.

If the *Early Sync Candidate Cell Information List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it as specified in TS 38.401 [4]. If the *UE Based TA Measurement Configuration* IE is contained in the *Early Sync Candidate Cell Information List* IE for some candidate cell, the gNB-DU shall, if supported, take them into account for UE based TA measurement during LTM cell switch as specified in TS 38.331 [8].

If the *Early Sync Serving Cell Information* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it as specified in TS 38.401 [4]. If the *UE Based TA Measurement Configuration* IE is contained in the *Early Sync Serving Cell Information* IE, the gNB-DU shall, if supported, take it into account for UE based TA measurement during LTM cell switch as specified in TS 38.331 [8].

If the *LTM CFRA Resource Config List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it for the LTM cell switch command as specified in TS 38.321 [16].

If the *LTM Configuration* IE is included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, consider it as the generated configuration for LTM from the accepted candidate cell in the gNB-DU.

If the *LTM Cells to be Released List* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, release the configured candidate cells in the list.

If the *LTM Reset Information* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, take them into account for L2 reset (i.e., RLC re-establishment) during an intra-DU LTM cell switch as specified in TS 38.331 [8].

If the *Complete Candidate Configuration Indicator* IE set to "complete" is contained in the *LTM Configuration* IE included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, consider that the LTM candidate configuration is a complete candidate configuration.

If the *LTM Security Information* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, store it and take it into account for supporting the UE's AS security continuation during an LTM cell switch and act as specified in TS 38.401 [4] and TS 38.321 [16].

If the *LTM with SCG Indicator* IE set to "true" is contained in the *LTM Information SN Modification* IE included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, consider that the UE Context Modification procedure has been triggered as part of an MCG LTM. The gNB-DU shall consider that the request concerns a PSCell change for the included *SpCell ID* IE and shall include it as the *Requested Target Cell ID* IE in the UE CONTEXT MODIFICATION RESPONSE message. The gNB-DU shall regard it as a reconfiguration with sync as defined in TS 38.331 [8].

If the *Inter-SN LTM MCG Information* IE is included in the UE CONTEXT MODIFICATION REQUEST message and the Inter-SN LTM Trigger is set to:

- "LTM-preparation": the gNB-DU shall, if supported, consider that the request concerns an inter-SN LTM. The gNB-DU takes the included *CG-Config* and/or *CG-ConfigInfo* IE into account, and may provide a corresponding *CellGroupConfig* IE for MCG configuration preparation in the UE CONTEXT MODIFICATION RESPONSE message. The UE CONTEXT MODIFICATION RESPONSE message also includes a *Requested Target Cell ID* IE corresponding to the *PSCell ID* IE in the UE CONTEXT MODIFICATION REQUEST message.

- "LTM-executed": the gNB-DU shall, if supported, consider that, for the included *PSCell ID* IE corresponding to the selected PSCell, the UE has successfully executed the inter-SN LTM cell switch. The gNB-DU shall apply the corresponding *CellGroupConfig* IE and/or the *Reference Configuration Information* IE for MCG configuration.

- "LTM-cancel": the gNB-DU shall, if supported, consider that the gNB-CU is about to release the prepared MCG configuration(s) corresponding to the PSCell(s) identified by the included NR CGI(s) in the *Candidate PSCells To Be Cancelled List* IE.

If the *Direct Path Addition* IE is contained in the *Path Addition Information* IE which is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, consider that the request concerns the direct path addition for the included *SpCell ID* IE as specified in TS 38.401 [4] and regard it as a reconfiguration with sync as defined in TS 38.331 [8]. If the *Indirect Path Addition* IE is contained in the *Path Addition Information* IE, the gNB-DU shall, if supported, consider that the request concerns the indirect path addition for the MP Remote UE using PC5 link and use it as specified in TS 38.401 [4]. If the *N3C Indirect Path Addition* IE is contained in the *Path Addition Information* IE, the gNB-DU shall, if supported, consider that the request concerns the indirect path addition for the MP Remote UE using N3C and use it as specified in TS 38.401 [4].

If the *S-NSSAI* IE is included within the *DRB to Be Modified Item* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, store the corresponding information and replace any existing information.

If the *S-CPAC Lower Layer Reference Config Request* IE set to "true" is contained in the *Conditional Intra-DU Mobility Information* IE included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, provide the lower layer configuration in the *Reference Configuration Information* IE in the *S-CPAC Configuration* IE in the UE CONTEXT MODIFICATION RESPONSE message for the gNB-CU to generate the S-CPAC reference configuration.

If the *Complete Candidate Configuration Indicator* IE set to "complete" is contained in the *S-CPAC Configuration* IE included in the UE CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall, if supported, consider that the S-CPAC candidate configuration is a complete candidate configuration.

If the *musim-CandidateBandList* IE is included in the *CU to DU RRC Information* IE in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use it for temporary capability restriction.

If the *DL LBT Failure Information Request* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, consider that the gNB-CU requests collection of DL LBT failure information for the analysis of the MRO events of the UE specified in TS 38.300 [6], , and act as specified in TS 38.401 [4].

If the *Ranging and Sidelink Positioning Service Information* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, update its service information for the UE accordingly. If the *Ranging and Sidelink Positioning Authorized* IE within the *Ranging and Sidelink Positioning Service Information* IE is set to "not authorized", the gNB-DU shall, if supported, initiate actions to ensure that the UE is no longer accessing the Ranging and Sidelink Positioning service.

For each DRB that has been successfully established or modified and for which the *Performance Delay Monitoring* IE was included in the *DRB to Be Setup List* IE or in the *DRB to Be Modified List* IE contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, store this information and use it to perform or update delay measurements on the successfully established or modified DRBs.

Interaction with UE Inactivity Notification procedure

If the *SDT Volume Threshold* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, use the information during an SDT transaction to inform the gNB-CU via the UE INACTIVITY NOTIFICATION message as specified in TS 38.401 [4].

For each GBR QoS flow whose DRB has been successfully established or modified and the *Monitoring Request on Available Bitrate* IE was included in the *GBR QoS Flow Information* IE contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, store this information and, perform available bitrate monitoring, as specified in TS 23.501 [21].

For each QoS flow whose DRB to be established or modified, if the *MMSID* IE was included in the *QoS Flow Level QoS Parameters* IE contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, consider that the QoS flow is related to a multi-modal service, as described in TS 23.501 [21] and TS 38.300 [6].

If the *Indication of Bitrate Adaptation* IE set to "uplink" is included in the *QoS Flow Level QoS Parameters* IE contained in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, store the received indication and use it as defined in TS 38.300 [6].

Interaction with UE Context Setup or UE Context Modification (gNB-CU initiated) procedures

If the UE CONTEXT MODIFICATION REQUEST message is sent for a UE context set up for S-CPAC and contains the *Transmission Action Indicator* IE set to "stop", the gNB-DU shall, if supported, reset the UE context for the included *SpCell ID* IE, prepare for subsequent CPAC. The gNB-DU shall include the *SpCell ID* IE as the *Requested Target Cell ID* IE in the UE CONTEXT MODIFICATION RESPONSE message.

If the UE CONTEXT MODIFICATION REQUEST message is sent for a UE context set up for conditional LTM and contains the *Transmission Action Indicator* IE set to "stop", the gNB-DU shall, if supported, reset the UE context for the included *SpCell ID* IE, prepare for conditional LTM. The gNB-DU shall include the *SpCell ID* IE as the *Requested Target Cell ID* IE and may include the *TA Remaining Information List* IE in the UE CONTEXT MODIFICATION RESPONSE message.

8.3.4.3 Unsuccessful Operation

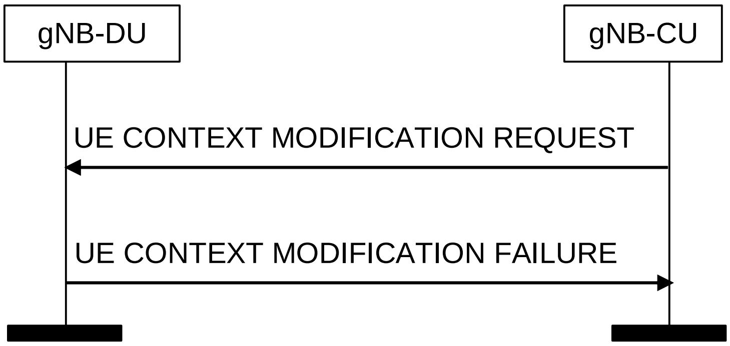

Figure 8.3.4.3-1: UE Context Modification procedure. Unsuccessful operation

In case none of the requested modifications of the UE context can be successfully performed, the gNB-DU shall respond with the UE CONTEXT MODIFICATION FAILURE message with an appropriate cause value. If the *Conditional Intra-DU Mobility Information* IE was included in the UE CONTEXT MODIFICATION REQUEST message and set to "CHO-initiation", the gNB-DU shall include the received *SpCell ID* IE as the *Requested Target Cell ID* IE in the UE CONTEXT MODIFICATION FAILURE message.

If the gNB-DU is not able to accept the *SpCell ID* IE in UE CONTEXT MODIFICATION REQUEST message, it shall reply with the UE CONTEXT MODIFICATION FAILURE message.

If the *Conditional Intra-DU Mobility Information* IE was included and set to "CHO-initiation" or "CHO-replace", but the *SpCell ID* IE was not included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall respond with the UE CONTEXT MODIFICATION FAILURE message with an appropriate cause value.

If the *LTM Information Modify* IE was included, but the *SpCell ID* IE and the *CSI Resource Configuration* IE were not included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall respond with the UE CONTEXT MODIFICATION FAILURE message with an appropriate cause value.

If the gNB-DU is not able to accept the UE CONTEXT MODIFICATION REQUEST message for mobility because an LTM command has been triggered to the UE, it shall reply with the UE CONTEXT MODIFICATION FAILURE message with an appropriate cause value.

8.3.4.4 Abnormal Conditions

If the gNB-DU receives a UE CONTEXT MODIFICATION REQUEST message containing a *E-UTRAN QoS* IE for a GBR QoS DRB but where the *GBR QoS Information* IE is not present, the gNB-DU shall report the establishment of the corresponding DRB as failed in the *DRB Failed to Setup List* IE of the UE CONTEXT MODIFICATION RESPONSE message with an appropriate cause value.

If the gNB-DU receives a UE CONTEXT MODIFICATION REQUEST message containing a *DRB QoS* IE for a GBR QoS DRB but where the *GBR QoS Flow Information* IE is not present, the gNB-DU shall report the establishment of the corresponding DRBs as failed in the *DRB Failed to Setup List* IE of the UE CONTEXT MODIFICATION RESPONSE message with an appropriate cause value.

If the *Delay Critical* IE is included in the *Dynamic 5QI Descriptor* IE within the *DRB QoS* IE in the UE CONTEXT MODIFICATION REQUEST message and is set to the value "delay critical" but the *Maximum Data Burst Volume* IE is not present, the gNB-DU shall report the establishment of the corresponding DRB as failed in the *DRB Failed to Setup List* IE of the of the UE CONTEXT MODIFICATION RESPONSE message with an appropriate cause value.

If one or more candidate cells in the *Candidate Cells To Be Cancelled List* IE included in the UE CONTEXT MODIFICATION REQUEST message were not prepared using the same UE-associated signaling connection, the gNB-DU shall ignore those non-associated candidate cells.

If more than one of the following IEs, i.e., the *Uplink TxDirectCurrentList Information* IE or the *Uplink TxDirectCurrentTwoCarrierList Information* IE or the *Uplink TxDirectCurrentMoreCarrierList Information* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall consider it as a logical error.

If one or more LTM cells in the *LTM Cells To Be Released List* IE included in the UE CONTEXT MODIFICATION REQUEST message were not prepared using the same UE-associated signaling connection, the gNB-DU shall ignore those non-associated LTM cells.

8.3.5 UE Context Modification Required (gNB-DU initiated)

8.3.5.1 General

The purpose of the UE Context Modification Required procedure is to modify the established UE Context, e.g., modifying and releasing radio bearer resources, or sidelink radio bearer resources or candidate cells in conditional handover, conditional PSCell addition, conditional PSCell change, subsequent CPAC, or LTM. The procedure uses UE-associated signalling.

8.3.5.2 Successful Operation

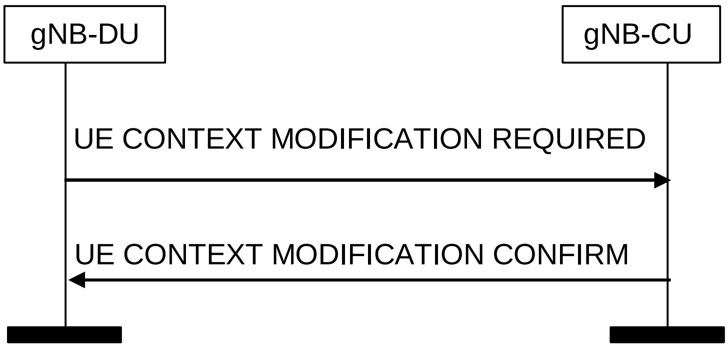

Figure 8.3.5.2-1: UE Context Modification Required procedure. Successful operation

The F1AP UE CONTEXT MODIFICATION REQUIRED message is initiated by the gNB-DU.

The gNB-CU reports the successful update of the UE context in the UE CONTEXT MODIFICATION CONFIRM message.

For a given bearer for which PDCP CA duplication or multi-path relay based duplication was already configured, if two *DL UP TNL Information* IEs are included in UE CONTEXT MODIFICATION REQUIRED message for a DRB, the gNB-CU shall include two *UL UP TNL Information* IEs in UE CONTEXT MODIFICATION CONFIRM message. The gNB-CU and gNB-DU use the *UL UP TNL Information* IEs and *DL UP TNL Information* IEs to support packet duplication for intra-gNB-DU CA and multi-path relay as defined in TS 38.470 [2], and the first *UP TNL Information* IE is still for the primary path.

For a given bearer for which PDCP CA duplication or multi-path relay based duplication was already configured, if one or two *Additional PDCP Duplication UP TNL Information* IEs are included in the UE CONTEXT MODIFICATION REQUIRED message for a DRB, the gNB-CU shall, if supported, include one or two *Additional PDCP Duplication UP TNL Information* IEs in the UE CONTEXT MODIFICATION CONFIRM message. The gNB-CU and gNB-DU use the *Additional PDCP Duplication UP TNL Information* IEs to support packet duplication for intra-gNB-DU CA and multi-path relay as defined in TS 38.470 [2].

If the *BH Information* IE is included in the *UL UP TNL Information to be setup List* IE or the *Additional PDCP Duplication TNL List* IE for a DRB, the gNB-DU shall, if supported, use the indicated BAP Routing ID and BH RLC channel for transmission of the corresponding GTP-U packets to the IAB-donor, as specified in TS 38.340 [30].

If the *Resource Coordination Transfer Container* IE is included in the UE CONTEXT MODIFICATION REQUIRED, the gNB-CU shall transparently transfer this information for the purpose of resource coordination as described in TS 36.423 [9], TS 38.423 [28].

For EN-DC operation, if the gNB-CU includes the *Resource Coordination Transfer Information* IE in the UE CONTEXT MODIFICATION CONFIRM message, the gNB-DU shall, if supported, use it for the purpose of resource coordination. If the gNB-CU received the MeNB Resource Coordination Information as defined in TS 36.423 [9], after completion of UE Context Modification Required procedures, the gNB-CU shall transparently transfer it to the gNB-DU via the *Resource Coordination Transfer Container* IE in the UE CONTEXT MODIFICATION CONFIRM message. The gNB-DU shall use the information received in the *Resource Coordination Transfer Container* IE for reception of MeNB Resource Coordination Information at the gNB acting as secondary node as described in TS 36.423 [9]. If the *Resource Coordination E-UTRA Cell Information* IE is included in the *Resource Coordination Transfer Information* IE, the gNB-DU shall store the information replacing previously received information for the same E-UTRA cell, and use the stored information for the purpose of resource coordination. If the *Ignore PRACH Configuration* IE is present and set to "true" the *E-UTRA PRACH Configuration* IE in the UE CONTEXT MODIFICATION CONFIRM message shall be ignored.

For NGEN-DC or NE-DC operation, if the gNB-CU includes the *Resource Coordination Transfer Information* IE in the UE CONTEXT MODIFICATION CONFIRM message, the gNB-DU shall, if supported, use it for the purpose of resource coordination. If the gNB-CU received the MR-DC Resource Coordination Information as defined in TS 38.423 [28], after completion of UE Context Modification Required procedures, the gNB-CU shall transparently transfer it to the gNB-DU via the *Resource Coordination Transfer Container* IE in the UE CONTEXT MODIFICATION CONFIRM message. The gNB-DU shall use the information received in the *Resource Coordination Transfer Container* IE for reception of MR-DC Resource Coordination Information at the gNB as described in TS 38.423 [28].

If the *DU to CU RRC Information* IE is included in the UE CONTEXT MODIFICATION REQUIRED message, the gNB-CU shall perform RRC Reconfiguration as described in TS 38.331 [8]. The *CellGroupConfig* IE shall transparently be signaled to the UE as specified in TS 38.331 [8].

If the *ServCellInfoList* IE is included in the *DU to CU RRC Information* IE contained in the UE CONTEXT MODIFICATION REQUIRED message, the gNB-CU shall take it into account to generate the content of inter-node message, i.e., *CG-Config* or *CG-ConfigInfo*, as described in TS 38.331 [8].

If the UE CONTEXT MODIFICATION CONFIRM message includes the *Execute Duplication* IE, the gNB-DU shall perform CA based duplication or multi-path relay based duplication, if configured, for the SRB for the included *RRC-Container* IE.

If the UE CONTEXT MODIFICATION REQUIRED message contains the *RLC Status* IE, the gNB-CU shall assume that RLC has been reestablished at the gNB-DU and may trigger PDCP data recovery.

If the *Candidate Cells To Be Cancelled List* IE is included in the UE CONTEXT MODIFICATION REQUIRED message, the gNB-CU shall consider that only the resources reserved for the candidate cells identified by the included NR CGIs and associated to the UE-associated signaling identified by the *gNB-CU UE F1AP ID* IE and the *gNB-CU UE F1AP ID* IE are about to be released by the gNB-DU.

If the *PC5 RLC Channel Required to be Modified List* IE or the *PC5 RLC Channel Required to be Released List* IE is included in the UE CONTEXT MODIFICATION REQUIRED message and the F1AP-IDs is associated with a U2N Relay UE, the *PC5 RLC Channel Required to be Modified List* IE or the *PC5 RLC Channel Required to be Released List* shall include the *Remote UE Local ID* in case of single-hop relay or for the PC5 RLC channel between the U2N Remote UE and the First U2N Relay UE in multi-hop relay and correspondingly, the *PC5 RLC Channel Modified Item IEs* in the UE CONTEXT MODIFICATION CONFIRM message shall include the *Remote UE Local ID* IE.

If the *UE Multicast MRB Required to Be Modified List* IE is included in the UE CONTEXT MODIFICATION REQUIRED message

- containing for an MRB the *MRB type reconfiguration* IE set to "true" the gNB-CU shall take the *MRB Reconfigured RLC mode* IE into account to reconfigure the UE and to decide whether to request a PDCP status report as specified in TS 38.300 [6] and include the *MBS PTP Retransmission Tunnel Required* IE in the *UE Multicast MRB Confirmed to Be Modified Item IEs* IE.

- containing for an MRB the *Multicast F1-U Context Reference CU* IE the gNB-CU shall, if supported, replace previously provided information by the newly received and take it into account when retrieving MRB progress information.

If the *LTM Cells To Be Released List* IE is included in the UE CONTEXT MODIFICATION REQUIRED message, the gNB-CU shall, if supported, consider that the configured candidate cells in the list are about to be released by the gNB-DU.

If the *LTM Configurations to Modify List* IE is contained in the UE CONTEXT MODIFICATION REQUIRED message, the gNB-CU shall, if supported, replace the previously received LTM configurations with the received information and act as specified in TS 38.401 [4].

Interaction with the Multicast Distribution Setup procedure:

If the UE CONTEXT MODIFICATION CONFIRM message contains for an MRB the *MBS PTP Retransmission Tunnel Required* IE in the *UE Multicast MRB Confirmed to Be Modified Item IEs* IE the gNB-DU shall, if supported, trigger the Multicast Distribution Setup procedure to setup requested F1-U resources, if applicable.

8.3.5.2A Unsuccessful Operation

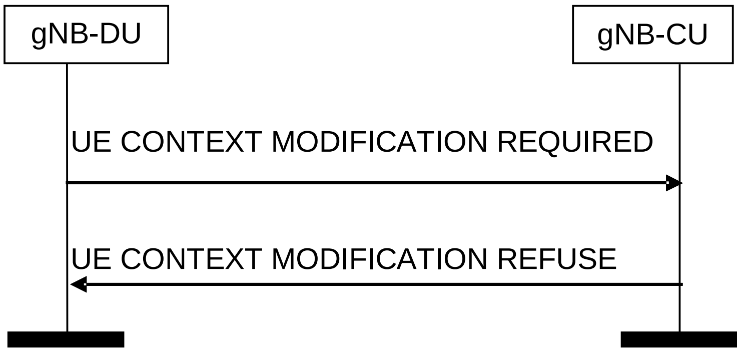

Figure 8.3.5.2A-1: UE Context Modification Required procedure. Unsuccessful operation.

In case none of the requested modifications of the UE context can be successfully performed, the gNB-CU shall respond with the UE CONTEXT MODIFICATION REFUSE message with an appropriate cause value.

8.3.5.3 Abnormal Conditions

If one or more candidate cells in the *Candidate Cells To Be Cancelled List* IE included in the UE CONTEXT MODIFICATION REQUIRED message were not prepared using the same UE-associated signaling connection, the gNB-CU shall ignore those non-associated candidate cells.

If one or more LTM cells in the *LTM Cells To Be Released List* IE included in the UE CONTEXT MODIFICATION REQUIRED message were not prepared using the same UE-associated signaling connection, the gNB-CU shall ignore those non-associated LTM cells.

8.3.6 UE Inactivity Notification

8.3.6.1 General

This procedure is initiated by the gNB-DU to indicate the UE activity event.

The procedure is also used to request the termination of SDT session.

The procedure uses UE-associated signalling.

8.3.6.2 Successful Operation

Figure 8.3.6.2-1: UE Inactivity Notification procedure.

The gNB-DU initiates the procedure by sending the UE INACTIVITY NOTIFICATION message to the gNB-CU.

If the *DRB ID* IE is included in the *DRB Activity Item* IE in the UE INACTIVITY NOTIFICATION message, the *DRB Activity* IE shall also be included

If the gNB-CU receives the *SDT Termination Request* IE in the UE INACTIVITY NOTIFICATION message, the gNB-CU shall, if supported, consider that the termination of the ongoing SDT transaction is requested from the gNB-DU for this UE and act as specified in TS 38.300 [6].

8.3.6.3 Abnormal Conditions

Not applicable.

8.3.7 Notify

8.3.7.1 General

The purpose of the Notify procedure is to enable the gNB-DU to inform the gNB-CU that the QoS of an already established GBR DRB cannot be fulfilled any longer or that it can be fulfilled again. The procedure uses UE-associated signalling.

8.3.7.2 Successful Operation

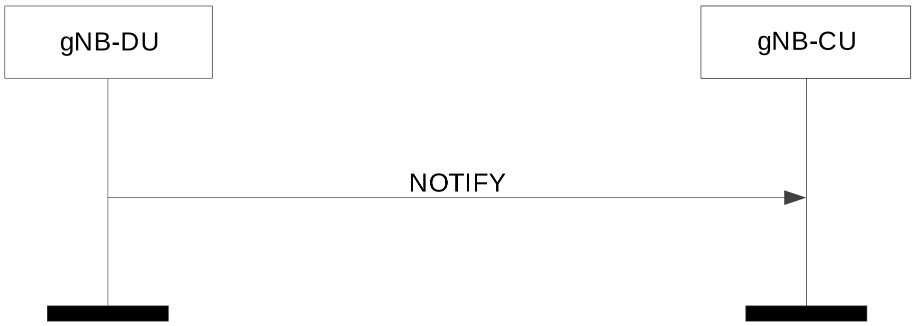

Figure 8.3.7.2-1: Notify procedure. Successful operation.

The gNB-DU initiates the procedure by sending a NOTIFY message.

The NOTIFY message shall contain the list of the GBR DRBs associated with notification control for which the QoS is not fulfilled anymore or for which the QoS is fulfilled again by the gNB-DU. The gNB-DU may also indicate an alternative QoS parameters set which it can currently fulfil in the *Current QoS Parameters Set Index* IE.The gNB-DU may also include the TSC feedback information in the *TSC Traffic Characteristics Feedback* IE.

Upon reception of the NOTIFY message, the gNB-CU may identify which are the affected PDU sessions and QoS flows. The gNB-CU may inform the 5GC that the QoS for these PDU sessions or QoS flows is not fulfilled any longer or it is fulfilled again.

8.3.7.3 Abnormal Conditions

Not applicable.

8.3.8 Access Success

8.3.8.1 General

The purpose of the Access Success procedure is to enable the gNB-DU to inform the gNB-CU of which cell the UE has successfully accessed during conditional handover, conditional PSCell addition, conditional PSCell change, LTM, conditional LTM, or subsequent CPAC. The procedure uses UE-associated signalling.

8.3.8.2 Successful Operation

Figure 8.3.8.2-1: Access Success procedure. Successful operation.

The gNB-DU initiates the procedure by sending a ACCESS SUCCESS message.

Upon reception of the ACCESS SUCCESS message, the gNB-CU shall consider that the UE successfully accessed the cell indicated by the included *NR CGI* IE in this gNB-DU and consider all the other CHO or conditional PSCell addition or conditional PSCell change preparations accepted for this UE under the same UE-associated signaling connection in this gNB-DU as cancelled. In case of subsequent mobility, the other preparations accepted for this UE under the same UE-associated signaling connection in this gNB-DU are kept.

Interaction with other procedure:

The gNB-CU may initiate UE Context Release procedure toward the other signalling connections or other candidate gNB-DUs for this UE, if any.

8.3.8.3 Abnormal Conditions

If the ACCESS SUCCESS message refers to a context that does not exist, the gNB-CU shall ignore the message.

8.3.9 DU-CU Cell Switch Notification

8.3.9.1 General

The purpose of the DU-CU Cell Switch Notification procedure is to enable the gNB-DU to inform the gNB-CU about the initiation of the cell switch command to the UE. This procedure is also used to transfer the selected TCI state from the gNB-DU to the gNB-CU. The procedure uses UE-associated signalling.

8.3.9.2 Successful Operation

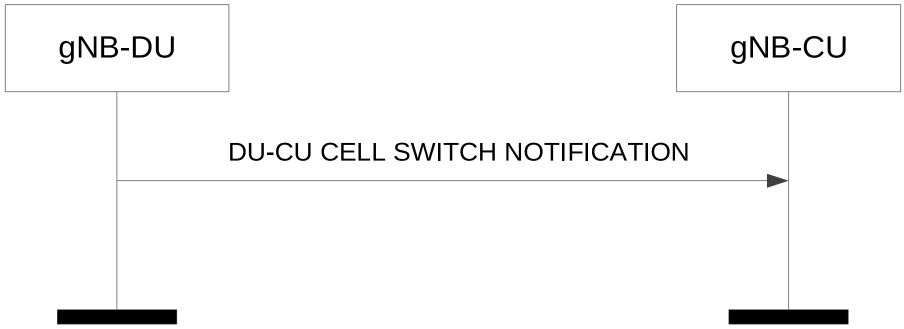

Figure 8.3.9.2-1: DU-CU Cell Switch Notification procedure. Successful operation.

The gNB-DU initiates the procedure by sending a DU-CU CELL SWITCH NOTIFICATION message.

Upon reception of the DU-CU CELL SWITCH NOTIFICATION message, the gNB-CU shall, if supported, consider that a cell switch command was sent to the UE where the target cell is indicated by the included *Cell ID* IE.

Interaction with CU-DU Cell Switch Notification procedure:

If the *LTM Cell Switch Information* IE is included in the DU-CU CELL SWITCH NOTIFICATION message, the gNB-CU shall, if supported, forward it to the target gNB-DU in the CU-DU CELL SWITCH NOTIFICATION message.

If the *TA Information List* IE is included in the DU-CU CELL SWITCH NOTIFICATION message, the gNB-CU shall, if supported, forward it to the target gNB-DU in the CU-DU CELL SWITCH NOTIFICATION message to transfer the TA value(s).

8.3.9.3 Unsuccessful Operation

Not applicable.

8.3.9.4 Abnormal Conditions

Not applicable.

8.3.10 CU-DU Cell Switch Notification

8.3.10.1 General

The purpose of the CU-DU Cell Switch Notification procedure is to enable the gNB-CU to inform the gNB-DU about the initiation of the cell switch command to the UE. This procedure is also used to transfer the selected TCI state from the gNB-CU to the gNB-DU. The procedure uses UE-associated signalling.

8.3.10.2 Successful Operation

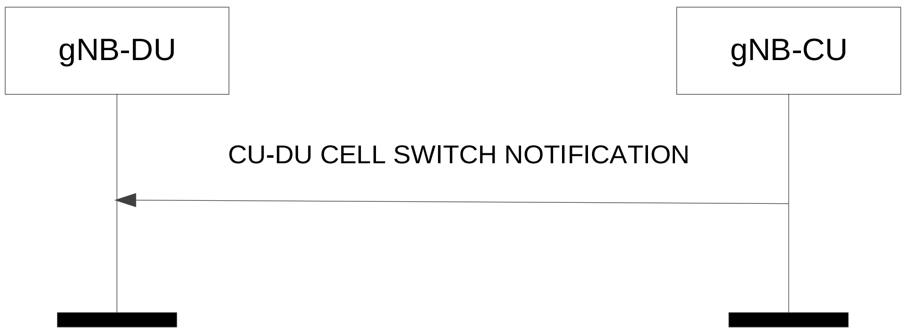

Figure 8.3.10.2-1: CU-DU Cell Switch Notification procedure. Successful operation.

The gNB-CU initiates the procedure by sending a CU-DU CELL SWITCH NOTIFICATION message.

Upon reception of the CU-DU CELL SWITCH NOTIFICATION message, the gNB-DU shall, if supported, consider that a cell switch command was sent to the UE where the target cell is indicated by the included *Cell ID* IE.

If the *LTM Cell Switch Information* IE is included in the CU-DU CELL SWITCH NOTIFICATION message, the gNB-DU shall, if supported, use it as specified in TS 38.401 [4].

If the *TA Information List* IE is included in the CU-DU CELL SWITCH NOTIFICATION message, the gNB-DU shall, if supported, use it as specified in TS 38.401 [4]. If the *Tag ID Pointer* IE is also included, the gNB-DU shall, if supported, consider it to determine the TAG corresponding to the received TA value.

If the *Mobility Issue* IE is included in the *Last Visited LTM Cells* IE in the CU-DU CELL SWITCH NOTIFICATION message, the gNB-DU may take this into account to resolve the potential mobility problems (e.g. ping-pong) detected by the gNB-CU.

8.3.10.3 Unsuccessful Operation

Not applicable.

8.3.10.4 Abnormal Conditions

Not applicable.

8.3.11 CU-DU Mobility Initiation

8.3.11.1 General

The purpose of the CU-DU Mobility Initiation procedure is to trigger cell switch command and/or early synchronization for the UE. The procedure uses UE-associated signalling.

For the case of conditional LTM, this procedure is used to trigger the early synchronization for the UE.

8.3.11.2 Successful Operation

Figure 8.3.11.2-1: CU-DU Mobility Initiation procedure. Successful operation.

The procedure is initiated with the CU-DU MOBILITY INITIATION REQUEST message sent from the gNB-CU to the gNB-DU. Upon receipt of this message,

- if the CU-DU MOBILITY INITIATION REQUEST message indicates Mobility Trigger, the gNB-DU shall trigger LTM procedure(s) accordingly.

- if the CU-DU MOBILITY INITIATION REQUEST message indicates Assistance Information, the gNB-DU shall take this information into account for triggering LTM procedure(s).

8.3.11.3 Unsuccessful Operation

Not applicable.

8.3.11.4 Abnormal Conditions

If the *Triggering Indication* IE bitmap is set to "0" (all bits are set to "0") in the CU-DU MOBILITY INITIATION REQUEST message, the gNB-DU shall initiate an ERROR INDICATION message with an appropriate cause value.

8.3.12 DU-CU CSI-RS Coordination

8.3.12.1 General

The purpose of the DU-CU CSI-RS Coordination procedure is e.g. to enable the gNB-DU to request the gNB-CU to activate/deactivate the SP CSI-RS transmissions from specific cells. The procedure uses UE-associated signalling.

8.3.12.2 Successful Operation

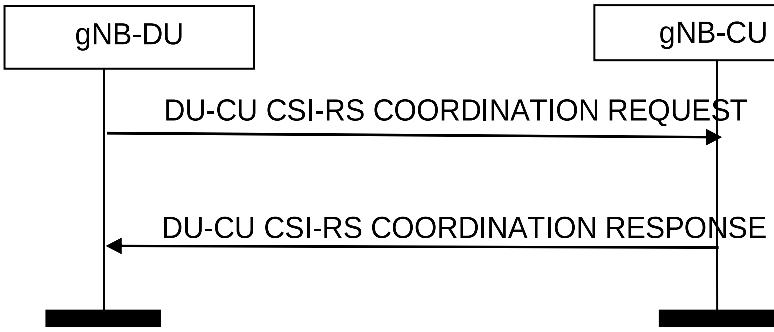

Figure 8.3.12.2-1: DU-CU CSI-RS Coordination procedure. Successful operation.

The gNB-DU initiates the procedure by sending a DU-CU CSI-RS COORDINATION REQUEST message.

If the *TCI State Information List* IE is included in the DU-CU CSI-RS COORDINATION REQUEST message, the gNB-CU shall, if supported, use it for Semi-Persistent CSI-RS activation.

8.3.12.3 Unsuccessful Operation

Not applicable.

8.3.12.4 Abnormal Conditions

Not applicable.

8.3.13 CU-DU CSI-RS Coordination

8.3.13.1 General

The purpose of the CU-DU CSI-RS Coordination procedure is e.g. to enable the gNB-CU to request the gNB-DU to activate/deactivate the SP CSI-RS transmission from specific cells. The procedure uses UE-associated signalling.

8.3.13.2 Successful Operation

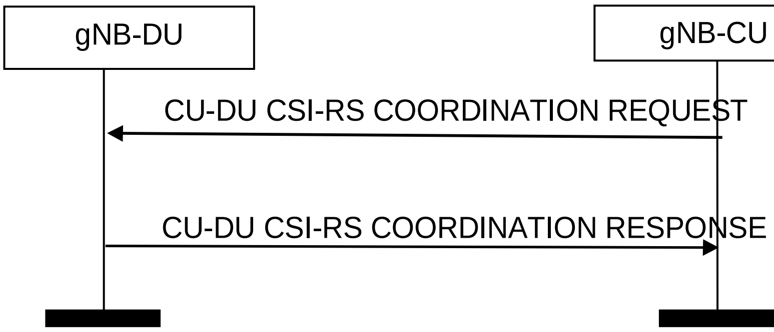

Figure 8.3.13.2-1: CU-DU CSI-RS COORDINATION procedure. Successful operation.

The gNB-CU initiates the procedure by sending a CU-DU CSI-RS COORDINATION REQUEST message.

If the *TCI State Information* IE is included in the CU-DU CSI-RS COORDINATION REQUEST message, the gNB-DU shall, if supported, use it for Semi-Persistent CSI-RS activation.

8.3.13.3 Unsuccessful Operation

Not applicable

8.3.13.4 Abnormal Conditions

Not applicable

8.4 RRC Message Transfer procedures

8.4.1 Initial UL RRC Message Transfer

8.4.1.1 General

The purpose of the Initial UL RRC Message Transfer procedure is to transfer the initial RRC message to the gNB-CU. The procedure uses non-UE-associated signaling.

8.4.1.2 Successful operation

Figure 8.4.1.2-1: Initial UL RRC Message Transfer procedure.

The gNB-DU initiates the procedure by sending an INITIAL UL RRC MESSAGE TRANSFER. The establishment of the UE-associated logical F1-connection shall be initiated as part of the procedure.

If neither the *DU to CU RRC Container* IE nor the *Sidelink Configuration Container* IE in the *Sidelink Relay Configuration* IE is included in the INITIAL UL RRC MESSAGE TRANSFER, the gNB-CU should reject the UE under the assumption that the gNB-DU is not able to serve such UE. If the gNB-DU is able to serve the UE, the gNB-

DU shall include the *DU to CU RRC Container* IE or the *Sidelink Configuration Container* IE in the *Sidelink Relay Configuration* IE and the gNB-CU shall configure the UE as specified in TS 38.331 [8]. The gNB-DU shall not include the *ReconfigurationWithSync* field in the *CellGroupConfig* IE as defined in TS 38.331 [8] of the *DU to CU RRC Container* IE.

If the *SUL Access Indication* IE is included in the INITIAL UL RRC MESSAGE TRANSFER, the gNB-CU shall consider that the UE has performed access on SUL carrier.

If the *RAN UE ID* IE is contained in the INITIAL UL RRC MESSAGE TRANSFER message, the gNB-CU shall, if supported, store it and use it as specified in TS 38.401 [4].

If the *RRC-Container-RRCSetupComplete* IE is included in the INITIAL UL RRC MESSAGE TRANSFER, the gNB-CU shall take it into account as specified in TS 38.401 [4].

If the *NR RedCap UE Indication* IE is included in the INITIAL UL RRC MESSAGE TRANSFER message, the gNB-CU shall, if supported, consider that the accessing UE is a RedCap UE.

If the *NR eRedCap UE Indication* IE is included in the INITIAL UL RRC MESSAGE TRANSFER message, the gNB-CU shall, if supported, consider that the accessing UE is an eRedCap UE.

If the *SDT Information* IE is included in the INITIAL UL RRC MESSAGE TRANSFER, the gNB-CU shall, if supported, consider that the UE is accessing for SDT as defined in TS 38.300 [6], and may use the information contained in the *SDT Assistant Information* IE, if any, for context retrieval.

If the *Sidelink Relay Configuration* IE is included in the INITIAL UL RRC MESSAGE TRANSFER, the gNB-CU shall, if supported, consider that the UE is a NR ProSe Layer-2 U2N Remote UE identified by the *Remote UE Local ID* IE, and it is connected to the U2N Relay UE indicated by the *gNB-DU UE F1AP ID of Relay UE* IE.

8.4.1.3 Abnormal Conditions

Not applicable.

8.4.2 DL RRC Message Transfer

8.4.2.1 General

The purpose of the DL RRC Message Transfer procedure is to transfer an RRC message The procedure uses UE-associated signalling.

8.4.2.2 Successful operation

Figure 8.4.2.2-1: DL RRC Message Transfer procedure

The gNB-CU initiates the procedure by sending a DL RRC MESSAGE TRANSFER message. If a UE-associated logical F1-connection exists, the DL RRC MESSAGE TRANSFER message shall contain the *gNB-DU UE F1AP ID* IE, which should be used by gNB-DU to lookup the stored UE context.If no UE-associated logical F1-connection exists, the UE-associated logical F1-connection shall be established at reception of the DL RRC MESSAGE TRANSFER message.

If the *Index to RAT/Frequency Selection Priority* IE is included in the DL RRC MESSAGE TRANSFER, the gNB-DU may use it for RRM purposes. If the *Additional RRM Policy Index* IE is included in the DL RRC MESSAGE TRANSFER, the gNB-DU may use it for RRM purposes.

The DL RRC MESSAGE TRANSFER message shall include, if available, the *old gNB-DU UE F1AP ID* IE so that the gNB-DU can retrieve the existing UE context in RRC connection reestablishment procedure, as defined in TS 38.401 [4].

The DL RRC MESSAGE TRANSFER message shall include, if SRB duplication is activated, the *Execute Duplication* IE, so that the gNB-DU can perform CA based duplication or multi-path relay based duplication for the SRB.

If the gNB-DU identifies the UE-associated logical F1-connection by the *gNB-DU UE F1AP ID* IE in the DL RRC MESSAGE TRANSFER message and the *old gNB-DU UE F1AP ID* IE is included, it shall release the old gNB-DU UE F1AP ID and the related configurations associated with the old gNB-DU UE F1AP ID.

If the *UE Context not retrievable* IE set to "true" is included in the DL RRC MESSAGE TRANSFER, the DL RRC MESSAGE TRANSFER may contain the *Redirected RRC message* IE and use it as specified in TS 38.401 [4].

If the *UE Context not retrievable* IE set to "true" is included in the DL RRC MESSAGE TRANSFER, the DL RRC MESSAGE TRANSFER may contain either the *PLMN Assistance Info for Network Sharing* IE or the *PLMN Index NR Assistance Info for Network Sharing* IE, if available at the gNB-CU and may use it as specified in TS 38.401 [4].

If the DL RRC MESSAGE TRANSFER message contains the *New gNB-CU UE F1AP ID* IE, the gNB-DU shall, if supported, replace the value received in the *gNB-CU UE F1AP ID* IE by the value of the *New gNB-CU UE F1AP ID* and use it for further signalling.

If the DL RRC MESSAGE TRANSFER contains the *SRB Mapping Info* IE, the gNB-DU shall, if supported, use it for the Remote UE’s SRB0 or SRB1 transfer.

If the DL RRC MESSAGE TRANSFER contains the *Area-specific Semi-persistent SRS Positioning Information* IE, the gNB-DU shall, if supported, take it into account to activate or deactivate the semi-persistent SRS transmission for area-specific SRS configuration.

Interactions with UE Context Release Request procedure:

If the *UE Context not retrievable* IE set to "true" is included in the DL RRC MESSAGE TRANSFER, the gNB-DU may trigger the UE Context Release Request procedure, as specified in TS 38.401 [4].

8.4.2.3 Abnormal Conditions

Not applicable.

8.4.3 UL RRC Message Transfer

8.4.3.1 General

The purpose of the UL RRC Message Transfer procedure is to transfer an RRC message as an UL PDCP-PDU to the gNB-CU. The procedure uses UE-associated signalling.

8.4.3.2 Successful operation

Figure 8.4.3.2-1: UL RRC Message Transfer procedure

The gNB-DU initiates the procedure by sending a UL RRC MESSAGE TRANSFER message. When the gNB-DU has received from the radio interface an RRC message to which a UE-associated logical F1-connection for the UE exists, the gNB-DU shall send the UL RRC MESSAGE TRANSFER message to the gNB-CU including the RRC message as a *RRC-Container* IE.

If the *Selected PLMN ID* IE is contained in the UL RRC MESSAGE TRANSFER message, the gNB-CU may use it as specified in TS 38.401 [4].

If the UL RRC MESSAGE TRANSFER message contains the *New gNB-DU UE F1AP ID* IE, the gNB-CU shall, if supported, replace the value received in the *gNB-DU UE F1AP ID* IE by the value of the *New gNB-DU UE F1AP ID* and use it for further signalling.

8.4.3.3 Abnormal Conditions

Not applicable.

8.4.4 RRC Delivery Report

8.4.4.1 General

The purpose of the RRC Delivery Report procedure is to transfer to the gNB-CU information about successful delivery of DL PDCP-PDUs including RRC messages. The procedure uses UE-associated signalling.

8.4.4.2 Successful operation

Figure 8.4.4.2-1: RRC Delivery Report procedure.

The gNB-DU initiates the procedure by sending an RRC DELIVERY REPORT message. When the gNB-DU has successfully delivered an RRC message to the UE for which the gNB-CU has requested a delivery report, the gNB-DU shall send the RRC DELIVERY REPORT message to the gNB-CU containng the *RRC Delivery Status* IE and the *SRB ID* IE.

8.4.4.3 Abnormal Conditions

Not applicable.

8.5 Warning Message Transmission Procedures

8.5.1 Write-Replace Warning

8.5.1.1 General

The purpose of Write-Replace Warning procedure is to start or overwrite the broadcasting of warning messages. The procedure uses non UE-associated signalling.

8.5.1.2 Successful Operation

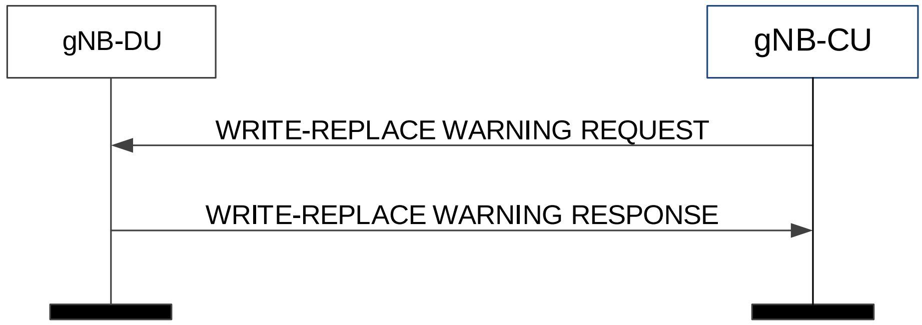

Figure 8.5.1.2-1: Write-Replace Warning procedure: successful operation

The gNB-CU initiates the procedure by sending a WRITE-REPLACE WARNING REQUEST message to the gNB-DU.

Upon receipt of the WRITE-REPLACE WARNING REQUEST message, the gNB-DU shall prioritise its resources to process the warning message.

The gNB-DU acknowledges the WRITE-REPLACE WARNING REQUEST message by sending a WRITE-REPLACE WARNING RESPONSE message to the gNB-CU.

Upon receipt of the WRITE-REPLACE WARNING REQUEST message, the gNB-DU shall include the *Dedicated SI Delivery Needed UE List* IE in the WRITE-REPLACE WARNING RESPONSE message for UEs that are unable to receive system information from broadcast.

If *Dedicated SI Delivery Needed UE List* IE is contained in the WRITE-REPLACE WARNING RESPONSE message, the gNB-CU should take it into account when informing the UE of the updated system information via the dedicated RRC message.

Upon reception of the *Notification Information* IE in the *PWS System Information* IE in the WRITE-REPLACE WARNING REQUEST message, the gNB-DU shall use this information to avoid that duplications trigger new broadcast or replace existing broadcast.

If the gNB-DU receives a WRITE-REPLACE WARNING REQUEST message with the *Notification Information* IE in the *PWS System Information* IE which are different from those of ongoing broadcast warning messages, and if the *SIB Type* IE is set to "8", the gNB-DU shall broadcast the received warning message concurrently with other ongoing messages.

If the gNB-DU receives a WRITE-REPLACE WARNING REQUEST message with the *Notification Information* IE in the *PWS System Information* IE which are different from those of ongoing broadcast warning messages, and if the *SIB Type* IE is set to the value other than ‘8’, the gNB-DU shall use the newly received one to replace the ongoing broadcast warning message with the same value of *SIB Type* IE.

If the *SIB Type* IE in the *PWS System Information* IE in the WRITE-REPLACE WARNING REQUEST message is set to "8" and if a value "0" is received in the *Number of Broadcast Requested* IE and if the *Repetition Period* IE is different from "0", the gNB-DU shall broadcast the received warning message indefinitely.

If *Additional SIB Message List* IE is included in *PWS System Information* IE, the gNB-DU shall store all SIB message(s) in *PWS System Information* IE, and consider that the first segment of public warning message is included in *SIB message* IE, and the remaining segments are listed in *Additional SIB Message List* IE in segmentation sequence order.

8.5.1.3 Unsuccessful Operation

Not applicable.

8.5.1.4 Abnormal Conditions

If the gNB-DU receives a WRITE-REPLACE WARNING REQUEST message which does not include the *Notification Information* IE in the *PWS System Information* IE, the gNB-DU shall consider it as a logical error.

8.5.2 PWS Cancel

8.5.2.1 General

The purpose of the PWS Cancel procedure is to cancel an already ongoing broadcast of a warning message. The procedure uses non UE-associated signalling.

8.5.2.2 Successful Operation

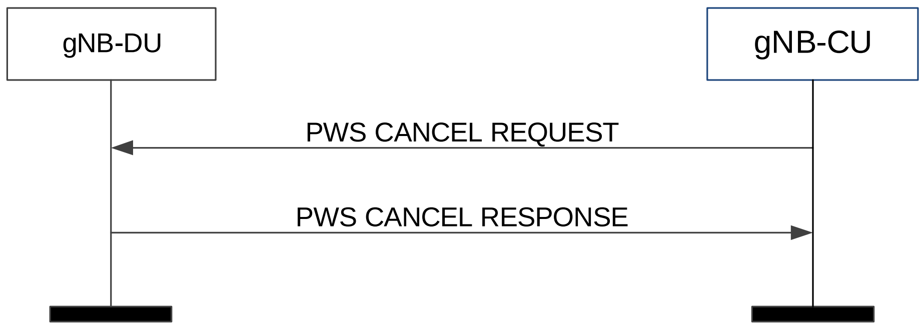

Figure 8.5.2.2-1: PWS Cancel procedure: successful operation

The gNB-CU initiates the procedure by sending a PWS CANCEL REQUEST message to the gNB-DU.

The gNB-DU shall acknowledge the PWS CANCEL REQUEST message by sending the PWS CANCEL RESPONSE message.

If the *Cancel-All Warning Messages Indicator* IE is present in the PWS CANCEL REQUEST message, then the gNB-DU shall stop broadcasting and discard all warning messages for the area as indicated in the *Cell Broadcast To Be Cancelled List* IE or in all the cells of the gNB-DU if the *Cell Broadcast To Be Cancelled List* IE is not included. The gNB-DU shall acknowledge the PWS CANCEL REQUEST message by sending the PWS CANCEL RESPONSE message, and shall, if there is area to report where an ongoing broadcast was stopped successfully, include the *Cell Broadcast Cancelled List* IE with the *Number of Broadcasts* IE set to 0.

If the *Cell Broadcast To Be Cancelled List* IE is not included in the PWS CANCEL REQUEST message, the gNB-DU shall stop broadcasting and discard the warning message identified by the *Message Identifier* IE and the *Serial Number* IE in the *Notification Information* IE in all of the cells in the gNB-DU.

If the *Notification Information* IE is included in the PWS CANCEL REQUEST, the gNB-DU shall cancel broadcast of the public warning message identified by the *Notification Information* IE.

If an area included in the *Cell Broadcast To Be Cancelled List* IE in the PWS CANCEL REQUEST message does not appear in the *Cell Broadcast Cancelled List* IE in the PWS CANCEL RESPONSE, the gNB-CU shall consider that the gNB-DU had no ongoing broadcast to stop for the public warning message identified, if present, by the *Notification Information* IE in that area.

If the *Cell Broadcast Cancelled List* IE is not included in the PWS CANCEL RESPONSE message, the gNB-CU shall consider that the gNB-DU had no ongoing broadcast to stop for the public warning message identified, if present, by the *Notification Information* IE.

8.5.2.3 Unsuccessful Operation

If the gNB-DU receives a PWS CANCEL REQUEST message which contains neither the *Cancel-all Warning Messages Indicator* IE nor the *Notification Information* IE, the gNB-DU shall consider it as a logical error.

8.5.2.4 Abnormal Conditions

Not applicable.

8.5.3 PWS Restart Indication

8.5.3.1 General

The purpose of PWS Restart Indication procedure is to inform the gNB-CU that PWS information for some or all cells of the gNB-DU are available for reloading from the CBC if needed. The procedure uses non UE-associated signalling.

8.5.3.2 Successful Operation

Figure 8.5.3.2-1: PWS restart indication

The gNB-DU initiates the procedure by sending a PWS RESTART INDICATION message to the gNB-CU.

8.5.3.3 Abnormal Conditions

Not applicable.

8.5.4 PWS Failure Indication

8.5.4.1 General

The purpose of the PWS Failure Indication procedure is to inform the gNB-CU that ongoing PWS operation for one or more cells of the gNB-DU has failed. The procedure uses non UE-associated signalling.

8.5.4.2 Successful Operation

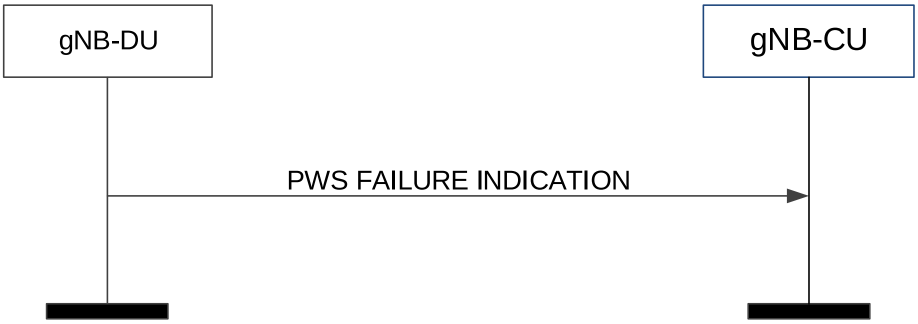

Figure 8.5.4.2-1: PWS failure indication

The gNB-DU initiates the procedure by sending a PWS FAILURE INDICATION message to the gNB-CU.

8.5.4.3 Abnormal Conditions

Not applicable.

8.6 System Information Procedures

8.6.1 System Information Delivery

8.6.1.1 General

The purpose of the System Information Delivery procedure is to command the gNB-DU to broadcast the requested one or several *SystemInformation* messages including the Other SI as requested by the gNB-CU. The procedure uses non-UE associated signalling.

8.6.1.2 Successful Operation

Figure 8.6.1.2-1: System Information Delivery procedure. Successful operation.

The gNB-CU initiates the procedure by sending a SYSTEM INFORMATION DELIVERY COMMAND message to the gNB-DU.

Upon reception of the SYSTEM INFORMATION DELIVERY COMMAND message, the gNB-DU shall broadcast the requested one or several *SystemInformation* messages, including the Other SI, indicated by the *SIType List* IE, and if the UE corresponding to the *confirmed UE ID* IE is not in RRC connected state, delete the UE context, if any.

Interactions with gNB-DU Configuration Update procedure:

Upon reception of SYSTEM INFORMATION DELIVERY COMMAND message, the gNB-DU Configuration Update procedure may be performed, and as part of such procedure the gNB-DU shall include the *Dedicated SI Delivery*

Needed UE List IE in GNB-DU CONFIGURATION UPDATE message for UEs that are unable to receive system information from broadcast.

8.6.1.3 Abnormal Conditions

Not applicable.

8.7 Paging procedures

8.7.1 Paging

8.7.1.1 General

The purpose of the Paging procedure is used to provide the paging information to enable the gNB-DU to page a UE. The procedure uses non-UE associated signalling.

8.7.1.2 Successful Operation

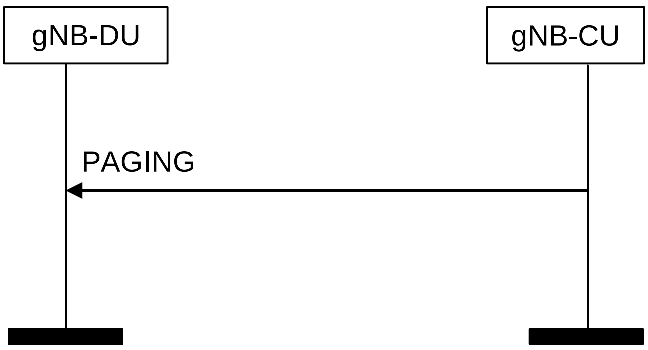

Figure 8.7.1.2-1: Paging procedure. Successful operation.

The gNB-CU initiates the procedure by sending a PAGING message.

The *Paging DRX* IE may be included in the PAGING message, and if present the gNB-DU may use it to determine the final paging cycle for the UE.

The *Paging Priority* IE may be included in the PAGING message, and if present the gNB-DU may use it according to TS 23.501 [21].

At the reception of the PAGING message, the gNB-DU shall perform paging of the UE in cells which belong to cells as indicated in the *Paging Cell List* IE.

The *Paging Origin* IE may be included in the PAGING message, and if present the gNB-DU shall transfer it to the UE.

The *RAN UE Paging DRX* IE may be included in the PAGING message, and if present the gNB-DU may use it according to TS 38.304 [24].

The *CN UE Paging DRX* IE may be included in the PAGING message, and if present the gNB-DU may use it according to TS 38.304 [24].

The *NR Paging eDRX Information* IE may be included in the PAGING message, and if present the gNB-DU may use it according to TS 38.304 [24].

The *NR Paging eDRX Information for RRC INACTIVE* IE may be included in the PAGING message, and if present the gNB-DU shall, if supported, use it according to TS 38.304 [24].

The *Paging Cause* IE may be included in the PAGING message. If present the gNB-DU shall, if supported, send it to UE according to TS 38.331 [8].

The *PEIPS Assistance Information* IE may be included in the PAGING message, and if present the gNB-DU shall, if supported, use it for paging subgrouping of the UE, as specified in TS 38.300 [6].

The *RedCap Indication* IE may be included in the *UE Paging Capability* IE in the PAGING message, and if present the gNB-DU shall, if supported, use it for paging of the RedCap UE or the eRedCap UE.

The *Paging Adaptation Indication* IE may be included in the *UE Paging Capability* IE in the PAGING message, and if present the gNB-DU shall, if supported, use it for paging adaptation to the UE.

The *Last Used Cell Indication* IE may be included in the *Paging Cell Item IEs* IE of the PAGING message, and if present the gNB-DU shall, if supported, consider the cell identified by the *NR CGI* IE as the last used cell of the paged UE, and use it as specified in TS 38.331 [8].

The *Recommended SSBs List* IE may be included in the *Paging Cell Item IEs* IE of the PAGING message, and if present the gNB-DU shall, if supported, use it to send the paging message over the indicated SSB beams.

The *PEI Subgrouping Support Indication* IE may be included in the *Paging Cell Item IEs* IE in the PAGING message, and if present the gNB-DU shall, if supported, consider that the cell identified by the *NR CGI* IE is supported by the UE to receive the paging early indication as described in TS 38.300 [6] and TS 38.304 [24].

The *PEI Subgrouping Support Indication – Paging Adaptation* IE may be included in the *Paging Cell Item IEs* IE in the PAGING message, and if present the gNB-DU shall, if supported, consider that the cell identified by the *NR CGI* IE is supported by the UE to receive the paging adaptation related paging early indication as described in TS 38.300 [6] and TS 38.304 [24].

The *UE Paging Capability* IE may be included in the PAGING message, and if present the gNB-DU shall, if supported, take it into account when paging the UE.

The *Extended UE Identity Index Value* IE may be included in the PAGING message, and if present the gNB-DU shall, if supported, use it according to TS 38.304 [24].

The *Hashed UE Identity Index Value* IE may be included in the PAGING message, and if present the gNB-DU shall, if supported, use it according to TS 38.304 [24].

The *MT-SDT Information* IE may be included in the PAGING message. If present the gNB-DU shall, if supported, use it for MT-SDT paging as specified in TS 38.331 [8].

The *NR Paging Long eDRX Information for RRC INACTIVE* IE may be included in the PAGING message, and if present, the gNB-DU shall, if supported, use it according to TS 38.304 [24].

The *LP-WUSPS Assistance Information* IE may be included in the PAGING message, and if present the gNB-DU shall, if supported, use it for LP-WUS paging subgrouping of the UE as specified in TS 38.300 [6].

The *LP-WUS Subgrouping Support Indication* IE may be included in the *Paging Cell Item IEs* IE in the PAGING message, and if present the gNB-DU shall, if supported, consider that the cell identified by the *NR CGI* IE is supported by the UE to receive the LP-WUS as described in TS 38.300 [6] and TS 38.304 [24].

The *LP-WUS Supported Band Information* IE may be included in the *Paging Cell Item IEs* IE in the PAGING message, and if present the gNB-DU shall, if supported, take it into account for LP-WUS paging the UE in the cell identified by the *NR CGI* IE.

The *Further Extended UE Identity Index Value* IE may be included in the PAGING message, and if present the gNB-DU shall, if supported, use it according to TS 38.304 [24].

8.7.1.3 Abnormal Conditions

Not applicable.

8.8 Trace Procedures

8.8.1 Trace Start

8.8.1.1 General

The purpose of the Trace Start procedure is to allow the gNB-CU to request the gNB-DU to initiate a trace session for a UE. The procedure uses UE-associated signalling.

8.8.1.2 Successful Operation

Figure 8.8.1.2-1: Trace start procedure: Successful Operation.

The gNB-CU initiates the procedure by sending a TRACE START message. Upon reception of the TRACE START message, the gNB-DU shall initiate the requested trace session for the requested UE, as described in TS 32.422 [29]. In particular, the gNB-DU shall, if supported:

- if the *Trace Activation* IE includes the *MDT Activation* IE set to "Immediate MDT and Trace" initiate the requested trace session and MDT session as described in TS 32.422 [29];
- if the *Trace Activation* IE includes the *MDT Activation* IE set to "Immediate MDT Only" initiate the requested MDT session as described in TS 32.422 [29] and the gNB-DU shall ignore *Interfaces To Trace* IE, and *Trace Depth* IE;

8.8.1.3 Abnormal Conditions

Void.

8.8.2 Deactivate Trace

8.8.2.1 General

The purpose of the Deactivate Trace procedure is to allow the gNB-CU to request the gNB-DU to stop the trace session for the indicated trace reference. The procedure uses UE-associated signalling.

8.8.2.2 Successful Operation

Figure 8.8.2.2-1: Deactivate trace procedure: Successful Operation

The gNB-CU initiates the procedure by sending a DEACTIVATE TRACE message. Upon reception of the DEACTIVATE TRACE message, the gNB-DU shall stop the trace session for the indicated trace reference contained in the *Trace ID* IE, as described in TS 32.422 [29].

8.8.2.3 Abnormal Conditions

Void.

8.8.3 Cell Traffic Trace

8.8.3.1 General

The purpose of the Cell Traffic Trace procedure is to send the allocated Trace Recording Session Reference and the Trace Reference to the gNB-CU. The procedure uses UE-associated signalling.

8.8.3.2 Successful Operation

Figure 8.8.3.2-1: Cell Traffic Trace procedure. Successful operation.

The procedure is initiated with a CELL TRAFFIC TRACE message sent from the gNB-DU to the gNB-CU.

If the *Privacy Indicator* IE is included in the message, the gNB-CU shall store the information so that it can be transferred towards the AMF.

8.8.3.3 Abnormal Conditions

Void.

8.9 Radio Information Transfer procedures

8.9.1 DU-CU Radio Information Transfer

8.9.1.1 General

The purpose of the DU-CU Radio Information Transfer procedure is to transfer radio-related information from the gNB-DU to the gNB-CU. The procedure uses non-UE-associated signalling.

8.9.1.2 Successful operation

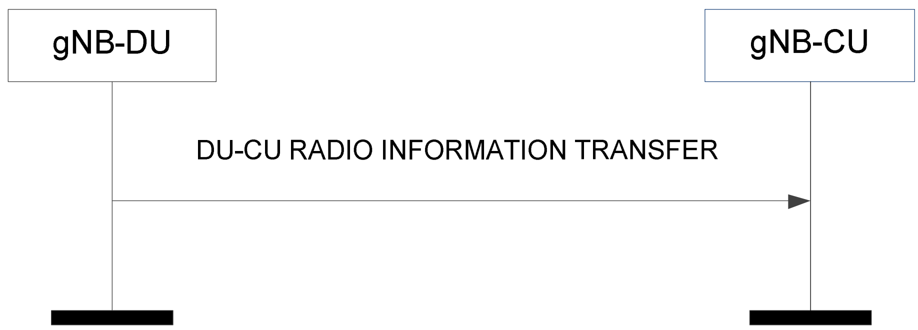

Figure 8.9.1.2-1: DU-CU Radio Information Transfer procedure.

The gNB-DU initiates the procedure by sending the DU-CU RADIO INFORMATION TRANSFER message to the gNB-CU.

The gNB-CU considers that the *RIM-RS Detection Status* IE indicates the RIM-RS detection status of the cell identified by *Aggressor Cell ID* IE.

8.9.1.3 Abnormal Conditions

Not applicable.

8.9.2 CU-DU Radio Information Transfer

8.9.2.1 General

The purpose of the CU-DU Radio Information Transfer procedure is to transfer radio-related information from the gNB-CU to the gNB-DU. The procedure uses non-UE-associated signalling.

8.9.2.2 Successful operation

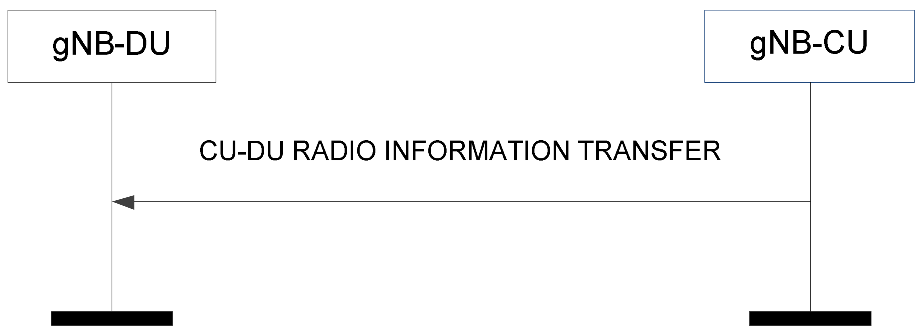

Figure 8.9.2.2-1: CU-DU Radio Information Transfer procedure.

The gNB-CU initiates the procedure by sending the CU-DU RADIO INFORMATION TRANSFER message to the gNB-DU. The gNB-DU considers that the *RIM-RS Detection Status* IE indicates the detection status of RIM-RS associated with *Victim gNB Set ID* IE.

8.9.2.3 Abnormal Conditions

Not applicable.

8.10 IAB Procedures

8.10.0 General

In this version of the specification, the IAB procedures are used to configure IAB-donor-DU or IAB-DU.

NOTE: The IAB procedures are applicable for IAB-nodes and IAB-donor-DU, where the term "gNB-DU" applies to IAB-DU and IAB-donor-DU, and the term "gNB-CU" applies to IAB-donor-CU, unless otherwise specified.

8.10.1 BAP Mapping Configuration

8.10.1.1 General

The BAP Mapping Configuration Procedure is initiated by the gNB-CU to configure the DL/UL routing information and/or traffic mapping information needed for the gNB-DU. The procedure uses non-UE associated signalling.

NOTE: Implementation shall ensure the avoidance of potential race conditions, i.e. it shall ensure that conflicting traffic mapping configurations are not concurrently performed using the non-UE-associated BAP Mapping Configuration procedure and the UE-associated UE Context Management procedures.

8.10.1.2 Successful Operation

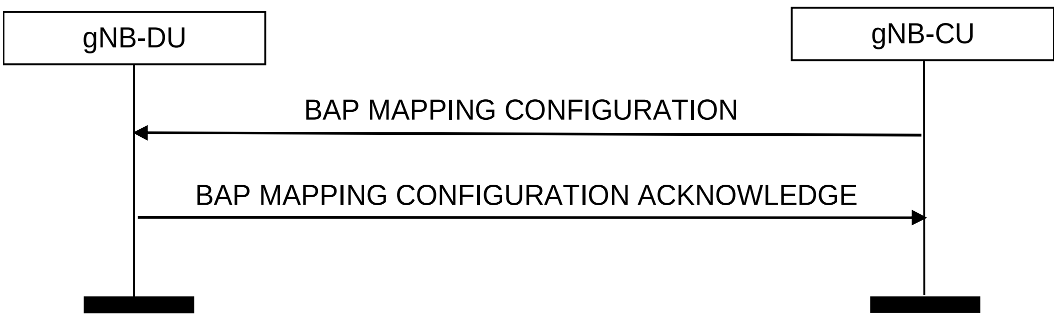

Figure 8.10.1.2-1: BAP Mapping Configuration procedure: Successful Operation

The gNB-CU initiates the procedure by sending BAP MAPPING CONFIGURATION message to the gNB-DU. The gNB-DU replies to the gNB-CU with BAP MAPPING CONFIGURATION ACKNOWLEDGE.

If *BH Routing Information Added List* IE is included in the BAP MAPPING CONFIGURATION message, the gNB-DU shall, if supported, store the BH routing information from this IE and use it for DL/UL traffic forwarding as specified in TS 38.340 [30]. If *BH Routing Information Added List* IE contains information for an existing BAP Routing ID, the gNB-DU shall, if supported, replace the previously stored routing information for this BAP Routing ID with the corresponding information in the *BH Routing Information Added List* IE.

If *BH Routing Information Removed List* IE is included in the BAP MAPPING CONFIGURATION message, the gNB-DU shall, if supported, remove the BH routing information according to such IE.

If the *Traffic Mapping Information* IE is included in the BAP MAPPING CONFIGURATION message, the gNB-DU shall, if supported, process the *Traffic Mapping Information* IE as follows:

- if the *IP to layer2 Traffic Mapping Info* IE is included, the gNB-DU shall store the mapping information contained in the *IP to layer2 Traffic Mapping Info To Add* IE, if present, and remove the previously stored mapping information as indicated by the *IP to layer2 Traffic Mapping Info To Remove* IE, if present. The gNB-DU shall use the mapping information stored for the mapping of IP traffic to layer 2, as specified in TS 38.340 [30].

- if the *BAP layer BH RLC channel Mapping Info* IE is included, the gNB-DU shall store the mapping information contained in the *BAP layer BH RLC channel Mapping Info To Add* IE, if present, and remove the previously stored mapping information as indicated by the *BAP layer BH RLC channel Mapping Info To Remove* IE, if present. The gNB-DU shall use the mapping information stored when forwarding traffic on BAP sublayer, as specified in TS 38.340 [30].

If the *Buffer Size Threshold* IE is included in the BAP MAPPING CONFIGURATION message, the gNB-DU shall, if supported, use it to determine the DL congestion based on the flow control feedback from child IAB-nodes as specified in TS 38.340 [30].

If *BAP Header Rewriting Added List* IE is included in the BAP MAPPING CONFIGURATION message, the gNB-DU shall, if supported, store the BAP header rewriting configuration from this IE, and use it as specified in TS 38.340 [30]. If *BAP Header Rewriting Added List* IE contains information for an existing ingress BAP Routing ID, the gNB-DU shall, if supported, replace the previously stored BAP header rewriting configuration for this ingress BAP Routing ID with the corresponding information in the *BAP Header Rewriting Added List* IE.

If *BAP Header Rewriting Removed List* IE is included in the BAP MAPPING CONFIGURATION message, the gNB-DU shall, if supported, remove the BAP header rewriting configuration according to such IE.

If the *Re-routing Enable Indicator* IE is included in the BAP MAPPING CONFIGURATION message, and the value is set as “false”, the gNB-DU shall, if supported, disable the inter-donor-DU re-routing. If the *Re-routing Enable Indicator* IE is included in the BAP MAPPING CONFIGURATION message, and the value is set as “true”, the gNB-DU shall, if supported, enable the inter-donor-DU re-routing, as specified in TS 38.340 [30].

8.10.1.A Unsuccessful Operation

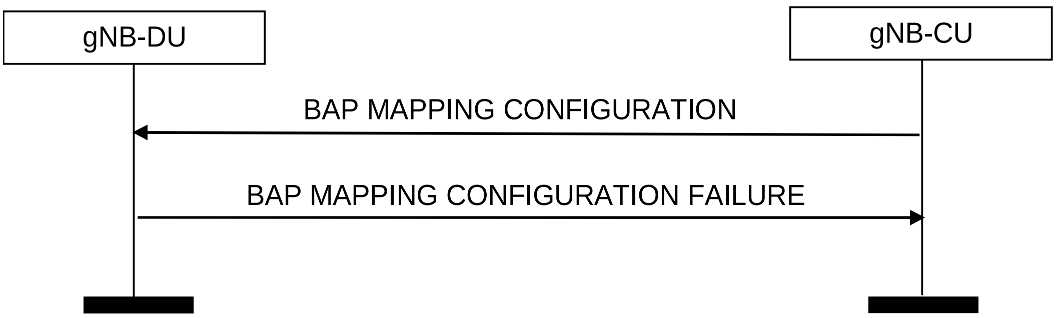

Figure 8.10.1.3-1: BAP Mapping Configuration procedure: Unsuccessful Operation

If the gNB-DU cannot accept the configuration, it shall respond with a BAP MAPPING CONFIGURATION FAILURE and appropriate cause value.

If the BAP MAPPING CONFIGURATION FAILURE message includes the *Time To Wait* IE, the gNB-CU shall wait at least for the indicated time before reinitiating the BAP MAPPING CONFIGURATION message towards the same gNB-DU.

8.10.1.3 Abnormal Conditions

Not applicable.

8.10.2 gNB-DU Resource Configuration

8.10.2.1 General

The gNB-DU Resource Configuration procedure is initiated by the gNB-CU in order to configure the resource usage for a gNB-DU. The procedure uses non-UE associated signalling.

In this version of the specification, this procedure is used to configure IAB resources.

8.10.2.2 Successful Operation

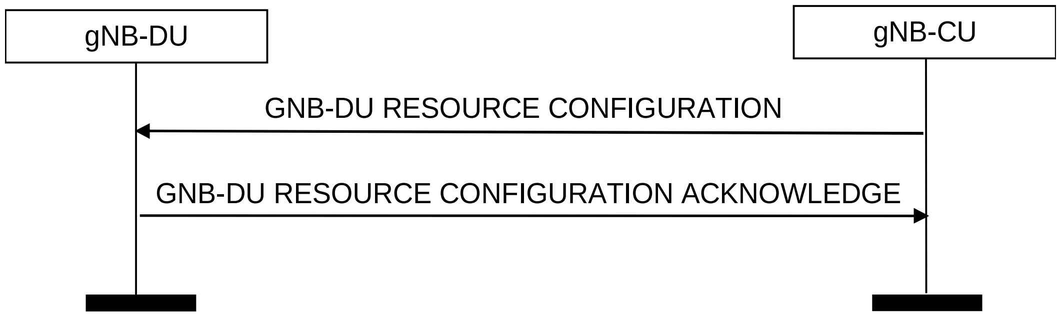

Figure 8.10.2.2-1: gNB-DU Resource Configuration procedure: Successful Operation

The gNB-CU initiates the procedure by sending the GNB-DU RESOURCE CONFIGURATION message to gNB-DU. The gNB-DU replies to the gNB-CU with the GNB-DU RESOURCE CONFIGURATION ACKNOWLEDGE message.

For each cell in the *Activated Cells to Be Updated List* IE of the GNB-DU RESOURCE CONFIGURATION message, the gNB-DU shall store the resource configuration contained in the *IAB-DU Cell Resource Configuration-Mode-Info* IE and use it when performing scheduling in compliance with TS 38.213 [31].

If the *Child-Node List* IE is included in the GNB-DU RESOURCE CONFIGURATION message, for each child-node indicated by the *gNB-CU UE F1AP ID* IE and *gNB-DU UE F1AP ID* IE, and for each cell served by this child node indicated by the *NR CGI* IE in the *Child-Node Cells List* IE, the gNB-DU shall store the received information and use this information for scheduling, in compliance with TS 38.213 [31], clause 14.

If the *Neighbour-Node Cells List* IE is included in the GNB-DU RESOURCE CONFIGURATION message, for each neighbour-node cell indicated by the *NR CGI* IE in the *Neighbour-Node Cells List* IE, the gNB-DU shall store the received information and use this information for cross-link interference management and/or semi-static resource coordination. If the *Peer Parent-Node Indicator* IE is included in the GNB-DU RESOURCE CONFIGURATION message and the value is set as "true", the gNB-DU shall, consider the cell indicated by the *NR CGI* IE is served by the peer parent node of the IAB-node indicated by the *gNB-CU UE F1AP ID* IE and the *gNB-DU UE F1AP ID* IE.

If the *Serving Cells List* IE is included in the GNB-DU RESOURCE CONFIGURATION message, the gNB-DU shall store the received information and use this information for scheduling, in compliance with TS 38.213 [31], clause 14.

8.10.2.B Unsuccessful Operation

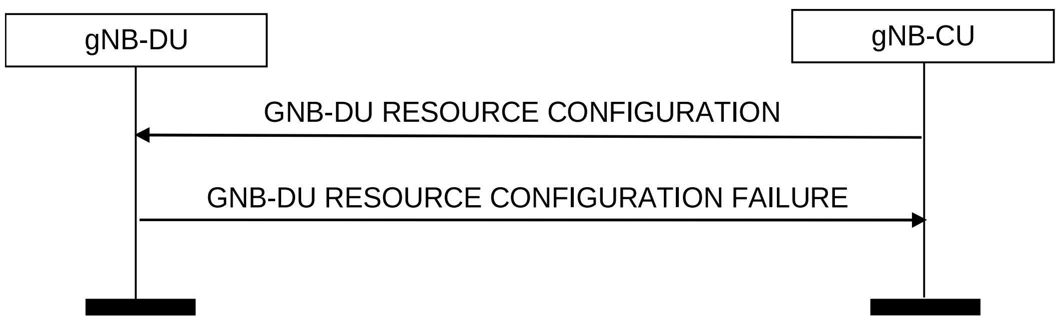

Figure 8.10.2.3-1: gNB-DU Resource Configuration procedure: Unsuccessful Operation

If the gNB-DU cannot accept the configuration, it shall respond with a GNB-DU RESOURCE CONFIGURATION FAILURE and appropriate cause value.

If the GNB-DU RESOURCE CONFIGURATION FAILURE message includes the *Time To Wait* IE, the gNB-CU shall wait at least for the indicated time before reinitiating the GNB-DU RESOURCE CONFIGURATION message towards the same gNB-DU.

8.10.2.3 Abnormal Conditions

Not applicable.

8.10.3 IAB TNL Address Allocation

8.10.3.1 General

The purpose of the IAB TNL Address Allocation procedure is to allocate TNL addresses to be used by the IAB-node(s). This procedure uses non-UE associated signalling.

NOTE: This procedure is applicable for IAB-donor-DU, where the term "gNB-DU" applies to IAB-donor-DU, and the term "gNB-CU" applies to IAB-donor-CU.

8.10.3.2 Successful Operation

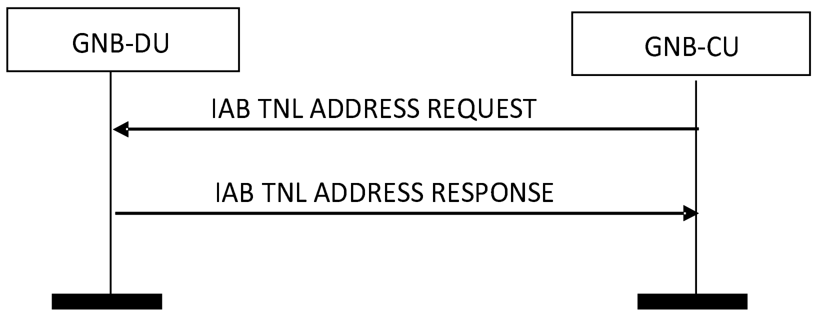

Figure 8.10.3.2-1: IAB TNL Address Allocation procedure: Successful Operation

The gNB-CU initiates the procedure by sending the IAB TNL ADDRESS REQUEST message to the gNB-DU.

If the IAB TNL ADDRESS REQUEST message contains the *IAB IPv4 Addresses Requested* IE, the gNB-DU shall allocate the individual TNL address(es) accordingly and include these IPv4 address(es) in the IAB TNL ADDRESS RESPONSE message.

If the IAB TNL ADDRESS REQUEST message contains the *IAB IPv6 Request Type* IE, the gNB-DU shall allocate the individual IPv6 address(es) or IPv6 address prefix(es) accordingly and include these IPv6 address(es) or IPv6 address prefix(es) in the IAB TNL ADDRESS RESPONSE message.

If the IAB TNL ADDRESS REQUEST message contains the *IAB TNL Addresses To Remove List* IE, the gNB-DU shall consider that the TNL address(es) and/or TNL address prefix(es) therein are no longer used by the IAB-node(s). In addition, if the IAB TNL ADDRESS REQUEST message only contains the *IAB TNL Addresses to Remove List* IE, the gNB-CU shall ignore the *IAB Allocated TNL Address List* IE in the IAB TNL ADDRESS RESPONSE message.

If the IAB TNL ADDRESS RESPONSE message contains the *IAB TNL Address Usage IE* in the *IAB Allocated TNL Address Item* IE, the gNB-CU shall consider the indicated TNL address usage when allocating a TNL address to an IAB-node. Otherwise, the gNB-CU shall consider that the TNL address can be used for all traffic when allocating the TNL address to an IAB-node.

If the *IAB TNL Address Exception* IE is included in the IAB TNL ADDRESS REQUEST message and the gNB-DU is an IAB-donor-DU, the gNB-DU shall, if supported, consider the IP address(es) therein as exempt from TNL address filtering, and forward the packets with the address(es) indicated by this IE, as specified in TS 38.401 [4].

8.10.3.C Unsuccessful Operation

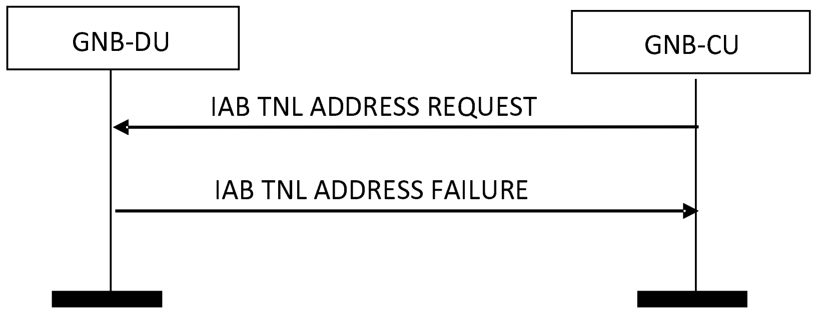

Figure 8.10.3.3-1: IAB TNL Address Allocation procedure: Unsuccessful Operation

If the gNB-DU cannot accept the request, it shall respond with an IAB TNL ADDRESS FAILURE and appropriate cause value.

If the IAB TNL ADDRESS FAILURE message includes the *Time To Wait* IE, the gNB-CU shall wait at least for the indicated time before reinitiating the IAB TNL ADDRESS REQUEST message towards the same gNB-DU.

8.10.3.3 Abnormal Conditions

Not applicable.

8.10.4 IAB UP Configuration Update

8.10.4.1 General

The purpose of the IAB UP Configuration Update procedure is to update the UP parameters including UL mapping configuration and the UL/DL UP TNL information between IAB-donor-CU and IAB-node. This procedure uses non-UE associated signalling.

NOTE: This procedure is applicable for IAB-nodes, where the term "gNB-DU" applies to IAB-DU, and the term "gNB-CU" applies to IAB-donor-CU.

NOTE: Implementation shall ensure the avoidance of potential race conditions, i.e. it shall ensure that the update of UP configuration (e.g. the UL/DL UP TNL information, UL mapping information) is not concurrently performed using the non-UE-associated IAB UP Configuration Update procedure and the UE-associated procedures for UE Context Management.

8.10.4.2 Successful Operation

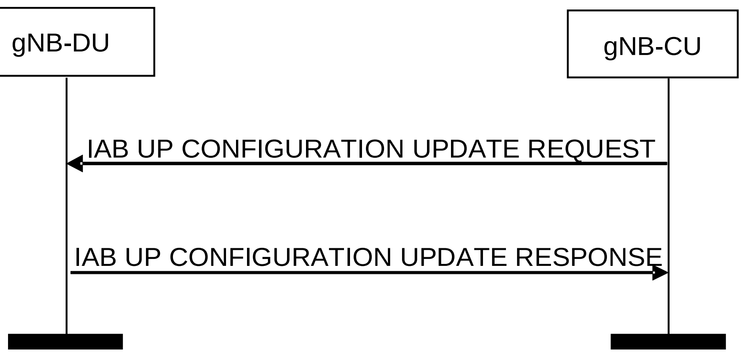

Figure 8.10.4.2-1: IAB UP Configuration Update procedure: Successful Operation

The gNB-CU initiates the procedure by sending the IAB UP CONFIGURATION UPDATE REQUEST message to the gNB-DU. The gNB-DU replies to the gNB-CU with the IAB UP CONFIGURATION UPDATE RESPONSE message.

If the *UL UP TNL Information to Update List* IE is included in the IAB UP CONFIGURATION UPDATE REQUEST message, the gNB-DU shall perform the mapping according to the new received *BH Information* IE for each F1-U GTP tunnel indicated by the *UL UP TNL Information* IE. If the *New UL UP TNL Information* IE is included in *UL UP TNL Information to Update List* IE, the gNB-DU shall use it to replace the information of UL F1-U GTP tunnel indicated by the *UL UP TNL Information* IE.

If the *UL UP TNL Address to Update List* IE is included in the IAB UP CONFIGURATION UPDATE REQUEST message, the gNB-DU shall replace the old TNL address with the new TNL address for all the maintained UL F1-U GTP tunnels corresponding to the old TNL address.

If the *DL UP TNL Address to Update List* IE is included in the IAB UP CONFIGURATION UPDATE RESPONSE message, the gNB-CU shall replace the old TNL address with the new TNL address for all the maintained DL F1-U GTP tunnels corresponding to the old TNL address.

8.10.4.3 Unsuccessful Operation

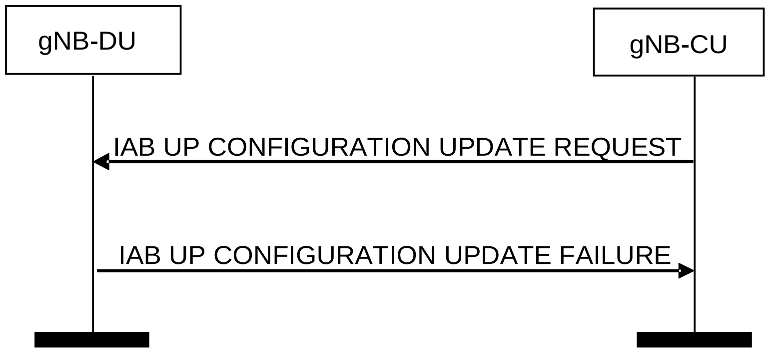

Figure 8.10.4.3-1: IAB UP Configuration Update procedure: Unsuccessful Operation

If the gNB-DU receives an IAB UP CONFIGURATION UPDATE REQUEST message and cannot perform any update accordingly, it shall consider the update procedure as failed and respond with an IAB UP CONFIGURATION UPDATE FAILURE message and an appropriate cause value.

If the IAB UP CONFIGURATION UPDATE FAILURE message includes the *Time To Wait* IE, the gNB-CU shall wait at least for the indicated time before reinitiating the IAB UP CONFIGURATION UPDATE REQUEST message towards the same gNB-DU.

8.10.4.4 Abnormal Conditions

Not applicable.

8.10.5 Mobile IAB F1 Setup Triggering

8.10.5.1 General

The purpose of the Mobile IAB F1 Setup Triggering procedure is to trigger F1 interface establishment between a target logical gNB-DU and a target F1-terminating IAB-donor-CU. The target logical gNB-DU is co-located with the gNB-DU that receives the triggering message. This procedure uses non-UE associated signalling.

NOTE: This procedure is applicable for mobile IAB-nodes, where the term "gNB-DU" applies to a mobile IAB-DU, and the term "gNB-CU" applies to a source F1-terminating IAB-donor-CU during mobile IAB-DU migration.

8.10.5.2 Successful Operation

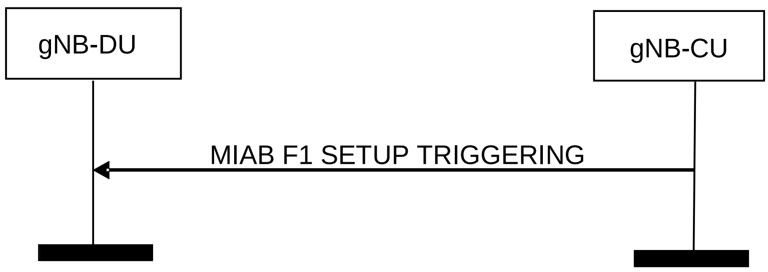

Figure 8.10.5.2-1: Mobile IAB F1 Setup Triggering: Successful Operation

The gNB-CU initiates the procedure by sending the MIAB F1 SETUP TRIGGERING message to the gNB-DU.

When the gNB-DU receives the MIAB F1 SETUP TRIGGERING message, the gNB-DU's co-located target logical gNB-DU shall, if supported, initiate the TNL connection establishment and F1 setup to a target F1-terminating IAB-donor-CU indicated by the *Target gNB ID* IE included in the MIAB F1 SETUP TRIGGERING message.

If the MIAB F1 SETUP TRIGGERING message received by the gNB-DU contains the *Target gNB IP address* IE, the gNB-DU's co-located target logical gNB-DU shall, if supported, store the IP address and use it for establishing the TNL connection towards a target F1-terminating IAB-donor-CU.

If the MIAB F1 SETUP TRIGGERING message received by the gNB-DU contains the *Target SeGW IP address* IE, the gNB-DU's co-located target logical gNB-DU shall, if supported, store the IP address and use it for establishing the security connection to protect the F1 interface towards the target F1-terminating IAB-donor-CU.

8.10.5.3 Abnormal Conditions

Not applicable.

8.10.6 Mobile IAB F1 Setup Outcome Notification

8.10.6.1 General

The purpose of the Mobile IAB F1 Setup Outcome Notification procedure is to report the outcome of the F1 interface setup between a target logical gNB-DU and a target F1-terminating IAB-donor-CU. The target logical gNB-DU is co-located with the gNB-DU that sends the notification message. This procedure uses non-UE associated signalling.

NOTE: This procedure is applicable for mobile IAB-nodes, where the term "gNB-DU" applies to mobile IAB-DU, and the term "gNB-CU" applies to source F1-terminating IAB-donor-CU during mobile IAB-DU migration.

8.10.6.2 Successful Operation

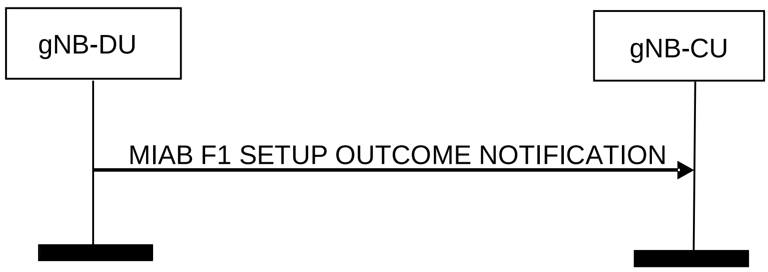

Figure 8.10.6.2-1: Mobile IAB F1 Setup Outcome Notification: Successful Operation

The gNB-DU initiates the procedure by sending the MIAB F1 SETUP OUTCOME NOTIFICATION message to the gNB-CU.

Upon the reception of the MIAB F1 SETUP OUTCOME NOTIFICATION message, the gNB-CU shall, if supported, consider the F1 setup outcome of the target logical gNB-DU co-located with the gNB-DU, for further mobile IAB-DU migration as specified in TS 38.401 [4].

If the *Activated Cells Mapping List* is included in the MIAB F1 SETUP OUTCOME NOTIFICATION message, the gNB-CU shall, if supported, take it into account for subsequent handover of the connected UEs from this gNB-DU to its co-located target logical gNB-DU.

If the *Target F1 Terminating IAB-Donor gNB ID* is included in the MIAB F1 SETUP OUTCOME NOTIFICATION message, the gNB-CU shall, if supported, take it into account for subsequent handover of the connected UEs from this gNB-DU to its co-located target logical gNB-DU.

8.10.6.3 Abnormal Conditions

Not applicable.

8.11 Self Optimisation Support procedures

8.11.1 Access and Mobility Indication

8.11.1.1 General

This procedure is initiated by gNB-CU to send the Access and Mobility related Information to gNB-DU.

The procedure uses non-UE-associated signalling.

8.11.1.2 Successful Operation

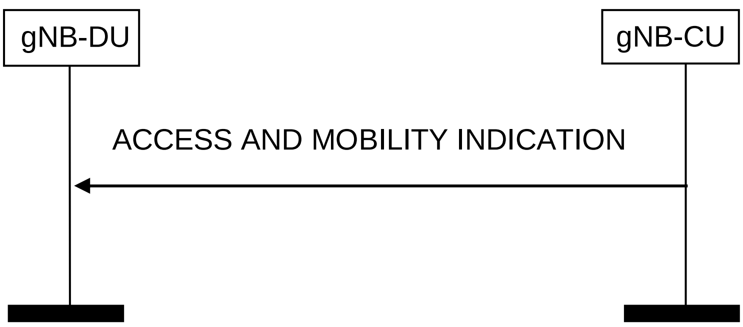

Figure 8.11.1.2-1: Access and Mobility Indication procedure. Successful operation

The Access and Mobility Indication procedure is initiated by ACCESS AND MOBILITY INDICATION message sent from gNB-CU to gNB-DU.

If the ACCESS AND MOBILITY INDICATION message contains the *RA Report List* IE the gNB-DU shall take it into account for optimisation of RACH access procedures.

If the ACCESS AND MOBILITY INDICATION message contains the *RLF Report Information List* IE the gNB-DU shall take it into account for optimisation of mobility parameters.

If the ACCESS AND MOBILITY INDICATION message contains the *Successful HO Report Information List* IE the gNB-DU may take it into account for optimisation of mobility parameters.

If the ACCESS AND MOBILITY INDICATION message contains the *Successful PSCell Change Report Information List* IE, the gNB-DU may take it into account for optimisation of PSCell change/addition related parameters.

If the ACCESS AND MOBILITY INDICATION message contains the *BFR SSB Index* IE within the *MRO for LTM Information* IE, the gNB-DU shall, if supported, consider that a Beam Failure Recovery after successful LTM Cell Switch has occurred in the target DU for the UE identified by the *gNB-DU UE F1AP ID* IE, and may take it into account for optimisation of target beam selection.

If the ACCESS AND MOBILITY INDICATION message contains the *Target SSB Index after Cell Switch Failure* IE within the *MRO for LTM Information* IE, the gNB-DU shall, if supported, consider that a successful recovery or re-establishment occurred in this SSB after an LTM Cell Switch Failure for the UE identified by the *gNB-DU UE F1AP ID* IE, and may take it into account for optimisation of target beam selection or optimisation of Timing Advance related parameters for the UE identified by the *gNB-DU UE F1AP ID* IE.

If the ACCESS AND MOBILITY INDICATION message contains the *TA Information IE* within the *MRO for LTM Information* IE, the gNB-DU may take it into account for optimisation of Timing Advance related parameters or optimisation of target beam selection for the UE identified by the *gNB-DU UE F1AP ID* IE.

8.11.1.3 Abnormal Conditions

Not applicable.

8.11.2 DU-CU Access and Mobility Indication

8.11.2.1 General

This procedure is initiated by the gNB-DU to send the Access and Mobility related Information to the gNB-CU.

The procedure uses non-UE-associated signalling.

8.11.2.2 Successful Operation

Figure 8.11.2.2-1: DU-CU Access and Mobility Indication procedure. Successful operation

The DU-CU Access and Mobility Indication procedure is initiated by DU-CU ACCESS AND MOBILITY INDICATION message sent from the gNB-DU to the gNB-CU.

If the DU-CU ACCESS AND MOBILITY INDICATION message contains the *DL LBT Failure Information List* IE, the gNB-CU shall take it into account for optimisation of mobility parameters.

8.11.2.3 Abnormal Conditions

Not applicable.

8.12 Reference Time Information Reporting procedures

8.12.1 Reference Time Information Reporting Control

8.12.1.1 General

The purpose of the Reference Time Information Reporting Control procedure is to command the gNB-DU to send the requested accurate reference time information to the gNB-CU. The procedure uses non-UE associated signalling.

8.12.1.2 Successful Operation

Figure 8.12.1.2-1: Reference Time Information Reporting Control

The gNB-CU initiates the procedure by sending REFERENCE TIME INFORMATION REPORTING CONTROL message to the gNB-DU. Upon reception of the REFERENCE TIME INFORMATION REPORTING CONTROL message, the gNB-DU shall, if supported, perform the requested reference time information reporting action.

The *Reporting Request Type* IE indicates to the gNB-DU whether:

- to report on demand;
- to report periodic, with a frequency as specified by the *Report Periodicity Value* IE;
- to stop periodic reporting.

8.12.1.3 Abnormal Conditions

Not applicable.

8.12.2 Reference Time Information Report

8.12.2.1 General

The purpose of the Reference Time Information Report procedure is to report the accurate reference time information from the gNB-DU to the gNB-CU. The procedure uses non-UE associated signalling.

8.12.2.2 Successful Operation

Figure 8.12.2-2-1: Reference Time Information Report

The gNB-DU initiates the procedure by sending a REFERENCE TIME INFORMATION REPORT message to the gNB-CU. The REFERENCE TIME INFORMATION REPORT message may be used as a response to the REFERENCE TIME INFORMATION REPORTING CONTROL message.

8.12.2.3 Abnormal Conditions

Not applicable.

8.13 Positioning Procedures

8.13.1 Positioning Assistance Information Control

8.13.1.1 General

The purpose of the Positioning Assistance Information Control procedure is to allow the gNB-CU to signal positioning assistance information to the gNB-DU for positioning assistance information broadcasting. The procedure uses non-UE-associated signalling.

8.13.1.2 Successful Operation

Figure 8.13.1.2-1: Positioning Assistance Information Control procedure

The gNB-CU initiates the procedure by sending a POSITIONING ASSISTANCE INFORMATION CONTROL message.

If the *Positioning Assistance Information* IE is included in the POSITIONING ASSISTANCE INFORMATION CONTROL message, the gNB-DU shall replace any previously received positioning assistance information with the newly received information, and use the newly received information to configure positioning assistance information broadcasting as specified in TS 38.455 [37].

If the *Broadcast* IE is included in the POSITIONING ASSISTANCE INFORMATION CONTROL message and set to "start", the gNB-DU may start broadcasting the positioning assistance information. If the *Broadcast* IE is included in the POSITIONING ASSISTANCE INFORMATION CONTROL message and set to "stop", the gNB-DU may stop broadcasting the positioning assistance information.

If the *Positioning Broadcast Cells* IE is included in the POSITIONING ASSISTANCE INFORMATION CONTROL message, the gNB-DU shall, if supported, consider that the procedure is applicable only to the indicated cells in this IE.

Interaction with the Positioning Assistance Information Feedback procedure:

If the *Routing ID* IE is included in the POSITIONING ASSISTANCE INFORMATION CONTROL message, the gNB-DU shall, if supported, store this information and include it in any future POSITIONING ASSISTANCE INFORMATION FEEDBACK messages associated to the requested positioning assistance information broadcasting.

8.13.1.3 Abnormal Conditions

If the *Broadcast* IE is set to "start", and the *Positioning Assistance Information* IE is not included, the gNB-DU shall consider the procedure as failed.

If the *Broadcast* IE is not included in the POSITIONING ASSISTANCE INFORMATION CONTROL message, the gNB-DU shall consider the procedure as failed.

8.13.2 Positioning Assistance Information Feedback

8.13.2.1 General

The purpose of the Positioning Assistance Information Feedback procedure is to allow the gNB-DU to give feedback to the gNB-CU on positioning assistance information broadcasting. The procedure uses non-UE-associated signalling.

8.13.2.2 Successful Operation

Figure 8.13.2.2-1: Positioning Assistance Information Feedback procedure

The gNB-DU initiates the procedure by sending a POSITIONING ASSISTANCE INFORMATION FEEDBACK message. If the *Positioning Assistance Information Failure List* IE is included in the POSITIONING ASSISTANCE INFORMATION FEEDBACK message, the gNB-CU shall consider that positioning assistance information broadcasting could not be configured for the relevant information.

If the *Positioning Broadcast Cells* IE is included in the POSITIONING ASSISTANCE INFORMATION FEEDBACK message, the gNB-CU shall consider that the feedback provided is applicable to the cells in this IE.

If the *Routing ID* IE is included in the POSITIONING ASSISTANCE INFORMATION FEEDBACK message, the gNB-CU may use this information to identify the positioning assistance information broadcasting for which feedback is provided.

8.13.2.3 Abnormal Conditions

Void.

8.13.3 Positioning Measurement

8.13.3.1 General

The purpose of the Positioning Measurement procedure is to allow the gNB-CU to request one or more TRPs in the gNB-DU to perform and report positioning measurements. The procedure uses non-UE-associated signalling.

8.13.3.2 Successful Operation

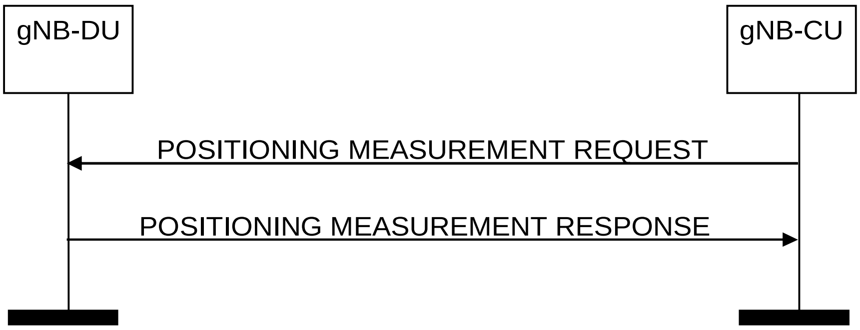

Figure 8.13.3.2-1: Positioning Measurement procedure: successful operation

The gNB-CU initiates the procedure by sending a POSITIONING MEASUREMENT REQUEST message to the gNB-DU, indicating in the *TRP Measurement Request List* IE the TRP(s) from which measurements are requested. The gNB-DU node shall use the included information to configure positioning measurements by the indicated TRP(s). If at least one of the requested measurements has been successful for at least one of the TRPs, the gNB-DU shall reply with the POSITIONING MEASUREMENT RESPONSE message including the *Positioning Measurement Response List* IE.

If the *Positioning Report Characteristics* IE is set to "OnDemand", the gNB-DU shall return the corresponding measurement results in the *Positioning Measurement Result List* IE in the POSITIONING MEASUREMENT RESPONSE message, and the gNB-CU shall consider that this reporting has been terminated by the gNB-DU.

If the *Measurement Beam Information Request* IE is included in the POSITIONING MEASUREMENT REQUEST message, the gNB-DU node shall, if supported, include the *Measurement Beam Information* IE in the *Positioning Measurement Result* IE of the POSITIONING MEASUREMENT RESPONSE message.

If the *Measurement Quality* IE is included in the *Measurement Result* IE in the POSITIONING MEASUREMENT RESPONSE message, the gNB-CU may use it for further signalling. If the *Measurement Quality* IE includes the *Zenith Quality* IE, the gNB-CU may use it for further signalling.

If the *System Frame Number* IE and/or the *Slot Number* IE are included in the POSITIONING MEASUREMENT REQUEST message, the gNB-DU node shall, if supported, consider that the respective information indicates the activation time of SRS transmission.

If the *Measurement Characteristics Request Indicator* IE is included in the POSITIONING MEASUREMENT REQUEST message, the gNB-DU shall, if supported, take the requested measurement characteristics into account when configuring measurements, and include the requested information, if available, in the POSITIONING MEASUREMENT RESPONSE message.

If the *Number of TRP Rx TEGs* IE is included in the POSITIONING MEASUREMENT REQUEST message, the gNB-DU shall, if supported, use it to measure the same SRS resource with different TRP Rx TEGs for the indicated TRP, and report the corresponding UL-RTOA and/or gNB Rx-Tx time difference measurements.

If the *Number of TRP RxTx TEGs* IE is included in the POSITIONING MEASUREMENT REQUEST message, the gNB-DU shall, if supported, use it to measure the same SRS resource with different TRP RxTx TEGs with the same TRP Tx TEG for the indicated TRP, and report the corresponding gNB Rx-Tx time difference measurements.

If the *Measurement Time Occasion* IE is included in the POSITIONING MEASUREMENT REQUEST message, the gNB-DU may take it into account as the number of SRS measurement time occasions for a measurement instance.

If the *Time Window Information Measurement List* IE is included in the POSITIONING MEASUREMENT REQUEST message, the gNB-DU shall, if supported, consider this as the time window(s) within which to perform the positioning measurements, as specified in TS 38.305 [42].

Interaction with the Positioning Measurement Report procedure:

If the *Positioning Report Characteristics* IE is set to "Periodic", the gNB-DU shall initiate the corresponding measurements, and it shall reply with the POSITIONING MEASUREMENT RESPONSE message without including any measurement results in the message. The gNB-DU shall then periodically initiate the Positioning Measurement Report procedure for the corresponding measurements, with the requested reporting periodicity.

If the *Report Characteristics* IE is set to "OnDemand" and the *Response Time* IE is included in the POSITIONING MEASUREMENT REQUEST message, the gNB-DU shall, if supported, return the corresponding measurement results in the POSITIONING MEASUREMENT RESPONSE message within the indicated time.

If the *Positioning Report Characteristics* IE is set to "Periodic" and the *Positioning Measurement Amount* IE is included in the POSITIONING MEASUREMENT REQUEST message, the gNB-DU shall, if supported, take it into account for sending the POSITIONING MEASUREMENT REPORT message.

8.13.3.3 Unsuccessful Operation

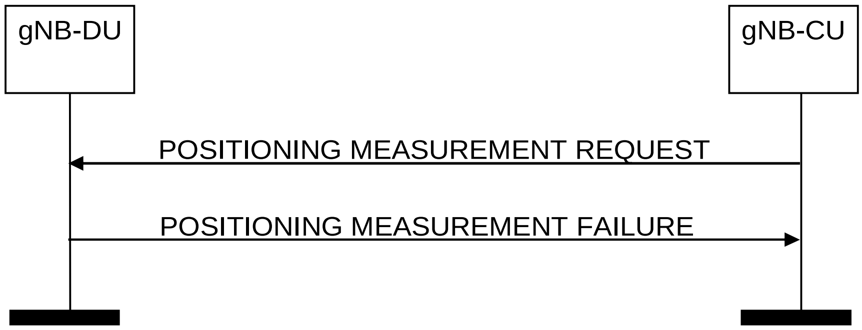

Figure 8.13.3.3-1: Positioning Measurement procedure: unsuccessful operation

If the gNB-DU is unable to configure any of the requested positioning measurements for any of the TRPs in the *TRP Measurement Request List* IE of the POSITIONING MEASUREMENT REQUEST message, it shall respond with a POSITIONING MEASUREMENT FAILURE message.

8.13.3.4 Abnormal Conditions

If the gNB-DU receives a POSITIONING MEASUREMENT REQUEST message containing an LMF Measurement ID corresponding to an ongoing positioning measurement, it shall consider the procedure as failed and initiate local error handling.

8.13.4 Positioning Measurement Report

8.13.4.1 General

The purpose of the Positioning Measurement Report procedure is for the gNB-DU to report positioning measurements to the gNB-CU. The procedure uses non-UE-associated signalling.

8.13.4.2 Successful Operation

Figure 8.13.4.2-1: Positioning Measurement Report procedure: successful operation

The gNB-DU initiates the procedure by sending a POSITIONING MEASUREMENT REPORT message. The POSITIONING MEASUREMENT REPORT message contains the positioning measurement results according to the associated measurement configuration.

8.13.4.3 Unsuccessful Operation

Not applicable.

8.13.4.4 Abnormal Conditions

Not applicable.

8.13.5 Positioning Measurement Abort

8.13.5.1 General

The purpose of the Positioning Measurement Abort procedure is to enable the gNB-CU to abort an on-going measurement. The procedure uses non-UE-associated signalling.

8.13.5.2 Successful Operation

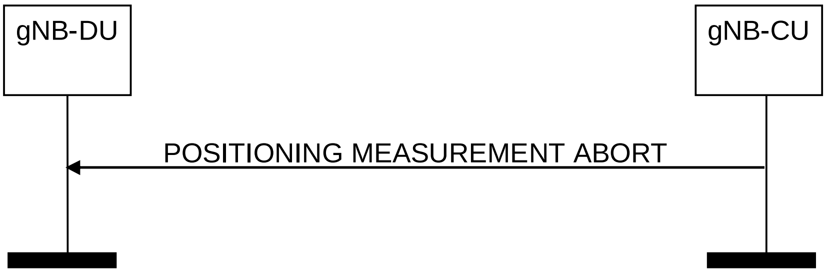

Figure 8.13.5.2-1: Positioning Measurement Abort procedure: successful operation

The gNB-CU initiates the procedure by generating a POSITIONING MEASUREMENT ABORT message. Upon receiving this message, the gNB-DU shall terminate the on-going measurement identified by the *RAN Measurement ID* IE and may release any resources previously allocated for the same measurement.

8.13.5.3 Unsuccessful Operation

Not applicable.

8.13.5.4 Abnormal Conditions

If the gNB-DU cannot identify the previously requested measurement to be aborted, it shall ignore the POSITIONING MEASUREMENT ABORT message.

8.13.6 Positioning Measurement Failure Indication

8.13.6.1 General

The purpose of the Positioning Measurement Failure Indication procedure is for the gNB-DU to notify the gNB-CU that the positioning measurements previously requested with the Positioning Measurement procedure can no longer be reported. The procedure uses non-UE-associated signalling.

8.13.6.2 Successful Operation

Figure 8.13.6.2-1: Positioning Measurement Failure Indication procedure: successful operation

The gNB-DU initiates the procedure by sending a POSITIONING MEASUREMENT FAILURE INDICATION message. Upon reception of the POSITIONING MEASUREMENT FAILURE INDICATION message, the gNB-CU shall consider that the indicated positioning measurements have been terminated by the gNB-DU.

8.13.6.3 Unsuccessful Operation

Not applicable.

8.13.6.4 Abnormal Conditions

Not applicable.

8.13.7 Positioning Measurement Update

8.13.7.1 General

The purpose of the Positioning Measurement Update procedure is to modify one or more periodic positioning measurements performed by the gNB-DU. The procedure uses non-UE-associated signalling.

8.13.7.2 Successful Operation

Figure 8.13.7.2-1: Positioning Measurement Update procedure: successful operation

The gNB-CU initiates the procedure by generating a POSITIONING MEASUREMENT UPDATE message. Upon receiving the message, the gNB-DU shall overwrite the previously received measurement configuration for the corresponding measurements.

If the *Number of TRP Rx TEGs* IE is included in the *TRP Measurement Update List* IE in the POSITIONING MEASUREMENT UPDATE message, the gNB-DU shall clear any previously stored information and store the newly received information.

If the *Number of TRP RxTx TEGs* IE is included in the *TRP Measurement Update List* IE in the POSITIONING MEASUREMENT UPDATE message, the gNB-DU shall clear any previously stored information and store the newly received information.

If the *Measurement Characteristics Request Indicator* IE is included in the POSITIONING MEASUREMENT UPDATE message, the gNB-DU shall clear any previously stored information and store the newly received information.

If the *Measurement Time Occasion* IE is included in the POSITIONING MEASUREMENT UPDATE message, the gNB-DU shall clear any previously stored information and store the newly received information.

8.13.7.3 Unsuccessful Operation

Not applicable.

8.13.7.4 Abnormal Conditions

If the gNB-DU cannot identify the given positioning measurements, it shall regard the procedure as failed and initiate local error handling.

8.13.8 TRP Information Exchange

8.13.8.1 General

The purpose of the TRP Information Exchange procedure is to allow the gNB-CU to request the gNB-DU to provide detailed information for TRPs hosted by the gNB-DU. The procedure uses non-UE-associated signalling.

8.13.8.2 Successful Operation

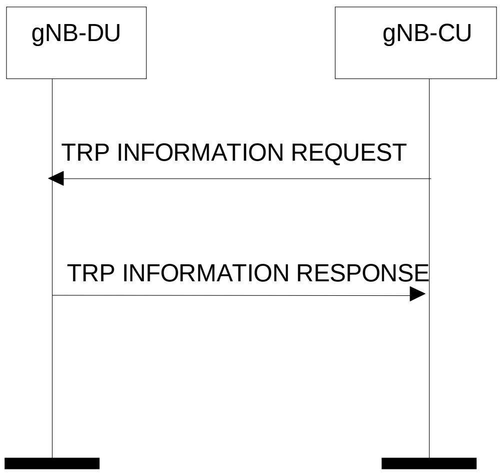

Figure 8.13.8.2-1: TRP Information Exchange procedure, successful operation

The gNB-CU initiates the procedure by sending a TRP INFORMATION REQUEST message. The gNB-DU responds with a TRP INFORMATION RESPONSE message that contains the requested TRP information.

If the *TRP List* IE is included in the TRP INFORMATION REQUEST message, the gNB-DU should include in the TRP INFORMATION RESPONSE message, the requested information for all TRPs included in the *TRP List* IE.

If the *TRP List* IE is not included in the TRP INFORMATION REQUEST message, the gNB-DU should include the requested information for all TRPs hosted by the gNB-DU in the TRP INFORMATION RESPONSE message.

If the *PRS Muting* IE is included in the *PRS Configuration* IE in the TRP INFORMATION RESPONSE message, the gNB-CU may use it for further signaling.

If the *QCL Info* IE is included in the *PRS Configuration* IE in the TRP INFORMATION RESPONSE message, the gNB-CU may use it for further signaling.

If the *Aggregated PRS Resource Set List* IE is included in the *PRS Configuration* IE in the TRP INFORMATION RESPONSE message, the gNB-CU may use it for further signaling.

If the *DL-PRS Resource Coordinates* IE is included in the *Geographical Coordinates* IE in the *TRP Information* IE in the TRP INFORMATION RESPONSE message, the gNB-CU may use it for further signaling.

If the *Mobile IAB-MT UE ID* IE is included in the *TRP Information* IE in the TRP INFORMATION RESPONSE message, the gNB-CU may use it for further signaling.

If the *TRP Information Type Item* IE is set to 'mobile TRP location info', the gNB-DU shall, if supported, derive the location of the Mobile TRP as specified in TS 23.273 [49] and include the *Mobile TRP Location Information* IE in the TRP INFORMATION RESPONSE message.

8.13.8.3 Unsuccessful Operation

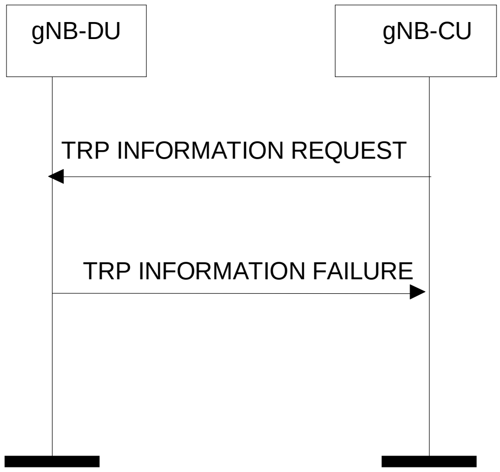

Figure 8.13.8.3-1: TRP Information Exchange procedure, unsuccessful operation

If the gNB-DU cannot provide any of the requested information, the gNB-DU shall respond with a TRP INFORMATION FAILURE message.

8.13.9 Positioning Information Exchange

8.13.9.1 General

The Positioning Information Exchange procedure is initiated by the gNB-CU to indicate to the gNB-DU the need to configure the UE to transmit SRS signals and to retrieve the SRS configuration from the gNB-DU. The procedure uses UE-associated signalling.

8.13.9.2 Successful Operation

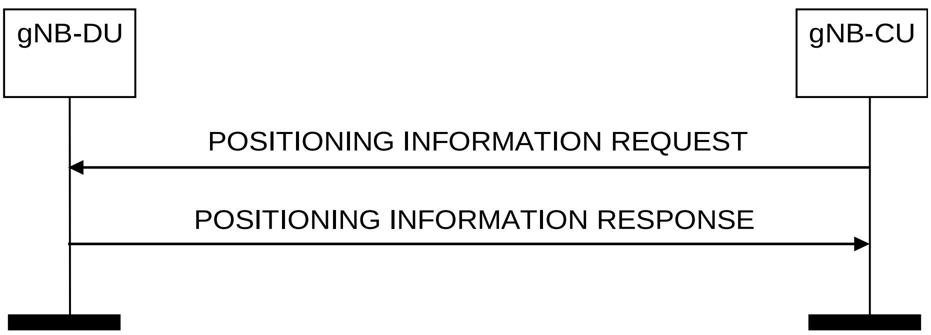

Figure 8.13.9.2-1: Positioning Information Exchange procedure, successful operation

The gNB-CU initiates the procedure by sending a POSITIONING INFORMATION REQUEST message to the gNB-DU.

If the *Requested SRS Transmission Characteristics* IE is included in the POSITIONING INFORMATION REQUEST message, the gNB-DU may take this information into account when configuring SRS transmissions for the UE, and it shall include the *SRS Configuration* IE and the *SFN Initialisation Time* IE in the POSITIONING INFORMATION RESPONSE message. If the *SRS Positioning INACTIVE Query Indication* IE is also included in the POSITIONING INFORMATION REQUEST message and set to 'true', the gNB-DU shall, if supported, include the *SRS-PosRRC-InactiveConfig* IE in the POSITIONING INFORMATION RESPONSE message.

If the *Spatial Relation Information per SRS Resource* IE and the *Periodicity List* IE are both included in the *Requested SRS Transmission Characteristics* IE, the gNB-DU shall consider that the *Spatial Relation per SRS Resource Item* IE and the *Periodicity List Item* IE have one-to-one mapping relation.

If the *UE Reporting Information* IE is included in the POSITIONING INFORMATION REQUEST message, the gNB-DU may take this information into account for allocating proper CG-SDT resources when positioning a UE.

If the *Time Window Information for SRS* IE is included in the POSITIONING INFORMATION REQUEST message, the gNB-DU shall, if supported, configure the UE to start transmitting its UL SRS transmission at the indicated time instance.

If the *Positioning Validity Area Cell List* IE and *Validity Area specific SRS Information* IE within the *Requested SRS Transmission Characteristics* IE are included in the POSITIONING INFORMATION REQUEST message, the gNB-DU shall, if supported, take this information into account for configuring SRS transmissions for the UE in the indicated validity area, and shall include the *SRS-PosRRC-InactiveValidityAreaConfig* IE, the *SFN Initialisation Time* IE and the *Positioning Validity Area Cell List* IE in the POSITIONING INFORMATION RESPONSE message.

If the *Requested SRS Preconfiguration Characteristics List* IE is included in the POSITIONING INFORMATION REQUEST message, the gNB-DU shall, if supported, take this information into account when preconfiguring area specific SRS configurations for the UE, and include the *SRS Preconfiguration List* IE in the POSITIONING INFORMATION RESPONSE message.

Interaction with the UE Context Modification Required (gNB-DU initiated) procedure:

The UE Context Modification Required (gNB-DU initiated) procedure may be performed before the POSITIONING INFORMATION RESPONSE message.

8.13.9.3 Unsuccessful Operation

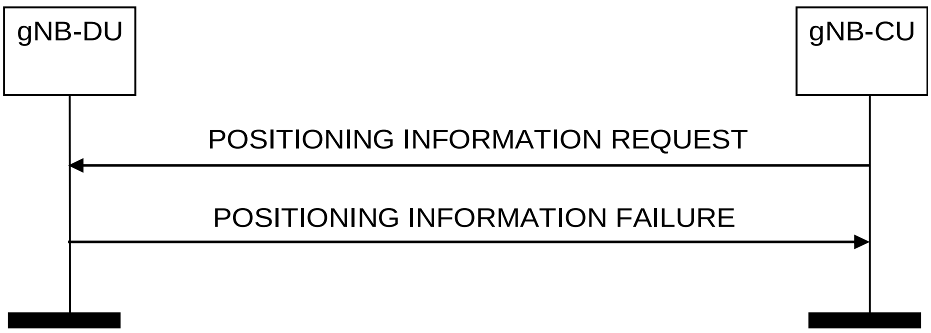

Figure 8.13.9.3-1: Positioning Information Exchange procedure, unsuccessful operation

If the *Requested SRS Transmission Characteristics* IE is included in the POSITIONING INFORMATION REQUEST message and the gNB-DU is unable to configure any SRS transmissions for the UE, the gNB-DU shall respond with a POSITIONING INFORMATION FAILURE message with an appropriate cause value.

If the gNB-DU is unable to provide any of the requested information, the gNB-DU shall respond with a POSITIONING INFORMATION FAILURE message with an appropriate cause value.

8.13.10 Positioning Activation

8.13.10.1 General

The Positioning Activation procedure is initiated by the gNB-CU to request the gNB-DU to activate semi-persistent or trigger aperiodic UL SRS transmission by the UE. The procedure uses UE-associated signalling.

8.13.10.2 Successful Operation

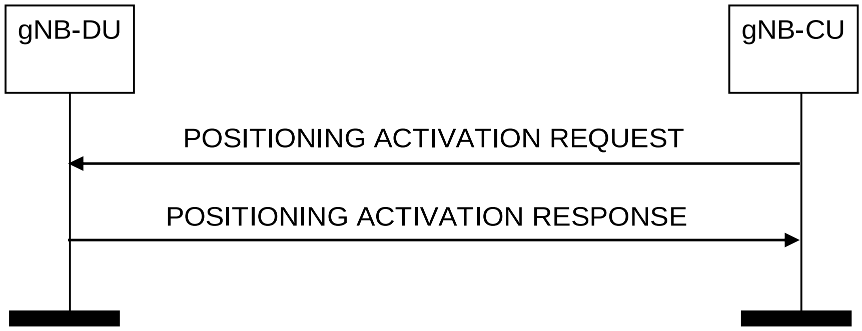

Figure 8.13.10.2-1: Positioning Activation procedure, successful operation

The gNB-CU initiates the procedure by sending a POSITIONING ACTIVATION REQUEST message to the gNB-DU.

For semi-persistent UL SRS, the POSITIONING ACTIVATION REQUEST message includes an indication of the UL SRS resource set to be activated, and may include the spatial relation for the semi-persistent UL SRS resource to be activated. For aperiodic UL SRS, if the *SRS Resource Trigger* IE is included in the POSITIONING ACTIVATION REQUEST message, the gNB-DU shall take the value of this IE into account when triggering aperiodic SRS transmission by the UE.

If the *Activation Time* IE is included in the POSITIONING ACTIVATION REQUEST message, the gNB-DU shall take the indicated value as the requested time for activation of the UE's SRS transmission.

Following successful activation of UL SRS transmission in the UE, the gNB-DU shall respond with a POSITIONING ACTIVATION RESPONSE message. If the POSITIONING ACTIVATION RESPONSE message includes the *System Frame Number* and/or the *Slot Number* IEs, the gNB-CU shall consider that the respective information indicates the activation time of SRS transmission by the UE.

If the *Aggregated Positioning SRS resource Set List* IE is included in POSITIONING ACTIVATION REQUEST message and SRS bandwidth aggregation is configured in the UE, the gNB-DU shall, if supported, activate the indicated SRS resource set(s) in the UE.

8.13.10.3 Unsuccessful Operation

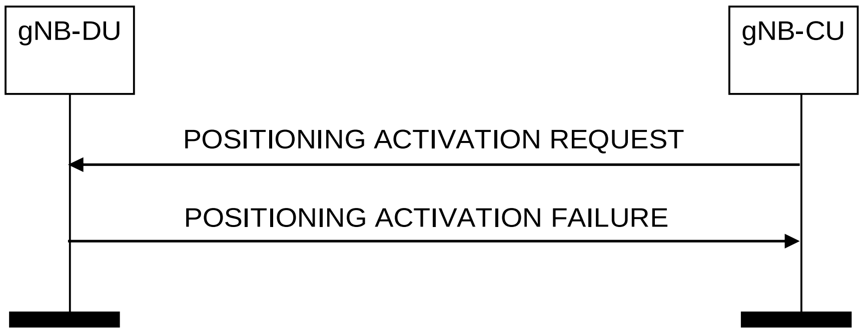

Figure 8.13.10.3-1: Positioning Activation procedure, unsuccessful operation

If the gNB-DU is unable to activate UL SRS transmission in the UE, it shall respond with a POSITIONING ACTIVATION FAILURE message.

If the gNB-DU is unable to trigger the aperiodic SRS transmission with the indicated *SRS Resource Trigger* IE, it shall respond with a POSITIONING ACTIVATION FAILURE message with an appropriate cause value

8.13.10.4 Abnormal Conditions

Void.

8.13.11 Positioning Deactivation

8.13.11.1 General

The Positioning Deactivation procedure is initiated by the gNB-CU to indicate to the gNB-DU node that UL SRS transmission should be deactivated in the UE. The procedure uses UE-associated signalling.

8.13.11.2 Successful Operation

Figure 8.13.11.2-1: Positioning Deactivation procedure, successful operation

The gNB-CU initiates the procedure by sending a POSITIONING DEACTIVATION message to the gNB-DU, including an indication of the UL SRS resource set(s) to be deactivated or release all the related resources.

8.13.11.3 Unsuccessful Operation

Not Applicable.

8.13.11.4 Abnormal Conditions

Void.

8.13.12 E-CID Measurement Initiation

8.13.12.1 General

The purpose of E-CID Measurement Initiation procedure is to allow the gNB-CU to request the gNB-DU to report E-CID measurements used by LMF to compute the location of the UE. The procedure uses UE-associated signalling.

8.13.12.2 Successful Operation

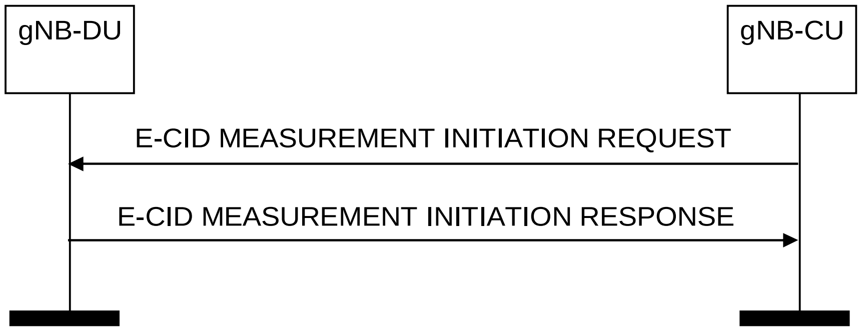

Figure 8.13.12.2-1: E-CID Measurement Initiation procedure, successful operation

The gNB-CU initiates the procedure by sending an E-CID MEASUREMENT INITIATION REQUEST message. If the gNB-DU is able to initiate the requested E-CID measurements, it shall reply with the E-CID MEASUREMENT INITIATION RESPONSE message.

If the *E-CID Report Characteristics* IE is set to "OnDemand", the gNB-DU shall return the result of the measurement in the E-CID MEASUREMENT INITIATION RESPONSE message including, if available, the *Geographical Coordinates* IE in the *E-CID Measurement Result* IE and the *Cell Portion ID* IE, and the gNB-CU shall consider that the E-CID measurements for the UE have been terminated by the gNB-DU. The *Measured Results List* IE shall be included in the *E-CID Measurement Result* IE of the E-CID MEASUREMENT INITIATION RESPONSE message when measurement quantities other than "Default" have been requested.

Interaction with the E-CID Measurement Report procedure:

If the *E-CID Report Characteristics* IE is set to "Periodic", the gNB-DU shall initiate the requested measurements and shall reply with the E-CID MEASUREMENT INITIATION RESPONSE message without including either the *E-CID Measurement Result* IE or the *Cell Portion ID* IE in this message. The gNB-DU shall then periodically initiate the E-CID Measurement Report procedure for the measurements, with the requested reporting periodicity.

8.13.12.3 Unsuccessful Operation

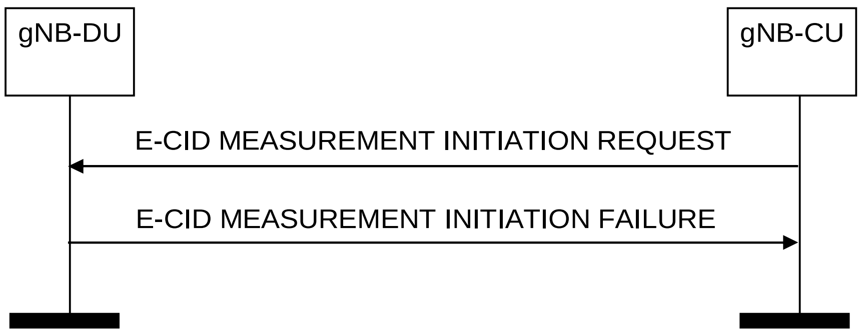

Figure 8.13.12.3-1: E-CID Measurement Initiation procedure, unsuccessful operation

If the gNB-DU is not able to initiate at least one of the requested E-CID measurements, the gNB-DU shall respond with an E-CID MEASUREMENT INITIATION FAILURE message.

8.13.13 E-CID Measurement Failure Indication

8.13.13.1 General

The purpose of the E-CID Measurement Failure Indication procedure is for the gNB-DU to notify the gNB-CU that the E-CID measurements previously requested with the E-CID Measurement Initiation procedure can no longer be reported. The procedure uses UE-associated signalling.

8.13.13.2 Successful Operation

Figure 8.13.13.2-1: E-CID Measurement Failure Indication, successful operation

The gNB-DU initiates the procedure by sending an E-CID MEASUREMENT FAILURE INDICATION message. Upon reception of the E-CID MEASUREMENT FAILURE INDICATION message, the gNB-CU shall consider that the E-CID measurements for the UE have been terminated by the gNB-DU.

8.13.13.3 Unsuccessful Operation

Not applicable.

8.13.14 E-CID Measurement Report

8.13.14.1 General

The purpose of E-CID Measurement Report procedure is for the gNB-DU to provide the E-CID measurements for the UE to the gNB-CU. The procedure uses UE-associated signalling.

8.13.14.2 Successful Operation

Figure 8.13.14.2-1: E-CID Measurement Report procedure, successful operation

The gNB-DU initiates the procedure by sending an E-CID MEASUREMENT REPORT message. The E-CID MEASUREMENT REPORT message contains the E-CID measurement results according to the measurement configuration in the respective E-CID MEASUREMENT INITIATION REQUEST message.

The *Measured Results List* IE shall be included in the *E-CID Measurement Result* IE of the E-CID MEASUREMENT REPORT message when measurement quantities other than "Default" have been requested.

If available, the gNB-DU shall include the *Geographical Coordinates* IE in the *E-CID Measurement Result* IE in the E-CID MEASUREMENT REPORT message.

If available, the gNB-DU shall include the *Cell Portion ID* IE in the E-CID MEASUREMENT REPORT message.

If available, the gNB-DU shall include the *Mobile Access Point Location Information* IE in the E-CID MEASUREMENT REPORT message.

8.13.14.3 Unsuccessful Operation

Not applicable.

8.13.15 E-CID Measurement Termination

8.13.15.1 General

The purpose of E-CID Measurement Termination procedure is to terminate periodical E-CID measurements for the UE performed by the gNB-DU. The procedure uses UE-associated signalling.

8.13.15.2 Successful Operation

Figure 8.13.15.2-1: E-CID Measurement Termination procedure, successful operation

The gNB-CU initiates the procedure by generating an E-CID MEASUREMENT TERMINATION COMMAND message.

8.13.15.3 Unsuccessful Operation

Not applicable.

8.13.16 Positioning Information Update

8.13.16.1 General

The Positioning Information Update procedure is initiated by the gNB-DU to indicate to the gNB-CU that a change has occurred in the SRS configuration. The procedure uses UE-associated signalling.

8.13.16.2 Successful Operation

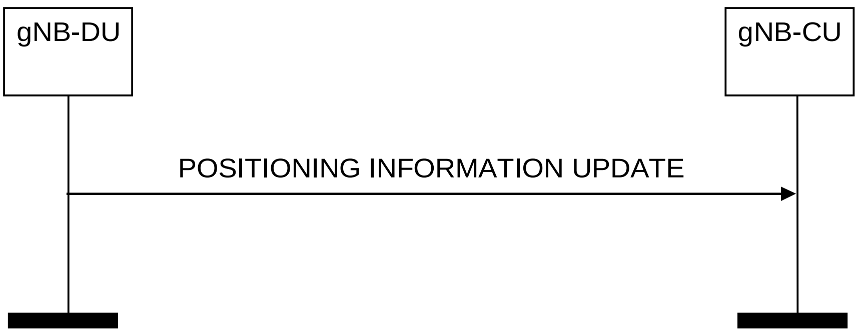

Figure 8.13.16.2-1: Positioning Information Update procedure, successful operation

The gNB-DU initiates the procedure by sending a POSITIONING INFORMATION UPDATE message to the gNB-CU.

If the *SRS Configuration* IE is included in the POSITIONING INFORMATION UPDATE message, the gNB-CU shall consider this information as the updated SRS Configuration for the UE. If the *SFN Initialisation Time* IE is included in the POSITIONING INFORMATION UPDATE message, the gNB-CU shall consider this information as the SFN Initialisation Time associated to the SRS Configuration.

8.13.16.3 Unsuccessful Operation

Not Applicable.

8.13.16.4 Abnormal Conditions

Void.

8.13.17 PRS Configuration Exchange

8.13.17.1 General

The PRS Configuration Exchange procedure is initiated by the gNB-CU to request the gNB-DU to configure or update (i.e., turn off) the PRS transmissions.

The procedure uses non-UE-associated signalling.

8.13.17.2 Successful Operation

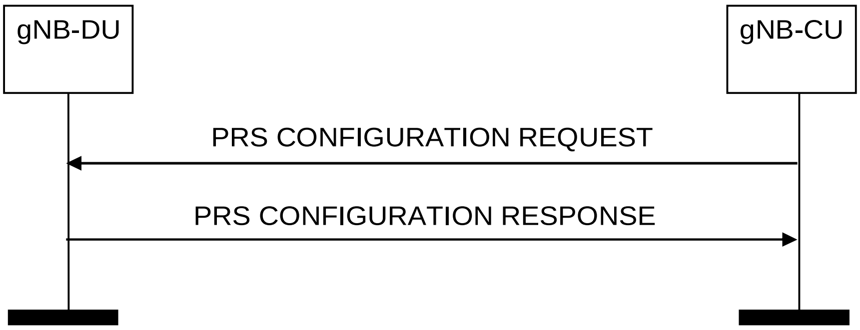

Figure 8.13.17.2-1: PRS Configuration Exchange procedure, successful operation

The gNB-CU initiates the procedure by sending a PRS CONFIGURATION REQUEST message to the gNB-DU.

If the *PRS Configuration Request Type* IE is set to "configure", the gNB-DU should use the information in the *Requested DL PRS Transmission Characteristics* IE to configure DL-PRS transmission by the indicated TRP(s).

If the *PRS Configuration Request Type* IE is set to "off", the gNB-DU should, if supported, use the information in the *PRS Transmission Off Information* IE to turn off the DL-PRS transmission for the indicated TRP(s), PRS Resource Set(s), or PRS Resource(s).

If DL-PRS transmission is successfully configured or updated for at least one of the TRPs, the gNB-DU shall respond with the PRS CONFIGURATION RESPONSE message.

8.13.17.3 Unsuccessful Operation

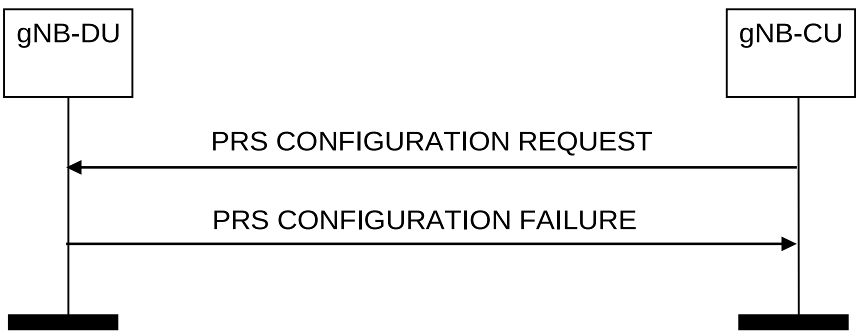

Figure 8.13.17.3-1: PRS Configuration Exchange procedure, unsuccessful operation

If the gNB-DU cannot configure or update DL-PRS transmission for any of the TRPs in the *PRS TRP List* IE of the PRS CONFIGURATION REQUEST message, it shall respond with a PRS CONFIGURATION FAILURE message with an appropriate cause value.

8.13.17.4 Abnormal Conditions

Void.

8.13.18 Measurement Preconfiguration

8.13.18.1 General

The Measurement Preconfiguration procedure allows the gNB-CU to provide necessary information to the serving gNB-DU and request the gNB-DU to preconfigure measurement gap and/or PRS processing window of the UE.

8.13.18.2 Successful Operation

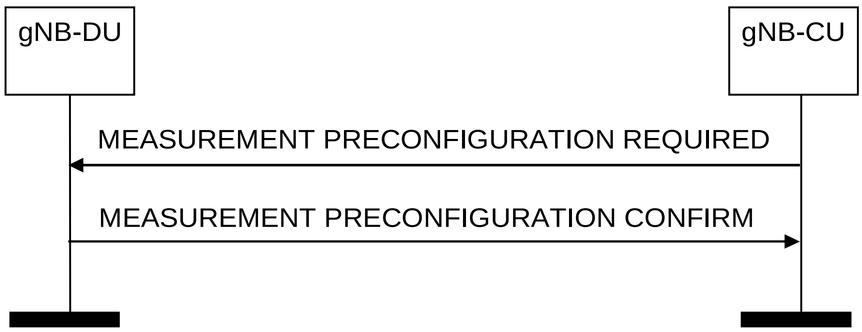

Figure 8.13.18.2-1: Measurement Preconfiguration procedure, successful operation

The gNB-CU initiates the procedure by sending a MEASUREMENT PRECONFIGURATION REQUIRED message.

If the gNB-DU is able to configure measurement gap or PRS processing window, it shall reply with the MEASUREMENT PRECONFIGURATION CONFIRM message.

If the *PosMeasGapPreConfigList* IE is included in the MEASUREMENT PRECONFIGURATION CONFIRM message, the gNB-CU shall, if supported, take the preconfigured measurement gaps information into account.

8.13.18.3 Unsuccessful Operation

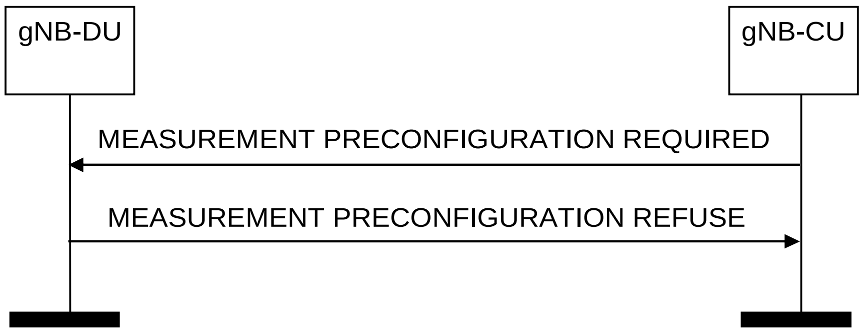

Figure 8.13.18.3-1: Measurement Preconfiguration procedure, unsuccessful operation

If the gNB-DU cannot configure any of the measurement gap or PRS processing window, the gNB-DU shall respond with a MEASUREMENT PRECONFIGURATION REFUSE message.

8.13.19 Measurement Activation

8.13.19.1 General

The Measurement Activation procedure is initiated by the gNB-CU to request the gNB-DU to activate or deactivate the preconfigured measurement gap or PRS processing window for the UE.

8.13.19.2 Successful Operation

Figure 8.13.19.2-1: Measurement Activation procedure, successful operation

The gNB-CU initiates the procedure by sending a MEASUREMENT ACTIVATION message.

If the *PRS Measurement Info List* IE is included in the MEASUREMENT ACTIVATION message, the gNB-DU may take it into account when activating pre-configured measurement gap in the UE.

8.13.19.3 Unsuccessful Operation

Not Applicable.

8.13.20 Positioning System Information Delivery

8.13.20.1 General

The purpose of the Positioning System Information Delivery procedure is to command the gNB-DU to broadcast the requested one or several Positioning SI messages indicated by the gNB-CU. The procedure uses non-UE associated signalling.

8.13.20.2 Successful Operation

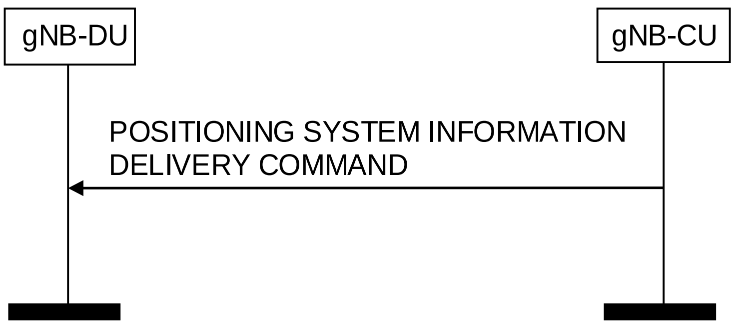

Figure 8.13.20.2-1: Positioning System Information Delivery procedure. Successful operation.

The gNB-CU initiates the procedure by sending a POSITIONING SYSTEM INFORMATION DELIVERY COMMAND message to the gNB-DU.

Upon reception of the POSITIONING SYSTEM INFORMATION DELIVERY COMMAND message, the gNB-DU shall broadcast the requested one or several Positioning SI messages, indicated by the *PosSITypeList* IE, and if the UE corresponding to the *Confirmed UE ID* IE is not in the RRC connected state, delete the UE context, if any.

If the *Delivery Status Request* IE is included in the POSITIONING SYSTEM INFORMATION DELIVERY COMMAND message, the gNB-DU shall, if supported, use it to start or stop broadcasting the Positioning SI message, indicated by the *PosSI Type* IE as described in TS 38.331 [8].

Interactions with gNB-DU Configuration Update procedure:

Upon reception of POSITIONING SYSTEM INFORMATION DELIVERY COMMAND message, the gNB-DU Configuration Update procedure may be performed, and as part of such procedure the gNB-DU shall include the *Dedicated SI Delivery Needed UE List* IE in GNB-DU CONFIGURATION UPDATE message for UEs that are unable to receive system information from broadcast.

8.13.20.3 Abnormal Conditions

Not applicable.

8.13.21 SRS Information Reservation Notification

8.13.21.1 General

The purpose of the SRS Information Reservation Notification procedure is to allow the gNB-CU to request the gNB-DU to reserve or release SRS resources in the positioning validity area.

8.13.21.2 Successful Operation

Figure 8.13.21.2-1: SRS Information Reservation Notification procedure, successful operation

The gNB-CU initiates the procedure by sending a SRS INFORMATION RESERVATION NOTIFICATION message to the gNB-DU

If the *SRS Reservation Type* IE is set to "reserve", the gNB-DU shall reserve the indicated (preconfigured) SRS information in the cells indicated by the *Positioning Validity Area Cell List* IE. If the *SRS Reservation Type* IE is set to "release", the gNB-DU shall release the indicated (preconfigured) SRS information in the cells indicated by the *Positioning Validity Area Cell List* IE.

8.13.21.3 Unsuccessful Operation

Not Applicable.

8.13.21.4 Abnormal Conditions

Void.

8.14 NR MBS Procedures

8.14.1 Broadcast Context Setup

8.14.1.1 General

The purpose of the Broadcast Context Setup procedure is to establish an MBS Session context for a broadcast session in the gNB-DU.

The procedure uses MBS-associated signalling.

8.14.1.2 Successful Operation

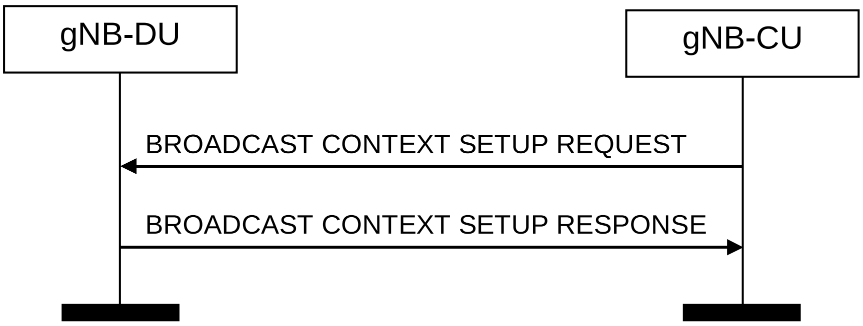

Figure 8.14.1.2-1: Broadcast Context Setup procedure: Successful Operation

The gNB-CU initiates the procedure by sending BROADCAST CONTEXT SETUP REQUEST message to the gNB-DU. If the gNB-DU succeeds to establish the broadcast MBS Session context, it replies to the gNB-CU with BROADCAST CONTEXT SETUP RESPONSE.

If the *MBS Service Area* IE is included in the BROADCAST CONTEXT SETUP REQUEST message, the gNB-DU shall take this information into account for shared F1-U tunnel assignment.

The gNB-DU shall report to the gNB-CU, in the BROADCAST CONTEXT SETUP RESPONSE message, the result of all the requested Broadcast MRBs in the following way:

- A list of MRBs which have been successfully established shall be included in the *Broadcast MRB Setup List* IE;
- A list of MRBs which failed to be established shall be included in the *Broadcast MRB Failed To Be Setup List* IE;

If the *Broadcast MRB Failed To Setup List* IE is contained in the BROADCAST CONTEXT SETUP RESPONSE message, the gNB-CU shall regard the Broadcast MRB(s) failed to be setup with an appropriate cause value for each Broadcast MRB failed to setup.

If

- either the *MBS Service Area* IE was included in the BROADCAST CONTEXT SETUP REQUEST message,
- or the the *MBS Service Area* IE was not included in the BROADCAST CONTEXT SETUP REQUEST message and the gNB-DU was not able to establish MBS Session Resources in all cells served by the gNB-DU,

the *Broadcast Area Scope* IE shall be included in the BROADCAST CONTEXT SETUP RESPONSE message to indicate the cells where MBS Session resources have been successfully established in the gNB-DU.

If the *Supported UE Type List* IE is included in the BROADCAST CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, store and use the information for configuring MBS session resources.

If the *Associated Session ID* IE is contained in the BROADCAST CONTEXT SETUP REQUEST message, the gNB-DU shall, if supported, take this information into account to determine whether MBS session resource sharing is possible, as specified in TS 38.401 [4]. If the gNB-DU decides to not establish F1-U tunnel towards the gNB-CU it shall include the *F1-U tunnel Not Established* IE set to "true" in the BROADCAST CONTEXT SETUP RESPONSE message.

If the *RAN Sharing Assistance Information* IE is included in the BROADCAST CONTEXT SETUP REQUEST message, the gNB-DU may store and use it for deciding whether to setup shared F1-U tunnel resources.

8.14.1.3 Unsuccessful Operation

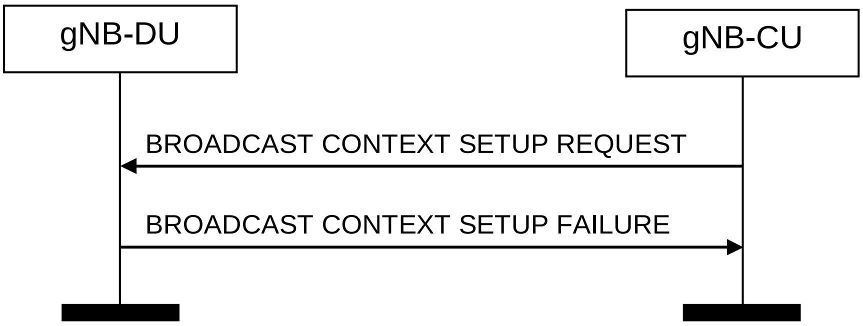

Figure 8.14.1.3-1: Broadcast Context Setup procedure: unsuccessful Operation

If the gNB-DU is not able to establish the requested MBS session context for all the MRBs in any of its cells it shall consider the procedure as failed and reply with the BROADCAST CONTEXT SETUP FAILURE message.

8.14.1.4 Abnormal Conditions

Not applicable.

8.14.2 Broadcast Context Release

8.14.2.1 General

The purpose of the Broadcast Context Release procedure is to enable the gNB-CU to order the release of an established MBS Session context for a broadcast session in the gNB-DU.

The procedure uses MBS-associated signalling.

8.14.2.2 Successful Operation

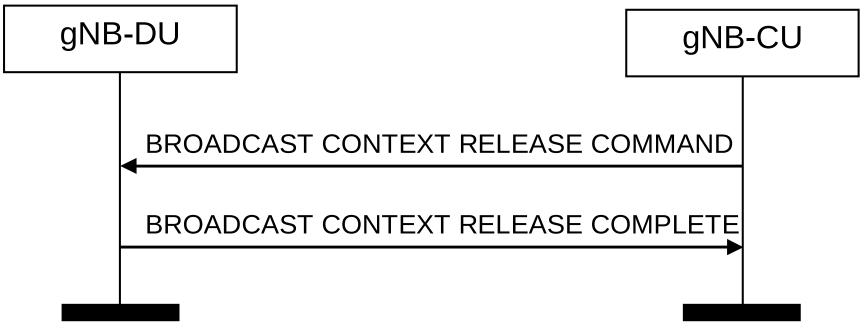

Figure 8.14.2.2-1: Broadcast Context Release procedure. Successful operation

The gNB-CU initiates the procedure by sending the BROADCAST CONTEXT RELEASE COMMAND message to the gNB-DU.

Upon reception of the BROADCAST CONTEXT RELEASE COMMAND message, the gNB-DU shall release all signalling and user data transport resources associated with the context and reply with the BROADCAST CONTEXT RELEASE COMPLETE message.

8.14.2.3 Unsuccessful Operation

Not applicable.

8.14.2.4 Abnormal Conditions

Not applicable.

8.14.3 Broadcast Context Release Request

8.14.3.1 General

The purpose of the Broadcast Context Release Request procedure is to request the gNB-CU to trigger the Broadcast Context Release procedure.

The procedure uses MBS-associated signalling.

8.14.3.2 Successful Operation

Figure 8.14.3.2-1: Broadcast Context Release Request procedure. Successful operation

The gNB-DU initiates the procedure by sending the BROADCAST CONTEXT RELEASE REQUEST message to the gNB-CU.

Interaction with the Broadcast Context Release procedure:

Upon reception of the BROADCAST CONTEXT RELEASE REQUEST message, the gNB-CU should trigger the Broadcast Context Release procedure.

8.14.3.3 Unsuccessful Operation

Not applicable.

8.14.3.4 Abnormal Conditions

Not applicable.

8.14.4 Broadcast Context Modification

8.14.4.1 General

The purpose of the Broadcast Context Modification procedure is to modify an established MBS Session context for a broadcast session in the gNB-DU.

The procedure uses MBS-associated signalling.

8.14.4.2 Successful Operation

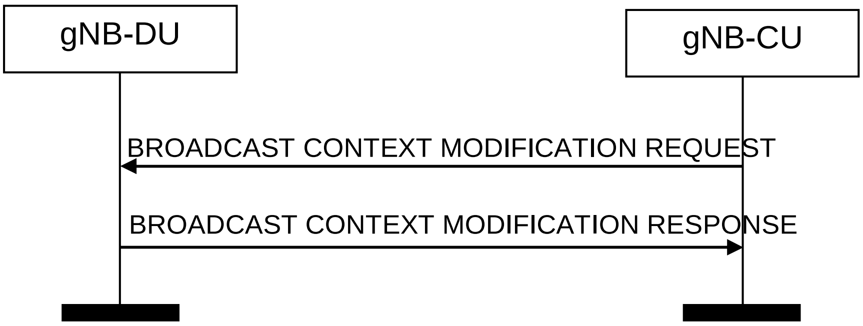

Figure 8.14.4.2-1: Broadcast Context Modification procedure. Successful operation

The BROADCAST CONTEXT MODIFICATION REQUEST message is initiated by the gNB-CU.

Upon reception of the BROADCAST CONTEXT MODIFICATION REQUEST message, the gNB-DU shall perform the modifications, and, if successful, report the update in the BROADCAST CONTEXT MODIFICATION RESPONSE message.

If the *Broadcast MRB To Be Setup List* IE is contained in the BROADCAST CONTEXT MODIFICATION REQUEST message, the gNB-DU shall setup the corresponding resources for the requested MRB(s), and report to the gNB-CU, in the BROADCAST CONTEXT MODIFICATION RESPONSE message, the result of all the requested Broadcast MRBs in the following way:

- A list of MRBs which have been successfully established shall be included in the *Broadcast MRB Setup List* IE;
- A list of MRBs which failed to be established shall be included in the *Broadcast MRB Failed To Be Setup List* IE;

If the *Broadcast MRB Failed To Be Setup List* IE is contained in the BROADCAST CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall regard the setup of the indicated MRB(s) as failed and indicate the reason for the failure with an appropriate cause value for each MRB failed to be setup.

If the *Broadcast MRB To Be Modified List* IE is contained in the BROADCAST CONTEXT MODIFICATION REQUEST message, the gNB-DU shall update the corresponding context and resources for the requested MRB(s), and report to the gNB-DU, in the BROADCAST CONTEXT MODIFICATION RESPONSE message, the modification result of all the requested Broadcast MRBs in the following way:

- A list of MRBs which have been successfully modified shall be included in the *Broadcast MRB Modified List* IE;
- A list of MRBs which failed to be modified shall be included in the *Broadcast MRB Failed To Be Modified List* IE;

If the *Broadcast MRB Failed To Be Modified List* IE is contained in the BROADCAST CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall regard the Broadcast MRB(s) failed to be modified with an appropriate cause value for each Broadcast MRB failed to modify.

If the *MBS Service Area* IE is included in the BROADCAST CONTEXT MODIFICATION REQUEST message, the gNB-DU shall

- release MBS Session Resources within cells not contained in the *MBS Service Area* IE, if any;
- establish MBS Session Resources within cells which have not been contained in MBS Service Area information previously received;
- replace MBS Service Area information previously received with information received in the *MBS Service Area* IE included in the BROADCAST CONTEXT MODIFICATION REQUEST message;
- include the *Broadcast Area Scope* IE in the BROADCAST CONTEXT MODIFICATION RESPONSE message to indicate the cells where MBS Session resources are currently established in the gNB-DU.

If the the *MBS Service Area* IE was not included in the BROADCAST CONTEXT MODIFICATION REQUEST message and the gNB-DU has released MBS Session Resources within at least one cell or has established MBS Session Resources within at least one cell the gNB-DU shall include the *Broadcast Area Scope* IE in the BROADCAST CONTEXT MODIFICATION RESPONSE message to indicate the cells where MBS Session resources are currently established in the gNB-DU.

If the *Supported UE Type List* IE is included in the BROADCAST CONTEXT MODIFICATION REQUEST message, the gNB-DU shall, if supported, store and use the information for configuring MBS session resources.

8.14.4.3 Unsuccessful Operation

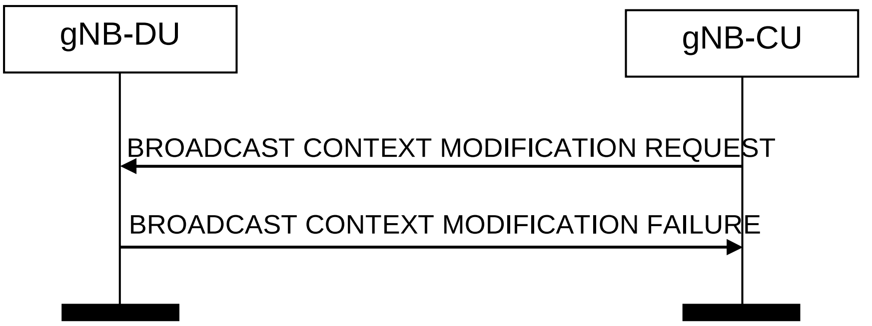

Figure 8.14.4.3-1: Broadcast Context Modification procedure. Unsuccessful operation

In case none of the requested modifications of the broadcast context can be successfully performed, the gNB-DU shall respond with the BROADCAST CONTEXT MODIFICATION FAILURE message with an appropriate cause value.

8.14.4.4 Abnormal Conditions

Not applicable.

8.14.5 Multicast Group Paging

8.14.5.1 General

The purpose of the Multicast Group Paging procedure is used to provide the paging information to enable the gNB-DU to multicast group page UEs which have joined an MBS Session and notify them about its activation. The procedure uses non-UE associated signalling.

8.14.5.2 Successful Operation

Figure 8.14.5.2-1: Multicast Group Paging

The gNB-CU initiates the Multicast Group Paging procedure by sending the MULTICAST GROUP PAGING message to the gNB-DU.

At the reception of the MULTICAST GROUP PAGING message, the gNB-DU shall perform paging of the MBS Session identified by the *MBS Session ID* IE.

If the *Paging DRX* IE is included in the MULTICAST GROUP PAGING message gNB-DU shall use it according to TS 38.304 [24].

If the *UE Identity List for Paging* IE is included in the MULTICAST GROUP PAGING message, the gNB-DU shall, if supported, use it according to TS 38.304 [24]. If absent, the gNB-DU shall perform multicast group paging of the MBS session in all paging occasions within at least one default paging cycle, as specified in TS 38.304 [24].

At the reception of the MULTICAST GROUP PAGING message, the gNB-DU shall perform paging of the MBS Session in all the served cells of the PLMN indicated in the *MBS Session ID* IE except if the *MC Paging Cell List* IE is included in which case the gNB-DU shall, if supported, perform multicast group paging of the MBS session in the indicated cells only.

8.14.5.3 Abnormal Conditions

Void.

8.14.6 Multicast Context Setup

8.14.6.1 General

The purpose of the Multicast Context Setup procedure is to establish an MBS Session context in the gNB-DU for a multicast session.

The procedure uses MBS-associated signalling.

8.14.6.2 Successful Operation

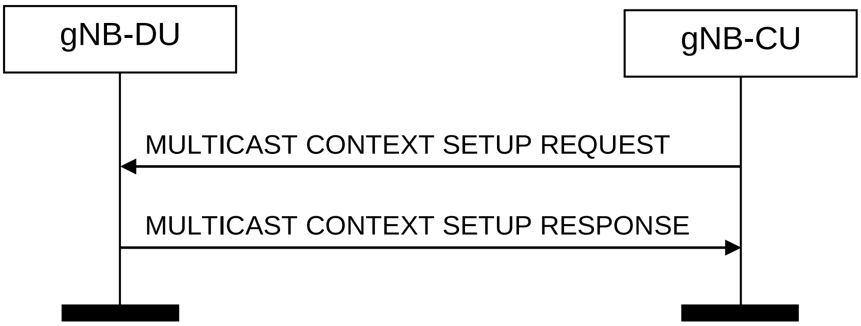

Figure 8.14.6.2-1: Multicast Context Setup procedure: Successful Operation

The gNB-CU initiates the procedure by sending MULTICAST CONTEXT SETUP REQUEST message to the gNB-DU. If the gNB-DU succeeds to establish the multicast MBS Session context, it replies to the gNB-CU with MULTICAST CONTEXT SETUP RESPONSE.

If the *MBS Service Area* IE is included in the MULTICAST CONTEXT SETUP REQUEST message, the gNB-DU shall take this information into account for shared F1-U tunnel assignment.

The gNB-DU shall report to the gNB-CU, in the MULTICAST CONTEXT SETUP RSPONSE message, the result of all the requested Multicast MRBs in the following way:

- A list of MRBs which have been successfully established shall be included in the *Multicast MRB Setup List* IE;
- A list of MRBs which failed to be established shall be included in the *Multicast MRB Failed To Be Setup List* IE;

If the *Multicast MRB Failed To Setup List* IE is contained in the MULTICAST CONTEXT SETUP RESPONSE message, the gNB-CU shall regard the Multicast MRB(s) failed to be setup with an appropriate cause value for each Multicast MRB failed to setup.

If the MULTICAST CONTEXT SETUP REQUEST message contains the *MBS Multicast Configuration Request* IE in the *Multicast CU to DU RRC Information* IE set to "query" and

- if the gNB-DU is able to provide information about the requested resources, the gNB-DU shall, if supported, include the *MBS Multicast Configuration* IE in the *MBS Multicast Configuration Response Information* IE in the *Multicast DU to CU RRC Information* IE,
- else if the gNB-DU is not able to provide information about the requested resources, the gNB-DU shall, if supported, include the *MBS Multicast Configuration not available* IE in the *MBS Multicast Configuration Response Information* IE in the *Multicast DU to CU RRC Information* IE set to "not available".

Interaction with the Multicast Distribution Context Setup procedure:

Upon reception of the MULTICAST CONTEXT SETUP REQUEST procedure, the gNB-DU shall trigger either per cell or per MBS Area Session ID or for the whole gNB-DU the Multicast Distribution Context Setup procedure to establish per cell or per MBS Area Session ID or the the whole gNB DU per accepted MRB a shared F1-U tunnel.

8.14.6.3 Unsuccessful Operation

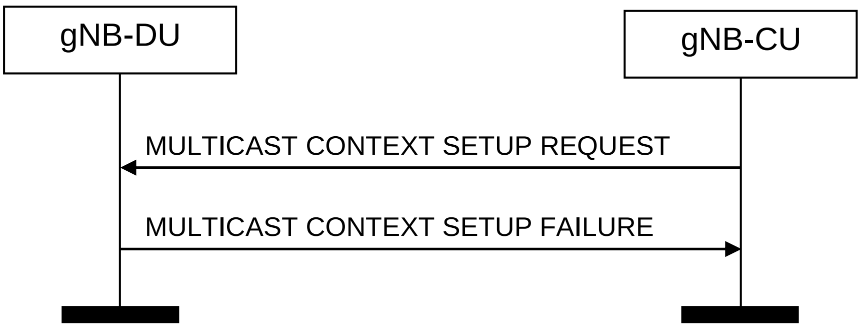

Figure 8.14.6.3-1: Multicast Context Setup procedure: unsuccessful Operation

If the gNB-DU is not able to establish the MBS session context it shall consider the procedure as failed and reply with the MULTICAST CONTEXT SETUP FAILURE message.

8.14.6.4 Abnormal Conditions

Not applicable.

8.14.7 Multicast Context Release

8.14.7.1 General

The purpose of the Multicast Context Release procedure is to enable the gNB-CU to order the release of an established MBS session context in the gNB-DU for a multicast session.

The procedure uses MBS-associated signalling.

8.14.7.2 Successful Operation

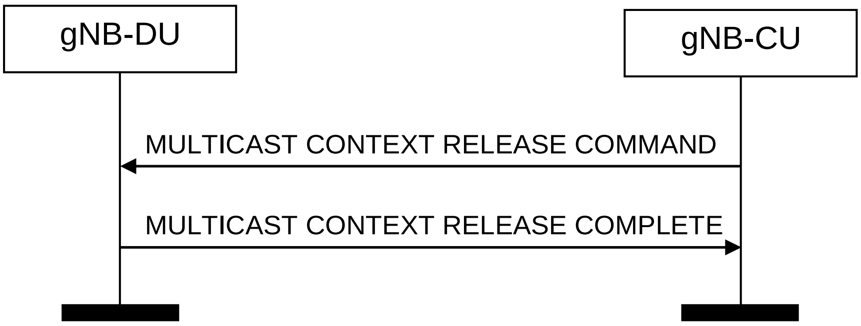

Figure 8.14.7.2-1: Multicast Context Release procedure. Successful operation

The gNB-CU initiates the procedure by sending the MULTICAST CONTEXT RELEASE COMMAND message to the gNB-DU.

Upon reception of the MULTICAST CONTEXT RELEASE COMMAND message, the gNB-DU shall release all signalling and user data transport resources associated with the context and reply with the MULTICAST CONTEXT RELEASE COMPLETE message.

8.14.7.3 Unsuccessful Operation

Not applicable.

8.14.7.4 Abnormal Conditions

Not applicable.

8.14.8 Multicast Context Release Request

8.14.8.1 General

The purpose of the Multicast Context Release Request procedure is to request the gNB-CU to trigger the Multicast Context Release procedure.

The procedure uses MBS-associated signalling.

8.14.8.2 Successful Operation

Figure 8.14.8.2-1: Multicast Context Release Request procedure. Successful operation

The gNB-DU initiates the procedure by sending the MULTICAST CONTEXT RELEASE REQUEST message to the gNB-CU.

Interaction with the Multicast Context Release procedure:

Upon reception of the MULTICAST CONTEXT RELEASE REQUEST message, the gNB-CU should trigger the Multicast Context Release procedure.

8.14.8.3 Unsuccessful Operation

Not applicable.

8.14.8.4 Abnormal Conditions

Not applicable.

8.14.9 Multicast Context Modification

8.14.9.1 General

The purpose of the Multicast Context Modification procedure is to modify an established MBS session context in the gNB-DU for a multicast session.

The procedure uses MBS-associated signalling.

8.14.9.2 Successful Operation

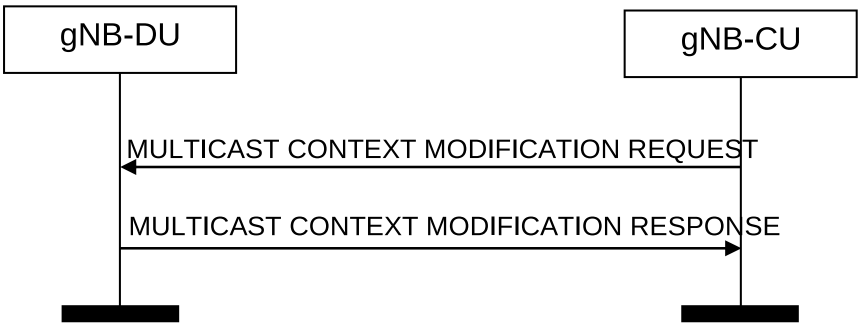

Figure 8.14.9.2-1: Multicast Context Modification procedure. Successful operation

The MULTICAST CONTEXT MODIFICATION REQUEST message is initiated by the gNB-CU.

Upon reception of the MULTICAST CONTEXT MODIFICATION REQUEST message, the gNB-DU shall perform the modifications, and, if successful, report the update in the MULTICAST CONTEXT MODIFICATION RESPONSE message.

If the *Multicast MRB To Be Setup List* IE is contained in the MULTICAST CONTEXT MODIFICATION REQUEST message, the gNB-DU shall setup the corresponding resources for the requested MRB(s), and report to the gNB-CU, in the MULTICAST CONTEXT MODIFICATION RESPONSE message, the result of all the requested Multicast MRBs in the following way:

- A list of MRBs which have been successfully established shall be included in the *Multicast MRB Setup List* IE;
- A list of MRBs which failed to be established shall be included in the *Multicast MRB Failed To Be Setup List* IE;

If the *Multicast MRB Failed To Be Setup List* IE is contained in the MULTICAST CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall regard the setup of the indicated MRB(s) as failed and indicated the resource for the failure with an appropriate cause value for each MRB failed to be setup.

If the *Multicast MRB To Be Modified List* IE is contained in the MULTICAST CONTEXT MODIFICATION REQUEST message, the gNB-DU shall update the corresponding context and resources for the requested MRB(s), and report to the gNB-CU, in the MULTICAST CONTEXT MODIFICATION RESPONSE message, the modification result of all the requested Multicast MRBs in the following way:

- A list of MRBs which have been successfully modified shall be included in the *Multicast MRB Modified List* IE;

- A list of MRBs which failed to be modified shall be included in the *Multicast MRB Failed To Be Modified List* IE;

If the *Multicast MRB Failed To Be Modified List* IE is contained in the MULTICAST CONTEXT MODIFICATION RESPONSE message, the gNB-CU shall regard the Multicast MRB(s) failed to be modified with an appropriate cause value for each Multicast MRB failed to modify.

If the MULTICAST CONTEXT MODIFICATION REQUEST message contains the *MBS Multicast Configuration Request* IE in the *Multicast CU to DU RRC Information* IE set to "query" and

- if the gNB-DU is able to provide information about the requested resources, the gNB-DU shall, if supported, include the *MBS Multicast Configuration* IE in the *MBS Multicast Configuration Response Information* IE in the *Multicast DU to CU RRC Information* IE,

- else if the gNB-DU is not able to provide information about the requested resources, the gNB-DU shall, if supported, include the *MBS Multicast Configuration not available* IE in the *MBS Multicast Configuration Response Information* IE in the *Multicast DU to CU RRC Information* IE set to "not available".

8.14.9.3 Unsuccessful Operation

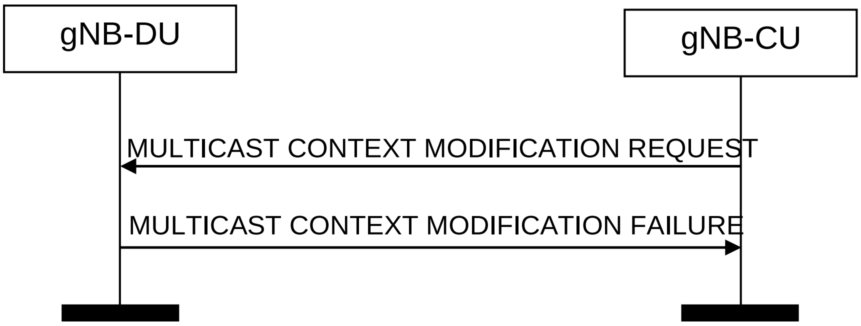

Figure 8.14.9.3-1: Multicast Context Modification procedure. Unsuccessful operation

In case none of the requested modifications of the multicast context can be successfully performed, the gNB-DU shall respond with the MULTICAST CONTEXT MODIFICATION FAILURE message with an appropriate cause value.

8.14.9.4 Abnormal Conditions

Not applicable.

8.14.10 Multicast Distribution Setup

8.14.10.1 General

The purpose of the Multicast Distribution Setup procedure is to establish F1-U bearers for the multicast MBS session.

The procedure uses MBS-associated signalling.

8.14.10.2 Successful Operation

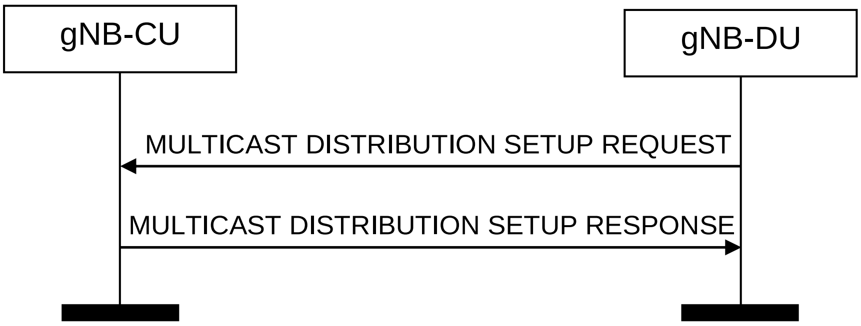

Figure 8.14.10.2-1: Multicast Distribution Setup procedure: Successful Operation

The gNB-DU initiates the procedure by sending MULTICAST DISTRIBUTION SETUP REQUEST message to the gNB-CU. If the gNB-CU succeeds to establish the multicast context, it replies to the gNB-DU with MULTICAST DISTRIBUTION SETUP RESPONSE.

The MULTICAST DISTRIBUTION SETUP REQUEST message shall contain F1-U TNL information for the MRBs accepted for the MBS Session by the gNB-DU and indicate in the *MBS Multicast F1-U Context Descriptor* IE, if the shared F1-U tunnel(s) for the MRB(s) are established on a per NR CGI or per MBS Area Session ID basis or for a ptp MRB leg.

Upon reception of the MULTICAST DISTRIBUTION SETUP REQUEST message the gNB-CU shall allocate F1-U resources and reply accordingly to the gNB-DU in the MULTICAST DISTRIBUTION SETUP RESPONSE message.

If the *MC F1-U Context usage* IE in the *MBS Multicast F1-U Context Descriptor* IE is set to "ptp forwarding" the gNB-CU shall, if supported, use the *MRB Progress Information* IE to determine at which PDCP SN to start transmitting multicast data to the gNB-DU.

8.14.10.3 Unsuccessful Operation

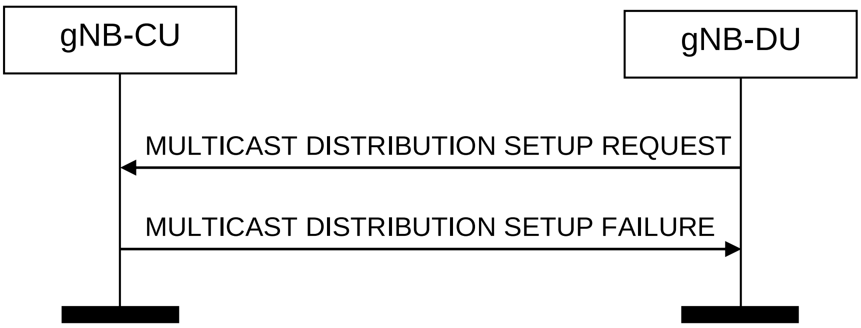

Figure 8.14.10.3-1: Multicast Distribution Setup procedure: unsuccessful Operation

If the gNB-CU is not able to provide the requested resources it shall consider the procedure as failed and reply with the MULTICAST CONTEXT SETUP FAILURE message.

8.14.10.4 Abnormal Conditions

Not applicable.

8.14.11 Multicast Distribution Release

8.14.11.1 General

The purpose of the Multicast Distribution Release procedure is to enable the gNB-DU to order the release of F1-U tunnels previously established using the Multicast Distribution Setup procedure.

The procedure uses MBS-associated signalling.

8.14.11.2 Successful Operation

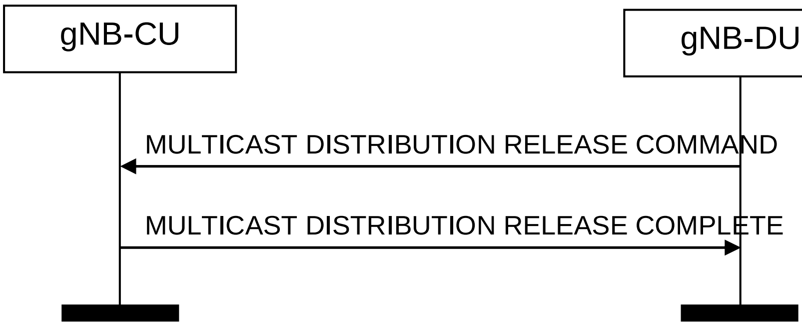

Figure 8.14.11.2-1: Multicast Distribution Release procedure. Successful operation

The gNB-DU initiates the procedure by sending the MULTICAST DISTRIBUTION RELEASE COMMAND message to the gNB-CU.

Upon reception of the MULTICAST DISTRIBUTION RELEASE COMMAND message, the gNB-CU shall release all signalling and user data transport resources associated with the context and reply with the MULTICAST DISTRIBUTION RELEASE COMPLETE message.

8.14.11.3 Unsuccessful Operation

Not applicable.

8.14.11.4 Abnormal Conditions

Not applicable.

8.14.12 Multicast Context Notification

8.14.12.1 General

The purpose of the Multicast Context Notification is to inform the gNB-CU about changes in the multicast context configuration.

The procedure uses MBS-associated signalling.

8.14.12.2 Successful Operation

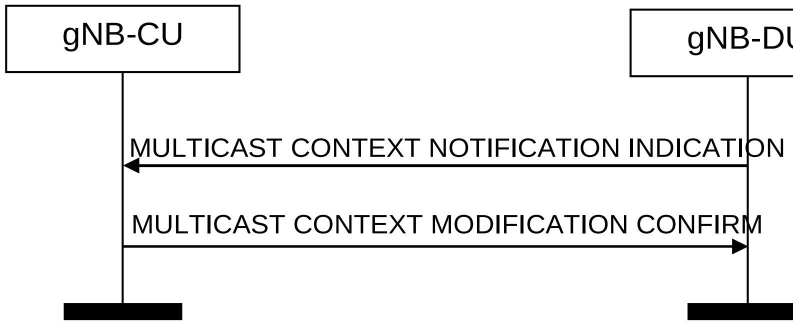

Figure 8.14.12.2-1: Multicast Context Notification. Successful operation

The gNB-DU initiates the procedure by sending the MULTICAST CONTEXT NOTIFICATION INDICATION message to the gNB-CU.

If the MULTICAST CONTEXT NOTIFICATION INDICATION message contains the *MBS Multicast Configuration Notification* IE within the *Multicast DU to CU RRC Information* IE, the gNB-CU shall replace, for the respective cell, the Multicast Configuration Information previously received with information received in the *MBS Multicast Configuration Notification* IE.

If the gNB-CU is able to execute the requested functions, the gNB-CU shall respond with the MULTICAST CONTEXT NOTIFICATION CONFIRM message to the gNB-DU.

8.14.12.3 Unsuccessful Operation

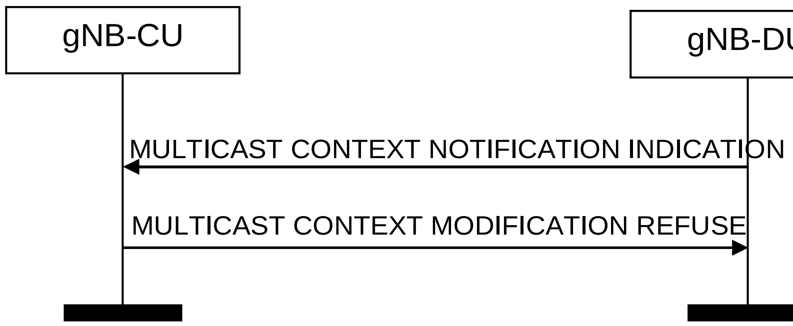

Figure 8.14.12.3-1: Multicast Context Notification. Unsuccessful operation

If the gNB-CU is not able to execute the requested functions, the gNB-CU shall respond with the MULTICAST CONTEXT NOTIFICATION REFUSE message to the gNB-DU.

8.14.12.4 Abnormal Conditions

Not applicable.

8.14.13 Multicast Common Configuration

8.14.13.1 General

The purpose of the Multicast Common Configuration procedure is to allow the gNB-CU to control the configuration of items common to all multicast contexts in the gNB-DU.

The procedure uses non UE-associated signalling.

8.14.13.2 Successful Operation

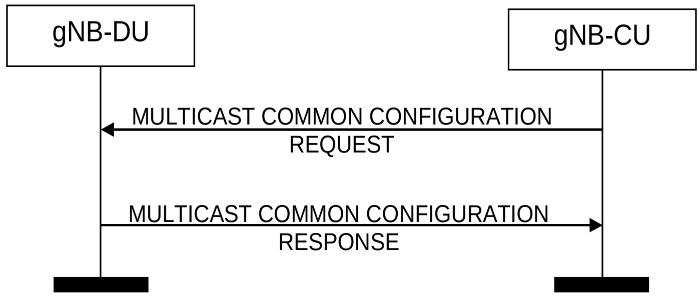

Figure 8.14.13.2-1: Multicast Common Configuration. Successful operation

The gNB-CU initiates the procedure by sending the MULTICAST COMMON CONFIGURATION REQUEST message to the gNB-DU.

If the *Multicast CU to DU Common RRC Information* IE is included in the MULTICAST COMMON CONFIGURATION REQUEST message and contains the *Multicast Common CU2DU Cell List* IE, the gNB-DU shall, if supported, use it to configure MBS session resources accordingly.

8.14.13.3 Unsuccessful Operation

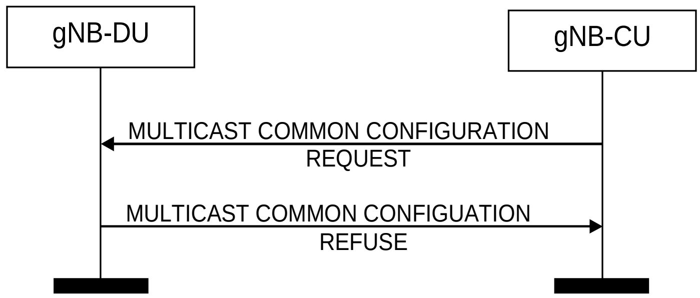

Figure 8.14.13.3-1: Multicast Common Configuration. Unsuccessful operation

If the gNB-DU is not able to execute the requested functions, the gNB-DU shall respond with the MULTICAST COMMON CONFIGURATION REFUSE message to the gNB-CU.

8.14.13.4 Abnormal Conditions

void.

8.14.14 Broadcast Transport Resource Request

8.14.14.1 General

The purpose of the Broadcast Transport Resource Request procedure is to request the gNB-CU to trigger the establishment of F1-U resources for the broadcast session.

The procedure uses MBS-associated signalling.

8.14.14.2 Successful Operation

Figure 8.14.14.2-1: Broadcast Transport Resource Request procedure. Successful operation

The gNB-DU initiates the procedure by sending the BROADCAST TRANSPORT RESOURCE REQUEST message to the gNB-CU.

Interaction with the Broadcast Context Modification procedure:

Upon reception of the BROADCAST TRANSPORT RESOURCE REQUEST message, the gNB-CU should trigger the Broadcast Context Modification procedure to establish the F1-U resources for the broadcast session. Besides, in case the *F1-U Path Failure indication* IE set to "true" is included in the BROADCAST TRANSPORT RESOURCE REQUEST message, the gNB-CU shall, if supported, consider that the previous established shared F1-U resources have been released in the gNB-DU and behave as specified in TS 38.401 [4].

8.14.14.3 Unsuccessful Operation

Not applicable.

8.14.14.4 Abnormal Conditions

Not applicable.

8.15 PDC Measurement Reporting procedures

8.15.1 PDC Measurement Initiation

8.15.1.1 General

The purpose of the PDC Measurement Initiation procedure is to enable the gNB-CU to request the gNB-DU to report measurements used for propagation delay compensation at the gNB-CU or UE. The procedure uses UE-associated signalling.

8.15.1.2 Successful Operation

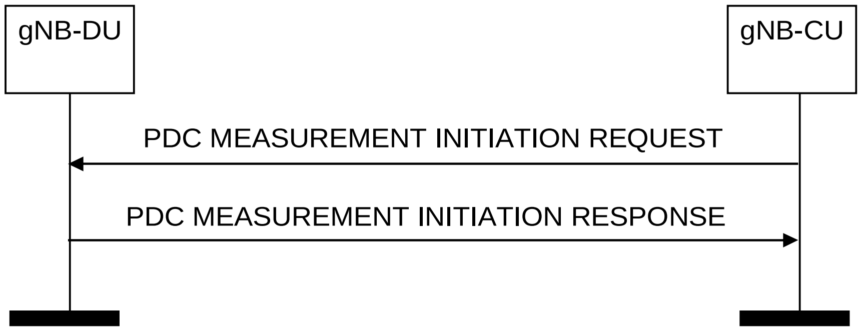

Figure 8.15.1.2-1: PDC Measurement Initiation procedure, successful operation

The gNB-CU initiates the procedure by sending a PDC MEASUREMENT INITIATION REQUEST message. If the gNB-DU is able to initiate the requested PDC measurements, it shall reply with the PDC MEASUREMENT INITIATION RESPONSE message.

If the *PDC Report Type* IE is set to "OnDemand", the gNB-DU shall return the result of the measurement in the PDC MEASUREMENT INITIATION RESPONSE message including the *PDC Measurement Result* IE, and the gNB-CU shall consider that the PDC measurements for the UE have been terminated by the gNB-DU.

Interaction with the PDC Measurement Report procedure:

If the *PDC Report Type* IE is set to "Periodic", the gNB-DU shall initiate the requested measurements and shall reply with the PDC MEASUREMENT INITIATION RESPONSE message without including the *PDC Measurement Result* IE in this message. The gNB-DU shall then periodically initiate the PDC Measurement Report procedure for the measurements, with the requested reporting periodicity.

8.15.1.3 Unsuccessful Operation

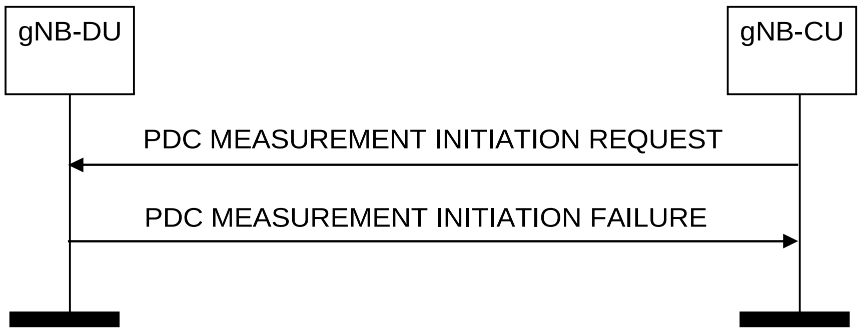

Figure 8.15.1.3-1: PDC Measurement Initiation procedure, unsuccessful operation

If the gNB-DU is not able to initiate at least one of the requested PDC measurements, the gNB-DU shall respond with a PDC MEASUREMENT INITIATION FAILURE message.

8.15.2 PDC Measurement Report

8.15.2.1 General

The purpose of the PDC Measurement Report procedure is for the gNB-DU to provide the PDC measurements for the UE to the gNB-CU. The procedure uses UE-associated signalling.

8.15.2.2 Successful Operation

Figure 8.15.2.2-1: PDC Measurement Report procedure, successful operation

The gNB-DU initiates the procedure by sending a PDC MEASUREMENT REPORT message. The PDC MEASUREMENT REPORT message contains the PDC measurement results according to the measurement configuration in the respective PDC MEASUREMENT INITIATION REQUEST message.

8.15.2.3 Unsuccessful Operation

Not applicable.

8.15.3 PDC Measurement Termination

8.15.3.1 General

The purpose of the PDC Measurement Termination procedure is to enable the gNB-CU to terminate an on-going periodical PDC measurement. The procedure uses UE-associated signalling.

8.15.3.2 Successful Operation

Figure 8.15.3.2-1: PDC Measurement Termination procedure: successful operation

The gNB-CU initiates the procedure by sending a PDC MEASUREMENT TERMINATION COMMAND message. Upon receiving this message, the gNB-DU shall terminate the ongoing PDC measurement and may release any resources previously allocated for the same measurement.

8.15.3.3 Unsuccessful Operation

Not applicable.

8.15.3.4 Abnormal Conditions

If the gNB-DU cannot identify the previously requested measurement to be terminated, it shall ignore the PDC MEASUREMENT TERMINATION COMMAND message.

8.15.4 PDC Measurement Failure Indication

8.15.4.1 General

The purpose of the PDC Measurement Failure Indication procedure is for the gNB-DU to notify the gNB-CU that the PDC measurements previously requested with the PDC Measurement Initiation procedure can no longer be reported. The procedure uses UE-associated signalling.

8.15.4.2 Successful Operation

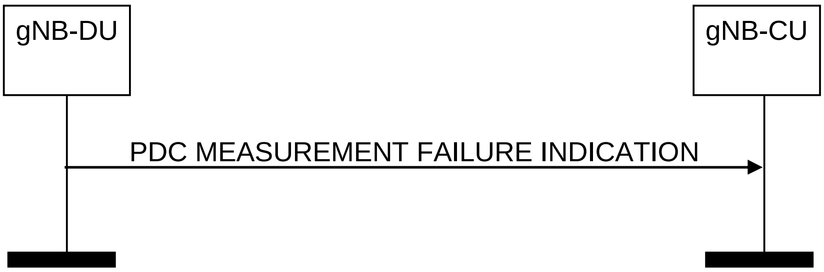

Figure 8.15.4.2-1: PDC Measurement Failure Indication procedure: successful operation

The gNB-DU initiates the procedure by sending a PDC MEASUREMENT FAILURE INDICATION message. Upon reception of the PDC MEASUREMENT FAILURE INDICATION message, the gNB-CU shall consider that the indicated PDC measurements have been terminated by the gNB-DU.

8.15.4.3 Unsuccessful Operation

Not applicable.

8.15.4.4 Abnormal Conditions

Void.

8.16 QMC Procedures

8.16.1 QoE Information Transfer

8.16.1.1 General

The purpose of the QoE Information Transfer procedure is to transfer RAN visible QoE information from the gNB-CU to the gNB-DU. The procedure uses UE-associated signalling.

8.16.1.2 Successful operation

Figure 8.16.1.2-1: QoE Information Transfer procedure

The gNB-CU initiates the procedure by sending the QOE INFORMATION TRANSFER message to the gNB-DU.

If the *QoE Information List* IE is included in QOE INFORMATION TRANSFER message, the gNB-DU may take it into account according to TS 38.300 [6].

8.16.1.3 Abnormal Conditions

Not applicable.

8.16.2 QoE Information Transfer Control

8.16.2.1 General

The purpose of the QoE Information Transfer Control procedure is to control the RAN visible QoE information transfer from the gNB-CU to the gNB-DU. The procedure uses non-UE associated signalling.

8.16.2.2 Successful operation

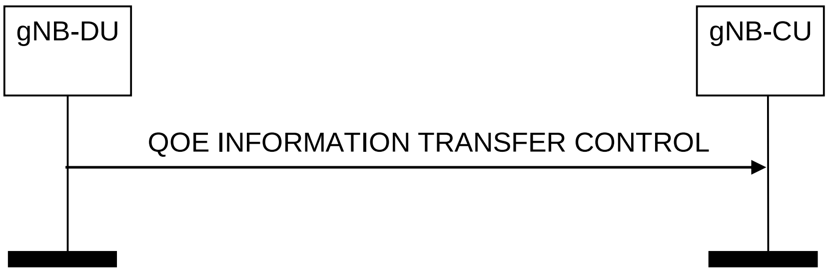

Figure 8.16.2.2-1: QoE Information Transfer Control procedure.

The gNB-DU initiates the procedure by sending the QOE INFORMATION TRANSFER CONTROL message to the gNB-CU.

If the *Deactivation Indication* IE is present in the message and set to 'Per UE', the gNB-CU shall, if supported, deactivate the QoE information transfer from gNB-CU to gNB-DU for the UEs indicated in the *Deactivation Indication List* IE.

If the *Deactivation Indication* IE is present in the message and set to 'Deactivate ALL', the gNB-CU shall, if supported, deactivate the QoE information transfer from the gNB-CU to the gNB-DU for all UEs served by the gNB-DU.

8.16.2.3 Abnormal Conditions

Not applicable.

8.17 Timing Synchronisation Status Reporting Procedures

8.17.1 Timing Synchronisation Status

8.17.1.1 General

The purpose of the Timing Synchronisation Status procedure is to enable the gNB-CU to request the gNB-DU to start or stop reporting of RAN timing synchronisation status information.

The procedure uses non-UE associated signalling.

8.17.1.2 Successful Operation

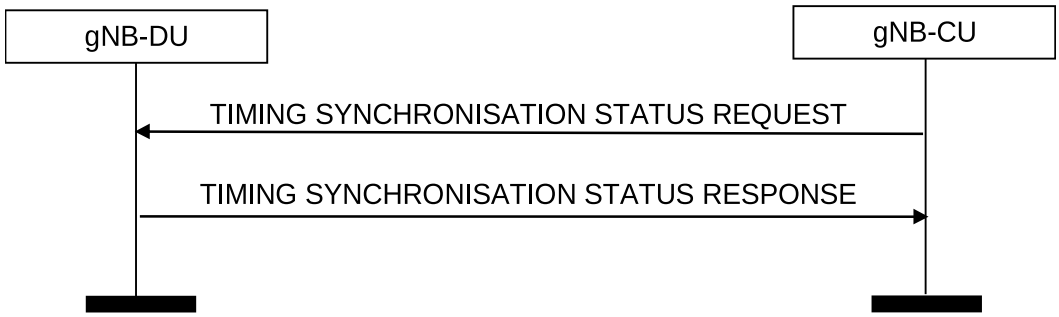

Figure 8.17.1.2-1: Timing synchronisation status procedure: successful operation

The gNB-CU initiates the procedure by sending a TIMING SYNCHRONISATION STATUS REQUEST message to the gNB-DU.

If the *RAN TSS Request Type* IE included in the TIMING SYNCHRONISATION STATUS REQUEST message is set to “start”, the gNB-DU shall start the reporting of RAN timing synchronization status information and reply with the TIMING SYNCHRONISATION STATUS RESPONSE message. If the *RAN TSS Request Type* IE is set to “stop”, the gNB-DU shall stop the reporting and reply with the TIMING SYNCHRONISATION STATUS RESPONSE message.

8.17.1.3 Unsuccessful Operation

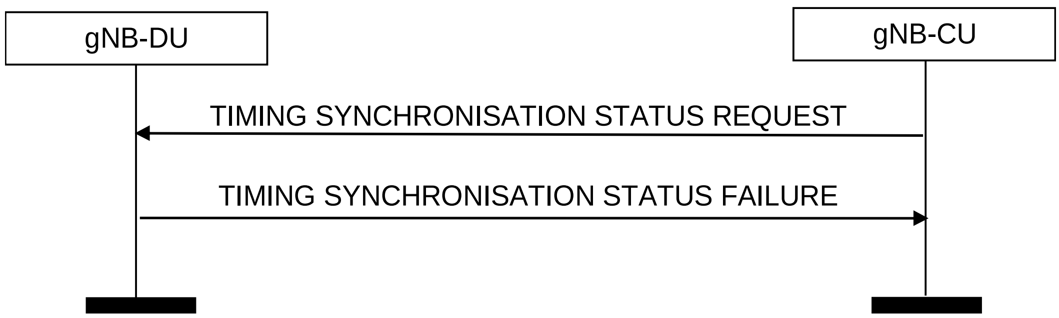

Figure 8.17.1.3-1: Timing synchronisation status procedure: unsuccessful operation

If the gNB-DU is not able to report timing synchronisation status, it shall consider the procedure as failed and reply with the TIMING SYNCHRONISATION STATUS FAILURE message.

8.17.1.4 Abnormal Conditions

Void.

8.17.2 Timing Synchronisation Status Report

8.17.2.1 General

The purpose of the Timing Synchronisation Status Report procedure is to enable the gNB-DU to provide RAN timing synchronisation status information to the gNB-CU.

The procedure uses non-UE associated signalling.

8.17.2.2 Successful Operation

Figure 8.17.2.2-1: Timing synchronisation status report

The gNB-DU initiates the procedure by sending a TIMING SYNCHRONISATION STATUS REPORT message to the gNB-CU.

8.17.2.3 Abnormal Conditions

Void.

9 Elements for F1AP Communication

9.1 General

Subclauses 9.2 and 9.3 present the F1AP message and IE definitions in tabular format. The corresponding ASN.1 definition is presented in subclause 9.4. In case there is contradiction between the tabular format and the ASN.1 definition, the ASN.1 shall take precedence, except for the definition of conditions for the presence of conditional IEs, where the tabular format shall take precedence.

The messages have been defined in accordance to the guidelines specified in TR 25.921 [14].

When specifying IEs which are to be represented by bitstrings, if not otherwise specifically stated in the semantics description of the concerned IE or elsewhere, the following principle applies with regards to the ordering of bits:

- The first bit (leftmost bit) contains the most significant bit (MSB);
- The last bit (rightmost bit) contains the least significant bit (LSB);
- When importing bitstrings from other specifications, the first bit of the bitstring contains the first bit of the concerned information;

The following attributes are used for the tabular description of the messages and information elements: Presence, Range Criticality and Assigned Criticality. Their definition and use can be found in TS 38.413 [3].

9.2 Message Functional Definition and Content

9.2.1 Interface Management messages

9.2.1.1 RESET

This message is sent by both the gNB-CU and the gNB-DU and is used to request that the F1 interface, or parts of the F1 interface, to be reset.

Direction: gNB-CU → gNB-DU and gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
CHOICE *Reset Type*	M				YES	reject
>F1 interface						
>>Reset All	M		ENUMERATED (Reset all,...)		-	
>Part of F1 interface						
>>UE-associated logical F1-connection list		*1*			-	
>>>UE-associated logical F1-connection Item		*1 .. <maxnoofIndividualF1ConnectionsToReset>*			EACH	reject
>>>>gNB-CU UE F1AP ID	O		9.3.1.4		-	
>>>>gNB-DU UE F1AP ID	O		9.3.1.5		-	

Range bound	Explanation
maxnoofIndividualF1ConnectionsToReset	Maximum no. of UE-associated logical F1-connections allowed to reset in one message. Value is 65536.

9.2.1.2 RESET ACKNOWLEDGE

This message is sent by both the gNB-CU and the gNB-DU as a response to a RESET message.

Direction: gNB-DU → gNB-CU and gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
UE-associated logical F1-connection list		*0..1*			YES	ignore
>UE-associated logical F1-connection Item		*1 .. <maxnoofIndividualF1ConnectionsToReset>*			EACH	ignore
>>gNB-CU UE F1AP ID	O		9.3.1.4		-	
>>gNB-DU UE F1AP ID	O		9.3.1.5		-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofIndividualF1ConnectionsToReset	Maximum no. of UE-associated logical F1-connections allowed to reset in one message. Value is 65536.

9.2.1.3 ERROR INDICATION

This message is sent by both the gNB-CU and the gNB-DU and is used to indicate that some error has been detected in the node.

Direction: gNB-CU → gNB-DU and gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23	This IE is ignored if received in UE associated signalling message.	YES	reject
gNB-CU UE F1AP ID	O		9.3.1.4		YES	ignore
gNB-DU UE F1AP ID	O		9.3.1.5		YES	ignore
Cause	O		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.1.4 F1 SETUP REQUEST

This message is sent by the gNB-DU to transfer information associated to an F1-C interface instance.

NOTE: If a TNL association is shared among several F1-C interface instances, several F1 Setup procedures are issued via the same TNL association after that TNL association has become operational.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
gNB-DU ID	M		9.3.1.9		YES	reject
gNB-DU Name	O		PrintableString(SIZE(1..150,...))		YES	ignore
gNB-DU Served Cells List		*0.. 1*		List of cells configured in the gNB-DU	YES	reject
>gNB-DU Served Cells Item		*1.. <maxCellingNBDU>*			EACH	reject
>>Served Cell Information	M		9.3.1.10	Information about the cells configured in the gNB-DU	-	
>>gNB-DU System Information	O		9.3.1.18	RRC container with system information owned by gNB-DU	-	
gNB-DU RRC version	M		RRC version 9.3.1.70		YES	reject
Transport Layer Address Info	O		9.3.2.5		YES	ignore
BAP Address	O		9.3.1.111	Indicates a BAP address assigned to the IAB-node.	YES	ignore
Extended gNB-DU Name	O		9.3.1.205		YES	ignore
RRC Terminating IAB-Donor gNB-ID	O		Global gNB ID 9.3.1.305	The Global gNB ID of a mobile IAB-node's RRC-terminating IAB donor. This IE is only present if the mobile IAB-node's RRC terminating IAB-donor-CU is different from the gNB-CU receiving	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				this message.		
Mobile IAB-MT User Location Information	O		9.3.1.307		YES	ignore

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.

9.2.1.5 F1 SETUP RESPONSE

This message is sent by the gNB-CU to transfer information associated to an F1-C interface instance.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
gNB-CU Name	O		PrintableString(SIZE(1..150,...))	Human readable name of the gNB-CU.	YES	ignore
Cells to be Activated List		*0.. 1*			YES	reject
>Cells to be Activated List Item		*1.. <maxCellingNBDU>*		List of cells to be activated	EACH	reject
>>NR CGI	M		9.3.1.12		-	
>>NR PCI	O		INTEGER (0..1007)	Physical Cell ID	-	
>>gNB-CU System Information	O		9.3.1.42	RRC container with system information owned by gNB-CU	YES	reject
>>Available PLMN List	O		9.3.1.65		YES	ignore
>>Extended Available PLMN List	O		9.3.1.76	This is included if *Available PLMN List* IE is included and if more than 6 Available PLMNs is to be signalled.	YES	ignore
>>IAB Info IAB-donor-CU	O		9.3.1.105	IAB-related configuration sent by the IAB-donor-CU.	YES	ignore
>>Available SNPN ID List	O		9.3.1.163	Indicates the available SNPN ID list. If this IE is included, the content of the *Available PLMN List* IE and *Extended Available PLMN List* IE if present in the *Cells to be Activated List Item* IE is ignored.	YES	ignore
>>MBS Broadcast Neighbour Cell List	O		9.3.1.226		YES	ignore
>>SSBs within the cell to be Activated List	O		9.3.1.326	This IE is not used in this version of the specification.	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
gNB-CU RRC version	M		RRC version 9.3.1.70		YES	reject
Transport Layer Address Info	O		9.3.2.5		YES	ignore
Uplink BH Non-UP Traffic Mapping	O		9.3.1.103		YES	reject
BAP Address	O		9.3.1.111	Indicates a BAP address assigned to the IAB-donor-DU.	YES	ignore
Extended gNB-CU Name	O		9.3.1.206		YES	ignore
NCGI to be Updated List		*0..1*			YES	reject
>NCGI to be Updated List Item		*1..<maxCellingNBDU>*		List of NCGIs to be updated.	EACH	reject
>>Old NCGI	M		NR CGI 9.3.1.12	Old NCGI of a cell served by the mobile IAB-DU	-	
>>New NCGI	M		NR CGI 9.3.1.12	New NCGI of a cell served by the mobile IAB-DU	-	

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.

9.2.1.6 F1 SETUP FAILURE

This message is sent by the gNB-CU to indicate F1 Setup failure.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to wait	O		9.3.1.13		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.1.7 GNB-DU CONFIGURATION UPDATE

This message is sent by the gNB-DU to transfer updated information associated to an F1-C interface instance.

NOTE: If F1-C signalling transport is shared among several F1-C interface instances, this message may transfer updated information associated to several F1-C interface instances.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Served Cells To Add List		*0..1*		Complete list of added cells served by the gNB-DU	YES	reject
>Served Cells To Add Item		*1 .. <maxCellingNBDU>*			EACH	reject
>>Served Cell	M		9.3.1.10	Information about	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Information				the cells configured in the gNB-DU		
>>gNB-DU System Information	O		9.3.1.18	RRC container with system information owned by gNB-DU	-	
Served Cells To Modify List		*0..1*		Complete list of modified cells served by the gNB-DU	YES	reject
>Served Cells To Modify Item		*1 .. <maxCellingNBDU>*			EACH	reject
>>Old NR CGI	M		NR CGI 9.3.1.12		-	
>>Served Cell Information	M		9.3.1.10	Information about the cells configured in the gNB-DU	-	
>>gNB-DU System Information	O		9.3.1.18	RRC container with system information owned by gNB-DU	-	
Served Cells To Delete List		*0..1*		Complete list of deleted cells served by the gNB-DU	YES	reject
>Served Cells To Delete Item		*1.. <maxCellingNBDU>*			EACH	reject
>>Old NR CGI	M		NR CGI 9.3.1.12		-	
Cells Status List		*0..1*		Complete list of active cells	YES	reject
>Cells Status Item		*0 .. <maxCellingNBDU>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	
>>Service Status	M		9.3.1.68		-	
Dedicated SI Delivery Needed UE List		*0..1*		List of UEs unable to receive system information from broadcast	YES	ignore
>Dedicated SI Delivery Needed UE Item		*1 .. <maxnoofUEIDs>*			EACH	ignore
>>gNB-CU UE F1AP ID	M		9.3.1.4		-	
>>NR CGI	M		9.3.1.12		-	
gNB-DU ID	O		9.3.1.9		YES	reject
gNB-DU TNL Association To Remove List		*0..1*			YES	reject
>gNB-DU TNL Association To Remove Item IEs		*1..<maxnoofTNLAssociation>*			EACH	reject
>>TNL Association Transport Layer Address	M		CP Transport Layer Information 9.3.2.4	Transport Layer Address of the gNB-DU.	-	
>>TNL Association Transport Layer Address gNB-CU	O		CP Transport Layer Information 9.3.2.4	Transport Layer Address of the gNB-CU	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Transport Layer Address Info	O		9.3.2.5		YES	ignore
Coverage Modification Notification	O		9.3.1.213		YES	ignore
gNB-DU Name	O		PrintableString(SIZE(1..150,...))	Human readable name of the gNB-DU.	YES	ignore
Extended gNB-DU Name	O		9.3.1.205		YES	ignore
RRC Terminating IAB-Donor Related Info	O		9.3.1.306	Indicates the information related to a mobile IAB-node's RRC-terminating IAB-donor.	YES	reject
Mobile IAB-MT User Location Information	O		9.3.1.307		YES	ignore
Future Coverage Modification Notification	O		9.3.1.368		YES	ignore

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.
maxnoofUEIDs	Maximum no. of UEs that can be served by a gNB-DU. Value is 65536.
maxnoofTNLAssociations	Maximum numbers of TNL Associations between the gNB-CU and the gNB-DU. Value is 32.

9.2.1.8 GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE

This message is sent by a gNB-CU to a gNB-DU to acknowledge update of information associated to an F1-C interface instance.

NOTE: If F1-C signalling transport is shared among several F1-C interface instances, this message may transfer updated information associated to several F1-C interface instances.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cells to be Activated List		*0.. 1*		List of cells to be activated	YES	reject
>Cells to be Activated List Item		*1.. <maxCellingNBDU>*			EACH	reject
>> NR CGI	M		9.3.1.12		-	
>> NR PCI	O		INTEGER (0..1007)	Physical Cell ID	-	
>> gNB-CU System Information	O		9.3.1.42	RRC container with system information owned by gNB-CU	YES	reject
>>Available PLMN List	O		9.3.1.65		YES	ignore
>>Extended Available PLMN List	O		9.3.1.76	This is included if *Available PLMN List* IE is included and if more than 6 Available PLMNs is to be signalled.	YES	ignore
>>IAB Info IAB-	O		9.3.1.105	IAB-related	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
donor-CU				configuration sent by the IAB-donor-CU.		
>>Available SNPN ID List	O		9.3.1.163	Indicates the available SNPN ID list. If this IE is included, the content of the *Available PLMN List* IE and *Extended Available PLMN List* IE if present in the *Cells to be Activated List Item* IE is ignored.	YES	ignore
>>MBS Broadcast Neighbour Cell List	O		9.3.1.226		YES	ignore
>>SSBs within the cell to be Activated List	O		9.3.1.326	This IE is not used in this version of the specification.	YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Cells to be Deactivated List		*0.. 1*		List of cells to be deactivated	YES	reject
>Cells to be Deactivated List Item		*1.. <maxCellingNBDU>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	
Transport Layer Address Info	O		9.3.2.5		YES	ignore
Uplink BH Non-UP Traffic Mapping	O		9.3.1.103		YES	reject
BAP Address	O		9.3.1.111	Indicates a BAP address assigned to the IAB-donor-DU.	YES	ignore
Cells for SON List	O		9.3.1.214		YES	ignore

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.

9.2.1.9 GNB-DU CONFIGURATION UPDATE FAILURE

This message is sent by the gNB-CU to indicate gNB-DU Configuration Update failure.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to wait	O		9.3.1.13		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.1.10 GNB-CU CONFIGURATION UPDATE

This message is sent by the gNB-CU to transfer updated information associated to an F1-C interface instance.

NOTE: If F1-C signalling transport is shared among several F1-C interface instances, this message may transfer updated information associated to several F1-C interface instances.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cells to be Activated List		*0..1*		List of cells to be activated or modified	YES	reject
>Cells to be Activated List Item		*1..<maxCellingNBDU>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	
>>NR PCI	O		INTEGER (0..1007)	Physical Cell ID	-	
>>gNB-CU System Information	O		9.3.1.42	RRC container with system information owned by gNB-CU	YES	reject
>>Available PLMN List	O		9.3.1.65		YES	ignore
>>Extended Available PLMN List	O		9.3.1.76	This is included if *Available PLMN List* IE is included and if more than 6 Available PLMNs is to be signalled.	YES	ignore
>>IAB Info IAB-donor-CU	O		9.3.1.105	IAB-related configuration sent by the IAB-donor-CU.	YES	ignore
>>Available SNPN ID List	O		9.3.1.163	Indicates the available SNPN ID list. If this IE is included, the content of the *Available PLMN List* IE and *Extended Available PLMN List* IE if present in the *Cells to be Activated List Item* IE is ignored.	YES	ignore
>>MBS Broadcast Neighbour Cell List	O		9.3.1.226		YES	ignore
>>SSBs within the cell to be Activated List	O		9.3.1.326	List of SSB beams within the cell requested to be activated.	YES	reject
Cells to be Deactivated List		*0..1*		List of cells to be deactivated	YES	reject
>Cells to be Deactivated List Item		*1..<maxCellingNBDU>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	
gNB-CU TNL Association To Add List		*0..1*			YES	ignore
>gNB-CU TNL Association To Add Item IEs		*1..<maxnoofTNLAssociations>*			EACH	ignore
>>TNL Association Transport Layer	M		CP Transport Layer	Transport Layer Address of the	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Information			Information 9.3.2.4	gNB-CU.		
>>TNL Association Usage	M		ENUMERATED (ue, non-ue, both, ...)	Indicates whether the TNL association is only used for UE-associated signalling, or non-UE-associated signalling, or both. For usage of this IE, refer to TS 38.472 [22].	-	
gNB-CU TNL Association To Remove List		*0..1*			YES	ignore
>gNB-CU TNL Association To Remove Item IEs		*1..<maxnoofTNLAssociation>*			EACH	ignore
>>TNL Association Transport Layer Address	M		CP Transport Layer Information 9.3.2.4	Transport Layer Address of the gNB-CU.	-	
>>TNL Association Transport Layer Address gNB-DU	O		CP Transport Layer Information 9.3.2.4	Transport Layer Address of the gNB-DU.	YES	reject
gNB-CU TNL Association To Update List		*0..1*			YES	ignore
>gNB-CU TNL Association To Update Item IEs		*1..<maxnoofTNLAssociations>*			EACH	ignore
>>TNL Association Transport Layer Address	M		CP Transport Layer Information 9.3.2.4	Transport Layer Address of the gNB-CU.	-	
>>TNL Association Usage	O		ENUMERATED (ue, non-ue, both, ...)	Indicates whether the TNL association is only used for UE-associated signalling, or non-UE-associated signalling, or both. For usage of this IE, refer to TS 38.472 [22].	-	
Cells to be barred List		*0..1*		List of cells to be barred.	YES	ignore
>Cells to be barred List Item		*1.. <maxCellingNBDU>*			EACH	ignore
>>NR CGI	M		9.3.1.12		-	
>>Cell Barred	M		ENUMERATED (barred, not-barred, ...)		-	
>>IAB Barred	O		ENUMERATED (barred, not-barred, ...)	Corresponds to information provided in the *iab-Support* contained in the *PLMN-IdentityInfo* IE or contained in the *NPN-*	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				IdentityInfo IE as defined in TS 38.331 [8]. The codepoint value "barred" indicates that the *iab-Support* is not sent in SIB1, and the codepoint value "not-barred" indicates that the *iab-Support* is sent in SIB1.		
>>Mobile IAB Barred	O		ENUMERATED (barred, not-barred, ...)	Corresponds to information provided in the *mobileIAB-Support* contained in the *PLMN-IdentityInfo* IE or contained in the *NPN-IdentityInfo* IE as defined in TS 38.331 [8]. The codepoint value "barred" indicates that the *mobileIAB-Support* is not sent in SIB1, and the codepoint value "not-barred" indicates that the *mobileIAB-Support* is sent in SIB1.	-	
Protected E-UTRA Resources List		*0..1*		List of Protected E-UTRA Resources.	YES	reject
>Protected E-UTRA Resources List Item		*1.. <maxCellineNB>*			EACH	reject
>>Spectrum Sharing Group ID	M		INTEGER (1.. maxCellineNB)	Indicates the E-UTRA cells involved in resource coordination with the NR cells affiliated with the same Spectrum Sharing Group ID.	-	
>>E-UTRA Cells List		*1*		List of applicable E-UTRA cells.	-	
>>>E-UTRA Cells List Item		*1 .. <maxCellineNB>*			-	
>>>>EUTRA Cell ID	M		BIT STRING (SIZE(28))	Indicates the E-UTRAN Cell Identifier IE contained in the ECGI as defined in subclause 9.2.14 in TS 36.423 [9].	-	
>>>>Served E-UTRA Cell Information	M		9.3.1.64		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Neighbour Cell Information List		*0..1*			YES	ignore
>Neighbour Cell Information List Item		*1 .. <maxCellingNBDU>*			EACH	ignore
>>NR CGI	M		9.3.1.12		-	
>>Intended TDD DL-UL Configuration	O		9.3.1.89		-	
>>SBFD Frequency Configuration	O		OCTET STRING	Includes the *SBFD-Subband-Allocation-r19* IE, as defined in TS 38.331[8].	YES	ignore
>>SSB Resource Configuration	O		OCTET STRING	Includes the *MeasTiming* contained in the MeasurementTimingConfiguration message as defined in TS 38.331 [8].	YES	ignore
>>NZP CSI-RS Resources Configuration	O		9.3.1.356		YES	ignore
>>SRS Resource Configuration	O		9.3.1.357		YES	ignore
Transport Layer Address Info	O		9.3.2.5		YES	ignore
Uplink BH Non-UP Traffic Mapping	O		9.3.1.103		YES	reject
BAP Address	O		9.3.1.111	Indicates a BAP address assigned to the IAB-donor-DU.	YES	ignore
CCO Assistance Information	O		9.3.1.211	Indicates CCO Assistance Information for cells and beams served by the gNB-DU of the same NG-RAN node or for cells and beams not served by the gNB-DU.	YES	ignore
Cells for SON List	O		9.3.1.214		YES	ignore
gNB-CU Name	O		PrintableString(SIZE(1..150,...))	Human readable name of the gNB-CU.	YES	ignore
Extended gNB-CU Name	O		9.3.1.206		YES	ignore
Cells Allowed to be Deactivated List		*0..1*			YES	ignore
>**Cells Allowed to be Deactivated List Item**		*1 .. <maxCellingNBDU>*			EACH	ignore
>>NR CGI	M		9.3.1.12		-	
On-demand SIB1 Cell		*0..1*			YES	ignore
>NR CGI	M		9.3.1.12		-	
>On-demand SIB1 Indicator	M		ENUMERATED (start, stop, ...)		-	
Predicted CCO Assistance Information	O		9.3.1.367	Indicates predicted CCO Assistance Information for cells and beams	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				served by the gNB-DU of the same NG-RAN node.		
Neighbour Future Coverage Modification Notification	O		9.3.1.369	Indicates Future Coverage Modifications for cells and beams not served by the gNB-DU.	YES	ignore

Range bound	Explanation
maxCellingNBDU	Maximum numbers of cells that can be served by a gNB-DU. Value is 512.
maxnoofTNLAssociations	Maximum numbers of TNL Associations between the gNB-CU and the gNB-DU. Value is 32.
maxCellineNB	Maximum no. cells that can be served by an eNB. Value is 256.
maxnoofSSBAreas	Maximum no. SSB Areas that can be served by a cell. Value is 64.

9.2.1.11 GNB-CU CONFIGURATION UPDATE ACKNOWLEDGE

This message is sent by a gNB-DU to a gNB-CU to acknowledge update of information associated to an F1-C interface instance.

NOTE: If F1-C signalling transport is shared among several F1-C interface instance, this message may transfer updated information associated to several F1-C interface instances.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cells Failed to be Activated List		*0..1*		List of cells which are failed to be activated	YES	reject
>Cells Failed to be Activated Item		*1..<maxCellingNBDU>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	
>>Cause	M		9.3.1.2		-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore
gNB-CU TNL Association Setup List		*0..1*			YES	ignore
>gNB-CU TNL Association Setup Item IEs		*1..<maxnoofTNLAssociations>*			EACH	ignore
>>TNL Association Transport Layer Address	M		CP Transport Layer Information 9.3.2.4	Transport Layer Address of the gNB-CU	-	
gNB-CU TNL Association Failed to Setup List		*0..1*			YES	ignore
>gNB-CU TNL Association Failed To Setup Item IEs		*1..<maxnoofTNLAssociations>*			EACH	ignore
>>TNL Association Transport Layer Address	M		CP Transport Layer Information 9.3.2.4	Transport Layer Address of the gNB-CU	-	
>>Cause	M		9.3.1.2		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Dedicated SI Delivery Needed UE List		*0..1*		List of UEs unable to receive system information from broadcast	YES	ignore
>Dedicated SI Delivery Needed UE List		*1 .. <maxnoofUEIDs>*			EACH	ignore
>>gNB-CU UE F1AP ID	M		9.3.1.4		-	
>>NR CGI	M		9.3.1.12		-	
Transport Layer Address Info	O		9.3.2.5		YES	ignore
Cells with SSBs Activated List		*0..1*			YES	ignore
>Cells with SSBs Activated List Item		*1.. <maxCellingNBDU>*			-	
>>NR CGI	M		9.3.1.12		-	
>>SSBs activated List		*1 .. < maxnoofSSBAreas >*			-	
>>>SSB Index	M		INTEGER (0..63)	Identifier of the SSB beam activated.	-	

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.
maxnoofTNLAssociations	Maximum no. of TNL Associations between the gNB-CU and the gNB-DU. Value is 32.
maxnoofUEIDs	Maximum no. of UEs that can be served by a gNB-DU. Value is 65536.
maxnoofSSBAreas	Maximum no. SSB Areas that can be served by a cell. Value is 64.

9.2.1.12 GNB-CU CONFIGURATION UPDATE FAILURE

This message is sent by the gNB-DU to indicate gNB-CU Configuration Update failure.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to wait	O		9.3.1.13		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.1.13 GNB-DU RESOURCE COORDINATION REQUEST

This message is sent by a gNB-CU to a gNB-DU, to express the desired resource allocation for data traffic, for the sake of resource coordination. The message triggers gNB-DU resource coordination (for NR-initiated resource coordination), to indicate an initial resource offer by the E-UTRA node (for E-UTRA-initiated gNB-DU Resource Coordination), or to indicate the agreed resource allocation that is to be executed.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Request type	M		ENUMERATED (offer, execution, ...)		YES	reject
E-UTRA – NR Cell Resource Coordination Request Container	M		OCTET STRING	In EN-DC case, includes the X2AP E-UTRA – NR CELL RESOURCE COORDINATION REQUEST message as defined in subclause 9.1.4.24 in TS 36.423 [9]. In NG-RAN cases, includes the XnAP E-UTRA – NR CELL RESOURCE COORDINATION REQUEST message as defined in subclause 9.1.2.23 in TS 38.423 [28].	YES	reject
Ignore Coordination Request Container	O		ENUMERATED (yes, ...)		YES	reject

9.2.1.14 GNB-DU RESOURCE COORDINATION RESPONSE

This message is sent by a gNB-DU to a gNB-CU, to express the desired resource allocation for data traffic, as a response to the GNB-DU RESOURCE COORDINATION REQUEST.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
E-UTRA – NR Cell Resource Coordination Response Container	M		OCTET STRING	In EN-DC case, includes the X2AP E-UTRA – NR CELL RESOURCE COORDINATION RESPONSE message as defined in subclause 9.1.4.25 in TS 36.423 [9]. In NG-RAN cases, includes the XnAP E-UTRA – NR CELL RESOURCE COORDINATION RESPONSE message as defined in subclause 9.1.2.24 in TS 38.423 [28].	YES	reject

9.2.1.15 GNB-DU STATUS INDICATION

This message is sent by the gNB-DU to indicate to the gNB-CU its status of overload.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
gNB-DU Overload Information	M		ENUMERATED (overloaded, not-overloaded)		YES	reject
IAB Congestion Indication	O		9.3.1.227		YES	ignore

9.2.1.16 F1 REMOVAL REQUEST

This message is sent by either the gNB-DU or the gNB-CU to intiate the removal of the interface instance and the related resources.

Direction: gNB-DU → gNB-CU, gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject

9.2.1.17 F1 REMOVAL RESPONSE

This message is sent by either the gNB-DU or the gNB-CU to acknowledge the initiation of removal of the interface instance and the related resources.

Direction: gNB-CU → gNB-DU, gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.1.18 F1 REMOVAL FAILURE

This message is sent by either the gNB-DU or the gNB-CU to indicate that removing the interface instance and the related resources cannot be accepted.

Direction: gNB-CU → gNB-DU, gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.1.19 NETWORK ACCESS RATE REDUCTION

This message is sent by the gNB-CU to indicate to the gNB-DU a need to reduce the rate at which UEs access the network.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
UAC Assistance Information	M		9.3.1.83		YES	reject

9.2.1.20 RESOURCE STATUS REQUEST

This message is sent by gNB-CU to gNB-DU to initiate the requested measurement according to the parameters given in the message.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
gNB-CU Measurement ID	M		INTEGER (0..4095,...)	Allocated by gNB-CU	YES	reject
gNB-DU Measurement ID	C-ifRegistrationRequestStoporAdd		INTEGER (0..4095,...)	Allocated by gNB-DU	YES	ignore
Registration Request	M		ENUMERATED (start, stop, add, ...)	Type of request for which the resource status is required.	YES	ignore
Report Characteristics	C-ifRegistrationRequestStart		BIT STRING (SIZE(32))	Each position in the bitmap indicates measurement object the gNB-DU is requested to report. First Bit = PRB Periodic, Second Bit = TNL Capacity Ind Periodic, Third Bit = Composite Available Capacity Periodic, Fourth Bit = HW LoadInd Periodic, Fifth Bit = Number of Active UEs Periodic, Sixth Bit = NR-U Channel List Periodic, Seventh Bit = Energy Cost. Other bits shall be ignored by the gNB-DU.	YES	ignore
Cell To Report List		*0..1*		Cell ID list to which the request applies.	YES	ignore
>Cell To Report Item		*1 .. <maxCellingNBDU>*			-	
>>Cell ID	M		NR CGI 9.3.1.12		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>SSB To Report List		*0..1*		SSB list to which the request applies.	-	
>>>SSB To Report Item		*1 .. < maxnoofSSBAreas>*			-	
>>>>SSB index	M		INTEGER (0..63)		-	
>>Slice To Report List		*0..1*		S-NSSAI list to which the request applies.	-	
>>>Slice To Report Item		*1..< maxnoofBPLMNsNR >*			-	
>>>>PLMN Identity	M		9.3.1.14	Broadcast PLMN	-	
>>>>S-NSSAI List		*1*			-	
>>>>>S-NSSAI Item		*1 .. < maxnoofSliceItems>*			-	
>>>>>>S-NSSAI	M		9.3.1.38		-	
Reporting Periodicity	O		ENUMERATED (500ms, 1000ms, 2000ms, 5000ms,10000 ms, ...)	Periodicity that can be used for reporting of indicated measurements. Also used as the averaging window length for all measurement object if supported. This IE is ignored if the *Registration Request* IE is set to "add".	YES	ignore

Condition	Explanation
ifRegistrationRequestStoporAdd	This IE shall be present if the *Registration Request* IE is set to the value "stop" or "add".
ifRegistrationRequestStart	This IE shall be present if the Registration Request IE is set to the value "start".

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.
maxnoofSSBAreas	Maximum no. SSB Areas that can be served by a gNB node cell. Value is 64.
maxnoofSliceItems	Maximum no. of signalled slice support items. Value is 1024.
maxnoofBPLMNsNR	Maximum no. of PLMN Ids.broadcast in a cell. Value is 12.

9.2.1.21 RESOURCE STATUS RESPONSE

This message is sent by gNB-DU to gNB-CU to indicate that the requested measurement is successfully initiated.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject

Transaction ID	M		9.3.1.23		YES	reject
gNB-CU Measurement ID	M		INTEGER (0..4095,...)	Allocated by gNB-CU	YES	reject
gNB-DU Measurement ID	M		INTEGER (0..4095,...)	Allocated by gNB-DU	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.1.22 RESOURCE STATUS FAILURE

This message is sent by gNB-DU to gNB-CU to indicate that for any of the requested measurement objects the measurement cannot be initiated.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
gNB-CU Measurement ID	M		INTEGER (0..4095,...)	Allocated by gNB-CU	YES	reject
gNB-DU Measurement ID	M		INTEGER (0..4095,...)	Allocated by gNB-DU	YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.1.23 RESOURCE STATUS UPDATE

This message is sent by gNB-DU to gNB-CU to report the results of the requested measurements.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
gNB-CU Measurement ID	M		INTEGER (0..4095,...)	Allocated by gNB-CU	YES	reject
gNB-DU Measurement ID	M		INTEGER (0..4095,...)	Allocated by gNB-DU	YES	ignore
Hardware Load Indicator	O		9.3.1.136		YES	ignore
TNL Capacity Indicator	O		9.3.1.128		YES	ignore
Cell Measurement Result		*0..1*			YES	ignore
>Cell Measurement Result Item		*1 .. <maxCellingNBDU>*			-	
>>Cell ID	M		NR CGI 9.3.1.12		-	
>>Radio Resource Status	O		9.3.1.129		-	
>>Composite Available Capacity Group	O		9.3.1.130		-	
>>Slice Available Capacity	O		9.3.1.134		-	
>>Number of Active UEs	O		9.3.1.135		-	
>>NR-U Channel List		*0..1*			YES	ignore
>>>NR-U Channel Item		*1..<maxnoofNR-*			-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
		UChannelIDs>				
>>>>NR-U Channel ID	M		INTEGER (1.. *maxnoofNR-UChannelIDs*)	Identifies a portion of the NR-U Channel Bandwidth on which channel access procedure in shared spectrum has been performed in the last reporting period.	-	
>>>>Channel Occupancy Time Percentage DL	M		INTEGER (0..100, ...)	The percentage of time for which the channel resources have been utilised for DL traffic served by the corresponding NR-U Channel of the serving cell. Value 100 indicates that the channel resources have been utilized for DL traffic served by the corresponding NR-U Channel of the serving cell for the whole duration between consecutive reporting.	-	
>>>>Energy Detection Threshold DL	M		INTEGER (-100..-50,...)	Average ED Threshold used for DL channel sensing at the gNB. Value is in dBm.	-	
>>>>Channel Occupancy Time Percentage UL	O		INTEGER (0..100, ...)	The percentage of time for which the channel resources have been utilised for UL traffic served by the corresponding NR-U Channel of the serving cell for UEs that transmit to the serving cell. Value 100 indicates that the channel resources have been utilized for UL traffic served by the corresponding NR-U Channel of the serving cell for the whole duration between consecutive reporting.	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>>>Radio Resource Status NR-U	O		9.3.1.295	Indicates the radio resource status per NR-U channel.	YES	ignore
Node Associated Info Result		*0..1*			YES	ignore
>Energy Cost	O		INTEGER (0..10000,...)	The node level Energy Consumption index measured at the gNB-DU. Value 0 indicates the minimum measured Energy Consumption and 10000 indicates the maximum measured Energy Consumption.	-	

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.
maxnoofNR-UChannelIDs	Maximum no. NR-U Channel IDs in a cell. Value is 16.

9.2.1.24 DU-CU TA INFORMATION TRANSFER

This message is sent by the gNB-DU to inform the gNB-CU about TA information.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
DU to CU TA Information List		*1*			YES	ignore
>DU to CU TA Information Item IEs		*1 .. <maxnoofTAList>*			EACH	ignore
>>Candidate Cell ID	M		NR CGI 9.3.1.12		-	
>>TA Value	M		INTEGER (0..4095)	Indicates the TA value as defined in TS 38.213 [31].	-	
>>Preamble Index	M		INTEGER (0..63)		-	
>>RA-RNTI	M		INTEGER (0..65535, ...)	RA-RNTI as defined in TS 38.321 [16].	-	
>>Source gNB-DU ID	M		gNB-DU ID 9.3.1.9		-	
>>Tag ID Pointer	O		OCTET STRING	Includes the *tag-Id-ptr* contained in the *TCI-UL-State* IE or the *TCI-State* IE, as defined in TS 38.331 [8].	-	
>>LTM gNB ID	O		Global gNB ID 9.3.1.305		YES	ignore

Range bound	Explanation
maxnoofTAList	Maximum no. of TA values to be sent, the maximum value is 8.

9.2.1.25 CU-DU TA INFORMATION TRANSFER

This message is sent by the gNB-CU to inform the gNB-DU about TA information.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
CU to DU TA Information List		*1*			YES	ignore
>CU to DU TA Information Item IEs		*1 .. <maxnoofTAList>*			EACH	ignore
>>Candidate Cell ID	M		NR CGI 9.3.1.12		-	
>>TA Value	M		INTEGER (0..4095)	Indicates the TA value as defined in TS 38.213 [31].	-	
>>Preamble Index	M		INTEGER (0..63)		-	
>>RA-RNTI	M		INTEGER (0..65535, ...)	RA-RNTI as defined in TS 38.321 [16].	-	
>>Tag ID Pointer	O		OCTET STRING	Includes the *tag-Id-ptr* contained in the *TCI-UL-State* IE or the *TCI-State* IE, as defined in TS 38.331 [8].	-	

Range bound	Explanation
maxnoofTAList	Maximum no. of TA values to be sent, the maximum value is 8.

9.2.1.26 RACH INDICATION

This message is sent by the gNB-DU to inform the gNB-CU about one or more random access procedures performed at the gNB-DU.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
RA Report Indication List		*1*			YES	reject
>RA Report Indication List Item		*1..<maxnoofUEsforRAReportIndications>*			-	
>>gNB-CU UE F1AP ID	M		9.3.1.4		-	

Range bound	Explanation
maxnoofUEsforRAReportIndications	Maximum number of UEs from which gNB-DU is interested to collect RA report. Value is 64.

9.2.2 UE Context Management messages

9.2.2.1 UE CONTEXT SETUP REQUEST

This message is sent by the gNB-CU to request the setup of a UE context.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	O		9.3.1.5		YES	ignore
SpCell ID	M		NR CGI 9.3.1.12	Special Cell as defined in TS 38.321 [16]. For handover case, this IE is considered as target cell.	YES	reject
ServCellIndex	M		INTEGER (0..31,...)		YES	reject
SpCell UL Configured	O		Cell UL Configured 9.3.1.33		YES	ignore
CU to DU RRC Information	M		9.3.1.25		YES	reject
Candidate SpCell List		*0..1*			YES	ignore
>Candidate SpCell Item IEs		*1 .. <maxnoofCandidateSpCells>*			EACH	ignore
>>Candidate SpCell ID	M		NR CGI 9.3.1.12	Special Cell as defined in TS 38.321 [16]	-	
DRX Cycle	O		9.3.1.24		YES	ignore
Resource Coordination Transfer Container	O		OCTET STRING	Includes the *MeNB Resource Coordination Information* IE as defined in subclause 9.2.116 of TS 36.423 [9] for EN-DC case or *MR-DC Resource Coordination Information* IE as defined in TS 38.423 [28] for NGEN-DC and NE-DC cases.	YES	ignore
SCell To Be Setup List		*0..1*			YES	ignore
>SCell to Be Setup Item IEs		*1.. <maxnoofSCells>*			EACH	ignore
>>SCell ID	M		NR CGI 9.3.1.12	SCell Identifier in gNB	-	
>>SCellIndex	M		INTEGER (1..31, ...)		-	
>>SCell UL Configured	O		Cell UL Configured 9.3.1.33		-	
>>servingCellMO	O		INTEGER (1..64, ...)		YES	ignore
>>servingCellMO-On-demand	O		INTEGER (1..64, ...)		YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
SRB to Be Setup List		*0..1*			YES	reject
>SRB to Be Setup Item IEs		*1 .. <maxnoofSRBs>*			EACH	reject
>>SRB ID	M		9.3.1.7		-	
>>Duplication Indication	O		ENUMERATED (true, ..., false)	If included, it should be set to true. This IE is ignored if the *Additional Duplication Indication* IE is present.	-	
>>Additional Duplication Indication	O		ENUMERATED (three, four, ...)		YES	ignore
>>SDT RLC Bearer Configuration	O		OCTET STRING	Includes the *RLC-BearerConfig* IE defined in subclause 6.3.2 of TS 38.331 [8]	YES	ignore
>>SRB Mapping Info	O		Uu RLC Channel ID 9.3.1.266	This IE contains the mapped Uu Relay RLC CH ID for the SRB	YES	ignore
DRB to Be Setup List		*0..1*			YES	reject
>DRB to Be Setup Item IEs		*1 .. <maxnoofDRBs>*			EACH	reject
>>DRB ID	M		9.3.1.8		-	
>>CHOICE *QoS Information*	M				-	
>>>E-UTRAN QoS						
>>>>E-UTRAN QoS	M		9.3.1.19	Shall be used for EN-DC case to convey E-RAB Level QoS Parameters	-	
>>>DRB Information						
>>>>DRB Information		*1*		Shall be used for NG-RAN cases	YES	ignore
>>>>>DRB QoS	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>>>>S-NSSAI	M		9.3.1.38		-	
>>>>>Notification Control	O		9.3.1.56		-	
>>>>>Flows Mapped to DRB Item		*1 .. <maxnoofQoSFlows>*			-	
>>>>>>QoS Flow Identifier	M		9.3.1.63		-	
>>>>>>QoS Flow Level QoS Parameters	M		9.3.1.45		-	
>>>>>>QoS Flow Mapping Indication	O		9.3.1.72		YES	ignore
>>>>>>TSC Traffic Characteristics	O		9.3.1.141	Traffic pattern information associated with the QFI. Details in	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				TS 23.501 [21].		
>>>>>ECN Marking or Congestion Information Reporting Request	O		9.3.1.321		YES	ignore
>>>>>PSI based SDU Discard UL	O		ENUMERATED (start, stop, ...)	Indicates whether UL PSI based SDU discard is (re)configured or released for the DRB. The codepoint “start” means that UL PSI based discarding is (re)configured, while the codepoint “stop” means that UL PSI based discarding is released. Up to 8 DRBs can be set as “start”.	YES	ignore
>>>>>PSI based SDU Discard DL	O		ENUMERATED (configured, not-configured, ...)	Indicates whether DL PSI based SDU discard is configured or not for the DRB.	YES	ignore
>>>>>UE Performance Delay Monitoring	O		9.3.1.370	Only the “UL and DL” codepoint value is used for this IE.	YES	ignore
>>UL UP TNL Information to be setup List		*1*			-	
>>>UL UP TNL Information to Be Setup Item IEs		*1 .. <maxnoofULUPTNLInformation>*			-	
>>>>UL UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-CU endpoint of the F1 transport bearer. For delivery of UL PDUs.	-	
>>>>BH Information	O		9.3.1.114		YES	ignore
>>>>DRB Mapping Info	O		Uu RLC Channel ID 9.3.1.266	This IE contains the mapped Uu Relay RLC CH ID of the DL tunnel corresponding to such UL tunnel	YES	ignore
>>RLC Mode	M		9.3.1.27		-	
>>UL Configuration	O		9.3.1.31	Information about UL usage in gNB-DU.	-	
>>Duplication Activation	O		9.3.1.36	Information on the initial state of CA based UL PDCP duplication. This IE is ignored if the *RLC Duplication*	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				Information IE is present.		
>>DC Based Duplication Configured	O		ENUMERATED (true, ..., false)	Indication on whether DC based PDCP duplication is configured or not. If included, it should be set to true. This IE is also applicable to multi-path relay.	YES	reject
>>DC Based Duplication Activation	O		Duplication Activation 9.3.1.36	Information on the initial state of DC based UL PDCP duplication. This IE is ignored if the *RLC Duplication Information* IE is present. This IE is also applicable to multi-path relay.	YES	reject
>>DL PDCP SN length	M		ENUMERATED (12bits, 18bits, ...)		YES	ignore
>>UL PDCP SN length	O		ENUMERATED (12bits, 18bits, ...)		YES	ignore
>>Additional PDCP Duplication TNL List		*0..1*			YES	ignore
>>>Additional PDCP Duplication TNL Items		*1 .. <maxnoofAdditionalPDCPDuplicationTNL>*			EACH	ignore
>>>>Additional PDCP Duplication UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-CU endpoint of the F1 transport bearer. For delivery of UL PDUs.	-	
>>>>BH Information	O		9.3.1.114		YES	ignore
>>RLC Duplication Information	O		9.3.1.146		YES	ignore
>>SDT RLC Bearer Configuration	O		OCTET STRING	RLC-BearerConfig IE defined in subclause 6.3.2 of TS 38.331 [8]	YES	ignore
Inactivity Monitoring Request	O		ENUMERATED (true, ...)		YES	reject
RAT-Frequency Priority Information	O		9.3.1.34		YES	reject
RRC-Container	O		9.3.1.6	Includes the *DL-DCCH-Message* message as defined in subclause 6.2 of TS 38.331 [8], encapsulated in a PDCP PDU.	YES	ignore
Masked IMEISV	O		9.3.1.55		YES	ignore
Serving PLMN	O		PLMN Identity 9.3.1.14	Indicates the PLMN serving the UE.	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
gNB-DU UE Aggregate Maximum Bit Rate Uplink	C-ifDRBSetup		Bit Rate 9.3.1.22	The gNB-DU UE Aggregate Maximum Bit Rate Uplink is to be enforced by the gNB-DU.	YES	ignore
RRC Delivery Status Request	O		ENUMERATED (true, ...)	Indicates whether RRC DELIVERY REPORT procedure is requested for the RRC message.	YES	ignore
Resource Coordination Transfer Information	O		9.3.1.73		YES	ignore
servingCellMO	O		INTEGER (1..64, ...)		YES	ignore
New gNB-CU UE F1AP ID	O		gNB-CU UE F1AP ID 9.3.1.4		YES	reject
RAN UE ID	O		OCTET STRING (SIZE (8))		YES	ignore
Trace Activation	O		9.3.1.88		YES	ignore
Additional RRM Policy Index	O		9.3.1.90		YES	ignore
BH RLC Channel to be Setup List		*0..1*			YES	reject
>BH RLC Channel to be Setup Item IEs		*1 .. <maxnoofBHRLCChannels>*			EACH	reject
>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113		-	
>>CHOICE *BH QoS Information*	M					
>>>BH RLC CH QoS						
>>>>BH RLC CH QoS	M		QoS Flow Level QoS Parameters 9.3.1.45	Shall be used for SA case.	-	
>>>E-UTRAN BH RLC CH QoS						
>>>>E-UTRAN BH RLC CH QoS	M		E-UTRAN QoS 9.3.1.19	Shall be used for EN-DC case.	-	
>>>Control Plane Traffic Type						
>>>>Control Plane Traffic Type	M		9.3.1.115		-	
>>RLC Mode	M		9.3.1.27		-	
>>BAP Control PDU Channel	O		ENUMERATED (true, ...)		-	
>>Traffic Mapping Information	O		9.3.1.95		-	
Configured BAP Address	O		BAP Address 9.3.1.111	The BAP address configured for the corresponding child IAB-node.	YES	reject
NR V2X Services Authorized	O		9.3.1.116		YES	ignore
LTE V2X Services Authorized	O		9.3.1.117		YES	ignore
NR UE Sidelink Aggregate Maximum Bit	O		9.3.1.119	This IE applies only if the UE is	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Rate				authorized for NR V2X services.		
LTE UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.118	This IE applies only if the UE is authorized for LTE V2X services.	YES	ignore
PC5 Link Aggregate Bit Rate	O		Bit Rate 9.3.1.22	Only applies for non-GBR and unicast QoS Flows.	YES	ignore
SL DRB to Be Setup List		*0..1*			YES	reject
>SL DRB to Be Setup Item IEs		*1 .. <maxnoofSLDRBs>*			EACH	reject
>>SL DRB ID	M		9.3.1.120		-	
>>SL DRB Information		*1*			-	
>>>SL DRB QoS	M		PC5 QoS Parameters 9.3.1.122		-	
>>>Flows Mapped to SL DRB Item		*1 .. <maxnoofPC5QoSFlows>*			-	
>>>>PC5 QoS Flow Identifier			9.3.1.121		-	
>>RLC mode	M		9.3.1.27		-	
>>Duplication Indication	O		ENUMERATED (true, ..., false)	If included, it should be set to true.	-	
Conditional Inter-DU Mobility Information	O				YES	reject
>CHO Trigger	M		ENUMERATED (CHO-initiation, CHO-replace, ...)		-	
>Target gNB-DU UE F1AP ID	C-ifCHOmod		gNB-DU UE F1AP ID 9.3.1.5	Allocated at the target gNB-DU	-	
>Estimated Arrival Probability	O		INTEGER (1..100)		YES	ignore
>S-CPAC Request	O		ENUMERATED (initiation, ...)	Indicates that SN change is for S-CPAC preparation.	YES	reject
>S-CPAC Lower Layer Reference Config Request	O		ENUMERATED (true, ...)		YES	reject
Management Based MDT PLMN List	O		MDT PLMN List 9.3.1.151		YES	ignore
Serving NID	O		NID 9.3.1.155		YES	reject
F1-C Transfer Path	O		9.3.1.207		YES	reject
F1-C Transfer Path NRDC	O		9.3.1.228		YES	reject
MDT Polluted Measurement Indicator	O		ENUMERATED (IDC,no-IDC, ...)	Indication on whether MDT Measurement affect (e.g. IDC) is undertaken or not.	YES	ignore
SCG Activation Request	O		9.3.1.233		YES	ignore
Old CG-SDT Session Info	O		CG-SDT Session Info 9.3.1.261		YES	ignore
5G ProSe Authorized	O		9.3.1.268		YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
5G ProSe UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.119	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
5G ProSe PC5 Link Aggregate Bit Rate	O		Bit Rate 9.3.1.22	This IE applies only if the UE is authorized for 5G ProSe services, and only applies for non-GBR and unicast QoS Flows.	YES	ignore
Uu RLC Channel to Be Setup List		*0..1*		This IE is not used in this version of the specification.	YES	reject
>Uu RLC Channel to be Setup Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266		-	
>>CHOICE *Uu RLC Channel QoS Information*	M				-	
>>>Uu RLC Channel QoS						
>>>>Uu RLC Channel QoS	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>>Uu Control Plane Traffic Type						
>>>>Uu Control Plane Traffic Type	M		ENUMERATED (SRB0, SRB1, SRB2, ...)	This IE indicates the type of SRB conveyed via the Uu Relay RLC Channel.	-	
>>RLC Mode	M		9.3.1.27		-	
PC5 RLC Channel to Be Setup List		*0..1*			YES	reject
>PC5 RLC Channel to be Setup Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267	This IE is not used in this version of the specification.	-	
>>CHOICE *PC5 RLC Channel QoS Information*	M				-	
>>>PC5 RLC Channel QoS						
>>>>PC5 RLC Channel QoS	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>>PC5 Control Plane Traffic Type						
>>>>PC5 Control Plane Traffic Type	M		ENUMERATED (SRB1, SRB2, ..., SRB0)	This IE indicates the type of SRB conveyed via the PC5 Relay RLC	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				Channel. This version of the specification does not use SRB0.		
>>>U2U RLC Channel QoS					YES	reject
>>>>U2U RLC Channel QoS	M		PC5 QoS Parameters 9.3.1.122		-	
>>RLC Mode	M		9.3.1.27		-	
>>Peer UE ID	O		BIT STRING (SIZE(24))	Corresponds to information provided in the *sl-DestinationIdentityL2-U2U* contained in the *SL-TxResourceReqL2-U2U* IE, defined in TS 38.331 [8]. This IE is included if the gNB-CU UE F1AP ID and/or gNB-DU UE F1AP ID are associated with a L2 U2U Remote UE or L2 U2U Relay UE.	YES	reject
Path Switch Configuration	O		9.3.1.263		YES	ignore
gNB-DU UE Slice Maximum Bit Rate List	O		9.3.1.271	The Slice Maximum Bit Rate List is the maximum aggregate UL bit rate per slice, to be enforced by the gNB-DU, if feasible. This IE is ignored if the *DRB to Be Setup List* IE is not present.	YES	ignore
Multicast MBS Session Setup List	O		Multicast MBS Session List 9.3.1.272	The list of MBS Session ID that UE has joined.	YES	reject
UE Multicast MRB to Be Setup List		*0..1*			YES	reject
>UE Multicast MRB to Be Setup Item IEs		*1 .. <maxnoofMRBsforUE>*			EACH	reject
>>MRB ID	M		9.3.1.224	MRB ID for the UE.	-	
>>MBS PTP Retransmission Tunnel Required	O		9.3.2.10		-	
>>MBS PTP Forwarding Tunnel Required Information	O		MRB Progress Information 9.3.2.12		-	
>>Source MRB ID	O		MRB ID 9.3.1.224	In case of inter-DU handover, indicates the MRB ID provided to the UE in the source cell.	YES	ignore
ServingCellMO List		*0..1*		For NCD-SSBs	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>ServingCellMO Item IEs		*1 .. <maxnoofServingCellMOs>*			EACH	ignore
>>servingCellMO	M		INTEGER (1..64, ...)		-	
>>SSB frequency	M		INTEGER (0..3279165)	ARFCN	-	
Network Controlled Repeater Authorized	O		9.3.1.288		YES	ignore
SDT Volume Threshold	O		INTEGER(1.. 192000,...)	Unit: byte.	YES	ignore
LTM Information Setup		*0..1*			YES	reject
>LTM Indicator	M		ENUMERATED (true, ..., C-LTM)		-	
>Reference Configuration	O		9.3.1.292		-	
>CSI Resource Configuration	O		9.3.1.330		-	
>Request for CSI-RS Resource Configuration for L1 measurements	O		ENUMERATED (true, ...)		YES	reject
>Request for L1 Execution Condition	O		9.3.1.361		YES	reject
LTM Configuration ID Mapping List	O		9.3.1.294		YES	reject
Early Sync Information Request		*0..1*			YES	ignore
>Request for RACH Configuration	M		ENUMERATED (true, ...)		-	
>LTM gNB-DUs List		*1*		This IE contains the IDs of the source gNB-DU and candidate gNB-DU(s). In case of inter-CU LTM, it contains the early RACH Resources Requester IDs of the source gNB and candidate gNB(s).	YES	reject
>>LTM gNB-DUs Item IEs		*1..<maxnoofLTMgNB-DUs>*			-	
>>>LTM gNB-DU ID	M		gNB-DU ID 9.3.1.9		-	
>>>LTM gNB ID	O		Global gNB ID 9.3.1.305		YES	reject
Path Addition Information	O		9.3.1.296		YES	reject
NR A2X Services Authorized	O		9.3.1.323		YES	ignore
LTE A2X Services Authorized	O		9.3.1.324		YES	ignore
NR UE Sidelink Aggregate Maximum Bit Rate for A2X	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.119	This IE applies only if the UE is authorized for NR A2X services.	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
LTE UE Sidelink Aggregate Maximum Bit Rate for A2X	O		LTE UE Sidelink Aggregate Maximum Bit Rate 9.3.1.118	This IE applies only if the UE is authorized for LTE A2X services.	YES	ignore
DL LBT Failure Information Request	O		ENUMERATED (inquiry, ...)		YES	ignore
Ranging and Sidelink Positioning Service Information	O		9.3.1.331	This IE applies only if the UE is authorized for NR V2X services and/or 5G ProSe services.	YES	ignore
Non-Integer DRX Cycle	O		9.3.1.344		YES	ignore
LTM Information SN Addition		*0..1*			YES	reject
>LTM with SCG Indicator	M		ENUMERATED (true, ...)		–	

Range bound	Explanation
maxnoofSCells	Maximum no. of SCells allowed towards one UE, the maximum value is 32.
maxnoofServingCellMOs	Maximum number of ServingCellMOs for NCD-SSB per cell. Maximum value is 16
maxnoofSRBs	Maximum no. of SRB allowed towards one UE, the maximum value is 8.
maxnoofDRBs	Maximum no. of DRB allowed towards one UE, the maximum value is 64.
maxnoofULUPTNLInformation	Maximum no. of ULUP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofCandidateSpCells	Maximum no. of SpCells allowed towards one UE, the maximum value is 64.
maxnoofQoSFlows	Maximum no. of flows allowed to be mapped to one DRB, the maximum value is 64.
maxnoofBHRLCChannels	Maximum no. of BH RLC channels allowed towards one IAB-node, the maximum value is 65536.
maxnoofSLDRBs	Maximum no. of SL DRB allowed for NR sidelink communication per UE, the maximum value is 512.
maxnoofPC5QoSFlows	Maximum no. of PC5 QoS flow allowed towards one UE for NR sidelink communication, the maximum value is 2048.
maxnoofAdditionalPDCPDuplicationTNL	Maximum no. of additional UP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofUuRLCChannels	Maximum no. of Uu Relay RLC channels for L2 U2N relaying per Relay UE, the maximum value is 32.
maxnoofPC5RLCChannels	Maximum no. of PC5 Relay RLC channels allowed for L2 U2N or U2U relaying per Remote UE or Relay UE, the maximum value is 512.
maxnoofMRBsforUE	Maximum no. of multicast MRB allowed towards one UE, the maximum value is 64.
maxnoofLTMgNB-DUs	Maximum no. of gNB-DUs allowed to be configured with LTM towards one UE, the maximum value is 8.

Condition	Explanation
ifDRBSetup	This IE shall be present only if the *DRB to Be Setup List* IE is present.
ifCHOmod	This IE shall be present if the *CHO Trigger* IE is present and set to "CHO-replace".

9.2.2.2 UE CONTEXT SETUP RESPONSE

This message is sent by the gNB-DU to confirm the setup of a UE context.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
DU To CU RRC Information	M		9.3.1.26		YES	reject
C-RNTI	O		9.3.1.32	C-RNTI allocated at the gNB-DU	YES	ignore
Resource Coordination Transfer Container	O		OCTET STRING	Includes the *SgNB Resource Coordination Information* IE as defined in subclause 9.2.117 of TS 36.423 [9] for EN-DC case or *MR-DC Resource Coordination Information* IE as defined in TS 38.423 [28] for NGEN-DC and NE-DC cases.	YES	ignore
Full Configuration	O		ENUMERATED (full, ...)		YES	reject
DRB Setup List		*0..1*		The List of DRBs which are successfully established.	YES	ignore
>DRB Setup Item list		*1 .. <maxnoofDRBs>*			EACH	ignore
>>DRB ID	M		9.3.1.8		-	
>>LCID	O		9.3.1.35	LCID for the primary path or for the split secondary path for fallback to split bearer if PDCP duplication is applied.	-	
>>DL UP TNL Information to be setup List		*1*			-	
>>>DL UP TNL Information to Be Setup Item IEs		*1 .. <maxnoofDLUPTNLInformation>*			-	
>>>>DL UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-DU endpoint of the F1 transport bearer. For delivery of DL PDUs.	-	
>>Additional PDCP Duplication TNL List		*0..1*			YES	ignore
>>>Additional PDCP Duplication TNL Items		*1 .. <maxnoofAdditionalPDCPDuplicationTNL>*			EACH	ignore
>>>>Additional	M		UP Transport	gNB-DU endpoint	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
PDCP Duplication UP TNL Information			Layer Information 9.3.2.1	of the F1 transport bearer. For delivery of DL PDUs.		
>>>>BH Information	O		9.3.1.114	This IE is not used in this version of the specification.	YES	ignore
>>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Index 9.3.1.123	Index to the currently fulfilled alternative QoS parameters set.	YES	ignore
>>TSC Traffic Characteristics Feedback	O		9.3.1.302		YES	ignore
>>ECN Marking or Congestion Information Reporting Status	O		9.3.1.322		YES	ignore
SRB Failed to Setup List		*0..1*			YES	ignore
>SRB Failed to Setup Item		*1 .. <maxnoofSRBs>*			EACH	ignore
>>SRB ID	M		9.3.1.7		-	
>>Cause	O		9.3.1.2		-	
DRB Failed to Setup List		*0..1*			YES	ignore
>DRB Failed to Setup Item		*1 .. <maxnoofDRBs>*			EACH	ignore
>>DRB ID	M		9.3.1.8		-	
>>Cause	O		9.3.1.2		-	
SCell Failed To Setup List		*0..1*			YES	ignore
>SCell Failed to Setup Item		*1 .. <maxnoofSCells>*			EACH	ignore
>>SCell ID	M		NR CGI 9.3.1.12	SCell Identifier in gNB	-	
>>Cause	O		9.3.1.2		-	
Inactivity Monitoring Response	O		ENUMERATED (not-supported, ...)		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore
SRB Setup List		*0..1*			YES	ignore
>SRB Setup Item		*1 .. <maxnoofSRBs>*			EACH	ignore
>>SRB ID	M		9.3.1.7		-	
>>LCID	M		9.3.1.35	LCID for the primary path if PDCP duplication is applied	-	
BH RLC Channel Setup List		*0..1*		The list of BH RLC channels which are successfully established.	YES	ignore
>BH RLC Channel Setup Item		*1 .. <maxnoofBHRLCChannels>*			EACH	ignore
>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113		-	
BH RLC Channel		*0..1*		The list of BH RLC	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Failed to be Setup List				channels whose setup has failed.		
>BH RLC Channel Failed to be Setup Item		*1 .. <maxnoofBHRLCChannels>*			EACH	ignore
>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113		-	
>>Cause	O		9.3.1.2		-	
SL DRB Setup List		*0..1*		The List of SL DRBs which are successfully established.	YES	ignore
>SL DRB Setup Item IEs		*1 .. <maxnoofSLDRBs>*			EACH	ignore
>>SL DRB ID	M		9.3.1.120		-	
SL DRB Failed To Setup List		*0..1*			EACH	ignore
>SL DRB Failed To Setup Item IE		*1 .. <maxnoofSLDRBs>*			EACH	ignore
>>SL DRB ID	M		9.3.1.120		-	
>>Cause	O		9.3.1.2		-	
Requested Target Cell ID	O		NR CGI 9.3.1.12	Special Cell indicated in the UE CONTEXT SETUP REQUEST message.	YES	reject
SCG Activation Status	O		9.3.1.234		YES	ignore
Uu RLC Channel Setup List		*0..1*		This IE is not used in this version of the specification.	YES	ignore
>Uu RLC Channel Setup Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266		-	
Uu RLC Channel Failed to be Setup List		*0..1*		This IE is not used in this version of the specification.	YES	ignore
>Uu RLC Channel Failed to be Setup Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266		-	
>>Cause	O		9.3.1.2		-	
PC5 RLC Channel Setup List		*0..1*			YES	ignore
>PC5 RLC Channel Setup Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267	This IE is not used in this version of the specification.		
>>Peer UE ID	O		BIT STRING (SIZE(24))	This IE is not used in this version of the specification.	YES	reject
PC5 RLC Channel		*0..1*			YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Failed to be Setup List						
>PC5 RLC Channel Failed to be Setup Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267	This IE is not used in this version of the specification.	-	
>>Cause	O		9.3.1.2		-	
>>Peer UE ID	O		BIT STRING (SIZE(24))	This IE is not used in this version of the specification.	YES	reject
ServingCellMO-encoded-in-CGC List		*0..1*			YES	ignore
>ServingCellMO-encoded-in-CGC Item IEs		*1 .. <maxNrofBWPs>*		The servingCellMO which has been encoded in *CellGroupConfig* IE.	EACH	ignore
>>servingCellMO	M		INTEGER (1..64, ...)		-	
>>BWP ID	M		INTEGER (0..4)		YES	ignore
UE Multicast MRB Setup List		*0..1*			YES	reject
>UE Multicast MRB Setup Item IEs		*1 .. <maxnoofMRBsforUE>*			EACH	reject
>>MRB ID	M		9.3.1.224	MRB ID for the UE.	-	
>>Multicast F1-U Context Reference CU	M		9.3.2.13		-	
Dedicated SI Delivery Indication	O		ENUMERATED (true, ...)		YES	ignore
Configured BWP List		0..1		This IE is present when the gNB-DU configures at least one BWP with NCD-SSB or without SSB.	YES	ignore
>Configured BWP Item IEs		*1 .. <maxNrofBWPs>*			EACH	ignore
>>BWP-Id	M		INTEGER (0..4)	The IE is used to refer to one BWP.	-	
>>BWP Location And Bandwidth	M		INTEGER (0..37949)	The IE type range is the same as the *locationAndBandwidth* IE in *BWP* IE as specified in TS 38.331 [8].	-	
Early Sync Information		*0..1*			YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>TCI States Configurations List	M		OCTET STRING	Includes the *LTM-TCI-Info* IE, as defined in TS 38.331 [8].	-	
>Early UL Sync Configuration	O		9.3.1.328		-	
>Early UL Sync Configuration for SUL	O		Early UL Sync Configuration 9.3.1.328	This IE applies for SUL carrier.	-	
LTM Configuration		*0..1*			YES	ignore
>SSB Information	M		9.3.1.202	Includes the SSB Information for the requested target cell.	-	
>Reference Configuration Information	O		OCTET STRING	Includes the *CellGroupConfig* IE, as defined in TS 38.331 [8].	-	
>Complete Candidate Configuration Indicator	O		ENUMERATED (complete, ...)		-	
>LTM CFRA Resource Configuration	O		OCTET STRING	Includes the *RACH-ConfigDedicated* IE, as defined in TS 38.331 [8].	-	
>LTM CFRA Resource Configuration for SUL	O		OCTET STRING	Includes the *RACH-ConfigDedicated* IE, as defined in TS 38.331 [8]. This IE applies for SUL carrier.	-	
>TCI States Configurations List	O		OCTET STRING	Includes the *LTM-TCI-Info* IE, as defined in TS 38.331 [8]. If present, this IE indicates the TCI States for the LTM candidate cell when early sync is not configured.	YES	reject
>L1 Execution Condition List	O		9.3.1.362		YES	ignore
>CSI-RS Resource Configuration for L1 measurement	O		CSI-RS Resource Configuration 9.3.1.360		YES	ignore
>CSI-RS Resource Configuration for Early CSI acquisition	O		CSI-RS Resource Configuration 9.3.1.360		YES	ignore
>CSI Report Configuration for Early CSI acquisition	O		OCTET STRING	Includes the ltm-CSI-ReportConfig-r19 IE, as defined in TS 38.331 [8]. If present, this IE is ignored.	YES	ignore
S-CPAC Configuration		*0..1*			YES	ignore
>Reference Configuration Information	O		OCTET STRING	Includes the *CellGroupConfig* IE, as defined in TS 38.331 [8].	-	
>Complete Candidate Configuration Indicator	O		ENUMERATED (complete, ...)		-	

Range bound	Explanation
maxnoofSCells	Maximum no. of SCells allowed towards one UE, the maximum value is 32.
maxnoofSRBs	Maximum no. of SRB allowed towards one UE, the maximum value is 8.
maxnoofDRBs	Maximum no. of DRB allowed towards one UE, the maximum value is 64.
maxnoofDLUPTNLInformation	Maximum no. of DL UP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofBHRLCChannels	Maximum no. of BH RLC channels allowed towards one IAB-node, the maximum value is 65536.
maxnoofSLDRBs	Maximum no. of SL DRB allowed for NR sidelink communication per UE, the maximum value is 512.
maxnoofAdditionalPDCPDuplicationTNL	Maximum no. of additional UP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofUuRLCChannels	Maximum no. of Uu Relay RLC channels for L2 U2N relaying per Relay UE, the maximum value is 32.
maxnoofPC5RLCChannels	Maximum no. of PC5 Relay RLC channels allowed for L2 U2N or L2 U2U relaying per Remote UE or Relay UE, the maximum value is 512.
maxNrofBWPs	Maximum number of BWPs per serving cell, the maximum value is 8.
maxnoofMRBsforUE	Maximum no. of multicast MRB allowed towards one UE, the maximum value is 64.

9.2.2.3 UE CONTEXT SETUP FAILURE

This message is sent by the gNB-DU to indicate that the setup of the UE context was unsuccessful.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	O		9.3.1.5		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Potential SpCell List		*0..1*			YES	ignore
>Potential SpCell Item IEs		*0 .. <maxnoofPotentialSpCells>*			EACH	ignore
>>Potential SpCell ID	M		NR CGI 9.3.1.12	Special Cell as defined in TS 38.321 [16]	-	
Requested Target Cell ID	O		NR CGI 9.3.1.12	Special Cell indicated in the UE CONTEXT SETUP REQUEST message.	YES	reject

Range bound	Explanation
maxnoofPotentialSpCells	Maximum no. of SpCells allowed towards one UE, the maximum value is 64.

9.2.2.4 UE CONTEXT RELEASE REQUEST

This message is sent by the gNB-DU to request the gNB-CU to release the UE-associated logical F1 connection or candidate cells in conditional handover, conditional PSCell addition, conditional PSCell change, LTM, or subsequent CPAC.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Cause	M		9.3.1.2		YES	ignore
Candidate Cells To Be Cancelled List		*0 .. <maxnoofCellsinCHO>*			YES	reject
>Target Cell ID	M		NR CGI 9.3.1.12		-	
LTM Cells To Be Released List	O		9.3.1.291		YES	reject

Range bound	Explanation
maxnoofCellsinCHO	Maximum no. cells that can be prepared for a conditional mobility. Value is 8.

9.2.2.5 UE CONTEXT RELEASE COMMAND

This message is sent by the gNB-CU to request the gNB-DU to release the UE-associated logical F1 connection or candidate cells in conditional handover, conditional PSCell addition, conditional PSCell change, LTM, or subsequent CPAC.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Cause	M		9.3.1.2		YES	ignore
RRC-Container	O		9.3.1.6	Includes the *DL-DCCH-Message* message as defined in subclause 6.2 of TS 38.331 [8] encapsulated in a PDCP PDU, or the *DL-CCCH-Message* message as defined in subclause 6.2 of TS 38.331 [8].	YES	ignore
SRB ID	C-ifRRCContainer		9.3.1.7	The gNB-DU sends the RRC message on the indicated SRB.	YES	ignore
old gNB-DU UE F1AP ID	O		gNB-DU UE F1AP ID 9.3.1.5	Include it if RRCReestablishmentRequest is not accepted	YES	ignore
Execute Duplication	O		ENUMERATED (true, ...)	This IE may be sent only if duplication has been configured for the UE.	YES	ignore
RRC Delivery Status Request	O		ENUMERATED (true, ...)	Indicates whether RRC DELIVERY REPORT	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				procedure is requested for the RRC message.		
Candidate Cells To Be Cancelled List		*0 .. <maxnoofCellsinCHO>*			YES	reject
>Target Cell ID	M		NR CGI 9.3.1.12		-	
Positioning Context Reservation Indication	O		ENUMERATED (true,...)		YES	ignore
CG-SDT Kept Indicator	O		ENUMERATED (true, ...)		YES	ignore
LTM Cells To Be Released List	O		9.3.1.291		YES	reject
DL LBT Failure Information Request	O		ENUMERATED (inquiry, ...)		YES	ignore

Range bound	Explanation
maxnoofCellsinCHO	Maximum no. cells that can be prepared for a conditional mobility. Value is 8.

Condition	Explanation
ifRRCContainer	This IE shall be present if the *RRC container* IE is present.

9.2.2.6 UE CONTEXT RELEASE COMPLETE

This message is sent by the gNB-DU to confirm the release of the UE-associated logical F1 connection or candidate cells in conditional handover, conditional PSCell addition, conditional PSCell change or subsequent CPAC.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Recommended SSBs for Paging List	O		9.3.1.297		YES	ignore

9.2.2.7 UE CONTEXT MODIFICATION REQUEST

This message is sent by the gNB-CU to provide UE Context information changes to the gNB-DU.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
SpCell ID	O		NR CGI 9.3.1.12	Special Cell as defined in TS 38.321 [16]. For handover case, this IE is considered as	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				target cell.		
ServCellIndex	O		INTEGER (0..31, ...)		YES	reject
SpCell UL Configured	O		Cell UL Configured 9.3.1.33		YES	ignore
DRX Cycle	O		9.3.1.24		YES	ignore
CU to DU RRC Information	O		9.3.1.25		YES	reject
Transmission Action Indicator	O		9.3.1.11		YES	ignore
Resource Coordination Transfer Container	O		OCTET STRING	Includes the *MeNB Resource Coordination Information* IE as defined in subclause 9.2.116 of TS 36.423 [9] for EN-DC case or *MR-DC Resource Coordination Information* IE as defined in TS 38.423 [28] for NGEN-DC and NE-DC cases.	YES	ignore
RRC Reconfiguration Complete Indicator	O		9.3.1.30		YES	ignore
RRC-Container	O		9.3.1.6	Includes the *DL-DCCH-Message* message as defined in subclause 6.2 of TS 38.331 [8], encapsulated in a PDCP PDU.	YES	reject
SCell To Be Setup List		*0..1*			YES	ignore
>SCell to Be Setup Item IEs		*1.. <maxnoofSCells>*			EACH	ignore
>>SCell ID	M		NR CGI 9.3.1.12	SCell Identifier in gNB	-	
>>SCellIndex	M		INTEGER (1..31, ...)		-	
>>SCell UL Configured	O		Cell UL Configured 9.3.1.33		-	
>>servingCellMO	O		INTEGER (1..64, ...)		YES	ignore
>>servingCellMO-On-demand	O		INTEGER (1..64, ...)		YES	ignore
SCell To Be Removed List		*0..1*			YES	ignore
>SCell to Be Removed Item IEs		*1 .. <maxnoofSCells>*			EACH	ignore
>>SCell ID	M		NR CGI 9.3.1.12	SCell Identifier in gNB	-	
SRB to Be Setup List		*0..1*			YES	reject
>SRB to Be Setup Item IEs		*1..<maxnoofSRBs>*			EACH	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>SRB ID	M		9.3.1.7		-	
>>Duplication Indication	O		ENUMERATED (true, ..., false)	This IE is ignored if the *Additional Duplication Indication* IE is present.	-	
>>Additional Duplication Indication	O		ENUMERATED (three, four, ...)		YES	ignore
>>SRB Mapping Info	O		Uu RLC Channel ID 9.3.1.266	This IE contains the mapped Uu Relay RLC CH ID for the SRB	YES	ignore
>>SDT Indicator Setup	O		ENUMERATED (true, ...)	Indicates SDT SRB.	YES	reject
DRB to Be Setup List		*0..1*			YES	reject
>DRB to Be Setup Item IEs		*1 .. <maxnoofDRBs>*			EACH	reject
>>DRB ID	M		9.3.1.8		-	
>>CHOICE QoS Information	M				-	
>>>E-UTRAN QoS						
>>>>E-UTRAN QoS	M		9.3.1.19	Shall be used for EN-DC case to convey E-RAB Level QoS Parameters		
>>>DRB Information						
>>>>DRB Information		*1*		Shall be used for NG-RAN cases	YES	ignore
>>>>>DRB QoS	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>>>>S-NSSAI	M		9.3.1.38		-	
>>>>>Notification Control	O		9.3.1.56		-	
>>>>>Flows Mapped to DRB Item		*1 .. <maxnoofQoSFlows>*			-	
>>>>>>QoS Flow Identifier	M		9.3.1.63		-	
>>>>>>QoS Flow Level QoS Parameters	M		9.3.1.45		-	
>>>>>>QoS Flow Mapping Indication	O		9.3.1.72		YES	ignore
>>>>>>TSC Traffic Characteristics	O		9.3.1.141	Traffic pattern information associated with the QFI. Details in TS 23.501 [21].	YES	ignore
>>>>>ECN Marking or Congestion Information Reporting Request	O		9.3.1.321		YES	ignore
>>>>>PSI based SDU Discard UL	O		ENUMERATED (start, stop, ...)	Indicates whether UL PSI based SDU discard is	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				(re)configured or released for the DRB. The codepoint “start” means that UL PSI based discarding is (re)configured, while the codepoint “stop” means that UL PSI based discarding is released. Up to 8 DRBs can be set as “start”.		
>>>>>PSI based SDU Discard DL	O		ENUMERATED (configured, not-configured, ...)	Indicates whether DL PSI based SDU discard is configured or not for the DRB.	YES	ignore
>>>>>UE Performance Delay Monitoring	O		9.3.1.370	Only the “UL and DL” codepoint value is used for this IE.	YES	ignore
>>UL UP TNL Information to be setup List		*1*			-	
>>>UL UP TNL Information to Be Setup Item IEs		*1 .. <maxnoofULUPTNLInformation>*			-	
>>>>UL UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-CU endpoint of the F1 transport bearer. For delivery of UL PDUs.	-	
>>>>BH Information	O		9.3.1.114		YES	ignore
>>>>DRB Mapping Info	O		Uu RLC Channel ID 9.3.1.266	This IE contains the mapped Uu Relay RLC CH ID of the DL tunnel corresponding to such UL tunnel	YES	ignore
>>RLC Mode	M		9.3.1.27		-	
>>UL Configuration	O		9.3.1.31	Information about UL usage in gNB-DU.	-	
>>Duplication Activation	O		9.3.1.36	Information on the initial state of CA based or multi-path relay based UL PDCP duplication. This IE is	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				ignored if the *RLC Duplication Information* IE is present.		
>>DC Based Duplication Configured	O		ENUMERATED (true, ..., false)	Indication on whether DC based PDCP duplication is configured or not. If included, it should be set to true. This IE is also applicable to multi-path relay.	YES	reject
>>DC Based Duplication Activation	O		Duplication Activation 9.3.1.36	Information on the initial state of DC based UL PDCP duplication. This IE is ignored if the *RLC Duplication Information* IE is present. This IE is also applicable to multi-path relay.	YES	reject
>>DL PDCP SN length	O		ENUMERATED (12bits, 18bits, ...)		YES	ignore
>>UL PDCP SN length	O		ENUMERATED (12bits, 18bits, ...)		YES	ignore
>>Additional PDCP Duplication TNL List		*0..1*			YES	ignore
>>>Additional PDCP Duplication TNL Items		*1 .. <maxnoofAdditionalPDCPDuplicationTNL>*			EACH	ignore
>>>>Additional PDCP Duplication UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-CU endpoint of the F1 transport bearer. For delivery of UL PDUs.	-	
>>>>BH Information	O		9.3.1.114		YES	ignore
>>RLC Duplication Information	O		9.3.1.146		YES	ignore
>>SDT Indicator Setup	O		ENUMERATED (true, ...)	Indicates SDT DRB.	YES	reject
DRB to Be Modified List		*0..1*			YES	reject
>DRB to Be Modified Item IEs		*1 .. <maxnoofDRBs>*			EACH	reject
>>DRB ID	M		9.3.1.8		-	
>>CHOICE *QoS Information*	O				-	
>>>E-UTRAN QoS						
>>>>E-UTRAN QoS	M		9.3.1.19	Used for EN-DC case to	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				convey E-RAB Level QoS Parameters		
>>>DRB Information						
>>>>DRB Information		*1*		Used for NG-RAN cases	YES	ignore
>>>>>DRB QoS	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>>>>S-NSSAI	M		9.3.1.38		-	
>>>>>Notification Control	O		9.3.1.56		-	
>>>>>Flows Mapped to DRB Item		*1 .. <maxnoofQoSFlows>*			-	
>>>>>>QoS Flow Identifier	M		9.3.1.63		-	
>>>>>>QoS Flow Level QoS Parameters	M		9.3.1.45		-	
>>>>>>QoS Flow Mapping Indication	O		9.3.1.72		YES	ignore
>>>>>>TSC Traffic Characteristics	O		9.3.1.141	Traffic pattern information associated with the QFI. Details in TS 23.501 [21].	YES	ignore
>>>>>ECN Marking or Congestion Information Reporting Request	O		9.3.1.321		YES	ignore
>>>>>PSI based SDU Discard UL	O		ENUMERATED (start, stop, ...)	Indicates whether UL PSI based SDU discard is (re)configured or released for the DRB. The codepoint "start" means that UL PSI based discarding is (re)configured, while the codepoint "stop" means that UL PSI based discarding is released. Up to 8 DRBs can be set as "start".	YES	ignore
>>>>>PSI based SDU Discard DL	O		ENUMERATED (configured, not-configured, ...)	Indicates whether DL PSI based SDU discard is configured or not for the DRB.	YES	ignore
>>>>>UE Performance Delay Monitoring	O		9.3.1.370		YES	ignore
>>UL UP TNL Information		*1*			-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
to be setup List						
>>>UL UP TNL Information to Be Setup Item IEs		*1 .. <maxnoofULUPTNLInformation>*			-	
>>>>UL UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-CU endpoint of the F1 transport bearer. For delivery of UL PDUs.	-	
>>>>BH Information	O		9.3.1.114		YES	ignore
>>>>DRB Mapping Info	O		Uu RLC Channel ID 9.3.1.266		YES	ignore
>>UL Configuration	O		9.3.1.31	Information about UL usage in gNB-DU.	-	
>>DL PDCP SN length	O		ENUMERATED(12bits,18bits, ...)		YES	ignore
>>UL PDCP SN length	O		ENUMERATED (12bits, 18bits, ...)		YES	ignore
>>Bearer Type Change	O		ENUMERATED (true, ...)		YES	ignore
>>RLC Mode	O		9.3.1.27		YES	ignore
>>Duplication Activation	O		9.3.1.36	Information on the initial state of CA based or multi-path relay based UL PDCP duplication. This IE is ignored if the *RLC Duplication Information* IE is present.	YES	reject
>>DC Based Duplication Configured	O		ENUMERATED (true, ..., false)	Indication on whether DC based PDCP duplication is configured or not. This IE is also applicable to multi-path relay.	YES	reject
>>DC Based Duplication Activation	O		Duplication activation 9.3.1.36	Information on the initial state of DC based UL PDCP duplication. This IE is ignored if the *RLC Duplication Information* IE is present. This IE is also applicable to multi-path relay.	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>Additional PDCP Duplication TNL List		*0..1*			YES	ignore
>>>Additional PDCP Duplication TNL Items		*1 .. <maxnoofAdditionalPDCPDuplicationTNL>*			EACH	ignore
>>>>Additional PDCP Duplication UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-CU endpoint of the F1 transport bearer. For delivery of UL PDUs.	-	
>>>>BH Information	O		9.3.1.114		YES	ignore
>>RLC Duplication Information	O		9.3.1.146		YES	ignore
>>Transmission Stop Indicator	O		9.3.1.209		YES	ignore
>>SDT Indicator Modify	O		ENUMTERATED (true, false, ...)	Indicates SDT DRB or not.	YES	reject
SRB To Be Released List		*0..1*			YES	reject
>SRB To Be Released Item IEs		*1.. <maxnoofSRBs>*			EACH	reject
>>SRB ID	M		9.3.1.7		-	
DRB to Be Released List		*0..1*			YES	reject
>DRB to Be Released Item IEs		*1 .. <maxnoofDRBs>*			EACH	reject
>>DRB ID	M		9.3.1.8		-	
Inactivity Monitoring Request	O		ENUMERATED (true, ...)		YES	reject
RAT-Frequency Priority Information	O		9.3.1.34		YES	reject
DRX configuration indicator	O		ENUMERATED(release,...)		YES	ignore
RLC Failure Indication	O		9.3.1.66		YES	ignore
Uplink TxDirectCurrentList Information	O		9.3.1.67		YES	ignore
GNB-DU Configuration Query	O		ENUMERATED (true, ...)	Used to request the gNB-DU to provide its configuration.	YES	reject
gNB-DU UE Aggregate Maximum Bit Rate Uplink	O		Bit Rate 9.3.1.22	The gNB-DU UE Aggregate Maximum Bit Rate Uplink is to be enforced by the gNB-DU.	YES	ignore
Execute Duplication	O		ENUMERATED (true, ...)	This IE may be sent only if duplication has been configured for the UE.	YES	ignore
RRC Delivery Status Request	O		ENUMERATED (true, ...)	Indicates whether RRC DELIVERY REPORT procedure is requested for the RRC message.	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Resource Coordination Transfer Information	O		9.3.1.73		YES	ignore
servingCellMO	O		INTEGER (1..64, ...)		YES	ignore
Need for Gap	O		ENUMERATED (true, ...)	Indicate gap for SeNB configured measurement is requested.It only applied to NE DC scenario.	YES	ignore
Full Configuration	O		ENUMERATED (full, ...)		YES	reject
Additional RRM Policy Index	O		9.3.1.90		YES	ignore
Lower Layer Presence Status Change	O		9.3.1.94		YES	ignore
BH RLC Channel to be Setup List		*0..1*			YES	reject
>BH RLC Channel to be Setup Item IEs		*1 .. <maxnoofBHRLCChannels>*			EACH	reject
>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113		-	
>>CHOICE *BH QoS information*	M					
>>>BH RLC CH QoS						
>>>>BH RLC CH QoS	M		QoS Flow Level QoS Parameters 9.3.1.45	Shall be used for SA case.		
>>>E-UTRAN BH RLC CH QoS						
>>>>E-UTRAN BH RLC CH QoS	M		E-UTRAN QoS 9.3.1.19	Shall be used for EN-DC case.		
>>>Control Plane Traffic Type						
>>>>Control Plane Traffic Type	M		9.3.1.115			
>>RLC Mode	M		9.3.1.27		-	
>>BAP Control PDU Channel	O		ENUMERATED (true, ...)		-	
>>Traffic Mapping Information	O		9.3.1.95		-	
BH RLC Channel to be Modified List		*0..1*			YES	reject
>BH RLC Channel to be Modified Item IEs		*1 .. <maxnoofBHRLCChannels>*			EACH	reject
>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113		-	
>>CHOICE *BH QoS information*	O					
>>>BH RLC CH QoS						
>>>>BH RLC CH QoS	M		QoS Flow Level QoS Parameters 9.3.1.45	Shall be used for SA case.	-	
>>>E-UTRAN BH RLC CH QoS						
>>>>E-UTRAN BH RLC CH QoS	M		E-UTRAN QoS	Shall be used for EN-DC	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			9.3.1.19	case.		
>>>Control Plane Traffic Type						
>>>>Control Plane Traffic Type	M		9.3.1.115		-	
>>RLC Mode	O		9.3.1.27		-	
>>BAP Control PDU Channel	O		ENUMERATED (true, ...)		-	
>>Traffic Mapping Information	O		9.3.1.95		-	
BH RLC Channel to be Released List		*0..1*			YES	reject
>BH RLC Channel to be Released Item IEs		*1 .. <maxnoofBHRLCChannels >*			EACH	reject
>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113		-	
NR V2X Services Authorized	O		9.3.1.116		YES	ignore
LTE V2X Services Authorized	O		9.3.1.117		YES	ignore
NR UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.119	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
LTE UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.118	This IE applies only if the UE is authorized for LTE V2X services.	YES	ignore
PC5 Link Aggregate Bit Rate	O		Bit Rate 9.3.1.22	Only applies for non-GBR and unicast QoS Flows.	YES	ignore
SL DRB to Be Setup List		*0..1*			YES	reject
>SL DRB to Be Setup Item IEs		*1 .. <maxnoofSLDRBs>*			EACH	reject
>>SL DRB ID	M		9.3.1.120		-	
>>SL DRB Information		*1*			-	
>>>SL DRB QoS	M		PC5 QoS Parameters 9.3.1.122		-	
>>>Flows Mapped to SL DRB Item		*1 .. <maxnoofPC5QoSFlows>*			-	
>>>>PC5 QoS Flow Identifier	M		9.3.1.121		-	
>>RLC mode	O		9.3.1.27		-	
>>Duplication Indication	O		ENUMERATED (true, ..., false)	If included, it should be set to true.	-	
SL DRB to Be Modified List		*0..1*			YES	reject
>SL DRB to Be Modified Item IEs		*1 .. <maxnoofSLDRBs>*			EACH	reject
>>SL DRB ID	M		9.3.1.120		-	
>>SL DRB Information		*1*			-	
>>>SL DRB QoS	M		PC5 QoS Parameters 9.3.1.122		-	
>>>Flows Mapped to SL DRB Item		*1 .. <maxnoofPC5QoSFlows>*			-	
>>>>PC5 QoS Flow	M		9.3.1.121		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Identifier						
>>RLC mode	O		9.3.1.27		-	
>>Duplication Indication	O		ENUMERATED (true, ..., false)		-	
SL DRB to Be Released List		*0..1*			YES	reject
>SL DRB to Be Released Item IEs		*1 .. <maxnoofSLDRBs>*			EACH	reject
>>SL DRB ID	M		9.3.1.120		-	
Conditional Intra-DU Mobility Information	O				YES	reject
>CHO Trigger	M		ENUMERATED (CHO-initiation, CHO-replace, CHO-cancel, ...)		-	-
>Candidate Cells To Be Cancelled List	C-ifCHOcancel	*0 .. <maxnoofCellsinCHO>*			-	-
>>Target Cell ID	M		NR CGI 9.3.1.12		-	-
>Estimated Arrival Probability	O		INTEGER (1..100)		YES	ignore
>S-CPAC Request	O		ENUMERATED (initiation, ...)	Indicates that SN change is for S-CPAC preparation.	YES	reject
>S-CPAC Lower Layer Reference Config Request	O		ENUMERATED (true, ...)		YES	reject
F1-C Transfer Path	O		9.3.1.207		YES	reject
SCG Indicator	O		ENUMERATED(released ,...)	This IE is used at the MN in NR-DC and NE-DC and it indicates the release of an SCG	YES	ignore
Uplink TxDirectCurrentTwoCarrierList Information	O		9.3.1.283		YES	ignore
IAB Conditional RRC Message Delivery Indication	O		ENUMERATED (true, ...)	Indicates whether the RRC message within should be withheld. This IE is only applicable if the UE is an IAB-MT, and the gNB-DU is an IAB-DU.	YES	reject
F1-C Transfer Path NRDC	O		9.3.1.228	This IE is only applicable if the UE is an IAB-MT.	YES	reject
MDT Polluted Measurement Indicator	O		ENUMERATED (IDC,no-IDC, ...)	Indication on whether MDT Measurement affect (e.g. IDC) is undertaken or not.	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
SCG Activation Request	O		9.3.1.233		YES	ignore
CG-SDT Query Indication	O		ENUMERATED (true, ...)		YES	ignore
5G ProSe Authorized	O		9.3.1.268		YES	ignore
5G ProSe UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.119	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
5G ProSe PC5 Link Aggregate Bit Rate	O		Bit Rate 9.3.1.22	This IE applies only if the UE is authorized for 5G ProSe services, and only applies for non-GBR and unicast QoS Flows.	YES	ignore
Updated Remote UE Local ID	O		Remote UE Local ID 9.3.1.267	This IE indicates the updated Remote UE Local ID for the U2N Remote UE associated with the F1AP-IDs	YES	ignore
Uu RLC Channel to Be Setup List		*0..1*			YES	reject
>Uu RLC Channel to be Setup Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266		-	
>>CHOICE *Uu RLC Channel QoS Information*	M				-	
>>>Uu RLC Channel QoS						
>>>>Uu RLC Channel QoS	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>>Uu Control Plane Traffic Type						
>>>>Uu Control Plane Traffic Type	M		ENUMERATED(SRB0, SRB1, SRB2, ...)	This IE indicates the type of SRB conveyed via the Uu Relay RLC Channel.	-	
>>RLC Mode	M		9.3.1.27		-	
Uu RLC Channel to Be Modified List		*0..1*			YES	reject
>Uu RLC Channel to be Modified Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266		-	
>>CHOICE *Uu RLC Channel QoS Information*	O				-	
>>>Uu RLC Channel QoS						
>>>>Uu RLC Channel QoS	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>>Uu Control Plane						

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Traffic Type						
>>>>Uu Control Plane Traffic Type	M		ENUMERATED(SRB0, SRB1, SRB2, ...)	This IE indicates the type of SRB conveyed via the Uu Relay RLC Channel.	-	
>>RLC Mode	O		9.3.1.27		-	
Uu RLC Channel to Be Released List		*0..1*			YES	reject
>Uu RLC Channel to Be Released Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC channel ID	M		9.3.1.266		-	
PC5 RLC Channel to Be Setup List		*0..1*			YES	reject
>PC5 RLC Channel to be Setup Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267		-	
>>CHOICE *PC5 RLC Channel QoS Information*	M				-	
>>>PC5 RLC Channel QoS						
>>>>PC5 RLC Channel QoS	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>>PC5 Control Plane Traffic Type						
>>>>PC5 Control Plane Traffic Type	M		ENUMERATED(SRB1, SRB2, ..., SRB0)	This IE indicates the type of SRB conveyed via the PC5 Relay RLC Channel.	-	
>>>U2U RLC Channel QoS					YES	reject
>>>>U2U RLC Channel QoS	M		PC5 QoS Parameters 9.3.1.122		-	
>>RLC Mode	M		9.3.1.27		-	
>>Peer UE ID	O		BIT STRING (SIZE(24))	Corresponds to information provided in the *sl-DestinationIdentityL2-U2U* contained in the *SL-TxResourceReqL2-U2U* IE, defined in TS 38.331 [8], or corresponds to the L2 ID of the parent UE or child UE in Multi-hop relay communication. This IE is included if the gNB-CU UE	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				F1AP ID and gNB-DU UE F1AP ID are associated with a L2 U2U Remote UE or L2 U2U Relay UE in U2U relay communication, or L2 U2N Relay UE in Multi-hop relay communication. This IE is ignored if the *Remote UE Local ID* IE is present.		
PC5 RLC Channel to Be Modified List		*0..1*			YES	reject
>PC5 RLC Channel to be Modified Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267			
>>CHOICE *PC5 RLC Channel QoS Information*	O				-	
>>>*PC5 RLC Channel QoS*						
>>>>PC5 RLC Channel QoS	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>>*PC5 Control Plane Traffic Type*						
>>>>PC5 Control Plane Traffic Type	M		ENUMERATED(SRB1, SRB2, ..., SRB0)	This IE indicate the type of SRB conveyed via the PC5 Relay RLC Channel.	-	
>>>*U2U RLC Channel QoS*					YES	reject
>>>>U2U RLC Channel QoS	M		PC5 QoS Parameters 9.3.1.122		-	
>>RLC Mode	O		9.3.1.27		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>Peer UE ID	O		BIT STRING (SIZE(24))	Corresponds to the L2 ID of the parent UE or child UE in Multi-hop relay communication. This IE is included if the gNB-CU UE F1AP ID and gNB-DU UE F1AP ID are associated with a L2 U2N Relay UE in Multi-hop relay communication. This IE is ignored if the *Remote UE Local ID* IE is present.	YES	reject
PC5 RLC Channel to Be Released List		*0..1*			YES	reject
>PC5 RLC Channel to be Released Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267		-	
>>Peer UE ID	O		BIT STRING (SIZE(24))	Corresponds to the L2 ID of the parent UE or child UE in Multi-hop relay communication. This IE is included if the gNB-CU UE F1AP ID and gNB-DU UE F1AP ID are associated with a L2 U2N Relay UE in Multi-hop relay communication. This IE is ignored if the *Remote UE Local ID* IE is present.	YES	reject
Path Switch Configuration	O		9.3.1.263		YES	ignore
gNB-DU UE Slice Maximum Bit Rate List	O		9.3.1.271	The Slice Maximum Bit Rate List is the maximum aggregate UL bit rate per slice, to be enforced by the gNB-DU, if feasible.	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Multicast MBS Session Setup List	O		Multicast MBS Session List 9.3.1.272	The list of MBS Session ID that UE has joined.	YES	reject
Multicast MBS Session Remove List	O		Multicast MBS Session List 9.3.1.272	The list of MBS Session ID that UE has left.	YES	reject
UE Multicast MRB to Be Setup at Modify List		*0..1*			YES	reject
>UE Multicast MRB to Be Setup at Modify Item IEs		*1 .. <maxnoofMRBsforUE>*			EACH	reject
>>MRB ID	M		9.3.1.224	MRB ID for the UE.	-	
>>MBS PTP Retransmission Tunnel Required	O		9.3.2.10		-	
>>MBS PTP Forwarding Tunnel Required Information	O		MRB Progress Information 9.3.2.12		-	
UE Multicast MRB to Be Released List		*0..1*			YES	reject
>UE Multicast MRB to Be Released Item IEs		*1 .. <maxnoofMRBsforUE>*			EACH	reject
>>MRB ID	M		9.3.1.224	MRB ID for the UE.	-	
SL DRX Cycle List		*0..1*			YES	ignore
>SL DRX Cycle Item IEs		*1 .. <maxnoofSLdestinations >*			EACH	ignore
>>RX UE ID	M		BIT STRING (SIZE(24))	Indicates the destination L2 ID of RX UE associated to this UE.	-	
>>CHOICE *SL DRX Information*	M				-	
>>>SL DRX Cycle						
>>>>SL DRX Cycle Length	M		ENUMERATED (ms10, ms20, ms32, ms40, ms60, ms64, ms70, ms80, ms128, ms160, ms256, ms320, ms512, ms640, ms1024, ms1280, ms2048, ms2560, ms5120, ms10240, ...)	Indicates the desired SL DRX cycle for RX UE associated to this UE.	-	
>>>No SL DRX					-	
>>>>SL DRX configuration indicator	M		ENUMERATED(release,...)		-	
Management Based MDT	O		MDT PLMN		YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
PLMN Modification List			Modification List 9.3.1.274			
SDT Bearer Configuration Query Indication	O		ENUMERATED (true, ...)		YES	ignore
DAPS HO status	O		ENUMERATED(initiation, ...)	This IE is used if DAPS HO is initiated.	YES	ignore
ServingCellMO List		*0..1*		For NCD-SSBs	YES	ignore
>ServingCellMO Item IEs		*1 .. <maxnoofServingCellMOs>*			EACH	ignore
>>servingCellMO	M		INTEGER (1..64, ...)		-	
>>SSB frequency	M		INTEGER (0..3279165)	ARFCN	-	
Uplink TxDirectCurrentMoreCarrierList Information	O		9.3.1.284		YES	ignore
CPAC MCG Information		*0..1*		This IE is used at the MN for MCG configuration as specified in TS 37.340 [7] for CPAC.	YES	ignore
>CPAC Trigger	M		ENUMERATED (CPAC-preparation, CPAC-executed, ..., CPAC-cancel)		-	
>PSCell ID	M		NR CGI 9.3.1.12	The PSCell corresponding to the included CG-Config IE at CPAC-preparation or the selected PSCell by the UE at CPAC-executed. This IE is ignored if the *CPAC Trigger* IE is set to “CPAC-cancel”.	-	
>Candidate PSCells To Be Cancelled List	C-ifCPACcancel				YES	ignore
>> Candidate PSCells To Be Cancelled Item IEs		*1 .. <maxnoofCellsinCHO>*			-	-
>>>PSCell ID	M		NR CGI 9.3.1.12	The corresponding PSCell cancelled at CPAC-cancel.	-	-
Network Controlled Repeater Authorized	O		9.3.1.288		YES	ignore
SDT Volume Threshold	O		INTEGER(1.. 192000,...)	Unit: byte.	YES	ignore
LTM Information Modify		*0..1*			YES	reject
>LTM Indicator	M		ENUMERAT		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			ED (true, ..., C-LTM)			
>Reference Configuration	O		9.3.1.292		-	
>CSI Resource Configuration	O		9.3.1.330		-	
>Request for CSI-RS Resource Configuration for L1 measurements	O		ENUMERATED (true, ...)		YES	reject
>Request for L1 Execution Condition	O		9.3.1.361		YES	reject
LTM CFRA Resource Config List		*0..1*			YES	ignore
>LTM CFRA Resource Config Item IEs		*1 .. <maxnoofLTMCells>*			EACH	ignore
>>Cell ID	M		NR CGI 9.3.1.12		-	
>>LTM CFRA Resource Configuration	O		OCTET STRING	Includes the *RACH-ConfigDedicated* IE, as defined in TS 38.331 [8].	-	
>>LTM CFRA Resource Configuration for SUL	O		OCTET STRING	Includes the *RACH-ConfigDedicated* IE, as defined in TS 38.331 [8]. This IE applies for SUL carrier.	-	
LTM Configuration ID Mapping List	O		9.3.1.294		YES	reject
Early Sync Information Request		*0..1*			YES	ignore
>Request for RACH Configuration	M		ENUMERATED (true, ...)		-	
>LTM gNB-DUs ID List		*1*		This IE contains the IDs of the source gNB-DU and candidate gNB-DU(s). In case of inter-CU LTM, it contains the early RACH Resources Requester IDs of the source gNB and candidate gNB(s).	YES	reject
>>LTM gNB-DUs Item IEs		*1..< maxnoofLTMgNB-DUs>*			-	
>>>LTM gNB-DU ID	M		gNB-DU ID 9.3.1.9		-	
>>>LTM gNB ID	O		Global gNB ID 9.3.1.305		YES	reject
Early Sync Candidate Cell Information List		*0..1*			YES	ignore
>Early Sync Candidate Cell Information Item IEs		*1 .. <maxnoofLTMCe*			EACH	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
		lls>				
>>Cell ID	M		NR CGI 9.3.1.12		-	
>>TCI States Configurations List	O		OCTET STRING	Includes the *LTM-TCI-Info* IE, as defined in TS 38.331 [8].	-	
>>Early UL Sync Configuration	O		9.3.1.328		-	
>>Early UL Sync Configuration for SUL	O		Early UL Sync Configuration 9.3.1.328	This IE applies for SUL carrier.	-	
>>TA Assistance Information	O		ENUMERATED (zero, ...)	The value "zero" corresponds to TA value of the cell being equal to zero.	-	
>>UE Based TA Measurement Configuration	O		OCTET STRING	Includes the *ltm-UE-MeasuredTA-ID* contained in the *LTM-Candidate* IE, as defined in TS 38.331 [8], for the LTM candidate cell identified by the *Cell ID* IE.	-	
>>SSB Positions In Burst	C-ifEarlyUL		9.3.1.138	This IE applies to early TA acquisition.	YES	ignore
>>LTM Residual TA Information List	O		9.3.1.363	This IE indicates the TA value and the remaining TA timers of the cell.	YES	ignore
Early Sync Serving Cell Information		*0..1*			YES	ignore
>UE Based TA Measurement Configuration	O		OCTET STRING	Includes the *ltm-ServingCellUE-MeasuredTA-ID* contained in the *LTM-Config* IE, as defined in TS 38.331 [8], for the current serving cell.	-	
LTM Cells To Be Released List	O		9.3.1.291		YES	reject
Path Addition Information	O		9.3.1.296		YES	reject
NR A2X Services Authorized	O		9.3.1.323		YES	ignore
LTE A2X Services Authorized	O		9.3.1.324		YES	ignore
NR UE Sidelink Aggregate Maximum Bit Rate for A2X	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.119	This IE applies only if the UE is authorized for NR A2X services.	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
LTE UE Sidelink Aggregate Maximum Bit Rate for A2X	O		LTE UE Sidelink Aggregate Maximum Bit Rate 9.3.1.118	This IE applies only if the UE is authorized for LTE A2X services.	YES	ignore
DL LBT Failure Information Request	O		ENUMERATED (inquiry, ...)		YES	ignore
Ranging and Sidelink Positioning Service Information	O		9.3.1.331	This IE applies only if the UE is authorized for NR V2X services and/or 5G ProSe services.	YES	ignore
Non-Integer DRX Cycle	O		9.3.1.344		YES	ignore
LTM Reset Information	O		9.3.1.346		YES	ignore
LTM TCI States Configurations List		*0..1*			YES	reject
>LTM TCI States Configurations Item IEs		*1 .. <maxnoofLTMCells>*			-	
>>Cell ID	M		NR CGI 9.3.1.12		-	
>>TCI States Configurations List	M		OCTET STRING	Includes the *LTM-TCI-Info* IE, as defined in TS 38.331 [8].	-	
LTM Security Information	O		9.3.1.359		YES	reject
LTM Information SN Modification		*0..1*			YES	reject
>LTM with SCG Indicator	M		ENUMERATED(true, ...)		–	
Inter-SN LTM MCG Information		*0..1*		This IE is used at the MN for MCG configuration as specified in TS 37.340 [7] for inter-SN LTM.	YES	ignore
>Inter-SN LTM Trigger	M		ENUMERATED (LTM-preparation, LTM-executed, LTM-cancel …,)		-	
>PSCell ID	C-ifInterSNLTMPrepOrExec		NR CGI 9.3.1.12	The PSCell corresponding to the included CG-Config IE at LTM-preparation or the selected PSCell for the UE at LTM-executed.	-	
>Candidate PSCells To Be Cancelled List	C-ifInterSNLTMCancel				-	
>> Candidate PSCells To Be Cancelled Item IEs		*1 .. <maxnoofLTMCe*			-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
		lls>				
>>>PSCell ID	M		NR CGI 9.3.1.12	The corresponding PSCell cancelled at LTM-cancel.	-	

Range bound	Explanation
maxnoofSCells	Maximum no. of SCells allowed towards one UE, the maximum value is 32.
maxnoofServingCellMOs	Maximum number of ServingCellMOs for NCD-SSB per cell. Maximum value is 16
maxnoofSRBs	Maximum no. of SRB allowed towards one UE, the maximum value is 8.
maxnoofDRBs	Maximum no. of DRB allowed towards one UE, the maximum value is 64.
maxnoofULUPTNLInformation	Maximum no. of UL UP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofQoSFlows	Maximum no. of flows allowed to be mapped to one DRB, the maximum value is 64.
maxnoofBHRLCChannels	Maximum no. of BH RLC channels allowed towards one IAB-node, the maximum value is 65536.
maxnoofSLDRBs	Maximum no. of SL DRB allowed for NR sidelink communication per UE, the maximum value is 512.
maxnoofPC5QoSFlows	Maximum no. of PC5 QoS flow allowed towards one UE for NR sidelink communication, the maximum value is 2048.
maxnoofAdditionalPDCPDuplicationTNL	Maximum no. of additional UP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofCellsinCHO	Maximum no. cells that can be prepared for a conditional mobility. Value is 8.
maxnoofUuRLCChannels	Maximum no. of Uu Relay RLC channels for L2 U2N relaying or L2 N3C relaying per Relay UE, the maximum value is 32.
maxnoofPC5RLCChannels	Maximum no. of PC5 Relay RLC channel allowed for L2 U2N or U2U relaying per Remote UE or Relay UE, the maximum value is 512.
maxnoofMRBsforUE	Maximum no. of multicast MRB allowed towards one UE, the maximum value is 64.
maxnoofSLdestinations	Maximum number of destination for NR sidelink communication, the maximum value is 32
maxnoofLTMCells	Maximum no. of Cells configured for LTM allowed towards one UE, the maximum value is 8.
maxnoofLTMgNB-DUs	Maximum no. of gNB-DUs allowed to be configured with LTM towards one UE, the maximum value is 8.

Condition	Explanation
ifCHOcancel	This IE shall be present if the *CHO Trigger* IE is present and set to "CHO-cancel".
ifEarlyUL	This IE shall be present if the *Early UL Sync Configuration* IE or the *Early UL Sync Configuration for SUL* IE is present.
ifCPACcancel	This IE may be present if the *CPAC Trigger* IE is present and set to "CPAC-cancel".
ifInterSNLTMPrepOrExec	This IE shall be present if the *Inter-SN LTM Trigger* IE is present and set to "LTM-preparation" or "LTM-executed".
ifInterSNLTMCancel	This IE shall be present if the *Inter-SN LTM Trigger* IE is present and set to "LTM-cancel".

9.2.2.8 UE CONTEXT MODIFICATION RESPONSE

This message is sent by the gNB-DU to confirm the modification of a UE context.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Resource Coordination Transfer Container	O		OCTET STRING	Includes the *SgNB Resource Coordination Information* IE as defined in subclause 9.2.117 of TS 36.423 [9] for EN-DC case or *MR-DC Resource Coordination Information* IE as defined in TS 38.423 [28] for NGEN-DC and NE-DC cases.	YES	ignore
DU To CU RRC Information	O		9.3.1.26		YES	reject
DRB Setup List		*0..1*		The List of DRBs which are successfully established.	YES	ignore
>DRB Setup Item IEs		*1 .. <maxnoofDRBs>*			EACH	ignore
>>DRB ID	M		9.3.1.8		-	
>>LCID	O		9.3.1.35	LCID for the primary path or for the split secondary path for fallback to split bearer if PDCP duplication is applied.	-	
>>DL UP TNL Information to be setup List		*1*			-	
>>>DL UP TNL Information to Be Setup Item IEs		*1 .. <maxnoofDLUPTNLInformation>*			-	
>>>>DL UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-DU endpoint of the F1 transport bearer. For delivery of DL PDUs.	-	
>>Additional PDCP Duplication TNL List		*0..1*			YES	ignore
>>>Additional PDCP Duplication TNL Items		*1 .. < maxnoofAdditionalPDCPDuplicationTNL >*			EACH	ignore
>>>>Additional PDCP Duplication UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-DU endpoint of the F1 transport bearer. For delivery of DL PDUs.	-	
>>>>BH Information	O		9.3.1.114	This IE is not used in this version of the specification.	YES	ignore
>>Current QoS	O		Alternative QoS	Index to the	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Parameters Set Index			Parameters Set Index 9.3.1.123	currently fulfilled alternative QoS parameters set.		
>>TSC Traffic Characteristics Feedback	O		9.3.1.302		YES	ignore
>>ECN Marking or Congestion Information Reporting Status	O		9.3.1.322		YES	ignore
DRB Modified List		*0..1*		The List of DRBs which are successfully modified.	YES	ignore
>DRB Modified Item IEs		*1 .. <maxnoofDRBs>*			EACH	ignore
>>DRB ID	M		9.3.1.8		-	
>>LCID	O		9.3.1.35	LCID for the primary path or for the split secondary path for fallback to split bearer if PDCP duplication is applied.	-	
>>DL UP TNL Information to be setup List		*1*			-	
>>>DL UP TNL Information to Be Setup Item IEs		*1 .. <maxnoofDLUPTNLInformation>*			-	
>>>>DL UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-DU endpoint of the F1 transport bearer. For delivery of DL PDUs.	-	
>>RLC Status	O		9.3.1.69	Indicates the RLC has been re-established at the gNB-DU.	YES	ignore
>>Additional PDCP Duplication TNL List		*0..1*			YES	ignore
>>>Additional PDCP Duplication TNL Items		*1 .. < maxnoofAdditionalPDCPDuplicationTNL>*			EACH	ignore
>>>>Additional PDCP Duplication UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-DU endpoint of the F1 transport bearer. For delivery of DL PDUs.	-	
>>>>BH Information	O		9.3.1.114	This IE is not used in this version of the specification.	YES	ignore
>>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Index 9.3.1.123	Index to the currently fulfilled alternative QoS parameters set.	YES	ignore
>>TSC Traffic Characteristics Feedback	O		9.3.1.302		YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>ECN Marking or Congestion Information Reporting Status	O		9.3.1.322		YES	ignore
SRB Failed to be Setup List		*0..1*		The List of SRBs which are failed to be established.	YES	ignore
>SRB Failed to be Setup Item IEs		*1 .. <maxnoofSRBs>*			EACH	ignore
>>SRB ID	M		9.3.1.7		-	
>>Cause	O		9.3.1.2		-	
DRB Failed to be Setup List		*0..1*		The List of DRBs which are failed to be setup.	YES	ignore
>DRB Failed to be Setup Item IEs		*1 .. <maxnoofDRBs>*			EACH	ignore
>>DRB ID	M		9.3.1.8		-	
>>Cause	O		9.3.1.2		-	
SCell Failed To Setup List		*0..1*			YES	ignore
>SCell Failed to Setup Item		*1 .. <maxnoofSCells>*			EACH	ignore
>>SCell ID	M		NR CGI 9.3.1.12	SCell Identifier in gNB	-	
>>Cause	O		9.3.1.2		-	
DRB Failed to be Modified List		0..1		The List of DRBs which are failed to be modified.	YES	ignore
>DRB Failed to be Modified Item IEs		*1 .. <maxnoofDRBs>*			EACH	ignore
>>DRB ID	M		9.3.1.8		-	
>>Cause	O		9.3.1.2		-	
Inactivity Monitoring Response	O		ENUMERATED (Not-supported, ...)		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore
C-RNTI	O		9.3.1.32	C-RNTI allocated at the gNB-DU	YES	ignore
Associated SCell List	O		9.3.1.77		YES	ignore
SRB Setup List		*0..1*			YES	ignore
>SRB Setup Item		*1 .. <maxnoofSRBs>*			EACH	ignore
>>SRB ID	M		9.3.1.7		-	
>>LCID	M		9.3.1.35	LCID for the primary path if PDCP duplication is applied	-	
SRB Modified List		*0..1*			YES	ignore
>SRB Modified Item		*1 .. <maxnoofSRBs>*			EACH	ignore
>>SRB ID	M		9.3.1.7		-	
>>LCID	M		9.3.1.35	LCID for the primary path if PDCP duplication is applied	-	
Full Configuration	O		ENUMERATED (full, ...)		YES	reject
BH RLC Channel Setup List		*0..1*		The list of BH RLC channels which	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				are successfully established.		
>BH RLC Channel Setup Item		*1 .. <maxnoofBHRLCChannels>*			EACH	ignore
>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113		-	
BH RLC Channel Modified List		*0..1*		The list of BH RLC channels which are successfully modified.	YES	ignore
>BH RLC Channel Modified Item		*1 .. <maxnoofBHRLCChannels>*			EACH	ignore
>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113		-	
BH RLC Channel Failed to be Setup List		*0..1*		The list of BH RLC channels whose setup has failed.	YES	ignore
>BH RLC Channel Failed to be Setup Item		*1 .. <maxnoofBHRLCChannels>*			EACH	ignore
>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113		-	
>>Cause	O		9.3.1.2		-	
BH RLC Channel Failed to be Modified List		*0..1*		The list of BH RLC channels whose modification has failed.	YES	ignore
>BH RLC Channel Failed to be Modified Item		*1 .. <maxnoofBHRLCChannels>*			EACH	ignore
>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113		-	
>>Cause	O		9.3.1.2		-	
SL DRB Setup List		*0..1*		The List of SL DRBs which are successfully established.	YES	ignore
>SL DRB Setup Item IEs		*1 .. <maxnoofSLDRBs>*			EACH	ignore
>>SL DRB ID	M		9.3.1.120		-	
SL DRB Modified List		*0..1*		The List of SL DRBs which are successfully modified.	YES	ignore
>SL DRB Modified Item IEs		*1 .. <maxnoofSLDRBs>*			EACH	ignore
>>SL DRB ID	M		9.3.1.120		-	
SL DRB Failed To Setup List		*0..1*		The List of SL DRBs which are failed to be setup.	YES	ignore
>SL DRB Failed To Setup Item		*1 .. <maxnoofSLDRBs>*			EACH	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>SL DRB ID	M		9.3.1.120		-	
>>Cause	O		9.3.1.2		-	
SL DRB Failed To be Modified List		*0..1*		The List of SL DRBs which are failed to be modified.	YES	ignore
>SL DRB Failed To be Modified Item		*1 .. <maxnoofSLDRBs>*			EACH	ignore
>>SL DRB ID	M		9.3.1.120		-	
>>cause	O		9.3.1.2		-	
Requested Target Cell ID	O		NR CGI 9.3.1.12	Special Cell, PSCell ID in the *CPAC MCG Information* IE, or PSCell ID in the *Inter-SN LTM MCG Information* IE indicated in the UE CONTEXT MODIFICATION REQUEST message.	YES	reject
SCG Activation Status	O		9.3.1.234		YES	ignore
Uu RLC Channel Setup List		*0..1*			YES	ignore
>Uu RLC Channel Setup Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266			
Uu RLC Channel Failed to be Setup List		*0..1*			YES	ignore
>Uu RLC Channel Failed to be Setup Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266		-	
>>Cause	O		9.3.1.2		-	
Uu RLC Channel Modified List		*0..1*			YES	ignore
>Uu RLC Channel Modified Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266		-	
Uu RLC Channel Failed to be Modified List		*0..1*			YES	ignore
>Uu RLC Channel Failed to be Modified Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266		-	
>>Cause	O		9.3.1.2		-	
PC5 RLC Channel Setup List		*0..1*			YES	ignore
>PC5 RLC Channel Setup Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267		-	
>>Peer UE ID	O		BIT STRING (SIZE(24))	Corresponds to the L2 ID of the parent UE or child UE in Multi-hop relay communication. This IE is included if the gNB-CU UE F1AP ID and gNB-DU UE F1AP ID are associated with a L2 U2N Relay UE in Multi-hop relay communication. This IE is ignored if the *Remote UE Local ID* IE is present.	YES	reject
PC5 RLC Channel Failed to be Setup List		*0..1*			YES	ignore
>PC5 RLC Channel Failed to be Setup Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267		-	
>>Cause	O		9.3.1.2		-	
>>Peer UE ID	O		BIT STRING (SIZE(24))	Corresponds to the L2 ID of the parent UE or child UE in Multi-hop relay communication. This IE is included if the gNB-CU UE F1AP ID and gNB-DU UE F1AP ID are associated with a L2 U2N Relay UE in Multi-hop relay communication. This IE is ignored if the *Remote UE Local ID* IE is present.	YES	reject
PC5 RLC Channel Modified List		*0..1*			YES	ignore
>PC5 RLC Channel Modified Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267		-	
>>Peer UE ID	O		BIT STRING (SIZE(24))	Corresponds to the L2 ID of the parent UE or child	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				UE in Multi-hop relay communication. This IE is included if the gNB-CU UE F1AP ID and gNB-DU UE F1AP ID are associated with a L2 U2N Relay UE in Multi-hop relay communication. This IE is ignored if the *Remote UE Local ID* IE is present.		
PC5 RLC Channel Failed to be Modified List		*0..1*			YES	ignore
>PC5 RLC Channel Failed to be Modified Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267		-	
>>Cause	O		9.3.1.2		-	
>>Peer UE ID	O		BIT STRING (SIZE(24))	Corresponds to the L2 ID of the parent UE or child UE in Multi-hop relay communication. This IE is included if the gNB-CU UE F1AP ID and gNB-DU UE F1AP ID are associated with a L2 U2N Relay UE in Multi-hop relay communication. This IE is ignored if the *Remote UE Local ID* IE is present.	YES	reject
SDT Bearer Configuration Info	O		9.3.1.277		YES	ignore
UE Multicast MRB Setup List		*0..1*			YES	reject
>UE Multicast MRB Setup Item IEs		*1 .. <maxnoofMRBsforUE>*			EACH	reject
>>MRB ID	M		9.3.1.224	MRB ID for the UE.	-	
>>Multicast F1-U Context Reference CU	M		9.3.2.13		-	
ServingCellMO-encoded-in-CGC List		*0..1*			YES	ignore
>ServingCellMO-encoded-in-CGC Item IEs		*1 .. <maxNrofBWPs>*		The servingCellMO which has been encoded in	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				CellGroupConfig IE.		
>>servingCellMO	M		INTEGER (1..64, ...)		-	
>>BWP ID	M		INTEGER (0..4)		YES	ignore
Dedicated SI Delivery Indication	O		ENUMERATED (true, ...)		YES	ignore
Configured BWP List		0..1		This IE is present when the gNB-DU configures at least one BWP with NCD-SSB or without SSB.	YES	ignore
>Configured BWP Item IEs		*1 .. <maxNrofBWPs>*			EACH	ignore
>>BWP-Id	M		INTEGER (0..4)	The IE is used to refer to one BWP.	-	
>>BWP Location And Bandwidth	M		INTEGER (0..37949)	The IE type range is the same as the *locationAndBandwidth* IE in *BWP* IE as specified in TS 38.331 [8].		
Early Sync Information		*0..1*			YES	ignore
>TCI States Configurations List	M		OCTET STRING	Includes the *LTM-TCI-Info* IE, as defined in TS 38.331 [8].	-	
>Early UL Sync Configuration	O		9.3.1.328		-	
>Early UL Sync Configuration for SUL	O		Early UL Sync Configuration 9.3.1.328	This IE applies for SUL carrier.	-	
LTM Configuration		*0..1*			YES	ignore
>SSB Information	M		9.3.1.202	Includes the SSB Information for the requested target cell	-	
>Reference Configuration Information	O		OCTET STRING	Includes the *CellGroupConfig* IE, as defined in TS 38.331 [8].	-	
>Complete Candidate Configuration Indicator	O		ENUMERATED (complete, ...)		-	
>LTM CFRA Resource Configuration	O		OCTET STRING	Includes the *RACH-ConfigDedicated* IE, as defined in TS 38.331 [8].	-	
>LTM CFRA Resource Configuration for SUL	O		OCTET STRING	Includes the *RACH-ConfigDedicated* IE, as defined in TS 38.331 [8]. This IE applies for SUL carrier.	-	
>TCI States Configurations List	O		OCTET STRING	Includes the *LTM-TCI-Info* IE, as defined in TS 38.331 [8]. If present, this IE indicates the TCI States for the LTM candidate cell	-	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				when early sync is not configured.		
>L1 Execution Condition List	O		9.3.1.362		YES	ignore
>CSI-RS Resource Configuration for L1 measurement	O		CSI-RS Resource Configuration 9.3.1.360		YES	ignore
>CSI-RS Resource Configuration for Early CSI acquisition	O		CSI-RS Resource Configuration 9.3.1.360		YES	ignore
>CSI Report Configuration for Early CSI acquisition	O		OCTET STRING	Includes the *ltm-CSI-ReportConfig-r19* IE, as defined in TS 38.331 [8].	YES	ignore
S-CPAC Configuration		*0..1*			YES	ignore
>Reference Configuration Information	O		OCTET STRING	Includes the *CellGroupConfig* IE, as defined in TS 38.331 [8].	-	
>Complete Candidate Configuration Indicator	O		ENUMERATED (complete, ...)		-	
TA Remaining Information List		*0..1*			YES	ignore
>TA Remaining Information Item IEs		*1 .. <maxnoofLTMCells>*			-	
>>Cell ID	M		NR CGI 9.3.1.12		-	
>>LTM Residual TA Information List	M		9.3.1.363	This IE indicates the TA value and the remaining TA timers of the cell.	-	

Range bound	Explanation
maxnoofSRBs	Maximum no. of SRB allowed towards one UE, the maximum value is 8.
maxnoofDRBs	Maximum no. of DRB allowed towards one UE, the maximum value is 64.
maxnoofDLUPTNLInformation	Maximum no. of DL UP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofSCells	Maximum no. of SCells allowed towards one UE, the maximum value is 32.
maxnoofBHRLCChannels	Maximum no. of BH RLC channels allowed towards one IAB-node, the maximum value is 65536.
maxnoofSLDRBs	Maximum no. of SL DRB allowed for NR sidelink communication per UE, the maximum value is 512.
maxnoofAdditionalPDCPDuplicationTNL	Maximum no. of additional UP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofUuRLCChannels	Maximum no. of Uu Relay RLC channels for L2 U2N relaying or L2 N3C relaying per Relay UE, the maximum value is 32.
maxnoofPC5RLCChannels	Maximum no. of PC5 Relay RLC channels allowed for L2 U2N or L2 U2U relaying per Remote UE or Relay UE, the maximum value is 512.
maxNrofBWPs	Maximum number of BWPs per serving cell, the maximum value is 8.
maxnoofMRBsforUE	Maximum no. of multicast MRB allowed towards one UE, the maximum value is 64.
maxnoofLTMCells	Maximum no. of Cells configured for LTM allowed towards one UE, the maximum value is 8.

9.2.2.9 UE CONTEXT MODIFICATION FAILURE

This message is sent by the gNB-DU to indicate a context modification failure.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Requested Target Cell ID	O		NR CGI 9.3.1.12	Special Cell indicated in the UE CONTEXT MODIFICATION REQUEST message.	YES	reject

9.2.2.10 UE CONTEXT MODIFICATION REQUIRED

This message is sent by the gNB-DU to request the modification of a UE context.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Resource Coordination Transfer Container	O		OCTET STRING	Includes the *SgNB Resource Coordination Information* IE as defined in subclause 9.2.117 of TS 36.423 [9] for EN-DC case or *MR-DC Resource Coordination Information* IE as defined in TS 38.423 [28] for NGEN-DC and NE-DC cases.	YES	ignore
DU To CU RRC Information	O		9.3.1.26		YES	reject
DRB Required to Be Modified List		*0..1*			YES	reject
>DRB Required to Be Modified Item IEs		*1 .. <maxnoofDRBs>*			EACH	reject
>>DRB ID	M		9.3.1.8		-	
>>DL UP TNL Information to be setup List		*1*			-	
>>>DL UP TNL Information to Be Setup Item IEs		*1 .. <maxnoofDLUPTNLInformation>*			-	
>>>>DL UP TNL Information	M		UP Transport Layer Information	gNB-DU endpoint of the F1 transport bearer. For	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			9.3.2.1	delivery of DL PDUs.		
>>RLC Status	O		9.3.1.69	Indicates the RLC has been re-established at the gNB-DU.	YES	ignore
>>Additional PDCP Duplication TNL List		*0..1*			YES	ignore
>>>Additional PDCP Duplication TNL Items		*1 .. <maxnoofAdditionalPDCPDuplicationTNL>*			EACH	ignore
>>>>Additional PDCP Duplication UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-CU endpoint of the F1 transport bearer. For delivery of DL PDUs.	-	
>>>>BH Information	O		9.3.1.114	This IE is not used in this version of the specification.	YES	ignore
SRB Required to be Released List		*0..1*			YES	reject
>SRB Required to be Released List Item IEs		*1 .. <maxnoofSRBs>*			EACH	reject
>>SRB ID	M		9.3.1.7		-	
DRB Required to be Released List		*0..1*			YES	reject
>DRB Required to be Released List Item IEs		*1 .. <maxnoofDRBs>*			EACH	reject
>>DRB ID	M		9.3.1.8		-	
Cause	M		9.3.1.2		YES	ignore
BH RLC Channel Required to be Released List		*0..1*			YES	reject
>BH RLC Channel Required to be Released Item IEs		*1 .. <maxnoofBHRLCChannels>*			EACH	reject
>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113		-	
SL DRB Required to Be Modified List		*0..1*			YES	reject
>SL DRB Required to Be Modified Item IEs		*1 .. <maxnoofSLDRBs>*			EACH	reject
>>SL DRB ID	M		9.3.1.120		-	
SL DRB Required to be Released List		*0..1*			YES	reject
>SL DRB Required to be Release Item IEs		*1 .. <maxnoofSLDRBs>*			EACH	reject
>>SL DRB ID	M		9.3.1.120		-	
Candidate Cells To Be Cancelled List		*0 .. <maxnoofCellsinCHO>*			YES	reject
>Target Cell ID	M		NR CGI 9.3.1.12		-	
Uu RLC Channel		*0..1*			YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Required to be Modified List						
>Uu RLC Channel Required to be Modified Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266		-	
Uu RLC Channel Required to be Released List		*0..1*			YES	reject
>Uu RLC Channel Required to be Released Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266		-	
PC5 RLC Channel Required to be Modified List		*0..1*			YES	reject
>PC5 RLC Channel Required to be Modified Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267		-	
>>Peer UE ID	O		BIT STRING (SIZE(24))	Corresponds to the L2 ID of the parent UE or child UE in Multi-hop relay communication. This IE is included if the gNB-CU UE F1AP ID and gNB-DU UE F1AP ID are associated with a L2 U2N Relay UE in Multi-hop relay communication. This IE is ignored if the *Remote UE Local ID* IE is present.	YES	reject
PC5 RLC Channel Required to be Released List		*0..1*			YES	reject
>PC5 RLC Channel Required to be Released Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267		-	
>>Peer UE ID	O		BIT STRING (SIZE(24))	Corresponds to the L2 ID of the parent UE or child UE in Multi-hop relay communication. This IE is included if the gNB-CU UE	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				F1AP ID and gNB-DU UE F1AP ID are associated with a L2 U2N Relay UE in Multi-hop relay communication. This IE is ignored if the *Remote UE Local ID* IE is present.		
UE Multicast MRB Required to Be Modified List		*0..1*			YES	reject
>UE Multicast MRB Required to Be Modified Item IEs		*1 .. <maxnoofMRBsforUE>*			EACH	reject
>>MRB ID	M		9.3.1.224	MRB ID for the UE.	-	
>>MRB type reconfiguration	O		ENUMERATED (true, ...)		-	
>>MRB Reconfigured RLC mode	C-ifMRBTypeReconf		MRB RLC Configuration 9.3.1.275		-	
>>Multicast F1-U Context Reference CU	O		9.3.2.13		YES	reject
UE Multicast MRB Required to Be Released List		*0..1*			YES	reject
>UE Multicast MRB Required to Be Released Item IEs		*1 .. <maxnoofMRBsforUE>*			EACH	reject
>>MRB ID	M		9.3.1.224	MRB ID for the UE.	-	
LTM Cells To Be Released List	O		9.3.1.291		YES	reject
LTM Configurations to Modify List		*0..1*			YES	ignore
>LTM Configurations to Modify Item		*1.. <maxnoofLTMCells>*			–	
>>Candidate Cell ID	M		NR CGI 9.3.1.12		–	
>>TCI States Configurations List	O		OCTET STRING	Includes the *LTM-TCI-Info* IE, as defined in TS 38.331 [8].	–	
>>CSI-RS Resource Configuration for L1 measurement	O		CSI-RS Resource Configuration 9.3.1.360		–	
>>CSI-RS Resource Configuration for Early CSI acquisition	O		CSI-RS Resource Configuration 9.3.1.360		–	
>>Complete Candidate Configuration Indicator	O		ENUMERATED (complete, ...)		–	

Range bound	Explanation
maxnoofSRBs	Maximum no. of SRB allowed towards one UE, the maximum value is 8.
maxnoofDRBs	Maximum no. of DRB allowed towards one UE, the maximum value is 64.
maxnoofDLUPTNLInformation	Maximum no. of DL UP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofBHRLCChannels	Maximum no. of BH RLC channels allowed towards one IAB-node, the maximum value is 65536.
maxnoofSLDRBs	Maximum no. of SL DRB allowed for NR sidelink communication per UE, the maximum value is 512.
maxnoofAdditionalPDCPDuplicationTNL	Maximum no. of additional UP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofCellsinCHO	Maximum no. cells that can be prepared for a conditional mobility. Value is 8.
maxnoofUuRLCChannels	Maximum no. of Uu Relay RLC channels for L2 U2N relaying or L2 N3C relaying per Relay UE, the maximum value is 32.
maxnoofPC5RLCChannels	Maximum no. of PC5 Relay RLC channels allowed for L2 U2N or L2 U2U relaying per Remote UE or Relay UE, the maximum value is 512.
maxnoofMRBsforUE	Maximum no. of multicast MRB allowed towards one UE, the maximum value is 64.

Condition	Explanation
ifMRBTypeReconf	This IE shall be present if the MRB Type Reconfiguration IE is present.

9.2.2.11 UE CONTEXT MODIFICATION CONFIRM

This message is sent by the gNB-CU to inform the gNB-DU the successful modification.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Resource Coordination Transfer Container	O		OCTET STRING	Includes the *MeNB Resource Coordination Information* IE as defined in subclause 9.2.116 of TS 36.423 [9] for EN-DC case or *MR-DC Resource Coordination Information* IE as defined in TS 38.423 [28] for NGEN-DC and NE-DC cases.	YES	ignore
DRB Modified List		*0..1*		The List of DRBs which are successfully modified.	YES	ignore
>DRB Modified Item IEs		*1 .. <maxnoofDRBs>*			EACH	ignore
>>DRB ID	M		9.3.1.8		-	
>>UL UP TNL Information to be setup List		*1*			-	
>>>UL UP TNL		*1 ..*			-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Information to Be Setup Item IEs		*<maxnoofULUPTNLInformation>*				
>>>>UL UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-CU endpoint of the F1 transport bearer. For delivery of UL PDUs.	-	
>>>>BH Information	O		9.3.1.114		YES	ignore
>>>>DRB Mapping Info	O		Uu RLC Channel ID 9.3.1.266	This IE is not used in this version of the specification.	YES	ignore
>>Additional PDCP Duplication TNL List		*0..1*			YES	ignore
>>>Additional PDCP Duplication TNL Items		*1 .. <maxnoofAdditionalPDCPDuplicationTNL>*			EACH	ignore
>>>>Additional PDCP Duplication UP TNL Information	M		UP Transport Layer Information 9.3.2.1	gNB-DU endpoint of the F1 transport bearer. For delivery of UL PDUs.	-	
>>>>BH Information	O		9.3.1.114		YES	ignore
RRC-Container	O		9.3.1.6	Includes the *DL-DCCH-Message* message as defined in subclause 6.2 of TS 38.331 [8], encapsulated in a PDCP PDU.	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Execute Duplication	O		ENUMERATED (true, ...)	This IE may be sent only if duplication has been configured for the UE.	YES	ignore
Resource Coordination Transfer Information	O		9.3.1.73		YES	ignore
SL DRB Modified List		*0..1*			YES	ignore
>SL DRB Modified Item IEs		*1 .. <maxnoofSLDRBs>*			EACH	reject
>>SL DRB ID	M		9.3.1.120		-	
Uu RLC Channel Modified List		*0..1*			YES	reject
>Uu RLC Channel Modified Item IEs		*1 .. <maxnoofUuRLCChannels>*			-	
>>Uu RLC Channel ID	M		9.3.1.266		-	
PC5 RLC Channel Modified List		*0..1*			YES	reject
>PC5 RLC Channel Modified Item IEs		*1 .. <maxnoofPC5RLCChannels>*			-	-

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>PC5 RLC Channel ID	M		9.3.1.265		-	
>>Remote UE Local ID	O		9.3.1.267		-	
>>Peer UE ID	O		BIT STRING (SIZE(24))	Corresponds to the L2 ID of the parent UE or child UE in Multi-hop relay communication. This IE is included if the gNB-CU UE F1AP ID and gNB-DU UE F1AP ID are associated with a L2 U2N Relay UE in Multi-hop relay communication. This IE is ignored if the *Remote UE Local ID* IE is present.	YES	reject
UE Multicast MRB Confirmed to Be Modified List		*0..1*			YES	reject
>UE Multicast MRB Confirmed to Be Modified Item IEs		*1 .. <maxnoofMRBsforUE>*			EACH	reject
>>MRB ID	M		9.3.1.224	MRB ID for the UE.	-	
>>MBS PTP Retransmission Tunnel Required	O		9.3.2.10		-	

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRB allowed towards one UE, the maximum value is 64.
maxnoofULUPTNLInformation	Maximum no. of UL UP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofSLDRBs	Maximum no. of SL DRB allowed for NR sidelink communication per UE, the maximum value is 512.
maxnoofAdditionalPDCPDuplicationTNL	Maximum no. of additional UP TNL Information allowed towards one DRB, the maximum value is 2.
maxnoofUuRLCChannels	Maximum no. of Uu Relay RLC channels for L2 U2N relaying or L2 N3C relaying per Relay UE, the maximum value is 32.
maxnoofPC5RLCChannels	Maximum no. of PC5 Relay RLC channels allowed for L2 U2N or L2 U2U relaying per Remote UE or Relay UE, the maximum value is 512.
maxnoofMRBsforUE	Maximum no. of multicast MRB allowed towards one UE, the maximum value is 64.

9.2.2.11A UE CONTEXT MODIFICATION REFUSE

This message is sent by the gNB-CU to indicate the UE context modification was unsuccessful.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.12 UE INACTIVITY NOTIFICATION

This message is sent by the gNB-DU to provide information about the UE activity to the gNB-CU.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
DRB Activity List		*1*			YES	reject
>DRB Activity Item		*1 .. <maxnoofDRBs>*			EACH	reject
>>DRB ID	M		9.3.1.8		-	
>>DRB Activity	O		ENUMERATED (Active, Not active)		-	
SDT Termination Request	O		ENUMERATED (radio link problem, normal, ..., SDT Volume Threshold Crossed)	Indicate the reason of request for termination of the ongoing SDT.	YES	ignore

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRB allowed towards one UE, the maximum value is 64.

9.2.2.13 NOTIFY

This message is sent by the gNB-DU to notify the gNB-CU that the QoS for already established DRBs associated with notification control is not fulfilled any longer or it is fulfilled again.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
DRB Notify List		*1*			YES	reject
>DRB Notify Item IEs		*<1 .. maxnoofDRBs>*			EACH	reject
>>DRB ID	M		9.3.1.8		-	
>>Notification Cause	M		ENUMERATED (Fulfilled, Not-Fulfilled, ..., not fulfilled DL, not fulfilled UL)		-	
>>Current QoS Parameters Set Index	O		Alternative QoS Parameters set Notify Index	Index to the currently fulfilled alternative QoS	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			9.3.1.124	parameters set. Value 0 indicates that NG-RAN cannot even fulfil the lowest alternative parameter set.		
>>TSC Traffic Characteristics Feedback	O		9.3.1.302		YES	ignore

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRB allowed towards one UE, the maximum value is 64.

9.2.2.14 ACCESS SUCCESS

This message is sent by the gNB-DU to inform the gNB-CU of which cell the UE has successfully accessed during conditional handover, conditional PSCell addition, conditional PSCell change, LTM, conditional LTM, or subsequent CPAC.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
NR CGI	M		9.3.1.12		YES	reject

9.2.2.15 DU-CU CELL SWITCH NOTIFICATION

This message is sent by the gNB-DU to inform the gNB-CU about the initiation of the cell switch command to the UE.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Cell ID	M		NR CGI 9.3.1.12		YES	reject
LTM Cell Switch Information		*0..1*			YES	ignore
>Joint or DL TCI State ID	M		OCTET STRING	Includes the *TCI-StateId* IE, as defined in TS 38.331 [8].	-	
>UL TCI State ID	O		OCTET STRING	Includes the *TCI-UL-StateId* IE, as defined in TS 38.331 [8].	-	
TA Information List		*0..1*			YES	ignore
>TA Information Item IEs		*1 .. <maxnoofTAList>*			EACH	ignore
>>Candidate Cell ID	M		NR CGI 9.3.1.12		-	
>>TA Value	M		INTEGER	Indicates the TA	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			(0..4095)	value as defined in TS 38.213 [31].		
>>Tag ID Pointer	O		OCTET STRING	Includes the *tag-Id-ptr* contained in the *TCI-UL-State* IE or the *TCI-State* IE, as defined in TS 38.331 [8].	YES	ignore

Range bound	Explanation
maxnoofTAList	Maximum no. of TA values to be sent, the maximum value is 8.

9.2.2.16 CU-DU CELL SWITCH NOTIFICATION

This message is sent by the gNB-CU to inform the gNB-DU about the initiation of the cell switch command to the UE.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Cell ID	M		NR CGI 9.3.1.12		YES	reject
LTM Cell Switch Information		*0..1*			YES	ignore
>Joint or DL TCI State ID	M		OCTET STRING	Includes the *TCI-StateId* IE, as defined in TS 38.331 [8].	-	
>UL TCI State ID	O		OCTET STRING	Includes the *TCI-UL-StateId* IE, as defined in TS 38.331 [8].	-	
TA Information List		*0..1*			YES	ignore
>TA Information Item IEs		*1 .. <maxnoofTAList>*			EACH	ignore
>>Candidate Cell ID	M		NR CGI 9.3.1.12		-	
>>TA Value	M		INTEGER (0..4095)	Indicates the TA value as defined in TS 38.213 [31].	-	
>>Tag ID Pointer	O		OCTET STRING	Includes the *tag-Id-ptr* contained in the *TCI-UL-State* IE or the *TCI-State* IE, as defined in TS 38.331 [8].	YES	ignore
Last Visited LTM Cells	O		9.3.1.358		YES	ignore

Range bound	Explanation
maxnoofTAList	Maximum no. of TA values to be sent, the maximum value is 8.

9.2.2.17 CU-DU MOBILITY INITIATION REQUEST

This message is sent by the gNB-CU to the gNB-DU to trigger cell switch command and/or early synchronization for the UE.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
CHOICE *Mobility Initiation*	M				YES	reject
>Mobility Trigger						
>>Triggering Indication	M		BIT STRING (SIZE(8))	Indicates the triggering of the CU-DU Mobility Initiation procedure. First bit = early UL synchronization, second bit = early DL synchronization, third bit = cell switch, other bits reserved for future use.	-	
>>Cell Switch Information	C-ifTriggerIndicationCellSwitch				YES	ignore
>>>Candidate Cell with Beam Information	M		9.3.1.348		-	
>>Early UL Sync Information	C-ifTriggerIndicationEarlyULSync				YES	ignore
>>>Candidate Cell with Beam Information List	M		9.3.1.349		-	
>>Early DL Sync Information	C-ifTriggerIndicationEarlyDLSync				YES	ignore
>>>Candidate Cell with Beam Information List	M		9.3.1.349		-	
>Assistance Information						
>>Serving Cell Measurements		*1*			-	
>>>Serving Cell ID	M		NR CGI 9.3.1.12		-	
>>>SSB Index with Measurements List		*1*			-	
>>>>SSB Index with Measurements Item		*1..<maxnoofSSBIndices>*				
>>>>>SSB Index	M		INTEGER (0..63)		-	
>>>>>Selected Measurement Quantities	M		Measurement Quantities 9.3.1.350		-	
>>>CSI-RS Measurements List		*0..1*			YES	ignore
>>>>CSI-RS Measurements Item		*1..<maxnoofCSI-RSs>*				
>>>>>CSI-RS	M		INTEGER		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Resource ID			(0..192)			
>>>>>Selected Measurement Quantities	M		Measurement Quantities 9.3.1.350		-	
>>Candidate Cell Measurements List		*1*			-	
>>>Candidate Cell Measurements Item		*1..<maxnoofCandidateCells>*				
>>>>Candidate Cell ID	M		NR CGI 9.3.1.12		-	
>>>>SSB Index with Measurements List		*1*			-	
>>>>>SSB Index with Measurements Item		*1..<maxnoofSSBIndices>*				
>>>>>>SSB Index	M		INTEGER (0..63)		-	
>>>>>>Selected Measurement Quantities	M		Measurement Quantities 9.3.1.350		-	
>>>>CSI-RS Measurements List		*0..1*			YES	ignore
>>>>>CSI-RS Measurements Item		*1..<maxnoofCSI-RSs>*				
>>>>>>CSI-RS Resource ID	M		INTEGER (0.. 192)		-	
>>>>>>Selected Measurement Quantities	M		Measurement Quantities 9.3.1.350		-	

Condition	Explanation
ifTriggeringIndicationCellSwitch	This IE shall be present if the *Triggering Indication* IE is set to the value "cell switch".
ifTriggeringIndicationEarlyULSync	This IE shall be present if the *Triggering Indication* IE is set to the value "early UL synchronization ".
ifTriggeringIndicationEarlyDLSync	This IE shall be present if the *Triggering Indication* IE is set to the value "early DL synchronization ".

Range bound	Explanation
maxnoofCandidateCells	Maximum no. of Cells configured towards one UE, the maximum value is 8.
maxnoofSSBsIndice	Maximum no. of SSBs that can be served by a NG-RAN node cell. Value is 64.
MaxnoofCSI-RSs	Maximum no. of CSI-RSs that can be served by a NG-RAN node cell. Value is 192.

9.2.2.18 DU-CU CSI-RS COORDINATION REQUEST

This message is sent by the gNB-DU to request the gNB-CU e.g. to activate/deactivate the SP CSI-RS transmissions from specific cells.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
CSI-RS Resource Coordination List		*1*			YES	ignore
>CSI-RS ResourceCoordination Request Item		*1 .. <maxnoofLTMCSI-RSResourceConfig>*			-	
>>LTM CSI Resource Configuration ID	M		INTEGER (0..111)		-	
>>Transmission Request	M		ENUMERATED (activate, deactivate)		-	
>>TCI State Information List		*0..1*		Indicates the TCI states where the semi persistent CSI-RS resource is transmited. The mapping between the CSI-RS Resource indicated by the LTM CSI Resource Configuration ID IE and the TCI state is defined in TS 38.321 [12].	-	
>>>TCI State Information Item		*1 .. < maxnoofLTM-CSI-ResourcesPerSet>*			-	
>>>>Joint or DL TCI State ID	M		OCTET STRING	Includes the *TCI-StateId* IE, as defined in TS 38.331 [8].	-	

Range bound	Explanation
maxnoofLTMCSI-RSResourceConfig	Maximum number of LTM CSI-Resource Configurations. Value is 112.
maxnoofLTM-CSI-ResourcesPerSet	Maximum number of LTM CSI-RS resource per set. Value is 512.

9.2.2.19 DU-CU CSI-RS COORDINATION RESPONSE

This message is sent by the gNB-CU e.g. to inform the gNB-DU about the SP CSI-RS transmissions activation/deactivation result.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
CSI-RS Coordination Result List		*1*			YES	ignore
>CSI-RS Coordination Result Item IEs		*1 .. < maxnoofLTMCSI-RSResour*			-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
		ceConfig >				
>>LTM CSI Resource Configuration ID	M		INTEGER (0..111)		-	
>>Transmission Status	M		ENUMERATED (activated, deactivated)		-	

Range bound	Explanation
maxnoofLTMCSI-RSResourceConfig	Maximum number of LTM CSI-RS Resource Configurations. Value is 112.

9.2.2.20 CU-DU CSI-RS COORDINATION REQUEST

This message is sent by the gNB-CU e.g. to coordinate the gNB-DU to activate/deactivate the SP CSI-RS transmissions from specific cells.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
CSI-RS Coordination Request List		*1*			YES	ignore
>CSI-RS Coordination Request Item		*1 .. <maxnoofLTMCSI-RSResourceConfig>*			-	
>>LTM CSI Resource Configuration ID	M		INTEGER (0..111)		-	
>>Transmission Request	M		ENUMERATED (activate, deactivate)		-	
>>TCI State Information List		*0..1*		Indicates the TCI states where the semi persistent CSI-RS resource is transmited. The mapping between the CSI-RS Resource indicated by the LTM CSI Resource Configuration ID IE and the TCI state is defined in TS 38.321 [12].	-	
>>>TCI State Information Item		*1 .. < maxnoofLTM-CSI-ResourcesPerSet>*			-	
>>>>Joint or DL TCI State ID	M		OCTET STRING	Includes the *TCI-StateId* IE, as defined in TS 38.331 [8].	-	

Range bound	Explanation

maxnoofLTMCSI-RSResourceConfig	Maximum number of LTM CSI-Resource Configurations. Value is 112.
maxnoofLTM-CSI-ResourcesPerSet	Maximum number of LTM CSI-RS resource per set. Value is 512.

9.2.2.21 CU-DU CSI-RS COORDINATION RESPONSE

This message is sent by the gNB-DU e.g. to coordinate the gNB-CU about the SP CSI-RS transmissions activation/deactivation result.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
CSI-RS Coordination Result List		*0..1*			YES	ignore
>CSI-RS Coordination Result Item IEs		*1 .. <maxnoofLTMCSI-RSResourceConfig>*			-	
>>LTM CSI Resource Configuration ID	M		INTEGER (0..111)		-	
>>Transmission Status	M		ENUMERATED (activated, deactivated)		-	

Range bound	Explanation
maxnoofLTMCSI-RSResourceConfig	Maximum number of LTM CSI-Resource Configurations. Value is 112.

9.2.3 RRC Message Transfer messages

9.2.3.1 INITIAL UL RRC MESSAGE TRANSFER

This message is sent by the gNB-DU to transfer the initial layer 3 message to the gNB-CU over the F1 interface.

Direction: gNB-DU →gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
NR CGI	M		9.3.1.12	NG-RAN Cell Global Identifier (NR CGI)	YES	reject
C-RNTI	M		9.3.1.32	C-RNTI allocated at the gNB-DU	YES	reject
RRC-Container	M		9.3.1.6	Includes the *UL-CCCH-Message* message or *UL-CCCH1-Message* message as defined in subclause 6.2 of TS 38.331 [8].	YES	reject
DU to CU RRC Container	O		OCTET STRING	Includes the *CellGroupConfig*	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				IE as defined in subclause 6.3.2 in TS 38.331 [8]. Required at least to carry SRB1 configuration. The ReconfigurationWithSync field is not included in the *CellGroupConfig* IE.		
SUL Access Indication	O		ENUMERATED (true, ...)		YES	ignore
Transaction ID	M		9.3.1.23		YES	ignore
RAN UE ID	O		OCTET STRING (SIZE (8))		YES	ignore
RRC-Container-RRCSetupComplete	O		RRC-Container 9.3.1.6	Includes the *UL-DCCH-Message* message including the RRCSetupComplete message, as defined in subclause 6.2 of TS 38.331 [8].	YES	ignore
NR RedCap UE Indication	O		ENUMERATED (true, ...)		YES	ignore
SDT Information	O		9.3.1.262		YES	ignore
Sidelink Relay Configuration	O		9.3.1.264		YES	ignore
NR eRedCap UE Indication	O		ENUMERATED (true, ...)		YES	ignore

9.2.3.2 DL RRC MESSAGE TRANSFER

This message is sent by the gNB-CU to transfer the layer 3 message to the gNB-DU over the F1 interface.

Direction: gNB-CU →gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
old gNB-DU UE F1AP ID	O		gNB-DU UE F1AP ID 9.3.1.5		YES	reject
SRB ID	M		9.3.1.7		YES	reject
Execute Duplication	O		ENUMERATED (true, ...)		YES	ignore
RRC-Container	M		9.3.1.6	Includes the *DL-DCCH-Message* message as defined in subclause 6.2 of TS 38.331 [8] encapsulated in a PDCP PDU, or the *DL-CCCH-Message* message as defined in subclause 6.2 of TS 38.331 [8].	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
RAT-Frequency Priority Information	O		9.3.1.34		YES	reject
RRC Delivery Status Request	O		ENUMERATED (true, ...)	Indicates whether RRC DELIVERY REPORT procedure is requested for the RRC message.	YES	ignore
UE Context not retrievable	O		ENUMERATED (true, ...)		YES	reject
Redirected RRC message	O		RRC-Container 9.3.1.6	Includes the *UL-CCCH-Message* message as defined in subclause 6.2 of TS 38.331 [8].	YES	reject
PLMN Assistance Info for Network Sharing	O		PLMN Identity 9.3.1.14	This IE is ignored if the *PLMN Index NR Assistance Info for Network Sharing* IE is present.	YES	ignore
New gNB-CU UE F1AP ID	O		gNB-CU UE F1AP ID 9.3.1.4		YES	reject
Additional RRM Policy Index	O		9.3.1.90		YES	ignore
SRB Mapping Info	O		Uu RLC Channel ID 9.3.1.266	This IE contains the mapped Uu Relay RLC CH ID for the Remote UE's SRB0 or SRB1.	YES	ignore
PLMN Index NR Assistance Info for Network Sharing	O		INTEGER (1..maxnoofBPLMNsNR)	Index of the PLMN or NPN in SIB1.	YES	ignore
Area-specific Semi-persistent SRS Positioning Information	O		9.3.1.372		YES	ignore

Range bound	Explanation
maxnoofBPLMNsNR	Maximum no. of PLMN Ids.broadcast in a cell. Value is 12.

9.2.3.3 UL RRC MESSAGE TRANSFER

This message is sent by the gNB-DU to transfer the layer 3 message to the gNB-CU over the F1 interface.

Direction: gNB-DU →gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
SRB ID	M		9.3.1.7		YES	reject
RRC-Container	M		9.3.1.6	Includes the *UL-DCCH-Message* message as defined in subclause 6.2 of TS 38.331 [8], encapsulated in a	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				PDCP PDU. In case of CG-SDT, may include the *UL-CCCH-Message* message or *UL-CCCH1-Message* message as defined in subclause 6.2 of TS 38.331 [8].		
Selected PLMN ID	O		PLMN Identity 9.3.1.14		YES	reject
New gNB-DU UE F1AP ID	O		gNB-DU UE F1AP ID 9.3.1.5		YES	reject

9.2.3.4 RRC DELIVERY REPORT

This message is sent by the gNB-DU to inform the gNB-CU about the delivery status of DL RRC messages.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
RRC Delivery Status	M		9.3.1.71		YES	ignore
SRB ID	M		9.3.1.7		YES	ignore

9.2.4 Warning Message Transmission Messages

9.2.4.1 WRITE-REPLACE WARNING REQUEST

This message is sent by the gNB-CU to request the start or overwrite of the broadcast of a warning message.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
PWS System Information	M		9.3.1.58	This IE includes the system information for public warning, as defined in TS 38.331 [8].	YES	reject
Repetition Period	M		9.3.1.59		YES	reject
Number of Broadcasts Requested	M		9.3.1.60		YES	reject
Cell To Be Broadcast List		*0..1*			YES	reject
>Cell to Be Broadcast Item IEs		*1..<maxCellingNBDU>*			EACH	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>NR CGI	M		9.3.1.12		-	

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.

9.2.4.2 WRITE-REPLACE WARNING RESPONSE

This message is sent by the gNB-DU to acknowledge the gNB-CU on the start or overwrite request of a warning message.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cell Broadcast Completed List		*0..1*			YES	reject
>Cell Broadcast Completed Item IEs		*1.. <maxCellingNBDU>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Dedicated SI Delivery Needed UE List		*0..1*		List of UEs unable to receive system information from broadcast	YES	ignore
>Dedicated SI Delivery Needed UE Item		*1 .. <maxnoofUEIDs>*			EACH	ignore
>>gNB-CU UE F1AP ID	M		9.3.1.4		-	
>>NR CGI	M		9.3.1.12		-	

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.
maxnoofUEIDs	Maximum no. of UEs that can be served by a gNB-DU. Value is 65536.

9.2.4.3 PWS CANCEL REQUEST

This message is forwarded by the gNB-CU to gNB-DU to cancel an already ongoing broadcast of a warning message

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Transaction ID	M		9.3.1.23		YES	reject
Number of Broadcasts Requested	M		9.3.1.60	This IE is not used in this version of the specification	YES	reject
Cell Broadcast To Be Cancelled List		*0..1*			YES	reject
>Cell Broadcast to Be Cancelled Item IEs		*1..<maxCellingNBDU>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	
Cancel-all Warning Messages Indicator	O		ENUMERATED (true, ...)		YES	reject
Notification Information	O			This IE is ignored If the *Cancel-all Warning Messages Indicator* IE is included.	YES	reject
>Message Identifier	M		9.3.1.81		-	
>Serial Number	M		9.3.1.82		-	

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.

9.2.4.4 PWS CANCEL RESPONSE

This message is sent by the gNB-DU to indicate the list of warning areas where cancellation of the broadcast of the identified message was successful and unsuccessful.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cell Broadcast Cancelled List		*0..1*			YES	reject
>Cell Broadcast Cancelled Item IEs		*1..<maxCellingNBDU>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	
>>Number of Broadcasts	M		INTEGER (0..65535)	This IE is set to '0' if valid results are not known or not available. It is set to 65535 if the counter results have overflowed.	-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxCellingNBDU	Maximum no. of cells that can be served by a gNB-DU. Value is 512.

9.2.4.5 PWS RESTART INDICATION

This message is sent by the gNB-DU to inform the gNB-CU that PWS information for some or all cells of the gNB-DU are available if needed.

Direction: gNB-DU →gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
NR CGI List for Restart List		*1*			YES	reject
>NR CGI List for Restart Item IEs		*1..<maxCellingNBDU>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	

Range bound	Explanation
maxCellingNBDU	Maximum no. of cells that can be served by a gNB-DU. Value is 512.

9.2.4.6 PWS FAILURE INDICATION

This message is sent by the gNB-DU to inform the gNB-CU that ongoing PWS operation for one or more cells of the gNB-DU has failed.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
PWS failed NR CGI List		*0..1*			YES	reject
>PWS failed NR CGI Item IEs		*1..<maxCellingNBDU>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	
>>Number of Broadcasts	M		INTEGER (0..65535)	This IE is not used in the specification and is ignored.	-	

Range bound	Explanation
maxCellingNBDU	Maximum no. of cells that can be served by a gNB-DU. Value is 512.

9.2.5 System Information messages

9.2.5.1 SYSTEM INFORMATION DELIVERY COMMAND

This message is sent by the gNB-CU and is used to request the gNB-DU to broadcast the requested *SystemInformation* messages including the Other SI.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
NR CGI	M		9.3.1.12	NR cell identifier	YES	reject
SIType List	M		9.3.1.62		YES	reject
Confirmed UE ID	M		gNB-DU UE F1AP ID 9.3.1.5		YES	reject

9.2.6 Paging messages

9.2.6.1 PAGING

This message is sent by the gNB-CU and is used to request the gNB-DU to page UEs.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
UE Identity Index value	M		9.3.1.39		YES	reject
CHOICE *Paging Identity*	M				YES	reject
>RAN UE Paging identity						
>>RAN UE Paging identity	M		9.3.1.43		-	
>CN UE paging identity						
>>CN UE paging identity	M		9.3.1.44		-	
Paging DRX	O		9.3.1.40	It is defined as the minimum between the RAN UE Paging DRX and CN UE Paging DRX	YES	ignore
Paging Priority	O		9.3.1.41		YES	ignore
Paging Cell List		*1*			YES	ignore
>Paging Cell Item IEs		*1 .. <maxnoofPagingCells>*			EACH	ignore
>>NR CGI	M		9.3.1.12		-	
>>Last Used Cell Indication	O		ENUMERATED (true, ...)		YES	ignore
>>PEI Subgrouping Support Indication	O		ENUMERATED (true, ...)		YES	ignore
>>Recommended SSBs List		*0 .. 1*			YES	ignore
>>>Recommended SSBs List Item		*1 .. < maxnoofSSBAreas >*			YES	ignore
>>>>SSB Index	M		INTEGER (0..63)	Identifier of the recommended SSB beam for paging.	-	
>>LP-WUS Subgrouping Support Indication	O		ENUMERATED (true, ...)		YES	ignore
>>PEI Subgrouping Support Indication – Paging Adaptation	O		ENUMERATED (true, ...)		YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>LP-WUS Supported Band Information	C-ifSubgroupingSupport		9.3.1.376		YES	ignore
Paging Origin	O		9.3.1.79		YES	ignore
RAN UE Paging DRX	O		Paging DRX 9.3.1.40	This IE indicates the RAN paging cycle as defined in TS 38.304 [24].	YES	ignore
CN UE Paging DRX	O		Paging DRX 9.3.1.40	This IE indicates the UE specific paging cycle as defined in TS 38.304 [24].	YES	ignore
NR Paging eDRX Information	O		9.3.1.258		YES	ignore
NR Paging eDRX Information for RRC INACTIVE	O		9.3.1.259		YES	ignore
Paging Cause	O		ENUMERATED (voice, ...)	This IE indicates the paging cause is IMS voice, refer to TS 23.501[21].	YES	ignore
PEIPS Assistance Information	O		9.3.1.269		YES	ignore
UE Paging Capability	O		9.3.1.270		YES	ignore
Extended UE Identity Index Value	O		9.3.1.285	This IE is ignored if the *Further Extended UE Identity Index Value* IE is present.	YES	ignore
Hashed UE Identity Index Value	O		9.3.1.286		YES	ignore
MT-SDT Information	O		9.3.1.289		YES	ignore
NR Paging Long eDRX Information for RRC INACTIVE	O		9.3.1.325		YES	ignore
LP-WUSPS Assistance Information	O		9.3.1.354		YES	ignore
Further Extended UE Identity Index Value	O		9.3.1.355		YES	ignore

Range bound	Explanation
maxnoofPagingCells	Maximum no. of paging cells, the maximum value is 512.
maxnoofSSBAreas	Maximum no. SSB Areas that can be served by a cell. Value is 64.

Condition	Explanation
C-ifSubgroupingSupport	This IE shall be present if the *LP-WUS Subgrouping Support Indication* IE is present.

9.2.7 Trace Messages

9.2.7.1 TRACE START

This message is sent by the gNB-CU to initiate a trace session for a UE.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Trace Activation	M		9.3.1.88		YES	ignore

9.2.7.2 DEACTIVATE TRACE

This message is sent by the gNB-CU to deactivate a trace session.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Trace ID	M		OCTET STRING (SIZE(8))	As per Trace ID in *Trace Activation* IE	YES	ignore

9.2.7.3 CELL TRAFFIC TRACE

This message is sent by the gNB-DU to to transfer trace specific information.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Trace ID	M		OCTET STRING (SIZE(8))	This IE is composed of the following: Trace Reference defined in TS 32.422 [29] (leftmost 6 octets, with PLMN information encoded as in 9.3.1.14), and Trace Recording Session Reference defined in TS 32.422 [29] (last 2 octets).	YES	ignore
Trace Collection Entity IP Address	M		Transport Layer Address 9.3.2.3	For File based Reporting. Defined in TS 32.422 [29]. Should be ignored if URI is present	YES	ignore
Privacy Indicator	O		ENUMERATED (Immediate MDT, Logged MDT, ...)		YES	ignore
Trace Collection Entity URI	O		URI 9.3.2.6	For Streaming based Reporting. Defined in TS 32.422 [29] Replaces Trace	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				Collection Entity IP Address if present		

9.2.8 Radio Information Transfer messages

9.2.8.1 DU-CU RADIO INFORMATION TRANSFER

This message is sent by a gNB-DU to a gNB-CU, to convey radio-related information.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
CHOICE *DU-CU Radio Information Type*	M				YES	ignore
>RIM						
>>DU-CU RIM Information	M		9.3.1.91		-	

9.2.8.2 CU-DU RADIO INFORMATION TRANSFER

This message is sent by a gNB-CU to a gNB-DU, to convey radio-related information.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
CHOICE *CU-DU Radio Information Type*	M				YES	ignore
>RIM						
>>CU-DU RIM Information	M		9.3.1.92		-	

9.2.9 IAB messages

9.2.9.1 BAP MAPPING CONFIGURATION

This message is sent by the gNB-CU to provide the backhaul routing information and/or traffic mapping information to the gNB-DU.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
BH Routing Information Added List		*0...1*			YES	ignore
>BH Routing Information Added List Item		*1.. <maxnoofRoutingEntries>*			EACH	ignore
>>BAP Routing ID	M		9.3.1.110		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>Next-Hop BAP Address	M		BAP Address 9.3.1.111	Indicates the BAP address of the next hop IAB-node or IAB-donor-DU.	-	
>>Non-F1-Terminating IAB-donor Topology Indicator	O		ENUMERATED (true, ...)	If present, indicates that the routing entry applies to the non-F1-terminating IAB-donor topology of the boundary IAB-node.	YES	ignore
BH Routing Information Removed List		*0...1*			YES	ignore
>BH Routing Information Removed List Item		*1.. <maxnoofRoutingEntries>*			EACH	ignore
>>BAP Routing ID	M		9.3.1.110		-	
Traffic Mapping Information	O		9.3.1.95		YES	ignore
Buffer Size Threshold	O		INTEGER ($0..2^{24}-1$)	The buffer size threshold (in kilobytes) for DL local rerouting, triggered by hop-by-hop flow control feedback.	YES	ignore
BAP Header Rewriting Added List		*0...1*			YES	ignore
>BAP Header Rewriting Added List Item		*1.. <maxnoofRoutingEntries>*			EACH	ignore
>>Ingress BAP Routing ID	M		BAP Routing ID 9.3.1.110		-	
>>Egress BAP Routing ID	M		BAP Routing ID 9.3.1.110		-	
>>Non-F1-terminating IAB-donor Topology Indicator	O		ENUMERATED (true, ...)	If present, indicates that the egress BAP Routing ID in the present BAP header rewriting entry pertains to the non-F1-terminating IAB-donor topology of the boundary IAB-node.	-	
Re-routing Enable Indicator	O		ENUMERATED (true, false, ...)		YES	ignore
BAP Header Rewriting Removed List		*0...1*			YES	ignore
>BAP Header Rewriting Removed List Item		*1.. <maxnoofRoutingEntries>*			EACH	ignore
>>Ingress BAP Routing ID	M		BAP Routing ID 9.3.1.110		-	

Range bound	Explanation
maxnoofRoutingEntries	Maximum no. of routing entries, the maximum value is 1024.

9.2.9.2 BAP MAPPING CONFIGURATION ACKNOWLEDGE

This message is sent by the gNB-DU as a response to a BAP MAPPING CONFIGURATION message.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.9.2A BAP MAPPING CONFIGURATION FAILURE

This message is sent by the gNB-DU to indicate a BAP Mapping Configuration Update failure.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to wait	O		9.3.1.13		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.9.3 GNB-DU RESOURCE CONFIGURATION

This message is sent by the gNB-CU to provide the resource configuration for an gNB-DU.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Activated Cells to Be Updated List		*0..1*		List of activated cells served by the IAB-DU or the IAB-donor-DU whose resource configuration is updated	YES	reject
>Activated Cells To Be Updated List Item		*1 .. <maxnoofServedCellsIAB>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	
>>IAB-DU Cell Resource Configuration-Mode-Info	M		9.3.1.279	In the current version of this specification, for FDD, this IE only contains the *gNB-DU Cell Resource Configuration-FDD-UL* IE and the *gNB-DU Cell*	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				Resource Configuration-FDD-DL IE, for TDD, this IE only contains the *gNB-DU Cell Resource Configuration-TDD* IE		
Child-Nodes List		*0..1*		List of child IAB-nodes served by the IAB-DU or IAB-donor-DU.	YES	reject
>Child-Nodes List Item		*1 .. <maxnoofChildIABNodes>*			EACH	reject
>>gNB-CU UE F1AP ID	M		9.3.1.4	Identifier of a descendant node IAB-MT at the IAB-donor-CU.	YES	reject
>>gNB-DU UE F1AP ID	M		9.3.1.5	Identifier of a child-node IAB-MT at an IAB-DU or IAB-donor-DU.	YES	reject
>>Child-Node Cells List		*0..1*		List of cells served by the child-node IAB-DU whose resource configuration is updated.	YES	reject
>>>Child-Node Cells List Item		*1 .. <maxnoofServedCellsIAB >*			EACH	reject
>>>>NR CGI	M		9.3.1.12		-	
>>>>IAB-DU Cell Resource Configuration-Mode-Info	O		9.3.1.279		-	
>>>>IAB STC Info	O		9.3.1.109	STC configuration of child-node IAB-DU's cell.	-	
>>>>RACH Config Common	O		OCTET STRING	Includes the *rach-ConfigCommon* contained in the *BWP-UplinkCommon* IE as defined in subclause 6.3.2 of TS 38.331 [8].	-	
>>>>RACH Config Common IAB	O		OCTET STRING	Includes the IAB-specific *rach-ConfigCommonIAB* contained in the *BWP-UplinkCommon* IE as defined in subclause 6.3.2 of TS 38.331 [8].	-	
>>>>CSI-RS Configuration	O		OCTET STRING	Includes the *NZP-CSI-RS-Resource* IE as defined in subclause 6.3.2 of TS 38.331 [8].	-	
>>>>SR	O		OCTET	Includes the	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Configuration			STRING	*SchedulingRequestResourceConfig* IE as defined in subclause 6.3.2 of TS 38.331 [8].		
>>>>PDCCH Configuration SIB1	O		OCTET STRING	Includes the *PDCCH-ConfigSIB1* IE as defined in subclause 6.3.2 of TS 38.331 [8].	-	
>>>>SCS Common	O		OCTET STRING	Includes the *subCarrierSpacingCommon* contained in the *MIB* message as defined in subclause 6.2.2 of TS 38.331 [8].	-	
>>>>Multiplexing Info	O		9.3.1.108	Contains information on multiplexing with cells configured for co-located IAB-MT.	-	
Neighbour-Node Cells List		*0..1*		List of neighbor node cells.	YES	reject
>Neighbour-Node Cells List Item		*1 .. <maxnoofNeighbourNodeCellsIAB>*			EACH	reject
>>NR CGI	M		9.3.1.12		-	
>>gNB-CU UE F1AP ID	O		9.3.1.4	Identifier of a child-node IAB-MT at an IAB-donor-CU.	-	
>>gNB-DU UE F1AP ID	O		9.3.1.5	Identifier of a child-node IAB-MT at an IAB-DU or IAB-donor-DU.	-	
>>Peer Parent-Node Indicator	O		ENUMERATED (true, ...)	Indicates if the cell belongs to the peer parent IAB-node of the dual connected IAB-node.	-	
>>IAB-DU Cell Resource Configuration-Mode-Info	O		9.3.1.279		-	
>>IAB STC Info	O		9.3.1.109	STC configuration of peer parent-node IAB-DU's cell.	-	
>>RACH Config Common	O		OCTET STRING	Common RACH Configuration of peer parent node IAB-DU's cell. Includes the *rach-ConfigCommon* contained in the *BWP-UplinkCommon* IE as defined in subclause 6.3.2 of	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				TS 38.331 [8].		
>>RACH Config Common IAB	O		OCTET STRING	IAB specific common RACH Configuration of peer parent node IAB-DU's cell. Includes the IAB-specific *rach-ConfigCommonIAB* contained in the *BWP-UplinkCommon* IE as defined in subclause 6.3.2 of TS 38.331 [8].	-	
>>CSI-RS Configuration	O		OCTET STRING	CSI-RS configuration of peer parent node IAB-DU's cell. Includes the *NZP-CSI-RS-Resource* as defined in subclause 6.3.2 of TS 38.331 [8].	-	
>>SR Configuration	O		OCTET STRING	SR configuration of peer parent node IAB-DU's cell. Includes the *SchedulingRequestResourceConfig* IE as defined in subclause 6.3.2 of TS 38.331 [8].	-	
>>PDCCH Configuration SIB1	O		OCTET STRING	PDCCH configuration SIB1 of peer parent node IAB-DU's cell. Includes the *PDCCH-ConfigSIB1* IE as defined in subclause 6.3.2 of TS 38.331 [8].	-	
>>SCS Common	O		OCTET STRING	SCS Common of peer parent node IAB-DU's cell. Includes the *subCarrierSpacingCommon* contained in the *MIB* message as defined in subclause 6.2.2 of TS 38.331 [8].	-	
Serving Cells List		*0..1*		List of serving cells of the co-located IAB-MT.	YES	reject
>Serving Cells List Item		1 .. <maxnoofServingCells>			EACH	reject
>>NR CGI	M		9.3.1.12		-	
>>CHOICE *IAB-MT Cell NA Resource Configuration-Mode-*	O				-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Info						
>>>FDD						
>>>>FDD Info		*1*			-	
>>>>>gNB-DU Cell NA Resource Configuration-FDD-UL	M		gNB-DU Cell Resource Configuration 9.3.1.107	Contains FDD UL NA resource configuration of parent IAB-node's cell for the co-located IAB-MT.	-	
>>>>>gNB-DU Cell NA Resource Configuration-FDD-DL	M		gNB-DU Cell Resource Configuration 9.3.1.107	Contains FDD DL NA resource configuration of parent IAB-node's cell for the co-located IAB-MT.	-	
>>>>>UL Frequency Info	O		NR Frequency Info 9.3.1.17		-	
>>>>>UL Transmission Bandwidth	O		Transmission Bandwidth 9.3.1.15		-	
>>>>>UL Carrier List	O		NR Carrier List 9.3.1.137	If included, the *UL Transmission Bandwidth* IE shall be ignored.	-	
>>>>>DL Frequency Info	O		NR Frequency Info 9.3.1.17		-	
>>>>>DL Transmission Bandwidth	O		Transmission Bandwidth 9.3.1.15		-	
>>>>>DL Carrier List	O		NR Carrier List 9.3.1.137	If included, the *DL Transmission Bandwidth* IE shall be ignored.	-	
>>>TDD					-	
>>>>TDD Info		*1*			-	
>>>>>gNB-DU Cell NA Resource Configuration-TDD	M		gNB-DU Cell Resource Configuration 9.3.1.107	Contains TDD NA resource configuration of parent IAB-node's cell for the co-located IAB-MT.	-	
>>>>>NR Frequency Info	O		9.3.1.17		-	
>>>>>Transmission Bandwidth	O		9.3.1.15		-	
>>>>>Carrier List	O		NR Carrier List 9.3.1.137	If included, the *Transmission Bandwidth* IE shall be ignored.	-	

Range bound	Explanation
maxnoofChildIABNodes	Maximum number of child nodes served by an IAB-DU or IAB-donor-DU. Value is 1024.
maxnoofServedCellsIAB	Maximum number of cells served by an IAB-DU or IAB-donor-DU. Value is 512.
maxnoofNeighbourNodeCellsIAB	Maximum no. of neighbour cells. Value is 1024.
MaxnoofServingCells	Maximum no. of serving cells for IAB-MT. Value is 32

9.2.9.4 GNB-DU RESOURCE CONFIGURATION ACKNOWLEDGE

This message is sent by the gNB-DU to acknowledge the reception of an GNB-DU RESOURCE CONFIGURATION message.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.9.4A GNB-DU RESOURCE CONFIGURATION FAILURE

This message is sent by the gNB-DU to indicate a gNB-DU Resource Configuration Update failure.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to wait	O		9.3.1.13		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.9.5 IAB TNL ADDRESS REQUEST

This message is sent by the gNB-CU to request the allocation of IP addresses for IAB-node(s).

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
IAB IPv4 Addresses Requested	O		IAB TNL Addresses Requested 9.3.1.101		YES	reject
CHOICE *IAB IPv6 Request Type*	O				YES	reject
>IPv6 Address					-	
>>IAB IPv6 Addresses Requested	M		IAB TNL Addresses Requested 9.3.1.101		-	
>IPv6 Prefix					-	
>>IAB IPv6 Address Prefixes Requested	M		IAB TNL Addresses Requested 9.3.1.101		-	
IAB TNL Addresses To Remove List		*0..1*			YES	reject
>IAB TNL Addresses To Remove Item		*1..<maxnoofTLAsIAB>*			EACH	reject
>>IAB TNL Address	M		9.3.1.102		-	
IAB TNL Address Exception	O		9.3.1.229		YES	reject

Range bound	Explanation
maxnoofTLAsIAB	Maximum no. of individual IPv4/IPv6 addresses or IPv6 address

Range bound	Explanation
	prefixes that can be allocated in one procedure execution. The value is 1024.

9.2.9.6 IAB TNL ADDRESS RESPONSE

This message is sent by the gNB-DU to indicate the TNL addresses allocated to IAB-node(s).

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
IAB Allocated TNL Address List		*1*			YES	reject
>IAB Allocated TNL Address Item		*1..<maxnoofTLAsIAB>*			EACH	reject
>>IAB TNL Address	M		9.3.1.102		-	
>>IAB TNL Address Usage	O		ENUMERATED (F1-C, F1-U, Non-F1, ...)	The usage of the allocated IPv4 or IPv6 address or IPv6 address prefix.	-	

Range bound	Explanation
maxnoofTLAsIAB	Maximum no. of IPv6 addresses or IPv6 address prefixes and/or individual IPv4 addresses that can be allocated in one procedure execution. The value is 1024.

9.2.9.6A IAB TNL ADDRESS FAILURE

This message is sent by the gNB-DU to indicate an IAB TNL Address Allocation failure.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to wait	O		9.3.1.13		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.9.7 IAB UP CONFIGURATION UPDATE REQUEST

This message is sent by the gNB-CU to provide the updated UL BH Information or the updated UL UP TNL Information/Address to the gNB-DU.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
UL UP TNL Information to Update List		*0..1*			YES	ignore
>UL UP TNL		*1.. <*			EACH	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Information to Update List Item IEs		*maxnoofULUPTNLInformationforIAB>*				
>>UL UP TNL Information	M		UP Transport Layer Information 9.3.2.1	This field indicates the UL UP TNL Information used before configuration update.	-	
>>New UL UP TNL Information	O		UP Transport Layer Information 9.3.2.1	If present, this field indicates the new UL UP TNL Information used after configuration update.	-	
>>BH Information	M		9.3.1.114		-	
UL UP TNL Address to Update List		*0..1*			YES	ignore
>UL UP TNL Address to Update List Item IEs		*1.. <maxnoofUPTNLAddresses>*			EACH	ignore
>>Old TNL Address	M		Transport Layer Address 9.3.2.3	The old UL UP Transport Layer Address of gNB-CU used for UL F1-U GTP Tunnel before the configuration update.	-	
>>New TNL Address	M		Transport Layer Address 9.3.2.3	The corresponding new UL UP Transport Layer Address that replaces the old one.	-	

Range bound	Explanation
maxnoofULUPTNLInformationforIAB	Maximum no. of UL UP TNL Information allowed towards one IAB node, the maximum value is 32768.
maxnoofUPTNLAddresses	Maximum no. of TNL addresses for F1-U. Value is 8.

9.2.9.8 IAB UP CONFIGURATION UPDATE RESPONSE

This message is sent by the gNB-DU to provide the updated TNL address(es) of the DL F1-U GTP tunnels to the gNB-CU.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore
DL UP TNL Address to Update List		*0..1*			YES	ignore
>DL UP TNL Address to Update List Item IEs		*1.. <maxnoofUPTNLAddresses>*			EACH	ignore
>>Old TNL Address	M		Transport Layer	The old DL UP	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			Address 9.3.2.3	Transport Layer Address of gNB-DU used for DL F1-U GTP tunnel before the configuration update.		
>>New TNL Address	M		Transport Layer Address 9.3.2.3	The corresponding new Transport Layer Address used to replace the old one.	-	

Range bound	Explanation
maxnoofUPTNLAddresses	Maximum no. of TNL addresses for F1-U. Value is 8.

9.2.9.9 IAB UP CONFIGURATION UPDATE FAILURE

This message is sent by the gNB-DU to indicate an IAB UP Configuration Update failure.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to wait	O		9.3.1.13		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.9.10 MIAB F1 SETUP TRIGGERING

This message is sent by the gNB-CU to trigger F1 interface setup from the gNB-DU's co-located target logical gNB-DU to the target F1-terminating IAB-donor-CU.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Target gNB ID	M		Global gNB ID 9.3.1.305		YES	reject
Target gNB IP Address	O		Transport Layer Address 9.3.2.3		YES	ignore
Target SeGW IP Address	O		Transport Layer Address 9.3.2.3		YES	ignore

9.2.9.11 MIAB F1 SETUP OUTCOME NOTIFICATION

This message is sent by the gNB-DU to notify the gNB-CU about the outcome of F1 interface setup between the gNB-DU's co-located target logical gNB-DU and a target F1-terminating IAB-donor-CU.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Transaction ID	M		9.3.1.23		YES	reject
F1 Setup Outcome	M		ENUMERATED (success, failure, ...)		YES	reject
Activated Cells Mapping List		*0..1*			YES	ignore
>Activated Cells List Mapping Item IEs		*1..<maxCellingNBDU>*		List of activated cells.	EACH	ignore
>>NR CGI for Target Logical gNB-DU	M		NR CGI 9.3.1.12	The identity of an activated cell belonging to the target logical gNB-DU of the mobile IAB-node	-	
>>NR CGI for Source Logical gNB-DU	M		NR CGI 9.3.1.12	The identity of an activated cell belonging to the source logical gNB-DU of the mobile IAB-node	-	
Target F1 Terminating IAB-Donor gNB ID	O		Global gNB ID 9.3.1.305	The Global gNB ID of an IAB donor terminates F1 connection towards the target logical gNB-DU of the mobile IAB-node. This IE is present if the mobile IAB-DU migration is triggerd by OAM.	YES	reject

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.

9.2.10 Self Optimisation Support Messages

9.2.10.1 ACCESS AND MOBILITY INDICATION

This message is sent by gNB-CU to gNB-DU to provide access and mobility information to the gNB-DU.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
RA Report List		*0..1*			YES	ignore
>RA Report Item		*1 .. <maxnoofRAReports>*			-	
>>RA Report Container	M		OCTET STRING	Includes the *RA-ReportList-r16* IE as defined in subclause 6.2.2 in TS 38.331 [8].	-	
>>UE Assistant Identifier	O		gNB-DU UE F1AP ID		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			9.3.1.5			
RLF Report Information List		*0..1*			YES	ignore
>RLF Report Information Item		*1 .. <maxnoofRLFReports>*			-	
>>NR UE RLF Report Container	M		OCTET STRING	Includes the *nr-RLF-Report-r16* IE contained in the *UEInformationResponse* message defined in TS 38.331 [8].	-	
>>UE Assistant Identifier	O		gNB-DU UE F1AP ID 9.3.1.5		-	
>>C-RNTI	O		9.3.1.32	C-RNTI allocated at the source gNB-DU. This IE is included in case the gNB-DU responsible for the LTM failure is not the gNB-DU serving the UE at the time of LTM failure.	YES	ignore
>>RLF Report Failure Type	O		ENUMERATED (too late LTM, too early LTM, LTM to wrong cell,...)		YES	ignore
Successful HO Report Information List		*0..1*			YES	ignore
>Successful HO Report Information Item		*1 .. <maxnoofSuccessfulHOReports>*			-	
>>Successful HO Report Container	M		OCTET STRING	Includes the *SuccessHO-Report* IE as defined in subclause 6.2.2 in TS 38.331 [8].	-	
Successful PSCell Change Report Information List		*0..1*			YES	ignore
>Successful PSCell Change Report information Item		*1..<maxnoofSuccessfulPSCellChangeReports>*			-	
>>Successful PSCell Change Report Container	M		OCTET STRING	Includes the *SuccessPSCell-Report* IE as defined in TS 38.331 [8].	-	
Failure Reporting without RLF report		*0..1*			YES	ignore
>gNB-DU UE F1AP ID	M		9.3.1.5		-	
>Re-establishment or Recovery cell ID	M		NR CGI 9.3.1.12	NR CGI for the re-establishment or recovery cell.	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>Failure Type	M		ENUMERATED (too late LTM, too early LTM, LTM to wrong cell,...)		-	
MRO for LTM Information		*0..1*			YES	ignore
>gNB-CU or gNB-DU UE F1AP ID	M		9.3.1.371	Contains the gNB-DU UE F1AP ID	-	
>BFR SSB Index	O		INTEGER (0..63)	SSB Index of the recovery beam or SSB Index of the beam which is QCLed with CSI-RS resources used at successful Beam Failure Recovery.	-	
>Target SSB Index after Cell Switch Failure	O		INTEGER (0..63)	SSB Index of the re-established or recovery beam after LTM Cell Switch Failure.	-	
>TA Information	O		INTEGER (0..4095)	Indicates the TA value, as defined in TS 38.213 [31], used at successful Random Access during LTM recovery or re-establishment after a Cell Switch failure in same cell.	-	

Range bound	Explanation
maxnoofRAReports	Maximum no. of RA Reports, the maximum value is 64.
maxnoofRLFReports	Maximum no. of RLF Reports, the maximum value is 64.
maxnoofSuccessfulHOReports	Maximum no. of Successful HO Reports, the maximum value is 64.
maxnoofSuccessfulPSCellChangeReports	Maximum no. of Successful PSCell Change Reports. Value is 64.

9.2.10.2 DU-CU ACCESS AND MOBILITY INDICATION

This message is sent by the gNB-DU to provide access and mobility information to the gNB-CU.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
DL LBT Failure Information List		*0..1*			YES	ignore
> DL LBT Failure Information Item		*1 .. <maxnoofLBTFailureInformation>*			-	
>>DL LBT Failure Information	M		9.3.1.327		-	
MRO for LTM Information		*0..1*			YES	ignore
>gNB-CU or gNB-DU	M		9.3.1.371	Contains the gNB-	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
UE F1AP ID				CU UE F1AP ID		
>BFR SSB Index	O		INTEGER (0..63)	SSB Index of the recovery beam or SSB Index of the beam which is QCLed with CSI-RS resources used at successful Beam Failure Recovery.	-	
>Target SSB Index after Cell Switch Failure	O		INTEGER (0..63)	SSB Index of the re-established or recovery beam after LTM Cell Switch Failure.	-	
>TA Information	O		INTEGER (0..4095)	Indicates the TA value, as defined in TS 38.213 [31], used at successful Random Access during LTM recovery or re-establishment after a Cell Switch failure in same cell.	-	

Range bound	Explanation
maxnoofLBTFailureInformation	Maximum no. of UEs for which LBT Failure Information is provided, the maximum value is 64.

9.2.11 Reference Time Information Reporting messages

9.2.11.1 REFERENCE TIME INFORMATION REPORTING CONTROL

This message is sent by the gNB-CU and is used to request the gNB-DU to deliver the accurate reference time information.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
Reporting Request Type	M		9.3.1.147		YES	reject

9.2.11.2 REFERENCE TIME INFORMATION REPORT

This message is sent by the gNB-DU and is used to report the accurate reference time information to the gNB-CU.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	ignore
Time Reference Information	M		9.3.1.148		YES	ignore

9.2.12 Messages for Positioning Procedures

9.2.12.1 POSITIONING ASSISTANCE INFORMATION CONTROL

This message is sent by the gNB-CU to transfer positioning assistance information.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
Positioning Assistance Information	O		OCTET STRING	Contains the *Assistance Information* IE as defined in TS 38.455 [37].	YES	reject
Broadcast	O		ENUMERATED (start, stop, ...)		YES	reject
Positioning Broadcast Cells	O		9.3.1.191		YES	reject
Routing ID	O		OCTET STRING		YES	reject

9.2.12.2 POSITIONING ASSISTANCE INFORMATION FEEDBACK

This message is sent by the gNB-DU to give feedback on positioning assistance information broadcasting.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
Positioning Assistance Information Failure List	O		OCTET STRING	Contains the *Assistance Information Failure List* IE as defined in TS 38.455 [37].	YES	reject
Positioning Broadcast Cells	O		9.3.1.191	The cells associated to the feedback provided in the *Positioning Assistance Information Failure List* IE.	YES	reject
Routing ID	O		OCTET STRING		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.12.3 POSITIONING MEASUREMENT REQUEST

This message is sent by the gNB-CU to request the gNB-DU to configure a positioning measurement.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
LMF Measurement ID	M		INTEGER (1..65536, ...)		YES	reject
RAN Measurement ID	M		INTEGER		YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			(1..65536, ...)			
TRP Measurement Request List		1			YES	reject
>TRP Measurement Request Item		1..<maxnoofMeasTRPs>			-	
>>TRP ID	M		9.3.1.197		-	
>>Search Window Information	O		9.3.1.204		-	
>>NR CGI	O		9.3.1.12	The Cell ID of the TRP identified by the *TRP ID* IE.	YES	ignore
>>AoA Search Window Information	O		UL-AoA Assistance Information 9.3.1.238		YES	ignore
>>Number of TRP Rx TEGs	O		ENUMERATED (2, 3, 4, 6, 8, ...)		YES	ignore
>>Number of TRP RxTx TEGs	O		ENUMERATED (2, 3, 4, 6, 8, ...)		YES	ignore
Positioning Report Characteristics	M		ENUMERATED (OnDemand, Periodic, ...)		YES	reject
Positioning Measurement Periodicity	C-ifReportCharacteristicsPeriodic		ENUMERATED (120ms, 240ms, 480ms, 640ms, 1024ms, 2048ms, 5120ms, 10240ms, 1min, 6min, 12min, 30min, ..., 20480ms, 40960ms, extended)	The codepoint 120ms, 240ms, 480ms, 1024ms, 2048ms, 1min, 6min, 12min, and 30min are not applicable.	YES	reject
Positioning Measurement Quantities		*1*			YES	reject
>Positioning Measurement Quantities Item		*1..<maxnoofPosMeas>*			EACH	
>>Positioning Measurement Type	M		ENUMERATED (gNB RX-TX, UL-SRS-RSRP, UL AoA, UL RTOA, ..., Multiple UL AoA, UL SRS-RSRPP, UL-RSCP, UL SRS-TDCT)	The UL-RSCP measurement is applicable only when the UL-RTOA and/or gNB-RxTxTimeDiff measurement(s) is also requested.	-	
>>Timing Reporting Granularity Factor	O		INTEGER (0..5)	TS 38.133 [38] This IE is ignored when the *Timing Reporting Granularity Factor Extended* IE is included.	-	
>>Timing Reporting Granularity Factor Extended	O		INTEGER (-6..-1, ...)	Value -6 corresponds to kminus6, value -5 corresponds to kminus5, and so on, see TS 38.133 [38].	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>Channel Response Information	O		9.3.1.364	Applicable to UL SRS-TDCT only.	YES	ignore
SFN Initialisation Time	O		Relative Time 1900 9.3.1.183	If this IE is not present, the TRP may assume that the value is same as its own SFN initialisation time.	YES	ignore
SRS Configuration	O		9.3.1.192		YES	ignore
Measurement Beam Information Request	O		ENUMERATED (true, ...)	This IE is ignored when the Measurement characteristics Request Indicator IE is included.	YES	ignore
System Frame Number	O		INTEGER(0..1023)		YES	ignore
Slot Number	O		INTEGER(0..79)		YES	ignore
Measurement Periodicity Extended	C-ifMeasPerExt		ENUMERATED (160ms, 320ms, 1280ms, 2560ms, 61440ms, 81920ms, 368640ms, 737280ms, 1843200ms, ...)		YES	reject
Response Time	O		9.3.1.242	This IE is ignored when the *Positioning Report Characteristics* IE is set to “Periodic”.	YES	ignore
Measurement Characteristics Request Indicator	O		9.3.1.254		YES	ignore
Measurement Time Occasion	O		ENUMERATED (o1, o4,...)		YES	ignore
Positioning Measurement Amount	O		ENUMERATED (0, 1, 2, 4, 8, 16, 32, 64)	This IE is ignored if the *Positioning Report Characteristics* IE is set to ‘OnDemand’. Value 0 represents an infinite number of periodic reporting.	YES	ignore
Time Window Information Measurement List	O		9.3.1.334		YES	ignore

Range bound	Explanation
maxnoofPosMeas	Maximum no. of measured quantities that can be configured and reported with one message. Value is 16384.
maxnoofMeasTRPs	Maximum no. of TRPs that can be included within one measurement message. Value is 64.

Condition	Explanation
ifReportCharacteristicsPeriodic	This IE shall be present if the *Positioning Report Characteristics* IE is set to the value "Periodic".
ifMeasPerExt	This IE shall be present if the *Positioning Measurement Periodicity* IE is set to the value "extended".

9.2.12.4 POSITIONING MEASUREMENT RESPONSE

This message is sent by the gNB-DU to report positioning measurements.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
LMF Measurement ID	M		INTEGER (1..65536, ...)		YES	reject
RAN Measurement ID	M		INTEGER (1..65536, ...)		YES	reject
Positioning Measurement Result List		0..1			YES	reject
>Positioning Measurement Result List Item		1..<maxnoofMeasTRPs>			-	
>>Positioning Measurement Result	M		9.3.1.166		-	
>>TRP ID	M		9.3.1.197		-	
>>NR CGI	O		9.3.1.12	The Cell ID of the TRP identified by the TRP ID IE.	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofMeasTRPs	Maximum no. of TRP measurements that can be included within one message. Value is 64.

9.2.12.5 POSITIONING MEASUREMENT FAILURE

This message is sent by the gNB-DU to report measurement failure.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
LMF Measurement ID	M		INTEGER (1..65536, ...)		YES	reject
RAN Measurement ID	M		INTEGER (1..65536, ...)		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.12.6 POSITIONING MEASUREMENT REPORT

This message is sent by the gNB-DU to report positioning measurements for the target UE.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
LMF Measurement ID	M		INTEGER (1..65536, ...)		YES	reject
RAN Measurement ID	M		INTEGER (1..65536, ...)		YES	reject
Positioning Measurement Result List		*1*			YES	reject
>Positioning Measurement Result List Item		*1..<maxnoofMeasTRPs>*			EACH	
>>Positioning Measurement Result	M		9.3.1.166		-	
>>TRP ID	M		9.3.1.197		-	
>>NR CGI	O		9.3.1.12	The Cell ID of the TRP identified by the *TRP ID* IE.	YES	ignore

Range bound	Explanation
maxnoofMeasTRPs	Maximum no. of TRP measurements that can be included within one message. Value is 64.

9.2.12.7 POSITIONING MEASUREMENT ABORT

This message is sent by the gNB-CU to request the gNB-DU to abort a positioning measurement.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
LMF Measurement ID	M		INTEGER (1.. 65536,...)		YES	reject
RAN Measurement ID	M		INTEGER (1..65536,...)		YES	reject

9.2.12.8 POSITIONING MEASUREMENT FAILURE INDICATION

This message is sent by the gNB-DU to indicate that the previously requested positioning measurements can no longer be reported.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
LMF Measurement ID	M		INTEGER (1.. 65536,...)		YES	reject
RAN Measurement ID	M		INTEGER (1..65536,...)		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.12.9 POSITIONING MEASUREMENT UPDATE

This message is sent by the gNB-CU to update a previously configured measurement.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
LMF Measurement ID	M		INTEGER (1..65536,...)		YES	reject
RAN Measurement ID	M		INTEGER (1..65536,...)		YES	reject
SRS Configuration	O		9.3.1.192		YES	ignore
TRP Measurement Update List		*0..1*			YES	reject
>TRP Measurement Update Item		*1..<maxnoofMeasTRPs>*			EACH	reject
>>TRP ID	M		9.3.1.197		-	
>>AoA Search Window Information	O		UL-AoA Assistance Information 9.3.1.238		-	
>>Number of TRP Rx TEGs	O		ENUMERATED (2, 3, 4, 6, 8, ...)		YES	ignore
>>Number of TRP RxTx TEGs	O		ENUMERATED (2, 3, 4, 6, 8, ...)		YES	ignore
Measurement Characteristics Request Indicator	O		9.3.1.254		YES	ignore
Measurement Time Occasion	O		ENUMERATED (o1, o4, ...)		YES	ignore

Range bound	Explanation
maxnoofMeasTRPs	Maxmum no. of TRPs that can be included within one message. Value is 64.

9.2.12.10 TRP INFORMATION REQUEST

This message is sent by a gNB-CU to request information for TRPs hosted by a gNB-DU.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
TRP list		*0..1*			YES	ignore
>TRP list Item		1..<maxnoofTRPs>			EACH	ignore
>>TRP ID	M		9.3.1.197		-	
TRP Information Type List		*1*			YES	reject
>TRP Information Type Item		*1 .. <maxnoofTRPInfoTypes>*			EACH	reject
>>TRP Information Type Item	M		ENUMERATED (nr pci, ng-ran cgi, nr arfcn, prs config, ssb config, sfn init time, spatial direction info, geo-coordinates, ..., trp type, on-demand prs, trp		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			Tx teg, beam antenna info, mobile TRP location info)			

Range bound	Explanation
maxnoofTRPInfoTypes	Maximum no of TRP information types that can be requested and reported with one message. Value is 64.
maxnoofTRPs	Maximum no. of TRPs in a gNB. Value is 65535.

9.2.12.11 TRP INFORMATION RESPONSE

This message is sent by a gNB-DU to convey TRP information to a gNB-CU.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
TRP Information List		*1*			YES	ignore
>TRP Information Item		*1 .. <maxnoofTRPs>*			EACH	ignore
>>TRP Information	M		9.3.1.176		-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofTRPs	Maximum no. of TRPs in a gNB-DU. Value is 65535.

9.2.12.12 TRP INFORMATION FAILURE

This message is sent by a gNB-DU node to indicate that the requested TRP information cannot be provided to a gNB-CU.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.12.13 POSITIONING INFORMATION REQUEST

This message is sent by the gNB-CU to indicate to the gNB-DU the need to configure the UE to transmit SRS signals for uplink positioning measurement and also to retrieve the SRS configuration from the gNB-DU.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Requested SRS	O		9.3.1.175		YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Transmission Characteristics						
UE Reporting Information	O		9.3.1.255		YES	ignore
SRS Positioning INACTIVE Query Indication	O		ENUMERATED (true, ...)	Applicable only if the *Requested SRS Transmission Characteristics* IE is present	YES	ignore
Time Window Information SRS List	O		9.3.1.333		YES	ignore
Requested SRS Preconfiguration Characteristics List	O		9.3.1.340		YES	ignore

9.2.12.14 POSITIONING INFORMATION RESPONSE

This message is sent by the gNB-DU to provide the configured SRS information to the gNB-CU.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
SRS Configuration	O		9.3.1.192		YES	ignore
SFN Initialisation Time	O		Relative Time 1900 9.3.1.183		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
SRS-PosRRC-InactiveConfig	O		OCTET STRING	Includes the *SRS-PosRRC-InactiveConfig* IE as defined in TS 38.331 [8]	YES	ignore
SRS-PosRRC-InactiveValidityAreaConfig	O		OCTET STRING	Includes the *SRS-PosRRC-InactiveValidityAreaConfig* IE as defined in TS 38.331 [8].	YES	ignore
SRS Preconfiguration List	O		9.3.1.341		YES	ignore

9.2.12.15 POSITIONING INFORMATION FAILURE

This message is sent by the gNB-DU to indicate that no SRS transmissions could be configured for the UE for uplink positioning measurement.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.12.16 POSITIONING ACTIVATION REQUEST

This message is sent by the gNB-CU to cause the gNB-DU to activate/trigger UL SRS transmission by the UE.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
CHOICE *SRS type*	M				YES	reject
>Semi-persistent						
>>SRS Resource Set ID	M		9.3.1.180	This IE is ignored if the *Aggregated Positioning SRS Resource Set List* IE below is present.	-	
>>SRS Spatial Relation	O		Spatial Relation Information 9.3.1.181	This IE is ignored if the *Spatial Relation Information per SRS Resource* IE is present.	-	
>>Spatial Relation Information per SRS Resource	O		9.3.1.210		YES	ignore
>>Aggregated Positioning SRS Resource Set List	O		9.3.1.337		YES	ignore
>Aperiodic						
>>Aperiodic	M		ENUMERATED (true, ...)		-	
>>SRS Resource Trigger	O		9.3.1.182		-	
>>Aggregated Positioning SRS Resource Set List	O		9.3.1.337		YES	ignore
Activation Time	O		Relative Time 1900 9.3.1.183	Indicates the start time when the SRS activation is requested	YES	ignore

9.2.12.17 POSITIONING ACTIVATION RESPONSE

This message is sent by the gNB-DU to confirm successful UL SRS activation in the UE.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
System Frame Number	O		INTEGER(0..1023)		YES	ignore
Slot Number	O		INTEGER(0..79)		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.12.18 POSITIONING ACTIVATION FAILURE

This message is sent by the gNB-DU to indicate that activation of UL SRS transmission in the UE was unsuccessful.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.12.19 POSITIONING DEACTIVATION

This message is sent by the gNB-CU to cause the NG RAN node to deactivate UL SRS transmission or release all the transmission by the UE.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
CHOICE *Abort Transmission*	M				YES	ignore
>SRS Resource Set ID deactivation						
>>SRS Resource Set ID	M		9.3.1.180		-	
>Release ALL			NULL			
>Deactivate aggregated SRS Resource Set						
>>Aggregated Positioning SRS Resource Set List	M		9.3.1.337		YES	ignore

9.2.12.20 E-CID MEASUREMENT INITIATION REQUEST

This message is sent by gNB-CU to initiate E-CID measurements.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
LMF UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject
RAN UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject
E-CID Report Characteristics	M		ENUMERATED (OnDemand, Periodic, ...)		YES	reject
E-CID Measurement Periodicity	C-ifReportCharacteristicsPeriodic		ENUMERATED (120ms, 240ms, 480ms, 640ms, 1024ms, 2048ms, 5120ms, 10240ms, 1min,	The codepoint “extended” is not applicable. This IE is not applicable to NR Angle of Arrival.	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			6min, 12min, 30min, ..., 20480ms, 40960ms, extended)			
E-CID Measurement Quantities		*1 .. <maxnoofMeasE-CID>*			EACH	reject
>E-CID Measurement Quantities Item	M		ENUMERATED (Default, NR Angle of Arrival, ..., NR Timing Advance, NR Angle of Arrival per TRP)	If "Default" is the only requested measurement quantity, it indicates that the *Measured Results List* IE need not be included in response or reporting messages.	-	
Measurement Periodicity NR-AoA	C-ifReportCharacteristicsPeriodicAndMeasQuantityItemAoA		ENUMERATED (160ms, 320ms, 640ms, 1280ms, 2560ms, 5120ms, 10240ms, 20480ms, 40960ms, 61440ms, 81920ms, 368640ms, 737280ms, 1843200ms, ...)		YES	reject

Range bound	Explanation
maxnoofMeasE-CID	Maximum no. of E-CID measured quantities that can be configured and reported with one message. Value is 64.

Condition	Explanation
ifReportCharacteristicsPeriodic	This IE shall be present if the E-CID Report Characteristics IE is set to the value "Periodic".
ifReportCharacteristicsPeriodicAndMeasQuantityItemAoA	This IE shall be present if the E-CID Report Characteristics IE is set to the value "Periodic" and the E-CID Measurement Quantities Item IE is set to the value "NR Angle of Arrival".

9.2.12.21 E-CID MEASUREMENT INITIATION RESPONSE

This message is sent by gNB-DU to indicate that the requested E-CID measurement is successfully initiated.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
LMF UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject
RAN UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
E-CID Measurement Result	O		9.3.1.199		YES	ignore
Cell Portion ID	O		9.3.1.200		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.12.22 E-CID MEASUREMENT INITIATION FAILURE

This message is sent by gNB-DU to indicate that the requested E-CID measurement cannot be initiated.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
LMF UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject
RAN UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.12.23 E-CID MEASUREMENT FAILURE INDICATION

This message is sent by gNB-DU to indicate that the previously requested E-CID measurement can no longer be reported.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
LMF UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject
RAN UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.12.24 E-CID MEASUREMENT REPORT

This message is sent by gNB-DU to report the results of the requested E-CID measurement.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
LMF UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject
RAN UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject
E-CID Measurement Result	M		9.3.1.199		YES	ignore
Cell Portion ID	O		9.3.1.200		YES	ignore

9.2.12.25 E-CID MEASUREMENT TERMINATION COMMAND

This message is sent by the gNB-CU to terminate the requested E-CID measurement.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
LMF UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject
RAN UE Measurement ID	M		INTEGER (1.. 256, ...)		YES	reject

9.2.12.26 POSITIONING INFORMATION UPDATE

This message is sent by the gNB-DU to indicate that a change in the SRS configuration has occurred.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
SRS configuration	O		9.3.1.192		YES	ignore
SFN Initialisation Time	O		Relative Time 1900 9.3.1.183		YES	ignore

9.2.12.27 PRS CONFIGURATION REQUEST

This message is sent by a gNB-CU to request a gNB-DU to configure or update PRS transmissions.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
PRS Configuration Request Type	M		ENUMERATED (configure, off, ...)		YES	reject
PRS TRP List		*1*			YES	ignore
>PRS TRP Item		*1 .. <maxnoofTRPs>*			EACH	ignore
>>TRP ID	M		9.3.1.197		-	
>>Requested DL PRS Transmission Characteristics	C-ifConf		9.3.1.235		-	
>>PRS Transmission Off Information	C-ifOff		9.3.1.237		-	

Range bound	Explanation
maxnoofTRPs	Maximum no. of TRPs in a gNB-DU Value is 65535

Condition	Explanation
ifConf	This IE shall be present if the *PRS Configuration Request Type* IE is set to the value "configure".
ifOff	This IE shall be present if the *PRS Configuration Request Type* IE is set to the value "off".

9.2.12.28 PRS CONFIGURATION RESPONSE

This message is sent by a gNB-DU to acknowledge configuring or updating the PRS transmissions.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
PRS Transmission TRP List		*0..1*			YES	ignore
>PRS Transmission TRP Item		*1 .. <maxnoofTRPs>*			EACH	ignore
>>TRP ID	M		9.3.1.197		-	
>>PRS Configuration	M		9.3.1.177		-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofTRPs	Maximum no. of TRPs in a gNB-DU Value is 65535.

9.2.12.29 PRS CONFIGURATION FAILURE

This message is sent by the gNB-DU to indicate that it cannot configure any PRS transmission.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.12.30 MEASUREMENT PRECONFIGURATION REQUIRED

This message is sent by a gNB-CU to provide the PRS configuration information of multiple TRPs to a gNB-DU and request to configure measurement gap or PRS processing window of the UE.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
TRP PRS Information List		*1*			YES	ignore
>TRP PRS Information Item		*1 .. <maxnoTRPs>*			EACH	ignore
>>TRP ID	M		9.3.1.197		-	
>>NR PCI	M		INTEGER (0..1007)		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>NR CGI	O		9.3.1.12		-	
>>PRS Configuration	M		9.3.1.177		-	

Range bound	Explanation
maxnoofTRPs	Maximum no. of TRPs for on-demand PRS in a gNB-DU Value is 256

9.2.12.31 MEASUREMENT PRECONFIGURATION CONFIRM

This message is sent by an gNB-DU to gNB-CU to confirm successful configuration of measurement gap or PRS processing window of the UE.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
PosMeasGapPreConfigList		*0..1*			YES	ignore
>PosMeasGapPreConfigToAddModList	O		OCTET STRING	Includes the *PosMeasGapPreConfigToAddModList* IE as defined in TS 38.331 [8]	YES	ignore
>PosMeasGapPreConfigToReleaseList	O		OCTET STRING	Includes the *PosMeasGapPreConfigToReleaseList* IE as defined in TS 38.331 [8]	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.12.32 MEASUREMENT PRECONFIGURATION REFUSE

This message is sent by gNB-DU to indicate configuration of measurement gap or PRS processing window of the UE was unsuccessful.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.12.33 MEASUREMENT ACTIVATION

This message is sent by the gNB-CU to request the gNB-DU to activate or deactivate the preconfigured measurement gap or PRS processing window for the UE.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
Request Type	M		ENUMERATED (activate, deactivate, ...)		YES	reject
PRS Measurement Info List		*0..1*			YES	ignore
>PRS Measurement Info Item		*1 .. <maxFreqLayers>*			-	
>>Point A	M		INTEGER (0..3279165)		-	
>>MeasPRS Periodicity	M		ENUMERATED (ms20, ms40, ms80, ms160, ...)	Measurement gap periodicity in units of ms	-	
>>MeasPRS Offset	M		INTEGER (0..159, ...)	Measurement gap offset in units of subframes	-	
>>Measurement PRS Length	M		ENUMERATED {ms1dot5, ms3, ms3dot5, ms4, ms5dot5, ms6, ms10, ms20}		-	

Range bound	Explanation
maxFreqLayers	Maximum no. of frequency layers. Value is 4

9.2.12.34 POSITIONING SYSTEM INFORMATION DELIVERY COMMAND

This message is sent by the gNB-CU and is used to request the gNB-DU to broadcast the indicated positioning SI message.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
NR CGI	M		9.3.1.12	NR cell identifier	YES	reject
PosSIType List	M		9.3.1.278		YES	reject
Confirmed UE ID	M		gNB-DU UE F1AP ID 9.3.1.5		YES	reject

9.2.12.35 SRS INFORMATION RESERVATION NOTIFICATION

This message is sent by the gNB-CU to notify the gNB-DU to reserve or release SRS resources in a Validity Area.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		-	
SRS Reservation Type	M		ENUMERATED (reserve, release, ...)		YES	reject
SRS Information	O		Requested SRS Transmission Characteristics 9.3.1.175		YES	ignore
Preconfigured SRS Information	O		Requested SRS Preconfiguration Characteristics List 9.3.1.340		YES	ignore

9.2.13 Broadcast Context Management messages

9.2.13.1 BROADCAST CONTEXT SETUP REQUEST

This message is sent by the gNB-CU to request the setup of an MBS session context for a broadcast session, and establish an MBS-associated logical F1-connection.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
MBS Session ID	M		9.3.1.218		YES	reject
MBS Service Area	O		9.3.1.222		YES	reject
MBS CU to DU RRC Information	M		9.3.1.225		YES	reject
S-NSSAI	M		9.3.1.38		YES	reject
Broadcast MRB To Be Setup List		*1*			YES	reject
>Broadcast MRB to Be Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>MRB QoS Information	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>MBS QoS Flows Mapped to MRB Item		*1 .. <maxnoofMBSQoSFlows>*			-	
>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.63		-	
>>>MBS QoS Flow Level QoS Parameters	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>BC Bearer Context F1-U TNL Info at CU	M		BC Bearer Context F1-U TNL Info 9.3.2.7	gNB-CU endpoint(s) of the F1 transport bearer(s). For delivery of F1-U PDU Type 1.	-	
Supported UE Type List	O		9.3.1.290		YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Associated Session ID	O		9.3.1.309		YES	ignore
RAN Sharing Assistance Information	O		9.3.1.345		YES	ignore

Range bound	Explanation
maxnoofMRBs	Maximum no. of MRB allowed to be setup for one MBS Session, the maximum value is 32.
maxnoofMBSQoSFlows	Maximum no. of flows allowed to be mapped to one MRB, the maximum value is 64.

9.2.13.2 BROADCAST CONTEXT SETUP RESPONSE

This message is sent by the gNB-DU to confirm the setup of a broadcast context.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Broadcast MRB Setup List		*1*			YES	reject
>Broadcast MRB Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>BC Bearer Context F1-U TNL Info at DU	M		BC Bearer Context F1-U TNL Info 9.3.2.7	gNB-DU endpoint(s) of the F1-U transport bearer(s). For delivery of DL PDUs.	-	
Broadcast MRB Failed To Be Setup List		*0..1*			YES	ignore
>Broadcast MRB Failed To Be Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	ignore
>>MRB ID	M		9.3.1.224		-	
>>Cause	O		9.3.1.2		-	
Broadcast Area Scope	O		9.3.1.287		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofMRBs	Maximum no. of MRB allowed to be setup for one MBS Session, the maximum value is 32.

9.2.13.3 BROADCAST CONTEXT SETUP FAILURE

This message is sent by the gNB-DU to indicate that the setup of the broadcast context was unsuccessful.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
gNB-DU MBS F1AP ID	O		9.3.1.220		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.13.4 BROADCAST CONTEXT RELEASE COMMAND

This message is sent by the gNB-CU to request the gNB-DU to release the broadcast context for a given broadcast service.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.13.5 BROADCAST CONTEXT RELEASE COMPLETE

This message is sent by the gNB-DU to confirm the release of the broadcast context for a given broadcast service.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.13.5a BROADCAST CONTEXT RELEASE REQUEST

This message is sent by the gNB-DU to request the gNB-CU to trigger the Broadcast Context Release procedure.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.13.6 BROADCAST CONTEXT MODIFICATION REQUEST

This message is sent by the gNB-CU to request the gNB-DU to modify broadcast context information.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
MBS Service Area	O		9.3.1.222	Overwrites any previously received MBS	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				Service Area information		
MBS CU to DU RRC Information	M		9.3.1.225		YES	reject
Broadcast MRB To Be Setup List		*0..1*			YES	reject
>Broadcast MRB to Be Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>MRB QoS Information	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>MBS QoS Flows Mapped to MRB Item		*1 .. <maxnoofMBSQoSFlows>*			-	
>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.63		-	
>>>MBS QoS Flow Level QoS Parameters	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>BC Bearer Context F1-U TNL Info at CU	M		BC Bearer Context F1-U TNL Info 9.3.2.7	gNB-CU endpoint(s) of the F1 transport bearer(s). For delivery of F1-U PDU Type 1.	-	
Broadcast MRB To Be Modified List		*0..1*			YES	reject
>Broadcast MRB to Be Modified Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>MRB QoS Information	O		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>MBS QoS Flows Mapped to MRB Item		*0 .. <maxnoofMBSQoSFlows>*			-	
>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.63		-	
>>>MBS QoS Flow Level QoS Parameters	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>BC Bearer Context F1-U TNL Info at CU	O		BC Bearer Context F1-U TNL Info 9.3.2.7	Updated gNB-CU endpoint(s) of the F1 transport bearer(s). For delivery of F1-U PDU Type 1.	-	
Broadcast MRB To Be Released List		*0..1*			YES	reject
>Broadcast MRB to Be Released Item IEs		*1 .. <maxnoofMRBs>*			YES	reject
>>MRB ID	M		9.3.1.224		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Supported UE Type List	O		9.3.1.290		YES	ignore

Range bound	Explanation
maxnoofMRBs	Maximum no. of MRB allowed to be setup for one MBS Session, the maximum value is 32.
maxnoofMBSQoSFlows	Maximum no. of flows allowed to be mapped to one MRB, the maximum value is 64.

9.2.13.7 BROADCAST CONTEXT MODIFICATION RESPONSE

This message is sent by the gNB-DU to confirm the modification of a broadcast context.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Broadcast MRB Setup List		*0..1*			YES	reject
>Broadcast MRB Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>BC Bearer Context F1-U TNL Info at DU	M		BC Bearer Context F1-U TNL Info 9.3.2.7	gNB-DU endpoint(s) of the F1-U transport bearer(s). For delivery of DL PDUs.	-	
Broadcast MRB Failed To Be Setup List		*0..1*			YES	ignore
>Broadcast MRB Failed To Be Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	ignore
>>MRB ID	M		9.3.1.224		-	
>>Cause	O		9.3.1.2		-	
Broadcast MRB Modified List		*0..1*			YES	reject
>Broadcast MRB Modified Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>BC Bearer Context F1-U TNL Info at DU	O		BC Bearer Context F1-U TNL Info 9.3.2.7	Updated gNB-DU endpoint(s) of the F1-U transport bearer(s). For delivery of DL PDUs.	-	
Broadcast MRB Failed To Be Modified List		*0..1*			YES	ignore
>Broadcast MRB Failed To Be Modified Item IEs		*1 .. <maxnoofMRBs>*			EACH	ignore
>>MRB ID	M		9.3.1.224		-	
>>Cause	O		9.3.1.2		-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Broadcast Area Scope	O		9.3.1.287		YES	ignore

Range bound	Explanation

maxnoofMRBs	Maximum no. of MRB allowed to be setup for one MBS Session, the maximum value is 32.

9.2.13.8 BROADCAST CONTEXT MODIFICATION FAILURE

This message is sent by the gNB-DU to indicate a broadcast context modification failure.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.13.9 BROADCAST TRANSPORT RESOURCE REQUEST

This message is sent by the gNB-DU to request the gNB-CU to establish the F1-U resources for the broadcast Session.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Broadcast MRB Transport Request List		*0..1*			YES	reject
>Broadcast MRB Transport Request Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>BC Bearer Context F1-U TNL Info at DU	M		BC Bearer Context F1-U TNL Info 9.3.2.7	gNB-DU endpoint(s) of the F1-U transport bearer(s). For delivery of DL PDUs.	-	
F1-U Path Failure	O		ENUMERATED (true, ...)		YES	ignore

Range bound	Explanation
maxnoofMRBs	Maximum no. of MRB allowed to be setup for one MBS Session, the maximum value is 32.

9.2.14 Multicast Context Management messages

9.2.14.1 MULTICAST GROUP PAGING

This message is sent by the gNB-CU and is used to request the gNB-DU to multicast group page UEs.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
MBS Session ID	M		9.3.1.218		YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
UE Identity List for Paging		*0..1*			YES	ignore
>UE Identity for Paging Item		*1..<maxnoofUEIDforPaging>*			EACH	ignore
>>UE Identity Index value	M		9.3.1.39		-	
>>Paging DRX	O		9.3.1.40		-	
MC Paging Cell List		*0..1*			YES	ignore
>MC Paging Cell Item IEs		*1 .. <maxnoofPagingCells>*			EACH	ignore
>>NR CGI	M		9.3.1.12		-	
Indication for Multicast RRC_INACTIVE Reception	O		ENUMERATED (true, ...)	Corresponds to information contained in the *inactiveReceptionAllowed* as specified in TS 38.331 [8].	YES	ignore

Range bound	Explanation
maxnoofUEIDforPaging	Maximum no. of UE ID for multicast group paging. Value is 4096.
maxnoofPagingCells	Maximum no. of paging cells, the maximum value is 512.

9.2.14.2 MULTICAST CONTEXT SETUP REQUEST

This message is sent by the gNB-CU to request the setup of an MBS session context for a multicast session, and establish an MBS-associated logical F1-connection.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
MBS Session ID	M		9.3.1.218		YES	reject
MBS Service Area	O		9.3.1.222		YES	reject
S-NSSAI	M		9.3.1.38		YES	reject
Multicast MRBs To Be Setup List		*1*			YES	reject
>Multicast MRBs to Be Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>MRB QoS Information	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>MBS QoS Flows Mapped to MRB Item		*1 .. <maxnoofMBSQoSFlows>*			-	
>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.63		-	
>>>MBS QoS Flow Level QoS Parameters	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>DL PDCP SN	M		ENUMERATED		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Length			(12bits, 18bits, ...)			
Multicast CU to DU RRC Information	O		9.3.1.310		YES	reject
MBS Multicast Session Reception State	O		9.3.1.317		YES	reject

Range bound	Explanation
maxnoofMRBs	Maximum no. of MRB allowed to be setup for one MBS Session, the maximum value is 32.
maxnoofMBSQoSFlows	Maximum no. of flows allowed to be mapped to one MRB, the maximum value is 64.

9.2.14.3 MULTICAST CONTEXT SETUP RESPONSE

This message is sent by the gNB-DU to confirm the setup of a multicast context.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Multicast MRB Setup List		*1*			YES	reject
>Multicast MRB Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
Multicast MRB Failed To Be Setup List		*0..1*			YES	ignore
>Multicast MRB Failed To Be Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	ignore
>>MRB ID	M		9.3.1.224		-	
>>Cause	O		9.3.1.2		-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Multicast DU to CU RRC Information	O		9.3.1.311		YES	reject

Range bound	Explanation
maxnoofMRBs	Maximum no. of MRB allowed to be setup for one MBS Session, the maximum value is 32.

9.2.14.4 MULTICAST CONTEXT SETUP FAILURE

This message is sent by the gNB-DU to indicate that the setup of the multicast context was unsuccessful.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	O		9.3.1.220		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.14.5 MULTICAST CONTEXT RELEASE COMMAND

This message is sent by the gNB-CU to request the gNB-DU to release the multicast context for a given multicast service.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.14.6 MULTICAST CONTEXT RELEASE COMPLETE

This message is sent by the gNB-DU to confirm the release of the multicast context for a given multicast service.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.14.6a MULTICAST CONTEXT RELEASE REQUEST

This message is sent by the gNB-DU to request the gNB-CU to trigger the Multicast Context Release procedure.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.14.7 MULTICAST CONTEXT MODIFICATION REQUEST

This message is sent by the gNB-CU to request the gNB-DU to modify multicast context information.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
MBS Service Area	O		9.3.1.222		YES	reject
Multicast MRB To Be Setup List		*0..1*			YES	reject
>Multicast MRB to Be Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>MRB QoS	M		QoS Flow Level		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Information			QoS Parameters 9.3.1.45			
>>MBS QoS Flows Mapped to MRB Item		*1 .. <maxnoofMBSQoSFlows>*			-	
>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.63		-	
>>>MBS QoS Flow Level QoS Parameters	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>DL PDCP SN Length	M		ENUMERATED (12bits, 18bits, ...)		-	
Multicast MRB To Be Modified List		*0..1*			YES	reject
>Multicast MRB to Be Modified Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>MRB QoS Information	O		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>MBS QoS Flows Mapped to MRB Item		*0 .. <maxnoofMBSQoSFlows>*			-	
>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.63		-	
>>>MBS QoS Flow Level QoS Parameters	M		QoS Flow Level QoS Parameters 9.3.1.45		-	
>>DL PDCP SN Length	O		ENUMERATED (12bits, 18bits, ...)		-	
Multicast MRB To Be Released List		*0..1*			YES	reject
>Multicast MRB to Be Released Item IEs		*1 .. <maxnoofMRBs>*			YES	reject
>>MRB ID	M		9.3.1.224		-	
Multicast CU to DU RRC Information	O		9.3.1.310		YES	reject
MBS Multicast Session Reception State	O		9.3.1.317		YES	reject

Range bound	Explanation
maxnoofMRBs	Maximum no. of MRB allowed to be setup for one MBS Session, the maximum value is 32.
maxnoofMBSQoSFlows	Maximum no. of flows allowed to be mapped to one MRB, the maximum value is 64.

9.2.14.8 MULTICAST CONTEXT MODIFICATION RESPONSE

This message is sent by the gNB-DU to confirm the modification of a multicast context.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Multicast MRB Setup List		*0..1*			YES	reject
>Multicast MRB Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
Multicast MRB Failed To Be Setup List		*0..1*			YES	ignore
>Multicast MRB Failed To Be Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	ignore
>>MRB ID	M		9.3.1.224		-	
>>Cause	O		9.3.1.2		-	
Multicast MRB Modified List		*0..1*			YES	reject
>Multicast MRB Modified Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
Multicast MRB Failed To Be Modified List		*0..1*			YES	ignore
>Multicast MRB Failed To Be Modified Item IEs		*1 .. <maxnoofMRBs>*			EACH	ignore
>>MRB ID	M		9.3.1.224		-	
>>Cause	O		9.3.1.2		-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Multicast DU to CU RRC Information	O		9.3.1.311		YES	reject

Range bound	Explanation
maxnoofMRBs	Maximum no. of MRB allowed to be setup for one MBS Session, the maximum value is 32.

9.2.14.9 MULTICAST CONTEXT MODIFICATION FAILURE

This message is sent by the gNB-DU to indicate a multicast context modification failure.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.14.10 MULTICAST DISTRIBUTION SETUP REQUEST

This message is sent by the gNB-DU to request the setup of a Multicast F1-U Context.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
MBS Multicast F1-U Context Descriptor	M		9.3.2.8		YES	reject
Multicast F1-U Context To Be Setup List		*1*			YES	reject
>Multicast F1-U Context To Be Setup Item		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>MRB F1-U TNL Info at DU	M		UP Transport Layer Information 9.3.2.1	gNB-DU endpoint of the F1-U transport bearer.	-	
>>MRB Progress Information	C-ifPTPForwarding		9.3.2.12		-	

Range bound	Explanation
maxnoofMRBs	Maximum no. of MRB allowed to be setup for one MBS Session, the maximum value is 32.
maxnoofMBSQoSFlows	Maximum no. of flows allowed to be mapped to one MRB, the maximum value is 64.

Condition	Explanation
ifPTPForwarding	This IE shall be present if the *MC F1-U Context usage* IE in the *MBS Multicast F1-U Context Descriptor* IE is set to "ptp forwarding".

9.2.14.11 MULTICAST DISTRIBUTION SETUP RESPONSE

This message is sent by the gNB-CU to confirm the setup of a Multicast F1-U Context.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
MBS Multicast F1-U Context Descriptor	M		9.3.2.8		YES	reject
Multicast F1-U Context Setup List		*1*			YES	reject
>Multicast F1-U Context Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	reject
>>MRB ID	M		9.3.1.224		-	
>>MRB F1-U TNL Info at CU	M		UP Transport Layer Information 9.3.2.1	gNB-CU endpoint of the F1-U transport bearer.	-	
Multicast F1-U Context Failed To Be Setup List		*0..1*			YES	ignore
>Multicast F1-U Context Failed To Be Setup Item IEs		*1 .. <maxnoofMRBs>*			EACH	ignore
>>MRB ID	M		9.3.1.224		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>Cause	O		9.3.1.2		-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Multicast F1-U Context Reference CU	M		9.3.2.13		YES	reject

Range bound	Explanation
maxnoofMRBs	Maximum no. of MRB allowed to be setup for one MBS Session, the maximum value is 32.

9.2.14.12 MULTICAST DISTRIBUTION SETUP FAILURE

This message is sent by the gNB-DU to indicate that the setup of the Multicast F1-U Context was was unsuccessful.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	O		9.3.1.220		YES	ignore
MBS Multicast F1-U Context Descriptor	M		9.3.2.8		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.14.13 MULTICAST DISTRIBUTION RELEASE COMMAND

This message is sent by the gNB-DU to request the gNB-CU to release the Multicast F1-U Context for a given multicast MBS Session.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
MBS Multicast F1-U Context Descriptor	M		9.3.2.8		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.14.14 MULTICAST DISTRIBUTION RELEASE COMPLETE

This message is sent by the gNB-CU to confirm the release of the Multicast F1-U Context.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
MBS Multicast F1-U Context Descriptor	M		9.3.2.8		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.14.15 MULTICAST CONTEXT NOTIFICATION INDICATION

This message is sent by the gNB-DU to notify the gNB-CU about changes of the multicast context.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Multicast DU to CU RRC Information	O		9.3.1.311		YES	reject

9.2.14.16 MULTICAST CONTEXT NOTIFICATION CONFIRM

This message is sent by the gNB-CU to notify the gNB-DU to confirm the execution of the requested functions.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.14.17 MULTICAST CONTEXT NOTIFICATION REFUSE

This message is sent by the gNB-CU to notify the gNB-DU that the execution of the requested functions was not successful.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU MBS F1AP ID	M		9.3.1.219		YES	reject
gNB-DU MBS F1AP ID	M		9.3.1.220		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Cause	M		9.3.1.2		YES	ignore

9.2.14.18 MULTICAST COMMON CONFIGURATION REQUEST

This message is sent by the gNB-CU to request the gNB-DU to configure common items in the gNB-DU.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Multicast CU to DU Common RRC Information	O		9.3.1.314		YES	reject

9.2.14.19 MULTICAST COMMON CONFIGURATION RESPONSE

This message is sent by the gNB-DU to notify the gNB-CU to confirm the execution of the requested functions.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.14.20 MULTICAST COMMON CONFIGURATION REFUSE

This message is sent by the gNB-DU to notify the gNB-CU that the execution of the requested functions was not successful.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.15 PDC Measurement Reporting messages

9.2.15.1 PDC MEASUREMENT INITIATION REQUEST

This message is sent by gNB-CU to initiate PDC measurements.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
RAN UE PDC Measurement ID	M		INTEGER (1..16, ...)		YES	reject
PDC Report Type	M		ENUMERATED (OnDemand, Periodic, ...)		YES	reject
PDC Measurement Periodicity	C-ifReportTypePeriodic		ENUMERATED (80ms, 120ms, 160ms, 240ms, 320ms, 480ms, 640ms, 1024ms, 1280ms, 2048ms, 2560ms, 5120ms, ...)		YES	reject
PDC Measurement Quantities		*1 .. <maxnoMeasPDC>*			EACH	reject
>PDC Measurement Quantities Item	M		ENUMERATED (NR PDC TADV, gNB RX-TX, ...)		-	

Range bound	Explanation
maxnoMeasPDC	Maximum no. of PDC measured quantities that can be configured and reported with one message. Value is 16. Maximum is 1 in this release.

Condition	Explanation
ifReportTypePeriodic	This IE shall be present if the PDC Report Type IE is set to the value "Periodic".

9.2.15.2 PDC MEASUREMENT INITIATION RESPONSE

This message is sent by gNB-DU to indicate that the requested PDC measurement is successfully initiated.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
RAN UE PDC Measurement ID	M		INTEGER (1..16, ...)		YES	reject
PDC Measurement Result	O		9.3.1.232		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.15.3 PDC MEASUREMENT INITIATION FAILURE

This message is sent by gNB-DU to indicate that the requested PDC measurement cannot be initiated.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
RAN UE PDC Measurement ID	M		INTEGER (1..16, ...)		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.15.4 PDC MEASUREMENT REPORT

This message is sent by gNB-DU to report the results of the requested PDC measurement.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
RAN UE PDC Measurement ID	M		INTEGER (1..16, ...)		YES	reject
PDC Measurement Result	M		9.3.1.232		YES	ignore

9.2.15.5 PDC MEASUREMENT TERMINATION COMMAND

This message is sent by the gNB-CU to request the gNB-DU to terminate an ongoing periodical PDC measurement.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
RAN UE PDC Measurement ID	M		INTEGER (1..16, ...)		YES	ignore

9.2.15.6 PDC MEASUREMENT FAILURE INDICATION

This message is sent by the gNB-DU to indicate that the previously requested PDC measurements can no longer be reported.

Direction: gNB-DU → gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
RAN UE PDC Measurement ID	M		INTEGER (1..16, ...)		YES	ignore
Cause	M		9.3.1.2		YES	ignore

9.2.16 QMC messages

9.2.16.1 QOE INFORMATION TRANSFER

This message is sent by a gNB-CU to a gNB-DU, to indicate information related to RAN visible QoE.

Direction: gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
gNB-CU UE F1AP ID	M		9.3.1.4		YES	reject
gNB-DU UE F1AP ID	M		9.3.1.5		YES	reject
QoE Information List		*0..1*			YES	ignore
>QoE Information Item		*1..<maxnoofQoEInformation>*			-	
>>QoE Metrics	O		9.3.1.260		-	
>>DRB List		*0..1*			YES	ignore
>>>DRB List Item		*1 .. <maxnoofDRBs>*		The List of DRBs corresponding to the QoE Information.		
>>>>DRB ID	M		9.3.1.8			

Range bound	Explanation
maxnoofQoEInformation	Maximum no. of QoE information for one UE, the maximum value is 16.
maxnoofDRBs	Maximum no. of DRBs allowed towards one UE, the maximum value is 64.

9.2.16.2 QOE INFORMATION TRANSFER CONTROL

This message is sent by a gNB-DU to the gNB-CU, to control the QoE information transfer.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
CHOICE *Deactivation Indication*	O				YES	ignore
>Per UE						
>>Deactivation Indication List		*1*			YES	ignore
>>>Deactivation Indication Item		*1..<maxnoofUEsInQMCTransferControlMessage>*			-	
>>>> gNB-CU UE F1AP ID	M		9.3.1.4		-	
>>>>gNB-DU UE F1AP ID	M		9.3.1.5		-	
>Deactivate ALL			NULL	This choice indicates that RAN visible QoE reporting pertaining to all the UEs served by the gNB-DU, should be deactivated.		

Range bound	Explanation
maxnoofUEsInQMCTransferControlMessage	Maximum no. of UEs for which QoE transfer control information is received, the maximum value is 512.

9.2.17 Timing Synchronisation Status Reporting Messages

9.2.17.1 TIMING SYNCHRONISATION STATUS REQUEST

This message is sent by the gNB-CU to request the gNB-DU to start or stop reporting of RAN timing synchronization status information.

Direction: gNB-CU → gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
RAN TSS Request Type	M		ENUMERATED (start, stop, ...)		YES	reject

9.2.17.2 TIMING SYNCHRONISATION STATUS RESPONSE

This message is sent by the gNB-DU to confirm the request to start or stop reporting of RAN timing synchronization status information.

Direction: gNB-DU→ gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject

Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.3 TIMING SYNCHRONISATION STATUS FAILURE

This message is sent by the gNB-DU to indicate that reporting of RAN timing synchronisation status information cannot be initiated.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Transaction ID	M		9.3.1.23		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.4 TIMING SYNCHRONISATION STATUS REPORT

This message is sent by the gNB-DU to report RAN timing synchronisation status information.

Direction: gNB-DU → gNB-CU

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
RAN Timing Synchronisation Status Information	M		9.3.1.298		YES	ignore

9.2.18 CLI Indication Messages

9.2.18.1 CLI INDICATION

This message is sent by gNB-DU or gNB-CU to report the results of the CLI measurements or to indicate the need for SRS Resource Configuration information.

Direction: gNB-DU → gNB-CU and gNB-CU → gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Transaction ID	M		9.3.1.23		YES	reject
CLI Measurement Result List		*0..1*			YES	ignore
>CLI Measurement Result Item		*1 .. <maxCellingNBDU>*			-	
>>Cell ID	M		NR CGI 9.3.1.12		-	
>>SSB Index	O		INTEGER (0..63, ...)	Strongest DL SSB beam information.	-	
>>NZP CSI-RS Resource Indication	O		INTEGER (1..64, ...)	Strongest DL NZP CSI-RS beam information. The value is a relative index of the CSI-RS resources within the set of resources signalled.	-	
>>CLI Mitigation Indication	O		ENUMERATED (true, ...)	Indicates the need for CLI mitigation.	-	
SRS Resource Indication	O		ENUMERATED (true, ...)	Indicates that SRS Resource configuration information is needed.	YES	ignore

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.

9.3 Information Element Definitions

9.3.1 Radio Network Layer Related IEs

9.3.1.1 Message Type

The *Message Type* IE uniquely identifies the message being sent. It is mandatory for all messages.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Message Type				
>Procedure Code	M		INTEGER (0..255)	
>Type of Message	M		CHOICE (Initiating Message, Successful Outcome, Unsuccessful Outcome, ...)	

9.3.1.2 Cause

The purpose of the *Cause* IE is to indicate the reason for a particular event for the F1AP protocol.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
CHOICE *Cause Group*	M			
>Radio Network Layer				
>>Radio Network Layer Cause	M		ENUMERATED (Unspecified, RL failure-RLC, Unknown or already allocated gNB-CU UE F1AP ID, Unknown or already allocated gNB-DU UE F1AP ID, Unknown or inconsistent pair of UE F1AP ID, Interaction with other procedure, Not supported QCI Value, Action Desirable for Radio Reasons, No Radio Resources Available, Procedure cancelled, Normal Release, ..., Cell not available, RL failure-others, UE rejection, Resources not available for the slice(s), AMF initiated abnormal release, Release due to Pre-Emption, PLMN not served by the gNB-CU, Multiple DRB ID Instances, Unknown DRB ID, Multiple BH RLC CH ID Instances, Unknown BH RLC CH ID, CHO-CPC resources to be changed, NPN not supported, NPN access denied, gNB-CU Cell Capacity Exceeded, Report Characteristics Empty, Existing Measurement ID, Measurement Temporarily not Available, Measurement not Supported For The Object, Unknown BAP address, Unknown BAP routing ID, Insufficient UE Capabilities, SCG activation deactivation failure,	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
			SCG deactivation failure due to data transmission, Requested Item not Supported on Time, Unknown or already allocated gNB-CU MBS F1AP ID, Unknown or already allocated gNB-DU MBS F1AP ID, Unknown or inconsistent pair of MBS F1AP ID, Unknown or inconsistent MRB ID, TAT-SDT expiry, LTM command triggered, SSB not Available)	
>Transport Layer				
>>Transport Layer Cause	M		ENUMERATED (Unspecified, Transport Resource Unavailable, ..., Unknown TNL address for IAB, Unknown UP TNL information for IAB)	
>Protocol				
>>Protocol Cause	M		ENUMERATED (Transfer Syntax Error, Abstract Syntax Error (Reject), Abstract Syntax Error (Ignore and Notify), Message not Compatible with Receiver State, Semantic Error, Abstract Syntax Error (Falsely Constructed Message), Unspecified, ...)	
>Misc				
>>Miscellaneous Cause	M		ENUMERATED (Control Processing Overload, Not enough User Plane Processing Resources, Hardware Failure, O&M Intervention, Unspecified, ...)	

The meaning of the different cause values is described in the following table. In general, "not supported" cause values indicate that the related capability is missing. On the other hand, "not available" cause values indicate that the related capability is present, but insufficient resources were available to perform the requested action.

Radio Network Layer cause	Meaning
Unspecified	Sent when none of the specified cause values applies but still the cause is Radio Network Layer related.

Radio Network Layer cause	Meaning
RL Failure-RLC	The action is due to an RL failure caused by exceeding the maximum number of ARQ retransmissions.
Unknown or already allocated gNB-CU UE F1AP ID	The action failed because the gNB-CU UE F1AP ID is either unknown, or (for a first message received at the gNB-DU) is known and already allocated to an existing context.
Unknown or already allocated gNB-DU UE F1AP ID	The action failed because the gNB-DU UE F1AP ID is either unknown, or (for a first message received at the gNB-CU) is known and already allocated to an existing context.
Unknown or inconsistent pair of UE F1AP ID	The action failed because both UE F1AP IDs are unknown, or are known but do not define a single UE context.
Interaction with other procedure	The action is due to an ongoing interaction with another procedure.
Not supported QCI Value	The action failed because the requested QCI is not supported.
Action Desirable for Radio Reasons	The reason for requesting the action is radio related.
No Radio Resources Available	The cell(s) in the requested node don't have sufficient radio resources available.
Procedure cancelled	The sending node cancelled the procedure due to other urgent actions to be performed.
Normal Release	The action is due to a normal release of the UE (e.g. because of mobility) and does not indicate an error.
Cell Not Available	The action failed due to no cell available in the requested node.
RL Failure-others	The action is due to an RL failure caused by other radio link failures than exceeding the maximum number of ARQ retransmissions.
UE rejection	The action is due to gNB-CU's rejection of a UE access request.
Resources not available for the slice(s)	The requested resources are not available for the slice(s).
AMF initiated abnormal release	The release is triggered by an error in the AMF or in the NAS layer.
Release due to Pre-Emption	Release is initiated due to pre-emption.
PLMN not served by the gNB-CU	The PLMN indicated by the UE is not served by the gNB-CU.
Multiple DRB ID Instances	The action failed because multiple instances of the same DRB had been provided.
Unknown DRB ID	The action failed because the DRB ID is unknow.
Multiple BH RLC CH ID Instances	The action failed because multiple instances of the same BH RLC CH ID had been provided. This cause value is only applicable to IAB.
Unknown BH RLC CH ID	The action failed because the BH RLC CH ID is unknown. This cause value is only applicable to IAB.
CHO-CPC resources to be changed	The gNB-DU requires gNB-CU to replace, i.e. overwrite the configuration of indicated candidate target cell.
NPN not supported	The action fails because the indicated SNPN is not supported in the node.
NPN access denied	The action is due to rejection of a UE access request for NPN.
gNB-CU Cell Capacity Exceeded	The number of cells requested to be added was exceeding maximum cell capacity in the gNB-CU.
Report Characteristics Empty	The action failed because there is no measurement object in the report characteristics.
Existing Measurement ID	The action failed because the measurement ID is already used.
Measurement Temporarily not Available	The gNB-DU can temporarily not provide the requested measurement object.
Measurement not Supported For The Object	At least one of the concerned object(s) does not support the requested measurement.
Unknown BAP address	The action failed because the BAP address is unknown. This cause value is only applicable to IAB.
Unknown BAP routing ID	The action failed because the BAP routing ID is unknown. This cause value is only applicable to IAB.
Insufficient UE Capabilities	The setup can't proceed due to insufficient UE capabilities.
SCG activation deactivation failure	The action failed due to rejection of the SCG activation deactivation request.
SCG deactivation failure due to data transmission	The SCG deactivation failure due to ongoing or arriving data transmission.
Requested Item not Supported on	The gNB-DU is unable to provide the measurement results on

Radio Network Layer cause	Meaning
Time	time.
Unknown or already allocated gNB-CU MBS F1AP ID	The action failed because the gNB-CU MBS F1AP ID is either unknown, or (for a first message received at the gNB-CU) is known and already allocated to an existing context.
Unknown or already allocated gNB-DU MBS F1AP ID	The action failed because the gNB-DU MBS F1AP ID is either unknown, or (for a first message received at the gNB-DU) is known and already allocated to an existing context.
Unknown or inconsistent pair of MBS F1AP ID	The action failed because both MBS F1AP IDs are unknown, or are known but do not define a single MBS context.
Unknown or inconsistent MRB ID	The action failed because the MRB ID is unknown or inconsistent.
TAT-SDT expiry	The UE context release is requested from the gNB-DU due to the expiry of the Timing Alignment timer for CG-SDT.
LTM command triggered	The action failed because the LTM command has been triggered.
SSB not Available	The action failed due to no SSB available in the requested node.

Transport Layer cause	Meaning
Unspecified	Sent when none of the specified cause cause values applies but still the cause is Transport Network Layer related.
Transport Resource Unavailable	The required transport resources are not available.
Unknown TNL address for IAB	The action failed because the TNL address is unknown. This cause value is only applicable to IAB.
Unknown UP TNL information for IAB	The action failed because the UP TNL information is unknown. This cause value is only applicable to IAB.

Protocol cause	Meaning
Transfer Syntax Error	The received message included a transfer syntax error.
Abstract Syntax Error (Reject)	The received message included an abstract syntax error and the concerning criticality indicated "reject".
Abstract Syntax Error (Ignore And Notify)	The received message included an abstract syntax error and the concerning criticality indicated "ignore and notify".
Message Not Compatible With Receiver State	The received message was not compatible with the receiver state.
Semantic Error	The received message included a semantic error.
Abstract Syntax Error (Falsely Constructed Message)	The received message contained IEs or IE groups in wrong order or with too many occurrences.
Unspecified	Sent when none of the specified cause cause values applies but still the cause is Protocol related.

Miscellaneous cause	Meaning
Control Processing Overload	Control processing overload.
Not Enough User Plane Processing Resources Available	No enough resources are available related to user plane processing.
Hardware Failure	Action related to hardware failure.
O&M Intervention	The action is due to O&M intervention.
Unspecified Failure	Sent when none of the specified cause cause values applies and the cause is not related to any of the categories Radio Network Layer, Transport Network Layer or Protocol.

9.3.1.3 Criticality Diagnostics

The *Criticality Diagnostics* IE is sent by the gNB-DU or the gNB-CU when parts of a received message have not been comprehended or were missing, or if the message contained logical errors. When applicable, it contains information about which IEs were not comprehended or were missing.

For further details on how to use the *Criticality Diagnostics* IE, (see clause 10). The conditions for inclusion of the *Transaction ID* IE are described in clause 10.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Procedure Code	O		INTEGER (0..255)	Procedure Code is to be used if Criticality Diagnostics is part of Error Indication procedure, and not within the response message of the same procedure that caused the error.
Triggering Message	O		ENUMERATED(initiating message, successful outcome, unsuccessful outcome)	The Triggering Message is used only if the Criticality Diagnostics is part of Error Indication procedure.
Procedure Criticality	O		ENUMERATED(reject, ignore, notify)	This Procedure Criticality is used for reporting the Criticality of the Triggering message (Procedure).
Transaction ID	O		9.3.1.23	
Information Element Criticality Diagnostics		*0 .. <maxnoofErrors>*		
>IE Criticality	M		ENUMERATED(reject, ignore, notify)	The IE Criticality is used for reporting the criticality of the triggering IE. The value 'ignore' is not applicable.
>IE ID	M		INTEGER (0..65535)	The IE ID of the not understood or missing IE.
>Type of Error	M		ENUMERATED(not understood, missing, ...)	

Range bound	Explanation
maxnoofErrors	Maximum no. of IE errors allowed to be reported with a single message. The value for maxnoofErrors is 256.

9.3.1.4 gNB-CU UE F1AP ID

The gNB-CU UE F1AP ID uniquely identifies the UE association over the F1 interface within the gNB-CU.

NOTE: If F1-C signalling transport is shared among multiple interface instances, the value of the gNB-CU UE F1AP ID is allocated so that it can be associated with the corresponding F1-C interface instance.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB-CU UE F1AP ID	M		INTEGER ($0 .. 2^{32} - 1$)	

9.3.1.5 gNB-DU UE F1AP ID

The gNB-DU UE F1AP ID uniquely identifies the UE association over the F1 interface within the gNB-DU.

NOTE: If F1-C signalling transport is shared among multiple interface instances, the value of the gNB-DU UE F1AP ID is allocated so that it can be associated with the corresponding F1-C interface instance.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB-DU UE F1AP ID	M		INTEGER ($0 .. 2^{32} - 1$)	

9.3.1.6 RRC-Container

This information element contains a gNB-CU→UE or a UE → gNB-CU message that is transferred without interpretation in the gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RRC-Container	M		OCTET STRING	

9.3.1.7 SRB ID

This IE uniquely identifies a SRB for a UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SRB ID	M		INTEGER (0..3, ..., 4 | 5 | 6)	Corresponds to the identities of SRB as defined in TS 38.331 [8]. Value 0 indicates SRB0, value 1 indicates SRB1, etc.

9.3.1.8 DRB ID

This IE uniquely identifies a DRB for a UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DRB ID	M		INTEGER (1.. 32, ...)	Corresponds to the *DRB-Identity* IE defined in TS 38.331 [8].

9.3.1.9 gNB-DU ID

The gNB-DU ID uniquely identifies the gNB-DU at least within a gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB-DU ID	M		INTEGER (0 .. 2^{36}-1)	The gNB-DU ID is independently configured from cell identifiers, i.e. no connection between gNB-DU ID and cell identifiers.

9.3.1.10 Served Cell Information

This IE contains cell configuration information of a cell in the gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
NR CGI	M		9.3.1.12		-	
NR PCI	M		INTEGER (0..1007)	Physical Cell ID	-	
5GS TAC	O		9.3.1.29	5GS Tracking Area Code	-	
Configured EPS TAC	O		9.3.1.29a		-	
Served PLMNs		*1..<maxnoofBPLMNs>*		Broadcast PLMNs in SIB 1 associated to the NR Cell Identity in the *NR CGI* IE	-	
>PLMN Identity	M		9.3.1.14		-	
>TAI Slice Support List	O		Slice Support List 9.3.1.37	Supported S-NSSAIs per PLMN or per SNPN.	YES	ignore
>NPN Support Information	O		9.3.1.156	Supported NPNs per PLMN.	YES	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>Extended TAI Slice Support List	O		Extended Slice Support List 9.3.1.165	Additional Supported S-NSSAIs per PLMN or per SNPN.	YES	reject
>TAI NSAG Support List	O		9.3.1.273	NSAG information associated with the slices per TAC, per PLMN or per SNPN.	YES	ignore
CHOICE *NR-Mode-Info*	M				-	
>FDD					-	
>>FDD Info		*1*			-	
>>>UL FreqInfo	M		NR Frequency Info 9.3.1.17	This IE is ignored if the *Cell Direction* IE is included and set to “dl-only”.	-	
>>>DL FreqInfo	M		NR Frequency Info 9.3.1.17	This IE is ignored if the *Cell Direction* IE is included and set to “ul-only”.	-	
>>>UL Transmission Bandwidth	M		Transmission Bandwidth 9.3.1.15	This IE is ignored if the *Cell Direction* IE is included and set to “dl-only”.	-	
>>>DL Transmission Bandwidth	M		Transmission Bandwidth 9.3.1.15	This IE is ignored if the *Cell Direction* IE is included and set to “ul-only”.	-	
>>>UL Carrier List	O		NR Carrier List 9.3.1.137	If included, the *UL Transmission Bandwidth* IE shall be ignored.	YES	ignore
>>>DL Carrier List	O		NR Carrier List 9.3.1.137	If included, the *DL Transmission Bandwidth* IE shall be ignored.	YES	ignore
>TDD					-	
>>TDD Info		*1*			-	
>>>NR FreqInfo	M		NR Frequency Info 9.3.1.17		-	
>>>Transmission Bandwidth	M		9.3.1.15	This IE is ignored if the *Transmission Bandwidth asymmetric* IE is present.	-	
>>>Intended TDD DL-UL Configuration	O		9.3.1.89		YES	ignore
>>>TDD UL-DL Configuration Common NR	O		OCTET STRING	Includes the *tdd-UL-DL-ConfigurationCommon* contained in the *ServingCellConfigCommon* IE as defined in TS 38.331 [8]	YES	ignore
>>>Carrier List	O		NR Carrier List 9.3.1.137	If included, the *Transmission Bandwidth* IE shall be ignored.	YES	ignore
>>>Transmission Bandwidth asymmetric		*0..1*		Indicates the asymmetric UL and DL transmission bandwidth.	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>>>UL Transmission Bandwidth	M		Transmission Bandwidth 9.3.1.15	.	–	
>>>>DL Transmission Bandwidth	M		Transmission Bandwidth 9.3.1.15		–	
>>>SBFD Frequency Configuration	O		OCTET STRING	Includes the *SBFD-Subband-Allocation-r19* IE, as defined in TS 38.331[8].	YES	ignore
>NR-U					YES	ignore
>>NR-U Channel Info List		*1..< maxnoofNR-UChannelIDs>*			-	
>>>NR-U Channel Info Item					-	
>>>>NR-U Channel ID	M		INTEGER (1.. maxnoofNR-UChannelIDs, .. .)	Index to uniquely identify the part of the NR-U Channel Bandwidth consisting of a contiguous set of resource blocks (RBs) on which a channel access procedure is performed in shared spectrum. Value 1 represents the first part of the NR-U Channel Bandwidth on which a channel access procedure is performed. Value 2 represents the second part of the NR-U Channel Bandwidth on which a channel access procedure is performed, and so on.	-	
>>>>NR-U ARFCN	M		INTEGER (0.. maxNRARFCN)	It represents the centre frequency of the NR-U Channel Bandwidth for NR bands restricted to operation with shared spectrum channel access, as defined in TS 37.213 [46]. Allowed values are specified in TS 38.101-1 [26] in Table 5.4.2.3-2, Table 5.4.2.3-3 and Table 5.4.2.3-4.	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>>>NR-U Channel Bandwidth	M		ENUMERATED (10MHz, 20MHz, 40MHz, 60 MHz, 80 MHz, ..., 100MHz)		-	
Measurement Timing Configuration	M		OCTET STRING	Includes the *MeasurementTimingConfiguration* inter-node message defined in TS 38.331 [8].	-	
RANAC	O		RAN Area Code 9.3.1.57		YES	ignore
Extended Served PLMNs List		*0..1*		This is included if more than 6 Served PLMNs is to be signalled.	YES	ignore
>Extended Served PLMNs Item		*1 ..<maxnoofExtendedBPLMNs>*			-	
>>PLMN Identity	M		9.3.1.14		-	
>>TAI Slice Support List	O		Slice Support List 9.3.1.37	Supported S-NSSAIs per PLMN or per SNPN.	-	
>>NPN Support Information	O		9.3.1.156	Supported NPNs per PLMN.	YES	reject
>>Extended TAI Slice Support List	O		Extended Slice Support List 9.3.1.165	Additional Supported S-NSSAIs per PLMN or per SNPN.	YES	reject
>>TAI NSAG Support List	O		9.3.1.273	NSAG information associated with the slices per TAC, per PLMN or per SNPN.	YES	ignore
Cell Direction	O		9.3.1.78		YES	ignore
Broadcast PLMN Identity Info List		*0..<maxnoofBPLMNsNR>*		This IE corresponds to the *PLMN-IdentityInfoList* IE and the *NPN-IdentityInfoList* IE (if available) in *SIB1* as specified in TS 38.331 [8]. All PLMN Identities and associated information contained in the *PLMN-IdentityInfoList* IE and NPN identities and associated information contained in the *NPN-IdentityInfoList* IE (if available) are included and provided in the same order as broadcast in SIB1. NOTE: In case of	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				NPN-only cell, the PLMN Identities and associated information contained in the *PLMN-IdentityInfoList* IE are not included.		
>PLMN Identity List	M		Available PLMN List 9.3.1.65	Broadcast PLMN IDs in SIB1 associated to the *NR Cell Identity* IE	-	
>Extended PLMN Identity List	O		Extended Available PLMN List 9.3.1.76		-	
>5GS-TAC	O		OCTET STRING (3)		-	
>NR Cell Identity	M		BIT STRING (36)		-	
>RANAC	O		RAN Area Code 9.3.1.57		-	
>Configured TAC Indication	O		9.3.1.87a	NOTE: This IE is associated with the 5GS TAC in the *Broadcast PLMN Identity Info List* IE	YES	ignore
>NPN Broadcast Information	O		9.3.1.157	If this IE is included the content of the *PLMN Identity List* IE and *Extended PLMN Identity List* IE if present in the *Broadcast PLMN Identity Info List* IE is ignored.	YES	reject
Cell Type	O		9.3.1.87		YES	ignore
Configured TAC Indication	O		9.3.1.87a	NOTE: This IE is associated with the 5GS TAC on top-level of the *Served Cell Information* IE	YES	ignore
Aggressor gNB Set ID	O		gNB Set ID 9.3.1.93	This IE indicates the associated aggressor gNB Set ID of the cell	YES	ignore
Victim gNB Set ID	O		gNB Set ID 9.3.1.93	This IE indicates the associated Victim gNB Set ID of the cell	YES	ignore
IAB Info IAB-DU	O		9.3.1.106		YES	ignore
SSB Positions In Burst	O		9.3.1.138		YES	ignore
NR PRACH Configuration	O		9.3.1.139		YES	ignore
SFN Offset	O		9.3.1.208		YES	ignore
NPN Broadcast Information	O		9.3.1.157		YES	reject
Supported MBS FSA ID List		*0..<maxnoofMBSFSAs>*		Shall contain all MBS Frequency Selection Area Identities associated with the NR CGI.	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>MBS Frequency Selection Area Identity	M		OCTET STRING(3)		–	
RedCap Broadcast Information	O		BIT STRING (SIZE(8))	The presence of this IE indicates that the intraFreqReselectionRedCap IE is broadcast in SIB1 of the corresponding cell, see TS 38.331 [8]. Each position in the bitmap indicates which RedCap UEs are allowed access, according to the setting of RedCap barring indicators in SIB1, see TS 38.331 [8]. First bit = 1Rx, second bit = 2Rx, third bit = halfDuplex, other bits reserved for future use. Value '1' indicates 'access allowed'. Value '0' indicates 'access not allowed".	YES	ignore
eRedCap Broadcast Information	O		BIT STRING (SIZE(8))	The presence of this IE indicates that the *intraFreqReselection-eRedCap* IE is broadcast in SIB1 of the corresponding cell, see TS 38.331 [8]. Each position in the bitmap indicates which eRedCap UEs are allowed access, according to the setting of the barring indicators in SIB1, see TS 38.331 [8]. First bit = 1Rx, second bit = 2Rx, third bit=half-duplex, other bits reserved for future use. Value '1' indicates 'access allowed'. Value '0' indicates 'access not allowed".	YES	ignore
XR Broadcast Information	O		ENUMERATED (true, ...)	Corresponds to information provided in the *cellBarred2RxXR*	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				contained in the *SIB1* message as defined in TS 38.331 [8].		
Barring Exemption for Emergency Call Information	O		ENUMERATED (true, ...)	Corresponds to information provided in the *barringExemptEmergencyCall* contained in the *SIB1* message as defined in TS 38.331 [8].	YES	ignore
NZP CSI-RS Resources Configuration	O		9.3.1.356		YES	ignore
SRS Resource Configuration	O		9.3.1.357		YES	ignore
CHOICE *on-demand SIB1*	O				YES	ignore
>Provision						
>>On-demand SIB1 Config	M		OCTET STRING	Includes the *OD-SIB1-Config* IE contained in the SIB26 message as defined in TS 38.331 [8].	-	
>Stop provision			NULL			

Range bound	Explanation
maxnoofBPLMNs	Maximum no. of Broadcast PLMN Ids. Value is 6.
maxnoofExtendedBPLMNs	Maximum no. of Extended Broadcast PLMN Ids. Value is 6.
maxnoofBPLMNsNR	Maximum no. of PLMN Ids.broadcast in an NR cell. Value is 12.
maxnoofNR-UChannelIDs	Maximum no. NR-U Channel IDs in a cell. Value is 16.
maxnoofMBSFSAs	Maximum no. of MBS FSAs by a cell. Value is 256.

9.3.1.11 Transmission Action Indicator

This IE indicates actions for the gNB-DU for the data transmission to the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Transmission Action Indicator	M		ENUMERATED (stop, ..., restart)	

9.3.1.12 NR CGI

The NR Cell Global Identifier (NR CGI) is used to globally identify a cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.1.14	
NR Cell Identity	M		BIT STRING (SIZE(36))	

9.3.1.13 Time To wait

This IE defines the minimum allowed waiting times.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Time to wait	M		ENUMERATED(1s, 2s, 5s, 10s, 20s, 60s, ...)	

9.3.1.14 PLMN Identity

This information element indicates the PLMN Identity.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		OCTET STRING (SIZE(3))	- digits 0 to 9, encoded 0000 to 1001, - 1111 used as filler digit, two digits per octet, - bits 4 to 1 of octet n encoding digit 2n-1 - bits 8 to 5 of octet n encoding digit 2n -The PLMN identity consists of 3 digits from MCC followed by either -a filler digit plus 2 digits from MNC (in case of 2 digit MNC) or -3 digits from MNC (in case of a 3 digit MNC).

9.3.1.15 Transmission Bandwidth

The *Transmission Bandwidth* IE is used to indicate the UL or DL transmission bandwidth.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
NR SCS	M		ENUMERATED (scs15, scs30, scs60, scs120, ..., scs480, scs960)	The values scs15, scs30, scs60, scs120, scs480 and scs960 corresponds to the sub carrier spacing in TS 38.104 [17].
NRB	M		ENUMERATED (nrb11, nrb18, nrb24, nrb25, nrb31, nrb32, nrb38, nrb51, nrb52, nrb65, nrb66, nrb78, nrb79, nrb93, nrb106, nrb107, nrb121, nrb132, nrb133, nrb135, nrb160, nrb162, nrb189, nrb216, nrb217, nrb245, nrb264, nrb270, nrb273, ..., nrb33, nrb62, nrb124, nrb148, nrb248, nrb44, nrb58, nrb92, nrb119, nrb188, nrb242, nrb15, nrb35)	This IE is used to indicate the UL or DL transmission bandwidth expressed in units of resource blocks "N_{RB}" (TS 38.104 [17]). The values nrb11, nrb18, etc. correspond to the number of resource blocks "N_{RB}" 11, 18, etc. NOTE: Handling of Asymmetric channel bandwidth is specified in TS 38.101-1 [26].

9.3.1.16 Void

Reserved for future use.

9.3.1.17 NR Frequency Info

The NR Frequency Info defines the carrier frequency used in a cell for a given direction (UL or DL) in FDD or for both UL and DL directions in TDD or for an SUL carrier.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
NR ARFCN	M		INTEGER (0.. maxNRARFCN)	RF Reference Frequency as defined in TS 38.104 [17] section 5.4.2.1. The frequency provided in this IE identifies the absolute frequency position of the reference resource block (Common RB 0) of the carrier. Its lowest subcarrier is also known as Point A.	–	
SUL Information	O		9.3.1.28		–	
Frequency Band List		*1*			–	
>Frequency Band Item		*1..<maxnoofNrCellBands>*			–	
>>NR Frequency Band	M		INTEGER (1.. 1024, ...)	Operating Band as defined in TS 38.104 [17] section 5.4.2.3. The value 1 corresponds to NR operating band n1, value 2 corresponds to NR operating band n2, etc.	–	
>>Supported SUL band List		*0..<maxnoofNrCellBands>*			–	
>>>Supported SUL band Item	M		INTEGER (1.. 1024, ...)	Supplementary NR Operating Band as defined in TS 38.104 [17] section 5.4.2.3 that can be used for SUL duplex mode as per TS 38.101-1 [26] table 5.2.-1. The value 80 corresponds to NR operating band n80, value 81 corresponds to NR operating band n81, etc.	–	
Frequency Shift 7p5khz	O		ENUMERATED (false, true, ...)	Indicate whether the value of Δ_{shift} is 0kHz or 7.5kHz when calculating $F_{REF,shift}$ as defined in Section 5.4.2.1 of TS 38.104 [17].	YES	ignore

Range bound	Explanation
maxNRARFCN	Maximum value of NR ARFCNs. Value is 3279165.
maxnoofNrCellBands	Maximum no. of frequency bands supported for a NR cell. Value is 32.

9.3.1.18 gNB-DU System Information

This IE contains the system information generated by the gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MIB message	M		OCTET STRING	Includes the *MIB* message, as defined in subclause 6.2.2 in TS 38.331 [8].	-	
SIB1 message	M		OCTET STRING	Includes the *SIB1* message, as defined in subclause 6.2.2 in TS 38.331 [8].	-	
SIB12 message	O		OCTET STRING	Includes the *SIB12* IE, as defined in subclause 6.3.1 in TS 38.331 [8].	YES	ignore
SIB13 message	O		OCTET STRING	Includes the *SIB13* IE, as defined in subclause 6.3.1 in TS 38.331 [8].	YES	ignore
SIB14 message	O		OCTET STRING	Includes the *SIB14* IE, as defined in subclause 6.3.1 in TS 38.331 [8].	YES	ignore
SIB10 message	O		OCTET STRING	Includes the *SIB10* IE, as defined in subclause 6.3.1 in TS 38.331 [8].	YES	ignore
SIB17 message	O		OCTET STRING	Includes the *SIB17* IE, as defined in subclause 6.3.1 in TS 38.331 [8]	YES	ignore
SIB20 message	O		OCTET STRING	Includes the *SIB20* IE, as defined in subclause 6.3.1 in TS 38.331 [8]	YES	ignore
SIB15 message	O		OCTET STRING	Includes the *SIB15* IE, as defined in subclause 6.3.1 in TS 38.331 [8].	YES	ignore
SIB24 message	O		OCTET STRING	Includes the *SIB24* IE, as defined in subclause 6.3.1 in TS 38.331 [8].	YES	ignore
SIB22 message	O		OCTET STRING	Includes the *SIB22* IE, as defined in subclause 6.3.1 in TS 38.331 [8].	YES	ignore
SIB23 message	O		OCTET STRING	Includes the *SIB23* IE, as defined in subclause 6.3.1 in TS 38.331 [8].	YES	ignore
SIB17bis message	O		OCTET STRING	Includes the *SIB17bis* IE, as defined in	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				subclause 6.3.1 in TS 38.331 [8].		

9.3.1.19 E-UTRAN QoS

This IE defines the QoS to be applied to a DRB or to a BH RLC channel for EN-DC case.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
QCI	M		INTEGER (0..255)	QoS Class Identifier defined in TS 23.401 [10]. Logical range and coding specified in TS 23.203 [11]. For a BH RLC channel, the Packet Delay Budget included in QCI defines the upper bound for the time that a packet may be delayed between the gNB-DU and its child IAB-MT.	-	
Allocation and Retention Priority	M		9.3.1.20		-	
GBR QoS Information	O		9.3.1.21	This IE shall be present for GBR bearers only and is ignored otherwise.	-	
ENB DL Transport Layer Address	O		Transport Layer Address 9.3.2.3	DL Transport Layer Address of node terminating PDCP. Included for MN-terminated SCG bearers.	YES	ignore

9.3.1.20 Allocation and Retention Priority

This IE specifies the relative importance compared to other E-RABs for allocation and retention of the E-UTRAN Radio Access Bearer.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Priority Level	M		INTEGER (0..15)	**Desc.:** This IE should be understood as "priority of allocation and retention" (see TS 23.401 [10]). **Usage:** Value 15 means "no priority". Values between 1 and 14 are ordered in decreasing order of priority, i.e. 1 is the highest and 14 the lowest. Value 0 shall be treated as a logical error if received.
Pre-emption Capability	M		ENUMERATED(shall not trigger pre-emption, may trigger	**Desc.:** This IE indicates the pre-emption capability of the request on other E-RABs (see TS 23.401

IE/Group Name	Presence	Range	IE type and reference	Semantics description
			pre-emption)	[10]). **Usage:** The E-RAB shall not pre-empt other E-RABs or, the E-RAB may pre-empt other E-RABs The Pre-emption Capability indicator applies to the allocation of resources for an E-RAB and as such it provides the trigger to the pre-emption procedures/processes of the eNB.
Pre-emption Vulnerability	M		ENUMERATED(not pre-emptable, pre-emptable)	**Desc.:** This IE indicates the vulnerability of the E-RAB to pre-emption of other E-RABs (see TS 23.401 [10]). **Usage**: The E-RAB shall not be pre-empted by other E-RABs or the E-RAB may be pre-empted by other RABs. Pre-emption Vulnerability indicator applies for the entire duration of the E-RAB, unless modified, and as such indicates whether the E-RAB is a target of the pre-emption procedures/processes of the eNB.

9.3.1.21 GBR QoS Information

This IE indicates the maximum and guaranteed bit rates of a GBR E-RAB for downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
E-RAB Maximum Bit Rate Downlink	M		Bit Rate 9.3.1.22	Maximum Bit Rate in DL (i.e. from EPC to E-UTRAN) for the bearer. Details in TS 23.401 [10].
E-RAB Maximum Bit Rate Uplink	M		Bit Rate 9.3.1.22	Maximum Bit Rate in UL (i.e. from E-UTRAN to EPC) for the bearer. Details in TS 23.401 [10].
E-RAB Guaranteed Bit Rate Downlink	M		Bit Rate 9.3.1.22	Guaranteed Bit Rate (provided that there is data to deliver) in DL (i.e. from EPC to E-UTRAN) for the bearer. Details in TS 23.401 [10].
E-RAB Guaranteed Bit Rate Uplink	M		Bit Rate 9.3.1.22	Guaranteed Bit Rate (provided that there is data to deliver) in UL (i.e. from E-UTRAN to EPC) for the bearer. Details in TS 23.401 [10].

9.3.1.22 Bit Rate

This IE indicates the number of bits delivered by NG-RAN in UL or to NG-RAN in DL within a period of time, divided by the duration of the period. It is used, for example, to indicate the maximum or guaranteed bit rate for a GBR QoS flow, or an aggregated maximum bit rate.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Bit Rate	M		INTEGER (0.. 4,000,000,000,000,...)	The unit is: bit/s

9.3.1.23 Transaction ID

The *Transaction ID* IE uniquely identifies a procedure among all ongoing parallel procedures of the same type initiated by the same protocol peer. Messages belonging to the same procedure use the same Transaction ID. The Transaction ID is determined by the initiating peer of a procedure.

NOTE: If F1-C signalling transport is shared among multiple interface instances, the Transaction ID is allocated so that it can be associated with an F1-C interface instance. The Transaction ID may identify more than one interface instance.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Transaction ID	M		INTEGER (0..255, ...)	

9.3.1.24 DRX Cycle

The *DRX Cycle* IE is to indicate the desired DRX cycle.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Long DRX Cycle Length	M		ENUMERATED (ms10, ms20, ms32, ms40, ms60, ms64, ms70, ms80, ms128, ms160, ms256, ms320, ms512, ms640, ms1024, ms1280, ms2048, ms2560, ms5120, ms10240, ...)	Corresponds to the *drx-LongCycle* which is the length of the *drx-LongCycleStartOffset* contained in the *DRX-Config* IE defined in TS 38.331 [8].
Short DRX Cycle Length	O		ENUMERATED (ms2, ms3, ms4, ms5, ms6, ms7, ms8, ms10, ms14, ms16, ms20, ms30, ms32, ms35, ms40, ms64, ms80, ms128, ms160, ms256, ms320, ms512, ms640, ...)	Corresponds to the *drx-ShortCycle* contained in the *DRX-Config* IE defined in TS 38.331 [8].
Short DRX Cycle Timer	O		INTEGER (1..16)	Corresponds to the *drx-ShortCycleTimer* contained in the *DRX-Config* IE defined in TS 38.331 [8].

9.3.1.25 CU to DU RRC Information

This IE contains the RRC Information that are sent from gNB-CU to gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CG-ConfigInfo	O		OCTET STRING	Includes the *CG-ConfigInfo* message, as	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				defined in TS 38.331 [8].		
UE-CapabilityRAT-ContainerList	O		OCTET STRING	This IE is used in the NG-RAN and it includes *the UE-CapabilityRAT-ContainerList* IE, as defined in TS 38.331 [8].	-	
MeasConfig	O		OCTET STRING	Includes the *MeasConfig* IE, as defined in TS 38.331 [8] (without the *MeasGapConfig* IE). For EN-DC/NGEN-DC operation, includes the list of FR2 frequencies for which the gNB-CU requests the gNB-DU to generate gaps. For NG-RAN, NE-DC and MN for NR-NR DC, includes the list of FR1 and/or FR2 frequencies, for which the gNB-CU requests the gNB-DU to generate gaps and the gap type (per-UE or per-FR).	-	
Handover Preparation Information	O		OCTET STRING	Includes the *HandoverPreparationInformation* message, as defined in TS 38.331 [8].	YES	ignore
CellGroupConfig	O		OCTET STRING	Includes the *CellGroupConfig* IE, as defined in TS 38.331 [8].	YES	ignore
Measurement Timing Configuration	O		OCTET STRING	Contains the *MeasurementTimingConfiguration* inter-node message defined in TS 38.331 [8]. In EN-DC/NGEN-DC, it is included when the gaps for FR2 are requested to be configured by the MeNB. For MN in NR-NR DC,it is included when the gaps for FR2 and/or FR1 are requested by the SgNB	YES	ignore
UEAssistanceInformatio	O		OCTET	Includes the	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
n			STRING	*UEAssistanceInformation* message, as defined in TS 38.331 [8].		
CG-Config	O		OCTET STRING	Includes the *CG-Config* message, as defined in TS 38.331 [8].	YES	ignore
UEAssistanceInformationEUTRA	O		OCTET STRING	Includes the *UEAssistanceInformation* message, as defined in TS 36.331 [41].	YES	ignore
Location Measurement Information	O		OCTET STRING	Includes the *LocationMeasurementInfo* IE, as defined in TS 38.331[8]	YES	ignore
MUSIM-GapConfig	O		OCTET STRING	Includes the *MUSIM-GapConfig* IE as defined in TS 38.331 [8].	YES	reject
SDT-MAC-PHY-CG-Config	O		OCTET STRING	Includes the *SDT-MAC-PHY-CG-Config* IE, as defined in TS 38.331 [8].	YES	ignore
MBSInterestIndication	O		OCTET STRING	Includes the *MBSInterestIndication* message as defined in TS 38.331 [8].	YES	ignore
NeedForGapsInfoNR	O		OCTET STRING	Includes the *NeedForGapsInfoNR* IE, as defined in TS 38.331 [8].	YES	ignore
NeedForGapNCSG-InfoNR	O		OCTET STRING	Includes the *NeedForGapNCSG-InfoNR* IE, as defined in TS 38.331 [8].	YES	ignore
NeedForGapNCSG-InfoEUTRA	O		OCTET STRING	Includes the *NeedForGapNCSG-InfoEUTRA* IE, as defined in TS 38.331 [8].	YES	ignore
ConfigRestrictInfoDAPS	O		OCTET STRING	Includes the *ConfigRestrictInfoDAPS-r16* IE as defined in TS 38.331 [8]. This IE is used at the source node if DAPS HO is configured.	YES	ignore
Preconfigured measurement GAP Request	O		ENUMERATED (true, ...)		YES	ignore
NeedForInterruptionInfoNR	O		OCTET STRING	Includes the *NeedForInterruptionInfoNR* IE, as defined in TS 38.331 [8].	YES	ignore
musim-CapabilityRestrictionIndi	O		ENUMERATED (true, ...)	Corresponds to the *musim-*	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
cation				*CapRestrictionInd* contained in the *RRCSetupComplete* message or the *RRCResumeComplete* message as defined in TS 38.331 [8].		
musim-CandidateBandList	O		OCTET STRING	Includes the *musim-CandidateBandList* contained in the *OtherConfig* IE, as defined in TS 38.331 [8].	YES	ignore

9.3.1.26 DU to CU RRC Information

This IE contains the RRC Information that are sent from the gNB-DU to the gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CellGroupConfig	M		OCTET STRING	Includes the *CellGroupConfig* IE, as defined in TS 38.331 [8].	-	
MeasGapConfig	O		OCTET STRING	Includes the *MeasGapConfig* IE as defined in TS 38.331 [8]. For EN-DC/NGEN-DC operation, includes the gap for FR2, as requested by the gNB-CU via *MeasConfig* IE. For NG-RAN, NE-DC and MN for NR-NR DC, includes the gap(s) for FR1 and/or FR2, as requested by the gNB-CU via *MeasConfig* IE. For pre-configured measurement GAP scenario, it includes the *gapToAddModList* and/or *gapToReleaseList* as defined in TS 38.331 [8].	-	
Requested P-MaxFR1	O		OCTET STRING	Includes the *requestedP-MaxFR1* contained in the *CG-Config* message, as defined in TS 38.331 [8].	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				For EN-DC, NGEN-DC and NR-DC operation, this IE should be included.		
DRX Long Cycle Start Offset	O		INTEGER (0..10239)	Corresponds to the *drx-LongCycleStartOffset* IE contained in the *DRX-Config* IE as defined in TS 38.331 [8]. This field is not used in NR-DC.	YES	ignore
Selected BandCombinationIndex	O		OCTET STRING	Includes the *BandCombinationIndex* IE, as defined in TS 38.331 [8]. For (NG)EN-DC and NR DC operation, this IE should be included so that gNB-CU is informed of the selected Band Combination; if this IE is included, the gNB-CU uses this information to deduce the selected band.	YES	ignore
Selected FeatureSetEntryIndex	O		OCTET STRING	Includes the *FeatureSetEntryIndex* IE, as defined in TS 38.331 [8]. For (NG)EN-DC and NR DC operation, this IE should be included so that gNB-CU is informed of the selected FeatureSet.	YES	ignore
Ph-InfoSCG	O		OCTET STRING	Includes the *PH-TypeListSCG* IE, as defined in TS 38.331 [8].For MR-DC, this IE should be included so that gNB-CU is informed of the Power Headroom type for each serving cell in SN.	YES	ignore
Requested BandCombinationIndex	O		OCTET STRING	Includes the *BandCombinationIndex* IE, as defined in TS 38.331 [8]. This IE is used for the gNB-DU to request a new Band Combination.	YES	ignore
Requested	O		OCTET	Includes the	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
FeatureSetEntryIndex			STRING	*FeatureSetEntryIndex* IE, as defined in TS 38.331 [8]. This IE is used for the gNB-DU to request a new Feature Set.		
DRX Config	O		OCTET STRING	Includes the *DRX-Config* IE, as defined in TS 38.331 [8]. This field is only used in NR-DC.	YES	ignore
PDCCH BlindDetectionSCG	O		OCTET STRING	Includes the *pdcch-BlindDetectionSCG* contained in the *CG-ConfigInfo* message, as defined in TS 38.331 [8]. This IE is used between the MgNB-DU and the MgNB-CU.	YES	ignore
Requested PDCCH BlindDetectionSCG	O		OCTET STRING	Includes the *requestedPDCCH-BlindDetectionSCG* contained in the *CG-Config* message, as defined in TS 38.331 [8]. This IE is used between the SgNB-DU and the SgNB-CU.	YES	ignore
Ph-InfoMCG	O		OCTET STRING	Includes the *PH-TypeListMCG* IE, as defined in TS 38.331 [8]. For MR-DC, this IE should be included so that gNB-CU is informed of the Power Headroom type for each serving cell in MCG.	YES	ignore
MeasGapSharingConfig	O		OCTET STRING	Includes the *MeasGapSharingConfig* IE as defined in TS 38.331 [8].	YES	ignore
SL-PHY-MAC-RLC-Config	O		OCTET STRING	Includes the *SL-PHY-MAC-RLC-Config-r16* IE as defined in TS 38.331 [8].	YES	ignore
SL-ConfigDedicatedEUTRA-Info	O		OCTET STRING	Includes the *SL-ConfigDedicatedEUTRA-Info* IE as defined in TS 38.331 [8].	YES	ignore
Requested P-MaxFR2	O		OCTET STRING	Includes the *requestedP-MaxFR2* contained in the *CG-Config*	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				message, as defined in TS 38.331 [8]. For NR-DC operation, this IE should be included.		
SDT-MAC-PHY-CG-Config	O		OCTET STRING	Includes the *SDT-MAC-PHY-CG-Config* IE, as defined in TS 38.331 [8].	YES	ignore
MUSIM-GapConfig	O		OCTET STRING	Includes the *MUSIM-GapConfig* IE as defined in TS 38.331 [8].	YES	ignore
SL-RLC-ChannelToAddModList	O		OCTET STRING	Includes the *sl-RLC-ChannelToAddModList-r17* contained in the *SL-ConfigDedicatedNR* IE, as defined in TS 38.331 [8].	YES	ignore
InterFrequencyConfig-NoGap	O		ENUMERATED (true, ...)	Corresponds to the *interFrequencyConfig-NoGap-r16* contained in the *MeasConfig* IE, as defined in TS 38.331 [8].	YES	ignore
ul-GapFR2-Config	O		OCTET STRING	Includes the *ul-GapFR2-Config* contained in the *RRCReconfiguration* message, as specifed in TS 38.331 [8].	YES	ignore
TwoPHRModeMCG	O		ENUMERATED (enabled, ...)	Corresponds to the *twoPHRModeMCG* contained in the *CG-ConfigInfo* message, as defined in TS 38.331 [8]. For NR-DC, this IE should be included so that gNB-CU is informed of the two PHR mode in the MN.	YES	ignore
TwoPHRModeSCG	O		ENUMERATED (enabled, ...)	Corresponds to the *twoPHRModeSCG* contained in the *CG-Config* message, as defined in TS 38.331 [8]. For NR-DC, this IE should be included so that gNB-CU is	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				informed of the two PHR mode in the SN.		
ncd-SSB-RedCapInitialBWP-SDT	O		OCTET STRING	Includes the *NonCellDefiningSSB* contained in the *RRCRelease* message, as specified in TS 38.331 [8].	YES	ignore
ServCellInfoList	O		OCTET STRING	Includes the *ServCellInfoListSCG-NR* IE or the *ServCellInfoListMCG-NR* IE, as defined in TS 38.331 [8]. This IE is used for inter-node message for MN and SN in case of split gNB architecture.	YES	ignore
Extended SL-PHY-MAC-RLC-Config	O		OCTET STRING	Includes the *SL-PHY-MAC-RLC-Config-v1700* IE as defined in TS 38.331 [8]. If this IE is present, the *SL-RLC-ChannelToAddModList* IE is ignored.	YES	ignore

9.3.1.27 RLC Mode

The *RLC Mode* IE indicates the RLC Mode used for a DRB or a BH RLC channel, or a Uu Relay RLC channel, or a PC5 Relay RLC channel.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
RLC Mode	M		ENUMERATED (RLC-AM, RLC-UM-Bidirectional, RLC-UM-Unidirectional-UL, RLC-UM-Unidirectional-DL, ...)	

9.3.1.28 SUL Information

This IE provides information about the SUL carrier.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
SUL ARFCN	M		INTEGER (0.. maxNRARFCN)	RF Reference Frequency as defined in TS 38.104 [17] section 5.4.2.1. The frequency provided in this IE	–	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				identifies the absolute frequency position of the reference resource block (Common RB 0) of the SUL carrier. Its lowest subcarrier is also known as Point A.		
SUL Transmission Bandwidth	M		Transmission Bandwidth 9.3.1.15		–	
Carrier List	O		NR Carrier List 9.3.1.137	If included, the *SUL Transmission Bandwidth* IE shall be ignored.	YES	ignore
Frequency Shift 7p5khz	O		ENUMERATED (false, true, ...)	Indicate whether the value of Δ_{shift} is 0kHz or 7.5kHz when calculating $F_{REF,shift}$ as defined in Section 5.4.2.1 of TS 38.104 [17].	YES	ignore

Range bound	Explanation
maxNRARFCN	Maximum value of NR ARFCNs. Value is 3279165.

9.3.1.29 5GS TAC

This information element is used to identify Tracking Area Code.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
5GS TAC	M		OCTET STRING (SIZE (3))	

9.3.1.29a Configured EPS TAC

This information element is used to identify a configured EPS Tracking Area Code in order to enable application of Roaming and Access Restrictions for EN-DC as specified in TS 37.340 [7]. This IE is configured for the cell, but not broadcast.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Configured EPS TAC	M		OCTET STRING (SIZE (2))	

9.3.1.30 RRC Reconfiguration Complete Indicator

This IE indicates the result of the reconfiguration performed towards the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RRC Reconfiguration Complete Indicator	M		ENUMERATED (true, ..., failure)	

9.3.1.31 UL Configuration

This IE indicates how the UL scheduling is configured at gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UL UE Configuration	M		ENUMERATED (no-data, shared, only, ...)	Indicates how the UE uses the UL at gNB-DU, for which "no-data" indicates that the UL scheduling is not performed at gNB-DU, "shared" indicates that the UL scheduling is performed at both gNB-DU and another node, and "only" indicates that the UL scheduling is only performed at the gNB-DU.

9.3.1.32 C-RNTI

This IE contains the C-RNTI information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
C-RNTI	M		INTEGER (0..65535, ...)	C-RNTI as defined in TS 38.331 [8].

9.3.1.33 Cell UL Configured

This IE indicates whether the gNB-CU requests the gNB-DU to configure the uplink as no UL, UL, SUL or UL+SUL for the indicated cell for the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell UL Configured	M		ENUMERATED (none, UL, SUL, UL and SUL, ...)	Further details are defined in TS 38.331 [8]

9.3.1.34 RAT-Frequency Priority Information

The RAT-Frequency Priority Information contains either the *Subscriber Profile ID for RAT/Frequency priority* IE or the *Index to RAT/Frequency Selection Priority* IE. These parameters are used to define local configuration for RRM strategies.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *RAT-Frequency Priority Information*	M			
>EN-DC				
>>Subscriber Profile ID for RAT/Frequency priority	M		INTEGER (1.. 256, ...)	
>NG-RAN				
>>Index to RAT/Frequency Selection Priority	M		INTEGER (1.. 256, ...)	

9.3.1.35 LCID

This IE uniquely identifies a LCID for the associated SRB or DRB.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
LCID	M		INTEGER (1..32, ...)	Corresponds to the *LogicalChannelIdentity* defined in TS 38.331 [8].

9.3.1.36 Duplication Activation

The *Duplication Activation* IE indicates whether UL PDCP Duplication is activated or not.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Duplication Activation	M		ENUMERATED (Active, Inactive, ...)	

9.3.1.37 Slice Support List

This IE indicates the list of supported slices.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Slice Support Item IEs		*1..<maxnoofSliceItems>*		
>S-NSSAI	M		9.3.1.38	

Range bound	Explanation
maxnoofSliceItems	Maximum no. of signalled slice support items. Value is 1024.

9.3.1.38 S-NSSAI

This IE indicates the S-NSSAI as defined in TS 23.003 [23].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SST	M		OCTET STRING (SIZE(1))	
SD	O		OCTET STRING (SIZE(3))	

9.3.1.39 UE Identity Index value

This IE is used by the gNB-DU to calculate the Paging Frame.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *UE Identity Index Value*	M			
>Length-10				
>>Index Length 10	M		BIT STRING (SIZE(10))	Coded as specified in TS 38.304 [24].

9.3.1.40 Paging DRX

This IE indicates the Paging DRX as defined in TS 38.304 [24].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging DRX	M		ENUMERATED(32,	Unit in radio frame.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
			64, 128, 256, ...)	

9.3.1.41 Paging Priority

This IE indicates the paging priority for paging a UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging Priority	M		ENUMERATED (PrioLevel1, PrioLevel2, PrioLevel3, PrioLevel4, PrioLevel5, PrioLevel6, PrioLevel7, PrioLevel8, ...)	Lower value codepoint indicates higher priority.

9.3.1.42 gNB-CU System Information

This IE contains the system information encoded by the gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
SIB type to Be Updated List		*1*			-	
>SIB type to Be Updated Item IEs		*1... <maxnoofSIBTypes>*			-	
>>SIB type	M		INTEGER (2..32, ...)	Indicates a certain SIB block, e.g. 2 means sibType2, 3 for sibType3, etc. Values for SIBs generated by the gNB-DU as defined subclause 5.2.2 in TS 38.470 [2], values 6, 7, 8 and values corresponding to not defined SIBs in TS 38.331 [8] are not applicable in this version of the specifications.	-	
>>SIB message	M		OCTET STRING	SIB as defined in subclause 6.3.1 in TS 38.331 [8].	-	
>>Value Tag	M		INTEGER (0..31, ...)		-	
>>areaScope	O		ENUMERATED (true, ...)	Indicates that a SIB is area specific. If the field is not present, the SIB is cell specific.	YES	ignore
SystemInformationAreaID	O		BIT STRING (SIZE (24))	Indicates the system information area that the cell belongs to, if any.	YES	ignore

Range bound	Explanation
maxnoofSIBTypes	Maximum no. of SIB types, the maximum value is 32.

9.3.1.43 RAN UE Paging identity

This IE indicates the RAN UE Paging identity.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
I-RNTI	M		BIT STRING (SIZE(40))	

9.3.1.44 CN UE Paging Identity

The 5G-S-TMSI is used as UE identifier for CN paging.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *CN UE paging identity*	M			
>5G-S-TMSI				
>>5G-S-TMSI	M		BIT STRING (SIZE(48))	Details defined in TS 38.413 [3]

9.3.1.45 QoS Flow Level QoS Parameters

This IE defines the QoS to be applied to a QoS flow, or to a DRB, or to a BH RLC channel, or to a Uu Relay RLC channel, or to a PC5 Relay RLC channel, or to a MRB.

NOTE: For a BH RLC channel, the listed mandatory IEs and the *GBR QoS Flow Information* IE are applicable, where *GBR QoS Flow Information* IE may be present if BH RLC channel conveys the traffic belonging to a GBR QoS Flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *QoS Characteristics*	M				-	
>Non-dynamic 5QI					-	
>>Non Dynamic 5QI Descriptor	M		9.3.1.49		-	
>Dynamic 5QI					-	
>>Dynamic 5QI Descriptor	M		9.3.1.47		-	
NG-RAN Allocation and Retention Priority	M		9.3.1.48		-	
GBR QoS Flow Information	O		9.3.1.46	This IE shall be present for GBR QoS Flows only and is ignored otherwise.	-	
Reflective QoS Attribute	O		ENUMERATED (subject to, ...)	Details in TS 23.501 [21]. This IE applies to non-GBR flows only and is ignored otherwise.	-	
PDU Session ID	O		INTEGER (0 ..255)	As specified in TS 23.501 [21].	YES	ignore
UL PDU Session Aggregate Maximum Bit Rate	O		Bit Rate 9.3.1.22	The PDU session Aggregate Maximum Bit Rate	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				Uplink which is associated with the involved PDU session.		
QoS Monitoring Request	O		ENUMERATED (UL, DL, Both, ..., stop)	Indicates to measure UL, or DL, or both UL/DL delays for the associated QoS flow or stop the corresponding QoS monitoring.	YES	ignore
PDCP Terminating Node DL Transport Layer Address	O		Transport Layer Address 9.3.2.3	DL Transport Layer Address of node terminating PDCP. Included for MN-terminated SCG bearers and SN-terminated MCG bearers.	YES	ignore
PDU Set QoS Parameters		0..1			YES	ignore
>UL PDU Set QoS Information	O		PDU Set QoS Information 9.3.1.319		-	
>DL PDU Set QoS Information	O		PDU Set QoS Information 9.3.1.319		-	
MMSID	O		OCTET STRING (SIZE (1))	Multi-modal service ID from the application, used to indicate QoS flows are related to a multi-modal service, as specified in TS 23.501 [21].	YES	ignore
Indication of Bitrate Adaptation	O		ENUMERATED (uplink, ...)	Indicates that the QoS Flow allows rate adaptation in the indicated direction.	YES	ignore
DL PDU Set Information Marking Support Indication	O		ENUMERATED (true, ...)		YES	ignore

9.3.1.46 GBR QoS Flow Information

This IE indicates QoS parameters for a GBR QoS flow or GBR bearer for downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Maximum Flow Bit Rate Downlink	M		Bit Rate 9.3.1.22	Maximum Bit Rate in DL. Details in TS 23.501 [21].	-	
Maximum Flow Bit Rate Uplink	M		Bit Rate 9.3.1.22	Maximum Bit Rate in UL. Details in TS 23.501 [21].	-	
Guaranteed Flow Bit Rate Downlink	M		Bit Rate 9.3.1.22	Guaranteed Bit Rate (provided there is data to deliver) in DL. Details in TS 23.501 [21].	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Guaranteed Flow Bit Rate Uplink	M		Bit Rate 9.3.1.22	Guaranteed Bit Rate (provided there is data to deliver). Details in TS 23.501 [21].	-	
Maximum Packet Loss Rate Downlink	O		Maximum Packet Loss Rate 9.3.1.50	Indicates the maximum rate for lost packets that can be tolerated in the downlink direction. Details in TS 23.501 [21].	-	
Maximum Packet Loss Rate Uplink	O		Maximum Packet Loss Rate 9.3.1.50	Indicates the maximum rate for lost packets that can be tolerated in the uplink direction. Details in TS 23.501 [21].	-	
Alternative QoS Parameters Set List	O		9.3.1.125	Indicates alternative sets of QoS Parameters for the QoS flow.	YES	ignore
Monitoring Request on Available Bitrate		*0..1*			YES	ignore
>Monitoring Request	M		ENUMERATED (ul, dl, both, stop, ...)	Indicates to monitor and report UL, DL, or both UL/DL available bitrate for the associated QoS flow as specified in TS 23.501 [21], or stop the corresponding monitoring.	-	
>DL Available Bitrate Report Thresholds	C-ifReportDL		Available Bitrate Reporting Threshold List 9.3.1.353	Indicates the DL report thresholds for available bitrate exposure, as specified in TS 23.501 [21].	-	
>UL Available Bitrate Report Thresholds	C-ifReportUL		Available Bitrate Reporting Threshold List 9.3.1.353	Indicates the UL report thresholds for available bitrate exposure, as specified in TS 23.501 [21].	-	

Condition	Explanation
ifReportDL	This IE shall be present if the *Monitoring Request* IE is set to the value “dl” or “both”.
ifReportUL	This IE shall be present if the *Monitoring Request* IE is set to the value “ul” or “both”.

9.3.1.47 Dynamic 5QI Descriptor

This IE indicates the QoS Characteristics for a Non-standardised or not pre-configured 5QI for downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
QoS Priority Level	M		INTEGER (1..127)	For details see TS 23.501 [21].	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Packet Delay Budget	M		9.3.1.51	For details see TS 23.501 [21]. For IAB, the Packet Delay Budget defines the upper bound for the time that a packet may be delayed between the IAB-DU/IAB-donor-DU and its child IAB-MT, or between the IAB-DU and its served UE. For a PC5 Relay RLC channel, the Packet Delay Budget defines the upper bound for the time that a packet may be delayed between the L2 U2N relay UE and L2 U2N remote UE. For a Uu Relay RLC channel, the Packet Delay Budget defines the upper bound for the time that a packet may be delayed between the gNB-DU and L2 U2N relay UE. This IE is ignored if the *Extended Packet Delay Budget* IE is present.	-	
Packet Error Rate	M		9.3.1.52	For details see TS 23.501 [21].	-	
5QI	O		INTEGER (0..255,...)	This IE contains the dynamically assigned 5QI as specified in TS 23.501 [21].	-	
Delay Critical	C-ifGBRflow		ENUMERATED (delay critical, non-delay critical)	For details see TS 23.501 [21].	-	
Averaging Window	C-ifGBRflow		9.3.1.53	For details see TS 23.501 [21].	-	
Maximum Data Burst Volume	O		9.3.1.54	For details see TS 23.501 [21]. This IE shall be included if the *Delay Critical* IE is set to "delay critical" and is ignored otherwise.	-	
Extended Packet Delay Budget	O		9.3.1.145	Packet Delay Budget is specified in TS 23.501 [21].	YES	ignore
CN Packet Delay Budget Downlink	O		Extended Packet Delay	Core Network Packet Delay	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			Budget 9.3.1.145	Budget is specified in TS 23.501 [21]. This IE may be present in case of GBR QoS flows and is ignored otherwise.		
CN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.145	Core Network Packet Delay Budget is specified in TS 23.501 [21]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore

Condition	Explanation
ifGBRflow	This IE shall be present if the *GBR QoS Flow Information* IE is present in the *QoS Flow Level QoS Parameters* IE.

9.3.1.48 NG-RAN Allocation and Retention Priority

This IE specifies the relative importance of a QoS flow or a DRB compared to other QoS flows or DRBs for allocation and retention of NG-RAN resources.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Priority Level	M		INTEGER (0..15)	**Desc**.: This IE defines the relative importance of a resource request (see TS 23.501 [21]). **Usage**: Values are ordered in decreasing order of priority, i.e., with 1 as the highest priority and 15 as the lowest priority. Further usage is defined in TS 23.501 [21].
Pre-emption Capability	M		ENUMERATED (shall not trigger pre-emption, may trigger pre-emption)	**Desc.:** This IE indicates the pre-emption capability of the request on other QoS flows (see TS 23.501 [21]). **Usage**: The QoS flow shall not pre-empt other QoS flows or, the QoS flow may pre-empt other QoS flows. Note: The Pre-emption Capability indicator applies to the allocation of resources for a QoS flow and as such it provides the trigger to the pre-emption procedures/processes of the gNB.
Pre-emption Vulnerability	M		ENUMERATED (not pre-emptable, pre-emptable)	**Desc.**: This IE indicates the vulnerability of the QoS flow to pre-emption of other QoS flows (see TS 23.501 [21]). **Usage**: The QoS flow shall not be pre-empted by other QoS flows or the QoS flow may be pre-empted by other QoS flows. Note: The Pre-emption Vulnerability indicator applies for the entire duration of the QoS

IE/Group Name	Presence	Range	IE type and reference	Semantics description
				flow, unless modified and as such indicates whether the QoS flow is a target of the pre-emption procedures/processes of thegNB.

9.3.1.49 Non Dynamic 5QI Descriptor

This IE indicates the QoS Characteristics for a standardized or pre-configured 5QI for downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
5QI	M		INTEGER (0..255,...)	This IE contains the standardized or pre-configured 5QI as specified in TS 23.501 [21]. For a BH RLC channel, the Packet Delay Budget included in 5QI defines the upper bound for the time that a packet may be delayed between the gNB-DU and its child IAB-MT.	-	
Priority Level	O		INTEGER (1..127)	For details see TS 23.501 [21]. When included overrides standardized or pre-configured value.	-	
Averaging Window	O		9.3.1.53	For details see TS 23.501 [21]. When included overrides standardized or pre-configured value.	-	
Maximum Data Burst Volume	O		9.3.1.54	For details see TS 23.501 [21]. When included overrides standardized or pre-configured value.	-	
CN Packet Delay Budget Downlink	O		Extended Packet Delay Budget 9.3.1.145	Core Network Packet Delay Budget is specified in TS 23.501 [21]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore
CN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.145	Core Network Packet Delay Budget is specified in TS 23.501 [21]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore

9.3.1.50 Maximum Packet Loss Rate

This IE indicates the Maximum Packet Loss Rate.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Maximum Packet Loss Rate	M		INTEGER(0..1000)	Ratio of lost packets per number of packets sent, expressed in tenth of percent.

9.3.1.51 Packet Delay Budget

This IE indicates the Packet Delay Budget for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Packet Delay Budget	M		INTEGER (0..1023, ...)	Upper bound value for the delay that a packet may experience expressed in unit of 0.5ms.

9.3.1.52 Packet Error Rate

This IE indicates the Packet Error Rate for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Scalar	M		INTEGER (0..9, ...)	The packet error rate is expressed as Scalar x 10-k where k is the Exponent.
Exponent	M		INTEGER (0..9, ...)	

9.3.1.53 Averaging Window

This IE indicates the Averaging Window for a QoS flow, and applies to GBR QoS Flows only.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Averaging Window	M		INTEGER (0..4095, ...)	Unit: ms. The default value is 2000ms.

9.3.1.54 Maximum Data Burst Volume

This IE indicates the Maximum Data Burst Volume for a QoS flow, and applies to delay critical GBR QoS flows only.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Maximum Data Burst Volume	M		INTEGER (0..4095, ..., 4096.. 2000000)	Unit: byte.

9.3.1.55 Masked IMEISV

This information element contains the IMEISV value with a mask, to identify a terminal model without identifying an individual Mobile Equipment.

IE/Group Name	Presence	Range	IE type and reference	Semantics description

Masked IMEISV	M		BIT STRING (SIZE (64))	Coded as the International Mobile station Equipment Identity and Software Version Number (IMEISV) defined in TS 23.003 [23] with the last 4 digits of the SNR masked by setting the corresponding bits to 1. The first to fourth bits correspond to the first digit of the IMEISV, the fifth to eighth bits correspond to the second digit of the IMEISV, and so on.

9.3.1.56 Notification Control

The *Notification Control* IE indicates whether the notification control for a given DRB is active or not-active.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Notification Control	M		ENUMERATED(Active, Not-Active, ...)	

9.3.1.57 RAN Area Code

This information element is used to uniquely identify a RAN Area Code.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RANAC	M		INTEGER (0..255)	RAN Area Code

9.3.1.58 PWS System Information

This IE contains the system information used for public warning.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
SIB type	M		INTEGER (6..8, ...)	Indicates a certain SIB block for public warning message, e.g. 6 means sibType6, 7 for sibType7, etc.	-	
SIB message	M		OCTET STRING	SIB message for public warning, as defined in TS 38.331 [8].	-	
Notification Information	O				YES	ignore
>Message Identifier	M		9.3.1.81		-	
>Serial Number	M		9.3.1.82		-	
Additional SIB Message List	O		9.3.1.86	Additional SIB messages containing different segments of a public warning message if segmentation is applied, as defined in TS 38.331 [8].	YES	reject

9.3.1.59 Repetition Period

This IE indicates the periodicity of the warning message to be broadcast.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Repetition Period	M		INTEGER (0..2^{17}-1, ...)	The unit of value 1 to 2^{17}-1 is [second].

9.3.1.60 Number of Broadcasts Requested

This IE indicates the number of times a message is to be broadcast.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Number of Broadcasts Requested	M		INTEGER (0..65535)	

9.3.1.61 Void

9.3.1.62 SIType List

This IE is used by the gNB-CU to indicate to the gNB-DU to broadcast one or several *SystemInformation* messages including the Other SI.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SI type item IEs		*1..<maxnoofSITypes>*		
>SI Type	M		INTEGER (1..32, ...)	Value "1" corresponds to the SI message identified by the first SI message indicated in the *SI-SchedulingInfo* IE in the *SIB1* message, value "2" to the SI message identified by the second SI message indicated in the *SI-SchedulingInfo* IE in the *SIB1* message, and so on, as defined in TS 38.331 [8].

Range bound	Explanation
maxnoofSITypes	Maximum no. of SI types, the maximum value is 32.

9.3.1.63 QoS Flow Identifier

This IE identifies a QoS Flow within a PDU Session, a MBS QoS flow within a MBS session. The definition and use of the QoS Flow Identifieris specified in TS 23.501 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoS Flow Identifier	M		INTEGER (0 ..63)	

9.3.1.64 Served E-UTRA Cell Information

This IE contains served cell information of an E-UTRA cell for spectrum sharing between E-UTRA and NR.

IE/Group Name	Presence	Range	IE type and	Semantics description

			reference	
CHOICE *EUTRA-Mode-Info*	M			
>FDD				
>>FDD Info		*1*		
>>>UL Offset to Point A	M		INTEGER (0..2199,...)	Indicates the offset to the center of the NR carrier for UL.
>>>DL Offset to Point A	M		INTEGER (0..2199,...)	Indicates the offset to the center of the NR carrier for DL.
>TDD				
>>TDD Info		*1*		
>>>Offset to Point A	M		INTEGER (0..2199,...)	Indicates the offset to the center of the NR carrier.
Protected E-UTRA Resource Indication	M		OCTET STRING	Indicates the Protected E-UTRA Resource Indication as defined in subclause 9.2.125 of TS 36.423 [9].

9.3.1.65 Available PLMN List

This IE indicates the list of available PLMN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Available PLMN Item IEs		*1..< maxnoofBPLMNs >*		
>PLMN Identity	M		9.3.1.14	

Range bound	Explanation
maxnoofBPLMNs	Maximum no. of Broadcast PLMN Ids. Value is 6.

9.3.1.66 RLC Failure Indication

This IE indicates the LCID associated with the RLC entity needing re-establishment.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Associated LCID	M		LCID 9.3.1.35	

9.3.1.67 Uplink TxDirectCurrentList Information

This IE contains the Uplink TxDirectCurrentList information that is configured by the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Uplink TxDirectCurrentList Information	M		OCTET STRING	Includes the *UplinkTxDirectCurrentList* IE as defined in TS 38.331 [8].

9.3.1.68 Service Status

This IE is used to indicate the service status of a cell by the gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Service State	M		ENUMERATED (In-Service, Out-Of-Service, ...)	Indicates the Service State of the cell. In-Service and Out-of-Service Service States are

				defined in TS 38.401 [4].
Switching Off Ongoing	O		ENUMERATED (True, ...)	This IE indicates that the gNB-DU will delete the cell after some time using a new gNB-DU Configuration Update procedure.

9.3.1.69 RLC Status

This IE indicates about the RLC configuration change included in the container towards the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Reestablishment Indication	M		ENUMERATED (reestablished, ...)	Indicates that following a change in the radio status, the RLC has been re-established.

9.3.1.70 RRC Version

This information element is used to identify RRC version corresponding to TS 38.331 [8].

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Latest RRC Version	M		BIT STRING (SIZE (3))	This IE is not used in this release.	-	
Latest RRC Version Enhanced	O		OCTET STRING (SIZE (3))	Latest supported RRC version in the release corresponding to TS 38.331 [8]. For a 3GPP specification version x.y.z, x is encoded by the leftmost byte, y by the middle byte, and z by the rightmost byte. If the RRC protocol is not supported in the gNB-DU, this IE is set to all ‘0’s.	YES	ignore

9.3.1.71 RRC Delivery Status

This IE provides information about the delivery status of RRC messages to the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Delivery Status	M		INTEGER ($0..2^{12}$-1)	Highest NR PDCP SN successfully delivered in sequence to the UE.
Triggering Message	M		INTEGER ($0..2^{12}$-1)	NR PDCP SN for the RRC message that triggered the report.

9.3.1.72 QoS Flow Mapping Indication

This IE is used to indicate only the uplink or downlink QoS flow is mapped to the DRB.

IE/Group Name	Presence	Range	IE type and	Semantics description

			reference	
QoS Flow Mapping Indication	M		ENUMERATED(ul, dl,...)	Indicates that only the uplink or downlink QoS flow is mapped to the DRB

9.3.1.73 Resource Coordination Transfer Information

This IE contains information for UE-associated E-UTRA – NR resource coordination.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MeNB Cell ID	M		BIT STRING (SIZE(28))	*E-UTRAN Cell Identifier* IE contained in the ECGI as defined in TS 36.423 [9] clause 9.2.14
Resource Coordination E-UTRA Cell Information	O		9.3.1.75	

9.3.1.74 E-UTRA PRACH Configuration

This IE indicates the PRACH resources used in E-UTRA cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RootSequenceIndex	M		INTEGER (0..837)	See section 5.7.2. in TS 36.211 [27]
ZeroCorrelationZoneConfiguration	M		INTEGER (0..15)	See section 5.7.2. in TS 36.211 [27]
HighSpeedFlag	M		BOOLEAN	TRUE corresponds to Restricted set and FALSE to Unrestricted set. See section 5.7.2 in TS 36.211 [27]
PRACH-FrequencyOffset	M		INTEGER (0..94)	See section 5.7.1 of TS 36.211 [27]
PRACH-ConfigurationIndex	C-ifTDD		INTEGER (0..63)	See section 5.7.1. in TS 36.211 [27]

Condition	Explanation
ifTDD	This IE shall be present if the *EUTRA-Mode-Info* IE in the *Resource Coordination E-UTRA Cell Information* IE is set to the value "TDD".

9.3.1.75 Resource Coordination E-UTRA Cell Information

This IE contains E-UTRA cell information for UE-associated E-UTRA – NR resource coordination.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *EUTRA-Mode-Info*	M				-	
>FDD					-	
>>FDD Info		*1*			-	
>>>UL EARFCN	O		INTEGER (0 .. maxExtendedEARFCN, ...)	The relation between EARFCN and carrier frequency (in MHz) is defined in TS 36.104 [25].	-	
>>>DL EARFCN	M		INTEGER (0 .. maxExtendedEARFCN, ...)	The relation between EARFCN and carrier frequency (in MHz) is defined in	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				TS 36.104 [25].		
>>>UL Transmission Bandwidth	O		E-UTRA Transmission Bandwidth 9.3.1.80	Present if *UL EARFCN* IE is present.	-	
>>>DL Transmission Bandwidth	M		E-UTRA Transmission Bandwidth 9.3.1.80		-	
>TDD					-	
>>TDD Info		*1*			-	
>>>EARFCN	M		INTEGER (0 .. maxExtendedEARFCN, ...)	The relation between EARFCN and carrier frequency (in MHz) is defined in TS 36.104 [25].	-	
>>>Transmission Bandwidth	M		E-UTRA Transmission Bandwidth 9.3.1.80		-	
>>>Subframe Assignment	M		ENUMERATED (sa0, sa1, sa2, sa3, sa4, sa5, sa6,...)	Uplink-downlink subframe configuration information defined in TS 36.211 [27]. In NB-IOT, sa0 and sa6 are not applicable.	-	
>>>Special Subframe Info		*1*		Special subframe configuration information defined in TS 36.211 [27]	-	
>>>>Special Subframe Patterns	M		ENUMERATED (ssp0, ssp1, ssp2, ssp3, ssp4, ssp5, ssp6, ssp7, ssp8, ssp9, ssp10, ...)		-	
>>>>Cyclic Prefix DL	M		ENUMERATED (Normal, Extended,...)		-	
>>>>Cyclic Prefix UL	M		ENUMERATED (Normal, Extended,...)		-	
E-UTRA PRACH Configuration	M		9.3.1.74		-	
Ignore PRACH Configuration	O		ENUMERATED (true,...)		YES	reject

Range bound	Explanation
maxExtendedEARFCN	Maximum value of extended EARFCN. Value is 262143.

9.3.1.76 Extended Available PLMN List

This IE indicates the list of available PLMN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description

Extended Available PLMN Item IEs		*1..< maxnoofExtendedBPLMNs >*		
>PLMN Identity	M		9.3.1.14	

Range bound	Explanation
maxnoofExtendedBPLMNs	Maximum no. of Extended Broadcast PLMN Ids. Value is 6.

9.3.1.77 Associated SCell List

This IE indicates the list of SCells associated with the RLC entity indicated by the *RLC Failure Indication* IE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Associated SCell Item IEs		*1..< maxnoofSCells >*			-	
>SCell ID	M		NR CGI 9.3.1.12		-	

Range bound	Explanation
maxnoofSCells	Maximum no. of SCells allowed towards one UE, the maximum value is 32.

9.3.1.78 Cell Direction

This IE indicates if the cell is either bidirectional or only DL or only UL.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell Direction	M		ENUMERATED (dl-only, ul-only)	

9.3.1.79 Paging Origin

This IE indicates whether Paging is originated due to the PDU sessions from the non-3GPP access.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging Origin	M		ENUMERATED (non-3GPP, ...)	

9.3.1.80 E-UTRA Transmission Bandwidth

This IE is used to indicate the E-UTRA UL or DL transmission bandwidth expressed in units of resource blocks " N_{RB} " (TS 36.104 [25]). The values bw1, bw6, bw15, bw25, bw50, bw75, bw100 correspond to the number of resource blocks "N_{RB}" 6, 15, 25, 50, 75, 100.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
E-UTRA Transmission Bandwidth	M		ENUMERATED (bw6, bw15, bw25, bw50, bw75, bw100,...)	

9.3.1.81 Message Identifier

This IE identifies the warning message.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Message Identifier	M		BIT STRING (SIZE(16))	This IE is set by the 5GC, transferred to the UE by the gNB node.

9.3.1.82 Serial Number

This IE identifies a particular message from the source and type indicated by the Message Identifier and is altered every time the message with a given Message Identifier is changed.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Serial Number	M		BIT STRING (SIZE(16))	

9.3.1.83 UAC Assistance Information

This information element contains assistance information helping the gNB-DU to set parameters for Unified Access Class barring.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
UAC PLMN List		*1*			-	
>UAC PLMN Item		*1..<maxnoofUACPLMNs>*			-	
>>PLMN Identity	M		9.3.1.14		-	
>>UAC Type List		*1*			-	
>>>UAC Type Item		*1..<maxnoofUACperPLMN>*			-	
>>>>UAC Reduction Indication	M		9.3.1.85		-	
>>>>CHOICE *UAC Category Type*	M				-	
>>>>>*UAC Standardized*					-	
>>>>>>UAC Action	M		9.3.1.84		-	
>>>>>*UAC Operator Defined*					-	
>>>>>>Access Category	M		INTEGER (32..63, ...)	Indicates the operator defined Access Category as defined in subclause 6.3.2 in TS 38.331 [8].	-	
>>>>>>Access Identity	M		BIT STRING (SIZE(7))	Indicates whether access attempt is allowed for each Access Identity as defined in subclause 6.3.2 in TS 38.331 [8].	-	
>>NID	O		9.3.1.155		YES	ignore

Range bound	Explanation
maxnoofUACPLMNs	Maximum no. of UAC PLMN Ids. Value is 12.

maxnoofUACperPLMN	Maximum no. of signalled categories per PLMN. Value is 64.

9.3.1.84 UAC Action

This IE indicates which signalling traffic is expected to be reduced by the gNB-CU, as defined in clause 8.7.7 of TS 38.413 [3]

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UAC Action	M		ENUMERATED (Reject RRC connection establishments for non-emergency MO DT, Reject RRC connection establishments for Signalling, Permit Emergency Sessions and mobile terminated services only, Permit High Priority Sessions and mobile terminated services only,...)	

9.3.1.85 UAC reduction Indication

This IE indicates the percentage of signalling traffic expected to be reduced by the gNB-CU, relative to the instantaneous incoming rate from the gNB-DU

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UAC reduction Indication	M		INTEGER (0..100)	Value 0 indicates that no access rate reduction is desired. In this version of specification, value 99 indicates the highest desired rate reduction.

9.3.1.86 Additional SIB Message List

This IE indicates the list of additional SIB messages containing all the remaining segments of a public warning message if segmentation is applied to such message.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Additional SIB Message List Item IEs		*1.. <maxnoofAdditionalSIBs >*		
>Additional SIB	M		OCTET STRING	SIB message containing one segment of a public warning message, as defined in TS 38.331 [8].

Range bound	Explanation
maxnoofAdditionalSIBs	Maximum no. of additional segments of a public warning message. Value is 63.

9.3.1.87 Cell Type

This IE provides the cell coverage area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell Size	M		ENUMERATED (verysmall, small, medium, large, ...)	

9.3.1.87a Configured TAC Indication

This IE indicates that the TAC with which this IE is associated, is only configured for the cell, but not broadcast.

NOTE: This IE is defined in accordance to the possibility foreseen in TS 38.331 [8] to not broadcast the TAC if the NR cell only supports PSCell/SCell functionality.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Configured TAC Indication	M		ENUMERATED (true, ...)	

9.3.1.88 Trace Activation

This IE defines parameters related to a trace session activation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Trace ID	M		OCTET STRING (SIZE(8))	This IE is composed of the following: Trace Reference defined in TS 32.422 [29] (leftmost 6 octets, with PLMN information encoded as in 9.3.1.14), and Trace Recording Session Reference defined in TS 32.422 [29] (last 2 octets).	-	
Interfaces To Trace	M		BIT STRING (SIZE(8))	Each position in the bitmap represents an NG-RAN node interface: first bit = NG-C, second bit = Xn-C, third bit = Uu, fourth bit = F1-C, fifth bit = E1: other bits reserved for future use. Value '1' indicates 'should be traced'. Value '0' indicates 'should not be traced'.	-	
Trace Depth	M		ENUMERATED (minimum, medium,	Defined in TS 32.422 [29].	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			maximum, minimumWithoutVendorSpecificExtension, mediumWithoutVendorSpecificExtension, maximumWithoutVendorSpecificExtension, ..., MinimumOnlyVendorSpecificTraceRecord, MediumOnlyVendorSpecificTraceRecord, MaximumOnlyVendorSpecificTraceRecord)			
Trace Collection Entity IP Address	M		Transport Layer Address 9.3.2.3	For File based Reporting. Defined in TS 32.422 [29]. Should be ignored if URI is present.	-	
MDT Configuration	O		9.3.1.150		YES	ignore
Trace Collection Entity URI	O		URI 9.3.2.6	For Streaming based Reporting. Defined in TS 32.422 [29] Replaces Trace Collection Entity IP Address if present	YES	ignore

9.3.1.89 Intended TDD DL-UL Configuration

This IE contains the subcarrier spacing, cyclic prefix and TDD DL-UL slot configuration of an NR cell that the receiving gNB needs to take into account for cross-link interference mitigation, and/or for NR-DC power coordination, when operating its own cells.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
NR SCS	M		ENUMERATED (scs15, scs30, scs60, scs120, ..., scs 480, scs960)	The values scs15, scs30, scs60, scs120, scs480 and scs960 corresponds to the sub carrier spacing in TS 38.104 [17].	-	
NR Cyclic Prefix	M		ENUMERATED (Normal, Extended, ...)	The type of cyclic prefix, which determines the number of symbols in a slot.	-	
NR DL-UL Transmission Periodicity	M		ENUMERATED (ms0p5, ms0p625, ms1, ms1p25, ms2, ms2p5, ms3, ms4, ms5, ms10, ms20, ms40, ms60, ms80, ms100,	The periodicity is expressed in the format msXpYZ, and equals X.YZ milliseconds.	-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
			ms120, ms140, ms160, ...)			
Slot Configuration List		1			-	
>Slot Configuration List Item		*1..<maxnoofslots>*			-	
>>Slot Index	M		INTEGER (0..5119, ...)		-	
>>CHOICE *Symbol Allocation in Slot*	M				-	
>>>All DL			NULL	This choice implies that all symbols in the slot are DL symbols.	-	
>>>All UL			NULL	This choice implies that all symbols in the slot are UL symbols.	-	
>>>Both DL and UL					-	
>>>>Number of DL Symbols	M		INTEGER (0..13, ...)	Number of consecutive DL symbols in the slot identified by Slot Index IE. If extended cyclic prefix is used, the maximum value is 11. The *Permutation* IE indicates the location of DL symbols in the slot.	-	
>>>>Number of UL Symbols	M		INTEGER (0..13, ...)	Number of consecutive UL symbols in the slot identified by Slot Index IE. If extended cyclic prefix is used, the maximum value is 11. The *Permutation* IE indicates the location of UL symbols in the slot.	-	
>>>>Permutation	O		ENUMERATED (DFU, UFD, ...)	If not present, the default value is DFU.	YES	ignore

Range bound	Explanation
maxnoofslots	Maximum length of number of slots in a 10-ms period. Value is 5120.

9.3.1.90 Additional RRM Policy Index

The *Additional RRM Policy Index* IE is used to provide additional information independent from the Subscriber Profile ID for RAT/Frequency priority as specified in TS 36.300 [20].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Additional RRM Policy Index	M		BIT STRING (32)	

9.3.1.91 DU-CU RIM Information

This IE conveys the Remote Interference Management message from the gNB-DU to the gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Victim gNB Set ID	M		gNB Set ID 9.3.1.93	
RIM-RS Detection Status	M		ENUMERATED(RS detected, RS disappeared, ...)	This IE indicates detection status of RIM-RS in gNB-DU
Aggressor Cell List		*1*		
>Aggressor Cell List Item		*1..< maxCellingNBDU >*		
>>Aggressor Cell ID	M		NR CGI 9.3.1.12	

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.

9.3.1.92 CU-DU RIM Information

This IE conveys the Remote Interference Management message from the gNB-CU to the gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Victim gNB Set ID	M		gNB Set ID 9.3.1.93	
RIM-RS Detection Status	M		ENUMERATED(RS detected, RS disappeared, ...)	This IE indicates detection status of RIM-RS in remote gNB(s).

9.3.1.93 gNB Set ID

The *gNB Set ID* IE is used to identify a group of gNBs which transmit the same RIM-RS.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB Set ID	M		BIT STRING (SIZE(22))	

9.3.1.94 Lower Layer Presence Status Change

This IE indicates lower layer resources' presence status shall be changed.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Lower Layer Presence Status Change	M		ENUMERATED (suspend lower layers, resume lower layers, ...)	"suspend lower layers" will store CellGroupConfig. From the parameters received within the ReconfigurationWithSync, only the sPCellConfigCommon is stored. "resume lower layers" shall restore SCG and it is set only after "suspend lower layers" has been indicated.

9.3.1.95 Traffic Mapping Information

This IE includes the information used by the gNB-DU to perform traffic mapping.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Traffic Mapping Information Type*	M			
>IP to layer2 Traffic Mapping Info				
>>IP to layer2 Traffic Mapping Info To Add	O		IP-to-layer-2 traffic mapping Information List 9.3.1.96	This IE indicates the mapping information for forwarding of IP traffic to layer-2 to be added.
>>IP to layer2 Traffic Mapping Info To Remove	O		Mapping Information to Remove 9.3.1.99	This IE indicates the mapping information for forwarding of IP traffic to layer 2 to be removed.
>BAP layer BH RLC channel Mapping Info				
>>BAP layer BH RLC channel Mapping Info To Add	O		BAP layer BH RLC channel mapping Information List 9.3.1.98	This IE indicates the mapping information for forwarding of traffic on BAP layer to be added.
>>BAP layer BH RLC channel Mapping Info To Remove	O		Mapping Information to Remove 9.3.1.99	This IE indicates the mapping information for forwarding of traffic on BAP layer to be removed.

9.3.1.96 IP-to-layer-2 traffic mapping Information List

This IE includes the information used by the IAB-donor-DU to perform the mapping from IP layer to layer-2. If this IE appears in the UE-associated F1AP signalling, the *BH Information* IE should only contain the *BAP Routing ID* IE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
IP-to-layer-2 mapping information Item		1.. *<maxnoofMappingEntries>*		
>Mapping Information Index	M		9.3.1.100	
>IP header information	M		9.3.1.97	
>BH Information	M		9.3.1.114	

Range bound	Explanation
maxnoofMappingEntries	Maximum no. of mapping entries, the maximum value is 67108864 (i.e. 2^26).

9.3.1.97 IP Header Information

This IE indicates the IP header information included in the *Traffic Mapping Information* IE for DL traffic.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Destination IAB TNL Address	M		9.3.1.102	This IE indicates the destination IPv4 address, or IPv6 address or IPv6 prefix of a DL packet.
DS Information List		*0.. <maxnoofDSInfo>*		
>DSCP	M		BIT STRING (SIZE(6))	This IE indicates the DS information of DL traffic.
IPv6 Flow Label	O		BIT STRING (SIZE(20))	This IE indicates the IPv6 Flow Label of DL traffic.

Range bound	Explanation
maxnoofDSInfo	Maximum no. of DSCP values related to a destination IP address that can be mapped to one BH RLC channel, the maximum value is 64.

9.3.1.98 BAP layer BH RLC channel mapping Information List

This IE includes the information used by the IAB-DU to perform the BH RLC channel mapping when forwarding traffic on BAP sublayer.

When this IE is included in the UE-associated F1AP signalling for setting up or modifying a BH RLC channel, it contains either the *Prior-Hop BAP Address* IE and the *Ingress BH RLC CH ID* IE to configure a mapping in downlink direction, or the *Next-Hop BAP address* IE and the *Egress BH RLC CH ID* IE to configure a mapping in uplink direction. This IE indicates the BH RLC channel served by the collocated IAB-MT.

When this IE is included in the non-UE-associated F1AP signalling, it shall contain the *Prior-Hop BAP Address* IE, the *Ingress BH RLC CH ID* IE, the *Next-Hop BAP address* IE and the *Egress BH RLC CH ID* IE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
BAP layer BH RLC channel mapping info Item		*1..<maxnoofMappingEntries>*			-	
>Mapping Information Index	M		9.3.1.100		-	
>Prior-Hop BAP Address	O		BAP Address 9.3.1.111		-	
>Ingress BH RLC CH ID	O		BH RLC Channel ID 9.3.1.113		-	
>Next-Hop BAP Address	O		BAP Address 9.3.1.111		-	
>Egress BH RLC CH ID	O		BH RLC Channel ID 9.3.1.113		-	
>Ingress Non-F1-terminating IAB-donor Topology Indicator	O		ENUMERATED (true, ...)	If present, indicates that the ingress topology for this entry is the non-F1-terminating IAB-donor topology of the boundary IAB-node.	YES	ignore
>Egress Non-F1-terminating IAB-donor Topology Indicator	O		ENUMERATED (true, ...)	If present, indicates that the egress topology for this entry is the non-F1-terminating IAB-donor topology of the boundary IAB-node.	YES	ignore

Range bound	Explanation
maxnoofMappingEntries	Maximum no. of mapping entries, the maximum value is 67108864 (i.e. 2^26).

9.3.1.99 Mapping Information to Remove

This IE includes a list of mapping information indexes corresponding to the mapping configuration which is to be removed.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Mapping Information to Remove List Item		*1.. <maxnoofMappingEntries>*		
>Mapping Information Index	M		9.3.1.100	

Range bound	Explanation
maxnoofMappingEntries	Maximum no. of mapping entries, the maximum value is 67108864 (i.e. 2^26).

9.3.1.100 Mapping Information Index

This IE includes an index of one mapping information entry at the IAB-donor-DU or an IAB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Mapping Information Index	M		BIT STRING (SIZE(26))	

9.3.1.101 IAB TNL Addresses Requested

The *IAB TNL Addresses Requested* IE indicates the number of IPv4 or IPv6 addresses or IPv6 address prefixes requested for the indicated usage.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
TNL Addresses or Prefixes Requested - All Traffic	O		INTEGER (1..256)	The number of TNL addresses/ IPv6 prefixes requested for all traffic.
TNL Addresses or Prefixes Requested - F1-C traffic	O		INTEGER (1..256)	The number of TNL addresses/IPv6 prefixes requested for F1-C traffic.
TNL Addresses or Prefixes Requested - F1-U traffic	O		INTEGER (1..256)	The number of TNL addresses/ IPv6 prefixes requested for F1-U traffic.
TNL Addresses or Prefixes Requested - Non-F1 traffic	O		INTEGER (1..256)	The number of TNL addresses/ IPv6 prefixes requested for non-F1 traffic.

9.3.1.102 IAB TNL Address

The *IAB TNL Address* IE indicates an IPv4 or IPv6 address or an IPv6 address prefix assigned to an IAB-node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *IAB TNL Address*	M			
>IPv4 Address				
>>IPv4 Address	M		BIT STRING (SIZE(32))	The IPv4 address allocated to an IAB-node.
>IPv6 Address				
>>IPv6 Address	M		BIT STRING (SIZE(128))	The IPv6 address allocated to an IAB-node.
>IPv6 Prefix				
>>IPv6 Prefix	M		BIT STRING (SIZE(64))	The IPv6 address prefix allocated to an IAB-node.

9.3.1.103 Uplink BH Non-UP Traffic Mapping

This IE indicates the mapping of uplink non-UP traffic to a BH RLC channel and BAP Routing ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Uplink Non-UP Traffic Mapping List		*1*		
>Uplink Non-UP Traffic Mapping List Item IEs		*1 .. <maxnoofNonUPTrafficMappings>*		
>>Non-UP Traffic Type	M		9.3.1.104	
>>BH Information	M		9.3.1.114	

Range bound	Explanation
maxnoofNonUPTrafficMappings	Maximum no. of non-UP traffic mappings. Value is 32.

9.3.1.104 Non-UP Traffic Type

This IE indicates the type of non-UP traffic.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Non-UP Traffic Type	M		ENUMERATED(UE-associated F1AP, non-UE-associated F1AP, non-F1, BAP control PDU, ...)	

9.3.1.105 IAB Info IAB-donor-CU

This IE contains cell-specific IAB-related information sent by an IAB-donor-CU to an IAB-DU or IAB-donor-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
IAB STC Info	O		9.3.1.109	Contains STC configuration of IAB-DU or IAB-donor-DU.

9.3.1.106 IAB Info IAB-DU

This IE contains cell-specific IAB-related information sent by an IAB-DU or IAB-donor-DU to an IAB-donor-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Multiplexing Info	O		9.3.1.108	Contains the information about multiplexing with cells configured for a collocated IAB-MT. Applicable for an IAB-DU.
IAB STC Info	O		9.3.1.109	Contains the information about STC configuration of IAB-DU or IAB-donor-DU.

9.3.1.107 gNB-DU Cell Resource Configuration

This IE contains the resource configuration of the cells served by a gNB-DU, i.e. the TDD/FDD resource parameters for each activated cell (TS 38.213 [31], clause 11.1.1).

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Subcarrier Spacing	M		ENUMERATED (kHz15, kHz30, kHz60, kHz120, kHz240, spare3, spare2, spare1, ...)	Subcarrier spacing used as reference for the TDD/FDD slot configuration.	-	
DUF Transmission Periodicity	O		ENUMERATED (ms0p5, ms0p625, ms1, ms1p25, ms2, ms2p5, ms5, ms10, ...)		-	
DUF Slot Configuration List		*0..1*			-	
>DUF Slot Configuration Item		*1..<maxnoofDUFSlots>*		The *maxNrofSlots* in TS 38.331 [8].	-	
>>CHOICE *DUF Slot Configuration*	M				-	
>>>Explicit Format						
>>>>Permutation	M		ENUMERATED (DFU, UFD, ...)		-	
>>>>Number of Downlink Symbols	O		INTEGER (0..14)		-	
>>>>Number of Uplink Symbols	O		INTEGER (0..14)		-	
>>>Implicit Format						
>>>>DUF Slot Format Index	M		INTEGER (0..254)	Index into Table 11.1.1-1 and Table 14-2 in TS 38.213 [31], excluding the last row in Table 14-2.	-	
HSNA Transmission Periodicity	M		ENUMERATED (ms0p5, ms0p625, ms1, ms1p25, ms2, ms2p5, ms5, ms10, ms20, ms40, ms80, ms160, ...)		-	
HSNA Slot Configuration List		*0..1*			-	
>HSNA Slot Configuration Item		*1..<maxnoofHSNASlots>*			-	
>>HSNA Downlink	O		ENUMERATED (HARD, SOFT, NOTAVAILABLE)	HSNA value for downlink symbols in a slot.	-	
>>HSNA Uplink	O		ENUMERATED (HARD, SOFT, NOTAVAILABLE)	HSNA value for uplink symbols in a slot.	-	
>>HSNA Flexible	O		ENUMERATED (HARD, SOFT, NOTAVAILABLE)	HSNA value for flexible symbols in a slot.	-	
RB Set Configuration	O		9.3.1.230		YES	reject
Frequency-Domain HSNA Configuration List		*0..1*			YES	reject
>Frequency-Domain HSNA Configuration Item		*1..<maxnoofRBsetsPerCell>*			EACH	reject

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>RB Set Index	M		INTEGER (0.. maxnoofRBsets PerCell-1, ...)	Refers to an RB set defined by RB Set Configuration. The RB set indices are consecutive (and increasing) starting at 0.	-	
>>Frequency-Domain HSNA Slot Configuration List		*1*			-	
>>>Frequency-Domain HSNA Slot Configuration Item		*1..<maxnoofHSNASlots>*			-	
>>>>Slot Index	O		INTEGER (0..5119)	Indicates an index to a slot within the HSNA Transmission Periodicity.	-	
>>>>HSNA Downlink	O		ENUMERATED (HARD, SOFT, NOTAVAILABLE)	HSNA value for downlink symbols in a slot, for an RB set.	-	
>>>>HSNA Uplink	O		ENUMERATED (HARD, SOFT, NOTAVAILABLE)	HSNA value for uplink symbols in a slot, for an RB set.	-	
>>>>HSNA Flexible	O		ENUMERATED (HARD, SOFT, NOTAVAILABLE)	HSNA value for flexible symbols in a slot, for an RB set.	-	
Child IAB-Nodes NA Resource List		*0..1*		List of child IAB-nodes served by the IAB-DU or IAB-donor-DU.	YES	reject
>Child IAB-Nodes NA Resource List Item		*1 .. <maxnoofChildIABNodes>*			EACH	reject
>>gNB-CU UE F1AP ID	M		9.3.1.4	Identifier of a child-node IAB-MT at the IAB-donor-CU.	-	
>>gNB-DU UE F1AP ID	M		9.3.1.5	Identifier of a child-node IAB-MT at an IAB-DU or IAB-donor-DU.	-	
>>NA Resource Configuration List		*0..1*		List of not-available resources of this cell for this child IAB-node	-	
>>>NA Resource Configuration Item		*1..<maxnoofHSNASlots>*			-	
>>>>NA Downlink	O		ENUMERATED (true, false, ...)	Indicates whether downlink symbols, in a slot, are available to serve the child IAB-node.	-	
>>>>NA Uplink	O		ENUMERATED (true, false, ...)	Indicates whether uplink symbols, in	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				a slot, are available to serve the child IAB-node.		
>>>>NA Flexible	O		ENUMERATED (true, false, ...)	Indicates whether flexible symbols, in a slot, are available to serve the child IAB-node.	-	
Parent IAB Nodes NA Resource Configuration List		*0..1*		List of unavailable resources of this cell for this IAB-node.	YES	reject
>Parent IAB Nodes NA Resource Configuration Item		*1..<maxnoofHSNASlots>*			EACH	reject
>>NA Downlink	O		ENUMERATED (true, false, ...)	Indicates whether downlink symbols, in a slot, are unavailable to serve the IAB-node.	-	
>>NA Uplink	O		ENUMERATED (true, false, ...)	Indicates whether uplink symbols, in a slot, are unavailable to serve the IAB-node.	-	
>>NA Flexible	O		ENUMERATED (true, false, ...)	Indicates whether flexible symbols, in a slot, are unavailable to serve the IAB-node.	-	

Range bound	Explanation
maxnoofDUFSlots	Maximum no. of slots in 10ms. Value is 320.
maxnoofHSNASlots	Maximum no of "Hard", "Soft" or "Not available" slots in 160ms. Value is 5120.
maxnoofRBsetsPerCell	Maximum no. of RB sets per IAB-DU cell. Value is 8
maxnoofRBsetsPerCell-1	Maximum no. of RB sets per IAB-DU cell minus 1. Value is 7
maxnoofChildIABNodes	Maximum number of child nodes served by an IAB-DU or an IAB-donor-DU. Value is 1024.

9.3.1.108 Multiplexing Info

This IE contains information about the multiplexing capabilities between the gNB-DU's cell and the cells configured on the co-located IAB-MT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
IAB-MT Cell List		*1*			-	
>IAB-MT Cell Item		*1 .. <maxnoofServingCells>*			-	
>>NR Cell Identity	M		BIT STRING (SIZE(36))	Cell identity of a serving cell configured for a co-located IAB-MT.	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>DU_RX/MT_RX	M		ENUMERATED (supported, not supported)	An indication of whether the IAB-node supports simultaneous reception at its DU and MT side.	-	
>>DU_TX/MT_TX	M		ENUMERATED (supported, not supported)	An indication of whether the IAB-node supports simultaneous transmission at its DU and MT side.	-	
>>DU_RX/MT_TX	M		ENUMERATED (supported, not supported)	An indication of whether the IAB-node supports simultaneous reception at its DU and transmission at its MT side.	-	
>>DU_TX/MT_RX	M		ENUMERATED (supported, not supported)	An indication of whether the IAB-node supports simultaneous transmission at its DU and reception at its MT side.	-	
>>DU_RX/MT_RX_extend	O		ENUMERATED (supported, not supported, supported and FDM required,...)	An indication of whether the IAB-node supports simultaneous reception at its DU and MT side. If present, the *DU_RX/MT_RX* IE shall be ignored.	YES	ignore
>>DU_TX/MT_TX_extend	O		ENUMERATED (supported, not supported, supported and FDM required,...)	An indication of whether the IAB-node supports simultaneous transmission at its DU and MT side. If present, the *DU_TX/MT_TX* IE shall be ignored.	YES	ignore
>>DU_RX/MT_TX_extend	O		ENUMERATED (supported, not supported, supported and FDM required,...)	An indication of whether the IAB-node supports simultaneous reception at its DU and transmission at its MT side. If present, the *DU_RX/MT_TX* IE shall be ignored.	YES	ignore
>>DU_TX/MT_RX_extend	O		ENUMERATED (supported, not supported, supported and FDM required,...)	An indication of whether the IAB-node supports simultaneous transmission at its DU and reception at its MT side. If present, the *DU_TX/MT_RX* IE shall be ignored.	YES	ignore

Range bound	Explanation
maxnoofServingCells	Maximum no. of serving cells for IAB-MT. Value is 32, as defined by the *maxNrofServingCells* in TS 38.331 [8].

9.3.1.109 IAB STC Info

This IE contains cell SSB Transmission Configuration (STC) information of an IAB-DU or IAB-donor-DU. The information is used by neighbour IAB-MTs for discovery and measurements of this IAB-DU or IAB-donor-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
IAB STC-Info List		*1*		
>IAB STC-Info Item		*1 ..<maxnoofIABSTCInfo>*		
>>SSB Frequency Info	M		INTEGER (0.. maxNRARFCN)	The SSB central frequency.
>>SSB Subcarrier Spacing	M		ENUMERATED (kHz15, kHz30, kHz120, kHz240, spare3, spare2, spare1, ...)	The SSB subcarrier spacing.
>>SSB Transmission Periodicity	M		ENUMERATED (sf10, sf20, sf40, sf80, sf160, sf320, sf640, ..., sf5)	
>>SSB Transmission Timing Offset	M		INTEGER (0.. 127, ...)	SSB transmission timing offset in number of half-frames.
>>CHOICE *SSB Transmission Bitmap*	M			The *SSB-ToMeasure* IE defined in TS 38.331 [8].
>>>Short Bitmap				
>>>>Short Bitmap	M		BIT STRING (SIZE (4))	
>>>Medium Bitmap				
>>>>Medium Bitmap	M		BIT STRING (SIZE (8))	
>>>Long Bitmap				
>>>>Long Bitmap	M		BIT STRING (SIZE (64))	

Range bound	Explanation
maxnoofIABSTCInfo	Maximum no. of STC configurations. Value is 5. This includes 1 STC configuration for access and 4 STC configurations for backhaul.
maxNRARFCN	Maximum value of NR ARFCNs. Value is 3279165.

9.3.1.110 BAP Routing ID

This IE indicates the BAP Routing ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
BAP Address	M		9.3.1.111	
Path ID	M		BAP Path ID	

			9.3.1.112	

9.3.1.111 BAP Address

This IE indicates the BAP address of an IAB-node or of an IAB-donor-DU, and it is part of the BAP Routing ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
BAP Address	M		BIT STRING (SIZE(10))	Corresponds to the *bap-Address* contained in the *RRCReconfiguration* message or contained in the *BAP-RoutingID* IE, *or the iab-donor-DU-BAP-Address* contained in the *RRCReconfiguration* message defined TS 38.331[8].

9.3.1.112 BAP Path ID

This IE indicates the BAP path ID, which is part of the BAP Routing ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
BAP Path ID	M		BIT STRING (SIZE(10))	Corresponds to the *bap-Pathid* contained in the *BAP-RoutingID* IE defined in subclause 6.3.2 of TS 38.331 [8].

9.3.1.113 BH RLC Channel ID

This IE uniquely identifies a BH RLC channel in the link between IAB-MT of the IAB-node and IAB-DU of the parent IAB-node or IAB-donor-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
BH RLC CH ID	M		BIT STRING (SIZE(16))	

9.3.1.114 BH Information

This IE includes the backhaul information for UL or DL.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
BAP Routing ID	O		9.3.1.110	This IE is not needed for the BAP control PDU. For UL F1-U traffic, the BAP address included in this IE also indicates the IAB-donor-DU via which the DL traffic is transmitted.	-	
Egress BH RLC CH List		*0..1*			-	
>Egress BH RLC CH List Item		*1..<maxnoofEgressLin*			-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
		ks>				
>>Next-Hop BAP Address	M		BAP Address 9.3.1.111	This IE identifies the next-hop node on the backhaul path to receive the packet. The value of this IE should be unique in the whole list.	-	
>>Egress BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113	This IE identifies the BH RLC channel in the link between the IAB node/IAB-donor-DU and the node identified by the *Next-Hop BAP Address* IE.	-	
Non-F1-Terminating IAB-donor Topology Indicator	O		ENUMERATED (true, ...)	If present, indicates that the Next-Hop BAP Address and Egress BH RLC CH ID contained in this IE pertain to the non-F1-terminating IAB-donor topology of the boundary IAB-node.	YES	ignore

Range bound	Explanation
maxnoofEgressLinks	Maximum no. of egress links. Value is 2.

9.3.1.115 Control Plane Traffic Type

This IE indicates the control plane traffic type carried over a BH RLC channel.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Control Plane Traffic Type	M		INTEGER (1..3, ...)	Control plane traffic types with different priorities are identified by the different codepoints in this IE, where 1 has the highest priority.

9.3.1.116 NR V2X Services Authorized

This IE provides information on the authorization status of the UE to use the NR sidelink for V2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Vehicle UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Vehicle UE.
Pedestrian UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Pedestrian UE.

9.3.1.117 LTE V2X Services Authorized

This IE provides information on the authorization status of the UE to use the LTE sidelink for V2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Vehicle UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Vehicle UE.
Pedestrian UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Pedestrian UE.

9.3.1.118 LTE UE Sidelink Aggregate Maximum Bit Rate

This IE provides information on the Aggregate Maximum Bitrate of the UE's communication over LTE sidelink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
LTE UE Sidelink Aggregate Maximum Bit Rate	M		Bit Rate 9.3.1.22	Value 0 shall be considered as a logical error by the receiving gNB-DU.

9.3.1.119 NR UE Sidelink Aggregate Maximum Bit Rate

This IE provides information on the Aggregate Maximum Bitrate of the UE's communication over NR sidelink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR UE Sidelink Aggregate Maximum Bit Rate	M		Bit Rate 9.3.1.22	Value 0 shall be considered as a logical error by the receiving gNB-DU.

9.3.1.120 SL DRB ID

This IE uniquely identifies a SL DRB for a UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SL DRB ID	M		INTEGER (1.. 512, ...)	Corresponds to the *SLRB-Uu-ConfigIndex* IE defined in TS 38.331 [8].

9.3.1.121 PC5 QoS Flow Identifier

This IE uniquely identifies one sidelink QoS flow between the UE and the network in the scope of UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PC5 QoS Flow Identifier	M		INTEGER (1.. 2048)	Corresponds to the *SL-QoS-FlowIdentity* IE defined in TS 38.331 [8].

9.3.1.122 PC5 QoS Parameters

This IE defines the QoS to be applied to a SL DRB or to a PC5 Relay RLC channel for L2 U2U relaying.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *PC5 QoS Characteristics*	M				-	
>Non-dynamic PQI					-	
>>Non Dynamic PQI Descriptor	M		9.3.1.126		-	
>Dynamic PQI					-	
>>Dynamic PQI Descriptor	M		9.3.1.127		-	
PC5 QoS Flow Bit Rates	O			Only applies for GBR QoS Flows.	-	
>Guaranteed Flow Bit Rate	M		Bit Rate 9.3.1.22	Guaranteed Bit Rate for the PC5 QoS flow. Details in TS 23.287 [40].	-	
>Maximum Flow Bit Rate	M		Bit Rate 9.3.1.22	Maximum Bit Rate for the PC5 QoS flow. Details in TS 23.287 [40].	-	

Range bound	Explanation
maxnoofPC5QoSFlows	Maximum no. of PC5 QoS flows allowed towards one UE for NR sidelink communication, the maximum value is 2048.

9.3.1.123 Alternative QoS Parameters Set Index

This IE indicates the index of alternative QoS parameters set.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Alternative QoS Parameters Set Index	M		INTEGER (1..8, ...)	Values are ordered in decreasing order of priority, i.e., with 1 as the highest priority and 8 as the lowest priority.

9.3.1.124 Alternative QoS Parameters Set Notify Index

This IE indicates the QoS parameters set which can currently be fulfilled.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Alternative QoS Parameters Set Notify Index	M		INTEGER (0..8, ...)	Indicates the index of the item within the the *Alternative QoS Parameters Set List* IE corresponding to the currently fulfilled alternative QoS parameters set. Value 0 indicates that NG-RAN cannot even fulfil the lowest priority alternative parameter set.

9.3.1.125 Alternative QoS Parameters Set List

This IE contains alternative sets of QoS parameters which the gNB can indicate to be fulfilled when notification control is enabled and it cannot fulfil the requested list of QoS parameters.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Alternative QoS Parameters Set Item		*1..<maxnoofQoSPar*			-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
		aSets>				
>Alternative QoS Parameters Set Index	M		9.3.1.123		-	
>Guaranteed Flow Bit Rate Downlink	O		Bit Rate 9.3.1.22		-	
>Guaranteed Flow Bit Rate Uplink	O		Bit Rate 9.3.1.22		-	
>Packet Delay Budget	O		9.3.1.51		-	
>Packet Error Rate	O		9.3.1.52		-	
>Maximum Data Burst Volume	O		9.3.1.54	Maximum Data Burst Volume is specified in TS 23.501 [21]. This IE may be included if the *Delay Critical* IE is set to “delay critical” and is ignored otherwise.	YES	ignore
>PDU Set Delay Budget Downlink	O		Extended Packet Delay Budget 9.3.1.145		YES	ignore
>PDU Set Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.145		YES	ignore
>PDU Set Error Rate Downlink	O		Packet Error Rate 9.3.1.52		YES	ignore
>PDU Set Error Rate Uplink	O		Packet Error Rate 9.3.1.52		YES	ignore

Range bound	Explanation
maxnoofQoSParaSets	Maximum no. of alternative sets of QoS Parameters allowed for the QoS profile. Value is 8.

9.3.1.126 Non Dynamic PQI Descriptor

This IE indicates the QoS Characteristics for a standardized or pre-configured PQI for sidelink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
5QI	M		INTEGER (0..255,...)	This IE contains the standardized or pre-configured PQI as specified in TS 23.287 [40]
QoS Priority Level	O		INTEGER (1..8,...)	For details see TS 23.501 [21]. When included overrides standardized or pre-configured value.
Averaging Window	O		9.3.1.53	For details see TS 23.501 [21]. When included overrides standardized or pre-configured value.
Maximum Data Burst Volume	O		9.3.1.54	For details see TS 23.501 [21]. When included overrides standardized or pre-configured value.

9.3.1.127 Dynamic PQI Descriptor

This IE indicates the QoS Characteristics for a Non-standardised or not pre-configured PQI for sidelink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Resource Type	O		ENUMERATED (GBR, non-GBR, delay critical GBR, ...)	
QoS Priority Level	M		INTEGER (1..8, ...)	For details see TS 23.501 [21].
Packet Delay Budget	M		9.3.1.51	For details see TS 23.501 [21]. For a PC5 Relay RLC channel, the Packet Delay Budget defines the upper bound for the time that a packet may be delayed between the L2 U2U Relay UE and L2 U2U Remote UE.
Packet Error Rate	M		9.3.1.52	For details see TS 23.501 [21].
Averaging Window	C-ifGBRflow		9.3.1.53	For details see TS 23.501 [21].
Maximum Data Burst Volume	O		9.3.1.54	For details see TS 23.501 [21]. This IE shall be included if the *Delay Critical* IE is set to "delay critical" and is ignored otherwise.

Condition	Explanation
ifGBRflow	This IE shall be present if the *PC5 QoS Flow Bit Rates* IE is present in the *PC5 QoS parameters* IE.

9.3.1.128 TNL Capacity Indicator

The *TNL Capacity Indicator* IE indicates the offered and available capacity of the Transport Network experienced by the gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DL TNL Offered Capacity	M		INTEGER (1.. 16777216,...)	Maximum capacity offered by the transport portion of the gNB-DU – gNB-CU in kbps
DL TNL Available Capacity	M		INTEGER (0.. 100,...)	Available capacity over the transport portion serving the node in percentage. Value 100 corresponds to the offered capacity
UL TNL Offered Capacity	M		INTEGER (1.. 16777216,...)	Maximum capacity offered by the transport portion of the gNB-DU – gNB-CU in kbps
UL TNL Available Capacity	M		INTEGER (0.. 100,...)	Available capacity over the transport portion serving the node in percentage. Value 100 corresponds to the offered capacity

9.3.1.129 Radio Resource Status

The *Radio Resource Status* IE indicates the usage of the PRBs per cell for MIMO, per SSB area and per slice for all traffic in Downlink and Uplink and the usage of PDCCH CCEs for Downlink and Uplink scheduling.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
SSB Area Radio Resource Status List		*1*			-	
>SSB Area Radio Resource Status Item		*1..<maxnoofSSBAreas>*			-	
>>SSB Index	M		INTEGER		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			(0..63)			
>>SSB Area DL GBR PRB usage	M		INTEGER (0..100)	Per SSB area DL GBR PRB usage in percentage of the cell total PRB number.	-	
>>SSB Area UL GBR PRB usage	M		INTEGER (0..100)	Per SSB area UL GBR PRB usage in percentage of the cell total PRB number.	-	
>>SSB Area DL non-GBR PRB usage	M		INTEGER (0..100)	Per SSB area DL non-GBR PRB usage in percentage of the cell total PRB number.	-	
>>SSB Area UL non-GBR PRB usage	M		INTEGER (0..100)	Per SSB area UL non-GBR PRB usage in percentage of the cell total PRB number.	-	
>>SSB Area DL Total PRB usage	M		INTEGER (0..100)	Per SSB area DL Total PRB usage in percentage of the cell total PRB number.	-	
>>SSB Area UL Total PRB usage	M		INTEGER (0..100)	Per SSB area UL Total PRB usage in percentage of the cell total PRB number.	-	
>>DL scheduling PDCCH CCE usage	O		INTEGER (0..100)		-	
>>UL scheduling PDCCH CCE usage	O		INTEGER (0..100)		-	
Slice Radio Resource List		*0..1*			YES	ignore
>Slice Radio Resource Item		*1..< maxnoofBPLMNsNR >*			-	
>>PLMN Identity	M		9.3.1.14	Broadcast PLMN	-	
>>S-NSSAI Radio Resource Status List		*1*			-	
>>>S-NSSAI Radio Resource Status Item		*1..<maxnoofSliceItems>*			-	
>>>>S-NSSAI	M		9.3.1.38		-	
>>>>S-NSSAI DL GBR PRB usage	M		INTEGER (0..100)	Per slice DL GBR PRB usage in percentage of the cell total PRB number.	-	
>>>>S-NSSAI UL GBR PRB usage	M		INTEGER (0..100)	Per slice UL GBR PRB usage for this slice in percentage of the cell total PRB number.	-	
>>>>S-NSSAI DL non-GBR PRB usage	M		INTEGER (0..100)	Per slice DL non-GBR PRB usage for this slice in percentage of the cell total PRB	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				number.		
>>>>S-NSSAI UL non-GBR PRB usage	M		INTEGER (0..100)	Per slice UL non-GBR PRB usage for this slice in percentage of the cell total PRB number.	-	
>>>>Slice DL Total PRB allocation	M		INTEGER (0..100)	Total amount of DL PRBs available per cell for this slice if all the resources the slice could access were usable.	-	
>>>>Slice UL Total PRB allocation	M		INTEGER (0..100)	Total amount of UL PRBs available per cell for this slice if all the resources the slice could access were usable.	-	
MIMO PRB usage Information	O				YES	ignore
>DL GBR PRB usage for MIMO	M		INTEGER (0..100)	Per cell DL GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [32].	-	
>UL GBR PRB usage for MIMO	M		INTEGER (0..100)	Per cell UL GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [32].	-	
>DL non-GBR PRB usage for MIMO	M		INTEGER (0..100)	Per cell DL non-GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [32].	-	
>UL non-GBR PRB usage for MIMO	M		INTEGER (0..100)	Per cell UL non-GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [32].	-	
>DL Total PRB usage for MIMO	M		INTEGER (0..100)	Per cell DL Total PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [32].	-	
>UL Total PRB usage for MIMO	M		INTEGER (0..100)	Per cell UL Total PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [32].	-	

Range bound	Explanation
maxnoofSSBAreas	Maximum no. SSB Areas that can be served by a cell. Value is 64.
maxnoofSliceItems	Maximum no. of signalled slice support items. Value is 1024.
maxnoofBPLMNsNR	Maximum no. of PLMN Ids.broadcast in a cell. Value is 12.

9.3.1.130 Composite Available Capacity Group

The *Composite Available Capacity Group* IE indicates the overall available resource level per cell and per SSB area in the cell in Downlink and Uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Composite Available Capacity Downlink	M		Composite Available Capacity 9.3.1.131	For the Downlink	-	
Composite Available Capacity Uplink	M		Composite Available Capacity 9.3.1.131	For the Uplink, including both NUL and SUL (if available)	-	
Composite Available Capacity Supplementary Uplink	O		Composite Available Capacity 9.3.1.131	For the SUL	YES	ignore

9.3.1.131 Composite Available Capacity

The *Composite Available Capacity* IE indicates the overall available resource level in the cell in either Downlink or Uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell Capacity Class Value	O		9.3.1.132	
Capacity Value	M		9.3.1.133	'0' indicates no resource is available, Measured on a linear scale.

9.3.1.132 Cell Capacity Class Value

The *Cell Capacity Class Value* IE indicates the value that classifies the cell capacity with regards to the other cells. The *Cell Capacity Class Value* IE only indicates resources that are configured for traffic purposes.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Capacity Class Value	M		INTEGER (1..100,...)	Value 1 shall indicate the minimum cell capacity, and 100 shall indicate the maximum cell capacity. There should be a linear relation between cell capacity and Cell Capacity Class Value.

9.3.1.133 Capacity Value

The *Capacity Value* IE indicates the amount of resources per cell and per SSB area that are available relative to the total gNB-DU resources. The capacity value should be measured and reported so that the minimum gNB-DU resource usage of existing services is reserved according to implementation. The *Capacity Value* IE can be weighted according to the ratio of cell capacity class values, if available.

IE/Group Name	Presence	Range	IE type and	Semantics description

			reference	
Capacity Value	M		INTEGER (0..100)	Value 0 shall indicate no available capacity, and 100 shall indicate maximum available capacity with respect to the whole cell. Capacity Value should be measured on a linear scale.
SSB Area Capacity Value List		*0..1*		
>SSB Area Capacity Value Item		*1..<maxnoofSSBAreas>*		
>>SSB Index	M		INTEGER (0..63)	
>>SSB Area Capacity Value	M		INTEGER (0..100)	Value 0 shall indicate no available capacity, and 100 shall indicate maximum available capacity . SSB Area Capacity Value should be measured on a linear scale.

Range bound	Explanation
maxnoofSSBAreas	Maximum no. SSB Areas that can be served by a cell. Value is 64.

9.3.1.134 Slice Available Capacity

The *Slice Available Capacity* IE indicates the amount of resources per network slice that are available per cell relative to the total gNB-DU resources per cell. The *Slice Available Capacity Value Downlink* IE and the *Slice Available Capacity Value Uplink* IE can be weighted according to the ratio of the corresponding cell capacity class values contained in the *Composite Available Capacity Group* IE, if available.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Slice Available Capacity List		*1*		
>Slice Available Capacity Item		*1..< maxnoofBPLMNsNR >*		
>>PLMN Identity	M		9.3.1.14	Broadcast PLMN
>>S-NSSAI Available Capacity List		*1*		
>>>S-NSSAI Available Capacity Item	M	*1 .. < maxnoofSliceItems>*		
>>>>S-NSSAI			9.3.1.38	
>>>>Slice Available Capacity Value Downlink	O		INTEGER (0..100)	Value 0 shall indicate no available capacity, and 100 shall indicate maximum available capacity . Slice Available Capacity Value Downlink should be measured on a linear scale.
>>>>Slice Available Capacity Value Uplink	O		INTEGER (0..100)	Value 0 shall indicate no available capacity, and 100 shall indicate maximum available capacity . Slice Available Capacity Value Uplink should be measured on a linear scale.

Range bound	Explanation
maxnoofSliceItems	Maximum no. of signalled slice support items. Value is 1024.
maxnoofBPLMNsNR	Maximum no. of PLMN Ids.broadcast in a cell. Value is 12.

9.3.1.135 Number of Active UEs

The *Number of Active UEs* IE indicates the mean number of active UEs as defined in TS 38.314 [32].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Mean number of Active UEs	M		INTEGER (0..16777215, ...)	As defined in TS 38.314 [32] and where value "1" is equivalent to 0.1 Active UEs, value "2" is equivalent to 0.2 Active UEs, value *n* is equivalent to n/10 Active UEs.

9.3.1.136 Hardware Load Indicator

The *Hardware Load Indicator* IE indicates the status of the Hardware Load.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DL Hardware Load Indicator	M		INTEGER (0..100, ...)	This indicates the load in percent
UL Hardware Load Indicator	M		INTEGER (0..100, ...)	This indicates the load in percent

9.3.1.137 NR Carrier List

This IE indicates the SCS-specific carriers per TDD, per DL, per UL or per SUL of an NR cell.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
NR Carrier Item		*1..<maxnoofNRSCSs>*		
>NR SCS	M		ENUMERATED (scs15, scs30, scs60, scs120, ..., scs480, scs960)	SCS for the corresponding carrier.
>Offset to Carrier	M		INTEGER (0.. 2199, ...)	Offset in frequency domain between Point A (lowest subcarrier of common RB 0) and the lowest usable subcarrier on this carrier in number of PRBs (using the *NR SCS* IE defined for this carrier). The maximum value corresponds to 275×8−1. See TS 38.211 [33], clause 4.4.2.
>Carrier Bandwidth	M		INTEGER (1.. maxnoofPhysicalResourceBlocks, ...)	Width of this carrier in number of PRBs (using the *NR SCS* IE defined for this carrier). See TS 38.211 [33], clause 4.4.2. For band n100 and specific GSCN values defined in Table 5.4.3.1-3 of TS 38.104 [17], the utilized transmission bandwidth is less than the maximum transmission bandwidth configuration N_{RB} specified in Table 5.3.2-1 of TS 38.104 [17].

Range bound	Explanation
maxnoofNRSCSs	Maximum no. of SCS-specific carriers per TDD, per DL, per UL or per SUL of an NR cell. Value is 5.
maxnoofPhysicalResourceBlocks	Maximum no. of Physical Resource Blocks. Value is 275.

9.3.1.138 SSB Positions In Burst

Indicates the time domain positions of the transmitted SS-blocks in a half frame with SS/PBCH blocks as defined in TS 38.213 [31], clause 4.1.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
CHOICE *ssb-PositionsInBurst*	M			The first/ leftmost bit corresponds to SS/PBCH block index 0, the second bit corresponds to SS/PBCH block index 1, and so on. Value 0 in the bitmap indicates that the corresponding SS/PBCH block is not transmitted while value 1 indicates that the corresponding SS/PBCH block is transmitted.
>ShortBitmap				
>>ShortBitmap	M		BIT STRING (SIZE(4))	
>MediumBitmap				
>>MediumBitmap	M		BIT STRING (SIZE(8))	
>LongBitmap				
>>LongBitmap	M		BIT STRING (SIZE(64))	

9.3.1.139 NR PRACH Configuration

This IE indicates the PRACH resources by a NR cell.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
UL PRACH Configuration	O		NR PRACH Configuration List 9.3.1.140	
SUL PRACH Configuration	O		NR PRACH Configuration List 9.3.1.140	

9.3.1.140 NR PRACH Configuration List

This IE indicates the PRACH resources used or reserved in the UL carrier(s) or SUL carrier(s) of the current NR cell.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
NR PRACH Configuration Item		*0..< maxnoofPrachConfiguration >*		Length=0 means releasing of all NR PRACH Configuration Items for this UL or SUL.	-	
>NR SCS	M		ENUMERATED (scs15, scs30, scs60, scs120, ..., scs 480, scs960)	The SCS of the carrier to which this *PRACH Configuration Item* relates, i.e. Δf in Section 5.3.2 in TS 38.211 [33]. The values scs15, scs30, scs60, scs120, scs 480, and scs960 corresponds to the	-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
				sub carrier spacing in TS 38.104 [17]. NOTE: Its value may not be identical to the SCS of PRACH.		
>PRACH Frequency Start from Carrier	M		INTEGER (0.. maxNrofPhysicalResourceBlocks-1, ...)	Lowest number of resource blocks which can be used to deliver MSG1 or the preamble part of MSGA, counting from the start number of the corresponding carrier. Identical to RB_{start} in Section 5.1.2.2.2 in TS 38.214 [34] plus *msg1-FrequencyStart* or *msgA-RO-FrequencyStart-r16* in TS 38.331 [8].	-	
>PRACH-FDM	M		ENUMERATED (one, two, four, eight, ...)	M in Section 6.3.3.2 in TS 38.211 [33].	-	
>PRACH Configuration Index	M		INTEGER (0.. 255, ..., 256..262)	See Section 6.3.3.2 in TS 38.211 [33].	-	
>SSB per RACH Occasion	M		ENUMERATED (oneEighth, oneFourth, oneHalf, one, two, four, eight, sixteen, ...)	Number of SSBs per RACH occasion. Value *oneEight* corresponds to one SSB associated with 8 RACH occasions, value *oneFourth* corresponds to one SSB associated with 4 RACH occasions, and so on.	-	
>CHOICE *FreqDomainLength*	M			For the case of PRACH resources reserved for BFR or MSG1-based SI Request, *L139* is always used.	-	
>>L839						
>>>L839 Info		*1*			-	
>>>>Root Sequence Index	M		INTEGER (0..837)	See Section 6.3.3.1 in TS 38.211 [33].	-	
>>>>Restricted Set Config	M		ENUMERATED (unrestrictedSet, restrictedSetTypeA, restrictedSetTypeB, ...)	See Section 6.3.3.1 in TS 38.211 [33].	-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
>>L139						
>>>L139 Info		*1*			-	
>>>>PRACH SCS	M		ENUMERATED (scs15, scs30, scs60, scs120, ..., scs480, scs960)	Subcarrier Spacing of PRACH, i.e. Δf_{RA} in Section 5.3.2 in TS 38.211 [33].	-	
>>>>Root Sequence Index	O		INTEGER (0..137)	See Section 6.3.3.1 in TS 38.211 [33].	-	
>>L571					YES	reject
>>>L571 Info		*1*			-	
>>>>PRACH SCS for L571	M		ENUMERATED (scs30, scs120, ..., scs480)	Subcarrier Spacing of PRACH, i.e. Δf_{RA} in Section 5.3.2 in TS 38.211 [33].	-	
>>>>Root Sequence Index	M		INTEGER (0..569)	See Section 6.3.3.1 in TS 38.211 [33].	-	
>>L1151					YES	reject
>>>L1151 Info		*1*			-	
>>>>PRACH SCS for L1151	M		ENUMERATED (scs15, scs120, ...)	Subcarrier Spacing of PRACH, i.e. Δf_{RA} in Section 5.3.2 in TS 38.211 [33].	-	
>>>>Root Sequence Index	M		INTEGER (0..1149)	See Section 6.3.3.1 in TS 38.211 [33].	-	
>Zero Correlation Zone Config	M		INTEGER (0..15)	See Section 6.3.3.1 in TS 38.211 [33].	-	
>SBFD Across Symbol Type	O		ENUMERATED (true, …, false)	If this IE is present, it indicates that the RACH configuration is for SBFD dual configuration and whether SBFD across symbol types is enabled.	YES	reject

Range bound	Explanation
maxnoofPhysicalResourceBlocks-1	Maximum no. of Physical Resource Blocks minus 1. Value is 274.
maxnoofPrachConfiguration	Maximum no. of PRACH Configuration. Value is 16.

9.3.1.141 TSC Traffic Characteristics

This IE provides the traffic characteristics of TSC QoS flows.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TSC Assistance Information Downlink	O		TSC Assistance Information 9.3.1.142	
TSC Assistance Information Uplink	O		TSC Assistance Information 9.3.1.142	

9.3.1.142 TSC Assistance Information

This IE provides the TSC assistance information for a TSC QoS flow in the uplink or downlink (see TS 23.501 [21]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Periodicity	M		9.3.1.143	Periodicity as specified in TS 23.501 [21].	-	
Burst Arrival Time	O		9.3.1.144	Burst Arrival Time as specified in TS 23.501 [21].	-	
Survival Time	O		9.3.1.231		YES	ignore
CHOICE *RAN Feedback Type*	O				YES	ignore
>proactive						
>>Burst Arrival Time Window	M		9.3.1.300		-	
>>Periodicity Range	O		9.3.1.301		-	
>reactive						
>>Capability for BAT Adaptation	M		ENUMERATED (true, ...)		-	
N6 Jitter Information	O		9.3.1.320	Indicates the jitter information associated with the Periodicity in downlink, as defined in TS 23.501[21].	YES	ignore

9.3.1.143 Periodicity

This IE indicates the Periodicity as defined in TS 23.501 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Periodicity	M		INTEGER (0..640000, ...)	Periodicity expressed in units of 1 us.

9.3.1.144 Burst Arrival Time

This IE indicates the Burst Arrival Time as defined in TS 23.501 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Burst Arrival Time	M		OCTET STRING	Encoded in the same format as the *ReferenceTime* IE as defined in TS 38.331 [8]. The value is provided with 1 us accuracy.

9.3.1.145 Extended Packet Delay Budget

This IE indicates the Packet Delay Budget for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Extended Packet Delay Budget	M		INTEGER (1..65535, ..., 65536..109999)	Upper bound value for the delay that a packet may experience expressed in unit of 0.01ms.

9.3.1.146 RLC Duplication Information

The IE contains the RLC duplication information in case that the indicated DRB is configured with more than two RLC entities as specified in TS 38.331 [8].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RLC Duplication State List		*1*		
>RLC Duplication State Items		*1 .. <maxnoofRLCDuplicationState>*		Each position in the list represents a secondary RLC entity in ascending order by the logical channel ID in the order of MCG and SCG. For Multi-path relay, each position in the list represents a secondary RLC entity in order of direct path (in ascending order of logical channel ID of secondary RLC entities) and indirect path.
>>Duplication State	M		ENUMERATED (Active, Inactive, ...)	
Primary Path Indication	O		ENUMERATED (True, False,...)	Indicates whether the primary path is located at the gNB-DU for DC based PDCP duplication. This IE is also applicable to multi-path relay.

Range bound	Explanation
maxnoofRLCDuplicationState	Maximum no of Secondary RLC entities. Value is 3.

9.3.1.147 Reporting Request Type

This IE indicates the type of accurate reference time information reporting to be handled by the gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Event Type	M		ENUMERATED (on demand, periodic, stop, ...)	
Report Periodicity Value	C-ifEventTypeisPeriodic		INTEGER (0..512, ...)	Indicates the periodicity of accurate reference time information report, Unit in radio frame.

Condition	Explanation
ifEventTypeisPeriodic	This IE shall be present if the *Event Type* IE is set to "periodic".

9.3.1.148 Time Reference Information

This IE contains the time reference information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Reference Time	M		9.3.1.149	
Reference SFN	M		INTEGER (0..1023)	Corresponds to the *referenceSFN* contained in the *ReferenceTimeInfo* IE, as defined in TS 38.331 [8]

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Uncertainty	O		INTEGER (0..32767, ...)	Corresponds to the *uncertainty* contained in the *ReferenceTimeInfo* IE, as defined in TS 38.331 [8].
Time Information Type	O		ENUMERATED (localClock)	Corresponds to the *timeInfoType* contained in the *ReferenceTimeInfo* IE, as defined in TS 38.331 [8].

9.3.1.149 Reference Time

This IE provides the accurate Reference Time information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Reference Time	M		OCTET STRING	Includes the *ReferenceTime* IE contained in the *ReferenceTimeInfo* IE as defined in TS 38.331 [8].

9.3.1.150 MDT Configuration

The IE defines the MDT configuration parameters.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT Activation	M		ENUMERATED(Immediate MDT only, Immediate MDT and Trace, ...)	
Measurements to Activate	M		BITSTRING (SIZE(8))	Each position in the bitmap indicates a MDT measurement, as defined in TS 37.320 [35]. Second Bit = M2, Fifth Bit = M5, Seventh Bit = M6, Eighth Bit = M7. Value "1" indicates "activate" and value "0" indicates "do not activate". This version of the specification does not use bits 1, bit 3, bit 4 and bit 6.
M2 Configuration	C-ifM2		ENUMERATED (true, ...)	
M5 Configuration	C-ifM5		9.3.1.152	
M6 Configuration	C-ifM6		9.3.1.153	
M7 Configuration	C-ifM7		9.3.1.154	

Condition	Explanation
ifM2	This IE shall be present if the *Measurements to Activate* IE has the second bit set to "1".
ifM5	This IE shall be present if the *Measurements to Activate* IE has the fifth bit set to "1".
ifM6	This IE shall be present if the *Measurements to Activate* IE has the seventh bit set to "1".
ifM7	This IE shall be present if the *Measurements to Activate* IE has the

	eighth bit set to "1".

9.3.1.151 MDT PLMN List

The purpose of the *MDT PLMN List* IE is to provide the list of PLMN allowed for MDT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT PLMN List		*1..<maxnoofMDTPLMNs>*		
>PLMN Identity	M		9.3.1.14	

Range bound	Explanation
maxnoofMDTPLMNs	Maximum no. of PLMNs in the MDT PLMN list. Value is 16.

9.3.1.152 M5 Configuration

This IE defines the parameters for M5 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M5 Collection Period	M		ENUMERATED (ms1024, ms2048, ms5120, ms10240, min1, ...)		-	
M5 Links to log	M		ENUMERATED (uplink, downlink, both-uplink-and-downlink, ...)		-	
M5 Report Amount	O		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity, ...)	Number of reports.	YES	ignore

9.3.1.153 M6 Configuration

This IE defines the parameters for M6 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M6 Report Interval	M		ENUMERATED (ms120, ms240, ms640, ms1024, ms2048, ms5120, ms10240, ms20480, ms40960, min1,min6, min12, min30,..., ms480)		-	
M6 Links to log	M		ENUMERATED (uplink, downlink, both-uplink-and-		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			downlink, ...)			
M6 Report Amount	O		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity, ...)	Number of reports.	YES	ignore

9.3.1.154 M7 Configuration

This IE defines the parameters for M7 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M7 Collection Period	M		INTEGER (1..60, ...)	Unit: minutes	-	
M7 Links to log	M		ENUMERATED (downlink, ...)		-	
M7 Report Amount	O		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity, ...)	Number of reports.	YES	ignore

9.3.1.155 NID

This IE is used to identify (together with a PLMN identifier) a Stand-alone Non-Public Network. The NID is specified in TS 23.003 [23].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NID	M		BIT STRING (SIZE(44))	

9.3.1.156 NPN Support Information

This IE contains NPN related information associated with Network Slicing information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *NPN Support Information*	M			
>SNPN Information				
>>NID	M		9.3.1.155	

9.3.1.157 NPN Broadcast Information

This IE contains NPN related broadcast information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *NPN Broadcast Information per PLMN*	M			
>SNPN Information				
>>Broadcast SNPN ID List	M		9.3.1.158	
>PNI-NPN Information				
>>Broadcast PNI-NPN ID List	M		Broadcast PNI-NPN ID Information 9.3.1.162	

9.3.1.158 Broadcast SNPN ID List

This IE contains SNPN related broadcast information associated with a set of PLMNs.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Broadcast SNPN ID List		*1..<maxnoofNIDs>*		
>PLMN Identity	M		9.3.1.14	
>Broadcast NID List	M		9.3.1.159	

Range bound	Explanation
maxnoofNIDs	Maximum no. of NIDs broadcast in a cell. Value is 12.

9.3.1.159 Broadcast NID List

This IE contains a list of NIDs.

IE/Group Name	Presence	RangeNIDsupported	IE type and reference	Semantics description
Broadcast NID		*1..<maxnoofNIDsupported*		
>NID	M		9.3.1.155	

Range bound	Explanation
maxnoofNIDsupported	Maximum no. of NIDs broadcast in a cell. Value is 12.

9.3.1.160 Broadcast CAG-Identifier List

This IE contains a list of CAG-Identifiers.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Broadcast CAG-Identifier List		*1..<maxnoofCAGsupported>*		
>CAG ID	M		9.3.1.161	

Range bound	Explanation
maxnoofCAGsupported	Maximum no. of CAG-Identifiers broadcast in a cell. Value is 12.

9.3.1.161 CAG ID

This IE is used to identify (together with a PLMN identifier) a Public Network Integrated NPN, as defined in TS 23.003 [23].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CAG ID	M		BIT STRING (SIZE (32))	Closed Access Group ID used in NR.

9.3.1.162 Broadcast PNI-NPN ID Information

This IE contains a list of PNI-NPN IDs.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Broadcast PNI-NPN ID		*1..<maxnoofB*		Broadcast PLMNs

Information		*PLMNs>*		
>PLMN Identity	M		9.3.1.14	
>Broadcast CAG-Identifier List	M		9.3.1.160	

Range bound	Explanation
maxnoofBPLMNs	Maximum no. of broadcast PLMNs by a cell. Value is 12.

9.3.1.163 Available SNPN ID List

This IE indicates the list of available SNPN ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Available SNPN ID List		*1..<maxnoofNIDs>*		
>PLMN Identity	M		9.3.1.14	
>Available NID List	M		Broadcast NID List 9.3.1.159	

Range bound	Explanation
maxnoofNIDs	Maximum no. of NIDs broadcast in a cell. Value is 12.

9.3.1.164 Void

9.3.1.165 Extended Slice Support List

This IE indicates a list of supported slices.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Slice Support Item IEs		*1..<maxnoofExtSliceItems>*		
>S-NSSAI	M		9.3.1.38	

Range bound	Explanation
maxnoofExtSliceItems	Maximum no. of signalled slice support items. Value is 65535.

9.3.1.166 Positioning Measurement Result

The purpose of this information element is to provide the measurement result(s).

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
Positioning Measured Result Item		*1 .. <maxnoofPosMeas>*			-	
>CHOICE *Measured Results Value*	M				-	
>>UL Angle of Arrival						
>>>UL Angle of Arrival	M		9.3.1.167		-	
>>UL SRS-RSRP						
>>>UL SRS-RSRP	M		INTEGER (0..126)		-	
>>UL RTOA						
>>>UL RTOA	M		UL RTOA		-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
			Measurement 9.3.1.168			
>>gNB Rx-Tx Time Difference						
>>>gNB Rx-Tx Time Difference	M		9.3.1.170		-	
>>Zenith Angle of Arrival Information					YES	reject
>>>Zenith Angle of Arrival Information	M		9.3.1.239		-	
>>Multiple UL AoA					YES	reject
>>>Multiple UL AoA	M		9.3.1.245		-	
>>UL SRS-RSRPP					YES	reject
>>>UL SRS-RSRPP	M		9.3.1.246		-	
>>UL RSCP					YES	reject
>>>UL RSCP	M		9.3.1.335		-	
>>UL SRS-TDCT					YES	reject
>>>UL SRS-TDCT	M		9.3.1.365		-	
>Time Stamp	M		9.3.1.171		-	
>Measurement Quality	O		TRP Measurement Quality 9.3.1.172		-	
>Measurement Beam Information	O		9.3.1.173		-	
>ARP ID	O		9.3.1.244		YES	ignore
>SRS Resource type	O		9.3.1.247		YES	ignore
>LoS/NLoS Information	O		9.3.1.249		YES	ignore
>Mobile TRP Location Information	O		9.3.1.304		YES	ignore
>Aggregated Positioning SRS Resource ID List		*0..1*		Indicates the used Positioning SRS resources across aggregated carriers.	YES	ignore

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
>>Aggregated Positioning SRS Resource ID Item		*2..<maxnoaggregatedSRS-Resources>*			-	
>>>Positioning SRS Resource ID	M		INTEGER (0..63)		-	
>>>Point A	M		INTEGER (0..3279165)		YES	ignore
>>>SCS Specific Carrier		*1*			YES	ignore
>>>>Offset To Carrier	M		INTEGER(0..2199,...)		-	
>>>>Subcarrier Spacing	M		ENUMERATED(kHz15, kHz30, kHz60, kHz120,..., kHz480, kHz960)		-	
>>>>Carrier Bandwidth	M		INTEGER (1..275,...)		-	
>>>PCI	O		INTEGER (0..1007)		YES	ignore
>Measured Frequency Hops	O		ENUMERATED (singleHop, multiHop, ...)		YES	ignore
>Measurement Based On Aggregated Resources	O		ENUMERATED(true, ...)		YES	ignore

Range bound	Explanation
maxnoofPosMeas	Maximum no. of measured quantities that can be configured and reported with one message. Value is 16384.
maxnoaggregatedSRS-Resources	Maximum no of aggregated SRS resources per UL BWP. Value is 3.

9.3.1.167 UL Angle of Arrival

This information element contains the uplink Angle of Arrival measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Azimuth Angle of Arrival	M		INTEGER(0..3599)	TS 38.133 [38]
Zenith Angle of Arrival	O		INTEGER(0..1799)	TS 38.133 [38]
LCS to GCS Translation AoA	O		LCS to GCS Translation 9.3.1.241	If absent, the azimuth and zenith are provided in GCS.

9.3.1.168 UL RTOA Measurement

This information element contains the uplink RTOA measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
CHOICE *UL RTOA Measurement*	M				-	
>k0						
>>k0	M		INTEGER (0.. 1970049)	TS 38.133 [38]	-	
>k1						
>>k1	M		INTEGER (0.. 985025)	TS 38.133 [38]	-	
>k2						

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
>>k2	M		INTEGER (0.. 492513)	TS 38.133 [38]	-	
>k3						
>>k3	M		INTEGER (0.. 246257)	TS 38.133 [38]	-	
>k4						
>>k4	M		INTEGER (0.. 123129)	TS 38.133 [38]	-	
>k5						
>>k5	M		INTEGER (0.. 61565)	TS 38.133 [38]	-	
>kminus1						
>>kminus1	M		INTEGER (0.. 3940097)	TS 38.133 [38]	YES	ignore
>kminus2						
>>kminus2	M		INTEGER (0.. 7880193)	TS 38.133 [38]	YES	ignore
>kminus3						
>>kminus3	M		INTEGER (0..15760385)	TS 38.133 [38]	YES	ignore
>kminus4						
>>kminus4	M		INTEGER (0..31520769)	TS 38.133 [38]	YES	ignore
>kminus5						
>>kminus5	M		INTEGER (0..63041537)	TS 38.133 [38]	YES	ignore
>kminus6						
>>kminus6	M		INTEGER (0..126083073)	TS 38.133 [38]	YES	ignore
Additional Path List	O		9.3.1.169	This IE is ignored if the *Extended Additional Path List* IE is included	-	
Extended Additional Path List	O		9.3.1.248		YES	ignore
TRP Rx TEG Information	O		9.3.1.280		YES	ignore

9.3.1.169 Additional Path List

This information element contains the additional path results of time measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
Additional Path Item		*1..<maxnoofPath>*			-	
>CHOICE *Relative Path Delay*	M				-	
>>k0						
>>>k0	M		INTEGER(0..16351)		-	
>>k1						
>>>k1	M		INTEGER(0..8176)		-	
>>k2						
>>>k2	M		INTEGER(0..4088)		-	
>>k3						
>>>k3	M		INTEGER(0..2044)		-	
>>k4						
>>>k4	M		INTEGER(0..1022)		-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
>>k5						
>>>k5	M		INTEGER(0..511)		-	
>>kminus1						
>>>kminus1	M		INTEGER (0.. 32701)	TS 38.133 [38]	YES	ignore
>>kminus2						
>>>kminus2	M		INTEGER (0.. 65401)	TS 38.133 [38]	YES	ignore
>>kminus3						
>>>kminus3	M		INTEGER (0..130801)	TS 38.133 [38]	YES	ignore
>>kminus4						
>>>kminus4	M		INTEGER (0..261601)	TS 38.133 [38]	YES	ignore
>>kminus5						
>>>kminus5	M		INTEGER (0..523201)	TS 38.133 [38]	YES	ignore
>>kminus6						
>>>kminus6	M		INTEGER (0..1046401)	TS 38.133 [38]	YES	ignore
>Path Quality	O		TRP Measurement Quality 9.3.1.172		-	
>Multiple UL AoA	O		9.3.1.245		YES	ignore
>Path Power	O		UL SRS-RSRPP 9.3.1.246		YES	ignore

Range bound	Explanation
maxnoofPath	Maximum no. of additional path measurements. Value is 2.

9.3.1.170 gNB Rx-Tx Time Difference

This information element contains the gNB Rx-Tx Time Difference measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
CHOICE *gNB Rx-Tx Time Difference Measurement*	M				-	
>k0						
>>k0	M		INTEGER (0.. 1970049)	TS 38.133 [38]	-	
>k1						
>>k1	M		INTEGER (0.. 985025)	TS 38.133 [38]	-	
>k2						
>>k2	M		INTEGER (0.. 492513)	TS 38.133 [38]	-	
>k3						
>>k3	M		INTEGER (0.. 246257)	TS 38.133 [38]	-	
>k4						
>>k4	M		INTEGER (0.. 123129)	TS 38.133 [38]	-	
>k5						
>>k5	M		INTEGER (0.. 61565)	TS 38.133 [38]	-	
>kminus1						
>>kminus1	M		INTEGER (0..	TS 38.133 [38]	YES	ignore

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
			3940097)			
>kminus2						
>>kminus2	M		INTEGER (0.. 7880193)	TS 38.133 [38]	YES	ignore
>kminus3						
>>kminus3	M		INTEGER (0..15760385)	TS 38.133 [38]	YES	ignore
>kminus4						
>>kminus4	M		INTEGER (0..31520769)	TS 38.133 [38]	YES	ignore
>kminus5						
>>kminus5	M		INTEGER (0..63041537)	TS 38.133 [38]	YES	ignore
>kminus6						
>>kminus6	M		INTEGER (0..126083073)	TS 38.133 [38]	YES	ignore
Additional Path List	O		9.3.1.169		-	
Extended Additional Path List	O		9.3.1.248		YES	ignore
TRP TEG Information	O		9.3.1.253		YES	ignore

9.3.1.171 Time Stamp

This information element contains the time stamp associated with the measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
System Frame Number	M		INTEGER(0..1023)		-	
CHOICE *Slot Index*	M				-	
>SCS-15						
>>SCS-15	M		INTEGER(0..9)		-	
>SCS-30						
>>SCS-30	M		INTEGER(0..19)		-	
>SCS-60						
>>SCS-60	M		INTEGER(0..39)		-	
>SCS-120						
>>SCS-120	M		INTEGER(0..79)		-	
>SCS-480						
>>SCS-480	M		INTEGER(0..319)		YES	reject
>SCS-960					-	
>>SCS-960	M		INTEGER(0..639)		YES	reject
Measurement Time	O		Relative Time 1900 9.3.1.183		-	
Symbol Index	O		INTEGER(0..13)	Applicable to UL RSCP measurement only.	YES	ignore

9.3.1.172 TRP Measurement Quality

This information element contains the TRP's best estimate of the quality of the measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
CHOICE *TRP Measurement Quality*	M				-	
>*Timing Measurement Quality*						
>>Measurement Quality	M		INTEGER(0..31)	TS 37.355 [39]	-	
>>Resolution	M		ENUMERATED (0.1m, 1m, 10m, 30m, ...)	TS 37.355 [39]	-	
>*Angle Measurement Quality*						
>>Azimuth Quality	M		INTEGER(0..255)		-	
>>Zenith Quality	O		INTEGER(0..255)		-	
>>Resolution	M		ENUMERATED (0.1deg, ...)		-	
>*Phase Quality*				Corresponds to information provided in *NR-PhaseQuality* IE as defined in TS 37.355 [39].	YES	ignore
>>Phase Quality Index	M		INTEGER(0..179)		-	
>>Phase Quality Resolution	M		ENUMERATED (0.1deg, 1deg, ...)		-	

9.3.1.173 Measurement Beam Information

This information element contains the receiving beam information when measuring UL signals.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
PRS Resource ID	O		INTEGER(0..63)	
PRS Resource Set ID	O		INTEGER(0..7)	
SSB Index	O		INTEGER(0..63)	

9.3.1.174 NG-RAN Access Point Position

This IE is used to identify the geographical position of an NG-RAN Access Point / TRP / TRP Antenna Reference Points. It is expressed as ellipsoid point with altitude and uncertainty ellipsoid according to TS 23.032 [36].

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Latitude Sign	M		ENUMERATED (North, South)	
Degrees Of Latitude	M		INTEGER ($0..2^{23}-1$)	The IE value (N) is derived by this formula: $N \le 2^{23} X /90 < N+1$ X being the latitude in degrees (0°.. 90°).
Degrees Of Longitude	M		INTEGER ($-2^{23}..2^{23}-1$)	The IE value (N) is derived by this formula: $N \le 2^{24} X /360 < N+1$ X being the longitude in degrees (-180°..+180°).
Direction of Altitude	M		ENUMERATED (Height, Depth)	
Altitude	M		INTEGER	The relation between the value

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
			(0..2^{15}-1)	(N) and the altitude (a) in meters it describes is N ≤ a < N+1, except for N=2^{15}-1 for which the range is extended to include all greater values of (a).
Uncertainty semi-major	M		INTEGER (0..127)	The uncertainty "r" is derived from the "uncertainty code" k by r = 10x(1.1^k-1).
Uncertainty semi-minor	M		INTEGER (0..127)	The uncertainty "r" is derived from the "uncertainty code" k by r = 10x(1.1^k-1).
Orientation of major axis	M		INTEGER (0..179)	
Uncertainty Altitude	M		INTEGER (0..127)	The uncertainty altitude "h" expressed in metres is derived from the "uncertainty code" k, by: h=45x(1.025^k-1).
Confidence	M		INTEGER (0..100)	In percentage

9.3.1.175 Requested SRS Transmission Characteristics

This IE contains the requested SRS configuration for the UE for positioning purposes.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
Number Of Periodic Transmissions	C-ifResourceTypePeriodic		INTEGER (0..500,...)	The number of periodic SRS transmissions requested. The value of '0' represents an infinite number of SRS transmissions.	-	
Resource Type	M		ENUMERATED (periodic, semi-persistent, aperiodic, ...)		-	
CHOICE *Bandwidth SRS*	M				-	
>FR1						
>>FR1 Bandwidth	M		ENUMERATED (5, 10, 20, 40, 50, 80, 100, ..., 160, 200, 15, 25, 30, 60, 35, 45, 70, 90)		-	
>FR2						
>>FR2 Bandwidth	M		ENUMERATED (50, 100, 200, 400,...,800,1600, 2000, 600)		-	
SRS Resource Set List		*0.. 1*			-	
>SRS Resource Set Item		*1..< maxnoSRS-Resource Sets>*			-	
>>Number of SRS Resources Per Set	O		INTEGER (1..16,...)	The number of SRS Resources per resource set for SRS transmission.	-	
>>Periodicity List		*0.. 1*			-	
>>>Periodicity List		1..<*maxno*			-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
Item		*SRS-ResourcePerSet>*				
>>>>PeriodicitySRS	M		ENUMERATED (0.125, 0.25, 0.5, 0.625, 1, 1.25, 2, 2.5, 4, 5, 8, 10, 16, 20, 32, 40, 64, 80, 160, 320, 640, 1280, 2560, 5120, 10240, ...)	Milli-seconds	-	
>>Spatial Relation Information	O		9.3.1.181	This IE is ignored if the *Spatial Relation Information per SRS Resource* IE is present.	-	
>>Pathloss Reference Information	O		9.3.1.201		-	
>>Spatial Relation Information per SRS Resource	O		9.3.1.210		YES	ignore
SSB Information	O		9.3.1.202		-	
SRS Frequency	O		INTEGER(0..3279165)	NR ARFCN The carrier frequency of SRS transmission bandwidth.	YES	ignore
Bandwidth Aggregation Request Indication	O		ENUMERATED (true, ...)		YES	ignore
Positioning Validity Area Cell List	O		9.3.1.336		YES	ignore
Validity Area specific SRS Information	O		9.3.1.339	This IE is ignored if the *Validity Area Specific SRS Information Extended* IE is present.	YES	ignore
Validity Area Specific SRS Information Extended	O		9.3.1.351		YES	ignore

Condition	Explanation
ifResourceTypePeriodic	This IE shall be present if the *Resource Type* IE is set to the value "Periodic".

Range bound	Explanation
maxnoSRS-ResourceSets	Maximum no of requested SRS Resource Sets for SRS transmission. Value is 16.
maxnoSRS-ResourcePerSet	Maximum no of SRS Resources per set. Value is 16.

9.3.1.176 TRP Information

The *TRP Information* IE contains information for one TRP within a gNB-DU.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
TRP ID	M		9.3.1.197		-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
TRP Information Type Response List		*1*			-	
>TRP Information Type Response Item		*1 .. <maxnoofTRPInfoTypes>*			-	
>>CHOICE *TRP Information Type Response Item*	M				-	
>>>NR PCI						
>>>>NR PCI	M		INTEGER (0..1007)	NR Physical Cell ID	-	
>>>NR CGI						
>>>>NR CGI			9.3.1.12		-	
>>>NR ARFCN						
>>>>NR ARFCN	M		INTEGER (0..3279165)		-	
>>>PRS Configuration						
>>>>PRS Configuration	M		9.3.1.177		-	
>>>SSB Information						
>>>>SSB Information	M		9.3.1.202		-	
>>>SFN Initialisation Time						
>>>>SFN Initialisation Time	M		Relative Time 1900 9.3.1.183		-	
>>>Spatial Direction Information						
>>>>Spatial Direction Information	M		9.3.1.179		-	
>>>Geographical Coordinates						
>>>>Geographical Coordinates	M		9.3.1.184		-	
>>>TRP Type					YES	reject
>>>>TRP Type	M		ENUMERATED (prs-only-tp, srs-only-rp, tp, rp, trp,..., mobile-trp)	TS 38.305 [42]	-	
>>>On-demand PRS TRP Information					YES	reject
>>>>On-demand PRS TRP Information	M		9.3.1.240		-	
>>>TRP Tx TEG Association					YES	reject
>>>>TRP Tx TEG Association	M		9.3.1.252		-	
>>>TRP Beam Antenna					YES	reject
>>>>TRP Beam Antenna Information	M		9.3.1.256		-	
>>>Mobile TRP Location						
>>>>Mobile TRP Location Information	M		9.3.1.304		YES	ignore
Mobile IAB-MT UE ID	C-		OCTET	The UE ID of the	YES	reject

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
	ifMobileTRP		STRING	IAB-MT associated with the mobile TRP, includes GPSI as defined in TS 29.571 [50]		

Range bound	Explanation
maxnoofTRPInfoTypes	Maximum no of TRP information types that can be requested and reported with one message. Value is 64.

Condition	Explanation
ifMobileTRP	This IE shall be present if the *TRP type* IE is set to the value 'mobile-trp'

9.3.1.177 PRS Configuration

This information element contains the DL PRS configuration for the TRP.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
PRS Resource Set List	M	*1..<maxnoofPRSresourceSets>*			-	
>PRS Resource Set ID	M		INTEGER(0..7)		-	
>Subcarrier Spacing	M		ENUMERATED (kHz15, kHz30, kHz60, kHz120, ...)		-	
>PRS bandwidth	M		INTEGER(1..63)	24,28,...,272 PRBs	-	
>Start PRB	M		INTEGER(0..2176)	Starting PRB to Point A	-	
>Point A	M		INTEGER (0..3279165)	NR ARFCN	-	
>Comb Size	M		ENUMERATED (2, 4, 6, 12, ...)		-	
>CP Type	M		ENUMERATED (normal, extended, ...)		-	
>Resource Set Periodicity	M		ENUMERATED (4,5,8,10,16,20,32,40,64,80,160,320,640,1280,2560,5120,10240,20480,40960,81920, ..., 128, 256, 512)	Slots	-	
>Resource Set Slot Offset	M		INTEGER(0..81919,...)		-	
>Resource Repetition Factor	M		ENUMERATED (rf1,rf2,rf4,rf6,rf8,rf16,rf32,...)		-	
>Resource Time Gap	M		ENUMERATED (tg1,tg2,tg4,tg8,tg16,tg32,...)		-	
>Resource Number of Symbols	M		ENUMERATED (n2,n4,n6,n12,...,n1)		-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
>PRS Muting	O				-	
>>Option1	O				-	
>>>Muting Pattern	M		DL-PRS Muting Pattern 9.3.1.178	Muting pattern option 1 is used to mute the whole PRS resource set (within a period)	-	
>>>Muting Bit Repetition Factor	M		ENUMERATED (rf1,rf2,rf4,rf8,...)		-	
>>Option2	O				-	
>>>Muting Pattern	M		DL-PRS Muting Pattern 9.3.1.178	Muting pattern option 2 is used to mute the selected repetition of the resource set (within the period)	-	
>PRS Resource Transmit Power	M		INTEGER(-60..50)		-	
>PRS Resource List	M	1..<maxnoofPRSresources>		*NR-DL-PRS-Resource-r16* as defined in TS 37.355 [39]	-	
>>PRS Resource ID	M		INTEGER(0..63)		-	
>>Sequence ID	M		INTEGER(0..4095)		-	
>>RE Offset	M		INTEGER(0..11, ...)		-	
>>Resource Slot Offset	M		INTEGER(0..511)		-	
>>Resource Symbol Offset	M		INTEGER(0..12)	This IE is ignored if the *Extended Resource Symbol Offset* IE is present.	-	
>>CHOICE *QCL Info*	O				-	
>>>SSB						
>>>>PCI	M		INTEGER (0..1007)		-	
>>>>SSB Index	O		INTEGER(0..63)		-	
>>>DL-PRS						
>>>>QCL Source PRS Resource Set ID	M		INTEGER(0..7)		-	
>>>>QCL Source PRS Resource ID	O		INTEGER(0..63)	If absent, the QCL source PRS resource ID is the same as the PRS resource ID	-	
>>Extended Resource Symbol Offset	O		INTEGER(0..13,...)		YES	ignore
Aggregated PRS Resource Set List	O		9.3.1.338		YES	ignore

Range bound	Explanation
maxnoofPRSresourceSets	Maximum no of PRS resource sets. Value is 8.
maxnoofPRSresources	Maximum no of PRS resources per PRS resource set. Value is 64.

9.3.1.178 DL-PRS Muting Pattern

This information element contains the DL-PRS muting pattern.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
CHOICE DL-*PRS Muting Pattern*	M			
>Two				
>>Two	M		BIT STRING (SIZE(2))	
>Four				
>>Four	M		BIT STRING (SIZE(4))	
>Six				
>>Six	M		BIT STRING (SIZE(6))	
>Eight				
>>Eight	M		BIT STRING (SIZE(8))	
>Sixteen				
>>Sixteen	M		BIT STRING (SIZE(16))	
>Thirty-two				
>>Thirty-two	M		BIT STRING (SIZE(32))	

9.3.1.179 Spatial Direction Information

This information element contains the spatial direction information of the DL PRS resources for the TRP.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
NR-PRS Beam Information	M		9.3.1.198	The spatial directions of DL-PRS Resources for TRP

9.3.1.180 SRS Resource Set ID

This information element indicates a resource set in the UE for UL SRS transmission.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
SRS Resource Set ID	M		INTEGER (0..15, ...)	According to TS 38.331 [8]

9.3.1.181 Spatial Relation Information

This information element indicates a spatial relation for transmission of UL SRS by a UE.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Spatial Relation for Resource ID		*1*		According to TS 38.321 [16] and and TS 38.331 [8]
>Spatial Relation for Resource ID Item		*1..<maxnoSpatialRelations>*		
>>CHOICE *Reference Signal*	M			
>>>NZP CSI-RS				
>>>>NZP CSI-RS Resource ID	M		INTEGER (0..191)	
>>>SSB				
>>>>PCI	M		INTEGER (0..1007)	
>>>>SSB Index	O		INTEGER (0..63)	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
>>>SRS				
>>>>SRS Resource ID	M		INTEGER (0..63)	
>>>Positioning SRS				
>>>>Positioning SRS Resource ID	M		INTEGER (0..63)	
>>>DL-PRS				
>>>>DL-PRS ID	M		INTEGER (0..255)	
>>>>DL-PRS Resource Set ID	M		INTEGER (0..7)	
>>>>DL PRS Resource ID	O		INTEGER (0..63)	

Range bound	Explanation
maxnoSpatialRelations	Maximum no. of Spatial Relations that can be configured. Value is 64.

9.3.1.182 SRS Resource Trigger

This information element indicates a DCI code point according to a SRS resource set configuration.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Aperiodic SRS Resource Trigger List		*1..<maxnoSRS-TriggerStates>*		According to TS 38.331 [8]
>Aperiodic SRS Resource Trigger			INTEGER (1..3)	

Range bound	Explanation
maxnoSRS-TriggerStates	Maximum no. of SRS trigger states. Value is 3.

9.3.1.183 Relative Time 1900

This information element indicates the initialisation time (e.g. SFN Initalisation Time for a cell, requested time for an action, etc).

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Relative Time 1900	M		BIT STRING (SIZE(64))	Time in seconds relative to 00:00:00 on 1 January 1900 (calculated as continuous time without leap seconds and traceable to a common time reference) where binary encoding of the integer part is in the first 32 bits and binary encoding of the fraction part in the last 32 bits. The fraction part is expressed with a granularity of 1 /2**32 second

9.3.1.184 Geographical Coordinates

This information element contains the geographical coordinates for the TRP and any associated ARP(s).

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
CHOICE *TRP Position Definition Type*	M				-	
>Direct						
>>CHOICE *Accuracy*	M				-	
>>>normal accuracy						
>>>>TRP Position	M		NG-RAN Access Point Position 9.3.1.174	The configured estimated geographical position of the antenna of the cell/TRP.	-	
>>>high accuracy						
>>>>TRP High Accuracy Access Position	M		NG-RAN High Accuracy Access Point Position 9.3.1.190	The configured estimated geographical high accuracy position of the antenna of the cell/TRP.	-	
>Referenced						
>>Reference Point	M		9.3.1.188	The reference point is used to derive the TRP position	-	
>>CHOICE *Type*	M				-	
>>>Geodetic						
>>>>TRP Position Relative Geodetic	M		Relative Geodetic Location 9.3.1.186	The configured estimated relative geodetic coordinate of the antenna of the cell/TRP	-	
>>>Cartesian						
>>>>TRP Position Relative Cartesian	M		Relative Cartesian Location 9.3.1.187	The configured estimated relative Cartesian coordinate of the antenna of the cell/TRP	-	
DL-PRS Resource Coordinates	O		9.3.1.185	DL-PRS Resource Coordinates relative to the TRP coordinate	-	
ARP Location Information	O		9.3.1.243		YES	ignore

9.3.1.185 DL-PRS Resource Coordinates

This information element contains the geographical coordinates of the antenna reference points (ARP) for the DL-PRS Resources of a TRP.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
DL-PRS Resource Set ARP List	M	*1..<maxnoofPRS-ResourceSets>*		
>DL-PRS Resource Set ID	M		INTEGER (0..7)	
>CHOICE *DL-PRS Resource Set ARP Location*	M			Relative to the geographical coordinates for the TRP. If this IE is absent, the Relative Location is zero for the indicated DL-PRS Resource Set ID.
>>Geodetic				

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
>>>Relative Geodetic Location	M		9.3.1.186	
>>*Cartesian*				
>>>Relative Cartesian Location	M		9.3.1.187	
>DL-PRS Resource ARP List	M	*1..<maxnoofPRS-ResourcesPerSet>*		
>>DL-PRS Resource ID	M		INTEGER (0..63)	
>>CHOICE *DL-PRS Resource ARP Location*	M			Relative to the DL-PRS Resource Set ARP Location. If this IE is absent, the Relative Location is zero for the indicated DL-PRS Resource ID.
>>>*Geodetic*				
>>>>Relative Geodetic Location	M		9.3.1.186	
>>>*Cartesian*				
>>>>Relative Cartesian Location	M		9.3.1.187	

Range bound	Explanation
maxnoofPRS-ResourceSets	Maximum no of DL-PRS resource sets per TRP. Value is 2.
maxnoofPRS-ResourcesPerSet	Maximum no of DL-PRS resources of the DL-PRS resource set of the TRP. Value is 64.

9.3.1.186 Relative Geodetic Location

This information element provides a location relative to some known reference location in a relative geodetic coordinate system.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Milli-Arc-Second Units	M		ENUMERATED (0.03, 0.3, 3, ...)	Units and scale factor for the delta-latitude and delta-longitude fields, TS 37.355 [39].
Height Units	M		ENUMERATED (mm, cm, m, ...)	Units and scale factor for the delta-height field, TS 37.355 [39].
Delta Latitude	M		INTEGER (-1024..1023)	Delta value in latitude in the unit provided in Milli-Arc-Second Units, TS 37.355 [39].
Delta Longitude	M		INTEGER (-1024..1023)	Delta value in longitude in the unit provided in Milli-Arc-Second Units, TS 37.355 [39].
Delta Height	M		INTEGER (-1024..1023)	Delta value in ellipsoidal height in the unit provided in Height Units, TS 37.355 [39].
Location uncertainty	M		9.3.1.189	

9.3.1.187 Relative Cartesian Location

This information element provides a location relative to some known reference location in a relative Cartesian coordinate.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
XYZ unit	M		ENUMERATED (mm, cm, dm,...)	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
X value	M		INTEGER (-2^{16}.. 2^{16}-1)	Positive value represents easting from reference point, in units of *XYZ Unit* IE.
Y value	M		INTEGER (-2^{16}.. 2^{16}-1)	Positive value represents northing from reference point in units of *XYZ Unit* IE.
Z value	M		INTEGER (-2^{15}.. 2^{15}-1)	Height with respect to reference point in units of *XYZ Unit* IE, where the XY-plane is horizontal and the Z-axis points up.
Location uncertainty	M		9.3.1.189	

9.3.1.188 Reference Point

This information element provides a reference point location information.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
CHOICE *ReferencePoint*	M			Reference point to which relative location information is related to	-	
>Coordinate ID						
>>Coordinate ID	M		INTEGER(0.. 2^9-1,..)	Referential ID mapped via OAM	-	
>Reference Point Coordinates						
>>Reference Point Position	M		NG-RAN Access Point Position 9.3.1.174		-	
>Reference Point Coordinates High Accuracy						
>>Reference Point High Accuracy Access Position	M		NG-RAN High Accuracy Access Point Position 9.3.1.190		-	
>LocalOrigin					YES	ignore
>>Coordinate ID	M		INTEGER(0.. 2^9-1,..)	Referential ID mapped via OAM	-	
>>Horizontal Axes Orientation	M		INTEGER (0..3599)	Indicates the local coordinate system horizontal orientation angle clockwise from northing. Scale factor 0.1 degrees.	-	
>>Reference Point High Accuracy Access Position	O		NG-RAN High Accuracy Access Point Position 9.3.1.190		-	

9.3.1.189 Location Uncertainty

This information element provides the location uncertainty information.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Horizontal Uncertainty	M		INTEGER (0..255)	Horizontal uncertainty of the ARP latitude/longitude. Corresponds to the encoded high accuracy uncertainty as defined in TS 23.032 [36]
Horizontal Confidence	M		INTEGER (0..100)	Corresponds to confidence as defined in TS 23.032 [36].
Vertical Uncertainty	M		INTEGER (0..255)	Vertical uncertainty of the ARP altitude. Corresponds to the encoded high accuracy uncertainty as defined in TS 23.032 [36]
Vertical Confidence	M		INTEGER (0..100)	Corresponds to confidence as defined in TS 23.032 [36].

9.3.1.190 NG-RAN High Accuracy Access Point Position

The *NG-RAN High Accuracy Access Point Position* IE is used to identify the geographical position of an NG-RAN Access Point. It is expressed as High Accuracy Ellipsoid point with altitude and uncertainty ellipsoid according to TS 23.032 [36].

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Degrees of Latitude	M		INTEGER(-2147483648..2147483647)	
Degrees of Longitude	M		INTEGER(-2147483648..2147483647)	
Altitude	M		INTEGER(-64000..1280000)	
Uncertainty Semi Major	M		INTEGER (0..255)	
Uncertainty Semi Minor	M		INTEGER (0..255)	
Orientation Major Axis	M		INTEGER (0..179)	
Horizontal Confidence	M		INTEGER (0..100)	
Uncertainty Altitude	M		INTEGER (0..255)	
Vertical Confidence	M		INTEGER (0..100)	

9.3.1.191 Positioning Broadcast Cells

This IE is used to indicate the cells that are requested to broadcast, or failed to broadcast, the associated posSIB(s).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Positioning Broadcast Cells		*1 .. <maxnoBcastCell>*		
>NR CGI	M		9.3.1.12	

Range bound	Explanation
maxnoBcastCells	Maximum no. of cells broadcasting a posSIB in a NB-DU. Value is 16384.

9.3.1.192 SRS Configuration

This information element contains the SRS configuration configured by the gNB-CU for the UE.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
SRS Carrier List		*1..<maxnoSRS-Carriers>*			-	
>Point A	M		INTEGER (0..3279165)	NR ARFCN	-	
>Uplink Channel BW-PerSCS-List		*1..<maxnoSCSs>*		SCS-SpecificCarrier TS 38.331 [8]	-	
>>Offset To Carrier	M		INTEGER(0..2199,...)	First usable RB to Point A in the number of PRBs	-	
>>Subcarrier Spacing	M		ENUMERATED (kHz15, kHz30, kHz60, kHz120,..., kHz480, kHz960)		-	
>>Carrier Bandwidth	M		INTEGER(1..275,...)		-	
>Active UL BWP	M			The configuration in the active UL BWP or the configuration of SRS for positioning frequency hopping outside of active BWP as defined in TS 38.331 [8] is needed.	-	
>>Location And Bandwidth	M		INTEGER(0..37949,...)	BWP TS 38.331 [8]	-	
>>Subcarrier Spacing	M		ENUMERATED (kHz15, kHz30, kHz60, kHz120,..., kHz480, kHz960)		-	
>>Cyclic Prefix	M		ENUMERATED (Normal, Extended)		-	
>>Tx Direct Current Location	M		INTEGER(0..3301,...)		-	
>>Shift7dot5kHz	O		ENUMERATED (true,...)		-	
>>SRS Config	M			*SRS-Config* as defined in TS 38.331 [8]	-	
>>>SRS Resource List		*0..<maxnoSRS-Resources>*			-	
>>>>SRS Resource	M		9.3.1.193	*SRS-Resource* as defined in TS 38.331 [8]	-	
>>>Positioning SRS Resource List		*0..<maxnoSRS-PosResources>*			-	
>>>>Positioning SRS Resource	M		9.3.1.194	*SRS-PosResource-r16* as defined in TS 38.331 [8]	-	
>>>SRS Resource Set List		*0..<maxnoSRS-Resource*			-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
		Sets>				
>>>>SRS Resource Set	M		9.3.1.195	*SRS-ResourceSet* as defined in TS 38.331 [8]	-	
>>>Positioning SRS Resource Set List		*0..<maxnoSRS-PosResourceSets>*			-	
>>>>Positioning SRS Resource Set	M		9.3.1.196	*SRS-PosResourceSet-r16* as defined in TS 38.331 [8]	-	
>PCI	O		INTEGER (0..1007)	Physical Cell ID of the cell that contains the SRS carrier	-	
Aggregated Positioning SRS Resource Set List	O		9.3.1.337		YES	ignore

Range bound	Explanation
maxnoSRS-Carriers	Maximum no of carriers for SRS. Value is 32.
maxnoSCSs	Maximum no of SCS spacings for a carrier. Value is 5.
maxnoSRS-Resources	Maximum no of SRS resources per UL BWP. Value is 64.
maxnoSRS-PosResources	Maximum no of positioning SRS resources per UL BWP. Value is 64.
maxnoSRS-ResourceSets	Maximum no of SRS resource sets. Value is 16.
maxnoSRS-PosResourceSets	Maximum no of positioning SRS resource sets per UL BWP. Value is 16.

9.3.1.193 SRS Resource

This information element contains the SRS resource.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
SRS Resource ID	M		INTEGER (0..63)		-	
Number of Ports	M		ENUMERATED (ports1, ports2, ports4)		-	
CHOICE *Transmission Comb*	M				-	
>Comb Two						
>>Comb Offset	M		INTEGER(0..1)		-	
>>Cyclic Shift	M		INTEGER(0..7)		-	
>Comb Four						
>>Comb Offset	M		INTEGER(0..3)		-	
>>Cyclic Shift	M		INTEGER(0..11)		-	
>Comb Eight	M				YES	reject
>>Comb Offset	M		INTEGER(0..7)		-	
>>Cyclic Shift	M		INTEGER(0..5)		-	
Start Position	M		INTEGER(0..13)		-	
Number of Symbols	M		ENUMERATED (1,2,4)	This IE is ignored if the *Number of Symbols Extended* IE is included.	-	
Repetition Factor	M		ENUMERATED (1,2,4)	This IE is ignored if the *Repetition Factor Extended* IE is included.	-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
Frequency Domain Position	M		INTEGER(0..67)		-	
Frequency Domain Shift	M		INTEGER(0..268)		-	
C-SRS	M		INTEGER(0..63)		-	
B-SRS	M		INTEGER(0..3)		-	
B-Hop	M		INTEGER(0..3)		-	
Group or Sequence Hopping	M		ENUMERATED (Neither, groupHopping, sequenceHopping)		-	
CHOICE *Resource Type*	M				-	
>Periodic						
>>Periodicity	M		ENUMERATED (slot1, slot2, slot4, slot5, slot8, slot10, slot16, slot20, slot32, slot40, slot64, slot80, slot160, slot320, slot640, slot1280, slot2560, ...)		-	
>>Offset	M		INTEGER(0..2559, ...)		-	
>Semi-persistent						
>>Periodicity	M		ENUMERATED (slot1, slot2, slot4, slot5, slot8, slot10, slot16, slot20, slot32, slot40, slot64, slot80, slot160, slot320, slot640, slot1280, slot2560, ...)		-	
>>Offset	M		INTEGER(0..2559, ...)		-	
>Aperiodic						
>>Aperiodic Resource Type	M		ENUMERATED (true,...)		-	
Sequence ID	M		INTEGER(0..1023)		-	
Number of Symbols Extended	O		ENUMERATED (n8,n10,n12, n14, ...)		YES	ignore
Repetition Factor Extended	O		ENUMERATED (r3, r5, r6, r7, r8, r10, r12, r14, ...)		YES	ignore
Start RB Hopping	O		ENUMERATED (enable)		YES	ignore
CHOICE Start RB Index	O				YES	ignore
>FreqScalingFactor2			INTEGER (0..1)		-	
>FreqScalingFactor4			INTEGER (0..3)		-	

9.3.1.194 Positioning SRS Resource

This information element contains the SRS resource for positioning.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
Positioning SRS Resource ID	M		INTEGER (0..63)		-	
CHOICE *Transmission Comb Positioning*	M				-	
>Comb Two						
>>Comb Offset	M		INTEGER(0..1)		-	
>>Cyclic Shift	M		INTEGER(0..7)		-	
>Comb Four						
>>Comb Offset	M		INTEGER(0..3)		-	
>>Cyclic Shift	M		INTEGER(0..11)		-	
>Comb Eight						
>>Comb Offset	M		INTEGER(0..7)		-	
>>Cyclic Shift	M		INTEGER(0..5)		-	
Start Position	M		INTEGER(0..13)		-	
Number of Symbols	M		ENUMERATED (1,2,4,8,12)		-	
Frequency Domain Shift	M		INTEGER(0..268)		-	
C-SRS	M		INTEGER(0..63)		-	
Group or Sequence Hopping	M		ENUMERATED (Neither, groupHopping, sequenceHopping)		-	
CHOICE *Resource Type Positioning*	M				-	
>Periodic						
>>Periodicity	M		SRS Periodicity 9.3.1.342		-	
>>Offset	M		INTEGER(0..81919,...)		-	
>>SRS PosPeriodicConfigHyperSFN Index	O		ENUMERATED (even0, odd1)	Indicates even or odd hyper SFN in which the SRS for positioning is transmitted for the periodicity value of 20480ms.	YES	ignore
>Semi-persistent						
>>Periodicity	M		SRS Periodicity 9.3.1.342		-	
>>Offset	M		INTEGER(0..81919,...)		-	
>>SRS PosPeriodicConfigHyperSFN Index	O		ENUMERATED (even0, odd1)	Indicates even or odd hyper SFN in which the SRS for positioning is transmitted for the periodicity value of 20480ms.	YES	ignore
>Aperiodic						
>>Slot offset	M		INTEGER(0..32)		-	
Sequence ID	M		INTEGER(0..65535)		-	
CHOICE *Spatial Relation Positioning*	O				-	
>SSB						

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
>>PCI	M		INTEGER (0..1007)		-	
>>SSB index	O		INTEGER(0..63)		-	
>PRS						
>>PRS ID	M		INTEGER(0..255)		-	
>>PRS Resource Set ID	M		INTEGER(0..7)		-	
>>PRS Resource ID	O		INTEGER(0..63)		-	
Tx Hopping Configuration	O		9.3.1.343		YES	ignore

9.3.1.195 SRS Resource Set

This information element indicates a SRS resource set in the UE for UL SRS transmission.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
SRS Resource Set ID	M		INTEGER(0..15, ...)	
SRS Resource ID List		1..<*maxnoSRS-ResourcePerSet*>		
>SRS Resource ID	M		INTEGER (0..63)	
CHOICE *Resource Set Type*	M			
>Periodic				
>>PeriodicSet	M		ENUMERATED(true,...)	
>Semi-persistent				
>>Semi-persistentSet	M		ENUMERATED(true,...)	
>Aperiodic				
>>SRS Resource Trigger List	M		INTEGER(1..3)	
>>Slot offset	M		INTEGER(0..32)	

Range bound	Explanation
maxnoSRS-ResourcePerSet	Maximum no of SRS resources per SRS resource set. Value is 16.

9.3.1.196 Positioning SRS Resource Set

This information element indicates a positioning SRS resource set in the UE for UL SRS transmission.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
Positioning SRS Resource Set ID	M		INTEGER(0..15)		-	
Positioning SRS Resource ID List		1..<*maxnoSRS-PosResourcePerSet*>			-	
>Positioning SRS Resource ID	M		INTEGER (0..63)		-	
CHOICE *Resource Type*	M				-	
>Periodic					-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
>>PosperiodicSet	M		ENUMERATED (true,...)		-	
>Semi-persistent					-	
>>Possemi-persistentSet	M		ENUMERATED (true,...)		-	
>Aperiodic					-	
>>SRS Resource Trigger List	M		INTEGER(1..3)		-	

Range bound	Explanation
maxnoSRS-PosResourcePerSet	Maximum no of positioning SRS resources per positioning SRS resource set. Value is 16.

9.3.1.197 TRP ID

The *TRP ID* IE is used to identify a TRP uniquely within a gNB-CU.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
TRP Identifier	M		INTEGER (1..65535,...)	Identifies a TRP within an gNB-CU

9.3.1.198 NR-PRS Beam Information

This IE contains spatial direction information of the DL-PRS Resources.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
NR-PRS Beam Information List		*1*			-	
>NR-PRS Beam Information Item		*1 .. < maxnoofPRS-Resource Sets >*			-	
>>PRS Resource Set ID	M		INTEGER (0..7)	The resource set in which the resources are associated with the angle.	-	
>>PRS Angle List		*1*			-	
>>>PRS Angle Item		*1..< maxnoofPRS-ResourcesPerSet >*			-	
>>>>NR PRS Azimuth	M		INTEGER (0..359)		-	
>>>>NR PRS Azimuth fine	O		INTEGER (0..9)	Fine angles	-	
>>>>NR PRS Elevation	O		INTEGER (0..180)		-	
>>>>NR PRS Elevation fine	O		INTEGER (0..9)	Fine angles	-	
>>>>PRS Resource ID	O		INTEGER(0..63)		YES	ignore
LCS to GCS Translation List		*0..1*		If absent, the azimuth and elevation are provided in GCS.	-	
>LCS to GCS		*1 ..*			-	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
Translation		*<maxnooflcs-gcs-translation>*				
>>Alpha	M		INTEGER (0..359)		-	
>>Alpha-fine	O		INTEGER (0..9)	Fine angles	-	
>>Beta	M		INTEGER (0..359)		-	
>>Beta-fine	O		INTEGER (0..9)	Fine angles	-	
>>Gamma	M		INTEGER (0..359)		-	
>>Gamma-fine	O		INTEGER (0..9)	Fine angles	-	

Range bound	Explanation
maxnoofPRS-ResourceSets	Maximum no of DL-PRS resource sets per TRP. Value is 2.
maxnoofPRS-ResourcesPerSet	Maximum no of DL-PRS resources of the DL-PRS resource set of the TRP. Value is 64.
maxnooflcs-gcs-translation	Maximum no. of LCS-GS-Translation-Parameters that can reported with one message. Value is 3. The current version of the specification supports 1.

9.3.1.199 E-CID Measurement Result

The purpose of this IE is to provide the E-CID measurement result.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
Geographical Coordinates	O		9.3.1.184	The configured estimated geographical position of the antenna of the cell. This IE is not applicable for per TRP measurements.	-	
Measured Results List		*0..1*			-	
>E-CID Measured Results Item		*1 .. <maxnoMeasE-CID>*			-	
>>CHOICE *Measured Results Value*	M				-	
>>>*Value Angle of Arrival NR*						
>>>>Value Angle of Arrival NR	M		UL Angle of Arrival 9.3.1.167		-	
>>>*Value Timing Advance NR*					YES	ignore
>>>>Value Timing Advance NR	M		INTEGER (0.. 7690)	As defined in TS 38.215 [43]	-	
>>>*Value Angle of Arrival NR per TRP*					YES	ignore
>>>>Value Angle of Arrival NR per TRP	M		E-CID Angle of Arrival per TRP 9.3.1.352		-	
Mobile Access Point Location Information	O		Mobile TRP Location Information	The location information of the mobile access	YES	ignore

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
			9.3.1.304	point of the cell that is associated to the mobile TRP.		
Measured Results Associated Information List		*0..1*		The *Measured Results Associated Information Item* IEs are in the same order as the *Measured Results* IEs.	YES	ignore
>Measured Results Associated Information Item		*1 .. <maxnoMeasE-CID>*			-	
>>Time Stamp	O		9.3.1.171	Time Stamp of the corresponding measured result.	-	
>>Measurement Quality	O		TRP Measurement Quality 9.3.1.172	Measurement Quality of the corresponding measured result.	-	

Range bound	Explanation
maxnoMeasE-CID	Maximum no. of measured quantities that can be configured and reported with one message. Value is 64.

9.3.1.200 Cell Portion ID

This IE gives the current Cell Portion associated with the target UE. The Cell Portion ID is the unique identifier for a cell portion within a cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell Portion ID	M		INTEGER (0..4095, ...)	

9.3.1.201 Pathloss Reference Information

This information element indicates a pathloss reference for transmission of UL SRS by a UE.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
CHOICE *Pathloss Reference Signal*	M			
>SSB				
>>PCI	M		INTEGER (0..1007)	
>>SSB Index	O		INTEGER (0..63)	
>DL-PRS				
>>DL-PRS ID	M		INTEGER (0..255)	
>>DL-PRS Resource Set ID	M		INTEGER (0..7)	
>>DL PRS Resource ID	O		INTEGER (0..63)	

9.3.1.202 SSB Information

This information element contains the SSB time/frequency information for the TRPs.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
SSB Information List		*1*		
>SSB Information Item		*1...<maxNoSSBs>*		
>>SSB Configuration	M		SSB Time/Frequency Configuration 9.3.1.203	
>>PCI	M		INTEGER (0..1007)	

Range bound	Explanation
maxNoSSBs	Maximum no of SSBs for which the configuration can be provided. Value is 255.

9.3.1.203 SSB Time/Frequency Configuration

This information element contains the time and frequency configuration of an SSB.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
SSB frequency	M		INTEGER (0..3279165)	ARFCN
SSB subcarrier spacing	M		ENUMERATED(kHz15, kHz30, kHz60, kHz120, kHz240,..., kHz480, kHz960)	The value 60kHz is not supported in this version of the specification.
SSB Transmit power	M		INTEGER (-60..50)	EPRE of SSS
SSB periodicity	M		ENUMERATED(ms5, ms10, ms20, ms40, ms80, ms160, ...)	
SSB half frame index	M		INTEGER(0..1)	
SSB SFN offset	M		INTEGER(0..15)	
SSB Positions in Burst	O		9.3.1.138	
SFN Initialisation Time	O		Relative Time 1900 9.3.1.183	

9.3.1.204 Search Window Information

This information element contains search window information for the TRP.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Expected Propagation Delay	M		INTEGER (-3841..3841,...)	Indicates when the SRS is expected to arrive in time at the TRP relative to the UL RTOA Reference Time. The UL RTOA Reference Time for a target SRS is defined as $T_0 + t_{\mathrm{SRS}}$, where - T_0 is the SFN Initialisation Time - $t_{\mathrm{SRS}} = (10n_{\mathrm{f}} + n_{\mathrm{sf}}) \times 10^{-3}$, where n_{f} and n_{sf} are the system frame number and the subframe number of the SRS, respectively. Granularity 4Ts, where Ts=1/(15·10^3 ·2048) seconds. Centre of the search window.
Delay Uncertainty	M		INTEGER (1..246,...)	Indicates the uncertainty of the expected SRS arrival time at the

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
				TRP Granularity 4Ts, where $Ts=1/(15 \cdot 10^{3} \cdot 2048)$ seconds. Single-sided search window.

9.3.1.205 Extended gNB-DU Name

This IE provides extended human readable name of the gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB-DU Name Visible	O		VisibleString (SIZE(1..150, ...))	
gNB-DU Name UTF8	O		UTF8String (SIZE(1..150, ...))	

9.3.1.206 Extended gNB-CU Name

This IE provides extended human readable name of the gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB-CU Name Visible	O		VisibleString (SIZE(1..150, ...))	
gNB-CU Name UTF8	O		UTF8String (SIZE(1..150, ...))	

9.3.1.207 F1-C Transfer Path

This IE indicates the transmission path of the F1-C traffic.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
F1-C Path NSA	M		ENUMERATED (lte, nr, both)	This IE indicates the transmission path of the F1-C traffic in EN-DC.

9.3.1.208 SFN Offset

This IE contains the time offset between an absolute time reference and the SFN0 start. The IE is calculated assuming that the SFN transmission started at the absolute time reference. The absolute time reference chosen is the 1980-01-06 T00:00:19 International Atomic Time (TAI).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SFN Time Offset	M		BIT STRING (SIZE(24))	Time offset in microseconds between the absolute time reference "1980-01-06 T00:00:19 International Atomic Time (TAI)" and the SFN0 start. The maximum usable value is (1024*10^4-1). Values higher than the maximum are discarded.

9.3.1.209 Transmission Stop Indicator

This IE indicates to stop the data transmission at gNB-DU side for an DRB not subject to DAPS Handover.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Transmission Stop Indicator	M		ENUMERATED (true, ...)	

9.3.1.210 Spatial Relation Information per SRS Resource

This information element indicates a spatial relation for transmission of each UL SRS resource recommened by LMF.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Spatial Relation per SRS Resource List		*1*		
>Spatial Relation per SRS Resource Item		*1..<maxnoSRS-ResourcePerSet>*		
>>CHOICE *Reference Signal*	M			
>>>NZP CSI-RS				
>>>>NZP CSI-RS Resource ID	M		INTEGER (0..191)	
>>>SSB				
>>>>NR PCI	M		INTEGER (0..1007)	
>>>>SSB Index	O		INTEGER (0..63)	
>>>SRS				
>>>>SRS Resource ID	M		INTEGER (0..63)	
>>>Positioning SRS				
>>>>Positioning SRS Resource ID	M		INTEGER (0..63)	
>>>DL-PRS				
>>>>DL-PRS ID	M		INTEGER (0..255)	
>>>>DL-PRS Resource Set ID	M		INTEGER (0..7)	
>>>>DL-PRS Resource ID	O		INTEGER (0..63)	

Range bound	Explanation
maxnoSRS-ResourcePerSet	Maximum no of SRS resources per SRS resource set. Value is 16.

9.3.1.211 CCO Assistance Information

This IE provides assistance information for the Capacity and Coverage (CCO) actions for specific CCO issues detected, and for network energy saving.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CCO issue detection	O		ENUMERATED (coverage, cell edge capacity, ..., network energy saving)	Indicates the type of CCO issue detected, or network energy saving cause.
Affected Cells and Beams	O		9.3.1.212	

9.3.1.212 Affected Cells and Beams

This IE includes a list of cells and/or SS/PBCH block indexes affected by the detected CCO issue or predicted to be affected by the predicted CCO issue.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Affected Cell List		*1 .. <maxAffectedCells>*		
>NR CGI	M		9.3.1.12	
>Affected SSB List		*0..<maxnoofSSBAreas>*		
>>SSB Index	M		INTEGER (0..63)	

Range bound	Explanation
maxAffectedCells	Maximum numbers of cells affected by a CCO issue. Value is 32.
maxnoofSSBAreas	Maximum no. SSB Areas that can be served by a NG-RAN node cell. Value is 64.

9.3.1.213 Coverage Modification Notification

This IE includes a list of cells and/or SS/PBCH block indexes with the corresponding coverage configuration selected by the gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Coverage Modification List		*1*			-	
>Coverage Modification Item		*1 .. <maxCellingNBDU>*			-	
>>NR CGI	M		9.3.1.12		-	
>>Cell Coverage State			INTEGER (0..63, ...)	Value '0' indicates that the cell is inactive. Other values Indicates that the cell is active and also indicates the coverage configuration of the concerned cell.	-	
>>SSB Coverage Modification List		*0..1*			-	
>>>SSB Coverage Modification Item		*1..<maxnoofSSBAreas>*			-	
>>>>SSB Index	M		INTEGER (0..63)		-	
>>>>SSB Coverage State	M		INTEGER (0..15, ...)	Value '0' indicates that the SS/PBCH block is inactive. Other values Indicates that the SS/PBCH block is active and also indicates the coverage configuration of the concerned SS/PBCH block.	-	
>>Coverage Modification Cause	O		ENUMERATED (coverage, cell edge capacity, ..., network energy saving)		YES	ignore

Range bound	Explanation
maxCellingNBDU	Maximum numbers of cells that can be served by a gNB-DU. Value is 512.
maxnoofSSBAreas	Maximum no. SSB Areas that can be served by a NG-RAN node cell. Value is 64.

9.3.1.214 Cells for SON List

This IE contains a list of served cells in potential PRACH conflict and it may contain neighbour cell PRACH configuration.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Cells for SON Item		*1 .. <maxServedCellforSON>*		
>NR CGI	M		9.3.1.12	
>Neighbour NR Cells for SON List	O		9.3.1.215	

Range bound	Explanation
maxServedCellforSON	Maximum numbers of served cells where PRACH conflict is possible. Value is 256.

9.3.1.215 Neighbour NR Cells for SON List

This IE contains the configuration of NR neighbour cells which the gNB-DU may take into consideration for SON purposes.

NOTE: If multiple served cells share a common neighbour cell and thus multiple copies of *Neighbour NR Cells for SON Item* IE for the neighbour cell are needed to be contained in an F1AP message, IEs other than the *NR CGI* IE may be omitted in the copies other than the first one present in the message.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Neighbour NR Cells for SON Item		*1 .. <maxNeighbourCellforSON>*		
>NR CGI	M		9.3.1.12	
>NR Mode Info Rel16	O		9.3.1.216	
>SSB Positions In Burst	O		9.3.1.138	
>NR Cell PRACH Configuration	O		9.3.1.139	

Range bound	Explanation
maxNeighbourCellforSON	Maximum numbers of neighbour cells which the gNB-DU may take into consideration for SON purposes on a given served cell. Value is 32.

9.3.1.216 NR Mode Info Rel16

This IE contains the information of a NR cell which needs to be encoded differently for FDD and TDD.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *NR-Mode-Info-Rel16*	M			
>FDD				

IE/Group Name	Presence	Range	IE type and reference	Semantics description
>>FDD Info Rel16		*1*		
>>>UL Frequency Info	O		Frequency Info Rel16 9.3.1.217	
>>>SUL Frequency Info	O		Frequency Info Rel16 9.3.1.217	
>TDD				
>>TDD Info Rel16		*1*		
>>>TDD Frequency Info	O		Frequency Info Rel16 9.3.1.217	
>>>SUL Frequency Info	O		Frequency Info Rel16 9.3.1.217	
>>>TDD DL-UL Configuration Common NR	O		OCTET STRING	Includes the *tdd-UL-DL-ConfigurationCommon* contained in the *ServingCellConfigCommon* IE as defined in TS 38.331 [8]

9.3.1.217 Frequency Info Rel16

This IE contains the information of a NR cell which should be encoded separately among FDD NDL, FDD NUL, TDD NDL+NUL and SUL.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR ARFCN	O		INTEGER (0.. maxNRARFCN)	RF Reference Frequency as defined in TS 38.104 [17] section 5.4.2.1. The frequency provided in this IE identifies the absolute frequency position of the reference resource block (Common RB 0) of the carrier. Its lowest subcarrier is also known as Point A.
Frequency Shift 7p5khz	O		ENUMERATED (false, true, ...)	Indicate whether the value of Δ_{shift} is 0kHz or 7.5kHz when calculating $F_{REF,shift}$ as defined in Section 5.4.2.1 of TS 38.104 [17].
Carrier List	O		NR Carrier List 9.3.1.137	

Range bound	Explanation
maxNRARFCN	Maximum value of NR ARFCNs. Value is 3279165.

9.3.1.218 MBS Session ID

This IE indicates the MBS Session ID uniquely identifies an MBS session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TMGI	M		OCTET STRING (SIZE(6))	Coded as Temporary Mobile Group Identity (TMGI) defined in TS 23.003 [23].
NID	O		9.3.1.155	

9.3.1.219 gNB-CU MBS F1AP ID

The gNB-CU MBS F1AP ID uniquely identifies the MBS association over the F1 interface within the gNB-CU.

NOTE: If F1-C signalling transport is shared among multiple interface instances, the value of the gNB-CU MBS F1AP ID is allocated so that it can be associated with the corresponding F1-C interface instance.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB-CU MBS F1AP ID	M		INTEGER (0 .. 2^{32} - 1)	

9.3.1.220 gNB-DU MBS F1AP ID

The gNB-DU MBS F1AP ID uniquely identifies the MBS association over the F1 interface within the gNB-DU.

NOTE: If F1-C signalling transport is shared among multiple interface instances, the value of the gNB-DU MBS F1AP ID is allocated so that it can be associated with the corresponding F1-C interface instance.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB-DU MBS F1AP ID	M		INTEGER (0 .. 2^{32} - 1)	

9.3.1.221 MBS Area Session ID

This IE indicates the Area Session ID for MBS Session with location dependent context.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Area Session ID	M		INTEGER (0 .. 65535, ...)	

9.3.1.222 MBS Service Area

This IE contains the MBS service area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Session Type*	M			
>location independent				
>>MBS Service Area Information	M		9.3.1.223	
>location dependent				
>>MBS Service Area Information Location Dependent List		*1..maxnoofMBSServiceAreaInformation*		
>>>MBS Area Session ID	M		9.3.1.221	
>>>MBS Service Area Information	M		9.3.1.223	

Range bound	Explanation
maxnoofMBSServiceAreaInformation	Maximum no. of MBS Service Area Information elements in the *MBS Service Area Information Location Dependent List* IE. Value is 256

9.3.1.223 MBS Service Area Information

This IE contains MBS service area information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Service Area Cell List		*0..<maxnoofCellsforMBS>*		
>NR CGI	M		9.3.1.12	
MBS Service Area TAI List		*0..<maxnoofTAIforMBS>*		
>PLMN Identity	M		9.3.1.14	
>5GS TAC	M		9.3.1.29	

Range bound	Explanation
maxnoofCellsforMBS	Maximum no. of cells allowed within one MBS Service Area. Value is 512.
maxnoofTAIforMBS	Maximum no. of TAs allowed within one MBS Service Area. Value is 512.

9.3.1.224 MRB ID

This IE indicates the MRB ID as specified in TS 38.401 [4].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MRB ID	M		INTEGER (1.. 512, ...)	

9.3.1.225 MBS CU to DU RRC Information

This IE indicates the MBS CU to DU RRC Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Broadcast Cell List	M			
>MBS Broadcast Cell Item		*1 .. <maxCellingNBDU>*		
>>NR CGI	M		9.3.1.12	
>>mtch-neighbourCell	O		OCTET STRING	Includes the *mtch-NeighbourCell-r17* contained in the *MBS-SessionInfoList* IE, as defined in TS 38.331[8]
MBS Broadcast MRB List		*1*		
>MBS Broadcast MRB Item		*1 .. <maxnoofMRBs>*		
>>MRB ID	M		9.3.1.224	
>>MRB PDCP Config Broadcast	M		OCTET STRING	Includes the *MRB-PDCP-ConfigBroadcast* IE, as defined in TS 38.331 [8].

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.
maxnoofMRBs	Maximum no. MRBs allowed to be setup for one MBS session, the maximum value is 32.

9.3.1.226 MBS Broadcast Neighbour Cell List

This IE indicates a list of neighbour cells where ongoing MBS sessions provided via broadcast MRB in the current cells are also provided.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Broadcast Neighbour	M		OCTET STRING	Includes the *MBS-*

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell List				*NeighbourCellList* IE, as defined in TS 38.331[8]

9.3.1.227 IAB Congestion Indication

This IE contains the IAB downlink congestion indication. This IE is only applicable if the gNB-DU is an IAB-DU or IAB-donor-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
IAB Congestion Indication List		*1*		
>IAB Congestion Indication Item		*1..<maxnoofIABCongInd>*		
>>Child Node Identifier	M		BAP Address 9.3.1.111	This IE identifies the child node, the link to which is congested.
>>BH RLC CH List		*0..1*		
>>>BH RLC CH Item		*1..<maxnoofBHRLCChannels>*		
>>>>BH RLC CH ID	M		BH RLC Channel ID 9.3.1.113	This IE identifies the congested BH RLC channel over the link towards the node identified by the *Child Node Identifier* IE.

Range bound	Explanation
maxnoofIABCongInd	Maximum no. of congestion indications, the maximum value is 1024.
maxnoofBHRLCChannels	Maximum no. of BH RLC channels allowed towards one IAB-node, the maximum value is 65536.

9.3.1.228 F1-C Transfer Path NRDC

This IE indicates the transmission path of the F1-C traffic in NR-DC. This IE is only applicable if the UE is an IAB-MT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
F1-C Path NRDC	M		ENUMERATED (mcg, scg, both)	This IE indicates the transmission path of the F1-C traffic in NR-DC.

9.3.1.229 IAB TNL Address Exception

This IE indicates the list of TNL addresses, pertaining to the packets to be forwarded via the inter-IAB-donor-DU tunnel by the IAB-donor-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
IAB TNL Address List		*1*		
>IAB IAB TNL Address Item		*1..<maxnoofTLAsIAB>*		
>>IAB TNL Address	M		9.3.1.102	

Range bound	Explanation
maxnoofTLAsIAB	Maximum no. of individual IPv4/IPv6 addresses or IPv6 address

Range bound	Explanation
	prefixes in one procedure execution. The value is 1024.

9.3.1.230 RB Set Configuration

This IE contains the RB Set Configuration. The IE is only applicable if the gNB-DU is an IAB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Subcarrier Spacing	M		ENUMERATED (kHz15, kHz30, kHz60, kHz120, kHz240, spare3, spare2, spare1, ...)	Subcarrier spacing used as reference for the RB set configuration.
RB Set Size	M		ENUMERATED (rb2, rb4, rb8, rb16, rb32, rb64)	Number of PRBs in each RB set. If the RB sets of IAB-DU H/S/NA resource configuration do not cover the entire carrier bandwidth, the remaining RBs not part of an RB set configuration are considered as included in the last RB set.
Number of RB Sets	M		INTEGER(1.. *maxnoofRBsetsPerCell)*	Number of configured RB sets. The RB sets are contiguous and non-overlapping. If *NR Carrier List* IE(9.3.1.137) is provided, the start RB index of the first RB set is the RB index of the lowest common RB with the SCS provided by *RB Set Configuration* IE, which overlaps with the lowest usable RB across all SCS-specific carriers provided by the *NR Carrier List* IE for the IAB-DU cell. Otherwise, the start RB of the first RB set is aligned with point A for the IAB-DU cell.

Range bound	Explanation
maxnoofRBsetsPerCell	Maximum no. of RB sets per IAB-DU cell. Value is 8.

9.3.1.231 Survival Time

This IE indicates the Survival Time of the TSC QoS flow as defined in TS 23.501 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Survival Time	M		INTEGER (0.. 1920000, ...)	Expressed in units of 1 us.

9.3.1.232 PDC Measurement Result

The purpose of this IE is to provide the PDC measurement result.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
PDC Measured Results List		*1*		
>PDC Measured Results Item		*1 .. <maxnoMeasPDC>*		

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
>>CHOICE *Measured Results Value*	M			
>>>NR PDC Timing Advance				
>>>>NR PDC Timing Advance	M		INTEGER (0..62500, ...)	Value is expressed in unit of [64* T_c] ns
>>>PDC gNB Rx-Tx Time Difference				
>>>>PDC gNB Rx-Tx Time Difference	M		INTEGER (0.. 61565, ...)	Report mapping as defined in TS 38.133 [38]

Range bound	Explanation
maxnoMeasPDC	Maximum no. of measured quantities that can be configured and reported with one message. Value is 16. Maximum is 1 in this release.

9.3.1.233 SCG Activation Request

This IE indicates whether the SCG resources are required to be activated or deactivated.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SCG Activation Request	M		ENUMERATED (activate SCG, deactivate SCG, ...)	

9.3.1.234 SCG Activation Status

This IE indicates the status of SCG resources.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SCG Activation Status	M		ENUMERATED (SCG activated, SCG deactivated, ...)	

9.3.1.235 Requested DL PRS Transmission Characteristics

This IE contains the requested PRS configuration for transmission by the gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Requested DL-PRS Resource Set List		*1*			-	
>Requested DL-PRS Resource Set Item		*1..<maxnoofPRSresourceSets>*			-	
>>PRS bandwidth	O		INTEGER(1..63)	24,28,...,272 PRBs	-	
>>Comb Size	O		ENUMERATED (2, 4, 6, 12, ...)		-	
>>Resource Set Periodicity	O		ENUMERATED (4,5,8,10,16,20, 32,40,64,80,160,320,640,1280,2560,5120,102	Slots	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			40,20480,40960,81920,..., 128, 256, 512)			
>>Resource Repetition Factor	O		ENUMERATED (rf1,rf2,rf4,rf6,rf8,rf16,rf32,...)		-	
>>Resource Number of Symbols	O		ENUMERATED (n2,n4,n6,n12,...,n1)		-	
>>Requested DL-PRS Resource List	O		9.3.1.250		-	
>>Resource Set Start Time and Duration	O		Start Time and Duration 9.3.1.236	This IE is ignored if the *Start Time and Duration* IE is present	-	
Number of Frequency Layers	O		INTEGER(1..4)		-	
Start Time and Duration	O		9.3.1.236		-	
PRS Bandwidth Aggregation Request Information List	O		9.3.1.347		YES	ignore

Range bound	Explanation
maxnoofPRSresourceSets	Maximum no of PRS resources set. Value is 8.

9.3.1.236 Start Time and Duration

This IE contains the start time and/or duration for the on-demand DL-PRS.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Start Time	O		Relative Time 1900 9.3.1.183	
Duration	O		INTEGER (0..90060, ...)	Unit: seconds

9.3.1.237 PRS Transmission Off Information

This IE contains the information to turn off particular PRS transmissions.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
CHOICE *level*	M			
>TRP level			NULL	
>PRS resource set level				
>>PRS Resource Set List		*1*		
>>>PRS Resource Set Item		*1..<maxnoofPRSresourceSet>*		
>>>>PRS Resource Set ID	M		INTEGER(0..7)	
>PRS resource level				
>>PRS Resource Set List		*1*		
>>>PRS Resource Set Item		*1..<maxnoofPRSresourceSet>*		
>>>>PRS Resource Set ID	M		INTEGER(0..7)	

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
>>>>PRS Resource List		*1*		
>>>>>PRS Resource Item		*1..<maxnoofPRSresource>*		
>>>>>>PRS Resource ID	M		INTEGER(0..63)	

Range bound	Explanation
maxnoofPRSresourceSet	Maximum no of PRS resources set. Value is 8.
maxnoofPRSresource	Maximum no of PRS resources per PRS resource set. Value is 64.

9.3.1.238 UL-AoA Assistance Information

This information element contains the expected uplink Angle of Arrival and uncertainty range.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
CHOICE *AngleMeasurement*	M			
>Expected UL Angle of Arrival				
>>Expected Azimuth AoA		*1*		Defined as (φ_{AOA} - $\Delta\varphi_{AOA}/2$, φ_{AOA} + $\Delta\varphi_{AOA}/2$)
>>>Expected Azimuth AoA Value	M		INTEGER(0..3599)	φ_{AOA} component of Expected Azimuth AoA
>>>Expected Azimuth AoA Uncertainty Range	M		INTEGER(0..3599)	$\Delta\varphi_{AOA}$ component of Expected Azimuth AoA
>>Expected Zenith AoA		*0..1*		Defined as (θ_{ZOA} – $\Delta\theta_{ZOA}/2$, θ_{ZOA} + $\Delta\theta_{ZOA}/2$)
>>>Expected Zenith AoA Value	M		INTEGER(0..1799)	θ_{ZOA} component of Expected Zenith AoA
>>>Expected Zenith AoA Uncertainty Range	M		INTEGER(0..1799)	$\Delta\theta_{ZOA}$ component of Expected Zenith AoA
>Expected UL Angle of Arrival Zenith Only				Defined as (θ_{ZOA} – $\Delta\theta_{ZOA}/2$, θ_{ZOA} + $\Delta\theta_{ZOA}/2$)
>>Expected Zenith AoA Value	M		INTEGER(0..1799)	θ_{ZOA} component of Expected Zenith AoA
>>Expected Zenith AoA Uncertainty Range	M		INTEGER(0..1799)	$\Delta\theta_{ZOA}$ component of Expected Zenith AoA
LCS to GCS Translation	O		9.3.1.241	If absent, the azimuth and zenith are provided in GCS. In case of zenith only, the z-axis of LCS is defined along the linear array axis.

9.3.1.239 Zenith Angle of Arrival Information

This information element contains the Zenith Angle of Arrival, which can correspond to linear array measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Zenith Angle of Arrival	M		INTEGER(0..1799)	TS 38.133 [38]
LCS to GCS Translation	O		9.3.1.241	If absent, the zenith is provided in GCS. the z-axis of LCS is defined along the linear array axis

9.3.1.240 On-demand PRS TRP Information

This IE contains on-demand PRS information for the TRP.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
On-demand PRS Request Allowed	M		BIT STRING (SIZE(16))	Each position in the bitmap represents an on-demand PRS transmission parameter: first bit: Resource Set Periodicity second bit: PRS Bandwidth third bit: Resource Repetition Factor fourth bit: Resource Number of Symbols fifth bit: Comb Size sixth bit: Number of Frequency Layers seventh bit: Start Time and Duration eighth bit: Off Indication ninth bit: QCL Information tenth bit: PRS Bandwidth Aggregation Other bits reserved for future use. Value '1' indicates 'request allowed', Value '0' indicates 'request not allowed'.
Allowed Resource Set Periodicity Values	O		BIT STRING (SIZE(24))	This IE applies only if the first bit of the *On-demand PRS Request Allowed* IE is set to '1'. Each position in the bitmap represents a value of the *Resource Set Periodicity* IE defined in subclause 9.2.235, first bit = 4 and so on. Bit 24 is reserved for future use. Value '1' indicates 'request allowed', Value '0' indicates 'request not allowed'. If this IE is absent, all Resource Set Periodicity values are allowed to be requested.
Allowed PRS Bandwidth Values	O		BIT STRING (SIZE(64))	This IE applies only if the second bit of the *On-demand PRS Request Allowed* IE is set to '1'. Each position in the bitmap represents a value of the *PRS Bandwidth* IE defined in subclause 9.3.1.235, first bit = 1 and so on. Bit 64 is reserved for future use. Value '1' indicates 'request allowed', Value '0' indicates 'request not allowed'. If this IE is absent, all PRS Bandwidth values are allowed to be requested.
Allowed Resource Repetition Factor Values	O		BIT STRING (SIZE(8))	This IE applies only if the third bit of the *On-demand PRS Request Allowed* IE is set to '1'. Each position in the bitmap represents a value of the *Resource Repetition Factor* IE defined in subclause 9.3.1.235, first bit = rf1 and so on. Bit 8 is reserved for future use. Value '1' indicates 'request allowed', Value '0' indicates 'request not

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
				allowed'. If this IE is absent, all Resource Repetition Factor values are allowed to be requested.
Allowed Resource Number of Symbols Values	O		BIT STRING (SIZE(8))	This IE applies only if the fourth bit of the *On-demand PRS Request Allowed* IE is set to '1'. Each position in the bitmap represents a value of the *Resource Number of Symbols* IE defined in subclause 9.3.1.235, first bit = n2 and so on. Bits 6-8 are reserved for future use. Value '1' indicates 'request allowed', Value '0' indicates 'request not allowed'. If this IE is absent, all Resource Number of Symbols values are allowed to be requested.
Allowed Comb Size Values	O		BIT STRING (SIZE(8))	This IE applies only if the fifth bit of the *On-demand PRS Request Allowed* IE is set to '1'. Each position in the bitmap represents a value of the *Comb Size* IE defined in subclause 9.3.1.235, first bit = 2 and so on. Bits 5-8 are reserved for future use. Value '1' indicates 'request allowed', Value '0' indicates 'request not allowed'. If this IE is absent, all Comb Size values are allowed to be requested.

9.3.1.241 LCS to GCS Translation

This IE contains the LCS to GCS Translation information.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Alpha	M		INTEGER (0..3599)	
Beta	M		INTEGER (0..3599)	
Gamma	M		INTEGER (0..3599)	

9.3.1.242 Response Time

This information element contains the response time of the measurement results reporting.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Time	M		INTEGER(1..128, ...)	
Time Unit	M		ENUMERATED(second, ten-seconds, ten-milliseconds, ...)	

9.3.1.243 ARP Location Information

This IE contains the relative position of ARP(s) to the TRP.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
ARP Location Information		*1*		
>ARP Location Information Item		*1..<maxnoARPs>*		
>>ARP ID	M		9.3.1.244	
>>CHOICE *ARP Location Type*	M			
>>>geodetic				
>>>>ARP Position Relative Geodetic	M		Relative Geodetic Location 9.3.1.186	
>>>cartesian				
>>>>ARP Position Relative Cartesian	M		Relative Cartesian Location 9.3.1.187	

Range bound	Explanation
maxnoARPs	Maximum no. of ARPs associated with a TRP. Value is 16.

9.3.1.244 ARP ID

This IE is used to uniquely identify an ARP associated with a TRP.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
ARP Identifier	M		INTEGER (1..16, ...)	

9.3.1.245 Multiple UL AoA

This information element contains the list of the multiple UL AOAs values.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
UL AoA List		*1*		
>UL AoA item		*1..<maxnoofULAoAs>*		
>>CHOICE *AngleMeasurement*	M			
>>>UL Angle of Arrival				
>>>>UL Angle of Arrival	M		9.3.1.167	
>>>UL Zenith Angle of Arrival				
>>>>Zenith Angle of Arrival Information	M		9.3.1.239	

Range bound	Explanation
maxnoofULAoAs	Maximum no of UL-AOAs values (pair of AOA & ZOA values) that can be reported. Value is 8

9.3.1.246 UL SRS-RSRPP

This information element contains the UL SRS RSRPP measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
First Path RSRP Power	M		INTEGER (0..126)	

9.3.1.247 SRS Resource type

This IE contains the SRS resource type.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
CHOICE *Reference Signal*	M				-	
>SRS						
>>SRS Resource ID	M		INTEGER(0..63)		-	
>Positioning SRS						
>>Positioning SRS Resource ID	M		INTEGER(0..63)		-	
SRS Port Index	O		ENUMERATED (id1000, id1001, id1002, id1003, ...)	This IE may be present if the *SRS Resource ID* IE is present, and is ignored otherwise.	YES	ignore

9.3.1.248 Extended Additional Path List

This IE contains the extended additional path results of time measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
Additional Path Item		*1..< maxNoPathExtended >*			-	
>CHOICE *Relative Path Delay*	M				-	
>>k0						
>>>k0	M		INTEGER(0..16351)		-	
>>k1						
>>>k1	M		INTEGER(0..8176)		-	
>>k2						
>>>k2	M		INTEGER(0..4088)		-	
>>k3						
>>>k3	M		INTEGER(0..2044)		-	
>>k4						
>>>k4	M		INTEGER(0..1022)		-	
>>k5						
>>>k5	M		INTEGER(0..511)		-	
>>kminus1						
>>>kminus1	M		INTEGER (0.. 32701)	TS 38.133 [38]	YES	ignore
>>kminus2						
>>>kminus2	M		INTEGER (0.. 65401)	TS 38.133 [38]	YES	ignore
>>kminus3						
>>>kminus3	M		INTEGER (0..130801)	TS 38.133 [38]	YES	ignore

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description	Criticality	Assigned Criticality
>>kminus4						
>>>kminus4	M		INTEGER (0..261601)	TS 38.133 [38]	YES	ignore
>>kminus5						
>>>kminus5	M		INTEGER (0..523201)	TS 38.133 [38]	YES	ignore
>>kminus6						
>>>kminus6	M		INTEGER (0..1046401)	TS 38.133 [38]	YES	ignore
>Path Quality	O		TRP Measurement Quality 9.3.1.172		-	
>Multiple UL AoA	O		9.3.1.245		-	
>Path Power	O		UL SRS-RSRPP 9.3.1.246		-	

Range bound	Explanation
maxNoPathExtended	Maximum no. of additional path measurement. Value is 8.

9.3.1.249 LoS/NLoS Information

This IE contains the LoS/NLoS information for UL measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
CHOICE *LoS/NLoS Indicator*	M			
>Soft Indicator				
>>LoS/NLoS Indicator Soft	M		INTEGER (0..10)	Values provide the likelihood of a LOS propagation path in the range between 0 and 1 with 0.1 steps resolution. Value '0' indicates NLOS and value '1' indicates LOS.
>Hard Indicator				
>>LoS/NLoS Indicator Hard	M		ENUMERATED (NLoS, LoS)	

9.3.1.250 Requested DL-PRS Resource List

This IE contains the requested DL-PRS resource list.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Requested DL-PRS Resource List		*1*		*NR-DL-PRS-Resource-r16* as defined in TS 37.355 [39]
>Requested DL-PRS Resource Item		*1..<maxnoofPRSresource>*		
>>CHOICE QCL Info	O			
>>>*SSB*				
>>>>NR PCI	M		INTEGER(0..1007)	
>>>>SSB Index	O		INTEGER(0..63)	
>>>*DL-PRS*				
>>>>QCL Source PRS Resource Set ID	M		INTEGER(0..7)	
>>>>QCL Source PRS Resource ID	O		INTEGER(0..63)	

Range bound	Explanation

maxnoofPRSresource	Maximum no of PRS resources per PRS resource set. Value is 64.

9.3.1.251 Void

9.3.1.252 TRP Tx TEG Association

This information element contains the TRP Tx TEG information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TRP TEG item		*1 .. <maxnoTRPTEGs>*		
>TRP Tx TEG Information	M		9.3.1.281	
>DL-PRS Resource Set ID	M		INTEGER (0..7)	
>DL-PRS Resource ID List		*0..1*		
>>DL-PRS Resource ID Item		*1..< maxnoofPRS-ResourcesPerSet >*		
>>>DL-PRS Resource ID	M		INTEGER (0..63)	

Range bound	Explanation
maxnoTRPTEGs	Maximum no of reported TRP Tx TEG association. Value is 8.
maxnoofPRS-ResourcesPerSet	Maximum no of DL-PRS resources of the DL-PRS resource set of the TRP. Value is 64.

9.3.1.253 TRP TEG Information

This information element contains the TRP TEG information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *TRP TEG*	M			
>RxTx TEG				
>>TRP RxTx TEG Information	M		9.3.1.282	
>>TRP Tx TEG Information	O		9.3.1.281	
>Rx TEG				
>>TRP Rx TEG Information	M		9.3.1.280	
>>TRP Tx TEG Information	M		9.3.1.281	

9.3.1.254 Measurement Characteristics Request Indicator

This IE contains the measurement characteristic information requested by the gNB-CU.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Measurement characteristic request indicator	M		BIT STRING (SIZE(16))	Each position in the bitmap represents a requested measurement characteristic: first bit: Measurement Beam Information

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
				Second bit: Extended Additional Path List Third bit: Additional Path Power Fourth Bit: Multiple UL AoA of Additional Path Fifth bit: LoS/NLoS Information Sixth bit: TRP Rx TEG association for UL-TDOA Seventh bit: TRP RxTxTEG-ID information for DL+UL positioning. Eighth bit: SRS Resource Type Ninth bit: Multiple Measurement Instances Tenth bit: Mobile TRP location information Eleventh bit: SRS bandwidth aggregation used for joint UL positioning measurement. Twelfth bit: Aggregated Positioning SRS resources IDs used for joint UL positioning measurement. Thirteenth bit: UL SRS-TDCP in UL SRS-TDCT. Other bits reserved for future use. Value ‘1’ indicates ‘requested measurement characteristic’, Value ‘0’ indicates ‘not requested’.

9.3.1.255 UE Reporting Information

This IE contains the UE Reporting Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Reporting Amount	M		ENUMERATED (0, 1, 2, 4, 8, 16, 32, 64)	Value 0 represents an infinite number of periodic reporting	-	
Reporting Interval	M		ENUMERATED (none, 1, 2, 4, 8, 10, 16, 20, 32, 64, ...)	Unit: seconds This IE is ignored when the *ReportingIntervalIMs* IE is present.	-	
Reporting IntervalIMs	O		INTEGER(1..999)	Indicates the interval between location information reports and the response	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				time requirement for the first location information report in milliseconds. Unit: milliseconds.		

9.3.1.256 TRP Beam Antenna Information

The IE provides the beam antenna information of the TRP. It includes either the explicit beam antenna information, or a reference to another TRP's signalled configuration, or the indication that no change has occurred with respect to previously signalled configuration.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *TRP Beam Antenna Info Item*	M			
>Reference				
>>Associated TRP ID	M		TRP ID 9.3.1.197	This IE specifies the TRP ID of the associated TRP from which the beam information parameters are adopted in Local Coordinate System (LCS).
>Explicit				
>>TRP Beam Antenna Angles	M		9.3.1.257	
>>LCS to GCS Translation	O		9.3.1.241	Included if the azimuth and elevation are not provided in GCS.
>No Change			NULL	No change compared to previously signalled configuration for this TRP.

9.3.1.257 TRP Beam Antenna Angles

The IE provides the beam antenna information of the TRP.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TRP Beam Antenna Angles Item		*1..<maxnoAzimuthAngles>*		
>TRP Azimuth Angle	M		INTEGER (0..3599)	For GCS, the azimuth angle is measured counter-clockwise from geographical North. For LCS, the azimuth angle is measured counter-clockwise from the x-axis of the LCS.
>TRP Azimuth Angle fine	O		INTEGER (0..9)	Fine angle
>TRP Elevation Angle List		*1*		
>>TRP Elevation Angle Item		*1..<maxnoElevationAngles>*		
>>>TRP Elevation Angle	M		INTEGER (0..1800)	For GCS, the elevation angle is measured relative to zenith and positive to the horizontal

IE/Group Name	Presence	Range	IE type and reference	Semantics description
				direction (elevation 0 deg. points to zenith, 90 deg to the horizon). For LCS, the elevation angle is measured relative to the z-axis of the LCS (elevation 0 deg. points to the z-axis, 90 deg to the x-y plane).
>>>TRP Elevation Angle fine	O		INTEGER (0..9)	Fine angle
>>>TRP Beam Power List		*1*		Relative power between DL-PRS Resources for the given Azimuth and Elevation Angle. The first Relative Power element in this list provides the peak power for this Azimuth/Elevation angle and is defined as 0dB power. All the remaining Relative Power Element's in this list provide the relative DL-PRS Resource power relative to this first element in the list.
>>>>TRP Beam Power Item		*2..<maxNumResourcesPerAngle>*		
>>>>>PRS Resource Set ID	O		INTEGER (0..7)	DL-PRS Resource Set ID of the DL-PRS Resource for which the Relative Power is provided. If this field is absent, the DL-PRS Resource Set ID for this instance of the Beam Power List is the same as the DL-PRS Resource Set ID of the previous instance in the Beam Power List. This field shall be included at least in the first instance of the Beam Power List.
>>>>>PRS Resource ID	M		INTEGER (0..63)	DL-PRS Resource for which the Relative Power is provided.
>>>>>TRP Beam Relative Power	M		INTEGER (0..30)	The power values span from -30 to 0dB
>>>>>TRP Beam Relative Power "fine"	O		INTEGER (0..9)	Relative Power with 0.1dB resolution. The power spans from -0.9 to 0dB

Range bound	Explanation
maxNumResourcesPerAngle	Maximum number of DL-PRS Resources per angle per TRP. Value is 24.
maxnoAzimuthAngles	Maximum number of azimuth angles per TRP. Value is 3600.
maxnoElevationAngles	Maximum number of elevation angles per azimuth angle/TRP. Value is 1801.

9.3.1.258 NR Paging eDRX Information

This IE indicates the NR Paging eDRX parameters for RRC_IDLE as defined in TS 38.304 [24].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR Paging eDRX Cycle Idle	M		ENUMERATED(hfquarter, hfhalf, hf1, hf2, hf4, hf8, hf16, hf32, hf64, hf128, hf256, hf512,	$T_{eDRX,CN}$ defined in TS 38.304 [24]. Unit: [number of hyperframes].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
			hf1024, ...)	
NR Paging Time Window	O		ENUMERATED(s1, s2, s3, s4, s5, s6, s7, s8, s9, s10, s11, s12, s13, s14, s15, s16, ..., s17, s18, s19, s20, s21, s22, s23, s24, s25, s26, s27, s28, s29, s30, s31, s32)	Unit: [1.28 second].

9.3.1.259 NR Paging eDRX Information for RRC INACTIVE

This IE indicates the NR Paging eDRX parameters for RRC_INACTIVE as defined in TS 38.304 [24].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR Paging eDRX Cycle Inactive	M		ENUMERATED (hfquarter, hfhalf, hf1, ...)	$T_{eDRX,RAN}$ defined in TS 38.304 [24]. Unit: [number of hyperframes].

9.3.1.260 QoE Metrics

This IE provides the RAN visible QoE measurement report to gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Application Layer Buffer Level List	O		OCTET STRING	As defined in TS 38.331 [8].
Playout Delay for Media Startup	O		OCTET STRING	As defined in TS 38.331 [8].

9.3.1.261 CG-SDT Session Info

This IE identifies an CG-SDT session for a UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB-CU UE F1AP ID	M		9.3.1.4	
gNB-DU UE F1AP ID	M		9.3.1.5	

9.3.1.262 SDT Information

This IE is used to indicate an SDT transaction and to provide the assistant information from the UE.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
SDT Indicator	M		ENUMERATED (true,...)	
SDT Assistant Information	O		ENUMERATED (single packet, multiple packets, ...)	“single packet” indicates no subsequent SDT transmission is expected. “multiple packets” indicates subsequent SDT transmission is expected.

9.3.1.263 Path Switch Configuration

This IE provides information for switching to a single-hop or multi-hop indirect path from a direct path or from a single-hop indirect path, or for switching to a single-hop indirect path from a multi-hop indirect path. This IE is also used for only releasing the direct path during MP.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target Relay UE ID	M		BIT STRING (SIZE(24))	Corresponds to the *targetRelayUE-Identity* contained in the *CellGroupConfig* IE, defined in TS 38.331 [8]. For path switching to a multi-hop indirect path, this IE corresponds to the L2 ID of the First U2N Relay UE.
Remote UE Local ID	M		9.3.1.267	
T420	M		ENUMERATED (ms50, ms100, ms150, ms200, ms500, ms1000, ms2000, ms10000)	Corresponds to the t*420* contained in the *CellGroupConfig* IE, defined in TS 38.331 [8]

9.3.1.264 Sidelink Relay Configuration

This IE provides information of a U2N Remote UE when accessing the network via a U2N Relay UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB-DU UE F1AP ID of Relay UE	M		gNB-DU UE F1AP ID 9.3.1.5	
Remote UE Local ID	M		9.3.1.267	
Sidelink Configuration Container	O		OCTET STRING	Includes the *SL-ConfigDedicatedNR* IE as defined in subclause 6.3.5 in TS 38.331 [8] to carry PC5 Relay RLC channel configuration for Remote UE's SRB1.

9.3.1.265 PC5 RLC Channel ID

This IE uniquely identifies a PC5 Relay RLC channel for a PC5 link between an L2 U2N Remote UE and an L2 U2N Relay UE, or between an L2 U2U Remote UE and an L2 U2U Relay UE, or between L2 U2N Relay UEs (in case of multi-hop L2 U2N relay communication).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PC5 RLC Channel ID	M		INTEGER (1.. 512, ...)	

9.3.1.266 Uu RLC Channel ID

This IE uniquely identifies a Uu Relay RLC channel for a L2 U2N Relay UE or a L2 MP Relay UE using N3C.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Uu RLC Channel ID	M		INTEGER (1..32)	Corresponds to information provided in the *Uu-RelayRLC-ChannelID* IE defined in TS 38.331 [8].

9.3.1.267 Remote UE Local ID

This IE uniquely identifies a L2 U2N Remote UE within the connected Relay UE for single-hop relay or within the Last U2N Relay UE for multi-hop relay.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Remote UE Local ID	M		INTEGER (0..255, ...)	Corresponds to the *sl-LocalIdentity* contained in the *SL-SRAP-Config* IE defined in TS 38.331 [8].

9.3.1.268 5G ProSe Authorized

This IE provides information on the authorization status of the UE for NR ProSe services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
5G ProSe Direct Discovery	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Direct Discovery	-	
5G ProSe Direct Communication	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Direct Communication	-	
5G ProSe Layer-2 UE-to-Network Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 UE-to-Network Relay	-	
5G ProSe Layer-3 UE-to-Network Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-3 UE-to-Network Relay	-	
5G ProSe Layer-2 Remote UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 Remote UE	-	
5G ProSe Layer-2 Multi-path	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the 5G ProSe Layer-2 Remote UE is authorized for 5G ProSe multi-path transmission.	YES	ignore
5G ProSe Layer-2 UE-to-UE Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 UE-to-UE Relay UE	YES	ignore
5G ProSe Layer-2 UE-to-UE Remote	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 UE-to-UE Remote UE.	YES	ignore
5G ProSe Layer-3 Multi-Hop UE-to-Network	O		ENUMERATED (authorized, not	Indicates whether the UE is	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Relay			authorized, ...)	authorized for 5G ProSe Layer-3 Multi-Hop UE-to-Network Relay.		
5G ProSe Layer-2 Multi-Hop UE-to-Network Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 Multi-Hop UE-to-Network Relay.	YES	ignore
5G ProSe Layer-2 Multi-Hop Intermediate UE-to-Network Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 Multi-Hop Intermediate UE-to-Network Relay.	YES	ignore
5G ProSe Layer-2 Multi-Hop Remote	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 Multi-Hop Remote.	YES	ignore

9.3.1.269 PEIPS Assistance Information

This IE provides the information related to CN paging subgrouping for a particular UE, as specified in TS 38.304 [24].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CN Subgroup ID	M		INTEGER (0..7, ...)	

9.3.1.270 UE Paging Capability

This IE provides the UE Paging Capability information needed for paging.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
INACTIVE State PO-Determination	O		ENUMERATED (supported,...)	Corresponds to the *inactiveStatePO-Determination* contained in the *UERadioPagingInformation* IE defined in TS 38.331 [8].	-	
RedCap Indication	O		ENUMERATED (true,...)	Indicates that the paged UE is a Redcap UE or an eRedCap UE.	YES	ignore
Paging Adaptation Indication	O		ENUMERATED (supported,...)	Corresponds to the *pagingAdaptation* contained in the *UERadioPagingInformation* IE defined in TS 38.331 [8].	YES	ignore

9.3.1.271 gNB-DU UE Slice Maximum Bit Rate List

This IE contains the UE Slice Maximum Bit Rate List as specified in TS 23.501 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Slice Maximum Bit Rate Item		*1..<maxnoofSMBRValues>*		
>S-NSSAI	M		9.3.1.38	
>UE Slice Maximum Bit Rate Uplink	M		Bit Rate 9.3.1.22	This IE indicates the UE-Slice-MBR as specified in TS 23.501 [21] in the uplink direction.

Range bound	Explanation
maxnoofSMBRValues	Maximum no. of SLICE MAXIMUM BIT RATE values for a UE. Value is 8.

9.3.1.272 Multicast MBS Session List

This IE indicates the Multicast MBS Sessions the UE has joined.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Multicast MBS Session Item		*1..<maxnoofMBSSessionsofUE>*		
>MBS Session ID	M		9.3.1.218	

Range bound	Explanation
maxnoofMBSSessionsofUE	Maximum no. of MBS sessions allowed towards one UE. Value is 256.

9.3.1.273 TAI NSAG Support List

This IE indicates the list of NSAGs configured at the gNB-DU and their associated S-NSSAIs as defined in TS 23.501 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NSAG Support Item		*1..<maxnoofNSAGs>*		
>NSAG ID	M		INTEGER (0.. 255, ...)	
>NSAG Slice Support List	M		Extended Slice Support List 9.3.1.165	Indicates the list of slices which belong to the NSAG.

Range bound	Explanation
maxnoofNSAGs	Maximum no. of signalled NSAGs. Value is 256.

9.3.1.274 MDT PLMN Modification List

The purpose of the *MDT PLMN List Modification* IE is to provide the modified list of PLMN allowed for MDT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT PLMN Modification List		*0..<maxnoofMDTPLMNs>*		An empty list indicates there is no PLMN allowed for MDT.
>PLMN Identity	M		9.3.1.14	

Range bound	Explanation
maxnoofMDTPLMNs	Maximum no. of PLMNs in the MDT PLMN list. Value is 16.

9.3.1.275 MRB RLC Configuration

This ID provides MRB RLC Configuration Information and is provided by the gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MRB RLC Configuration	M		ENUMERATED (rlc-um-ptp, rlc-am-ptp, rlc-um-dl-ptm, two-rlc-um-dl-ptp-and-dl-ptm, three-rlc-um-dl-ptp-ul-ptp-dl-ptm, two-rlc-am-ptp-um-dl-ptm, ...)	The various codepoints correspond to MRB configurations specified in TS 38.300 [6] as follows: "rlc-um-ptp ": Multicast MRB with DL only RLC-UM or bidirectional RLC-UM configuration for PTP transmission; "rlc-am-ptp " Multicast MRB with RLC-AM entity configuration for PTP transmission; " rlc-um-dl-ptm " Multicast MRB with DL only RLC-UM entity for PTM transmission; "two-rlc-um-dl-ptp-and-dl-ptm ": Multicast MRB with two RLC-UM entities, one DL only RLC-UM entity for PTP transmission and the other DL only RLC-UM entity for PTM transmission; "three-rlc-um-dl-ptp-ul-ptp-dl-ptm ": Multicast MRB with three RLC-UM entities, one DL RLC-UM entity and one UL RLC-UM entity for PTP transmission and the other DL only RLC-UM entity for PTM transmission; "two-rlc-am-ptp-um-dl-ptm "; Multicast MRB with two RLC entities, one RLC-AM entity for PTP transmission and the other DL only RLC-UM entity for PTM transmission;

9.3.1.276 Timing Error Margin

This information element contains the Timing error margin for the TRP Rx TEG, or TRP Tx TEG.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Timing Error Margin	M		ENUMERATED(Tc0, Tc2, Tc4, Tc6, Tc8, Tc12, Tc16, Tc20, Tc24, Tc32, Tc40, Tc48, Tc56, Tc64, Tc72, Tc80,...)	

9.3.1.277 SDT Bearer Configuration Info

This IE contains RLC bearer configuration of each SDT bearer.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SDT Bearer Config List		*1*		
>SDT Bearer Config Item IEs		*1 .. <maxnoofSDTBearers>*		
>>CHOICE *SDT Bearer Type*	M			
>>>*SRB*				
>>>>SRB ID	M		9.3.1.7	
>>>*DRB*				
>>>>DRB ID	M		9.3.1.8	
>>SDT RLC Bearer Configuration	M		OCTET STRING	Includes the *RLC-BearerConfig* IE defined in subclause 6.3.2 of TS 38.331 [8]

Range bound	Explanation
maxnoofSDTBearers	Maximum no. of SDT bearers. Value is the summation of maximum numbers of DRBs and SRBs, i.e., 72.

9.3.1.278 PosSIType List

This IE is used to indicate the list of positioning SI message to be broadcast.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
PosSI type item IEs		*1.. <maxnoofPosSITypes>*			-	
>PosSI Type	M		INTEGER (1..32, ...)	Value "1" corresponds to the positioning SI message identified by the first SI message indicated in the *posSI-SchedulingInfo* IE in the *SIB1* message, value "2" to the positioning SI message identified by the second SI message indicated in the *posSI-SchedulingInfo* IE in the *SIB1* message, and so on, as defined in TS 38.331 [8].	-	
>Delivery Status Request	O		ENUMERATED (start, stop, ...)	Corresponds to the *deliveryStatusReq* included in the DedicatedSIBRequest message as defined in TS 38.331 [8].	YES	reject

Range bound	Explanation
maxnoofPosSITypes	Maximum no. of positioning SI types, the maximum value is 32.

9.3.1.279 IAB-DU Cell Resource Configuration-Mode-Info

This IE contains the IAB-DU Cell Resource Configuration-Mode-Info.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *IAB-DU Cell Resource Configuration-Mode-Info*	M				-	
>FDD						
>>FDD Info		*1*			-	
>>>gNB-DU Cell Resource Configuration-FDD-UL	M		gNB-DU Cell Resource Configuration 9.3.1.107	Contains FDD UL resource configuration of the gNB-DU cell. Only applicable if the gNB-DU is an IAB-DU or an IAB-donor-DU.	-	
>>>gNB-DU Cell Resource Configuration-FDD-DL	M		gNB-DU Cell Resource Configuration 9.3.1.107	Contains FDD DL resource configuration of the gNB-DU cell. Only applicable if the gNB-DU is an IAB-DU or an IAB-donor-DU.	-	
>>>UL Frequency Info	O		NR Frequency Info 9.3.1.17		YES	reject
>>>UL Transmission Bandwidth	O		Transmission Bandwidth 9.3.1.15		YES	reject
>>>UL Carrier List	O		NR Carrier List 9.3.1.137	If included, the *UL Transmission Bandwidth* IE shall be ignored.	YES	reject
>>>DL Frequency Info	O		NR Frequency Info 9.3.1.17		YES	reject
>>>DL Transmission Bandwidth	O		Transmission Bandwidth 9.3.1.15		YES	reject
>>>DL Carrier List	O		NR Carrier List 9.3.1.137	If included, the *DLTransmission Bandwidth* IE shall be ignored.	YES	reject
>TDD						
>>TDD Info		*1*			-	
>>>gNB-DU Cell Resource Configuration-TDD	M		gNB-DU Cell Resource Configuration 9.3.1.107	Contains TDD resource configuration of the gNB-DU cell. Only applicable if the gNB-DU is an IAB-DU or an IAB-donor-DU.	-	
>>>NR Frequency Info	O		9.3.1.17		YES	reject
>>>Transmission Bandwidth	O		9.3.1.15		YES	reject
>>>Carrier List	O		NR Carrier List 9.3.1.137	If included, the *Transmission Bandwidth* IE shall be ignored.	YES	reject

9.3.1.280 TRP Rx TEG Information

This information element contains the TRP Rx TEG information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TRP Rx TEG ID	M		INTEGER (0..31)	
TRP Rx Timing Error Margin	M		Timing Error Margin 9.3.1.276	Timing error margin associated to the TRP Rx TEG ID.

9.3.1.281 TRP Tx TEG Information

This information element contains the TRP Tx TEG information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TRP Tx TEG ID	M		INTEGER (0..7)	
TRP Tx Timing Error Margin	M		Timing Error Margin 9.3.1.276	Timing error margin associated to the TRP Tx TEG ID.

9.3.1.282 TRP RxTx TEG Information

This information element contains the TRP RxTx TEG information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TRP RxTx TEG ID	M		INTEGER (0..255)	
TRP RxTx Timing Error Margin	M		ENUMERATED(Tc0dot5, Tc1, Tc2, Tc4, Tc8, Tc12, Tc16, Tc20, Tc24, Tc32, Tc40, Tc48, Tc64, Tc80, Tc96, Tc128, ...)	Timing error margin associated to the TRP RxTx TEG ID.

9.3.1.283 Uplink TxDirectCurrentTwoCarrierList Information

This IE contains the Uplink TxDirectCurrentTwoCarrierList information that is configured by the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Uplink TxDirectCurrentTwoCarrierList Information	M		OCTET STRING	Includes the *UplinkTxDirectCurrentTwoCarrierList* IE as defined in TS 38.331 [8].

9.3.1.284 Uplink TxDirectCurrentMoreCarrierList Information

This IE contains the Uplink TxDirectCurrentMoreCarrierList information that is configured by the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Uplink TxDirectCurrentMoreCarrierList Information	M		OCTET STRING	Includes the *UplinkTxDirectCurrentMoreCarrierList* IE as defined in TS 38.331 [8].

9.3.1.285 Extended UE Identity Index Value

This IE is used by the gNB-DU to calculate the Paging Frame and Paging Occasion for eDRX, and the UE_ID based subgroup ID for PEI as specified in TS 38.304 [24].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Extended UE Identity Index Value	M		BIT STRING (SIZE(16))	

9.3.1.286 Hashed UE Identity Index Value

This IE is the 13 Most Significant Bits (MSBs) of the Hashed ID defined in TS 38.304 [24].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Hashed UE Identity Index Value	M		BIT STRING (SIZE(13, ...))	

9.3.1.287 Broadcast Area Scope

This IE contains the Broadcast Area where the broadcast session is delivered.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
CHOICE *BroadcastAreaScope*	M			
>CompleteSuccess			NULL	
>PartialSuccess				
>>Broadcast Cell List		*1*		
>>>Broadcast Cell Item		*1 .. <maxcellingNBDU>*		
>>>>Cell ID	M		NR CGI 9.3.1.12	Identifier of the cells that establish the broadcast service successfully.

Range bound	Explanation
maxcellingNBDU	Maximum no. of cells which establish the MRBs successfully in one DU, the maximum value is 512.

9.3.1.288 Network Controlled Repeater Authorized

This IE provides the authorization status of the Network Controlled Repeater.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Network Controlled Repeater Authorized	M		ENUMERATED (authorized, not authorized, ...)	

9.3.1.289 MT-SDT Information

This IE indicates MT-SDT information.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
MT-SDT Indicator	M		ENUMERATED (true, ...)	

9.3.1.290 Supported UE Type List

This IE indicates the supported UE Type list for MBS session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Supported UE Type List Item IEs		*1..< maxnoofUETypes >*		
>Supported UE type	M		ENUMERATED (Non-(e)RedCap-UE, (e)RedCap-UE, ...)	

Range bound	Explanation
maxnoofUETypes	Maximum no. of associated UE types. Value is 8.

9.3.1.291 LTM Cells To Be Released List

This IE indicates a list of LTM cells to be released.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
LTM Cells To Be Released Item IEs		*1 .. <maxnoofLTMCells >*			EACH	reject
>LTM Cell ID	M		NR CGI 9.3.1.12		-	

Range bound	Explanation
maxnoofLTMCells	Maximum no. of Cells configured for LTM allowed towards one UE, the maximum value is 8.

9.3.1.292 Reference Configuration

This IE contains either the request for lower layer reference configuration or the generated lower layer reference configuration used for LTM preparation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Reference Configuration*	M			
>Request for Lower Layer Configuration				
>>Request for Lower Layer Configuration	M		ENUMERATED (true, ...)	
>Reference Configuration Information				
>>LTM Reference Configuration	M		OCTET STRING	Includes the *CellGroupConfig* IE, as defined in TS 38.331 [8].

9.3.1.293 Void

9.3.1.294 LTM Configuration ID Mapping List

This IE indicates the list of LTM cells associated with its configuration IDs.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Configuration ID Mapping Item IEs		*1..<maxnoofLTMCells>*			-	
>LTM Cell ID	M		NR CGI 9.3.1.12		-	
>LTM Configuration ID	M		INTEGER (1..8)	Corresponds to the *LTM-CandidateId* IE, as defined in TS 38.331 [8].	-	

Range bound	Explanation
maxnoofLTMCells	Maximum no. of Cells configured LTM allowed towards one UE, the maximum value is 8.

9.3.1.295 Radio Resource Status NR-U

The *Radio Resource Status NR-U* IE indicates the usage of the PRBs per NR-U channel for all traffic in Downlink and Uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DL Total PRB Usage	M		INTEGER (0..100)	Per NR-U Channel DL Total PRB usage in percentage of the cell total PRB number.
UL Total PRB Usage	M		INTEGER (0..100)	Per NR-U Channel UL Total PRB usage in percentage of the cell total PRB number.

9.3.1.296 Path Addition Information

This IE provides information for path addition in case of MP.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Path Addition Information*	M			
>Indirect Path Addition				
>>Target Relay UE ID	M		BIT STRING (SIZE(24))	Corresponds to information provided in the *SL-SourceIdentity* IE, defined in TS 38.331 [8]
>>Remote UE Local ID	M		9.3.1.267	
>Direct Path Addition			NULL	
>N3C Indirect Path Addition				
>>Target Relay UE ID	M		gNB-DU UE F1AP ID 9.3.1.5	Indicates the *gNB-DU UE F1AP ID* IE of MP Relay UE using N3C.

9.3.1.297 Recommended SSBs for Paging List

This IE indicates the recommended SSBs for paging list.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Recommended SSBs for Paging List Item		*1 .. <maxCellingNBDU>*		

IE/Group Name	Presence	Range	IE type and reference	Semantics description
>NR CGI	M		9.3.1.12	
>SSBs for Paging List		*1 .. < maxnoofSSBAreas >*		
>>SSB Index	M		INTEGER (0..63)	Identifier of the recommended SSB beam for paging

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.
maxnoofSSBAreas	Maximum no. SSB Areas that can be served by a cell. Value is 64.

9.3.1.298 RAN Timing Synchronisation Status Information

This IE indicates the RAN timing synchronisation status information provided from the gNB-DU to the gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Synchronisation State	O		ENUMERATED (locked, holdover, freeRun, ...)	
Traceable to UTC	O		ENUMERATED (true, false, ...)	
Traceable to GNSS	O		ENUMERATED (true, false, ...)	
Clock Frequency Stability	O		BIT STRING (SIZE (16))	Indicates the offsetScaledLogVariance as specified in TS 23.501 [21].
Clock Accuracy	O		9.3.1.299	
Parent Time Source	O		ENUMERATED (syncE, pTP, gNSS, atomicClock, terrestrialRadio, serialTimeCode, nTP, handSet, other, ...)	

9.3.1.299 Clock Accuracy

This IE indicates the clock accuracy as defined in TS 23.501 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Clock Accuracy*	M			
>Value				
>>Clock Accuracy Value	M		INTEGER (1.. 40000000, ...)	Indicates the absolute clock accuracy value expressed in units of 25 ns .
>Index				
>>Clock Accuracy Index	M		INTEGER (32..47, ...)	Indicates the clockAccuracy enumeration value specified in Table 5 of clause 7.6.2.6 of IEEE Std 1588 [48].

9.3.1.300 Burst Arrival Time Window

This IE indicates the Burst Arrival Time Window of the TSC QoS flow as defined in TS 23.501 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Earliest Burst Arrival Time	M		INTEGER (0..640000, ...)	Start of the burst arrival time window calculated with reference to the *Burst Arrival Time* IE, expressed in units of 1 us. Integer values are negative.
Latest Burst Arrival Time	M		INTEGER (0..640000, ...)	End of the burst arrival time window calculated with reference to the *Burst Arrival Time* IE, expressed in units of 1 us. Integer values are positive.

9.3.1.301 Periodicity Range

This IE indicates the periodicity range for the TSC QoS flow as defined in TS 23.501 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Periodicity Range*	M			
>Periodicity Bound				
>>Periodicity Lower Bound	M		Periodicity 9.3.1.143	
>>Periodicity Upper Bound	M		Periodicity 9.3.1.143	
>Periodicity List				
>>Allowed Periodicity List		*1..<maxnoofPeriodicities>*		
>>>Allowed Periodicity	M		Periodicity 9.3.1.143	

Range bound	Explanation
maxnoofPeriodicities	Maximum no. of allowed periodicities. Value is 8.

9.3.1.302 TSC Traffic Characteristics Feedback

This IE provides the TSC traffic characteristics feedback of a TSC QoS flow (see TS 23.501 [21]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TSC Feedback Information Downlink	O		TSC Feedback Information 9.3.1.303	
TSC Feedback Information Uplink	O		TSC Feedback Information 9.3.1.303	

9.3.1.303 TSC Feedback Information

This IE provides the TSC feedback information for a TSC QoS flow in the uplink or downlink (see TS 23.501 [21]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Burst Arrival Time Offset	M		INTEGER (-640000..640000, ...)	Burst arrival time offset expressed in units of 1 us.
Adjusted Periodicity	O		Periodicity 9.3.1.143	Not applicable to reactive RAN feedback.

9.3.1.304 Mobile TRP Location Information

This IE contains location information for a mobile TRP.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Location Information	M		OCTET STRING	Location of the mobile TRP, includes the *locationEstimate* IE as defined in TS 37.355 [39]
Velocity Information	O		OCTET STRING	Velocity of the mobile TRP, includes the *velocityEstimate* IE as defined in TS 37.355 [39]
Location Time Stamp	O		Time Stamp 9.3.1.171	Time stamp, indicates the time when the mobile TRP location information is generated

9.3.1.305 Global gNB ID

This IE is used to globally identify a gNB (see TS 38.300 [6]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.1.14	
CHOICE *gNB ID*	M			
>gNB ID				
>>gNB ID	M		BIT STRING (SIZE(22..32))	Equal to the leftmost bits of the *NR Cell Identity* IE contained in the *NR CGI* IE of each cell served by the gNB.

9.3.1.306 RRC Terminating IAB-Donor Related Info

This IE contains the information related to a mobile IAB-node's RRC-terminating IAB-donor.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RRC Terminating IAB-Donor gNB-ID	M		Global gNB ID 9.3.1.305	The Global gNB ID of a mobile IAB-node's RRC-terminating IAB donor.
Mobile IAB-MT BAP Address	M		BAP Address 9.3.1.111	The BAP address assigned to the mobile IAB-node by the RRC-terminating IAB-donor.

9.3.1.307 Mobile IAB-MT User Location Information

This IE contains the user location information of mobile IAB-MT which is co-located with the mobile IAB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR CGI	M		9.3.1.12	The NR CGI of the cell, which is the serving cell of the mobile IAB-MT co-located with the mobile IAB-DU that serves the UE.
TAI	M		9.3.1.308	The TAI supported by the cell, which is the serving cell of the mobile IAB-MT co-located with the mobile IAB-DU which serves the UE.

9.3.1.308 TAI

This IE is used to uniquely identify a Tracking Area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.1.14	
5GS TAC	M		9.3.1.29	

9.3.1.309 Associated Session ID

This IE is used to associate MBS Session IDs providing identical user data.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Associated Session ID	M		OCTET STRING	Coded as *AssociatedSessionId* defined in TS 29.571 [50]. The gNB-DU does not interpret the content of the *Associated Session ID* IE.

9.3.1.310 Multicast CU to DU RRC Information

This IE indicates the multicast specific CU to DU RRC Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Multicast Cell List		*0..1*		
>MBS Multicast Cell Item		*1 .. <maxCellingNBDU>*		
>>NR CGI	M		NR CGI 9.3.1.12	
>>Multicast RRC_INACTIVE Reception Mode	O		9.3.1.318	
>>MBS Multicast Configuration Request	O		ENUMERATED (query, ...)	
MBS Multicast MRB List		*0..1*		
>MBS Multicast MRB Item		*1 .. <maxnoofMRBs>*		
>>MRB ID	M		9.3.1.224	
>>MRB PDCP Config Broadcast	M		OCTET STRING	Includes the *MRB-PDCP-ConfigBroadcast* IE, as defined in TS 38.331 [8].

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.
maxnoofMRBs	Maximum no. MRBs allowed to be setup for one MBS session, the maximum value is 32.

9.3.1.311 Multicast DU to CU RRC Information

This IE indicates the multicast specific DU to CU RRC Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Multicast Cell List		*0..1*		

IE/Group Name	Presence	Range	IE type and reference	Semantics description
>MBS Multicast Cell Item		*1 .. <maxCellingNBDU>*		
>>NR CGI	M		NR CGI 9.3.1.12	
>>MBS Multicast Configuration Response Information	O		9.3.1.312	
>>MBS Multicast Configuration Notification	O		9.3.1.313	

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.

9.3.1.312 MBS Multicast Configuration Response Information

This IE contains information on the gNB-DU's response to the requested multicast configuration for reception RRC_INACTIVE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *MBS Multicast Configuration Response Information*	O			
>MBS Multicast Configuration available				
>>MBS Multicast Configuration	M		OCTET STRING	Includes the *MBSMulticastConfiguration* message as defined in TS 38.331 [8].
>MBS Multicast Configuration not available				
>>MBS Multicast Configuration not available	M		ENUMERATED (not available, ...)	

9.3.1.313 MBS Multicast Configuration Notification

This IE contains information on the gNB-DU's notification of MBS Multicast Configuration information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *MBS Multicast Configuration Notification Information*	O			
>MBS Multicast Configuration changed				
>>MBS Multicast Configuration changed	M		OCTET STRING	Includes the *MBSMulticastConfiguration* message as defined in TS 38.331 [8].
>MBS Multicast Configuration removed			NULL	

9.3.1.314 Multicast CU to DU Common RRC Information

This IE includes multicast specific CU to DU common RRC information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Multicast Common CU2DU Cell List		*0..1*		
>Multicast COMMON CU2DU Cell Item		*1 .. <maxCellingNBDU>*		
>>NR CGI	M		NR CGI 9.3.1.12	
>>Multicast Common CU2DU Cell Information		*1*		
>>>CHOICE *MBS Multicast Neighbour Cell List Item*	O			
>>>>MBS Multicast Neighbour Cell List Information provided				
>>>>>MBS Multicast Neighbour Cell List Information provided	M		Update MBS Multicast Neighbour Cell List Information 9.3.1.315	
>>>>No MBS Multicast Neighbour Cell List provided			NULL	
>>>CHOICE *ThresholdMBS-List Item*	O			
>>>>ThresholdMBS-List Information provided				
>>>>>ThresholdMBS-List Information provided	M		Update ThresholdMBS-List Information 9.3.1.316	
>>>>No ThresholdMBS-List provided			NULL	

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.

9.3.1.315 Update MBS Multicast Neighbour Cell List Information

This IE includes MBS multicast neighbour cell related information provided in the multicast MCCH.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS-NeighbourCellList	O		OCTET STRING	Includes *mbs-NeighbourCellList-r18* as defined in TS 38.331[8]
MTCH-NeighbourCell Session List		*0..1*		
>MTCH-NeighbourCell Session Item		*1 .. <maxMBSSessionsinSessionInfoList>*		
>>MBS Session ID	M		9.3.1.218	
>>CHOICE *MTCH-NeighbourCell Information*	M			
>>>MTCH-NeighbourCell provided				
>>>>MTCH-NeighbourCell provided	M		OCTET STRING	Includes the *mtch-NeighbourCell-r18* in the *MBS-SessionInfoListMulticast* IE as specified in TS 38.331 [8].
>>>MTCH-			NULL	Indicates that the the

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NeighbourCell not provided				*thresholdIndex* as defined in TS 38.331 [8] is not provided for the respective multicast MBS session.

Range bound	Explanation
maxMBSSessionsinSessionInfoList	Maximum no. multicast MBS sessions contained in the *MBS-SessionInfoListMulticast* IE as specified in TS 38.331 [8]. Value is 1024.

9.3.1.316 Update ThresholdMBS-List Information

This IE includes thereshold MBS related list information provided in the multicast MCCH.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
ThresholdMBS List	O		OCTET STRING	Includes *thresholdMBS-List* as specified in TS 38.331[8]
ThresholdIndex Session List		*0..1*		
>ThresholdIndex Session Item		*1 .. <maxMBSSessionsinSessionInfoList>*		
>>MBS Session ID	M		9.3.1.218	
>>CHOICE *ThresholdIndex Information*	M			
>>>ThresholdIndex				
>>>>ThresholdIndex	M		INTEGER (0..*maxnoofThresholdMBS-1*)	Corresponds to the *thresholdIndex* as specified in TS 38.331 [8].
>>>ThresholdIndex not provided			NULL	Indicates that the the *thresholdIndex* as defined in TS 38.331 [8] is not provided for the respective multicast MBS session.

Range bound	Explanation
maxMBSSessionsinSessionInfoList	Maximum no. multicast MBS sessions contained in the *MBS-SessionInfoListMulticast* IE as specified in TS 38.331 [8]. Value is 1024.
maxnoofThresholdMBS-1	Maximum no. thresholds configured in a cell minus 1. Value is 7.

9.3.1.317 MBS Multicast Session Reception State

This IE indicates the reception state of MBS Multicast Session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Multicast Session Reception State	M		ENUMERATED (start monitoring G-RNTI, stop monitoring G-RNTI, ...)	

9.3.1.318 Multicast RRC_INACTIVE Reception Mode

This IE indicates the activation or deactivation of the multicast RRC_INACTIVE reception mode for a multicast MBS session in a particular cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Multicast RRC_INACTIVE Reception Mode	M		ENUMERATED (activated, deactivated, ...)	

9.3.1.319 PDU Set QoS Information

This IE defines the PDU Set QoS Information to be applied to a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Set Delay Budget	O		Extended Packet Delay Budget 9.3.1.145	PDU Set Delay Budget as defined in TS 23.501 [21].
PDU Set Error Rate	O		Packet Error Rate 9.3.1.52	PDU Set Error Rate as defined in TS 23.501 [21].
PDU Set Integrated Handling Information	O		ENUMERATED(true, false, ...)	PDU Set Integrated Handling Information as defined in TS 23.501 [21].

9.3.1.320 N6 Jitter Information

This IE indicates the N6 jitter information associated with the Periodicity in downlink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
N6 Jitter Lower Bound	M		INTEGER (-127..127)	Indicates the lower bound of the N6 jitter. The unit is: 0.5ms.
N6 Jitter Upper Bound	M		INTEGER (-127..127)	Indicates the upper bound of the N6 jitter. The unit is: 0.5ms.

9.3.1.321 ECN Marking or Congestion Information Reporting Request

This IE indicates to the gNB-DU to report information for ECN marking or to report congestion information for a DRB.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE ECN Marking or Congestion Information Request	M			
>ECN Marking				
>>ECN Marking Request	M		ENUMERATED (ul, dl, both, stop, ...)	
>*Congestion* Information				
>>Congestion Information Request	M		ENUMERATED (ul, dl, both, stop, ...)	

9.3.1.322 ECN Marking or Congestion Information Reporting Status

This IE indicates the status of information reporting for ECN marking or congestion information reporting for a DRB.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
ECN Marking or Congestion Information Reporting Status	O		ENUMERATED (active, not active, ...)	Indicates whether information reporting for ECN marking or congestion information reporting

IE/Group Name	Presence	Range	IE type and reference	Semantics description
				is active or not active.

9.3.1.323 NR A2X Services Authorized

This IE provides information on the authorization status of the UE to use the NR sidelink for A2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Aerial UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Aerial UE.
Aerial Controller UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Aerial Controller UE.

9.3.1.324 LTE A2X Services Authorized

This IE provides information on the authorization status of the UE to use the LTE sidelink for A2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Aerial UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Aerial UE.
Aerial Controller UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Aerial Controller UE.

9.3.1.325 NR Paging Long eDRX Information for RRC INACTIVE

This IE indicates the NR Paging long eDRX parameters for RRC INACTIVE as defined in TS 38.304 [24].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR Paging Long eDRX Cycle for RRC INACTIVE	M		ENUMERATED (hf2, hf4, hf8, hf16, hf32, hf64, hf128, hf256, hf512, hf1024, ...)	$T_{eDRX, RAN}$ defined in TS 38.304 [24]. Unit: [number of hyperframes].
NR Paging Time Window for RRC INACTIVE	M		ENUMERATED (s1, s2, s3, s4, s5, s6, s7, s8, s9, s10, s11, s12, s13, s14, s15, s16, s17, s18, s19, s20, s21, s22, s23, s24, s25, s26, s27, s28, s29, s30, s31, s32, ...)	Unit: [1.28 second].

9.3.1.326 SSBs within the cell to be Activated List

This IE indicates the SSBs within the cell requested to be activated.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SSBs within the cell to be Activated List Item		*1 .. <maxnoofSSBAreas>*		

IE/Group Name	Presence	Range	IE type and reference	Semantics description
>SSB Index	M		INTEGER (0..63)	Identifier of SSB beam requested to be activated.

Range bound	Explanation
maxnoofSSBAreas	Maximum no. SSB Areas that can be served by a cell. Value is 64.

9.3.1.327 DL LBT Failure Information

This IE contains information on DL LBT Failures at the target gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Assistant Identifier	M		gNB-CU UE AP ID 9.3.1.4	
Number of DL LBT Failures	O		INTEGER (1..1000,...)	This IE indicates the number of DL LBT Failures, if available, occurring at the target gNB-DU.

9.3.1.328 Early UL Sync Configuration

This IE indicates the early UL sync configurations for the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
RACH Configuration	M		OCTET STRING	Includes the *EarlyUL-SyncConfig* IE, as defined in TS 38.331 [8].	-	
LTM gNB-DUs List		*0..1*		This IE contains the IDs of the source gNB-DU and candidate gNB-DU(s).	-	
>LTM gNB-DUs Item IEs		*1..<maxnoofLTMgNBDUs>*			-	
>>LTM gNB-DU ID	M		gNB-DU ID 9.3.1.9		-	
>>Preamble Index List	O		9.3.1.329		-	
>>LTM gNB ID	O		Global gNB ID 9.3.1.305		YES	reject

Range bound	Explanation
maxnoofLTMgNB-DUs	Maximum no. of gNB-DUs allowed to be configured with LTM towards one UE, the maximum value is 8.

9.3.1.329 Preamble Index List

This IE indicates the list of preamble indexes to be used for the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality

Preamble Index Item IEs		*1..<maxnoofPreambleIndex>*			-	
>Preamble Index	M		INTEGER (0..63)		-	

Range bound	Explanation
maxnoofPreambleIndex	Maximum no. of preamble index, the maximum value is 64.

9.3.1.330 CSI Resource Configuration

This IE contains the CSI resource configuration used for LTM.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CSI Resource Configuration To AddMod List	O		OCTET STRING	Includes the *ltm-CSI-ResourceConfigToAddModList* contained in the *LTM-Config* IE as defined in TS 38.331 [8].
CSI Resource Configuration To Release List	O		OCTET STRING	Includes the *ltm-CSI-ResourceConfigToReleaseList* contained in the *LTM-Config* IE as defined in TS 38.331 [8].

9.3.1.331 Ranging and Sidelink Positioning Service Information

This IE provides information for the UE's Ranging and Sidelink Positioning service.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Sidelink Positioning and Ranging Authorized	M		ENUMERATED (authorized, not authorized, ...)	This IE indicates whether the UE is authorized to use RSPP communication resources and SL-PRS resources.
RSPP Transport QoS Parameters	O		9.3.1.332	This IE applies only if the UE is authorized for Ranging and Sidelink Positioning service.

9.3.1.332 RSPP Transport QoS Parameters

This IE provides information on the RSPP Transport QoS Parameters.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RSPP Transport QoS Flow List		*1*		
>RSPP Transport QoS Flow Item		*1..<maxnoofRSPPQoSFlows>*		
>>PQI	M		INTEGER (0..255, ...)	PQI is a special 5QI as specified in TS 23.501 [21].
>>RSPP Transport Bit Rates		*0..1*		Only applies for GBR QoS flows.
>>>Guaranteed Flow Bit Rate	M		Bit Rate 9.3.1.22	Guaranteed Bit Rate for the RSPP QoS flow. Details in TS 23.501 [21].
>>>Maximum Flow Bit Rate	M		Bit Rate 9.3.1.22	Maximum Bit Rate for the RSPP QoS flow. Details in TS 23.501 [21].
>>Range	O		ENUMERATED (m50, m80, m180, m200, m350, m400,	Only applies for groupcast.

			m500, m700, m1000, ...)	
RSPP Transport Link Aggregate Bit Rates	O		Bit Rate 9.3.1.22	Only applies for Non-GBR QoS flows.

Range bound	Explanation
maxnoofRSPPQoSFlows	Maximum no. of RSPP QoS flows allowed towards one UE for NR Ranging and Positioning sidelink communication, the maximum value is 2048.

9.3.1.333 Time Window Information SRS List

This IE contains the time window(s) when UL SRS transmission is requested.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Time Window Information SRS List		*1*		
>Time Window Information SRS Item		*1..<maxnoofTimeWindowSRS>*		
>>Time Window Start		*1*		
>>>System Frame Number	M		INTEGER(0..1023)	
>>>Slot Number	M		INTEGER(0..79)	
>>>Symbol Index	M		INTEGER(0..13)	
>>CHOICE *Time Window Duration*	M			
>>>*Symbols*				
>>>>Duration in Symbols	M		ENUMERATED (1, 2, 4, 8, 12, ...)	
>>>*Slots*				
>>>>Duration in Slots	M		ENUMERATED (1, 2, 4, 6, 8, 12, 16, ...)	
>>Time Window Type	M		ENUMERATED (single, periodic, ...)	
>>Time Window Periodicity	C-ifTimeWindowTypePeriodic		ENUMERATED (0.125, 0.25, 0.5, 0.625, 1, 1.25, 2, 2.5, 4, 5, 8, 10, 16, 20, 32, 40, 64, 80, 160, 320, 640, 1280, 2560, 5120, 10240, ...)	Unit: Milli-seconds.

Condition	Explanation
ifTimeWindowTypePeriodic	This IE shall be present if the *Time Window Type* IE is set to the value “periodic”.

Range bound	Explanation
maxnoofTimeWindowSRS	Maximum no of Time Window of SRS. Value is 16.

9.3.1.334 Time Window Information Measurement List

This IE contains the time window(s) when UL SRS measurement is requested.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Time Window Information Measurement List		*1*		

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
>Time Window Information Measurement Item		*1..<maxnoofTimeWindowMeas>*		
>>CHOICE *Time Window Duration*	M			
>>>*Slots*				
>>>>Duration in Slots	M		ENUMERATED (1, 2, 4, 6, 8, 12, 16, ...)	
>>Time Window Type	M		ENUMERATED (single, periodic, ...)	
>>Time Window Periodicity	C-ifTimeWindowTypePeriodic		ENUMERATED (160, 320, 640, 1280, 2560, 5120, 10240, 20480, 40960, 61440, 81920, 368640, 737280, 1843200, ...)	Unit: Milli-seconds
>>Time Window Start		*1*		
>>>System Frame Number	M		INTEGER(0..1023)	
>>>Slot Number	M		INTEGER(0..79)	
>>>Symbol Index	M		INTEGER(0..13)	

Condition	Explanation
ifTimeWindowTypePeriodic	This IE shall be present if the *Time Window Type* IE is set to the value "periodic".

Range bound	Explanation
maxnoofTimeWindowMeas	Maximum no of Time Window of Measurements. Value is 16.

9.3.1.335 UL RSCP

This IE contains the UL Reference Signal Carrier Phase (RSCP) measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
UL RSCP	M		INTEGER (0..3599)	TS 38.133 [38]

9.3.1.336 Positioning Validity Area Cell List

This IE is used to indicate the cells belong to the validity area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Positioning Validity Area Cell List		*1*		
>Positioning Validity Area Cell List Item		*1 .. < maxnoVACell>*		
>>NR CGI	M		9.3.1.12	
>>NR PCI	O		INTEGER (0..1007)	

Range bound	Explanation
maxnoVACell	Maximum number of cells in a Validity Area, Number is 32.

9.3.1.337 Aggregated Positioning SRS Resource Set List

This information element is used to indicate the aggregated Positioning SRS Resource Set List.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Aggregated SRS Positioning Resource Set List		*1*		
>Aggregated SRS Positioning Resource Set Item		*1..<maxnoAggregatedPosSRSCombinations>*		
>>Combined Positioning SRS Resource Set List		*1*		
>>>Combined Positioning SRS Resource Set Item		*2..< maxnoAggregatedPosSRSResourceSets>*		
>>>>Point A	M		INTEGER (0..3279165)	NR ARFCN
>>>>NR PCI	O		INTEGER(0..1007)	
>>>>Positioning SRS Resource Set ID	M		INTEGER(0..15)	
>>>>SCS Specific Carrier		*1*		
>>>>>Offset To Carrier	M		INTEGER (0..2199, ...)	
>>>>>Subcarrier Spacing	M		ENUMERATED(kHz15, kHz30, kHz60, kHz120, ... , kHz480, kHz960)	
>>>>>Carrier Bandwidth	M		INTEGER (1..275, ...)	

Range bound	Explanation
maxnoAggregatedPosSRSResourceSets	Maximum no of aggregated SRS Positioning Resource Sets among 2~3 Carriers. Value is 3.
maxnoAggregatedPosSRSCombinations	Maximum no of Aggregated SRS Positioning Resource Set Combinations. Value is 32.

9.3.1.338 Aggregated PRS Resource Set List

This information element is used to indicate the aggregated PRS Resource Set List.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Aggregated PRS Resource Set Item		*1..<maxnoAggCombinations>*		
>DL-PRS Resource Set List		*1*		
>>DL-PRS Resource Set Item		*1.. < maxnoAggregatedPosPRSResourceSets>*		
>>>DL-PRS Resource Set Index			INTEGER(1..8)	This IE specifies the PRS Resource Set List's that are linked for DL-PRS bandwidth aggregation. The Integer Value defines an index to the *PRS Resource Set List* IE within the *PRS Configuration* IE (9.3.1.177). I.e., Integer value 1 corresponds to the first entry in *PRS Resource*

IE/Group Name	Presence	Range	IE type and reference	Semantics description
				Set List IE, Integer value 2 corresponds to the second entry in *PRS Resource Set List* IE and so on.

Range bound	Explanation
maxnoAggCombinations	Maximum number of aggregated frequency layer (carrier) combinations. Value is 2.
maxnoAggregatedPosPRSResourceSets	Maximum no of PRS resource sets aggregated. Value is 3.

9.3.1.339 Validity Area specific SRS Information

This information element is used to indicate the SRS information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *Transmission Comb*	O				-	
>Comb Two						
>>Comb Offset	M		INTEGER(0..1)		-	
>>Cyclic Shift	M		INTEGER(0..7)		-	
>Comb Four						
>>Comb Offset	M		INTEGER(0..3)		-	
>>Cyclic Shift	M		INTEGER(0..11)		-	
>Comb Eight						
>>Comb Offset	M		INTEGER(0..7)		-	
>>Cyclic Shift	M		INTEGER(0..5)		-	
Resource Mapping		*0..1*				
>Start Position	M		INTEGER(0..13)		-	
>Number of Symbols	M		ENUMERATED (n1,n2,n4, n8, n12}		-	
Frequency Domain Shift	O		INTEGER(0..268)		-	
C-SRS	O		INTEGER(0..63)		-	
CHOICE *Resource Type Positioning*	O				-	
>periodic						
>>Periodicity	M		SRS Periodicity 9.3.1.342		-	
>>Offset	M		INTEGER(0..81919,...)		-	
>>SRS PosPeriodicConfigHyperSFN Index	O		ENUMERATED (even0, odd1)	Indicates even or odd hyper SFN in which the SRS for positioning is transmitted for the periodicity value of 20480ms.	YES	ignore
>semi-persistent						
>>Periodicity	M		SRS Periodicity 9.3.1.342		-	
>>Offset	M		INTEGER(0..81919,...)		-	
>>SRS PosPeriodicConfigHyperSFN Index	O		ENUMERATED (even0, odd1)	Indicates even or odd hyper SFN in which the SRS for positioning is transmitted for the	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				periodicity value of 20480ms.		
>aperiodic				Not applicable if the *Positioning Validity Area Cell List* IE is included.		
>>slot offset	M		INTEGER(0..32)		-	
Sequence ID	O		INTEGER(0..65535)		-	

9.3.1.340 Requested SRS Preconfiguration Characteristics List

This information element is used to indicate the requested SRS Preconfiguration list.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Requested SRS Preconfiguration List		*1*		
>Requested SRS Preconfiguration Item		*1..<maxnoPreconfiguredSRS>*		
>>Requested SRS Transmission Characteristics	M		9.3.1.175	

Range bound	Explanation
maxnoPreconfiguredSRS	Maximum number of validity areas that can be configured. Value is 16.

9.3.1.341 SRS Preconfiguration List

This information element is used to indicate the SRS Preconfiguration list.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SRS Preconfiguration List		*1*		
> SRS Preconfiguration Item		*1..<maxnoPreconfiguredSRS>*		
>>SRS-PosRRC-InactiveValidityAreaConfig	M		OCTET STRING	Includes the *SRS-PosRRC-InactiveValidityAreaConfig* IE as defined in TS 38.331 [8].
>>Positioning Validity Area Cell List	M		9.3.1.336	

Range bound	Explanation
maxnoPreconfiguredSRS	Maximum number of validity areas that can be configured. Value is 16.

9.3.1.342 SRS Periodicity

This information element indicates the SRS periodicity.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
SRS Periodicity	M		ENUMERATED(slot1, slot2, slot4, slot5, slot8, slot10, slot16, slot20, slot32, slot40, slot64, slot80, slot160, slot320, slot640, slot1280, slot2560, slot5120, slot10240, slot40960, slot81920,..., slot128, slot256, slot512, slot20480)	

9.3.1.343 Tx Hopping Configuration

This information element indicates the SRS Tx frequency hopping configuration.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Overlap Value	M		ENUMERATED(rb0, rb1, rb2, rb4)	
Number of Hops	M		INTEGER(2..6)	
Slot Offset for Remaining Hops List		*1*		
>Slot Offset for Remaining Hops Item		*1..<maxnoofHopsMinusOne>*		
>>CHOICE *slot offset remaining hops*	M			
>>>aperiodic				
>>>>Slot Offset	O		INTEGER(1..32)	
>>>>Start Position	O		INTEGER(0..13)	In symbols
>>>semi-persistent				
>>>>SRS Periodicity	M		9.3.1.342	
>>>>Offset	M		INTEGER(0..81919, ...)	In slots
>>>>Start Position	O		INTEGER(0..13)	In symbols
>>>periodic				
>>>>SRS Periodicity	M		9.3.1.342	
>>>>Offset	M		INTEGER(0..81919, ...)	In slots
>>>>Start Position	O		INTEGER(0..13)	In symbols

Range bound	Explanation
maxnoofHopsMinusOne	Maximum no of hops that can be configured for positioning SRS transmission minus one. Value is 5.

9.3.1.344 Non-Integer DRX Cycle

The *Non-Integer DRX Cycle* IE is used to indicate the desired non-integer DRX cycle.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Long Non-Integer DRX Cycle Length	M		ENUMERATED (ms1001over240, ms25over6, ms25over3, ms1001over120, ms100over9, ms25over2,	Corresponds to the *drx-NonIntegerLongCycle* which is the length of the *drx-NonIntegerLongCycleStartOffset* contained in the *DRX-Config* IE defined in TS 38.331 [8]

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
			ms40over3, ms125over9, ms50over3, ms1001over60, ms125over6, ms200over9, ms250over9, ms100over3, ms1001over30, ms75over2, ms125over3, ms1001over24, ms200over3, ms1001over15, ms250over3, ms1001over12, ms400over3, ...)	
Short Non-Integer DRX Cycle Length	O		ENUMERATED (ms1001over240, ms25over6, ms25over3, ms1001over120, ms100over9, ms25over2, ms40over3, ms125over9, ms50over3, ms1001over60, ms125over6, ms200over9, ms100over3, ms1001over30, ms125over3, ms1001over24, ms200over3, ...)	Corresponds to the *drx-NonIntegerShortCycle* contained in the *DRX-Config* IE defined in TS 38.331 [8].
Short DRX Cycle Timer	O		INTEGER (1..16)	Corresponds to the *drx-ShortCycleTimer* contained in the *DRX-Config* IE defined in TS 38.331 [8].

9.3.1.345 RAN Sharing Assistance Information

The *RAN Sharing Assistance Information* IE assists the gNB-DU in deciding whether to setup F1-U tunnel resources for an F1 MBS session context.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
RAN Sharing Assistance Information	M		ENUMERATED (MBS session in non-shared cell resources, ...)	"MBS session in non-shared cell resources" indicates that the gNB-CU-CP has already established MBS session resources utilising cell resources for which RAN sharing by means of RAN sharing with multiple cell ID broadcast is not applied.

9.3.1.346 LTM Reset Information

This IE contains the L2 reset configuration for the serving cell and LTM candidate cell(s).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Serving Cell L2 Reset Configuration	O		OCTET STRING	Includes the *ltm-ServingCellNoResetID* contained in the *LTM-Config* IE, as defined in TS 38.331 [8], for the current serving cell.
LTM L2 Reset Configuration List		*0..1*		
>LTM L2 Reset Configuration Item		*1..<maxnoofLTMCells>*		
>>Cell ID	M		NR CGI 9.3.1.12	
>>LTM L2 Reset Configuration	M		OCTET STRING	Includes the *ltm-NoResetID* IE as defined in TS 38.331 [8], for the LTM candidate cell identified by the *Cell ID* IE.

Range bound	Explanation
maxnoofLTMCells	Maximum no. of Cells configured for LTM allowed towards one UE, the maximum value is 8.

9.3.1.347 PRS Bandwidth Aggregation Request Information List

This IE contains the aggregated DL-PRS configuration information requested.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
PRS Bandwidth Aggregation Request Information List		*1*		
>PRS Bandwidth Aggregation Request Information Item		*1..<maxnoAggCombinations>*		
>>DL-PRS Bandwidth Aggregation Request Information List		*1*		
>>>DL-PRS Bandwidth Aggregation Request Information Item		*2..<maxnoAggregatedPosPRSResourceSets>*		
>>>>Requested DL-PRS Resource Set Index	M		INTEGER(1..8)	This IE specifies which of the indicated *Requested DL-PRS Resource Set Item*'s are requested for aggregation. The Integer Value defines an index to the *Requested DL-PRS Resource Set Item* in IE *Requested DL PRS Transmission Characteristics* (9.3.1.235). Integer value 1 defines the first entry in *Requested DL-PRS Resource Set Item,* Integer value 2 defines the second entry in *Requested DL-PRS Resource Set Item* and so on.

Range bound	Explanation
maxnoAggCombinations	Maximum number of aggregated frequency layer (carrier) combinations. Value is 2.

maxnoAggregatedPosPRSResourceSets	Maximum no of PRS resources sets aggregated. Value is 3.

9.3.1.348 Candidate Cell with Beam Information

This IE includes a candidate cell and associated SSB index information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Candidate Cell ID	M		NR CGI 9.3.1.12	
SSB Index	M		INTEGER (0..63)	

9.3.1.349 Candidate Cell with Beam Information List

This IE includes a list of candidate cells and associated SSB indexes information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Candidate Cell with Beam Information Item		*1 .. <maxnoofCandidateCells>*		
>Candidate Cell ID	M		NR CGI 9.3.1.12	
>SSB Index List		*1*		
>>SSB Index Item		*1..<maxnoofSSBIndices>*		
>>>SSB Index	M		INTEGER (0..63)	

Range bound	Explanation
maxnoofCandidateCells	Maximum no. of Cells configured as candidates allowed towards one UE, the maximum value is 8.
maxnoofSSBIndices	Maximum no. of SSBs that can be served by a NG-RAN node cell. Value is 64.

9.3.1.350 Measurement Quantities

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RSRP	M		INTEGER (0..127)	Corresponds to information provided in the *RSRP-Range* IE as defined in TS 38.331 [8].
RSRQ	M		INTEGER (0..127)	Corresponds to information provided in the *RSRQ-Range* IE as defined in TS 38.331 [8].
SINR	O		INTEGER (0..127)	Corresponds to information provided in the *SINR-Range* IE as defined in TS 38.331 [8].

9.3.1.351 Validity Area specific SRS Information Extended

This information element is used to indicate the extended SRS information.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Positioning SRS Resource List		*0..1*		
>Positioning SRS Resource Item		*1..< maxnoSRS-PosResources >*		
>>Positioning SRS Resource	M		9.3.1.194	Corresponds to information provided in *SRS-PosResource* contained in *SRS-Config* IE as defined in TS 38.331 [8]
Positioning SRS Resource Set List		*0..1*		
>Positioning SRS Resource Set Item		*1..<maxnoSRS-PosResourceSets>*		
>>Positioning SRS Resource Set	M		9.3.1.196	Corresponds to information provided in *SRS-PosResourceSet* contained in *SRS-Config* IE as defined in TS 38.331 [8]

Range bound	Explanation
maxnoSRS-PosResources	Maximum no of positioning SRS resources per UL BWP. Value is 64.
maxnoSRS-PosResourceSets	Maximum no of positioning SRS resource sets per UL BWP. Value is 16.

9.3.1.352 E-CID Angle of Arrival per TRP

This information element contains the UL Angle of Arrival measurement per TRP.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Angle of Arrival NR per TRP List		*1*		
>Angle of Arrival NR per TRP Item		*1..<maxnoofMeasTRPs>*		
>>TRP ID	M		9.3.1.197	
>>TRP Geographical Coordinates	O		Geographical Coordinates 9.3.1.184	
>>UL Angle of Arrival	M		9.3.1.167	
>>Time Stamp	M		9.3.1.171	
>>Measurement Quality	O		9.3.1.172	

Range bound	Explanation
maxnoofMeasTRPs	Maximum no. of TRPs that can be included within one message. Value is 64.

9.3.1.353 Available Bitrate Reporting Threshold List

This IE contains a list of available bitrate report thresholds. It is used for available bitrate report for UL and/or DL as specified in TS 23.501 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Available Bitrate Report Threshold Item		*1..<maxnoofThresholds>*		
>Reporting Threshold	M		INTEGER (0.. 4,000,000,000, ...)	This IE indicates the Reporting threshold as specified in TS 23.501 [21]. The unit is Kbps

Range bound	Explanation
maxnoofThresholds	Maximum no. of reporting thresholds allowed for one QoS flow on one direction. Value is 8.

9.3.1.354 LP-WUSPS Assistance Information

This IE provides the LP-WUS Paging Subgrouping information related to CN assigned subgrouping for a particular UE as specified in TS 38.304 [24].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
LP-WUS CN Subgroup ID	M		INTEGER (0..30, ...)	

9.3.1.355 Further Extended UE Identity Index Value

This IE is used by the gNB-DU to calculate the Paging Frame and Paging Occasion for eDRX, and the UE_ID based subgroup ID for PEI and LP-WUS as specified in TS 38.304 [24].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Further Extended UE Identity Index Value	M		BIT STRING (SIZE(20))	Encoded as 5G-S-TMSI mod 1048576.

9.3.1.356 NZP CSI-RS Resources Configuration

This IE contains the NZP CSI-RS resources configuration of an NR cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NZP-CSI-RS-ResourceSet	M		OCTET STRING	Includes the *NZP-CSI-RS-ResourceSet* IE, as defined in TS 38.331 [8].
NZP-CSI-RS-Resource List		*1*		
>NZP-CSI-RS-Resource Item		*1..<maxnoofNZP-CSI-RS-ResourcesPerSet>*		
>>NZP-CSI-RS-Resource	M		OCTET STRING	Includes the *NZP-CSI-RS-Resource* IE, as defined in TS 38.331 [8].

Range bound	Explanation
maxnoofNZP-CSI-RS-ResourcesPerSet	Maximum no. of NZP CSI-RS resources per resource set. Value is 64.

9.3.1.357 SRS Resource Configuration

This IE contains a list of SRS Resource of UEs in the current cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SRS Resource Configuration List		*1*		
>SRS Resource Configuration Item		*1..<maxnoofSRS-Resource>*		
>>SRS Resource	M		OCTET STRING	Includes the *SRS-Resource* IE as defined in TS 38.331 [8].

Range bound	Explanation
maxnoofSRS-Resource	Maximum number of SRS Resource. Value is 64

9.3.1.358 Last visited LTM Cells

This IE contains information about the last consecutive LTM cell switch entries where a UE was served by a cell which initiated an LTM cell switch.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Last Visited LTM Cell List		*1*		
>Last Visited LTM Cell Item		*1..<maxnoofCellsinUEHistoryInfo>*		Most recent information is added to the top of this list
>>Cell ID	M		NR CGI 9.3.1.12	
>>Time UE Stayed in Cell	M		INTEGER (0..40950)	The duration of time the UE stayed in the cell in 1/10 seconds. If the duration is more than 4095s, this IE is set to 40950.
Mobility Issue	O		ENUMERATED(ping-pong, ...)	

Range bound	Explanation
maxnoofCellsinUEHistoryInfo	Maximum number of last visited cell information records that can be reported in the IE. Value is 16.

9.3.1.359 LTM Security Information

This IE contains the security related information for LTM candidate cell(s) to support the UE in generating the key material for AS security during an LTM cell switch.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Next Hop Chaining Count	M		INTEGER (0..7)	Next Hop Chaining Count (NCC) defined in TS 33.501 [51]
Serving Cell No Security Change ID	O		OCTET STRING	Includes the *ltm-ServingCellNoSecurityChangeID* IE as defined in TS 38.331 [8], for the current serving cell.
Security Change Candidate Cell Information List		*0..1*		
>Security Change Candidate Cell Information Item		*1..<maxnoofLTMCells>*		

IE/Group Name	Presence	Range	IE type and reference	Semantics description
>>Cell ID	M		NR CGI 9.3.1.12	
>>No Security Change ID	M		OCTET STRING	Includes the *ltm-NoSecurityChangeID* IE as defined in TS 38.331 [8], for the LTM candidate cell identified by the Cell ID IE.

Range bound	Explanation
maxnoofLTMCells	Maximum no. of Cells configured for LTM allowed towards one UE, the maximum value is 8.

9.3.1.360 CSI-RS Resource Configuration

This IE contains the CSI-RS resource configuration used for LTM.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Periodic CSI-RS Resource Configuration	O		NZP CSI-RS Resource Configuration 9.3.1.373	
Semi Persistent CSI-RS Resource Configuration	O		NZP CSI-RS Resource Configuration 9.3.1.373	
CSI-RS Resource Configuration To Release List	O		OCTET STRING	Includes the *ltm-NZP-CSI-RS-ResourceToReleaseList* contained in the *LTM-Config* IE as defined in TS 38.331 [8].
Periodic CSI-RS Resource Set Configuration	O		NZP CSI-RS Resource Set Configuration 9.3.1.374	
Semi-Persistent CSI-RS Resource Set Configuration	O		NZP CSI-RS Resource Set Configuration 9.3.1.374	
Periodic CSI-IM Resource Configuration	O		CSI-IM Resource Configuration 9.3.1.375	
Semi-Persistent CSI-IM Resource Configuration	O		CSI-IM Resource Configuration 9.3.1.375	

9.3.1.361 Request for L1 Execution Condition

This IE indicates the list of LTM candidate cells requested for generating conditional LTM L1 execution conditions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Request for L1 Execution Condition Candidate Cell List Item IEs		*1..<maxnoofLTMCells>*		
>Candidate Cell ID	M		NR CGI 9.3.1.12	

Range bound	Explanation
maxnoofLTMCells	Maximum no. of Cells configured LTM allowed towards one UE, the maximum value is 8.

9.3.1.362 L1 Execution Condition List

This IE indicates the list of conditional LTM L1 execution conditions to be used by the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
L1 Execution Condition Item IEs		*1..< maxnoofLTMCells>*		
>LTM Cell ID	M		NR CGI 9.3.1.12	
>Execution Condition	M		OCTET STRING	Includes the *LTM-CSI-ReportConfigId-r18* IE as defined in subclause 6.3.2 in TS 38.331 [8].

Range bound	Explanation
maxnoofLTMCells	Maximum no. of Cells configured LTM allowed towards one UE, the maximum value is 8.

9.3.1.363 LTM Residual TA Information List

This IE indicates the LTM Residual TA information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TAT Information Item		*1 .. <maxnoofTAs>*		
>TA Value	M		INTEGER (0..4095)	Indicates the TA value as defined in TS 38.213 [31].
>TAT Remaining Time	M		INTEGER (0..10240)	Unit: ms.
>TAG ID Pointer	O		OCTET STRING	Includes the *tag-Id-ptr* contained in the *TCI-UL-State* IE or the *TCI-State* IE, as defined in TS 38.331 [8].

Range bound	Explanation
maxnoofTAs	Maximum no. of TAs, the maximum value is 2.

9.3.1.364 Channel Response Information

This IE contains information about the requested UL SRS-TDCT measurement.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Channel Response Window Size	M		ENUMERATED (32, 64, 128, ...)	Represents the window size over which channel response values are selected.
Channel Response Number	M		ENUMERATED (8, 16, 24, ...)	Represents the number of channel responses selected over the channel response window size.

9.3.1.365 UL SRS-TDCT

This information element contains the UL SRS time domain channel timing.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
UL SRS-TDCT List		*1*		
>UL SRS-TDCT Item		*1..<maxnoofChannelRes>*		
>>CHOICE *Timing Information*	M			
>>>k0				
>>>>k0	M		INTEGER (0.. 1970049)	TS 38.133 [38]
>>>k1				
>>>>k1	M		INTEGER (0.. 985025)	TS 38.133 [38]
>>>k2				
>>>>k2	M		INTEGER (0.. 492513)	TS 38.133 [38]
>>>k3				
>>>>k3	M		INTEGER (0.. 246257)	TS 38.133 [38]
>>>k4				
>>>>k4	M		INTEGER (0.. 123129)	TS 38.133 [38]
>>>k5				
>>>>k5	M		INTEGER (0.. 61565)	TS 38.133 [38]
>>Power Information	O		UL SRS-TDCP Item 9.3.1.366	

Range bound	Explanation
maxnoofChannelRes	Maximum no of channel response. Value is 24

9.3.1.366 UL SRS-TDCP Item

This IE contains an item of the UL SRS time domain channel power for a given channel response.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
UL SRS-TDCP Item	M		INTEGER (0..126)	TS 38.133 [38]

9.3.1.367 Predicted CCO Assistance Information

This IE provides predicted assistance information for the future Coverage and Capacity Optimisation (CCO) actions for predicted CCO issues.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Predicted CCO Issue	M		ENUMERATED (coverage, cell edge capacity, cancel, ...)	Indicates the type of predicted CCO issue or that the previously sent predicted CCO issue is cancelled.
Predicted Affected Cells and Beams	M		Affected Cells and Beams 9.3.1.212	
Time for Predicted CCO Issue	O		INTEGER (1..60, ...)	Indicates the time when the predicted CCO issue will happen from the time of receiving this

IE/Group Name	Presence	Range	IE type and reference	Semantics description
				information, in seconds. The IE is ignored if the *Predicted CCO Issue* IE is set to "cancel".

9.3.1.368 Future Coverage Modification Notification

This IE includes a list of cells and/or SS/PBCH block indexes with the corresponding future coverage configuration selected by a gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Future Coverage Modification Notification List		*1*		
>Future Coverage Modification Notification Item		*1..<maxCellingNBDU>*		
>>NR CGI	M		9.3.1.12	
>>Future Cell Coverage State	M		INTEGER (0..63, ...)	Value '0' indicates that the cell will be inactive. Other values indicate that the cell will be active and also indicate the future coverage configuration of the concerned cell.
>>Future SSB Modification Notification List		*0..1*		
>>>Future SSB Modification Notification Item		*1..<maxnoofSSBAreas>*		
>>>>SSB Index	M		INTEGER (0..63)	
>>>>Future SSB Coverage State	M		INTEGER (0..15, ...)	Value '0' indicates that the SSB beam will be inactive. Other values indicate that the SSB beams will be active and also indicate the future coverage configuration of the concerned SSB beams.
>>Time for Future Coverage Modification	M		INTEGER (1..60, ...)	Indicates the time when the Future Cell Coverage State(s) and/or the Future SSB Coverage State(s) will be applied by the gNB-DU relative to the time of receiving this information, in seconds. The IE is ignored if the Future Coverage Modification Cause is set to "cancel"
>>Future Coverage Modification Cause	O		ENUMERATED(coverage, cell edge capacity, cancel, ...)	

Range bound	Explanation
maxCellingNBDU	Maximum no. cells that can be served by a gNB-DU. Value is 512.
maxnoofSSBAreas	Maximum numbers of SSB Areas that can be served by a NG-RAN node cell. Value is 64.

9.3.1.369 Neighbour Future Coverage Modification Notification

This IE includes a list of cells and/or SS/PBCH block indexes with the corresponding future coverage configuration selected by one or more neighbour node(s).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Neighbour Future Coverage Modification Notification List		*1*		
>Neighbour Future Coverage Modification Notification Item		*1..< maxnoofCellsinNG-RANnode>*		
>>NR CGI	M		9.3.1.12	
>>Neighbour Future Cell Coverage State	M		INTEGER (0..63, ...)	Value ‘0’ indicates that the cell will be inactive. Other values indicate that the cell will be active and also indicate the future coverage configuration of the concerned cell.
>>Neighbour Future SSB Modification Notification List		*0..1*		
>>>Neighbour Future SSB Modification Notification Item		*1..<maxnoofSSBAreas>*		
>>>>SSB Index	M		INTEGER (0..63)	
>>>>Neighbour Future SSB Coverage State	M		INTEGER (0..15, ...)	Value ‘0’ indicates that the SSB beam will be inactive. Other values indicate that the SSB beams will be active and also indicate the future coverage configuration of the concerned SSB beams.
>>Time for Neighbour Future Coverage Modification	C-ifFutureCoverageModificationCauseCoverageorCapacity		INTEGER (1..60, ...)	Indicates the time when the Future Cell Coverage State(s) and/or the Future SSB Coverage State(s) will be applied by the gNB-DU relative to the time of receiving this information, in seconds.
Future Coverage Modification Cause	M		ENUMERATED (coverage, cell edge capacity, cancel, …)	Indicates the same Future Coverage Modification Cause that triggered the future coverage modification at the neighbor node.

Condition	Explanation
ifFutureCoverageModificationCauseCoverageorCapacity	This IE shall be present if the *Future Coverage Modification Cause* IE is set to the value "coverage" or "cell edge capacity".

Range bound	Explanation
maxnoofCellsinNG-RANnode	Maximum no. cells that can be served by a NG-RAN node. Value is 16384.
maxnoofSSBAreas	Maximum numbers of SSB Areas that can be served by a NG-RAN node cell. Value is 64.

9.3.1.370 UE Performance Delay Monitoring

This IE defines the parameters for UE performance delay measurements, and whether to stop an ongoing measurement.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Performance Delay Monitoring Request	M		ENUMERATED (UL and DL, stop,...)	Indicates to measure UL and DL delay for the DRB, or to stop the ongoing measurement.
UE Performance Delay Monitoring Reporting	O		ENUMERATED(ms500, ms1000,	Periodicity of reporting of UL and DL delay for the DRB.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Periodicity			ms2000, ms5000, ms10000, ...)	

9.3.1.371 gNB-CU or cNB-DU UE F1AP ID

The gNB-CU or gNB-DU UE F1AP ID uniquely identifies the UE association over the F1 interface within the gNB-CU or the gNB-DU.

NOTE: If F1-C signalling transport is shared among multiple interface instances, the value of the gNB-CU UE F1AP ID is allocated so that it can be associated with the corresponding F1-C interface instance.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB-CU of gNB-DU UE F1AP ID	M		INTEGER (0 .. 2^{32} - 1)	

9.3.1.372 Area-specific Semi-persistent SRS Positioning Information

This IE contains area-specific semi-persistent SRS positioning information to notify the activation or deactivation of semi-persistent SRS transmission for area-specific SRS configuration.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SRS Transmission Type	M	ENUMERATED (activate, deactivate, …)		
SRS Resource Set ID	M	9.3.1.180		
Spatial Relation Information	O	Spatial Relation Information 9.3.1.181		Applicable only if the SRS Transmission Type IE is set to ‘activate’.
Spatial Relation Information per SRS Resource	O	9.3.1.210		Applicable only if the SRS Transmission Type IE is set to ‘activate’.

9.3.1.373 NZP CSI-RS Resource Configuration

This IE contains the CSI-RS resource configuration used for LTM.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CSI-RS Resource to AddMod List	O		OCTET STRING	Contains the *ltm-NZP-CSI-RS-ResourceToAddModList* as defined in TS 38.331 [8].

9.3.1.374 NZP CSI-RS Resource Set Configuration

This IE contains the CSI-RS resource set configuration used for LTM.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CSI-RS Resource Set to AddMod List	O		OCTET STRING	Includes *ltm-NZP-CSI-RS-ResourceSetToAddModList* contained in the *LTM-Config* IE as defined in TS 38.331 [8].

9.3.1.375 CSI-IM Resource Configuration

This IE contains the CSI-IM resource configuration used for LTM.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CSI-IM Resource to AddMod List	O		OCTET STRING	Includes ltm-CSI-IM-ResourceToAddModList contained in the LTM-Config IE as defined in TS 38.331 [8].
CSI-IM ResourceSet to AddMod List	O		OCTET STRING	Includes *ltm-CSI-IM-*ResourceSetToAddModList contained in the *LTM-Config* IE as defined in TS 38.331 [8].

9.3.1.376 LP-WUS Supported Band Information

This IE provides the LP-WUS supported band information needed for paging.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Minimum Time Gap Indication	M		ENUMERATED (cap1, cap2, cap3, ...)	Corresponds to the *minimumTimeGap* contained in the *UERadioPagingInformation* IE defined in TS 38.331 [8].
OOK Indication	O		ENUMERATED (supported, ...)	Corresponds to the *lpwus-OOK* contained in the *UERadioPagingInformation* IE defined in TS 38.331 [8].
OFDM Indication	O		ENUMERATED (supported, ...)	Corresponds to the *lpwus-OFDM* contained in the *UERadioPagingInformation* IE defined in TS 38.331 [8].
LP-SS Indication	O		ENUMERATED (supported, ...)	Corresponds to the *lpwus-LP-SS* contained in the *UERadioPagingInformation* IE defined in TS 38.331 [8].

9.3.2 Transport Network Layer Related IEs

9.3.2.1 UP Transport Layer Information

The *UP Transport Layer Information* IE identifies an F1 transport bearer associated to a DRB. It contains a Transport Layer Address and a GTP Tunnel Endpoint Identifier. The Transport Layer Address is an IP address to be used for the F1 user plane transport. The GTP Tunnel Endpoint Identifier is to be used for the user plane transport between gNB-CU and gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE Transport Layer Information	M			
>GTP Tunnel				
>>Transport Layer Address	M		9.3.2.3	
>>GTP-TEID	M		9.3.2.2	

9.3.2.2 GTP-TEID

The *GTP-TEID* IE is the GTP Tunnel Endpoint Identifier to be used for the user plane transport between the gNB-CU and gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
GTP-TEID	M		OCTET STRING (SIZE(4))	For details and range, see TS 29.281 [18].

9.3.2.3 Transport Layer Address

This *Transport Layer Address* IE is an IP address.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Transport Layer Address	M		BIT STRING (SIZE(1..160, ...))	The Radio Network Layer is not supposed to interpret the address information. It should pass it to the Transport Layer for interpretation. For details, see TS 38.414 [19].

9.3.2.4 CP Transport Layer Information

This IE is used to provide the F1 control plane transport layer information associated with a gNB-CU – gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE CP Transport Layer Information					-	
>Endpoint-IP-address						
>>Endpoint IP address	M		Transport Layer Address 9.3.2.3		-	
>Endpoint-IP-address-and-port						
>>Endpoint IP address	M		Transport Layer Address 9.3.2.3		-	
>>Port Number	O		BIT STRING (SIZE(16))		YES	reject

9.3.2.5 Transport Layer Address Info

This IE is used for signalling TNL Configuration information for IPSec tunnel over which GTP traffic is transmitted.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Transport UP Layer Address Info to Add List		*0..1*		
>Transport UP Layer Address Info to Add Item		*1..<maxnoofTLAs>*		
>>IP-Sec Transport Layer Address	M		Transport Layer Address 9.3.2.3	Transport Layer Address for IP-Sec endpoint.
>>GTP Transport Layer Address To Add List		*0..1*		
>>>GTP Transport Layer Address To Add Item		*1..<maxnoofGTPTLAs>*		
>>>>GTP Transport Layer Address Info	M		Transport Layer Address 9.3.2.3	GTP Transport Layer Address for GTP end-points.
Transport UP Layer Address Info to Remove List		*0..1*		
>Transport UP Layer		*1..<maxnoofTL*		

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Address Info to Remove Item		*As>*		
>>IP-Sec Transport Layer Address	M		Transport Layer Address 9.3.2.3	Transport Layer Address for IP-Sec endpoint.
>>GTP Transport Layer Address To Remove List		*0..1*		
>>>GTP Transport Layer Address To Remove Item		*1..<maxnoofGTPTLAs>*		
>>>>GTP Transport Layer Address Info	M		Transport Layer Address 9.3.2.3	GTP Transport Layer Address for GTP end-points.

Range bound	Explanation
maxnoofTLAs	Maximum no. of F1 Transport Layer Address in the message. Value is 16.
maxnoofGTPTLAs	Maximum no. of F1 GTP Transport Layer Address for a GTP end-point in the message. Value is 16.

9.3.2.6 URI

This IE is an URI.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
URI	M		VisibleString	String representing URI (Uniform Resource Identifier)

9.3.2.7 BC Bearer Context F1-U TNL Info

This IE contains F1-U TNL information for an MBS Session. In case of location dependent MBS sessions, it also contains per Area Session ID F1-U TNL information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *MBS Session Type*	M				-	
>location independent						
>>MBS F1-U Information	M		UP Transport Layer Information 9.3.2.1		-	
>>F1-U Tunnel Not Established	O		ENUMERATED (true, ...)	Indicates F1-U tunnel not established for this MBS Session.	YES	ignore
>location dependent						
>>Location dependent MBS F1-U Information Item		*1..<maxnoofMBSAreaSessionIDs>*			-	
>>>MBS Area Session ID	M		9.3.1.221		-	
>>>MBS F1-U Information	M		UP Transport Layer Information 9.3.2.1		-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>>>F1-U Tunnel Not Established	O		ENUMERATED (true, ...)	Indicates F1-U tunnel not established for this MBS Session.	YES	ignore

Range bound	Explanation
maxnoofMBSAreaSessionIDs	Maximum no. of MBS Area Session IDs. Value is 256.

9.3.2.8 MBS Multicast F1-U Context Descriptor

This IE contains a reference to a Multicast F1-U Context, information about the usage of the MBS Multicast F1-U Context and may contain an MBS Area Session ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Multicast F1-U Context Reference F1	M		9.3.2.11	
MC F1-U Context usage	M		ENUMERATED (ptm, ptp, ptp retransmission, ptp forwarding, ...)	"ptm" indicates that the Multicast F1-U Context is setup for ptm transmissions; decided by the DU. "ptp" indicates that the Multicast F1-U Context is setup for ptp transmissions; decided by the DU. "ptp retransmission" indicates that the Multicast F1-U Context is setup for ptp retransmissions (based on PDCP Status Report); requested by the CU "ptp forwarding" indicates that the Multicast F1-U Context is setup for transmitting from a defined MBS Progress Information status onwards; requested by the CU.
MBS Area Session ID	O		9.3.1.221	

9.3.2.9 Void

Void

9.3.2.10 MBS PTP Retransmission Tunnel Required

This IE indicates the request to establishment of a PTP Retransmission F1-U Tunnel for retransmitting user data for a multicast MBS Session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS PTP Retransmission Tunnel Required	M		ENUMERATED (true, ...)	

9.3.2.11 Multicast F1-U Context Reference F1

This IE contains a reference to a Multicast F1-U Context associated with an MBS Session context in a gNB-DU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Multicast F1-U Bearer	M		OCTET STRING	This value is allocated to

Context Reference			(SIZE(4))	uniquely denote an Multicast F1-U Context within an MBS-associated logical F1-connection.

9.3.2.12 MRB Progress Information

This IE contains the MRB progress Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *MRB Progress Information*	M			
>12bits				
>>PDCP SN Length 12	M		INTEGER (0..4095)	
>18bits				
>>PDCP SN Length 18	M		INTEGER (0..262143)	

9.3.2.13 Multicast F1-U Context Reference CU

This IE contains a reference to a Multicast F1-U Context associated with MBS session resources allocated in the gNB-CU.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Multicast F1-U Bearer Context Reference CU	M		OCTET STRING (SIZE(4))	This value is allocated to uniquely denote a Multicast F1-U Context associated with multicast MBS session resources allocated in the gNB-CU. NOTE: If E1 is deployed, the *Multicast F1-U Bearer Context Reference CU* IE refers to the *Multicast F1-U Context ReferenceE1* IE as specified in TS 37.483 [47].

9.4 Message and Information Element Abstract Syntax (with ASN.1)

9.4.1 General

F1AP ASN.1 definition conforms to ITU-T Recommendation X.691 [5], ITU-T Recommendation X.680 [12] and ITU-T Recommendation X.681 [13].

The ASN.1 definition specifies the structure and content of F1AP messages. F1AP messages can contain any IEs specified in the object set definitions for that message without the order or number of occurrence being restricted by ASN.1. However, for this version of the standard, a sending entity shall construct an F1AP message according to the PDU definitions module and with the following additional rules:

- IEs shall be ordered (in an IE container) in the order they appear in object set definitions.

- Object set definitions specify how many times IEs may appear. An IE shall appear exactly once if the presence field in an object has value "mandatory". An IE may appear at most once if the presence field in an object has value "optional" or "conditional". If in a tabular format there is multiplicity specified for an IE (i.e., an IE list) then in the corresponding ASN.1 definition the list definition is separated into two parts. The first part defines an IE container list where the list elements reside. The second part defines list elements. The IE container list appears as an IE of its own. For this version of the standard an IE container list may contain only one kind of list elements.

NOTE: In the above "IE" means an IE in the object set with an explicit ID. If one IE needs to appear more than once in one object set, then the different occurrences will have different IE IDs.

If an F1AP message that is not constructed as defined above is received, this shall be considered as Abstract Syntax Error, and the message shall be handled as defined for Abstract Syntax Error in clause 10.

9.4.2 Usage of private message mechanism for non-standard use

The private message mechanism for non-standard use may be used:

- for special operator- (and/or vendor) specific features considered not to be part of the basic functionality, i.e., the functionality required for a complete and high-quality specification in order to guarantee multivendor interoperability;
- by vendors for research purposes, e.g., to implement and evaluate new algorithms/features before such features are proposed for standardisation.

The private message mechanism shall not be used for basic functionality. Such functionality shall be standardised.

9.4.3 Elementary Procedure Definitions

```
-- ASN1START
-- **************************************************************
--
-- Elementary Procedure definitions
--
-- **************************************************************

F1AP-PDU-Descriptions  {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
ngran-access (22) modules (3) f1ap (3) version1 (1) f1ap-PDU-Descriptions (0)}

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- **************************************************************
--
-- IE parameter types from other modules.
--
-- **************************************************************

IMPORTS
    Criticality,
    ProcedureCode

FROM F1AP-CommonDataTypes
    Reset,
    ResetAcknowledge,
    F1SetupRequest,
    F1SetupResponse,
    F1SetupFailure,
    GNBDUConfigurationUpdate,
    GNBDUConfigurationUpdateAcknowledge,
    GNBDUConfigurationUpdateFailure,
    GNBCUConfigurationUpdate,
    GNBCUConfigurationUpdateAcknowledge,
    GNBCUConfigurationUpdateFailure,
    UEContextSetupRequest,
    UEContextSetupResponse,
    UEContextSetupFailure,
    UEContextReleaseCommand,
    UEContextReleaseComplete,
    UEContextModificationRequest,
    UEContextModificationResponse,
    UEContextModificationFailure,
    UEContextModificationRequired,
    UEContextModificationConfirm,
    ErrorIndication,
    UEContextReleaseRequest,
    DLRRCMessageTransfer,
    ULRRCMessageTransfer,
```

```
	GNBDUResourceCoordinationRequest,
	GNBDUResourceCoordinationResponse,
	PrivateMessage,
	UEInactivityNotification,
	InitialULRRCMessageTransfer,
	SystemInformationDeliveryCommand,
	Paging,
	Notify,
	WriteReplaceWarningRequest,
	WriteReplaceWarningResponse,
	PWSCancelRequest,
	PWSCancelResponse,
	PWSRestartIndication,
	PWSFailureIndication,
	GNBDUStatusIndication,
	RRCDeliveryReport,
	UEContextModificationRefuse,
	F1RemovalRequest,
	F1RemovalResponse,
	F1RemovalFailure,
	NetworkAccessRateReduction,
	TraceStart,
	DeactivateTrace,
	DUCURadioInformationTransfer,
	CUDURadioInformationTransfer,
	BAPMappingConfiguration,
	BAPMappingConfigurationAcknowledge,
	BAPMappingConfigurationFailure,
	GNBDUResourceConfiguration,
	GNBDUResourceConfigurationAcknowledge,
	GNBDUResourceConfigurationFailure,
	IABTNLAddressRequest,
	IABTNLAddressResponse,
	IABTNLAddressFailure,
	IABUPConfigurationUpdateRequest,
	IABUPConfigurationUpdateResponse,
	IABUPConfigurationUpdateFailure,
	ResourceStatusRequest,
	ResourceStatusResponse,
	ResourceStatusFailure,
	ResourceStatusUpdate,
	AccessAndMobilityIndication,
	ReferenceTimeInformationReportingControl,
	ReferenceTimeInformationReport,
	AccessSuccess,
	CellTrafficTrace,
	PositioningMeasurementRequest,
	PositioningMeasurementResponse,
	PositioningMeasurementFailure,
	PositioningAssistanceInformationControl,
	PositioningAssistanceInformationFeedback,
	PositioningMeasurementReport,
	PositioningMeasurementAbort,
	PositioningMeasurementFailureIndication,
```

```
	PositioningMeasurementUpdate,
	TRPInformationRequest,
	TRPInformationResponse,
	TRPInformationFailure,
	PositioningInformationRequest,
	PositioningInformationResponse,
	PositioningInformationFailure,
	PositioningActivationRequest,
	PositioningActivationResponse,
	PositioningActivationFailure,
	PositioningDeactivation,
	PositioningInformationUpdate,
	E-CIDMeasurementInitiationRequest,
	E-CIDMeasurementInitiationResponse,
	E-CIDMeasurementInitiationFailure,
	E-CIDMeasurementFailureIndication,
	E-CIDMeasurementReport,
	E-CIDMeasurementTerminationCommand,
	BroadcastContextSetupRequest,
	BroadcastContextSetupResponse,
	BroadcastContextSetupFailure,
	BroadcastContextReleaseCommand,
	BroadcastContextReleaseComplete,
	BroadcastContextReleaseRequest,
	BroadcastContextModificationRequest,
	BroadcastContextModificationResponse,
	BroadcastContextModificationFailure,
	MulticastGroupPaging,
	MulticastContextSetupRequest,
	MulticastContextSetupResponse,
	MulticastContextSetupFailure,
	MulticastContextReleaseCommand,
	MulticastContextReleaseComplete,
	MulticastContextReleaseRequest,
	MulticastContextModificationRequest,
	MulticastContextModificationResponse,
	MulticastContextModificationFailure,
	MulticastDistributionSetupRequest,
	MulticastDistributionSetupResponse,
	MulticastDistributionSetupFailure,
	MulticastDistributionReleaseCommand,
	MulticastDistributionReleaseComplete,
	PDCMeasurementInitiationRequest,
	PDCMeasurementInitiationResponse,
	PDCMeasurementInitiationFailure,
	PDCMeasurementReport,
	PDCMeasurementTerminationCommand,
	PDCMeasurementFailureIndication,
	PRSConfigurationRequest,
	PRSConfigurationResponse,
	PRSConfigurationFailure,
	MeasurementPreconfigurationRequired,
	MeasurementPreconfigurationConfirm,
	MeasurementPreconfigurationRefuse,
```

```
	MeasurementActivation,
	QoEInformationTransfer,
	PosSystemInformationDeliveryCommand,
	DUCUCellSwitchNotification,
	CUDUCellSwitchNotification,
	DUCUTAInformationTransfer,
	CUDUTAInformationTransfer,
	QoEInformationTransferControl,
	RachIndication,
	TimingSynchronisationStatusRequest,
	TimingSynchronisationStatusResponse,
	TimingSynchronisationStatusFailure,
	TimingSynchronisationStatusReport,
	MIABF1SetupTriggering,
	MIABF1SetupOutcomeNotification,
	MulticastContextNotificationIndication,
	MulticastContextNotificationConfirm,
	MulticastContextNotificationRefuse,
	MulticastCommonConfigurationRequest,
	MulticastCommonConfigurationResponse,
	MulticastCommonConfigurationRefuse,
	BroadcastTransportResourceRequest,
	DUCUAccessAndMobilityIndication,
	SRSInformationReservationNotification,
	CUDUMobilityInitiationRequest,
	CLI-Indication,
	DUCUCSIRSCoordinationRequest,
	DUCUCSIRSCoordinationResponse,
	CUDUCSIRSCoordinationRequest,
	CUDUCSIRSCoordinationResponse




FROM F1AP-PDU-Contents
	id-Reset,
	id-F1Setup,
	id-gNBDUConfigurationUpdate,
	id-gNBCUConfigurationUpdate,
	id-UEContextSetup,
	id-UEContextRelease,
	id-UEContextModification,
	id-UEContextModificationRequired,
	id-DUCUAccessAndMobilityIndication,
	id-ErrorIndication,
	id-UEContextReleaseRequest,
	id-DLRRCMessageTransfer,
	id-ULRRCMessageTransfer,
	id-GNBDUResourceCoordination,
	id-privateMessage,
	id-UEInactivityNotification,
	id-InitialULRRCMessageTransfer,
	id-SystemInformationDeliveryCommand,
```

```
	id-Paging,
	id-Notify,
	id-WriteReplaceWarning,
	id-PWSCancel,
	id-PWSRestartIndication,
	id-PWSFailureIndication,
	id-GNBDUStatusIndication,
	id-RRCDeliveryReport,
	id-F1Removal,
	id-NetworkAccessRateReduction,
	id-TraceStart,
	id-DeactivateTrace,
	id-DUCURadioInformationTransfer,
	id-CUDURadioInformationTransfer,
	id-BAPMappingConfiguration,
	id-GNBDUResourceConfiguration,
	id-IABTNLAddressAllocation,
	id-IABUPConfigurationUpdate,
	id-resourceStatusReportingInitiation,
	id-resourceStatusReporting,
	id-accessAndMobilityIndication,
	id-ReferenceTimeInformationReportingControl,
	id-ReferenceTimeInformationReport,
	id-accessSuccess,
	id-cellTrafficTrace,
	id-PositioningMeasurementExchange,
	id-PositioningAssistanceInformationControl,
	id-PositioningAssistanceInformationFeedback,
	id-PositioningMeasurementReport,
	id-PositioningMeasurementAbort,
	id-PositioningMeasurementFailureIndication,
	id-PositioningMeasurementUpdate,
	id-TRPInformationExchange,
	id-PositioningInformationExchange,
	id-PositioningActivation,
	id-PositioningDeactivation,
	id-PositioningInformationUpdate,
	id-E-CIDMeasurementInitiation,
	id-E-CIDMeasurementFailureIndication,
	id-E-CIDMeasurementReport,
	id-E-CIDMeasurementTermination,
	id-BroadcastContextSetup,
	id-BroadcastContextRelease,
	id-BroadcastContextReleaseRequest,
	id-BroadcastContextModification,
	id-MulticastGroupPaging,
	id-MulticastContextSetup,
	id-MulticastContextRelease,
	id-MulticastContextReleaseRequest,
	id-MulticastContextModification,
	id-MulticastDistributionSetup,
	id-MulticastDistributionRelease,
	id-PDCMeasurementInitiation,
	id-PDCMeasurementTerminationCommand,
```

```
    id-PDCMeasurementFailureIndication,
    id-PDCMeasurementReport,
    id-pRSConfigurationExchange,
    id-measurementPreconfiguration,
    id-measurementActivation,
    id-QoEInformationTransfer,
    id-PosSystemInformationDeliveryCommand,
    id-DUCUCellSwitchNotification,
    id-CUDUCellSwitchNotification,
    id-DUCUTAInformationTransfer,
    id-CUDUTAInformationTransfer,
    id-QoEInformationTransferControl,
    id-RachIndication,
    id-TimingSynchronisationStatus,
    id-TimingSynchronisationStatusReport,
    id-MIABF1SetupTriggering,
    id-MIABF1SetupOutcomeNotification,
    id-MulticastContextNotification,
    id-MulticastCommonConfiguration,
    id-BroadcastTransportResourceRequest,
    id-SRSInformationReservationNotification,
    id-CUDUMobilityInitiationRequest,
    id-CLI-Indication,
    id-DUCUCSIRSCoordination,
    id-CUDUCSIRSCoordination




FROM F1AP-Constants

    ProtocolIE-SingleContainer{},
    F1AP-PROTOCOL-IES

FROM F1AP-Containers;


-- **************************************************************
--
-- Interface Elementary Procedure Class
--
-- **************************************************************

F1AP-ELEMENTARY-PROCEDURE ::= CLASS {
    &InitiatingMessage              ,
    &SuccessfulOutcome                          OPTIONAL,
    &UnsuccessfulOutcome                        OPTIONAL,
    &procedureCode              ProcedureCode   UNIQUE,
    &criticality                Criticality     DEFAULT ignore
}
WITH SYNTAX {
    INITIATING MESSAGE          &InitiatingMessage
    [SUCCESSFUL OUTCOME         &SuccessfulOutcome]
```

```
    [UNSUCCESSFUL OUTCOME       &UnsuccessfulOutcome]
    PROCEDURE CODE              &procedureCode
    [CRITICALITY                &criticality]
}

-- **************************************************************
--
-- Interface PDU Definition
--
-- **************************************************************

F1AP-PDU ::= CHOICE {
    initiatingMessage   InitiatingMessage,
    successfulOutcome   SuccessfulOutcome,
    unsuccessfulOutcome UnsuccessfulOutcome,
    choice-extension    ProtocolIE-SingleContainer { { F1AP-PDU-ExtIEs} }
}

F1AP-PDU-ExtIEs F1AP-PROTOCOL-IES ::= { -- this extension is not used
    ...
}

InitiatingMessage ::= SEQUENCE {
    procedureCode   F1AP-ELEMENTARY-PROCEDURE.&procedureCode        ({F1AP-ELEMENTARY-PROCEDURES}),
    criticality     F1AP-ELEMENTARY-PROCEDURE.&criticality          ({F1AP-ELEMENTARY-PROCEDURES}{@procedureCode}),
    value           F1AP-ELEMENTARY-PROCEDURE.&InitiatingMessage    ({F1AP-ELEMENTARY-PROCEDURES}{@procedureCode})
}

SuccessfulOutcome ::= SEQUENCE {
    procedureCode   F1AP-ELEMENTARY-PROCEDURE.&procedureCode        ({F1AP-ELEMENTARY-PROCEDURES}),
    criticality     F1AP-ELEMENTARY-PROCEDURE.&criticality          ({F1AP-ELEMENTARY-PROCEDURES}{@procedureCode}),
    value           F1AP-ELEMENTARY-PROCEDURE.&SuccessfulOutcome    ({F1AP-ELEMENTARY-PROCEDURES}{@procedureCode})
}

UnsuccessfulOutcome ::= SEQUENCE {
    procedureCode   F1AP-ELEMENTARY-PROCEDURE.&procedureCode        ({F1AP-ELEMENTARY-PROCEDURES}),
    criticality     F1AP-ELEMENTARY-PROCEDURE.&criticality          ({F1AP-ELEMENTARY-PROCEDURES}{@procedureCode}),
    value           F1AP-ELEMENTARY-PROCEDURE.&UnsuccessfulOutcome  ({F1AP-ELEMENTARY-PROCEDURES}{@procedureCode})
}

-- **************************************************************
--
-- Interface Elementary Procedure List
--
-- **************************************************************

F1AP-ELEMENTARY-PROCEDURES F1AP-ELEMENTARY-PROCEDURE ::= {
    F1AP-ELEMENTARY-PROCEDURES-CLASS-1          |
    F1AP-ELEMENTARY-PROCEDURES-CLASS-2,
    ...
}


F1AP-ELEMENTARY-PROCEDURES-CLASS-1 F1AP-ELEMENTARY-PROCEDURE ::= {
```

```
    reset                            |
    f1Setup                          |
    gNBDUConfigurationUpdate         |
    gNBCUConfigurationUpdate         |
    uEContextSetup                   |
    uEContextRelease                 |
    uEContextModification            |
    uEContextModificationRequired    |
    writeReplaceWarning              |
    pWSCancel                        |
    gNBDUResourceCoordination        |
    f1Removal                        |
    bAPMappingConfiguration          |
    gNBDUResourceConfiguration       |
    iABTNLAddressAllocation          |
    iABUPConfigurationUpdate         |
    resourceStatusReportingInitiation    |
    positioningMeasurementExchange   |
    tRPInformationExchange           |
    positioningInformationExchange   |
    positioningActivation            |
    e-CIDMeasurementInitiation       |
    broadcastContextSetup            |
    broadcastContextRelease          |
    broadcastContextModification     |
    multicastContextSetup            |
    multicastContextRelease          |
    multicastContextModification     |
    multicastDistributionSetup       |
    multicastDistributionRelease     |
    pDCMeasurementInitiation         |
    pRSConfigurationExchange         |
    measurementPreconfiguration |
    timingSynchronisationStatus |
    multicastContextNotification     |
    multicastCommonConfiguration     |
    cUDUCSIRSCoordination|
    dUCUCSIRSCoordination,
    ...
}

F1AP-ELEMENTARY-PROCEDURES-CLASS-2 F1AP-ELEMENTARY-PROCEDURE ::= {
    errorIndication                          |
    uEContextReleaseRequest                  |
    dLRRCMessageTransfer                     |
    uLRRCMessageTransfer                     |
    uEInactivityNotification                 |
    privateMessage                           |
    initialULRRCMessageTransfer              |
    systemInformationDelivery                |
    paging                                   |
    notify                                   |
    pWSRestartIndication                     |
    pWSFailureIndication                     |
```

```
    gNBDUStatusIndication                   |
    rRCDeliveryReport                       |
    networkAccessRateReduction              |
    traceStart                              |
    deactivateTrace                         |
    dUCURadioInformationTransfer            |
    cUDURadioInformationTransfer            |
    resourceStatusReporting                 |
    accessAndMobilityIndication             |
    referenceTimeInformationReportingControl|
    referenceTimeInformationReport          |
    accessSuccess                           |
    cellTrafficTrace                        |
    positioningAssistanceInformationControl     |
    positioningAssistanceInformationFeedback    |
    positioningMeasurementReport                |
    positioningMeasurementAbort                 |
    positioningMeasurementFailureIndication     |
    positioningMeasurementUpdate                |
    positioningDeactivation                     |
    e-CIDMeasurementFailureIndication           |
    e-CIDMeasurementReport                      |
    e-CIDMeasurementTermination                 |
    positioningInformationUpdate                |
    multicastGroupPaging                        |
    broadcastContextReleaseRequest              |
    multicastContextReleaseRequest              |
    pDCMeasurementReport                        |
    pDCMeasurementTerminationCommand            |
    pDCMeasurementFailureIndication             |
    measurementActivation                       |
    qoEInformationTransfer                      |
    posSystemInformationDelivery                    |
    dUCUCellSwitchNotification                  |
    cUDUCellSwitchNotification                  |
    dUCUTAInformationTransfer                   |
    cUDUTAInformationTransfer                   |
    qoEInformationTransferControl                   |
    rachIndication                              |
    timingSynchronisationStatusReport               |
    mIABF1SetupTriggering                       |
    mIABF1SetupOutcomeNotification              |
    broadcastTransportResourceRequest           |
    dUCUAccessAndMobilityIndication         |
    sRSInformationReservationNotification       |
    cUDUMobilityInitiation|
    cLI-Indication,
    ...
}
-- **************************************************************
--
-- Interface Elementary Procedures
--
-- **************************************************************
```

```
reset F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      Reset
    SUCCESSFUL OUTCOME      ResetAcknowledge
    PROCEDURE CODE          id-Reset
    CRITICALITY             reject
}

f1Setup F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      F1SetupRequest
    SUCCESSFUL OUTCOME      F1SetupResponse
    UNSUCCESSFUL OUTCOME    F1SetupFailure
    PROCEDURE CODE          id-F1Setup
    CRITICALITY             reject
}

gNBDUConfigurationUpdate F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      GNBDUConfigurationUpdate
    SUCCESSFUL OUTCOME      GNBDUConfigurationUpdateAcknowledge
    UNSUCCESSFUL OUTCOME    GNBDUConfigurationUpdateFailure
    PROCEDURE CODE          id-gNBDUConfigurationUpdate
    CRITICALITY             reject
}

gNBCUConfigurationUpdate F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      GNBCUConfigurationUpdate
    SUCCESSFUL OUTCOME      GNBCUConfigurationUpdateAcknowledge
    UNSUCCESSFUL OUTCOME    GNBCUConfigurationUpdateFailure
    PROCEDURE CODE          id-gNBCUConfigurationUpdate
    CRITICALITY             reject
}

uEContextSetup F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UEContextSetupRequest
    SUCCESSFUL OUTCOME      UEContextSetupResponse
    UNSUCCESSFUL OUTCOME    UEContextSetupFailure
    PROCEDURE CODE          id-UEContextSetup
    CRITICALITY             reject
}

uEContextRelease F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UEContextReleaseCommand
    SUCCESSFUL OUTCOME      UEContextReleaseComplete
    PROCEDURE CODE          id-UEContextRelease
    CRITICALITY             reject
}

uEContextModification F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UEContextModificationRequest
    SUCCESSFUL OUTCOME      UEContextModificationResponse
    UNSUCCESSFUL OUTCOME    UEContextModificationFailure
    PROCEDURE CODE          id-UEContextModification
    CRITICALITY             reject
}
```

```
uEContextModificationRequired F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UEContextModificationRequired
    SUCCESSFUL OUTCOME      UEContextModificationConfirm
    UNSUCCESSFUL OUTCOME    UEContextModificationRefuse
    PROCEDURE CODE          id-UEContextModificationRequired
    CRITICALITY             reject
}

writeReplaceWarning F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      WriteReplaceWarningRequest
    SUCCESSFUL OUTCOME      WriteReplaceWarningResponse
    PROCEDURE CODE          id-WriteReplaceWarning
    CRITICALITY             reject
}

pWSCancel F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PWSCancelRequest
    SUCCESSFUL OUTCOME      PWSCancelResponse
    PROCEDURE CODE          id-PWSCancel
    CRITICALITY             reject
}

errorIndication F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ErrorIndication
    PROCEDURE CODE          id-ErrorIndication
    CRITICALITY             ignore
}

uEContextReleaseRequest F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UEContextReleaseRequest
    PROCEDURE CODE          id-UEContextReleaseRequest
    CRITICALITY             ignore
}


initialULRRCMessageTransfer F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      InitialULRRCMessageTransfer
    PROCEDURE CODE          id-InitialULRRCMessageTransfer
    CRITICALITY             ignore
}

dLRRCMessageTransfer F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      DLRRCMessageTransfer
    PROCEDURE CODE          id-DLRRCMessageTransfer
    CRITICALITY             ignore
}

uLRRCMessageTransfer F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ULRRCMessageTransfer
    PROCEDURE CODE          id-ULRRCMessageTransfer
    CRITICALITY             ignore
}
```

```
uEInactivityNotification  F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UEInactivityNotification
    PROCEDURE CODE          id-UEInactivityNotification
    CRITICALITY             ignore
}

gNBDUResourceCoordination F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      GNBDUResourceCoordinationRequest
    SUCCESSFUL OUTCOME      GNBDUResourceCoordinationResponse
    PROCEDURE CODE          id-GNBDUResourceCoordination
    CRITICALITY             reject
}

privateMessage F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PrivateMessage
    PROCEDURE CODE          id-privateMessage
    CRITICALITY             ignore
}

systemInformationDelivery F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SystemInformationDeliveryCommand
    PROCEDURE CODE          id-SystemInformationDeliveryCommand
    CRITICALITY             ignore
}


paging F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      Paging
    PROCEDURE CODE          id-Paging
    CRITICALITY             ignore
}

notify F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      Notify
    PROCEDURE CODE          id-Notify
    CRITICALITY             ignore
}

networkAccessRateReduction F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      NetworkAccessRateReduction
    PROCEDURE CODE          id-NetworkAccessRateReduction
    CRITICALITY             ignore
}


pWSRestartIndication F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PWSRestartIndication
    PROCEDURE CODE          id-PWSRestartIndication
    CRITICALITY             ignore
}

pWSFailureIndication F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PWSFailureIndication
```

```
    PROCEDURE CODE          id-PWSFailureIndication
    CRITICALITY             ignore
}

gNBDUStatusIndication   F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      GNBDUStatusIndication
    PROCEDURE CODE          id-GNBDUStatusIndication
    CRITICALITY             ignore
}


rRCDeliveryReport F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RRCDeliveryReport
    PROCEDURE CODE          id-RRCDeliveryReport
    CRITICALITY             ignore
}

f1Removal F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      F1RemovalRequest
    SUCCESSFUL OUTCOME      F1RemovalResponse
    UNSUCCESSFUL OUTCOME    F1RemovalFailure
    PROCEDURE CODE          id-F1Removal
    CRITICALITY             reject
}

traceStart F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      TraceStart
    PROCEDURE CODE          id-TraceStart
    CRITICALITY             ignore
}

deactivateTrace F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      DeactivateTrace
    PROCEDURE CODE          id-DeactivateTrace
    CRITICALITY             ignore
}

dUCURadioInformationTransfer F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      DUCURadioInformationTransfer
    PROCEDURE CODE          id-DUCURadioInformationTransfer
    CRITICALITY             ignore
}

cUDURadioInformationTransfer F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      CUDURadioInformationTransfer
    PROCEDURE CODE          id-CUDURadioInformationTransfer
    CRITICALITY             ignore
}

bAPMappingConfiguration F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      BAPMappingConfiguration
    SUCCESSFUL OUTCOME      BAPMappingConfigurationAcknowledge
    UNSUCCESSFUL OUTCOME    BAPMappingConfigurationFailure
    PROCEDURE CODE          id-BAPMappingConfiguration
```

```
    CRITICALITY             reject
}

gNBDUResourceConfiguration F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      GNBDUResourceConfiguration
    SUCCESSFUL OUTCOME      GNBDUResourceConfigurationAcknowledge
    UNSUCCESSFUL OUTCOME    GNBDUResourceConfigurationFailure
    PROCEDURE CODE          id-GNBDUResourceConfiguration
    CRITICALITY             reject
}

iABTNLAddressAllocation F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      IABTNLAddressRequest
    SUCCESSFUL OUTCOME      IABTNLAddressResponse
    UNSUCCESSFUL OUTCOME    IABTNLAddressFailure
    PROCEDURE CODE          id-IABTNLAddressAllocation
    CRITICALITY             reject
}

iABUPConfigurationUpdate F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      IABUPConfigurationUpdateRequest
    SUCCESSFUL OUTCOME      IABUPConfigurationUpdateResponse
    UNSUCCESSFUL OUTCOME    IABUPConfigurationUpdateFailure
    PROCEDURE CODE          id-IABUPConfigurationUpdate
    CRITICALITY             reject
}

resourceStatusReportingInitiation F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ResourceStatusRequest
    SUCCESSFUL OUTCOME      ResourceStatusResponse
    UNSUCCESSFUL OUTCOME    ResourceStatusFailure
    PROCEDURE CODE          id-resourceStatusReportingInitiation
    CRITICALITY             reject
}

resourceStatusReporting F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ResourceStatusUpdate
    PROCEDURE CODE          id-resourceStatusReporting
    CRITICALITY             ignore
}

accessAndMobilityIndication F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      AccessAndMobilityIndication
    PROCEDURE CODE          id-accessAndMobilityIndication
    CRITICALITY             ignore
}

referenceTimeInformationReportingControl F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ReferenceTimeInformationReportingControl
    PROCEDURE CODE          id-ReferenceTimeInformationReportingControl
    CRITICALITY             ignore
}

referenceTimeInformationReport F1AP-ELEMENTARY-PROCEDURE ::= {
```

```
    INITIATING MESSAGE      ReferenceTimeInformationReport
    PROCEDURE CODE          id-ReferenceTimeInformationReport
    CRITICALITY             ignore
}

accessSuccess F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      AccessSuccess
    PROCEDURE CODE          id-accessSuccess
    CRITICALITY             ignore
}

cellTrafficTrace F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      CellTrafficTrace
    PROCEDURE CODE          id-cellTrafficTrace
    CRITICALITY             ignore
}

positioningAssistanceInformationControl F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PositioningAssistanceInformationControl
    PROCEDURE CODE          id-PositioningAssistanceInformationControl
    CRITICALITY             ignore
}

positioningAssistanceInformationFeedback F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PositioningAssistanceInformationFeedback
    PROCEDURE CODE          id-PositioningAssistanceInformationFeedback
    CRITICALITY             ignore
}

positioningMeasurementExchange F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PositioningMeasurementRequest
    SUCCESSFUL OUTCOME      PositioningMeasurementResponse
    UNSUCCESSFUL OUTCOME    PositioningMeasurementFailure
    PROCEDURE CODE          id-PositioningMeasurementExchange
    CRITICALITY             reject
}

positioningMeasurementReport F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PositioningMeasurementReport
    PROCEDURE CODE          id-PositioningMeasurementReport
    CRITICALITY             ignore
}

positioningMeasurementAbort F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PositioningMeasurementAbort
    PROCEDURE CODE          id-PositioningMeasurementAbort
    CRITICALITY             ignore
}

positioningMeasurementFailureIndication F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PositioningMeasurementFailureIndication
    PROCEDURE CODE          id-PositioningMeasurementFailureIndication
    CRITICALITY             ignore
}
```

```
positioningMeasurementUpdate F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PositioningMeasurementUpdate
    PROCEDURE CODE          id-PositioningMeasurementUpdate
    CRITICALITY             ignore
}


tRPInformationExchange F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      TRPInformationRequest
    SUCCESSFUL OUTCOME      TRPInformationResponse
    UNSUCCESSFUL OUTCOME    TRPInformationFailure
    PROCEDURE CODE          id-TRPInformationExchange
    CRITICALITY             reject
}

positioningInformationExchange F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PositioningInformationRequest
    SUCCESSFUL OUTCOME      PositioningInformationResponse
    UNSUCCESSFUL OUTCOME    PositioningInformationFailure
    PROCEDURE CODE          id-PositioningInformationExchange
    CRITICALITY             reject
}

positioningActivation F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PositioningActivationRequest
    SUCCESSFUL OUTCOME      PositioningActivationResponse
    UNSUCCESSFUL OUTCOME    PositioningActivationFailure
    PROCEDURE CODE          id-PositioningActivation
    CRITICALITY             reject
}

positioningDeactivation F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PositioningDeactivation
    PROCEDURE CODE          id-PositioningDeactivation
    CRITICALITY             ignore
}

e-CIDMeasurementInitiation F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      E-CIDMeasurementInitiationRequest
    SUCCESSFUL OUTCOME      E-CIDMeasurementInitiationResponse
    UNSUCCESSFUL OUTCOME    E-CIDMeasurementInitiationFailure
    PROCEDURE CODE          id-E-CIDMeasurementInitiation
    CRITICALITY             reject
}

e-CIDMeasurementFailureIndication F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      E-CIDMeasurementFailureIndication
    PROCEDURE CODE          id-E-CIDMeasurementFailureIndication
    CRITICALITY             ignore
}

e-CIDMeasurementReport F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      E-CIDMeasurementReport
```

```
    PROCEDURE CODE          id-E-CIDMeasurementReport
    CRITICALITY             ignore
}

e-CIDMeasurementTermination F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      E-CIDMeasurementTerminationCommand
    PROCEDURE CODE          id-E-CIDMeasurementTermination
    CRITICALITY             ignore
}

positioningInformationUpdate F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PositioningInformationUpdate
    PROCEDURE CODE          id-PositioningInformationUpdate
    CRITICALITY             ignore
}

broadcastContextSetup F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      BroadcastContextSetupRequest
    SUCCESSFUL OUTCOME      BroadcastContextSetupResponse
    UNSUCCESSFUL OUTCOME    BroadcastContextSetupFailure
    PROCEDURE CODE          id-BroadcastContextSetup
    CRITICALITY             reject
}

broadcastContextRelease F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      BroadcastContextReleaseCommand
    SUCCESSFUL OUTCOME      BroadcastContextReleaseComplete
    PROCEDURE CODE          id-BroadcastContextRelease
    CRITICALITY             reject
}

broadcastContextReleaseRequest F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      BroadcastContextReleaseRequest
    PROCEDURE CODE          id-BroadcastContextReleaseRequest
    CRITICALITY             reject
}

broadcastContextModification F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      BroadcastContextModificationRequest
    SUCCESSFUL OUTCOME      BroadcastContextModificationResponse
    UNSUCCESSFUL OUTCOME    BroadcastContextModificationFailure
    PROCEDURE CODE          id-BroadcastContextModification
    CRITICALITY             reject
}

multicastGroupPaging F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MulticastGroupPaging
    PROCEDURE CODE          id-MulticastGroupPaging
    CRITICALITY             ignore
}


multicastContextSetup F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MulticastContextSetupRequest
```

```
    SUCCESSFUL OUTCOME      MulticastContextSetupResponse
    UNSUCCESSFUL OUTCOME    MulticastContextSetupFailure
    PROCEDURE CODE          id-MulticastContextSetup
    CRITICALITY             reject
}

multicastContextRelease F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MulticastContextReleaseCommand
    SUCCESSFUL OUTCOME      MulticastContextReleaseComplete
    PROCEDURE CODE          id-MulticastContextRelease
    CRITICALITY             reject
}

multicastContextReleaseRequest F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MulticastContextReleaseRequest
    PROCEDURE CODE          id-MulticastContextReleaseRequest
    CRITICALITY             reject
}

multicastContextModification F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MulticastContextModificationRequest
    SUCCESSFUL OUTCOME      MulticastContextModificationResponse
    UNSUCCESSFUL OUTCOME    MulticastContextModificationFailure
    PROCEDURE CODE          id-MulticastContextModification
    CRITICALITY             reject
}

multicastDistributionSetup F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MulticastDistributionSetupRequest
    SUCCESSFUL OUTCOME      MulticastDistributionSetupResponse
    UNSUCCESSFUL OUTCOME    MulticastDistributionSetupFailure
    PROCEDURE CODE          id-MulticastDistributionSetup
    CRITICALITY             reject
}

multicastDistributionRelease F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MulticastDistributionReleaseCommand
    SUCCESSFUL OUTCOME      MulticastDistributionReleaseComplete
    PROCEDURE CODE          id-MulticastDistributionRelease
    CRITICALITY             reject
}


pDCMeasurementInitiation F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PDCMeasurementInitiationRequest
    SUCCESSFUL OUTCOME      PDCMeasurementInitiationResponse
    UNSUCCESSFUL OUTCOME    PDCMeasurementInitiationFailure
    PROCEDURE CODE          id-PDCMeasurementInitiation
    CRITICALITY             reject
}

pDCMeasurementReport F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PDCMeasurementReport
    PROCEDURE CODE          id-PDCMeasurementReport
```

```
    CRITICALITY             ignore
}

pDCMeasurementTerminationCommand F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PDCMeasurementTerminationCommand
    PROCEDURE CODE          id-PDCMeasurementTerminationCommand
    CRITICALITY             ignore
}

pDCMeasurementFailureIndication F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PDCMeasurementFailureIndication
    PROCEDURE CODE          id-PDCMeasurementFailureIndication
    CRITICALITY             ignore
}

pRSConfigurationExchange F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PRSConfigurationRequest
    SUCCESSFUL OUTCOME      PRSConfigurationResponse
    UNSUCCESSFUL OUTCOME    PRSConfigurationFailure
    PROCEDURE CODE          id-pRSConfigurationExchange
    CRITICALITY             reject
}

measurementPreconfiguration F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MeasurementPreconfigurationRequired
    SUCCESSFUL OUTCOME      MeasurementPreconfigurationConfirm
    UNSUCCESSFUL OUTCOME    MeasurementPreconfigurationRefuse
    PROCEDURE CODE          id-measurementPreconfiguration
    CRITICALITY             reject
}


measurementActivation       F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MeasurementActivation
    PROCEDURE CODE          id-measurementActivation
    CRITICALITY             ignore
}

qoEInformationTransfer F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      QoEInformationTransfer
    PROCEDURE CODE          id-QoEInformationTransfer
    CRITICALITY             ignore
}

posSystemInformationDelivery F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PosSystemInformationDeliveryCommand
    PROCEDURE CODE          id-PosSystemInformationDeliveryCommand
    CRITICALITY             ignore
}

dUCUCellSwitchNotification  F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      DUCUCellSwitchNotification
    PROCEDURE CODE          id-DUCUCellSwitchNotification
    CRITICALITY             ignore
```

```
}

cUDUCellSwitchNotification  F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      CUDUCellSwitchNotification
    PROCEDURE CODE          id-CUDUCellSwitchNotification
    CRITICALITY             ignore
}


dUCUTAInformationTransfer   F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      DUCUTAInformationTransfer
    PROCEDURE CODE          id-DUCUTAInformationTransfer
    CRITICALITY             ignore
}

cUDUTAInformationTransfer   F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      CUDUTAInformationTransfer
    PROCEDURE CODE          id-CUDUTAInformationTransfer
    CRITICALITY             ignore
}

qoEInformationTransferControl F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      QoEInformationTransferControl
    PROCEDURE CODE          id-QoEInformationTransferControl
    CRITICALITY             ignore
}

rachIndication F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RachIndication
    PROCEDURE CODE          id-RachIndication
    CRITICALITY             ignore
}

timingSynchronisationStatus F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      TimingSynchronisationStatusRequest
    SUCCESSFUL OUTCOME      TimingSynchronisationStatusResponse
    UNSUCCESSFUL OUTCOME    TimingSynchronisationStatusFailure
    PROCEDURE CODE          id-TimingSynchronisationStatus
    CRITICALITY             reject
}

timingSynchronisationStatusReport F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      TimingSynchronisationStatusReport
    PROCEDURE CODE          id-TimingSynchronisationStatusReport
    CRITICALITY             ignore
}

mIABF1SetupTriggering F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MIABF1SetupTriggering
    PROCEDURE CODE          id-MIABF1SetupTriggering
    CRITICALITY             reject
}

mIABF1SetupOutcomeNotification F1AP-ELEMENTARY-PROCEDURE ::= {
```

```
    INITIATING MESSAGE      MIABF1SetupOutcomeNotification
    PROCEDURE CODE          id-MIABF1SetupOutcomeNotification
    CRITICALITY             reject
}

multicastContextNotification F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MulticastContextNotificationIndication
    SUCCESSFUL OUTCOME      MulticastContextNotificationConfirm
    UNSUCCESSFUL OUTCOME    MulticastContextNotificationRefuse
    PROCEDURE CODE          id-MulticastContextNotification
    CRITICALITY             reject
}

multicastCommonConfiguration F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MulticastCommonConfigurationRequest
    SUCCESSFUL OUTCOME      MulticastCommonConfigurationResponse
    UNSUCCESSFUL OUTCOME    MulticastCommonConfigurationRefuse
    PROCEDURE CODE          id-MulticastCommonConfiguration
    CRITICALITY             reject
}


broadcastTransportResourceRequest F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      BroadcastTransportResourceRequest
    PROCEDURE CODE          id-BroadcastTransportResourceRequest
    CRITICALITY             reject
}

dUCUAccessAndMobilityIndication F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      DUCUAccessAndMobilityIndication
    PROCEDURE CODE          id-DUCUAccessAndMobilityIndication
    CRITICALITY             ignore
}

sRSInformationReservationNotification F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SRSInformationReservationNotification
    PROCEDURE CODE          id-SRSInformationReservationNotification
    CRITICALITY             reject
}

cUDUMobilityInitiation      F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      CUDUMobilityInitiationRequest
    PROCEDURE CODE          id-CUDUMobilityInitiationRequest
    CRITICALITY             ignore
}

cLI-Indication  F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      CLI-Indication
    PROCEDURE CODE          id-CLI-Indication
    CRITICALITY             ignore
}


dUCUCSIRSCoordination F1AP-ELEMENTARY-PROCEDURE ::= {
```

```
    INITIATING MESSAGE      DUCUCSIRSCoordinationRequest
    SUCCESSFUL OUTCOME      DUCUCSIRSCoordinationResponse
    PROCEDURE CODE          id-DUCUCSIRSCoordination
    CRITICALITY             reject
}


cUDUCSIRSCoordination F1AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      CUDUCSIRSCoordinationRequest
    SUCCESSFUL OUTCOME      CUDUCSIRSCoordinationResponse
    PROCEDURE CODE          id-CUDUCSIRSCoordination
    CRITICALITY             reject
}


END
-- ASN1STOP
```

9.4.4 PDU Definitions

```
-- ASN1START
-- **************************************************************
--
-- PDU definitions for F1AP.
--
-- **************************************************************

F1AP-PDU-Contents {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
ngran-access (22) modules (3) f1ap (3) version1 (1) f1ap-PDU-Contents (1) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- **************************************************************
--
-- IE parameter types from other modules.
--
-- **************************************************************

IMPORTS
    AssociatedSessionID,
    BroadcastMRBs-FailedToBeModified-Item,
    BroadcastMRBs-FailedToBeSetup-Item,
    BroadcastMRBs-FailedToBeSetupMod-Item,
    BroadcastMRBs-Modified-Item,
    BroadcastMRBs-Setup-Item,
    BroadcastMRBs-SetupMod-Item,
    BroadcastMRBs-ToBeModified-Item,
    BroadcastMRBs-ToBeReleased-Item,
    BroadcastMRBs-ToBeSetup-Item,
    BroadcastMRBs-ToBeSetupMod-Item,
```

```
    Candidate-SpCell-Item,
    Cause,
    Cells-Allowed-to-be-Deactivated-List-Item,
    Cells-Failed-to-be-Activated-List-Item,
    Cells-Status-Item,
    Cells-to-be-Activated-List-Item,
    Cells-to-be-Deactivated-List-Item,
    CellULConfigured,
    CriticalityDiagnostics,
    C-RNTI,
    CUtoDURRCInformation,
    DRB-Activity-Item,
    DRBs-FailedToBeModified-Item,
    DRBs-FailedToBeSetup-Item,
    DRBs-FailedToBeSetupMod-Item,
    DRB-Notify-Item,
    DRBs-ModifiedConf-Item,
    DRBs-Modified-Item,
    DRBs-Required-ToBeModified-Item,
    DRBs-Required-ToBeReleased-Item,
    DRBs-Setup-Item,
    DRBs-SetupMod-Item,
    DRBs-ToBeModified-Item,
    DRBs-ToBeReleased-Item,
    DRBs-ToBeSetup-Item,
    DRBs-ToBeSetupMod-Item,
    DRXCycle,
    DRXConfigurationIndicator,
    DUtoCURRCInformation,
    ExecuteDuplication,
    FullConfiguration,
    GNB-CU-MBS-F1AP-ID,
    GNB-CU-UE-F1AP-ID,
    GNB-DU-MBS-F1AP-ID,
    GNB-DU-UE-F1AP-ID,
    GNB-DU-ID,
    GNB-DU-Served-Cells-Item,
    GNB-CU-Name,
    GNB-DU-Name,
    InactivityMonitoringRequest,
    InactivityMonitoringResponse,
    LowerLayerPresenceStatusChange,
    MBS-CUtoDURRCInformation,
    MBSMulticastF1UContextDescriptor,
    MBS-Session-ID,
    MBS-ServiceArea,
    MulticastF1UContextReferenceCU,
    MulticastF1UContext-ToBeSetup-Item,
    MulticastF1UContext-Setup-Item,
    MulticastF1UContext-FailedToBeSetup-Item,
    MulticastMBSSessionList,
    MulticastMRBs-ToBeSetup-Item,
    MulticastMRBs-Setup-Item,
    MulticastMRBs-FailedToBeSetup-Item,
```

```
    MulticastMRBs-ToBeSetupMod-Item,
    MulticastMRBs-ToBeModified-Item,
    MulticastMRBs-ToBeReleased-Item,
    MulticastMRBs-SetupMod-Item,
    MulticastMRBs-FailedToBeSetupMod-Item,
    MulticastMRBs-Modified-Item,
    MulticastMRBs-FailedToBeModified-Item,
    BroadcastAreaScope,
    NetworkControlledRepeaterAuthorized,
    NodeAssociatedInfoResult,
    NRCGI,
    UEContextNotRetrievable,
    Potential-SpCell-Item,
    RANSharingAssistanceInformation,
    RAT-FrequencyPriorityInformation,
    RequestedSRSTransmissionCharacteristics,
    ResourceCoordinationTransferContainer,
    RRCContainer,
    RRCContainer-RRCSetupComplete,
    RRCReconfigurationCompleteIndicator,
    SCell-ToBeRemoved-Item,
    SCell-ToBeSetup-Item,
    SCell-ToBeSetupMod-Item,
    SCell-FailedtoSetup-Item,
    SCell-FailedtoSetupMod-Item,
    SDT-Volume-Threshold,
    ServCellIndex,
    Served-Cells-To-Add-Item,
    Served-Cells-To-Delete-Item,
    Served-Cells-To-Modify-Item,
    ServingCellMO,
    SNSSAI,
    SRBID,
    SRBs-FailedToBeSetup-Item,
    SRBs-FailedToBeSetupMod-Item,
    SRBs-Required-ToBeReleased-Item,
    SRBs-ToBeReleased-Item,
    SRBs-ToBeSetup-Item,
    SRBs-ToBeSetupMod-Item,
    SRBs-Modified-Item,
    SRBs-Setup-Item,
    SRBs-SetupMod-Item,
    SupportedUETypeList,
    TimeToWait,
    TransactionID,
    TransmissionActionIndicator,
    UE-associatedLogicalF1-ConnectionItem,
    UEIdentity-List-For-Paging-Item,
    DUtoCURRCContainer,
    PagingCell-Item,
    SItype-List,
    UEIdentityIndexValue,
    GNB-CU-TNL-Association-Setup-Item,
    GNB-CU-TNL-Association-Failed-To-Setup-Item,
```

```
	GNB-CU-TNL-Association-To-Add-Item,
	GNB-CU-TNL-Association-To-Remove-Item,
	GNB-CU-TNL-Association-To-Update-Item,
	MaskedIMEISV,
	PagingDRX,
	PagingPriority,
	PagingIdentity,
	Cells-to-be-Barred-Item,
	PWSSystemInformation,
	Broadcast-To-Be-Cancelled-Item,
	Cells-Broadcast-Cancelled-Item,
	NR-CGI-List-For-Restart-Item,
	PWS-Failed-NR-CGI-Item,
	RepetitionPeriod,
	NumberofBroadcastRequest,
	Cells-To-Be-Broadcast-Item,
	Cells-Broadcast-Completed-Item,
	Cancel-all-Warning-Messages-Indicator,
	EUTRA-NR-CellResourceCoordinationReq-Container,
	EUTRA-NR-CellResourceCoordinationReqAck-Container,
	RequestType,
	PLMN-Identity,
	RLCFailureIndication,
	UplinkTxDirectCurrentListInformation,
	SULAccessIndication,
	Protected-EUTRA-Resources-Item,
	GNB-DUConfigurationQuery,
	BitRate,
	RRC-Version,
	GNBDUOverloadInformation,
	RRCDeliveryStatusRequest,
	NeedforGap,
	RRCDeliveryStatus,
	ResourceCoordinationTransferInformation,
	Dedicated-SIDelivery-NeededUE-Item,
	Associated-SCell-Item,
	IgnoreResourceCoordinationContainer,
	PagingOrigin,
	UAC-Assistance-Info,
	RANUEID,
	GNB-DU-TNL-Association-To-Remove-Item,
	NotificationInformation,
	TraceActivation,
	TraceID,
	Neighbour-Cell-Information-Item,
	AdditionalRRMPriorityIndex,
	DUCURadioInformationType,
	CUDURadioInformationType,
	Transport-Layer-Address-Info,
	BHChannels-ToBeSetup-Item,
	BHChannels-Setup-Item,
	BHChannels-FailedToBeSetup-Item,
	BHChannels-ToBeModified-Item,
	BHChannels-ToBeReleased-Item,
```

```
	BHChannels-ToBeSetupMod-Item,
	BHChannels-FailedToBeModified-Item,
	BHChannels-FailedToBeSetupMod-Item,
	BHChannels-Modified-Item,
	BHChannels-SetupMod-Item,
	BHChannels-Required-ToBeReleased-Item,
	BAPAddress,
	BH-Routing-Information-Added-List-Item,
	BH-Routing-Information-Removed-List-Item,
	Child-Nodes-List,
	Activated-Cells-to-be-Updated-List,
	UL-BH-Non-UP-Traffic-Mapping,
	IABIPv6RequestType,
	IAB-TNL-Addresses-To-Remove-Item,
	IABTNLAddress,
	IAB-Allocated-TNL-Address-Item,
	IABv4AddressesRequested,
	TrafficMappingInfo,
	UL-UP-TNL-Information-to-Update-List-Item,
	UL-UP-TNL-Address-to-Update-List-Item,
	DL-UP-TNL-Address-to-Update-List-Item,
	NRV2XServicesAuthorized,
	LTEV2XServicesAuthorized,
	NRUESidelinkAggregateMaximumBitrate,
	LTEUESidelinkAggregateMaximumBitrate,
	SLDRBs-SetupMod-Item,
	SLDRBs-ModifiedConf-Item,
	SLDRBs-FailedToBeModified-Item,
	SLDRBs-FailedToBeSetup-Item,
	SLDRBs-FailedToBeSetupMod-Item,
	SLDRBs-Modified-Item,
	SLDRBs-Required-ToBeModified-Item,
	SLDRBs-Required-ToBeReleased-Item,
	SLDRBs-Setup-Item,
	SLDRBs-ToBeModified-Item,
	SLDRBs-ToBeReleased-Item,
	SLDRBs-ToBeSetup-Item,
	SLDRBs-ToBeSetupMod-Item,
	GNBCUMeasurementID,
	GNBDUMeasurementID,
	RegistrationRequest,
	ReportCharacteristics,
	CellToReportList,
	HardwareLoadIndicator,
	CellMeasurementResultList,
	ReportingPeriodicity,
	TNLCapacityIndicator,
	RAReportList,
	RLFReportInformationList,
	ReportingRequestType,
	TimeReferenceInformation,
	ConditionalInterDUMobilityInformation,
	ConditionalIntraDUMobilityInformation,
	TargetCellList,
```

```
	MDTPLMNList,
	PrivacyIndicator,
	TransportLayerAddress,
	URI-address,
	NID,
	PosAssistance-Information,
	PosBroadcast,
	PositioningBroadcastCells,
	RoutingID,
	PosAssistanceInformationFailureList,
	PosMeasurementQuantities,
	PosMeasurementResultList,
	PosReportCharacteristics,
	TRPInformationTypeItem,
	TRPInformationItem,
	LMF-MeasurementID,
	RAN-MeasurementID,
	SDT-Termination-Request,
	SRSResourceSetID,
	SpatialRelationInfo,
	SRSResourceTrigger,
	SRSConfiguration,
	TRPList,
	E-CID-MeasurementQuantities,
	MeasurementPeriodicity,
	E-CID-MeasurementResult,
	Cell-Portion-ID,
	LMF-UE-MeasurementID,
	RAN-UE-MeasurementID,
	RelativeTime1900,
	SystemFrameNumber,
	SlotNumber,
	AbortTransmission,
	TRP-MeasurementRequestList,
	MeasurementBeamInfoRequest,
	E-CID-ReportCharacteristics,
	Extended-GNB-CU-Name,
	Extended-GNB-DU-Name,
	F1CTransferPath,
	SCGIndicator,
	SpatialRelationPerSRSResource,
	MeasurementPeriodicityExtended,
	SuccessfulHOReportInformationList,
	Coverage-Modification-Notification,
	CCO-Assistance-Information,
	CellsForSON-List,
	IABCongestionIndication,
	IABConditionalRRCMessageDeliveryIndication,
	F1CTransferPathNRDC,
	BufferSizeThresh,
	IAB-TNL-Addresses-Exception,
	BAP-Header-Rewriting-Added-List-Item,
	Re-routingEnableIndicator,
	Neighbour-Node-Cells-List,
```

```
	Serving-Cells-List,
	RBSetConfiguration,
	PDCMeasurementPeriodicity,
	PDCMeasurementQuantities,
	PDCMeasurementResult,
	PDCReportType,
	RAN-UE-PDC-MeasID,
	SCGActivationRequest,
	SCGActivationStatus,
	TRP-MeasurementUpdateList,
	PRSTRPList,
	PRSTransmissionTRPList,
	ResponseTime,
	TRP-PRS-Info-List,
	PRS-Measurement-Info-List,
	PRSConfigRequestType,
	MeasurementCharacteristicsRequestIndicator,
	MeasurementTimeOccasion,
	UEReportingInformation,
	PosContextRevIndication,
	NRRedCapUEIndication,
	NRPagingeDRXInformation,
	NRPagingeDRXInformationforRRCINACTIVE,
	QoEInformation,
	CG-SDTQueryIndication,
	CG-SDTKeptIndicator,
	CG-SDTSessionInfo,
	SDTInformation,
	FiveG-ProSeAuthorized,
	UuRLCChannelToBeSetupList,
	UuRLCChannelToBeModifiedList,
	UuRLCChannelToBeReleasedList,
	UuRLCChannelSetupList,
	UuRLCChannelFailedToBeSetupList,
	UuRLCChannelModifiedList,
	UuRLCChannelFailedToBeModifiedList,
	UuRLCChannelRequiredToBeModifiedList,
	UuRLCChannelRequiredToBeReleasedList,
	PC5RLCChannelToBeSetupList,
	PC5RLCChannelToBeModifiedList,
	PC5RLCChannelToBeReleasedList,
	PC5RLCChannelSetupList,
	PC5RLCChannelFailedToBeSetupList,
	PC5RLCChannelFailedToBeModifiedList,
	PC5RLCChannelRequiredToBeModifiedList,
	PC5RLCChannelRequiredToBeReleasedList,
	PC5RLCChannelModifiedList,
	RemoteUELocalID,
	PathSwitchConfiguration,
	SidelinkRelayConfiguration,
	PagingCause,
	PEIPSAssistanceInfo,
	UEPagingCapability,
	GNBDUUESliceMaximumBitRateList,
```

```
    MDTPollutedMeasurementIndicator,
    UE-MulticastMRBs-ConfirmedToBeModified-Item,
    UE-MulticastMRBs-RequiredToBeModified-Item,
    UE-MulticastMRBs-RequiredToBeReleased-Item,
    UE-MulticastMRBs-Setup-Item,
    UE-MulticastMRBs-Setupnew-Item,
    UE-MulticastMRBs-ToBeReleased-Item,
    UE-MulticastMRBs-ToBeSetup-Item,
    UE-MulticastMRBs-ToBeSetup-atModify-Item,
    PosMeasurementAmount,
    BAP-Header-Rewriting-Removed-List-Item,
    SLDRXCycleList,
    MDTPLMNModificationList,
    ActivationRequestType,
    PosMeasGapPreConfigList,
    PosMeasurementPeriodicityNR-AoA,
    SRSPosRRCInactiveConfig,
    SDTBearerConfigurationQueryIndication,
    SDTBearerConfigurationInfo,
    ServingCellMO-List-Item,
    ServingCellMO-encoded-in-CGC-List,
    PosSItypeList,
    DAPS-HO-Status,
    UuRLCChannelID,
    UplinkTxDirectCurrentTwoCarrierListInfo,
    SRSPosRRCInactiveQueryIndication,
    MC-PagingCell-Item,
    UlTxDirectCurrentMoreCarrierInformation,
    CPACMCGInformation,
    ExtendedUEIdentityIndexValue,
    HashedUEIdentityIndexValue,
    DedicatedSIDeliveryIndication,
    Configured-BWP-List,
    MT-SDT-Information,
    LTMInformation-Setup,
    LTMConfigurationIDMappingList,
    LTMInformation-Modify,
    LTMCells-ToBeReleased-List,
    LTMCFRAResourceConfig-List,
    LTMConfiguration,
    EarlySyncInformation-Request,
    EarlySyncInformation,
    EarlySyncCandidateCellInformation-List,
    EarlySyncServingCellInformation,
    LTMCellSwitchInformation,
    DUtoCUTAInformation-List,
    CUtoDUTAInformation-List,
    DeactivationIndication,
    RAReportIndicationList,
    SuccessfulPSCellChangeReportInformationList,
    PathAdditionInformation,
    RANTSSRequestType,
    RANTimingSynchronisationStatusInfo,
    GlobalGNB-ID,
```

```
    Activated-Cells-Mapping-List-Item,
    RRC-Terminating-IAB-Donor-Related-Info,
    NCGI-to-be-Updated-List-Item,
    Mobile-IAB-MTUserLocationInformation,
    TAI,
    IndicationMCInactiveReception,
    MulticastCU2DURRCInfo,
    MulticastDU2CURRCInfo,
    MBSMulticastSessionReceptionState,
    MulticastCU2DUCommonRRCInfo,
    NRA2XServicesAuthorized,
    LTEA2XServicesAuthorized,
    NReRedCapUEIndication,
    NRPaginglongeDRXInformationforRRCINACTIVE,
    Cells-With-SSBs-Activated-List,
    Recommended-SSBs-for-Paging-List,
    S-CPAC-Configuration,
    DLLBTFailureInformationRequest,
    DLLBTFailureInformationList,
    SLPositioning-Ranging-Service-Info,
    TimeWindowInformation-SRS-List,
    TimeWindowInformation-Measurement-List,
    SRSPosRRCInactiveValidityAreaConfig,
    SRSReservationType,
    RequestedSRSPreconfigurationCharacteristics-List,
    SRSPreconfiguration-List,
    Broadcast-MRBs-Transport-Request-Item,
    TAInformation-List,
    NonIntegerDRXCycle,
    AggregatedPosSRSResourceSetList,
    F1U-PathFailure,
    LTMResetInformation,
    MobilityInitiation,
    PLMNIndexNR,
    LTMTCIStatesConfigurationsList,
    LPWUSPSAssistanceInfo,
    FurtherExtendedUEIdentityIndexValue,
    CLI-MeasurementResult-List,
    SRS-Resource-Indication,
    FailureReportingWithoutRLFReport,
    MROForLTM-Information,
    LastVisitedLTMCells,
    OnDemand-SIB1-Cell,
    LTMSecurityInformation,
    LTMInformationSCGAdd,
    LTMInformationSCGMod,
    TARemainingInfoList,
    CSI-RSCoordinationRequestList,
    CSI-RSCoordinationResultList,
    Future-Coverage-Modification-Notification,
    Predicted-CCO-Assistance-Information,
    Neighbour-Future-Coverage-Modification-Notification,
    AreaSpecificSemiPersistentSRSPosInfo,
    LTMConfig-to-Modify-List,
```

```
    InterSNLTMMCGInformation



FROM F1AP-IEs

    PrivateIE-Container{},
    ProtocolExtensionContainer{},
    ProtocolIE-Container{},
    ProtocolIE-ContainerPair{},
    ProtocolIE-SingleContainer{},
    F1AP-PRIVATE-IES,
    F1AP-PROTOCOL-EXTENSION,
    F1AP-PROTOCOL-IES,
    F1AP-PROTOCOL-IES-PAIR

FROM F1AP-Containers

    id-AssociatedSessionID,
    id-BroadcastMRBs-FailedToBeModified-List,
    id-BroadcastMRBs-FailedToBeModified-Item,
    id-BroadcastMRBs-FailedToBeSetup-List,
    id-BroadcastMRBs-FailedToBeSetup-Item,
    id-BroadcastMRBs-FailedToBeSetupMod-List,
    id-BroadcastMRBs-FailedToBeSetupMod-Item,
    id-BroadcastMRBs-Modified-List,
    id-BroadcastMRBs-Modified-Item,
    id-BroadcastMRBs-Setup-List,
    id-BroadcastMRBs-Setup-Item,
    id-BroadcastMRBs-SetupMod-List,
    id-BroadcastMRBs-SetupMod-Item,
    id-BroadcastMRBs-ToBeModified-List,
    id-BroadcastMRBs-ToBeModified-Item,
    id-BroadcastMRBs-ToBeReleased-List,
    id-BroadcastMRBs-ToBeReleased-Item,
    id-BroadcastMRBs-ToBeSetup-List,
    id-BroadcastMRBs-ToBeSetup-Item,
    id-BroadcastMRBs-ToBeSetupMod-List,
    id-BroadcastMRBs-ToBeSetupMod-Item,
    id-Candidate-SpCell-Item,
    id-Candidate-SpCell-List,
    id-Cause,
    id-Cancel-all-Warning-Messages-Indicator,
    id-Cells-Failed-to-be-Activated-List,
    id-Cells-Failed-to-be-Activated-List-Item,
    id-Cells-Status-Item,
    id-Cells-Status-List,
    id-Cells-to-be-Activated-List,
    id-Cells-to-be-Activated-List-Item,
    id-Cells-to-be-Deactivated-List,
    id-Cells-to-be-Deactivated-List-Item,
    id-Cells-Allowed-to-be-Deactivated-List,
    id-Cells-Allowed-to-be-Deactivated-List-Item,
```

```
	id-Cells-With-SSBs-Activated-List,
	id-Recommended-SSBs-for-Paging-List,
	id-ConfirmedUEID,
	id-CriticalityDiagnostics,
	id-C-RNTI,
	id-CUtoDURRCInformation,
	id-DRB-Activity-Item,
	id-DRB-Activity-List,
	id-DRBs-FailedToBeModified-Item,
	id-DRBs-FailedToBeModified-List,
	id-DRBs-FailedToBeSetup-Item,
	id-DRBs-FailedToBeSetup-List,
	id-DRBs-FailedToBeSetupMod-Item,
	id-DRBs-FailedToBeSetupMod-List,
	id-DRBs-ModifiedConf-Item,
	id-DRBs-ModifiedConf-List,
	id-DRBs-Modified-Item,
	id-DRBs-Modified-List,
	id-DRB-Notify-Item,
	id-DRB-Notify-List,
	id-DRBs-Required-ToBeModified-Item,
	id-DRBs-Required-ToBeModified-List,
	id-DRBs-Required-ToBeReleased-Item,
	id-DRBs-Required-ToBeReleased-List,
	id-DRBs-Setup-Item,
	id-DRBs-Setup-List,
	id-DRBs-SetupMod-Item,
	id-DRBs-SetupMod-List,
	id-DRBs-ToBeModified-Item,
	id-DRBs-ToBeModified-List,
	id-DRBs-ToBeReleased-Item,
	id-DRBs-ToBeReleased-List,
	id-DRBs-ToBeSetup-Item,
	id-DRBs-ToBeSetup-List,
	id-DRBs-ToBeSetupMod-Item,
	id-DRBs-ToBeSetupMod-List,
	id-DRXCycle,
	id-DUtoCURRCInformation,
	id-ExecuteDuplication,
	id-FullConfiguration,
	id-gNB-CU-MBS-F1AP-ID,
	id-gNB-CU-UE-F1AP-ID,
	id-gNB-DU-MBS-F1AP-ID,
	id-gNB-DU-UE-F1AP-ID,
	id-gNB-DU-ID,
	id-GNB-DU-Served-Cells-Item,
	id-gNB-DU-Served-Cells-List,
	id-gNB-CU-Name,
	id-gNB-DU-Name,
	id-Extended-GNB-CU-Name,
	id-Extended-GNB-DU-Name,
	id-InactivityMonitoringRequest,
	id-InactivityMonitoringResponse,
	id-MBS-CUtoDURRCInformation,
```

```
	id-MBS-Session-ID,
	id-MBS-ServiceArea,
	id-MBSMulticastF1UContextDescriptor,
	id-MC-PagingCell-Item,
	id-MC-PagingCell-List,
	id-MulticastF1UContextReferenceCU,
	id-MulticastMBSSessionSetupList,
	id-MulticastMBSSessionRemoveList,
	id-MulticastMRBs-FailedToBeModified-List,
	id-MulticastMRBs-FailedToBeModified-Item,
	id-MulticastMRBs-FailedToBeSetup-List,
	id-MulticastMRBs-FailedToBeSetup-Item,
	id-MulticastMRBs-FailedToBeSetupMod-List,
	id-MulticastMRBs-FailedToBeSetupMod-Item,
	id-MulticastMRBs-Modified-List,
	id-MulticastMRBs-Modified-Item,
	id-MulticastMRBs-Setup-List,
	id-MulticastMRBs-Setup-Item,
	id-MulticastMRBs-SetupMod-List,
	id-MulticastMRBs-SetupMod-Item,
	id-MulticastMRBs-ToBeModified-List,
	id-MulticastMRBs-ToBeModified-Item,
	id-MulticastMRBs-ToBeReleased-List,
	id-MulticastMRBs-ToBeReleased-Item,
	id-MulticastMRBs-ToBeSetup-List,
	id-MulticastMRBs-ToBeSetup-Item,
	id-MulticastMRBs-ToBeSetupMod-List,
	id-MulticastMRBs-ToBeSetupMod-Item,
	id-MulticastF1UContext-ToBeSetup-List,
	id-MulticastF1UContext-ToBeSetup-Item,
	id-MulticastF1UContext-Setup-List,
	id-MulticastF1UContext-Setup-Item,
	id-MulticastF1UContext-FailedToBeSetup-List,
	id-MulticastF1UContext-FailedToBeSetup-Item,
	id-BroadcastAreaScope,
	id-new-gNB-CU-UE-F1AP-ID,
	id-new-gNB-DU-UE-F1AP-ID,
	id-NodeAssociatedInfoResult,
	id-oldgNB-DU-UE-F1AP-ID,
	id-PLMNAssistanceInfoForNetShar,
	id-Potential-SpCell-Item,
	id-Potential-SpCell-List,
	id-RAT-FrequencyPriorityInformation,
	id-RedirectedRRCmessage,
	id-ResetType,
	id-RequestedSRSTransmissionCharacteristics,
	id-ResourceCoordinationTransferContainer,
	id-RRCContainer,
	id-RRCContainer-RRCSetupComplete,
	id-RRCReconfigurationCompleteIndicator,
	id-SCell-FailedtoSetup-List,
	id-SCell-FailedtoSetup-Item,
	id-SCell-FailedtoSetupMod-List,
	id-SCell-FailedtoSetupMod-Item,
```

```
	id-SCell-ToBeRemoved-Item,
	id-SCell-ToBeRemoved-List,
	id-SCell-ToBeSetup-Item,
	id-SCell-ToBeSetup-List,
	id-SCell-ToBeSetupMod-Item,
	id-SCell-ToBeSetupMod-List,
	id-SDT-Termination-Request,
	id-SDT-Volume-Threshold,
	id-SelectedPLMNID,
	id-Served-Cells-To-Add-Item,
	id-Served-Cells-To-Add-List,
	id-Served-Cells-To-Delete-Item,
	id-Served-Cells-To-Delete-List,
	id-Served-Cells-To-Modify-Item,
	id-Served-Cells-To-Modify-List,
	id-ServCellIndex,
	id-ServingCellMO,
	id-SNSSAI,
	id-SpCell-ID,
	id-SpCellULConfigured,
	id-SRBID,
	id-SRBs-FailedToBeSetup-Item,
	id-SRBs-FailedToBeSetup-List,
	id-SRBs-FailedToBeSetupMod-Item,
	id-SRBs-FailedToBeSetupMod-List,
	id-SRBs-Required-ToBeReleased-Item,
	id-SRBs-Required-ToBeReleased-List,
	id-SRBs-ToBeReleased-Item,
	id-SRBs-ToBeReleased-List,
	id-SRBs-ToBeSetup-Item,
	id-SRBs-ToBeSetup-List,
	id-SRBs-ToBeSetupMod-Item,
	id-SRBs-ToBeSetupMod-List,
	id-SRBs-Modified-Item,
	id-SRBs-Modified-List,
	id-SRBs-Setup-Item,
	id-SRBs-Setup-List,
	id-SRBs-SetupMod-Item,
	id-SRBs-SetupMod-List,
	id-SupportedUETypeList,
	id-TimeToWait,
	id-TransactionID,
	id-TransmissionActionIndicator,
	id-UEContextNotRetrievable,
	id-UE-associatedLogicalF1-ConnectionItem,
	id-UE-associatedLogicalF1-ConnectionListResAck,
	id-UEIdentity-List-For-Paging-List,
	id-UEIdentity-List-For-Paging-Item,
	id-UE-MulticastMRBs-ConfirmedToBeModified-List,
	id-UE-MulticastMRBs-ConfirmedToBeModified-Item,
	id-UE-MulticastMRBs-RequiredToBeModified-List,
	id-UE-MulticastMRBs-RequiredToBeModified-Item,
	id-UE-MulticastMRBs-RequiredToBeReleased-List,
	id-UE-MulticastMRBs-RequiredToBeReleased-Item,
```

```
	id-UE-MulticastMRBs-Setup-List,
	id-UE-MulticastMRBs-Setup-Item,
	id-UE-MulticastMRBs-Setupnew-List,
	id-UE-MulticastMRBs-Setupnew-Item,
	id-UE-MulticastMRBs-ToBeReleased-List,
	id-UE-MulticastMRBs-ToBeReleased-Item,
	id-UE-MulticastMRBs-ToBeSetup-atModify-List,
	id-UE-MulticastMRBs-ToBeSetup-atModify-Item,
	id-UE-MulticastMRBs-ToBeSetup-List,
	id-UE-MulticastMRBs-ToBeSetup-Item,
	id-DUtoCURRCContainer,
	id-NRCGI,
	id-PagingCell-Item,
	id-PagingCell-List,
	id-PagingDRX,
	id-PagingPriority,
	id-SItype-List,
	id-UEIdentityIndexValue,
	id-GNB-CU-TNL-Association-Setup-List,
	id-GNB-CU-TNL-Association-Setup-Item,
	id-GNB-CU-TNL-Association-Failed-To-Setup-List,
	id-GNB-CU-TNL-Association-Failed-To-Setup-Item,
	id-GNB-CU-TNL-Association-To-Add-Item,
	id-GNB-CU-TNL-Association-To-Add-List,
	id-GNB-CU-TNL-Association-To-Remove-Item,
	id-GNB-CU-TNL-Association-To-Remove-List,
	id-GNB-CU-TNL-Association-To-Update-Item,
	id-GNB-CU-TNL-Association-To-Update-List,
	id-MaskedIMEISV,
	id-PagingIdentity,
	id-Cells-to-be-Barred-List,
	id-Cells-to-be-Barred-Item,
	id-PWSSystemInformation,
	id-RepetitionPeriod,
	id-NumberofBroadcastRequest,
	id-Cells-To-Be-Broadcast-List,
	id-Cells-To-Be-Broadcast-Item,
	id-Cells-Broadcast-Completed-List,
	id-Cells-Broadcast-Completed-Item,
	id-Broadcast-To-Be-Cancelled-List,
	id-Broadcast-To-Be-Cancelled-Item,
	id-Cells-Broadcast-Cancelled-List,
	id-Cells-Broadcast-Cancelled-Item,
	id-NR-CGI-List-For-Restart-List,
	id-NR-CGI-List-For-Restart-Item,
	id-PWS-Failed-NR-CGI-List,
	id-PWS-Failed-NR-CGI-Item,
	id-EUTRA-NR-CellResourceCoordinationReq-Container,
	id-EUTRA-NR-CellResourceCoordinationReqAck-Container,
	id-Protected-EUTRA-Resources-List,
	id-RequestType,
	id-ServingPLMN,
	id-DRXConfigurationIndicator,
	id-RLCFailureIndication,
```

```
	id-UplinkTxDirectCurrentListInformation,
	id-SULAccessIndication,
	id-Protected-EUTRA-Resources-Item,
	id-GNB-DUConfigurationQuery,
	id-GNB-DU-UE-AMBR-UL,
	id-GNB-CU-RRC-Version,
	id-GNB-DU-RRC-Version,
	id-GNBDUOverloadInformation,
	id-NeedforGap,
	id-RRCDeliveryStatusRequest,
	id-RRCDeliveryStatus,
	id-Dedicated-SIDelivery-NeededUE-List,
	id-Dedicated-SIDelivery-NeededUE-Item,
	id-ResourceCoordinationTransferInformation,
	id-Associated-SCell-List,
	id-Associated-SCell-Item,
	id-IgnoreResourceCoordinationContainer,
	id-UAC-Assistance-Info,
	id-RANUEID,
	id-PagingOrigin,
	id-GNB-DU-TNL-Association-To-Remove-Item,
	id-GNB-DU-TNL-Association-To-Remove-List,
	id-NotificationInformation,
	id-TraceActivation,
	id-TraceID,
	id-Neighbour-Cell-Information-List,
	id-Neighbour-Cell-Information-Item,
	id-AdditionalRRMPriorityIndex,
	id-DUCURadioInformationType,
	id-CUDURadioInformationType,
	id-LowerLayerPresenceStatusChange,
	id-Transport-Layer-Address-Info,
	id-BHChannels-ToBeSetup-List,
	id-BHChannels-ToBeSetup-Item,
	id-BHChannels-Setup-List,
	id-BHChannels-Setup-Item,
	id-BHChannels-ToBeModified-Item,
	id-BHChannels-ToBeModified-List,
	id-BHChannels-ToBeReleased-Item,
	id-BHChannels-ToBeReleased-List,
	id-BHChannels-ToBeSetupMod-Item,
	id-BHChannels-ToBeSetupMod-List,
	id-BHChannels-FailedToBeSetup-Item,
	id-BHChannels-FailedToBeSetup-List,
	id-BHChannels-FailedToBeModified-Item,
	id-BHChannels-FailedToBeModified-List,
	id-BHChannels-FailedToBeSetupMod-Item,
	id-BHChannels-FailedToBeSetupMod-List,
	id-BHChannels-Modified-Item,
	id-BHChannels-Modified-List,
	id-BHChannels-SetupMod-Item,
	id-BHChannels-SetupMod-List,
	id-BHChannels-Required-ToBeReleased-Item,
	id-BHChannels-Required-ToBeReleased-List,
```

```
	id-BAPAddress,
	id-ConfiguredBAPAddress,
	id-BH-Routing-Information-Added-List,
	id-BH-Routing-Information-Added-List-Item,
	id-BH-Routing-Information-Removed-List,
	id-BH-Routing-Information-Removed-List-Item,
	id-UL-BH-Non-UP-Traffic-Mapping,
	id-Child-Nodes-List,
	id-Activated-Cells-to-be-Updated-List,
	id-IABIPv6RequestType,
	id-IAB-TNL-Addresses-To-Remove-List,
	id-IAB-TNL-Addresses-To-Remove-Item,
	id-IAB-Allocated-TNL-Address-List,
	id-IAB-Allocated-TNL-Address-Item,
	id-IABv4AddressesRequested,
	id-TrafficMappingInformation,
	id-UL-UP-TNL-Information-to-Update-List,
	id-UL-UP-TNL-Information-to-Update-List-Item,
	id-UL-UP-TNL-Address-to-Update-List,
	id-UL-UP-TNL-Address-to-Update-List-Item,
	id-DL-UP-TNL-Address-to-Update-List,
	id-DL-UP-TNL-Address-to-Update-List-Item,
	id-NRV2XServicesAuthorized,
	id-LTEV2XServicesAuthorized,
	id-NRUESidelinkAggregateMaximumBitrate,
	id-LTEUESidelinkAggregateMaximumBitrate,
	id-PC5LinkAMBR,
	id-SLDRBs-FailedToBeModified-Item,
	id-SLDRBs-FailedToBeModified-List,
	id-SLDRBs-FailedToBeSetup-Item,
	id-SLDRBs-FailedToBeSetup-List,
	id-SLDRBs-Modified-Item,
	id-SLDRBs-Modified-List,
	id-SLDRBs-Required-ToBeModified-Item,
	id-SLDRBs-Required-ToBeModified-List,
	id-SLDRBs-Required-ToBeReleased-Item,
	id-SLDRBs-Required-ToBeReleased-List,
	id-SLDRBs-Setup-Item,
	id-SLDRBs-Setup-List,
	id-SLDRBs-ToBeModified-Item,
	id-SLDRBs-ToBeModified-List,
	id-SLDRBs-ToBeReleased-Item,
	id-SLDRBs-ToBeReleased-List,
	id-SLDRBs-ToBeSetup-Item,
	id-SLDRBs-ToBeSetup-List,
	id-SLDRBs-ToBeSetupMod-Item,
	id-SLDRBs-ToBeSetupMod-List,
	id-SLDRBs-SetupMod-List,
	id-SLDRBs-FailedToBeSetupMod-List,
	id-SLDRBs-SetupMod-Item,
	id-SLDRBs-FailedToBeSetupMod-Item,
	id-SLDRBs-ModifiedConf-List,
	id-SLDRBs-ModifiedConf-Item,
	id-gNBCUMeasurementID,
```

```
    id-gNBDUMeasurementID,
    id-RegistrationRequest,
    id-ReportCharacteristics,
    id-CellToReportList,
    id-CellMeasurementResultList,
    id-HardwareLoadIndicator,
    id-ReportingPeriodicity,
    id-TNLCapacityIndicator,
    id-RAReportList,
    id-RLFReportInformationList,
    id-ReportingRequestType,
    id-TimeReferenceInformation,
    id-ConditionalInterDUMobilityInformation,
    id-ConditionalIntraDUMobilityInformation,
    id-targetCellsToCancel,
    id-requestedTargetCellGlobalID,
    id-TraceCollectionEntityIPAddress,
    id-ManagementBasedMDTPLMNList,
    id-PrivacyIndicator,
    id-TraceCollectionEntityURI,
    id-ServingNID,
    id-PosAssistance-Information,
    id-PosBroadcast,
    id-PositioningBroadcastCells,
    id-RoutingID,
    id-PosAssistanceInformationFailureList,
    id-PosMeasurementQuantities,
    id-PosMeasurementResultList,
    id-PosMeasurementPeriodicity,
    id-PosReportCharacteristics,
    id-TRPInformationTypeListTRPReq,
    id-TRPInformationTypeItem,
    id-TRPInformationListTRPResp,
    id-TRPInformationItem,
    id-LMF-MeasurementID,
    id-RAN-MeasurementID,
    id-SRSType,
    id-ActivationTime,
    id-AbortTransmission,
    id-SRSConfiguration,
    id-TRPList,
    id-E-CID-MeasurementQuantities,
    id-E-CID-MeasurementPeriodicity,
    id-E-CID-MeasurementResult,
    id-Cell-Portion-ID,
    id-LMF-UE-MeasurementID,
    id-RAN-UE-MeasurementID,
    id-SFNInitialisationTime,
    id-SystemFrameNumber,
    id-SlotNumber,
    id-TRP-MeasurementRequestList,
    id-MeasurementBeamInfoRequest,
    id-E-CID-ReportCharacteristics,
    id-F1CTransferPath,
```

```
	id-SCGIndicator,
	id-SRSSpatialRelationPerSRSResource,
	id-PosMeasurementPeriodicityExtended,
	id-SuccessfulHOReportInformationList,
	id-Coverage-Modification-Notification,
	id-CCO-Assistance-Information,
	id-CellsForSON-List,
	id-IABCongestionIndication,
	id-IABConditionalRRCMessageDeliveryIndication,
	id-F1CTransferPathNRDC,
	id-BufferSizeThresh,
	id-IAB-TNL-Addresses-Exception,
	id-BAP-Header-Rewriting-Added-List,
	id-BAP-Header-Rewriting-Added-List-Item,
	id-Re-routingEnableIndicator,
	id-Neighbour-Node-Cells-List,
	id-Serving-Cells-List,
	id-MDTPollutedMeasurementIndicator,
	id-PDCMeasurementPeriodicity,
	id-PDCMeasurementQuantities,
	id-PDCMeasurementResult,
	id-PDCReportType,
	id-RAN-UE-PDC-MeasID,
	id-SCGActivationRequest,
	id-SCGActivationStatus,
	id-TRP-MeasurementUpdateList,
	id-PRSTRPList,
	id-PRSTransmissionTRPList,
	id-ResponseTime,
	id-TRP-PRS-Info-List,
	id-PRS-Measurement-Info-List,
	id-PRSConfigRequestType,
	id-MeasurementCharacteristicsRequestIndicator,
	id-MeasurementTimeOccasion,
	id-UEReportingInformation,
	id-PosContextRevIndication,
	id-NRRedCapUEIndication,
	id-RANUEPagingDRX,
	id-CNUEPagingDRX,
	id-NRPagingeDRXInformation,
	id-NRPagingeDRXInformationforRRCINACTIVE,
	id-QoEInformation,
	id-CG-SDTQueryIndication,
	id-CG-SDTKeptIndicator,
	id-CG-SDTSessionInfoOld,
	id-SDTInformation,
	id-FiveG-ProSeAuthorized,
	id-FiveG-ProSePC5LinkAMBR,
	id-FiveG-ProSeUEPC5AggregateMaximumBitrate,
	id-UuRLCChannelToBeSetupList,
	id-UuRLCChannelToBeModifiedList,
	id-UuRLCChannelToBeReleasedList,
	id-UuRLCChannelSetupList,
	id-UuRLCChannelFailedToBeSetupList,
```

```
	id-UuRLCChannelModifiedList,
	id-UuRLCChannelFailedToBeModifiedList,
	id-UuRLCChannelRequiredToBeModifiedList,
	id-UuRLCChannelRequiredToBeReleasedList,
	id-PC5RLCChannelToBeSetupList,
	id-PC5RLCChannelToBeModifiedList,
	id-PC5RLCChannelToBeReleasedList,
	id-PC5RLCChannelSetupList,
	id-PC5RLCChannelFailedToBeSetupList,
	id-PC5RLCChannelModifiedList,
	id-PC5RLCChannelFailedToBeModifiedList,
	id-PC5RLCChannelRequiredToBeModifiedList,
	id-PC5RLCChannelRequiredToBeReleasedList,
	id-SidelinkRelayConfiguration,
	id-UpdatedRemoteUELocalID,
	id-PathSwitchConfiguration,
	id-PagingCause,
	id-PEIPSAssistanceInfo,
	id-UEPagingCapability,
	id-GNBDUUESliceMaximumBitRateList,
	id-PosMeasurementAmount,
	id-BAP-Header-Rewriting-Removed-List,
	id-BAP-Header-Rewriting-Removed-List-Item,
	id-SLDRXCycleList,
	id-ManagementBasedMDTPLMNModificationList,
	id-ActivationRequestType,
	id-PosMeasGapPreConfigList,
	id-PosMeasurementPeriodicityNR-AoA,
	id-SRSPosRRCInactiveConfig,
	id-SDTBearerConfigurationQueryIndication,
	id-SDTBearerConfigurationInfo,
	id-ServingCellMO-List,
	id-ServingCellMO-List-Item,
	id-ServingCellMO-encoded-in-CGC-List,
	id-PosSItypeList,
	id-DAPS-HO-Status,
	id-SRBMappingInfo,
	id-UplinkTxDirectCurrentTwoCarrierListInfo,
	id-SRSPosRRCInactiveQueryIndication,
	id-UlTxDirectCurrentMoreCarrierInformation,
	id-CPACMCGInformation,
	id-ExtendedUEIdentityIndexValue,
	id-HashedUEIdentityIndexValue,
	id-DedicatedSIDeliveryIndication,
	id-Configured-BWP-List,
	id-NetworkControlledRepeaterAuthorized,
	id-MT-SDT-Information,
	id-LTMInformation-Setup,
	id-LTMConfigurationIDMappingList,
	id-LTMInformation-Modify,
	id-LTMCells-ToBeReleased-List,
	id-LTMConfiguration,
	id-LTMCFRAResourceConfig-List,
	id-EarlySyncInformation-Request,
```

```
	id-EarlySyncInformation,
	id-EarlySyncCandidateCellInformation-List,
	id-EarlySyncServingCellInformation,
	id-LTMCellSwitchInformation,
	id-DUtoCUTAInformation-List,
	id-CUtoDUTAInformation-List,
	id-DeactivationIndication,
	id-RAReportIndicationList,
	id-SuccessfulPSCellChangeReportInformationList,
	id-PathAdditionInformation,
	id-RANTSSRequestType,
	id-RANTimingSynchronisationStatusInfo,
	id-Target-gNB-ID,
	id-Target-gNB-IP-address,
	id-Target-SeGW-IP-address,
	id-Activated-Cells-Mapping-List,
	id-Activated-Cells-Mapping-List-Item,
	id-F1SetupOutcome,
	id-RRC-Terminating-IAB-Donor-Related-Info,
	id-RRC-Terminating-IAB-Donor-gNB-ID,
	id-NCGI-to-be-Updated-List,
	id-NCGI-to-be-Updated-List-Item,
	id-Mobile-IAB-MTUserLocationInformation,
	id-IndicationMCInactiveReception,
	id-MulticastCU2DURRCInfo,
	id-MulticastDU2CURRCInfo,
	id-MBSMulticastSessionReceptionState,
	id-MulticastCU2DUCommonRRCInfo,
	id-NRA2XServicesAuthorized,
	id-LTEA2XServicesAuthorized,
	id-NRUESidelinkAggregateMaximumBitrateForA2X,
	id-LTEUESidelinkAggregateMaximumBitrateForA2X,
	id-NReRedCapUEIndication,
	id-NRPaginglongeDRXInformationforRRCINACTIVE,
	id-Target-F1-Terminating-Donor-gNB-ID,
	id-Broadcast-MRBs-Transport-Request-List,
	id-Broadcast-MRBs-Transport-Request-Item,
	id-S-CPAC-Configuration,
	id-DLLBTFailureInformationRequest,
	id-DLLBTFailureInformationList,
	id-SLPositioning-Ranging-Service-Info,
	id-TimeWindowInformation-SRS-List,
	id-TimeWindowInformation-Measurement-List,
	id-SRSPosRRCInactiveValidityAreaConfig,
	id-SRSReservationType,
	id-RequestedSRSPreconfigurationCharacteristics-List,
	id-SRSPreconfiguration-List,
	id-SRSInformation,
	id-TAInformation-List,
	id-NonIntegerDRXCycle,
	id-AggregatedPosSRSResourceSetList,
	id-RANSharingAssistanceInformation,
	id-F1U-PathFailure,
	id-LTMResetInformation,
```

```
    id-PreconfiguredSRSInformation,
    id-MobilityInitiation,
    id-PLMNIndexNRAssistanceInfoForNetShar,
    id-LTMTCIStatesConfigurationsList,
    id-LPWUSPSAssistanceInfo,
    id-FurtherExtendedUEIdentityIndexValue,
    id-CLI-MeasurementResult-List,
    id-SRS-Resource-Indication,
    id-FailureReportingWithoutRLFReport,
    id-MROForLTM-Information,
    id-LastVisitedLTMCells,
    id-OnDemand-SIB1-Cell,
    id-LTMSecurityInformation,
    id-LTMInformationSCGAdd,
    id-LTMInformationSCGMod,
    id-TARemainingInfoList,
    id-CSI-RSCoordinationRequestList,
    id-CSI-RSCoordinationResultList,
    id-Future-Coverage-Modification-Notification,
    id-Predicted-CCO-Assistance-Information,
    id-NeighbourFutureCoverageModNotification,
    id-AreaSpecificSemiPersistentSRSPosInfo,
    id-LTMConfig-to-Modify-List,
    id-InterSNLTMMCGInformation,
    maxCellingNBDU,
    maxnoofCandidateSpCells,
    maxnoofDRBs,
    maxnoofIndividualF1ConnectionsToReset,
    maxnoofPotentialSpCells,
    maxnoofSCells,
    maxnoofSRBs,
    maxnoofPagingCells,
    maxnoofTNLAssociations,
    maxCellineNB,
    maxnoofUEIDs,
    maxnoofBHRLCChannels,
    maxnoofRoutingEntries,
    maxnoofTLAsIAB,
    maxnoofULUPTNLInformationforIAB,
    maxnoofUPTNLAddresses,
    maxnoofSLDRBs,
    maxnoofTRPInfoTypes,
    maxnoofTRPs,
    maxnoofMRBs,
    maxnoofUEIDforPaging,
    maxnoofMRBsforUE,
    maxnoofServingCellMOs


FROM F1AP-Constants;
```

```
-- **************************************************************
--
-- RESET ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Reset
--
-- **************************************************************

Reset ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { {ResetIEs} },
    ...
}

ResetIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID               CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory   }|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory   }|
    { ID id-ResetType                   CRITICALITY reject  TYPE ResetType                  PRESENCE mandatory   },
    ...
}

ResetType ::= CHOICE {
    f1-Interface                    ResetAll,
    partOfF1-Interface              UE-associatedLogicalF1-ConnectionListRes,
    choice-extension                ProtocolIE-SingleContainer { { ResetType-ExtIEs} }
}

ResetType-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}


ResetAll ::= ENUMERATED {
    reset-all,
    ...
}

UE-associatedLogicalF1-ConnectionListRes ::= SEQUENCE (SIZE(1.. maxnoofIndividualF1ConnectionsToReset)) OF ProtocolIE-SingleContainer { { UE-
associatedLogicalF1-ConnectionItemRes } }

UE-associatedLogicalF1-ConnectionItemRes F1AP-PROTOCOL-IES ::= {
    { ID id-UE-associatedLogicalF1-ConnectionItem   CRITICALITY reject  TYPE UE-associatedLogicalF1-ConnectionItem  PRESENCE mandatory},
    ...
}


-- **************************************************************
--
-- Reset Acknowledge
--
-- **************************************************************
```

```
ResetAcknowledge ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ {ResetAcknowledgeIEs} },
	...
}

ResetAcknowledgeIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID									CRITICALITY reject	TYPE TransactionID										PRESENCE mandatory	}|
	{ ID id-UE-associatedLogicalF1-ConnectionListResAck		CRITICALITY ignore	TYPE UE-associatedLogicalF1-ConnectionListResAck		PRESENCE optional	}|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics			PRESENCE optional	},
	...
}

UE-associatedLogicalF1-ConnectionListResAck ::= SEQUENCE (SIZE(1.. maxnoofIndividualF1ConnectionsToReset)) OF ProtocolIE-SingleContainer { { UE-associatedLogicalF1-ConnectionItemResAck } }

UE-associatedLogicalF1-ConnectionItemResAck 	F1AP-PROTOCOL-IES ::= {
	{ ID id-UE-associatedLogicalF1-ConnectionItem	CRITICALITY ignore 	TYPE UE-associatedLogicalF1-ConnectionItem 	PRESENCE mandatory },
	...
}

-- **************************************************************
--
-- ERROR INDICATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Error Indication
--
-- **************************************************************

ErrorIndication ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ErrorIndicationIEs}},
	...
}

ErrorIndicationIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID				CRITICALITY reject	TYPE TransactionID				PRESENCE mandatory}|
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY ignore	TYPE GNB-CU-UE-F1AP-ID			PRESENCE optional }|
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY ignore	TYPE GNB-DU-UE-F1AP-ID			PRESENCE optional }|
	{ ID id-Cause						CRITICALITY ignore	TYPE Cause						PRESENCE optional }|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional },
	...
}

-- **************************************************************
--
-- F1 SETUP ELEMENTARY PROCEDURE
--
-- **************************************************************
```

```
-- **************************************************************
--
-- F1 Setup Request
--
-- **************************************************************

F1SetupRequest ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ {F1SetupRequestIEs} },
	...
}

F1SetupRequestIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID					CRITICALITY reject	TYPE TransactionID					PRESENCE mandatory	}|
	{ ID id-gNB-DU-ID						CRITICALITY reject	TYPE GNB-DU-ID						PRESENCE mandatory	}|
	{ ID id-gNB-DU-Name						CRITICALITY ignore	TYPE GNB-DU-Name					PRESENCE optional }|
	{ ID id-gNB-DU-Served-Cells-List		CRITICALITY reject	TYPE GNB-DU-Served-Cells-List		PRESENCE optional }|
	{ ID id-GNB-DU-RRC-Version				CRITICALITY reject	TYPE RRC-Version					PRESENCE mandatory	}|
	{ ID id-Transport-Layer-Address-Info	CRITICALITY ignore	TYPE Transport-Layer-Address-Info	PRESENCE optional }|
	{ ID id-BAPAddress						CRITICALITY ignore	TYPE BAPAddress						PRESENCE optional }|
	{ ID id-Extended-GNB-DU-Name			CRITICALITY ignore	TYPE Extended-GNB-DU-Name			PRESENCE optional }|
	{ ID id-RRC-Terminating-IAB-Donor-gNB-ID	CRITICALITY reject	TYPE GlobalGNB-ID				PRESENCE optional }|
	{ ID id-Mobile-IAB-MTUserLocationInformation	CRITICALITY ignore	TYPE Mobile-IAB-MTUserLocationInformation		PRESENCE optional },
	...
}


GNB-DU-Served-Cells-List	::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { GNB-DU-Served-Cells-ItemIEs } }

GNB-DU-Served-Cells-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-GNB-DU-Served-Cells-Item		CRITICALITY reject	TYPE		GNB-DU-Served-Cells-Item PRESENCE mandatory	},
	...
}


-- **************************************************************
--
-- F1 Setup Response
--
-- **************************************************************

F1SetupResponse ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ {F1SetupResponseIEs} },
	...
}


F1SetupResponseIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID					CRITICALITY reject	TYPE TransactionID					PRESENCE mandatory	}|
	{ ID id-gNB-CU-Name						CRITICALITY ignore	TYPE GNB-CU-Name					PRESENCE optional }|
	{ ID id-Cells-to-be-Activated-List		CRITICALITY reject	TYPE Cells-to-be-Activated-List		PRESENCE optional }|
	{ ID id-GNB-CU-RRC-Version				CRITICALITY reject	TYPE RRC-Version					PRESENCE mandatory	}|
	{ ID id-Transport-Layer-Address-Info	CRITICALITY ignore	TYPE Transport-Layer-Address-Info	PRESENCE optional }|
	{ ID id-UL-BH-Non-UP-Traffic-Mapping	CRITICALITY reject	TYPE UL-BH-Non-UP-Traffic-Mapping	PRESENCE optional }|
```

```
	{ ID id-BAPAddress				CRITICALITY ignore	TYPE BAPAddress					PRESENCE optional }|
	{ ID id-Extended-GNB-CU-Name			CRITICALITY ignore	TYPE Extended-GNB-CU-Name			PRESENCE optional }|
	{ ID id-NCGI-to-be-Updated-List		CRITICALITY reject	TYPE NCGI-to-be-Updated-List		PRESENCE optional },
	...
}


Cells-to-be-Activated-List  ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Cells-to-be-Activated-List-ItemIEs } }

Cells-to-be-Activated-List-ItemIEs  F1AP-PROTOCOL-IES::= {
	{ ID id-Cells-to-be-Activated-List-Item		CRITICALITY reject	TYPE Cells-to-be-Activated-List-Item		PRESENCE mandatory},
	...
}


NCGI-to-be-Updated-List ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { NCGI-to-be-Updated-List-ItemIEs } }

NCGI-to-be-Updated-List-ItemIEs F1AP-PROTOCOL-IES::= {
	{ ID id-NCGI-to-be-Updated-List-Item		CRITICALITY reject	TYPE NCGI-to-be-Updated-List-Item		PRESENCE mandatory},
	...
}


-- **************************************************************
--
-- F1 Setup Failure
--
-- **************************************************************

F1SetupFailure ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ {F1SetupFailureIEs} },
	...
}

F1SetupFailureIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID			CRITICALITY reject	TYPE TransactionID			PRESENCE mandatory	}|
	{ ID id-Cause				CRITICALITY ignore	TYPE Cause				PRESENCE mandatory	}|
	{ ID id-TimeToWait			CRITICALITY ignore	TYPE TimeToWait			PRESENCE optional	}|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional	},
	...
}

-- **************************************************************
--
-- GNB-DU CONFIGURATION UPDATE ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- GNB-DU CONFIGURATION UPDATE
--
-- **************************************************************
```

```
GNBDUConfigurationUpdate::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { {GNBDUConfigurationUpdateIEs} },
    ...
}

GNBDUConfigurationUpdateIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                           CRITICALITY reject  TYPE TransactionID                                  PRESENCE mandatory   }|
    { ID id-Served-Cells-To-Add-List                CRITICALITY reject  TYPE Served-Cells-To-Add-List                       PRESENCE optional }|
    { ID id-Served-Cells-To-Modify-List             CRITICALITY reject  TYPE Served-Cells-To-Modify-List                    PRESENCE optional }|
    { ID id-Served-Cells-To-Delete-List             CRITICALITY reject  TYPE Served-Cells-To-Delete-List                    PRESENCE optional }|
    { ID id-Cells-Status-List                       CRITICALITY reject  TYPE Cells-Status-List                              PRESENCE optional }|
    { ID id-Dedicated-SIDelivery-NeededUE-List      CRITICALITY ignore  TYPE Dedicated-SIDelivery-NeededUE-List             PRESENCE optional }|
    { ID id-gNB-DU-ID                               CRITICALITY reject  TYPE GNB-DU-ID                                      PRESENCE optional }|
    { ID id-GNB-DU-TNL-Association-To-Remove-List   CRITICALITY reject  TYPE GNB-DU-TNL-Association-To-Remove-List          PRESENCE optional }|
    { ID id-Transport-Layer-Address-Info            CRITICALITY ignore  TYPE Transport-Layer-Address-Info                   PRESENCE optional }|
    { ID id-Coverage-Modification-Notification      CRITICALITY ignore  TYPE Coverage-Modification-Notification             PRESENCE optional }|
    { ID id-gNB-DU-Name                             CRITICALITY ignore  TYPE GNB-DU-Name                                    PRESENCE optional }|
    { ID id-Extended-GNB-DU-Name                    CRITICALITY ignore  TYPE Extended-GNB-DU-Name                           PRESENCE optional }|
    { ID id-RRC-Terminating-IAB-Donor-Related-Info  CRITICALITY reject  TYPE RRC-Terminating-IAB-Donor-Related-Info         PRESENCE optional }|
    { ID id-Mobile-IAB-MTUserLocationInformation    CRITICALITY ignore  TYPE Mobile-IAB-MTUserLocationInformation           PRESENCE optional }|
    { ID id-Future-Coverage-Modification-Notification       CRITICALITY ignore  TYPE Future-Coverage-Modification-Notification             PRESENCE
optional    },
    ...
}

Served-Cells-To-Add-List        ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Served-Cells-To-Add-ItemIEs } }
Served-Cells-To-Modify-List ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Served-Cells-To-Modify-ItemIEs } }
Served-Cells-To-Delete-List ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Served-Cells-To-Delete-ItemIEs } }
Cells-Status-List   ::= SEQUENCE (SIZE(0.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Cells-Status-ItemIEs } }

Dedicated-SIDelivery-NeededUE-List::= SEQUENCE (SIZE(1.. maxnoofUEIDs)) OF ProtocolIE-SingleContainer { { Dedicated-SIDelivery-NeededUE-ItemIEs } }

GNB-DU-TNL-Association-To-Remove-List   ::= SEQUENCE (SIZE(1.. maxnoofTNLAssociations)) OF ProtocolIE-SingleContainer { { GNB-DU-TNL-Association-
To-Remove-ItemIEs } }


Served-Cells-To-Add-ItemIEs F1AP-PROTOCOL-IES   ::= {
    { ID id-Served-Cells-To-Add-Item        CRITICALITY reject  TYPE    Served-Cells-To-Add-Item          PRESENCE mandatory   },
    ...
}

Served-Cells-To-Modify-ItemIEs F1AP-PROTOCOL-IES    ::= {
    { ID id-Served-Cells-To-Modify-Item         CRITICALITY reject  TYPE        Served-Cells-To-Modify-Item                    PRESENCE mandatory
    },
    ...
}

Served-Cells-To-Delete-ItemIEs F1AP-PROTOCOL-IES    ::= {
    { ID id-Served-Cells-To-Delete-Item             CRITICALITY reject  TYPE        Served-Cells-To-Delete-Item             PRESENCE mandatory   },
    ...
}

Cells-Status-ItemIEs F1AP-PROTOCOL-IES  ::= {
```

```
    { ID id-Cells-Status-Item               CRITICALITY reject  TYPE        Cells-Status-Item              PRESENCE mandatory   },
    ...
}

Dedicated-SIDelivery-NeededUE-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-Dedicated-SIDelivery-NeededUE-Item      CRITICALITY ignore  TYPE    Dedicated-SIDelivery-NeededUE-Item          PRESENCE mandatory   },
    ...
}

GNB-DU-TNL-Association-To-Remove-ItemIEs F1AP-PROTOCOL-IES  ::= {
    { ID id-GNB-DU-TNL-Association-To-Remove-Item       CRITICALITY reject  TYPE     GNB-DU-TNL-Association-To-Remove-Item        PRESENCE 
mandatory   },
    ...
}


-- **************************************************************
--
-- GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE
--
-- **************************************************************

GNBDUConfigurationUpdateAcknowledge ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { {GNBDUConfigurationUpdateAcknowledgeIEs} },
    ...
}


GNBDUConfigurationUpdateAcknowledgeIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                       CRITICALITY reject  TYPE TransactionID                      PRESENCE mandatory   }|
    { ID id-Cells-to-be-Activated-List          CRITICALITY reject  TYPE Cells-to-be-Activated-List          PRESENCE optional }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics              PRESENCE optional }|
    { ID id-Cells-to-be-Deactivated-List        CRITICALITY reject  TYPE Cells-to-be-Deactivated-List        PRESENCE optional }|
    { ID id-Transport-Layer-Address-Info        CRITICALITY ignore  TYPE Transport-Layer-Address-Info        PRESENCE optional }|
    { ID id-UL-BH-Non-UP-Traffic-Mapping        CRITICALITY reject  TYPE UL-BH-Non-UP-Traffic-Mapping        PRESENCE optional }|
    { ID id-BAPAddress                          CRITICALITY ignore  TYPE BAPAddress                          PRESENCE optional }|
    { ID id-CellsForSON-List                    CRITICALITY ignore  TYPE CellsForSON-List                    PRESENCE optional },
    ...
}

-- **************************************************************
--
-- GNB-DU CONFIGURATION UPDATE FAILURE
--
-- **************************************************************

GNBDUConfigurationUpdateFailure ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { {GNBDUConfigurationUpdateFailureIEs} },
    ...
}

GNBDUConfigurationUpdateFailureIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID               CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory   }|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory   }|
```

```
	{ ID id-TimeToWait					CRITICALITY ignore	TYPE TimeToWait					PRESENCE optional }|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional },
	...
}

-- **************************************************************
--
-- GNB-CU CONFIGURATION UPDATE ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- GNB-CU CONFIGURATION UPDATE
--
-- **************************************************************

GNBCUConfigurationUpdate ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { GNBCUConfigurationUpdateIEs} },
	...
}

GNBCUConfigurationUpdateIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID						CRITICALITY reject	TYPE	TransactionID								PRESENCE mandatory	}|
	{ ID id-Cells-to-be-Activated-List			CRITICALITY reject	TYPE	Cells-to-be-Activated-List					PRESENCE optional }|
	{ ID id-Cells-to-be-Deactivated-List		CRITICALITY reject	TYPE	Cells-to-be-Deactivated-List				PRESENCE optional }|
	{ ID id-GNB-CU-TNL-Association-To-Add-List		CRITICALITY ignore	TYPE	GNB-CU-TNL-Association-To-Add-List		PRESENCE optional }|
	{ ID id-GNB-CU-TNL-Association-To-Remove-List	CRITICALITY ignore	TYPE	GNB-CU-TNL-Association-To-Remove-List	PRESENCE optional }|
	{ ID id-GNB-CU-TNL-Association-To-Update-List	CRITICALITY ignore	TYPE	GNB-CU-TNL-Association-To-Update-List	PRESENCE optional }|
	{ ID id-Cells-to-be-Barred-List				CRITICALITY ignore	TYPE	Cells-to-be-Barred-List						PRESENCE optional }|
	{ ID id-Protected-EUTRA-Resources-List		CRITICALITY reject	TYPE	Protected-EUTRA-Resources-List				PRESENCE optional }|
	{ ID id-Neighbour-Cell-Information-List		CRITICALITY ignore	TYPE	Neighbour-Cell-Information-List				PRESENCE optional }|
	{ ID id-Transport-Layer-Address-Info		CRITICALITY ignore	TYPE	Transport-Layer-Address-Info				PRESENCE optional }|
	{ ID id-UL-BH-Non-UP-Traffic-Mapping		CRITICALITY reject	TYPE	UL-BH-Non-UP-Traffic-Mapping				PRESENCE optional }|
	{ ID id-BAPAddress							CRITICALITY ignore	TYPE	BAPAddress									PRESENCE optional }|
	{ ID id-CCO-Assistance-Information			CRITICALITY ignore	TYPE	CCO-Assistance-Information					PRESENCE optional }|
	{ ID id-CellsForSON-List					CRITICALITY ignore	TYPE	CellsForSON-List							PRESENCE optional }|
	{ ID id-gNB-CU-Name							CRITICALITY ignore	TYPE	GNB-CU-Name									PRESENCE optional }|
	{ ID id-Extended-GNB-CU-Name				CRITICALITY ignore	TYPE	Extended-GNB-CU-Name						PRESENCE optional }|
	{ ID id-Cells-Allowed-to-be-Deactivated-List		CRITICALITY ignore	TYPE	Cells-Allowed-to-be-Deactivated-List	PRESENCE optional }|
	{ ID id-OnDemand-SIB1-Cell					CRITICALITY ignore	TYPE	OnDemand-SIB1-Cell						PRESENCE optional }|
	{ ID id-Predicted-CCO-Assistance-Information			CRITICALITY ignore	TYPE	Predicted-CCO-Assistance-Information
	PRESENCE optional	}|
	{ ID id-NeighbourFutureCoverageModNotification	CRITICALITY ignore	TYPE	Neighbour-Future-Coverage-Modification-Notification		PRESENCE
optional	},
	...
}

Cells-to-be-Deactivated-List	::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Cells-to-be-Deactivated-List-ItemIEs } }
GNB-CU-TNL-Association-To-Add-List		::= SEQUENCE (SIZE(1.. maxnoofTNLAssociations)) OF ProtocolIE-SingleContainer { { GNB-CU-TNL-Association-
To-Add-ItemIEs } }
GNB-CU-TNL-Association-To-Remove-List	::= SEQUENCE (SIZE(1.. maxnoofTNLAssociations)) OF ProtocolIE-SingleContainer { { GNB-CU-TNL-Association-
To-Remove-ItemIEs } }
```

```
GNB-CU-TNL-Association-To-Update-List   ::= SEQUENCE (SIZE(1.. maxnoofTNLAssociations)) OF ProtocolIE-SingleContainer { { GNB-CU-TNL-Association-To-Update-ItemIEs } }
Cells-to-be-Barred-List         ::= SEQUENCE(SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Cells-to-be-Barred-ItemIEs } }

Cells-Allowed-to-be-Deactivated-List    ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Cells-Allowed-to-be-Deactivated-List-ItemIEs } }

Cells-Allowed-to-be-Deactivated-List-ItemIEs F1AP-PROTOCOL-IES  ::= {
    { ID id-Cells-Allowed-to-be-Deactivated-List-Item   CRITICALITY ignore  TYPE    Cells-Allowed-to-be-Deactivated-List-Item  PRESENCE mandatory },
    ...
}


Cells-to-be-Deactivated-List-ItemIEs F1AP-PROTOCOL-IES  ::= {
    { ID id-Cells-to-be-Deactivated-List-Item                       CRITICALITY reject  TYPE    Cells-to-be-Deactivated-List-Item
    PRESENCE mandatory  },
    ...
}


GNB-CU-TNL-Association-To-Add-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-GNB-CU-TNL-Association-To-Add-Item      CRITICALITY ignore  TYPE     GNB-CU-TNL-Association-To-Add-Item      PRESENCE mandatory   },
    ...
}

GNB-CU-TNL-Association-To-Remove-ItemIEs F1AP-PROTOCOL-IES  ::= {
    { ID id-GNB-CU-TNL-Association-To-Remove-Item       CRITICALITY ignore  TYPE     GNB-CU-TNL-Association-To-Remove-Item        PRESENCE mandatory   },
    ...
}

GNB-CU-TNL-Association-To-Update-ItemIEs F1AP-PROTOCOL-IES  ::= {
    { ID id-GNB-CU-TNL-Association-To-Update-Item       CRITICALITY ignore  TYPE     GNB-CU-TNL-Association-To-Update-Item        PRESENCE mandatory   },
    ...
}

Cells-to-be-Barred-ItemIEs F1AP-PROTOCOL-IES    ::= {
    { ID id-Cells-to-be-Barred-Item     CRITICALITY ignore  TYPE     Cells-to-be-Barred-Item           PRESENCE mandatory   },
    ...
}

Protected-EUTRA-Resources-List ::= SEQUENCE (SIZE(1.. maxCellineNB))    OF ProtocolIE-SingleContainer { { Protected-EUTRA-Resources-ItemIEs } }
Protected-EUTRA-Resources-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-Protected-EUTRA-Resources-Item                  CRITICALITY reject  TYPE Protected-EUTRA-Resources-Item                  PRESENCE mandatory},
    ...
}

Neighbour-Cell-Information-List ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Neighbour-Cell-Information-ItemIEs } }
Neighbour-Cell-Information-ItemIEs F1AP-PROTOCOL-IES    ::= {
    { ID id-Neighbour-Cell-Information-Item                    CRITICALITY ignore  TYPE Neighbour-Cell-Information-Item                   PRESENCE mandatory},
```

```
	...
}

-- **************************************************************
--
-- GNB-CU CONFIGURATION UPDATE ACKNOWLEDGE
--
-- **************************************************************

GNBCUConfigurationUpdateAcknowledge ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { GNBCUConfigurationUpdateAcknowledgeIEs} },
	...
}


GNBCUConfigurationUpdateAcknowledgeIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID							CRITICALITY reject	TYPE TransactionID								PRESENCE mandatory	}|
	{ ID id-Cells-Failed-to-be-Activated-List			CRITICALITY reject	TYPE Cells-Failed-to-be-Activated-List			PRESENCE optional}|
	{ ID id-CriticalityDiagnostics						CRITICALITY ignore	TYPE CriticalityDiagnostics						PRESENCE optional
	}|
	{ ID id-GNB-CU-TNL-Association-Setup-List			CRITICALITY ignore	TYPE GNB-CU-TNL-Association-Setup-List			PRESENCE optional }|
	{ ID id-GNB-CU-TNL-Association-Failed-To-Setup-List CRITICALITY ignore	TYPE GNB-CU-TNL-Association-Failed-To-Setup-List PRESENCE optional }|
	{ ID id-Dedicated-SIDelivery-NeededUE-List				CRITICALITY ignore	TYPE Dedicated-SIDelivery-NeededUE-List			PRESENCE optional
	}|
	{ ID id-Transport-Layer-Address-Info				CRITICALITY ignore	TYPE Transport-Layer-Address-Info				PRESENCE optional }|
	{ ID id-Cells-With-SSBs-Activated-List				CRITICALITY ignore	TYPE Cells-With-SSBs-Activated-List				PRESENCE optional },
	...
}

Cells-Failed-to-be-Activated-List	::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Cells-Failed-to-be-Activated-List-
ItemIEs } }
GNB-CU-TNL-Association-Setup-List ::= SEQUENCE (SIZE(1.. maxnoofTNLAssociations))	OF ProtocolIE-SingleContainer { { GNB-CU-TNL-Association-Setup-
ItemIEs } }
GNB-CU-TNL-Association-Failed-To-Setup-List ::= SEQUENCE (SIZE(1.. maxnoofTNLAssociations)) OF ProtocolIE-SingleContainer { { GNB-CU-TNL-
Association-Failed-To-Setup-ItemIEs } }

Cells-Failed-to-be-Activated-List-ItemIEs F1AP-PROTOCOL-IES	::= {
	{ ID id-Cells-Failed-to-be-Activated-List-Item		CRITICALITY reject	TYPE Cells-Failed-to-be-Activated-List-Item		PRESENCE mandatory	},
	...
}

GNB-CU-TNL-Association-Setup-ItemIEs F1AP-PROTOCOL-IES	::= {
	{ ID id-GNB-CU-TNL-Association-Setup-Item		CRITICALITY ignore	TYPE	GNB-CU-TNL-Association-Setup-Item		PRESENCE mandatory	},
	...
}


GNB-CU-TNL-Association-Failed-To-Setup-ItemIEs F1AP-PROTOCOL-IES	::= {
	{ ID id-GNB-CU-TNL-Association-Failed-To-Setup-Item		CRITICALITY ignore	TYPE	GNB-CU-TNL-Association-Failed-To-Setup-Item		PRESENCE
mandatory	},
	...
}
```

```
-- **************************************************************
--
-- GNB-CU CONFIGURATION UPDATE FAILURE
--
-- **************************************************************

GNBCUConfigurationUpdateFailure ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { GNBCUConfigurationUpdateFailureIEs} },
    ...
}

GNBCUConfigurationUpdateFailureIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID               CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory   }|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory   }|
    { ID id-TimeToWait                  CRITICALITY ignore  TYPE TimeToWait                 PRESENCE optional }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional },
    ...
}


-- **************************************************************
--
-- GNB-DU RESOURCE COORDINATION REQUEST
--
-- **************************************************************

GNBDUResourceCoordinationRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{GNBDUResourceCoordinationRequest-IEs}},
    ...
}

GNBDUResourceCoordinationRequest-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                                   CRITICALITY reject  TYPE TransactionID                              PRESENCE mandatory   }|
    { ID id-RequestType                                     CRITICALITY reject  TYPE RequestType                                PRESENCE mandatory   }|
    { ID id-EUTRA-NR-CellResourceCoordinationReq-Container  CRITICALITY reject  TYPE EUTRA-NR-CellResourceCoordinationReq-Container  PRESENCE 
mandatory}|
    { ID id-IgnoreResourceCoordinationContainer             CRITICALITY reject  TYPE IgnoreResourceCoordinationContainer    PRESENCE optional },
    ...
}


-- **************************************************************
--
-- GNB-DU RESOURCE COORDINATION RESPONSE
--
-- **************************************************************

GNBDUResourceCoordinationResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{GNBDUResourceCoordinationResponse-IEs}},
    ...
}

GNBDUResourceCoordinationResponse-IEs F1AP-PROTOCOL-IES ::= {
```

```
	{ ID id-TransactionID							CRITICALITY reject	TYPE TransactionID					PRESENCE mandatory	}|
	{ ID id-EUTRA-NR-CellResourceCoordinationReqAck-Container	CRITICALITY reject	TYPE EUTRA-NR-CellResourceCoordinationReqAck-Container	PRESENCE mandatory},
	...
}

-- **************************************************************
--
-- UE Context Setup ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- UE CONTEXT SETUP REQUEST
--
-- **************************************************************

UEContextSetupRequest ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { UEContextSetupRequestIEs} },
	...
}

UEContextSetupRequestIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID					CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID						PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID					CRITICALITY ignore	TYPE GNB-DU-UE-F1AP-ID						PRESENCE optional	}|
	{ ID id-SpCell-ID							CRITICALITY reject	TYPE NRCGI								PRESENCE mandatory	}|
	{ ID id-ServCellIndex						CRITICALITY reject	TYPE ServCellIndex							PRESENCE mandatory	}|
	{ ID id-SpCellULConfigured					CRITICALITY ignore	TYPE CellULConfigured						PRESENCE optional }|
	{ ID id-CUtoDURRCInformation				CRITICALITY reject	TYPE CUtoDURRCInformation					PRESENCE mandatory}|
	{ ID id-Candidate-SpCell-List				CRITICALITY ignore	TYPE Candidate-SpCell-List					PRESENCE optional }|
	{ ID id-DRXCycle							CRITICALITY ignore	TYPE DRXCycle								PRESENCE optional }|
	{ ID id-ResourceCoordinationTransferContainer	CRITICALITY ignore	TYPE ResourceCoordinationTransferContainer		PRESENCE optional }|
	{ ID id-SCell-ToBeSetup-List				CRITICALITY ignore	TYPE SCell-ToBeSetup-List					PRESENCE optional }|
	{ ID id-SRBs-ToBeSetup-List					CRITICALITY reject	TYPE SRBs-ToBeSetup-List					PRESENCE optional }|
	{ ID id-DRBs-ToBeSetup-List					CRITICALITY reject	TYPE DRBs-ToBeSetup-List					PRESENCE optional }|
	{ ID id-InactivityMonitoringRequest			CRITICALITY reject	TYPE InactivityMonitoringRequest				PRESENCE optional }|
	{ ID id-RAT-FrequencyPriorityInformation		CRITICALITY reject	TYPE RAT-FrequencyPriorityInformation			PRESENCE optional }|
	{ ID id-RRCContainer						CRITICALITY ignore	TYPE RRCContainer							PRESENCE optional }|
	{ ID id-MaskedIMEISV						CRITICALITY ignore	TYPE MaskedIMEISV							PRESENCE optional }|
	{ ID id-ServingPLMN						CRITICALITY ignore	TYPE PLMN-Identity							PRESENCE optional }|
	{ ID id-GNB-DU-UE-AMBR-UL					CRITICALITY ignore	TYPE BitRate								PRESENCE conditional }|
	-- The above IE shall be present only if the DRB to Be Setup List IE is present.
	{ ID id-RRCDeliveryStatusRequest				CRITICALITY ignore	TYPE RRCDeliveryStatusRequest				PRESENCE optional }|
	{ ID id-ResourceCoordinationTransferInformation CRITICALITY ignore	TYPE ResourceCoordinationTransferInformation	PRESENCE optional }|
	{ ID id-ServingCellMO						CRITICALITY ignore	TYPE ServingCellMO							PRESENCE optional }|
	{ ID id-new-gNB-CU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID						PRESENCE optional }|
	{ ID id-RANUEID							CRITICALITY ignore	TYPE RANUEID								PRESENCE optional }|
	{ ID id-TraceActivation						CRITICALITY ignore	TYPE TraceActivation						PRESENCE optional }|
	{ ID id-AdditionalRRMPriorityIndex			CRITICALITY ignore	TYPE AdditionalRRMPriorityIndex				PRESENCE optional }|
	{ ID id-BHChannels-ToBeSetup-List			CRITICALITY reject	TYPE BHChannels-ToBeSetup-List				PRESENCE optional }|
	{ ID id-ConfiguredBAPAddress				CRITICALITY reject	TYPE BAPAddress							PRESENCE optional }|
	{ ID id-NRV2XServicesAuthorized				CRITICALITY ignore	TYPE NRV2XServicesAuthorized				PRESENCE optional }|
	{ ID id-LTEV2XServicesAuthorized				CRITICALITY ignore	TYPE LTEV2XServicesAuthorized				PRESENCE optional }|
```

```
	{ ID id-NRUESidelinkAggregateMaximumBitrate	CRITICALITY ignore	TYPE NRUESidelinkAggregateMaximumBitrate	PRESENCE optional }|
	{ ID id-LTEUESidelinkAggregateMaximumBitrate	CRITICALITY ignore	TYPE LTEUESidelinkAggregateMaximumBitrate	PRESENCE optional }|
	{ ID id-PC5LinkAMBR	CRITICALITY ignore	TYPE BitRate	PRESENCE optional}|
	{ ID id-SLDRBs-ToBeSetup-List	CRITICALITY reject	TYPE SLDRBs-ToBeSetup-List	PRESENCE optional }|
	{ ID id-ConditionalInterDUMobilityInformation	CRITICALITY reject	TYPE ConditionalInterDUMobilityInformation	PRESENCE optional}|
	{ ID id-ManagementBasedMDTPLMNList	CRITICALITY ignore	TYPE	MDTPLMNList	PRESENCE optional }|
	{ ID id-ServingNID	CRITICALITY reject	TYPE NID	PRESENCE optional }|
	{ ID id-F1CTransferPath	CRITICALITY reject	TYPE F1CTransferPath	PRESENCE optional }|
	{ ID id-F1CTransferPathNRDC	CRITICALITY reject	TYPE F1CTransferPathNRDC	PRESENCE optional }|
	{ ID id-MDTPollutedMeasurementIndicator	CRITICALITY ignore	TYPE MDTPollutedMeasurementIndicator	PRESENCE optional }|
	{ ID id-SCGActivationRequest	CRITICALITY ignore	TYPE SCGActivationRequest	PRESENCE optional }|
	{ ID id-CG-SDTSessionInfoOld	CRITICALITY ignore	TYPE CG-SDTSessionInfo	PRESENCE optional }|
	{ ID id-FiveG-ProSeAuthorized	CRITICALITY ignore	TYPE FiveG-ProSeAuthorized	PRESENCE optional }|
	{ ID id-FiveG-ProSeUEPC5AggregateMaximumBitrate CRITICALITY ignore	TYPE NRUESidelinkAggregateMaximumBitrate	PRESENCE optional }|
	{ ID id-FiveG-ProSePC5LinkAMBR	CRITICALITY ignore	TYPE BitRate	PRESENCE optional}|
	{ ID id-UuRLCChannelToBeSetupList	CRITICALITY reject	TYPE UuRLCChannelToBeSetupList	PRESENCE optional}|
	{ ID id-PC5RLCChannelToBeSetupList	CRITICALITY reject	TYPE PC5RLCChannelToBeSetupList	PRESENCE optional}|
	{ ID id-PathSwitchConfiguration	CRITICALITY ignore	TYPE PathSwitchConfiguration	PRESENCE optional }|
	{ ID id-GNBDUUESliceMaximumBitRateList	CRITICALITY ignore	TYPE GNBDUUESliceMaximumBitRateList	PRESENCE optional }|
	{ ID id-MulticastMBSSessionSetupList	CRITICALITY reject	TYPE MulticastMBSSessionList	PRESENCE optional }|
	{ ID id-UE-MulticastMRBs-ToBeSetup-List	CRITICALITY reject	TYPE UE-MulticastMRBs-ToBeSetup-List	PRESENCE optional }|
	{ ID id-ServingCellMO-List	CRITICALITY ignore	TYPE ServingCellMO-List	PRESENCE optional }|
	{ ID id-NetworkControlledRepeaterAuthorized	CRITICALITY ignore	TYPE NetworkControlledRepeaterAuthorized	PRESENCE optional }|
	{ ID id-SDT-Volume-Threshold	CRITICALITY ignore	TYPE SDT-Volume-Threshold	PRESENCE optional	}|
	{ ID id-LTMInformation-Setup	CRITICALITY reject	TYPE LTMInformation-Setup	PRESENCE optional }|
	{ ID id-LTMConfigurationIDMappingList	CRITICALITY reject	TYPE LTMConfigurationIDMappingList	PRESENCE optional }|
	{ ID id-EarlySyncInformation-Request	CRITICALITY ignore	TYPE EarlySyncInformation-Request	PRESENCE optional }|
	{ ID id-PathAdditionInformation	CRITICALITY reject	TYPE PathAdditionInformation	PRESENCE optional}|
	{ ID id-NRA2XServicesAuthorized	CRITICALITY ignore	TYPE NRA2XServicesAuthorized	PRESENCE optional }|
	{ ID id-LTEA2XServicesAuthorized	CRITICALITY ignore	TYPE LTEA2XServicesAuthorized	PRESENCE optional }|
	{ ID id-NRUESidelinkAggregateMaximumBitrateForA2X	CRITICALITY ignore	TYPE NRUESidelinkAggregateMaximumBitrate	PRESENCE optional }|
	{ ID id-LTEUESidelinkAggregateMaximumBitrateForA2X	CRITICALITY ignore	TYPE LTEUESidelinkAggregateMaximumBitrate	PRESENCE optional }|
	{ ID id-DLLBTFailureInformationRequest	CRITICALITY ignore	TYPE DLLBTFailureInformationRequest	PRESENCE optional }|
	{ ID id-SLPositioning-Ranging-Service-Info	CRITICALITY ignore	TYPE SLPositioning-Ranging-Service-Info	PRESENCE optional }|
	{ ID id-NonIntegerDRXCycle	CRITICALITY ignore	TYPE NonIntegerDRXCycle	PRESENCE optional }|
	{ ID id-LTMInformationSCGAdd	CRITICALITY reject	TYPE LTMInformationSCGAdd	PRESENCE optional },
	...
}

Candidate-SpCell-List::= SEQUENCE (SIZE(1..maxnoofCandidateSpCells)) OF ProtocolIE-SingleContainer { { Candidate-SpCell-ItemIEs} }
SCell-ToBeSetup-List::= SEQUENCE (SIZE(1..maxnoofSCells)) OF ProtocolIE-SingleContainer { { SCell-ToBeSetup-ItemIEs} }
SRBs-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofSRBs)) OF ProtocolIE-SingleContainer { { SRBs-ToBeSetup-ItemIEs} }
DRBs-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-ToBeSetup-ItemIEs} }
BHChannels-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF ProtocolIE-SingleContainer { { BHChannels-ToBeSetup-ItemIEs} }
SLDRBs-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-ToBeSetup-ItemIEs} }
UE-MulticastMRBs-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofMRBsforUE)) OF ProtocolIE-SingleContainer { { UE-MulticastMRBs-ToBeSetup-ItemIEs} }
ServingCellMO-List ::= SEQUENCE (SIZE(1..maxnoofServingCellMOs)) OF ProtocolIE-SingleContainer { { ServingCellMO-List-ItemIEs} }

Candidate-SpCell-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-Candidate-SpCell-Item	CRITICALITY ignore	TYPE Candidate-SpCell-Item	PRESENCE mandatory	},
	...
}
```

```
SCell-ToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SCell-ToBeSetup-Item			CRITICALITY ignore	TYPE SCell-ToBeSetup-Item		PRESENCE mandatory	},
	...
}

SRBs-ToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SRBs-ToBeSetup-Item		CRITICALITY reject		TYPE SRBs-ToBeSetup-Item		PRESENCE mandatory},
	...
}

DRBs-ToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-DRBs-ToBeSetup-Item			CRITICALITY reject	TYPE DRBs-ToBeSetup-Item			PRESENCE mandatory},
	...
}

BHChannels-ToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BHChannels-ToBeSetup-Item			CRITICALITY reject	TYPE BHChannels-ToBeSetup-Item			PRESENCE mandatory},
	...
}

SLDRBs-ToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SLDRBs-ToBeSetup-Item			CRITICALITY reject	TYPE SLDRBs-ToBeSetup-Item			PRESENCE mandatory},
	...
}

UE-MulticastMRBs-ToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-UE-MulticastMRBs-ToBeSetup-Item		CRITICALITY reject	TYPE UE-MulticastMRBs-ToBeSetup-Item		PRESENCE mandatory},
	...
}

ServingCellMO-List-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-ServingCellMO-List-Item		CRITICALITY reject	TYPE ServingCellMO-List-Item	PRESENCE mandatory},
	...
}
-- **************************************************************
--
-- UE CONTEXT SETUP RESPONSE
--
-- **************************************************************

UEContextSetupResponse ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { UEContextSetupResponseIEs} },
	...
}


UEContextSetupResponseIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-DUtoCURRCInformation			CRITICALITY reject	TYPE DUtoCURRCInformation			PRESENCE mandatory }|
	{ ID id-C-RNTI					CRITICALITY ignore	TYPE C-RNTI					PRESENCE optional }|
	{ ID id-ResourceCoordinationTransferContainer	CRITICALITY ignore	TYPE ResourceCoordinationTransferContainer	PRESENCE optional }|
	{ ID id-FullConfiguration				CRITICALITY reject	TYPE FullConfiguration				PRESENCE optional }|
	{ ID id-DRBs-Setup-List				CRITICALITY ignore	TYPE DRBs-Setup-List				PRESENCE optional }|
```

```
    { ID id-SRBs-FailedToBeSetup-List               CRITICALITY ignore  TYPE SRBs-FailedToBeSetup-List            PRESENCE optional }|
    { ID id-DRBs-FailedToBeSetup-List               CRITICALITY ignore  TYPE DRBs-FailedToBeSetup-List            PRESENCE optional }|
    { ID id-SCell-FailedtoSetup-List                CRITICALITY ignore  TYPE SCell-FailedtoSetup-List             PRESENCE optional }|
    { ID id-InactivityMonitoringResponse            CRITICALITY reject  TYPE InactivityMonitoringResponse         PRESENCE optional }|
    { ID id-CriticalityDiagnostics                  CRITICALITY ignore  TYPE CriticalityDiagnostics               PRESENCE optional }|
    { ID id-SRBs-Setup-List                         CRITICALITY ignore  TYPE SRBs-Setup-List                      PRESENCE optional }|
    { ID id-BHChannels-Setup-List                   CRITICALITY ignore  TYPE BHChannels-Setup-List                PRESENCE optional }|
    { ID id-BHChannels-FailedToBeSetup-List         CRITICALITY ignore  TYPE BHChannels-FailedToBeSetup-List      PRESENCE optional }|
    { ID id-SLDRBs-Setup-List                       CRITICALITY ignore  TYPE SLDRBs-Setup-List                    PRESENCE optional }|
    { ID id-SLDRBs-FailedToBeSetup-List             CRITICALITY ignore  TYPE SLDRBs-FailedToBeSetup-List          PRESENCE optional }|
    { ID id-requestedTargetCellGlobalID             CRITICALITY reject  TYPE NRCGI                                PRESENCE optional}|
    { ID id-SCGActivationStatus                     CRITICALITY ignore  TYPE SCGActivationStatus                  PRESENCE optional }|
    { ID id-UuRLCChannelSetupList                   CRITICALITY ignore  TYPE UuRLCChannelSetupList                PRESENCE optional}|
    { ID id-UuRLCChannelFailedToBeSetupList         CRITICALITY ignore  TYPE UuRLCChannelFailedToBeSetupList      PRESENCE optional}|
    { ID id-PC5RLCChannelSetupList                  CRITICALITY ignore  TYPE PC5RLCChannelSetupList               PRESENCE optional}|
    { ID id-PC5RLCChannelFailedToBeSetupList        CRITICALITY ignore  TYPE PC5RLCChannelFailedToBeSetupList     PRESENCE optional}|
    { ID id-ServingCellMO-encoded-in-CGC-List       CRITICALITY ignore  TYPE ServingCellMO-encoded-in-CGC-List       PRESENCE optional}|
    { ID id-UE-MulticastMRBs-Setupnew-List          CRITICALITY reject  TYPE UE-MulticastMRBs-Setupnew-List          PRESENCE optional}|
    { ID id-DedicatedSIDeliveryIndication           CRITICALITY ignore  TYPE DedicatedSIDeliveryIndication        PRESENCE optional}|
    { ID id-Configured-BWP-List                     CRITICALITY ignore  TYPE Configured-BWP-List                     PRESENCE optional}|
    { ID id-EarlySyncInformation                    CRITICALITY ignore  TYPE EarlySyncInformation                 PRESENCE optional }|
    { ID id-LTMConfiguration                        CRITICALITY ignore  TYPE LTMConfiguration                     PRESENCE optional }|
    { ID id-S-CPAC-Configuration                    CRITICALITY ignore  TYPE S-CPAC-Configuration                 PRESENCE optional },
    ...
}

DRBs-Setup-List ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-Setup-ItemIEs} }


SRBs-FailedToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofSRBs)) OF ProtocolIE-SingleContainer { { SRBs-FailedToBeSetup-ItemIEs} }
DRBs-FailedToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-FailedToBeSetup-ItemIEs} }
SCell-FailedtoSetup-List ::= SEQUENCE (SIZE(1..maxnoofSCells)) OF ProtocolIE-SingleContainer { { SCell-FailedtoSetup-ItemIEs} }
SRBs-Setup-List ::= SEQUENCE (SIZE(1..maxnoofSRBs)) OF ProtocolIE-SingleContainer { { SRBs-Setup-ItemIEs} }
BHChannels-Setup-List ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF ProtocolIE-SingleContainer { { BHChannels-Setup-ItemIEs} }
BHChannels-FailedToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF ProtocolIE-SingleContainer { { BHChannels-FailedToBeSetup-ItemIEs}
}

DRBs-Setup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-DRBs-Setup-Item                     CRITICALITY ignore  TYPE DRBs-Setup-Item                  PRESENCE mandatory},
    ...
}

SRBs-Setup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SRBs-Setup-Item                     CRITICALITY ignore  TYPE SRBs-Setup-Item                  PRESENCE mandatory},
    ...
}

SRBs-FailedToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SRBs-FailedToBeSetup-Item       CRITICALITY ignore      TYPE SRBs-FailedToBeSetup-Item     PRESENCE mandatory},
    ...
}


DRBs-FailedToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
```

```
	{ ID id-DRBs-FailedToBeSetup-Item		CRITICALITY ignore	TYPE DRBs-FailedToBeSetup-Item		PRESENCE mandatory},
	...
}

SCell-FailedtoSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SCell-FailedtoSetup-Item			CRITICALITY ignore	TYPE SCell-FailedtoSetup-Item		PRESENCE mandatory},
	...
}

BHChannels-Setup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BHChannels-Setup-Item				CRITICALITY ignore	TYPE BHChannels-Setup-Item				PRESENCE mandatory},
	...
}

BHChannels-FailedToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BHChannels-FailedToBeSetup-Item				CRITICALITY ignore	TYPE BHChannels-FailedToBeSetup-Item	PRESENCE mandatory},
	...
}

SLDRBs-Setup-List ::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-Setup-ItemIEs} }

SLDRBs-FailedToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-FailedToBeSetup-ItemIEs} }

SLDRBs-Setup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SLDRBs-Setup-Item				CRITICALITY ignore	TYPE SLDRBs-Setup-Item				PRESENCE mandatory},
	...
}

SLDRBs-FailedToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SLDRBs-FailedToBeSetup-Item		CRITICALITY ignore	TYPE SLDRBs-FailedToBeSetup-Item		PRESENCE mandatory},
	...
}

UE-MulticastMRBs-Setupnew-List ::= SEQUENCE (SIZE(1..maxnoofMRBsforUE)) OF ProtocolIE-SingleContainer { { UE-MulticastMRBs-Setupnew-ItemIEs } }

UE-MulticastMRBs-Setupnew-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-UE-MulticastMRBs-Setupnew-Item		CRITICALITY reject	TYPE UE-MulticastMRBs-Setupnew-Item	PRESENCE mandatory},
	...
}


-- **************************************************************
--
-- UE CONTEXT SETUP FAILURE
--
-- **************************************************************

UEContextSetupFailure ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { UEContextSetupFailureIEs} },
	...
}

UEContextSetupFailureIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID			PRESENCE mandatory	}|
```

```
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY ignore	TYPE GNB-DU-UE-F1AP-ID			PRESENCE optional }|
	{ ID id-Cause					CRITICALITY ignore	TYPE Cause					PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional }|
	{ ID id-Potential-SpCell-List		CRITICALITY ignore	TYPE Potential-SpCell-List		PRESENCE optional }|
	{ ID id-requestedTargetCellGlobalID CRITICALITY reject	TYPE NRCGI					PRESENCE optional},
	...
}

Potential-SpCell-List::= SEQUENCE (SIZE(0..maxnoofPotentialSpCells)) OF ProtocolIE-SingleContainer { { Potential-SpCell-ItemIEs} }

Potential-SpCell-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-Potential-SpCell-Item			CRITICALITY ignore	TYPE Potential-SpCell-Item			PRESENCE mandatory	},
	...
}

-- **************************************************************
--
-- UE Context Release Request ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- UE Context Release Request
--
-- **************************************************************

UEContextReleaseRequest ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{{ UEContextReleaseRequestIEs}},
	...
}

UEContextReleaseRequestIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-Cause						CRITICALITY ignore	TYPE Cause						PRESENCE mandatory	}|
	{ ID id-targetCellsToCancel			CRITICALITY reject	TYPE TargetCellList				PRESENCE optional	}|
	{ ID id-LTMCells-ToBeReleased-List		CRITICALITY reject	TYPE LTMCells-ToBeReleased-List	PRESENCE optional },
	...
}


-- **************************************************************
--
-- UE Context Release (gNB-CU initiated) ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- UE CONTEXT RELEASE COMMAND
--
-- **************************************************************
```

```
UEContextReleaseCommand ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { UEContextReleaseCommandIEs} },
    ...
}

UEContextReleaseCommandIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID               CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID           PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID               CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID           PRESENCE mandatory   }|
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                       PRESENCE mandatory   }|
    { ID id-RRCContainer                    CRITICALITY ignore  TYPE RRCContainer                PRESENCE optional }|
    { ID id-SRBID                           CRITICALITY ignore  TYPE SRBID                       PRESENCE conditional }|
    -- The above IE shall be present if the RRC container IE is present.
    { ID id-oldgNB-DU-UE-F1AP-ID            CRITICALITY ignore  TYPE GNB-DU-UE-F1AP-ID           PRESENCE optional }|
    { ID id-ExecuteDuplication              CRITICALITY ignore  TYPE ExecuteDuplication          PRESENCE optional}|
    { ID id-RRCDeliveryStatusRequest        CRITICALITY ignore  TYPE RRCDeliveryStatusRequest    PRESENCE optional }|
    { ID id-targetCellsToCancel             CRITICALITY reject  TYPE TargetCellList              PRESENCE optional}|
    { ID id-PosContextRevIndication         CRITICALITY ignore  TYPE PosContextRevIndication        PRESENCE optional}|
    { ID id-CG-SDTKeptIndicator             CRITICALITY ignore  TYPE CG-SDTKeptIndicator         PRESENCE optional}|
    { ID id-LTMCells-ToBeReleased-List      CRITICALITY reject  TYPE LTMCells-ToBeReleased-List  PRESENCE optional }|
    { ID id-DLLBTFailureInformationRequest  CRITICALITY ignore  TYPE DLLBTFailureInformationRequest  PRESENCE optional},
    ...
}

-- **************************************************************
--
-- UE CONTEXT RELEASE COMPLETE
--
-- **************************************************************

UEContextReleaseComplete ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { UEContextReleaseCompleteIEs} },
    ...
}


UEContextReleaseCompleteIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID          PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID          PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional }|
    { ID id-Recommended-SSBs-for-Paging-List        CRITICALITY ignore  TYPE Recommended-SSBs-for-Paging-List   PRESENCE optional },
    ...
}

-- **************************************************************
--
-- UE Context Modification ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- UE CONTEXT MODIFICATION REQUEST
--
-- **************************************************************
```

```
UEContextModificationRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { UEContextModificationRequestIEs} },
    ...
}

UEContextModificationRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID                      CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID                          PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID                      CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID                          PRESENCE mandatory   }|
    { ID id-SpCell-ID                              CRITICALITY ignore  TYPE NRCGI                                      PRESENCE optional }|
    { ID id-ServCellIndex                          CRITICALITY reject  TYPE ServCellIndex                              PRESENCE optional }|
    { ID id-SpCellULConfigured                     CRITICALITY ignore  TYPE CellULConfigured                           PRESENCE optional }|
    { ID id-DRXCycle                               CRITICALITY ignore  TYPE DRXCycle                                   PRESENCE optional }|
    { ID id-CUtoDURRCInformation                   CRITICALITY reject  TYPE CUtoDURRCInformation                       PRESENCE optional }|
    { ID id-TransmissionActionIndicator            CRITICALITY ignore  TYPE TransmissionActionIndicator                PRESENCE optional }|
    { ID id-ResourceCoordinationTransferContainer  CRITICALITY ignore  TYPE ResourceCoordinationTransferContainer      PRESENCE optional }|
    { ID id-RRCReconfigurationCompleteIndicator    CRITICALITY ignore  TYPE RRCReconfigurationCompleteIndicator        PRESENCE optional }|
    { ID id-RRCContainer                           CRITICALITY reject  TYPE RRCContainer                               PRESENCE optional }|
    { ID id-SCell-ToBeSetupMod-List                CRITICALITY ignore  TYPE SCell-ToBeSetupMod-List                    PRESENCE optional }|
    { ID id-SCell-ToBeRemoved-List                 CRITICALITY ignore  TYPE SCell-ToBeRemoved-List                     PRESENCE optional }|
    { ID id-SRBs-ToBeSetupMod-List                 CRITICALITY reject  TYPE SRBs-ToBeSetupMod-List                     PRESENCE optional }|
    { ID id-DRBs-ToBeSetupMod-List                 CRITICALITY reject  TYPE DRBs-ToBeSetupMod-List                     PRESENCE optional }|
    { ID id-DRBs-ToBeModified-List                 CRITICALITY reject  TYPE DRBs-ToBeModified-List                     PRESENCE optional }|
    { ID id-SRBs-ToBeReleased-List                 CRITICALITY reject  TYPE SRBs-ToBeReleased-List                     PRESENCE optional }|
    { ID id-DRBs-ToBeReleased-List                 CRITICALITY reject  TYPE DRBs-ToBeReleased-List                     PRESENCE optional }|
    { ID id-InactivityMonitoringRequest            CRITICALITY reject  TYPE InactivityMonitoringRequest                PRESENCE optional }|
    { ID id-RAT-FrequencyPriorityInformation       CRITICALITY reject  TYPE RAT-FrequencyPriorityInformation           PRESENCE optional }|
    { ID id-DRXConfigurationIndicator              CRITICALITY ignore  TYPE DRXConfigurationIndicator                  PRESENCE optional }|
    { ID id-RLCFailureIndication                   CRITICALITY ignore  TYPE RLCFailureIndication                       PRESENCE optional }|
    { ID id-UplinkTxDirectCurrentListInformation   CRITICALITY ignore  TYPE UplinkTxDirectCurrentListInformation       PRESENCE optional }|
    { ID id-GNB-DUConfigurationQuery               CRITICALITY reject  TYPE GNB-DUConfigurationQuery                   PRESENCE optional }|
    { ID id-GNB-DU-UE-AMBR-UL                      CRITICALITY ignore  TYPE BitRate                                    PRESENCE optional }|
    { ID id-ExecuteDuplication                     CRITICALITY ignore  TYPE ExecuteDuplication                         PRESENCE optional}|
    { ID id-RRCDeliveryStatusRequest               CRITICALITY ignore  TYPE RRCDeliveryStatusRequest                   PRESENCE optional }|
    { ID id-ResourceCoordinationTransferInformation CRITICALITY ignore  TYPE ResourceCoordinationTransferInformation   PRESENCE optional }|
    { ID id-ServingCellMO                          CRITICALITY ignore  TYPE ServingCellMO                              PRESENCE optional }|
    { ID id-NeedforGap                             CRITICALITY ignore  TYPE NeedforGap                                 PRESENCE optional }|
    { ID id-FullConfiguration                      CRITICALITY reject  TYPE FullConfiguration                          PRESENCE optional }|
    { ID id-AdditionalRRMPriorityIndex             CRITICALITY ignore  TYPE AdditionalRRMPriorityIndex                 PRESENCE optional }|
    { ID id-LowerLayerPresenceStatusChange         CRITICALITY ignore  TYPE LowerLayerPresenceStatusChange             PRESENCE optional }|
    { ID id-BHChannels-ToBeSetupMod-List           CRITICALITY reject  TYPE BHChannels-ToBeSetupMod-List               PRESENCE optional }|
    { ID id-BHChannels-ToBeModified-List           CRITICALITY reject  TYPE BHChannels-ToBeModified-List               PRESENCE optional }|
    { ID id-BHChannels-ToBeReleased-List           CRITICALITY reject  TYPE BHChannels-ToBeReleased-List               PRESENCE optional }|
    { ID id-NRV2XServicesAuthorized                CRITICALITY ignore  TYPE NRV2XServicesAuthorized                    PRESENCE optional }|
    { ID id-LTEV2XServicesAuthorized               CRITICALITY ignore  TYPE LTEV2XServicesAuthorized                   PRESENCE optional }|
    { ID id-NRUESidelinkAggregateMaximumBitrate    CRITICALITY ignore  TYPE NRUESidelinkAggregateMaximumBitrate        PRESENCE optional }|
    { ID id-LTEUESidelinkAggregateMaximumBitrate   CRITICALITY ignore  TYPE LTEUESidelinkAggregateMaximumBitrate       PRESENCE optional }|
    { ID id-PC5LinkAMBR                            CRITICALITY ignore  TYPE BitRate                                    PRESENCE optional}|
    { ID id-SLDRBs-ToBeSetupMod-List               CRITICALITY reject  TYPE SLDRBs-ToBeSetupMod-List                   PRESENCE optional }|
    { ID id-SLDRBs-ToBeModified-List               CRITICALITY reject  TYPE SLDRBs-ToBeModified-List                   PRESENCE optional }|
    { ID id-SLDRBs-ToBeReleased-List               CRITICALITY reject  TYPE SLDRBs-ToBeReleased-List                   PRESENCE optional }|
    { ID id-ConditionalIntraDUMobilityInformation  CRITICALITY reject  TYPE ConditionalIntraDUMobilityInformation      PRESENCE optional}|
    { ID id-F1CTransferPath                        CRITICALITY reject  TYPE F1CTransferPath                            PRESENCE optional }|
    { ID id-SCGIndicator                           CRITICALITY ignore  TYPE SCGIndicator                               PRESENCE optional }|
```

```
	{ ID id-UplinkTxDirectCurrentTwoCarrierListInfo	CRITICALITY ignore	TYPE UplinkTxDirectCurrentTwoCarrierListInfo	PRESENCE optional }|
	{ ID id-IABConditionalRRCMessageDeliveryIndication		CRITICALITY reject	TYPE IABConditionalRRCMessageDeliveryIndication
		PRESENCE optional	}|
	{ ID id-F1CTransferPathNRDC	CRITICALITY reject	TYPE F1CTransferPathNRDC	PRESENCE optional }|
	{ ID id-MDTPollutedMeasurementIndicator	CRITICALITY ignore	TYPE MDTPollutedMeasurementIndicator	PRESENCE optional }|
	{ ID id-SCGActivationRequest	CRITICALITY ignore	TYPE SCGActivationRequest	PRESENCE optional }|
	{ ID id-CG-SDTQueryIndication	CRITICALITY ignore	TYPE CG-SDTQueryIndication	PRESENCE optional }|
	{ ID id-FiveG-ProSeAuthorized	CRITICALITY ignore	TYPE FiveG-ProSeAuthorized	PRESENCE optional }|
	{ ID id-FiveG-ProSeUEPC5AggregateMaximumBitrate	CRITICALITY ignore	TYPE NRUESidelinkAggregateMaximumBitrate	PRESENCE optional }|
	{ ID id-FiveG-ProSePC5LinkAMBR	CRITICALITY ignore	TYPE BitRate	PRESENCE optional}|
	{ ID id-UpdatedRemoteUELocalID	CRITICALITY ignore	TYPE RemoteUELocalID	PRESENCE optional }|
	{ ID id-UuRLCChannelToBeSetupList	CRITICALITY reject	TYPE UuRLCChannelToBeSetupList	PRESENCE optional}|
	{ ID id-UuRLCChannelToBeModifiedList	CRITICALITY reject	TYPE UuRLCChannelToBeModifiedList	PRESENCE optional}|
	{ ID id-UuRLCChannelToBeReleasedList	CRITICALITY reject	TYPE UuRLCChannelToBeReleasedList	PRESENCE optional}|
	{ ID id-PC5RLCChannelToBeSetupList	CRITICALITY reject	TYPE PC5RLCChannelToBeSetupList	PRESENCE optional}|
	{ ID id-PC5RLCChannelToBeModifiedList	CRITICALITY reject	TYPE PC5RLCChannelToBeModifiedList	PRESENCE optional}|
	{ ID id-PC5RLCChannelToBeReleasedList	CRITICALITY reject	TYPE PC5RLCChannelToBeReleasedList	PRESENCE optional}|
	{ ID id-PathSwitchConfiguration	CRITICALITY ignore	TYPE PathSwitchConfiguration	PRESENCE optional }|
	{ ID id-GNBDUUESliceMaximumBitRateList	CRITICALITY ignore	TYPE GNBDUUESliceMaximumBitRateList	PRESENCE optional }|
	{ ID id-MulticastMBSSessionSetupList	CRITICALITY reject	TYPE MulticastMBSSessionList	PRESENCE optional }|
	{ ID id-MulticastMBSSessionRemoveList	CRITICALITY reject	TYPE MulticastMBSSessionList	PRESENCE optional }|
	{ ID id-UE-MulticastMRBs-ToBeSetup-atModify-List	CRITICALITY reject	TYPE UE-MulticastMRBs-ToBeSetup-atModify-List	PRESENCE optional }|
	{ ID id-UE-MulticastMRBs-ToBeReleased-List	CRITICALITY reject	TYPE UE-MulticastMRBs-ToBeReleased-List	PRESENCE optional }|
	{ ID id-SLDRXCycleList	CRITICALITY ignore	TYPE SLDRXCycleList	PRESENCE optional }|
	{ ID id-ManagementBasedMDTPLMNModificationList	CRITICALITY ignore	TYPE MDTPLMNModificationList	PRESENCE optional }|
	{ ID id-SDTBearerConfigurationQueryIndication	CRITICALITY ignore	TYPE SDTBearerConfigurationQueryIndication	PRESENCE optional }|
	{ ID id-DAPS-HO-Status	CRITICALITY ignore	TYPE DAPS-HO-Status	PRESENCE optional }|
	{ ID id-ServingCellMO-List	CRITICALITY ignore	TYPE ServingCellMO-List	PRESENCE optional }|
	{ ID id-UlTxDirectCurrentMoreCarrierInformation	CRITICALITY ignore	TYPE UlTxDirectCurrentMoreCarrierInformation	PRESENCE optional }|
	{ ID id-CPACMCGInformation	CRITICALITY ignore	TYPE CPACMCGInformation	PRESENCE optional }|
	{ ID id-NetworkControlledRepeaterAuthorized	CRITICALITY ignore	TYPE NetworkControlledRepeaterAuthorized	PRESENCE optional }|
	{ ID id-SDT-Volume-Threshold		CRITICALITY ignore	TYPE SDT-Volume-Threshold		PRESENCE optional
}|
	{ ID id-LTMInformation-Modify	CRITICALITY reject	TYPE LTMInformation-Modify	PRESENCE optional }|
	{ ID id-LTMCFRAResourceConfig-List	CRITICALITY ignore	TYPE LTMCFRAResourceConfig-List	PRESENCE optional }|
	{ ID id-LTMConfigurationIDMappingList	CRITICALITY reject	TYPE LTMConfigurationIDMappingList	PRESENCE optional }|
	{ ID id-EarlySyncInformation-Request	CRITICALITY ignore	TYPE EarlySyncInformation-Request	PRESENCE optional }|
	{ ID id-EarlySyncCandidateCellInformation-List	CRITICALITY ignore	TYPE EarlySyncCandidateCellInformation-List	PRESENCE optional }|
	{ ID id-EarlySyncServingCellInformation	CRITICALITY ignore	TYPE EarlySyncServingCellInformation	PRESENCE optional }|
	{ ID id-LTMCells-ToBeReleased-List	CRITICALITY reject	TYPE LTMCells-ToBeReleased-List	PRESENCE optional }|
	{ ID id-PathAdditionInformation	CRITICALITY reject	TYPE PathAdditionInformation	PRESENCE optional}|
	{ ID id-NRA2XServicesAuthorized	CRITICALITY ignore	TYPE NRA2XServicesAuthorized	PRESENCE optional }|
	{ ID id-LTEA2XServicesAuthorized		CRITICALITY ignore	TYPE LTEA2XServicesAuthorized		PRESENCE optional }|
	{ ID id-NRUESidelinkAggregateMaximumBitrateForA2X		CRITICALITY ignore	TYPE NRUESidelinkAggregateMaximumBitrate		PRESENCE optional }|
	{ ID id-LTEUESidelinkAggregateMaximumBitrateForA2X	CRITICALITY ignore	TYPE LTEUESidelinkAggregateMaximumBitrate	PRESENCE optional }|
	{ ID id-DLLBTFailureInformationRequest	CRITICALITY ignore	TYPE DLLBTFailureInformationRequest	PRESENCE optional }|
	{ ID id-SLPositioning-Ranging-Service-Info	CRITICALITY ignore	TYPE SLPositioning-Ranging-Service-Info		PRESENCE optional }|
	{ ID id-NonIntegerDRXCycle	CRITICALITY ignore	TYPE NonIntegerDRXCycle	PRESENCE optional }|
	{ ID id-LTMResetInformation	CRITICALITY ignore	TYPE LTMResetInformation	PRESENCE optional }|
	{ ID id-LTMTCIStatesConfigurationsList	CRITICALITY reject	TYPE LTMTCIStatesConfigurationsList	PRESENCE optional }|
	{ ID id-LTMSecurityInformation	CRITICALITY reject	TYPE LTMSecurityInformation	PRESENCE optional }|
	{ ID id-LTMInformationSCGMod	CRITICALITY reject	TYPE LTMInformationSCGMod	PRESENCE optional }|
	{ ID id-InterSNLTMMCGInformation	CRITICALITY ignore	TYPE InterSNLTMMCGInformation	PRESENCE optional },
	...
```

```
}

SCell-ToBeSetupMod-List::= SEQUENCE (SIZE(1..maxnoofSCells)) OF ProtocolIE-SingleContainer { { SCell-ToBeSetupMod-ItemIEs} }
SCell-ToBeRemoved-List::= SEQUENCE (SIZE(1..maxnoofSCells)) OF ProtocolIE-SingleContainer { { SCell-ToBeRemoved-ItemIEs} }
SRBs-ToBeSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofSRBs)) OF ProtocolIE-SingleContainer { { SRBs-ToBeSetupMod-ItemIEs} }
DRBs-ToBeSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-ToBeSetupMod-ItemIEs} }
BHChannels-ToBeSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF ProtocolIE-SingleContainer { { BHChannels-ToBeSetupMod-ItemIEs} }

DRBs-ToBeModified-List ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-ToBeModified-ItemIEs} }
BHChannels-ToBeModified-List ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF ProtocolIE-SingleContainer { { BHChannels-ToBeModified-ItemIEs} }
SRBs-ToBeReleased-List ::= SEQUENCE (SIZE(1..maxnoofSRBs)) OF ProtocolIE-SingleContainer { { SRBs-ToBeReleased-ItemIEs} }
DRBs-ToBeReleased-List ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-ToBeReleased-ItemIEs} }
BHChannels-ToBeReleased-List ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF ProtocolIE-SingleContainer { { BHChannels-ToBeReleased-ItemIEs} }
UE-MulticastMRBs-ToBeSetup-atModify-List ::= SEQUENCE (SIZE(1..maxnoofMRBsforUE)) OF
                                        ProtocolIE-SingleContainer { { UE-MulticastMRBs-ToBeSetup-atModify-ItemIEs} }

UE-MulticastMRBs-ToBeReleased-List ::= SEQUENCE (SIZE(1..maxnoofMRBsforUE)) OF ProtocolIE-SingleContainer { { UE-MulticastMRBs-ToBeReleased-
ItemIEs} }

SCell-ToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SCell-ToBeSetupMod-Item         CRITICALITY ignore  TYPE SCell-ToBeSetupMod-Item        PRESENCE mandatory   },
    ...
}

SCell-ToBeRemoved-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SCell-ToBeRemoved-Item          CRITICALITY ignore  TYPE SCell-ToBeRemoved-Item      PRESENCE mandatory   },
    ...
}


SRBs-ToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SRBs-ToBeSetupMod-Item      CRITICALITY reject  TYPE SRBs-ToBeSetupMod-Item     PRESENCE mandatory},
    ...
}


DRBs-ToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-DRBs-ToBeSetupMod-Item      CRITICALITY reject  TYPE DRBs-ToBeSetupMod-Item     PRESENCE mandatory},
    ...
}

DRBs-ToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-DRBs-ToBeModified-Item      CRITICALITY reject  TYPE DRBs-ToBeModified-Item         PRESENCE mandatory},
    ...
}


SRBs-ToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SRBs-ToBeReleased-Item  CRITICALITY reject  TYPE SRBs-ToBeReleased-Item     PRESENCE mandatory},
    ...
}

DRBs-ToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-DRBs-ToBeReleased-Item      CRITICALITY reject  TYPE DRBs-ToBeReleased-Item     PRESENCE mandatory},
```

```
	...
}

BHChannels-ToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BHChannels-ToBeSetupMod-Item		CRITICALITY reject	TYPE BHChannels-ToBeSetupMod-Item		PRESENCE mandatory},
	...
}

BHChannels-ToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BHChannels-ToBeModified-Item		CRITICALITY reject	TYPE BHChannels-ToBeModified-Item		PRESENCE mandatory},
	...
}

BHChannels-ToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BHChannels-ToBeReleased-Item		CRITICALITY reject	TYPE BHChannels-ToBeReleased-Item		PRESENCE mandatory},
	...
}

SLDRBs-ToBeSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-ToBeSetupMod-ItemIEs} }
SLDRBs-ToBeModified-List ::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-ToBeModified-ItemIEs} }
SLDRBs-ToBeReleased-List ::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-ToBeReleased-ItemIEs} }

SLDRBs-ToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SLDRBs-ToBeSetupMod-Item		CRITICALITY reject	TYPE SLDRBs-ToBeSetupMod-Item		PRESENCE mandatory},
	...
}

SLDRBs-ToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SLDRBs-ToBeModified-Item		CRITICALITY reject	TYPE SLDRBs-ToBeModified-Item		PRESENCE mandatory},
	...
}

SLDRBs-ToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SLDRBs-ToBeReleased-Item		CRITICALITY reject	TYPE SLDRBs-ToBeReleased-Item		PRESENCE mandatory},
	...
}

UE-MulticastMRBs-ToBeSetup-atModify-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-UE-MulticastMRBs-ToBeSetup-atModify-Item	CRITICALITY reject	TYPE UE-MulticastMRBs-ToBeSetup-atModify-Item		PRESENCE mandatory},
	...
}


UE-MulticastMRBs-ToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-UE-MulticastMRBs-ToBeReleased-Item		CRITICALITY reject	TYPE UE-MulticastMRBs-ToBeReleased-Item		PRESENCE mandatory},
	...
}

-- **************************************************************
--
-- UE CONTEXT MODIFICATION RESPONSE
--
-- **************************************************************
```

```
UEContextModificationResponse ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { UEContextModificationResponseIEs} },
    ...
}


UEContextModificationResponseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID                     CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID                      PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID                     CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID                      PRESENCE mandatory   }|
    { ID id-ResourceCoordinationTransferContainer   CRITICALITY ignore  TYPE ResourceCoordinationTransferContainer  PRESENCE optional }|
    { ID id-DUtoCURRCInformation                  CRITICALITY reject  TYPE DUtoCURRCInformation                   PRESENCE optional}|
    { ID id-DRBs-SetupMod-List                    CRITICALITY ignore  TYPE DRBs-SetupMod-List                     PRESENCE optional}|
    { ID id-DRBs-Modified-List                    CRITICALITY ignore  TYPE DRBs-Modified-List                     PRESENCE optional}|
    { ID id-SRBs-FailedToBeSetupMod-List          CRITICALITY ignore  TYPE SRBs-FailedToBeSetupMod-List           PRESENCE optional }|
    { ID id-DRBs-FailedToBeSetupMod-List          CRITICALITY ignore  TYPE DRBs-FailedToBeSetupMod-List           PRESENCE optional }|
    { ID id-SCell-FailedtoSetupMod-List           CRITICALITY ignore  TYPE SCell-FailedtoSetupMod-List            PRESENCE optional }|
    { ID id-DRBs-FailedToBeModified-List          CRITICALITY ignore  TYPE DRBs-FailedToBeModified-List           PRESENCE optional }|
    { ID id-InactivityMonitoringResponse          CRITICALITY reject  TYPE InactivityMonitoringResponse           PRESENCE optional }|
    { ID id-CriticalityDiagnostics                CRITICALITY ignore  TYPE CriticalityDiagnostics                 PRESENCE optional }|
    { ID id-C-RNTI                                CRITICALITY ignore  TYPE C-RNTI                                 PRESENCE optional }|
    { ID id-Associated-SCell-List                 CRITICALITY ignore  TYPE Associated-SCell-List                  PRESENCE optional }|
    { ID id-SRBs-SetupMod-List                    CRITICALITY ignore  TYPE SRBs-SetupMod-List                     PRESENCE optional }|
    { ID id-SRBs-Modified-List                    CRITICALITY ignore  TYPE SRBs-Modified-List                     PRESENCE optional }|
    { ID id-FullConfiguration                     CRITICALITY reject  TYPE FullConfiguration                      PRESENCE optional }|
    { ID id-BHChannels-SetupMod-List              CRITICALITY ignore  TYPE BHChannels-SetupMod-List               PRESENCE optional}|
    { ID id-BHChannels-Modified-List              CRITICALITY ignore  TYPE BHChannels-Modified-List               PRESENCE optional}|
    { ID id-BHChannels-FailedToBeSetupMod-List    CRITICALITY ignore  TYPE BHChannels-FailedToBeSetupMod-List     PRESENCE optional }|
    { ID id-BHChannels-FailedToBeModified-List    CRITICALITY ignore  TYPE BHChannels-FailedToBeModified-List     PRESENCE optional }|
    { ID id-SLDRBs-SetupMod-List                  CRITICALITY ignore  TYPE SLDRBs-SetupMod-List                   PRESENCE optional }|
    { ID id-SLDRBs-Modified-List                  CRITICALITY ignore  TYPE SLDRBs-Modified-List                   PRESENCE optional }|
    { ID id-SLDRBs-FailedToBeSetupMod-List        CRITICALITY ignore  TYPE SLDRBs-FailedToBeSetupMod-List         PRESENCE optional }|
    { ID id-SLDRBs-FailedToBeModified-List        CRITICALITY ignore  TYPE SLDRBs-FailedToBeModified-List         PRESENCE optional }|
    { ID id-requestedTargetCellGlobalID           CRITICALITY reject  TYPE NRCGI                                  PRESENCE optional}|
    { ID id-SCGActivationStatus                   CRITICALITY ignore  TYPE SCGActivationStatus                    PRESENCE optional }|
    { ID id-UuRLCChannelSetupList                 CRITICALITY ignore  TYPE UuRLCChannelSetupList                  PRESENCE optional}|
    { ID id-UuRLCChannelFailedToBeSetupList       CRITICALITY ignore  TYPE UuRLCChannelFailedToBeSetupList        PRESENCE optional}|
    { ID id-UuRLCChannelModifiedList              CRITICALITY ignore  TYPE UuRLCChannelModifiedList               PRESENCE optional}|
    { ID id-UuRLCChannelFailedToBeModifiedList    CRITICALITY ignore  TYPE UuRLCChannelFailedToBeModifiedList     PRESENCE optional}|
    { ID id-PC5RLCChannelSetupList                CRITICALITY ignore  TYPE PC5RLCChannelSetupList                 PRESENCE optional}|
    { ID id-PC5RLCChannelFailedToBeSetupList      CRITICALITY ignore  TYPE PC5RLCChannelFailedToBeSetupList       PRESENCE optional}|
    { ID id-PC5RLCChannelModifiedList             CRITICALITY ignore  TYPE PC5RLCChannelModifiedList              PRESENCE optional}|
    { ID id-PC5RLCChannelFailedToBeModifiedList   CRITICALITY ignore  TYPE PC5RLCChannelFailedToBeModifiedList    PRESENCE optional}|
    { ID id-SDTBearerConfigurationInfo            CRITICALITY ignore  TYPE SDTBearerConfigurationInfo             PRESENCE optional}|
    { ID id-UE-MulticastMRBs-Setup-List           CRITICALITY reject  TYPE UE-MulticastMRBs-Setup-List            PRESENCE optional}|
    { ID id-ServingCellMO-encoded-in-CGC-List     CRITICALITY ignore  TYPE ServingCellMO-encoded-in-CGC-List      PRESENCE optional}|
    { ID id-DedicatedSIDeliveryIndication         CRITICALITY ignore  TYPE DedicatedSIDeliveryIndication          PRESENCE optional}|
    { ID id-Configured-BWP-List                   CRITICALITY ignore  TYPE Configured-BWP-List                    PRESENCE optional}|
    { ID id-EarlySyncInformation                  CRITICALITY ignore  TYPE EarlySyncInformation                   PRESENCE optional}|
    { ID id-LTMConfiguration                      CRITICALITY ignore  TYPE LTMConfiguration                       PRESENCE optional}|
    { ID id-S-CPAC-Configuration                  CRITICALITY ignore  TYPE S-CPAC-Configuration                   PRESENCE optional }|
    { ID id-TARemainingInfoList                   CRITICALITY ignore  TYPE TARemainingInfoList                    PRESENCE optional },
    ...
}
```

```
DRBs-SetupMod-List ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-SetupMod-ItemIEs} }
DRBs-Modified-List::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-Modified-ItemIEs } }
SRBs-SetupMod-List ::= SEQUENCE (SIZE(1..maxnoofSRBs)) OF ProtocolIE-SingleContainer { { SRBs-SetupMod-ItemIEs} }
SRBs-Modified-List ::= SEQUENCE (SIZE(1..maxnoofSRBs)) OF ProtocolIE-SingleContainer { { SRBs-Modified-ItemIEs } }
DRBs-FailedToBeModified-List ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-FailedToBeModified-ItemIEs} }
SRBs-FailedToBeSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofSRBs)) OF ProtocolIE-SingleContainer { { SRBs-FailedToBeSetupMod-ItemIEs} }
DRBs-FailedToBeSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-FailedToBeSetupMod-ItemIEs} }
SCell-FailedtoSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofSCells)) OF ProtocolIE-SingleContainer { { SCell-FailedtoSetupMod-ItemIEs} }
BHChannels-SetupMod-List ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF ProtocolIE-SingleContainer { { BHChannels-SetupMod-ItemIEs} }
BHChannels-Modified-List ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF ProtocolIE-SingleContainer { { BHChannels-Modified-ItemIEs } }
BHChannels-FailedToBeModified-List ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF ProtocolIE-SingleContainer { { BHChannels-FailedToBeModified-
ItemIEs} }
BHChannels-FailedToBeSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF ProtocolIE-SingleContainer { { BHChannels-FailedToBeSetupMod-
ItemIEs} }

Associated-SCell-List ::= SEQUENCE (SIZE(1.. maxnoofSCells)) OF ProtocolIE-SingleContainer { { Associated-SCell-ItemIEs} }

DRBs-SetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-DRBs-SetupMod-Item      CRITICALITY ignore      TYPE DRBs-SetupMod-Item     PRESENCE mandatory},
    ...
}


DRBs-Modified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-DRBs-Modified-Item          CRITICALITY ignore  TYPE DRBs-Modified-Item     PRESENCE mandatory},
    ...
}

SRBs-SetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SRBs-SetupMod-Item      CRITICALITY ignore      TYPE SRBs-SetupMod-Item     PRESENCE mandatory},
    ...
}


SRBs-Modified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SRBs-Modified-Item          CRITICALITY ignore  TYPE SRBs-Modified-Item     PRESENCE mandatory},
    ...
}

SRBs-FailedToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SRBs-FailedToBeSetupMod-Item        CRITICALITY ignore  TYPE SRBs-FailedToBeSetupMod-Item     PRESENCE mandatory},
    ...
}


DRBs-FailedToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-DRBs-FailedToBeSetupMod-Item        CRITICALITY ignore  TYPE DRBs-FailedToBeSetupMod-Item     PRESENCE mandatory},
    ...
}


DRBs-FailedToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-DRBs-FailedToBeModified-Item        CRITICALITY ignore  TYPE DRBs-FailedToBeModified-Item     PRESENCE mandatory},
```

```
	...
}

SCell-FailedtoSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SCell-FailedtoSetupMod-Item		CRITICALITY ignore	TYPE SCell-FailedtoSetupMod-Item		PRESENCE mandatory},
	...
}

Associated-SCell-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-Associated-SCell-Item			CRITICALITY ignore	TYPE Associated-SCell-Item		PRESENCE mandatory},
	...
}

BHChannels-SetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BHChannels-SetupMod-Item		CRITICALITY ignore		TYPE BHChannels-SetupMod-Item	PRESENCE mandatory},
	...
}


BHChannels-Modified-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BHChannels-Modified-Item		CRITICALITY ignore	TYPE BHChannels-Modified-Item		PRESENCE mandatory},
	...
}

BHChannels-FailedToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BHChannels-FailedToBeSetupMod-Item		CRITICALITY ignore	TYPE BHChannels-FailedToBeSetupMod-Item		PRESENCE mandatory},
	...
}

BHChannels-FailedToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BHChannels-FailedToBeModified-Item		CRITICALITY ignore	TYPE BHChannels-FailedToBeModified-Item		PRESENCE mandatory},
	...
}

SLDRBs-SetupMod-List			::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-SetupMod-ItemIEs} }
SLDRBs-Modified-List				::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-Modified-ItemIEs } }
SLDRBs-FailedToBeModified-List	::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-FailedToBeModified-ItemIEs} }
SLDRBs-FailedToBeSetupMod-List	::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-FailedToBeSetupMod-ItemIEs} }

SLDRBs-SetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SLDRBs-SetupMod-Item		CRITICALITY ignore		TYPE SLDRBs-SetupMod-Item		PRESENCE mandatory},
	...
}

SLDRBs-Modified-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SLDRBs-Modified-Item			CRITICALITY ignore	TYPE SLDRBs-Modified-Item		PRESENCE mandatory},
	...
}

SLDRBs-FailedToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SLDRBs-FailedToBeSetupMod-Item		CRITICALITY ignore	TYPE SLDRBs-FailedToBeSetupMod-Item	PRESENCE mandatory},
	...
}
```

```
SLDRBs-FailedToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SLDRBs-FailedToBeModified-Item      CRITICALITY ignore  TYPE SLDRBs-FailedToBeModified-Item    PRESENCE mandatory},
    ...
}

UE-MulticastMRBs-Setup-List ::= SEQUENCE (SIZE(1..maxnoofMRBsforUE)) OF ProtocolIE-SingleContainer { { UE-MulticastMRBs-Setup-ItemIEs } }

UE-MulticastMRBs-Setup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-UE-MulticastMRBs-Setup-Item     CRITICALITY reject  TYPE UE-MulticastMRBs-Setup-Item     PRESENCE mandatory},
    ...
}


-- **************************************************************
--
-- UE CONTEXT MODIFICATION FAILURE
--
-- **************************************************************

UEContextModificationFailure ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { UEContextModificationFailureIEs} },
    ...
}

UEContextModificationFailureIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID               CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID           PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID               CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID           PRESENCE mandatory   }|
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                       PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics      PRESENCE optional }|
    { ID id-requestedTargetCellGlobalID     CRITICALITY reject  TYPE NRCGI                       PRESENCE optional},
    ...
}


-- **************************************************************
--
-- UE Context Modification Required (gNB-DU initiated) ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- UE CONTEXT MODIFICATION REQUIRED
--
-- **************************************************************

UEContextModificationRequired ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { UEContextModificationRequiredIEs} },
    ...
}

UEContextModificationRequiredIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID                           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID                          PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID                           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID                          PRESENCE mandatory   }|
```

```
	{ ID id-ResourceCoordinationTransferContainer		CRITICALITY ignore	TYPE ResourceCoordinationTransferContainer		PRESENCE optional }|
	{ ID id-DUtoCURRCInformation		CRITICALITY reject	TYPE DUtoCURRCInformation		PRESENCE optional}|
	{ ID id-DRBs-Required-ToBeModified-List		CRITICALITY reject	TYPE DRBs-Required-ToBeModified-List		PRESENCE optional}|
	{ ID id-SRBs-Required-ToBeReleased-List		CRITICALITY reject	TYPE SRBs-Required-ToBeReleased-List		PRESENCE optional}|
	{ ID id-DRBs-Required-ToBeReleased-List		CRITICALITY reject	TYPE DRBs-Required-ToBeReleased-List		PRESENCE optional}|
	{ ID id-Cause		CRITICALITY ignore	TYPE Cause		PRESENCE mandatory	}|
	{ ID id-BHChannels-Required-ToBeReleased-List		CRITICALITY reject	TYPE BHChannels-Required-ToBeReleased-List		PRESENCE optional}|
	{ ID id-SLDRBs-Required-ToBeModified-List		CRITICALITY reject	TYPE SLDRBs-Required-ToBeModified-List		PRESENCE optional}|
	{ ID id-SLDRBs-Required-ToBeReleased-List		CRITICALITY reject	TYPE SLDRBs-Required-ToBeReleased-List		PRESENCE optional}|
	{ ID id-targetCellsToCancel		CRITICALITY reject	TYPE TargetCellList		PRESENCE optional}|
	{ ID id-UuRLCChannelRequiredToBeModifiedList		CRITICALITY reject	TYPE UuRLCChannelRequiredToBeModifiedList		PRESENCE optional}|
	{ ID id-UuRLCChannelRequiredToBeReleasedList		CRITICALITY reject	TYPE UuRLCChannelRequiredToBeReleasedList		PRESENCE optional}|
	{ ID id-PC5RLCChannelRequiredToBeModifiedList		CRITICALITY reject	TYPE PC5RLCChannelRequiredToBeModifiedList		PRESENCE optional}|
	{ ID id-PC5RLCChannelRequiredToBeReleasedList		CRITICALITY reject	TYPE PC5RLCChannelRequiredToBeReleasedList		PRESENCE optional}|
	{ ID id-UE-MulticastMRBs-RequiredToBeModified-List	CRITICALITY reject	TYPE UE-MulticastMRBs-RequiredToBeModified-List	PRESENCE optional	}|
	{ ID id-UE-MulticastMRBs-RequiredToBeReleased-List	CRITICALITY reject	TYPE UE-MulticastMRBs-RequiredToBeReleased-List	PRESENCE optional	}|
	{ ID id-LTMCells-ToBeReleased-List		CRITICALITY reject	TYPE LTMCells-ToBeReleased-List		PRESENCE optional }|
	{ ID id-LTMConfig-to-Modify-List		CRITICALITY ignore	TYPE LTMConfig-to-Modify-List		PRESENCE optional },
	...
}

DRBs-Required-ToBeModified-List::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-Required-ToBeModified-ItemIEs } }
DRBs-Required-ToBeReleased-List::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-Required-ToBeReleased-ItemIEs } }

SRBs-Required-ToBeReleased-List::= SEQUENCE (SIZE(1..maxnoofSRBs)) OF ProtocolIE-SingleContainer { { SRBs-Required-ToBeReleased-ItemIEs } }

BHChannels-Required-ToBeReleased-List ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF ProtocolIE-SingleContainer { { BHChannels-Required-
ToBeReleased-ItemIEs } }

DRBs-Required-ToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-DRBs-Required-ToBeModified-Item		CRITICALITY reject	TYPE DRBs-Required-ToBeModified-Item		PRESENCE mandatory},
	...
}

DRBs-Required-ToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-DRBs-Required-ToBeReleased-Item		CRITICALITY reject	TYPE DRBs-Required-ToBeReleased-Item		PRESENCE mandatory},
	...
}

SRBs-Required-ToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SRBs-Required-ToBeReleased-Item		CRITICALITY reject	TYPE SRBs-Required-ToBeReleased-Item			PRESENCE mandatory},
	...
}

BHChannels-Required-ToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BHChannels-Required-ToBeReleased-Item			CRITICALITY reject	TYPE BHChannels-Required-ToBeReleased-Item		PRESENCE mandatory},
	...
}

SLDRBs-Required-ToBeModified-List::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-Required-ToBeModified-ItemIEs } }
SLDRBs-Required-ToBeReleased-List::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-Required-ToBeReleased-ItemIEs } }

SLDRBs-Required-ToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SLDRBs-Required-ToBeModified-Item			CRITICALITY reject	TYPE SLDRBs-Required-ToBeModified-Item		PRESENCE mandatory},
```

```
    ...
}

SLDRBs-Required-ToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SLDRBs-Required-ToBeReleased-Item           CRITICALITY reject  TYPE SLDRBs-Required-ToBeReleased-Item    PRESENCE mandatory},
    ...
}

UE-MulticastMRBs-RequiredToBeModified-List ::= SEQUENCE (SIZE(1..maxnoofMRBsforUE)) OF
                            ProtocolIE-SingleContainer { { UE-MulticastMRBs-RequiredToBeModified-ItemIEs} }

UE-MulticastMRBs-RequiredToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-UE-MulticastMRBs-RequiredToBeModified-Item  CRITICALITY reject  TYPE UE-MulticastMRBs-RequiredToBeModified-Item    PRESENCE mandatory},
    ...
}

UE-MulticastMRBs-RequiredToBeReleased-List ::= SEQUENCE (SIZE(1..maxnoofMRBsforUE)) OF
                            ProtocolIE-SingleContainer { { UE-MulticastMRBs-RequiredToBeReleased-ItemIEs} }

UE-MulticastMRBs-RequiredToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-UE-MulticastMRBs-RequiredToBeReleased-Item      CRITICALITY reject  TYPE UE-MulticastMRBs-RequiredToBeReleased-Item   PRESENCE 
mandatory},
    ...
}


-- **************************************************************
--
-- UE CONTEXT MODIFICATION CONFIRM
--
-- **************************************************************

UEContextModificationConfirm::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { UEContextModificationConfirmIEs} },
    ...
}


UEContextModificationConfirmIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID                           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID                          PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID                           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID                          PRESENCE mandatory   }|
    { ID id-ResourceCoordinationTransferContainer       CRITICALITY ignore  TYPE ResourceCoordinationTransferContainer      PRESENCE optional    }|
    { ID id-DRBs-ModifiedConf-List                      CRITICALITY ignore  TYPE DRBs-ModifiedConf-List                     PRESENCE optional    }|
    { ID id-RRCContainer                                CRITICALITY ignore  TYPE RRCContainer                               PRESENCE optional    }|
    { ID id-CriticalityDiagnostics                      CRITICALITY ignore  TYPE CriticalityDiagnostics                     PRESENCE optional    }|
    { ID id-ExecuteDuplication                          CRITICALITY ignore  TYPE ExecuteDuplication                         PRESENCE optional    }|
    { ID id-ResourceCoordinationTransferInformation     CRITICALITY ignore  TYPE ResourceCoordinationTransferInformation    PRESENCE optional    }|
    { ID id-SLDRBs-ModifiedConf-List                    CRITICALITY ignore  TYPE SLDRBs-ModifiedConf-List                   PRESENCE optional    }|
    { ID id-UuRLCChannelModifiedList                    CRITICALITY reject  TYPE UuRLCChannelModifiedList                   PRESENCE optional    }|
    { ID id-PC5RLCChannelModifiedList                   CRITICALITY reject  TYPE PC5RLCChannelModifiedList                  PRESENCE optional    }|
    { ID id-UE-MulticastMRBs-ConfirmedToBeModified-List CRITICALITY reject  TYPE UE-MulticastMRBs-ConfirmedToBeModified-ListPRESENCE optional    },
    ...
}
```

```
DRBs-ModifiedConf-List::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRBs-ModifiedConf-ItemIEs } }

DRBs-ModifiedConf-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-DRBs-ModifiedConf-Item		CRITICALITY ignore	TYPE DRBs-ModifiedConf-Item		PRESENCE mandatory},
	...
}

SLDRBs-ModifiedConf-List::= SEQUENCE (SIZE(1..maxnoofSLDRBs)) OF ProtocolIE-SingleContainer { { SLDRBs-ModifiedConf-ItemIEs } }

SLDRBs-ModifiedConf-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-SLDRBs-ModifiedConf-Item		CRITICALITY ignore	TYPE SLDRBs-ModifiedConf-Item		PRESENCE mandatory},
	...
}

UE-MulticastMRBs-ConfirmedToBeModified-List ::= SEQUENCE (SIZE(1..maxnoofMRBsforUE)) OF
							ProtocolIE-SingleContainer { { UE-MulticastMRBs-ConfirmedToBeModified-ItemIEs} }

UE-MulticastMRBs-ConfirmedToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-UE-MulticastMRBs-ConfirmedToBeModified-Item CRITICALITY reject	TYPE UE-MulticastMRBs-ConfirmedToBeModified-Item	PRESENCE mandatory},
	...
}


-- **************************************************************
--
-- UE CONTEXT MODIFICATION REFUSE
--
-- **************************************************************

UEContextModificationRefuse::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { UEContextModificationRefuseIEs} },
	...
}


UEContextModificationRefuseIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-Cause					CRITICALITY ignore	TYPE Cause					PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional },
	...
}


-- **************************************************************
--
-- WRITE-REPLACE WARNING ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Write-Replace Warning Request
--
```

```
-- **************************************************************

WriteReplaceWarningRequest ::= SEQUENCE {
    protocolIEs ProtocolIE-Container { {WriteReplaceWarningRequestIEs} },
    ...
}

WriteReplaceWarningRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                       CRITICALITY reject  TYPE TransactionID                       PRESENCE mandatory   }|
    { ID id-PWSSystemInformation                CRITICALITY reject  TYPE PWSSystemInformation                PRESENCE mandatory }|
    { ID id-RepetitionPeriod                    CRITICALITY reject  TYPE RepetitionPeriod                    PRESENCE mandatory }|
    { ID id-NumberofBroadcastRequest            CRITICALITY reject  TYPE NumberofBroadcastRequest            PRESENCE mandatory }|
    { ID id-Cells-To-Be-Broadcast-List          CRITICALITY reject  TYPE Cells-To-Be-Broadcast-List          PRESENCE optional },
    ...
}

Cells-To-Be-Broadcast-List      ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Cells-To-Be-Broadcast-List-ItemIEs } }

Cells-To-Be-Broadcast-List-ItemIEs F1AP-PROTOCOL-IES    ::= {
    { ID id-Cells-To-Be-Broadcast-Item      CRITICALITY reject  TYPE    Cells-To-Be-Broadcast-Item     PRESENCE mandatory   },
    ...
}

-- **************************************************************
--
-- Write-Replace Warning Response
--
-- **************************************************************

WriteReplaceWarningResponse ::= SEQUENCE {
    protocolIEs ProtocolIE-Container { {WriteReplaceWarningResponseIEs} },
    ...
}

WriteReplaceWarningResponseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                           CRITICALITY reject  TYPE TransactionID                           PRESENCE mandatory   }|
    { ID id-Cells-Broadcast-Completed-List          CRITICALITY reject  TYPE Cells-Broadcast-Completed-List          PRESENCE optional }|
    { ID id-CriticalityDiagnostics                  CRITICALITY ignore  TYPE CriticalityDiagnostics                  PRESENCE optional }|
    { ID id-Dedicated-SIDelivery-NeededUE-List      CRITICALITY ignore  TYPE Dedicated-SIDelivery-NeededUE-List      PRESENCE optional },
    ...
}

Cells-Broadcast-Completed-List      ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Cells-Broadcast-Completed-List-
ItemIEs } }

Cells-Broadcast-Completed-List-ItemIEs F1AP-PROTOCOL-IES    ::= {
    { ID id-Cells-Broadcast-Completed-Item      CRITICALITY reject  TYPE    Cells-Broadcast-Completed-Item   PRESENCE mandatory   },
    ...
}


-- **************************************************************
--
-- PWS CANCEL ELEMENTARY PROCEDURE
```

```
--
-- **************************************************************

-- **************************************************************
--
-- PWS Cancel Request
--
-- **************************************************************

PWSCancelRequest ::= SEQUENCE {
    protocolIEs ProtocolIE-Container { {PWSCancelRequestIEs} },
    ...
}

PWSCancelRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                          CRITICALITY reject TYPE TransactionID                          PRESENCE mandatory   }|
    { ID id-NumberofBroadcastRequest               CRITICALITY reject TYPE NumberofBroadcastRequest               PRESENCE mandatory }|
    { ID id-Broadcast-To-Be-Cancelled-List         CRITICALITY reject TYPE Broadcast-To-Be-Cancelled-List         PRESENCE optional }|
    { ID id-Cancel-all-Warning-Messages-Indicator  CRITICALITY reject TYPE Cancel-all-Warning-Messages-Indicator  PRESENCE optional }|
    { ID id-NotificationInformation                CRITICALITY reject TYPE NotificationInformation                PRESENCE optional},
    ...
}

Broadcast-To-Be-Cancelled-List      ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Broadcast-To-Be-Cancelled-List-
ItemIEs } }

Broadcast-To-Be-Cancelled-List-ItemIEs F1AP-PROTOCOL-IES    ::= {
    { ID id-Broadcast-To-Be-Cancelled-Item      CRITICALITY reject  TYPE    Broadcast-To-Be-Cancelled-Item   PRESENCE mandatory   },
    ...
}

-- **************************************************************
--
-- PWS Cancel Response
--
-- **************************************************************

PWSCancelResponse ::= SEQUENCE {
    protocolIEs ProtocolIE-Container { {PWSCancelResponseIEs} },
    ...
}

PWSCancelResponseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                  CRITICALITY reject  TYPE TransactionID                  PRESENCE mandatory   }|
    { ID id-Cells-Broadcast-Cancelled-List  CRITICALITY reject  TYPE Cells-Broadcast-Cancelled-List  PRESENCE optional }|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics          PRESENCE optional },
    ...
}

Cells-Broadcast-Cancelled-List      ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { Cells-Broadcast-Cancelled-List-
ItemIEs } }

Cells-Broadcast-Cancelled-List-ItemIEs F1AP-PROTOCOL-IES    ::= {
    { ID id-Cells-Broadcast-Cancelled-Item      CRITICALITY reject  TYPE    Cells-Broadcast-Cancelled-Item   PRESENCE mandatory   },
```

```
    ...
}

-- **************************************************************
--
-- UE Inactivity Notification ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- UE Inactivity Notification
--
-- **************************************************************

UEInactivityNotification ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ UEInactivityNotificationIEs}},
    ...
}

UEInactivityNotificationIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID                       CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID                    PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID                       CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID                    PRESENCE mandatory   }|
    { ID id-DRB-Activity-List                       CRITICALITY reject  TYPE DRB-Activity-List                    PRESENCE mandatory   }|
    { ID id-SDT-Termination-Request                 CRITICALITY ignore  TYPE SDT-Termination-Request              PRESENCE optional    },
    ...
}

DRB-Activity-List::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRB-Activity-ItemIEs } }

DRB-Activity-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-DRB-Activity-Item           CRITICALITY reject  TYPE DRB-Activity-Item      PRESENCE mandatory},
    ...
}
-- **************************************************************
--
-- Initial UL RRC Message Transfer ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- INITIAL UL RRC Message Transfer
--
-- **************************************************************

InitialULRRCMessageTransfer ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ InitialULRRCMessageTransferIEs}},
    ...
}

InitialULRRCMessageTransferIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-DU-UE-F1AP-ID                   CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID                PRESENCE mandatory   }|
    { ID id-NRCGI                               CRITICALITY reject  TYPE NRCGI                            PRESENCE mandatory   }|
```

```
    { ID id-C-RNTI                          CRITICALITY reject  TYPE C-RNTI                         PRESENCE mandatory   }|
    { ID id-RRCContainer                    CRITICALITY reject  TYPE RRCContainer                   PRESENCE mandatory   }|
    { ID id-DUtoCURRCContainer              CRITICALITY reject  TYPE DUtoCURRCContainer             PRESENCE optional }|
    { ID id-SULAccessIndication             CRITICALITY ignore  TYPE SULAccessIndication            PRESENCE optional }|
    { ID id-TransactionID                   CRITICALITY ignore  TYPE TransactionID                  PRESENCE mandatory   }|
    { ID id-RANUEID                         CRITICALITY ignore  TYPE RANUEID                        PRESENCE optional }|
    { ID id-RRCContainer-RRCSetupComplete   CRITICALITY ignore  TYPE RRCContainer-RRCSetupComplete     PRESENCE optional }|
    { ID id-NRRedCapUEIndication            CRITICALITY ignore  TYPE NRRedCapUEIndication           PRESENCE optional }|
    { ID id-SDTInformation                  CRITICALITY ignore  TYPE SDTInformation                 PRESENCE optional }|
    { ID id-SidelinkRelayConfiguration      CRITICALITY ignore  TYPE SidelinkRelayConfiguration     PRESENCE optional }|
{ ID id-NReRedCapUEIndication               CRITICALITY ignore  TYPE NReRedCapUEIndication          PRESENCE optional },
    ...
}


-- **************************************************************
--
-- DL RRC Message Transfer ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- DL RRC Message Transfer
--
-- **************************************************************

DLRRCMessageTransfer ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ DLRRCMessageTransferIEs}},
    ...
}

DLRRCMessageTransferIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID                       CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID                      PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID                       CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID                      PRESENCE mandatory   }|
    { ID id-oldgNB-DU-UE-F1AP-ID                    CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID                      PRESENCE optional }|
    { ID id-SRBID                                   CRITICALITY reject  TYPE SRBID                                  PRESENCE mandatory   }|
    { ID id-ExecuteDuplication                      CRITICALITY ignore  TYPE ExecuteDuplication                     PRESENCE optional}|
    { ID id-RRCContainer                            CRITICALITY reject  TYPE RRCContainer                           PRESENCE mandatory   }|
    { ID id-RAT-FrequencyPriorityInformation        CRITICALITY reject  TYPE RAT-FrequencyPriorityInformation       PRESENCE optional }|
    { ID id-RRCDeliveryStatusRequest                CRITICALITY ignore  TYPE RRCDeliveryStatusRequest               PRESENCE optional }|
    { ID id-UEContextNotRetrievable                 CRITICALITY reject  TYPE UEContextNotRetrievable                PRESENCE optional }|
    { ID id-RedirectedRRCmessage                    CRITICALITY reject  TYPE OCTET STRING                           PRESENCE optional }|
    { ID id-PLMNAssistanceInfoForNetShar            CRITICALITY ignore  TYPE PLMN-Identity                          PRESENCE optional }|
    { ID id-new-gNB-CU-UE-F1AP-ID                   CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID                      PRESENCE optional }|
    { ID id-AdditionalRRMPriorityIndex              CRITICALITY ignore  TYPE AdditionalRRMPriorityIndex             PRESENCE optional }|
    { ID id-SRBMappingInfo                          CRITICALITY ignore  TYPE UuRLCChannelID                         PRESENCE optional }|
    { ID id-PLMNIndexNRAssistanceInfoForNetShar                 CRITICALITY ignore  TYPE PLMNIndexNR                            PRESENCE optional
}|
    { ID id-AreaSpecificSemiPersistentSRSPosInfo    CRITICALITY ignore  TYPE AreaSpecificSemiPersistentSRSPosInfo PRESENCE optional },
    ...
}
-- **************************************************************
--
```

```
-- UL RRC Message Transfer ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- UL RRC Message Transfer
--
-- **************************************************************

ULRRCMessageTransfer ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ ULRRCMessageTransferIEs}},
    ...
}

ULRRCMessageTransferIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID               CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID           PRESENCE mandatory  }|
    { ID id-gNB-DU-UE-F1AP-ID               CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID           PRESENCE mandatory  }|
    { ID id-SRBID                           CRITICALITY reject  TYPE SRBID                       PRESENCE mandatory  }|
    { ID id-RRCContainer                    CRITICALITY reject  TYPE RRCContainer                PRESENCE mandatory  }|
    { ID id-SelectedPLMNID                  CRITICALITY reject  TYPE PLMN-Identity               PRESENCE optional   }|
    { ID id-new-gNB-DU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID           PRESENCE optional   },
    ...
}

-- **************************************************************
--
-- PRIVATE MESSAGE
--
-- **************************************************************

PrivateMessage ::= SEQUENCE {
    privateIEs      PrivateIE-Container {{PrivateMessage-IEs}},
    ...
}

PrivateMessage-IEs F1AP-PRIVATE-IES ::= {
    ...
}


-- **************************************************************
--
-- System Information ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- System information Delivery Command
--
-- **************************************************************

SystemInformationDeliveryCommand ::= SEQUENCE {
```

```
	protocolIEs			ProtocolIE-Container		{{ SystemInformationDeliveryCommandIEs}},
	...
}

SystemInformationDeliveryCommandIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID			CRITICALITY reject	TYPE TransactionID					PRESENCE mandatory	}|
	{ ID id-NRCGI					CRITICALITY reject	TYPE NRCGI							PRESENCE mandatory	}|
	{ ID id-SItype-List				CRITICALITY reject	TYPE SItype-List					PRESENCE mandatory	}|
	{ ID id-ConfirmedUEID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID				PRESENCE mandatory	},
	...
}


-- **************************************************************
--
-- Paging PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Paging
--
-- **************************************************************

Paging ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ PagingIEs}},
	...
}

PagingIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-UEIdentityIndexValue	CRITICALITY reject	TYPE UEIdentityIndexValue		PRESENCE mandatory	}|
	{ ID id-PagingIdentity			CRITICALITY reject	TYPE PagingIdentity				PRESENCE mandatory	}|
	{ ID id-PagingDRX				CRITICALITY ignore	TYPE PagingDRX					PRESENCE optional	}|
	{ ID id-PagingPriority			CRITICALITY ignore	TYPE PagingPriority				PRESENCE optional	}|
	{ ID id-PagingCell-List			CRITICALITY ignore	TYPE PagingCell-list			PRESENCE mandatory	}|
	{ ID id-PagingOrigin			CRITICALITY ignore	TYPE PagingOrigin				PRESENCE optional	}|
	{ ID id-RANUEPagingDRX			CRITICALITY ignore	TYPE PagingDRX					PRESENCE optional	}|
	{ ID id-CNUEPagingDRX			CRITICALITY ignore	TYPE PagingDRX					PRESENCE optional	}|
	{ ID id-NRPagingeDRXInformation CRITICALITY ignore	TYPE NRPagingeDRXInformation	PRESENCE optional	}|
	{ ID id-NRPagingeDRXInformationforRRCINACTIVE	CRITICALITY ignore	TYPE NRPagingeDRXInformationforRRCINACTIVE	PRESENCE optional }|
	{ ID id-PagingCause			CRITICALITY ignore	TYPE PagingCause				PRESENCE optional	}|
	{ ID id-PEIPSAssistanceInfo		CRITICALITY ignore	TYPE PEIPSAssistanceInfo		PRESENCE optional	}|
	{ ID id-UEPagingCapability		CRITICALITY ignore	TYPE UEPagingCapability			PRESENCE optional	}|
	{ ID id-ExtendedUEIdentityIndexValue	CRITICALITY ignore	TYPE ExtendedUEIdentityIndexValue	PRESENCE optional}|
	{ ID id-HashedUEIdentityIndexValue		CRITICALITY ignore	TYPE HashedUEIdentityIndexValue		PRESENCE optional}|
	{ ID id-MT-SDT-Information		CRITICALITY ignore	TYPE MT-SDT-Information			PRESENCE optional	}|
	{ ID id-NRPaginglongeDRXInformationforRRCINACTIVE		CRITICALITY ignore		TYPE NRPaginglongeDRXInformationforRRCINACTIVE			PRESENCE 
optional }|
	{ ID id-LPWUSPSAssistanceInfo	CRITICALITY ignore	TYPE LPWUSPSAssistanceInfo		PRESENCE optional	}|
	{ ID id-FurtherExtendedUEIdentityIndexValue CRITICALITY ignore	TYPE FurtherExtendedUEIdentityIndexValue	PRESENCE optional },
	...
}
```

```
PagingCell-list::= SEQUENCE (SIZE(1.. maxnoofPagingCells)) OF ProtocolIE-SingleContainer { { PagingCell-ItemIEs } }

PagingCell-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-PagingCell-Item		CRITICALITY ignore	TYPE PagingCell-Item			PRESENCE mandatory} ,
	...
}



-- **************************************************************
--
-- Notify
--
-- **************************************************************

Notify ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ NotifyIEs}},
	...
}

NotifyIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID					CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID					CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-DRB-Notify-List						CRITICALITY reject	TYPE DRB-Notify-List				PRESENCE mandatory	},
	...
}

DRB-Notify-List::= SEQUENCE (SIZE(1.. maxnoofDRBs)) OF ProtocolIE-SingleContainer { { DRB-Notify-ItemIEs } }

DRB-Notify-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-DRB-Notify-Item			CRITICALITY reject	TYPE DRB-Notify-Item		PRESENCE mandatory},
	...
}


-- **************************************************************
--
-- NETWORK ACCESS RATE REDUCTION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Network Access Rate Reduction
--
-- **************************************************************

NetworkAccessRateReduction ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ NetworkAccessRateReductionIEs }},
	...
}

NetworkAccessRateReductionIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID					CRITICALITY reject	TYPE TransactionID				PRESENCE mandatory	}|
```

```
    { ID id-UAC-Assistance-Info             CRITICALITY reject  TYPE UAC-Assistance-Info         PRESENCE mandatory   },
    ...
}

-- **************************************************************
--
-- PWS RESTART INDICATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- PWS Restart Indication
--
-- **************************************************************

PWSRestartIndication ::= SEQUENCE {
    protocolIEs ProtocolIE-Container { { PWSRestartIndicationIEs} },
    ...
}

PWSRestartIndicationIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                   CRITICALITY reject  TYPE TransactionID               PRESENCE mandatory   }|
    { ID id-NR-CGI-List-For-Restart-List    CRITICALITY reject  TYPE NR-CGI-List-For-Restart-List PRESENCE mandatory   },
    ...
}

NR-CGI-List-For-Restart-List        ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { NR-CGI-List-For-Restart-List-ItemIEs
} }

NR-CGI-List-For-Restart-List-ItemIEs F1AP-PROTOCOL-IES  ::= {
    { ID id-NR-CGI-List-For-Restart-Item        CRITICALITY reject  TYPE    NR-CGI-List-For-Restart-Item     PRESENCE mandatory   },
    ...
}

-- **************************************************************
--
-- PWS FAILURE INDICATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- PWS Failure Indication
--
-- **************************************************************

PWSFailureIndication ::= SEQUENCE {
    protocolIEs ProtocolIE-Container { { PWSFailureIndicationIEs} },
    ...
}

PWSFailureIndicationIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID           CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory }|
```

```
	{ ID id-PWS-Failed-NR-CGI-List	CRITICALITY reject	TYPE PWS-Failed-NR-CGI-List		PRESENCE optional	},
	...
}

PWS-Failed-NR-CGI-List		::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF ProtocolIE-SingleContainer { { PWS-Failed-NR-CGI-List-ItemIEs } }

PWS-Failed-NR-CGI-List-ItemIEs F1AP-PROTOCOL-IES	::= {
	{ ID id-PWS-Failed-NR-CGI-Item		CRITICALITY reject	TYPE	PWS-Failed-NR-CGI-Item		PRESENCE mandatory	},
	...
}


-- **************************************************************
--
-- gNB-DU STATUS INDICATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- gNB-DU Status Indication
--
-- **************************************************************

GNBDUStatusIndication ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ {GNBDUStatusIndicationIEs} },
	...
}

GNBDUStatusIndicationIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID					CRITICALITY reject	TYPE TransactionID				PRESENCE mandatory	}|
	{ ID id-GNBDUOverloadInformation		CRITICALITY reject	TYPE GNBDUOverloadInformation		PRESENCE mandatory	}|
	{ ID id-IABCongestionIndication		CRITICALITY ignore	TYPE IABCongestionIndication		PRESENCE optional	},
	...
}



-- **************************************************************
--
-- RRC Delivery Report ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- RRC Delivery Report
--
-- **************************************************************

RRCDeliveryReport ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ RRCDeliveryReportIEs}},
	...
}
```

```
RRCDeliveryReportIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID	CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID	PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID	CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID	PRESENCE mandatory	}|
	{ ID id-RRCDeliveryStatus	CRITICALITY ignore	TYPE RRCDeliveryStatus	PRESENCE mandatory	}|
	{ ID id-SRBID				CRITICALITY ignore	TYPE SRBID				PRESENCE mandatory	},
	...
}

-- **************************************************************
--
-- F1 Removal ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- F1 Removal Request
--
-- **************************************************************

F1RemovalRequest ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ F1RemovalRequestIEs }},
	...
}

F1RemovalRequestIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID			CRITICALITY reject	TYPE TransactionID					PRESENCE mandatory	},
	...
}

-- **************************************************************
--
-- F1 Removal Response
--
-- **************************************************************

F1RemovalResponse ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ F1RemovalResponseIEs }},
	...
}

F1RemovalResponseIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID				CRITICALITY reject	TYPE TransactionID					PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics			PRESENCE optional	},

	...
}

-- **************************************************************
--
-- F1 Removal Failure
--
-- **************************************************************
```

```
F1RemovalFailure ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ F1RemovalFailureIEs }},
    ...
}

F1RemovalFailureIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID               CRITICALITY reject  TYPE TransactionID                  PRESENCE mandatory }|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                          PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics         PRESENCE optional  },

    ...
}


-- **************************************************************
--
-- TRACE ELEMENTARY PROCEDURES
--
-- **************************************************************

-- **************************************************************
--
-- TRACE START
--
-- **************************************************************

TraceStart ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        { {TraceStartIEs} },
    ...
}

TraceStartIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID              PRESENCE mandatory }|
    { ID id-gNB-DU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID              PRESENCE mandatory }|
    { ID id-TraceActivation             CRITICALITY ignore  TYPE TraceActivation                PRESENCE mandatory },
    ...
}

-- **************************************************************
--
-- DEACTIVATE TRACE
--
-- **************************************************************

DeactivateTrace ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        { {DeactivateTraceIEs} },
    ...
}

DeactivateTraceIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID              PRESENCE mandatory }|
    { ID id-gNB-DU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID              PRESENCE mandatory }|
    { ID id-TraceID                     CRITICALITY ignore  TYPE TraceID                        PRESENCE mandatory },
```

```
	...
}

-- **************************************************************
--
-- CELL TRAFFIC TRACE
--
-- **************************************************************

CellTrafficTrace ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ {CellTrafficTraceIEs} },
	...
}

CellTrafficTraceIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ID id-TraceID					CRITICALITY ignore	TYPE TraceID					PRESENCE mandatory	}|
	{ID id-TraceCollectionEntityIPAddress	CRITICALITY ignore	TYPE TransportLayerAddress		PRESENCE mandatory	}|
	{ID id-PrivacyIndicator				CRITICALITY ignore	TYPE PrivacyIndicator			PRESENCE optional }|
	{ID id-TraceCollectionEntityURI CRITICALITY ignore	TYPE URI-address		PRESENCE optional },
	...
}

-- **************************************************************
--
-- DU-CU Radio Information Transfer ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- DU-CU Radio Information Transfer
--
-- **************************************************************

DUCURadioInformationTransfer ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ DUCURadioInformationTransferIEs}},
	...
}

DUCURadioInformationTransferIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID				CRITICALITY reject	TYPE TransactionID					PRESENCE mandatory	}|
	{ ID id-DUCURadioInformationType		CRITICALITY ignore	TYPE DUCURadioInformationType			PRESENCE mandatory	},
	...
}


-- **************************************************************
--
-- CU-DU Radio Information Transfer ELEMENTARY PROCEDURE
--
-- **************************************************************
```

```
-- **************************************************************
--
-- CU-DU Radio Information Transfer
--
-- **************************************************************

CUDURadioInformationTransfer ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ CUDURadioInformationTransferIEs}},
    ...
}

CUDURadioInformationTransferIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                   CRITICALITY reject  TYPE TransactionID                  PRESENCE mandatory   }|
    { ID id-CUDURadioInformationType    CRITICALITY ignore  TYPE CUDURadioInformationType           PRESENCE mandatory   },
    ...
}

-- **************************************************************
--
-- IAB PROCEDURES
--
-- **************************************************************
-- **************************************************************
--
-- BAP Mapping Configuration ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- BAP MAPPING CONFIGURATION
--
-- **************************************************************


BAPMappingConfiguration ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container    { {BAPMappingConfiguration-IEs} },
    ...
 }

BAPMappingConfiguration-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                       CRITICALITY reject  TYPE    TransactionID                       PRESENCE mandatory}|
    { ID id-BH-Routing-Information-Added-List       CRITICALITY ignore  TYPE    BH-Routing-Information-Added-List   PRESENCE optional}|
    { ID id-BH-Routing-Information-Removed-List     CRITICALITY ignore  TYPE    BH-Routing-Information-Removed-List PRESENCE optional}|
    { ID id-TrafficMappingInformation               CRITICALITY ignore  TYPE    TrafficMappingInfo                  PRESENCE optional}|
    { ID id-BufferSizeThresh                        CRITICALITY ignore  TYPE    BufferSizeThresh                    PRESENCE optional}|
    { ID id-BAP-Header-Rewriting-Added-List         CRITICALITY ignore  TYPE    BAP-Header-Rewriting-Added-List     PRESENCE optional}|
    { ID id-Re-routingEnableIndicator               CRITICALITY ignore  TYPE    Re-routingEnableIndicator           PRESENCE optional}|
    { ID id-BAP-Header-Rewriting-Removed-List       CRITICALITY ignore  TYPE    BAP-Header-Rewriting-Removed-List   PRESENCE optional},
    ...
}
```

```
BH-Routing-Information-Added-List ::= SEQUENCE (SIZE(1.. maxnoofRoutingEntries))    OF ProtocolIE-SingleContainer { { BH-Routing-Information-Added-List-ItemIEs } }
BH-Routing-Information-Removed-List ::= SEQUENCE (SIZE(1.. maxnoofRoutingEntries))  OF ProtocolIE-SingleContainer { { BH-Routing-Information-Removed-List-ItemIEs } }

BH-Routing-Information-Added-List-ItemIEs   F1AP-PROTOCOL-IES ::= {
    { ID id-BH-Routing-Information-Added-List-Item              CRITICALITY ignore  TYPE BH-Routing-Information-Added-List-Item
    PRESENCE mandatory},
    ...
}

BH-Routing-Information-Removed-List-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BH-Routing-Information-Removed-List-Item                CRITICALITY ignore  TYPE BH-Routing-Information-Removed-List-Item
    PRESENCE mandatory},
    ...
}

BAP-Header-Rewriting-Added-List ::= SEQUENCE (SIZE(1.. maxnoofRoutingEntries))  OF ProtocolIE-SingleContainer { { BAP-Header-Rewriting-Added-List-ItemIEs } }

BAP-Header-Rewriting-Added-List-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BAP-Header-Rewriting-Added-List-Item    CRITICALITY ignore  TYPE BAP-Header-Rewriting-Added-List-Item PRESENCE mandatory},
    ...
}

BAP-Header-Rewriting-Removed-List ::= SEQUENCE (SIZE(1.. maxnoofRoutingEntries))    OF ProtocolIE-SingleContainer { { BAP-Header-Rewriting-Removed-List-ItemIEs } }

BAP-Header-Rewriting-Removed-List-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BAP-Header-Rewriting-Removed-List-Item  CRITICALITY ignore  TYPE BAP-Header-Rewriting-Removed-List-Item PRESENCE mandatory},
    ...
}

-- **************************************************************
--
-- BAP MAPPING CONFIGURATION ACKNOWLEDGE
--
-- **************************************************************

BAPMappingConfigurationAcknowledge ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        { {BAPMappingConfigurationAcknowledge-IEs} },
    ...
}

BAPMappingConfigurationAcknowledge-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID           CRITICALITY reject  TYPE    TransactionID           PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics  CRITICALITY ignore  TYPE    CriticalityDiagnostics  PRESENCE optional},
    ...
}

-- **************************************************************
--
-- BAP MAPPING CONFIGURATION FAILURE
--
```

```
-- **************************************************************

BAPMappingConfigurationFailure ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { BAPMappingConfigurationFailureIEs} },
	...
}

BAPMappingConfigurationFailureIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID				CRITICALITY reject	TYPE TransactionID				PRESENCE mandatory	}|
	{ ID id-Cause						CRITICALITY ignore	TYPE Cause						PRESENCE mandatory	}|
	{ ID id-TimeToWait					CRITICALITY ignore	TYPE TimeToWait					PRESENCE optional }|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional },
	...
}


-- **************************************************************
--
-- GNB-DU Configuration ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- GNB-DU RESOURCE CONFIGURATION
--
-- **************************************************************


GNBDUResourceConfiguration ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ GNBDUResourceConfigurationIEs}},
	...
}


GNBDUResourceConfigurationIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID							CRITICALITY reject	TYPE TransactionID						PRESENCE mandatory	}|
	{ ID id-Activated-Cells-to-be-Updated-List		CRITICALITY reject	TYPE Activated-Cells-to-be-Updated-List	PRESENCE optional}|
	{ ID id-Child-Nodes-List						CRITICALITY reject	TYPE Child-Nodes-List					PRESENCE optional}|
	{ ID id-Neighbour-Node-Cells-List				CRITICALITY reject	TYPE Neighbour-Node-Cells-List			PRESENCE optional}|
	{ ID id-Serving-Cells-List						CRITICALITY reject	TYPE Serving-Cells-List					PRESENCE optional},
	...
}



-- **************************************************************
--
-- GNB-DU RESOURCE CONFIGURATION ACKNOWLEDGE
--
-- **************************************************************
```

```
GNBDUResourceConfigurationAcknowledge ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container        { { GNBDUResourceConfigurationAcknowledgeIEs} },
    ...
}


GNBDUResourceConfigurationAcknowledgeIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                       CRITICALITY reject  TYPE TransactionID                      PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics             PRESENCE optional },
    ...
}

-- **************************************************************
--
-- GNB-DU RESOURCE CONFIGURATION FAILURE
--
-- **************************************************************

GNBDUResourceConfigurationFailure ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container        { { GNBDUResourceConfigurationFailureIEs} },
    ...
}

GNBDUResourceConfigurationFailureIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID               CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory   }|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory   }|
    { ID id-TimeToWait                  CRITICALITY ignore  TYPE TimeToWait                 PRESENCE optional }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional },
    ...
}


-- **************************************************************
--
-- IAB TNL Address Allocation ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- IAB TNL ADDRESS REQUEST
--
-- **************************************************************



IABTNLAddressRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container        { {IABTNLAddressRequestIEs} },
    ...
}

IABTNLAddressRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                       CRITICALITY reject  TYPE TransactionID                          PRESENCE mandatory   }|
    { ID id-IABv4AddressesRequested             CRITICALITY reject  TYPE IABv4AddressesRequested                PRESENCE optional }|
```

```
	{ ID id-IABIPv6RequestType					CRITICALITY reject	TYPE IABIPv6RequestType					PRESENCE optional }|
	{ ID id-IAB-TNL-Addresses-To-Remove-List	CRITICALITY reject	TYPE IAB-TNL-Addresses-To-Remove-List	PRESENCE optional }|
	{ ID id-IAB-TNL-Addresses-Exception			CRITICALITY reject	TYPE IAB-TNL-Addresses-Exception		PRESENCE optional },
	...
}


IAB-TNL-Addresses-To-Remove-List	::= SEQUENCE (SIZE(1..maxnoofTLAsIAB))	OF ProtocolIE-SingleContainer { { IAB-TNL-Addresses-To-Remove-ItemIEs }
}

IAB-TNL-Addresses-To-Remove-ItemIEs F1AP-PROTOCOL-IES::= {
	{ ID id-IAB-TNL-Addresses-To-Remove-Item			CRITICALITY reject	TYPE IAB-TNL-Addresses-To-Remove-Item			PRESENCE mandatory},
	...
}


-- **************************************************************
--
-- IAB TNL ADDRESS RESPONSE
--
-- **************************************************************


IABTNLAddressResponse ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ {IABTNLAddressResponseIEs} },
	...
}


IABTNLAddressResponseIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID							CRITICALITY reject	TYPE TransactionID						PRESENCE mandatory	}|
	{ ID id-IAB-Allocated-TNL-Address-List			CRITICALITY reject	TYPE IAB-Allocated-TNL-Address-List		PRESENCE mandatory	},
	...
}


IAB-Allocated-TNL-Address-List ::= SEQUENCE (SIZE(1.. maxnoofTLAsIAB))	OF ProtocolIE-SingleContainer { { IAB-Allocated-TNL-Address-List-ItemIEs }
}


IAB-Allocated-TNL-Address-List-ItemIEs	F1AP-PROTOCOL-IES::= {
	{ ID id-IAB-Allocated-TNL-Address-Item			CRITICALITY reject	TYPE IAB-Allocated-TNL-Address-Item			PRESENCE mandatory},
	...
}

-- **************************************************************
--
-- IAB TNL ADDRESS FAILURE
--
-- **************************************************************

IABTNLAddressFailure ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { IABTNLAddressFailureIEs} },
	...
```

```
}

IABTNLAddressFailureIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID				CRITICALITY reject	TYPE TransactionID				PRESENCE mandatory	}|
	{ ID id-Cause						CRITICALITY ignore	TYPE Cause						PRESENCE mandatory	}|
	{ ID id-TimeToWait					CRITICALITY ignore	TYPE TimeToWait					PRESENCE optional }|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional },
	...
}

-- **************************************************************
--
-- IAB UP Configuration Update ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- IAB UP Configuration Update Request
--
-- **************************************************************

IABUPConfigurationUpdateRequest ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { IABUPConfigurationUpdateRequestIEs} },
	...
}

IABUPConfigurationUpdateRequestIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID								CRITICALITY reject	TYPE TransactionID								PRESENCE mandatory	}|
	{ ID id-UL-UP-TNL-Information-to-Update-List	CRITICALITY ignore	TYPE UL-UP-TNL-Information-to-Update-List		PRESENCE optional }|
	{ ID id-UL-UP-TNL-Address-to-Update-List		CRITICALITY ignore	TYPE UL-UP-TNL-Address-to-Update-List			PRESENCE optional },
	...
}

UL-UP-TNL-Information-to-Update-List ::= SEQUENCE (SIZE(1.. maxnoofULUPTNLInformationforIAB))	OF ProtocolIE-SingleContainer { { UL-UP-TNL-Information-to-Update-List-ItemIEs } }

UL-UP-TNL-Information-to-Update-List-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-UL-UP-TNL-Information-to-Update-List-Item	CRITICALITY ignore	TYPE UL-UP-TNL-Information-to-Update-List-Item PRESENCE mandatory },
	...
}

UL-UP-TNL-Address-to-Update-List ::= SEQUENCE (SIZE(1.. maxnoofUPTNLAddresses)) OF ProtocolIE-SingleContainer { { UL-UP-TNL-Address-to-Update-List-ItemIEs } }

UL-UP-TNL-Address-to-Update-List-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-UL-UP-TNL-Address-to-Update-List-Item	CRITICALITY ignore	TYPE UL-UP-TNL-Address-to-Update-List-Item PRESENCE mandatory },
	...
}


-- **************************************************************
--
-- IAB UP Configuration Update Response
```

```
--
-- **************************************************************

IABUPConfigurationUpdateResponse ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container        { { IABUPConfigurationUpdateResponseIEs} },
    ...
}

IABUPConfigurationUpdateResponseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                       CRITICALITY reject  TYPE TransactionID                      PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics             PRESENCE optional }|
    { ID id-DL-UP-TNL-Address-to-Update-List    CRITICALITY reject  TYPE DL-UP-TNL-Address-to-Update-List  PRESENCE optional },
    ...
}

DL-UP-TNL-Address-to-Update-List ::= SEQUENCE (SIZE(1.. maxnoofUPTNLAddresses)) OF ProtocolIE-SingleContainer { { DL-UP-TNL-Address-to-Update-List-
ItemIEs } }

DL-UP-TNL-Address-to-Update-List-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-DL-UP-TNL-Address-to-Update-List-Item   CRITICALITY ignore  TYPE DL-UP-TNL-Address-to-Update-List-Item  PRESENCE mandatory },
    ...
}

-- **************************************************************
--
-- IAB UP Configuration Update Failure
--
-- **************************************************************

IABUPConfigurationUpdateFailure ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container        { { IABUPConfigurationUpdateFailureIEs} },
    ...
}

IABUPConfigurationUpdateFailureIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID               CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory   }|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory   }|
    { ID id-TimeToWait                  CRITICALITY ignore  TYPE TimeToWait                 PRESENCE optional }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional },
    ...
}

-- MIAB F1 SETUP TRIGGERING PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- MIAB F1 SETUP TRIGGERING
--
-- **************************************************************

MIABF1SetupTriggering ::= SEQUENCE {
```

```
	protocolIEs			ProtocolIE-Container		{{ MIABF1SetupTriggeringIEs}},
	...
}

MIABF1SetupTriggeringIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID					CRITICALITY reject	TYPE TransactionID					PRESENCE mandatory	}|
	{ ID id-Target-gNB-ID					CRITICALITY reject	TYPE GlobalGNB-ID					PRESENCE mandatory	}|
	{ ID id-Target-gNB-IP-address			CRITICALITY ignore	TYPE TransportLayerAddress			PRESENCE optional }|
	{ ID id-Target-SeGW-IP-address			CRITICALITY ignore	TYPE TransportLayerAddress			PRESENCE optional },
	...
}


-- MIAB F1 SETUP OUTCOME NOTIFICATION PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- MIAB F1 SETUP OUTCOME NOTIFICATION
--
-- **************************************************************

MIABF1SetupOutcomeNotification ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ MIABF1SetupOutcomeNotificationIEs}},
	...
}

MIABF1SetupOutcomeNotificationIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID					CRITICALITY reject	TYPE TransactionID						PRESENCE mandatory	}|
	{ ID id-F1SetupOutcome					CRITICALITY reject	TYPE F1SetupOutcome						PRESENCE mandatory	}|
	{ ID id-Activated-Cells-Mapping-List		CRITICALITY ignore	TYPE Activated-Cells-Mapping-List		PRESENCE optional }|
	{ ID id-Target-F1-Terminating-Donor-gNB-ID	CRITICALITY reject	TYPE GlobalGNB-ID						PRESENCE optional },
	...
}

F1SetupOutcome ::= ENUMERATED {success, failure,...}

Activated-Cells-Mapping-List ::= SEQUENCE (SIZE(1.. maxCellingNBDU))	OF ProtocolIE-SingleContainer { { Activated-Cells-Mapping-List-ItemIEs } }

Activated-Cells-Mapping-List-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-Activated-Cells-Mapping-List-Item	CRITICALITY ignore	TYPE Activated-Cells-Mapping-List-Item PRESENCE mandatory },
	...
}


-- **************************************************************
--
-- Resource Status Reporting Initiation ELEMENTARY PROCEDURE
--
-- **************************************************************
```

```
-- **************************************************************
--
-- Resource Status Request
--
-- **************************************************************

ResourceStatusRequest::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ {ResourceStatusRequestIEs} },
	...
}

ResourceStatusRequestIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID			CRITICALITY reject	TYPE TransactionID			PRESENCE mandatory	}|
	{ ID id-gNBCUMeasurementID		CRITICALITY reject	TYPE GNBCUMeasurementID		PRESENCE mandatory	}|
	{ ID id-gNBDUMeasurementID		CRITICALITY ignore	TYPE GNBDUMeasurementID		PRESENCE conditional	}|
	-- The above IE shall be present if the Registration Request IE is set to the value "stop" or "add".
	{ ID id-RegistrationRequest		CRITICALITY ignore	TYPE RegistrationRequest	PRESENCE mandatory	}|
	{ ID id-ReportCharacteristics	CRITICALITY ignore	TYPE ReportCharacteristics	PRESENCE conditional	}|
	-- The above IE shall be present if the Registration Request IE is set to the value "start".
	{ ID id-CellToReportList		CRITICALITY ignore	TYPE CellToReportList		PRESENCE optional	}|
	{ ID id-ReportingPeriodicity	CRITICALITY ignore	TYPE ReportingPeriodicity	PRESENCE  optional },
	...
}


-- **************************************************************
--
-- Resource Status Response
--
-- **************************************************************

ResourceStatusResponse ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { ResourceStatusResponseIEs} },
	...
}


ResourceStatusResponseIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID			CRITICALITY reject	TYPE TransactionID			PRESENCE mandatory	}|
	{ ID id-gNBCUMeasurementID		CRITICALITY reject	TYPE GNBCUMeasurementID		PRESENCE mandatory	}|
	{ ID id-gNBDUMeasurementID		CRITICALITY ignore	TYPE GNBDUMeasurementID		PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics	CRITICALITY ignore	TYPE CriticalityDiagnostics PRESENCE optional	},
	...
}

-- **************************************************************
--
-- Resource Status Failure
--
-- **************************************************************

ResourceStatusFailure ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { ResourceStatusFailureIEs} },
```

```
	...
}

ResourceStatusFailureIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID			CRITICALITY reject	TYPE TransactionID			PRESENCE mandatory	}|
	{ ID id-gNBCUMeasurementID		CRITICALITY reject	TYPE GNBCUMeasurementID		PRESENCE mandatory	}|
	{ ID id-gNBDUMeasurementID		CRITICALITY ignore	TYPE GNBDUMeasurementID		PRESENCE mandatory	}|
	{ ID id-Cause					CRITICALITY ignore	TYPE Cause					PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics	CRITICALITY ignore	TYPE CriticalityDiagnostics	PRESENCE optional	},
	...
}

-- **************************************************************
--
-- Resource Status Reporting ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Resource Status Update
--
-- **************************************************************

ResourceStatusUpdate ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ ResourceStatusUpdateIEs}},
	...
}

ResourceStatusUpdateIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID					CRITICALITY reject	TYPE TransactionID			PRESENCE mandatory	}|
	{ ID id-gNBCUMeasurementID			CRITICALITY reject	TYPE GNBCUMeasurementID			PRESENCE mandatory }|
	{ ID id-gNBDUMeasurementID			CRITICALITY ignore	TYPE GNBDUMeasurementID		PRESENCE mandatory	}|
	{ ID id-HardwareLoadIndicator			CRITICALITY ignore	TYPE HardwareLoadIndicator		PRESENCE optional }|
	{ ID id-TNLCapacityIndicator			CRITICALITY ignore	TYPE TNLCapacityIndicator		PRESENCE optional	}|
	{ ID id-CellMeasurementResultList		CRITICALITY ignore	TYPE CellMeasurementResultList	PRESENCE optional	}|
	{ ID id-NodeAssociatedInfoResult		CRITICALITY ignore	TYPE NodeAssociatedInfoResult	PRESENCE optional	},
	...
}

-- **************************************************************
--
--  Access And Mobility Indication ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Access And Mobility Indication
--
-- **************************************************************

AccessAndMobilityIndication ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { AccessAndMobilityIndicationIEs} },
```

```
    ...
}

AccessAndMobilityIndicationIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                           CRITICALITY reject  TYPE TransactionID                PRESENCE mandatory }|
    { ID id-RAReportList                CRITICALITY ignore  TYPE RAReportList           PRESENCE optional }|
    { ID id-RLFReportInformationList                CRITICALITY ignore  TYPE RLFReportInformationList           PRESENCE optional }|
    { ID id-SuccessfulHOReportInformationList       CRITICALITY ignore  TYPE SuccessfulHOReportInformationList  PRESENCE optional }|
    { ID id-SuccessfulPSCellChangeReportInformationList CRITICALITY ignore  TYPE SuccessfulPSCellChangeReportInformationList   PRESENCE optional }|
    { ID id-FailureReportingWithoutRLFReport        CRITICALITY ignore  TYPE FailureReportingWithoutRLFReport   PRESENCE optional }|
    { ID id-MROForLTM-Information                   CRITICALITY ignore  TYPE MROForLTM-Information        PRESENCE optional },
    ...
}


-- **************************************************************
--
-- REFERENCE TIME INFORMATION REPORTING CONTROL ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- REFERENCE TIME INFORMATION REPORTING CONTROL
--
-- **************************************************************

ReferenceTimeInformationReportingControl::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        { { ReferenceTimeInformationReportingControlIEs} },
    ...
}

ReferenceTimeInformationReportingControlIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID               CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory   }|
    { ID id-ReportingRequestType        CRITICALITY reject  TYPE ReportingRequestType       PRESENCE mandatory   },
    ...
}


-- **************************************************************
--
-- REFERENCE TIME INFORMATION REPORT ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- REFERENCE TIME INFORMATION REPORT
--
-- **************************************************************

ReferenceTimeInformationReport::= SEQUENCE {
```

```
	protocolIEs		ProtocolIE-Container		{ { ReferenceTimeInformationReportIEs} },
	...
}

ReferenceTimeInformationReportIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID				CRITICALITY ignore	TYPE TransactionID				PRESENCE mandatory }|
	{ ID id-TimeReferenceInformation	CRITICALITY ignore	TYPE TimeReferenceInformation		PRESENCE mandatory },
	...
}


-- **************************************************************
--
-- ACCESS SUCCESS ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Access Success
--
-- **************************************************************

AccessSuccess ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ AccessSuccessIEs}},
	...
}

AccessSuccessIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-NRCGI							CRITICALITY reject	TYPE NRCGI							PRESENCE mandatory	},
	...
}


-- **************************************************************
--
-- POSITIONING ASSISTANCE INFORMATION CONTROL ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Positioning Assistance Information Control
--
-- **************************************************************

PositioningAssistanceInformationControl ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ PositioningAssistanceInformationControlIEs}},
	...
}

PositioningAssistanceInformationControlIEs F1AP-PROTOCOL-IES ::= {
```

```
        { ID id-TransactionID                   CRITICALITY reject  TYPE TransactionID           PRESENCE mandatory   }|
        { ID id-PosAssistance-Information       CRITICALITY reject  TYPE PosAssistance-Information     PRESENCE optional}|
        { ID id-PosBroadcast                    CRITICALITY reject  TYPE PosBroadcast            PRESENCE optional}|
        { ID id-PositioningBroadcastCells       CRITICALITY reject  TYPE PositioningBroadcastCells     PRESENCE optional}|
        { ID id-RoutingID                       CRITICALITY reject  TYPE RoutingID              PRESENCE optional},
    ...
}

-- **************************************************************
--
-- POSITIONING ASSISTANCE INFORMATION FEEDBACK ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Positioning Assistance Information Feedback
--
-- **************************************************************

PositioningAssistanceInformationFeedback ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ PositioningAssistanceInformationFeedbackIEs}},
    ...
}

PositioningAssistanceInformationFeedbackIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                       CRITICALITY reject  TYPE TransactionID                   PRESENCE mandatory   }|
    { ID id-PosAssistanceInformationFailureList CRITICALITY reject  TYPE PosAssistanceInformationFailureList  PRESENCE optional}|
    { ID id-PositioningBroadcastCells               CRITICALITY reject  TYPE PositioningBroadcastCells          PRESENCE optional}|
    { ID id-RoutingID                               CRITICALITY reject  TYPE RoutingID                          PRESENCE optional}|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics              PRESENCE optional},
    ...
}

-- **************************************************************
--
-- POSITONING MEASUREMENT EXCHANGE ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Positioning Measurement Request
--
-- **************************************************************

PositioningMeasurementRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { PositioningMeasurementRequestIEs} },
    ...
}

PositioningMeasurementRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                           CRITICALITY reject  TYPE TransactionID                          PRESENCE mandatory}|
    { ID id-LMF-MeasurementID                       CRITICALITY reject  TYPE LMF-MeasurementID                      PRESENCE mandatory}|
```

```
	{ ID id-RAN-MeasurementID					CRITICALITY reject	TYPE RAN-MeasurementID					PRESENCE mandatory}|
	{ ID id-TRP-MeasurementRequestList			CRITICALITY reject	TYPE TRP-MeasurementRequestList			PRESENCE mandatory}|
	{ ID id-PosReportCharacteristics			CRITICALITY reject	TYPE PosReportCharacteristics			PRESENCE mandatory}|
	{ ID id-PosMeasurementPeriodicity			CRITICALITY reject	TYPE MeasurementPeriodicity				PRESENCE conditional }|
	-- The above IE shall be present if the PosReportCharacteristics IE is set to “periodic” --
	{ ID id-PosMeasurementQuantities			CRITICALITY reject	TYPE PosMeasurementQuantities			PRESENCE mandatory}|
	{ ID id-SFNInitialisationTime				CRITICALITY ignore	TYPE RelativeTime1900	PRESENCE optional	}|
	{ ID id-SRSConfiguration					CRITICALITY ignore	TYPE SRSConfiguration					PRESENCE optional}|
	{ ID id-MeasurementBeamInfoRequest			CRITICALITY ignore	TYPE MeasurementBeamInfoRequest PRESENCE optional }|
	{ ID id-SystemFrameNumber					CRITICALITY ignore	TYPE SystemFrameNumber		PRESENCE optional}|
	{ ID id-SlotNumber							CRITICALITY ignore	TYPE SlotNumber				PRESENCE optional}|
	{ ID id-PosMeasurementPeriodicityExtended	CRITICALITY reject	TYPE MeasurementPeriodicityExtended		PRESENCE conditional }|
	-- The IE shall be present the MeasurementPeriodicity IE is set to the value "extended"

	{ ID id-ResponseTime						CRITICALITY ignore	TYPE ResponseTime						PRESENCE optional}|
	{ ID id-MeasurementCharacteristicsRequestIndicator		CRITICALITY ignore	TYPE MeasurementCharacteristicsRequestIndicator	PRESENCE 
optional}|
	{ ID id-MeasurementTimeOccasion				CRITICALITY ignore	TYPE MeasurementTimeOccasion PRESENCE optional }|
	{ ID id-PosMeasurementAmount				CRITICALITY ignore	TYPE PosMeasurementAmount PRESENCE optional }|
	{ ID id-TimeWindowInformation-Measurement-List		CRITICALITY ignore	TYPE TimeWindowInformation-Measurement-List	PRESENCE optional },
	...
}


-- **************************************************************
--
-- Positioning Measurement Response
--
-- **************************************************************

PositioningMeasurementResponse ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { PositioningMeasurementResponseIEs} },
	...
}


PositioningMeasurementResponseIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID						CRITICALITY reject	TYPE TransactionID						PRESENCE mandatory}|
	{ ID id-LMF-MeasurementID					CRITICALITY reject	TYPE LMF-MeasurementID					PRESENCE mandatory}|
	{ ID id-RAN-MeasurementID					CRITICALITY reject	TYPE RAN-MeasurementID					PRESENCE mandatory}|
	{ ID id-PosMeasurementResultList			CRITICALITY reject	TYPE PosMeasurementResultList			PRESENCE optional }|
	{ ID id-CriticalityDiagnostics				CRITICALITY ignore	TYPE CriticalityDiagnostics				PRESENCE optional },
	...
}


-- **************************************************************
--
-- Positioning Measurement Failure
--
-- **************************************************************

PositioningMeasurementFailure ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { PositioningMeasurementFailureIEs} },
```

```
	...
}

PositioningMeasurementFailureIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID					CRITICALITY reject	TYPE TransactionID				PRESENCE mandatory }|
	{ ID id-LMF-MeasurementID				CRITICALITY reject	TYPE LMF-MeasurementID			PRESENCE mandatory }|
	{ ID id-RAN-MeasurementID				CRITICALITY reject	TYPE RAN-MeasurementID			PRESENCE mandatory	}|
	{ ID id-Cause							CRITICALITY ignore	TYPE Cause						PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics			CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional },
	...
}


-- **************************************************************
--
-- POSITIONING MEASUREMENT REPORT ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Positioning Measurement Report
--
-- **************************************************************

PositioningMeasurementReport ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { PositioningMeasurementReportIEs} },
	...
}

PositioningMeasurementReportIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID				CRITICALITY reject	TYPE TransactionID				PRESENCE mandatory	}|
	{ ID id-LMF-MeasurementID			CRITICALITY reject	TYPE LMF-MeasurementID			PRESENCE mandatory	}|
	{ ID id-RAN-MeasurementID			CRITICALITY reject	TYPE RAN-MeasurementID			PRESENCE mandatory	}|
	{ ID id-PosMeasurementResultList	CRITICALITY reject	TYPE PosMeasurementResultList	PRESENCE mandatory	},
	...
}

-- **************************************************************
--
-- POSITIONING MEASUREMENT ABORT ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Positioning Measurement Abort
--
-- **************************************************************

PositioningMeasurementAbort ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { PositioningMeasurementAbortIEs} },
	...
}
```

```
PositioningMeasurementAbortIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                   CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory   }|
    { ID id-LMF-MeasurementID               CRITICALITY reject  TYPE LMF-MeasurementID          PRESENCE mandatory   }|
    { ID id-RAN-MeasurementID               CRITICALITY reject  TYPE RAN-MeasurementID          PRESENCE mandatory   },
    ...
}

-- **************************************************************
--
-- POSITIONING MEASUREMENT FAILURE INDICATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Positioning Measurement Failure Indication
--
-- **************************************************************

PositioningMeasurementFailureIndication ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        { { PositioningMeasurementFailureIndicationIEs} },
    ...
}

PositioningMeasurementFailureIndicationIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID               CRITICALITY reject  TYPE TransactionID                  PRESENCE mandatory }|
    { ID id-LMF-MeasurementID           CRITICALITY reject  TYPE LMF-MeasurementID              PRESENCE mandatory }|
    { ID id-RAN-MeasurementID           CRITICALITY reject  TYPE RAN-MeasurementID              PRESENCE mandatory }|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                          PRESENCE mandatory },
    ...
}

-- **************************************************************
--
-- POSITIONING MEASUREMENT UPDATE ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Positioning Measurement Update
--
-- **************************************************************

PositioningMeasurementUpdate ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        { { PositioningMeasurementUpdateIEs} },
    ...
}

PositioningMeasurementUpdateIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID               CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory   }|
    { ID id-LMF-MeasurementID           CRITICALITY reject  TYPE LMF-MeasurementID          PRESENCE mandatory   }|
    { ID id-RAN-MeasurementID           CRITICALITY reject  TYPE RAN-MeasurementID          PRESENCE mandatory   }|
```

```
    { ID id-SRSConfiguration            CRITICALITY ignore  TYPE SRSConfiguration           PRESENCE optional}|
    { ID id-TRP-MeasurementUpdateList   CRITICALITY reject  TYPE TRP-MeasurementUpdateList  PRESENCE optional}|
    { ID id-MeasurementCharacteristicsRequestIndicator  CRITICALITY ignore  TYPE    MeasurementCharacteristicsRequestIndicator  PRESENCE optional}|
    { ID id-MeasurementTimeOccasion     CRITICALITY ignore  TYPE MeasurementTimeOccasion    PRESENCE optional },
    ...
}


-- **************************************************************
--
-- TRP INFORMATION EXCHANGE ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- TRP Information Request
--
-- **************************************************************

TRPInformationRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        { { TRPInformationRequestIEs} },
    ...
}

TRPInformationRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                       CRITICALITY reject  TYPE TransactionID                          PRESENCE mandatory   }|
    { ID id-TRPList                             CRITICALITY ignore  TYPE TRPList                                PRESENCE optional }|
    { ID id-TRPInformationTypeListTRPReq        CRITICALITY reject  TYPE TRPInformationTypeListTRPReq           PRESENCE mandatory   },
    ...
}

TRPInformationTypeListTRPReq ::= SEQUENCE (SIZE(1.. maxnoofTRPInfoTypes)) OF ProtocolIE-SingleContainer { { TRPInformationTypeItemTRPReq } }

TRPInformationTypeItemTRPReq    F1AP-PROTOCOL-IES ::= {
    { ID id-TRPInformationTypeItem   CRITICALITY reject     TYPE TRPInformationTypeItem     PRESENCE mandatory },
    ...
}


-- **************************************************************
--
-- TRP Information Response
--
-- **************************************************************

TRPInformationResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        { { TRPInformationResponseIEs} },
    ...
}

TRPInformationResponseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                   CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory   }|
    { ID id-TRPInformationListTRPResp       CRITICALITY ignore  TYPE TRPInformationListTRPResp  PRESENCE mandatory   }|
```

```
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional },
	...
}

TRPInformationListTRPResp ::= SEQUENCE (SIZE(1.. maxnoofTRPs)) OF ProtocolIE-SingleContainer { { TRPInformationItemTRPResp } }

TRPInformationItemTRPResp	F1AP-PROTOCOL-IES ::= {
	{ ID id-TRPInformationItem	CRITICALITY ignore	TYPE TRPInformationItem		PRESENCE mandatory },
	...
}


-- **************************************************************
--
-- TRP Information Failure
--
-- **************************************************************

TRPInformationFailure ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { TRPInformationFailureIEs} },
	...
}

TRPInformationFailureIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID				CRITICALITY reject	TYPE TransactionID				PRESENCE mandatory	}|
	{ ID id-Cause						CRITICALITY ignore	TYPE Cause						PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional },
	...
}


-- **************************************************************
--
-- POSITIONING INFORMATION EXCHANGE ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Positioning Information Request
--
-- **************************************************************

PositioningInformationRequest ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { PositioningInformationRequestIEs} },
	...
}

PositioningInformationRequestIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID						CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID							PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID						CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID							PRESENCE mandatory	}|
	{ ID id-RequestedSRSTransmissionCharacteristics	CRITICALITY ignore	TYPE RequestedSRSTransmissionCharacteristics	PRESENCE optional}|
	{ ID id-UEReportingInformation					CRITICALITY ignore	TYPE UEReportingInformation						PRESENCE optional}|
	{ ID id-SRSPosRRCInactiveQueryIndication		CRITICALITY ignore	TYPE SRSPosRRCInactiveQueryIndication			PRESENCE optional}|
```

```
	{ ID id-TimeWindowInformation-SRS-List		CRITICALITY ignore	TYPE TimeWindowInformation-SRS-List			PRESENCE optional}|
	{ ID id-RequestedSRSPreconfigurationCharacteristics-List	CRITICALITY ignore	TYPE RequestedSRSPreconfigurationCharacteristics-List	PRESENCE optional	},
	...
}


-- **************************************************************
--
-- Positioning Information Response
--
-- **************************************************************

PositioningInformationResponse ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { PositioningInformationResponseIEs} },
	...
}


PositioningInformationResponseIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-SRSConfiguration			CRITICALITY ignore	TYPE SRSConfiguration			PRESENCE optional}|
	{ ID id-SFNInitialisationTime		CRITICALITY ignore	TYPE RelativeTime1900			PRESENCE optional}|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional }|
	{ ID id-SRSPosRRCInactiveConfig		CRITICALITY ignore	TYPE SRSPosRRCInactiveConfig		PRESENCE optional}|
	{ ID id-SRSPosRRCInactiveValidityAreaConfig	CRITICALITY ignore	TYPE SRSPosRRCInactiveValidityAreaConfig PRESENCE optional}|
	{ ID id-SRSPreconfiguration-List	CRITICALITY ignore	TYPE SRSPreconfiguration-List	PRESENCE optional},
	...
}


-- **************************************************************
--
-- Positioning Information Failure
--
-- **************************************************************

PositioningInformationFailure ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { PositioningInformationFailureIEs} },
	...
}

PositioningInformationFailureIEs F1AP-PROTOCOL-IES ::= {

	{ ID id-gNB-CU-UE-F1AP-ID		CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID			PRESENCE mandatory }|
	{ ID id-gNB-DU-UE-F1AP-ID		CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID			PRESENCE mandatory }|
	{ ID id-Cause				CRITICALITY ignore	TYPE Cause					PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional },
	...
}


-- **************************************************************
```

```
--
-- POSITIONING ACTIVATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Positioning Activation Request
--
-- **************************************************************

PositioningActivationRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { PositioningActivationRequestIEs} },
    ...
}

PositioningActivationRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID          PRESENCE mandatory }|
    { ID id-gNB-DU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID          PRESENCE mandatory }|
    { ID id-SRSType                 CRITICALITY reject  TYPE SRSType                    PRESENCE mandatory }|
    { ID id-ActivationTime          CRITICALITY ignore  TYPE RelativeTime1900       PRESENCE optional   },
    ...
}

SRSType ::= CHOICE {
    semipersistentSRS               SemipersistentSRS,
    aperiodicSRS                    AperiodicSRS,
    choice-extension                ProtocolIE-SingleContainer { { SRSType-ExtIEs} }
}

SRSType-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

SemipersistentSRS ::= SEQUENCE {
    sRSResourceSetID            SRSResourceSetID,
    sRSSpatialRelation          SpatialRelationInfo OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {SemipersistentSRS-ExtIEs} } OPTIONAL,
    ...
}

SemipersistentSRS-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-SRSSpatialRelationPerSRSResource    CRITICALITY ignore  EXTENSION SpatialRelationPerSRSResource PRESENCE optional}|
    { ID id-AggregatedPosSRSResourceSetList CRITICALITY ignore EXTENSION AggregatedPosSRSResourceSetList   PRESENCE optional},
    ...
}

AperiodicSRS ::= SEQUENCE {
    aperiodic                   ENUMERATED {true, ...},
    sRSResourceTrigger          SRSResourceTrigger      OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {AperiodicSRS-ExtIEs} } OPTIONAL,
    ...
}
```

```
AperiodicSRS-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-AggregatedPosSRSResourceSetList CRITICALITY ignore EXTENSION AggregatedPosSRSResourceSetList   PRESENCE optional},
    ...
}


-- **************************************************************
--
-- Positioning Activation Response
--
-- **************************************************************

PositioningActivationResponse ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { PositioningActivationResponseIEs} },
    ...
}


PositioningActivationResponseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID          PRESENCE mandatory }|
    { ID id-gNB-DU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID          PRESENCE mandatory }|
    { ID id-SystemFrameNumber       CRITICALITY ignore  TYPE SystemFrameNumber          PRESENCE optional }|
    { ID id-SlotNumber              CRITICALITY ignore  TYPE SlotNumber                 PRESENCE optional }|
    { ID id-CriticalityDiagnostics  CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional },
    ...
}




-- **************************************************************
--
-- Positioning Activation Failure
--
-- **************************************************************

PositioningActivationFailure ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { PositioningActivationFailureIEs} },
    ...
}

PositioningActivationFailureIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID              PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID              PRESENCE mandatory   }|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                          PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional },
    ...
}

-- **************************************************************
--
-- POSITIONING DEACTIVATION ELEMENTARY PROCEDURE
--
```

```
-- **************************************************************

-- **************************************************************
--
-- Positioning Deactivation
--
-- **************************************************************

PositioningDeactivation ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { PositioningDeactivationIEs} },
    ...
}

PositioningDeactivationIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID          PRESENCE mandatory }|
    { ID id-gNB-DU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID          PRESENCE mandatory }|
    { ID id-AbortTransmission       CRITICALITY ignore  TYPE AbortTransmission          PRESENCE mandatory },
    ...
}

-- **************************************************************
--
-- POSITIONING INFORMATION UPDATE ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Positioning Information Update
--
-- **************************************************************

PositioningInformationUpdate ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { PositioningInformationUpdateIEs} },
    ...
}


PositioningInformationUpdateIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID          PRESENCE mandatory }|
    { ID id-gNB-DU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID          PRESENCE mandatory }|
    { ID id-SRSConfiguration        CRITICALITY ignore  TYPE SRSConfiguration           PRESENCE optional}|
    { ID id-SFNInitialisationTime   CRITICALITY ignore  TYPE RelativeTime1900           PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SRS Information Reservation Notification
--
-- **************************************************************

SRSInformationReservationNotification ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ SRSInformationReservationNotificationIEs}},
```

```
	...
}

SRSInformationReservationNotificationIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID			CRITICALITY reject	TYPE TransactionID					PRESENCE mandatory	}|
	{ ID id-SRSReservationType		CRITICALITY reject	TYPE SRSReservationType			PRESENCE mandatory }|
	{ ID id-SRSInformation			CRITICALITY ignore	TYPE RequestedSRSTransmissionCharacteristics			PRESENCE optional }|
	{ ID id-PreconfiguredSRSInformation CRITICALITY ignore	TYPE RequestedSRSPreconfigurationCharacteristics-List			PRESENCE optional },
	...
}


-- **************************************************************
--
-- E-CID MEASUREMENT ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- E-CID Measurement Initiation Request
--
-- **************************************************************

E-CIDMeasurementInitiationRequest ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container	{{E-CIDMeasurementInitiationRequest-IEs}},
	...
}

E-CIDMeasurementInitiationRequest-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-LMF-UE-MeasurementID			CRITICALITY reject	TYPE LMF-UE-MeasurementID			PRESENCE mandatory	}|
	{ ID id-RAN-UE-MeasurementID			CRITICALITY reject	TYPE RAN-UE-MeasurementID			PRESENCE mandatory	}|
	{ ID id-E-CID-ReportCharacteristics		CRITICALITY reject	TYPE E-CID-ReportCharacteristics		PRESENCE mandatory	}|
	{ ID id-E-CID-MeasurementPeriodicity	CRITICALITY reject	TYPE MeasurementPeriodicity		PRESENCE conditional	}|
	-- The above IE shall be present if the E-CID-ReportCharacteristics IE is set to “periodic” –-
	{ ID id-E-CID-MeasurementQuantities		CRITICALITY reject	TYPE E-CID-MeasurementQuantities	PRESENCE mandatory}|
	{ ID id-PosMeasurementPeriodicityNR-AoA CRITICALITY reject	TYPE PosMeasurementPeriodicityNR-AoA	PRESENCE conditional},
	-- The IE shall be present if the E-CID-ReportCharacteristics IE is set to “periodic” and the E-CID-MeasurementQuantities-Item IE in the E-CID-
MeasurementQuantities IE is set to the value "angleOfArrivalNR"--
	...
}

-- **************************************************************
--
-- E-CID Measurement Initiation Response
--
-- **************************************************************

E-CIDMeasurementInitiationResponse ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container	{{E-CIDMeasurementInitiationResponse-IEs}},
	...
}
```

```
E-CIDMeasurementInitiationResponse-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID              PRESENCE mandatory  }|
    { ID id-gNB-DU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID              PRESENCE mandatory  }|
    { ID id-LMF-UE-MeasurementID        CRITICALITY reject  TYPE LMF-UE-MeasurementID           PRESENCE mandatory  }|
    { ID id-RAN-UE-MeasurementID        CRITICALITY reject  TYPE RAN-UE-MeasurementID           PRESENCE mandatory  }|
    { ID id-E-CID-MeasurementResult     CRITICALITY ignore  TYPE E-CID-MeasurementResult        PRESENCE optional}|
    { ID id-Cell-Portion-ID             CRITICALITY ignore  TYPE Cell-Portion-ID                PRESENCE optional}|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics         PRESENCE optional},
    ...
}

-- **************************************************************
--
-- E-CID Measurement Initiation Failure
--
-- **************************************************************

E-CIDMeasurementInitiationFailure ::= SEQUENCE {
    protocolIEs                     ProtocolIE-Container        {{E-CIDMeasurementInitiationFailure-IEs}},
    ...
}


E-CIDMeasurementInitiationFailure-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID              PRESENCE mandatory  }|
    { ID id-gNB-DU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID              PRESENCE mandatory  }|
    { ID id-LMF-UE-MeasurementID        CRITICALITY reject  TYPE LMF-UE-MeasurementID           PRESENCE mandatory  }|
    { ID id-RAN-UE-MeasurementID        CRITICALITY reject  TYPE RAN-UE-MeasurementID           PRESENCE mandatory  }|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                          PRESENCE mandatory  }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics         PRESENCE optional},
    ...
}

-- **************************************************************
--
-- E-CID MEASUREMENT FAILURE INDICATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- E-CID Measurement Failure Indication
--
-- **************************************************************

E-CIDMeasurementFailureIndication ::= SEQUENCE {
    protocolIEs                     ProtocolIE-Container        {{E-CIDMeasurementFailureIndication-IEs}},
    ...
}


E-CIDMeasurementFailureIndication-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID          PRESENCE mandatory   }|
```

```
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-LMF-UE-MeasurementID		CRITICALITY reject	TYPE LMF-UE-MeasurementID		PRESENCE mandatory	}|
	{ ID id-RAN-UE-MeasurementID		CRITICALITY reject	TYPE RAN-UE-MeasurementID		PRESENCE mandatory	}|
	{ ID id-Cause						CRITICALITY ignore	TYPE Cause						PRESENCE mandatory},
	...
}

-- **************************************************************
--
-- E-CID MEASUREMENT REPORT ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- E-CID Measurement Report
--
-- **************************************************************

E-CIDMeasurementReport ::= SEQUENCE {
	protocolIEs					ProtocolIE-Container		{{E-CIDMeasurementReport-IEs}},
	...
}


E-CIDMeasurementReport-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID				PRESENCE mandatory }|
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID				PRESENCE mandatory }|
	{ ID id-LMF-UE-MeasurementID		CRITICALITY reject	TYPE LMF-UE-MeasurementID			PRESENCE mandatory }|
	{ ID id-RAN-UE-MeasurementID		CRITICALITY reject	TYPE RAN-UE-MeasurementID			PRESENCE mandatory }|
	{ ID id-E-CID-MeasurementResult		CRITICALITY ignore	TYPE E-CID-MeasurementResult		PRESENCE mandatory }|
	{ ID id-Cell-Portion-ID				CRITICALITY ignore	TYPE Cell-Portion-ID				PRESENCE optional},

	...
}

-- **************************************************************
--
-- E-CID MEASUREMENT TERMINATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- E-CID Measurement Termination Command
--
-- **************************************************************


E-CIDMeasurementTerminationCommand ::= SEQUENCE {
	protocolIEs					ProtocolIE-Container		{{E-CIDMeasurementTerminationCommand-IEs}},
	...
}
```

```
E-CIDMeasurementTerminationCommand-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-LMF-UE-MeasurementID		CRITICALITY reject	TYPE LMF-UE-MeasurementID			PRESENCE mandatory	}|
	{ ID id-RAN-UE-MeasurementID		CRITICALITY reject	TYPE RAN-UE-MeasurementID			PRESENCE mandatory	},
	...
}

-- **************************************************************
--
-- BROADCAST CONTEXT SETUP ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- BROADCAST CONTEXT SETUP REQUEST
--
-- **************************************************************

BroadcastContextSetupRequest ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { BroadcastContextSetupRequestIEs} },
	...
}

BroadcastContextSetupRequestIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-MBS-F1AP-ID					CRITICALITY reject	TYPE	GNB-CU-MBS-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-MBS-Session-ID						CRITICALITY reject	TYPE	MBS-Session-ID					PRESENCE mandatory	}|
	{ ID id-MBS-ServiceArea						CRITICALITY reject	TYPE	MBS-ServiceArea					PRESENCE optional	}|
	{ ID id-MBS-CUtoDURRCInformation			CRITICALITY reject	TYPE	MBS-CUtoDURRCInformation		PRESENCE mandatory	}|
	{ ID id-SNSSAI								CRITICALITY reject	TYPE	SNSSAI							PRESENCE mandatory	}|
	{ ID id-BroadcastMRBs-ToBeSetup-List		CRITICALITY reject	TYPE	BroadcastMRBs-ToBeSetup-List	PRESENCE mandatory	}|
	{ ID id-SupportedUETypeList					CRITICALITY ignore	TYPE	SupportedUETypeList				PRESENCE optional	}|
	{ ID id-AssociatedSessionID					CRITICALITY ignore	TYPE	AssociatedSessionID				PRESENCE optional	}|
	{ ID id-RANSharingAssistanceInformation		CRITICALITY ignore	TYPE	RANSharingAssistanceInformation	PRESENCE optional	},
	...
}

BroadcastMRBs-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { BroadcastMRBs-ToBeSetup-ItemIEs} }


BroadcastMRBs-ToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-BroadcastMRBs-ToBeSetup-Item	CRITICALITY reject	TYPE	BroadcastMRBs-ToBeSetup-Item PRESENCE mandatory	},
	...
}


-- **************************************************************
--
-- BROADCAST CONTEXT SETUP RESPONSE
--
-- **************************************************************
```

```
BroadcastContextSetupResponse ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { BroadcastContextSetupResponseIEs} },
    ...
}

BroadcastContextSetupResponseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject TYPE GNB-CU-MBS-F1AP-ID                PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject TYPE GNB-DU-MBS-F1AP-ID                PRESENCE mandatory   }|
    { ID id-BroadcastMRBs-Setup-List            CRITICALITY reject TYPE BroadcastMRBs-Setup-List          PRESENCE mandatory   }|
    { ID id-BroadcastMRBs-FailedToBeSetup-List  CRITICALITY ignore TYPE BroadcastMRBs-FailedToBeSetup-List PRESENCE optional }|
    { ID id-BroadcastAreaScope                  CRITICALITY ignore TYPE BroadcastAreaScope                PRESENCE optional }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore TYPE CriticalityDiagnostics            PRESENCE optional },
    ...
}

BroadcastMRBs-Setup-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { BroadcastMRBs-Setup-ItemIEs} }

BroadcastMRBs-FailedToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { BroadcastMRBs-FailedToBeSetup-ItemIEs} }

BroadcastMRBs-Setup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BroadcastMRBs-Setup-Item            CRITICALITY reject  TYPE BroadcastMRBs-Setup-Item      PRESENCE mandatory},
    ...
}

BroadcastMRBs-FailedToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BroadcastMRBs-FailedToBeSetup-Item  CRITICALITY ignore  TYPE BroadcastMRBs-FailedToBeSetup-Item   PRESENCE mandatory}, ...
}


-- **************************************************************
--
-- BROADCAST CONTEXT SETUP FAILURE
--
-- **************************************************************

BroadcastContextSetupFailure ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { BroadcastContextSetupFailureIEs} },
    ...
}

BroadcastContextSetupFailureIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID          CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID          PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID          CRITICALITY ignore  TYPE GNB-DU-UE-F1AP-ID          PRESENCE optional }|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional },
    ...
}

-- **************************************************************
--
-- BROADCAST CONTEXT RELEASE ELEMENTARY PROCEDURE
--
-- **************************************************************
```

```
-- **************************************************************
--
-- BROADCAST CONTEXT RELEASE COMMAND
--
-- **************************************************************

BroadcastContextReleaseCommand ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { BroadcastContextReleaseCommandIEs} },
    ...
}

BroadcastContextReleaseCommandIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID              CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID           PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID              CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID           PRESENCE mandatory   }|
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                        PRESENCE mandatory   },
    ...
}

-- **************************************************************
--
-- BROADCAST CONTEXT RELEASE COMPLETE
--
-- **************************************************************

BroadcastContextReleaseComplete ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { BroadcastContextReleaseCompleteIEs} },
    ...
}
BroadcastContextReleaseCompleteIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID          CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID         PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID          CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID         PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional },
    ...
}


-- **************************************************************
--
-- BROADCAST CONTEXT RELEASE REQUEST ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- BROADCAST CONTEXT RELEASE REQUEST
--
-- **************************************************************

BroadcastContextReleaseRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ BroadcastContextReleaseRequestIEs}},
    ...
}
```

```
BroadcastContextReleaseRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                       PRESENCE mandatory   },
    ...
}


-- **************************************************************
--
-- BROADCAST CONTEXT MODIFICATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- BROADCAST CONTEXT MODIFICATION REQUEST
--
-- **************************************************************

BroadcastContextModificationRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { BroadcastContextModificationRequestIEs} },
    ...
}

BroadcastContextModificationRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID                PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID                PRESENCE mandatory   }|
    { ID id-MBS-ServiceArea                     CRITICALITY reject  TYPE MBS-ServiceArea                   PRESENCE optional    }|
    { ID id-MBS-CUtoDURRCInformation                CRITICALITY reject  TYPE MBS-CUtoDURRCInformation          PRESENCE mandatory   }|
    { ID id-BroadcastMRBs-ToBeSetupMod-List     CRITICALITY reject  TYPE BroadcastMRBs-ToBeSetupMod-List     PRESENCE optional }|
    { ID id-BroadcastMRBs-ToBeModified-List     CRITICALITY reject  TYPE BroadcastMRBs-ToBeModified-List     PRESENCE optional }|
    { ID id-BroadcastMRBs-ToBeReleased-List     CRITICALITY reject  TYPE BroadcastMRBs-ToBeReleased-List     PRESENCE optional }|
    { ID id-SupportedUETypeList                 CRITICALITY ignore  TYPE SupportedUETypeList               PRESENCE optional },
    ...
}

BroadcastMRBs-ToBeSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { BroadcastMRBs-ToBeSetupMod-ItemIEs} }
BroadcastMRBs-ToBeModified-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { BroadcastMRBs-ToBeModified-ItemIEs} }
BroadcastMRBs-ToBeReleased-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { BroadcastMRBs-ToBeReleased-ItemIEs} }

BroadcastMRBs-ToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BroadcastMRBs-ToBeSetupMod-Item     CRITICALITY reject  TYPE BroadcastMRBs-ToBeSetupMod-Item     PRESENCE mandatory},
    ...
}

BroadcastMRBs-ToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BroadcastMRBs-ToBeModified-Item     CRITICALITY reject  TYPE BroadcastMRBs-ToBeModified-Item     PRESENCE mandatory},
    ...
}

BroadcastMRBs-ToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BroadcastMRBs-ToBeReleased-Item     CRITICALITY reject  TYPE BroadcastMRBs-ToBeReleased-Item     PRESENCE mandatory},
    ...
```

```
}


-- **************************************************************
--
-- BROADCAST CONTEXT MODIFICATION RESPONSE
--
-- **************************************************************

BroadcastContextModificationResponse ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { BroadcastContextModificationResponseIEs} },
    ...
}


BroadcastContextModificationResponseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                      CRITICALITY reject TYPE GNB-CU-MBS-F1AP-ID                     PRESENCE mandatory}|
    { ID id-gNB-DU-MBS-F1AP-ID                      CRITICALITY reject TYPE GNB-DU-MBS-F1AP-ID                     PRESENCE mandatory}|

    { ID id-BroadcastMRBs-SetupMod-List             CRITICALITY reject TYPE BroadcastMRBs-SetupMod-List            PRESENCE optional}|
    { ID id-BroadcastMRBs-FailedToBeSetupMod-List   CRITICALITY ignore TYPE BroadcastMRBs-FailedToBeSetupMod-List  PRESENCE optional}|
    { ID id-BroadcastMRBs-Modified-List             CRITICALITY reject TYPE BroadcastMRBs-Modified-List            PRESENCE optional}|
    { ID id-BroadcastMRBs-FailedToBeModified-List   CRITICALITY ignore TYPE BroadcastMRBs-FailedToBeModified-List  PRESENCE optional}|
    { ID id-CriticalityDiagnostics                  CRITICALITY ignore  TYPE CriticalityDiagnostics                PRESENCE optional}|
    { ID id-BroadcastAreaScope                      CRITICALITY ignore TYPE BroadcastAreaScope                     PRESENCE optional},
    ...
}

BroadcastMRBs-SetupMod-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { BroadcastMRBs-SetupMod-ItemIEs} }

BroadcastMRBs-FailedToBeSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { BroadcastMRBs-FailedToBeSetupMod-
ItemIEs} }

BroadcastMRBs-Modified-List::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { BroadcastMRBs-Modified-ItemIEs } }

BroadcastMRBs-FailedToBeModified-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { BroadcastMRBs-FailedToBeModified-
ItemIEs} }


BroadcastMRBs-SetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BroadcastMRBs-SetupMod-Item             CRITICALITY     reject  TYPE BroadcastMRBs-SetupMod-Item       PRESENCE mandatory},
    ...
}

BroadcastMRBs-FailedToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BroadcastMRBs-FailedToBeSetupMod-Item   CRITICALITY     ignore  TYPE BroadcastMRBs-FailedToBeSetupMod-Item   PRESENCE mandatory},
    ...
}

BroadcastMRBs-Modified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BroadcastMRBs-Modified-Item             CRITICALITY     reject  TYPE BroadcastMRBs-Modified-Item       PRESENCE mandatory},
    ...
}
```

```
BroadcastMRBs-FailedToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-BroadcastMRBs-FailedToBeModified-Item   CRITICALITY     ignore  TYPE BroadcastMRBs-FailedToBeModified-Item   PRESENCE mandatory},
    ...
}

-- **************************************************************
--
-- BROADCAST CONTEXT MODIFICATION FAILURE
--
-- **************************************************************

BroadcastContextModificationFailure ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { BroadcastContextModificationFailureIEs} },
    ...
}

BroadcastContextModificationFailureIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID              CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID              CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                       PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics      PRESENCE optional },
    ...
}

-- **************************************************************
--
-- BROADCAST TRANSPORT RESOURCE REQUEST ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- BROADCAST TRANSPORT RESOURCE REQUEST
--
-- **************************************************************

BroadcastTransportResourceRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ BroadcastTransportResourceRequestIEs}},
    ...
}

BroadcastTransportResourceRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                          CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID                     PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                          CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID                     PRESENCE mandatory   }|
    { ID id-Broadcast-MRBs-Transport-Request-List       CRITICALITY reject  TYPE Broadcast-MRBs-Transport-Request-List PRESENCE optional    }|
    { ID id-F1U-PathFailure                             CRITICALITY ignore  TYPE F1U-PathFailure                        PRESENCE optional    },
    ...
}

Broadcast-MRBs-Transport-Request-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { Broadcast-MRBs-Transport-Request-
ItemIEs} }

Broadcast-MRBs-Transport-Request-ItemIEs F1AP-PROTOCOL-IES ::= {
```

```
    { ID id-Broadcast-MRBs-Transport-Request-Item           CRITICALITY reject  TYPE Broadcast-MRBs-Transport-Request-Item        PRESENCE 
mandatory},
    ...
}


-- **************************************************************
--
-- Multicast Group Paging ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- Multicast Group Paging
--
-- **************************************************************

MulticastGroupPaging ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ MulticastGroupPagingIEs}},
    ...
}

MulticastGroupPagingIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MBS-Session-ID                 CRITICALITY reject  TYPE MBS-Session-ID                    PRESENCE mandatory   }|
    { ID id-UEIdentity-List-For-Paging-List CRITICALITY ignore  TYPE UEIdentity-List-For-Paging-List   PRESENCE optional    }|
    { ID id-MC-PagingCell-List             CRITICALITY ignore  TYPE MC-PagingCell-list                PRESENCE optional    }|
    { ID id-IndicationMCInactiveReception       CRITICALITY ignore  TYPE IndicationMCInactiveReception    PRESENCE optional    },
    ...
}

UEIdentity-List-For-Paging-List  ::= SEQUENCE (SIZE(1.. maxnoofUEIDforPaging)) OF ProtocolIE-SingleContainer { { UEIdentity-List-For-Paging-ItemIEs 
} }


UEIdentity-List-For-Paging-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-UEIdentity-List-For-Paging-Item CRITICALITY ignore  TYPE UEIdentity-List-For-Paging-Item         PRESENCE mandatory } ,
    ...
}

MC-PagingCell-list::= SEQUENCE (SIZE(1.. maxnoofPagingCells)) OF ProtocolIE-SingleContainer { { MC-PagingCell-ItemIEs } }

MC-PagingCell-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MC-PagingCell-Item      CRITICALITY ignore  TYPE MC-PagingCell-Item         PRESENCE mandatory}   ,
    ...
}



-- **************************************************************
--
-- MULTICAST CONTEXT SETUP ELEMENTARY PROCEDURE
--
```

```
-- **************************************************************


-- **************************************************************
--
-- MULTICAST CONTEXT SETUP REQUEST
--
-- **************************************************************

MulticastContextSetupRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ MulticastContextSetupRequestIEs}},
    ...
}

MulticastContextSetupRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE    GNB-CU-MBS-F1AP-ID                      PRESENCE mandatory  }|
    { ID id-MBS-Session-ID                      CRITICALITY reject  TYPE    MBS-Session-ID                          PRESENCE mandatory  }|
    { ID id-MBS-ServiceArea                     CRITICALITY reject  TYPE    MBS-ServiceArea                         PRESENCE optional   }|
    { ID id-SNSSAI                              CRITICALITY reject  TYPE    SNSSAI                                  PRESENCE mandatory  }|
    { ID id-MulticastMRBs-ToBeSetup-List            CRITICALITY reject  TYPE    MulticastMRBs-ToBeSetup-List            PRESENCE mandatory  }|
    { ID id-MulticastCU2DURRCInfo                   CRITICALITY reject  TYPE    MulticastCU2DURRCInfo                   PRESENCE optional   }|
    { ID id-MBSMulticastSessionReceptionState       CRITICALITY reject  TYPE    MBSMulticastSessionReceptionState           PRESENCE optional   },
    ...
}

MulticastMRBs-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { MulticastMRBs-ToBeSetup-ItemIEs} }


MulticastMRBs-ToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastMRBs-ToBeSetup-Item    CRITICALITY reject  TYPE    MulticastMRBs-ToBeSetup-Item PRESENCE mandatory   },
    ...
}


-- **************************************************************
--
-- MULTICAST CONTEXT SETUP RESPONSE
--
-- **************************************************************

MulticastContextSetupResponse ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ MulticastContextSetupResponseIEs}},
    ...
}

MulticastContextSetupResponseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID                  PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID                  PRESENCE mandatory   }|
    { ID id-MulticastMRBs-Setup-List            CRITICALITY reject TYPE MulticastMRBs-Setup-List            PRESENCE mandatory   }|
    { ID id-MulticastMRBs-FailedToBeSetup-List  CRITICALITY ignore TYPE MulticastMRBs-FailedToBeSetup-List   PRESENCE optional }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore TYPE CriticalityDiagnostics              PRESENCE optional }|
    { ID id-MulticastDU2CURRCInfo               CRITICALITY reject TYPE MulticastDU2CURRCInfo               PRESENCE optional },
    ...
}
```

```
MulticastMRBs-Setup-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { MulticastMRBs-Setup-ItemIEs} }

MulticastMRBs-FailedToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { MulticastMRBs-FailedToBeSetup-ItemIEs} }

MulticastMRBs-Setup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastMRBs-Setup-Item            CRITICALITY reject  TYPE MulticastMRBs-Setup-Item      PRESENCE mandatory},
    ...
}

MulticastMRBs-FailedToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastMRBs-FailedToBeSetup-Item      CRITICALITY ignore  TYPE MulticastMRBs-FailedToBeSetup-Item PRESENCE mandatory},
    ...
}


-- **************************************************************
--
-- MULTICAST CONTEXT SETUP FAILURE
--
-- **************************************************************

MulticastContextSetupFailure ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ MulticastContextSetupFailureIEs}},
    ...
}

MulticastContextSetupFailureIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY ignore  TYPE GNB-DU-MBS-F1AP-ID          PRESENCE optional }|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                       PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics      PRESENCE optional },
    ...
}


-- **************************************************************
--
-- MULTICAST CONTEXT RELEASE ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- MULTICAST CONTEXT RELEASE COMMAND
--
-- **************************************************************

MulticastContextReleaseCommand ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ MulticastContextReleaseCommandIEs}},
    ...
}
```

```
MulticastContextReleaseCommandIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                       PRESENCE mandatory   },
    ...
}


-- **************************************************************
--
-- MULTICAST CONTEXT RELEASE COMPLETE
--
-- **************************************************************

MulticastContextReleaseComplete ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ MulticastContextReleaseCompleteIEs}},
    ...
}

MulticastContextReleaseCompleteIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore TYPE CriticalityDiagnostics      PRESENCE optional },
    ...
}


-- **************************************************************
--
-- MULTICAST CONTEXT RELEASE REQUEST ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- MULTICAST CONTEXT RELEASE REQUEST
--
-- **************************************************************

MulticastContextReleaseRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ MulticastContextReleaseRequestIEs}},
    ...
}

MulticastContextReleaseRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                       PRESENCE mandatory   },
    ...
}


-- **************************************************************
```

```
--
-- MULTICAST CONTEXT MODIFICATION ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- MULTICAST CONTEXT MODIFICATION REQUEST
--
-- **************************************************************

MulticastContextModificationRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ MulticastContextModificationRequestIEs}},
    ...
}

MulticastContextModificationRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID               PRESENCE mandatory }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID               PRESENCE mandatory }|
    { ID id-MBS-ServiceArea                     CRITICALITY reject  TYPE MBS-ServiceArea                     PRESENCE optional  }|
    { ID id-MulticastMRBs-ToBeSetupMod-List     CRITICALITY reject  TYPE MulticastMRBs-ToBeSetupMod-List     PRESENCE optional  }|
    { ID id-MulticastMRBs-ToBeModified-List     CRITICALITY reject  TYPE MulticastMRBs-ToBeModified-List     PRESENCE optional  }|
    { ID id-MulticastMRBs-ToBeReleased-List     CRITICALITY reject  TYPE MulticastMRBs-ToBeReleased-List     PRESENCE optional  }|
    { ID id-MulticastCU2DURRCInfo                   CRITICALITY reject  TYPE        MulticastCU2DURRCInfo            PRESENCE optional  }|
    { ID id-MBSMulticastSessionReceptionState       CRITICALITY reject  TYPE MBSMulticastSessionReceptionState     PRESENCE optional  },
    ...
}

MulticastMRBs-ToBeSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { MulticastMRBs-ToBeSetupMod-ItemIEs} }
MulticastMRBs-ToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastMRBs-ToBeSetupMod-Item     CRITICALITY reject  TYPE MulticastMRBs-ToBeSetupMod-Item     PRESENCE mandatory},
    ...
}

MulticastMRBs-ToBeModified-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { MulticastMRBs-ToBeModified-ItemIEs} }
MulticastMRBs-ToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastMRBs-ToBeModified-Item     CRITICALITY reject  TYPE MulticastMRBs-ToBeModified-Item     PRESENCE mandatory},
    ...
}


MulticastMRBs-ToBeReleased-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { MulticastMRBs-ToBeReleased-ItemIEs} }
MulticastMRBs-ToBeReleased-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastMRBs-ToBeReleased-Item     CRITICALITY reject  TYPE MulticastMRBs-ToBeReleased-Item     PRESENCE mandatory},
    ...
}


-- **************************************************************
--
-- MULTICAST CONTEXT MODIFICATION RESPONSE
--
-- **************************************************************
```

```
MulticastContextModificationResponse ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ MulticastContextModificationResponseIEs}},
    ...
}

MulticastContextModificationResponseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                      CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID                     PRESENCE mandatory }|
    { ID id-gNB-DU-MBS-F1AP-ID                      CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID                     PRESENCE mandatory }|
    { ID id-MulticastMRBs-SetupMod-List         CRITICALITY reject TYPE MulticastMRBs-SetupMod-List                 PRESENCE optional  }|
    { ID id-MulticastMRBs-FailedToBeSetupMod-List   CRITICALITY ignore TYPE MulticastMRBs-FailedToBeSetupMod-List  PRESENCE optional  }|
    { ID id-MulticastMRBs-Modified-List         CRITICALITY reject TYPE MulticastMRBs-Modified-List                 PRESENCE optional  }|
    { ID id-MulticastMRBs-FailedToBeModified-List   CRITICALITY ignore TYPE MulticastMRBs-FailedToBeModified-List  PRESENCE optional  }|
    { ID id-CriticalityDiagnostics                  CRITICALITY ignore  TYPE CriticalityDiagnostics                 PRESENCE optional  }|
    { ID id-MulticastDU2CURRCInfo                   CRITICALITY reject TYPE MulticastDU2CURRCInfo                   PRESENCE optional },
    ...
}

MulticastMRBs-SetupMod-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { MulticastMRBs-SetupMod-ItemIEs} }
MulticastMRBs-SetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastMRBs-SetupMod-Item             CRITICALITY     reject  TYPE MulticastMRBs-SetupMod-Item       PRESENCE mandatory},
    ...
}

MulticastMRBs-FailedToBeSetupMod-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { MulticastMRBs-FailedToBeSetupMod-
ItemIEs} }
MulticastMRBs-FailedToBeSetupMod-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastMRBs-FailedToBeSetupMod-Item   CRITICALITY     ignore  TYPE MulticastMRBs-FailedToBeSetupMod-Item   PRESENCE mandatory},
    ...
}

MulticastMRBs-Modified-List::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { MulticastMRBs-Modified-ItemIEs } }
MulticastMRBs-Modified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastMRBs-Modified-Item             CRITICALITY     reject  TYPE MulticastMRBs-Modified-Item       PRESENCE mandatory},
    ...
}

MulticastMRBs-FailedToBeModified-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { MulticastMRBs-FailedToBeModified-
ItemIEs} }
MulticastMRBs-FailedToBeModified-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastMRBs-FailedToBeModified-Item   CRITICALITY     ignore  TYPE MulticastMRBs-FailedToBeModified-Item   PRESENCE mandatory},
    ...
}


-- **************************************************************
--
-- MULTICAST CONTEXT MODIFICATION FAILURE
--
-- **************************************************************

MulticastContextModificationFailure ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ MulticastContextModificationFailureIEs}},
    ...
```

```
}

MulticastContextModificationFailureIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID          PRESENCE mandatory   }|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                       PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics      PRESENCE optional },
    ...
}


-- **************************************************************
--
-- MULTICAST CONTEXT NOTIFICATION ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- MULTICAST CONTEXT NOTIFICATION INDICATION
--
-- **************************************************************

MulticastContextNotificationIndication ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{MulticastContextNotificationIndicationIEs}},
    ...
}

MulticastContextNotificationIndicationIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID                  PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID                  PRESENCE mandatory   }|
    { ID id-MulticastDU2CURRCInfo               CRITICALITY reject  TYPE MulticastDU2CURRCInfo               PRESENCE optional    },
    ...
}


-- **************************************************************
--
-- MULTICAST CONTEXT NOTIFICATION CONFIRM
--
-- **************************************************************

MulticastContextNotificationConfirm ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{MulticastContextNotificationConfirmIEs}},
    ...
}

MulticastContextNotificationConfirmIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID                  PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID                  PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics              PRESENCE optional    },
    ...
}
```

```
-- **************************************************************
--
-- MULTICAST CONTEXT NOTIFICATION REFUSE
--
-- **************************************************************

MulticastContextNotificationRefuse ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{MulticastContextNotificationRefuseIEs}},
    ...
}

MulticastContextNotificationRefuseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID                 PRESENCE mandatory   }|
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID                 PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics             PRESENCE optional    }|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                              PRESENCE mandatory   },
    ...
}


-- **************************************************************
--
-- MULTICAST COMMON CONFIGURATION ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- MULTICAST COMMON CONFIGURATION REQUEST
--
-- **************************************************************

MulticastCommonConfigurationRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{MulticastCommonConfigurationRequestIEs}},
    ...
}

MulticastCommonConfigurationRequestIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                           CRITICALITY reject  TYPE TransactionID                              PRESENCE mandatory   }|
    { ID id-MulticastCU2DUCommonRRCInfo             CRITICALITY reject  TYPE MulticastCU2DUCommonRRCInfo                PRESENCE optional    },
    ...
}


-- **************************************************************
--
-- MULTICAST COMMON CONFIGURATION RESPONSE
--
-- **************************************************************

MulticastCommonConfigurationResponse ::= SEQUENCE {
```

```
	protocolIEs			ProtocolIE-Container		{{MulticastCommonConfigurationResponseIEs}},
	...
}

MulticastCommonConfigurationResponseIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID						CRITICALITY reject	TYPE TransactionID						PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics				CRITICALITY ignore	TYPE CriticalityDiagnostics				PRESENCE optional	},
	...
}


-- **************************************************************
--
-- MULTICAST COMMON CONFIGURATION REFUSE
--
-- **************************************************************

MulticastCommonConfigurationRefuse ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{MulticastCommonConfigurationRefuseIEs}},
	...
}

MulticastCommonConfigurationRefuseIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID						CRITICALITY reject	TYPE TransactionID						PRESENCE mandatory	}|
	{ ID id-Cause								CRITICALITY ignore	TYPE Cause								PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics				CRITICALITY ignore	TYPE CriticalityDiagnostics				PRESENCE optional	},
	...
}


-- **************************************************************
--
-- MULTICAST DISTRIBUTION SETUP ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- MULTICAST DISTRIBUTION SETUP REQUEST
--
-- **************************************************************

MulticastDistributionSetupRequest ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ MulticastDistributionSetupRequestIEs}},
	...
}

MulticastDistributionSetupRequestIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-MBS-F1AP-ID					CRITICALITY reject	TYPE GNB-CU-MBS-F1AP-ID					PRESENCE mandatory	}|
	{ ID id-gNB-DU-MBS-F1AP-ID					CRITICALITY reject	TYPE GNB-DU-MBS-F1AP-ID					PRESENCE mandatory	}|
	{ ID id-MBSMulticastF1UContextDescriptor	CRITICALITY reject	TYPE MBSMulticastF1UContextDescriptor	PRESENCE mandatory	}|
	{ ID id-MulticastF1UContext-ToBeSetup-List	CRITICALITY reject	TYPE MulticastF1UContext-ToBeSetup-List	PRESENCE mandatory	},
	...
```

```
}

MulticastF1UContext-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF
                                                                    ProtocolIE-SingleContainer { { MulticastF1UContext-ToBeSetup-ItemIEs} }
MulticastF1UContext-ToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastF1UContext-ToBeSetup-Item              CRITICALITY     reject  TYPE MulticastF1UContext-ToBeSetup-Item       PRESENCE
mandatory},
    ...
}


-- **************************************************************
--
-- MULTICAST DISTRIBUTION SETUP RESPONSE
--
-- **************************************************************

MulticastDistributionSetupResponse ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ MulticastDistributionSetupResponseIEs}},
    ...
}

MulticastDistributionSetupResponseIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-MBS-F1AP-ID                          CRITICALITY reject  TYPE GNB-CU-MBS-F1AP-ID                         PRESENCE mandatory}|
    { ID id-gNB-DU-MBS-F1AP-ID                          CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID                         PRESENCE mandatory}|
    { ID id-MBSMulticastF1UContextDescriptor            CRITICALITY reject  TYPE MBSMulticastF1UContextDescriptor           PRESENCE mandatory}|
    { ID id-MulticastF1UContext-Setup-List              CRITICALITY reject  TYPE MulticastF1UContext-Setup-List             PRESENCE mandatory}|
    { ID id-MulticastF1UContext-FailedToBeSetup-List    CRITICALITY ignore  TYPE MulticastF1UContext-FailedToBeSetup-List   PRESENCE optional}|
    { ID id-CriticalityDiagnostics                      CRITICALITY ignore  TYPE CriticalityDiagnostics                     PRESENCE optional }|
    { ID id-MulticastF1UContextReferenceCU              CRITICALITY reject  TYPE MulticastF1UContextReferenceCU             PRESENCE mandatory},
    ...
}

MulticastF1UContext-Setup-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF ProtocolIE-SingleContainer { { MulticastF1UContext-Setup-ItemIEs} }
MulticastF1UContext-Setup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastF1UContext-Setup-Item              CRITICALITY     reject  TYPE MulticastF1UContext-Setup-Item      PRESENCE mandatory},
    ...
}

MulticastF1UContext-FailedToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF
                                                                ProtocolIE-SingleContainer { { MulticastF1UContext-FailedToBeSetup-ItemIEs} }
MulticastF1UContext-FailedToBeSetup-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-MulticastF1UContext-FailedToBeSetup-Item    CRITICALITY  ignore TYPE MulticastF1UContext-FailedToBeSetup-Item  PRESENCE mandatory},
    ...
}


-- **************************************************************
--
-- MULTICAST DISTRIBUTION SETUP FAILURE
--
-- **************************************************************

MulticastDistributionSetupFailure ::= SEQUENCE {
```

```
	protocolIEs		ProtocolIE-Container		{{ MulticastDistributionSetupFailureIEs}},
	...
}

MulticastDistributionSetupFailureIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-MBS-F1AP-ID					CRITICALITY reject	TYPE GNB-CU-MBS-F1AP-ID					PRESENCE mandatory	}|
	{ ID id-gNB-DU-MBS-F1AP-ID					CRITICALITY ignore	TYPE GNB-DU-MBS-F1AP-ID					PRESENCE optional }|
	{ ID id-MBSMulticastF1UContextDescriptor	CRITICALITY reject	TYPE MBSMulticastF1UContextDescriptor	PRESENCE mandatory	}|
	{ ID id-Cause								CRITICALITY ignore	TYPE Cause								PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics				CRITICALITY ignore	TYPE CriticalityDiagnostics				PRESENCE optional },
	...
}


-- **************************************************************
--
-- MULTICAST DISTRIBUTION RELEASE ELEMENTARY PROCEDURE
--
-- **************************************************************


-- **************************************************************
--
-- MULTICAST DISTRIBUTION RELEASE COMMAND
--
-- **************************************************************

MulticastDistributionReleaseCommand ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{{ MulticastDistributionReleaseCommandIEs}},
	...
}

MulticastDistributionReleaseCommandIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-MBS-F1AP-ID					CRITICALITY reject	TYPE GNB-CU-MBS-F1AP-ID					PRESENCE mandatory	}|
	{ ID id-gNB-DU-MBS-F1AP-ID					CRITICALITY reject	TYPE GNB-DU-MBS-F1AP-ID					PRESENCE mandatory	}|
	{ ID id-MBSMulticastF1UContextDescriptor	CRITICALITY reject	TYPE MBSMulticastF1UContextDescriptor	PRESENCE mandatory	}|
	{ ID id-Cause								CRITICALITY ignore	TYPE Cause								PRESENCE mandatory	},
	...
}


-- **************************************************************
--
-- MULTICAST DISTRIBUTION RELEASE COMPLETE
--
-- **************************************************************

MulticastDistributionReleaseComplete ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{{ MulticastDistributionReleaseCompleteIEs}},
	...
}

MulticastDistributionReleaseCompleteIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-MBS-F1AP-ID					CRITICALITY reject	TYPE GNB-CU-MBS-F1AP-ID					PRESENCE mandatory	}|
```

```
    { ID id-gNB-DU-MBS-F1AP-ID                  CRITICALITY reject  TYPE GNB-DU-MBS-F1AP-ID                  PRESENCE mandatory   }|
    { ID id-MBSMulticastF1UContextDescriptor    CRITICALITY reject  TYPE MBSMulticastF1UContextDescriptor    PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics              PRESENCE optional },
    ...
}


-- **************************************************************
--
-- PDC MEASUREMENT ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- PDC Measurement Initiation Request
--
-- **************************************************************

PDCMeasurementInitiationRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{PDCMeasurementInitiationRequest-IEs}},
    ...
}

PDCMeasurementInitiationRequest-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID               CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID               PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID               CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID               PRESENCE mandatory   }|
    { ID id-RAN-UE-PDC-MeasID               CRITICALITY reject  TYPE RAN-UE-PDC-MeasID               PRESENCE mandatory   }|
    { ID id-PDCReportType                   CRITICALITY reject  TYPE PDCReportType                   PRESENCE mandatory   }|
    { ID id-PDCMeasurementPeriodicity       CRITICALITY reject  TYPE PDCMeasurementPeriodicity       PRESENCE conditional }|
    -- The above IE shall be present if the PDCReportType IE is set to “periodic” –-
    { ID id-PDCMeasurementQuantities        CRITICALITY reject  TYPE PDCMeasurementQuantities        PRESENCE mandatory   },
    ...
}

-- **************************************************************
--
-- PDC Measurement Initiation Response
--
-- **************************************************************

PDCMeasurementInitiationResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{PDCMeasurementInitiationResponse-IEs}},
    ...
}

PDCMeasurementInitiationResponse-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID          PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID          PRESENCE mandatory   }|
    { ID id-RAN-UE-PDC-MeasID           CRITICALITY reject  TYPE RAN-UE-PDC-MeasID          PRESENCE mandatory   }|
    { ID id-PDCMeasurementResult        CRITICALITY ignore  TYPE PDCMeasurementResult       PRESENCE optional    }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional    },
    ...
}
```

```
-- **************************************************************
--
-- PDC Measurement Initiation Failure
--
-- **************************************************************

PDCMeasurementInitiationFailure ::= SEQUENCE {
	protocolIEs					ProtocolIE-Container		{{PDCMeasurementInitiationFailure-IEs}},
	...
}

PDCMeasurementInitiationFailure-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-RAN-UE-PDC-MeasID			CRITICALITY ignore	TYPE RAN-UE-PDC-MeasID				PRESENCE mandatory	}|
	{ ID id-Cause						CRITICALITY ignore	TYPE Cause							PRESENCE mandatory	}|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics			PRESENCE optional	},
	...
}

-- **************************************************************
--
-- PDC MEASUREMENT REPORT ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- PDC Measurement Report
--
-- **************************************************************

PDCMeasurementReport ::= SEQUENCE {
	protocolIEs					ProtocolIE-Container		{{PDCMeasurementReport-IEs}},
	...
}

PDCMeasurementReport-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-RAN-UE-PDC-MeasID			CRITICALITY reject	TYPE RAN-UE-PDC-MeasID			PRESENCE mandatory	}|
	{ ID id-PDCMeasurementResult		CRITICALITY ignore	TYPE PDCMeasurementResult		PRESENCE mandatory	},
	...
}

-- **************************************************************
--
-- PDC MEASUREMENT TERMINATION PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
```

```
-- PDC Measurement Termination
--
-- **************************************************************

PDCMeasurementTerminationCommand ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { PDCMeasurementTerminationCommand-IEs} },
	...
}


PDCMeasurementTerminationCommand-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-RAN-UE-PDC-MeasID			CRITICALITY ignore	TYPE RAN-UE-PDC-MeasID			PRESENCE mandatory	},
	...
}


-- **************************************************************
--
-- PDC MEASUREMENT FAILURE INDICATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- PDC Measurement Failure Indication
--
-- **************************************************************


PDCMeasurementFailureIndication ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { PDCMeasurementFailureIndication-IEs} },
	...
}

PDCMeasurementFailureIndication-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-RAN-UE-PDC-MeasID			CRITICALITY ignore	TYPE RAN-UE-PDC-MeasID			PRESENCE mandatory	}|
	{ ID id-Cause						CRITICALITY ignore	TYPE Cause						PRESENCE mandatory	},
	...
}

-- **************************************************************
--
-- PPS CONFIGURATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- PRS CONFIGURATION REQUEST
--
```

```
-- **************************************************************

PRSConfigurationRequest ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container	{{PRSConfigurationRequest-IEs}},
	...
}

PRSConfigurationRequest-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID			CRITICALITY reject	TYPE TransactionID					PRESENCE mandatory }|
	{ ID id-PRSConfigRequestType	CRITICALITY reject	TYPE PRSConfigRequestType			PRESENCE mandatory }|
	{ ID id-PRSTRPList				CRITICALITY ignore	TYPE PRSTRPList						PRESENCE mandatory },
	...
}

-- **************************************************************
--
-- PRS CONFIGURATION RESPONSE
--
-- **************************************************************

PRSConfigurationResponse ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container	{{ PRSConfigurationResponse-IEs}},
	...
}

PRSConfigurationResponse-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID			CRITICALITY reject	TYPE TransactionID				PRESENCE mandatory}|
	{ ID id-PRSTransmissionTRPList	CRITICALITY ignore	TYPE PRSTransmissionTRPList		PRESENCE optional}|
	{ ID id-CriticalityDiagnostics	CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional},
	...
}

-- **************************************************************
--
-- PRS CONFIGURATION FAILURE
--
-- **************************************************************

PRSConfigurationFailure ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container	{{ PRSConfigurationFailure-IEs}},
	...
}

PRSConfigurationFailure-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID			CRITICALITY reject	TYPE TransactionID			PRESENCE mandatory}|
	{ ID id-Cause	CRITICALITY ignore	TYPE Cause	PRESENCE mandatory}|
	{ ID id-CriticalityDiagnostics	CRITICALITY ignore	TYPE CriticalityDiagnostics		PRESENCE optional},
	...
}


-- **************************************************************
--
-- MEASUREMENT PRECONFIGURATION ELEMENTARY PROCEDURE
```

```
--
-- **************************************************************

-- **************************************************************
--
-- Positioning Preconfiguration Required
--
-- **************************************************************

MeasurementPreconfigurationRequired ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container	{{ MeasurementPreconfigurationRequired-IEs}},
	...
}

MeasurementPreconfigurationRequired-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID	CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID	PRESENCE mandatory}|
	{ ID id-gNB-DU-UE-F1AP-ID	CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID	PRESENCE mandatory}|
	{ ID id-TRP-PRS-Info-List	CRITICALITY ignore	TYPE TRP-PRS-Info-List	PRESENCE mandatory},
	...
}


-- **************************************************************
--
-- Positioning Preconfiguration Confirm
--
-- **************************************************************

MeasurementPreconfigurationConfirm ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { MeasurementPreconfigurationConfirm-IEs} },
	...
}


MeasurementPreconfigurationConfirm-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID			CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID				PRESENCE mandatory	}|
	{ ID id-PosMeasGapPreConfigList		CRITICALITY ignore	TYPE PosMeasGapPreConfigList		PRESENCE optional }|
	{ ID id-CriticalityDiagnostics		CRITICALITY ignore	TYPE CriticalityDiagnostics			PRESENCE optional },
	...
}



-- **************************************************************
--
-- Positioning Preconfiguration Refuse
--
-- **************************************************************

MeasurementPreconfigurationRefuse ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ { MeasurementPreconfigurationRefuse-IEs} },
	...
```

```
}

MeasurementPreconfigurationRefuse-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID          PRESENCE mandatory  }|
    { ID id-gNB-DU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID          PRESENCE mandatory  }|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory  }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional },
    ...
}


-- **************************************************************
--
-- MEASUREMENT ACTIVATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Measurement Activation
--
-- **************************************************************

MeasurementActivation ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { MeasurementActivation-IEs} },
    ...
}

MeasurementActivation-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID              PRESENCE mandatory }|
    { ID id-gNB-DU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID              PRESENCE mandatory }|
    { ID id-ActivationRequestType       CRITICALITY reject  TYPE ActivationRequestType          PRESENCE mandatory}|
    { ID id-PRS-Measurement-Info-List   CRITICALITY ignore  TYPE PRS-Measurement-Info-List      PRESENCE optional},
    ...
}

-- **************************************************************
--
-- QOE INFORMATION TRANSFER ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- QoE Information Transfer
--
-- **************************************************************


QoEInformationTransfer ::= SEQUENCE {
    protocolIEs                     ProtocolIE-Container        {{QoEInformationTransfer-IEs}},
    ...
}
```

```
QoEInformationTransfer-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID               PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID               PRESENCE mandatory   }|
    { ID id-QoEInformation              CRITICALITY ignore  TYPE QoEInformation                  PRESENCE optional },
    ...
}


-- **************************************************************
--
-- POSITIONING SYSTEM INFORMATION DELIVERY ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- Positioning System information Delivery Command
--
-- **************************************************************

PosSystemInformationDeliveryCommand ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ PosSystemInformationDeliveryCommandIEs}},
    ...
}

PosSystemInformationDeliveryCommandIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID           CRITICALITY reject  TYPE TransactionID                  PRESENCE mandatory   }|
    { ID id-NRCGI                   CRITICALITY reject  TYPE NRCGI                          PRESENCE mandatory   }|
    { ID id-PosSItypeList           CRITICALITY reject  TYPE PosSItypeList                  PRESENCE mandatory   }|
    { ID id-ConfirmedUEID           CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID              PRESENCE mandatory   },
    ...
}

-- **************************************************************
--
-- DU-CU CELL SWITCH NOTIFICATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- DU-CU Cell Switch Notification
--
-- **************************************************************

DUCUCellSwitchNotification ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ DUCUCellSwitchNotificationIEs}},
    ...
}

DUCUCellSwitchNotificationIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID              PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID              PRESENCE mandatory   }|
```

```
    { ID id-NRCGI                   CRITICALITY reject  TYPE NRCGI                          PRESENCE mandatory   }|
    { ID id-LTMCellSwitchInformation     CRITICALITY ignore TYPE LTMCellSwitchInformation   PRESENCE optional    }|
    { ID id-TAInformation-List          CRITICALITY ignore  TYPE TAInformation-List         PRESENCE optional    },
    ...
}


-- **************************************************************
--
-- CU-DU CELL SWITCH NOTIFICATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- CU-DU Cell Switch Notification
--
-- **************************************************************

CUDUCellSwitchNotification ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ CUDUCellSwitchNotificationIEs}},
    ...
}

CUDUCellSwitchNotificationIEs F1AP-PROTOCOL-IES ::= {
    { ID id-gNB-CU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-CU-UE-F1AP-ID              PRESENCE mandatory   }|
    { ID id-gNB-DU-UE-F1AP-ID       CRITICALITY reject  TYPE GNB-DU-UE-F1AP-ID              PRESENCE mandatory   }|
    { ID id-NRCGI                   CRITICALITY reject  TYPE NRCGI                          PRESENCE mandatory   }|
    { ID id-LTMCellSwitchInformation     CRITICALITY ignore TYPE LTMCellSwitchInformation   PRESENCE optional}|
    { ID id-TAInformation-List          CRITICALITY ignore  TYPE TAInformation-List         PRESENCE optional    }|
    { ID id-LastVisitedLTMCells         CRITICALITY ignore  TYPE LastVisitedLTMCells            PRESENCE optional     },
    ...
}


-- **************************************************************
--
-- DU-CU TA INFORMATION TRANSFER ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- DU-CU TA Information Transfer
--
-- **************************************************************

DUCUTAInformationTransfer ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       {{ DUCUTAInformationTransferIEs}},
    ...
}

DUCUTAInformationTransferIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                   CRITICALITY reject  TYPE TransactionID                      PRESENCE mandatory   }|
```

```
	{ ID id-DUtoCUTAInformation-List		CRITICALITY ignore	TYPE DUtoCUTAInformation-List			PRESENCE mandatory	},
	...
}

-- **************************************************************
--
-- CU-DU TA INFORMATION TRANSFER ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- CU-DU TA Information Transfer
--
-- **************************************************************

CUDUTAInformationTransfer ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{{ CUDUTAInformationTransferIEs}},
	...
}

CUDUTAInformationTransferIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID					CRITICALITY reject	TYPE TransactionID				PRESENCE mandatory	}|
	{ ID id-CUtoDUTAInformation-List		CRITICALITY ignore	TYPE CUtoDUTAInformation-List			PRESENCE mandatory	},
	...
}

-- **************************************************************
--
-- QOE INFORMATION TRANSFER CONTROL ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- QoE Information Transfer Control
--
-- **************************************************************


QoEInformationTransferControl ::= SEQUENCE {
	protocolIEs						ProtocolIE-Container {{QoEInformationTransferControl-IEs}},
	...
}


QoEInformationTransferControl-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID						CRITICALITY reject	TYPE TransactionID						PRESENCE mandatory	}|
	{ ID id-DeactivationIndication				CRITICALITY ignore	TYPE DeactivationIndication				PRESENCE optional },
	...
}


-- **************************************************************
```

```
--
-- RACH Indication ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- RACH Indication
--
-- **************************************************************


RachIndication ::= SEQUENCE {
    protocolIEs                     ProtocolIE-Container        {{ RachIndication-IEs}},
    ...
}


RachIndication-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID               CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory   }|
    { ID id-RAReportIndicationList              CRITICALITY reject  TYPE RAReportIndicationList              PRESENCE mandatory   },
    ...
}

-- **************************************************************
--
-- Timing Synchronisation Status Elementary Procedure
--
-- **************************************************************

-- **************************************************************
--
-- TIMING SYNCHRONISATION STATUS REQUEST
--
-- **************************************************************

TimingSynchronisationStatusRequest::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{TimingSynchronisationStatusRequest-IEs}},
    ...
}

TimingSynchronisationStatusRequest-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                   CRITICALITY reject  TYPE TransactionID                          PRESENCE mandatory   }|
    { ID id-RANTSSRequestType               CRITICALITY reject  TYPE RANTSSRequestType                      PRESENCE mandatory },
    ...
}

-- **************************************************************
--
-- TIMING SYNCHRONISATION STATUS RESPONSE
--
-- **************************************************************

TimingSynchronisationStatusResponse::= SEQUENCE {
```

```
    protocolIEs     ProtocolIE-Container    {{TimingSynchronisationStatusResponse-IEs}},
    ...
}

TimingSynchronisationStatusResponse-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID               CRITICALITY reject  TYPE TransactionID                  PRESENCE mandatory  }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics         PRESENCE optional   },
    ...
}

-- **************************************************************
--
-- TIMING SYNCHRONISATION STATUS FAILURE
--
-- **************************************************************

TimingSynchronisationStatusFailure::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{TimingSynchronisationStatusFailure-IEs}},
    ...
}

TimingSynchronisationStatusFailure-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                       CRITICALITY reject  TYPE TransactionID              PRESENCE mandatory   }|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional },
    ...
}
-- **************************************************************
--
-- Timing Synchronisation Status Reporting Elementary Procedure
--
-- **************************************************************

-- **************************************************************
--
-- TIMING SYNCHRONISATION STATUS REPORT
--
-- **************************************************************

TimingSynchronisationStatusReport::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ TimingSynchronisationStatusReport-IEs}},
    ...
}

TimingSynchronisationStatusReport-IEs F1AP-PROTOCOL-IES ::= {
    { ID id-TransactionID                                   CRITICALITY reject  TYPE TransactionID                  PRESENCE mandatory   }|
    { ID id-RANTimingSynchronisationStatusInfo              CRITICALITY ignore  TYPE RANTimingSynchronisationStatusInfo                    PRESENCE 
mandatory },
    ...
}

-- **************************************************************
--
--  DU-CU Access And Mobility Indication ELEMENTARY PROCEDURE
```

```
--
-- **************************************************************

-- **************************************************************
--
-- DU-CU Access And Mobility Indication
--
-- **************************************************************

DUCUAccessAndMobilityIndication ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{ { DUCUAccessAndMobilityIndicationIEs} },
	...
}

DUCUAccessAndMobilityIndicationIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID				CRITICALITY reject	TYPE TransactionID			PRESENCE mandatory }|
	{ ID id-DLLBTFailureInformationList		CRITICALITY ignore	TYPE DLLBTFailureInformationList	PRESENCE optional}|
	{ ID id-MROForLTM-Information		CRITICALITY ignore	TYPE MROForLTM-Information	PRESENCE optional },
	...
}


-- **************************************************************
--
-- CU-DU MOBILITY INITIATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- CU-DU Mobility Initiation Request
--
-- **************************************************************

CUDUMobilityInitiationRequest ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container		{{ CUDUMobilityInitiationRequestIEs }},
	...
}

CUDUMobilityInitiationRequestIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID		CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID		CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID			PRESENCE mandatory	}|
	{ ID id-MobilityInitiation		CRITICALITY reject	TYPE MobilityInitiation			PRESENCE mandatory	},
	...
}

-- **************************************************************
--
-- CLI Indication
--
-- **************************************************************

CLI-Indication ::= SEQUENCE {
	protocolIEs		ProtocolIE-Container	{{CLI-Indication-IEs}},
```

```
	...
}

CLI-Indication-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-TransactionID					CRITICALITY reject	TYPE TransactionID				PRESENCE mandatory }|
	{ ID id-CLI-MeasurementResult-List			CRITICALITY ignore	TYPE CLI-MeasurementResult-List			PRESENCE optional }|
	{ ID id-SRS-Resource-Indication				CRITICALITY ignore	TYPE SRS-Resource-Indication		PRESENCE optional },
	...
}



-- **************************************************************
--
-- DU-CU CSI-RS COORDINATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- DU-CU CSI-RS COORDINATION REQUEST
--
-- **************************************************************
DUCUCSIRSCoordinationRequest ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ {DUCUCSIRSCoordinationRequest-IEs} },
	...
}

DUCUCSIRSCoordinationRequest-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID					PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID					PRESENCE mandatory	}|
	{ ID id-CSI-RSCoordinationRequestList			CRITICALITY ignore	TYPE CSI-RSCoordinationRequestList			PRESENCE mandatory},
	...
}


-- **************************************************************
--
-- DU-CU CSI-RS COORDINATION RESPONSE
--
-- **************************************************************

DUCUCSIRSCoordinationResponse ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ {DUCUCSIRSCoordinationResponse-IEs} },
	...
}


DUCUCSIRSCoordinationResponse-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID					PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID				CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID					PRESENCE mandatory	}|
	{ ID id-CSI-RSCoordinationResultList			CRITICALITY ignore	TYPE CSI-RSCoordinationResultList			PRESENCE mandatory},
	...
}
```

```
-- **************************************************************
--
-- CU-DU CSI-RS COORDINATION ELEMENTARY PROCEDURE
--
-- **************************************************************

-- **************************************************************
--
-- CU-DU CSI-RS COORDINATION REQUEST
--
-- **************************************************************

CUDUCSIRSCoordinationRequest::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ {CUDUCSIRSCoordinationRequest-IEs} },
	...
}

CUDUCSIRSCoordinationRequest-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID						CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID							PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID						CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID							PRESENCE mandatory	}|
	{ ID id-CSI-RSCoordinationRequestList			CRITICALITY ignore	TYPE CSI-RSCoordinationRequestList			PRESENCE mandatory},
	...
}


-- **************************************************************
--
-- CU-DU CSI-RS COORDINATION RESPONSE
--
-- **************************************************************

CUDUCSIRSCoordinationResponse ::= SEQUENCE {
	protocolIEs			ProtocolIE-Container		{ {CUDUCSIRSCoordinationResponse-IEs} },
	...
}


CUDUCSIRSCoordinationResponse-IEs F1AP-PROTOCOL-IES ::= {
	{ ID id-gNB-CU-UE-F1AP-ID						CRITICALITY reject	TYPE GNB-CU-UE-F1AP-ID							PRESENCE mandatory	}|
	{ ID id-gNB-DU-UE-F1AP-ID						CRITICALITY reject	TYPE GNB-DU-UE-F1AP-ID							PRESENCE mandatory	}|
	{ ID id-CSI-RSCoordinationResultList			CRITICALITY ignore	TYPE CSI-RSCoordinationResultList			PRESENCE mandatory},
	...
}


END
-- ASN1STOP
```

9.4.5 Information Element Definitions

```
-- ASN1START
-- **************************************************************
--
-- Information Element Definitions
--
-- **************************************************************

F1AP-IEs {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
ngran-access (22) modules (3) f1ap (3) version1 (1) f1ap-IEs (2) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

IMPORTS
    id-gNB-CUSystemInformation,
    id-HandoverPreparationInformation,
    id-TAISliceSupportList,
    id-RANAC,
    id-BearerTypeChange,
    id-Coverage-Modification-Cause,
    id-Cell-Direction,
    id-Cell-Type,
    id-CellGroupConfig,
    id-AvailablePLMNList,
    id-PDUSessionID,
    id-ULPDUSessionAggregateMaximumBitRate,
    id-DC-Based-Duplication-Configured,
    id-DC-Based-Duplication-Activation,
    id-Duplication-Activation,
    id-DLPDCPSNLength,
    id-ULPDCPSNLength,
    id-RLC-Status,
    id-MeasurementTimingConfiguration,
    id-DRB-Information,
    id-QoSFlowMappingIndication,
    id-ServingCellMO,
    id-RLCMode,
    id-ExtendedServedPLMNs-List,
    id-ExtendedAvailablePLMN-List,
    id-DRX-LongCycleStartOffset,
    id-SelectedBandCombinationIndex,
    id-SelectedFeatureSetEntryIndex,
    id-Ph-InfoSCG,
    id-latest-RRC-Version-Enhanced,
    id-RequestedBandCombinationIndex,
    id-RequestedFeatureSetEntryIndex,
    id-DRX-Config,
    id-UEAssistanceInformation,
    id-PDCCH-BlindDetectionSCG,
```

```
	id-Requested-PDCCH-BlindDetectionSCG,
	id-BPLMN-ID-Info-List,
	id-NotificationInformation,
	id-TNLAssociationTransportLayerAddressgNBDU,
	id-portNumber,
	id-AdditionalSIBMessageList,
	id-IgnorePRACHConfiguration,
	id-CG-Config,
	id-Ph-InfoMCG,
	id-AggressorgNBSetID,
	id-VictimgNBSetID,
	id-MeasGapSharingConfig,
	id-systemInformationAreaID,
	id-areaScope,
	id-IntendedTDD-DL-ULConfig,
	id-QosMonitoringRequest,
	id-BHInfo,
	id-IAB-Info-IAB-DU,
	id-IAB-Info-IAB-donor-CU,
	id-IAB-Barred,
	id-SIB12-message,
	id-SIB13-message,
	id-SIB14-message,
	id-UEAssistanceInformationEUTRA,
	id-SL-PHY-MAC-RLC-Config,
	id-SL-ConfigDedicatedEUTRA-Info,
	id-AlternativeQoSParaSetList,
	id-CurrentQoSParaSetIndex,
	id-CarrierList,
	id-ULCarrierList,
	id-FrequencyShift7p5khz,
	id-SSB-PositionsInBurst,
	id-NRPRACHConfig,
	id-TDD-UL-DLConfigCommonNR,
	id-CNPacketDelayBudgetDownlink,
	id-CNPacketDelayBudgetUplink,
	id-ExtendedPacketDelayBudget,
	id-TSCTrafficCharacteristics,
	id-AdditionalPDCPDuplicationTNL-List,
	id-RLCDuplicationInformation,
	id-AdditionalDuplicationIndication,
	id-mdtConfiguration,
	id-TraceCollectionEntityURI,
	id-NID,
	id-NPNSupportInfo,
	id-NPNBroadcastInformation,
	id-AvailableSNPN-ID-List,
	id-SIB10-message,
	id-RequestedP-MaxFR2,
	id-DLCarrierList,
	id-ExtendedTAISliceSupportList,
	id-E-CID-MeasurementQuantities-Item,
	id-ConfiguredTACIndication,
	id-NRCGI,
```

```
	id-SFN-Offset,
	id-TransmissionStopIndicator,
	id-SrsFrequency,
	id-EstimatedArrivalProbability,
	id-Supported-MBS-FSA-ID-List,
	id-TRPType,
	id-SRSSpatialRelationPerSRSResource,
	id-MBS-Broadcast-NeighbourCellList,
	id-PDCPTerminatingNodeDLTNLAddrInfo,
	id-ENBDLTNLAddress,
	id-PRS-Resource-ID,
	id-LocationMeasurementInformation,
	id-SliceRadioResourceStatus,
	id-CompositeAvailableCapacity-SUL,
	id-NR-U,
	id-NR-U-Channel-List,
	id-MIMOPRBusageInformation,
	id-IngressNonF1terminatingTopologyIndicator,
	id-NonF1terminatingTopologyIndicator,
	id-EgressNonF1terminatingTopologyIndicator,
	id-rBSetConfiguration,
	id-frequency-Domain-HSNA-Configuration-List,
	id-child-IAB-Nodes-NA-Resource-List,
	id-Parent-IAB-Nodes-NA-Resource-Configuration-List,
	id-uL-FreqInfo,
	id-uL-Transmission-Bandwidth,
	id-dL-FreqInfo,
	id-dL-Transmission-Bandwidth,
	id-uL-NR-Carrier-List,
	id-dL-NR-Carrier-List,
	id-nRFreqInfo,
	id-transmission-Bandwidth,
	id-nR-Carrier-List,
	id-permutation,
	id-M5ReportAmount,
	id-M6ReportAmount,
	id-M7ReportAmount,
	id-SurvivalTime,
	id-PDCMeasurementQuantities-Item,
	id-OnDemandPRS,
	id-AoA-SearchWindow,
	id-ZoAInformation,
	id-ARPLocationInfo,
	id-ARP-ID,
	id-MultipleULAoA,
	id-UL-SRS-RSRPP,
	id-SRSResourcetype,
	id-ExtendedAdditionalPathList,
	id-LoS-NLoSInformation,
	id-NumberOfTRPRxTEG,
	id-NumberOfTRPRxTxTEG,
	id-TRPTxTEGAssociation,
	id-TRPTEGInformation,
	id-TRPRx-TEGInformation,
```

```
	id-TRPBeamAntennaInformation,
	id-Redcap-Bcast-Information,
	id-NR-TADV,
	id-SDT-MAC-PHY-CG-Config,
	id-CG-SDTindicatorSetup,
	id-CG-SDTindicatorMod,
	id-SDTRLCBearerConfiguration,
	id-SRBMappingInfo,
	id-DRBMappingInfo,
	id-LastUsedCellIndication,
	id-Recommended-SSBs-List,
	id-SSBs-withinTheCell-tobe-Activated-List,
	id-SIB17-message,
	id-MUSIM-GapConfig,
	id-SIB20-message,
	id-pathPower,
	id-DU-RX-MT-RX-Extend,
	id-DU-TX-MT-TX-Extend,
	id-DU-RX-MT-TX-Extend,
	id-DU-TX-MT-RX-Extend,
	id-TAINSAGSupportList,
	id-SL-RLC-ChannelToAddModList,
	id-SIB15-message,
	id-InterFrequencyConfig-NoGap,
	id-MBSInterestIndication,
	id-L571Info,
	id-L1151Info,
	id-SCS-480,
	id-SCS-960,
	id-SRSPortIndex,
	id-PEISubgroupingSupportIndication,
	id-NeedForGapsInfoNR,
	id-NeedForGapNCSGInfoNR,
	id-NeedForGapNCSGInfoEUTRA,
	id-Source-MRB-ID,
	id-RedCapIndication,
	id-UL-GapFR2-Config,
	id-ConfigRestrictInfoDAPS,
	id-MulticastF1UContextReferenceCU,
	id-TwoPHRModeMCG,
	id-TwoPHRModeSCG,
	id-ncd-SSB-RedCapInitialBWP-SDT,
	id-nrofSymbolsExtended,
	id-repetitionFactorExtended,
	id-startRBHopping,
	id-startRBIndex,
	id-transmissionCombn8,
	id-ServCellInfoList,
	id-Preconfigured-measurement-GAP-Request,
	id-BWP-Id,
	id-ExtendedResourceSymbolOffset,
	id-MusimCapabilityRestrictionIndication,
	id-duplicationIndication,
	id-dRB-List,
```

```
	id-ChannelOccupancyTimePercentageUL,
	id-RadioResourceStatusNR-U,
	id-FiveG-ProSeLayer2Multipath,
	id-FiveG-ProSeLayer2UEtoUERelay,
	id-FiveG-ProSeLayer2UEtoUERemote,
	id-TSCTrafficCharacteristicsFeedback,
	id-RANfeedbacktype,
	id-Mobile-TRP-LocationInformation,
	id-Mobile-IAB-MT-UE-ID,
	id-MobileAccessPointLocation,
	id-SIB24-message,
	id-PDUSetQoSParameters,
	id-N6JitterInformation,
	id-ECNMarkingorCongestionInformationReportingRequest,
	id-ECNMarkingorCongestionInformationReportingStatus,
	id-ERedcap-Bcast-Information,
	id-NeedForInterruptionInfoNR,
	id-SCPAC-Request,
	id-MobileIAB-Barred,
	id-F1UTunnelNotEstablished,
	id-S-CPACLowerLayerReferenceConfigRequest,
	id-MusimCandidateBandList,
	id-PSIbasedSDUdiscardUL,
	id-SIB22-message,
	id-U2URLCChannelQoS,
	id-SL-PHY-MAC-RLC-ConfigExt,
	id-UL-RSCP,
	id-BW-Aggregation-Request-Indication,
	id-ReportingGranularitykminus1,
	id-ReportingGranularitykminus1additionalpath,
	id-ReportingGranularitykminus2,
	id-ReportingGranularitykminus2additionalpath,
	id-ReportingGranularitykminus3,
	id-ReportingGranularitykminus3additionalpath,
	id-ReportingGranularitykminus4,
	id-ReportingGranularitykminus4additionalpath,
	id-ReportingGranularitykminus5,
	id-ReportingGranularitykminus5additionalpath,
	id-ReportingGranularitykminus6,
	id-ReportingGranularitykminus6additionalpath,
	id-TimingReportingGranularityFactorExtended,
	id-PosValidityAreaCellList,
	id-SymbolIndex,
	id-AggregatedPosSRSResourceIDList,
	id-PhaseQuality,
	id-PRSBWAggregationRequestInfoList,
	id-AggregatedPRSResourceSetList,
	id-MeasuredFrequencyHops,
	id-TxHoppingConfiguration,
	id-AggregatedPosSRSResourceSetList,
	id-ValidityAreaSpecificSRSInformation,
	id-PeerUE-ID,
	id-MeasBasedOnAggregatedResources,
	id-SIB23-message,
```

```
	id-PointA,
	id-SCS-SpecificCarrier,
	id-NR-PCI,
	id-E-CID-MeasuredResultsAssociatedInfoList,
	id-XR-Bcast-Information,
	id-MaxDataBurstVolume,
	id-BarringExemptionforEmerCallInfo,
	id-SIB17bis-message,
	id-ReportingIntervalIMs,
	id-Transmission-Bandwidth-asymmetric,
	id-TagIDPointer,
	id-LocalOrigin,
	id-SRSPosPeriodicConfigHyperSFNIndex,
	id-candidatePSCellsToCancel,
	id-ValidityAreaSpecificSRSInformationExtended,
	id-TCIStatesConfigurationsList,
	id-E-CID-AoA-NR-per-TRP,
	id-PSIbasedSDUdiscardDL,
	id-PduSetDelayBudgetDownlink,
	id-PduSetDelayBudgetUplink,
	id-PduSetErrorRateDownlink,
	id-PduSetErrorRateUplink,
	id-MonitoringRequestonAvailableBitrate,
	id-MMSID,
	id-Indication-of-Bitrate-Adaptation,
	id-DLPDUSetInformationMarkingSupportIndication,
	id-FiveGProSeLayer3MHUEtoNetworkRelay,
	id-FiveGProSeLayer2MHUEtoNetworkRelay,
	id-FiveGProSeLayer2MHIntermediateUEtoNetworkRelay,
	id-FiveGProSeLayer2MHRemote,
	id-LPWUSSubgroupingSupportIndication,
	id-SBFD-Frequency-Configuration,
	id-SSB-resource-config,
	id-NZP-CSI-RS-Resources-Config,
	id-SRS-Resource-Configuration,
	id-rLFReportFailureType,
	id-C-RNTI,
	id-OnDemandSIB1,
	id-PagingAdaptationIndication,
	id-PEISubgroupingSupportIndication-PagingAdaptation,
	id-ServingCellMO-Ondemand,
	id-LTMgNB-ID,
	id-L1ExecutionConditionList,
	id-LTMSecurityInformation,
	id-RequestforCSI-RSResourceConfigforL1Measure,
	id-CSI-RSResourceConfigforL1Measure,
	id-CSI-RSResourceConfigforCSIAcquisition,
	id-CSIReportConfigforCSIAcquisition,
	id-RequestforL1ExecutionCondition,
	id-CSI-RSMeasurementsList,
	id-LTMResidualTAInfoList,
	id-ChannelResponseInformation,
	id-UL-SRS-TDCT,
	id-UEPerformanceDelayMonitoring,
```

```
	id-SBFD-AcrossSymbolType,
	id-LPWUSSupportedBandInfo,
	id-DeliveryStatusReq,
	maxNRARFCN,
	maxnoofErrors,
	maxnoofBPLMNs,
	maxnoofBPLMNsNR,
	maxnoofDLUPTNLInformation,
	maxnoofNrCellBands,
	maxnoofULUPTNLInformation,
	maxnoofQoSFlows,
	maxnoofSliceItems,
	maxnoofSIBTypes,
	maxnoofSITypes,
	maxCellineNB,
	maxnoofExtendedBPLMNs,
	maxnoofAdditionalSIBs,
	maxnoofUACPLMNs,
	maxnoofUACperPLMN,
	maxCellingNBDU,
	maxnoofTLAs,
	maxnoofGTPTLAs,
	maxnoofslots,
	maxnoofNonUPTrafficMappings,
	maxnoofServingCells,
	maxnoofServedCellsIAB,
	maxnoofChildIABNodes,
	maxnoofIABSTCInfo,
	maxnoofDUFSlots,
	maxnoofHSNASlots,
	maxnoofEgressLinks,
	maxnoofMappingEntries,
	maxnoofDSInfo,
	maxnoofQoSParaSets,
	maxnoofPC5QoSFlows,
	maxnoofSSBAreas,
	maxnoofNRSCSs,
	maxnoofPhysicalResourceBlocks,
	maxnoofPhysicalResourceBlocks-1,
	maxnoofPRACHconfigs,
	maxnoofRAReports,
	maxnoofRLFReports,
	maxnoofAdditionalPDCPDuplicationTNL,
	maxnoofRLCDuplicationState,
	maxnoofCHOcells,
	maxnoofMDTPLMNs,
	maxnoofCAGsupported,
	maxnoofNIDsupported,
	maxnoofExtSliceItems,
	maxnoofPosMeas,
	maxnoofTRPInfoTypes,
	maxnoofSRSTriggerStates,
	maxnoofSpatialRelations,
	maxnoBcastCell,
```

```
	maxnoofTRPs,
	maxnooflcs-gcs-translation,
	maxnoofPath,
	maxnoofMeasE-CID,
	maxnoofSSBs,
	maxnoSRS-ResourceSets,
	maxnoSRS-ResourcePerSet,
	maxnoSRS-Carriers,
	maxnoSCSs,
	maxnoSRS-Resources,
	maxnoSRS-PosResources,
	maxnoSRS-PosResourceSets,
	maxnoSRS-PosResourcePerSet,
	maxnoofPRS-ResourceSets,
	maxnoofPRS-ResourcesPerSet,
	maxNoOfMeasTRPs,
	maxnoofPRSresourceSets,
	maxnoofPRSresources,
	maxnoofSuccessfulHOReports,
	maxnoofNR-UChannelIDs,
	maxServedCellforSON,
	maxNeighbourCellforSON,
	maxAffectedCells,
	maxnoofMBSQoSFlows,
	maxnoofMBSFSAs,
	maxnoofMBSAreaSessionIDs,
	maxnoofMBSServiceAreaInformation,
	maxnoofTAIforMBS,
	maxnoofCellsforMBS,
	maxnoofIABCongInd,
	maxnoofBHRLCChannels,
	maxnoofTLAsIAB,
	maxnoofRBsetsPerCell,
	maxnoofRBsetsPerCell-1,
	maxnoofNeighbourNodeCellsIAB,
	maxnoofMeasPDC,
	maxnoARPs,
	maxnoofULAoAs,
	maxNoPathExtended,
	maxnoTRPTEGs,
	maxFreqLayers,
	maxNumResourcesPerAngle,
	maxnoAzimuthAngles,
	maxnoElevationAngles,
	maxnoofPRSTRPs,
	maxnoofQoEInformation,
	maxnoofUuRLCChannels,
	maxnoofPC5RLCChannels,
	maxnoofSMBRValues,
	maxnoofMBSSessionsofUE,
	maxnoofSLdestinations,
	maxnoofNSAGs,
	maxnoofSDTBearers,
	maxnoofPosSITypes,
```

```
	maxnoofMRBs,
	maxNrofBWPs,
	maxnoofUETypes,
	maxnoofLTMCells,
	maxnoofLTMgNB-DUs,
	maxnoofTAList,
	maxnoofDRBs,
	maxnoofUEsInQMCTransferControlMessage,
	maxnoofUEsforRAReportIndications,
	maxnoofSuccessfulPSCellChangeReports,
	maxnoofPeriodicities,
	maxnoofThresholdMBS-1,
	maxMBSSessionsinSessionInfoList,
	maxnoofLBTFailureInformation,
	maxnoofRSPPQoSFlows,
	maxnoVACell,
	maxnoAggregatedSRS-Resources,
	maxnoAggregatedPosSRSResourceSets,
	maxnoAggregatedPosPRSResourceSets,
	maxnoofTimeWindowSRS,
	maxnoofTimeWindowMea,
	maxnoPreconfiguredSRS,
	maxnoHopsMinusOne,
	maxnoAggCombinations,
	maxnoAggregatedPosSRSCombinations,
	maxnoofCandidateCells,
	maxnoofSSBIndices,
	maxnoofPreambleIndex,
	maxnoofThresholds,
	maxnoofNZP-CSI-RS-ResourcesPerSet,
	maxnoofSRS-Resources,
	maxnoofCellsinUEHistoryInfo,
	maxnoofTAs,
	maxnoofLTMCSI-RSResourceConfig,
	maxnoofCSI-RSs,
	maxnoofChannelRes,
	maxnoofCellsinNG-RANnode,
	maxnoofLTM-CSI-ResourcesPerSet




FROM F1AP-Constants

	Criticality,
	ProcedureCode,
	ProtocolIE-ID,
	TriggeringMessage
```

```
FROM F1AP-CommonDataTypes

    ProtocolExtensionContainer{},
    F1AP-PROTOCOL-EXTENSION,
    ProtocolIE-SingleContainer{},
    F1AP-PROTOCOL-IES

FROM F1AP-Containers;

-- A

AbortTransmission ::= CHOICE {
    sRSResourceSetID        SRSResourceSetID,
    releaseALL              NULL,
    choice-extension        ProtocolIE-SingleContainer { { AbortTransmission-ExtIEs } }
}

AbortTransmission-ExtIEs F1AP-PROTOCOL-IES ::= {
    { ID id-AggregatedPosSRSResourceSetList CRITICALITY ignore TYPE AggregatedPosSRSResourceSetList  PRESENCE mandatory },
    ...
}

AccessPointPosition ::= SEQUENCE {
    latitudeSign                ENUMERATED {north, south},
    latitude                    INTEGER (0..8388607),
    longitude                   INTEGER (-8388608..8388607),
    directionOfAltitude         ENUMERATED {height, depth},
    altitude                    INTEGER (0..32767),
    uncertaintySemi-major       INTEGER (0..127),
    uncertaintySemi-minor       INTEGER (0..127),
    orientationOfMajorAxis      INTEGER (0..179),
    uncertaintyAltitude         INTEGER (0..127),
    confidence                  INTEGER (0..100),
    iE-Extensions               ProtocolExtensionContainer { { AccessPointPosition-ExtIEs} } OPTIONAL
}

AccessPointPosition-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Activated-Cells-Mapping-List-Item   ::= SEQUENCE {
    nRCGIforTargetLogicalDU                     NRCGI,
    nRCGIforSourceLogicalDU                     NRCGI,
    iE-Extensions   ProtocolExtensionContainer { { Activated-Cells-Mapping-List-ItemExtIEs } }  OPTIONAL,
    ...
}

Activated-Cells-Mapping-List-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Activated-Cells-to-be-Updated-List ::= SEQUENCE (SIZE(1..maxnoofServedCellsIAB)) OF Activated-Cells-to-be-Updated-List-Item

Activated-Cells-to-be-Updated-List-Item ::= SEQUENCE{
```

```
    nRCGI                               NRCGI,
    iAB-DU-Cell-Resource-Configuration-Mode-Info    IAB-DU-Cell-Resource-Configuration-Mode-Info,
    iE-Extensions                       ProtocolExtensionContainer { { Activated-Cells-to-be-Updated-List-Item-ExtIEs} } OPTIONAL
}

Activated-Cells-to-be-Updated-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ActivationRequestType ::= ENUMERATED {activate, deactivate, ...}

ActiveULBWP  ::= SEQUENCE {
    locationAndBandwidth        INTEGER (0..37949,...),
    subcarrierSpacing           ENUMERATED {kHz15, kHz30, kHz60, kHz120,..., kHz480, kHz960},
    cyclicPrefix                ENUMERATED {normal, extended},
    txDirectCurrentLocation     INTEGER (0..3301,...),
    shift7dot5kHz               ENUMERATED {true, ...} OPTIONAL,
    sRSConfig                   SRSConfig,
    iE-Extensions                   ProtocolExtensionContainer { { ActiveULBWP-ExtIEs} } OPTIONAL
}

ActiveULBWP-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AdditionalDuplicationIndication ::= ENUMERATED {
    three,
    four,
    ...
}


AdditionalPath-List::= SEQUENCE (SIZE(1..maxnoofPath)) OF AdditionalPath-Item

AdditionalPath-Item ::=SEQUENCE {
    relativePathDelay   RelativePathDelay,
    pathQuality         TRPMeasurementQuality   OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { AdditionalPath-Item-ExtIEs } }   OPTIONAL
}

AdditionalPath-Item-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-MultipleULAoA   CRITICALITY ignore EXTENSION MultipleULAoA      PRESENCE optional}|
    { ID id-pathPower       CRITICALITY ignore EXTENSION UL-SRS-RSRPP       PRESENCE optional},
    ...
}

ExtendedAdditionalPathList ::= SEQUENCE (SIZE (1.. maxNoPathExtended)) OF ExtendedAdditionalPathList-Item


ExtendedAdditionalPathList-Item ::= SEQUENCE {
    relativeTimeOfPath  RelativePathDelay,
    pathQuality         TRPMeasurementQuality   OPTIONAL,
    multipleULAoA       MultipleULAoA           OPTIONAL,
    pathPower           UL-SRS-RSRPP            OPTIONAL,
```

```
    iE-Extensions       ProtocolExtensionContainer { { ExtendedAdditionalPathList-Item-ExtIEs} } OPTIONAL,
    ...
}

ExtendedAdditionalPathList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AdditionalPDCPDuplicationTNL-List ::= SEQUENCE (SIZE(1..maxnoofAdditionalPDCPDuplicationTNL)) OF AdditionalPDCPDuplicationTNL-Item

AdditionalPDCPDuplicationTNL-Item ::=SEQUENCE {
    additionalPDCPDuplicationUPTNLInformation       UPTransportLayerInformation,
    iE-Extensions   ProtocolExtensionContainer { { AdditionalPDCPDuplicationTNL-ItemExtIEs } }  OPTIONAL,
    ...
}

AdditionalPDCPDuplicationTNL-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
{ ID id-BHInfo      CRITICALITY ignore  EXTENSION BHInfo        PRESENCE optional   },
    ...
}

AdditionalSIBMessageList ::= SEQUENCE (SIZE(1..maxnoofAdditionalSIBs)) OF AdditionalSIBMessageList-Item

AdditionalSIBMessageList-Item ::= SEQUENCE {
    additionalSIB           OCTET STRING,
    iE-Extensions       ProtocolExtensionContainer { { AdditionalSIBMessageList-Item-ExtIEs} } OPTIONAL
}

AdditionalSIBMessageList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AdditionalRRMPriorityIndex ::= BIT STRING (SIZE(32))

AffectedCellsAndBeams-List ::= SEQUENCE (SIZE (1.. maxAffectedCells)) OF AffectedCellsAndBeams-Item

AffectedCellsAndBeams-Item::= SEQUENCE {
    nRCGI                   NRCGI,
    affectedSSB-List        AffectedSSB-List OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { AffectedCellsAndBeams-Item-ExtIEs} } OPTIONAL,
    ...
}
AffectedCellsAndBeams-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


AffectedSSB-List::= SEQUENCE (SIZE (1..maxnoofSSBAreas)) OF AffectedSSB-Item

AffectedSSB-Item::= SEQUENCE {
    sSB-Index   INTEGER(0..63),
    iE-Extensions       ProtocolExtensionContainer { { AffectedSSB-Item-ExtIEs} } OPTIONAL,
    ...
}
```

```
AffectedSSB-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AggregatedPosSRSResourceIDList ::= SEQUENCE (SIZE(2..maxnoAggregatedSRS-Resources)) OF Aggregated-PosSRS-Resource-ID-Item

Aggregated-PosSRS-Resource-ID-Item ::= SEQUENCE {
    positioningSRS      SRSPosResourceID,
    iE-Extensions       ProtocolExtensionContainer { { Aggregated-PosSRS-Resource-ID-Item-ExtIEs} } OPTIONAL,
    ...
}

Aggregated-PosSRS-Resource-ID-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-PointA              CRITICALITY ignore  EXTENSION PointA                    PRESENCE mandatory}|
    { ID id-SCS-SpecificCarrier CRITICALITY ignore  EXTENSION SCS-SpecificCarrier       PRESENCE mandatory}|
    { ID id-NR-PCI              CRITICALITY ignore  EXTENSION NRPCI                     PRESENCE optional},
    ...
}

AggregatedPosSRSResourceSetList ::= SEQUENCE (SIZE(1.. maxnoAggregatedPosSRSCombinations)) OF AggregatedPosSRSResourceSet-Item

AggregatedPosSRSResourceSet-Item ::= SEQUENCE {
    combined-posSRSResourceSet-List         Combined-PosSRSResourceSet-List,
    iE-Extensions                   ProtocolExtensionContainer { { AggregatedPosSRSResourceSet-Item-ExtIEs} } OPTIONAL,
    ...
}

AggregatedPosSRSResourceSet-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Combined-PosSRSResourceSet-List ::= SEQUENCE (SIZE (2..maxnoAggregatedPosSRSResourceSets)) OF Combined-PosSRSResourceSet-Item


Combined-PosSRSResourceSet-Item::= SEQUENCE {
    pointA                          INTEGER (0..3279165),
    nRPCI                           NRPCI               OPTIONAL,
    posSRSResourceSetID             INTEGER(0..15),
    scs-specificCarrier             SCS-SpecificCarrier,
    iE-Extensions                   ProtocolExtensionContainer { { Combined-PosSRSResourceSet-Item-ExtIEs} } OPTIONAL,
    ...
}

Combined-PosSRSResourceSet-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}



AggregatedPRSResourceSetList ::= SEQUENCE (SIZE (1..maxnoAggCombinations)) OF AggregatedPRSResourceSet-Item

AggregatedPRSResourceSet-Item ::= SEQUENCE {
    dl-PRS-ResourceSet-List         DL-PRS-ResourceSet-List,
    iE-Extensions                   ProtocolExtensionContainer { { AggregatedPRSResourceSet-Item-ExtIEs} } OPTIONAL,
```

```
    ...
}

AggregatedPRSResourceSet-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


DL-PRS-ResourceSet-List ::= SEQUENCE (SIZE (1..maxnoAggregatedPosPRSResourceSets)) OF DL-PRS-ResourceSet-Item

DL-PRS-ResourceSet-Item ::= SEQUENCE {
    dl-prs-ResourceSetIndex             INTEGER (1..8),
    iE-Extensions                       ProtocolExtensionContainer { { DL-PRS-ResourceSet-Item-ExtIEs} } OPTIONAL,
    ...
}

DL-PRS-ResourceSet-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AggressorCellList ::= SEQUENCE (SIZE(1..maxCellingNBDU)) OF AggressorCellList-Item

AggressorCellList-Item ::= SEQUENCE {
    aggressorCell-ID        NRCGI,
    iE-Extensions   ProtocolExtensionContainer { { AggressorCellList-Item-ExtIEs } }        OPTIONAL
}

AggressorCellList-Item-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AggressorgNBSetID ::= SEQUENCE {
    aggressorgNBSetID       GNBSetID,
    iE-Extensions   ProtocolExtensionContainer { { AggressorgNBSetID-ExtIEs } } OPTIONAL
}

AggressorgNBSetID-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AllocationAndRetentionPriority ::= SEQUENCE {
    priorityLevel               PriorityLevel,
    pre-emptionCapability       Pre-emptionCapability,
    pre-emptionVulnerability    Pre-emptionVulnerability,
    iE-Extensions               ProtocolExtensionContainer { {AllocationAndRetentionPriority-ExtIEs} } OPTIONAL,
    ...
}

AllocationAndRetentionPriority-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AlternativeQoSParaSetList ::= SEQUENCE (SIZE(1..maxnoofQoSParaSets)) OF AlternativeQoSParaSetItem
```

```
AlternativeQoSParaSetItem ::= SEQUENCE {
    alternativeQoSParaSetIndex          QoSParaSetIndex,
    guaranteedFlowBitRateDL             BitRate                 OPTIONAL,
    guaranteedFlowBitRateUL             BitRate                 OPTIONAL,
    packetDelayBudget                   PacketDelayBudget       OPTIONAL,
    packetErrorRate                     PacketErrorRate         OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { {AlternativeQoSParaSetItem-ExtIEs} }  OPTIONAL,
    ...
}

AlternativeQoSParaSetItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-MaxDataBurstVolume  CRITICALITY ignore  EXTENSION MaxDataBurstVolume        PRESENCE optional    }|
    { ID id-PduSetDelayBudgetDownlink   CRITICALITY ignore  EXTENSION ExtendedPacketDelayBudget   PRESENCE optional    }|
    { ID id-PduSetDelayBudgetUplink     CRITICALITY ignore  EXTENSION ExtendedPacketDelayBudget   PRESENCE optional    }|
    { ID id-PduSetErrorRateDownlink     CRITICALITY ignore  EXTENSION PacketErrorRate             PRESENCE optional    }|
    { ID id-PduSetErrorRateUplink       CRITICALITY ignore  EXTENSION PacketErrorRate             PRESENCE optional    },
    ...
}


AngleMeasurementQuality ::= SEQUENCE {
    azimuthQuality  INTEGER(0..255),
    zenithQuality   INTEGER(0..255) OPTIONAL,
    resolution      ENUMERATED{deg0dot1,...},
    iE-Extensions   ProtocolExtensionContainer { { AngleMeasurementQuality-ExtIEs } }   OPTIONAL
}

AngleMeasurementQuality-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


AperiodicSRSResourceTriggerList ::= SEQUENCE (SIZE(1..maxnoofSRSTriggerStates)) OF AperiodicSRSResourceTrigger

AperiodicSRSResourceTrigger ::= INTEGER (1..3)

Associated-SCell-Item ::= SEQUENCE {
    sCell-ID        NRCGI,
    iE-Extensions   ProtocolExtensionContainer { { Associated-SCell-ItemExtIEs } }  OPTIONAL
}

Associated-SCell-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AssociatedSessionID ::= OCTET STRING


AvailablePLMNList ::= SEQUENCE (SIZE(1..maxnoofBPLMNs)) OF AvailablePLMNList-Item

AvailablePLMNList-Item ::= SEQUENCE {
    pLMNIdentity            PLMN-Identity,
    iE-Extensions       ProtocolExtensionContainer { { AvailablePLMNList-Item-ExtIEs} } OPTIONAL
}
```

```
AvailablePLMNList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AvailableSNPN-ID-List ::= SEQUENCE (SIZE(1..maxnoofNIDsupported)) OF AvailableSNPN-ID-List-Item

AvailableSNPN-ID-List-Item ::= SEQUENCE {
    pLMN-Identity               PLMN-Identity,
    availableNIDList            BroadcastNIDList,
    iE-Extensions               ProtocolExtensionContainer { { AvailableSNPN-ID-List-ItemExtIEs} } OPTIONAL,
    ...
}

AvailableSNPN-ID-List-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AveragingWindow  ::= INTEGER (0..4095, ...)

AreaScope ::= ENUMERATED {true, ...}

AreaSpecificSemiPersistentSRSPosInfo ::= SEQUENCE {
    sRSTransmissionType     ENUMERATED{activate, deactivate,...},
    sRSResourceSetID        SRSResourceSetID,
    spatialRelationInfo     SpatialRelationInfo                                                     OPTIONAL,
    spatialRelationPerSRSResource   SpatialRelationPerSRSResource                                   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { AreaSpecificSemiPersistentSRSPosInfo-ExtIEs } }  OPTIONAL
}

AreaSpecificSemiPersistentSRSPosInfo-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


AoA-AssistanceInfo ::= SEQUENCE {
    angleMeasurement            AngleMeasurementType,
    lCS-to-GCS-Translation  LCS-to-GCS-Translation      OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { AoA-AssistanceInfo-ExtIEs } } OPTIONAL,
    ...
}

AoA-AssistanceInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AngleMeasurementType ::= CHOICE {
    expected-ULAoA      Expected-UL-AoA,
    expected-ZoA        Expected-ZoA-only,
    choice-extension ProtocolIE-SingleContainer { { AngleMeasurementType-ExtIEs } }
}

AngleMeasurementType-ExtIEs F1AP-PROTOCOL-IES ::= {
...
```

```
}

AppLayerBufferLevelList ::= OCTET STRING

ARP-ID ::= INTEGER (1..16, ...)

ARPLocationInformation ::= SEQUENCE (SIZE (1..maxnoARPs)) OF ARPLocationInformation-Item

ARPLocationInformation-Item ::= SEQUENCE {
    aRP-ID              ARP-ID,
    aRPLocationType     ARPLocationType,
    iE-Extensions       ProtocolExtensionContainer { { ARPLocationInformation-ExtIEs} } OPTIONAL,
...
}

ARPLocationInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ARPLocationType ::= CHOICE {
    aRPPositionRelativeGeodetic         RelativeGeodeticLocation,
    aRPPositionRelativeCartesian        RelativeCartesianLocation,
    choice-extension                    ProtocolIE-SingleContainer { { ARPLocationType-ExtIEs } }
}

ARPLocationType-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

AvailableBitrateReportThresholdList ::= SEQUENCE (SIZE(1..maxnoofThresholds)) OF AvailableBitrateReportThresholdItem

AvailableBitrateReportThresholdItem ::= SEQUENCE {
    reportingThreshold              ReportingThreshold,
    iE-Extensions                   ProtocolExtensionContainer { { AvailableBitrateReportThresholdItem-ExtIEs} } OPTIONAL,
    ...
}

AvailableBitrateReportThresholdItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- B

BAP-Header-Rewriting-Added-List-Item::= SEQUENCE {
    ingressBAPRoutingID     BAPRoutingID,
    egressBAPRoutingID      BAPRoutingID,
    nonF1terminatingTopologyIndicator       NonF1terminatingTopologyIndicator       OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { BAP-Header-Rewriting-Added-List-Item-ExtIEs} }  OPTIONAL
}

BAP-Header-Rewriting-Added-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
BAP-Header-Rewriting-Removed-List-Item::= SEQUENCE {
    ingressBAPRoutingID     BAPRoutingID,
    iE-Extensions           ProtocolExtensionContainer { { BAP-Header-Rewriting-Removed-List-Item-ExtIEs} } OPTIONAL
}

BAP-Header-Rewriting-Removed-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


BandwidthSRS ::= CHOICE {
    fR1                             FR1-Bandwidth,
    fR2                             FR2-Bandwidth,
    choice-extension                ProtocolIE-SingleContainer {{ BandwidthSRS-ExtIEs }}
}

BandwidthSRS-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}


BAPAddress ::= BIT STRING (SIZE(10))

BAPCtrlPDUChannel ::= ENUMERATED {true, ...}

BAPlayerBHRLCchannelMappingInfo ::= SEQUENCE {
    bAPlayerBHRLCchannelMappingInfoToAdd            BAPlayerBHRLCchannelMappingInfoList         OPTIONAL,
    bAPlayerBHRLCchannelMappingInfoToRemove         MappingInformationtoRemove                  OPTIONAL,
    iE-Extensions                                   ProtocolExtensionContainer { { BAPlayerBHRLCchannelMappingInfo-ExtIEs} } OPTIONAL,
    ...
}

BAPlayerBHRLCchannelMappingInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BAPlayerBHRLCchannelMappingInfoList ::= SEQUENCE (SIZE(1..maxnoofMappingEntries)) OF BAPlayerBHRLCchannelMappingInfo-Item

BAPlayerBHRLCchannelMappingInfo-Item ::= SEQUENCE {
    mappingInformationIndex         MappingInformationIndex,
    priorHopBAPAddress              BAPAddress      OPTIONAL,
    ingressbHRLCChannelID           BHRLCChannelID      OPTIONAL,
    nextHopBAPAddress               BAPAddress      OPTIONAL,
    egressbHRLCChannelID            BHRLCChannelID      OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { BAPlayerBHRLCchannelMappingInfo-ItemExtIEs} } OPTIONAL,
    ...
}

BAPlayerBHRLCchannelMappingInfo-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-IngressNonF1terminatingTopologyIndicator    CRITICALITY ignore  EXTENSION IngressNonF1terminatingTopologyIndicator    PRESENCE 
optional}|
    { ID id-EgressNonF1terminatingTopologyIndicator CRITICALITY ignore  EXTENSION EgressNonF1terminatingTopologyIndicator   PRESENCE optional},
    ...
```

```
}

BAPPathID ::= BIT STRING (SIZE(10))

BAPRoutingID ::= SEQUENCE {
    bAPAddress      BAPAddress,
    bAPPathID       BAPPathID,
    iE-Extensions   ProtocolExtensionContainer { { BAPRoutingIDExtIEs } }   OPTIONAL
}

BAPRoutingIDExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BarringExemptionforEmerCallInfo ::= ENUMERATED {true, ...}

BCBearerContextF1U-TNLInfo ::= CHOICE {
    locationindpendent              MBSF1UInformation,
    locationdependent               LocationDependentMBSF1UInformation,
    choice-extension    ProtocolIE-SingleContainer  {{BCBearerContextF1U-TNLInfo-ExtIEs}}
}

BCBearerContextF1U-TNLInfo-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

BitRate ::= INTEGER (0..4000000000000,...)

BearerTypeChange ::= ENUMERATED {true, ...}

BHRLCChannelID ::= BIT STRING (SIZE(16))

BHChannels-FailedToBeModified-Item ::= SEQUENCE {
    bHRLCChannelID      BHRLCChannelID,
    cause       Cause       OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { BHChannels-FailedToBeModified-ItemExtIEs } } OPTIONAL
}

BHChannels-FailedToBeModified-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BHChannels-FailedToBeSetup-Item ::= SEQUENCE {
    bHRLCChannelID      BHRLCChannelID,
    cause   Cause   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { BHChannels-FailedToBeSetup-ItemExtIEs } }    OPTIONAL
}

BHChannels-FailedToBeSetup-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BHChannels-FailedToBeSetupMod-Item ::= SEQUENCE {
    bHRLCChannelID      BHRLCChannelID,
```

```
    cause       Cause           OPTIONAL ,
    iE-Extensions   ProtocolExtensionContainer { { BHChannels-FailedToBeSetupMod-ItemExtIEs } } OPTIONAL
}

BHChannels-FailedToBeSetupMod-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BHChannels-Modified-Item ::= SEQUENCE {
    bHRLCChannelID          BHRLCChannelID,
    iE-Extensions   ProtocolExtensionContainer { { BHChannels-Modified-ItemExtIEs } }   OPTIONAL
}

BHChannels-Modified-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BHChannels-Required-ToBeReleased-Item ::= SEQUENCE {
    bHRLCChannelID      BHRLCChannelID,
    iE-Extensions   ProtocolExtensionContainer { { BHChannels-Required-ToBeReleased-ItemExtIEs } }   OPTIONAL
}

BHChannels-Required-ToBeReleased-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BHChannels-Setup-Item ::= SEQUENCE {
    bHRLCChannelID                          BHRLCChannelID,
    iE-Extensions   ProtocolExtensionContainer { { BHChannels-Setup-ItemExtIEs } }  OPTIONAL
}

BHChannels-Setup-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BHChannels-SetupMod-Item ::= SEQUENCE {
    bHRLCChannelID                          BHRLCChannelID,
    iE-Extensions   ProtocolExtensionContainer { { BHChannels-SetupMod-ItemExtIEs } }   OPTIONAL
}

BHChannels-SetupMod-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BHChannels-ToBeModified-Item ::= SEQUENCE {
    bHRLCChannelID                  BHRLCChannelID,
    bHQoSInformation                BHQoSInformation,
    rLCmode             RLCMode OPTIONAL,
    bAPCtrlPDUChannel   BAPCtrlPDUChannel       OPTIONAL,
    trafficMappingInfo  TrafficMappingInfo      OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { BHChannels-ToBeModified-ItemExtIEs } }   OPTIONAL
}

BHChannels-ToBeModified-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

BHChannels-ToBeReleased-Item ::= SEQUENCE {
    bHRLCChannelID      BHRLCChannelID,
    iE-Extensions   ProtocolExtensionContainer { { BHChannels-ToBeReleased-ItemExtIEs } }   OPTIONAL
}

BHChannels-ToBeReleased-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BHChannels-ToBeSetup-Item ::= SEQUENCE  {
    bHRLCChannelID                      BHRLCChannelID,
    bHQoSInformation                    BHQoSInformation,
    rLCmode                             RLCMode,
    bAPCtrlPDUChannel                   BAPCtrlPDUChannel       OPTIONAL,
    trafficMappingInfo                  TrafficMappingInfo      OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { BHChannels-ToBeSetup-ItemExtIEs } } OPTIONAL
}

BHChannels-ToBeSetup-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BHChannels-ToBeSetupMod-Item ::= SEQUENCE {
    bHRLCChannelID              BHRLCChannelID,
    bHQoSInformation            BHQoSInformation,
    rLCmode             RLCMode,
    bAPCtrlPDUChannel   BAPCtrlPDUChannel       OPTIONAL,
    trafficMappingInfo  TrafficMappingInfo      OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { BHChannels-ToBeSetupMod-ItemExtIEs } }   OPTIONAL
}

BHChannels-ToBeSetupMod-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BHInfo ::= SEQUENCE {
    bAProutingID            BAPRoutingID    OPTIONAL,
    egressBHRLCCHList       EgressBHRLCCHList   OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { BHInfo-ExtIEs} } OPTIONAL
}

BHInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-NonF1terminatingTopologyIndicator   CRITICALITY ignore  EXTENSION NonF1terminatingTopologyIndicator     PRESENCE optional },
    ...
}

BHQoSInformation ::= CHOICE {
    bHRLCCHQoS                  QoSFlowLevelQoSParameters,
    eUTRANBHRLCCHQoS            EUTRANQoS,
    cPTrafficType               CPTrafficType,
    choice-extension            ProtocolIE-SingleContainer { { BHQoSInformation-ExtIEs} }
```

```
}

BHQoSInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

BHRLCCHList ::= SEQUENCE (SIZE(1..maxnoofBHRLCChannels)) OF BHRLCCHItem

BHRLCCHItem ::= SEQUENCE {
    bHRLCChannelID          BHRLCChannelID,
    iE-Extensions           ProtocolExtensionContainer {{BHRLCCHItemExtIEs }}    OPTIONAL
}

BHRLCCHItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BH-Routing-Information-Added-List-Item ::= SEQUENCE {
    bAPRoutingID                BAPRoutingID,
    nextHopBAPAddress           BAPAddress,
    iE-Extensions               ProtocolExtensionContainer { { BH-Routing-Information-Added-List-ItemExtIEs} }   OPTIONAL
}

BH-Routing-Information-Added-List-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-NonF1terminatingTopologyIndicator  CRITICALITY ignore EXTENSION   NonF1terminatingTopologyIndicator     PRESENCE optional},
    ...
}

BH-Routing-Information-Removed-List-Item ::= SEQUENCE {
    bAPRoutingID                BAPRoutingID,
    iE-Extensions               ProtocolExtensionContainer { { BH-Routing-Information-Removed-List-ItemExtIEs} } OPTIONAL
}

BH-Routing-Information-Removed-List-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BPLMN-ID-Info-List ::= SEQUENCE (SIZE(1..maxnoofBPLMNsNR)) OF BPLMN-ID-Info-Item

BPLMN-ID-Info-Item ::= SEQUENCE {
    pLMN-Identity-List          AvailablePLMNList,
    extended-PLMN-Identity-List ExtendedAvailablePLMN-List  OPTIONAL,
    fiveGS-TAC                  FiveGS-TAC                  OPTIONAL,
    nr-cell-ID                  NRCellIdentity,
    ranac                       RANAC                       OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { BPLMN-ID-Info-ItemExtIEs} } OPTIONAL,
    ...
}

BPLMN-ID-Info-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {   ID id-ConfiguredTACIndication       CRITICALITY ignore  EXTENSION ConfiguredTACIndication    PRESENCE optional }|
    {   ID id-NPNBroadcastInformation       CRITICALITY reject EXTENSION NPNBroadcastInformation     PRESENCE optional},
    ...
}
```

```
ServedPLMNs-List ::= SEQUENCE (SIZE(1..maxnoofBPLMNs)) OF ServedPLMNs-Item

ServedPLMNs-Item ::= SEQUENCE {
    pLMN-Identity               PLMN-Identity,
    iE-Extensions               ProtocolExtensionContainer { { ServedPLMNs-ItemExtIEs} } OPTIONAL,
    ...
}

ServedPLMNs-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
{ ID id-TAISliceSupportList         CRITICALITY ignore  EXTENSION SliceSupportList              PRESENCE optional  }|
{ ID id-NPNSupportInfo              CRITICALITY reject  EXTENSION NPNSupportInfo                PRESENCE optional  }|
{ ID id-ExtendedTAISliceSupportList CRITICALITY reject  EXTENSION ExtendedSliceSupportList      PRESENCE optional  }|
{ ID id-TAINSAGSupportList          CRITICALITY ignore  EXTENSION NSAGSupportList               PRESENCE optional},
    ...
}

BroadcastCAGList ::= SEQUENCE (SIZE(1..maxnoofCAGsupported)) OF CAGID


BroadcastMRBs-FailedToBeModified-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    cause                       Cause                           OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { BroadcastMRBs-FailedtoBeModified-Item-ExtIEs} } OPTIONAL,
    ...
}

BroadcastMRBs-FailedtoBeModified-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastMRBs-FailedToBeSetup-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    cause                       Cause                           OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { BroadcastMRBs-FailedToBeSetup-Item-ExtIEs} } OPTIONAL,
    ...
}

BroadcastMRBs-FailedToBeSetup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastMRBs-FailedToBeSetupMod-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    cause                       Cause                           OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { BroadcastMRBs-FailedToBeSetupMod-Item-ExtIEs} } OPTIONAL,
    ...
}

BroadcastMRBs-FailedToBeSetupMod-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastMRBs-Modified-Item ::= SEQUENCE {
```

```
    mRB-ID                        MRB-ID,
    bcBearerCtxtF1U-TNLInfoatDU BCBearerContextF1U-TNLInfo      OPTIONAL,
    iE-Extensions                 ProtocolExtensionContainer { { BroadcastMRBs-Modified-Item-ExtIEs} } OPTIONAL,
    ...
}

BroadcastMRBs-Modified-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastMRBs-Setup-Item ::= SEQUENCE {
    mRB-ID                        MRB-ID,
    bcBearerCtxtF1U-TNLInfoatDU BCBearerContextF1U-TNLInfo,
    iE-Extensions                 ProtocolExtensionContainer { { BroadcastMRBs-Setup-Item-ExtIEs} } OPTIONAL,
    ...
}

BroadcastMRBs-Setup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastMRBs-SetupMod-Item ::= SEQUENCE {
    mRB-ID                        MRB-ID,
    bcBearerCtxtF1U-TNLInfoatDU BCBearerContextF1U-TNLInfo,
    iE-Extensions                 ProtocolExtensionContainer { { BroadcastMRBs-SetupMod-Item-ExtIEs} } OPTIONAL,
    ...
}

BroadcastMRBs-SetupMod-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastMRBs-ToBeModified-Item ::= SEQUENCE {
    mRB-ID                            MRB-ID,
    mRB-QoSInformation                QoSFlowLevelQoSParameters       OPTIONAL,
    mBS-Flows-Mapped-To-MRB-List      MBS-Flows-Mapped-To-MRB-List    OPTIONAL,
    bcBearerCtxtF1U-TNLInfoatCU       BCBearerContextF1U-TNLInfo      OPTIONAL,
    iE-Extensions                     ProtocolExtensionContainer { { BroadcastMRBs-ToBeModified-Item-ExtIEs} } OPTIONAL,
    ...
}

BroadcastMRBs-ToBeModified-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastMRBs-ToBeReleased-Item ::= SEQUENCE {
    mRB-ID                        MRB-ID,
    iE-Extensions                 ProtocolExtensionContainer { { BroadcastMRBs-ToBeReleased-ItemExtIEs } }   OPTIONAL,
    ...
}

BroadcastMRBs-ToBeReleased-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
BroadcastMRBs-ToBeSetup-Item ::= SEQUENCE {
    mRB-ID                          MRB-ID,
    mRB-QoSInformation              QoSFlowLevelQoSParameters,
    mBS-Flows-Mapped-To-MRB-List    MBS-Flows-Mapped-To-MRB-List,
    bcBearerCtxtF1U-TNLInfoatCU     BCBearerContextF1U-TNLInfo  ,
    iE-Extensions                   ProtocolExtensionContainer { { BroadcastMRBs-ToBeSetup-Item-ExtIEs} } OPTIONAL,
    ...
}

BroadcastMRBs-ToBeSetup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastMRBs-ToBeSetupMod-Item ::= SEQUENCE {
    mRB-ID                          MRB-ID,
    mRB-QoSInformation              QoSFlowLevelQoSParameters,
    mBS-Flows-Mapped-To-MRB-List    MBS-Flows-Mapped-To-MRB-List,
    bcBearerCtxtF1U-TNLInfoatCU     BCBearerContextF1U-TNLInfo,
    iE-Extensions                   ProtocolExtensionContainer { { BroadcastMRBs-ToBeSetupMod-Item-ExtIEs} } OPTIONAL,
    ...
}

BroadcastMRBs-ToBeSetupMod-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


BroadcastNIDList ::= SEQUENCE (SIZE(1..maxnoofNIDsupported)) OF NID

BroadcastSNPN-ID-List ::= SEQUENCE (SIZE(1..maxnoofNIDsupported)) OF BroadcastSNPN-ID-List-Item

BroadcastSNPN-ID-List-Item ::= SEQUENCE {
    pLMN-Identity               PLMN-Identity,
    broadcastNIDList            BroadcastNIDList,
    iE-Extensions               ProtocolExtensionContainer { { BroadcastSNPN-ID-List-ItemExtIEs} } OPTIONAL,
    ...
}

BroadcastSNPN-ID-List-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastPNI-NPN-ID-List ::= SEQUENCE (SIZE(1..maxnoofCAGsupported)) OF BroadcastPNI-NPN-ID-List-Item

BroadcastPNI-NPN-ID-List-Item ::= SEQUENCE {
    pLMN-Identity               PLMN-Identity,
    broadcastCAGList            BroadcastCAGList,
    iE-Extensions               ProtocolExtensionContainer { { BroadcastPNI-NPN-ID-List-ItemExtIEs} } OPTIONAL,
    ...
}

BroadcastPNI-NPN-ID-List-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}

BroadcastAreaScope ::= CHOICE {
    completeSuccess         NULL,
    partialSuccess          PartialSuccessCell,
    choice-extension        ProtocolIE-SingleContainer { { BroadcastAreaScope-ExtIEs } }
}

BroadcastAreaScope-ExtIEs F1AP-PROTOCOL-IES::={
    ...
}

BroadcastCellList ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF Broadcast-Cell-List-Item
Broadcast-Cell-List-Item ::= SEQUENCE {
    cellID              NRCGI,
    iE-Extensions               ProtocolExtensionContainer { { Broadcast-Cell-List-ItemExtIEs} } OPTIONAL,
    ...
}

Broadcast-Cell-List-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

BufferSizeThresh ::= INTEGER(0..16777215)

BurstArrivalTime ::= OCTET STRING

BW-Aggregation-Request-Indication ::= ENUMERATED  {true, ...}


BWP-Id ::= INTEGER (0..4)


BurstArrivalTimeWindow ::= SEQUENCE {
    burstArrivalTimeWindowStart             INTEGER (0..640000, ...),
    burstArrivalTimeWindowEnd               INTEGER (0..640000, ...),
    iE-Extension            ProtocolExtensionContainer { {BurstArrivalTimeWindow-ExtIEs} } OPTIONAL,
    ...
}

BurstArrivalTimeWindow-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Broadcast-MRBs-Transport-Request-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    bcBearerCtxtF1U-TNLInfoatDU BCBearerContextF1U-TNLInfo,
    iE-Extensions               ProtocolExtensionContainer { {Broadcast-MRBs-Transport-Request-Item-ExtIEs} } OPTIONAL,
    ...
}

Broadcast-MRBs-Transport-Request-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
-- C
CAGID ::= BIT STRING (SIZE(32))

Cancel-all-Warning-Messages-Indicator ::= ENUMERATED {true, ...}

Candidate-SpCell-Item ::= SEQUENCE {
    candidate-SpCell-ID         NRCGI   ,
    iE-Extensions   ProtocolExtensionContainer { { Candidate-SpCell-ItemExtIEs } }  OPTIONAL,
    ...
}

Candidate-SpCell-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CandidateCellwithBeamInfo   ::= SEQUENCE {
    candidateCellID             NRCGI,
    sSBIndex            SSBIndex,
    iE-Extensions       ProtocolExtensionContainer { { CandidateCellwithBeamInfo-ExtIEs } } OPTIONAL
}

CandidateCellwithBeamInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CandidateCellwithBeamInfoList ::= SEQUENCE (SIZE(1..maxnoofCandidateCells)) OF CandidateCellwithBeamInfo-Item

CandidateCellwithBeamInfo-Item ::= SEQUENCE {
    candidateCellID             NRCGI,
    sSBIndexList        SSBIndexList,
    iE-Extensions       ProtocolExtensionContainer { { CandidateCellwithBeamInfo-Item-ExtIEs } }  OPTIONAL
}

CandidateCellwithBeamInfo-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CandidateCellwithMeasurementsList ::= SEQUENCE (SIZE(1..maxnoofCandidateCells)) OF CandidateCellwithMeasurements-Item

CandidateCellwithMeasurements-Item  ::= SEQUENCE {
    candidateCellID             NRCGI,
    sSBIndexwithMeasurementsList        SSBIndexwithMeasurementsList,
    iE-Extensions       ProtocolExtensionContainer { { CandidateCellwithMeasurements-Item-ExtIEs } } OPTIONAL
}

CandidateCellwithMeasurements-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-CSI-RSMeasurementsList      CRITICALITY ignore      EXTENSION CSI-RSMeasurementsList           PRESENCE optional },
    ...
}


CapacityValue::= SEQUENCE {
```

```
    capacityValue                INTEGER (0..100),
    sSBAreaCapacityValueList     SSBAreaCapacityValueList        OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { { CapacityValue-ExtIEs} } OPTIONAL
}

CapacityValue-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Cause ::= CHOICE {
    radioNetwork        CauseRadioNetwork,
    transport           CauseTransport,
    protocol            CauseProtocol,
    misc                CauseMisc,
    choice-extension    ProtocolIE-SingleContainer { { Cause-ExtIEs} }
}

Cause-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

CauseMisc ::= ENUMERATED {
    control-processing-overload,
    not-enough-user-plane-processing-resources,
    hardware-failure,
    om-intervention,
    unspecified,
    ...
}

CauseProtocol ::= ENUMERATED {
    transfer-syntax-error,
    abstract-syntax-error-reject,
    abstract-syntax-error-ignore-and-notify,
    message-not-compatible-with-receiver-state,
    semantic-error,
    abstract-syntax-error-falsely-constructed-message,
    unspecified,
    ...
}

CauseRadioNetwork ::= ENUMERATED {
    unspecified,
    rl-failure-rlc,
    unknown-or-already-allocated-gnb-cu-ue-f1ap-id,
    unknown-or-already-allocated-gnb-du-ue-f1ap-id,
    unknown-or-inconsistent-pair-of-ue-f1ap-id,
    interaction-with-other-procedure,
    not-supported-qci-Value,
    action-desirable-for-radio-reasons,
    no-radio-resources-available,
    procedure-cancelled,
    normal-release,
    ...,
```

```
    cell-not-available,
    rl-failure-others,
    ue-rejection,
    resources-not-available-for-the-slice,
    amf-initiated-abnormal-release,
    release-due-to-pre-emption,
    plmn-not-served-by-the-gNB-CU,
    multiple-drb-id-instances,
    unknown-drb-id,
    multiple-bh-rlc-ch-id-instances,
    unknown-bh-rlc-ch-id,
    cho-cpc-resources-tobechanged,
    nPN-not-supported,
    nPN-access-denied,
    gNB-CU-Cell-Capacity-Exceeded,
    report-characteristics-empty,
    existing-measurement-ID,
    measurement-temporarily-not-available,
    measurement-not-supported-for-the-object,
    unknown-bh-address,
    unknown-bap-routing-id,
    insufficient-ue-capabilities,
    scg-activation-deactivation-failure,
    scg-deactivation-failure-due-to-data-transmission,
    requested-item-not-supported-on-time,
    unknown-or-already-allocated-gNB-CU-MBS-F1AP-ID,
    unknown-or-already-allocated-gNB-DU-MBS-F1AP-ID,
    unknown-or-inconsistent-pair-of-MBS-F1AP-ID,
    unknown-or-inconsistent-MRB-ID,
    tat-sdt-expiry,
    lTM-command-triggered,
    sSB-not-available

}

CauseTransport ::= ENUMERATED {
    unspecified,
    transport-resource-unavailable,
    ...,
    unknown-TNL-address-for-IAB,
    unknown-UP-TNL-information-for-IAB
}

CellGroupConfig ::= OCTET STRING

CellCapacityClassValue ::= INTEGER (1..100,...)

Cell-Direction ::= ENUMERATED {dl-only, ul-only}

CellMeasurementResultList ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF CellMeasurementResultItem

CellMeasurementResultItem ::= SEQUENCE {
    cellID                          NRCGI,
    radioResourceStatus             RadioResourceStatus             OPTIONAL,
```

```
    compositeAvailableCapacityGroup CompositeAvailableCapacityGroup OPTIONAL,
    sliceAvailableCapacity          SliceAvailableCapacity          OPTIONAL,
    numberofActiveUEs               NumberofActiveUEs               OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { CellMeasurementResultItem-ExtIEs} } OPTIONAL
}

CellMeasurementResultItem-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-NR-U-Channel-List   CRITICALITY ignore  EXTENSION NR-U-Channel-List PRESENCE optional },
    ...
}

Cell-Portion-ID ::= INTEGER (0..4095,...)

CellsForSON-List ::= SEQUENCE (SIZE(1.. maxServedCellforSON)) OF CellsForSON-Item

CellsForSON-Item ::= SEQUENCE {
    nRCGI                                       NRCGI,
    neighbourNR-CellsForSON-List                NeighbourNR-CellsForSON-List                             OPTIONAL,
    iE-Extensions                               ProtocolExtensionContainer { { CellsForSON-Item-ExtIEs} } OPTIONAL,
    ...
}

CellsForSON-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Cells-Failed-to-be-Activated-List-Item ::= SEQUENCE {
    nRCGI               NRCGI,
    cause               Cause,
    iE-Extensions       ProtocolExtensionContainer { { Cells-Failed-to-be-Activated-List-ItemExtIEs } } OPTIONAL,
    ...
}

Cells-Failed-to-be-Activated-List-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Cells-Status-Item ::= SEQUENCE {
    nRCGI           NRCGI,
    service-status      Service-Status,
    iE-Extensions               ProtocolExtensionContainer { { Cells-Status-ItemExtIEs } }  OPTIONAL,
    ...
}

Cells-Status-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Cells-To-Be-Broadcast-Item ::= SEQUENCE {
    nRCGI               NRCGI,
    iE-Extensions       ProtocolExtensionContainer { { Cells-To-Be-Broadcast-ItemExtIEs } } OPTIONAL,
    ...
}
```

```
Cells-To-Be-Broadcast-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Cells-Broadcast-Completed-Item ::= SEQUENCE {
    nRCGI               NRCGI,
    iE-Extensions       ProtocolExtensionContainer { { Cells-Broadcast-Completed-ItemExtIEs } } OPTIONAL,
    ...
}

Cells-Broadcast-Completed-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Broadcast-To-Be-Cancelled-Item ::= SEQUENCE {
    nRCGI               NRCGI,
    iE-Extensions       ProtocolExtensionContainer { { Broadcast-To-Be-Cancelled-ItemExtIEs } } OPTIONAL,
    ...
}

Broadcast-To-Be-Cancelled-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


Cells-Broadcast-Cancelled-Item ::= SEQUENCE {
    nRCGI               NRCGI,
    numberOfBroadcasts  NumberOfBroadcasts,
    iE-Extensions       ProtocolExtensionContainer { { Cells-Broadcast-Cancelled-ItemExtIEs } } OPTIONAL,
    ...
}

Cells-Broadcast-Cancelled-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Cells-to-be-Activated-List-Item ::= SEQUENCE {
    nRCGI       NRCGI,
    nRPCI       NRPCI       OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { Cells-to-be-Activated-List-ItemExtIEs} } OPTIONAL,
    ...
}

Cells-to-be-Activated-List-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-gNB-CUSystemInformation         CRITICALITY reject  EXTENSION GNB-CUSystemInformation          PRESENCE optional }|
    { ID id-AvailablePLMNList               CRITICALITY ignore  EXTENSION AvailablePLMNList                PRESENCE optional }|
    { ID id-ExtendedAvailablePLMN-List      CRITICALITY ignore  EXTENSION ExtendedAvailablePLMN-List       PRESENCE optional }|
    { ID id-IAB-Info-IAB-donor-CU           CRITICALITY ignore  EXTENSION IAB-Info-IAB-donor-CU            PRESENCE optional}|
    { ID id-AvailableSNPN-ID-List           CRITICALITY ignore  EXTENSION AvailableSNPN-ID-List            PRESENCE optional }|
    { ID id-MBS-Broadcast-NeighbourCellList CRITICALITY ignore  EXTENSION MBS-Broadcast-NeighbourCellList  PRESENCE optional }|
    { ID id-SSBs-withinTheCell-tobe-Activated-List          CRITICALITY reject  EXTENSION SSBs-toBeActivated-List           PRESENCE optional },
    ...
}
Cells-With-SSBs-Activated-List ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF Cells-With-SSBs-Activated-List-Item
```

```
Cells-With-SSBs-Activated-List-Item::= SEQUENCE {
    nRCGI                       NRCGI,
    sSBs-activated-List         SSBs-activated-List,
    iE-Extensions               ProtocolExtensionContainer { { Cells-With-SSBs-Activated-List-Item-ExtIEs} } OPTIONAL
}

Cells-With-SSBs-Activated-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Cells-Allowed-to-be-Deactivated-List-Item ::= SEQUENCE {
    nRCGI       NRCGI,
    iE-Extensions               ProtocolExtensionContainer { { Cells-Allowed-to-be-Deactivated-List-ItemExtIEs} }  OPTIONAL,
    ...
}


Cells-Allowed-to-be-Deactivated-List-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
        ...
}

Cells-to-be-Deactivated-List-Item ::= SEQUENCE {
    nRCGI           NRCGI   ,
    iE-Extensions               ProtocolExtensionContainer { { Cells-to-be-Deactivated-List-ItemExtIEs } } OPTIONAL,
    ...
}

Cells-to-be-Deactivated-List-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Cells-to-be-Barred-Item::= SEQUENCE {
    nRCGI           NRCGI   ,
    cellBarred      CellBarred,
    iE-Extensions               ProtocolExtensionContainer { { Cells-to-be-Barred-Item-ExtIEs } } OPTIONAL
}

Cells-to-be-Barred-Item-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-IAB-Barred  CRITICALITY ignore  EXTENSION IAB-Barred        PRESENCE optional }|
    { ID id-MobileIAB-Barred    CRITICALITY ignore  EXTENSION MobileIAB-Barred      PRESENCE optional },
    ...
}


CellBarred  ::= ENUMERATED {barred, not-barred, ...}

CellSize ::= ENUMERATED {verysmall, small, medium, large, ...}

CellToReportList ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF CellToReportItem

CellToReportItem ::= SEQUENCE {
    cellID      NRCGI,
    sSBToReportList     SSBToReportList      OPTIONAL,
```

```
    sliceToReportList   SliceToReportList    OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { CellToReportItem-ExtIEs} } OPTIONAL
}

CellToReportItem-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CellType ::= SEQUENCE {
    cellSize        CellSize,
    iE-Extensions       ProtocolExtensionContainer { {CellType-ExtIEs} }    OPTIONAL,
    ...
}

CellType-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CellULConfigured ::=  ENUMERATED {none, ul, sul, ul-and-sul, ...}

CG-SDTQueryIndication ::=  ENUMERATED {true, ...}

CG-SDTKeptIndicator ::= ENUMERATED {true, ...}

CG-SDTindicatorSetup ::= ENUMERATED {true, ...}

CG-SDTindicatorMod ::= ENUMERATED {true, false, ...}

CG-SDTSessionInfo ::= SEQUENCE {
    gNB-CU-UE-F1AP-ID                   GNB-CU-UE-F1AP-ID,
    gNB-DU-UE-F1AP-ID                   GNB-DU-UE-F1AP-ID,
    iE-Extensions                       ProtocolExtensionContainer {{CG-SDTSessionInfo-ExtIEs}}   OPTIONAL,
    ...
}


CG-SDTSessionInfo-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ChannelOccupancyTimePercentage ::= INTEGER (0..100,...)

ChannelResponseInformation ::= SEQUENCE {
    channelResponseWindowSize       ENUMERATED {ws32, ws64, ws128, ...},
    channelResponseNumber           ENUMERATED {n8, n16, n24, ...},
    iE-Extension            ProtocolExtensionContainer { { ChannelResponseInformation-ExtIEs} } OPTIONAL,
    ...
}

ChannelResponseInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Child-IAB-Nodes-NA-Resource-List ::= SEQUENCE (SIZE(1..maxnoofChildIABNodes)) OF Child-IAB-Nodes-NA-Resource-List-Item
```

```
Child-IAB-Nodes-NA-Resource-List-Item::= SEQUENCE {
    gNB-CU-UE-F1AP-ID   GNB-CU-UE-F1AP-ID,
    gNB-DU-UE-F1AP-ID   GNB-DU-UE-F1AP-ID,
    nA-Resource-Configuration-List      NA-Resource-Configuration-List      OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { Child-IAB-Nodes-NA-Resource-List-Item-ExtIEs} } OPTIONAL
}

Child-IAB-Nodes-NA-Resource-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


Child-Node-Cells-List ::= SEQUENCE (SIZE(1..maxnoofChildIABNodes)) OF Child-Node-Cells-List-Item

Child-Node-Cells-List-Item ::=  SEQUENCE{
    nRCGI                               NRCGI,
    iAB-DU-Cell-Resource-Configuration-Mode-Info    IAB-DU-Cell-Resource-Configuration-Mode-Info  OPTIONAL,
    iAB-STC-Info                        IAB-STC-Info            OPTIONAL,
    rACH-Config-Common                  RACH-Config-Common      OPTIONAL,
    rACH-Config-Common-IAB              RACH-Config-Common-IAB  OPTIONAL,
    cSI-RS-Configuration                OCTET STRING            OPTIONAL,
    sR-Configuration                    OCTET STRING            OPTIONAL,
    pDCCH-ConfigSIB1                    OCTET STRING            OPTIONAL,
    sCS-Common                          OCTET STRING    OPTIONAL,
    multiplexingInfo                    MultiplexingInfo    OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer {{Child-Node-Cells-List-Item-ExtIEs}}     OPTIONAL
}

Child-Node-Cells-List-Item-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Child-Nodes-List ::= SEQUENCE (SIZE(1..maxnoofChildIABNodes)) OF Child-Nodes-List-Item

Child-Nodes-List-Item ::= SEQUENCE{
    gNB-CU-UE-F1AP-ID   GNB-CU-UE-F1AP-ID,
    gNB-DU-UE-F1AP-ID   GNB-DU-UE-F1AP-ID,
    child-Node-Cells-List   Child-Node-Cells-List   OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer {{Child-Nodes-List-Item-ExtIEs}}     OPTIONAL
}

Child-Nodes-List-Item-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CHOtrigger-InterDU ::= ENUMERATED {
    cho-initiation,
    cho-replace,
    ...
}

CHOtrigger-IntraDU ::= ENUMERATED {
    cho-initiation,
```

```
    cho-replace,
    cho-cancel,
    ...
}

CLI-MeasurementResult-List ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF CLI-MeasurementResult-Item

CLI-MeasurementResult-Item ::= SEQUENCE {
    cellID                                          NRCGI,
    ssbIndex                                        INTEGER(0..63,...)                  OPTIONAL,
    nZP-CSI-RS-ResourceIndication                   INTEGER(1..64,...)                  OPTIONAL,
    cLI-MitigationIndication                        CLI-MitigationIndication            OPTIONAL,
    iE-Extensions                                   ProtocolExtensionContainer { {CLI-MeasurementResult-Item-ExtIEs} } OPTIONAL,
    ...
}

CLI-MeasurementResult-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CLI-MitigationIndication ::= ENUMERATED {true,...}

CNSubgroupID ::= INTEGER (0..7, ...)

CNSubgroupID-LP-WUS ::= INTEGER (0..30, ...)

CNUEPagingIdentity ::= CHOICE {
    fiveG-S-TMSI            BIT STRING (SIZE(48)),
    choice-extension            ProtocolIE-SingleContainer { { CNUEPagingIdentity-ExtIEs } }
}

CNUEPagingIdentity-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

CompositeAvailableCapacityGroup ::= SEQUENCE {
    compositeAvailableCapacityDownlink  CompositeAvailableCapacity,
    compositeAvailableCapacityUplink    CompositeAvailableCapacity,
    iE-Extensions   ProtocolExtensionContainer { { CompositeAvailableCapacityGroup-ExtIEs} } OPTIONAL
}

CompositeAvailableCapacityGroup-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-CompositeAvailableCapacity-SUL      CRITICALITY ignore  EXTENSION CompositeAvailableCapacity     PRESENCE optional },
    ...
}

CompositeAvailableCapacity ::= SEQUENCE {
    cellCapacityClassValue  CellCapacityClassValue      OPTIONAL,
    capacityValue           CapacityValue,
    iE-Extensions   ProtocolExtensionContainer { { CompositeAvailableCapacity-ExtIEs} } OPTIONAL
}

CompositeAvailableCapacity-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}

CHO-Probability ::= INTEGER (1..100)

ConditionalInterDUMobilityInformation ::= SEQUENCE {
    cho-trigger                     CHOtrigger-InterDU,
    targetgNB-DUUEF1APID            GNB-DU-UE-F1AP-ID                           OPTIONAL
    -- The above IE shall be present if the cho-trigger IE is present and set to "cho-replace" --,
    iE-Extensions                   ProtocolExtensionContainer { { ConditionalInterDUMobilityInformation-ExtIEs} }  OPTIONAL,
    ...
}

ConditionalInterDUMobilityInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::={
    { ID id-EstimatedArrivalProbability     CRITICALITY ignore      EXTENSION CHO-Probability           PRESENCE optional }|
    { ID id-SCPAC-Request                   CRITICALITY reject      EXTENSION SCPAC-Request       PRESENCE optional }|
    { ID id-S-CPACLowerLayerReferenceConfigRequest  CRITICALITY reject      EXTENSION S-CPACLowerLayerReferenceConfigRequest   PRESENCE optional },
    ...
}

ConditionalIntraDUMobilityInformation ::= SEQUENCE {
    cho-trigger                     CHOtrigger-IntraDU,
    targetCellsTocancel             TargetCellList                              OPTIONAL,
    -- The above IE shall be present if the cho-trigger IE is present and set to "cho-cancel"
    iE-Extensions                   ProtocolExtensionContainer { { ConditionalIntraDUMobilityInformation-ExtIEs} }  OPTIONAL,
    ...
}

ConditionalIntraDUMobilityInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::={
    { ID id-EstimatedArrivalProbability     CRITICALITY ignore      EXTENSION CHO-Probability           PRESENCE optional }|
    { ID id-SCPAC-Request                   CRITICALITY reject      EXTENSION SCPAC-Request       PRESENCE optional }|
    { ID id-S-CPACLowerLayerReferenceConfigRequest  CRITICALITY reject      EXTENSION S-CPACLowerLayerReferenceConfigRequest   PRESENCE optional },
    ...
}

ConfigRestrictInfoDAPS ::= OCTET STRING

ConfiguredTACIndication ::= ENUMERATED {
    true,
    ...
}

Configured-BWP-List ::= SEQUENCE (SIZE(1.. maxNrofBWPs)) OF Configured-BWP-Item

Configured-BWP-Item ::= SEQUENCE {
    bWP-Id                                  BWP-Id,
    bWP-Location-and-bandwidth              INTEGER (0..37949),
    iE-Extensions                   ProtocolExtensionContainer { { Configured-BWP-Item-ExtIEs } } OPTIONAL,
    ...
}

Configured-BWP-Item-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
CoordinateID ::= INTEGER (0..511, ...)

Coverage-Modification-Notification ::= SEQUENCE {
    coverage-Modification-List              Coverage-Modification-List,
    iE-Extensions                               ProtocolExtensionContainer { { Coverage-Modification-Notification-ExtIEs} } OPTIONAL,
    ...
}

Coverage-Modification-Notification-ExtIEs F1AP-PROTOCOL-EXTENSION ::={
    ...
}

Coverage-Modification-List ::= SEQUENCE (SIZE (1..maxCellingNBDU)) OF Coverage-Modification-Item

Coverage-Modification-Item ::= SEQUENCE {
    nRCGI                               NRCGI,
    cellCoverageState                   CellCoverageState,
    sSBCoverageModificationList         SSBCoverageModification-List OPTIONAL,
    iE-Extension            ProtocolExtensionContainer { { Coverage-Modification-Item-ExtIEs} }           OPTIONAL,
    ...
}

Coverage-Modification-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-Coverage-Modification-Cause CRITICALITY ignore  EXTENSION CCO-issue-detection       PRESENCE optional },
    ...
}

CellCoverageState ::= INTEGER (0..63, ...)


CCO-Assistance-Information ::= SEQUENCE {
    cCO-issue-detection             CCO-issue-detection     OPTIONAL,
    affectedCellsAndBeams-List      AffectedCellsAndBeams-List      OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { CCO-Assistance-Information-ExtIEs} } OPTIONAL,
    ...
}

CCO-Assistance-Information-ExtIEs F1AP-PROTOCOL-EXTENSION ::={
    ...
}


CCO-issue-detection ::= ENUMERATED {
    coverage,
    cell-edge-capacity,
    ...,
    network-energy-saving}


CP-TransportLayerAddress ::= CHOICE {
    endpoint-IP-address             TransportLayerAddress,
    endpoint-IP-address-and-port    Endpoint-IP-address-and-port,
    choice-extension                ProtocolIE-SingleContainer { { CP-TransportLayerAddress-ExtIEs } }
```

```
}

CP-TransportLayerAddress-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

CPACMCGInformation ::= SEQUENCE {
    cpac-trigger                    CPAC-trigger,
    pscellid                        NRCGI,
    iE-Extensions                   ProtocolExtensionContainer { { CPACMCGInformation-ExtIEs} } OPTIONAL,
    ...
}

CPACMCGInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-candidatePSCellsToCancel        CRITICALITY ignore  EXTENSION PSCellList        PRESENCE optional },
    -- The above IE shall be present if the cpac-trigger IE is present and set to "cpac-cancel"
    ...
}

CPAC-trigger ::= ENUMERATED {
    cpac-preparation,
    cpac-executed,
    ... ,
    cpac-cancel
}

CPTrafficType ::= INTEGER (1..3,...)

CriticalityDiagnostics ::= SEQUENCE {
    procedureCode                   ProcedureCode                                                   OPTIONAL,
    triggeringMessage               TriggeringMessage                                               OPTIONAL,
    procedureCriticality            Criticality                                                     OPTIONAL,
    transactionID                   TransactionID                                                   OPTIONAL,
    iEsCriticalityDiagnostics       CriticalityDiagnostics-IE-List                                  OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer {{CriticalityDiagnostics-ExtIEs}}    OPTIONAL,
    ...
}

CriticalityDiagnostics-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CriticalityDiagnostics-IE-List ::= SEQUENCE (SIZE (1.. maxnoofErrors)) OF CriticalityDiagnostics-IE-Item

CriticalityDiagnostics-IE-Item ::= SEQUENCE {
    iECriticality           Criticality,
    iE-ID                   ProtocolIE-ID,
    typeOfError             TypeOfError,
    iE-Extensions           ProtocolExtensionContainer {{CriticalityDiagnostics-IE-Item-ExtIEs}}  OPTIONAL,
    ...
}

CriticalityDiagnostics-IE-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}

C-RNTI ::= INTEGER (0..65535, ...)

CUDURadioInformationType ::= CHOICE {
    rIM                             CUDURIMInformation,
    choice-extension                ProtocolIE-SingleContainer { { CUDURadioInformationType-ExtIEs} }
}

CUDURadioInformationType-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

CUDURIMInformation ::= SEQUENCE {
    victimgNBSetID          GNBSetID,
    rIMRSDetectionStatus    RIMRSDetectionStatus,
    iE-Extensions           ProtocolExtensionContainer { { CUDURIMInformation-ExtIEs} } OPTIONAL
}

CUDURIMInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CUtoDURRCInformation ::= SEQUENCE {
    cG-ConfigInfo                       CG-ConfigInfo                       OPTIONAL,
    uE-CapabilityRAT-ContainerList      UE-CapabilityRAT-ContainerList      OPTIONAL,
    measConfig                          MeasConfig                          OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { CUtoDURRCInformation-ExtIEs} } OPTIONAL,
    ...
}

CUtoDURRCInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-HandoverPreparationInformation  CRITICALITY ignore  EXTENSION HandoverPreparationInformation     PRESENCE optional }|
    { ID id-CellGroupConfig                 CRITICALITY ignore  EXTENSION CellGroupConfig                    PRESENCE optional }|
    { ID id-MeasurementTimingConfiguration  CRITICALITY ignore  EXTENSION MeasurementTimingConfiguration     PRESENCE optional }|
    { ID id-UEAssistanceInformation         CRITICALITY ignore  EXTENSION UEAssistanceInformation            PRESENCE optional }|
    { ID id-CG-Config                       CRITICALITY ignore  EXTENSION CG-Config                          PRESENCE optional }|
    { ID id-UEAssistanceInformationEUTRA    CRITICALITY ignore  EXTENSION UEAssistanceInformationEUTRA       PRESENCE optional }|
    { ID id-LocationMeasurementInformation  CRITICALITY ignore  EXTENSION LocationMeasurementInformation     PRESENCE optional }|
    { ID id-MUSIM-GapConfig                 CRITICALITY reject  EXTENSION MUSIM-GapConfig                    PRESENCE optional }|
    { ID id-SDT-MAC-PHY-CG-Config           CRITICALITY ignore  EXTENSION SDT-MAC-PHY-CG-Config              PRESENCE optional }|
    { ID id-MBSInterestIndication           CRITICALITY ignore  EXTENSION MBSInterestIndication              PRESENCE optional }|
    { ID id-NeedForGapsInfoNR               CRITICALITY ignore  EXTENSION NeedForGapsInfoNR                  PRESENCE optional }|
    { ID id-NeedForGapNCSGInfoNR            CRITICALITY ignore  EXTENSION NeedForGapNCSGInfoNR               PRESENCE optional }|
    { ID id-NeedForGapNCSGInfoEUTRA         CRITICALITY ignore  EXTENSION NeedForGapNCSGInfoEUTRA            PRESENCE optional }|
    { ID id-ConfigRestrictInfoDAPS          CRITICALITY ignore  EXTENSION ConfigRestrictInfoDAPS             PRESENCE optional }|
    { ID id-Preconfigured-measurement-GAP-Request           CRITICALITY ignore  EXTENSION Preconfigured-measurement-GAP-Request            PRESENCE 
optional }|
    { ID id-NeedForInterruptionInfoNR       CRITICALITY ignore  EXTENSION NeedForInterruptionInfoNR          PRESENCE optional }|
    { ID id-MusimCapabilityRestrictionIndication        CRITICALITY ignore  EXTENSION MusimCapabilityRestrictionIndication           PRESENCE 
optional }|
    { ID id-MusimCandidateBandList          CRITICALITY ignore  EXTENSION MusimCandidateBandList             PRESENCE optional },
    ...
}
```

```
CUtoDUTAInformation-List ::= SEQUENCE (SIZE(1.. maxnoofTAList)) OF CUtoDUTAInformation-Item

CUtoDUTAInformation-Item    ::= SEQUENCE {
    nRCGI                   NRCGI,
    tAValue                 TAValue,
    preambleIndex           PreambleIndex,
    rA-RNTI                 RA-RNTI,
    tagIDPointer            TagIDPointer        OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { CUtoDUTAInformation-Item-ExtIEs} }   OPTIONAL,
    ...
}

CUtoDUTAInformation-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CSIResourceConfiguration ::= SEQUENCE  {
    cSIResourceConfigToAddModList           OCTET STRING        OPTIONAL,
    cSIResourceConfigToReleaseList          OCTET STRING        OPTIONAL,
    iE-Extensions                           ProtocolExtensionContainer { { CSIResourceConfiguration-ExtIEs} } OPTIONAL
}

CSIResourceConfiguration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CSI-RSResourceConfig ::= SEQUENCE  {
    periodicCSI-RSResourceConfiguration     NZP-CSI-RS-ResourceConfiguration    OPTIONAL,
    spCSI-RSResourceConfiguration           NZP-CSI-RS-ResourceConfiguration    OPTIONAL,
    cSI-RSResourceConfigurationToReleaseList            OCTET STRING    OPTIONAL,
    periodicCSI-RSResourceSetConfiguration              NZP-CSI-RS-ResourceSetConfiguration OPTIONAL,
    spCSI-RSResourceSetConfiguration                    NZP-CSI-RS-ResourceSetConfiguration OPTIONAL,
    periodicCSI-IMResourceConfiguration                 CSI-IM-ResourceConfiguration    OPTIONAL,
    spCSI-IMResourceConfiguration                       CSI-IM-ResourceConfiguration    OPTIONAL,
    iE-Extensions                                       ProtocolExtensionContainer { { CSI-RSResourceConfig-ExtIEs} }  OPTIONAL,
    ...
}

CSI-RSResourceConfig-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CSIReportConfig ::= OCTET STRING


CSI-RSCoordinationRequestList   ::= SEQUENCE (SIZE(1.. maxnoofLTMCSI-RSResourceConfig)) OF CSI-RSCoordinationRequest-Item

CSI-RSCoordinationRequest-Item  ::= SEQUENCE {
    ltmCSIResourceConfigurationID       INTEGER (0..111),
    transmissionRequest                 ENUMERATED{activate, deactivate},
    tci-State-InformationList           Tci-State-InformationList   OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { CSI-RSCoordinationRequest-Item-ExtIEs} }  OPTIONAL,
    ...
```

```
}

CSI-RSCoordinationRequest-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CSI-RSCoordinationResultList    ::= SEQUENCE (SIZE(1.. maxnoofLTMCSI-RSResourceConfig)) OF CSI-RSCoordinationResult-Item

CSI-RSCoordinationResult-Item   ::= SEQUENCE {
    ltmCSIResourceConfigurationID       INTEGER (0..111),
    transmissionStatus                  ENUMERATED{activated, deactivated},
    iE-Extensions                       ProtocolExtensionContainer { { CSI-RSCoordinationResult-Item-ExtIEs} }  OPTIONAL,
    ...
}

CSI-RSCoordinationResult-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

CSI-RSMeasurementsList  ::=  SEQUENCE (SIZE(1..maxnoofCSI-RSs)) OF CSI-RSMeasurements-Item

CSI-RSMeasurements-Item ::= SEQUENCE {
    csi-rsResourceID            INTEGER (0..192),
    selectedMeasurementQuantities   SelectedMeasurementQuantities,
    iE-Extensions       ProtocolExtensionContainer { { CSI-RSMeasurements-Item-ExtIEs } }   OPTIONAL,
    ...
}

CSI-RSMeasurements-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- D

DAPS-HO-Status::= ENUMERATED{initiation,... }

DCBasedDuplicationConfigured::= ENUMERATED{true,..., false}

DeactivationIndication ::= CHOICE {
    perUE                       DeactivationIndicationList,
    deactivateAll               NULL,
    choice-extension            ProtocolIE-SingleContainer { { DeactivationIndication-ExtIEs} }
}

DeactivationIndication-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

DeactivationIndicationList ::= SEQUENCE (SIZE(1..maxnoofUEsInQMCTransferControlMessage)) OF DeactivationIndicationList-Item

DeactivationIndicationList-Item ::= SEQUENCE {
    gNB-CU-UE-F1AP-ID                       GNB-CU-UE-F1AP-ID,
    gNB-DU-UE-F1AP-ID                       GNB-DU-UE-F1AP-ID,
```

```
    iE-Extensions                           ProtocolExtensionContainer { { DeactivationIndicationList-Item-ExtIEs} } OPTIONAL,
    ...
}

DeactivationIndicationList-Item-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Dedicated-SIDelivery-NeededUE-Item ::= SEQUENCE {
    gNB-CU-UE-F1AP-ID                       GNB-CU-UE-F1AP-ID,
    nRCGI                                   NRCGI,
    iE-Extensions                           ProtocolExtensionContainer { { DedicatedSIDeliveryNeededUE-Item-ExtIEs} } OPTIONAL,
    ...
}

DedicatedSIDeliveryNeededUE-Item-ExtIEs F1AP-PROTOCOL-EXTENSION::={
    ...
}

DedicatedSIDeliveryIndication::= ENUMERATED{true, ...}

DeliveryStatusReq ::= ENUMERATED{start, stop, ...}

DL-PRS ::= SEQUENCE {
    prsid                   INTEGER (0..255),
    dl-PRSResourceSetID     PRS-Resource-Set-ID,
    dl-PRSResourceID        PRS-Resource-ID OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { {DL-PRS-ExtIEs} }  OPTIONAL
}

DL-PRS-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DL-PRSMutingPattern ::= CHOICE {
    two                 BIT STRING (SIZE(2)),
    four                BIT STRING (SIZE(4)),
    six                 BIT STRING (SIZE(6)),
    eight               BIT STRING (SIZE(8)),
    sixteen             BIT STRING (SIZE(16)),
    thirty-two          BIT STRING (SIZE(32)),
    choice-extension                            ProtocolIE-SingleContainer { { DL-PRSMutingPattern-ExtIEs } }
}

DL-PRSMutingPattern-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

DLPRSResourceCoordinates ::= SEQUENCE {
    listofDL-PRSResourceSetARP      SEQUENCE (SIZE(1.. maxnoofPRS-ResourceSets)) OF DLPRSResourceSetARP,
    iE-Extensions                   ProtocolExtensionContainer { { DLPRSResourceCoordinates-ExtIEs } } OPTIONAL
}

DLPRSResourceCoordinates-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

DLPRSResourceSetARP ::= SEQUENCE {
    dl-PRSResourceSetID             PRS-Resource-Set-ID,
    dL-PRSResourceSetARPLocation    DL-PRSResourceSetARPLocation,
    listofDL-PRSResourceARP         SEQUENCE (SIZE(1.. maxnoofPRS-ResourcesPerSet)) OF DLPRSResourceARP,
    iE-Extensions                   ProtocolExtensionContainer { { DLPRSResourceSetARP-ExtIEs } } OPTIONAL
}

DLPRSResourceSetARP-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


DL-PRSResourceSetARPLocation ::= CHOICE {
    relativeGeodeticLocation            RelativeGeodeticLocation,
    relativeCartesianLocation           RelativeCartesianLocation,
    choice-Extension                    ProtocolIE-SingleContainer { { DL-PRSResourceSetARPLocation-ExtIEs } }
}

DL-PRSResourceSetARPLocation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}


DLPRSResourceARP ::= SEQUENCE {
    dl-PRSResourceID            PRS-Resource-ID,
    dL-PRSResourceARPLocation   DL-PRSResourceARPLocation,
    iE-Extensions               ProtocolExtensionContainer { { DLPRSResourceARP-ExtIEs } } OPTIONAL
}

DLPRSResourceARP-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DL-PRSResourceARPLocation ::= CHOICE {
    relativeGeodeticLocation            RelativeGeodeticLocation,
    relativeCartesianLocation           RelativeCartesianLocation,
    choice-Extension                    ProtocolIE-SingleContainer { { DL-PRSResourceARPLocation-ExtIEs } }
}

DL-PRSResourceARPLocation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

DLPDUSetInformationMarkingSupportIndication ::= ENUMERATED { true,...}

DL-UP-TNL-Address-to-Update-List-Item   ::= SEQUENCE {
    oldIPAdress                     TransportLayerAddress,
    newIPAdress                     TransportLayerAddress,
    iE-Extensions   ProtocolExtensionContainer { { DL-UP-TNL-Address-to-Update-List-ItemExtIEs } }  OPTIONAL,
    ...
}
```

```
DL-UP-TNL-Address-to-Update-List-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DLUPTNLInformation-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofDLUPTNLInformation)) OF DLUPTNLInformation-ToBeSetup-Item

DLUPTNLInformation-ToBeSetup-Item ::= SEQUENCE {
    dLUPTNLInformation  UPTransportLayerInformation ,
    iE-Extensions   ProtocolExtensionContainer { { DLUPTNLInformation-ToBeSetup-ItemExtIEs } }  OPTIONAL,
    ...
}

DLUPTNLInformation-ToBeSetup-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DRB-Activity-Item ::= SEQUENCE {
    dRBID           DRBID,
    dRB-Activity    DRB-Activity        OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { DRB-Activity-ItemExtIEs } }  OPTIONAL,
    ...
}

DRB-Activity-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DRB-Activity ::= ENUMERATED {active, not-active}

DRBID ::= INTEGER (1..32, ...)

DRBs-FailedToBeModified-Item    ::= SEQUENCE {
    dRBID       DRBID       ,
    cause       Cause       OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { DRBs-FailedToBeModified-ItemExtIEs } }   OPTIONAL,
    ...
}

DRBs-FailedToBeModified-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DRBs-FailedToBeSetup-Item   ::= SEQUENCE {
    dRBID   DRBID,
    cause   Cause   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { DRBs-FailedToBeSetup-ItemExtIEs } }  OPTIONAL,
    ...
}

DRBs-FailedToBeSetup-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
DRBs-FailedToBeSetupMod-Item    ::= SEQUENCE {
    dRBID       DRBID   ,
    cause       Cause           OPTIONAL ,
    iE-Extensions   ProtocolExtensionContainer { { DRBs-FailedToBeSetupMod-ItemExtIEs } }   OPTIONAL,
    ...
}

DRBs-FailedToBeSetupMod-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DRB-Information ::= SEQUENCE {
    dRB-QoS     QoSFlowLevelQoSParameters,
    sNSSAI      SNSSAI,
    notificationControl     NotificationControl     OPTIONAL,
    flows-Mapped-To-DRB-List    Flows-Mapped-To-DRB-List,
    iE-Extensions   ProtocolExtensionContainer { { DRB-Information-ItemExtIEs } }   OPTIONAL
}

DRB-Information-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    {   ID id-ECNMarkingorCongestionInformationReportingRequest     CRITICALITY ignore  EXTENSION ECNMarkingorCongestionInformationReportingRequest
        PRESENCE optional   }|
    {   ID id-PSIbasedSDUdiscardUL      CRITICALITY ignore  EXTENSION PSIbasedSDUdiscardUL      PRESENCE optional   }|
    {   ID id-PSIbasedSDUdiscardDL      CRITICALITY ignore  EXTENSION PSIbasedSDUdiscardDL      PRESENCE optional   }|
    {   ID id-UEPerformanceDelayMonitoring  CRITICALITY ignore  EXTENSION UEPerformanceDelayMonitoring    PRESENCE optional},
    ...
}

DRBs-Modified-Item  ::= SEQUENCE {
    dRBID                           DRBID,
    lCID                                LCID        OPTIONAL,
    dLUPTNLInformation-ToBeSetup-List       DLUPTNLInformation-ToBeSetup-List,
    iE-Extensions   ProtocolExtensionContainer { { DRBs-Modified-ItemExtIEs } } OPTIONAL,
    ...
}

DRBs-Modified-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-RLC-Status                          CRITICALITY ignore  EXTENSION RLC-Status                            PRESENCE optional }|
    { ID id-AdditionalPDCPDuplicationTNL-List   CRITICALITY ignore  EXTENSION AdditionalPDCPDuplicationTNL-List    PRESENCE optional }|
    { ID id-CurrentQoSParaSetIndex              CRITICALITY ignore  EXTENSION QoSParaSetIndex                       PRESENCE optional }|
    { ID id-TSCTrafficCharacteristicsFeedback   CRITICALITY ignore  EXTENSION TSCTrafficCharacteristicsFeedback    PRESENCE optional }|
    { ID id-ECNMarkingorCongestionInformationReportingStatus    CRITICALITY ignore  EXTENSION ECNMarkingorCongestionInformationReportingStatus
            PRESENCE optional },
    ...
}

DRBs-ModifiedConf-Item  ::= SEQUENCE {
    dRBID                           DRBID,
    uLUPTNLInformation-ToBeSetup-List       ULUPTNLInformation-ToBeSetup-List   ,
    iE-Extensions   ProtocolExtensionContainer { { DRBs-ModifiedConf-ItemExtIEs } } OPTIONAL,
    ...
}
```

```
DRBs-ModifiedConf-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalPDCPDuplicationTNL-List   CRITICALITY ignore  EXTENSION AdditionalPDCPDuplicationTNL-List       PRESENCE optional },
    ...
}

DRB-Notify-Item ::= SEQUENCE {
    dRBID           DRBID,
    notification-Cause  Notification-Cause,
    iE-Extensions   ProtocolExtensionContainer { { DRB-Notify-ItemExtIEs } }    OPTIONAL,
    ...
}

DRB-Notify-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-CurrentQoSParaSetIndex  CRITICALITY ignore  EXTENSION QoSParaSetNotifyIndex PRESENCE optional  }|
    { ID id-TSCTrafficCharacteristicsFeedback   CRITICALITY ignore  EXTENSION TSCTrafficCharacteristicsFeedback  PRESENCE optional },
    ...
}

DRBs-Required-ToBeModified-Item ::= SEQUENCE {
    dRBID                           DRBID,
    dLUPTNLInformation-ToBeSetup-List       DLUPTNLInformation-ToBeSetup-List   ,
    iE-Extensions   ProtocolExtensionContainer { { DRBs-Required-ToBeModified-ItemExtIEs } }    OPTIONAL,
    ...
}

DRBs-Required-ToBeModified-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-RLC-Status          CRITICALITY ignore          EXTENSION RLC-Status                PRESENCE optional }|
    { ID id-AdditionalPDCPDuplicationTNL-List   CRITICALITY ignore  EXTENSION AdditionalPDCPDuplicationTNL-List       PRESENCE optional },
    ...
}

DRBs-Required-ToBeReleased-Item ::= SEQUENCE {
    dRBID       DRBID,
    iE-Extensions   ProtocolExtensionContainer { { DRBs-Required-ToBeReleased-ItemExtIEs } }    OPTIONAL,
    ...
}

DRBs-Required-ToBeReleased-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DRBs-Setup-Item ::= SEQUENCE {
    dRBID                           DRBID,
    lCID                                LCID        OPTIONAL,
    dLUPTNLInformation-ToBeSetup-List       DLUPTNLInformation-ToBeSetup-List   ,
    iE-Extensions   ProtocolExtensionContainer { { DRBs-Setup-ItemExtIEs } }    OPTIONAL,
    ...
}

DRBs-Setup-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalPDCPDuplicationTNL-List   CRITICALITY ignore  EXTENSION AdditionalPDCPDuplicationTNL-List       PRESENCE optional }|
    { ID id-CurrentQoSParaSetIndex              CRITICALITY ignore  EXTENSION QoSParaSetIndex                         PRESENCE optional }|
    { ID id-TSCTrafficCharacteristicsFeedback   CRITICALITY ignore  EXTENSION TSCTrafficCharacteristicsFeedback       PRESENCE optional }|
```

```
	{ ID id-ECNMarkingorCongestionInformationReportingStatus	CRITICALITY ignore	EXTENSION ECNMarkingorCongestionInformationReportingStatus
		PRESENCE optional },
	...
}

DRBs-SetupMod-Item	::= SEQUENCE {
	dRBID							DRBID,
	lCID								LCID		OPTIONAL,
	dLUPTNLInformation-ToBeSetup-List		DLUPTNLInformation-ToBeSetup-List	,
	iE-Extensions	ProtocolExtensionContainer { { DRBs-SetupMod-ItemExtIEs } } OPTIONAL,
	...
}

DRBs-SetupMod-ItemExtIEs	F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-AdditionalPDCPDuplicationTNL-List	CRITICALITY ignore	EXTENSION AdditionalPDCPDuplicationTNL-List		PRESENCE optional }|
	{ ID id-CurrentQoSParaSetIndex			CRITICALITY ignore	EXTENSION QoSParaSetIndex				PRESENCE optional }|
	{ ID id-TSCTrafficCharacteristicsFeedback	CRITICALITY ignore	EXTENSION TSCTrafficCharacteristicsFeedback		PRESENCE optional }|
	{ ID id-ECNMarkingorCongestionInformationReportingStatus	CRITICALITY ignore	EXTENSION ECNMarkingorCongestionInformationReportingStatus
	PRESENCE optional },
	...
}


DRBs-ToBeModified-Item	::= SEQUENCE {
	dRBID						DRBID,
	qoSInformation				QoSInformation	OPTIONAL,
	uLUPTNLInformation-ToBeSetup-List	ULUPTNLInformation-ToBeSetup-List	,
	uLConfiguration				ULConfiguration OPTIONAL,
	iE-Extensions	ProtocolExtensionContainer { { DRBs-ToBeModified-ItemExtIEs } } OPTIONAL,
	...
}

DRBs-ToBeModified-ItemExtIEs	F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-DLPDCPSNLength					CRITICALITY ignore	EXTENSION PDCPSNLength						PRESENCE optional }|
	{ ID id-ULPDCPSNLength					CRITICALITY ignore	EXTENSION PDCPSNLength						PRESENCE optional }|
	{ID id-BearerTypeChange					CRITICALITY ignore	EXTENSION BearerTypeChange					PRESENCE optional}|
	{ ID id-RLCMode						CRITICALITY ignore	EXTENSION RLCMode						PRESENCE optional }|
	{ ID id-Duplication-Activation				CRITICALITY reject	EXTENSION DuplicationActivation				PRESENCE optional }|
	{ ID id-DC-Based-Duplication-Configured		CRITICALITY reject	EXTENSION DCBasedDuplicationConfigured			PRESENCE optional }|
	{ ID id-DC-Based-Duplication-Activation		CRITICALITY reject	EXTENSION DuplicationActivation				PRESENCE optional }|
	{ ID id-AdditionalPDCPDuplicationTNL-List	CRITICALITY ignore	EXTENSION AdditionalPDCPDuplicationTNL-List		PRESENCE optional }|
	{ ID id-RLCDuplicationInformation			CRITICALITY ignore	EXTENSION RLCDuplicationInformation			PRESENCE optional}|
	{ ID id-TransmissionStopIndicator			CRITICALITY ignore	EXTENSION TransmissionStopIndicator			PRESENCE optional}|
	{ ID id-CG-SDTindicatorMod				CRITICALITY reject	EXTENSION CG-SDTindicatorMod				PRESENCE optional },
	...
}

DRBs-ToBeReleased-Item	::= SEQUENCE {
	dRBID	DRBID,
	iE-Extensions	ProtocolExtensionContainer { { DRBs-ToBeReleased-ItemExtIEs } } OPTIONAL,
	...
}

DRBs-ToBeReleased-ItemExtIEs	F1AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

DRBs-ToBeSetup-Item ::= SEQUENCE    {
    dRBID                       DRBID,
    qoSInformation              QoSInformation,
    uLUPTNLInformation-ToBeSetup-List   ULUPTNLInformation-ToBeSetup-List   ,
    rLCMode                     RLCMode,
    uLConfiguration             ULConfiguration OPTIONAL,
    duplicationActivation       DuplicationActivation   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { DRBs-ToBeSetup-ItemExtIEs } }    OPTIONAL,
    ...
}

DRBs-ToBeSetup-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-DC-Based-Duplication-Configured     CRITICALITY reject  EXTENSION DCBasedDuplicationConfigured          PRESENCE optional }|
    { ID id-DC-Based-Duplication-Activation     CRITICALITY reject  EXTENSION DuplicationActivation                 PRESENCE optional }|
    { ID id-DLPDCPSNLength                      CRITICALITY ignore  EXTENSION PDCPSNLength                          PRESENCE mandatory }|
    { ID id-ULPDCPSNLength                      CRITICALITY ignore  EXTENSION PDCPSNLength                          PRESENCE optional }|
    { ID id-AdditionalPDCPDuplicationTNL-List   CRITICALITY ignore  EXTENSION AdditionalPDCPDuplicationTNL-List     PRESENCE optional }|
    { ID id-RLCDuplicationInformation           CRITICALITY ignore  EXTENSION RLCDuplicationInformation             PRESENCE optional}|
    { ID id-SDTRLCBearerConfiguration           CRITICALITY ignore  EXTENSION SDTRLCBearerConfiguration             PRESENCE optional },
    ...
}


DRBs-ToBeSetupMod-Item  ::= SEQUENCE {
    dRBID                       DRBID,
    qoSInformation              QoSInformation,
    uLUPTNLInformation-ToBeSetup-List       ULUPTNLInformation-ToBeSetup-List,
    rLCMode                     RLCMode,
    uLConfiguration             ULConfiguration OPTIONAL,
    duplicationActivation       DuplicationActivation   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { DRBs-ToBeSetupMod-ItemExtIEs } } OPTIONAL,
    ...
}

DRBs-ToBeSetupMod-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-DC-Based-Duplication-Configured     CRITICALITY reject  EXTENSION DCBasedDuplicationConfigured          PRESENCE optional }|
    { ID id-DC-Based-Duplication-Activation     CRITICALITY reject  EXTENSION DuplicationActivation                 PRESENCE optional }|
    { ID id-DLPDCPSNLength                      CRITICALITY ignore  EXTENSION PDCPSNLength                          PRESENCE optional }|
    { ID id-ULPDCPSNLength                      CRITICALITY ignore  EXTENSION PDCPSNLength                          PRESENCE optional }|
    { ID id-AdditionalPDCPDuplicationTNL-List   CRITICALITY ignore  EXTENSION AdditionalPDCPDuplicationTNL-List     PRESENCE optional }|
    { ID id-RLCDuplicationInformation           CRITICALITY ignore  EXTENSION RLCDuplicationInformation             PRESENCE optional}|
    { ID id-CG-SDTindicatorSetup                CRITICALITY reject  EXTENSION CG-SDTindicatorSetup                  PRESENCE optional },
    ...
}

DRB-List ::= SEQUENCE (SIZE(1.. maxnoofDRBs)) OF DRB-List-Item

DRB-List-Item ::= SEQUENCE {
    dRBID                       DRBID,
    iE-Extensions               ProtocolExtensionContainer { { DRB-List-Item-ExtIEs} } OPTIONAL
}
```

```
DRB-List-Item-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DRXCycle    ::= SEQUENCE {
    longDRXCycleLength  LongDRXCycleLength,
    shortDRXCycleLength     ShortDRXCycleLength OPTIONAL,
    shortDRXCycleTimer  ShortDRXCycleTimer OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { DRXCycle-ExtIEs} } OPTIONAL,
    ...
}

DRXCycle-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NonIntegerDRXCycle  ::= SEQUENCE {
    longNonIntegerDRXCycleLength    LongNonIntegerDRXCycleLength,
    shortNonIntegerDRXCycleLength   ShortNonIntegerDRXCycleLength   OPTIONAL,
    shortDRXCycleTimer              ShortDRXCycleTimer OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { NonIntegerDRXCycle-ExtIEs} } OPTIONAL,
    ...
}

NonIntegerDRXCycle-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DRX-Config ::= OCTET STRING

DRXConfigurationIndicator   ::= ENUMERATED{ release, ...}

DRX-LongCycleStartOffset ::= INTEGER (0..10239)

DSInformationList ::= SEQUENCE (SIZE(0..maxnoofDSInfo)) OF DSCP

DSCP ::= BIT STRING (SIZE (6))

DUtoCURRCContainer ::= OCTET STRING

DUCURadioInformationType ::= CHOICE {
    rIM                             DUCURIMInformation,
    choice-extension                ProtocolIE-SingleContainer { { DUCURadioInformationType-ExtIEs} }
}

DUCURadioInformationType-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

DUCURIMInformation ::= SEQUENCE {
    victimgNBSetID              GNBSetID,
    rIMRSDetectionStatus        RIMRSDetectionStatus,
    aggressorCellList           AggressorCellList,
```

```
	iE-Extensions				ProtocolExtensionContainer { { DUCURIMInformation-ExtIEs} }		OPTIONAL
}

DUCURIMInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

DUF-Slot-Config-Item	::= CHOICE {
	explicitFormat				ExplicitFormat,
	implicitFormat				ImplicitFormat,
	choice-extension				ProtocolIE-SingleContainer { { DUF-Slot-Config-Item-ExtIEs} }
}

DUF-Slot-Config-Item-ExtIEs F1AP-PROTOCOL-IES ::= {
	...
}
DUF-Slot-Config-List	::= SEQUENCE (SIZE(1..maxnoofDUFSlots)) OF DUF-Slot-Config-Item

DUFSlotformatIndex ::= INTEGER(0..254)

DUFTransmissionPeriodicity ::= ENUMERATED { ms0p5, ms0p625, ms1, ms1p25, ms2, ms2p5, ms5, ms10, ...}

DU-RX-MT-RX ::= ENUMERATED {supported, not-supported }

DU-TX-MT-TX ::= ENUMERATED {supported, not-supported }

DU-RX-MT-TX ::= ENUMERATED {supported, not-supported }

DU-TX-MT-RX ::= ENUMERATED {supported, not-supported }

DU-RX-MT-RX-Extend ::= ENUMERATED {supported, not-supported, supported-and-FDM-required, ...}

DU-TX-MT-TX-Extend ::= ENUMERATED {supported, not-supported, supported-and-FDM-required, ...}

DU-RX-MT-TX-Extend ::= ENUMERATED {supported, not-supported, supported-and-FDM-required, ...}

DU-TX-MT-RX-Extend ::= ENUMERATED {supported, not-supported, supported-and-FDM-required, ...}

DUtoCURRCInformation ::= SEQUENCE {
	cellGroupConfig		CellGroupConfig,
	measGapConfig			MeasGapConfig	OPTIONAL,
	requestedP-MaxFR1				OCTET STRING				OPTIONAL,
	iE-Extensions				ProtocolExtensionContainer { { DUtoCURRCInformation-ExtIEs} } OPTIONAL,
	...
}

DUtoCURRCInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-DRX-LongCycleStartOffset			CRITICALITY ignore	EXTENSION DRX-LongCycleStartOffset			PRESENCE optional }|
	{ ID id-SelectedBandCombinationIndex		CRITICALITY ignore	EXTENSION SelectedBandCombinationIndex		PRESENCE optional }|
	{ ID id-SelectedFeatureSetEntryIndex		CRITICALITY ignore	EXTENSION SelectedFeatureSetEntryIndex		PRESENCE optional }|
	{ ID id-Ph-InfoSCG						CRITICALITY ignore	EXTENSION Ph-InfoSCG						PRESENCE optional }|
	{ ID id-RequestedBandCombinationIndex		CRITICALITY ignore	EXTENSION RequestedBandCombinationIndex		PRESENCE optional }|
	{ ID id-RequestedFeatureSetEntryIndex		CRITICALITY ignore	EXTENSION RequestedFeatureSetEntryIndex		PRESENCE optional }|
	{ ID id-DRX-Config						CRITICALITY ignore	EXTENSION DRX-Config						PRESENCE optional }|
```

```
    { ID id-PDCCH-BlindDetectionSCG            CRITICALITY ignore  EXTENSION PDCCH-BlindDetectionSCG           PRESENCE optional }|
    { ID id-Requested-PDCCH-BlindDetectionSCG  CRITICALITY ignore  EXTENSION Requested-PDCCH-BlindDetectionSCG PRESENCE optional }|
    { ID id-Ph-InfoMCG                         CRITICALITY ignore  EXTENSION Ph-InfoMCG                        PRESENCE optional }|
    { ID id-MeasGapSharingConfig               CRITICALITY ignore  EXTENSION MeasGapSharingConfig              PRESENCE optional }|
    { ID id-SL-PHY-MAC-RLC-Config              CRITICALITY ignore  EXTENSION SL-PHY-MAC-RLC-Config             PRESENCE optional }|
    { ID id-SL-ConfigDedicatedEUTRA-Info       CRITICALITY ignore  EXTENSION SL-ConfigDedicatedEUTRA-Info      PRESENCE optional }|
    { ID id-RequestedP-MaxFR2                  CRITICALITY ignore  EXTENSION RequestedP-MaxFR2                 PRESENCE optional }|
    { ID id-SDT-MAC-PHY-CG-Config              CRITICALITY ignore  EXTENSION SDT-MAC-PHY-CG-Config             PRESENCE optional }|
    { ID id-MUSIM-GapConfig                    CRITICALITY ignore  EXTENSION MUSIM-GapConfig                   PRESENCE optional }|
    { ID id-SL-RLC-ChannelToAddModList         CRITICALITY ignore  EXTENSION SL-RLC-ChannelToAddModList        PRESENCE optional }|
    { ID id-InterFrequencyConfig-NoGap         CRITICALITY ignore  EXTENSION InterFrequencyConfig-NoGap        PRESENCE optional }|
    { ID id-UL-GapFR2-Config                   CRITICALITY ignore  EXTENSION UL-GapFR2-Config                  PRESENCE optional }|
    { ID id-TwoPHRModeMCG                      CRITICALITY ignore  EXTENSION TwoPHRModeMCG                     PRESENCE optional }|
    { ID id-TwoPHRModeSCG                      CRITICALITY ignore  EXTENSION TwoPHRModeSCG                     PRESENCE optional }|
    { ID id-ncd-SSB-RedCapInitialBWP-SDT       CRITICALITY ignore  EXTENSION Ncd-SSB-RedCapInitialBWP-SDT      PRESENCE optional }|
    { ID id-ServCellInfoList                   CRITICALITY ignore  EXTENSION ServCellInfoList                  PRESENCE optional }|
    { ID id-SL-PHY-MAC-RLC-ConfigExt           CRITICALITY ignore  EXTENSION SL-PHY-MAC-RLC-ConfigExt          PRESENCE optional },
    ...
}

DUtoCUTAInformation-List ::= SEQUENCE (SIZE(1.. maxnoofTAList)) OF DUtoCUTAInformation-Item

DUtoCUTAInformation-Item    ::= SEQUENCE {
    nRCGI                   NRCGI,
    tAValue                 TAValue,
    preambleIndex           PreambleIndex,
    rA-RNTI                 RA-RNTI,
    sourceGNB-DU-ID         GNB-DU-ID,
    tagIDPointer            TagIDPointer        OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { DUtoCUTAInformation-Item-ExtIEs} }   OPTIONAL,
    ...
}

DUtoCUTAInformation-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-LTMgNB-ID       CRITICALITY ignore  EXTENSION GlobalGNB-ID  PRESENCE optional },
    ...
}

DuplicationActivation ::= ENUMERATED{active,inactive,... }

DuplicationIndication ::= ENUMERATED {true, ... , false }

DuplicationState ::= ENUMERATED {
    active,
    inactive,
    ...
}

Dynamic5QIDescriptor    ::= SEQUENCE {
    qoSPriorityLevel                INTEGER (1..127),
    packetDelayBudget               PacketDelayBudget,
    packetErrorRate                 PacketErrorRate,
    fiveQI                          INTEGER (0..255, ...)                           OPTIONAL,
    delayCritical                   ENUMERATED {delay-critical, non-delay-critical}     OPTIONAL,
```

```
    -- The above IE shall be present if the GBR QoS Flow Information IE is present in the QoS Flow Level QoS Parameters IE.
    averagingWindow                     AveragingWindow                                     OPTIONAL,
    -- The above IE shall be present if the GBR QoS Flow Information IE is present in the QoS Flow Level QoS Parameters IE.
    maxDataBurstVolume                  MaxDataBurstVolume                                  OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { Dynamic5QIDescriptor-ExtIEs } } OPTIONAL
}

Dynamic5QIDescriptor-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-ExtendedPacketDelayBudget           CRITICALITY ignore  EXTENSION ExtendedPacketDelayBudget   PRESENCE optional    }|
    { ID id-CNPacketDelayBudgetDownlink         CRITICALITY ignore  EXTENSION ExtendedPacketDelayBudget   PRESENCE optional    }|
    { ID id-CNPacketDelayBudgetUplink           CRITICALITY ignore  EXTENSION ExtendedPacketDelayBudget   PRESENCE optional    },
    ...
}

DynamicPQIDescriptor    ::= SEQUENCE {
    resourceType                        ENUMERATED {gbr, non-gbr, delay-critical-grb, ...}  OPTIONAL,
    qoSPriorityLevel                    INTEGER (1..8, ...),
    packetDelayBudget                   PacketDelayBudget,
    packetErrorRate                     PacketErrorRate,
    averagingWindow                     AveragingWindow                                     OPTIONAL,
    -- The above IE shall be present if the GBR QoS Flow Information IE is present in the QoS Flow Level QoS Parameters IE.
    maxDataBurstVolume                  MaxDataBurstVolume                                  OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { DynamicPQIDescriptor-ExtIEs } } OPTIONAL
}

DynamicPQIDescriptor-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DLLBTFailureInformationRequest  ::= ENUMERATED {inquiry, ...}
DLLBTFailureInformationList     ::= SEQUENCE (SIZE(1.. maxnoofLBTFailureInformation)) OF DLLBTFailureInformationList-Item

DLLBTFailureInformationList-Item::= SEQUENCE {
    uEAssistantIdentifier       GNB-CU-UE-F1AP-ID,
    numberOfDLLBTFailures       INTEGER (1..1000,...)                               OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { DLLBTFailureInformationList-Item-ExtIEs} }  OPTIONAL,
    ...
}

DLLBTFailureInformationList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- E


EarlyULSyncConfig ::= SEQUENCE {
    rACH                                RACHConfiguration,
    lTMgNB-DU-IDs-PreambleIndexList     LTMgNB-DU-IDs-PreambleIndexList                         OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { EarlyULSyncConfig-ExtIEs} } OPTIONAL,
    ...
}
```

```
EarlyULSyncConfig-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

EarlySyncInformation-Request  ::= SEQUENCE {
    requestforRACHConfiguration                RequestforRACHConfiguration,
    lTMgNB-DU-IDsList                          LTMgNB-DU-IDsList,
    iE-Extensions                              ProtocolExtensionContainer { { EarlySyncInformation-Request-ExtIEs} } OPTIONAL,
    ...
}


EarlySyncInformation-Request-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


EarlySyncInformation  ::= SEQUENCE {
    tCIStatesConfigurationsList     TCIStatesConfigurationsList,
    earlyULSyncConfig               EarlyULSyncConfig                                   OPTIONAL,
    earlyULSyncConfigSUL            EarlyULSyncConfig                                   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { EarlySyncInformation-ExtIEs} } OPTIONAL,
    ...
}


EarlySyncInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

EarlySyncCandidateCellInformation-List ::= SEQUENCE (SIZE (1.. maxnoofLTMCells)) OF EarlySyncCandidateCellInformation-Item

EarlySyncCandidateCellInformation-Item ::= SEQUENCE {
    nRCGI               NRCGI,
    tCIStatesConfigurationsList     TCIStatesConfigurationsList                         OPTIONAL,
    earlyULSyncConfig               EarlyULSyncConfig                                   OPTIONAL,
    earlyULSyncConfigSUL            EarlyULSyncConfig                                   OPTIONAL,
    tAAssistanceInfo                TAAssistanceInfo                                    OPTIONAL,
    uEbasedTAmeasurementConfig      OCTET STRING                                        OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { EarlySyncCandidateCellInformation-Item-ExtIEs } }  OPTIONAL,
    ...
}

EarlySyncCandidateCellInformation-Item-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-SSB-PositionsInBurst    CRITICALITY ignore  EXTENSION SSB-PositionsInBurst      PRESENCE optional }|
-- The above IE shall be present if the earlyULSyncConfig IE or the earlyULSyncConfigSUL IE is present
    {   ID id-LTMResidualTAInfoList CRITICALITY ignore  EXTENSION   LTMResidualTAInfoList       PRESENCE optional},
    ...
}

EarlySyncServingCellInformation ::= SEQUENCE {
    uEbasedTAmeasurementConfig      OCTET STRING                                        OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { EarlySyncServingCellInformation-ExtIEs } } OPTIONAL,
```

```
	...
}

EarlySyncServingCellInformation-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
	...
}


E-CID-MeasurementQuantities ::= SEQUENCE (SIZE (1.. maxnoofMeasE-CID)) OF ProtocolIE-SingleContainer { {E-CID-MeasurementQuantities-ItemIEs} }

E-CID-MeasurementQuantities-ItemIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-E-CID-MeasurementQuantities-Item	CRITICALITY reject  TYPE E-CID-MeasurementQuantities-Item	 PRESENCE mandatory}
}

E-CID-MeasurementQuantities-Item ::= SEQUENCE {
	e-CIDmeasurementQuantitiesValue			E-CID-MeasurementQuantitiesValue,
	iE-Extensions					ProtocolExtensionContainer { { E-CID-MeasurementQuantitiesValue-ExtIEs} } OPTIONAL
}

E-CID-MeasurementQuantitiesValue-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

E-CID-MeasurementQuantitiesValue ::= ENUMERATED {
	default,
	angleOfArrivalNR,
	... ,
	timingAdvanceNR,
	angleOfArrivalNR-per-TRP
}

E-CID-MeasurementResult ::= SEQUENCE {
	geographicalCoordinates		GeographicalCoordinates		OPTIONAL,
	measuredResults-List		E-CID-MeasuredResults-List	OPTIONAL,
	iE-Extensions				ProtocolExtensionContainer { { E-CID-MeasurementResult-ExtIEs} } OPTIONAL
}

E-CID-MeasurementResult-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-MobileAccessPointLocation   CRITICALITY ignore EXTENSION Mobile-TRP-LocationInformation  PRESENCE optional }|
	{ ID id-E-CID-MeasuredResultsAssociatedInfoList CRITICALITY ignore  EXTENSION E-CID-MeasuredResultsAssociatedInfoList   PRESENCE optional},
	...
}

E-CID-MeasuredResults-List ::= SEQUENCE (SIZE(1..maxnoofMeasE-CID)) OF E-CID-MeasuredResults-Item

E-CID-MeasuredResults-Item ::= SEQUENCE {
	e-CID-MeasuredResults-Value	E-CID-MeasuredResults-Value,
	iE-Extensions			ProtocolExtensionContainer {{ E-CID-MeasuredResults-Item-ExtIEs }}	OPTIONAL
}

E-CID-MeasuredResults-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}
```

```
E-CID-MeasuredResults-Value ::= CHOICE {
    valueAngleofArrivalNR   UL-AoA,
    choice-extension        ProtocolIE-SingleContainer { { E-CID-MeasuredResults-Value-ExtIEs} }
}

E-CID-MeasuredResults-Value-ExtIEs F1AP-PROTOCOL-IES ::= {
    { ID id-NR-TADV         CRITICALITY ignore  TYPE NR-TADV    PRESENCE mandatory  }|
    { ID id-E-CID-AoA-NR-per-TRP CRITICALITY ignore TYPE E-CID-AoA-NR-per-TRP       PRESENCE mandatory  },
    ...
}

E-CID-AoA-NR-per-TRP ::= SEQUENCE (SIZE (1..maxNoOfMeasTRPs)) OF E-CID-AoA-NR-per-TRP-Item

E-CID-AoA-NR-per-TRP-Item ::= SEQUENCE {
    tRP-ID                          TRPID,
    geographicalCoordinates         GeographicalCoordinates     OPTIONAL,
    uL-AngleOfArrival               UL-AoA,
    timeStamp                       TimeStamp,
    measurementQuality              TRPMeasurementQuality       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer {{ E-CID-AoA-NR-per-TRP-Item-ExtIEs}} OPTIONAL,
    ...
}

E-CID-AoA-NR-per-TRP-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-CID-MeasuredResultsAssociatedInfoList ::= SEQUENCE (SIZE (1..maxnoofMeasE-CID)) OF E-CID-MeasuredResultsAssociatedInfoItem

E-CID-MeasuredResultsAssociatedInfoItem ::= SEQUENCE {
    timeStamp                   TimeStamp               OPTIONAL,
    measurementQuality          TRPMeasurementQuality   OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { E-CID-MeasuredResultsAssociatedInfoItem-ExtIEs} } OPTIONAL,
    ...
}

E-CID-MeasuredResultsAssociatedInfoItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-CID-ReportCharacteristics ::= ENUMERATED {
    onDemand,
    periodic,
    ...
}

EgressBHRLCCHList ::= SEQUENCE (SIZE(1..maxnoofEgressLinks)) OF EgressBHRLCCHItem

EgressBHRLCCHItem ::= SEQUENCE {
    nextHopBAPAddress       BAPAddress,
    bHRLCChannelID          BHRLCChannelID,
    iE-Extensions           ProtocolExtensionContainer {{EgressBHRLCCHItemExtIEs }}  OPTIONAL
}
```

```
EgressBHRLCCHItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

EgressNonF1terminatingTopologyIndicator ::= ENUMERATED {true, ...}

Endpoint-IP-address-and-port ::=SEQUENCE {
    endpointIPAddress TransportLayerAddress,
    iE-Extensions                   ProtocolExtensionContainer { { Endpoint-IP-address-and-port-ExtIEs} } OPTIONAL
}

Endpoint-IP-address-and-port-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-portNumber  CRITICALITY reject  EXTENSION PortNumber        PRESENCE optional},
    ...
}

EnergyDetectionThreshold ::= INTEGER (-100..-50, ...)

ExtendedAvailablePLMN-List ::= SEQUENCE (SIZE(1..maxnoofExtendedBPLMNs)) OF ExtendedAvailablePLMN-Item

ExtendedAvailablePLMN-Item ::= SEQUENCE {
    pLMNIdentity            PLMN-Identity,
    iE-Extensions       ProtocolExtensionContainer { { ExtendedAvailablePLMN-Item-ExtIEs} } OPTIONAL
}

ExplicitFormat ::=  SEQUENCE {
    permutation         Permutation,
    noofDownlinkSymbols NoofDownlinkSymbols     OPTIONAL,
    noofUplinkSymbols   NoofUplinkSymbols       OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { ExplicitFormat-ExtIEs} } OPTIONAL
}

ExplicitFormat-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ExtendedAvailablePLMN-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ExtendedServedPLMNs-List ::= SEQUENCE (SIZE(1.. maxnoofExtendedBPLMNs)) OF ExtendedServedPLMNs-Item

ExtendedServedPLMNs-Item ::= SEQUENCE {
    pLMN-Identity               PLMN-Identity,
    tAISliceSupportList         SliceSupportList    OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { ExtendedServedPLMNs-ItemExtIEs} } OPTIONAL,
    ...
}

ExtendedServedPLMNs-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-NPNSupportInfo              CRITICALITY reject  EXTENSION NPNSupportInfo              PRESENCE optional }|
    { ID id-ExtendedTAISliceSupportList CRITICALITY reject  EXTENSION ExtendedSliceSupportList    PRESENCE optional }|
    { ID id-TAINSAGSupportList          CRITICALITY ignore  EXTENSION NSAGSupportList             PRESENCE optional},
    ...
```

```
}

ExtendedSliceSupportList ::= SEQUENCE (SIZE(1.. maxnoofExtSliceItems)) OF SliceSupportItem

ExtendedUEIdentityIndexValue ::= BIT STRING (SIZE(16))

EUTRACells-List  ::= SEQUENCE (SIZE (1.. maxCellineNB)) OF EUTRACells-List-item

EUTRACells-List-item ::= SEQUENCE {
    eUTRA-Cell-ID                  EUTRA-Cell-ID,
    served-EUTRA-Cells-Information  Served-EUTRA-Cells-Information,
    iE-Extensions ProtocolExtensionContainer { { EUTRACells-List-itemExtIEs } }    OPTIONAL
}

EUTRACells-List-itemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


EUTRA-Cell-ID ::= BIT STRING (SIZE(28))

EUTRA-Coex-FDD-Info ::= SEQUENCE {
    uL-EARFCN                       ExtendedEARFCN                   OPTIONAL,
    dL-EARFCN                       ExtendedEARFCN,
    uL-Transmission-Bandwidth       EUTRA-Transmission-Bandwidth    OPTIONAL,
    dL-Transmission-Bandwidth       EUTRA-Transmission-Bandwidth,
    iE-Extensions                   ProtocolExtensionContainer { {EUTRA-Coex-FDD-Info-ExtIEs} } OPTIONAL,
    ...
}

EUTRA-Coex-FDD-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

EUTRA-Coex-Mode-Info ::= CHOICE {
    fDD     EUTRA-Coex-FDD-Info,
    tDD     EUTRA-Coex-TDD-Info,
    ...
}

EUTRA-Coex-TDD-Info ::= SEQUENCE {
    eARFCN                          ExtendedEARFCN,
    transmission-Bandwidth          EUTRA-Transmission-Bandwidth,
    subframeAssignment              EUTRA-SubframeAssignment,
    specialSubframe-Info            EUTRA-SpecialSubframe-Info,
    iE-Extensions                   ProtocolExtensionContainer { {EUTRA-Coex-TDD-Info-ExtIEs} } OPTIONAL,
    ...
}
EUTRA-Coex-TDD-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
EUTRA-CyclicPrefixDL ::= ENUMERATED {
    normal,
    extended,
```

```
	...
}

EUTRA-CyclicPrefixUL ::= ENUMERATED {
	normal,
	extended,
	...
}

EUTRA-PRACH-Configuration ::= SEQUENCE {
	rootSequenceIndex					INTEGER (0..837),
	zeroCorrelationIndex				INTEGER (0..15),
	highSpeedFlag						BOOLEAN,
	prach-FreqOffset					INTEGER (0..94),
	prach-ConfigIndex					INTEGER (0..63)		OPTIONAL,
	-- The above IE shall be present if the EUTRA-Mode-Info IE in the Resource Coordination E-UTRA Cell Information IE is set to the value "TDD"
	iE-Extensions						ProtocolExtensionContainer { {EUTRA-PRACH-Configuration-ExtIEs} } OPTIONAL,
	...
}

EUTRA-PRACH-Configuration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}


EUTRA-SpecialSubframe-Info ::= SEQUENCE {
	specialSubframePatterns		EUTRA-SpecialSubframePatterns,
	cyclicPrefixDL				EUTRA-CyclicPrefixDL,
	cyclicPrefixUL				EUTRA-CyclicPrefixUL,
	iE-Extensions				ProtocolExtensionContainer { { EUTRA-SpecialSubframe-Info-ExtIEs} } OPTIONAL,
	...
}

EUTRA-SpecialSubframe-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

EUTRA-SpecialSubframePatterns ::= ENUMERATED {
	ssp0,
	ssp1,
	ssp2,
	ssp3,
	ssp4,
	ssp5,
	ssp6,
	ssp7,
	ssp8,
	ssp9,
	ssp10,
	...
}

EUTRA-SubframeAssignment ::= ENUMERATED {
	sa0,
```

```
    sa1,
    sa2,
    sa3,
    sa4,
    sa5,
    sa6,
    ...
}

EUTRA-Transmission-Bandwidth ::= ENUMERATED {
    bw6,
    bw15,
    bw25,
    bw50,
    bw75,
    bw100,
    ...
}

EUTRANQoS   ::= SEQUENCE {
    qCI                             QCI,
    allocationAndRetentionPriority  AllocationAndRetentionPriority,
    gbrQosInformation               GBR-QosInformation                                OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { EUTRANQoS-ExtIEs} }  OPTIONAL,
    ...
}

EUTRANQoS-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
{ ID id-ENBDLTNLAddress     CRITICALITY ignore  EXTENSION TransportLayerAddress PRESENCE optional },
    ...
}

ExecuteDuplication ::= ENUMERATED{true,...}

ExtendedEARFCN ::= INTEGER (0..262143)

EUTRA-Mode-Info ::= CHOICE {
    eUTRAFDD        EUTRA-FDD-Info,
    eUTRATDD        EUTRA-TDD-Info,
    choice-extension    ProtocolIE-SingleContainer { { EUTRA-Mode-Info-ExtIEs} }
}

EUTRA-Mode-Info-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

EUTRA-NR-CellResourceCoordinationReq-Container  ::= OCTET STRING

EUTRA-NR-CellResourceCoordinationReqAck-Container   ::= OCTET STRING

EUTRA-FDD-Info ::= SEQUENCE {
    uL-offsetToPointA               OffsetToPointA,
    dL-offsetToPointA               OffsetToPointA,
    iE-Extensions                   ProtocolExtensionContainer { {EUTRA-FDD-Info-ExtIEs} } OPTIONAL,
```

```
	...
}

EUTRA-FDD-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

EUTRA-TDD-Info ::= SEQUENCE {
	offsetToPointA				OffsetToPointA,
	iE-Extensions				ProtocolExtensionContainer { {EUTRA-TDD-Info-ExtIEs} } OPTIONAL,
	...
}

EUTRA-TDD-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

EventType ::= ENUMERATED {
	on-demand,
	periodic,
	stop,
	...
}

ExtendedPacketDelayBudget ::= INTEGER (1..65535, ..., 65536..109999)

Expected-UL-AoA ::= SEQUENCE {
	expected-Azimuth-AoA	Expected-Azimuth-AoA,
	expected-Zenith-AoA		Expected-Zenith-AoA		OPTIONAL,
	iE-extensions			ProtocolExtensionContainer { { Expected-UL-AoA-ExtIEs } }	OPTIONAL,
	...
}
Expected-UL-AoA-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

Expected-ZoA-only ::= SEQUENCE {
	expected-ZoA-only	Expected-Zenith-AoA,
	iE-extensions		ProtocolExtensionContainer { { Expected-ZoA-only-ExtIEs } } OPTIONAL,
	...
}

Expected-ZoA-only-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

Expected-Azimuth-AoA ::= SEQUENCE {
	expected-Azimuth-AoA-value			Expected-Value-AoA,
	expected-Azimuth-AoA-uncertainty	Uncertainty-range-AoA,
	iE-Extensions						ProtocolExtensionContainer { { Expected-Azimuth-AoA-ExtIEs } }	OPTIONAL,
	...
}

Expected-Azimuth-AoA-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}
Expected-Zenith-AoA ::= SEQUENCE {
    expected-Zenith-AoA-value           Expected-Value-ZoA,
    expected-Zenith-AoA-uncertainty     Uncertainty-range-ZoA,
    iE-Extensions                       ProtocolExtensionContainer { { Expected-Zenith-AoA-ExtIEs } }     OPTIONAL,
    ...
}

Expected-Zenith-AoA-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Expected-Value-AoA ::= INTEGER (0..3599)

Expected-Value-ZoA ::= INTEGER (0..1799)

ECNMarkingorCongestionInformationReportingRequest ::= CHOICE {
    ecnMarking          ECNmarkingRequest,
    congestionInformation   CongestionInformationRequest,
    choice-extension        ProtocolIE-SingleContainer { { ECNMarkingorCongestionInformationReportingRequest-ExtIEs } }
}

ECNMarkingorCongestionInformationReportingRequest-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

ECNmarkingRequest ::= ENUMERATED { ul, dl, both, stop, ... }
CongestionInformationRequest ::= ENUMERATED { ul, dl, both, stop, ... }
ECNMarkingorCongestionInformationReportingStatus ::= ENUMERATED { active, not-active, ...}

EnergyCost ::= INTEGER (0..10000, ...)

-- F

F1CPathNSA ::= ENUMERATED {lte, nr, both}

F1CTransferPath ::= SEQUENCE {
    f1CPathNSA                      F1CPathNSA,
    iE-Extensions                   ProtocolExtensionContainer { { F1CTransferPath-ExtIEs} } OPTIONAL,
    ...
}

F1CTransferPath-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

F1CPathNRDC ::= ENUMERATED {mcg, scg, both}

F1CTransferPathNRDC ::= SEQUENCE {
    f1CPathNRDC                     F1CPathNRDC,
    iE-Extensions                   ProtocolExtensionContainer { { F1CTransferPathNRDC-ExtIEs} } OPTIONAL,
    ...
}
```

```
F1CTransferPathNRDC-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

F1U-PathFailure ::= ENUMERATED {
	true,
	...
}


F1UTunnelNotEstablished ::= ENUMERATED {
	true,
	...
}

FDD-Info ::= SEQUENCE {
	uL-NRFreqInfo						NRFreqInfo,
	dL-NRFreqInfo						NRFreqInfo,
	uL-Transmission-Bandwidth			Transmission-Bandwidth,
	dL-Transmission-Bandwidth			Transmission-Bandwidth,
	iE-Extensions						ProtocolExtensionContainer { {FDD-Info-ExtIEs} } OPTIONAL,
	...
}

FDD-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-ULCarrierList				CRITICALITY ignore	EXTENSION NRCarrierList				PRESENCE optional }|
	{ ID id-DLCarrierList				CRITICALITY ignore EXTENSION NRCarrierList				PRESENCE optional },
	...
}

FDD-InfoRel16 ::= SEQUENCE {
	uL-FreqInfo							FreqInfoRel16										OPTIONAL,
	sUL-FreqInfo						FreqInfoRel16										OPTIONAL,
	iE-Extensions						ProtocolExtensionContainer { {FDD-InfoRel16-ExtIEs} }	OPTIONAL,
	...
}

FDD-InfoRel16-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

FiveG-ProSeAuthorized ::= SEQUENCE {
	fiveG-proSeDirectDiscovery				FiveG-ProSeDirectDiscovery						OPTIONAL,
	fiveG-proSeDirectCommunication			FiveG-ProSeDirectCommunication					OPTIONAL,
	fiveG-ProSeLayer2UEtoNetworkRelay		FiveG-ProSeLayer2UEtoNetworkRelay				OPTIONAL,
	fiveG-ProSeLayer3UEtoNetworkRelay		FiveG-ProSeLayer3UEtoNetworkRelay				OPTIONAL,
	fiveG-ProSeLayer2RemoteUE				FiveG-ProSeLayer2RemoteUE						OPTIONAL,
	iE-Extensions							ProtocolExtensionContainer { {FiveG-ProSeAuthorized-ExtIEs} } OPTIONAL,
	...
}

FiveG-ProSeAuthorized-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-FiveG-ProSeLayer2Multipath	CRITICALITY ignore	EXTENSION FiveG-ProSeLayer2Multipath	PRESENCE optional	}|
```

```
    { ID id-FiveG-ProSeLayer2UEtoUERelay    CRITICALITY ignore  EXTENSION FiveG-ProSeLayer2UEtoUERelay     PRESENCE optional    }|
    { ID id-FiveG-ProSeLayer2UEtoUERemote   CRITICALITY ignore  EXTENSION FiveG-ProSeLayer2UEtoUERemote    PRESENCE optional    }|
    { ID id-FiveGProSeLayer3MHUEtoNetworkRelay              CRITICALITY ignore  EXTENSION FiveGProSeLayer3MHUEtoNetworkRelay   PRESENCE optional}|
    { ID id-FiveGProSeLayer2MHUEtoNetworkRelay              CRITICALITY ignore  EXTENSION FiveGProSeLayer2MHUEtoNetworkRelay   PRESENCE optional}|
    { ID id-FiveGProSeLayer2MHIntermediateUEtoNetworkRelay  CRITICALITY ignore  EXTENSION FiveGProSeLayer2MHIntermediateUEtoNetworkRelay   PRESENCE 
optional}|
    { ID id-FiveGProSeLayer2MHRemote                        CRITICALITY ignore  EXTENSION FiveGProSeLayer2MHRemote             PRESENCE optional},
    ...
}

FiveG-ProSeDirectDiscovery ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

FiveG-ProSeDirectCommunication ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

FiveG-ProSeLayer2UEtoNetworkRelay ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

FiveG-ProSeLayer3UEtoNetworkRelay ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

FiveG-ProSeLayer2RemoteUE ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

FiveG-ProSeLayer2Multipath ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}


FiveG-ProSeLayer2UEtoUERelay ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

FiveG-ProSeLayer2UEtoUERemote ::= ENUMERATED {
```

```
    authorized,
    not-authorized,
    ...
}

FiveGProSeLayer3MHUEtoNetworkRelay  ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

FiveGProSeLayer2MHUEtoNetworkRelay  ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

FiveGProSeLayer2MHIntermediateUEtoNetworkRelay  ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

FiveGProSeLayer2MHRemote    ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}


FiveQI ::= INTEGER (0..255, ...)

Flows-Mapped-To-DRB-List    ::= SEQUENCE (SIZE(1.. maxnoofQoSFlows)) OF Flows-Mapped-To-DRB-Item

Flows-Mapped-To-DRB-Item    ::= SEQUENCE {
    qoSFlowIdentifier                           QoSFlowIdentifier,
    qoSFlowLevelQoSParameters               QoSFlowLevelQoSParameters,
    iE-Extensions                           ProtocolExtensionContainer { { Flows-Mapped-To-DRB-ItemExtIEs} } OPTIONAL
}

Flows-Mapped-To-DRB-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-QoSFlowMappingIndication     CRITICALITY ignore  EXTENSION QoSFlowMappingIndication      PRESENCE optional}|
    {ID id-TSCTrafficCharacteristics    CRITICALITY ignore  EXTENSION TSCTrafficCharacteristics     PRESENCE optional},
    ...
}

FR1-Bandwidth ::= ENUMERATED {bw5, bw10, bw20, bw40, bw50, bw80, bw100, ..., bw160, bw200, bw15, bw25, bw30, bw60, bw35, bw45, bw70, bw90}

FR2-Bandwidth ::= ENUMERATED {bw50, bw100, bw200, bw400, ..., bw800, bw1600, bw2000, bw600}

FurtherExtendedUEIdentityIndexValue ::= BIT STRING (SIZE(20))

FreqBandNrItem ::= SEQUENCE {
    freqBandIndicatorNr         INTEGER (1..1024,...),
```

```
    supportedSULBandList        SEQUENCE (SIZE(0..maxnoofNrCellBands)) OF SupportedSULFreqBandItem,
    iE-Extensions               ProtocolExtensionContainer { {FreqBandNrItem-ExtIEs} } OPTIONAL,
    ...
}

FreqBandNrItem-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

FreqDomainLength ::= CHOICE {
    l839                            L839Info,
    l139                            L139Info,
    choice-extension                ProtocolIE-SingleContainer { {FreqDomainLength-ExtIEs} }
}

FreqDomainLength-ExtIEs F1AP-PROTOCOL-IES ::= {
    { ID id-L571Info    CRITICALITY reject  TYPE L571Info PRESENCE mandatory}|
    { ID id-L1151Info   CRITICALITY reject  TYPE L1151Info PRESENCE mandatory},
    ...
}

FreqInfoRel16 ::=  SEQUENCE {
    nRARFCN                                         INTEGER (0..maxNRARFCN)                             OPTIONAL,
    frequencyShift7p5khz                            FrequencyShift7p5khz                                OPTIONAL,
    carrierList                                     NRCarrierList                                       OPTIONAL,
    iE-Extensions                                   ProtocolExtensionContainer { { FreqInfoRel16-ExtIEs} }   OPTIONAL,
    ...
}

FreqInfoRel16-ExtIEs        F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

FrequencyShift7p5khz ::= ENUMERATED {false, true, ...}

Frequency-Domain-HSNA-Configuration-List ::= SEQUENCE (SIZE(1..maxnoofRBsetsPerCell)) OF Frequency-Domain-HSNA-Configuration-Item

Frequency-Domain-HSNA-Configuration-Item::= SEQUENCE {
    rBSetIndex          INTEGER (0..maxnoofRBsetsPerCell-1, ...),
    frequency-Domain-HSNA-Slot-Configuration-List               Frequency-Domain-HSNA-Slot-Configuration-List,
    iE-Extensions       ProtocolExtensionContainer { { Frequency-Domain-HSNA-Configuration-Item-ExtIEs} } OPTIONAL
}


Frequency-Domain-HSNA-Configuration-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Frequency-Domain-HSNA-Slot-Configuration-List ::= SEQUENCE (SIZE(1..maxnoofHSNASlots)) OF Frequency-Domain-HSNA-Slot-Configuration-Item

Frequency-Domain-HSNA-Slot-Configuration-Item::= SEQUENCE {
    slotIndex               INTEGER (0..5119)       OPTIONAL,
    hSNADownlink            HSNADownlink            OPTIONAL,
    hSNAUplink              HSNAUplink              OPTIONAL,
```

```
    hSNAFlexible            HSNAFlexible            OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { Frequency-Domain-HSNA-Slot-Configuration-Item-ExtIEs } } OPTIONAL
}

Frequency-Domain-HSNA-Slot-Configuration-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

FullConfiguration ::= ENUMERATED {full, ...}

FlowsMappedToSLDRB-List ::= SEQUENCE (SIZE(1.. maxnoofPC5QoSFlows)) OF FlowsMappedToSLDRB-Item

FlowsMappedToSLDRB-Item ::= SEQUENCE {
    pc5QoSFlowIdentifier            PC5QoSFlowIdentifier,
    iE-Extensions                   ProtocolExtensionContainer { {FlowsMappedToSLDRB-Item-ExtIEs} } OPTIONAL,
    ...
}

FlowsMappedToSLDRB-Item-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Future-Coverage-Modification-Notification-Cause ::= ENUMERATED {coverage, cell-edge-capacity, cancel, ...}

Future-Coverage-Modification-Notification ::= SEQUENCE {
    future-coverage-Modification-List           Future-Coverage-Modification-List,
    iE-Extensions                               ProtocolExtensionContainer { { Future-Coverage-Modification-Notification-ExtIEs} }OPTIONAL,
    ...
}

Future-Coverage-Modification-Notification-ExtIEs F1AP-PROTOCOL-EXTENSION ::={
    ...
}

Future-Coverage-Modification-List ::= SEQUENCE (SIZE (1..maxCellingNBDU)) OF Future-Coverage-Modification-Item

Future-Coverage-Modification-Item ::= SEQUENCE {
    nRCGI                                   NRCGI,
    futurecellCoverageState                 FutureCellCoverageState,
    futureSSBCoverageModificationList       FutureSSBCoverageModification-List  OPTIONAL,
    timeforFutureCoverageModification       TimeforFutureCoverageModification,
    futureCoverageModificationCause         Future-Coverage-Modification-Notification-Cause     OPTIONAL,
    iE-Extension            ProtocolExtensionContainer { { Future-Coverage-Modification-Item-ExtIEs} }          OPTIONAL,
    ...
}

Future-Coverage-Modification-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

FutureCellCoverageState ::= INTEGER (0..63, ...)


FutureSSBCoverageModification-List ::= SEQUENCE (SIZE (1..maxnoofSSBAreas)) OF FutureSSBCoverageModification-Item
```

```
FutureSSBCoverageModification-Item::= SEQUENCE {
    sSBIndex                        INTEGER(0..63),
    futureSSBCoverageState          FutureSSBCoverageState,
    iE-Extensions       ProtocolExtensionContainer { { FutureSSBCoverageModification-Item-ExtIEs} }  OPTIONAL,
...
}

FutureSSBCoverageModification-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


FutureSSBCoverageState ::= INTEGER (0..15, ...)

-- G


GBR-QosInformation ::= SEQUENCE {
    e-RAB-MaximumBitrateDL          BitRate,
    e-RAB-MaximumBitrateUL          BitRate,
    e-RAB-GuaranteedBitrateDL       BitRate,
    e-RAB-GuaranteedBitrateUL       BitRate,
    iE-Extensions                   ProtocolExtensionContainer { { GBR-QosInformation-ExtIEs} } OPTIONAL,
    ...
}

GBR-QosInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

GBR-QoSFlowInformation::= SEQUENCE {
    maxFlowBitRateDownlink          BitRate,
    maxFlowBitRateUplink            BitRate,
    guaranteedFlowBitRateDownlink   BitRate,
    guaranteedFlowBitRateUplink     BitRate,
    maxPacketLossRateDownlink       MaxPacketLossRate       OPTIONAL,
    maxPacketLossRateUplink         MaxPacketLossRate       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { GBR-QosFlowInformation-ExtIEs} } OPTIONAL,
    ...
}

GBR-QosFlowInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {   ID id-AlternativeQoSParaSetList CRITICALITY ignore  EXTENSION AlternativeQoSParaSetList PRESENCE optional  }|
    {   ID id-MonitoringRequestonAvailableBitrate   CRITICALITY ignore  EXTENSION MonitoringRequestonAvailableBitrate PRESENCE optional },
    ...
}

CG-Config ::= OCTET STRING

GeographicalCoordinates ::= SEQUENCE {
    tRPPositionDefinitionType   TRPPositionDefinitionType,
    dLPRSResourceCoordinates    DLPRSResourceCoordinates    OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { GeographicalCoordinates-ExtIEs } } OPTIONAL
```

```
}

GeographicalCoordinates-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-ARPLocationInfo     CRITICALITY ignore EXTENSION ARPLocationInformation PRESENCE optional},
    ...
}

GlobalGNB-ID ::= SEQUENCE {
    pLMNIdentity        PLMN-Identity,
    gNB-ID              GNB-ID,
    iE-Extensions       ProtocolExtensionContainer { {GlobalGNB-ID-ExtIEs} } OPTIONAL,
    ...
}

GlobalGNB-ID-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
GNB-ID ::= CHOICE {
    gNB-ID                  BIT STRING (SIZE(22..32)),
    choice-Extensions       ProtocolIE-SingleContainer { {GNB-ID-ExtIEs} }
}

GNB-ID-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}


GNB-CU-MBS-F1AP-ID      ::= INTEGER (0..4294967295)

GNBCUMeasurementID ::= INTEGER (0.. 4095, ...)

GNBDUMeasurementID ::= INTEGER (0.. 4095, ...)

GNB-CUSystemInformation::= SEQUENCE {
    sibtypetobeupdatedlist  SEQUENCE (SIZE(1.. maxnoofSIBTypes)) OF SibtypetobeupdatedListItem,
    iE-Extensions                   ProtocolExtensionContainer { { GNB-CUSystemInformation-ExtIEs} } OPTIONAL,
    ...
}

GNB-CUSystemInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-systemInformationAreaID  CRITICALITY ignore  EXTENSION SystemInformationAreaID PRESENCE optional},
    ...
}

GNB-CU-TNL-Association-Setup-Item::= SEQUENCE {
    tNLAssociationTransportLayerAddress     CP-TransportLayerAddress    ,
    iE-Extensions                           ProtocolExtensionContainer { { GNB-CU-TNL-Association-Setup-Item-ExtIEs} } OPTIONAL
}

GNB-CU-TNL-Association-Setup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

GNB-CU-TNL-Association-Failed-To-Setup-Item ::= SEQUENCE {
```

```
    tNLAssociationTransportLayerAddress     CP-TransportLayerAddress    ,
    cause                                   Cause,
    iE-Extensions                           ProtocolExtensionContainer { { GNB-CU-TNL-Association-Failed-To-Setup-Item-ExtIEs} } OPTIONAL
}

GNB-CU-TNL-Association-Failed-To-Setup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


GNB-CU-TNL-Association-To-Add-Item ::= SEQUENCE {
    tNLAssociationTransportLayerAddress     CP-TransportLayerAddress    ,
    tNLAssociationUsage                     TNLAssociationUsage,
    iE-Extensions                           ProtocolExtensionContainer { { GNB-CU-TNL-Association-To-Add-Item-ExtIEs} } OPTIONAL
}

GNB-CU-TNL-Association-To-Add-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

GNB-CU-TNL-Association-To-Remove-Item::= SEQUENCE {
    tNLAssociationTransportLayerAddress     CP-TransportLayerAddress    ,
    iE-Extensions                           ProtocolExtensionContainer { { GNB-CU-TNL-Association-To-Remove-Item-ExtIEs} } OPTIONAL
}

GNB-CU-TNL-Association-To-Remove-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-TNLAssociationTransportLayerAddressgNBDU CRITICALITY reject  EXTENSION CP-TransportLayerAddress PRESENCE optional},
    ...
}


GNB-CU-TNL-Association-To-Update-Item::= SEQUENCE {
    tNLAssociationTransportLayerAddress     CP-TransportLayerAddress    ,
    tNLAssociationUsage                     TNLAssociationUsage OPTIONAL,
    iE-Extensions                           ProtocolExtensionContainer { { GNB-CU-TNL-Association-To-Update-Item-ExtIEs} } OPTIONAL
}

GNB-CU-TNL-Association-To-Update-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

GNB-CU-UE-F1AP-ID       ::= INTEGER (0..4294967295)

GNB-DU-Cell-Resource-Configuration  ::= SEQUENCE {
    subcarrierSpacing               SubcarrierSpacing,
    dUFTransmissionPeriodicity      DUFTransmissionPeriodicity  OPTIONAL,
    dUF-Slot-Config-List            DUF-Slot-Config-List    OPTIONAL,
    hSNATransmissionPeriodicity     HSNATransmissionPeriodicity,
    hsNSASlotConfigList             HSNASlotConfigList  OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { GNB-DU-Cell-Resource-Configuration-ExtIEs } } OPTIONAL
}

GNB-DU-Cell-Resource-Configuration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-rBSetConfiguration       CRITICALITY reject  EXTENSION       RBSetConfiguration  PRESENCE optional}|
```

```
    {ID id-frequency-Domain-HSNA-Configuration-List  CRITICALITY reject EXTENSION    Frequency-Domain-HSNA-Configuration-List   PRESENCE optional}|
    {ID id-child-IAB-Nodes-NA-Resource-List CRITICALITY reject  EXTENSION Child-IAB-Nodes-NA-Resource-List    PRESENCE optional}|
    {ID id-Parent-IAB-Nodes-NA-Resource-Configuration-List   CRITICALITY reject EXTENSION  Parent-IAB-Nodes-NA-Resource-Configuration-List
PRESENCE optional},
    ...
}

GNB-DU-MBS-F1AP-ID      ::= INTEGER (0..4294967295)


GNB-DU-UE-F1AP-ID       ::= INTEGER (0..4294967295)

GNB-CUorDU-UE-F1AP-ID       ::= INTEGER (0..4294967295)


GNB-DU-ID           ::= INTEGER (0..68719476735)

GNB-CU-Name ::= PrintableString(SIZE(1..150,...))

GNB-DU-Name ::= PrintableString(SIZE(1..150,...))

Extended-GNB-CU-Name     ::= SEQUENCE {
    gNB-CU-NameVisibleString        GNB-CU-NameVisibleString                    OPTIONAL,
    gNB-CU-NameUTF8String           GNB-CU-NameUTF8String                       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { Extended-GNB-CU-Name-ExtIEs } } OPTIONAL,
    ...
}

Extended-GNB-CU-Name-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

GNB-CU-NameVisibleString ::= VisibleString(SIZE(1..150,...))

GNB-CU-NameUTF8String ::= UTF8String(SIZE(1..150,...))

Extended-GNB-DU-Name     ::= SEQUENCE {
    gNB-DU-NameVisibleString        GNB-DU-NameVisibleString                    OPTIONAL,
    gNB-DU-NameUTF8String           GNB-DU-NameUTF8String                       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { Extended-GNB-DU-Name-ExtIEs } } OPTIONAL,
    ...
}

Extended-GNB-DU-Name-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

GNB-DU-NameVisibleString ::= VisibleString(SIZE(1..150,...))

GNB-DU-NameUTF8String ::= UTF8String(SIZE(1..150,...))


GNB-DU-Served-Cells-Item ::= SEQUENCE {
    served-Cell-Information     Served-Cell-Information,
```

```
    gNB-DU-System-Information   GNB-DU-System-Information   OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { GNB-DU-Served-Cells-ItemExtIEs} }  OPTIONAL,
    ...
}

GNB-DU-Served-Cells-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

GNB-DU-System-Information ::= SEQUENCE {
    mIB-message     MIB-message,
    sIB1-message        SIB1-message,
    iE-Extensions                   ProtocolExtensionContainer { { GNB-DU-System-Information-ExtIEs } } OPTIONAL,
    ...
}

GNB-DU-System-Information-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-SIB12-message       CRITICALITY ignore  EXTENSION SIB12-message     PRESENCE optional}|
    { ID id-SIB13-message       CRITICALITY ignore  EXTENSION SIB13-message     PRESENCE optional}|
    { ID id-SIB14-message       CRITICALITY ignore  EXTENSION SIB14-message     PRESENCE optional}|
    { ID id-SIB10-message       CRITICALITY ignore  EXTENSION SIB10-message     PRESENCE optional}|
    { ID id-SIB17-message       CRITICALITY ignore  EXTENSION SIB17-message     PRESENCE optional}|
    { ID id-SIB20-message       CRITICALITY ignore  EXTENSION SIB20-message     PRESENCE optional}|
    { ID id-SIB15-message       CRITICALITY ignore  EXTENSION SIB15-message     PRESENCE optional}|
    { ID id-SIB24-message       CRITICALITY ignore  EXTENSION SIB24-message     PRESENCE optional}|
    { ID id-SIB22-message       CRITICALITY ignore  EXTENSION SIB22-message     PRESENCE optional}|
    { ID id-SIB23-message       CRITICALITY ignore  EXTENSION SIB23-message     PRESENCE optional}|
    { ID id-SIB17bis-message    CRITICALITY ignore  EXTENSION SIB17bis-message  PRESENCE optional},
    ...
}

GNB-DUConfigurationQuery ::= ENUMERATED {true, ...}

GNBDUOverloadInformation ::= ENUMERATED {overloaded, not-overloaded}

GNB-DU-TNL-Association-To-Remove-Item::= SEQUENCE {
    tNLAssociationTransportLayerAddress             CP-TransportLayerAddress    ,
    tNLAssociationTransportLayerAddressgNBCU        CP-TransportLayerAddress        OPTIONAL,
    iE-Extensions                                   ProtocolExtensionContainer { { GNB-DU-TNL-Association-To-Remove-Item-ExtIEs} } OPTIONAL
}

GNB-DU-TNL-Association-To-Remove-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

GNBDUUESliceMaximumBitRateList::= SEQUENCE (SIZE(1.. maxnoofSMBRValues)) OF GNBDUUESliceMaximumBitRateItem

GNBDUUESliceMaximumBitRateItem::= SEQUENCE {
    sNSSAI                      SNSSAI,
    uESliceMaximumBitRateUL     BitRate,
    iE-Extensions               ProtocolExtensionContainer { { GNBDUUESliceMaximumBitRateItem-ExtIEs} } OPTIONAL,
    ...
}
```

```
GNBDUUESliceMaximumBitRateItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

GNB-RxTxTimeDiff ::= SEQUENCE {
    rxTxTimeDiff            GNBRxTxTimeDiffMeas,
    additionalPath-List     AdditionalPath-List     OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { GNB-RxTxTimeDiff-ExtIEs} }  OPTIONAL
}

GNB-RxTxTimeDiff-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-ExtendedAdditionalPathList      CRITICALITY ignore EXTENSION ExtendedAdditionalPathList  PRESENCE optional}|
    { ID id-TRPTEGInformation               CRITICALITY ignore EXTENSION TRPTEGInformation       PRESENCE optional },
    ...
}

GNBRxTxTimeDiffMeas ::= CHOICE {
    k0          INTEGER (0.. 1970049),
    k1          INTEGER (0.. 985025),
    k2          INTEGER (0.. 492513),
    k3          INTEGER (0.. 246257),
    k4          INTEGER (0.. 123129),
    k5          INTEGER (0.. 61565),
    choice-extension        ProtocolIE-SingleContainer { { GNBRxTxTimeDiffMeas-ExtIEs } }
}

GNBRxTxTimeDiffMeas-ExtIEs      F1AP-PROTOCOL-IES ::= {
    {ID id-ReportingGranularitykminus1  CRITICALITY ignore TYPE ReportingGranularitykminus1 PRESENCE mandatory}|
    {ID id-ReportingGranularitykminus2  CRITICALITY ignore TYPE ReportingGranularitykminus2 PRESENCE mandatory }|
    {ID id-ReportingGranularitykminus3  CRITICALITY ignore TYPE ReportingGranularitykminus3 PRESENCE mandatory}|
    {ID id-ReportingGranularitykminus4  CRITICALITY ignore TYPE ReportingGranularitykminus4 PRESENCE mandatory }|
    {ID id-ReportingGranularitykminus5  CRITICALITY ignore TYPE ReportingGranularitykminus5 PRESENCE mandatory}|
    {ID id-ReportingGranularitykminus6  CRITICALITY ignore TYPE ReportingGranularitykminus6 PRESENCE mandatory },
    ...
}

GNBSetID ::= BIT STRING (SIZE(22))

GTP-TEID                ::= OCTET STRING (SIZE (4))

GTPTLAs ::= SEQUENCE (SIZE(1.. maxnoofGTPTLAs)) OF  GTPTLA-Item


GTPTLA-Item ::= SEQUENCE {
    gTPTransportLayerAddress                TransportLayerAddress,
    iE-Extensions   ProtocolExtensionContainer { { GTPTLA-Item-ExtIEs } }           OPTIONAL
}

GTPTLA-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

GTPTunnel               ::= SEQUENCE {
    transportLayerAddress       TransportLayerAddress,
```

```
    gTP-TEID        GTP-TEID,
    iE-Extensions               ProtocolExtensionContainer { { GTPTunnel-ExtIEs } } OPTIONAL,
    ...
}

GTPTunnel-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- H

HandoverPreparationInformation ::= OCTET STRING

HardwareLoadIndicator ::= SEQUENCE {
    dLHardwareLoadIndicator         INTEGER (0..100, ...),
    uLHardwareLoadIndicator         INTEGER (0..100, ...),
    iE-Extensions                   ProtocolExtensionContainer { { HardwareLoadIndicator-ExtIEs } }  OPTIONAL,
    ...
}

HardwareLoadIndicator-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

HSNASlotConfigList ::= SEQUENCE (SIZE(1..maxnoofHSNASlots)) OF HSNASlotConfigItem

HSNASlotConfigItem  ::= SEQUENCE {
    hSNADownlink            HSNADownlink        OPTIONAL,
    hSNAUplink              HSNAUplink          OPTIONAL,
    hSNAFlexible            HSNAFlexible        OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { HSNASlotConfigItem-ExtIEs } } OPTIONAL
}

HSNASlotConfigItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
HSNADownlink ::= ENUMERATED { hard, soft, notavailable }

HSNAFlexible ::= ENUMERATED { hard, soft, notavailable }

HSNAUplink ::= ENUMERATED { hard, soft, notavailable }

HSNATransmissionPeriodicity ::= ENUMERATED { ms0p5, ms0p625, ms1, ms1p25, ms2, ms2p5, ms5, ms10, ms20, ms40, ms80, ms160, ...}

HashedUEIdentityIndexValue ::= BIT STRING (SIZE(13, ...))


-- I

IAB-Barred  ::= ENUMERATED {barred, not-barred, ...}

IABConditionalRRCMessageDeliveryIndication ::= ENUMERATED {true, ...}

IABCongestionIndication ::= SEQUENCE {
```

```
    iAB-Congestion-Indication-List          IAB-Congestion-Indication-List,
    iE-Extensions   ProtocolExtensionContainer { { IAB-Congestion-Indication-List-ExtIEs } } OPTIONAL
}

IAB-Congestion-Indication-List-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

IAB-Congestion-Indication-List ::= SEQUENCE (SIZE(1..maxnoofIABCongInd)) OF IAB-Congestion-Indication-Item

IAB-Congestion-Indication-Item ::= SEQUENCE {
    childNodeIdentifier             BAPAddress,
    bHRLCCHList                     BHRLCCHList     OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { IAB-Congestion-Indication-ItemExtIEs } }  OPTIONAL
}

IAB-Congestion-Indication-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


IAB-Info-IAB-donor-CU ::=   SEQUENCE{
    iAB-STC-Info    IAB-STC-Info    OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { IAB-Info-IAB-donor-CU-ExtIEs } } OPTIONAL
}

IAB-Info-IAB-donor-CU-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

IAB-Info-IAB-DU ::= SEQUENCE{
    multiplexingInfo        MultiplexingInfo    OPTIONAL,
    iAB-STC-Info        IAB-STC-Info    OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { IAB-Info-IAB-DU-ExtIEs } } OPTIONAL
}

IAB-Info-IAB-DU-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

IAB-MT-Cell-List ::= SEQUENCE (SIZE(1..maxnoofServingCells)) OF IAB-MT-Cell-List-Item

IAB-MT-Cell-List-Item ::=   SEQUENCE {
    nRCellIdentity              NRCellIdentity,
    dU-RX-MT-RX                 DU-RX-MT-RX,
    dU-TX-MT-TX                 DU-TX-MT-TX,
    dU-RX-MT-TX                 DU-RX-MT-TX,
    dU-TX-MT-RX                 DU-TX-MT-RX,
    iE-Extensions               ProtocolExtensionContainer { { IAB-MT-Cell-List-Item-ExtIEs } } OPTIONAL
}

IAB-MT-Cell-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-DU-RX-MT-RX-Extend          CRITICALITY ignore  EXTENSION DU-RX-MT-RX-Extend    PRESENCE optional }|
    { ID id-DU-TX-MT-TX-Extend          CRITICALITY ignore  EXTENSION DU-TX-MT-TX-Extend    PRESENCE optional }|
```

```
	{ ID id-DU-RX-MT-TX-Extend			CRITICALITY ignore	EXTENSION DU-RX-MT-TX-Extend	PRESENCE optional }|
	{ ID id-DU-TX-MT-RX-Extend			CRITICALITY ignore	EXTENSION DU-TX-MT-RX-Extend	PRESENCE optional },
	...
}



IAB-MT-Cell-NA-Resource-Configuration-Mode-Info ::= CHOICE {
	fDD		IAB-MT-Cell-NA-Resource-Configuration-FDD-Info,
	tDD		IAB-MT-Cell-NA-Resource-Configuration-TDD-Info,
	choice-extension			ProtocolIE-SingleContainer { { IAB-MT-Cell-NA-Resource-Configuration-Mode-Info-ExtIEs} }
}

IAB-MT-Cell-NA-Resource-Configuration-Mode-Info-ExtIEs F1AP-PROTOCOL-IES ::= {
	...
}

IAB-MT-Cell-NA-Resource-Configuration-FDD-Info ::= SEQUENCE {
	gNB-DU-Cell-NA-Resource-Configuration-FDD-UL				GNB-DU-Cell-Resource-Configuration,
	gNB-DU-Cell-NA-Resource-Configuration-FDD-DL				GNB-DU-Cell-Resource-Configuration,
	uL-FreqInfo						NRFreqInfo					OPTIONAL,
	uL-Transmission-Bandwidth			Transmission-Bandwidth		OPTIONAL,
	uL-NR-Carrier-List				NRCarrierList				OPTIONAL,
	dL-FreqInfo						NRFreqInfo					OPTIONAL,
	dL-Transmission-Bandwidth			Transmission-Bandwidth		OPTIONAL,
	dL-NR-Carrier-List				NRCarrierList				OPTIONAL,
	iE-Extensions					ProtocolExtensionContainer { {IAB-MT-Cell-NA-Resource-Configuration-FDD-Info-ExtIEs} } OPTIONAL,
	...
}

IAB-MT-Cell-NA-Resource-Configuration-FDD-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

IAB-MT-Cell-NA-Resource-Configuration-TDD-Info ::= SEQUENCE {
	gNB-DU-Cell-NA-Resourc-Configuration-TDD				GNB-DU-Cell-Resource-Configuration,
	nRFreqInfo						NRFreqInfo		OPTIONAL,
	transmission-Bandwidth			Transmission-Bandwidth		OPTIONAL,
	nR-Carrier-List					NRCarrierList	OPTIONAL,
	iE-Extensions					ProtocolExtensionContainer { {IAB-MT-Cell-NA-Resource-Configuration-TDD-Info-ExtIEs} } OPTIONAL,
	...
}

IAB-MT-Cell-NA-Resource-Configuration-TDD-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

IAB-STC-Info	::= SEQUENCE{
	iAB-STC-Info-List	IAB-STC-Info-List,
	iE-Extensions		ProtocolExtensionContainer { { IAB-STC-Info-ExtIEs } } OPTIONAL
}

IAB-STC-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
```

```
}

IAB-STC-Info-List ::=   SEQUENCE (SIZE(1..maxnoofIABSTCInfo)) OF IAB-STC-Info-Item

IAB-STC-Info-Item::=    SEQUENCE {
    sSB-freqInfo                        SSB-freqInfo,
    sSB-subcarrierSpacing               SSB-subcarrierSpacing,
    sSB-transmissionPeriodicity         SSB-transmissionPeriodicity,
    sSB-transmissionTimingOffset        SSB-transmissionTimingOffset,
    sSB-transmissionBitmap              SSB-transmissionBitmap,
    iE-Extensions       ProtocolExtensionContainer { { IAB-STC-Info-Item-ExtIEs } } OPTIONAL
}

IAB-STC-Info-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

IAB-Allocated-TNL-Address-Item  ::= SEQUENCE {
    iABTNLAddress               IABTNLAddress,
    iABTNLAddressUsage          IABTNLAddressUsage      OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { IAB-Allocated-TNL-Address-Item-ExtIEs } } OPTIONAL
}

IAB-Allocated-TNL-Address-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

IAB-DU-Cell-Resource-Configuration-Mode-Info    ::= CHOICE {
    fDD     IAB-DU-Cell-Resource-Configuration-FDD-Info,
    tDD     IAB-DU-Cell-Resource-Configuration-TDD-Info,
    choice-extension            ProtocolIE-SingleContainer { { IAB-DU-Cell-Resource-Configuration-Mode-Info-ExtIEs} }
}

IAB-DU-Cell-Resource-Configuration-Mode-Info-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

IAB-DU-Cell-Resource-Configuration-FDD-Info ::= SEQUENCE {
    gNB-DU-Cell-Resource-Configuration-FDD-UL               GNB-DU-Cell-Resource-Configuration,
    gNB-DU-Cell-Resource-Configuration-FDD-DL               GNB-DU-Cell-Resource-Configuration,
    iE-Extensions                   ProtocolExtensionContainer { {IAB-DU-Cell-Resource-Configuration-FDD-Info-ExtIEs} } OPTIONAL,
    ...
}

IAB-DU-Cell-Resource-Configuration-FDD-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-uL-FreqInfo                  CRITICALITY reject  EXTENSION   NRFreqInfo                PRESENCE optional}|
    {ID id-uL-Transmission-Bandwidth    CRITICALITY reject  EXTENSION   Transmission-Bandwidth    PRESENCE optional}|
    {ID id-uL-NR-Carrier-List           CRITICALITY reject  EXTENSION   NRCarrierList             PRESENCE optional}|
    {ID id-dL-FreqInfo                  CRITICALITY reject  EXTENSION   NRFreqInfo                PRESENCE optional}|
    {ID id-dL-Transmission-Bandwidth    CRITICALITY reject  EXTENSION   Transmission-Bandwidth    PRESENCE optional}|
    {ID id-dL-NR-Carrier-List           CRITICALITY reject  EXTENSION   NRCarrierList             PRESENCE optional},
    ...
}
```

```
IAB-DU-Cell-Resource-Configuration-TDD-Info ::= SEQUENCE {
    gNB-DU-Cell-Resourc-Configuration-TDD               GNB-DU-Cell-Resource-Configuration,
    iE-Extensions                   ProtocolExtensionContainer { {IAB-DU-Cell-Resource-Configuration-TDD-Info-ExtIEs} } OPTIONAL,
    ...
}

IAB-DU-Cell-Resource-Configuration-TDD-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-nRFreqInfo               CRITICALITY reject  EXTENSION  NRFreqInfo               PRESENCE optional}|
    {ID id-transmission-Bandwidth   CRITICALITY reject  EXTENSION  Transmission-Bandwidth   PRESENCE optional}|
    {ID id-nR-Carrier-List          CRITICALITY reject  EXTENSION  NRCarrierList            PRESENCE optional},
    ...
}

IABIPv6RequestType   ::= CHOICE {
    iPv6Address                     IABTNLAddressesRequested,
    iPv6Prefix                      IABTNLAddressesRequested,
    choice-extension                ProtocolIE-SingleContainer { { IABIPv6RequestType-ExtIEs} }
}

IABIPv6RequestType-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

IABTNLAddress ::= CHOICE {
    iPv4Address                     BIT STRING (SIZE(32)),
    iPv6Address                     BIT STRING (SIZE(128)),
    iPv6Prefix                      BIT STRING (SIZE(64)),
    choice-extension                ProtocolIE-SingleContainer { { IABTNLAddress-ExtIEs} }
}

IABTNLAddress-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

IABTNLAddressesRequested ::= SEQUENCE {
    tNLAddressesOrPrefixesRequestedAllTraffic   INTEGER (1..256)    OPTIONAL,
    tNLAddressesOrPrefixesRequestedF1-C         INTEGER (1..256)    OPTIONAL,
    tNLAddressesOrPrefixesRequestedF1-U         INTEGER (1..256)    OPTIONAL,
    tNLAddressesOrPrefixesRequestedNoNF1        INTEGER (1..256)    OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { IABTNLAddressesRequested-ExtIEs } } OPTIONAL
}

IABTNLAddressesRequested-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

IAB-TNL-Addresses-To-Remove-Item ::= SEQUENCE {
    iABTNLAddress           IABTNLAddress,
    iE-Extensions           ProtocolExtensionContainer { { IAB-TNL-Addresses-To-Remove-Item-ExtIEs} } OPTIONAL
}

IAB-TNL-Addresses-To-Remove-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
IAB-TNL-Addresses-Exception ::=     SEQUENCE {
    iABTNLAddressList           IABTNLAddressList,
    iE-Extensions               ProtocolExtensionContainer { { IAB-TNL-Addresses-Exception-ExtIEs} } OPTIONAL
}

IAB-TNL-Addresses-Exception-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

IABTNLAddressList ::= SEQUENCE (SIZE(1.. maxnoofTLAsIAB)) OF IABTNLAddress-Item

IABTNLAddress-Item ::= SEQUENCE {
    iABTNLAddress       IABTNLAddress   ,
    iE-Extensions   ProtocolExtensionContainer { { IABTNLAddress-ItemExtIEs } } OPTIONAL
}

IABTNLAddress-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

IABTNLAddressUsage ::= ENUMERATED {
    f1-c,
    f1-u,
    non-f1,
    ...
}


IABv4AddressesRequested ::= SEQUENCE {
    iABv4AddressesRequested         IABTNLAddressesRequested,
    iE-Extensions       ProtocolExtensionContainer { { IABv4AddressesRequested-ExtIEs} } OPTIONAL
}

IABv4AddressesRequested-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Mobile-IAB-MTUserLocationInformation ::= SEQUENCE {
    nRCGI                           NRCGI,
    tAI                             TAI,
    iE-Extensions                   ProtocolExtensionContainer { { Mobile-IAB-MTUserLocationInformation-ExtIEs} }  OPTIONAL
}

Mobile-IAB-MTUserLocationInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


ImplicitFormat  ::= SEQUENCE    {
    dUFSlotformatIndex          DUFSlotformatIndex,
    iE-Extensions       ProtocolExtensionContainer { { ImplicitFormat-ExtIEs } } OPTIONAL
}
```

```
ImplicitFormat-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

IgnorePRACHConfiguration::= ENUMERATED { true,...}

IgnoreResourceCoordinationContainer ::= ENUMERATED { yes,...}
InactivityMonitoringRequest ::= ENUMERATED { true,...}
InactivityMonitoringResponse ::= ENUMERATED { not-supported,...}

Indication-of-Bitrate-Adaptation ::= ENUMERATED {uplink, ...}

IndirectPathAddition ::= SEQUENCE {
    targetRelayUEID         BIT STRING(SIZE(24)),
    remoteUELocalID         RemoteUELocalID,
    iE-Extensions           ProtocolExtensionContainer { { IndirectPathAddition-ExtIEs } }      OPTIONAL,
    ...
}

IndirectPathAddition-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
InterfacesToTrace ::= BIT STRING (SIZE(8))

IntendedTDD-DL-ULConfig ::= SEQUENCE {
    nRSCS                       ENUMERATED { scs15, scs30, scs60, scs120,..., scs480, scs960},
    nRCP                        ENUMERATED { normal, extended,...},
    nRDLULTxPeriodicity         ENUMERATED { ms0p5, ms0p625, ms1, ms1p25, ms2, ms2p5, ms3, ms4, ms5, ms10, ms20, ms40, ms60, ms80, ms100, ms120,
ms140, ms160, ...},
    slot-Configuration-List     Slot-Configuration-List,
    iE-Extensions                       ProtocolExtensionContainer { {IntendedTDD-DL-ULConfig-ExtIEs} } OPTIONAL
}

InterFrequencyConfig-NoGap ::= ENUMERATED {
    true,
    ...
}

InterSNLTMMCGInformation ::= SEQUENCE {
    interSNLTM-trigger              LTM-trigger,
    pscellid                        NRCGI                                                       OPTIONAL,
    -- The above IE shall be present if the interSNLTM-trigger IE is set to "LTM-preparation" or "LTM-executed"
    candidatePSCellsToCancel        PSCellList                                                  OPTIONAL,
    -- The above IE shall be present if the interSNLTM-trigger IE is set to "LTM-cancel"
    iE-Extensions                   ProtocolExtensionContainer { {InterSNLTMMCGInformation-ExtIEs} } OPTIONAL,
    ...
}

InterSNLTMMCGInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

LTM-trigger ::= ENUMERATED {
    ltm-preparation,
```

```
    ltm-executed,
    ltm-cancel,
    ...
}

IngressNonF1terminatingTopologyIndicator ::= ENUMERATED {true, ...}

IntendedTDD-DL-ULConfig-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

IndicationMCInactiveReception ::= ENUMERATED {true, ...}

LTMResetInformation ::= SEQUENCE {
    servingCellL2ResetConfiguration                     OCTET STRING        OPTIONAL,
    lTML2ResetConfigurationList                         LTML2ResetConfigurationList     OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { LTMResetInformation-ItemExtIEs} } OPTIONAL,
    ...
}

LTMResetInformation-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

LTML2ResetConfigurationList ::= SEQUENCE (SIZE(1.. maxnoofLTMCells)) OF LTML2ResetConfiguration-Item

LTML2ResetConfiguration-Item ::= SEQUENCE {
    cellID                      NRCGI,
    ltmL2ResetConfiguration     OCTET STRING,
    iE-Extensions           ProtocolExtensionContainer { { LTML2ResetConfiguration-ItemExtIEs } } OPTIONAL
}

LTML2ResetConfiguration-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}



IPHeaderInformation ::= SEQUENCE {
    destinationIABTNLAddress            IABTNLAddress,
    dsInformationList                   DSInformationList   OPTIONAL,
    iPv6FlowLabel                       BIT STRING (SIZE (20))  OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { IPHeaderInformation-ItemExtIEs} } OPTIONAL,
    ...
}

IPHeaderInformation-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

IPtolayer2TrafficMappingInfo ::= SEQUENCE {
    iPtolayer2TrafficMappingInfoToAdd                   IPtolayer2TrafficMappingInfoList        OPTIONAL,
    iPtolayer2TrafficMappingInfoToRemove                MappingInformationtoRemove              OPTIONAL,
    iE-Extensions                                       ProtocolExtensionContainer { { IPtolayer2TrafficMappingInfo-ItemExtIEs} } OPTIONAL,
```

```
	...
}

IPtolayer2TrafficMappingInfoList ::= SEQUENCE (SIZE(1..maxnoofMappingEntries)) OF IPtolayer2TrafficMappingInfo-Item

IPtolayer2TrafficMappingInfo-Item ::= SEQUENCE {
	mappingInformationIndex		MappingInformationIndex,
	iPHeaderInformation			IPHeaderInformation,
	bHInfo						BHInfo, iE-Extensions				ProtocolExtensionContainer { { IPtolayer2TrafficMappingInfo-ItemExtIEs} }
OPTIONAL,
	...
}

IPtolayer2TrafficMappingInfo-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

-- J

JointorDLTCIStateID  ::= OCTET STRING



-- K

-- L

LTEA2XServicesAuthorized ::= SEQUENCE {
	aerialUE			AerialUE													OPTIONAL,
	aerialControllerUE		AerialControllerUE												OPTIONAL,
	iE-Extensions		ProtocolExtensionContainer { {LTEA2XServicesAuthorized-ExtIEs} }		OPTIONAL
}

LTEA2XServicesAuthorized-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

L139Info ::= SEQUENCE {
	prachSCS						ENUMERATED {scs15, scs30, scs60, scs120, ..., scs480, scs960},
	rootSequenceIndex			INTEGER (0..137)								OPTIONAL,
	iE-Extension				ProtocolExtensionContainer { {L139Info-ExtIEs} }		OPTIONAL,
	...
}

L139Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

L839Info ::= SEQUENCE {
	rootSequenceIndex			INTEGER (0..837),
	restrictedSetConfig			ENUMERATED {unrestrictedSet, restrictedSetTypeA,
											restrictedSetTypeB, ...},
	iE-Extension		ProtocolExtensionContainer { {L839Info-ExtIEs} }		OPTIONAL,
	...
```

```
}

L839Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

L571Info ::= SEQUENCE {
    prachSCSForL571             ENUMERATED { scs30, scs120, ... , scs480},
    rootSequenceIndex           INTEGER (0..569),
    iE-Extension                ProtocolExtensionContainer { {L571Info-ExtIEs} }        OPTIONAL,
    ...
}

L571Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

L1151Info ::= SEQUENCE {
    prachSCSForL1151                ENUMERATED { scs15, scs120,...},
    rootSequenceIndex               INTEGER (0..1149),
    iE-Extension                    ProtocolExtensionContainer { {L1151Info-ExtIEs} }       OPTIONAL,
    ...
}

L1151Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


LastUsedCellIndication ::= ENUMERATED {true, ...}

LastVisitedLTMCells ::= SEQUENCE {
    lastVisitedLTMCellList          LastVisitedLTMCellList          ,
    mobilityissue               MobilityIssue           OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { LastVisitedCells-ExtIEs } }      OPTIONAL,
    ...
}

LastVisitedCells-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}



LastVisitedLTMCellList ::= SEQUENCE (SIZE(1.. maxnoofCellsinUEHistoryInfo)) OF LastVisitedLTMCellItem

LastVisitedLTMCellItem ::= SEQUENCE {
    cellID                                  NRCGI,
    timeUEStayedInCell                      INTEGER(0..40950),
    iE-Extensions           ProtocolExtensionContainer { { LastVisitedLTMCellItem-ExtIEs } }      OPTIONAL,
    ...
}

LastVisitedLTMCellItem-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

LCID ::= INTEGER (1..32, ...)


LCS-to-GCS-Translation::= SEQUENCE {
    alpha               INTEGER (0..3599),
    beta                INTEGER (0..3599),
    gamma               INTEGER (0..3599),
    iE-Extensions       ProtocolExtensionContainer { { LCS-to-GCS-Translation-ExtIEs} } OPTIONAL,
    ...
}

LCS-to-GCS-Translation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

LCStoGCSTranslationList ::= SEQUENCE (SIZE (1.. maxnooflcs-gcs-translation)) OF LCStoGCSTranslation

LCStoGCSTranslation ::= SEQUENCE {
    alpha           INTEGER (0..359),
    alpha-fine      INTEGER (0..9)      OPTIONAL,
    beta            INTEGER (0..359),
    beta-fine       INTEGER (0..9)      OPTIONAL,
    gamma           INTEGER (0..359),
    gamma-fine      INTEGER (0..9)      OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {LCStoGCSTranslation-ExtIEs} } OPTIONAL
}

LCStoGCSTranslation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

LMF-MeasurementID ::= INTEGER (1.. 65536, ...)

LMF-UE-MeasurementID ::= INTEGER (1.. 256, ...)

LocationDependentMBSF1UInformation ::= SEQUENCE (SIZE(1..maxnoofMBSAreaSessionIDs)) OF LocationDependentMBSF1UInformation-Item
LocationDependentMBSF1UInformation-Item ::= SEQUENCE {
    mbsAreaSession-ID               MBS-Area-Session-ID,
    mbs-f1u-info-at-CU              UPTransportLayerInformation,
    iE-Extensions                   ProtocolExtensionContainer  { { LocationDependentMBSF1UInformation-Item-ExtIEs } } OPTIONAL,
    ...
}

LocationDependentMBSF1UInformation-Item-ExtIEs      F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-F1UTunnelNotEstablished CRITICALITY     ignore      EXTENSION       F1UTunnelNotEstablished   PRESENCE    optional },
    ...
}

LocationMeasurementInformation ::= OCTET STRING

LocationUncertainty ::= SEQUENCE {
```

```
    horizontalUncertainty       INTEGER (0..255),
    horizontalConfidence        INTEGER (0..100),
    verticalUncertainty         INTEGER (0..255),
    verticalConfidence          INTEGER (0..100),
    iE-Extensions               ProtocolExtensionContainer { { LocationUncertainty-ExtIEs} } OPTIONAL
}

LocationUncertainty-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

LongDRXCycleLength ::=  ENUMERATED
{ms10, ms20, ms32, ms40, ms60, ms64, ms70, ms80, ms128, ms160, ms256, ms320, ms512, ms640, ms1024, ms1280, ms2048, ms2560, ms5120, ms10240, ...}

LongNonIntegerDRXCycleLength ::=    ENUMERATED
{ ms1001over240, ms25over6, ms25over3, ms1001over120, ms100over9, ms25over2, ms40over3, ms125over9, ms50over3, ms1001over60, ms125over6,
ms200over9, ms250over9, ms100over3, ms1001over30, ms75over2, ms125over3, ms1001over24, ms200over3, ms1001over15, ms250over3, ms1001over12,
ms400over3, ...}

LowerLayerPresenceStatusChange ::= ENUMERATED {
    suspend-lower-layers,
    resume-lower-layers,
    ...

}

LoS-NLoSIndicatorHard ::= ENUMERATED {nLoS, loS}

LoS-NLoSIndicatorSoft ::= INTEGER (0..10)

LoS-NLoSInformation ::= CHOICE {
    loS-NLoSIndicatorSoft       LoS-NLoSIndicatorSoft,
    loS-NLoSIndicatorHard       LoS-NLoSIndicatorHard,
    choice-Extension            ProtocolIE-SingleContainer {{ LoS-NLoSInformation-ExtIEs}}
}

LoS-NLoSInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

LPWUSSupportedBandInfo ::= SEQUENCE {
    minimumTimeGapIndication            ENUMERATED {cap1, cap2, cap3, ...}  ,
    oOKIndication                       ENUMERATED {supported, ...}     OPTIONAL,
    oFDMIndication                      ENUMERATED {supported, ...}     OPTIONAL,
    lP-SSIndication                     ENUMERATED {supported, ...}     OPTIONAL,
    iE-Extension                ProtocolExtensionContainer { { LPWUSSupportedBandInfo-ExtIEs} }      OPTIONAL,
    ...
}

LPWUSSupportedBandInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
LPWUSPSAssistanceInfo ::= SEQUENCE {
    cNSubgroupID-LP-WUS     CNSubgroupID-LP-WUS,
    iE-Extensions           ProtocolExtensionContainer { { LPWUSPSAssistanceInfo-ExtIEs } } OPTIONAL
}

LPWUSPSAssistanceInfo-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

LPWUSSubgroupingSupportIndication ::= ENUMERATED {true, ...}

LTEUESidelinkAggregateMaximumBitrate ::= SEQUENCE {
    uELTESidelinkAggregateMaximumBitrate        BitRate,
    iE-Extensions                   ProtocolExtensionContainer { {LTEUESidelinkAggregateMaximumBitrate-ExtIEs} } OPTIONAL
}

LTEUESidelinkAggregateMaximumBitrate-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

LTEV2XServicesAuthorized ::= SEQUENCE {
    vehicleUE           VehicleUE                                                     OPTIONAL,
    pedestrianUE        PedestrianUE                                                  OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {LTEV2XServicesAuthorized-ExtIEs} }        OPTIONAL
}

LTEV2XServicesAuthorized-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


LTMCells-ToBeReleased-List ::= SEQUENCE (SIZE(1..maxnoofLTMCells)) OF  LTMCells-ToBeReleased-Item



LTMCells-ToBeReleased-Item ::= SEQUENCE {
    nRCGI               NRCGI,
    iE-Extensions       ProtocolExtensionContainer { { LTMCells-ToBeReleased-ItemExtIEs } } OPTIONAL,
    ...
}

LTMCells-ToBeReleased-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

LTMInformation-Setup ::= SEQUENCE {
    lTMIndicator                        LTMIndicator,
    referenceConfiguration              ReferenceConfiguration                              OPTIONAL,
    cSIResourceConfiguration            CSIResourceConfiguration                            OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { LTMInformation-Setup-ExtIEs} }   OPTIONAL,
    ...
}

LTMInformation-Setup-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
```

```
  { ID id-RequestforCSI-RSResourceConfigforL1Measure CRITICALITY reject  EXTENSION       RequestforCSI-RSResourceConfig    PRESENCE    optional } |
  { ID id-RequestforL1ExecutionCondition     CRITICALITY     reject      EXTENSION       RequestedforL1ExecutionCondition     PRESENCE    optional },
    ...
}

LTMConfigurationIDMappingList   ::= SEQUENCE (SIZE(1..maxnoofLTMCells)) OF LTMConfigurationIDMapping-Item

LTMConfigurationIDMapping-Item::= SEQUENCE{
    lTMCellID           NRCGI,
    lTMConfigurationID  LTMConfigurationID,
    iE-Extensions       ProtocolExtensionContainer {{ LTMConfigurationIDMapping-Item-ExtIEs}}    OPTIONAL
}

LTMConfigurationIDMapping-Item-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


LTMInformation-Modify   ::= SEQUENCE {
    lTMIndicator                        LTMIndicator,
    referenceConfiguration              ReferenceConfiguration                                OPTIONAL,
    cSIResourceConfiguration            CSIResourceConfiguration                              OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { LTMInformation-Modify-ExtIEs} }  OPTIONAL,
    ...
}

LTMInformation-Modify-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
{ ID id-RequestforCSI-RSResourceConfigforL1Measure  CRITICALITY reject  EXTENSION       RequestforCSI-RSResourceConfig    PRESENCE    optional }|
{ ID id-RequestforL1ExecutionCondition  CRITICALITY     reject      EXTENSION       RequestedforL1ExecutionCondition     PRESENCE    optional },
    ...
}

RequestforCSI-RSResourceConfig ::= ENUMERATED {true, ...}

RequestedforL1ExecutionCondition ::= SEQUENCE (SIZE(1.. maxnoofLTMCells)) OF RequestedforL1ExecutionConditionCandidateCellList-Item
RequestedforL1ExecutionConditionCandidateCellList-Item ::= SEQUENCE {
    candidateCellID         NRCGI,
    iE-Extensions           ProtocolExtensionContainer { { RequestedforL1ExecutionConditionCandidateCellList-ExtIEs } }  OPTIONAL,
    ...
}

RequestedforL1ExecutionConditionCandidateCellList-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


LTMIndicator ::= ENUMERATED {true, ..., c-ltm }

CompleteCandidateConfigurationIndicator ::= ENUMERATED {complete, ...}

LTMConfigurationID ::=  INTEGER (1..8)
ReferenceConfigurationInformation ::= OCTET STRING

LTMConfiguration    ::= SEQUENCE {
```

```
    sSBInformation                  SSBInformation,
    referenceConfigurationInformation   ReferenceConfigurationInformation           OPTIONAL,
    completeCandidateConfigurationIndicator     CompleteCandidateConfigurationIndicator             OPTIONAL,
    lTMCFRAResourceConfig               LTMCFRAResourceConfig                           OPTIONAL,
    lTMCFRAResourceConfigSUL            LTMCFRAResourceConfig                           OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { LTMConfiguration-ExtIEs } }  OPTIONAL,
    ...
}

LTMConfiguration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-TCIStatesConfigurationsList CRITICALITY     reject      EXTENSION       TCIStatesConfigurationsList     PRESENCE    optional }|
    { ID id-L1ExecutionConditionList                    CRITICALITY ignore  EXTENSION L1ExecutionConditionList  PRESENCE optional }|
    { ID id-CSI-RSResourceConfigforL1Measure            CRITICALITY ignore  EXTENSION CSI-RSResourceConfig      PRESENCE optional }|
    { ID id-CSI-RSResourceConfigforCSIAcquisition       CRITICALITY ignore  EXTENSION CSI-RSResourceConfig      PRESENCE optional }|
    { ID id-CSIReportConfigforCSIAcquisition                CRITICALITY ignore  EXTENSION CSIReportConfig          PRESENCE optional },
    ...
}

LTMCellSwitchInformation    ::= SEQUENCE {
    jointorDLTCIStateID         JointorDLTCIStateID,
    uLTCIStateID                    ULTCIStateID                OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { LTMCellSwitchInformation-ExtIEs } }  OPTIONAL,
    ...
}

LTMCellSwitchInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

LTMgNB-DU-IDsList   ::= SEQUENCE (SIZE(1..maxnoofLTMgNB-DUs)) OF LTMgNB-DU-IDs-Item

LTMgNB-DU-IDs-Item  ::= SEQUENCE{
    lTMgNB-DU-ID            GNB-DU-ID,
    iE-Extensions           ProtocolExtensionContainer {{ LTMgNB-DU-IDs-Item-ExtIEs}}       OPTIONAL
}

LTMgNB-DU-IDs-Item-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-LTMgNB-ID       CRITICALITY reject  EXTENSION GlobalGNB-ID  PRESENCE optional },
    ...
}


LTMgNB-DU-IDs-PreambleIndexList ::= SEQUENCE (SIZE(1..maxnoofLTMgNB-DUs)) OF LTMgNB-DU-IDs-PreambleIndex-Item

LTMgNB-DU-IDs-PreambleIndex-Item    ::= SEQUENCE{
    lTMgNB-DU-ID            GNB-DU-ID,
    preambleIndexList       PreambleIndexList                                                   OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer {{ LTMgNB-DU-IDs-PreambleIndex-Item-ExtIEs}}    OPTIONAL
}

LTMgNB-DU-IDs-PreambleIndex-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-LTMgNB-ID       CRITICALITY reject  EXTENSION GlobalGNB-ID  PRESENCE optional },
    ...
}
```

```
LTMCFRAResourceConfig-List ::= SEQUENCE (SIZE (1.. maxnoofLTMCells)) OF LTMCFRAResourceConfig-Item

LTMCFRAResourceConfig-Item ::= SEQUENCE {
    nRCGI               NRCGI,
    lTMCFRAResourceConfig           LTMCFRAResourceConfig                           OPTIONAL,
    lTMCFRAResourceConfigSUL        LTMCFRAResourceConfig                           OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { LTMCFRAResourceConfig-Item-ExtIEs } }   OPTIONAL,
    ...
}

LTMCFRAResourceConfig-Item-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

LTMCFRAResourceConfig ::= OCTET STRING

LTMTCIStatesConfigurationsList ::= SEQUENCE (SIZE (1.. maxnoofLTMCells)) OF LTMTCIStatesConfigurations-Item

LTMTCIStatesConfigurations-Item::= SEQUENCE {
    nRCGI                           NRCGI,
    tCIStatesConfigurationsList     TCIStatesConfigurationsList,
    iE-Extensions                   ProtocolExtensionContainer { { LTMTCIStatesConfigurations-Item-ExtIEs } } OPTIONAL,
    ...
}

LTMTCIStatesConfigurations-Item-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

L1ExecutionConditionList    ::= SEQUENCE (SIZE(1..maxnoofLTMCells)) OF L1ExecutionCondition-Item

L1ExecutionCondition-Item   ::= SEQUENCE{
    ltmCellID               NRCGI,
    executionCondition      OCTET STRING,
    iE-Extensions           ProtocolExtensionContainer {{ L1ExecutionCondition-Item-ExtIEs}}      OPTIONAL,
    ...
}

L1ExecutionCondition-Item-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


SecurityChangeCandidateCellInfoList ::= SEQUENCE (SIZE(1..maxnoofLTMCells)) OF SecurityChangeCandidateCellInfo-Item

SecurityChangeCandidateCellInfo-Item ::=    SEQUENCE{
    cellID                              NRCGI,
    noSecurityChangeID                  OCTET STRING,
    iE-Extensions                       ProtocolExtensionContainer { { SecurityChangeCandidateCellInfo-Item-ExtIEs} } OPTIONAL
}

SecurityChangeCandidateCellInfo-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}


LTMInformationSCGAdd    ::= SEQUENCE {
    lTMwSCGIndicator    LTMwSCGIndicator,
    iE-Extensions       ProtocolExtensionContainer { { LTMInformationSCGAdd-ExtIEs} }   OPTIONAL,
    ...
}

LTMInformationSCGAdd-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

LTMInformationSCGMod    ::= SEQUENCE {
    lTMwSCGIndicator    LTMwSCGIndicator,
    iE-Extensions       ProtocolExtensionContainer { { LTMInformationSCGMod-ExtIEs} }   OPTIONAL,
    ...
}

LTMInformationSCGMod-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


LTMwSCGIndicator ::= ENUMERATED {true, ...}

LTMSecurityInformation  ::= SEQUENCE{
    nextHopChainingCount    INTEGER (0..7),
    servingCellNoSecurityChangeID       OCTET STRING    OPTIONAL,
    securityChangeCandidateCellInfoList SecurityChangeCandidateCellInfoList OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer {{ LTMSecurityInformation-ExtIEs}}       OPTIONAL,
    ...
}

LTMSecurityInformation-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
LTMResidualTAInfoList   ::= SEQUENCE (SIZE(1..maxnoofTAs)) OF LTMResidualTAInfo-Item
LTMResidualTAInfo-Item ::=  SEQUENCE {
    tAValue                 TAValue,
    tATRemainingTime            INTEGER (0..10240),
    tagIDPointer                TagIDPointer        OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { LTMResidualTAInfo-Item-ExtIEs} }   OPTIONAL,
    ...

}

LTMResidualTAInfo-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


LTMConfig-to-Modify-List ::= SEQUENCE (SIZE(1..maxnoofLTMCells)) OF LTMConfig-to-Modify-Item
```

```
LTMConfig-to-Modify-Item ::= SEQUENCE {
    candidateCellID                           NRCGI,
    tCIStatesConfigurationsList               TCIStatesConfigurationsList                OPTIONAL,
    cSI-RSResourceConfigforL1Measurement      CSI-RSResourceConfig                       OPTIONAL,
    cSI-RSResourceConfigforCSIAcquisition     CSI-RSResourceConfig                       OPTIONAL,
    completeCandidateConfigurationIndicator   CompleteCandidateConfigurationIndicator OPTIONAL,
    iE-Extensions                             ProtocolExtensionContainer { { LTMConfig-to-Modify-Item-ExtIEs} } OPTIONAL,
    ...
}

LTMConfig-to-Modify-Item-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- M

MappingInformationIndex ::= BIT STRING (SIZE (26))

MappingInformationtoRemove  ::= SEQUENCE (SIZE(1..maxnoofMappingEntries)) OF MappingInformationIndex

MaskedIMEISV ::=    BIT STRING (SIZE (64))

MaxDataBurstVolume  ::= INTEGER (0..4095, ..., 4096.. 2000000)
MaxPacketLossRate ::= INTEGER (0..1000)

MBS-Broadcast-NeighbourCellList ::= OCTET STRING

MBS-Flows-Mapped-To-MRB-List    ::= SEQUENCE (SIZE(1.. maxnoofMBSQoSFlows)) OF MBS-Flows-Mapped-To-MRB-Item

MBS-Flows-Mapped-To-MRB-Item    ::= SEQUENCE {
    mBS-QoSFlowIdentifier                     QoSFlowIdentifier,
    mbs-QoSFlowLevelQoSParameters             QoSFlowLevelQoSParameters,
    iE-Extensions                             ProtocolExtensionContainer { { MBS-Flows-Mapped-To-MRB-Item-ExtIEs} } OPTIONAL
}

MBS-Flows-Mapped-To-MRB-Item-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


MBSF1UInformation ::= SEQUENCE {
    mbs-f1u-info                UPTransportLayerInformation,
    iE-Extensions                   ProtocolExtensionContainer  { { MBSF1UInformation-ExtIEs } }  OPTIONAL,
    ...
}

MBSF1UInformation-ExtIEs        F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-F1UTunnelNotEstablished CRITICALITY     ignore      EXTENSION       F1UTunnelNotEstablished   PRESENCE    optional },
    ...
}

MBSInterestIndication ::= OCTET STRING
```

```
MBS-Session-ID ::= SEQUENCE {
    tMGI                        TMGI,
    nID                         NID                 OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { MBS-Session-ID-ExtIEs} } OPTIONAL,
    ...
}

MBS-Session-ID-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-Area-Session-ID  ::= INTEGER (0..65535, ...)


MBS-CUtoDURRCInformation        ::= SEQUENCE {
    mBS-Broadcast-Cell-List     MBS-Broadcast-Cell-List,
    mBS-Broadcast-MRB-List      MBS-Broadcast-MRB-List,
    iE-Extensions               ProtocolExtensionContainer { { MBS-CUtoDURRCInformation-ExtIEs } } OPTIONAL,
    ...
}

MBS-CUtoDURRCInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-Broadcast-Cell-List ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF  MBS-Broadcast-Cell-Item

MBS-Broadcast-Cell-Item ::= SEQUENCE {
    nRCGI                       NRCGI,
    mtch-neighbourCell          OCTET STRING        OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { MBS-Broadcast-Cell-Item-ExtIEs} } OPTIONAL,
    ...
}

MBS-Broadcast-Cell-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-Broadcast-MRB-List  ::= SEQUENCE (SIZE(1.. maxnoofMRBs))    OF  MBS-Broadcast-MRB-Item

MBS-Broadcast-MRB-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    mRB-PDCP-Config-Broadcast   OCTET STRING,
    iE-Extensions               ProtocolExtensionContainer { { MBS-Broadcast-MRB-Item-ExtIEs} } OPTIONAL,
    ...
}

MBS-Broadcast-MRB-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBSMulticastF1UContextDescriptor ::= SEQUENCE {
    multicastF1UContextReferenceF1      MulticastF1UContextReferenceF1,
    mc-F1UCtxtusage     ENUMERATED {ptm, ptp, ptp-retransmission, ptp-forwarding, ...},
```

```
    mbsAreaSession                      MBS-Area-Session-ID             OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer  {{MBSMulticastF1UContextDescriptor-ExtIEs}} OPTIONAL,
    ...
}

MBSMulticastF1UContextDescriptor-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MT-SDT-Information ::= SEQUENCE {
    mt-SDT-Indicator    MT-SDT-Indicator,
    iE-Extensions       ProtocolExtensionContainer { { MT-SDT-Information-ExtIEs } } OPTIONAL,
    ...
}

MT-SDT-Information-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MT-SDT-Indicator ::= ENUMERATED {true, ...}

MBSMulticastSessionReceptionState ::= ENUMERATED {start-monitoring-G-RNTI, stop-monitoring-G-RNTI, ...}

MulticastCU2DURRCInfo       ::= SEQUENCE {
    mBS-Multicast-CU2DU-Cell-List       MBS-Multicast-CU2DU-Cell-List   OPTIONAL,
    mBS-Multicast-MRB-List      MBS-Multicast-MRB-List                  OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { MulticastCU2DURRCInfo-ExtIEs } } OPTIONAL,
    ...
}

MulticastCU2DURRCInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-Multicast-CU2DU-Cell-List   ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF  MBS-Multicast-CU2DU-Cell-Item

MBS-Multicast-CU2DU-Cell-Item ::= SEQUENCE {
    nRCGI                                   NRCGI,
    mbsMulticastRRC-INACTIVEReceptionMode   MBSMulticastRRCINACTIVEReceptionMode    OPTIONAL,
    mbsMulticastConfigurationRequest        ENUMERATED {query, ...}                 OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { MBS-Multicast-CU2DU-Cell-Item-ExtIEs} } OPTIONAL,
    ...
}

MBS-Multicast-CU2DU-Cell-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBSMulticastRRCINACTIVEReceptionMode ::= ENUMERATED {activated, deactivated, ...}

MBS-Multicast-MRB-List  ::= SEQUENCE (SIZE(1.. maxnoofMRBs))    OF  MBS-Multicast-MRB-Item

MBS-Multicast-MRB-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
```

```
    mRB-PDCP-Config-Broadcast   OCTET STRING,
    iE-Extensions               ProtocolExtensionContainer { { MBS-Multicast-MRB-Item-ExtIEs} } OPTIONAL,
    ...
}

MBS-Multicast-MRB-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastCU2DUCommonRRCInfo ::= SEQUENCE {
    multicastCommonCU2DUCellList        MulticastCommonCU2DUCellList        OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {MulticastCU2DUCommonRRCInfo-ExtIEs} } OPTIONAL,
    ...
}

MulticastCU2DUCommonRRCInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastCommonCU2DUCellList ::= SEQUENCE (SIZE(1.. maxCellingNBDU))    OF MulticastCommonCU2DUCell-Item

MulticastCommonCU2DUCell-Item ::= SEQUENCE {
    nRCGI                                   NRCGI,
    multicastCommonCu2DUCellInformation     MulticastCommonCu2DUCellInformation,
    iE-Extensions               ProtocolExtensionContainer { {MulticastCommonCU2DUCell-Item-ExtIEs} } OPTIONAL,
    ...
}

MulticastCommonCU2DUCell-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastCommonCu2DUCellInformation ::= SEQUENCE {
    mBSMulticastNeighbourCellListItem   MBSMulticastNeighbourCellListItem       OPTIONAL,
    thresholdMBS-ListItem               ThresholdMBS-ListItem                   OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {MulticastCommonCu2DUCellInformation-ExtIEs} } OPTIONAL,
    ...
}

MulticastCommonCu2DUCellInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBSMulticastNeighbourCellListItem ::= CHOICE {
    mbsMulticastNeighbourCellListInformationprovided        UpdateMBSMulticastNeighbourCellListInformation,
    nombsMulticastNeighbourCellListInformationprovided      NULL,
    choice-extension        ProtocolIE-SingleContainer { {MBSMulticastNeighbourCellListItem-ExtIEs} }
}

MBSMulticastNeighbourCellListItem-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}
ThresholdMBS-ListItem ::= CHOICE {
    thresholdMBS-ListInformationprovided        UpdateThresholdMBS-ListInformation,
```

```
    nothresholdMBSListInformationprovided       NULL,
    choice-extension        ProtocolIE-SingleContainer { {ThresholdMBS-ListItem-ExtIEs} }
}

ThresholdMBS-ListItem-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

UpdateMBSMulticastNeighbourCellListInformation ::= SEQUENCE {
    mbs-NeighbourCellList           OCTET STRING                        OPTIONAL,
    mbs-MulticastSessionList        MTCH-NeighbourCellSessionList       OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {UpdateMBSMulticastNeighbourCellListInformation-ExtIEs} } OPTIONAL,
    ...
}

UpdateMBSMulticastNeighbourCellListInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MTCH-NeighbourCellSessionList ::= SEQUENCE (SIZE(1..maxMBSSessionsinSessionInfoList))   OF MTCH-NeighbourCellSession-Item
MTCH-NeighbourCellSession-Item ::= SEQUENCE {
    mbsSessionID                        MBS-Session-ID,
    mtch-NeighbourCellInformation           MTCH-NeighbourCellInformation,
        iE-Extensions               ProtocolExtensionContainer { {MTCH-NeighbourCellSession-Item-ExtIEs} } OPTIONAL,
    ...
}

MTCH-NeighbourCellSession-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MTCH-NeighbourCellInformation ::= CHOICE {
    mtch-NeighbourCellprovided              OCTET STRING,
    mtch-NeighbourCellnotprovided           NULL,
    choice-extension        ProtocolIE-SingleContainer { {MTCH-NeighbourCellInformation-ExtIEs} }
}

MTCH-NeighbourCellInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

UpdateThresholdMBS-ListInformation ::= SEQUENCE {
    thresholdMBSList                OCTET STRING                OPTIONAL,
    thresholdIndexSessionList       ThresholdIndexSessionList   OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {UpdateThresholdMBS-ListInformation-ExtIEs} } OPTIONAL,
    ...
}

UpdateThresholdMBS-ListInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ThresholdIndexSessionList ::= SEQUENCE (SIZE(1..maxMBSSessionsinSessionInfoList))   OF ThresholdIndexSession-Item
ThresholdIndexSession-Item ::= SEQUENCE {
```

```
    mbsSessionID                        MBS-Session-ID,
    thresholdIndexInformation           ThresholdIndexInformation,
    iE-Extensions               ProtocolExtensionContainer { {ThresholdIndexSession-Item-ExtIEs} } OPTIONAL,
    ...
}

ThresholdIndexSession-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ThresholdIndexInformation ::= CHOICE {
    thresholdIndexprovided              ThresholdIndex,
    thresholdIndexnotprovided           NULL,
    choice-extension        ProtocolIE-SingleContainer { {ThresholdIndexInformation-ExtIEs} }
}

ThresholdIndexInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

ThresholdIndex ::= INTEGER (0..maxnoofThresholdMBS-1)

MulticastDU2CURRCInfo       ::= SEQUENCE {
    mBS-Multicast-DU2CU-Cell-List       MBS-Multicast-DU2CU-Cell-List           OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { MulticastDU2CURRCInfo-ExtIEs } } OPTIONAL,
    ...
}

MulticastDU2CURRCInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-Multicast-DU2CU-Cell-List   ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF  MBS-Multicast-DU2CU-Cell-Item

MBS-Multicast-DU2CU-Cell-Item ::= SEQUENCE {
    nRCGI                                   NRCGI,
    mbsMulticastConfigurationResponseInfo   MBSMulticastConfigurationResponseInfo       OPTIONAL,
    mbsMulticastConfigurationNotification   MBSMulticastConfigurationNotification                        OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { MBS-Multicast-DU2CU-Cell-Item-ExtIEs} } OPTIONAL,
    ...
}

MBS-Multicast-DU2CU-Cell-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBSMulticastConfigurationResponseInfo ::= CHOICE {
    mbsMulticastConfiguration-available         MBSMulticastConfiguration-available,
    mbsMulticastConfiguration-notavailable      MBSMulticastConfiguration-notavailable,
    choice-extension        ProtocolIE-SingleContainer { {MBSMulticastConfigurationResponseInfo-ExtIEs} }
}

MBSMulticastConfigurationResponseInfo-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
```

```
}

MBSMulticastConfiguration-available ::= SEQUENCE {
    mBSMulticastConfiguration               OCTET STRING,
    iE-Extensions               ProtocolExtensionContainer { { MBSMulticastConfiguration-available-ExtIEs} } OPTIONAL,
    ...
}

MBSMulticastConfiguration-available-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBSMulticastConfiguration-notavailable ::= SEQUENCE {
    mBSMulticastConfiguration-notavailable          ENUMERATED {not-available, ...},
    iE-Extensions               ProtocolExtensionContainer { { MBSMulticastConfiguration-notavailable-ExtIEs} } OPTIONAL,
    ...
}

MBSMulticastConfiguration-notavailable-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBSMulticastConfigurationNotification ::= SEQUENCE {
    mbsMulticastConfigurationNotificationInfo           MBSMulticastConfigurationNotificationInfo   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {MBSMulticastConfigurationNotification-ExtIEs} }  OPTIONAL,
    ...
}

MBSMulticastConfigurationNotification-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MBSMulticastConfigurationNotificationInfo ::= CHOICE {
    mbsMulticastConfigurationChanged            OCTET STRING,
    mbsMulticastConfigurationRemoved            NULL,
    choice-extension        ProtocolIE-SingleContainer { {MBSMulticastConfigurationNotificationInfo-ExtIEs} }
}

MBSMulticastConfigurationNotificationInfo-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}


MulticastF1UContext-ToBeSetup-Item ::= SEQUENCE {
   mRB-ID                  MRB-ID,
   mbs-f1u-info-at-DU      UPTransportLayerInformation,
   mbsProgressInformation       MRB-ProgressInformation                     OPTIONAL,
    -- The above IE shall be present if the MC F1-U Context usage IE in the MBS Multicast F1-U Context Descriptor IE is set to "ptp forwarding".
   iE-Extensions           ProtocolExtensionContainer { {MulticastF1UContext-ToBeSetup-Item-ExtIEs} }  OPTIONAL,
   ...
}

MulticastF1UContext-ToBeSetup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
   ...
```

```
}

MulticastF1UContext-Setup-Item ::= SEQUENCE {
   mRB-ID                  MRB-ID,
   mbs-f1u-info-at-CU      UPTransportLayerInformation,
   iE-Extensions           ProtocolExtensionContainer { {MulticastF1UContext-Setup-Item-ExtIEs} } OPTIONAL,
   ...
}

MulticastF1UContext-Setup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
   ...
}


MulticastF1UContext-FailedToBeSetup-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    cause                       Cause                           OPTIONAL,
   iE-Extensions           ProtocolExtensionContainer { {MulticastF1UContext-FailedToBeSetup-Item-ExtIEs} }   OPTIONAL,
   ...
}

MulticastF1UContext-FailedToBeSetup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
   ...
}



MBSPTPRetransmissionTunnelRequired ::= ENUMERATED {true,    ...}


MBS-ServiceArea ::= CHOICE {
   locationindependent     MBS-ServiceAreaInformation,
   locationdependent    MBS-ServiceAreaInformationList,
   choice-Extensions    ProtocolIE-SingleContainer { {MBSServiceArea-ExtIEs} }
}

MBSServiceArea-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}


MBS-ServiceAreaInformation ::= SEQUENCE {
   mBS-ServiceAreaCellList    MBS-ServiceAreaCellList                                              OPTIONAL,
   mBS-ServiceAreaTAIList     MBS-ServiceAreaTAIList                                               OPTIONAL,
   iE-Extensions              ProtocolExtensionContainer { {MBS-ServiceAreaInformation-ExtIEs} }  OPTIONAL,
   ...
}

MBS-ServiceAreaInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
   ...
}

MBS-ServiceAreaCellList ::= SEQUENCE (SIZE(1.. maxnoofCellsforMBS)) OF NRCGI
```

```
MBS-ServiceAreaTAIList ::= SEQUENCE (SIZE(1.. maxnoofTAIforMBS)) OF MBS-ServiceAreaTAIList-Item
MBS-ServiceAreaTAIList-Item ::= SEQUENCE {
    plmn-ID                         PLMN-Identity,
    fiveGS-TAC                      FiveGS-TAC,
   iE-Extensions              ProtocolExtensionContainer { {MBS-ServiceAreaTAIList-Item-ExtIEs} } OPTIONAL,
   ...
}

MBS-ServiceAreaTAIList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
   ...
}


MBS-ServiceAreaInformationList ::= SEQUENCE (SIZE(1..maxnoofMBSServiceAreaInformation)) OF MBS-ServiceAreaInformationItem

MBS-ServiceAreaInformationItem ::= SEQUENCE {
    mBS-AreaSessionID                   MBS-Area-Session-ID,
    mBS-ServiceAreaInformation          MBS-ServiceAreaInformation,
    iE-Extensions                       ProtocolExtensionContainer { { MBS-ServiceAreaInformationItem-ExtIEs} }  OPTIONAL,
    ...
}
MBS-ServiceAreaInformationItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MC-PagingCell-Item ::= SEQUENCE {
    nRCGI           NRCGI,
    iE-Extensions   ProtocolExtensionContainer { { MC-PagingCell-ItemExtIEs } } OPTIONAL
}

MC-PagingCell-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


MIB-message ::= OCTET STRING

MeasConfig ::= OCTET STRING

MeasGapConfig ::= OCTET STRING

MeasGapSharingConfig ::= OCTET STRING

PosMeasurementAmount ::= ENUMERATED {ma0, ma1, ma2, ma4, ma8, ma16, ma32, ma64}

MeasurementBeamInfoRequest ::= ENUMERATED {true, ...}

MeasurementBeamInfo  ::= SEQUENCE {
    pRS-Resource-ID             PRS-Resource-ID     OPTIONAL,
    pRS-Resource-Set-ID         PRS-Resource-Set-ID OPTIONAL,
    sSB-Index                   SSB-Index           OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { MeasurementBeamInfo-ExtIEs} } OPTIONAL
}
```

```
MeasurementBeamInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


MeasurementTimingConfiguration ::= OCTET STRING

MessageIdentifier ::= BIT STRING (SIZE (16))


MeasurementTimeOccasion ::= ENUMERATED {o1, o4, ...}

MeasurementCharacteristicsRequestIndicator ::= BIT STRING (SIZE (16))

MRB-ProgressInformation ::= CHOICE {
    pdcp-SN12           INTEGER (0..4095),
    pdcp-SN18           INTEGER (0..262143),
    choice-extension        ProtocolIE-SingleContainer { { MRB-ProgressInformation-ExtIEs} }
}

MRB-ProgressInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

MulticastF1UContextReferenceF1 ::= OCTET STRING (SIZE(4))

MulticastF1UContextReferenceCU ::= OCTET STRING (SIZE(4))

MultipleULAoA ::= SEQUENCE {
    multipleULAoA                   MultipleULAoA-List,
    iE-Extensions                   ProtocolExtensionContainer { { MultipleULAoA-ExtIEs} } OPTIONAL,
    ...
}

MultipleULAoA-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MultipleULAoA-List ::= SEQUENCE (SIZE(1.. maxnoofULAoAs)) OF MultipleULAoA-Item

MultipleULAoA-Item ::= CHOICE {
    uL-AoA      UL-AoA,
    ul-ZoA      ZoAInformation,
    choice-extension ProtocolIE-SingleContainer { { MultipleULAoA-Item-ExtIEs } }
}

MultipleULAoA-Item-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

MDTPollutedMeasurementIndicator ::= ENUMERATED {iDC,no-IDC, ...}

MRB-ID ::= INTEGER (1..512, ...)
```

```
MulticastMBSSessionList ::= SEQUENCE (SIZE(1..maxnoofMBSSessionsofUE)) OF MulticastMBSSessionList-Item
MulticastMBSSessionList-Item ::= SEQUENCE {
    mbsSessionId                MBS-Session-ID,
    iE-Extensions               ProtocolExtensionContainer { { MulticastMBSSessionList-Item-ExtIEs } } OPTIONAL,
    ...
}

MulticastMBSSessionList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastMRBs-FailedToBeModified-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    cause                       Cause                           OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { MulticastMRBs-FailedtoBeModified-Item-ExtIEs} } OPTIONAL,
    ...
}

MulticastMRBs-FailedtoBeModified-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastMRBs-FailedToBeSetup-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    cause                       Cause                           OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { MulticastMRBs-FailedToBeSetup-Item-ExtIEs} } OPTIONAL,
    ...
}

MulticastMRBs-FailedToBeSetup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastMRBs-FailedToBeSetupMod-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    cause                       Cause                           OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { MulticastMRBs-FailedToBeSetupMod-Item-ExtIEs} } OPTIONAL,
    ...
}

MulticastMRBs-FailedToBeSetupMod-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastMRBs-Modified-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    iE-Extensions               ProtocolExtensionContainer { { MulticastMRBs-Modified-Item-ExtIEs} } OPTIONAL,
    ...
}

MulticastMRBs-Modified-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
MulticastMRBs-Setup-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    iE-Extensions               ProtocolExtensionContainer { { MulticastMRBs-Setup-Item-ExtIEs} } OPTIONAL,
    ...
}

MulticastMRBs-Setup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastMRBs-SetupMod-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    iE-Extensions               ProtocolExtensionContainer { { MulticastMRBs-SetupMod-Item-ExtIEs} } OPTIONAL,
    ...
}

MulticastMRBs-SetupMod-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastMRBs-ToBeModified-Item ::= SEQUENCE {
    mRB-ID                          MRB-ID,
    mRB-QoSInformation              QoSFlowLevelQoSParameters       OPTIONAL,
    mBS-Flows-Mapped-To-MRB-List    MBS-Flows-Mapped-To-MRB-List    OPTIONAL,
    mBS-DL-PDCP-SN-Length           PDCPSNLength                    OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { MulticastMRBs-ToBeModified-Item-ExtIEs} } OPTIONAL,
    ...
}

MulticastMRBs-ToBeModified-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastMRBs-ToBeReleased-Item ::= SEQUENCE {
    mRB-ID                      MRB-ID,
    iE-Extensions               ProtocolExtensionContainer { { MulticastMRBs-ToBeReleased-ItemExtIEs } }  OPTIONAL,
    ...
}

MulticastMRBs-ToBeReleased-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastMRBs-ToBeSetup-Item ::= SEQUENCE {
    mRB-ID                          MRB-ID,
    mRB-QoSInformation              QoSFlowLevelQoSParameters,
    mBS-Flows-Mapped-To-MRB-List    MBS-Flows-Mapped-To-MRB-List,
    mBS-DL-PDCP-SN-Length           PDCPSNLength,
    iE-Extensions                   ProtocolExtensionContainer { { MulticastMRBs-ToBeSetup-Item-ExtIEs} } OPTIONAL,
    ...
}

MulticastMRBs-ToBeSetup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}

MulticastMRBs-ToBeSetupMod-Item ::= SEQUENCE {
    mRB-ID                          MRB-ID,
    mRB-QoSInformation              QoSFlowLevelQoSParameters,
    mBS-Flows-Mapped-To-MRB-List    MBS-Flows-Mapped-To-MRB-List,
    mBS-DL-PDCP-SN-Length           PDCPSNLength,
    iE-Extensions                   ProtocolExtensionContainer { { MulticastMRBs-ToBeSetupMod-Item-ExtIEs} } OPTIONAL,
    ...
}

MulticastMRBs-ToBeSetupMod-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MultiplexingInfo    ::= SEQUENCE{
    iAB-MT-Cell-List    IAB-MT-Cell-List,
    iE-Extensions       ProtocolExtensionContainer { {MultiplexingInfo-ExtIEs} } OPTIONAL
}

MultiplexingInfo-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MusimCapabilityRestrictionIndication ::= ENUMERATED {true, ...}

MusimCandidateBandList ::= OCTET STRING

M2Configuration ::= ENUMERATED {true, ...}


M5Configuration ::= SEQUENCE {
    m5period            M5period,
    m5-links-to-log     M5-Links-to-log,
    iE-Extensions       ProtocolExtensionContainer { { M5Configuration-ExtIEs} } OPTIONAL,
    ...
}

M5Configuration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-M5ReportAmount   CRITICALITY ignore  EXTENSION M5ReportAmount PRESENCE optional },
    ...
}

M5period ::= ENUMERATED { ms1024, ms2048, ms5120, ms10240, min1, ... }

M5ReportAmount  ::= ENUMERATED { r1, r2, r4, r8, r16, r32, r64, infinity, ... }

M5-Links-to-log ::= ENUMERATED {uplink, downlink, both-uplink-and-downlink, ...}


M6Configuration ::= SEQUENCE {
    m6report-Interval   M6report-Interval,
    m6-links-to-log     M6-Links-to-log,
    iE-Extensions       ProtocolExtensionContainer { { M6Configuration-ExtIEs} } OPTIONAL,
```

```
    ...
}

M6Configuration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-M6ReportAmount   CRITICALITY ignore  EXTENSION M6ReportAmount PRESENCE optional },
    ...
}

M6report-Interval ::= ENUMERATED { ms120, ms240, ms640, ms1024, ms2048, ms5120, ms10240, ms20480, ms40960, min1, min6, min12, min30, ..., ms480}

M6ReportAmount  ::= ENUMERATED { r1, r2, r4, r8, r16, r32, r64, infinity, ... }


M6-Links-to-log ::= ENUMERATED {uplink, downlink, both-uplink-and-downlink, ...}


M7Configuration ::= SEQUENCE {
    m7period            M7period,
    m7-links-to-log     M7-Links-to-log,
    iE-Extensions       ProtocolExtensionContainer { { M7Configuration-ExtIEs} } OPTIONAL,
    ...
}

M7Configuration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-M7ReportAmount   CRITICALITY ignore  EXTENSION M7ReportAmount PRESENCE optional},
    ...
}

M7period    ::= INTEGER(1..60, ...)

M7ReportAmount  ::= ENUMERATED { r1, r2, r4, r8, r16, r32, r64, infinity, ... }

M7-Links-to-log ::= ENUMERATED {downlink, ...}

MDT-Activation ::= ENUMERATED {
    immediate-MDT-only,
    immediate-MDT-and-Trace,
    ...
}

MDTConfiguration ::= SEQUENCE {
    mdt-Activation              MDT-Activation,
    measurementsToActivate      MeasurementsToActivate,
    m2Configuration             M2Configuration     OPTIONAL,
    --  The above IE shall be present if the Measurements to Activate IE has the second bit set to "1".
    m5Configuration             M5Configuration     OPTIONAL,
    -- The above IE shall be present if the Measurements to Activate IE has the fifth bit set to "1".
    m6Configuration             M6Configuration     OPTIONAL,
    --  The above IE shall be present if the Measurements to Activate IE has the seventh bit set to "1".
    m7Configuration             M7Configuration     OPTIONAL,
    --  The above IE shall be present if the Measurements to Activate IE has the eighth bit set to "1".
    iE-Extensions               ProtocolExtensionContainer { { MDTConfiguration-ExtIEs} } OPTIONAL,
    ...
}
```

```
MDTConfiguration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


MDTPLMNList ::= SEQUENCE (SIZE(1..maxnoofMDTPLMNs)) OF PLMN-Identity

MDTPLMNModificationList ::= SEQUENCE (SIZE(0..maxnoofMDTPLMNs)) OF PLMN-Identity

MeasuredFrequencyHops ::= ENUMERATED {singleHop, multiHop, ...}

MeasuredResultsValue ::= CHOICE {
    uL-AngleOfArrival   UL-AoA,
    uL-SRS-RSRP         UL-SRS-RSRP,
    uL-RTOA             UL-RTOA-Measurement,
    gNB-RxTxTimeDiff    GNB-RxTxTimeDiff,
    choice-extension    ProtocolIE-SingleContainer { { MeasuredResultsValue-ExtIEs } }
}

MeasuredResultsValue-ExtIEs F1AP-PROTOCOL-IES ::= {
    { ID id-ZoAInformation  CRITICALITY reject TYPE ZoAInformation  PRESENCE mandatory}|
    { ID id-MultipleULAoA   CRITICALITY reject TYPE MultipleULAoA   PRESENCE mandatory}|
    { ID id-UL-SRS-RSRPP    CRITICALITY reject TYPE UL-SRS-RSRPP    PRESENCE mandatory}|
    { ID id-UL-RSCP         CRITICALITY reject TYPE UL-RSCP         PRESENCE mandatory}|
    { ID id-UL-SRS-TDCT     CRITICALITY reject TYPE UL-SRS-TDCT     PRESENCE mandatory},
    ...
}

MeasurementsToActivate ::= BIT STRING (SIZE (8))

Mobile-TRP-LocationInformation ::= SEQUENCE {
    location-Information            OCTET STRING,
    velocity-Information            OCTET STRING    OPTIONAL,
    location-time-stamp             TimeStamp       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { Mobile-TRP-LocationInformation-ExtIEs} } OPTIONAL,
    ...
}

Mobile-TRP-LocationInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Mobile-IAB-MT-UE-ID ::= OCTET STRING


MUSIM-GapConfig ::= OCTET STRING

MobileIAB-Barred    ::= ENUMERATED {barred, not-barred, ...}

MeasBasedOnAggregatedResources ::= ENUMERATED { true, ... }

MMSID ::= OCTET STRING (SIZE (1))

MonitoringRequestonAvailableBitrate ::= SEQUENCE{
```

```
    monitoringRequest                       MonitoringRequest,
    dlAvailableBitrateReportThresholds      AvailableBitrateReportThresholdList     OPTIONAL,
-- The above IE shall be present if the Monitoring Request IE is set to the value “dl” or “both”
    ulAvailableBitrateReportThresholds      AvailableBitrateReportThresholdList     OPTIONAL,
-- The above IE shall be present if the Monitoring Request IE is set to the value “ul” or “both”
    iE-Extensions                           ProtocolExtensionContainer { { MonitoringRequestonAvailableBitrate-ExtIEs} } OPTIONAL,
    ...
}

MonitoringRequestonAvailableBitrate-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MonitoringRequest ::= ENUMERATED {ul, dl, both, stop,...}

MobilityInitiation  ::= CHOICE {
    mobilityTrigger                     MobilityTrigger,
    mobilityInitiation-AssistanceInfo   MobilityInitiation-AssistanceInfo,
    choice-extension                    ProtocolIE-SingleContainer { { MobilityInitiation-ExtIEs} }
}

MobilityInitiation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

MobilityIssue ::= ENUMERATED {ping-pong, ...}


MobilityTrigger ::= SEQUENCE {
    mobilityTriggeringIndication        MobilityTriggeringIndication,
    mobilityInitiation-CellSwitchInfo   MobilityInitiation-CellSwitchInfo   OPTIONAL,
    mobilityInitiation-EarlyULSyncInfo  MobilityInitiation-EarlyULSyncInfo  OPTIONAL,
    mobilityInitiation-EarlyDLSyncInfo  MobilityInitiation-EarlyDLSyncInfo  OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { MobilityTrigger-ExtIEs} }
}

MobilityTrigger-ExtIEs F1AP-PROTOCOL-EXTENSION::= {
    ...
}

MobilityTriggeringIndication ::=  BIT STRING (SIZE(8))

MobilityInitiation-CellSwitchInfo ::= SEQUENCE {
    candidateCellwithBeamInfo       CandidateCellwithBeamInfo,
    iE-Extensions                   ProtocolExtensionContainer { { MobilityInitiation-CellSwitchInfo-ExtIEs } } OPTIONAL,
    ...
}

MobilityInitiation-CellSwitchInfo-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MobilityInitiation-EarlyULSyncInfo ::= SEQUENCE {
    candidateCellwithBeamInfoList       CandidateCellwithBeamInfoList,
```

```
    iE-Extensions                       ProtocolExtensionContainer { { MobilityInitiation-EarlyULSyncInfo-ExtIEs } }  OPTIONAL,
    ...
}

MobilityInitiation-EarlyULSyncInfo-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MobilityInitiation-EarlyDLSyncInfo  ::= SEQUENCE {
    candidateCellwithBeamInfoList       CandidateCellwithBeamInfoList,
    iE-Extensions                       ProtocolExtensionContainer { { MobilityInitiation-EarlyDLSyncInfo-ExtIEs } }  OPTIONAL,
    ...
}

MobilityInitiation-EarlyDLSyncInfo-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MobilityInitiation-AssistanceInfo   ::= SEQUENCE {
    servingCellMeasurements             ServingCellMeasurements,
    candidateCellwithMeasurementsList   CandidateCellwithMeasurementsList,
    iE-Extensions                       ProtocolExtensionContainer { { MobilityInitiation-AssistanceInfo-ExtIEs } } OPTIONAL,
    ...
}

MobilityInitiation-AssistanceInfo-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}



MROForLTM-Information ::= SEQUENCE {
    gNB-CUorDU-UE-F1AP-ID       GNB-CUorDU-UE-F1AP-ID,
    bFR-SSB-Index               SSB-Index                       OPTIONAL,
    target-SSB-Index-Failure    SSB-Index                       OPTIONAL,
    tA-Value                    TAValue                         OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { MROForLTM-Information-ExtIEs } } OPTIONAL,
    ...
}

MROForLTM-Information-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- N

NRA2XServicesAuthorized ::= SEQUENCE {
    aerialUE            AerialUE                                                            OPTIONAL,
    aerialCcontrollerUE     AerialControllerUE                                                      OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {NRA2XServicesAuthorized-ExtIEs} } OPTIONAL
}

NRA2XServicesAuthorized-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

AerialUE ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

AerialControllerUE ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}


N3CIndirectPathAddition::= SEQUENCE {
    targetRelayUEID         GNB-DU-UE-F1AP-ID,
    iE-Extensions           ProtocolExtensionContainer { { N3CIndirectPathAddition-ExtIEs } }     OPTIONAL,
    ...
}

N3CIndirectPathAddition-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NA-Resource-Configuration-List ::= SEQUENCE (SIZE(1.. maxnoofHSNASlots)) OF NA-Resource-Configuration-Item

NA-Resource-Configuration-Item ::= SEQUENCE {
    nADownlink                  NADownlink      OPTIONAL,
    nAUplink                    NAUplink        OPTIONAL,
    nAFlexible                  NAFlexible      OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { NA-Resource-Configuration-Item-ExtIEs} } OPTIONAL
}

NA-Resource-Configuration-Item-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NADownlink ::= ENUMERATED { true, false, ...}
NAFlexible ::= ENUMERATED { true, false, ...}
NAUplink ::= ENUMERATED { true, false, ...}

Ncd-SSB-RedCapInitialBWP-SDT ::= OCTET STRING

NetworkControlledRepeaterAuthorized ::= ENUMERATED { authorized, not-authorized, ...}

NCGI-to-be-Updated-List-Item ::= SEQUENCE {
    oLDNCGI     NRCGI,
    nEWNCGI     NRCGI,
    iE-Extensions               ProtocolExtensionContainer { { NCGI-to-be-Updated-List-ItemExtIEs} } OPTIONAL,
    ...
}
```

```
NCGI-to-be-Updated-List-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Neighbour-Node-Cells-List ::= SEQUENCE (SIZE(1..maxnoofNeighbourNodeCellsIAB)) OF Neighbour-Node-Cells-List-Item

Neighbour-Node-Cells-List-Item ::= SEQUENCE{
    nRCGI                                         NRCGI,
    gNB-CU-UE-F1AP-ID   GNB-CU-UE-F1AP-ID         OPTIONAL,
    gNB-DU-UE-F1AP-ID   GNB-DU-UE-F1AP-ID         OPTIONAL,
    peer-Parent-Node-Indicator                    ENUMERATED {true, ...}  OPTIONAL,
    iAB-DU-Cell-Resource-Configuration-Mode-Info    IAB-DU-Cell-Resource-Configuration-Mode-Info  OPTIONAL,
    iAB-STC-Info                                  IAB-STC-Info    OPTIONAL,
    rACH-Config-Common                            RACH-Config-Common  OPTIONAL,
    rACH-Config-Common-IAB                        RACH-Config-Common-IAB  OPTIONAL,
    cSI-RS-Configuration                          OCTET STRING    OPTIONAL,
    sR-Configuration                              OCTET STRING    OPTIONAL,
    pDCCH-ConfigSIB1                              OCTET STRING    OPTIONAL,
    sCS-Common                                    OCTET STRING    OPTIONAL,
    iE-Extensions                                 ProtocolExtensionContainer {{Neighbour-Node-Cells-List-Item-ExtIEs}}    OPTIONAL
}

Neighbour-Node-Cells-List-Item-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NeedforGap::= ENUMERATED {true, ...}

NeedForGapsInfoNR ::= OCTET STRING

NeedForGapNCSGInfoNR ::= OCTET STRING

NeedForGapNCSGInfoEUTRA ::= OCTET STRING

NeedForInterruptionInfoNR ::= OCTET STRING

Neighbour-Cell-Information-Item ::= SEQUENCE {
    nRCGI               NRCGI,
    intendedTDD-DL-ULConfig     IntendedTDD-DL-ULConfig OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { Neighbour-Cell-Information-ItemExtIEs } }    OPTIONAL
}

Neighbour-Cell-Information-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-SBFD-Frequency-Configuration         CRITICALITY ignore  EXTENSION   SBFD-Frequency-Configuration       PRESENCE optional}|
    {ID id-SSB-resource-config                  CRITICALITY ignore  EXTENSION   SSB-resource-config          PRESENCE optional}|
    {ID id-NZP-CSI-RS-Resources-Config          CRITICALITY ignore  EXTENSION   NZP-CSI-RS-Resources-Config        PRESENCE optional}|
    {ID id-SRS-Resource-Configuration           CRITICALITY ignore  EXTENSION   SRS-Resource-Configuration      PRESENCE optional},
    ...
}

NeighbourNR-CellsForSON-List ::= SEQUENCE (SIZE(1.. maxNeighbourCellforSON)) OF NeighbourNR-CellsForSON-Item

NeighbourNR-CellsForSON-Item ::= SEQUENCE {
    nRCGI                                         NRCGI,
```

```
    nR-ModeInfoRel16                                NR-ModeInfoRel16                                        OPTIONAL,
    sSB-PositionsInBurst                            SSB-PositionsInBurst                                    OPTIONAL,
    nRPRACHConfig                                   NRPRACHConfig                                           OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { NeighbourNR-CellsForSON-Item-ExtIEs} } OPTIONAL,
    ...
}

NeighbourNR-CellsForSON-Item-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NGRANAllocationAndRetentionPriority ::= SEQUENCE {
    priorityLevel               PriorityLevel,
    pre-emptionCapability       Pre-emptionCapability,
    pre-emptionVulnerability    Pre-emptionVulnerability,
    iE-Extensions               ProtocolExtensionContainer { {NGRANAllocationAndRetentionPriority-ExtIEs} } OPTIONAL
}

NGRANAllocationAndRetentionPriority-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


NGRANHighAccuracyAccessPointPosition ::= SEQUENCE {
    latitude                    INTEGER (-2147483648.. 2147483647),
    longitude                   INTEGER (-2147483648.. 2147483647),
    altitude                    INTEGER (-64000..1280000),
    uncertaintySemi-major       INTEGER (0..255),
    uncertaintySemi-minor       INTEGER (0..255),
    orientationOfMajorAxis      INTEGER (0..179),
    horizontalConfidence        INTEGER (0..100),
    uncertaintyAltitude         INTEGER (0..255),
    verticalConfidence          INTEGER (0..100),

    iE-Extensions               ProtocolExtensionContainer { { NGRANHighAccuracyAccessPointPosition-ExtIEs} } OPTIONAL
}

NGRANHighAccuracyAccessPointPosition-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NID ::= BIT STRING (SIZE(44))

NonF1terminatingTopologyIndicator ::= ENUMERATED {
    true,
    ...
}

NR-CGI-List-For-Restart-Item ::= SEQUENCE {
    nRCGI               NRCGI,
    iE-Extensions       ProtocolExtensionContainer { { NR-CGI-List-For-Restart-ItemExtIEs } }   OPTIONAL,
    ...
}
```

```
NR-CGI-List-For-Restart-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NrofSymbolsExtended ::=  ENUMERATED {n8, n10, n12, n14, ...}

NR-PRSBeamInformation ::= SEQUENCE {
    nR-PRSBeamInformationList       NR-PRSBeamInformationList,
    lCStoGCSTranslationList         LCStoGCSTranslationList     OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { NR-PRSBeamInformation-ExtIEs } } OPTIONAL
}

NR-PRSBeamInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NR-PRSBeamInformationList ::= SEQUENCE (SIZE(1.. maxnoofPRS-ResourceSets)) OF NR-PRSBeamInformationItem

NR-PRSBeamInformationItem ::= SEQUENCE {
    pRSResourceSetID    PRS-Resource-Set-ID,
    pRSAngleList        PRSAngleList,
    iE-Extensions       ProtocolExtensionContainer { { NR-PRSBeamInformationItem-ExtIEs } } OPTIONAL
}

NR-PRSBeamInformationItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NR-TADV ::= INTEGER (0.. 7690)

NReRedCapUEIndication ::= ENUMERATED {true, ...}

ERedcap-Bcast-Information ::= BIT STRING(SIZE(8))

NRRedCapUEIndication ::= ENUMERATED {true, ...}

NRPagingeDRXInformation ::= SEQUENCE {
    nrpaging-eDRX-Cycle-Idle        NRPaging-eDRX-Cycle-Idle,
    nrpaging-Time-Window            NRPaging-Time-Window                                            OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {NRPagingeDRXInformation-ExtIEs} }  OPTIONAL,
    ...
}

NRPagingeDRXInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRPaging-eDRX-Cycle-Idle ::= ENUMERATED {
    hfquarter, hfhalf, hf1, hf2, hf4,
    hf8, hf16, hf32, hf64, hf128, hf256, hf512, hf1024,
    ...
}
```

```
NRPaging-Time-Window ::= ENUMERATED {
    s1, s2, s3, s4, s5,
    s6, s7, s8, s9, s10,
    s11, s12, s13, s14, s15, s16,
    ...,
    s17, s18, s19, s20, s21,
    s22, s23, s24, s25, s26,
    s27, s28, s29, s30, s31, s32
}

NRPagingeDRXInformationforRRCINACTIVE ::= SEQUENCE {
    nrpaging-eDRX-Cycle-Inactive        NRPaging-eDRX-Cycle-Inactive,
    iE-Extensions           ProtocolExtensionContainer { { NRPagingeDRXInformationforRRCINACTIVE-ExtIEs} } OPTIONAL,
    ...
}

NRPagingeDRXInformationforRRCINACTIVE-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {

    ...
}

NRPaging-eDRX-Cycle-Inactive ::= ENUMERATED {
    hfquarter, hfhalf, hf1,
    ...
}


NRPaginglongeDRXInformationforRRCINACTIVE ::= SEQUENCE {
    nRPaging-long-eDRX-Cycle-Inactive       NRPaging-long-eDRX-Cycle-Inactive,
    nRPaging-Time-Window-Inactive           NRPaging-Time-Window-Inactive,
    iE-Extensions                           ProtocolExtensionContainer { { NRPaginglongeDRXInformationforRRCINACTIVE-ExtIEs} } OPTIONAL,
    ...
}

NRPaginglongeDRXInformationforRRCINACTIVE-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRPaging-long-eDRX-Cycle-Inactive ::= ENUMERATED {
    hf2, hf4, hf8, hf16, hf32, hf64, hf128, hf256, hf512, hf1024, ...
}


NRPaging-Time-Window-Inactive ::= ENUMERATED {
    s1, s2, s3, s4, s5,
    s6, s7, s8, s9, s10,
    s11, s12, s13, s14, s15, s16,
    s17, s18, s19, s20, s21, s22,
    s23, s24, s25, s26, s27, s28, s29,
    s30, s31, s32, ...
}

NonDynamic5QIDescriptor ::= SEQUENCE {
    fiveQI                      INTEGER (0..255, ...),
```

```
    qoSPriorityLevel            INTEGER (1..127)                OPTIONAL,
    averagingWindow             AveragingWindow                 OPTIONAL,
    maxDataBurstVolume          MaxDataBurstVolume              OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { NonDynamic5QIDescriptor-ExtIEs } } OPTIONAL
}

NonDynamic5QIDescriptor-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-CNPacketDelayBudgetDownlink CRITICALITY ignore  EXTENSION ExtendedPacketDelayBudget   PRESENCE optional }|
    { ID id-CNPacketDelayBudgetUplink   CRITICALITY ignore  EXTENSION ExtendedPacketDelayBudget   PRESENCE optional },
    ...
}

NonDynamicPQIDescriptor ::= SEQUENCE {
    fiveQI                      INTEGER (0..255, ...),
    qoSPriorityLevel            INTEGER (1..8, ...)             OPTIONAL,
    averagingWindow             AveragingWindow                 OPTIONAL,
    maxDataBurstVolume          MaxDataBurstVolume              OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { NonDynamicPQIDescriptor-ExtIEs } } OPTIONAL
}

NonDynamicPQIDescriptor-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NonUPTrafficType ::=    ENUMERATED {ue-associated, non-ue-associated, non-f1, bap-control-pdu,...}

NoofDownlinkSymbols ::= INTEGER (0..14)

NoofUplinkSymbols   ::= INTEGER (0..14)

Notification-Cause ::= ENUMERATED {fulfilled, not-fulfilled, ..., not-fulfilled-dl, not-fulfilled-ul}

NotificationControl ::= ENUMERATED {active, not-active, ...}

NotificationInformation ::= SEQUENCE {
    message-Identifier  MessageIdentifier,
    serialNumber        SerialNumber,
    iE-Extensions   ProtocolExtensionContainer { { NotificationInformationExtIEs} } OPTIONAL,
    ...
}

NotificationInformationExtIEs       F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NPNBroadcastInformation ::= CHOICE {
    sNPN-Broadcast-Information                  NPN-Broadcast-Information-SNPN,
    pNI-NPN-Broadcast-Information               NPN-Broadcast-Information-PNI-NPN,
    choice-extension                    ProtocolIE-SingleContainer { {NPNBroadcastInformation-ExtIEs} }
}

NPNBroadcastInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}
```

```
NPN-Broadcast-Information-SNPN ::= SEQUENCE {
    broadcastSNPNID-List        BroadcastSNPN-ID-List,
    iE-Extension                ProtocolExtensionContainer { {NPN-Broadcast-Information-SNPN-ExtIEs} }  OPTIONAL,
    ...
}

NPN-Broadcast-Information-SNPN-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
NPN-Broadcast-Information-PNI-NPN ::= SEQUENCE {
    broadcastPNI-NPN-ID-Information     BroadcastPNI-NPN-ID-List,
    iE-Extension                            ProtocolExtensionContainer { {NPN-Broadcast-Information-PNI-NPN-ExtIEs} } OPTIONAL,
    ...
}

NPN-Broadcast-Information-PNI-NPN-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


NPNSupportInfo ::= CHOICE {
    sNPN-Information        NID,
    choice-extension        ProtocolIE-SingleContainer { { NPNSupportInfo-ExtIEs } }
}

NPNSupportInfo-ExtIEs       F1AP-PROTOCOL-IES ::= {
    ...
}

NRCarrierList ::= SEQUENCE (SIZE(1..maxnoofNRSCSs)) OF NRCarrierItem

NRCarrierItem ::= SEQUENCE {
    carrierSCS                      NRSCS,
    offsetToCarrier                 INTEGER (0..2199, ...),
    carrierBandwidth                INTEGER (0..maxnoofPhysicalResourceBlocks, ...),
    iE-Extension                    ProtocolExtensionContainer { {NRCarrierItem-ExtIEs} }           OPTIONAL,
    ...
}

NRCarrierItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRFreqInfo ::=  SEQUENCE {
    nRARFCN         INTEGER (0..maxNRARFCN),
    sul-Information SUL-Information     OPTIONAL,
    freqBandListNr  SEQUENCE (SIZE(1..maxnoofNrCellBands)) OF FreqBandNrItem,
    iE-Extensions   ProtocolExtensionContainer { { NRFreqInfoExtIEs} } OPTIONAL,
    ...
}

NRFreqInfoExtIEs        F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-FrequencyShift7p5khz    CRITICALITY ignore  EXTENSION FrequencyShift7p5khz  PRESENCE optional },
```

```
	...
}

NRCGI ::= SEQUENCE {
	pLMN-Identity			PLMN-Identity,
	nRCellIdentity			NRCellIdentity,
	iE-Extensions			ProtocolExtensionContainer { {NRCGI-ExtIEs} } OPTIONAL,
	...
}

NRCGI-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

NR-Mode-Info ::= CHOICE {
	fDD		FDD-Info,
	tDD		TDD-Info,
	choice-extension			ProtocolIE-SingleContainer { { NR-Mode-Info-ExtIEs} }
}

NR-Mode-Info-ExtIEs F1AP-PROTOCOL-IES ::= {
	{ ID id-NR-U		CRITICALITY ignore	TYPE NR-U-Channel-Info-List PRESENCE mandatory},
	...
}

NR-ModeInfoRel16 ::= CHOICE {
	fDD								FDD-InfoRel16,
	tDD								TDD-InfoRel16,
	choice-extension				ProtocolIE-SingleContainer { { NR-ModeInfoRel16-ExtIEs} }
}

NR-ModeInfoRel16-ExtIEs F1AP-PROTOCOL-IES ::= {
	...
}


NRPRACHConfig ::= SEQUENCE {
	ulPRACHConfigList			NRPRACHConfigList								OPTIONAL,
	sulPRACHConfigList			NRPRACHConfigList								OPTIONAL,
	iE-Extension				ProtocolExtensionContainer { {NRPRACHConfig-ExtIEs} }	OPTIONAL,
	...
}

NRPRACHConfig-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

NRCellIdentity ::= BIT STRING (SIZE(36))

NRNRB ::= ENUMERATED { nrb11, nrb18, nrb24, nrb25, nrb31, nrb32, nrb38, nrb51, nrb52, nrb65, nrb66, nrb78, nrb79, nrb93, nrb106, nrb107, nrb121,
nrb132, nrb133, nrb135, nrb160, nrb162, nrb189, nrb216, nrb217, nrb245, nrb264, nrb270, nrb273, ..., nrb33, nrb62, nrb124, nrb148, nrb248, nrb44,
nrb58, nrb92, nrb119, nrb188, nrb242, nrb15, nrb35}

NRPCI ::= INTEGER(0..1007)
```

```
NRPRACHConfigList ::= SEQUENCE (SIZE(0..maxnoofPRACHconfigs)) OF NRPRACHConfigItem

NRPRACHConfigItem ::= SEQUENCE {
    nRSCS                   NRSCS,
    prachFreqStartfromCarrier   INTEGER (0..maxnoofPhysicalResourceBlocks-1, ...),
    prachFDM                    ENUMERATED {one, two, four, eight, ...},
    prachConfigIndex            INTEGER (0..255, ..., 256..262),
    ssb-perRACH-Occasion        ENUMERATED {oneEighth, oneFourth, oneHalf, one,
                                            two, four, eight, sixteen, ...},
    freqDomainLength            FreqDomainLength,
    zeroCorrelZoneConfig        INTEGER (0..15),
    iE-Extension        ProtocolExtensionContainer { { NRPRACHConfigItem-ExtIEs} }      OPTIONAL,
    ...
}

NRPRACHConfigItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
        { ID id-SBFD-AcrossSymbolType               CRITICALITY reject EXTENSION SBFD-AcrossSymbolType           PRESENCE optional},
    ...
}

NRSCS ::= ENUMERATED { scs15, scs30, scs60, scs120, ..., scs480, scs960}

NRUERLFReportContainer ::= OCTET STRING


NR-U-Channel-Info-List ::= SEQUENCE (SIZE (1..maxnoofNR-UChannelIDs)) OF NR-U-Channel-Info-Item

NR-U-Channel-Info-Item ::= SEQUENCE {
    nr-U-channel-ID         INTEGER(1.. maxnoofNR-UChannelIDs,...),
    nR-ARFCN                INTEGER (0..maxNRARFCN),
    bandwidth               ENUMERATED{mHz-10,mHz-20,mHz-40, mHz-60, mHz-80,..., mHz-100},
    iE-Extensions           ProtocolExtensionContainer { { NR-U-Channel-Info-List-ExtIEs } }     OPTIONAL,
    ...
}

NR-U-Channel-Info-List-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


NR-U-Channel-List ::= SEQUENCE (SIZE (1..maxnoofNR-UChannelIDs)) OF NR-U-Channel-Item

NR-U-Channel-Item ::= SEQUENCE {
    nR-U-ChannelID                      INTEGER(1..maxnoofNR-UChannelIDs),
    channelOccupancyTimePercentageDL        ChannelOccupancyTimePercentage,
    energyDetectionThreshold                EnergyDetectionThreshold,
    iE-Extensions           ProtocolExtensionContainer { { NR-U-Channel-Item-ExtIEs} } OPTIONAL,
    ...
}

NR-U-Channel-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-ChannelOccupancyTimePercentageUL    CRITICALITY ignore EXTENSION ChannelOccupancyTimePercentage PRESENCE optional}|
```

```
	{ ID id-RadioResourceStatusNR-U					CRITICALITY ignore EXTENSION RadioResourceStatusNR-U PRESENCE optional},
	...
}


NumberofActiveUEs ::= INTEGER(0..16777215, ...)

NumberOfBroadcasts ::= INTEGER (0..65535)

NumberofBroadcastRequest ::= INTEGER (0..65535)


NumberOfTRPRxTEG ::= ENUMERATED {two, three, four, six, eight, ...}

NumberOfTRPRxTxTEG ::= ENUMERATED {wo, three, four, six, eight, ...}

NumDLULSymbols ::= SEQUENCE {
	numDLSymbols	INTEGER (0..13, ...),
	numULSymbols	INTEGER (0..13, ...),
	iE-Extensions			ProtocolExtensionContainer { { NumDLULSymbols-ExtIEs} } OPTIONAL
}

NumDLULSymbols-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-permutation	CRITICALITY ignore	EXTENSION Permutation		PRESENCE optional },
	...
}

NRV2XServicesAuthorized ::= SEQUENCE {
	vehicleUE			VehicleUE											OPTIONAL,
	pedestrianUE		PedestrianUE										OPTIONAL,
	iE-Extensions		ProtocolExtensionContainer { {NRV2XServicesAuthorized-ExtIEs} } OPTIONAL
}

NRV2XServicesAuthorized-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

NRUESidelinkAggregateMaximumBitrate ::= SEQUENCE {
	uENRSidelinkAggregateMaximumBitrate		BitRate,
	iE-Extensions					ProtocolExtensionContainer { {NRUESidelinkAggregateMaximumBitrate-ExtIEs} } OPTIONAL
}

NRUESidelinkAggregateMaximumBitrate-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

NZP-CSI-RS-ResourceID::= INTEGER  (0..191)

NZP-CSI-RS-Resources-Config ::= SEQUENCE {
	nZP-CSI-RS-ResourceSet				OCTET STRING,
	nZP-CSI-RS-Resource-List			NZP-CSI-RS-Resource-List,
	iE-Extensions						ProtocolExtensionContainer { {NZP-CSI-RS-Resources-Config-ExtIEs} } OPTIONAL,
	...
}
```

```
NZP-CSI-RS-Resources-Config-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NZP-CSI-RS-Resource-List ::= SEQUENCE (SIZE(1..maxnoofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-Resource-Item

NZP-CSI-RS-Resource-Item ::= SEQUENCE {
    nZP-CSI-RS-Resource                     OCTET STRING,
    iE-Extensions                           ProtocolExtensionContainer { {NZP-CSI-RS-Resource-Item-ExtIEs} } OPTIONAL,
    ...
}

NZP-CSI-RS-Resource-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

N6JitterInformation ::= SEQUENCE {
    n6JitterLowerBound      INTEGER (-127..127),
    n6JitterUpperBound      INTEGER (-127..127),
    iE-Extensions           ProtocolExtensionContainer { { N6JitterInformationExtIEs } }    OPTIONAL,
    ...}

N6JitterInformationExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


Neighbour-Future-Coverage-Modification-Notification ::= SEQUENCE {
    neighbour-future-coverage-Modification-List         Neighbour-Future-Coverage-Modification-List,
    iE-Extensions                               ProtocolExtensionContainer { { Neighbour-Future-Coverage-Modification-Notification-ExtIEs} }
    OPTIONAL,
    ...
}

Neighbour-Future-Coverage-Modification-Notification-ExtIEs F1AP-PROTOCOL-EXTENSION ::={
    ...
}

Neighbour-Future-Coverage-Modification-List ::= SEQUENCE (SIZE (1.. maxnoofCellsinNG-RANnode)) OF Neighbour-Future-Coverage-Modification-Item

Neighbour-Future-Coverage-Modification-Item ::= SEQUENCE {
    nRCGI                               NRCGI,
    neighbourfuturecellCoverageState                NeighbourFutureCellCoverageState,
    neighbourfutureSSBCoverageModificationList  NeighbourFutureSSBCoverageModification-List OPTIONAL,
    timeforneighbourFutureCoverageModification  TimeforNeighbourFutureCoverageModification  OPTIONAL,
-- The above IE shall be present if the Future Coverage Modification Cause IE is set to the value "coverage" or "cell edge capacity".
    futureCoverageModificationCause             Future-Coverage-Modification-Notification-Cause,  iE-Extension        ProtocolExtensionContainer {
{ Neighbour-Future-Coverage-Modification-Item-ExtIEs} }                 OPTIONAL,
    ...
}

Neighbour-Future-Coverage-Modification-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}

NeighbourFutureCellCoverageState ::= INTEGER (0..63, ...)

NeighbourFutureSSBCoverageModification-List ::= SEQUENCE (SIZE (1..maxnoofSSBAreas)) OF NeighbourFutureSSBCoverageModification-Item

NeighbourFutureSSBCoverageModification-Item::= SEQUENCE {
    sSBIndex                        INTEGER(0..63),
    neighbourfutureSSBCoverageState         NeighbourFutureSSBCoverageState,
    iE-Extensions       ProtocolExtensionContainer { { NeighbourFutureSSBCoverageModification-Item-ExtIEs} }  OPTIONAL,
...
}

NeighbourFutureSSBCoverageModification-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NeighbourFutureSSBCoverageState ::= INTEGER (0..15, ...)


NodeAssociatedInfoResult ::= SEQUENCE {
    energyCost              EnergyCost          OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { NodeAssociatedInfoResult-ExtIEs} } OPTIONAL,
    ...
}

NodeAssociatedInfoResult-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NZP-CSI-RS-ResourceConfiguration ::= SEQUENCE {
    cSI-RSResourceToAddModList          OCTET STRING OPTIONAL,
    cSI-RSResourceToReleaseList         OCTET STRING OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { NZP-CSI-RS-ResourceConfiguration-ExtIEs} } OPTIONAL,
    ...
}


NZP-CSI-RS-ResourceConfiguration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


NZP-CSI-RS-ResourceSetConfiguration ::= SEQUENCE {
    cSI-RSResourceSetToAddModList               OCTET STRING OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { NZP-CSI-RS-ResourceSetConfiguration-ExtIEs} } OPTIONAL,
...
}


NZP-CSI-RS-ResourceSetConfiguration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
...
}
```

```
CSI-IM-ResourceConfiguration ::= SEQUENCE {
    cSI-IMResourceToAddModList                  OCTET STRING OPTIONAL,
    cSI-IMResourceSetToAddModList               OCTET STRING OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { CSI-IM-ResourceConfiguration-ExtIEs} } OPTIONAL,
...
}


CSI-IM-ResourceConfiguration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- O

OffsetToPointA  ::= INTEGER (0..2199,...)

OnDemandPRS-Info ::= SEQUENCE {
    onDemandPRSRequestAllowed               BIT STRING (SIZE (16)),
    allowedResourceSetPeriodicityValues     BIT STRING (SIZE (24))  OPTIONAL,
    allowedPRSBandwidthValues               BIT STRING (SIZE (64))  OPTIONAL,
    allowedResourceRepetitionFactorValues   BIT STRING (SIZE (8))   OPTIONAL,
    allowedResourceNumberOfSymbolsValues    BIT STRING (SIZE (8))   OPTIONAL,
    allowedCombSizeValues                   BIT STRING (SIZE (8))   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { OnDemandPRS-Info-ExtIEs} } OPTIONAL,
    ...
}

OnDemandPRS-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

OnDemandSIB1 ::= CHOICE {
    provision                       OndemandSIB1Config,
    stopProvison                    NULL,
    choice-extension                ProtocolIE-SingleContainer { { OnDemandSIB1-ExtIEs} }
}

OnDemandSIB1-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

OnDemand-SIB1-Cell ::= SEQUENCE {
    nRCGI                           NRCGI,
    on-demandSIBIndindicator        ENUMERATED{start, stop, ...},
    iE-Extensions   ProtocolExtensionContainer { { OnDemand-SIB1-Cell-ExtIEs} }     OPTIONAL
}

OnDemand-SIB1-Cell-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

OndemandSIB1Config  ::= OCTET STRING
```

```
-- P

PacketDelayBudget ::= INTEGER (0..1023, ...)

PacketErrorRate ::= SEQUENCE {
    pER-Scalar          PER-Scalar,
    pER-Exponent        PER-Exponent,
    iE-Extensions       ProtocolExtensionContainer { {PacketErrorRate-ExtIEs} } OPTIONAL,
    ...
}

PacketErrorRate-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PathAdditionInformation ::= CHOICE {
    indirectPathAddition            IndirectPathAddition,
    directPathAddition          NULL,
    n3C-indirectPathAddition        N3CIndirectPathAddition,
    choice-extension                ProtocolIE-SingleContainer { { PathAdditionInformation-ExtIEs} }
}

PathAdditionInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}



PagingAdaptationIndication ::= ENUMERATED {supported, ...}

PER-Scalar ::= INTEGER (0..9, ...)
PER-Exponent ::= INTEGER (0..9, ...)

PagingCell-Item ::= SEQUENCE {
    nRCGI       NRCGI   ,
    iE-Extensions   ProtocolExtensionContainer { { PagingCell-ItemExtIEs } }    OPTIONAL
}

PagingCell-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    {   ID id-LastUsedCellIndication                    CRITICALITY ignore  EXTENSION LastUsedCellIndication                PRESENCE optional }|
    {   ID id-PEISubgroupingSupportIndication           CRITICALITY ignore  EXTENSION PEISubgroupingSupportIndication       PRESENCE optional }|
    {   ID id-Recommended-SSBs-List                     CRITICALITY ignore  EXTENSION Recommended-SSBs-List                 PRESENCE optional }|
    {   ID id-LPWUSSubgroupingSupportIndication         CRITICALITY ignore  EXTENSION LPWUSSubgroupingSupportIndication     PRESENCE optional }|
    {   ID id-PEISubgroupingSupportIndication-PagingAdaptation          CRITICALITY ignore  EXTENSION PEISubgroupingSupportIndication-
PagingAdaptation            PRESENCE optional }|
    {   ID id-LPWUSSupportedBandInfo                    CRITICALITY ignore  EXTENSION LPWUSSupportedBandInfo          PRESENCE optional },
        -- The above IE shall be present if the LPWUSSubgroupingSupportIndication IE is present --
    ...
}

Recommended-SSBs-List ::= SEQUENCE (SIZE(1.. maxnoofSSBAreas)) OF RecommendedSSBItem-List-Item
```

```
RecommendedSSBItem-List-Item::= SEQUENCE {
    sSB-Index           SSB-Index,
    iE-Extensions       ProtocolExtensionContainer { { RecommendedSSBItem-List-Item-ExtIEs} } OPTIONAL
}

RecommendedSSBItem-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PagingDRX ::= ENUMERATED {
    v32,
    v64,
    v128,
    v256,
    ...
}

PagingIdentity ::=  CHOICE {
    rANUEPagingIdentity RANUEPagingIdentity,
    cNUEPagingIdentity  CNUEPagingIdentity,
    choice-extension            ProtocolIE-SingleContainer { { PagingIdentity-ExtIEs } }
}

PagingCause ::= ENUMERATED { voice, ...}

PagingIdentity-ExtIEs F1AP-PROTOCOL-IES::= {
    ...
}

PagingOrigin ::= ENUMERATED { non-3gpp, ...}

PagingPriority ::= ENUMERATED { priolevel1, priolevel2, priolevel3, priolevel4, priolevel5, priolevel6, priolevel7, priolevel8,...}

ParentTImeSource ::= ENUMERATED {synce, ptp, gnss, atomicclock, terrestrialradio, serialtimecode, ntp, handset, other, ...}

PEIPSAssistanceInfo ::= SEQUENCE {
    cNSubgroupID        CNSubgroupID,
    iE-Extensions   ProtocolExtensionContainer { { PEIPSAssistanceInfo-ExtIEs } }   OPTIONAL
}

PEIPSAssistanceInfo-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RelativePathDelay ::= CHOICE {
    k0                  INTEGER (0..16351),
    k1                  INTEGER (0..8176),
    k2                  INTEGER (0..4088),
    k3                  INTEGER (0..2044),
    k4                  INTEGER (0..1022),
    k5                  INTEGER (0..511),
    choice-extension            ProtocolIE-SingleContainer { { RelativePathDelay-ExtIEs } }
}
```

```
RelativePathDelay-ExtIEs F1AP-PROTOCOL-IES ::= {
    {ID id-ReportingGranularitykminus1additionalpath    CRITICALITY ignore TYPE ReportingGranularitykminus1AdditionalPath PRESENCE mandatory}|
    {ID id-ReportingGranularitykminus2additionalpath    CRITICALITY ignore TYPE ReportingGranularitykminus2AdditionalPath PRESENCE mandatory }|
    {ID id-ReportingGranularitykminus3additionalpath    CRITICALITY ignore TYPE ReportingGranularitykminus3AdditionalPath PRESENCE mandatory}|
    {ID id-ReportingGranularitykminus4additionalpath    CRITICALITY ignore TYPE ReportingGranularitykminus4AdditionalPath PRESENCE mandatory }|
    {ID id-ReportingGranularitykminus5additionalpath    CRITICALITY ignore TYPE ReportingGranularitykminus5AdditionalPath PRESENCE mandatory}|
    {ID id-ReportingGranularitykminus6additionalpath    CRITICALITY ignore TYPE ReportingGranularitykminus6AdditionalPath PRESENCE mandatory },
    ...
}

Parent-IAB-Nodes-NA-Resource-Configuration-List ::= SEQUENCE (SIZE(1..maxnoofHSNASlots)) OF Parent-IAB-Nodes-NA-Resource-Configuration-Item

Parent-IAB-Nodes-NA-Resource-Configuration-Item::= SEQUENCE {
    nADownlink                  NADownlink      OPTIONAL,
    nAUplink                    NAUplink        OPTIONAL,
    nAFlexible                  NAFlexible      OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { Parent-IAB-Nodes-NA-Resource-Configuration-Item-ExtIEs} } OPTIONAL
}

Parent-IAB-Nodes-NA-Resource-Configuration-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PartialSuccessCell ::= SEQUENCE {
    broadcastCellList           BroadcastCellList,
    iE-Extensions               ProtocolExtensionContainer { { PartialSuccessCell-ExtIEs} } OPTIONAL,
    ...
}
PartialSuccessCell-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PathlossReferenceInfo ::= SEQUENCE {
    pathlossReferenceSignal         PathlossReferenceSignal,
    iE-Extensions                   ProtocolExtensionContainer { {PathlossReferenceInfo-ExtIEs} } OPTIONAL
}

PathlossReferenceInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PathlossReferenceSignal ::= CHOICE {
    sSB                                     SSB,
    dL-PRS                                  DL-PRS,
    choice-extension                        ProtocolIE-SingleContainer {{PathlossReferenceSignal-ExtIEs }}
}

PathlossReferenceSignal-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

PathSwitchConfiguration ::= SEQUENCE {
    targetRelayUEID         BIT STRING(SIZE(24)),
    remoteUELocalID         RemoteUELocalID,
```

```
    t420                    ENUMERATED {ms50, ms100, ms150, ms200, ms500, ms1000, ms2000, ms10000},
    iE-Extensions           ProtocolExtensionContainer { { PathSwitchConfiguration-ExtIEs } }    OPTIONAL,
    ...
}

PathSwitchConfiguration-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PC5QoSFlowIdentifier ::= INTEGER (1..2048)

PC5-QoS-Characteristics ::= CHOICE {
    non-Dynamic-PQI             NonDynamicPQIDescriptor,
    dynamic-PQI                 DynamicPQIDescriptor,
    choice-extension            ProtocolIE-SingleContainer { { PC5-QoS-Characteristics-ExtIEs } }
}

PC5-QoS-Characteristics-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}


PC5QoSParameters    ::= SEQUENCE {
    pC5-QoS-Characteristics             PC5-QoS-Characteristics,
    pC5-QoS-Flow-Bit-Rates              PC5FlowBitRates             OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { PC5QoSParameters-ExtIEs } }  OPTIONAL,
    ...
}

PC5QoSParameters-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PC5FlowBitRates ::= SEQUENCE {
    guaranteedFlowBitRate       BitRate,
    maximumFlowBitRate          BitRate,
    iE-Extensions               ProtocolExtensionContainer { { PC5FlowBitRates-ExtIEs } }   OPTIONAL,
    ...
}

PC5FlowBitRates-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PC5RLCChannelID ::= INTEGER (1..512, ...)

PC5RLCChannelQoSInformation ::= CHOICE {
    pC5RLCChannelQoS                QoSFlowLevelQoSParameters,
    pC5ControlPlaneTrafficType      ENUMERATED {srb1,srb2,...,srb0},
    choice-extension                ProtocolIE-SingleContainer { { PC5RLCChannelQoSInformation-ExtIEs} }
}

PC5RLCChannelQoSInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    {   ID id-U2URLCChannelQoS      CRITICALITY reject TYPE PC5QoSParameters        PRESENCE mandatory},
```

```
	...
}

PC5RLCChannelToBeSetupList ::= SEQUENCE (SIZE(1.. maxnoofPC5RLCChannels)) OF PC5RLCChannelToBeSetupItem

PC5RLCChannelToBeSetupItem ::= SEQUENCE {
	pC5RLCChannelID				PC5RLCChannelID,
	remoteUELocalID				RemoteUELocalID		OPTIONAL,
	pC5RLCChannelQoSInformation	PC5RLCChannelQoSInformation,
	rLCMode						RLCMode,
	iE-Extensions				ProtocolExtensionContainer { { PC5RLCChannelToBeSetupItem-ExtIEs } }	OPTIONAL,
	...
}

PC5RLCChannelToBeSetupItem-ExtIEs	F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-PeerUE-ID		CRITICALITY reject		EXTENSION BIT STRING (SIZE (24))		PRESENCE optional },
	...
}

PC5RLCChannelToBeModifiedList ::= SEQUENCE (SIZE(1.. maxnoofPC5RLCChannels)) OF PC5RLCChannelToBeModifiedItem

PC5RLCChannelToBeModifiedItem ::= SEQUENCE {
	pC5RLCChannelID				PC5RLCChannelID,
	remoteUELocalID				RemoteUELocalID		OPTIONAL,
	pC5RLCChannelQoSInformation	PC5RLCChannelQoSInformation		OPTIONAL,
	rLCMode						RLCMode		OPTIONAL,
	iE-Extensions				ProtocolExtensionContainer { { PC5RLCChannelToBeModifiedItem-ExtIEs } }	OPTIONAL,
	...
}

PC5RLCChannelToBeModifiedItem-ExtIEs	F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-PeerUE-ID		CRITICALITY reject		EXTENSION BIT STRING (SIZE (24))		PRESENCE optional },
	...
}

PC5RLCChannelToBeReleasedList ::= SEQUENCE (SIZE(1.. maxnoofPC5RLCChannels)) OF PC5RLCChannelToBeReleasedItem

PC5RLCChannelToBeReleasedItem ::= SEQUENCE {
	pC5RLCChannelID				PC5RLCChannelID,
	remoteUELocalID				RemoteUELocalID		OPTIONAL,
	iE-Extensions				ProtocolExtensionContainer { { PC5RLCChannelToBeReleasedItem-ExtIEs } }	OPTIONAL,
	...
}

PC5RLCChannelToBeReleasedItem-ExtIEs	F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-PeerUE-ID		CRITICALITY reject		EXTENSION BIT STRING (SIZE (24))		PRESENCE optional },
	...
}

PC5RLCChannelSetupList ::= SEQUENCE (SIZE(1.. maxnoofPC5RLCChannels)) OF PC5RLCChannelSetupItem

PC5RLCChannelSetupItem ::= SEQUENCE {
	pC5RLCChannelID				PC5RLCChannelID,
```

```
    remoteUELocalID                 RemoteUELocalID         OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { PC5RLCChannelSetupItem-ExtIEs } } OPTIONAL,
    ...
}

PC5RLCChannelSetupItem-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-PeerUE-ID       CRITICALITY reject      EXTENSION BIT STRING (SIZE (24))        PRESENCE optional },
    ...
}

PC5RLCChannelFailedToBeSetupList ::= SEQUENCE (SIZE(1.. maxnoofPC5RLCChannels)) OF PC5RLCChannelFailedToBeSetupItem

PC5RLCChannelFailedToBeSetupItem ::= SEQUENCE {
    pC5RLCChannelID                 PC5RLCChannelID,
    remoteUELocalID                 RemoteUELocalID         OPTIONAL,
    cause                           Cause                   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { PC5RLCChannelFailedToBeSetupItem-ExtIEs } }  OPTIONAL,
    ...
}

PC5RLCChannelFailedToBeSetupItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-PeerUE-ID       CRITICALITY reject      EXTENSION BIT STRING (SIZE (24))        PRESENCE optional },
    ...
}

PC5RLCChannelModifiedList ::= SEQUENCE (SIZE(1.. maxnoofPC5RLCChannels)) OF PC5RLCChannelModifiedItem

PC5RLCChannelModifiedItem ::= SEQUENCE {
    pC5RLCChannelID                 PC5RLCChannelID,
    remoteUELocalID                 RemoteUELocalID         OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { PC5RLCChannelModifiedItem-ExtIEs } } OPTIONAL,
    ...
}

PC5RLCChannelModifiedItem-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-PeerUE-ID       CRITICALITY reject      EXTENSION BIT STRING (SIZE (24))        PRESENCE optional },
    ...
}

PC5RLCChannelFailedToBeModifiedList ::= SEQUENCE (SIZE(1.. maxnoofPC5RLCChannels)) OF PC5RLCChannelFailedToBeModifiedItem

PC5RLCChannelFailedToBeModifiedItem ::= SEQUENCE {
    pC5RLCChannelID                 PC5RLCChannelID,
    remoteUELocalID                 RemoteUELocalID         OPTIONAL,
    cause                           Cause   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { PC5RLCChannelFailedToBeModifiedItem-ExtIEs } }  OPTIONAL,
    ...
}

PC5RLCChannelFailedToBeModifiedItem-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-PeerUE-ID       CRITICALITY reject      EXTENSION BIT STRING (SIZE (24))        PRESENCE optional },
    ...
}
```

```
PC5RLCChannelRequiredToBeModifiedList ::= SEQUENCE (SIZE(1.. maxnoofPC5RLCChannels)) OF PC5RLCChannelRequiredToBeModifiedItem

PC5RLCChannelRequiredToBeModifiedItem ::= SEQUENCE {
    pC5RLCChannelID                 PC5RLCChannelID,
    remoteUELocalID                 RemoteUELocalID         OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { PC5RLCChannelRequiredToBeModifiedItem-ExtIEs } } OPTIONAL,
    ...
}

PC5RLCChannelRequiredToBeModifiedItem-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-PeerUE-ID       CRITICALITY reject      EXTENSION BIT STRING (SIZE (24))        PRESENCE optional },
    ...
}

PC5RLCChannelRequiredToBeReleasedList ::= SEQUENCE (SIZE(1.. maxnoofPC5RLCChannels)) OF PC5RLCChannelRequiredToBeReleasedItem

PC5RLCChannelRequiredToBeReleasedItem ::= SEQUENCE {
    pC5RLCChannelID                 PC5RLCChannelID,
    remoteUELocalID                 RemoteUELocalID         OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { PC5RLCChannelRequiredToBeReleasedItem-ExtIEs } } OPTIONAL,
    ...
}

PC5RLCChannelRequiredToBeReleasedItem-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-PeerUE-ID       CRITICALITY reject      EXTENSION BIT STRING (SIZE (24))        PRESENCE optional },
    ...
}

PDCCH-BlindDetectionSCG ::= OCTET STRING

PDCMeasurementPeriodicity ::= ENUMERATED
{ms80, ms120, ms160, ms240, ms320, ms480, ms640, ms1024, ms1280, ms2048, ms2560, ms5120, ...}

PDCMeasurementQuantities ::= SEQUENCE (SIZE (1.. maxnoofMeasPDC)) OF ProtocolIE-SingleContainer { {PDCMeasurementQuantities-ItemIEs} }

PDCMeasurementQuantities-ItemIEs F1AP-PROTOCOL-IES ::= {
    { ID id-PDCMeasurementQuantities-Item   CRITICALITY reject  TYPE PDCMeasurementQuantities-Item      PRESENCE mandatory}
}

PDCMeasurementQuantities-Item ::= SEQUENCE {
    pDCmeasurementQuantitiesValue               PDCMeasurementQuantitiesValue,
    iE-Extensions                               ProtocolExtensionContainer { { PDCMeasurementQuantitiesValue-ExtIEs} } OPTIONAL
}

PDCMeasurementQuantitiesValue-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PDCMeasurementQuantitiesValue ::= ENUMERATED {
    nr-pdc-tadv,
    gNB-rx-tx,
    ...
}
```

```
PDCMeasurementResult ::= SEQUENCE {
    pDCMeasuredResultsList          PDCMeasuredResultsList,
    iE-Extensions                   ProtocolExtensionContainer { { PDCMeasurementResult-ExtIEs} } OPTIONAL
}

PDCMeasurementResult-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PDCMeasuredResultsList ::= SEQUENCE (SIZE(1..maxnoofMeasPDC)) OF PDCMeasuredResults-Item

PDCMeasuredResults-Item ::= SEQUENCE {
    pDCMeasuredResults-Value    PDCMeasuredResults-Value,
    iE-Extensions               ProtocolExtensionContainer {{ PDCMeasuredResults-Item-ExtIEs }}  OPTIONAL
}

PDCMeasuredResults-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PDCMeasuredResults-Value ::= CHOICE {
    pDC-TADV-NR             PDC-TADV-NR,
    pDC-RxTxTimeDiff        PDC-RxTxTimeDiff,
    choice-extension        ProtocolIE-SingleContainer { { PDCMeasuredResults-Value-ExtIEs} }
}

PDCMeasuredResults-Value-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

PDCReportType ::= ENUMERATED {
    onDemand,
    periodic,
    ...
}

PDC-RxTxTimeDiff ::= INTEGER (0..61565, ...)

PDC-TADV-NR ::= INTEGER (0..62500, ...)

PDCP-SN ::= INTEGER (0..4095)

PDCPSNLength    ::= ENUMERATED { twelve-bits,eighteen-bits,...}

PDUSessionID ::= INTEGER (0..255)

PEISubgroupingSupportIndication ::= ENUMERATED {true, ...}

PEISubgroupingSupportIndication-PagingAdaptation ::= ENUMERATED {true, ...}

ReportingPeriodicityValue ::= INTEGER (0..512, ...)

UEPerformanceDelayMonitoring ::= SEQUENCE {
    uEPerformanceDelayMonitoringRequest         UEPerformanceDelayMonitoringRequest,
```

```
    uEPerformanceDelayMonitoringPeriodicity     UEPerformanceDelayMonitoringPeriodicity OPTIONAL,
    iE-Extensions                               ProtocolExtensionContainer { { UEPerformanceDelayMonitoring-ExtIEs} } OPTIONAL,
    ...
}

UEPerformanceDelayMonitoring-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UEPerformanceDelayMonitoringPeriodicity ::= ENUMERATED  {
    ms500,
    ms1000,
    ms2000,
    ms5000,
    ms10000,
    ...
}

UEPerformanceDelayMonitoringRequest ::=ENUMERATED   {ul-and-dl, stop, ...}

Periodicity ::= INTEGER (0..640000, ...)

PeriodicitySRS ::= ENUMERATED { ms0p125, ms0p25, ms0p5, ms0p625, ms1, ms1p25, ms2, ms2p5, ms4, ms5, ms8, ms10, ms16, ms20, ms32, ms40, ms64, ms80,
ms160, ms320, ms640, ms1280, ms2560, ms5120, ms10240, ...}

Predicted-CCO-Assistance-Information ::= SEQUENCE {
    predicted-CCO-issue-detection           Predicted-CCO-issue-detection,
    predictedAffectedCellsAndBeams-List     AffectedCellsAndBeams-List,
    timeforPredictedCCOIssue                TimeforPredictedCCOIssue    OPTIONAL,
    iE-Extensions                           ProtocolExtensionContainer { {Predicted-CCO-Assistance-Information-ExtIEs} } OPTIONAL,
    ...
}

Predicted-CCO-Assistance-Information-ExtIEs F1AP-PROTOCOL-EXTENSION ::={
    ...
}

Predicted-CCO-issue-detection ::= ENUMERATED {coverage, cell-edge-capacity, cancel, ...}



PeriodicityList ::= SEQUENCE (SIZE(1.. maxnoSRS-ResourcePerSet)) OF PeriodicityList-Item

PeriodicityList-Item ::= SEQUENCE {
    periodicitySRS              PeriodicitySRS,
    iE-Extensions               ProtocolExtensionContainer { { PeriodicityList-ItemExtIEs} } OPTIONAL
}

PeriodicityList-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PeriodicityBound ::= SEQUENCE {
    periodicityLowerBound                   Periodicity,
```

```
    periodicityUpperBound                   Periodicity,
    iE-Extensions               ProtocolExtensionContainer { {PeriodicityBound-ExtIEs} } OPTIONAL,
    ...
}

PeriodicityBound-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

AllowedPeriodicityList ::= SEQUENCE (SIZE(1..maxnoofPeriodicities)) OF Periodicity

PeriodicityRange ::= CHOICE {
    periodicityBound                PeriodicityBound,
    periodicityList                 AllowedPeriodicityList,
    choice-extensions               ProtocolIE-SingleContainer { {PeriodicityRange-ExtIEs} }
}

PeriodicityRange-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

Permutation ::= ENUMERATED {dfu, ufd, ...}

Ph-InfoMCG  ::= OCTET STRING

Ph-InfoSCG  ::= OCTET STRING

PLMN-Identity ::= OCTET STRING (SIZE(3))

PLMNIndexNR ::= INTEGER (1..maxnoofBPLMNsNR)

PlayoutDelayForMediaStartup ::= OCTET STRING

PortNumber ::= BIT STRING (SIZE (16))


PosAssistance-Information ::= OCTET STRING

PosAssistanceInformationFailureList ::= OCTET STRING

PosBroadcast ::= ENUMERATED {
    start,
    stop,
    ...
}

PosContextRevIndication ::= ENUMERATED {true, ...}

PositioningBroadcastCells ::= SEQUENCE (SIZE (1..maxnoBcastCell)) OF NRCGI


PosMeasGapPreConfigList ::= SEQUENCE {
    posMeasGapPreConfigToAddModList             OCTET STRING                    OPTIONAL,
    posMeasGapPreConfigToReleaseList            OCTET STRING                    OPTIONAL,
```

```
    iE-Extensions   ProtocolExtensionContainer { { PosMeasGapPreConfigList-ExtIEs} } OPTIONAL
}

PosMeasGapPreConfigList-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MeasurementPeriodicity ::= ENUMERATED
{ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, min1, min6, min12, min30, ..., ms20480, ms40960, extended }


MeasurementPeriodicityExtended ::= ENUMERATED {ms160, ms320, ms1280, ms2560, ms61440, ms81920,  ms368640, ms737280, ms1843200, ...}

PosMeasurementPeriodicityNR-AoA ::= ENUMERATED {
    ms160,
    ms320,
    ms640,
    ms1280,
    ms2560,
    ms5120,
    ms10240,
    ms20480,
    ms40960,
    ms61440,
    ms81920,
    ms368640,
    ms737280,
    ms1843200,
    ...

}

PosMeasurementQuantities ::= SEQUENCE (SIZE(1.. maxnoofPosMeas)) OF PosMeasurementQuantities-Item

PosMeasurementQuantities-Item ::= SEQUENCE {
    posMeasurementType                  PosMeasurementType,
    timingReportingGranularityFactor    INTEGER (0..5) OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { PosMeasurementQuantities-ItemExtIEs} } OPTIONAL
}

PosMeasurementQuantities-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-TimingReportingGranularityFactorExtended CRITICALITY ignore EXTENSION TimingReportingGranularityFactorExtended PRESENCE optional}|
    {ID id-ChannelResponseInformation               CRITICALITY ignore EXTENSION ChannelResponseInformation                PRESENCE optional},
    ...
}

PosMeasurementResult ::= SEQUENCE (SIZE (1.. maxnoofPosMeas)) OF PosMeasurementResultItem

PosMeasurementResultItem ::= SEQUENCE {
    measuredResultsValue                MeasuredResultsValue,
    timeStamp                           TimeStamp,
    measurementQuality                  TRPMeasurementQuality   OPTIONAL,
    measurementBeamInfo                 MeasurementBeamInfo     OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { PosMeasurementResultItemExtIEs } }   OPTIONAL
```

```
}

PosMeasurementResultItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-ARP-ID              CRITICALITY ignore EXTENSION ARP-ID                     PRESENCE optional}|
    { ID id-SRSResourcetype     CRITICALITY ignore EXTENSION SRSResourcetype            PRESENCE optional}|
    { ID id-LoS-NLoSInformation CRITICALITY ignore EXTENSION LoS-NLoSInformation        PRESENCE optional }|
    { ID id-Mobile-TRP-LocationInformation  CRITICALITY ignore EXTENSION Mobile-TRP-LocationInformation   PRESENCE optional }|
    { ID id-AggregatedPosSRSResourceIDList CRITICALITY ignore EXTENSION AggregatedPosSRSResourceIDList PRESENCE optional }|
    { ID id-MeasuredFrequencyHops           CRITICALITY ignore EXTENSION MeasuredFrequencyHops PRESENCE optional }|
    { ID id-MeasBasedOnAggregatedResources          CRITICALITY ignore EXTENSION MeasBasedOnAggregatedResources PRESENCE optional },
    ...
}

PosMeasurementResultList ::= SEQUENCE (SIZE(1.. maxNoOfMeasTRPs)) OF PosMeasurementResultList-Item

PosMeasurementResultList-Item ::= SEQUENCE {
    posMeasurementResult            PosMeasurementResult,
    tRPID                           TRPID,
    iE-Extensions                   ProtocolExtensionContainer { { PosMeasurementResultList-ItemExtIEs} } OPTIONAL
}

PosMeasurementResultList-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-NRCGI   CRITICALITY ignore EXTENSION NRCGI      PRESENCE optional },
    ...
}

PosMeasurementType ::= ENUMERATED {
    gnb-rx-tx,
    ul-srs-rsrp,
    ul-aoa,
    ul-rtoa,
    ... ,
    multiple-ul-aoa,
    ul-srs-rsrpp,
    ul-rscp,
    ul-SRS-TDCT
}

PosReportCharacteristics ::= ENUMERATED {
    ondemand,
    periodic,
    ...
}

PosResourceSetType  ::= CHOICE {
    periodic            PosResourceSetTypePR,
    semi-persistent     PosResourceSetTypeSP,
    aperiodic           PosResourceSetTypeAP,
    choice-extension    ProtocolIE-SingleContainer {{ PosResourceSetType-ExtIEs }}
}

PosResourceSetType-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}
```

```
PosResourceSetTypePR ::= SEQUENCE {
    posperiodicSet      ENUMERATED{true, ...},
    iE-Extensions       ProtocolExtensionContainer { { PosResourceSetTypePR-ExtIEs} }   OPTIONAL
}

PosResourceSetTypePR-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PosResourceSetTypeSP ::= SEQUENCE {
    possemi-persistentSet       ENUMERATED{true, ...},
    iE-Extensions               ProtocolExtensionContainer { { PosResourceSetTypeSP-ExtIEs} }   OPTIONAL
}

PosResourceSetTypeSP-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PosResourceSetTypeAP ::= SEQUENCE {
    sRSResourceTrigger-List     INTEGER(1..3),
    iE-Extensions               ProtocolExtensionContainer { { PosResourceSetTypeAP-ExtIEs} }   OPTIONAL
}

PosResourceSetTypeAP-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PosSItypeList ::= SEQUENCE (SIZE(1.. maxnoofPosSITypes)) OF PosSItype-Item
PosSItype-Item ::= SEQUENCE {
    posItype            PosSItype   ,
    iE-Extensions   ProtocolExtensionContainer { { PosSItype-ItemExtIEs } } OPTIONAL
}

PosSItype-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-DeliveryStatusReq           CRITICALITY reject EXTENSION DeliveryStatusReq PRESENCE optional },
    ...
}

PosSItype ::= INTEGER (1..32, ...)

PosSRSResourceID-List ::= SEQUENCE (SIZE (1..maxnoSRS-PosResourcePerSet)) OF SRSPosResourceID

PosSRSResource-Item ::= SEQUENCE {
    srs-PosResourceId               SRSPosResourceID,
    transmissionCombPos             TransmissionCombPos,
    startPosition                   INTEGER (0..13),
    nrofSymbols                     ENUMERATED {n1, n2, n4, n8, n12},
    freqDomainShift                 INTEGER (0..268),
    c-SRS                           INTEGER (0..63),
    groupOrSequenceHopping          ENUMERATED { neither, groupHopping, sequenceHopping },
    resourceTypePos                 ResourceTypePos,
    sequenceId                      INTEGER (0.. 65535),
    spatialRelationPos              SpatialRelationPos  OPTIONAL,
```

```
    iE-Extensions                   ProtocolExtensionContainer { { PosSRSResource-Item-ExtIEs} }  OPTIONAL
}

PosSRSResource-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-TxHoppingConfiguration  CRITICALITY ignore  EXTENSION TxHoppingConfiguration    PRESENCE optional},
    ...
}

PosSRSResource-List ::= SEQUENCE (SIZE (1..maxnoSRS-PosResources)) OF PosSRSResource-Item

PosSRSResourceSet-Item ::= SEQUENCE {
    possrsResourceSetID             INTEGER(0..15),
    possRSResourceID-List           PosSRSResourceID-List,
    posresourceSetType              PosResourceSetType,
    iE-Extensions       ProtocolExtensionContainer { { PosSRSResourceSet-Item-ExtIEs} } OPTIONAL
}

PosSRSResourceSet-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PosValidityAreaCellList ::= SEQUENCE (SIZE(1.. maxnoVACell)) OF PosValidityAreaCellList-Item

PosValidityAreaCellList-Item ::= SEQUENCE {
    nRCGI                       NRCGI,
    nRPCI                       INTEGER (0..1007)   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { PosValidityAreaCellList-Item-ExtIEs } }  OPTIONAL
}

PosValidityAreaCellList-Item-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
PosSRSResourceSet-List ::= SEQUENCE (SIZE (1..maxnoSRS-PosResourceSets)) OF PosSRSResourceSet-Item

PrimaryPathIndication ::= ENUMERATED {
    true,
    false,
    ...
}

PreambleIndexList ::= SEQUENCE (SIZE (1.. maxnoofPreambleIndex)) OF PreambleIndexList-Item

PreambleIndexList-Item::= SEQUENCE {
    preambleIndex       INTEGER (0..63),
    iE-Extensions       ProtocolExtensionContainer { { PreambleIndex-Item-ExtIEs} } OPTIONAL
}

PreambleIndex-Item-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Pre-emptionCapability ::= ENUMERATED {
    shall-not-trigger-pre-emption,
    may-trigger-pre-emption
```

```
}

Pre-emptionVulnerability ::= ENUMERATED {
    not-pre-emptable,
    pre-emptable
}

Preconfigured-measurement-GAP-Request ::= ENUMERATED {true, ...}

PriorityLevel   ::= INTEGER { spare (0), highest (1), lowest (14), no-priority (15) } (0..15)

ProtectedEUTRAResourceIndication        ::= OCTET STRING

Protected-EUTRA-Resources-Item ::= SEQUENCE {
    spectrumSharingGroupID                  SpectrumSharingGroupID,
    eUTRACells-List     EUTRACells-List,
    iE-Extensions       ProtocolExtensionContainer { { Protected-EUTRA-Resources-ItemExtIEs } } OPTIONAL
}

Protected-EUTRA-Resources-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PRSConfiguration ::= SEQUENCE {
    pRSResourceSet-List         PRSResourceSet-List,
    iE-Extensions               ProtocolExtensionContainer { { PRSConfiguration-ExtIEs } }  OPTIONAL
}

PRSConfiguration-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-AggregatedPRSResourceSetList    CRITICALITY     ignore  EXTENSION AggregatedPRSResourceSetList   PRESENCE    optional },
    ...
}

PRSInformationPos  ::= SEQUENCE {
    pRS-IDPos                   INTEGER(0..255),
    pRS-Resource-Set-IDPos      INTEGER(0..7),
    pRS-Resource-IDPos          INTEGER(0..63)  OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { PRSInformationPos-ExtIEs} } OPTIONAL
}

PRSInformationPos-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PRS-Measurement-Info-List ::= SEQUENCE (SIZE(1..maxFreqLayers)) OF PRS-Measurement-Info-List-Item

PRS-Measurement-Info-List-Item ::= SEQUENCE {
    pointA                      INTEGER (0..3279165),
    measPRSPeriodicity          ENUMERATED {ms20, ms40, ms80, ms160, ...},
    measPRSOffset               INTEGER (0..159, ...),
    measurementPRSLength        ENUMERATED {ms1dot5, ms3, ms3dot5, ms4, ms5dot5, ms6, ms10, ms20},
    iE-Extensions               ProtocolExtensionContainer { { PRS-Measurement-Info-List-Item-ExtIEs} } OPTIONAL,
        ...
}
```

```
PRS-Measurement-Info-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


Potential-SpCell-Item ::= SEQUENCE {
    potential-SpCell-ID         NRCGI   ,
    iE-Extensions               ProtocolExtensionContainer { { Potential-SpCell-ItemExtIEs } }  OPTIONAL,
    ...
}

Potential-SpCell-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


PRSAngleList ::= SEQUENCE (SIZE(1.. maxnoofPRS-ResourcesPerSet)) OF PRSAngleItem

PRSAngleItem ::= SEQUENCE {
    nR-PRS-Azimuth          INTEGER (0..359),
    nR-PRS-Azimuth-fine     INTEGER (0..9) OPTIONAL,
    nR-PRS-Elevation        INTEGER (0..180) OPTIONAL,
    nR-PRS-Elevation-fine   INTEGER (0..9) OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { PRSAngleItem-ItemExtIEs } }  OPTIONAL
}

PRSAngleItem-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-PRS-Resource-ID     CRITICALITY ignore EXTENSION PRS-Resource-ID        PRESENCE optional },
    ...
}

PRSConfigRequestType ::= ENUMERATED {configure, off, ...}

PRSMuting::= SEQUENCE {
    pRSMutingOption1            PRSMutingOption1        OPTIONAL,
    pRSMutingOption2            PRSMutingOption2        OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { PRSMuting-ExtIEs} } OPTIONAL
}

PRSMuting-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PRSMutingOption1 ::= SEQUENCE {
    mutingPattern                   DL-PRSMutingPattern,
    mutingBitRepetitionFactor       ENUMERATED{rf1,rf2,rf4,rf8,...},
    iE-Extensions                   ProtocolExtensionContainer { { PRSMutingOption1-ExtIEs} } OPTIONAL
}

PRSMutingOption1-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
PRSMutingOption2 ::= SEQUENCE {
    mutingPattern                   DL-PRSMutingPattern,
    iE-Extensions                   ProtocolExtensionContainer { { PRSMutingOption2-ExtIEs} } OPTIONAL
}

PRSMutingOption2-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PRS-Resource-ID ::= INTEGER (0..63)

PRSResource-List::= SEQUENCE (SIZE (1..maxnoofPRSresources)) OF PRSResource-Item

PRSResource-Item  ::= SEQUENCE {
    pRSResourceID           PRS-Resource-ID,
    sequenceID              INTEGER(0..4095),
    rEOffset                INTEGER(0..11,...),
    resourceSlotOffset      INTEGER(0..511),
    resourceSymbolOffset    INTEGER(0..12),
    qCLInfo                 PRSResource-QCLInfo     OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { PRSResource-Item-ExtIEs} } OPTIONAL
}

PRSResource-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-ExtendedResourceSymbolOffset        CRITICALITY ignore EXTENSION ExtendedResourceSymbolOffset  PRESENCE optional},
    ...
}



PRSBWAggregationRequestInfoList ::= SEQUENCE (SIZE (1..maxnoAggCombinations)) OF PRSBWAggregationRequestInfo-Item

PRSBWAggregationRequestInfo-Item ::= SEQUENCE {
    dl-PRSBWAggregationRequestInfo-List DL-PRSBWAggregationRequestInfo-List,
    iE-Extensions                       ProtocolExtensionContainer { { PRSBWAggregationRequestInfo-Item-ExtIEs} } OPTIONAL,
    ...
}

PRSBWAggregationRequestInfo-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DL-PRSBWAggregationRequestInfo-List ::= SEQUENCE (SIZE (2..maxnoAggregatedPosPRSResourceSets)) OF DL-PRSBWAggregationRequestInfo-Item

DL-PRSBWAggregationRequestInfo-Item ::= SEQUENCE {
    dl-prs-ResourceSetIndex             INTEGER (1..8),
    iE-Extensions                       ProtocolExtensionContainer { {DL-PRSBWAggregationRequestInfo-Item-ExtIEs} } OPTIONAL,
    ...
}

DL-PRSBWAggregationRequestInfo-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
ExtendedResourceSymbolOffset ::= INTEGER (0..13,...)


PRSResource-QCLInfo  ::= CHOICE {
    qCLSourceSSB        PRSResource-QCLSourceSSB,
    qCLSourcePRS        PRSResource-QCLSourcePRS,
    choice-extension    ProtocolIE-SingleContainer { { PRSResource-QCLInfo-ExtIEs } }
}
PRSResource-QCLInfo-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

PRSResource-QCLSourceSSB ::= SEQUENCE {
    pCI-NR              INTEGER(0..1007),
    sSB-Index           SSB-Index OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { PRSResource-QCLSourceSSB-ExtIEs} } OPTIONAL,
    ...
}

PRSResource-QCLSourceSSB-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PRSResource-QCLSourcePRS ::= SEQUENCE {
    qCLSourcePRSResourceSetID       PRS-Resource-Set-ID,
    qCLSourcePRSResourceID          PRS-Resource-ID OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { PRSResource-QCLSourcePRS-ExtIEs} } OPTIONAL
}

PRSResource-QCLSourcePRS-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PRS-Resource-Set-ID ::= INTEGER(0..7)

PRSResourceSet-List ::= SEQUENCE (SIZE (1.. maxnoofPRSresourceSets)) OF PRSResourceSet-Item
PRSResourceSet-Item ::= SEQUENCE {
    pRSResourceSetID                PRS-Resource-Set-ID,
    subcarrierSpacing               ENUMERATED{kHz15, kHz30, kHz60, kHz120, ...},
    pRSbandwidth                    INTEGER(1..63),
    startPRB                        INTEGER(0..2176),
    pointA                          INTEGER (0..3279165),
    combSize                        ENUMERATED{n2, n4, n6, n12, ...},
    cPType                          ENUMERATED{normal, extended, ...},
    resourceSetPeriodicity          ENUMERATED{n4,n5,n8,n10,n16,n20,n32,n40,n64,n80,n160,n320,n640,n1280,n2560,n5120,n10240,n20480,n40960,
n81920,..., n128, n256, n512},
    resourceSetSlotOffset           INTEGER(0..81919,...),
    resourceRepetitionFactor        ENUMERATED{rf1,rf2,rf4,rf6,rf8,rf16,rf32,...},
    resourceTimeGap                 ENUMERATED{tg1,tg2,tg4,tg8,tg16,tg32,...},
    resourceNumberofSymbols         ENUMERATED{n2,n4,n6,n12,...,n1},
    pRSMuting                       PRSMuting       OPTIONAL,
    pRSResourceTransmitPower        INTEGER(-60..50),
    pRSResource-List                PRSResource-List,
    iE-Extensions                   ProtocolExtensionContainer { { PRSResourceSet-Item-ExtIEs} } OPTIONAL
```

```
}

PRSResourceSet-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

PRSTransmissionOffIndication ::= CHOICE {
	pRSTransmissionOffPerTRP				NULL,
	pRSTransmissionOffPerResourceSet		PRSTransmissionOffPerResourceSet,
	pRSTransmissionOffPerResource			PRSTransmissionOffPerResource,
	choice-extension		ProtocolIE-SingleContainer { { PRSTransmissionOffIndication-ExtIEs } }
}

PRSTransmissionOffIndication-ExtIEs F1AP-PROTOCOL-IES ::= {
	...
}

PRSTransmissionOffPerResource ::= SEQUENCE (SIZE (1..maxnoofPRSresourceSets)) OF PRSTransmissionOffPerResource-Item

PRSTransmissionOffPerResource-Item	::= SEQUENCE {
	pRSResourceSetID								PRS-Resource-Set-ID,
	pRSTransmissionOffIndicationPerResourceList		SEQUENCE (SIZE(1.. maxnoofPRSresources)) OF PRSTransmissionOffIndicationPerResource-Item,
	iE-Extensions		ProtocolExtensionContainer { { PRSTransmissionOffPerResource-Item-ExtIEs } } OPTIONAL,
	...
}

PRSTransmissionOffPerResource-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

PRSTransmissionOffIndicationPerResource-Item	::= SEQUENCE {
	pRSResourceID		PRS-Resource-ID,
	iE-Extensions		ProtocolExtensionContainer { { PRSTransmissionOffIndicationPerResource-Item-ExtIEs } } OPTIONAL,
	...
}

PRSTransmissionOffIndicationPerResource-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

PRSTransmissionOffInformation ::= SEQUENCE {
	pRSTransmissionOffIndication	PRSTransmissionOffIndication,
	iE-Extensions					ProtocolExtensionContainer { { PRSTransmissionOffInformation-ExtIEs } } OPTIONAL,
	...
}

PRSTransmissionOffInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

PRSTransmissionOffPerResourceSet ::= SEQUENCE (SIZE (1..maxnoofPRSresourceSets)) OF PRSTransmissionOffPerResourceSet-Item

PRSTransmissionOffPerResourceSet-Item	::= SEQUENCE {
	pRSResourceSetID		PRS-Resource-Set-ID,
```

```
    iE-Extensions           ProtocolExtensionContainer { { PRSTransmissionOffPerResourceSet-Item-ExtIEs } } OPTIONAL,
    ...
}

PRSTransmissionOffPerResourceSet-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


PWS-Failed-NR-CGI-Item ::= SEQUENCE {
    nRCGI               NRCGI,
    numberOfBroadcasts  NumberOfBroadcasts,
    iE-Extensions       ProtocolExtensionContainer { { PWS-Failed-NR-CGI-ItemExtIEs } } OPTIONAL,
    ...
}

PWS-Failed-NR-CGI-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PWSSystemInformation ::= SEQUENCE {
    sIBtype                 SIBType-PWS,
    sIBmessage              OCTET STRING,
    iE-Extensions           ProtocolExtensionContainer { { PWSSystemInformationExtIEs } }   OPTIONAL,
    ...
}

PWSSystemInformationExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-NotificationInformation      CRITICALITY ignore  EXTENSION NotificationInformation     PRESENCE optional}|
    { ID id-AdditionalSIBMessageList    CRITICALITY reject  EXTENSION AdditionalSIBMessageList    PRESENCE optional},
    ...
}

PrivacyIndicator ::= ENUMERATED {immediate-MDT, logged-MDT, ...}


PRSTRPList ::= SEQUENCE (SIZE(1.. maxnoofTRPs)) OF PRSTRPItem

PRSTRPItem ::= SEQUENCE {
    tRP-ID      TRPID,
    requestedDLPRSTransmissionCharacteristics   RequestedDLPRSTransmissionCharacteristics       OPTIONAL,
    -- The IE shall be present if the PRS Configuration Request Type IE is set to “configure” --
    pRSTransmissionOffInformation       PRSTransmissionOffInformation               OPTIONAL,
    -- The IE shall be present if the PRS Configuration Request Type IE is set to “off” --

    iE-Extensions   ProtocolExtensionContainer { { PRSTRPItem-ExtIEs} } OPTIONAL,
    ...
}

PRSTRPItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RequestedDLPRSTransmissionCharacteristics ::= SEQUENCE {
```

```
    requestedDLPRSResourceSet-List      RequestedDLPRSResourceSet-List,
    numberofFrequencyLayers         INTEGER(1..4)                                           OPTIONAL,
    startTimeAndDuration            StartTimeAndDuration                                    OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { RequestedDLPRSTransmissionCharacteristics-ExtIEs} } OPTIONAL,
    ...
}
RequestedDLPRSTransmissionCharacteristics-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-PRSBWAggregationRequestInfoList CRITICALITY ignore EXTENSION PRSBWAggregationRequestInfoList PRESENCE optional},
    ...
}

RequestedDLPRSResourceSet-List ::= SEQUENCE (SIZE (1..maxnoofPRSresourceSets)) OF RequestedDLPRSResourceSet-Item

RequestedDLPRSResourceSet-Item ::= SEQUENCE {
    pRSbandwidth                    INTEGER(1..63) OPTIONAL,
    combSize                        ENUMERATED{n2, n4, n6, n12, ...}        OPTIONAL,
    resourceSetPeriodicity          ENUMERATED{n4,n5,n8,n10,n16,n20,n32,n40,n64,n80,n160,n320,n640,n1280,n2560,n5120,n10240,n20480,n40960,
n81920,..., n128, n256, n512}   OPTIONAL,
    resourceRepetitionFactor        ENUMERATED{rf1,rf2,rf4,rf6,rf8,rf16,rf32,...}       OPTIONAL,
    resourceNumberofSymbols         ENUMERATED{n2,n4,n6,n12,...,n1}                     OPTIONAL,
    requestedDLPRSResource-List     RequestedDLPRSResource-List                         OPTIONAL,
    resourceSetStartTimeAndDuration StartTimeAndDuration                                OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { RequestedDLPRSResourceSet-Item-ExtIEs} } OPTIONAL,
    ...
}

RequestedDLPRSResourceSet-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RequestedDLPRSResource-List::= SEQUENCE (SIZE (1..maxnoofPRSresources)) OF RequestedDLPRSResource-Item

RequestedDLPRSResource-Item  ::= SEQUENCE {
    qCLInfo                 PRSResource-QCLInfo     OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { RequestedDLPRSResource-Item-ExtIEs} } OPTIONAL,
    ...
}

RequestedDLPRSResource-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


PRSTransmissionTRPList ::= SEQUENCE (SIZE(1.. maxnoofTRPs)) OF PRSTransmissionTRPItem

PRSTransmissionTRPItem ::= SEQUENCE {
        tRP-ID              TRPID,
        pRSConfiguration    PRSConfiguration,
    iE-Extensions           ProtocolExtensionContainer { { PRSTransmissionTRPItem-ExtIEs} } OPTIONAL,
        ...
}

PRSTransmissionTRPItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}

PreambleIndex ::= INTEGER(0..63)

PDUSetQoSParameters ::= SEQUENCE {
    ulPDUSetQoSInformation                  PDUSetQoSInformation    OPTIONAL,
    dlPDUSetQoSInformation                  PDUSetQoSInformation    OPTIONAL,
    iE-Extensions                           ProtocolExtensionContainer { { PDUSetQoSParameters-ExtIEs } }  OPTIONAL
}

PDUSetQoSParameters-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSetQoSInformation    ::= SEQUENCE {
    pduSetDelayBudget                       ExtendedPacketDelayBudget       OPTIONAL,
    pduSetErrorRate                         PacketErrorRate                 OPTIONAL,
    pduSetIntegratedHandlingInformation     ENUMERATED {true, false, ...}   OPTIONAL,
    iE-Extensions                           ProtocolExtensionContainer { { PDUSetQoSInformation-ExtIEs } } OPTIONAL
}

PDUSetQoSInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

PSIbasedSDUdiscardUL ::= ENUMERATED {start, stop, ...}

PointA  ::= INTEGER (0..3279165)

PSIbasedSDUdiscardDL ::= ENUMERATED {configured, not-configured, ...}

PSCellList ::= SEQUENCE (SIZE(1..maxnoofCHOcells)) OF PSCellList-Item

PSCellList-Item ::= SEQUENCE {
    pscell                              NRCGI,
    iE-Extensions                           ProtocolExtensionContainer { { PSCellList-Item-ExtIEs} } OPTIONAL
}

PSCellList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}



-- Q

QCI ::= INTEGER (0..255)

QoEInformation ::= SEQUENCE {
    qoEInformationList      QoEInformationList,
    iE-Extensions           ProtocolExtensionContainer { { QoEInformation-ExtIEs} } OPTIONAL
}

QoEInformation-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
```

```
	...
}

QoEInformationList ::= SEQUENCE (SIZE(1.. maxnoofQoEInformation)) OF QoEInformationList-Item

QoEInformationList-Item ::= SEQUENCE {
	qoEMetrics			QoEMetrics	OPTIONAL,
	iE-Extensions		ProtocolExtensionContainer { { QoEInformationList-Item-ExtIEs} }	OPTIONAL
}

QoEInformationList-Item-ExtIEs	F1AP-PROTOCOL-EXTENSION ::= {
	{ID id-dRB-List CRITICALITY ignore EXTENSION DRB-List PRESENCE optional},
	...
}

QoEMetrics ::= SEQUENCE {
	appLayerBufferLevelList				AppLayerBufferLevelList	OPTIONAL,
	playoutDelayForMediaStartup			PlayoutDelayForMediaStartup OPTIONAL,
	iE-Extensions						ProtocolExtensionContainer { { QoEMetrics-ExtIEs} } OPTIONAL,
	...
}

QoEMetrics-ExtIEs	F1AP-PROTOCOL-EXTENSION ::= {
	...
}

QoS-Characteristics ::= CHOICE {
	non-Dynamic-5QI				NonDynamic5QIDescriptor,
	dynamic-5QI					Dynamic5QIDescriptor,
	choice-extension			ProtocolIE-SingleContainer { { QoS-Characteristics-ExtIEs } }
}

QoS-Characteristics-ExtIEs F1AP-PROTOCOL-IES ::= {
	...
}

QoSFlowIdentifier ::= INTEGER (0..63)

QoSFlowLevelQoSParameters	::= SEQUENCE {
	qoS-Characteristics						QoS-Characteristics,
	nGRANallocationRetentionPriority		NGRANAllocationAndRetentionPriority,
	gBR-QoS-Flow-Information				GBR-QoSFlowInformation				OPTIONAL,
	reflective-QoS-Attribute				ENUMERATED {subject-to, ...}				OPTIONAL,
	iE-Extensions							ProtocolExtensionContainer { { QoSFlowLevelQoSParameters-ExtIEs } }	OPTIONAL
}

QoSFlowLevelQoSParameters-ExtIEs	F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-PDUSessionID								CRITICALITY ignore	EXTENSION PDUSessionID			PRESENCE optional}|
	{ ID id-ULPDUSessionAggregateMaximumBitRate			CRITICALITY ignore	EXTENSION BitRate				PRESENCE optional}|
	{ ID id-QosMonitoringRequest						CRITICALITY ignore	EXTENSION QosMonitoringRequest	PRESENCE optional}|
	{ ID id-PDCPTerminatingNodeDLTNLAddrInfo			CRITICALITY ignore	EXTENSION TransportLayerAddress	PRESENCE optional }|
	{ ID id-PDUSetQoSParameters							CRITICALITY ignore	EXTENSION PDUSetQoSParameters	PRESENCE optional}|
	{ ID id-MMSID										CRITICALITY ignore	EXTENSION MMSID					PRESENCE optional}|
	{ ID id-Indication-of-Bitrate-Adaptation			CRITICALITY ignore	EXTENSION Indication-of-Bitrate-Adaptation	PRESENCE optional}|
```

```
    { ID id-DLPDUSetInformationMarkingSupportIndication CRITICALITY ignore  EXTENSION DLPDUSetInformationMarkingSupportIndication  PRESENCE 
optional},
    ...
}

QoSFlowMappingIndication ::= ENUMERATED {ul,dl,...}

QoSInformation  ::= CHOICE {
    eUTRANQoS                   EUTRANQoS,
    choice-extension            ProtocolIE-SingleContainer { { QoSInformation-ExtIEs} }
}

QoSInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    {   ID id-DRB-Information       CRITICALITY ignore TYPE DRB-Information     PRESENCE mandatory},
    ...
}

QosMonitoringRequest ::= ENUMERATED {ul, dl, both, ..., stop}

QoSParaSetIndex ::= INTEGER (1..8, ...)

QoSParaSetNotifyIndex ::= INTEGER (0..8, ...)

-- R

RACH-Config-Common  ::= OCTET STRING

RACH-Config-Common-IAB  ::= OCTET STRING

Range ::= ENUMERATED {m50, m80, m180, m200, m350, m400, m500, m700, m1000, ...}

RAReportContainer::= OCTET STRING

RAReportList    ::= SEQUENCE (SIZE(1.. maxnoofRAReports)) OF RAReportItem

RAReportItem    ::= SEQUENCE {
    rAReportContainer               RAReportContainer,
    uEAssitantIdentifier            GNB-DU-UE-F1AP-ID       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { RAReportItem-ExtIEs} }   OPTIONAL,
    ...
}

RAReportItem-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RAReportIndicationList ::= SEQUENCE (SIZE(1..maxnoofUEsforRAReportIndications)) OF RAReportIndicationList-Item

RAReportIndicationList-Item ::= SEQUENCE {
    gNB-CU-UE-F1AP-ID                   GNB-CU-UE-F1AP-ID,
    iE-Extensions                       ProtocolExtensionContainer { { RAReportIndicationList-Item-ExtIEs} } OPTIONAL,
    ...

}
```

```
RAReportIndicationList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}



RadioResourceStatus ::= SEQUENCE {
    sSBAreaRadioResourceStatusList      SSBAreaRadioResourceStatusList,
    iE-Extensions   ProtocolExtensionContainer { { RadioResourceStatus-ExtIEs} } OPTIONAL
}

RadioResourceStatus-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-SliceRadioResourceStatus        CRITICALITY ignore  EXTENSION SliceRadioResourceStatus      PRESENCE optional }|
    { ID id-MIMOPRBusageInformation         CRITICALITY ignore  EXTENSION MIMOPRBusageInformation       PRESENCE optional },
    ...
}

RadioResourceStatusNR-U ::= SEQUENCE {
    dl-Total-PRB-usage  INTEGER (0..100),
    ul-Total-PRB-usage  INTEGER (0..100),
    iE-Extensions                   ProtocolExtensionContainer { { RadioResourceStatusNR-U-ExtIEs} } OPTIONAL,
    ...
}
RadioResourceStatusNR-U-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

MIMOPRBusageInformation ::= SEQUENCE {
    dl-GBR-PRB-usage-for-MIMO       INTEGER (0..100),
    ul-GBR-PRB-usage-for-MIMO       INTEGER (0..100),
    dl-non-GBR-PRB-usage-for-MIMO   INTEGER (0..100),
    ul-non-GBR-PRB-usage-for-MIMO   INTEGER (0..100),
    dl-Total-PRB-usage-for-MIMO     INTEGER (0..100),
    ul-Total-PRB-usage-for-MIMO     INTEGER (0..100),
    iE-Extensions                   ProtocolExtensionContainer { { MIMOPRBusageInformation-ExtIEs} } OPTIONAL,
    ...
}

MIMOPRBusageInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RANfeedbacktype ::= CHOICE {
    proactive                   RANfeedbacktype-proactive,
    reactive                    RANfeedbacktype-reactive,
    choice-extensions           ProtocolIE-SingleContainer { {RANfeedbacktype-ExtIEs} }
}

RANfeedbacktype-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

RANfeedbacktype-proactive ::= SEQUENCE {
```

```
    burstArrivalTimeWindow  BurstArrivalTimeWindow,
    periodicityRange        PeriodicityRange    OPTIONAL,
    iE-Extension            ProtocolExtensionContainer { {RANfeedbacktype-proactive-ExtIEs} }   OPTIONAL,
    ...
}

RANfeedbacktype-proactive-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RANfeedbacktype-reactive ::= SEQUENCE {
    capabilityForBATAdaptation  ENUMERATED {true, ...},
    iE-Extension            ProtocolExtensionContainer { {RANfeedbacktype-reactive-ExtIEs} }    OPTIONAL,
    ...
}

RANfeedbacktype-reactive-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RANSharingAssistanceInformation ::= ENUMERATED {
    mbs-session-in-non-shared-cell-resources,
    ...
}

RANTSSRequestType ::= ENUMERATED {start, stop, ...}

RANTimingSynchronisationStatusInfo ::= SEQUENCE {
    synchronisationstate            ENUMERATED {locked, holdover, freeRun, ...}      OPTIONAL,
    traceabletoUTC                  ENUMERATED { true, false, ...}                  OPTIONAL,
    traceabletoGNSS                 ENUMERATED { true, false, ...}                  OPTIONAL,
    clockFrequencyStability         BIT STRING (SIZE(16))                           OPTIONAL,
    clockAccuracy                   ClockAccuracy                                   OPTIONAL,
    parentTimeSource                ParentTImeSource                                OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { RANTimingSynchronisationStatusInfo-ExtIEs} }  OPTIONAL,
...
}

RANTimingSynchronisationStatusInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
        ...
    }

ClockAccuracy ::= CHOICE {
    clockAccuracyValue      INTEGER (1..40000000, ...),
    clockAccuracyIndex      INTEGER (32..47, ...),
    choice-Extensions       ProtocolIE-SingleContainer { { ClockAccuracy-ExtIEs} }
}

ClockAccuracy-ExtIEs F1AP-PROTOCOL-IES ::= {
        ...
}

RANAC ::= INTEGER (0..255)
```

```
RAN-MeasurementID ::= INTEGER (1.. 65536, ...)

RAN-UE-MeasurementID ::= INTEGER (1.. 256, ...)

RAN-UE-PDC-MeasID ::= INTEGER (1..16, ...)

RANUEID ::= OCTET STRING (SIZE (8))

RANUEPagingIdentity ::= SEQUENCE    {
    iRNTI                       BIT STRING (SIZE(40)),
    iE-Extensions               ProtocolExtensionContainer { { RANUEPagingIdentity-ExtIEs } }   OPTIONAL}

RANUEPagingIdentity-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RAT-FrequencyPriorityInformation::= CHOICE {
    eNDC        SubscriberProfileIDforRFP,
    nGRAN       RAT-FrequencySelectionPriority,
    choice-extension            ProtocolIE-SingleContainer { { RAT-FrequencyPriorityInformation-ExtIEs} }
}

RAT-FrequencyPriorityInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

RAT-FrequencySelectionPriority::= INTEGER (1.. 256, ...)

RBSetConfiguration ::= SEQUENCE {
    subcarrierSpacing               SubcarrierSpacing,
    rBSetSize                       RBSetSize,
    nUmberRBsets                    INTEGER(1..maxnoofRBsetsPerCell),
    iE-Extensions                   ProtocolExtensionContainer { { RBSetConfiguration-ExtIEs} } OPTIONAL
}

RBSetConfiguration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RBSetSize ::=   ENUMERATED { rb2, rb4, rb8, rb16, rb32, rb64}



Re-routingEnableIndicator ::= ENUMERATED {
    true,
    false,
    ...
}

Recommended-SSBs-for-Paging-List ::= SEQUENCE (SIZE(1.. maxCellingNBDU)) OF Recommended-SSBs-for-Paging-List-Item

Recommended-SSBs-for-Paging-List-Item::= SEQUENCE {
    nRCGI                       NRCGI,
    sSBs-forPaging-List         SSBs-forPaging-List,
```

```
	iE-Extensions				ProtocolExtensionContainer { { Recommended-SSBs-for-Paging-List-Item-ExtIEs} } OPTIONAL
}

Recommended-SSBs-for-Paging-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}



Redcap-Bcast-Information ::= BIT STRING(SIZE(8))

RedCapIndication ::= ENUMERATED {true, ...}

Reestablishment-Indication  ::= ENUMERATED  {
	reestablished,
	...
}

ReferencePoint ::= CHOICE {
	coordinateID					CoordinateID,
	referencePointCoordinate		AccessPointPosition,
	referencePointCoordinateHA		NGRANHighAccuracyAccessPointPosition,
	choice-Extension				ProtocolIE-SingleContainer { { ReferencePoint-ExtIEs} }
}

ReferencePoint-ExtIEs F1AP-PROTOCOL-IES ::= {
	{ID id-LocalOrigin  CRITICALITY ignore TYPE LocalOrigin PRESENCE mandatory},
	...
}

LocalOrigin ::= SEQUENCE {
	relativeCoordinateID			CoordinateID,
	horizontalAxesOrientation		INTEGER (0..3599),
	referencePointCoordinateHA		NGRANHighAccuracyAccessPointPosition		OPTIONAL,
	iE-Extensions				ProtocolExtensionContainer { { LocalOrigin-ExtIEs} } OPTIONAL,
	...
}

LocalOrigin-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

ReferenceSFN ::= INTEGER (0..1023)

ReferenceSignal ::= CHOICE {
	nZP-CSI-RS						NZP-CSI-RS-ResourceID,
	sSB								SSB,
	sRS								SRSResourceID,
	positioningSRS					SRSPosResourceID,
	dL-PRS							DL-PRS,
	choice-extension				ProtocolIE-SingleContainer {{ReferenceSignal-ExtIEs }}
}

ReferenceSignal-ExtIEs F1AP-PROTOCOL-IES ::= {
```

```
    ...
}


RA-RNTI ::= INTEGER (0..65535, ...)

ReferenceConfiguration ::= CHOICE {
    rEQUESTforLowerLayerConfiguration           RequestforLowerLayerConfiguration,
    referenceConfiguration                  ReferenceConfigurationInformation,
    choice-extension ProtocolIE-SingleContainer { { ReferenceConfiguration-ExtIEs } }
}

ReferenceConfiguration-ExtIEs F1AP-PROTOCOL-IES ::= {
...
}


RelativeCartesianLocation ::= SEQUENCE {
    xYZunit                     ENUMERATED {mm, cm, dm, ...},
    xvalue                      INTEGER (-65536..65535),
    yvalue                      INTEGER (-65536..65535),
    zvalue                      INTEGER (-32768..32767),
    locationUncertainty         LocationUncertainty,
    iE-Extensions               ProtocolExtensionContainer { { RelativeCartesianLocation-ExtIEs} } OPTIONAL
}

RelativeCartesianLocation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RelativeGeodeticLocation ::= SEQUENCE {
    milli-Arc-SecondUnits       ENUMERATED {zerodot03, zerodot3, three, ...},
    heightUnits                 ENUMERATED {mm, cm, m, ...},
    deltaLatitude               INTEGER (-1024.. 1023),
    deltaLongitude              INTEGER (-1024.. 1023),
    deltaHeight                 INTEGER (-1024.. 1023),
    locationUncertainty         LocationUncertainty,
    iE-extensions               ProtocolExtensionContainer {{RelativeGeodeticLocation-ExtIEs }} OPTIONAL
}

RelativeGeodeticLocation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RemoteUELocalID ::= INTEGER (0..255, ...)


ReferenceTime ::= OCTET STRING

RegistrationRequest ::= ENUMERATED{start, stop, add, ...}

ReportCharacteristics ::= BIT STRING (SIZE(32))

ReportingGranularitykminus1 ::= INTEGER(0..3940097)
```

```
ReportingGranularitykminus2 ::= INTEGER(0..7880193)


ReportingGranularitykminus3 ::= INTEGER(0..15760385)

ReportingGranularitykminus4 ::= INTEGER(0..31520769)

ReportingGranularitykminus5 ::= INTEGER(0..63041537)

ReportingGranularitykminus6 ::= INTEGER(0..126083073)


ReportingGranularitykminus1AdditionalPath ::= INTEGER(0..32701)

ReportingGranularitykminus2AdditionalPath ::= INTEGER(0..65401)

ReportingGranularitykminus3AdditionalPath ::= INTEGER(0..130801)

ReportingGranularitykminus4AdditionalPath ::= INTEGER(0..261601)

ReportingGranularitykminus5AdditionalPath ::= INTEGER(0..523201)

ReportingGranularitykminus6AdditionalPath ::= INTEGER(0..1046401)


ReportingPeriodicity ::= ENUMERATED{ms500, ms1000, ms2000, ms5000, ms10000, ...}

FailureReportingWithoutRLFReport ::= SEQUENCE {
    gNB-DU-UE-F1AP-ID       GNB-DU-UE-F1AP-ID,
    reestRecoveryCGI        NRCGI,
    failureType             RLFReportFailureType,
    iE-Extensions           ProtocolExtensionContainer { { FailureReportingWithoutRLFReport-ExtIEs } }    OPTIONAL,
    ...
}

FailureReportingWithoutRLFReport-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


RequestedBandCombinationIndex ::= OCTET STRING

RequestedFeatureSetEntryIndex ::= OCTET STRING

RequestedP-MaxFR2 ::= OCTET STRING

Requested-PDCCH-BlindDetectionSCG ::= OCTET STRING

RequestedSRSPreconfigurationCharacteristics-List ::= SEQUENCE (SIZE (1.. maxnoPreconfiguredSRS)) OF RequestedSRSPreconfigurationCharacteristics-
Item

RequestedSRSPreconfigurationCharacteristics-Item ::= SEQUENCE {
```

```
	requestedSRSTransmissionCharacteristics		RequestedSRSTransmissionCharacteristics,
	iE-Extensions			ProtocolExtensionContainer {{ RequestedSRSPreconfigurationCharacteristics-Item-ExtIEs}} OPTIONAL,
	...
}

RequestedSRSPreconfigurationCharacteristics-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

RequestedSRSTransmissionCharacteristics ::= SEQUENCE {
	numberOfTransmissions		INTEGER (0..500, ...)		OPTIONAL,
	-- The above IE shall be present if the Resource Type IE is set to "periodic" --
	resourceType				ENUMERATED	{periodic, semi-persistent, aperiodic,...},
	bandwidthSRS				BandwidthSRS,
	sRSResourceSetList			SRSResourceSetList				OPTIONAL,
	sSBInformation				SSBInformation				OPTIONAL,
	iE-Extensions			ProtocolExtensionContainer { { RequestedSRSTransmissionCharacteristics-ExtIEs} } OPTIONAL
}

RequestedSRSTransmissionCharacteristics-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-SrsFrequency		CRITICALITY ignore EXTENSION SrsFrequency		PRESENCE optional }|
	{ ID id-BW-Aggregation-Request-Indication		CRITICALITY ignore EXTENSION BW-Aggregation-Request-Indication	PRESENCE optional }|
	{ ID id-PosValidityAreaCellList					CRITICALITY ignore EXTENSION PosValidityAreaCellList		PRESENCE optional }|
	{ ID id-ValidityAreaSpecificSRSInformation		CRITICALITY ignore EXTENSION ValidityAreaSpecificSRSInformation PRESENCE optional }|
	{ ID id-ValidityAreaSpecificSRSInformationExtended	CRITICALITY ignore EXTENSION ValidityAreaSpecificSRSInformationExtended PRESENCE optional },
	...
}


RequestType ::= ENUMERATED {offer, execution, ...}

ResourceCoordinationEUTRACellInfo ::= SEQUENCE {
	eUTRA-Mode-Info							EUTRA-Coex-Mode-Info,
	eUTRA-PRACH-Configuration				EUTRA-PRACH-Configuration,
	iE-Extensions	ProtocolExtensionContainer { { ResourceCoordinationEUTRACellInfo-ExtIEs } } OPTIONAL,
	...
}

ResourceCoordinationEUTRACellInfo-ExtIEs	F1AP-PROTOCOL-EXTENSION ::= {
	{ID id-IgnorePRACHConfiguration		CRITICALITY reject EXTENSION IgnorePRACHConfiguration		PRESENCE optional },
	...
}

ResourceCoordinationTransferInformation ::= SEQUENCE {
	meNB-Cell-ID								EUTRA-Cell-ID,
	resourceCoordinationEUTRACellInfo		ResourceCoordinationEUTRACellInfo	OPTIONAL,
	iE-Extensions	ProtocolExtensionContainer { { ResourceCoordinationTransferInformation-ExtIEs } }	OPTIONAL,
	...
}

ResourceCoordinationTransferInformation-ExtIEs	F1AP-PROTOCOL-EXTENSION ::= {
	...
}
```

```
ResourceCoordinationTransferContainer ::= OCTET STRING


ResourceMapping ::= SEQUENCE {
    startPosition           INTEGER (0..13),
    nrofSumbols             ENUMERATED {n1, n2, n4, n8, n12},
    iE-Extensions       ProtocolExtensionContainer { { ResourceMapping-ExtIEs} }    OPTIONAL,
    ...
}

ResourceMapping-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


ResourceSetType  ::= CHOICE {
    periodic            ResourceSetTypePeriodic,
    semi-persistent     ResourceSetTypeSemi-persistent,
    aperiodic           ResourceSetTypeAperiodic,
    choice-extension                ProtocolIE-SingleContainer {{ ResourceSetType-ExtIEs }}
}

ResourceSetType-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

ResourceSetTypePeriodic ::= SEQUENCE {
    periodicSet         ENUMERATED{true, ...},
    iE-Extensions       ProtocolExtensionContainer { { ResourceSetTypePeriodic-ExtIEs} }    OPTIONAL
}

ResourceSetTypePeriodic-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ResourceSetTypeSemi-persistent ::= SEQUENCE {
    semi-persistentSet  ENUMERATED{true, ...},
    iE-Extensions       ProtocolExtensionContainer { { ResourceSetTypeSemi-persistent-ExtIEs} } OPTIONAL
}

ResourceSetTypeSemi-persistent-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ResourceSetTypeAperiodic ::= SEQUENCE {
    sRSResourceTrigger-List     INTEGER(1..3),
    slotoffset                  INTEGER(0..32),
    iE-Extensions               ProtocolExtensionContainer { { ResourceSetTypeAperiodic-ExtIEs} } OPTIONAL
}

ResourceSetTypeAperiodic-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
RepetitionFactorExtended ::=  ENUMERATED {n3, n5, n6, n7, n8, n10, n12, n14, ...}
RepetitionPeriod ::= INTEGER (0..131071, ...)

ReportingRequestType ::= SEQUENCE {
    eventType                       EventType,
    reportingPeriodicityValue                       ReportingPeriodicityValue       OPTIONAL,
    -- The above IE shall be present if the Event Type IE is set to "periodic" in the Event Type IE.
    iE-Extensions                   ProtocolExtensionContainer { {ReportingRequestType-ExtIEs} }  OPTIONAL
}

ReportingRequestType-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ResourceType ::= CHOICE {
    periodic            ResourceTypePeriodic,
    semi-persistent     ResourceTypeSemi-persistent,
    aperiodic           ResourceTypeAperiodic,
    choice-extension                ProtocolIE-SingleContainer {{ ResourceType-ExtIEs }}
}

ResourceType-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

ResourceTypePeriodic ::= SEQUENCE {
    periodicity        ENUMERATED{slot1, slot2, slot4, slot5, slot8, slot10, slot16, slot20, slot32, slot40, slot64, slot80, slot160, slot320,
slot640, slot1280, slot2560, ...},
    offset              INTEGER(0..2559, ...),
    iE-Extensions       ProtocolExtensionContainer { { ResourceTypePeriodic-ExtIEs} }   OPTIONAL
}

ResourceTypePeriodic-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ResourceTypeSemi-persistent ::= SEQUENCE {
    periodicity        ENUMERATED{slot1, slot2, slot4, slot5, slot8, slot10, slot16, slot20, slot32, slot40, slot64, slot80, slot160, slot320,
slot640, slot1280, slot2560, ...},
    offset              INTEGER(0..2559, ...),
    iE-Extensions       ProtocolExtensionContainer { { ResourceTypeSemi-persistent-ExtIEs} }    OPTIONAL
}

ResourceTypeSemi-persistent-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ResourceTypeAperiodic ::= SEQUENCE {
    aperiodicResourceType      ENUMERATED{true, ...},
    iE-Extensions       ProtocolExtensionContainer { { ResourceTypeAperiodic-ExtIEs} }  OPTIONAL
}

ResourceTypeAperiodic-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}

ResourceTypePos ::= CHOICE {
    periodic            ResourceTypePeriodicPos,
    semi-persistent     ResourceTypeSemi-persistentPos,
    aperiodic           ResourceTypeAperiodicPos,
    choice-extension    ProtocolIE-SingleContainer {{ ResourceTypePos-ExtIEs }}
}

ResourceTypePos-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

ResourceTypePeriodicPos ::= SEQUENCE {
    periodicity         SRS-Periodicity,
    offset              INTEGER(0..81919, ...),
    iE-Extensions       ProtocolExtensionContainer { { ResourceTypePeriodicPos-ExtIEs} }    OPTIONAL
}

ResourceTypePeriodicPos-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-SRSPosPeriodicConfigHyperSFNIndex           CRITICALITY ignore  EXTENSION SRSPosPeriodicConfigHyperSFNIndex     PRESENCE optional },
    ...
}

ResourceTypeSemi-persistentPos ::= SEQUENCE {
    periodicity         SRS-Periodicity,
    offset              INTEGER(0..81919, ...),
    iE-Extensions       ProtocolExtensionContainer { { ResourceTypeSemi-persistentPos-ExtIEs} } OPTIONAL
}

ResourceTypeSemi-persistentPos-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-SRSPosPeriodicConfigHyperSFNIndex           CRITICALITY ignore  EXTENSION SRSPosPeriodicConfigHyperSFNIndex     PRESENCE optional },
    ...
}

ResourceTypeAperiodicPos ::= SEQUENCE {
    slotOffset          INTEGER (0..32),
    iE-Extensions       ProtocolExtensionContainer { { ResourceTypeAperiodicPos-ExtIEs} }   OPTIONAL
}

ResourceTypeAperiodicPos-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ReportingThreshold ::= INTEGER(0..4000000000, ...)

RLCDuplicationInformation ::= SEQUENCE {
    rLCDuplicationStateList             RLCDuplicationStateList,
    primaryPathIndication           PrimaryPathIndication   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {RLCDuplicationInformation-ExtIEs} }   OPTIONAL
}

RLCDuplicationInformation-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}

RLCDuplicationStateList ::= SEQUENCE (SIZE(1..maxnoofRLCDuplicationState)) OF RLCDuplicationState-Item

RLCDuplicationState-Item ::=SEQUENCE {
    duplicationState        DuplicationState,
    iE-Extensions   ProtocolExtensionContainer { {RLCDuplicationState-Item-ExtIEs } }   OPTIONAL,
    ...
}


RLCDuplicationState-Item-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RLCFailureIndication ::= SEQUENCE {
    assocatedLCID               LCID,
    iE-Extensions               ProtocolExtensionContainer { {RLCFailureIndication-ExtIEs} } OPTIONAL
}

RLCFailureIndication-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RLCMode ::= ENUMERATED {
    rlc-am,
    rlc-um-bidirectional,
    rlc-um-unidirectional-ul,
    rlc-um-unidirectional-dl,
    ...
}

RLC-Status ::= SEQUENCE {
    reestablishment-Indication  Reestablishment-Indication,
    iE-Extensions               ProtocolExtensionContainer { { RLC-Status-ExtIEs } } OPTIONAL,
    ...
}

RLC-Status-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RLFReportInformationList    ::= SEQUENCE (SIZE(1.. maxnoofRLFReports)) OF RLFReportInformationItem

RLFReportInformationItem    ::= SEQUENCE {
    nRUERLFReportContainer      NRUERLFReportContainer,
    uEAssitantIdentifier        GNB-DU-UE-F1AP-ID       OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { RLFReportInformationItem-ExtIEs} } OPTIONAL,
    ...
}

RLFReportInformationItem-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-C-RNTI               CRITICALITY ignore EXTENSION C-RNTI                 PRESENCE optional }|
    {ID id-rLFReportFailureType CRITICALITY ignore EXTENSION RLFReportFailureType       PRESENCE optional },
```

```
    ...
}

RLFReportFailureType ::= ENUMERATED{ toolateLTM, tooearlyLTM, lTMtowrongcell,...}

RIMRSDetectionStatus ::= ENUMERATED {rs-detected, rs-disappeared, ...}

RRCContainer ::= OCTET STRING

RRCContainer-RRCSetupComplete ::= OCTET STRING

RRCDeliveryStatus ::= SEQUENCE  {
    delivery-status             PDCP-SN,
    triggering-message          PDCP-SN,
    iE-Extensions               ProtocolExtensionContainer { { RRCDeliveryStatus-ExtIEs } } OPTIONAL}

RRCDeliveryStatus-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


RRCDeliveryStatusRequest ::= ENUMERATED {true, ...}

RRCReconfigurationCompleteIndicator ::= ENUMERATED {
    true,
     ...,
    failure
}

RRC-Terminating-IAB-Donor-Related-Info  ::= SEQUENCE {
    rRC-TerminatingIAB-Donor-gNB-ID         GlobalGNB-ID,
    mobileIAB-MT-BAP-Address                BAPAddress,
    iE-Extensions                   ProtocolExtensionContainer { { RRC-Terminating-IAB-Donor-Related-Info-ExtIEs} } OPTIONAL,
    ...
}

RRC-Terminating-IAB-Donor-Related-Info-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


RRC-Version ::= SEQUENCE    {
    latest-RRC-Version          BIT STRING (SIZE(3)),
    iE-Extensions               ProtocolExtensionContainer { { RRC-Version-ExtIEs } }   OPTIONAL}

RRC-Version-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-latest-RRC-Version-Enhanced      CRITICALITY ignore EXTENSION OCTET STRING (SIZE(3))   PRESENCE optional },
    ...
}

RoutingID ::= OCTET STRING

ResponseTime ::= SEQUENCE {
    time                INTEGER (1..128,...),
```

```
    timeUnit    ENUMERATED {second, ten-seconds, ten-milliseconds,...},
    iE-Extensions       ProtocolExtensionContainer { { ResponseTime-ExtIEs} }   OPTIONAL,
    ...
}

ResponseTime-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RACHConfiguration ::= OCTET STRING

RequestforRACHConfiguration  ::= ENUMERATED {true, ...}

RequestforLowerLayerConfiguration::= ENUMERATED {true, ...}



RxTxTimingErrorMargin ::= ENUMERATED {tc0dot5, tc1, tc2, tc4, tc8, tc12, tc16, tc20, tc24, tc32, tc40, tc48, tc64, tc80, tc96, tc128, ...}

ReportingIntervalIMs  ::= INTEGER (1.. 999)

-- S

SBFD-AcrossSymbolType ::= ENUMERATED {true, ..., false}


SBFD-Frequency-Configuration ::= OCTET STRING

SCell-FailedtoSetup-Item    ::= SEQUENCE {
    sCell-ID            NRCGI   ,
    cause       Cause           OPTIONAL ,
    iE-Extensions   ProtocolExtensionContainer { { SCell-FailedtoSetup-ItemExtIEs } }   OPTIONAL,
    ...
}

SCell-FailedtoSetup-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SCell-FailedtoSetupMod-Item ::= SEQUENCE {
    sCell-ID            NRCGI   ,
    cause       Cause           OPTIONAL ,
    iE-Extensions   ProtocolExtensionContainer { { SCell-FailedtoSetupMod-ItemExtIEs } }    OPTIONAL,
    ...
}

SCell-FailedtoSetupMod-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SCell-ToBeRemoved-Item  ::= SEQUENCE {
    sCell-ID            NRCGI   ,
    iE-Extensions   ProtocolExtensionContainer { { SCell-ToBeRemoved-ItemExtIEs } } OPTIONAL,
    ...
```

```
}

SCell-ToBeRemoved-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SCell-ToBeSetup-Item ::= SEQUENCE {
    sCell-ID            NRCGI   ,
    sCellIndex          SCellIndex,
    sCellULConfigured       CellULConfigured    OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { SCell-ToBeSetup-ItemExtIEs } }   OPTIONAL,
    ...
}

SCell-ToBeSetup-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-ServingCellMO       CRITICALITY ignore  EXTENSION ServingCellMO     PRESENCE optional }|
    { ID id-ServingCellMO-Ondemand      CRITICALITY ignore  EXTENSION ServingCellMO     PRESENCE optional  },
    ...
}

SCell-ToBeSetupMod-Item ::= SEQUENCE {
    sCell-ID            NRCGI   ,
    sCellIndex          SCellIndex,
    sCellULConfigured       CellULConfigured    OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { SCell-ToBeSetupMod-ItemExtIEs } }    OPTIONAL,
    ...
}

SCell-ToBeSetupMod-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-ServingCellMO       CRITICALITY ignore  EXTENSION ServingCellMO     PRESENCE optional }|
    { ID id-ServingCellMO-Ondemand      CRITICALITY ignore  EXTENSION ServingCellMO     PRESENCE optional  },
    ...
}

SCellIndex ::=INTEGER (1..31, ...)

SCGActivationRequest ::= ENUMERATED {activate-scg, deactivate-scg, ...}

SCGActivationStatus ::= ENUMERATED {scg-activated, scg-deactivated, ...}

SCGIndicator    ::= ENUMERATED{released, ...}

SCPAC-Request ::= ENUMERATED {initiation, ...}

S-CPAC-Configuration    ::= SEQUENCE {
    referenceConfigurationInformation   ReferenceConfigurationInformation           OPTIONAL,
    completeCandidateConfigurationIndicator     CompleteCandidateConfigurationIndicator     OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { S-CPAC-Configuration-ExtIEs } }  OPTIONAL,
    ...
}

S-CPAC-Configuration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
S-CPACLowerLayerReferenceConfigRequest  ::= ENUMERATED{true, ...}

SCS-480 ::= INTEGER(0..319)

SCS-960 ::= INTEGER(0..639)

SCS-SpecificCarrier ::=                 SEQUENCE {
    offsetToCarrier                     INTEGER (0..2199,...),
    subcarrierSpacing                   ENUMERATED {kHz15, kHz30, kHz60, kHz120,..., kHz480, kHz960},
    carrierBandwidth                    INTEGER (1..275,...),
    iE-Extensions                       ProtocolExtensionContainer { { SCS-SpecificCarrier-ExtIEs } } OPTIONAL
}

SCS-SpecificCarrier-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SDTBearerConfigurationQueryIndication ::=   ENUMERATED {true, ...}

SDTBearerConfigurationInfo ::= SEQUENCE {
    sDTBearerConfig-List            SDTBearerConfig-List,
    iE-Extensions                   ProtocolExtensionContainer { { SDTBearerConfigurationInfo-ExtIEs } } OPTIONAL
}

SDTBearerConfigurationInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SDTBearerConfig-List ::= SEQUENCE (SIZE(1..maxnoofSDTBearers)) OF SDTBearerConfig-List-Item

SDTBearerConfig-List-Item ::= SEQUENCE{
    sDTBearerType                           SDTBearerType,
    sDTRLCBearerConfiguration               SDTRLCBearerConfiguration,
    iE-Extensions           ProtocolExtensionContainer {{ SDTBearerConfig-List-Item-ExtIEs}}      OPTIONAL
}

SDTBearerConfig-List-Item-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SDTBearerType ::= CHOICE {
    sRB                 SRBID,
    dRB                 DRBID,
    choice-extension                ProtocolIE-SingleContainer {{ SDTBearerType-ExtIEs }}
}

SDTBearerType-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

SDT-MAC-PHY-CG-Config ::= OCTET STRING

SDTInformation ::= SEQUENCE {
```

```
    sdtIndicator                    ENUMERATED {true,...},
    sdtAssistantInformation         ENUMERATED {singlepacket, multiplepackets,...}  OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { SDTInformation-ExtIEs } } OPTIONAL
}

SDTInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SDTRLCBearerConfiguration ::= OCTET STRING

SDT-Termination-Request ::= ENUMERATED {radio-link-problem, normal, ...,sdt-volume-threshold-crossed}

SDT-Volume-Threshold ::= INTEGER(1.. 192000,...)

Search-window-information ::= SEQUENCE {
    expectedPropagationDelay        INTEGER (-3841..3841,...),
    delayUncertainty                INTEGER (1..246,...),
    iE-Extensions                   ProtocolExtensionContainer { { Search-window-information-ExtIEs } } OPTIONAL
}

Search-window-information-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SerialNumber ::= BIT STRING (SIZE (16))

SIBType-PWS ::=INTEGER (6..8, ...)

SelectedBandCombinationIndex ::= OCTET STRING

SelectedFeatureSetEntryIndex ::= OCTET STRING

CG-ConfigInfo ::= OCTET STRING

ServCellInfoList ::= OCTET STRING

ServCellIndex ::= INTEGER (0..31, ...)

ServingCellMO ::= INTEGER (1..64, ...)

ServingCellMO-List-Item ::= SEQUENCE {
    servingCellMO                   ServingCellMO,
    sSB-Frequency                   INTEGER (0..3279165),
    iE-Extensions                   ProtocolExtensionContainer { { ServingCellMO-List-Item-ExtIEs } } OPTIONAL
}

ServingCellMO-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ServingCellMO-encoded-in-CGC-List ::= SEQUENCE (SIZE(1.. maxNrofBWPs)) OF ServingCellMO-encoded-in-CGC-Item

ServingCellMO-encoded-in-CGC-Item ::= SEQUENCE {
```

```
    servingCellMO                   ServingCellMO,
    iE-Extensions                   ProtocolExtensionContainer { { ServingCellMO-encoded-in-CGC-Item-ExtIEs } }  OPTIONAL,
    ...
}

ServingCellMO-encoded-in-CGC-Item-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-BWP-Id  CRITICALITY ignore  EXTENSION BWP-Id    PRESENCE optional },
    ...
}

Served-Cell-Information ::= SEQUENCE {
    nRCGI                           NRCGI,
    nRPCI                           NRPCI,
    fiveGS-TAC                      FiveGS-TAC          OPTIONAL,
    configured-EPS-TAC              Configured-EPS-TAC      OPTIONAL,
    servedPLMNs                     ServedPLMNs-List,
    nR-Mode-Info                    NR-Mode-Info,
    measurementTimingConfiguration  OCTET STRING,
    iE-Extensions       ProtocolExtensionContainer { {Served-Cell-Information-ExtIEs} } OPTIONAL,
    ...
}

Served-Cell-Information-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {   ID id-RANAC                         CRITICALITY ignore  EXTENSION RANAC                         PRESENCE optional }|
    {   ID id-ExtendedServedPLMNs-List      CRITICALITY ignore  EXTENSION ExtendedServedPLMNs-List   PRESENCE optional }|
    {   ID id-Cell-Direction                CRITICALITY ignore  EXTENSION Cell-Direction             PRESENCE optional }|
    {   ID id-BPLMN-ID-Info-List            CRITICALITY ignore  EXTENSION BPLMN-ID-Info-List         PRESENCE optional }|
    {   ID id-Cell-Type                     CRITICALITY ignore  EXTENSION CellType                   PRESENCE optional}|
    {   ID id-ConfiguredTACIndication       CRITICALITY ignore  EXTENSION ConfiguredTACIndication    PRESENCE optional }|
    {   ID id-AggressorgNBSetID             CRITICALITY ignore  EXTENSION AggressorgNBSetID          PRESENCE optional}|
    {   ID id-VictimgNBSetID                CRITICALITY ignore  EXTENSION VictimgNBSetID             PRESENCE optional}|
    {   ID id-IAB-Info-IAB-DU               CRITICALITY ignore  EXTENSION IAB-Info-IAB-DU            PRESENCE optional}|
    {   ID id-SSB-PositionsInBurst          CRITICALITY ignore  EXTENSION SSB-PositionsInBurst       PRESENCE optional }|
    {   ID id-NRPRACHConfig                 CRITICALITY ignore  EXTENSION NRPRACHConfig              PRESENCE optional }|
    {   ID id-SFN-Offset                    CRITICALITY ignore  EXTENSION SFN-Offset                 PRESENCE optional }|
    {   ID id-NPNBroadcastInformation       CRITICALITY reject  EXTENSION NPNBroadcastInformation    PRESENCE optional }|
    {   ID id-Supported-MBS-FSA-ID-List             CRITICALITY ignore  EXTENSION Supported-MBS-FSA-ID-List                 PRESENCE optional }|
    {   ID id-Redcap-Bcast-Information      CRITICALITY ignore  EXTENSION Redcap-Bcast-Information   PRESENCE optional }|
    {   ID id-ERedcap-Bcast-Information     CRITICALITY ignore  EXTENSION ERedcap-Bcast-Information  PRESENCE optional }|
    {   ID id-XR-Bcast-Information          CRITICALITY ignore  EXTENSION XR-Bcast-Information       PRESENCE optional }|
    {   ID id-BarringExemptionforEmerCallInfo               CRITICALITY ignore  EXTENSION BarringExemptionforEmerCallInfo         PRESENCE
optional }|
    {   ID id-NZP-CSI-RS-Resources-Config   CRITICALITY ignore  EXTENSION NZP-CSI-RS-Resources-Config   PRESENCE optional }|
    {   ID id-SRS-Resource-Configuration    CRITICALITY ignore  EXTENSION SRS-Resource-Configuration PRESENCE optional }|
    {   ID id-OnDemandSIB1                  CRITICALITY ignore  EXTENSION   OnDemandSIB1             PRESENCE optional },
    ...
}

Serving-Cells-List ::= SEQUENCE (SIZE(1..maxnoofServingCells)) OF Serving-Cells-List-Item

Serving-Cells-List-Item ::= SEQUENCE{
    nRCGI                               NRCGI,
    iAB-MT-Cell-NA-Resource-Configuration-Mode-Info       IAB-MT-Cell-NA-Resource-Configuration-Mode-Info     OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer {{Serving-Cells-List-Item-ExtIEs}}       OPTIONAL
```

```
}

Serving-Cells-List-Item-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Supported-MBS-FSA-ID-List::= SEQUENCE (SIZE(1.. maxnoofMBSFSAs)) OF MBS-FrequencySelectionArea-Identity

MBS-FrequencySelectionArea-Identity::= OCTET STRING (SIZE(3))

SFN-Offset ::= SEQUENCE {
    sFN-Time-Offset                 BIT STRING (SIZE(24)),
    iE-Extensions       ProtocolExtensionContainer { {SFN-Offset-ExtIEs} } OPTIONAL,
    ...
}

SFN-Offset-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
        ...
}

Served-Cells-To-Add-Item ::= SEQUENCE {
    served-Cell-Information     Served-Cell-Information,
    gNB-DU-System-Information   GNB-DU-System-Information    OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { Served-Cells-To-Add-ItemExtIEs} }  OPTIONAL,
    ...
}

Served-Cells-To-Add-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Served-Cells-To-Delete-Item ::= SEQUENCE {
    oldNRCGI                    NRCGI   ,
    iE-Extensions               ProtocolExtensionContainer { { Served-Cells-To-Delete-ItemExtIEs } } OPTIONAL,
    ...
}

Served-Cells-To-Delete-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Served-Cells-To-Modify-Item ::= SEQUENCE {
    oldNRCGI                    NRCGI                              ,
    served-Cell-Information     Served-Cell-Information     ,
    gNB-DU-System-Information   GNB-DU-System-Information   OPTIONAL    ,
    iE-Extensions               ProtocolExtensionContainer { { Served-Cells-To-Modify-ItemExtIEs } } OPTIONAL,
    ...
}

Served-Cells-To-Modify-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Served-EUTRA-Cells-Information::= SEQUENCE {
```

```
    eUTRA-Mode-Info                         EUTRA-Mode-Info,
    protectedEUTRAResourceIndication        ProtectedEUTRAResourceIndication,
    iE-Extensions                           ProtocolExtensionContainer { {Served-EUTRA-Cell-Information-ExtIEs} } OPTIONAL,
    ...
}

Served-EUTRA-Cell-Information-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Service-State ::= ENUMERATED {
    in-service,
    out-of-service,
    ...
}

Service-Status ::= SEQUENCE {
    service-state               Service-State,
    switchingOffOngoing         ENUMERATED {true, ...}  OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { Service-Status-ExtIEs } }    OPTIONAL,
    ...
}

Service-Status-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SelectedMeasurementQuantities ::= SEQUENCE {
    rSRP            INTEGER (0..127),
    rSRQ            INTEGER (0..127),
    sINR            INTEGER (0..127)    OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { SelectedMeasurementQuantities-ExtIEs } } OPTIONAL,
    ...
}

SelectedMeasurementQuantities-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ServingCellMeasurements ::= SEQUENCE {
    servingCellID                               NRCGI,
    sSBIndexwithMeasurementsList        SSBIndexwithMeasurementsList,
    iE-Extensions                           ProtocolExtensionContainer { { ServingCellMeasurements-ExtIEs } }  OPTIONAL,
    ...
}

ServingCellMeasurements-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
{ ID id-CSI-RSMeasurementsList          CRITICALITY ignore  EXTENSION CSI-RSMeasurementsList      PRESENCE optional },
    ...
}


SSBIndex  ::= INTEGER(0..63)
```

```
SSBIndexList    ::=  SEQUENCE (SIZE(1..maxnoofSSBIndices)) OF SSBIndex-Item

SSBIndex-Item   ::= SEQUENCE {
    sSBIndex            SSBIndex,
    iE-Extensions       ProtocolExtensionContainer { { SSBIndex-Item-ExtIEs } } OPTIONAL}

SSBIndex-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SSBIndexwithMeasurementsList    ::=  SEQUENCE (SIZE(1..maxnoofSSBIndices)) OF SSBIndexwithMeasurements-Item

SSBIndexwithMeasurements-Item   ::= SEQUENCE {
    sSBIndex            SSBIndex,
    selectedMeasurementQuantities   SelectedMeasurementQuantities,
    iE-Extensions       ProtocolExtensionContainer { { SSBIndexwithMeasurements-Item-ExtIEs } } OPTIONAL}

SSBIndexwithMeasurements-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SSB-resource-config ::= OCTET STRING

RelativeTime1900 ::=    BIT STRING (SIZE (64))

ShortDRXCycleLength ::=  ENUMERATED {ms2, ms3, ms4, ms5, ms6, ms7, ms8, ms10, ms14, ms16, ms20, ms30, ms32, ms35, ms40, ms64, ms80, ms128, ms160,
ms256, ms320, ms512, ms640, ...}

ShortNonIntegerDRXCycleLength ::=  ENUMERATED { ms1001over240, ms25over6, ms25over3, ms1001over120, ms100over9, ms25over2, ms40over3, ms125over9,
ms50over3, ms1001over60, ms125over6, ms200over9, ms100over3, ms1001over30, ms125over3, ms1001over24, ms200over3, ...}

ShortDRXCycleTimer ::= INTEGER (1..16)

SIB1-message ::= OCTET STRING

SIB10-message ::= OCTET STRING

SIB12-message ::= OCTET STRING

SIB13-message ::= OCTET STRING

SIB14-message ::= OCTET STRING

SIB15-message ::= OCTET STRING

SIB17-message ::= OCTET STRING

SIB20-message ::= OCTET STRING

SIB24-message ::= OCTET STRING

SIB22-message ::= OCTET STRING

SIB23-message ::= OCTET STRING
```

```
SIB17bis-message ::= OCTET STRING

SItype ::= INTEGER (1..32, ...)

SItype-List ::= SEQUENCE (SIZE(1.. maxnoofSITypes)) OF SItype-Item

SItype-Item ::= SEQUENCE {
    sItype      SItype  ,
    iE-Extensions   ProtocolExtensionContainer { { SItype-ItemExtIEs } }    OPTIONAL
}

SItype-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SibtypetobeupdatedListItem ::= SEQUENCE {
    sIBtype             INTEGER (2..32,...),
    sIBmessage          OCTET STRING,
    valueTag            INTEGER (0..31,...),
    iE-Extensions       ProtocolExtensionContainer { { SibtypetobeupdatedListItem-ExtIEs } }    OPTIONAL,
    ...
}

SibtypetobeupdatedListItem-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-areaScope    CRITICALITY ignore  EXTENSION   AreaScope   PRESENCE optional},
    ...
}

SidelinkRelayConfiguration ::= SEQUENCE {
    gNB-DU-UE-F1APIDofRelayUE           GNB-DU-UE-F1AP-ID,
    remoteUELocalID                     RemoteUELocalID,
    sidelinkConfigurationContainer      SidelinkConfigurationContainer      OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { SidelinkRelayConfiguration-ExtIEs } }    OPTIONAL,
    ...
}

SidelinkRelayConfiguration-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


SidelinkConfigurationContainer ::= OCTET STRING

SLDRBID ::= INTEGER (1..512, ...)

SLDRBInformation ::= SEQUENCE {
    sLDRB-QoS               PC5QoSParameters,
    flowsMappedToSLDRB-List FlowsMappedToSLDRB-List,
    ...
}
SLDRBs-FailedToBeModified-Item  ::= SEQUENCE {
    sLDRBID     SLDRBID     ,
    cause       Cause       OPTIONAL,
```

```
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-FailedToBeModified-ItemExtIEs } } OPTIONAL
}

SLDRBs-FailedToBeModified-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SLDRBs-FailedToBeSetup-Item ::= SEQUENCE {
    sLDRBID SLDRBID,
    cause   Cause   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-FailedToBeSetup-ItemExtIEs } }        OPTIONAL
}

SLDRBs-FailedToBeSetup-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SLDRBs-FailedToBeSetupMod-Item  ::= SEQUENCE {
    sLDRBID         SLDRBID ,
    cause           Cause           OPTIONAL ,
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-FailedToBeSetupMod-ItemExtIEs } } OPTIONAL
}

SLDRBs-FailedToBeSetupMod-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SLDRBs-Modified-Item    ::= SEQUENCE {
    sLDRBID                         SLDRBID,
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-Modified-ItemExtIEs } }   OPTIONAL
}

SLDRBs-Modified-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SLDRBs-ModifiedConf-Item    ::= SEQUENCE {
    sLDRBID                         SLDRBID,
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-ModifiedConf-ItemExtIEs } }   OPTIONAL
}

SLDRBs-ModifiedConf-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SLDRBs-Required-ToBeModified-Item   ::= SEQUENCE {
    sLDRBID                         SLDRBID,
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-Required-ToBeModified-ItemExtIEs } }  OPTIONAL
}

SLDRBs-Required-ToBeModified-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
SLDRBs-Required-ToBeReleased-Item   ::= SEQUENCE {
    sLDRBID     SLDRBID,
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-Required-ToBeReleased-ItemExtIEs } }  OPTIONAL
}

SLDRBs-Required-ToBeReleased-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SLDRBs-Setup-Item ::= SEQUENCE {
    sLDRBID                         SLDRBID,
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-Setup-ItemExtIEs } }  OPTIONAL
}

SLDRBs-Setup-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SLDRBs-SetupMod-Item    ::= SEQUENCE {
    sLDRBID                         SLDRBID,
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-SetupMod-ItemExtIEs } }   OPTIONAL
}

SLDRBs-SetupMod-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SLDRBs-ToBeModified-Item    ::= SEQUENCE {
    sLDRBID                     SLDRBID,
    sLDRBInformation            SLDRBInformation        OPTIONAL,
    rLCMode                     RLCMode                 OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-ToBeModified-ItemExtIEs } }   OPTIONAL
}

SLDRBs-ToBeModified-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-duplicationIndication  CRITICALITY ignore EXTENSION   DuplicationIndication      PRESENCE optional},
    ...
}

SLDRBs-ToBeReleased-Item    ::= SEQUENCE {
    sLDRBID         SLDRBID,
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-ToBeReleased-ItemExtIEs } }   OPTIONAL
}

SLDRBs-ToBeReleased-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SLDRBs-ToBeSetup-Item ::= SEQUENCE  {
    sLDRBID                     SLDRBID,
    sLDRBInformation            SLDRBInformation,
    rLCMode                     RLCMode,

    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-ToBeSetup-ItemExtIEs } }  OPTIONAL
```

```
}

SLDRBs-ToBeSetup-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-duplicationIndication  CRITICALITY ignore EXTENSION   DuplicationIndication      PRESENCE optional},
    ...
}

SLDRBs-ToBeSetupMod-Item    ::= SEQUENCE {
    sLDRBID                     SLDRBID,
    sLDRBInformation            SLDRBInformation,
    rLCMode                     RLCMode         OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { SLDRBs-ToBeSetupMod-ItemExtIEs } }   OPTIONAL
}

SLDRBs-ToBeSetupMod-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-duplicationIndication  CRITICALITY ignore EXTENSION   DuplicationIndication      PRESENCE optional},
    ...
}

SLDRXCycleList ::= SEQUENCE (SIZE(1.. maxnoofSLdestinations)) OF SLDRXCycleItem
SLDRXCycleItem ::= SEQUENCE {
    rXUEID                  BIT STRING (SIZE(24)),
    sLDRXInformation        SLDRXInformation,
    iE-Extensions           ProtocolExtensionContainer { { SLDRXCycleItem-ExtIEs } }    OPTIONAL,
    ...
}

SLDRXCycleItem-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SLDRXInformation    ::= CHOICE {
    sLDRXCycle          SLDRXCycleLength,
    nosLDRX             SLDRXConfigurationIndicator,
    choice-extension    ProtocolIE-SingleContainer { { SLDRXInformation-ExtIEs} }
}

SLDRXCycleLength ::= ENUMERATED{ms10, ms20, ms32, ms40, ms60, ms64, ms70, ms80, ms128, ms160, ms256, ms320, ms512, ms640, ms1024, ms1280, ms2048,
ms2560, ms5120, ms10240, ...}

SLDRXConfigurationIndicator ::= ENUMERATED{ release, ...}


SLDRXInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

SL-PHY-MAC-RLC-Config ::= OCTET STRING

SL-PHY-MAC-RLC-ConfigExt ::= OCTET STRING

SL-RLC-ChannelToAddModList::= OCTET STRING

SL-ConfigDedicatedEUTRA-Info ::= OCTET STRING
```

```
SliceAvailableCapacity ::= SEQUENCE {
    sliceAvailableCapacityList  SliceAvailableCapacityList,
    iE-Extensions               ProtocolExtensionContainer { { SliceAvailableCapacity-ExtIEs} } OPTIONAL
}

SliceAvailableCapacity-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SliceAvailableCapacityList ::= SEQUENCE (SIZE(1.. maxnoofBPLMNsNR)) OF SliceAvailableCapacityItem

SliceAvailableCapacityItem ::= SEQUENCE {
    pLMNIdentity                    PLMN-Identity,
    sNSSAIAvailableCapacity-List    SNSSAIAvailableCapacity-List,
    iE-Extensions   ProtocolExtensionContainer { { SliceAvailableCapacityItem-ExtIEs} } OPTIONAL
}

SliceAvailableCapacityItem-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SNSSAIAvailableCapacity-List ::= SEQUENCE (SIZE(1.. maxnoofSliceItems)) OF SNSSAIAvailableCapacity-Item

SNSSAIAvailableCapacity-Item ::= SEQUENCE {
    sNSSAI      SNSSAI,
    sliceAvailableCapacityValueDownlink INTEGER (0..100)    OPTIONAL,
    sliceAvailableCapacityValueUplink   INTEGER (0..100)    OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { SNSSAIAvailableCapacity-Item-ExtIEs } }  OPTIONAL
}

SNSSAIAvailableCapacity-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SliceRadioResourceStatus ::= SEQUENCE {
    sliceRadioResourceStatus    SliceRadioResourceStatus-List,
    iE-Extensions               ProtocolExtensionContainer { { SliceRadioResourceStatus-ExtIEs} } OPTIONAL
}

SliceRadioResourceStatus-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


SliceRadioResourceStatus-List ::= SEQUENCE (SIZE(1..maxnoofBPLMNsNR)) OF SliceRadioResourceStatus-Item

SliceRadioResourceStatus-Item::= SEQUENCE {
    pLMNIdentity                    PLMN-Identity,
    sNSSAIRadioResourceStatus-List  SNSSAIRadioResourceStatus-List,
    iE-Extensions                   ProtocolExtensionContainer { { SliceRadioResourceStatus-Item-ExtIEs} } OPTIONAL
}

SliceRadioResourceStatus-Item-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

SNSSAIRadioResourceStatus-List ::= SEQUENCE (SIZE(1.. maxnoofSliceItems)) OF SNSSAIRadioResourceStatus-Item

SNSSAIRadioResourceStatus-Item ::= SEQUENCE {
    sNSSAI      SNSSAI,
    sNSSAIdlGBRPRBusage         INTEGER (0..100),
    sNSSAIulGBRPRBusage         INTEGER (0..100),
    sNSSAIdlNonGBRPRBusage      INTEGER (0..100),
    sNSSAIulNonGBRPRBusage      INTEGER (0..100),
    sNSSAIdlTotalPRBallocation  INTEGER (0..100),
    sNSSAIulTotalPRBallocation  INTEGER (0..100),
    iE-Extensions               ProtocolExtensionContainer { { SNSSAIRadioResourceStatus-Item-ExtIEs } }  OPTIONAL
}

SNSSAIRadioResourceStatus-Item-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SliceSupportList ::= SEQUENCE (SIZE(1.. maxnoofSliceItems)) OF SliceSupportItem

SliceSupportItem ::= SEQUENCE {
    sNSSAI  SNSSAI,
    iE-Extensions               ProtocolExtensionContainer { { SliceSupportItem-ExtIEs } }  OPTIONAL
}

SliceSupportItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SliceToReportList ::= SEQUENCE (SIZE(1.. maxnoofBPLMNsNR)) OF SliceToReportItem

SliceToReportItem ::= SEQUENCE {
    pLMNIdentity                PLMN-Identity,
    sNSSAIlist                  SNSSAI-list,
    iE-Extensions               ProtocolExtensionContainer { { SliceToReportItem-ExtIEs} } OPTIONAL
}

SliceToReportItem-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SlotNumber ::= INTEGER (0..79)

SLPositioning-Ranging-Service-Info ::= SEQUENCE{
    sLPositioning-Ranging-Authorized        SLPositioning-Ranging-Authorized,
    rSPP-transport-QoS-parameters           RSPP-transport-QoS-parameters       OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { SLPositioning-Ranging-Service-Info-ExtIEs} }  OPTIONAL,
    ...
}

SLPositioning-Ranging-Service-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}


SLPositioning-Ranging-Authorized ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

RSPP-transport-QoS-parameters ::= SEQUENCE {
    rSPPQoSFlowList             RSPPQoSFlowList,
    rSPPLinkAggregateBitRates   BitRate                                             OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { RSPP-transport-QoS-parameters-ExtIEs} }  OPTIONAL,
    ...
}

RSPP-transport-QoS-parameters-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
RSPPQoSFlowList ::= SEQUENCE (SIZE(1..maxnoofRSPPQoSFlows)) OF RSPPQoSFlowItem

RSPPQoSFlowItem ::= SEQUENCE {
    pQI                 FiveQI,
    rSPPFlowBitRates    RSPPFlowBitRates                                            OPTIONAL,
    range               Range                                                       OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { RSPPQoSFlowItem-ExtIEs} }    OPTIONAL,
    ...
}

RSPPQoSFlowItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RSPPFlowBitRates ::= SEQUENCE {
    guaranteedFlowBitRate       BitRate,
    maximumFlowBitRate          BitRate,
    iE-Extensions       ProtocolExtensionContainer { { RSPPFlowBitRates-ExtIEs} }   OPTIONAL,
    ...
}

RSPPFlowBitRates-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


SNSSAI-list ::= SEQUENCE (SIZE(1.. maxnoofSliceItems)) OF SNSSAI-Item

SNSSAI-Item ::= SEQUENCE {
    sNSSAI      SNSSAI,
    iE-Extensions               ProtocolExtensionContainer { { SNSSAI-Item-ExtIEs } }   OPTIONAL
}

SNSSAI-Item-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}

Slot-Configuration-List ::= SEQUENCE (SIZE(1.. maxnoofslots)) OF Slot-Configuration-Item

Slot-Configuration-Item ::= SEQUENCE {
    slotIndex               INTEGER (0..5119, ...),
    symbolAllocInSlot       SymbolAllocInSlot,
    iE-Extensions           ProtocolExtensionContainer { { Slot-Configuration-ItemExtIEs } }    OPTIONAL
}

Slot-Configuration-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SlotOffsetForRemainingHopsList ::= SEQUENCE (SIZE (1..maxnoHopsMinusOne)) OF SlotOffsetForRemainingHopsItem

SlotOffsetForRemainingHopsItem ::= SEQUENCE {
    slotOffsetRemainingHops         SlotOffsetRemainingHops,
    iE-Extensions                   ProtocolExtensionContainer { { SlotOffsetForRemainingHopsItem-ExtIEs} } OPTIONAL,
    ...
}

SlotOffsetForRemainingHopsItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SlotOffsetRemainingHops ::= CHOICE {
    aperiodic           SlotOffsetRemainingHopsAperiodic,
    semi-persistent     SlotOffsetRemainingHopsSemiPersistent,
    periodic            SlotOffsetRemainingHopsPeriodic,
    choice-extension        ProtocolIE-SingleContainer {{ SlotOffsetRemainingHops-ExtIEs }}
}

SlotOffsetRemainingHops-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

SlotOffsetRemainingHopsAperiodic ::= SEQUENCE {
    slotOffset          INTEGER (1..32)         OPTIONAL,
    startPosition       INTEGER (0..13)         OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { SlotOffsetRemainingHopsAperiodic-ExtIEs} } OPTIONAL,
    ...
}

SlotOffsetRemainingHopsAperiodic-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SlotOffsetRemainingHopsSemiPersistent ::= SEQUENCE {
    sRSperiodicity      SRS-Periodicity,
    offset              INTEGER(0..81919, ...),
    startPosition       INTEGER (0..13)         OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { SlotOffsetRemainingHopsSemiPersistent-ExtIEs} }  OPTIONAL,
    ...
```

```
}

SlotOffsetRemainingHopsSemiPersistent-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SlotOffsetRemainingHopsPeriodic ::= SEQUENCE {
    sRSperiodicity      SRS-Periodicity,
    offset              INTEGER(0..81919, ...),
    startPosition       INTEGER (0..13)         OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { SlotOffsetRemainingHopsSemiPeriodic-ExtIEs} } OPTIONAL,
    ...
}

SlotOffsetRemainingHopsSemiPeriodic-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SNSSAI ::= SEQUENCE {
    sST         OCTET STRING (SIZE(1)),
    sD          OCTET STRING (SIZE(3))  OPTIONAL    ,
    iE-Extensions               ProtocolExtensionContainer { { SNSSAI-ExtIEs } }    OPTIONAL
}

SNSSAI-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SpatialDirectionInformation ::= SEQUENCE {
    nR-PRSBeamInformation           NR-PRSBeamInformation,
    iE-Extensions                   ProtocolExtensionContainer { { SpatialDirectionInformation-ExtIEs } } OPTIONAL
}

SpatialDirectionInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SpatialRelationInfo ::= SEQUENCE {
    spatialRelationforResourceID                    SpatialRelationforResourceID,
    iE-Extensions       ProtocolExtensionContainer { {SpatialRelationInfo-ExtIEs} } OPTIONAL
}

SpatialRelationInfo-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SpatialRelationforResourceID ::= SEQUENCE (SIZE(1..maxnoofSpatialRelations)) OF SpatialRelationforResourceIDItem

SpatialRelationforResourceIDItem ::= SEQUENCE {
    referenceSignal     ReferenceSignal,
    iE-Extensions       ProtocolExtensionContainer { {SpatialRelationforResourceIDItem-ExtIEs} }  OPTIONAL
}

SpatialRelationforResourceIDItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

SpatialRelationPerSRSResource ::= SEQUENCE {
    spatialRelationPerSRSResource-List  SpatialRelationPerSRSResource-List,
    iE-Extensions       ProtocolExtensionContainer { { SpatialRelationPerSRSResource-ExtIEs} }  OPTIONAL,
    ...
}

SpatialRelationPerSRSResource-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SpatialRelationPerSRSResource-List::= SEQUENCE(SIZE (1.. maxnoSRS-ResourcePerSet)) OF SpatialRelationPerSRSResourceItem

SpatialRelationPerSRSResourceItem ::= SEQUENCE {
    referenceSignal     ReferenceSignal,
    iE-Extensions       ProtocolExtensionContainer { { SpatialRelationPerSRSResourceItem-ExtIEs} }   OPTIONAL,
    ...
}

SpatialRelationPerSRSResourceItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SpatialRelationPos ::= CHOICE {
    sSBPos                  SSB,
    pRSInformationPos       PRSInformationPos,
    choice-extension        ProtocolIE-SingleContainer {{ SpatialInformationPos-ExtIEs }}
}

SpatialInformationPos-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

SpectrumSharingGroupID ::= INTEGER (1..maxCellineNB)

SRBID ::= INTEGER (0..3, ..., 4 | 5 | 6)

SRBs-FailedToBeSetup-Item   ::= SEQUENCE {
    sRBID       SRBID   ,
    cause       Cause   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { SRBs-FailedToBeSetup-ItemExtIEs } }  OPTIONAL,
    ...
}

SRBs-FailedToBeSetup-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRBs-FailedToBeSetupMod-Item    ::= SEQUENCE {
    sRBID       SRBID       ,
    cause       Cause       OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { SRBs-FailedToBeSetupMod-ItemExtIEs } }   OPTIONAL,
```

```
    ...
}

SRBs-FailedToBeSetupMod-ItemExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRBs-Modified-Item ::= SEQUENCE {
    sRBID                           SRBID,
    lCID                            LCID,
    iE-Extensions   ProtocolExtensionContainer { { SRBs-Modified-ItemExtIEs } } OPTIONAL,
    ...
}

SRBs-Modified-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRBs-Required-ToBeReleased-Item ::= SEQUENCE {
    sRBID   SRBID,
    iE-Extensions   ProtocolExtensionContainer { { SRBs-Required-ToBeReleased-ItemExtIEs } }    OPTIONAL,
    ...
}

SRBs-Required-ToBeReleased-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRBs-Setup-Item ::= SEQUENCE {
    sRBID                           SRBID,
    lCID                                LCID,
    iE-Extensions   ProtocolExtensionContainer { { SRBs-Setup-ItemExtIEs } }    OPTIONAL,
    ...
}

SRBs-Setup-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRBs-SetupMod-Item ::= SEQUENCE {
    sRBID                       SRBID,
    lCID                            LCID,
    iE-Extensions   ProtocolExtensionContainer { { SRBs-SetupMod-ItemExtIEs } } OPTIONAL,
    ...
}

SRBs-SetupMod-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRBs-ToBeReleased-Item  ::= SEQUENCE {
    sRBID       SRBID,
    iE-Extensions   ProtocolExtensionContainer { { SRBs-ToBeReleased-ItemExtIEs } } OPTIONAL,
    ...
```

```
}

SRBs-ToBeReleased-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRBs-ToBeSetup-Item ::= SEQUENCE {
    sRBID    SRBID  ,
    duplicationIndication   DuplicationIndication   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { SRBs-ToBeSetup-ItemExtIEs } }    OPTIONAL,
    ...
}

SRBs-ToBeSetup-ItemExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalDuplicationIndication CRITICALITY ignore  EXTENSION AdditionalDuplicationIndication    PRESENCE optional }|
    { ID id-SDTRLCBearerConfiguration       CRITICALITY ignore  EXTENSION SDTRLCBearerConfiguration          PRESENCE optional }|
    { ID id-SRBMappingInfo                  CRITICALITY ignore  EXTENSION UuRLCChannelID                     PRESENCE optional },
    ...
}

SRBs-ToBeSetupMod-Item  ::= SEQUENCE {
    sRBID   SRBID,
    duplicationIndication   DuplicationIndication   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { SRBs-ToBeSetupMod-ItemExtIEs } } OPTIONAL,
    ...
}

SRBs-ToBeSetupMod-ItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalDuplicationIndication CRITICALITY ignore  EXTENSION AdditionalDuplicationIndication    PRESENCE optional }|
    { ID id-SRBMappingInfo                  CRITICALITY ignore  EXTENSION UuRLCChannelID                     PRESENCE optional }|
    { ID id-CG-SDTindicatorSetup            CRITICALITY reject  EXTENSION CG-SDTindicatorSetup               PRESENCE optional },
    ...
}

SRSCarrier-List ::= SEQUENCE (SIZE(1.. maxnoSRS-Carriers)) OF SRSCarrier-List-Item

SRSCarrier-List-Item ::= SEQUENCE {
    pointA                          INTEGER (0..3279165),
    uplinkChannelBW-PerSCS-List     UplinkChannelBW-PerSCS-List,
    activeULBWP                     ActiveULBWP,
    pci                             NRPCI       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { SRSCarrier-List-Item-ExtIEs } } OPTIONAL
}

SRSCarrier-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRSConfig  ::= SEQUENCE {
    sRSResource-List            SRSResource-List        OPTIONAL,
    posSRSResource-List         PosSRSResource-List     OPTIONAL,
    sRSResourceSet-List         SRSResourceSet-List     OPTIONAL,
    posSRSResourceSet-List      PosSRSResourceSet-List  OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { SRSConfig-ExtIEs } } OPTIONAL
```

```
}

SRSConfig-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRSConfiguration ::= SEQUENCE {
    sRSCarrier-List     SRSCarrier-List,
    iE-Extensions       ProtocolExtensionContainer { { SRSConfiguration-ExtIEs } } OPTIONAL
}

SRSConfiguration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-AggregatedPosSRSResourceSetList CRITICALITY ignore EXTENSION AggregatedPosSRSResourceSetList   PRESENCE optional},
    ...
}

SRS-Resource-Configuration ::= SEQUENCE {
    sRS-Resource-Configuration-List                     SRS-Resource-Configuration-List,
    iE-Extensions                       ProtocolExtensionContainer { {SRS-Resource-Configuration-ExtIEs} } OPTIONAL,
    ...
}

SRS-Resource-Configuration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRS-Resource-Configuration-List ::= SEQUENCE (SIZE(1..maxnoofSRS-Resources)) OF SRS-Resource-Configuration-Item

SRS-Resource-Configuration-Item ::= SEQUENCE {
    sRS-Resource                        OCTET STRING,
    iE-Extensions                       ProtocolExtensionContainer { {SRS-Resource-Configuration-Item-ExtIEs} } OPTIONAL,
    ...
}

SRS-Resource-Configuration-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRS-Resource-Indication ::= ENUMERATED {true, ...}

SrsFrequency ::= INTEGER (0..3279165)

SRSPortIndex ::= ENUMERATED {id1000, id1001, id1002, id1003,...}

SRSPosPeriodicConfigHyperSFNIndex ::=ENUMERATED {even0, odd1}

SRSPosResourceID ::= INTEGER (0..63)

SRSPreconfiguration-List ::= SEQUENCE (SIZE (1.. maxnoPreconfiguredSRS)) OF SRSPreconfiguration-Item

SRSPreconfiguration-Item ::= SEQUENCE {
    sRSPosRRCInactiveValidityAreaConfig     SRSPosRRCInactiveValidityAreaConfig,
    posValidityAreaCellList                 PosValidityAreaCellList,
    iE-Extensions           ProtocolExtensionContainer {{ SRSPreconfiguration-Item-ExtIEs}}       OPTIONAL,
```

```
    ...
}

SRSPreconfiguration-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRSResource::= SEQUENCE {
    sRSResourceID                   SRSResourceID,
    nrofSRS-Ports                   ENUMERATED {port1, ports2, ports4},
    transmissionComb                TransmissionComb,
    startPosition                   INTEGER (0..13),
    nrofSymbols                     ENUMERATED {n1, n2, n4},
    repetitionFactor                ENUMERATED {n1, n2, n4},
    freqDomainPosition              INTEGER (0..67),
    freqDomainShift                 INTEGER (0..268),
    c-SRS                           INTEGER (0..63),
    b-SRS                           INTEGER (0..3),
    b-hop                           INTEGER (0..3),
    groupOrSequenceHopping          ENUMERATED { neither, groupHopping, sequenceHopping },
    resourceType                    ResourceType,
    sequenceId                      INTEGER (0..1023),
    iE-Extensions                   ProtocolExtensionContainer { { SRSResource-ExtIEs } } OPTIONAL
}

SRSResource-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-nrofSymbolsExtended         CRITICALITY ignore EXTENSION NrofSymbolsExtended          PRESENCE optional}|
    { ID id-repetitionFactorExtended        CRITICALITY ignore EXTENSION RepetitionFactorExtended    PRESENCE optional}|
    { ID id-startRBHopping          CRITICALITY ignore EXTENSION StartRBHopping             PRESENCE optional}|
    { ID id-startRBIndex            CRITICALITY ignore EXTENSION StartRBIndex           PRESENCE optional},
    ...
}

SRSResourceID ::= INTEGER (0..63)

SRSResourceID-List::= SEQUENCE (SIZE (1..maxnoSRS-ResourcePerSet)) OF SRSResourceID

SRSResource-List ::= SEQUENCE (SIZE (1..maxnoSRS-Resources)) OF SRSResource

SRSResourceSet::= SEQUENCE {
    sRSResourceSetID                SRSResourceSetID,
    sRSResourceID-List              SRSResourceID-List,
    resourceSetType                 ResourceSetType,
    iE-Extensions                   ProtocolExtensionContainer { { SRSResourceSet-ExtIEs } } OPTIONAL
}

SRSResourceSet-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRSResourceSetID ::= INTEGER (0..15, ...)

SRSResourceSetList ::= SEQUENCE (SIZE(1.. maxnoSRS-ResourceSets)) OF SRSResourceSetItem
```

```
SRSResourceSetItem ::= SEQUENCE {
    numSRSresourcesperset       INTEGER (1..16, ...)    OPTIONAL,
    periodicityList             PeriodicityList         OPTIONAL,
    spatialRelationInfo         SpatialRelationInfo     OPTIONAL,
    pathlossReferenceInfo       PathlossReferenceInfo   OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { SRSResourceSetItemExtIEs } } OPTIONAL
}

SRSResourceSetItemExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-SRSSpatialRelationPerSRSResource    CRITICALITY ignore  EXTENSION SpatialRelationPerSRSResource PRESENCE optional},
    ...
}

SRSResourceSet-List ::= SEQUENCE (SIZE (1..maxnoSRS-ResourceSets)) OF SRSResourceSet

SRSResourceTrigger ::= SEQUENCE {
    aperiodicSRSResourceTriggerList                 AperiodicSRSResourceTriggerList,
    iE-Extensions       ProtocolExtensionContainer { {SRSResourceTrigger-ExtIEs} }  OPTIONAL
}

SRSResourceTrigger-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


SRSResourcetype ::= SEQUENCE {
    sRSResourceTypeChoice                   SRSResourceTypeChoice,
    iE-Extensions       ProtocolExtensionContainer { { SRSResourcetype-ExtIEs} }    OPTIONAL,
    ...
}

SRSResourcetype-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-SRSPortIndex            CRITICALITY ignore EXTENSION SRSPortIndex   PRESENCE optional },
    ...
}

SRSResourceTypeChoice ::= CHOICE {
    sRSResourceInfo             SRSInfo,
    posSRSResourceInfo          PosSRSInfo,
    choice-extension            ProtocolIE-SingleContainer { { SRSResourceTypeChoice-ExtIEs} }
}

SRSResourceTypeChoice-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

SRSInfo ::= SEQUENCE {
    sRSResource         SRSResourceID,
    ...
}

SRS-Periodicity ::= ENUMERATED{slot1, slot2, slot4, slot5, slot8, slot10, slot16, slot20, slot32, slot40, slot64, slot80, slot160, slot320,
slot640, slot1280, slot2560, slot5120, slot10240, slot40960, slot81920, ..., slot128, slot256, slot512, slot20480}
```

```
SRSPosRRCInactiveConfig ::= OCTET STRING

SRSPosRRCInactiveValidityAreaConfig ::= OCTET STRING

SRSPosRRCInactiveQueryIndication ::= ENUMERATED {true, ...}

PosSRSInfo ::= SEQUENCE {
    posSRSResourceID        SRSPosResourceID,
    ...
}

SRSReservationType ::= ENUMERATED {reserve, release, ...}


SSB ::= SEQUENCE {
    pCI-NR              NRPCI,
    ssb-index           SSB-Index   OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {SSB-ExtIEs} } OPTIONAL
}


SSBCoverageModification-List ::= SEQUENCE (SIZE (1..maxnoofSSBAreas)) OF SSBCoverageModification-Item

SSBCoverageModification-Item::= SEQUENCE {
    sSBIndex                        INTEGER(0..63),
    sSBCoverageState                SSBCoverageState,
    iE-Extensions       ProtocolExtensionContainer { { SSBCoverageModification-Item-ExtIEs} }   OPTIONAL,
...
}

SSBCoverageModification-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


SSBCoverageState ::= INTEGER (0..15, ...)


SSB-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SSB-freqInfo ::= INTEGER (0..maxNRARFCN)

SSB-Index ::= INTEGER(0..63)

SSB-subcarrierSpacing ::=  ENUMERATED {kHz15, kHz30, kHz120, kHz240, spare3, spare2, spare1, ...}

SSB-transmissionPeriodicity ::= ENUMERATED {sf10, sf20, sf40, sf80, sf160, sf320, sf640, ..., sf5}

SSB-transmissionTimingOffset ::= INTEGER (0..127, ...)

SSB-transmissionBitmap ::= CHOICE {
    shortBitmap         BIT STRING (SIZE (4)),
```

```
    mediumBitmap        BIT STRING (SIZE (8)),
    longBitmap          BIT STRING (SIZE (64)),
    choice-extension    ProtocolIE-SingleContainer { { SSB-transmisisonBitmap-ExtIEs} }
}

SSB-transmisisonBitmap-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

SSBAreaCapacityValueList ::= SEQUENCE (SIZE(1.. maxnoofSSBAreas)) OF    SSBAreaCapacityValueItem

SSBAreaCapacityValueItem ::= SEQUENCE {
    sSBIndex                INTEGER(0..63),
    sSBAreaCapacityValue    INTEGER (0..100),
    iE-Extensions           ProtocolExtensionContainer { { SSBAreaCapacityValueItem-ExtIEs} } OPTIONAL
}

SSBAreaCapacityValueItem-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SSBAreaRadioResourceStatusList::= SEQUENCE (SIZE(1.. maxnoofSSBAreas)) OF   SSBAreaRadioResourceStatusItem

SSBAreaRadioResourceStatusItem::= SEQUENCE {
    sSBIndex                    INTEGER(0..63),
    sSBAreaDLGBRPRBusage        INTEGER (0..100),
    sSBAreaULGBRPRBusage        INTEGER (0..100),
    sSBAreaDLnon-GBRPRBusage    INTEGER (0..100),
    sSBAreaULnon-GBRPRBusage    INTEGER (0..100),
    sSBAreaDLTotalPRBusage      INTEGER (0..100),
    sSBAreaULTotalPRBusage      INTEGER (0..100),
    dLschedulingPDCCHCCEusage   INTEGER (0..100)        OPTIONAL,
    uLschedulingPDCCHCCEusage   INTEGER (0..100)        OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { SSBAreaRadioResourceStatusItem-ExtIEs} } OPTIONAL
}

SSBAreaRadioResourceStatusItem-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SSBInformation ::= SEQUENCE {
    sSBInformationList  SSBInformationList,
    iE-Extensions       ProtocolExtensionContainer { { SSBInformation-ExtIEs } }    OPTIONAL
}

SSBInformation-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SSBInformationList ::= SEQUENCE (SIZE(1.. maxnoofSSBs)) OF SSBInformationItem

SSBInformationItem ::= SEQUENCE {
    sSB-Configuration   SSB-TF-Configuration,
    pCI-NR              NRPCI,
```

```
    iE-Extensions       ProtocolExtensionContainer { { SSBInformationItem-ExtIEs } }    OPTIONAL
}

SSBInformationItem-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SSB-PositionsInBurst ::= CHOICE {
    shortBitmap                     BIT STRING (SIZE (4)),
    mediumBitmap                    BIT STRING (SIZE (8)),
    longBitmap                      BIT STRING (SIZE (64)),
    choice-extension                ProtocolIE-SingleContainer { {SSB-PositionsInBurst-ExtIEs} }
}

SSB-PositionsInBurst-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

SSBs-activated-List ::= SEQUENCE (SIZE(1.. maxnoofSSBAreas)) OF SSB-Index

SSBs-forPaging-List ::= SEQUENCE (SIZE(1.. maxnoofSSBAreas)) OF SSB-Index

SSBs-toBeActivated-List ::= SEQUENCE (SIZE(1.. maxnoofSSBAreas)) OF SSB-Index


SSB-TF-Configuration ::= SEQUENCE {
    sSB-frequency               INTEGER (0..3279165),
    sSB-subcarrier-spacing      ENUMERATED {kHz15, kHz30, kHz60, kHz120, kHz240, ..., kHz480, kHz960},
    -- The value kHz60 is not supported in this version of the specification.
    sSB-Transmit-power          INTEGER (-60..50),
    sSB-periodicity             ENUMERATED {ms5, ms10, ms20, ms40, ms80, ms160, ...},
    sSB-half-frame-offset       INTEGER(0..1),
    sSB-SFN-offset              INTEGER(0..15),
    sSB-position-in-burst       SSB-PositionsInBurst        OPTIONAL,
    sFNInitialisationTime       RelativeTime1900        OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { SSB-TF-Configuration-ExtIEs} } OPTIONAL
}

SSB-TF-Configuration-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


SSBToReportList ::= SEQUENCE (SIZE(1.. maxnoofSSBAreas)) OF SSBToReportItem

SSBToReportItem ::= SEQUENCE {
    sSBIndex                    INTEGER(0..63),
    iE-Extensions               ProtocolExtensionContainer { { SSBToReportItem-ExtIEs} } OPTIONAL
}

SSBToReportItem-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
StartRBIndex  ::= CHOICE{
    freqScalingFactor2   INTEGER(0..1),
    freqScalingFactor4   INTEGER(0..3),
    choice-extension     ProtocolIE-SingleContainer { { StartRBIndex-ExtIEs} }
}
StartRBIndex-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

StartRBHopping  ::= ENUMERATED {enable}

StartTimeAndDuration ::= SEQUENCE {
    startTime           RelativeTime1900                OPTIONAL,
    duration            INTEGER (0..90060, ...)         OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { StartTimeAndDuration-ExtIEs } }  OPTIONAL,
    ...
}

StartTimeAndDuration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SUL-Information ::= SEQUENCE {
    sUL-NRARFCN                         INTEGER (0..maxNRARFCN),
    sUL-transmission-Bandwidth          Transmission-Bandwidth,
    iE-Extensions                       ProtocolExtensionContainer { { SUL-InformationExtIEs} } OPTIONAL,
    ...
}

SUL-InformationExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-CarrierList             CRITICALITY ignore  EXTENSION NRCarrierList         PRESENCE optional }|
    { ID id-FrequencyShift7p5khz    CRITICALITY ignore  EXTENSION FrequencyShift7p5khz  PRESENCE optional },
    ...
}

SubcarrierSpacing ::=   ENUMERATED { kHz15, kHz30, kHz60, kHz120, kHz240, spare3, spare2, spare1, ...}

SubscriberProfileIDforRFP ::= INTEGER (1..256, ...)

SuccessfulHOReportInformationList::= SEQUENCE (SIZE(1.. maxnoofSuccessfulHOReports)) OF SuccessfulHOReportInformation-Item

SuccessfulHOReportInformation-Item ::= SEQUENCE {
    successfulHOReportContainer             OCTET STRING,
    iE-Extensions   ProtocolExtensionContainer { { SuccessfulHOReportInformation-Item-ExtIEs } }  OPTIONAL
}

SuccessfulHOReportInformation-Item-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SuccessfulPSCellChangeReportInformationList::= SEQUENCE (SIZE(1.. maxnoofSuccessfulPSCellChangeReports)) OF
SuccessfulPSCellChangeReportInformation-Item

SuccessfulPSCellChangeReportInformation-Item ::= SEQUENCE {
```

```
    successfulPSCellChangeReportContainer               OCTET STRING,
    iE-Extensions   ProtocolExtensionContainer { { SuccessfulPSCellChangeReportInformation-Item-ExtIEs } } OPTIONAL
}

SuccessfulPSCellChangeReportInformation-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


SULAccessIndication ::= ENUMERATED {true,...}


SupportedSULFreqBandItem ::= SEQUENCE {
    freqBandIndicatorNr             INTEGER (1..1024,...),
    iE-Extensions                   ProtocolExtensionContainer { { SupportedSULFreqBandItem-ExtIEs} } OPTIONAL,
    ...
}

SupportedSULFreqBandItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SupportedUETypeList ::= SEQUENCE (SIZE(1.. maxnoofUETypes)) OF SupportedUETypeList-Item

SupportedUETypeList-Item ::= SEQUENCE {
    supportedUEtype             ENUMERATED {non-redcap-eredcap-ue, redcap-eredcap-ue, ...},
    iE-Extensions               ProtocolExtensionContainer { { SupportedUETypeList-Item-ExtIEs } } OPTIONAL,
    ...
}

SupportedUETypeList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

SurvivalTime ::= INTEGER (0.. 1920000,...)

SymbolAllocInSlot ::= CHOICE {
    all-DL                  NULL,
    all-UL                  NULL,
    both-DL-and-UL          NumDLULSymbols,
    choice-extension        ProtocolIE-SingleContainer { { SymbolAllocInSlot-ExtIEs } }
}

SymbolAllocInSlot-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

SymbolIndex ::= INTEGER (0..13)


SystemFrameNumber ::= INTEGER (0..1023)

SystemInformationAreaID ::=BIT STRING (SIZE (24))
```

```
-- T

TAI ::= SEQUENCE {
    pLMN-Identity                           PLMN-Identity,
    fiveGS-TAC                              FiveGS-TAC,
    iE-Extensions                      ProtocolExtensionContainer { {TAI-ExtIEs} } OPTIONAL,
    ...
}

TAI-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TAAssistanceInfo ::=  ENUMERATED{zero, ...}

FiveGS-TAC ::= OCTET STRING (SIZE(3))

Configured-EPS-TAC ::= OCTET STRING (SIZE(2))

TagIDPointer ::= OCTET STRING

TargetCellList ::= SEQUENCE (SIZE(1..maxnoofCHOcells)) OF TargetCellList-Item

TargetCellList-Item ::= SEQUENCE {
    target-cell                                NRCGI,
    iE-Extensions                              ProtocolExtensionContainer { { TargetCellList-Item-ExtIEs} } OPTIONAL
}

TargetCellList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NSAGSupportList ::= SEQUENCE (SIZE(1.. maxnoofNSAGs)) OF NSAGSupportItem

NSAGSupportItem ::= SEQUENCE {
    nSAG-ID                 NSAG-ID,
    nSAGSliceSupport        ExtendedSliceSupportList,
    iE-Extensions           ProtocolExtensionContainer { {NSAGSupportItem-ExtIEs} } OPTIONAL,
    ...
}

NSAGSupportItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

NSAG-ID ::= INTEGER (0..255, ...)


TCIStatesConfigurationsList ::= OCTET STRING

TAValue ::= INTEGER (0..4095)


TDD-Info ::= SEQUENCE {
```

```
    nRFreqInfo                      NRFreqInfo,
    transmission-Bandwidth          Transmission-Bandwidth,
    iE-Extensions                   ProtocolExtensionContainer { {TDD-Info-ExtIEs} } OPTIONAL,
    ...
}

TDD-Info-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-IntendedTDD-DL-ULConfig  CRITICALITY ignore  EXTENSION   IntendedTDD-DL-ULConfig PRESENCE optional}|
    {ID id-TDD-UL-DLConfigCommonNR  CRITICALITY ignore  EXTENSION TDD-UL-DLConfigCommonNR   PRESENCE optional }|
    {ID id-CarrierList              CRITICALITY ignore  EXTENSION NRCarrierList            PRESENCE optional }|
    {ID id-Transmission-Bandwidth-asymmetric    CRITICALITY ignore  EXTENSION Transmission-Bandwidth-asymmetric    PRESENCE optional }|
    {ID id-SBFD-Frequency-Configuration                 CRITICALITY ignore  EXTENSION SBFD-Frequency-Configuration    PRESENCE optional },
    ...
}

TDD-InfoRel16 ::= SEQUENCE {
    tDD-FreqInfo                                FreqInfoRel16                                           OPTIONAL,
    sUL-FreqInfo                                FreqInfoRel16                                           OPTIONAL,
    tDD-UL-DLConfigCommonNR                     TDD-UL-DLConfigCommonNR                                 OPTIONAL,
    iE-Extensions                               ProtocolExtensionContainer { {TDD-InfoRel16-ExtIEs} }    OPTIONAL,
    ...
}

TDD-InfoRel16-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


TDD-UL-DLConfigCommonNR ::= OCTET STRING

TRPTEGInformation ::= CHOICE {
    rxTx-TEG            RxTxTEG,
    rx-TEG              RxTEG,
    choice-extension        ProtocolIE-SingleContainer { { TRPTEGInformation-ExtIEs} }
}

TRPTEGInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

RxTxTEG ::= SEQUENCE {
    tRP-RxTx-TEGInformation         TRP-RxTx-TEGInformation,
    tRP-Tx-TEGInformation           TRP-Tx-TEGInformation       OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { RxTxTEG-ExtIEs } }       OPTIONAL,
    ...
}

RxTxTEG-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

RxTEG ::= SEQUENCE {
    tRP-Rx-TEGInformation       TRP-Rx-TEGInformation,
    tRP-Tx-TEGInformation       TRP-Tx-TEGInformation,
```

```
    iE-Extensions           ProtocolExtensionContainer { { RxTEG-ExtIEs } }     OPTIONAL,
    ...
}

RxTEG-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TimingInformation ::= CHOICE {
    k0                  INTEGER (0..1970049),
    k1                  INTEGER (0..985025),
    k2                  INTEGER (0..492513),
    k3                  INTEGER (0..246257),
    k4                  INTEGER (0..123129),
    k5                  INTEGER (0..61565),
    choice-extension            ProtocolIE-SingleContainer { { TimingInformation-ExtIEs } }
}

TimingInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}


TimeReferenceInformation ::= SEQUENCE {
    referenceTime                   ReferenceTime,
    referenceSFN                    ReferenceSFN,
    uncertainty                     Uncertainty             OPTIONAL,
    timeInformationType             TimeInformationType     OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {TimeReferenceInformation-ExtIEs} }    OPTIONAL
}

TimeReferenceInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TimeInformationType ::= ENUMERATED {localClock}

TimeStamp ::= SEQUENCE {
    systemFrameNumber       SystemFrameNumber,
    slotIndex               TimeStampSlotIndex,
    measurementTime         RelativeTime1900    OPTIONAL,
    iE-Extension            ProtocolExtensionContainer { { TimeStamp-ExtIEs} }  OPTIONAL
}

TimeStamp-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-SymbolIndex  CRITICALITY ignore EXTENSION SymbolIndex   PRESENCE optional },
    ...
}

TimeStampSlotIndex ::= CHOICE {
    sCS-15          INTEGER(0..9),
    sCS-30          INTEGER(0..19),
    sCS-60          INTEGER(0..39),
    sCS-120         INTEGER(0..79),
```

```
    choice-extension        ProtocolIE-SingleContainer { { TimeStampSlotIndex-ExtIEs} }
}

TimeStampSlotIndex-ExtIEs F1AP-PROTOCOL-IES ::= {
    { ID id-SCS-480     CRITICALITY reject  TYPE SCS-480 PRESENCE mandatory}|
    { ID id-SCS-960     CRITICALITY reject  TYPE SCS-960 PRESENCE mandatory},
    ...
}

TimeToWait ::= ENUMERATED {v1s, v2s, v5s, v10s, v20s, v60s, ...}

TimingErrorMargin ::= ENUMERATED {m0Tc, m2Tc, m4Tc, m6Tc, m8Tc, m12Tc, m16Tc, m20Tc, m24Tc, m32Tc, m40Tc, m48Tc, m56Tc, m64Tc, m72Tc, m80Tc, ...}

TimingMeasurementQuality ::= SEQUENCE {
    measurementQuality      INTEGER(0..31),
    resolution              ENUMERATED{m0dot1, m1, m10, m30, ...},
    iE-Extensions           ProtocolExtensionContainer { { TimingMeasurementQuality-ExtIEs} }   OPTIONAL
}

TimingMeasurementQuality-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TimingReportingGranularityFactorExtended ::=INTEGER(-6..-1,...)

TimeWindowStart ::= SEQUENCE {
    systemFrameNumber       SystemFrameNumber,
    slotNumber              SlotNumber,
    symbolIndex             INTEGER (0..13),
    iE-Extension            ProtocolExtensionContainer { { TimeWindowStart-ExtIEs} }    OPTIONAL,
    ...
}

TimeWindowStart-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TimeWindowInformation-Measurement-List ::= SEQUENCE (SIZE (1.. maxnoofTimeWindowMea)) OF TimeWindowInformation-Measurement-Item


TimeWindowInformation-Measurement-Item ::= SEQUENCE {
    timeWindowDurationMeasurement       TimeWindowDurationMeasurement,
    timeWindowType                      ENUMERATED {single, periodic, ...},
    timeWindowPeriodicityMeasurement    TimeWindowPeriodicityMeasurement        OPTIONAL,
    -- This IE shall be present if the Time Window Type IE is set to the value “periodic”. --
    timeWindowStart                     TimeWindowStart,
    iE-Extension            ProtocolExtensionContainer { { TimeWindowInformation-Measurement-Item-ExtIEs} }  OPTIONAL,
    ...}

TimeWindowInformation-Measurement-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TimeWindowInformation-SRS-List ::= SEQUENCE (SIZE (1.. maxnoofTimeWindowSRS)) OF TimeWindowInformation-SRS-Item
```

```
TimeWindowInformation-SRS-Item ::= SEQUENCE {
    timeWindowStartSRS                  TimeWindowStartSRS,
    timeWindowDurationSRS               TimeWindowDurationSRS,
    timeWindowType                      ENUMERATED {single, periodic, ...},
    timeWindowPeriodicitySRS            TimeWindowPeriodicitySRS                OPTIONAL,
    -- The above IE shall be present if the Time Window Type IE is set to the value "periodic".
    iE-Extension            ProtocolExtensionContainer { { TimeWindowInformation-SRS-ExtIEs} }  OPTIONAL,
    ...
}

TimeWindowInformation-SRS-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TimeWindowDurationMeasurement ::= CHOICE {
    durationSlots       ENUMERATED {n1, n2, n4, n6, n8, n12, n16, ...},
    choice-extension        ProtocolIE-SingleContainer { { TimeWindowDurationMeasurement-ExtIEs} }
}

TimeWindowDurationMeasurement-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

TimeWindowDurationSRS ::= CHOICE {
    durationSymbols     ENUMERATED {n1, n2, n4, n8, n12, ...},
    durationSlots       ENUMERATED {n1, n2, n4, n6, n8, n12, n16, ...},
    choice-extension        ProtocolIE-SingleContainer { { TimeWindowDurationSRS-ExtIEs} }
}

TimeWindowDurationSRS-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

TimeWindowPeriodicityMeasurement ::= ENUMERATED {ms160, ms320, ms640, ms1280, ms2560, ms5120, ms10240, ms20480, ms40960, ms61440, ms81920,
ms368640, ms737280, ms1843200, ...}

TimeWindowPeriodicitySRS ::= ENUMERATED {ms0dot125, ms0dot25, ms0dot5, ms0dot625, ms1, ms1dot25, ms2, ms2dot5, ms4, ms5, ms8, ms10, ms16, ms20,
ms32, ms40, ms64, ms80, ms160, ms320, ms640, ms1280, ms2560, ms5120, ms10240, ...}

TimeWindowStartSRS ::= SEQUENCE {
    systemFrameNumber       SystemFrameNumber,
    slotNumber              SlotNumber,
    symbolIndex             SymbolIndex,
    iE-Extension            ProtocolExtensionContainer { { TimeWindowStartSRS-ExtIEs} } OPTIONAL,
    ...
}

TimeWindowStartSRS-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


TMGI ::=  OCTET STRING (SIZE(6))
```

```
TNLAssociationUsage ::= ENUMERATED {
    ue,
    non-ue,
    both,
    ...
}

TNLCapacityIndicator::= SEQUENCE {
    dLTNLOfferedCapacity        INTEGER (1.. 16777216,...),
    dLTNLAvailableCapacity      INTEGER (0.. 100,...),
    uLTNLOfferedCapacity        INTEGER (1.. 16777216,...),
    uLTNLAvailableCapacity      INTEGER (0.. 100,...),
    iE-Extensions   ProtocolExtensionContainer { { TNLCapacityIndicator-ExtIEs} } OPTIONAL
}

TNLCapacityIndicator-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TraceActivation ::= SEQUENCE {
    traceID                             TraceID,
    interfacesToTrace                   InterfacesToTrace,
    traceDepth                          TraceDepth,
    traceCollectionEntityIPAddress      TransportLayerAddress,
    iE-Extensions       ProtocolExtensionContainer { {TraceActivation-ExtIEs} } OPTIONAL
}

TraceActivation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    {ID id-mdtConfiguration         CRITICALITY ignore  EXTENSION   MDTConfiguration        PRESENCE optional}|
    {ID id-TraceCollectionEntityURI CRITICALITY ignore  EXTENSION URI-address               PRESENCE optional },
    ...
}

TraceDepth ::= ENUMERATED {
    minimum,
    medium,
    maximum,
    minimumWithoutVendorSpecificExtension,
    mediumWithoutVendorSpecificExtension,
    maximumWithoutVendorSpecificExtension,
    ... ,
    minimumOnlyVendorSpecificTraceRecord,
    mediumOnlyVendorSpecificTraceRecord,
    maximumOnlyVendorSpecificTraceRecord
}

TraceID ::= OCTET STRING (SIZE(8))

TrafficMappingInfo  ::= CHOICE {
    iPtolayer2TrafficMappingInfo                    IPtolayer2TrafficMappingInfo,
    bAPlayerBHRLCchannelMappingInfo                 BAPlayerBHRLCchannelMappingInfo,
    choice-extension                                ProtocolIE-SingleContainer { { TrafficMappingInfo-ExtIEs} }
}
```

```
TrafficMappingInfo-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

TransportLayerAddress       ::= BIT STRING (SIZE(1..160, ...))

TransactionID               ::= INTEGER (0..255, ...)

Transmission-Bandwidth ::= SEQUENCE {
    nRSCS   NRSCS,
    nRNRB   NRNRB,
    iE-Extensions               ProtocolExtensionContainer { { Transmission-Bandwidth-ExtIEs} } OPTIONAL,
    ...
}

Transmission-Bandwidth-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Transmission-Bandwidth-asymmetric ::= SEQUENCE {
    ul-Transmission-Bandwidth   Transmission-Bandwidth,
    dl-Transmission-Bandwidth   Transmission-Bandwidth,
    iE-Extensions               ProtocolExtensionContainer { { Transmission-Bandwidth-asymmetric-ExtIEs} } OPTIONAL,
    ...
}

Transmission-Bandwidth-asymmetric-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TransmissionComb ::= CHOICE {
    n2    SEQUENCE {
            combOffset-n2              INTEGER (0..1),
            cyclicShift-n2             INTEGER (0..7)
        },
    n4    SEQUENCE {
            combOffset-n4              INTEGER (0..3),
            cyclicShift-n4             INTEGER (0..11)
        },
    choice-extension                ProtocolIE-SingleContainer { { TransmissionComb-ExtIEs} }
}
TransmissionComb-ExtIEs F1AP-PROTOCOL-IES ::= {
    { ID id-transmissionCombn8  CRITICALITY reject TYPE TransmissionCombn8 PRESENCE mandatory},
    ...
}

TransmissionCombn8 ::= SEQUENCE {
    combOffset-n8              INTEGER (0..7),
    cyclicShift-n8             INTEGER (0..5),
    iE-Extensions              ProtocolExtensionContainer { { TransmissionCombn8-ExtIEs} } OPTIONAL
}

TransmissionCombn8-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
```

```
	...
}

TransmissionCombPos ::= CHOICE {
	n2	SEQUENCE {
			combOffset-n2				INTEGER (0..1),
			cyclicShift-n2				INTEGER (0..7)
		},
	n4	SEQUENCE {
			combOffset-n4				INTEGER (0..3),
			cyclicShift-n4				INTEGER (0..11)
		},
	n8	SEQUENCE {
			combOffset-n8				INTEGER (0..7),
			cyclicShift-n8				INTEGER (0..5)
		},

	choice-extension				ProtocolIE-SingleContainer { { TransmissionCombPos-ExtIEs} }
}
TransmissionCombPos-ExtIEs F1AP-PROTOCOL-IES ::= {
	...
}

TransmissionStopIndicator ::= ENUMERATED {true, ... }

Transport-UP-Layer-Address-Info-To-Add-List ::= SEQUENCE (SIZE(1.. maxnoofTLAs)) OF Transport-UP-Layer-Address-Info-To-Add-Item

Transport-UP-Layer-Address-Info-To-Add-Item ::= SEQUENCE {
	iP-SecTransportLayerAddress		TransportLayerAddress,
	gTPTransportLayerAddressToAdd			GTPTLAs							OPTIONAL,
	iE-Extensions					ProtocolExtensionContainer { { Transport-UP-Layer-Address-Info-To-Add-ItemExtIEs } }	OPTIONAL
}

Transport-UP-Layer-Address-Info-To-Add-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

Transport-UP-Layer-Address-Info-To-Remove-List	::= SEQUENCE (SIZE(1.. maxnoofTLAs)) OF Transport-UP-Layer-Address-Info-To-Remove-Item

Transport-UP-Layer-Address-Info-To-Remove-Item ::= SEQUENCE {
	iP-SecTransportLayerAddress		TransportLayerAddress,
	gTPTransportLayerAddressToRemove			GTPTLAs							OPTIONAL,
	iE-Extensions					ProtocolExtensionContainer { { Transport-UP-Layer-Address-Info-To-Remove-ItemExtIEs } }	OPTIONAL
}

Transport-UP-Layer-Address-Info-To-Remove-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

TransmissionActionIndicator ::= ENUMERATED {stop, ..., restart }

TRPBeamAntennaInformation ::= SEQUENCE {
	choice-TRP-Beam-Antenna-Info-Item		Choice-TRP-Beam-Antenna-Info-Item							,
	iE-Extensions			ProtocolExtensionContainer {{ TRPBeamAntennaInformation-ExtIEs}}			OPTIONAL,
```

```
    ...
}

TRPBeamAntennaInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

Choice-TRP-Beam-Antenna-Info-Item ::= CHOICE {
    reference                   TRPID,
    explicit                    TRP-BeamAntennaExplicitInformation,
    noChange                    NULL,
    choice-extension            ProtocolIE-SingleContainer { { Choice-TRP-Beam-Info-Item-ExtIEs } }
}

Choice-TRP-Beam-Info-Item-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

TRP-BeamAntennaExplicitInformation ::= SEQUENCE {
    trp-BeamAntennaAngles                   TRP-BeamAntennaAngles,
    lcs-to-gcs-translation                  LCS-to-GCS-Translation                                  OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer {{ TRP-BeamAntennaExplicitInformation-ExtIEs}}           OPTIONAL,
    ...
}

TRP-BeamAntennaExplicitInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}



TRP-BeamAntennaAngles ::= SEQUENCE (SIZE (1.. maxnoAzimuthAngles)) OF TRP-BeamAntennaAnglesList-Item

TRP-BeamAntennaAnglesList-Item ::= SEQUENCE {
    trp-azimuth-angle                   INTEGER (0..359),
    trp-azimuth-angle-fine              INTEGER (0..9)  OPTIONAL,
    trp-elevation-angle-list            SEQUENCE (SIZE (1.. maxnoElevationAngles)) OF TRP-ElevationAngleList-Item,
    iE-Extensions           ProtocolExtensionContainer {{ TRP-BeamAntennaAnglesList-Item-ExtIEs}}      OPTIONAL,
    ...
}

TRP-BeamAntennaAnglesList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TRP-ElevationAngleList-Item ::= SEQUENCE {
    trp-elevation-angle                 INTEGER (0..180),
    trp-elevation-angle-fine            INTEGER (0..9)  OPTIONAL,
    trp-beam-power-list                 SEQUENCE (SIZE (2..maxNumResourcesPerAngle)) OF TRP-Beam-Power-Item,
    iE-Extensions           ProtocolExtensionContainer {{ TRP-ElevationAngleList-Item-ExtIEs}}      OPTIONAL,
...
}
TRP-ElevationAngleList-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}

TRP-Beam-Power-Item ::= SEQUENCE {
    pRSResourceSetID                PRS-Resource-Set-ID                                             OPTIONAL,
    pRSResourceID                   PRS-Resource-ID,
    relativePower                   INTEGER (0..30), --negative value
    relativePowerFine               INTEGER (0..9)          OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer {{ TRP-Beam-Power-Item-ExtIEs}}              OPTIONAL,
    ...
}

TRP-Beam-Power-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


TRPID ::= INTEGER (0.. maxnoofTRPs, ...)

TRPInformation ::= SEQUENCE {
    tRPID                           TRPID,
    tRPInformationTypeResponseList  TRPInformationTypeResponseList,
    iE-Extensions                   ProtocolExtensionContainer { { TRPInformation-ExtIEs } }     OPTIONAL
}

TRPInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-Mobile-IAB-MT-UE-ID         CRITICALITY reject EXTENSION Mobile-IAB-MT-UE-ID        PRESENCE optional},
    -- The above IE shall be present if the TRP type IE is set to the value "mobile-trp"
    ...
}

TRPInformationItem ::= SEQUENCE {
    tRPInformation                  TRPInformation,
    iE-Extensions                   ProtocolExtensionContainer { { TRPInformationItem-ExtIEs } }  OPTIONAL
}

TRPInformationItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TRPInformationTypeItem ::= ENUMERATED {
        nrPCI,
        nG-RAN-CGI,
        arfcn,
        pRSConfig,
        sSBConfig,
        sFNInitTime,
        spatialDirectInfo,
        geoCoord,
        ...,
        trp-type,
        ondemandPRS,
        trpTxTeg,
        beam-antenna-info,
        mobile-trp-location-info
```

```
}


TRPInformationTypeResponseList ::= SEQUENCE (SIZE(1.. maxnoofTRPInfoTypes)) OF TRPInformationTypeResponseItem

TRPInformationTypeResponseItem ::= CHOICE {
    pCI-NR                              NRPCI,
    nG-RAN-CGI                          NRCGI,
    nRARFCN                             INTEGER (0..maxNRARFCN),
    pRSConfiguration                    PRSConfiguration,
    sSBinformation                      SSBInformation,
    sFNInitialisationTime               RelativeTime1900,
    spatialDirectionInformation         SpatialDirectionInformation,
    geographicalCoordinates             GeographicalCoordinates,
    choice-extension                    ProtocolIE-SingleContainer { { TRPInformationTypeResponseItem-ExtIEs} }
}

TRPInformationTypeResponseItem-ExtIEs F1AP-PROTOCOL-IES ::= {
    { ID id-TRPType                     CRITICALITY reject      TYPE TRPType                    PRESENCE mandatory }|
    { ID id-OnDemandPRS                 CRITICALITY reject      TYPE OnDemandPRS-Info           PRESENCE mandatory}|
    { ID id-TRPTxTEGAssociation         CRITICALITY reject      TYPE TRPTxTEGAssociation        PRESENCE mandatory}|
    { ID id-TRPBeamAntennaInformation   CRITICALITY reject      TYPE TRPBeamAntennaInformation  PRESENCE mandatory }|
    { ID id-Mobile-TRP-LocationInformation      CRITICALITY ignore      TYPE Mobile-TRP-LocationInformation     PRESENCE mandatory },
    ...
}


TRPList ::= SEQUENCE (SIZE(1.. maxnoofTRPs)) OF TRPListItem

TRPListItem ::= SEQUENCE {
    tRPID                           TRPID,
    iE-Extensions                   ProtocolExtensionContainer { { TRPListItem-ExtIEs } }   OPTIONAL
}

TRPListItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TRPMeasurementQuality ::= SEQUENCE {
    tRPmeasurementQuality-Item  TRPMeasurementQuality-Item,
    iE-Extensions               ProtocolExtensionContainer { {TRPMeasurementQuality-ExtIEs} } OPTIONAL
}

TRPMeasurementQuality-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TRPMeasurementQuality-Item ::= CHOICE {
    timingMeasurementQuality    TimingMeasurementQuality,
    angleMeasurementQuality     AngleMeasurementQuality,
    choice-extension            ProtocolIE-SingleContainer { { TRPMeasurementQuality-Item-ExtIEs } }
}
```

```
TRPMeasurementQuality-Item-ExtIEs F1AP-PROTOCOL-IES ::= {
    {ID id-PhaseQuality             CRITICALITY ignore TYPE PhaseQuality        PRESENCE mandatory},
    ...
}

PhaseQuality ::= SEQUENCE {
    phaseQualityIndex           INTEGER(0..179),
    phaseQualityResolution      ENUMERATED {deg0dot1, deg1, ...},
    iE-Extensions   ProtocolExtensionContainer { { PhaseQuality-ExtIEs } }  OPTIONAL
}

PhaseQuality-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TRP-MeasurementRequestList ::= SEQUENCE (SIZE (1..maxNoOfMeasTRPs)) OF TRP-MeasurementRequestItem

TRP-MeasurementRequestItem ::= SEQUENCE {
    tRPID                           TRPID,
    search-window-information       Search-window-information   OPTIONAL,
    iE-extensions       ProtocolExtensionContainer { { TRP-MeasurementRequestItem-ExtIEs } } OPTIONAL
}

TRP-MeasurementRequestItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-NRCGI                   CRITICALITY ignore EXTENSION NRCGI                  PRESENCE optional }|
    { ID id-AoA-SearchWindow        CRITICALITY ignore EXTENSION AoA-AssistanceInfo     PRESENCE optional }|
    { ID id-NumberOfTRPRxTEG        CRITICALITY ignore EXTENSION NumberOfTRPRxTEG       PRESENCE optional }|
    { ID id-NumberOfTRPRxTxTEG      CRITICALITY ignore EXTENSION NumberOfTRPRxTxTEG     PRESENCE optional },
    ...
}

TRP-PRS-Info-List ::= SEQUENCE (SIZE(1.. maxnoofPRSTRPs)) OF TRP-PRS-Info-List-Item

TRP-PRS-Info-List-Item ::= SEQUENCE {
        tRP-ID              TRPID,
        nR-PCI              NRPCI,
        cGI-NR              NRCGI                   OPTIONAL,
        pRSConfiguration                            PRSConfiguration,
        iE-Extensions   ProtocolExtensionContainer { { TRP-PRS-Info-List-Item-ExtIEs} } OPTIONAL,
        ...
}

TRP-PRS-Info-List-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


TRPPositionDefinitionType ::= CHOICE {
    direct      TRPPositionDirect,
    referenced  TRPPositionReferenced,
    choice-extension                            ProtocolIE-SingleContainer { { TRPPositionDefinitionType-ExtIEs } }
}

TRPPositionDefinitionType-ExtIEs F1AP-PROTOCOL-IES ::= {
```

```
	...
}

TRPPositionDirect ::= SEQUENCE {
	accuracy	TRPPositionDirectAccuracy,
	iE-extensions		ProtocolExtensionContainer { { TRPPositionDirect-ExtIEs } } OPTIONAL
}

TRPPositionDirect-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

TRPPositionDirectAccuracy ::= CHOICE {
	tRPPosition			AccessPointPosition,
	tRPHAposition			NGRANHighAccuracyAccessPointPosition,
	choice-extension		ProtocolIE-SingleContainer { { TRPPositionDirectAccuracy-ExtIEs } }
}

TRPPositionDirectAccuracy-ExtIEs F1AP-PROTOCOL-IES ::= {
	...
}

TRPPositionReferenced ::= SEQUENCE {
	referencePoint				ReferencePoint,
	referencePointType			TRPReferencePointType,
	iE-extensions				ProtocolExtensionContainer { { TRPPositionReferenced-ExtIEs } }  OPTIONAL
}

TRPPositionReferenced-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

TRPReferencePointType ::= CHOICE {
	tRPPositionRelativeGeodetic			RelativeGeodeticLocation,
	tRPPositionRelativeCartesian		RelativeCartesianLocation,
	choice-extension					ProtocolIE-SingleContainer { { TRPReferencePointType-ExtIEs } }
}

TRPReferencePointType-ExtIEs F1AP-PROTOCOL-IES ::= {
	...
}

TRP-Rx-TEGInformation ::= SEQUENCE {
	tRP-Rx-TEGID				INTEGER (0..31),
	tRP-Rx-TimingErrorMargin		TimingErrorMargin,
	iE-Extensions		ProtocolExtensionContainer { { TRP-Rx-TEGInformation-ExtIEs } } OPTIONAL,
	...
}

TRP-Rx-TEGInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}

TRP-RxTx-TEGInformation ::= SEQUENCE {
```

```
    tRP-RxTx-TEGID                  INTEGER (0..255),
    tRP-RxTx-TimingErrorMargin      RxTxTimingErrorMargin,
    iE-Extensions       ProtocolExtensionContainer { { TRP-RxTx-TEGInformation-ExtIEs } } OPTIONAL,
    ...
}

TRP-RxTx-TEGInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


TRP-Tx-TEGInformation ::= SEQUENCE {
    tRP-Tx-TEGID                    INTEGER (0..7),
    tRP-Tx-TimingErrorMargin        TimingErrorMargin,
    iE-Extensions       ProtocolExtensionContainer { { TRP-Tx-TEGInformation-ExtIEs } } OPTIONAL,
    ...
}

TRP-Tx-TEGInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TRPTxTEGAssociation ::= SEQUENCE (SIZE(1.. maxnoTRPTEGs)) OF TRPTEG-Item

TRPTEG-Item ::= SEQUENCE {
        tRP-Tx-TEGInformation       TRP-Tx-TEGInformation,
        dl-PRSResourceSetID         PRS-Resource-Set-ID,
        dl-PRSResourceID-List   SEQUENCE (SIZE(1.. maxnoofPRS-ResourcesPerSet)) OF DLPRSResourceID-Item OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { TRPTEGItem-ExtIEs } } OPTIONAL,
    ...
}

TRPTEGItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

DLPRSResourceID-Item ::= SEQUENCE {
    dl-PRSResourceID        PRS-Resource-ID,
    iE-Extensions           ProtocolExtensionContainer { { DLPRSResource-Item-ExtIEs} } OPTIONAL,
    ...
}

DLPRSResource-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


TypeOfError ::= ENUMERATED {
    not-understood,
    missing,
    ...
}

Transport-Layer-Address-Info ::= SEQUENCE {
```

```
    transport-UP-Layer-Address-Info-To-Add-List     Transport-UP-Layer-Address-Info-To-Add-List                 OPTIONAL,
    transport-UP-Layer-Address-Info-To-Remove-List  Transport-UP-Layer-Address-Info-To-Remove-List              OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { Transport-Layer-Address-Info-ExtIEs } }                      OPTIONAL
}

Transport-Layer-Address-Info-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TRPType ::= ENUMERATED {
    prsOnlyTP,
    srsOnlyRP,
    tp,
    rp,
    trp,
    ...,
    mobile-trp
}

TSCAssistanceInformation ::= SEQUENCE {
    periodicity             Periodicity,
    burstArrivalTime        BurstArrivalTime                                                    OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { {TSCAssistanceInformation-ExtIEs} }    OPTIONAL,
    ...
}

TSCAssistanceInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-SurvivalTime    CRITICALITY ignore  EXTENSION SurvivalTime  PRESENCE optional }|
    { ID id-RANfeedbacktype CRITICALITY ignore  EXTENSION RANfeedbacktype       PRESENCE optional}|
    { ID id-N6JitterInformation CRITICALITY ignore  EXTENSION N6JitterInformation   PRESENCE optional },
    ...
}

TSCTrafficCharacteristics ::= SEQUENCE {
    tSCAssistanceInformationDL      TSCAssistanceInformation                                OPTIONAL,
    tSCAssistanceInformationUL      TSCAssistanceInformation                                OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {TSCTrafficCharacteristics-ExtIEs} }   OPTIONAL,
    ...
}

TSCTrafficCharacteristics-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TSCTrafficCharacteristicsFeedback ::= SEQUENCE {
    tSCFeedbackInformationDL        TSCFeedbackInformation                              OPTIONAL,
    tSCFeedbackInformationUL        TSCFeedbackInformation                              OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { TSCTrafficCharacteristicsFeedback-ExtIEs} }  OPTIONAL,
    ...
}

TSCTrafficCharacteristicsFeedback-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
TSCFeedbackInformation ::= SEQUENCE {
    burstArrivalTimeOffset                          INTEGER (-640000..640000, ...),
    adjustedPeriodicity                             Periodicity                         OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { TSCFeedbackInformation-ExtIEs} } OPTIONAL,
    ...
}

TSCFeedbackInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TRP-MeasurementUpdateList ::= SEQUENCE (SIZE (1..maxNoOfMeasTRPs)) OF TRP-MeasurementUpdateItem

TRP-MeasurementUpdateItem ::= SEQUENCE {
    tRP-ID                      TRPID,
    aoA-window-information      AoA-AssistanceInfo  OPTIONAL,
    iE-extensions               ProtocolExtensionContainer { { TRP-MeasurementUpdateItem-ExtIEs } } OPTIONAL,
    ...
}

TRP-MeasurementUpdateItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-NumberOfTRPRxTEG        CRITICALITY ignore EXTENSION NumberOfTRPRxTEG       PRESENCE optional }|
    { ID id-NumberOfTRPRxTxTEG      CRITICALITY ignore EXTENSION NumberOfTRPRxTxTEG     PRESENCE optional },
    ...
}

TwoPHRModeMCG ::= ENUMERATED {enabled, ...}

TwoPHRModeSCG ::= ENUMERATED {enabled, ...}

TxHoppingConfiguration ::= SEQUENCE {
    overlapValue                        ENUMERATED {rb0, rb1, rb2, rb4},
    numberOfHops                        INTEGER (2..6),
    slotOffsetForRemainingHopsList      SlotOffsetForRemainingHopsList,
    iE-extensions       ProtocolExtensionContainer { { TxHoppingConfiguration-ExtIEs } }    OPTIONAL,
    ...
}

TxHoppingConfiguration-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

TAInformation-List  ::= SEQUENCE (SIZE(1.. maxnoofTAList)) OF TAInformation-Item

TAInformation-Item  ::= SEQUENCE {
    nRCGI                   NRCGI,
    tAValue                 TAValue,
    iE-Extensions           ProtocolExtensionContainer { { TAInformation-Item-ExtIEs} } OPTIONAL,
    ...
}

TAInformation-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID    id-TagIDPointer CRITICALITY ignore  EXTENSION   TagIDPointer    PRESENCE optional},
```

```
...
}

TARemainingInfoList ::= SEQUENCE (SIZE(1..maxnoofLTMCells)) OF TARemainingInfo-Item

TARemainingInfo-Item    ::= SEQUENCE {
    cellID                  NRCGI,
    ltmResidualTAInfoList   LTMResidualTAInfoList,
    iE-Extensions           ProtocolExtensionContainer { { TARemainingInfo-Item-ExtIEs} }   OPTIONAL,
    ...
}

TARemainingInfo-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


TimeforFutureCoverageModification ::= INTEGER (1..60, ...)

TimeforPredictedCCOIssue ::= INTEGER (1..60, ...)

TimeforNeighbourFutureCoverageModification ::= INTEGER (1..60, ...)


Tci-State-InformationList   ::=  SEQUENCE (SIZE(1.. maxnoofLTM-CSI-ResourcesPerSet)) OF Tci-State-Information-Item

Tci-State-Information-Item  ::= SEQUENCE {
    jointorDLTCIStateID         JointorDLTCIStateID,
    iE-Extensions       ProtocolExtensionContainer { { Tci-State-Information-Item-ExtIEs } }    OPTIONAL,
    ...
}

Tci-State-Information-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- U
UAC-Assistance-Info ::= SEQUENCE {
    uACPLMN-List        UACPLMN-List,
    iE-Extensions       ProtocolExtensionContainer { { UAC-Assistance-InfoExtIEs} } OPTIONAL
}

UAC-Assistance-InfoExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UACPLMN-List ::= SEQUENCE (SIZE(1..maxnoofUACPLMNs)) OF UACPLMN-Item

UACPLMN-Item::= SEQUENCE {
    pLMNIdentity                PLMN-Identity,
    uACType-List                UACType-List,   iE-Extensions       ProtocolExtensionContainer { { UACPLMN-Item-ExtIEs} } OPTIONAL
}
```

```
UACPLMN-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-NID CRITICALITY ignore  EXTENSION NID   PRESENCE optional },
    ...
}

UACType-List ::= SEQUENCE (SIZE(1..maxnoofUACperPLMN)) OF UACType-Item

UACType-Item::= SEQUENCE {
    uACReductionIndication      UACReductionIndication,
    uACCategoryType             UACCategoryType,
    iE-Extensions       ProtocolExtensionContainer { { UACType-Item-ExtIEs } } OPTIONAL
}

UACType-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UACCategoryType ::= CHOICE {
    uACstandardized             UACAction,
    uACOperatorDefined          UACOperatorDefined,
    choice-extension            ProtocolIE-SingleContainer { { UACCategoryType-ExtIEs } }
}

UACCategoryType-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

UACOperatorDefined ::= SEQUENCE {
    accessCategory                  INTEGER (32..63,...),
    accessIdentity                  BIT STRING (SIZE(7)),
    iE-Extensions       ProtocolExtensionContainer { { UACOperatorDefined-ExtIEs} } OPTIONAL
}

UACOperatorDefined-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


UACAction ::= ENUMERATED {
    reject-non-emergency-mo-dt,
    reject-rrc-cr-signalling,
    permit-emergency-sessions-and-mobile-terminated-services-only,
    permit-high-priority-sessions-and-mobile-terminated-services-only,
    ...
}

UACReductionIndication ::= INTEGER (0..100)


UE-associatedLogicalF1-ConnectionItem ::= SEQUENCE {
    gNB-CU-UE-F1AP-ID       GNB-CU-UE-F1AP-ID    OPTIONAL,
    gNB-DU-UE-F1AP-ID       GNB-DU-UE-F1AP-ID    OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { UE-associatedLogicalF1-ConnectionItemExtIEs} } OPTIONAL,
    ...
```

```
}

UEAssistanceInformation ::= OCTET STRING

UEAssistanceInformationEUTRA ::= OCTET STRING

UE-associatedLogicalF1-ConnectionItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UE-CapabilityRAT-ContainerList::= OCTET STRING

UEContextNotRetrievable ::= ENUMERATED {true, ...}

UEIdentityIndexValue ::= CHOICE {
    indexLength10           BIT STRING (SIZE (10)),
    choice-extension        ProtocolIE-SingleContainer { {UEIdentityIndexValueChoice-ExtIEs} }
}

UEIdentityIndexValueChoice-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

UEIdentity-List-For-Paging-Item     ::= SEQUENCE {
    uEIdentityIndexValue                        UEIdentityIndexValue,
    pagingDRX                                   PagingDRX       OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { UEIdentity-List-For-Paging-Item-ExtIEs} } OPTIONAL
}

UEIdentity-List-For-Paging-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UE-MulticastMRBs-ConfirmedToBeModified-Item::= SEQUENCE {
    mRB-ID                      MRB-ID,
    mrb-type-reconfiguration    MBSPTPRetransmissionTunnelRequired          OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { UE-MulticastMRBs-ConfirmedToBeModified-Item-ExtIEs } } OPTIONAL
}

UE-MulticastMRBs-ConfirmedToBeModified-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
UE-MulticastMRBs-RequiredToBeModified-Item::= SEQUENCE {
    mRB-ID                          MRB-ID,
    mrb-type-reconfiguration        ENUMERATED {true, ...}                          OPTIONAL,
    mrb-reconfigured-RLCtype        ENUMERATED {
                                                rlc-um-ptp,
                                                rlc-am-ptp,
                                                rlc-um-dl-ptm,
                                                two-rlc-um-dl-ptp-and-dl-ptm,
                                                three-rlc-um-dl-ptp-ul-ptp-dl-ptm,
                                                two-rlc-am-ptp-um-dl-ptm,
                                                ...}                                OPTIONAL,
    -- The above IE shall be present if the MRB Type Reconfiguration IE is present.
```

```
    iE-Extensions                   ProtocolExtensionContainer { { UE-MulticastMRBs-RequiredToBeModified-Item-ExtIEs } } OPTIONAL
}

UE-MulticastMRBs-RequiredToBeModified-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-MulticastF1UContextReferenceCU              CRITICALITY reject  EXTENSION MulticastF1UContextReferenceCU              PRESENCE 
optional},
    ...
}

UE-MulticastMRBs-RequiredToBeReleased-Item::= SEQUENCE {
    mRB-ID                  MRB-ID,
    iE-Extensions           ProtocolExtensionContainer { { UE-MulticastMRBs-RequiredToBeReleased-Item-ExtIEs } } OPTIONAL
}

UE-MulticastMRBs-RequiredToBeReleased-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UE-MulticastMRBs-Setup-Item ::= SEQUENCE {
    mRB-ID                          MRB-ID,
    multicastF1UContextReferenceCU  MulticastF1UContextReferenceCU,
    iE-Extensions                   ProtocolExtensionContainer { { UE-MulticastMRBs-Setup-Item-ExtIEs } } OPTIONAL
}

UE-MulticastMRBs-Setup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UE-MulticastMRBs-Setupnew-Item  ::= SEQUENCE {
    mRB-ID                          MRB-ID,
    multicastF1UContextReferenceCU  MulticastF1UContextReferenceCU,
    iE-Extensions                   ProtocolExtensionContainer { { UE-MulticastMRBs-Setupnew-Item-ExtIEs } } OPTIONAL
}

UE-MulticastMRBs-Setupnew-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UE-MulticastMRBs-ToBeReleased-Item  ::= SEQUENCE {
    mRB-ID                  MRB-ID,
    iE-Extensions           ProtocolExtensionContainer { { UE-MulticastMRBs-ToBeReleased-Item-ExtIEs } } OPTIONAL
}

UE-MulticastMRBs-ToBeReleased-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UE-MulticastMRBs-ToBeSetup-Item ::= SEQUENCE {
    mRB-ID                                  MRB-ID,
    mbsPTPRetransmissionTunnelRequired      MBSPTPRetransmissionTunnelRequired          OPTIONAL,
    mbsPTPForwardingRequiredInformation     MRB-ProgressInformation                     OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { UE-MulticastMRBs-ToBeSetup-Item-ExtIEs } } OPTIONAL
}
```

```
UE-MulticastMRBs-ToBeSetup-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-Source-MRB-ID		CRITICALITY ignore EXTENSION MRB-ID		PRESENCE optional },
	...
}

UE-MulticastMRBs-ToBeSetup-atModify-Item	::= SEQUENCE {
	mRB-ID									MRB-ID,
	mbsPTPRetransmissionTunnelRequired		MBSPTPRetransmissionTunnelRequired			OPTIONAL,
	mbsPTPForwardingRequiredInformation		MRB-ProgressInformation						OPTIONAL,
	iE-Extensions			ProtocolExtensionContainer { { UE-MulticastMRBs-ToBeSetup-atModify-Item-ExtIEs } } OPTIONAL
}

UE-MulticastMRBs-ToBeSetup-atModify-Item-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	...
}


UEPagingCapability ::= SEQUENCE {
	iNACTIVEStatePODetermination						ENUMERATED {supported, ...}		OPTIONAL,
	iE-Extension				ProtocolExtensionContainer { { UEPagingCapability-ExtIEs} }		OPTIONAL,
	...
}

UEPagingCapability-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	{	ID id-RedCapIndication		CRITICALITY ignore	EXTENSION RedCapIndication		PRESENCE optional }|
	{	ID id-PagingAdaptationIndication	CRITICALITY ignore	EXTENSION PagingAdaptationIndication PRESENCE optional },
	...
}

UEReportingInformation::= SEQUENCE {
	reportingAmount				ENUMERATED {ma0, ma1, ma2, ma4, ma8, ma16, ma32, ma64},
	reportingInterval			ENUMERATED {none, one, two, four, eight, ten, sixteen, twenty, thirty-two, sixty-four, ...},
	iE-extensions			ProtocolExtensionContainer { { UEReportingInformation-ExtIEs } }	OPTIONAL,
	...
}


UEReportingInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
	{ ID id-ReportingIntervalIMs			CRITICALITY ignore EXTENSION ReportingIntervalIMs	PRESENCE optional },
	...
}


UlTxDirectCurrentMoreCarrierInformation::= OCTET STRING

UL-AoA ::= SEQUENCE {
	azimuthAoA					INTEGER (0..3599),
	zenithAoA					INTEGER (0..1799)		OPTIONAL,
	lCS-to-GCS-Translation	LCS-to-GCS-Translation		OPTIONAL,
	iE-extensions			ProtocolExtensionContainer { { UL-AoA-ExtIEs } }	OPTIONAL,
	...
}

UL-AoA-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

UL-BH-Non-UP-Traffic-Mapping ::= SEQUENCE {
    uL-BH-Non-UP-Traffic-Mapping-List           UL-BH-Non-UP-Traffic-Mapping-List,
    iE-Extensions   ProtocolExtensionContainer { { UL-BH-Non-UP-Traffic-Mapping-ExtIEs } } OPTIONAL
}

UL-BH-Non-UP-Traffic-Mapping-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UL-BH-Non-UP-Traffic-Mapping-List ::= SEQUENCE (SIZE(1..maxnoofNonUPTrafficMappings)) OF UL-BH-Non-UP-Traffic-Mapping-Item

UL-BH-Non-UP-Traffic-Mapping-Item ::= SEQUENCE {
    nonUPTrafficType                NonUPTrafficType,
    bHInfo                      BHInfo,
    iE-Extensions                   ProtocolExtensionContainer { { UL-BH-Non-UP-Traffic-Mapping-ItemExtIEs } }  OPTIONAL
}

UL-BH-Non-UP-Traffic-Mapping-ItemExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ULConfiguration ::= SEQUENCE    {
    uLUEConfiguration       ULUEConfiguration,
    iE-Extensions   ProtocolExtensionContainer { { ULConfigurationExtIEs } }    OPTIONAL,
    ...
}
ULConfigurationExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UL-GapFR2-Config ::= OCTET STRING

UL-RTOA-Measurement ::= SEQUENCE {
    uL-RTOA-MeasurementItem     UL-RTOA-MeasurementItem,
    additionalPath-List         AdditionalPath-List OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { UL-RTOA-Measurement-ExtIEs } }   OPTIONAL
}

UL-RTOA-Measurement-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-ExtendedAdditionalPathList  CRITICALITY ignore EXTENSION ExtendedAdditionalPathList PRESENCE optional}|
    { ID id-TRPRx-TEGInformation        CRITICALITY ignore EXTENSION TRP-Rx-TEGInformation      PRESENCE optional},
    ...
}

UL-RTOA-MeasurementItem ::= CHOICE {
    k0                  INTEGER (0..1970049),
    k1                  INTEGER (0..985025),
    k2                  INTEGER (0..492513),
    k3                  INTEGER (0..246257),
    k4                  INTEGER (0..123129),
    k5                  INTEGER (0..61565),
```

```
    choice-extension            ProtocolIE-SingleContainer { { UL-RTOA-MeasurementItem-ExtIEs } }
}

UL-RTOA-MeasurementItem-ExtIEs F1AP-PROTOCOL-IES ::= {
    {ID id-ReportingGranularitykminus1  CRITICALITY ignore TYPE ReportingGranularitykminus1 PRESENCE mandatory}|
    {ID id-ReportingGranularitykminus2  CRITICALITY ignore TYPE ReportingGranularitykminus2 PRESENCE mandatory }|
    {ID id-ReportingGranularitykminus3  CRITICALITY ignore TYPE ReportingGranularitykminus3 PRESENCE mandatory}|
    {ID id-ReportingGranularitykminus4  CRITICALITY ignore TYPE ReportingGranularitykminus4 PRESENCE mandatory}|
    {ID id-ReportingGranularitykminus5  CRITICALITY ignore TYPE ReportingGranularitykminus5 PRESENCE mandatory}|
    {ID id-ReportingGranularitykminus6  CRITICALITY ignore TYPE ReportingGranularitykminus6 PRESENCE mandatory},
    ...
}

UL-SRS-RSRP ::= INTEGER (0..126)

UL-SRS-RSRPP ::= SEQUENCE {
    firstPathRSRPP              INTEGER (0..126),
    iE-extensions           ProtocolExtensionContainer { { UL-SRS-RSRPP-ExtIEs } }  OPTIONAL,
    ...
}

UL-SRS-RSRPP-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UL-SRS-TDCT ::= SEQUENCE {
    ul-SRS-TDCT-List            UL-SRS-TDCT-List,
    iE-Extensions               ProtocolExtensionContainer { { UL-SRS-TDCT-ExtIEs } }   OPTIONAL,
    ...
}

UL-SRS-TDCT-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UL-SRS-TDCT-List::= SEQUENCE (SIZE(1..maxnoofChannelRes)) OF UL-SRS-TDCT-Item

UL-SRS-TDCT-Item ::=SEQUENCE {
    timingInformation   TimingInformation,
    powerInformation    UL-SRS-TDCP-Item    OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { UL-SRS-TDCT-Item-ExtIEs } }  OPTIONAL,
    ...
}

UL-SRS-TDCT-Item-ExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UL-SRS-TDCP-Item ::= INTEGER (0..126)

UL-RSCP ::= SEQUENCE {
    uLRSCP                  INTEGER (0..3599),
    iE-extensions           ProtocolExtensionContainer { { UL-RSCP-ExtIEs } }   OPTIONAL,
    ...
```

```
}

UL-RSCP-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


ULUEConfiguration ::= ENUMERATED {no-data, shared, only, ...}

UL-UP-TNL-Information-to-Update-List-Item   ::= SEQUENCE {
    uLUPTNLInformation      UPTransportLayerInformation,
    newULUPTNLInformation   UPTransportLayerInformation     OPTIONAL,
    bHInfo  BHInfo,
    iE-Extensions   ProtocolExtensionContainer { { UL-UP-TNL-Information-to-Update-List-ItemExtIEs } }  OPTIONAL,
    ...
}

UL-UP-TNL-Information-to-Update-List-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UL-UP-TNL-Address-to-Update-List-Item   ::= SEQUENCE {
    oldIPAdress                     TransportLayerAddress,
    newIPAdress                     TransportLayerAddress,
    iE-Extensions   ProtocolExtensionContainer { { UL-UP-TNL-Address-to-Update-List-ItemExtIEs } }   OPTIONAL,
    ...
}

UL-UP-TNL-Address-to-Update-List-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ULUPTNLInformation-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofULUPTNLInformation)) OF ULUPTNLInformation-ToBeSetup-Item

ULUPTNLInformation-ToBeSetup-Item ::=SEQUENCE {
    uLUPTNLInformation      UPTransportLayerInformation,
    iE-Extensions   ProtocolExtensionContainer { { ULUPTNLInformation-ToBeSetup-ItemExtIEs } }  OPTIONAL,
    ...
}

ULUPTNLInformation-ToBeSetup-ItemExtIEs     F1AP-PROTOCOL-EXTENSION ::= {
    { ID id-BHInfo          CRITICALITY ignore  EXTENSION BHInfo            PRESENCE optional   }|
    { ID id-DRBMappingInfo  CRITICALITY ignore  EXTENSION UuRLCChannelID    PRESENCE optional   },
    ...
}

Uncertainty ::= INTEGER (0..32767, ...)

UplinkChannelBW-PerSCS-List ::= SEQUENCE (SIZE (1..maxnoSCSs)) OF SCS-SpecificCarrier

UplinkTxDirectCurrentListInformation ::= OCTET STRING

UplinkTxDirectCurrentTwoCarrierListInfo ::= OCTET STRING
```

```
ULTCIStateID  ::= OCTET STRING

UPTransportLayerInformation     ::= CHOICE {
    gTPTunnel       GTPTunnel,
    choice-extension            ProtocolIE-SingleContainer { { UPTransportLayerInformation-ExtIEs} }
}

UPTransportLayerInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

URI-address ::= VisibleString

Uncertainty-range-AoA ::= INTEGER (0..3599)

Uncertainty-range-ZoA ::= INTEGER (0..1799)


UuRLCChannelID ::= INTEGER (1..32)

UuRLCChannelQoSInformation ::= CHOICE {
    uuRLCChannelQoS                 QoSFlowLevelQoSParameters,
    uuControlPlaneTrafficType       ENUMERATED {srb0,srb1,srb2,...},
    choice-extension        ProtocolIE-SingleContainer { { UuRLCChannelQoSInformation-ExtIEs} }
}

UuRLCChannelQoSInformation-ExtIEs F1AP-PROTOCOL-IES ::= {
    ...
}

UuRLCChannelToBeSetupList ::= SEQUENCE (SIZE(1.. maxnoofUuRLCChannels)) OF UuRLCChannelToBeSetupItem

UuRLCChannelToBeSetupItem ::= SEQUENCE {
    uuRLCChannelID                  UuRLCChannelID,
    uuRLCChannelQoSInformation      UuRLCChannelQoSInformation,
    rLCMode                         RLCMode,
    iE-Extensions                   ProtocolExtensionContainer { { UuRLCChannelToBeSetupItem-ExtIEs } } OPTIONAL,
    ...
}

UuRLCChannelToBeSetupItem-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UuRLCChannelToBeModifiedList ::= SEQUENCE (SIZE(1.. maxnoofUuRLCChannels)) OF UuRLCChannelToBeModifiedItem

UuRLCChannelToBeModifiedItem ::= SEQUENCE {
    uuRLCChannelID                  UuRLCChannelID,
    uuRLCChannelQoSInformation      UuRLCChannelQoSInformation          OPTIONAL,
    rLCMode                         RLCMode         OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { UuRLCChannelToBeModifiedItem-ExtIEs } } OPTIONAL,
    ...
}
```

```
UuRLCChannelToBeModifiedItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UuRLCChannelToBeReleasedList ::= SEQUENCE (SIZE(1.. maxnoofUuRLCChannels)) OF UuRLCChannelToBeReleasedItem

UuRLCChannelToBeReleasedItem ::= SEQUENCE {
    uuRLCChannelID          UuRLCChannelID,
    iE-Extensions           ProtocolExtensionContainer { { UuRLCChannelToBeReleasedItem-ExtIEs } }  OPTIONAL,
    ...
}

UuRLCChannelToBeReleasedItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UuRLCChannelSetupList ::= SEQUENCE (SIZE(1.. maxnoofUuRLCChannels)) OF UuRLCChannelSetupItem

UuRLCChannelSetupItem ::= SEQUENCE {
    uuRLCChannelID          UuRLCChannelID,
    iE-Extensions           ProtocolExtensionContainer { { UuRLCChannelSetupItem-ExtIEs } } OPTIONAL,
    ...
}

UuRLCChannelSetupItem-ExtIEs    F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UuRLCChannelFailedToBeSetupList ::= SEQUENCE (SIZE(1.. maxnoofUuRLCChannels)) OF UuRLCChannelFailedToBeSetupItem

UuRLCChannelFailedToBeSetupItem ::= SEQUENCE {
    uuRLCChannelID          UuRLCChannelID,
    cause                   Cause   OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { UuRLCChannelFailedToBeSetupItem-ExtIEs } }  OPTIONAL,
    ...
}

UuRLCChannelFailedToBeSetupItem-ExtIEs  F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UuRLCChannelModifiedList ::= SEQUENCE (SIZE(1.. maxnoofUuRLCChannels)) OF UuRLCChannelModifiedItem

UuRLCChannelModifiedItem ::= SEQUENCE {
    uuRLCChannelID          UuRLCChannelID,
    iE-Extensions           ProtocolExtensionContainer { { UuRLCChannelModifiedItem-ExtIEs } }  OPTIONAL,
    ...
}

UuRLCChannelModifiedItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UuRLCChannelFailedToBeModifiedList ::= SEQUENCE (SIZE(1.. maxnoofUuRLCChannels)) OF UuRLCChannelFailedToBeModifiedItem
```

```
UuRLCChannelFailedToBeModifiedItem ::= SEQUENCE {
    uuRLCChannelID          UuRLCChannelID,
    cause                   Cause   OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { UuRLCChannelFailedToBeModifiedItem-ExtIEs } }  OPTIONAL,
    ...
}

UuRLCChannelFailedToBeModifiedItem-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UuRLCChannelRequiredToBeModifiedList ::= SEQUENCE (SIZE(1.. maxnoofUuRLCChannels)) OF UuRLCChannelRequiredToBeModifiedItem

UuRLCChannelRequiredToBeModifiedItem ::= SEQUENCE {
    uuRLCChannelID          UuRLCChannelID,
    iE-Extensions           ProtocolExtensionContainer { { UuRLCChannelRequiredToBeModifiedItem-ExtIEs } } OPTIONAL,
    ...
}

UuRLCChannelRequiredToBeModifiedItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

UuRLCChannelRequiredToBeReleasedList ::= SEQUENCE (SIZE(1.. maxnoofUuRLCChannels)) OF UuRLCChannelRequiredToBeReleasedItem

UuRLCChannelRequiredToBeReleasedItem ::= SEQUENCE {
    uuRLCChannelID          UuRLCChannelID,
    iE-Extensions           ProtocolExtensionContainer { { UuRLCChannelRequiredToBeReleasedItem-ExtIEs } } OPTIONAL,
    ...
}

UuRLCChannelRequiredToBeReleasedItem-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- V

VictimgNBSetID ::= SEQUENCE {
    victimgNBSetID      GNBSetID,
    iE-Extensions       ProtocolExtensionContainer { { VictimgNBSetID-ExtIEs } }        OPTIONAL
}

VictimgNBSetID-ExtIEs   F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

VehicleUE ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}
```

```
PedestrianUE ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

-- V

ValidityAreaSpecificSRSInformation ::= SEQUENCE {
    transmissionCombPos             TransmissionCombPos     OPTIONAL,
    resourceMapping                 ResourceMapping         OPTIONAL,
    freqDomainShift                 INTEGER (0..268)        OPTIONAL,
    c-SRS                           INTEGER (0..63)         OPTIONAL,
    resourceTypePos                 ResourceTypePos         OPTIONAL,
    sequenceIDPos                   INTEGER (0..65535)      OPTIONAL,
    iE-extensions       ProtocolExtensionContainer { { ValidityAreaSpecificSRSInformation-ExtIEs } } OPTIONAL,
    ...
}

ValidityAreaSpecificSRSInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}

ValidityAreaSpecificSRSInformationExtended ::= SEQUENCE {
    posSRSResource-List         PosSRSResource-List     OPTIONAL,
    posSRSResourceSet-List      PosSRSResourceSet-List  OPTIONAL,
    iE-extensions       ProtocolExtensionContainer { { ValidityAreaSpecificSRSInformationExtended-ExtIEs } }  OPTIONAL,
    ...
}

ValidityAreaSpecificSRSInformationExtended-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- W

-- X
XR-Bcast-Information ::= ENUMERATED {true, ...}

-- Y

-- Z

ZoAInformation ::= SEQUENCE {
    zenithAoA                   INTEGER (0..1799),
    lCS-to-GCS-Translation  LCS-to-GCS-Translation      OPTIONAL,
    iE-extensions           ProtocolExtensionContainer { { ZoAInformation-ExtIEs } }    OPTIONAL,
    ...
}

ZoAInformation-ExtIEs F1AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
END
-- ASN1STOP
```

9.4.6 Common Definitions

```
-- ASN1START
-- **************************************************************
--
-- Common definitions
--
-- **************************************************************

F1AP-CommonDataTypes {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
ngran-access (22) modules (3) f1ap (3) version1 (1) f1ap-CommonDataTypes (3) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

Criticality     ::= ENUMERATED { reject, ignore, notify }

Presence        ::= ENUMERATED { optional, conditional, mandatory }

PrivateIE-ID    ::= CHOICE {
    local               INTEGER (0..65535),
    global              OBJECT IDENTIFIER
}

ProcedureCode       ::= INTEGER (0..255)

ProtocolExtensionID ::= INTEGER (0..65535)

ProtocolIE-ID       ::= INTEGER (0..65535)

TriggeringMessage   ::= ENUMERATED { initiating-message, successful-outcome, unsuccessful-outcome }

END
-- ASN1STOP
```

9.4.7 Constant Definitions

```
-- ASN1START
-- **************************************************************
--
-- Constant definitions
--
-- **************************************************************
```

```
F1AP-Constants {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
ngran-access (22) modules (3) f1ap (3) version1 (1) f1ap-Constants (4) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- **************************************************************
--
-- IE parameter types from other modules.
--
-- **************************************************************

IMPORTS
    ProcedureCode,
    ProtocolIE-ID

FROM F1AP-CommonDataTypes;


-- **************************************************************
--
-- Elementary Procedures
--
-- **************************************************************

id-Reset                                ProcedureCode ::= 0
id-F1Setup                              ProcedureCode ::= 1
id-ErrorIndication                      ProcedureCode ::= 2
id-gNBDUConfigurationUpdate             ProcedureCode ::= 3
id-gNBCUConfigurationUpdate             ProcedureCode ::= 4
id-UEContextSetup                       ProcedureCode ::= 5
id-UEContextRelease                     ProcedureCode ::= 6
id-UEContextModification                ProcedureCode ::= 7
id-UEContextModificationRequired        ProcedureCode ::= 8
id-procedure-code-9-not-to-be-used      ProcedureCode ::= 9
id-UEContextReleaseRequest              ProcedureCode ::= 10
id-InitialULRRCMessageTransfer          ProcedureCode ::= 11
id-DLRRCMessageTransfer                 ProcedureCode ::= 12
id-ULRRCMessageTransfer                 ProcedureCode ::= 13
id-privateMessage                       ProcedureCode ::= 14
id-UEInactivityNotification             ProcedureCode ::= 15
id-GNBDUResourceCoordination            ProcedureCode ::= 16
id-SystemInformationDeliveryCommand     ProcedureCode ::= 17
id-Paging                               ProcedureCode ::= 18
id-Notify                               ProcedureCode ::= 19
id-WriteReplaceWarning                  ProcedureCode ::= 20
id-PWSCancel                            ProcedureCode ::= 21
id-PWSRestartIndication                 ProcedureCode ::= 22
id-PWSFailureIndication                 ProcedureCode ::= 23
id-GNBDUStatusIndication                ProcedureCode ::= 24
id-RRCDeliveryReport                    ProcedureCode ::= 25
id-F1Removal                            ProcedureCode ::= 26
```

```
id-NetworkAccessRateReduction             ProcedureCode ::= 27
id-TraceStart                             ProcedureCode ::= 28
id-DeactivateTrace                        ProcedureCode ::= 29
id-DUCURadioInformationTransfer           ProcedureCode ::= 30
id-CUDURadioInformationTransfer           ProcedureCode ::= 31
id-BAPMappingConfiguration                ProcedureCode ::= 32
id-GNBDUResourceConfiguration             ProcedureCode ::= 33
id-IABTNLAddressAllocation                ProcedureCode ::= 34
id-IABUPConfigurationUpdate               ProcedureCode ::= 35
id-resourceStatusReportingInitiation      ProcedureCode ::= 36
id-resourceStatusReporting                ProcedureCode ::= 37
id-accessAndMobilityIndication            ProcedureCode ::= 38
id-accessSuccess                          ProcedureCode ::= 39
id-cellTrafficTrace                       ProcedureCode ::= 40
id-PositioningMeasurementExchange         ProcedureCode ::= 41
id-PositioningAssistanceInformationControl  ProcedureCode ::= 42
id-PositioningAssistanceInformationFeedback ProcedureCode ::= 43
id-PositioningMeasurementReport           ProcedureCode ::= 44
id-PositioningMeasurementAbort            ProcedureCode ::= 45
id-PositioningMeasurementFailureIndication  ProcedureCode ::= 46
id-PositioningMeasurementUpdate           ProcedureCode ::= 47
id-TRPInformationExchange                 ProcedureCode ::= 48
id-PositioningInformationExchange         ProcedureCode ::= 49
id-PositioningActivation                  ProcedureCode ::= 50
id-PositioningDeactivation                ProcedureCode ::= 51
id-E-CIDMeasurementInitiation             ProcedureCode ::= 52
id-E-CIDMeasurementFailureIndication      ProcedureCode ::= 53
id-E-CIDMeasurementReport                 ProcedureCode ::= 54
id-E-CIDMeasurementTermination            ProcedureCode ::= 55
id-PositioningInformationUpdate           ProcedureCode ::= 56
id-ReferenceTimeInformationReport         ProcedureCode ::= 57
id-ReferenceTimeInformationReportingControl ProcedureCode ::= 58
id-BroadcastContextSetup                  ProcedureCode ::= 59
id-BroadcastContextRelease                ProcedureCode ::= 60
id-BroadcastContextReleaseRequest         ProcedureCode ::= 61
id-BroadcastContextModification           ProcedureCode ::= 62
id-MulticastGroupPaging                   ProcedureCode ::= 63
id-MulticastContextSetup                  ProcedureCode ::= 64
id-MulticastContextRelease                ProcedureCode ::= 65
id-MulticastContextReleaseRequest         ProcedureCode ::= 66
id-MulticastContextModification           ProcedureCode ::= 67
id-MulticastDistributionSetup             ProcedureCode ::= 68
id-MulticastDistributionRelease           ProcedureCode ::= 69
id-PDCMeasurementInitiation               ProcedureCode ::= 70
id-PDCMeasurementReport                   ProcedureCode ::= 71
id-procedure-code-72-not-to-be-used       ProcedureCode ::= 72
id-procedure-code-73-not-to-be-used       ProcedureCode ::= 73
id-procedure-code-74-not-to-be-used       ProcedureCode ::= 74
id-pRSConfigurationExchange               ProcedureCode ::= 75
id-measurementPreconfiguration            ProcedureCode ::= 76
id-measurementActivation                  ProcedureCode ::= 77
id-QoEInformationTransfer                 ProcedureCode ::= 78
id-PDCMeasurementTerminationCommand       ProcedureCode ::= 79
id-PDCMeasurementFailureIndication        ProcedureCode ::= 80
```

```
id-PosSystemInformationDeliveryCommand      ProcedureCode ::= 81
id-DUCUCellSwitchNotification               ProcedureCode ::= 82
id-CUDUCellSwitchNotification               ProcedureCode ::= 83
id-DUCUTAInformationTransfer                ProcedureCode ::= 84
id-CUDUTAInformationTransfer                ProcedureCode ::= 85
id-QoEInformationTransferControl            ProcedureCode ::= 86
id-RachIndication                           ProcedureCode ::= 87
id-TimingSynchronisationStatus              ProcedureCode ::= 88
id-TimingSynchronisationStatusReport        ProcedureCode ::= 89
id-MIABF1SetupTriggering                    ProcedureCode ::= 90
id-MIABF1SetupOutcomeNotification           ProcedureCode ::= 91
id-MulticastContextNotification             ProcedureCode ::= 92
id-MulticastCommonConfiguration             ProcedureCode ::= 93
id-BroadcastTransportResourceRequest        ProcedureCode ::= 94
id-DUCUAccessAndMobilityIndication          ProcedureCode ::= 95
id-SRSInformationReservationNotification    ProcedureCode ::= 96
id-CUDUMobilityInitiationRequest            ProcedureCode ::= 97
id-CLI-Indication                           ProcedureCode ::= 98
id-DUCUCSIRSCoordination                    ProcedureCode ::= 99
id-CUDUCSIRSCoordination                    ProcedureCode ::= 100




-- **************************************************************
--
-- Extension constants
--
-- **************************************************************

maxPrivateIEs                           INTEGER ::= 65535
maxProtocolExtensions                   INTEGER ::= 65535
maxProtocolIEs                          INTEGER ::= 65535
-- **************************************************************
--
-- Lists
--
-- **************************************************************

maxNRARFCN                              INTEGER ::= 3279165
maxnoofErrors                           INTEGER ::= 256
maxnoofIndividualF1ConnectionsToReset   INTEGER ::= 65536
maxCellingNBDU                          INTEGER ::= 512
maxnoofSCells                           INTEGER ::= 32
maxnoofSRBs                             INTEGER ::= 8
maxnoofDRBs                             INTEGER ::= 64
maxnoofULUPTNLInformation               INTEGER ::= 2
maxnoofDLUPTNLInformation               INTEGER ::= 2
maxnoofBPLMNs                           INTEGER ::= 6
maxnoofCandidateSpCells                 INTEGER ::= 64
maxnoofPotentialSpCells                 INTEGER ::= 64
maxnoofNrCellBands                      INTEGER ::= 32
maxnoofSIBTypes                         INTEGER ::= 32
```

```
maxnoofSITypes                          INTEGER ::= 32
maxnoofPagingCells                      INTEGER ::= 512
maxnoofTNLAssociations                  INTEGER ::= 32
maxnoofQoSFlows                         INTEGER ::= 64
maxnoofSliceItems                       INTEGER ::= 1024
maxCellineNB                            INTEGER ::= 256
maxnoofExtendedBPLMNs                   INTEGER ::= 6
maxnoofUEIDs                            INTEGER ::= 65536
maxnoofBPLMNsNR                         INTEGER ::= 12
maxnoofUACPLMNs                         INTEGER ::= 12
maxnoofUACperPLMN                       INTEGER ::= 64
maxnoofAdditionalSIBs                   INTEGER ::= 63
maxnoofslots                            INTEGER ::= 5120
maxnoofTLAs                             INTEGER ::= 16
maxnoofGTPTLAs                          INTEGER ::= 16
maxnoofBHRLCChannels                    INTEGER ::= 65536
maxnoofRoutingEntries                   INTEGER ::= 1024
maxnoofIABSTCInfo                       INTEGER ::= 45
maxnoofSymbols                          INTEGER ::= 14
maxnoofServingCells                     INTEGER ::= 32
maxnoofDUFSlots                         INTEGER ::= 320
maxnoofHSNASlots                        INTEGER ::= 5120
maxnoofServedCellsIAB                   INTEGER ::= 512
maxnoofSSBarea                          INTEGER ::=64
maxnoofChildIABNodes                    INTEGER ::= 1024
maxnoofNonUPTrafficMappings             INTEGER ::= 32
maxnoofTLAsIAB                          INTEGER ::= 1024
maxnoofMappingEntries                   INTEGER ::= 67108864
maxnoofDSInfo                           INTEGER ::= 64
maxnoofEgressLinks                      INTEGER ::= 2
maxnoofULUPTNLInformationforIAB         INTEGER ::= 32678
maxnoofUPTNLAddresses                   INTEGER ::= 8
maxnoofSLDRBs                           INTEGER ::= 512
maxnoofQoSParaSets                      INTEGER ::= 8
maxnoofPC5QoSFlows                      INTEGER ::= 2048
maxnoofSSBAreas                         INTEGER ::= 64
maxnoofPhysicalResourceBlocks           INTEGER ::= 275
maxnoofPhysicalResourceBlocks-1         INTEGER ::= 274
maxnoofPRACHconfigs                     INTEGER ::= 16
maxnoofRAReports                        INTEGER ::= 64
maxnoofRLFReports                       INTEGER ::= 64
maxnoofAdditionalPDCPDuplicationTNL     INTEGER ::= 2
maxnoofRLCDuplicationState              INTEGER ::= 3
maxnoofCHOcells                         INTEGER ::= 8
maxnoofMDTPLMNs                         INTEGER ::= 16
maxnoofCAGsupported                     INTEGER ::= 12
maxnoofNIDsupported                     INTEGER ::= 12
maxnoofNRSCSs                           INTEGER ::= 5
maxnoofExtSliceItems                    INTEGER ::= 65535
maxnoofPosMeas                          INTEGER ::= 16384
maxnoofTRPInfoTypes                     INTEGER ::= 64
maxnoofTRPs                             INTEGER ::= 65535
maxnoofSRSTriggerStates                 INTEGER ::= 3
maxnoofSpatialRelations                 INTEGER ::= 64
```

```
maxnoBcastCell                          INTEGER ::= 16384
maxnoofAngleInfo                        INTEGER ::= 65535
maxnooflcs-gcs-translation              INTEGER ::= 3
maxnoofPath                             INTEGER ::= 2
maxnoofMeasE-CID                        INTEGER ::= 64
maxnoofSSBs                             INTEGER ::= 255
maxnoSRS-ResourceSets                   INTEGER ::= 16
maxnoSRS-ResourcePerSet                 INTEGER ::= 16
maxnoSRS-Carriers                       INTEGER ::= 32
maxnoSCSs                               INTEGER ::= 5
maxnoSRS-Resources                      INTEGER ::= 64
maxnoSRS-PosResources                   INTEGER ::= 64
maxnoSRS-PosResourceSets                INTEGER ::= 16
maxnoSRS-PosResourcePerSet              INTEGER ::= 16
maxnoofPRS-ResourceSets                 INTEGER ::= 2
maxnoofPRS-ResourcesPerSet              INTEGER ::= 64
maxNoOfMeasTRPs                         INTEGER ::= 64
maxnoofPRSresourceSets                  INTEGER ::= 8
maxnoofPRSresources                     INTEGER ::= 64
maxnoofSuccessfulHOReports              INTEGER ::= 64
maxnoofNR-UChannelIDs                   INTEGER ::= 16
maxServedCellforSON                     INTEGER ::= 256
maxNeighbourCellforSON                  INTEGER ::= 32
maxAffectedCells                        INTEGER ::= 32
maxnoofMRBs                             INTEGER ::= 32
maxnoofMBSQoSFlows                      INTEGER ::= 64
maxnoofMBSFSAs                          INTEGER ::= 256
maxnoofUEIDforPaging                    INTEGER ::= 4096
maxnoofCellsforMBS                      INTEGER ::= 512
maxnoofTAIforMBS                        INTEGER ::= 512
maxnoofMBSAreaSessionIDs                INTEGER ::= 256
maxnoofMBSServiceAreaInformation        INTEGER ::= 256
maxnoofIABCongInd                       INTEGER ::= 1024
maxnoofNeighbourNodeCellsIAB            INTEGER ::= 1024
maxnoofRBsetsPerCell                    INTEGER ::= 8
maxnoofRBsetsPerCell-1                  INTEGER ::= 7
maxnoofMeasPDC                          INTEGER ::= 16
maxnoARPs                               INTEGER ::= 16
maxnoofULAoAs                           INTEGER ::= 8
maxNoPathExtended                       INTEGER ::= 8
maxnoTRPTEGs                            INTEGER ::= 8
maxFreqLayers                           INTEGER ::= 4
maxNumResourcesPerAngle                 INTEGER ::= 24
maxnoAzimuthAngles                      INTEGER ::= 3600
maxnoElevationAngles                    INTEGER ::= 1801
maxnoofPRSTRPs                          INTEGER ::= 256
maxnoofQoEInformation                   INTEGER ::= 16
maxnoofUuRLCChannels                    INTEGER ::= 32
maxnoofPC5RLCChannels                   INTEGER ::= 512
maxnoofSMBRValues                       INTEGER ::= 8
maxnoofMRBsforUE                        INTEGER ::= 64
maxnoofMBSSessionsofUE                  INTEGER ::= 256
maxnoofSLdestinations                   INTEGER ::= 32
maxnoofNSAGs                            INTEGER ::= 256
```

```
maxnoofSDTBearers                      INTEGER ::= 72
maxnoofServingCellMOs                  INTEGER ::= 16
maxNrofBWPs                            INTEGER ::= 8
maxnoofPosSITypes                      INTEGER ::= 32
maxnoofUETypes                         INTEGER ::= 8
maxnoofLTMCells                        INTEGER ::= 8
maxnoofTAList                          INTEGER ::= 8
maxnoofLTMgNB-DUs                      INTEGER ::= 8
maxnoofUEsInQMCTransferControlMessage  INTEGER ::= 512
maxnoofUEsforRAReportIndications       INTEGER ::= 64
maxnoofSuccessfulPSCellChangeReports   INTEGER ::= 64
maxnoofPeriodicities                   INTEGER ::= 8
maxnoofThresholdMBS-1                  INTEGER ::= 7
maxMBSSessionsinSessionInfoList        INTEGER ::= 1024
maxnoofLBTFailureInformation           INTEGER ::= 64
maxnoofRSPPQoSFlows                    INTEGER ::= 2048
maxnoVACell                            INTEGER ::= 32
maxnoAggregatedSRS-Resources           INTEGER ::= 3
maxnoAggregatedPosSRSResourceSets      INTEGER ::= 3
maxnoAggregatedPosPRSResourceSets      INTEGER ::= 3
maxnoofTimeWindowSRS                   INTEGER ::= 16
maxnoofTimeWindowMea                   INTEGER ::= 16
maxnoPreconfiguredSRS                  INTEGER ::= 16
maxnoHopsMinusOne                      INTEGER ::= 5
maxnoAggCombinations                   INTEGER ::= 2
maxnoAggregatedPosSRSCombinations      INTEGER ::= 32
maxnoofCandidateCells                  INTEGER ::= 8
maxnoofSSBIndices                      INTEGER ::= 64
maxnoofPreambleIndex                   INTEGER ::= 64
maxnoofThresholds                      INTEGER ::= 8
maxnoofNZP-CSI-RS-ResourcesPerSet      INTEGER ::= 64
maxnoofSRS-Resources                   INTEGER ::= 64
maxnoofCellsinUEHistoryInfo            INTEGER ::= 16
maxnoofLTMCSI-RSResourceConfig         INTEGER ::= 112
maxnoofCSI-RSs                         INTEGER ::= 192
maxnoofTAs                             INTEGER ::= 2
maxnoofChannelRes                      INTEGER ::= 24
maxnoofCellsinNG-RANnode                   INTEGER ::= 16384
maxnoofLTM-CSI-ResourcesPerSet         INTEGER ::= 512



-- **************************************************************
--
-- IEs
--
-- **************************************************************

id-Cause                                           ProtocolIE-ID ::= 0
id-Cells-Failed-to-be-Activated-List               ProtocolIE-ID ::= 1
```

```
id-Cells-Failed-to-be-Activated-List-Item          ProtocolIE-ID ::= 2
id-Cells-to-be-Activated-List                      ProtocolIE-ID ::= 3
id-Cells-to-be-Activated-List-Item                 ProtocolIE-ID ::= 4
id-Cells-to-be-Deactivated-List                    ProtocolIE-ID ::= 5
id-Cells-to-be-Deactivated-List-Item               ProtocolIE-ID ::= 6
id-CriticalityDiagnostics                          ProtocolIE-ID ::= 7
id-CUtoDURRCInformation                            ProtocolIE-ID ::= 9
id-DRBs-FailedToBeModified-Item                    ProtocolIE-ID ::= 12
id-DRBs-FailedToBeModified-List                    ProtocolIE-ID ::= 13
id-DRBs-FailedToBeSetup-Item                       ProtocolIE-ID ::= 14
id-DRBs-FailedToBeSetup-List                       ProtocolIE-ID ::= 15
id-DRBs-FailedToBeSetupMod-Item                    ProtocolIE-ID ::= 16
id-DRBs-FailedToBeSetupMod-List                    ProtocolIE-ID ::= 17
id-DRBs-ModifiedConf-Item                          ProtocolIE-ID ::= 18
id-DRBs-ModifiedConf-List                          ProtocolIE-ID ::= 19
id-DRBs-Modified-Item                              ProtocolIE-ID ::= 20
id-DRBs-Modified-List                              ProtocolIE-ID ::= 21
id-DRBs-Required-ToBeModified-Item                 ProtocolIE-ID ::= 22
id-DRBs-Required-ToBeModified-List                 ProtocolIE-ID ::= 23
id-DRBs-Required-ToBeReleased-Item                 ProtocolIE-ID ::= 24
id-DRBs-Required-ToBeReleased-List                 ProtocolIE-ID ::= 25
id-DRBs-Setup-Item                                 ProtocolIE-ID ::= 26
id-DRBs-Setup-List                                 ProtocolIE-ID ::= 27
id-DRBs-SetupMod-Item                              ProtocolIE-ID ::= 28
id-DRBs-SetupMod-List                              ProtocolIE-ID ::= 29
id-DRBs-ToBeModified-Item                          ProtocolIE-ID ::= 30
id-DRBs-ToBeModified-List                          ProtocolIE-ID ::= 31
id-DRBs-ToBeReleased-Item                          ProtocolIE-ID ::= 32
id-DRBs-ToBeReleased-List                          ProtocolIE-ID ::= 33
id-DRBs-ToBeSetup-Item                             ProtocolIE-ID ::= 34
id-DRBs-ToBeSetup-List                             ProtocolIE-ID ::= 35
id-DRBs-ToBeSetupMod-Item                          ProtocolIE-ID ::= 36
id-DRBs-ToBeSetupMod-List                          ProtocolIE-ID ::= 37
id-DRXCycle                                        ProtocolIE-ID ::= 38
id-DUtoCURRCInformation                            ProtocolIE-ID ::= 39
id-gNB-CU-UE-F1AP-ID                               ProtocolIE-ID ::= 40
id-gNB-DU-UE-F1AP-ID                               ProtocolIE-ID ::= 41
id-gNB-DU-ID                                       ProtocolIE-ID ::= 42
id-GNB-DU-Served-Cells-Item                        ProtocolIE-ID ::= 43
id-gNB-DU-Served-Cells-List                        ProtocolIE-ID ::= 44
id-gNB-DU-Name                                     ProtocolIE-ID ::= 45
id-ProtocolIE-ID-46-not-to-be-used                 ProtocolIE-ID ::= 46
id-oldgNB-DU-UE-F1AP-ID                            ProtocolIE-ID ::= 47
id-ResetType                                       ProtocolIE-ID ::= 48
id-ResourceCoordinationTransferContainer           ProtocolIE-ID ::= 49
id-RRCContainer                                    ProtocolIE-ID ::= 50
id-SCell-ToBeRemoved-Item                          ProtocolIE-ID ::= 51
id-SCell-ToBeRemoved-List                          ProtocolIE-ID ::= 52
id-SCell-ToBeSetup-Item                            ProtocolIE-ID ::= 53
id-SCell-ToBeSetup-List                            ProtocolIE-ID ::= 54
id-SCell-ToBeSetupMod-Item                         ProtocolIE-ID ::= 55
id-SCell-ToBeSetupMod-List                         ProtocolIE-ID ::= 56
id-Served-Cells-To-Add-Item                        ProtocolIE-ID ::= 57
id-Served-Cells-To-Add-List                        ProtocolIE-ID ::= 58
```

```
id-Served-Cells-To-Delete-Item                          ProtocolIE-ID ::= 59
id-Served-Cells-To-Delete-List                          ProtocolIE-ID ::= 60
id-Served-Cells-To-Modify-Item                          ProtocolIE-ID ::= 61
id-Served-Cells-To-Modify-List                          ProtocolIE-ID ::= 62
id-SpCell-ID                                            ProtocolIE-ID ::= 63
id-SRBID                                                ProtocolIE-ID ::= 64
id-SRBs-FailedToBeSetup-Item                            ProtocolIE-ID ::= 65
id-SRBs-FailedToBeSetup-List                            ProtocolIE-ID ::= 66
id-SRBs-FailedToBeSetupMod-Item                         ProtocolIE-ID ::= 67
id-SRBs-FailedToBeSetupMod-List                         ProtocolIE-ID ::= 68
id-SRBs-Required-ToBeReleased-Item                      ProtocolIE-ID ::= 69
id-SRBs-Required-ToBeReleased-List                      ProtocolIE-ID ::= 70
id-SRBs-ToBeReleased-Item                               ProtocolIE-ID ::= 71
id-SRBs-ToBeReleased-List                               ProtocolIE-ID ::= 72
id-SRBs-ToBeSetup-Item                                  ProtocolIE-ID ::= 73
id-SRBs-ToBeSetup-List                                  ProtocolIE-ID ::= 74
id-SRBs-ToBeSetupMod-Item                               ProtocolIE-ID ::= 75
id-SRBs-ToBeSetupMod-List                               ProtocolIE-ID ::= 76
id-TimeToWait                                           ProtocolIE-ID ::= 77
id-TransactionID                                        ProtocolIE-ID ::= 78
id-TransmissionActionIndicator                          ProtocolIE-ID ::= 79
id-UE-associatedLogicalF1-ConnectionItem                ProtocolIE-ID ::= 80
id-UE-associatedLogicalF1-ConnectionListResAck          ProtocolIE-ID ::= 81
id-gNB-CU-Name                                          ProtocolIE-ID ::= 82
id-SCell-FailedtoSetup-List                             ProtocolIE-ID ::= 83
id-SCell-FailedtoSetup-Item                             ProtocolIE-ID ::= 84
id-SCell-FailedtoSetupMod-List                          ProtocolIE-ID ::= 85
id-SCell-FailedtoSetupMod-Item                          ProtocolIE-ID ::= 86
id-RRCReconfigurationCompleteIndicator                  ProtocolIE-ID ::= 87
id-Cells-Status-Item                                    ProtocolIE-ID ::= 88
id-Cells-Status-List                                    ProtocolIE-ID ::= 89
id-Candidate-SpCell-List                                ProtocolIE-ID ::= 90
id-Candidate-SpCell-Item                                ProtocolIE-ID ::= 91
id-Potential-SpCell-List                                ProtocolIE-ID ::= 92
id-Potential-SpCell-Item                                ProtocolIE-ID ::= 93
id-FullConfiguration                                    ProtocolIE-ID ::= 94
id-C-RNTI                                               ProtocolIE-ID ::= 95
id-SpCellULConfigured                                   ProtocolIE-ID ::= 96
id-InactivityMonitoringRequest                          ProtocolIE-ID ::= 97
id-InactivityMonitoringResponse                         ProtocolIE-ID ::= 98
id-DRB-Activity-Item                                    ProtocolIE-ID ::= 99
id-DRB-Activity-List                                    ProtocolIE-ID ::= 100
id-EUTRA-NR-CellResourceCoordinationReq-Container       ProtocolIE-ID ::= 101
id-EUTRA-NR-CellResourceCoordinationReqAck-Container        ProtocolIE-ID ::= 102
id-Protected-EUTRA-Resources-List                       ProtocolIE-ID ::= 105
id-RequestType                                          ProtocolIE-ID ::= 106
id-ServCellIndex                                        ProtocolIE-ID ::= 107
id-RAT-FrequencyPriorityInformation                     ProtocolIE-ID ::= 108
id-ExecuteDuplication                                   ProtocolIE-ID ::= 109
id-NRCGI                                                ProtocolIE-ID ::= 111
id-PagingCell-Item                                      ProtocolIE-ID ::= 112
id-PagingCell-List                                      ProtocolIE-ID ::= 113
id-PagingDRX                                            ProtocolIE-ID ::= 114
id-PagingPriority                                       ProtocolIE-ID ::= 115
```

```
id-SItype-List                                  ProtocolIE-ID ::= 116
id-UEIdentityIndexValue                         ProtocolIE-ID ::= 117
id-gNB-CUSystemInformation                      ProtocolIE-ID ::= 118
id-HandoverPreparationInformation               ProtocolIE-ID ::= 119
id-GNB-CU-TNL-Association-To-Add-Item           ProtocolIE-ID ::= 120
id-GNB-CU-TNL-Association-To-Add-List           ProtocolIE-ID ::= 121
id-GNB-CU-TNL-Association-To-Remove-Item        ProtocolIE-ID ::= 122
id-GNB-CU-TNL-Association-To-Remove-List        ProtocolIE-ID ::= 123
id-GNB-CU-TNL-Association-To-Update-Item        ProtocolIE-ID ::= 124
id-GNB-CU-TNL-Association-To-Update-List        ProtocolIE-ID ::= 125
id-MaskedIMEISV                                 ProtocolIE-ID ::= 126
id-PagingIdentity                               ProtocolIE-ID ::= 127
id-DUtoCURRCContainer                           ProtocolIE-ID ::= 128
id-Cells-to-be-Barred-List                      ProtocolIE-ID ::= 129
id-Cells-to-be-Barred-Item                      ProtocolIE-ID ::= 130
id-TAISliceSupportList                          ProtocolIE-ID ::= 131
id-GNB-CU-TNL-Association-Setup-List            ProtocolIE-ID ::= 132
id-GNB-CU-TNL-Association-Setup-Item            ProtocolIE-ID ::= 133
id-GNB-CU-TNL-Association-Failed-To-Setup-List  ProtocolIE-ID ::= 134
id-GNB-CU-TNL-Association-Failed-To-Setup-Item  ProtocolIE-ID ::= 135
id-DRB-Notify-Item                              ProtocolIE-ID ::= 136
id-DRB-Notify-List                              ProtocolIE-ID ::= 137
id-ProtocolIE-ID-138-not-to-be-used             ProtocolIE-ID ::= 138
id-RANAC                                        ProtocolIE-ID ::= 139
id-PWSSystemInformation                         ProtocolIE-ID ::= 140
id-RepetitionPeriod                             ProtocolIE-ID ::= 141
id-NumberofBroadcastRequest                     ProtocolIE-ID ::= 142
id-Cells-To-Be-Broadcast-List                   ProtocolIE-ID ::= 144
id-Cells-To-Be-Broadcast-Item                   ProtocolIE-ID ::= 145
id-Cells-Broadcast-Completed-List               ProtocolIE-ID ::= 146
id-Cells-Broadcast-Completed-Item               ProtocolIE-ID ::= 147
id-Broadcast-To-Be-Cancelled-List               ProtocolIE-ID ::= 148
id-Broadcast-To-Be-Cancelled-Item               ProtocolIE-ID ::= 149
id-Cells-Broadcast-Cancelled-List               ProtocolIE-ID ::= 150
id-Cells-Broadcast-Cancelled-Item               ProtocolIE-ID ::= 151
id-NR-CGI-List-For-Restart-List                 ProtocolIE-ID ::= 152
id-NR-CGI-List-For-Restart-Item                 ProtocolIE-ID ::= 153
id-PWS-Failed-NR-CGI-List                       ProtocolIE-ID ::= 154
id-PWS-Failed-NR-CGI-Item                       ProtocolIE-ID ::= 155
id-ConfirmedUEID                                ProtocolIE-ID ::= 156
id-Cancel-all-Warning-Messages-Indicator        ProtocolIE-ID ::= 157
id-GNB-DU-UE-AMBR-UL                            ProtocolIE-ID ::= 158
id-DRXConfigurationIndicator                    ProtocolIE-ID ::= 159
id-RLC-Status                                   ProtocolIE-ID ::= 160
id-DLPDCPSNLength                               ProtocolIE-ID ::= 161
id-GNB-DUConfigurationQuery                     ProtocolIE-ID ::= 162
id-MeasurementTimingConfiguration               ProtocolIE-ID ::= 163
id-DRB-Information                              ProtocolIE-ID ::= 164
id-ServingPLMN                                  ProtocolIE-ID ::= 165
id-Protected-EUTRA-Resources-Item               ProtocolIE-ID ::= 168
id-GNB-CU-RRC-Version                           ProtocolIE-ID ::= 170
id-GNB-DU-RRC-Version                           ProtocolIE-ID ::= 171
id-GNBDUOverloadInformation                     ProtocolIE-ID ::= 172
id-CellGroupConfig                              ProtocolIE-ID ::= 173
```

```
id-RLCFailureIndication                     ProtocolIE-ID ::= 174
id-UplinkTxDirectCurrentListInformation     ProtocolIE-ID ::= 175
id-DC-Based-Duplication-Configured          ProtocolIE-ID ::= 176
id-DC-Based-Duplication-Activation          ProtocolIE-ID ::= 177
id-SULAccessIndication                      ProtocolIE-ID ::= 178
id-AvailablePLMNList                        ProtocolIE-ID ::= 179
id-PDUSessionID                             ProtocolIE-ID ::= 180
id-ULPDUSessionAggregateMaximumBitRate      ProtocolIE-ID ::= 181
id-ServingCellMO                            ProtocolIE-ID ::= 182
id-QoSFlowMappingIndication                 ProtocolIE-ID ::= 183
id-RRCDeliveryStatusRequest                 ProtocolIE-ID ::= 184
id-RRCDeliveryStatus                        ProtocolIE-ID ::= 185
id-BearerTypeChange                         ProtocolIE-ID ::= 186
id-RLCMode                                  ProtocolIE-ID ::= 187
id-Duplication-Activation                   ProtocolIE-ID ::= 188
id-Dedicated-SIDelivery-NeededUE-List       ProtocolIE-ID ::= 189
id-Dedicated-SIDelivery-NeededUE-Item       ProtocolIE-ID ::= 190
id-DRX-LongCycleStartOffset                 ProtocolIE-ID ::= 191
id-ULPDCPSNLength                           ProtocolIE-ID ::= 192
id-SelectedBandCombinationIndex             ProtocolIE-ID ::= 193
id-SelectedFeatureSetEntryIndex             ProtocolIE-ID ::= 194
id-ResourceCoordinationTransferInformation  ProtocolIE-ID ::= 195
id-ExtendedServedPLMNs-List                 ProtocolIE-ID ::= 196
id-ExtendedAvailablePLMN-List               ProtocolIE-ID ::= 197
id-Associated-SCell-List                    ProtocolIE-ID ::= 198
id-latest-RRC-Version-Enhanced              ProtocolIE-ID ::= 199
id-Associated-SCell-Item                    ProtocolIE-ID ::= 200
id-Cell-Direction                           ProtocolIE-ID ::= 201
id-SRBs-Setup-List                          ProtocolIE-ID ::= 202
id-SRBs-Setup-Item                          ProtocolIE-ID ::= 203
id-SRBs-SetupMod-List                       ProtocolIE-ID ::= 204
id-SRBs-SetupMod-Item                       ProtocolIE-ID ::= 205
id-SRBs-Modified-List                       ProtocolIE-ID ::= 206
id-SRBs-Modified-Item                       ProtocolIE-ID ::= 207
id-Ph-InfoSCG                               ProtocolIE-ID ::= 208
id-RequestedBandCombinationIndex            ProtocolIE-ID ::= 209
id-RequestedFeatureSetEntryIndex            ProtocolIE-ID ::= 210
id-RequestedP-MaxFR2                        ProtocolIE-ID ::= 211
id-DRX-Config                               ProtocolIE-ID ::= 212
id-IgnoreResourceCoordinationContainer      ProtocolIE-ID ::= 213
id-UEAssistanceInformation                  ProtocolIE-ID ::= 214
id-NeedforGap                               ProtocolIE-ID ::= 215
id-PagingOrigin                             ProtocolIE-ID ::= 216
id-new-gNB-CU-UE-F1AP-ID                    ProtocolIE-ID ::= 217
id-RedirectedRRCmessage                     ProtocolIE-ID ::= 218
id-new-gNB-DU-UE-F1AP-ID                    ProtocolIE-ID ::= 219
id-NotificationInformation                  ProtocolIE-ID ::= 220
id-PLMNAssistanceInfoForNetShar             ProtocolIE-ID ::= 221
id-UEContextNotRetrievable                  ProtocolIE-ID ::= 222
id-BPLMN-ID-Info-List                       ProtocolIE-ID ::= 223
id-SelectedPLMNID                           ProtocolIE-ID ::= 224
id-UAC-Assistance-Info                      ProtocolIE-ID ::= 225
id-RANUEID                                  ProtocolIE-ID ::= 226
id-GNB-DU-TNL-Association-To-Remove-Item    ProtocolIE-ID ::= 227
```

```
id-GNB-DU-TNL-Association-To-Remove-List                    ProtocolIE-ID ::= 228
id-TNLAssociationTransportLayerAddressgNBDU                 ProtocolIE-ID ::= 229
id-portNumber                                               ProtocolIE-ID ::= 230
id-AdditionalSIBMessageList                                 ProtocolIE-ID ::= 231
id-Cell-Type                                                ProtocolIE-ID ::= 232
id-IgnorePRACHConfiguration                                 ProtocolIE-ID ::= 233
id-CG-Config                                                ProtocolIE-ID ::= 234
id-PDCCH-BlindDetectionSCG                                  ProtocolIE-ID ::= 235
id-Requested-PDCCH-BlindDetectionSCG                        ProtocolIE-ID ::= 236
id-Ph-InfoMCG                                               ProtocolIE-ID ::= 237
id-MeasGapSharingConfig                                     ProtocolIE-ID ::= 238
id-systemInformationAreaID                                  ProtocolIE-ID ::= 239
id-areaScope                                                ProtocolIE-ID ::= 240
id-RRCContainer-RRCSetupComplete                            ProtocolIE-ID ::= 241
id-TraceActivation                                          ProtocolIE-ID ::= 242
id-TraceID                                                  ProtocolIE-ID ::= 243
id-Neighbour-Cell-Information-List                          ProtocolIE-ID ::= 244
id-ProtocolIE-ID-246-not-to-be-used                         ProtocolIE-ID ::= 246
id-ProtocolIE-ID-247-not-to-be-used                         ProtocolIE-ID ::= 247
id-AdditionalRRMPriorityIndex                               ProtocolIE-ID ::= 248
id-DUCURadioInformationType                                 ProtocolIE-ID ::= 249
id-CUDURadioInformationType                                 ProtocolIE-ID ::= 250
id-AggressorgNBSetID                                        ProtocolIE-ID ::= 251
id-VictimgNBSetID                                           ProtocolIE-ID ::= 252
id-LowerLayerPresenceStatusChange                           ProtocolIE-ID ::= 253
id-Transport-Layer-Address-Info                             ProtocolIE-ID ::= 254
id-Neighbour-Cell-Information-Item                          ProtocolIE-ID ::= 255
id-IntendedTDD-DL-ULConfig                                  ProtocolIE-ID ::= 256
id-QosMonitoringRequest                                     ProtocolIE-ID ::= 257
id-BHChannels-ToBeSetup-List                                ProtocolIE-ID ::= 258
id-BHChannels-ToBeSetup-Item                                ProtocolIE-ID ::= 259
id-BHChannels-Setup-List                                    ProtocolIE-ID ::= 260
id-BHChannels-Setup-Item                                    ProtocolIE-ID ::= 261
id-BHChannels-ToBeModified-Item                             ProtocolIE-ID ::= 262
id-BHChannels-ToBeModified-List                             ProtocolIE-ID ::= 263
id-BHChannels-ToBeReleased-Item                             ProtocolIE-ID ::= 264
id-BHChannels-ToBeReleased-List                             ProtocolIE-ID ::= 265
id-BHChannels-ToBeSetupMod-Item                             ProtocolIE-ID ::= 266
id-BHChannels-ToBeSetupMod-List                             ProtocolIE-ID ::= 267
id-BHChannels-FailedToBeModified-Item                       ProtocolIE-ID ::= 268
id-BHChannels-FailedToBeModified-List                       ProtocolIE-ID ::= 269
id-BHChannels-FailedToBeSetupMod-Item                       ProtocolIE-ID ::= 270
id-BHChannels-FailedToBeSetupMod-List                       ProtocolIE-ID ::= 271
id-BHChannels-Modified-Item                                 ProtocolIE-ID ::= 272
id-BHChannels-Modified-List                                 ProtocolIE-ID ::= 273
id-BHChannels-SetupMod-Item                                 ProtocolIE-ID ::= 274
id-BHChannels-SetupMod-List                                 ProtocolIE-ID ::= 275
id-BHChannels-Required-ToBeReleased-Item                    ProtocolIE-ID ::= 276
id-BHChannels-Required-ToBeReleased-List                    ProtocolIE-ID ::= 277
id-BHChannels-FailedToBeSetup-Item                          ProtocolIE-ID ::= 278
id-BHChannels-FailedToBeSetup-List                          ProtocolIE-ID ::= 279
id-BHInfo                                                   ProtocolIE-ID ::= 280
id-BAPAddress                                               ProtocolIE-ID ::= 281
id-ConfiguredBAPAddress                                     ProtocolIE-ID ::= 282
```

```
id-BH-Routing-Information-Added-List                ProtocolIE-ID ::= 283
id-BH-Routing-Information-Added-List-Item           ProtocolIE-ID ::= 284
id-BH-Routing-Information-Removed-List              ProtocolIE-ID ::= 285
id-BH-Routing-Information-Removed-List-Item         ProtocolIE-ID ::= 286
id-UL-BH-Non-UP-Traffic-Mapping                     ProtocolIE-ID ::= 287
id-Activated-Cells-to-be-Updated-List               ProtocolIE-ID ::= 288
id-Child-Nodes-List                                 ProtocolIE-ID ::= 289
id-IAB-Info-IAB-DU                                  ProtocolIE-ID ::= 290
id-IAB-Info-IAB-donor-CU                            ProtocolIE-ID ::= 291
id-IAB-TNL-Addresses-To-Remove-List                 ProtocolIE-ID ::= 292
id-IAB-TNL-Addresses-To-Remove-Item                 ProtocolIE-ID ::= 293
id-IAB-Allocated-TNL-Address-List                   ProtocolIE-ID ::= 294
id-IAB-Allocated-TNL-Address-Item                   ProtocolIE-ID ::= 295
id-IABIPv6RequestType                               ProtocolIE-ID ::= 296
id-IABv4AddressesRequested                          ProtocolIE-ID ::= 297
id-IAB-Barred                                       ProtocolIE-ID ::= 298
id-TrafficMappingInformation                        ProtocolIE-ID ::= 299
id-UL-UP-TNL-Information-to-Update-List             ProtocolIE-ID ::= 300
id-UL-UP-TNL-Information-to-Update-List-Item        ProtocolIE-ID ::= 301
id-UL-UP-TNL-Address-to-Update-List                 ProtocolIE-ID ::= 302
id-UL-UP-TNL-Address-to-Update-List-Item            ProtocolIE-ID ::= 303
id-DL-UP-TNL-Address-to-Update-List                 ProtocolIE-ID ::= 304
id-DL-UP-TNL-Address-to-Update-List-Item            ProtocolIE-ID ::= 305
id-NRV2XServicesAuthorized                          ProtocolIE-ID ::= 306
id-LTEV2XServicesAuthorized                         ProtocolIE-ID ::= 307
id-NRUESidelinkAggregateMaximumBitrate              ProtocolIE-ID ::= 308
id-LTEUESidelinkAggregateMaximumBitrate             ProtocolIE-ID ::= 309
id-SIB12-message                                    ProtocolIE-ID ::= 310
id-SIB13-message                                    ProtocolIE-ID ::= 311
id-SIB14-message                                    ProtocolIE-ID ::= 312
id-SLDRBs-FailedToBeModified-Item                   ProtocolIE-ID ::= 313
id-SLDRBs-FailedToBeModified-List                   ProtocolIE-ID ::= 314
id-SLDRBs-FailedToBeSetup-Item                      ProtocolIE-ID ::= 315
id-SLDRBs-FailedToBeSetup-List                      ProtocolIE-ID ::= 316
id-SLDRBs-Modified-Item                             ProtocolIE-ID ::= 317
id-SLDRBs-Modified-List                             ProtocolIE-ID ::= 318
id-SLDRBs-Required-ToBeModified-Item                ProtocolIE-ID ::= 319
id-SLDRBs-Required-ToBeModified-List                ProtocolIE-ID ::= 320
id-SLDRBs-Required-ToBeReleased-Item                ProtocolIE-ID ::= 321
id-SLDRBs-Required-ToBeReleased-List                ProtocolIE-ID ::= 322
id-SLDRBs-Setup-Item                                ProtocolIE-ID ::= 323
id-SLDRBs-Setup-List                                ProtocolIE-ID ::= 324
id-SLDRBs-ToBeModified-Item                         ProtocolIE-ID ::= 325
id-SLDRBs-ToBeModified-List                         ProtocolIE-ID ::= 326
id-SLDRBs-ToBeReleased-Item                         ProtocolIE-ID ::= 327
id-SLDRBs-ToBeReleased-List                         ProtocolIE-ID ::= 328
id-SLDRBs-ToBeSetup-Item                            ProtocolIE-ID ::= 329
id-SLDRBs-ToBeSetup-List                            ProtocolIE-ID ::= 330
id-SLDRBs-ToBeSetupMod-Item                         ProtocolIE-ID ::= 331
id-SLDRBs-ToBeSetupMod-List                         ProtocolIE-ID ::= 332
id-SLDRBs-SetupMod-List                             ProtocolIE-ID ::= 333
id-SLDRBs-FailedToBeSetupMod-List                   ProtocolIE-ID ::= 334
id-SLDRBs-SetupMod-Item                             ProtocolIE-ID ::= 335
id-SLDRBs-FailedToBeSetupMod-Item                   ProtocolIE-ID ::= 336
```

```
id-SLDRBs-ModifiedConf-List                                 ProtocolIE-ID ::= 337
id-SLDRBs-ModifiedConf-Item                                 ProtocolIE-ID ::= 338
id-UEAssistanceInformationEUTRA                             ProtocolIE-ID ::= 339
id-PC5LinkAMBR                                              ProtocolIE-ID ::= 340
id-SL-PHY-MAC-RLC-Config                                    ProtocolIE-ID ::= 341
id-SL-ConfigDedicatedEUTRA-Info                             ProtocolIE-ID ::= 342
id-AlternativeQoSParaSetList                                ProtocolIE-ID ::= 343
id-CurrentQoSParaSetIndex                                   ProtocolIE-ID ::= 344
id-gNBCUMeasurementID                                       ProtocolIE-ID ::= 345
id-gNBDUMeasurementID                                       ProtocolIE-ID ::= 346
id-RegistrationRequest                                      ProtocolIE-ID ::= 347
id-ReportCharacteristics                                    ProtocolIE-ID ::= 348
id-CellToReportList                                         ProtocolIE-ID ::= 349
id-CellMeasurementResultList                                ProtocolIE-ID ::= 350
id-HardwareLoadIndicator                                    ProtocolIE-ID ::= 351
id-ReportingPeriodicity                                     ProtocolIE-ID ::= 352
id-TNLCapacityIndicator                                     ProtocolIE-ID ::= 353
id-CarrierList                                              ProtocolIE-ID ::= 354
id-ULCarrierList                                            ProtocolIE-ID ::= 355
id-FrequencyShift7p5khz                                     ProtocolIE-ID ::= 356
id-SSB-PositionsInBurst                                     ProtocolIE-ID ::= 357
id-NRPRACHConfig                                            ProtocolIE-ID ::= 358
id-RAReportList                                             ProtocolIE-ID ::= 359
id-RLFReportInformationList                                 ProtocolIE-ID ::= 360
id-TDD-UL-DLConfigCommonNR                                  ProtocolIE-ID ::= 361
id-CNPacketDelayBudgetDownlink                              ProtocolIE-ID ::= 362
id-ExtendedPacketDelayBudget                                ProtocolIE-ID ::= 363
id-TSCTrafficCharacteristics                                ProtocolIE-ID ::= 364
id-ReportingRequestType                                     ProtocolIE-ID ::= 365
id-TimeReferenceInformation                                 ProtocolIE-ID ::= 366
id-CNPacketDelayBudgetUplink                                ProtocolIE-ID ::= 369
id-AdditionalPDCPDuplicationTNL-List                        ProtocolIE-ID ::= 370
id-RLCDuplicationInformation                                ProtocolIE-ID ::= 371
id-AdditionalDuplicationIndication                          ProtocolIE-ID ::= 372
id-ConditionalInterDUMobilityInformation                    ProtocolIE-ID ::= 373
id-ConditionalIntraDUMobilityInformation                    ProtocolIE-ID ::= 374
id-targetCellsToCancel                                      ProtocolIE-ID ::= 375
id-requestedTargetCellGlobalID                              ProtocolIE-ID ::= 376
id-ManagementBasedMDTPLMNList                               ProtocolIE-ID ::= 377
id-TraceCollectionEntityIPAddress                           ProtocolIE-ID ::= 378
id-PrivacyIndicator                                         ProtocolIE-ID ::= 379
id-TraceCollectionEntityURI                                 ProtocolIE-ID ::= 380
id-mdtConfiguration                                         ProtocolIE-ID ::= 381
id-ServingNID                                               ProtocolIE-ID ::= 382
id-NPNBroadcastInformation                                  ProtocolIE-ID ::= 383
id-NPNSupportInfo                                           ProtocolIE-ID ::= 384
id-NID                                                      ProtocolIE-ID ::= 385
id-AvailableSNPN-ID-List                                    ProtocolIE-ID ::= 386
id-SIB10-message                                            ProtocolIE-ID ::= 387
id-DLCarrierList                                            ProtocolIE-ID ::= 389
    id-ExtendedTAISliceSupportList                          ProtocolIE-ID ::= 390
id-RequestedSRSTransmissionCharacteristics                  ProtocolIE-ID ::= 391
id-PosAssistance-Information                                ProtocolIE-ID ::= 392
id-PosBroadcast                                             ProtocolIE-ID ::= 393
```

```
id-RoutingID                                        ProtocolIE-ID ::= 394
id-PosAssistanceInformationFailureList              ProtocolIE-ID ::= 395
id-PosMeasurementQuantities                         ProtocolIE-ID ::= 396
id-PosMeasurementResultList                         ProtocolIE-ID ::= 397
id-TRPInformationTypeListTRPReq                     ProtocolIE-ID ::= 398
id-TRPInformationTypeItem                           ProtocolIE-ID ::= 399
id-TRPInformationListTRPResp                        ProtocolIE-ID ::= 400
id-TRPInformationItem                               ProtocolIE-ID ::= 401
id-LMF-MeasurementID                                ProtocolIE-ID ::= 402
id-SRSType                                          ProtocolIE-ID ::= 403
id-ActivationTime                                   ProtocolIE-ID ::= 404
id-AbortTransmission                                ProtocolIE-ID ::= 405
id-PositioningBroadcastCells                        ProtocolIE-ID ::= 406
id-SRSConfiguration                                 ProtocolIE-ID ::= 407
id-PosReportCharacteristics                         ProtocolIE-ID ::= 408
id-PosMeasurementPeriodicity                        ProtocolIE-ID ::= 409
id-TRPList                                          ProtocolIE-ID ::= 410
id-RAN-MeasurementID                                ProtocolIE-ID ::= 411
id-LMF-UE-MeasurementID                             ProtocolIE-ID ::= 412
id-RAN-UE-MeasurementID                             ProtocolIE-ID ::= 413
id-E-CID-MeasurementQuantities                      ProtocolIE-ID ::= 414
id-E-CID-MeasurementQuantities-Item                 ProtocolIE-ID ::= 415
id-E-CID-MeasurementPeriodicity                     ProtocolIE-ID ::= 416
id-E-CID-MeasurementResult                          ProtocolIE-ID ::= 417
id-Cell-Portion-ID                                  ProtocolIE-ID ::= 418
id-SFNInitialisationTime                            ProtocolIE-ID ::= 419
id-SystemFrameNumber                                ProtocolIE-ID ::= 420
id-SlotNumber                                       ProtocolIE-ID ::= 421
id-TRP-MeasurementRequestList                       ProtocolIE-ID ::= 422
id-MeasurementBeamInfoRequest                       ProtocolIE-ID ::= 423
id-E-CID-ReportCharacteristics                      ProtocolIE-ID ::= 424
id-ConfiguredTACIndication                          ProtocolIE-ID ::= 425
id-Extended-GNB-CU-Name                             ProtocolIE-ID ::= 426
id-Extended-GNB-DU-Name                             ProtocolIE-ID ::= 427
id-F1CTransferPath                                  ProtocolIE-ID ::= 428
id-SFN-Offset                                       ProtocolIE-ID ::= 429
id-TransmissionStopIndicator                        ProtocolIE-ID ::= 430
id-SrsFrequency                                     ProtocolIE-ID ::= 431
id-SCGIndicator                                     ProtocolIE-ID ::= 432
id-EstimatedArrivalProbability                      ProtocolIE-ID ::= 433
id-TRPType                                          ProtocolIE-ID ::= 434
id-SRSSpatialRelationPerSRSResource                 ProtocolIE-ID ::= 435
id-PDCPTerminatingNodeDLTNLAddrInfo                 ProtocolIE-ID ::= 436
id-ENBDLTNLAddress                                  ProtocolIE-ID ::= 437
id-PosMeasurementPeriodicityExtended                ProtocolIE-ID ::= 438
id-PRS-Resource-ID                                  ProtocolIE-ID ::= 439
id-LocationMeasurementInformation                   ProtocolIE-ID ::= 440
id-SliceRadioResourceStatus                         ProtocolIE-ID ::= 441
id-CompositeAvailableCapacity-SUL                   ProtocolIE-ID ::= 442
id-SuccessfulHOReportInformationList                ProtocolIE-ID ::= 443
id-NR-U-Channel-List                                ProtocolIE-ID ::= 444
id-NR-U                                             ProtocolIE-ID ::= 445
id-Coverage-Modification-Notification               ProtocolIE-ID ::= 446
id-CCO-Assistance-Information                       ProtocolIE-ID ::= 447
```

```
id-ProtocolIE-ID-448-not-to-be-used             ProtocolIE-ID ::= 448
id-CellsForSON-List                                 ProtocolIE-ID ::= 449
id-MIMOPRBusageInformation                          ProtocolIE-ID ::= 450
id-gNB-CU-MBS-F1AP-ID                               ProtocolIE-ID ::= 451
id-gNB-DU-MBS-F1AP-ID                               ProtocolIE-ID ::= 452
id-ProtocolIE-ID-453-not-to-be-used             ProtocolIE-ID ::= 453
id-MBS-CUtoDURRCInformation                         ProtocolIE-ID ::= 454
id-MBS-Session-ID                                   ProtocolIE-ID ::= 455
id-SNSSAI                                           ProtocolIE-ID ::= 456
id-MBS-Broadcast-NeighbourCellList                  ProtocolIE-ID ::= 457
id-BroadcastMRBs-FailedToBeModified-List            ProtocolIE-ID ::= 458
id-BroadcastMRBs-FailedToBeModified-Item            ProtocolIE-ID ::= 459
id-BroadcastMRBs-FailedToBeSetup-List               ProtocolIE-ID ::= 460
id-BroadcastMRBs-FailedToBeSetup-Item               ProtocolIE-ID ::= 461
id-BroadcastMRBs-FailedToBeSetupMod-List            ProtocolIE-ID ::= 462
id-BroadcastMRBs-FailedToBeSetupMod-Item            ProtocolIE-ID ::= 463
id-BroadcastMRBs-Modified-List                      ProtocolIE-ID ::= 464
id-BroadcastMRBs-Modified-Item                      ProtocolIE-ID ::= 465
id-BroadcastMRBs-Setup-List                         ProtocolIE-ID ::= 466
id-BroadcastMRBs-Setup-Item                         ProtocolIE-ID ::= 467
id-BroadcastMRBs-SetupMod-List                      ProtocolIE-ID ::= 468
id-BroadcastMRBs-SetupMod-Item                      ProtocolIE-ID ::= 469
id-BroadcastMRBs-ToBeModified-List                  ProtocolIE-ID ::= 470
id-BroadcastMRBs-ToBeModified-Item                  ProtocolIE-ID ::= 471
id-BroadcastMRBs-ToBeReleased-List                  ProtocolIE-ID ::= 472
id-BroadcastMRBs-ToBeReleased-Item                  ProtocolIE-ID ::= 473
id-BroadcastMRBs-ToBeSetup-List                     ProtocolIE-ID ::= 474
id-BroadcastMRBs-ToBeSetup-Item                     ProtocolIE-ID ::= 475
id-BroadcastMRBs-ToBeSetupMod-List                  ProtocolIE-ID ::= 476
id-BroadcastMRBs-ToBeSetupMod-Item                  ProtocolIE-ID ::= 477
id-Supported-MBS-FSA-ID-List                        ProtocolIE-ID ::= 478
id-UEIdentity-List-For-Paging-List                  ProtocolIE-ID ::= 479
id-UEIdentity-List-For-Paging-Item                  ProtocolIE-ID ::= 480
id-MBS-ServiceArea                                  ProtocolIE-ID ::= 481
id-MulticastMRBs-FailedToBeModified-List            ProtocolIE-ID ::= 482
id-MulticastMRBs-FailedToBeModified-Item            ProtocolIE-ID ::= 483
id-MulticastMRBs-FailedToBeSetup-List               ProtocolIE-ID ::= 484
id-MulticastMRBs-FailedToBeSetup-Item               ProtocolIE-ID ::= 485
id-MulticastMRBs-FailedToBeSetupMod-List            ProtocolIE-ID ::= 486
id-MulticastMRBs-FailedToBeSetupMod-Item            ProtocolIE-ID ::= 487
id-MulticastMRBs-Modified-List                      ProtocolIE-ID ::= 488
id-MulticastMRBs-Modified-Item                      ProtocolIE-ID ::= 489
id-MulticastMRBs-Setup-List                         ProtocolIE-ID ::= 490
id-MulticastMRBs-Setup-Item                         ProtocolIE-ID ::= 491
id-MulticastMRBs-SetupMod-List                      ProtocolIE-ID ::= 492
id-MulticastMRBs-SetupMod-Item                      ProtocolIE-ID ::= 493
id-MulticastMRBs-ToBeModified-List                  ProtocolIE-ID ::= 494
id-MulticastMRBs-ToBeModified-Item                  ProtocolIE-ID ::= 495
id-MulticastMRBs-ToBeReleased-List                  ProtocolIE-ID ::= 496
id-MulticastMRBs-ToBeReleased-Item                  ProtocolIE-ID ::= 497
id-MulticastMRBs-ToBeSetup-List                     ProtocolIE-ID ::= 498
id-MulticastMRBs-ToBeSetup-Item                     ProtocolIE-ID ::= 499
id-MulticastMRBs-ToBeSetupMod-List                  ProtocolIE-ID ::= 500
id-MulticastMRBs-ToBeSetupMod-Item                  ProtocolIE-ID ::= 501
```

```
id-MBSMulticastF1UContextDescriptor                 ProtocolIE-ID ::= 502
id-MulticastF1UContext-ToBeSetup-List               ProtocolIE-ID ::= 503
id-MulticastF1UContext-ToBeSetup-Item               ProtocolIE-ID ::= 504
id-MulticastF1UContext-Setup-List                   ProtocolIE-ID ::= 505
id-MulticastF1UContext-Setup-Item                   ProtocolIE-ID ::= 506
id-MulticastF1UContext-FailedToBeSetup-List         ProtocolIE-ID ::= 507
id-MulticastF1UContext-FailedToBeSetup-Item         ProtocolIE-ID ::= 508
id-IABCongestionIndication                          ProtocolIE-ID ::= 509
id-IABConditionalRRCMessageDeliveryIndication       ProtocolIE-ID ::= 510
id-F1CTransferPathNRDC                              ProtocolIE-ID ::= 511
id-BufferSizeThresh                                 ProtocolIE-ID ::= 512
id-IAB-TNL-Addresses-Exception                      ProtocolIE-ID ::= 513
id-BAP-Header-Rewriting-Added-List                  ProtocolIE-ID ::= 514
id-BAP-Header-Rewriting-Added-List-Item             ProtocolIE-ID ::= 515
id-Re-routingEnableIndicator                        ProtocolIE-ID ::= 516
id-NonF1terminatingTopologyIndicator                ProtocolIE-ID ::= 517
id-EgressNonF1terminatingTopologyIndicator          ProtocolIE-ID ::= 518
id-IngressNonF1terminatingTopologyIndicator         ProtocolIE-ID ::= 519
id-rBSetConfiguration                               ProtocolIE-ID ::= 520
id-frequency-Domain-HSNA-Configuration-List         ProtocolIE-ID ::= 521
id-child-IAB-Nodes-NA-Resource-List                 ProtocolIE-ID ::= 522
id-Parent-IAB-Nodes-NA-Resource-Configuration-List  ProtocolIE-ID ::= 523
id-uL-FreqInfo                                      ProtocolIE-ID ::= 524
id-uL-Transmission-Bandwidth                        ProtocolIE-ID ::= 525
id-dL-FreqInfo                                      ProtocolIE-ID ::= 526
id-dL-Transmission-Bandwidth                        ProtocolIE-ID ::= 527
id-uL-NR-Carrier-List                               ProtocolIE-ID ::= 528
id-dL-NR-Carrier-List                               ProtocolIE-ID ::= 529
id-nRFreqInfo                                       ProtocolIE-ID ::= 530
id-transmission-Bandwidth                           ProtocolIE-ID ::= 531
id-nR-Carrier-List                                  ProtocolIE-ID ::= 532
id-Neighbour-Node-Cells-List                        ProtocolIE-ID ::= 533
id-Serving-Cells-List                               ProtocolIE-ID ::= 534
id-permutation                                      ProtocolIE-ID ::= 535
id-MDTPollutedMeasurementIndicator                  ProtocolIE-ID ::= 536
id-M5ReportAmount                                   ProtocolIE-ID ::= 537
id-M6ReportAmount                                   ProtocolIE-ID ::= 538
id-M7ReportAmount                                   ProtocolIE-ID ::= 539
id-SurvivalTime                                     ProtocolIE-ID ::= 540
id-PDCMeasurementPeriodicity                        ProtocolIE-ID ::= 541
id-PDCMeasurementQuantities                         ProtocolIE-ID ::= 542
id-PDCMeasurementQuantities-Item                    ProtocolIE-ID ::= 543
id-PDCMeasurementResult                             ProtocolIE-ID ::= 544
id-PDCReportType                                    ProtocolIE-ID ::= 545
id-RAN-UE-PDC-MeasID                                ProtocolIE-ID ::= 546
id-SCGActivationRequest                             ProtocolIE-ID ::= 547
id-SCGActivationStatus                              ProtocolIE-ID ::= 548
id-PRSTRPList                                       ProtocolIE-ID ::= 549
id-PRSTransmissionTRPList                           ProtocolIE-ID ::= 550
id-OnDemandPRS                                      ProtocolIE-ID ::= 551
id-AoA-SearchWindow                                 ProtocolIE-ID ::= 552
id-TRP-MeasurementUpdateList                        ProtocolIE-ID ::= 553
id-ZoAInformation                                   ProtocolIE-ID ::= 554
id-ResponseTime                                     ProtocolIE-ID ::= 555
```

```
id-ARPLocationInfo                              ProtocolIE-ID ::= 556
id-ARP-ID                                       ProtocolIE-ID ::= 557
id-MultipleULAoA                                ProtocolIE-ID ::= 558
id-UL-SRS-RSRPP                                 ProtocolIE-ID ::= 559
id-SRSResourcetype                              ProtocolIE-ID ::= 560
id-ExtendedAdditionalPathList                   ProtocolIE-ID ::= 561
id-LoS-NLoSInformation                          ProtocolIE-ID ::= 562
id-NumberOfTRPRxTEG                             ProtocolIE-ID ::= 564
id-NumberOfTRPRxTxTEG                           ProtocolIE-ID ::= 565
id-TRPTxTEGAssociation                          ProtocolIE-ID ::= 566
id-TRPTEGInformation                            ProtocolIE-ID ::= 567
id-TRPRx-TEGInformation                         ProtocolIE-ID ::= 568
id-TRP-PRS-Info-List                            ProtocolIE-ID ::= 569
id-PRS-Measurement-Info-List                    ProtocolIE-ID ::= 570
id-PRSConfigRequestType                         ProtocolIE-ID ::= 571
id-MeasurementTimeOccasion                      ProtocolIE-ID ::= 573
id-MeasurementCharacteristicsRequestIndicator   ProtocolIE-ID ::= 574
id-UEReportingInformation                       ProtocolIE-ID ::= 575
id-PosContextRevIndication                      ProtocolIE-ID ::= 576
id-TRPBeamAntennaInformation                    ProtocolIE-ID ::= 577
id-NRRedCapUEIndication                         ProtocolIE-ID ::= 578
id-Redcap-Bcast-Information                     ProtocolIE-ID ::= 579
id-RANUEPagingDRX                               ProtocolIE-ID ::= 580
id-CNUEPagingDRX                                ProtocolIE-ID ::= 581
id-NRPagingeDRXInformation                      ProtocolIE-ID ::= 582
id-NRPagingeDRXInformationforRRCINACTIVE        ProtocolIE-ID ::= 583
id-NR-TADV                                      ProtocolIE-ID ::= 584
id-QoEInformation                               ProtocolIE-ID ::= 585
id-CG-SDTQueryIndication                        ProtocolIE-ID ::= 586
id-SDT-MAC-PHY-CG-Config                        ProtocolIE-ID ::= 587
id-CG-SDTKeptIndicator                          ProtocolIE-ID ::= 588
id-CG-SDTindicatorSetup                         ProtocolIE-ID ::= 589
id-CG-SDTindicatorMod                           ProtocolIE-ID ::= 590
id-CG-SDTSessionInfoOld                         ProtocolIE-ID ::= 591
id-SDTInformation                               ProtocolIE-ID ::= 592
id-SDTRLCBearerConfiguration                    ProtocolIE-ID ::= 593
id-FiveG-ProSeAuthorized                        ProtocolIE-ID ::= 594
id-FiveG-ProSeUEPC5AggregateMaximumBitrate      ProtocolIE-ID ::= 595
id-FiveG-ProSePC5LinkAMBR                       ProtocolIE-ID ::= 596
id-SRBMappingInfo                               ProtocolIE-ID ::= 597
id-DRBMappingInfo                               ProtocolIE-ID ::= 598
id-UuRLCChannelToBeSetupList                    ProtocolIE-ID ::= 599
id-UuRLCChannelToBeModifiedList                 ProtocolIE-ID ::= 600
id-UuRLCChannelToBeReleasedList                 ProtocolIE-ID ::= 601
id-UuRLCChannelSetupList                        ProtocolIE-ID ::= 602
id-UuRLCChannelFailedToBeSetupList              ProtocolIE-ID ::= 603
id-UuRLCChannelModifiedList                     ProtocolIE-ID ::= 604
id-UuRLCChannelFailedToBeModifiedList           ProtocolIE-ID ::= 605
id-UuRLCChannelRequiredToBeModifiedList         ProtocolIE-ID ::= 606
id-UuRLCChannelRequiredToBeReleasedList         ProtocolIE-ID ::= 607
id-PC5RLCChannelToBeSetupList                   ProtocolIE-ID ::= 608
id-PC5RLCChannelToBeModifiedList                ProtocolIE-ID ::= 609
id-PC5RLCChannelToBeReleasedList                ProtocolIE-ID ::= 610
id-PC5RLCChannelSetupList                       ProtocolIE-ID ::= 611
```

```
id-PC5RLCChannelFailedToBeSetupList                ProtocolIE-ID ::= 612
id-PC5RLCChannelFailedToBeModifiedList             ProtocolIE-ID ::= 613
id-PC5RLCChannelRequiredToBeModifiedList           ProtocolIE-ID ::= 614
id-PC5RLCChannelRequiredToBeReleasedList           ProtocolIE-ID ::= 615
id-PC5RLCChannelModifiedList                       ProtocolIE-ID ::= 616
id-SidelinkRelayConfiguration                      ProtocolIE-ID ::= 617
id-UpdatedRemoteUELocalID                          ProtocolIE-ID ::= 618
id-PathSwitchConfiguration                         ProtocolIE-ID ::= 619
id-PagingCause                                     ProtocolIE-ID ::= 620
id-MUSIM-GapConfig                                 ProtocolIE-ID ::= 621
id-PEIPSAssistanceInfo                             ProtocolIE-ID ::= 622
id-UEPagingCapability                              ProtocolIE-ID ::= 623
id-LastUsedCellIndication                          ProtocolIE-ID ::= 624
id-SIB17-message                                   ProtocolIE-ID ::= 625
id-GNBDUUESliceMaximumBitRateList                  ProtocolIE-ID ::= 626
id-SIB20-message                                   ProtocolIE-ID ::= 627
id-UE-MulticastMRBs-ToBeReleased-List              ProtocolIE-ID ::= 628
id-UE-MulticastMRBs-ToBeReleased-Item              ProtocolIE-ID ::= 629
id-UE-MulticastMRBs-ToBeSetup-List                 ProtocolIE-ID ::= 630
id-UE-MulticastMRBs-ToBeSetup-Item                 ProtocolIE-ID ::= 631
id-MulticastMBSSessionSetupList                    ProtocolIE-ID ::= 632
id-MulticastMBSSessionRemoveList                   ProtocolIE-ID ::= 633
id-PosMeasurementAmount                            ProtocolIE-ID ::= 634
id-SDT-Termination-Request                         ProtocolIE-ID ::= 635
id-pathPower                                       ProtocolIE-ID ::= 636
id-DU-RX-MT-RX-Extend                              ProtocolIE-ID ::= 637
id-DU-TX-MT-TX-Extend                              ProtocolIE-ID ::= 638
id-DU-RX-MT-TX-Extend                              ProtocolIE-ID ::= 639
id-DU-TX-MT-RX-Extend                              ProtocolIE-ID ::= 640
id-BAP-Header-Rewriting-Removed-List               ProtocolIE-ID ::= 641
id-BAP-Header-Rewriting-Removed-List-Item          ProtocolIE-ID ::= 642
id-SLDRXCycleList                                  ProtocolIE-ID ::= 643
id-TAINSAGSupportList                              ProtocolIE-ID ::= 644
id-SL-RLC-ChannelToAddModList                      ProtocolIE-ID ::= 645
id-BroadcastAreaScope                              ProtocolIE-ID ::= 646
id-ManagementBasedMDTPLMNModificationList          ProtocolIE-ID ::= 647
id-SIB15-message                                   ProtocolIE-ID ::= 648
id-ActivationRequestType                           ProtocolIE-ID ::= 649
id-PosMeasGapPreConfigList                         ProtocolIE-ID ::= 650
id-InterFrequencyConfig-NoGap                      ProtocolIE-ID ::= 651
id-MBSInterestIndication                           ProtocolIE-ID ::= 652
id-UE-MulticastMRBs-ConfirmedToBeModified-List     ProtocolIE-ID ::= 653
id-UE-MulticastMRBs-ConfirmedToBeModified-Item     ProtocolIE-ID ::= 654
id-UE-MulticastMRBs-RequiredToBeModified-List      ProtocolIE-ID ::= 655
id-UE-MulticastMRBs-RequiredToBeModified-Item      ProtocolIE-ID ::= 656
id-UE-MulticastMRBs-RequiredToBeReleased-List      ProtocolIE-ID ::= 657
id-UE-MulticastMRBs-RequiredToBeReleased-Item      ProtocolIE-ID ::= 658
id-L571Info                                        ProtocolIE-ID ::= 659
id-L1151Info                                       ProtocolIE-ID ::= 660
id-SCS-480                                         ProtocolIE-ID ::= 661
id-SCS-960                                         ProtocolIE-ID ::= 662
id-SRSPortIndex                                    ProtocolIE-ID ::= 663
id-PEISubgroupingSupportIndication                 ProtocolIE-ID ::= 664
id-NeedForGapsInfoNR                               ProtocolIE-ID ::= 665
```

```
id-NeedForGapNCSGInfoNR                                 ProtocolIE-ID ::= 666
id-NeedForGapNCSGInfoEUTRA                              ProtocolIE-ID ::= 667
id-ProtocolIE-ID-668-not-to-be-used                     ProtocolIE-ID ::= 668
id-ProtocolIE-ID-669-not-to-be-used                     ProtocolIE-ID ::= 669
id-ProtocolIE-ID-670-not-to-be-used                     ProtocolIE-ID ::= 670
id-Source-MRB-ID                                        ProtocolIE-ID ::= 671
id-PosMeasurementPeriodicityNR-AoA                      ProtocolIE-ID ::= 672
id-RedCapIndication                                     ProtocolIE-ID ::= 673
id-SRSPosRRCInactiveConfig                              ProtocolIE-ID ::= 674
id-SDTBearerConfigurationQueryIndication                ProtocolIE-ID ::= 675
id-SDTBearerConfigurationInfo                           ProtocolIE-ID ::= 676
id-UL-GapFR2-Config                                     ProtocolIE-ID ::= 677
id-ConfigRestrictInfoDAPS                               ProtocolIE-ID ::= 678
id-UE-MulticastMRBs-Setup-List                          ProtocolIE-ID ::= 679
id-UE-MulticastMRBs-Setup-Item                          ProtocolIE-ID ::= 680
id-MulticastF1UContextReferenceCU                       ProtocolIE-ID ::= 681
id-PosSItypeList                                        ProtocolIE-ID ::= 682
id-DAPS-HO-Status                                       ProtocolIE-ID ::= 683
id-UplinkTxDirectCurrentTwoCarrierListInfo              ProtocolIE-ID ::= 684
id-UE-MulticastMRBs-ToBeSetup-atModify-List             ProtocolIE-ID ::= 685
id-UE-MulticastMRBs-ToBeSetup-atModify-Item             ProtocolIE-ID ::= 686
id-MC-PagingCell-List                                   ProtocolIE-ID ::= 687
id-MC-PagingCell-Item                                   ProtocolIE-ID ::= 688
id-SRSPosRRCInactiveQueryIndication                     ProtocolIE-ID ::= 689
id-UlTxDirectCurrentMoreCarrierInformation              ProtocolIE-ID ::= 690
id-CPACMCGInformation                                   ProtocolIE-ID ::= 691
id-TwoPHRModeMCG                                        ProtocolIE-ID ::= 692
id-TwoPHRModeSCG                                        ProtocolIE-ID ::= 693
id-ExtendedUEIdentityIndexValue                         ProtocolIE-ID ::= 694
id-ServingCellMO-List                                   ProtocolIE-ID ::= 695
id-ServingCellMO-List-Item                              ProtocolIE-ID ::= 696
id-ServingCellMO-encoded-in-CGC-List                    ProtocolIE-ID ::= 697
id-HashedUEIdentityIndexValue                           ProtocolIE-ID ::= 698
id-UE-MulticastMRBs-Setupnew-List                       ProtocolIE-ID ::= 699
id-UE-MulticastMRBs-Setupnew-Item                       ProtocolIE-ID ::= 700
id-ncd-SSB-RedCapInitialBWP-SDT                         ProtocolIE-ID ::= 701
id-nrofSymbolsExtended                                  ProtocolIE-ID ::= 702
id-repetitionFactorExtended                             ProtocolIE-ID ::= 703
id-startRBHopping                                       ProtocolIE-ID ::= 704
id-startRBIndex                                         ProtocolIE-ID ::= 705
id-transmissionCombn8                                   ProtocolIE-ID ::= 706
id-ServCellInfoList                                     ProtocolIE-ID ::= 707
id-DedicatedSIDeliveryIndication                        ProtocolIE-ID ::= 708
id-Configured-BWP-List                                  ProtocolIE-ID ::= 709
id-Preconfigured-measurement-GAP-Request                ProtocolIE-ID ::= 710
id-BWP-Id                                               ProtocolIE-ID ::= 711
id-NetworkControlledRepeaterAuthorized                  ProtocolIE-ID ::= 712
id-MT-SDT-Information                                   ProtocolIE-ID ::= 713
id-ExtendedResourceSymbolOffset                         ProtocolIE-ID ::= 714
id-NeedForInterruptionInfoNR                            ProtocolIE-ID ::= 715
id-SDT-Volume-Threshold                                 ProtocolIE-ID ::= 716
id-SupportedUETypeList                                  ProtocolIE-ID ::= 717
id-MusimCapabilityRestrictionIndication                 ProtocolIE-ID ::= 718
id-duplicationIndication                                ProtocolIE-ID ::= 719
```

```
id-LTMInformation-Setup                         ProtocolIE-ID ::= 720
id-LTMConfigurationIDMappingList                ProtocolIE-ID ::= 721
id-LTMInformation-Modify                        ProtocolIE-ID ::= 722
id-LTMCells-ToBeReleased-List                   ProtocolIE-ID ::= 723
id-ProtocolIE-ID-724-not-to-be-used             ProtocolIE-ID ::= 724
id-LTMConfiguration                             ProtocolIE-ID ::= 725
id-EarlySyncInformation-Request                 ProtocolIE-ID ::= 726
id-EarlySyncInformation                         ProtocolIE-ID ::= 727
id-EarlySyncCandidateCellInformation-List       ProtocolIE-ID ::= 728
id-LTMCellSwitchInformation                     ProtocolIE-ID ::= 729
id-DUtoCUTAInformation-List                     ProtocolIE-ID ::= 730
id-ProtocolIE-ID-731-not-to-be-used             ProtocolIE-ID ::= 731
id-dRB-List                                     ProtocolIE-ID ::= 732
id-DeactivationIndication                       ProtocolIE-ID ::= 733
id-RAReportIndicationList                       ProtocolIE-ID ::= 734
id-ChannelOccupancyTimePercentageUL             ProtocolIE-ID ::= 735
id-SuccessfulPSCellChangeReportInformationList  ProtocolIE-ID ::= 736
id-RadioResourceStatusNR-U                      ProtocolIE-ID ::= 737
id-FiveG-ProSeLayer2Multipath                   ProtocolIE-ID ::= 738
id-FiveG-ProSeLayer2UEtoUERelay                 ProtocolIE-ID ::= 739
id-FiveG-ProSeLayer2UEtoUERemote                ProtocolIE-ID ::= 740
id-PathAdditionInformation                      ProtocolIE-ID ::= 741
id-Recommended-SSBs-List                        ProtocolIE-ID ::= 742
id-Recommended-SSBs-for-Paging-List             ProtocolIE-ID ::= 743
id-SSBs-withinTheCell-tobe-Activated-List       ProtocolIE-ID ::= 744
id-Cells-With-SSBs-Activated-List               ProtocolIE-ID ::= 745
id-Cells-Allowed-to-be-Deactivated-List         ProtocolIE-ID ::= 746
id-Cells-Allowed-to-be-Deactivated-List-Item    ProtocolIE-ID ::= 747
id-Coverage-Modification-Cause                  ProtocolIE-ID ::= 748
id-RANTSSRequestType                            ProtocolIE-ID ::= 749
id-RANTimingSynchronisationStatusInfo           ProtocolIE-ID ::= 750
id-TSCTrafficCharacteristicsFeedback            ProtocolIE-ID ::= 751
id-RANfeedbacktype                              ProtocolIE-ID ::= 752
id-Mobile-TRP-LocationInformation               ProtocolIE-ID ::= 753
id-Mobile-IAB-MT-UE-ID                          ProtocolIE-ID ::= 754
id-Target-gNB-ID                                ProtocolIE-ID ::= 755
id-Target-gNB-IP-address                        ProtocolIE-ID ::= 756
id-Target-SeGW-IP-address                       ProtocolIE-ID ::= 757
id-Activated-Cells-Mapping-List                 ProtocolIE-ID ::= 758
id-Activated-Cells-Mapping-List-Item            ProtocolIE-ID ::= 759
id-F1SetupOutcome                               ProtocolIE-ID ::= 760
id-RRC-Terminating-IAB-Donor-Related-Info       ProtocolIE-ID ::= 761
id-RRC-Terminating-IAB-Donor-gNB-ID             ProtocolIE-ID ::= 762
id-NCGI-to-be-Updated-List                      ProtocolIE-ID ::= 763
id-NCGI-to-be-Updated-List-Item                 ProtocolIE-ID ::= 764
id-Mobile-IAB-MTUserLocationInformation         ProtocolIE-ID ::= 765
id-MobileAccessPointLocation                    ProtocolIE-ID ::= 766
id-AssociatedSessionID                          ProtocolIE-ID ::= 767
id-IndicationMCInactiveReception                ProtocolIE-ID ::= 768
id-MulticastCU2DURRCInfo                        ProtocolIE-ID ::= 769
id-MBSMulticastSessionReceptionState            ProtocolIE-ID ::= 770
id-F1UTunnelNotEstablished                      ProtocolIE-ID ::= 771
id-MulticastDU2CURRCInfo                        ProtocolIE-ID ::= 772
id-SIB24-message                                ProtocolIE-ID ::= 773
```

```
id-MulticastCU2DUCommonRRCInfo                    ProtocolIE-ID ::= 774
id-PDUSetQoSParameters                            ProtocolIE-ID ::= 775
id-N6JitterInformation                            ProtocolIE-ID ::= 776
id-ECNMarkingorCongestionInformationReportingRequest    ProtocolIE-ID ::= 777
id-ECNMarkingorCongestionInformationReportingStatus ProtocolIE-ID ::= 778
id-NRA2XServicesAuthorized                        ProtocolIE-ID ::= 779
id-LTEA2XServicesAuthorized                       ProtocolIE-ID ::= 780
id-NRUESidelinkAggregateMaximumBitrateForA2X      ProtocolIE-ID ::= 781
id-LTEUESidelinkAggregateMaximumBitrateForA2X     ProtocolIE-ID ::= 782
id-NReRedCapUEIndication                          ProtocolIE-ID ::= 783
id-ERedcap-Bcast-Information                      ProtocolIE-ID ::= 784
id-NRPaginglongeDRXInformationforRRCINACTIVE      ProtocolIE-ID ::= 785
id-SCPAC-Request                                  ProtocolIE-ID ::= 786
id-Target-F1-Terminating-Donor-gNB-ID             ProtocolIE-ID ::= 787
id-MobileIAB-Barred                               ProtocolIE-ID ::= 788
id-Broadcast-MRBs-Transport-Request-List          ProtocolIE-ID ::= 789
id-Broadcast-MRBs-Transport-Request-Item          ProtocolIE-ID ::= 790
id-S-CPACLowerLayerReferenceConfigRequest         ProtocolIE-ID ::= 791
id-S-CPAC-Configuration                           ProtocolIE-ID ::= 792
id-MusimCandidateBandList                         ProtocolIE-ID ::= 793
id-DLLBTFailureInformationRequest                 ProtocolIE-ID ::= 794
id-DLLBTFailureInformationList                    ProtocolIE-ID ::= 795
id-PSIbasedSDUdiscardUL                           ProtocolIE-ID ::= 796
id-SIB22-message                                  ProtocolIE-ID ::= 797
id-CUtoDUTAInformation-List                       ProtocolIE-ID ::= 798
id-U2URLCChannelQoS                               ProtocolIE-ID ::= 799
id-SL-PHY-MAC-RLC-ConfigExt                       ProtocolIE-ID ::= 800
id-SLPositioning-Ranging-Service-Info             ProtocolIE-ID ::= 801
id-TimeWindowInformation-SRS-List                 ProtocolIE-ID ::= 802
id-TimeWindowInformation-Measurement-List         ProtocolIE-ID ::= 803
id-UL-RSCP                                        ProtocolIE-ID ::= 804
id-BW-Aggregation-Request-Indication              ProtocolIE-ID ::= 805
id-ReportingGranularitykminus1                    ProtocolIE-ID ::= 806
id-ReportingGranularitykminus2                    ProtocolIE-ID ::= 807
id-ReportingGranularitykminus1additionalpath      ProtocolIE-ID ::= 808
id-ReportingGranularitykminus2additionalpath      ProtocolIE-ID ::= 809
id-TimingReportingGranularityFactorExtended       ProtocolIE-ID ::= 810
id-SRSPosRRCInactiveValidityAreaConfig            ProtocolIE-ID ::= 811
id-PosValidityAreaCellList                        ProtocolIE-ID ::= 812
id-SRSReservationType                             ProtocolIE-ID ::= 813
id-SymbolIndex                                    ProtocolIE-ID ::= 814
id-PRSBWAggregationRequestInfoList                ProtocolIE-ID ::= 815
id-AggregatedPosSRSResourceIDList                 ProtocolIE-ID ::= 816
id-AggregatedPRSResourceSetList                   ProtocolIE-ID ::= 817
id-PhaseQuality                                   ProtocolIE-ID ::= 818
id-MeasuredFrequencyHops                          ProtocolIE-ID ::= 819
id-TxHoppingConfiguration                         ProtocolIE-ID ::= 820
id-ReportingGranularitykminus3                    ProtocolIE-ID ::= 821
id-ReportingGranularitykminus4                    ProtocolIE-ID ::= 822
id-ReportingGranularitykminus5                    ProtocolIE-ID ::= 823
id-ReportingGranularitykminus6                    ProtocolIE-ID ::= 824
id-ReportingGranularitykminus3additionalpath      ProtocolIE-ID ::= 825
id-ReportingGranularitykminus4additionalpath      ProtocolIE-ID ::= 826
id-ReportingGranularitykminus5additionalpath      ProtocolIE-ID ::= 827
```

```
id-ReportingGranularitykminus6additionalpath        ProtocolIE-ID ::= 828
id-AggregatedPosSRSResourceSetList                  ProtocolIE-ID ::= 829
id-RequestedSRSPreconfigurationCharacteristics-List ProtocolIE-ID ::= 830
id-SRSPreconfiguration-List                         ProtocolIE-ID ::= 831
id-SRSInformation                                   ProtocolIE-ID ::= 832
id-ValidityAreaSpecificSRSInformation               ProtocolIE-ID ::= 833
id-E-CID-MeasuredResultsAssociatedInfoList          ProtocolIE-ID ::= 834
id-XR-Bcast-Information                             ProtocolIE-ID ::= 835
id-MaxDataBurstVolume                               ProtocolIE-ID ::= 836
id-TAInformation-List                               ProtocolIE-ID ::= 837
id-NonIntegerDRXCycle                               ProtocolIE-ID ::= 838
id-PointA                                           ProtocolIE-ID ::= 839
id-SCS-SpecificCarrier                              ProtocolIE-ID ::= 840
id-NR-PCI                                           ProtocolIE-ID ::= 841
id-PeerUE-ID                                            ProtocolIE-ID ::= 842
id-EarlySyncServingCellInformation                  ProtocolIE-ID ::= 843
id-RANSharingAssistanceInformation                  ProtocolIE-ID ::= 844
id-LTMCFRAResourceConfig-List                       ProtocolIE-ID ::= 845
id-F1U-PathFailure                                  ProtocolIE-ID ::= 846
id-MeasBasedOnAggregatedResources                   ProtocolIE-ID ::= 847
id-SIB23-message                                    ProtocolIE-ID ::= 848
id-BarringExemptionforEmerCallInfo                  ProtocolIE-ID ::= 849
id-SIB17bis-message                                 ProtocolIE-ID ::= 850
id-ReportingIntervalIMs                             ProtocolIE-ID ::= 851
id-Transmission-Bandwidth-asymmetric                ProtocolIE-ID ::= 852
id-TagIDPointer                                     ProtocolIE-ID ::= 853
id-LocalOrigin                                      ProtocolIE-ID ::= 854
id-LTMResetInformation                              ProtocolIE-ID ::= 855
id-SRSPosPeriodicConfigHyperSFNIndex                ProtocolIE-ID ::= 856
id-PreconfiguredSRSInformation                      ProtocolIE-ID ::= 857
id-candidatePSCellsToCancel                         ProtocolIE-ID ::= 858
id-MobilityInitiation                               ProtocolIE-ID ::= 859
id-ValidityAreaSpecificSRSInformationExtended       ProtocolIE-ID ::= 860
id-PLMNIndexNRAssistanceInfoForNetShar                      ProtocolIE-ID ::= 861
id-TCIStatesConfigurationsList                      ProtocolIE-ID ::= 862
id-LTMTCIStatesConfigurationsList                   ProtocolIE-ID ::= 863
id-E-CID-AoA-NR-per-TRP                             ProtocolIE-ID ::= 864
id-PSIbasedSDUdiscardDL                             ProtocolIE-ID ::= 865
id-PduSetDelayBudgetDownlink                        ProtocolIE-ID ::= 866
id-PduSetDelayBudgetUplink                          ProtocolIE-ID ::= 867
id-PduSetErrorRateDownlink                          ProtocolIE-ID ::= 868
id-PduSetErrorRateUplink                            ProtocolIE-ID ::= 869
id-MonitoringRequestonAvailableBitrate              ProtocolIE-ID ::= 870
id-MMSID                                            ProtocolIE-ID ::= 871
id-Indication-of-Bitrate-Adaptation                 ProtocolIE-ID ::= 872
id-DLPDUSetInformationMarkingSupportIndication      ProtocolIE-ID ::= 873
id-FiveGProSeLayer3MHUEtoNetworkRelay               ProtocolIE-ID ::= 874
id-FiveGProSeLayer2MHUEtoNetworkRelay               ProtocolIE-ID ::= 875
id-FiveGProSeLayer2MHIntermediateUEtoNetworkRelay   ProtocolIE-ID ::= 876
id-FiveGProSeLayer2MHRemote                         ProtocolIE-ID ::= 877
id-LPWUSPSAssistanceInfo                            ProtocolIE-ID ::= 878
id-LPWUSSubgroupingSupportIndication                ProtocolIE-ID ::= 879
id-FurtherExtendedUEIdentityIndexValue              ProtocolIE-ID ::= 880
id-CLI-MeasurementResult-List                       ProtocolIE-ID ::= 881
```

```
id-SBFD-Frequency-Configuration                         ProtocolIE-ID ::= 882
id-SSB-resource-config                                  ProtocolIE-ID ::= 883
id-NZP-CSI-RS-Resources-Config                          ProtocolIE-ID ::= 884
id-SRS-Resource-Indication                              ProtocolIE-ID ::= 885
id-SRS-Resource-Configuration                           ProtocolIE-ID ::= 886
id-rLFReportFailureType                                 ProtocolIE-ID ::= 887
id-FailureReportingWithoutRLFReport                     ProtocolIE-ID ::= 888
id-MROForLTM-Information                                    ProtocolIE-ID ::= 889
id-LastVisitedLTMCells                                  ProtocolIE-ID ::= 890
id-OnDemandSIB1                                         ProtocolIE-ID ::= 891
id-PagingAdaptationIndication                           ProtocolIE-ID ::= 892
id-OnDemand-SIB1-Cell                                   ProtocolIE-ID ::= 893
id-PEISubgroupingSupportIndication-PagingAdaptation         ProtocolIE-ID ::= 894
id-ServingCellMO-Ondemand                               ProtocolIE-ID ::= 895
id-LTMgNB-ID                                            ProtocolIE-ID ::= 896
id-L1ExecutionConditionList                             ProtocolIE-ID ::= 897
id-LTMSecurityInformation                               ProtocolIE-ID ::= 898
id-RequestforCSI-RSResourceConfigforL1Measure           ProtocolIE-ID ::= 899
id-CSI-RSResourceConfigforL1Measure                     ProtocolIE-ID ::= 901
id-CSI-RSResourceConfigforCSIAcquisition                ProtocolIE-ID ::= 902
id-CSIReportConfigforCSIAcquisition                     ProtocolIE-ID ::= 903
id-RequestforL1ExecutionCondition                       ProtocolIE-ID ::= 904
id-LTMInformationSCGAdd                                 ProtocolIE-ID ::= 905
id-LTMInformationSCGMod                                 ProtocolIE-ID ::= 906
id-TARemainingInfoList                                  ProtocolIE-ID ::= 907
id-CSI-RSCoordinationRequestList                        ProtocolIE-ID ::= 908
id-CSI-RSCoordinationResultList                         ProtocolIE-ID ::= 909
id-CSI-RSMeasurementsList                               ProtocolIE-ID ::= 910
id-LTMResidualTAInfoList                                ProtocolIE-ID ::= 911
id-ChannelResponseInformation                           ProtocolIE-ID ::= 912
id-UL-SRS-TDCT                                          ProtocolIE-ID ::= 913
id-Future-Coverage-Modification-Notification            ProtocolIE-ID ::= 914
id-Predicted-CCO-Assistance-Information                 ProtocolIE-ID ::= 915
id-UEPerformanceDelayMonitoring                         ProtocolIE-ID ::= 916
id-NodeAssociatedInfoResult                             ProtocolIE-ID ::= 917
id-NeighbourFutureCoverageModNotification               ProtocolIE-ID ::= 918
id-AreaSpecificSemiPersistentSRSPosInfo                 ProtocolIE-ID ::= 919
id-SBFD-AcrossSymbolType                                ProtocolIE-ID ::= 920
id-LTMConfig-to-Modify-List                             ProtocolIE-ID ::= 921
id-InterSNLTMMCGInformation                             ProtocolIE-ID ::= 922
id-LPWUSSupportedBandInfo                               ProtocolIE-ID ::= 923
id-DeliveryStatusReq                                    ProtocolIE-ID ::= 924


END
-- ASN1STOP
```

9.4.8 Container Definitions

```
-- ASN1START
-- **************************************************************
```

```
--
-- Container definitions
--
-- **************************************************************

F1AP-Containers {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
ngran-access (22) modules (3) f1ap (3) version1 (1) f1ap-Containers (5) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- **************************************************************
--
-- IE parameter types from other modules.
--
-- **************************************************************

IMPORTS
    Criticality,
    Presence,
    PrivateIE-ID,
    ProtocolExtensionID,
    ProtocolIE-ID

FROM F1AP-CommonDataTypes
    maxPrivateIEs,
    maxProtocolExtensions,
    maxProtocolIEs

FROM F1AP-Constants;

-- **************************************************************
--
-- Class Definition for Protocol IEs
--
-- **************************************************************

F1AP-PROTOCOL-IES ::= CLASS {
    &id             ProtocolIE-ID                   UNIQUE,
    &criticality    Criticality,
    &Value,
    &presence       Presence
}
WITH SYNTAX {
    ID              &id
    CRITICALITY     &criticality
    TYPE            &Value
    PRESENCE        &presence
}

-- **************************************************************
--
```

```
-- Class Definition for Protocol IEs
--
-- **************************************************************

F1AP-PROTOCOL-IES-PAIR ::= CLASS {
    &id                 ProtocolIE-ID               UNIQUE,
    &firstCriticality   Criticality,
    &FirstValue,
    &secondCriticality  Criticality,
    &SecondValue,
    &presence           Presence
}
WITH SYNTAX {
    ID              &id
    FIRST CRITICALITY       &firstCriticality
    FIRST TYPE              &FirstValue
    SECOND CRITICALITY      &secondCriticality
    SECOND TYPE             &SecondValue
    PRESENCE                &presence
}

-- **************************************************************
--
-- Class Definition for Protocol Extensions
--
-- **************************************************************

F1AP-PROTOCOL-EXTENSION ::= CLASS {
    &id             ProtocolExtensionID         UNIQUE,
    &criticality    Criticality,
    &Extension,
    &presence       Presence
}
WITH SYNTAX {
    ID              &id
    CRITICALITY     &criticality
    EXTENSION       &Extension
    PRESENCE        &presence
}

-- **************************************************************
--
-- Class Definition for Private IEs
--
-- **************************************************************

F1AP-PRIVATE-IES ::= CLASS {
    &id             PrivateIE-ID,
    &criticality    Criticality,
    &Value,
    &presence       Presence
}
WITH SYNTAX {
    ID              &id
```

```
    CRITICALITY     &criticality
    TYPE            &Value
    PRESENCE        &presence
}

-- **************************************************************
--
-- Container for Protocol IEs
--
-- **************************************************************

ProtocolIE-Container {F1AP-PROTOCOL-IES : IEsSetParam} ::=
    SEQUENCE (SIZE (0..maxProtocolIEs)) OF
    ProtocolIE-Field {{IEsSetParam}}

ProtocolIE-SingleContainer {F1AP-PROTOCOL-IES : IEsSetParam} ::=
    ProtocolIE-Field {{IEsSetParam}}

ProtocolIE-Field {F1AP-PROTOCOL-IES : IEsSetParam} ::= SEQUENCE {
    id              F1AP-PROTOCOL-IES.&id               ({IEsSetParam}),
    criticality     F1AP-PROTOCOL-IES.&criticality      ({IEsSetParam}{@id}),
    value           F1AP-PROTOCOL-IES.&Value            ({IEsSetParam}{@id})
}

-- **************************************************************
--
-- Container for Protocol IE Pairs
--
-- **************************************************************

ProtocolIE-ContainerPair {F1AP-PROTOCOL-IES-PAIR : IEsSetParam} ::=
    SEQUENCE (SIZE (0..maxProtocolIEs)) OF
    ProtocolIE-FieldPair {{IEsSetParam}}

ProtocolIE-FieldPair {F1AP-PROTOCOL-IES-PAIR : IEsSetParam} ::= SEQUENCE {
    id                  F1AP-PROTOCOL-IES-PAIR.&id                  ({IEsSetParam}),
    firstCriticality    F1AP-PROTOCOL-IES-PAIR.&firstCriticality    ({IEsSetParam}{@id}),
    firstValue          F1AP-PROTOCOL-IES-PAIR.&FirstValue          ({IEsSetParam}{@id}),
    secondCriticality   F1AP-PROTOCOL-IES-PAIR.&secondCriticality   ({IEsSetParam}{@id}),
    secondValue         F1AP-PROTOCOL-IES-PAIR.&SecondValue         ({IEsSetParam}{@id})
}

-- **************************************************************
--
-- Container for Protocol Extensions
--
-- **************************************************************

ProtocolExtensionContainer {F1AP-PROTOCOL-EXTENSION : ExtensionSetParam} ::=
    SEQUENCE (SIZE (1..maxProtocolExtensions)) OF
    ProtocolExtensionField {{ExtensionSetParam}}

ProtocolExtensionField {F1AP-PROTOCOL-EXTENSION : ExtensionSetParam} ::= SEQUENCE {
    id                  F1AP-PROTOCOL-EXTENSION.&id             ({ExtensionSetParam}),
```

```
	criticality			F1AP-PROTOCOL-EXTENSION.&criticality	({ExtensionSetParam}{@id}),
	extensionValue		F1AP-PROTOCOL-EXTENSION.&Extension		({ExtensionSetParam}{@id})
}

-- **************************************************************
--
-- Container for Private IEs
--
-- **************************************************************

PrivateIE-Container {F1AP-PRIVATE-IES : IEsSetParam } ::=
	SEQUENCE (SIZE (1.. maxPrivateIEs)) OF
	PrivateIE-Field {{IEsSetParam}}

PrivateIE-Field {F1AP-PRIVATE-IES : IEsSetParam} ::= SEQUENCE {
	id					F1AP-PRIVATE-IES.&id				({IEsSetParam}),
	criticality			F1AP-PRIVATE-IES.&criticality		({IEsSetParam}{@id}),
	value				F1AP-PRIVATE-IES.&Value				({IEsSetParam}{@id})
}

END
-- ASN1STOP
```

9.5 Message Transfer Syntax

F1AP shall use the ASN.1 Basic Packed Encoding Rules (BASIC-PER) Aligned Variant as transfer syntax, as specified in ITU-T Recommendation X.691 [5].

9.6 Timers

10 Handling of unknown, unforeseen and erroneous protocol data

Clause 10 of TS 38.413 [3] is applicable for the purposes of the present document, with the following additions for non-UE-associated procedures:

- In case of Abstract Syntax Error, when reporting the *Criticality Diagnostics* IE for not comprehended IE/IEgroups or missing IE/IE groups, the *Transaction ID* IE shall also be included;
- In case of Logical Error, when reporting the *Criticality Diagnostics* IE, the *Transaction ID* IE shall also be included;
- In case of Logical Error in a response message of a Class 1 procedure, or failure to comprehend *Transaction ID* IE from a received message, the procedure shall be considered as unsuccessfully terminated or not terminated (e.g., transaction ID unknown in response message), and local error handling shall be initiated.

Annex A (informative): Change History

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2017-06	R3 NR#2	R3-172493	-	-	-	First version	0.1.0
2017-07	R3 NR#2	R3-172640	-	-	-	Incorporated agreed TPs from R3 NR#2 Adhoc	0.2.0
2017-08	R3#97	R3-173451	-	-	-	Incorporated agreed TPs from R3#97	0.3.0
2017-10	R3#97b	R3-174247	-	-	-	Incorporated agreed TPs from R3#97b	0.4.0
2017-12	R3#98	R3-175062	-	-	-	Incorporated agreed TPs from R3#98	0.5.0
2017-12	RAN#78	RP-172287				Submitted for approval to RAN	1.0.0
2017-12	RAN#78					TR approved by RAN plenary	15.0.0
2018-03	RP-79	RP-180468	0001	2	B	Baseline CR for March version of TS 38.473 covering agreements of RAN3#99	15.1.0
2018-04						Editorial correction to ASN.1 (correction to id-TimeToWait ProtocolIE-ID)	15.1.1
2018-06	RP-80	RP-181237	0011	6	B	Introduction of SA NR (38.473 Baseline CR covering RAN3 agreements)	15.2.0
2018-06	RP-80	RP-181239	0043	3	F	Essential corrections of EN-DC for NSA NR (38.473 Baseline CR covering RAN3 agreements)	15.2.0
2018-06	RP-80	RP-181237	0045	-	B	F1 support for LTE - NR coexistence	15.2.0
2018-06	RP-80					Correction to ASN.1 and to Change History table	15.2.1
2018-09	RP-81	RP-181920	0055	2	F	Introduction of DU Configuration Query	15.3.0
2018-09	RP-81	RP-181921	0056	4	F	CR to 38.473 on further clarifications on System information transfer over F1	15.3.0
2018-09	RP-81	RP-181921	0058	4	F	CR to 38.473 on corrections to System information delivery	15.3.0
2018-09	RP-81	RP-181920	0059	1	F	CR to 38.473 on corrections to PWS transfer over F1	15.3.0
2018-09	RP-81	RP-181921	0063	3	F	CR to 38.473 on PDCP SN over F1 interface	15.3.0
2018-09	RP-81	RP-181922	0064	3	F	NR Corrections (38.473 Baseline CR covering RAN3-101 agreements)	15.3.0
2018-09	RP-81	RP-181997	0068	-	F	Introduction of UL AMBR on F1	15.3.0
2018-09	RP-81	RP-181921	0072	3	F	Correction on cell management	15.3.0
2018-09	RP-81	RP-181921	0073	2	F	RLC Mode Indication over F1	15.3.0
2018-09	RP-81	RP-181921	0076	3	F	CR to 38.473 on UE Identity Index value	15.3.0
2018-09	RP-81	RP-181920	0077	1	F	Correction for UE Context Modification on presence of ServCellIndex IE	15.3.0
2018-09	RP-81	RP-181920	0078	-	F	Executing duplication for RRC-container	15.3.0
2018-09	RP-81	RP-181921	0079	1	F	Indication of RLC re-establishment at the gNB-DU	15.3.0
2018-09	RP-81	RP-181920	0080	-	F	Exchange of SMTC over F1	15.3.0
2018-09	RP-81	RP-181920	0081	-	F	Solving remaining issues with QoS parameters – TS 38.473	15.3.0
2018-09	RP-81	RP-181921	0090		F	Correction of 5GS TAC	15.3.0
2018-09	RP-81	RP-181921	0095	1	F	Extend the RANAC size to 8bits	15.3.0
2018-09	RP-81	RP-181921	0097	-	F	Corrections of Choice	15.3.0
2018-09	RP-81	RP-181921	0098	1	F	Correction of TNL criticality	15.3.0
2018-09	RP-81	RP-181921	0099	1	F	Corrections of usage of single container	15.3.0
2018-09	RP-81	RP-181921	0105	2	B	RRC version handling	15.3.0
2018-09	RP-81	RP-181921	0106	1	B	Introduction of Overload Handling in F1-C	15.3.0
2018-09	RP-81	RP-181921	0113	-	F	CR to 38.473 on presence of QoS information	15.3.0
2018-09	RP-81	RP-181921	0114	1	F	Correction C-RNTI format	15.3.0
2018-09	RP-81	RP-181921	0115	-	F	Correction of QoS Parameters	15.3.0
2018-09	RP-81	RP-181921	0116	1	F	Correction on F1 Setup Request	15.3.0
2018-12	RP-82	RP-182446	0070	3	F	RRC Delivery Indication	15.4.0
2018-12	RP-82	RP-182446	0117	1	F	Correction of AMBR Enforcement	15.4.0
2018-12	RP-82	RP-182446	0138	-	F	CR for correction on Initial UL RRC message transfer	15.4.0
2018-12	RP-82	RP-182446	0140	1	F	CR to 38.473 on bearer type change indication	15.4.0
2018-12	RP-82	RP-182446	0142	1	F	CR to 38.473 on correction to PWS System Information	15.4.0
2018-12	RP-82	RP-182446	0144	2	F	CR to 38.473 on asymmetric mapping for UL and DL QoS flow	15.4.0
2018-12	RP-82	RP-182447	0145	4	F	Corrections on UE-associated LTE/NR resource coordination	15.4.0
2018-12	RP-82	RP-182446	0147	2	F	CR for F1 Cell Management	15.4.0
2018-12	RP-82	RP-182447	0150	1	F	Missing Transaction ID in non-UE-associated procedures	15.4.0
2018-12	RP-82	RP-182446	0157	1	F	CR to 38.473 on mapping of servingCellMO and Serving Cell	15.4.0
2018-12	RP-82	RP-182446	0160	1	F	CR to 38.473 on UE context modification required procedure	15.4.0
2018-12	RP-82	RP-182447	0165	1	F	Addition of the RLC Mode information for bearer modification	15.4.0
2018-12	RP-82	RP-182448	0167	2	F	Rapporteur CR to align tabular	15.4.0
2018-12	RP-82	RP-182448	0168	2	F	Rapporteur CR to align ASN.1	15.4.0
2018-12	RP-82	RP-182447	0169	2	F	Correction of MaxnoofBPLMNs	15.4.0
2018-12	RP-82	RP-182351	0174	2	F	Correction on PDCP SN length on F1	15.4.0
2018-12	RP-82	RP-182447	0178	2	F	CR for TS 38.473 for MR-DC coordination	15.4.0

2018-12	RP-82	RP-182447	0179	2	F	Support of system information update for active UE without CSS	15.4.0
2018-12	RP-82	RP-182447	0187	1	F	CR to 38.473 on clarification to the presence of UE AMBR	15.4.0
2018-12	RP-82	RP-182506	0195	2	F	CR on Scell release for RLC failure	15.4.0
2018-12	RP-82	RP-182447	0205	1	F	About bandcombinationindex and featureSetEntryIndex	15.4.0
2018-12	RP-82	RP-182447	0211	1	F	CR to 38.473 on DRB PDCP duplication	15.4.0
2018-12	RP-82	RP-182447	0216	1	F	CR to 38.473 on clarifications on system information update over F1	15.4.0
2018-12	RP-82	RP-182448	0219	-	F	Correction of RRC version handling and UE inactivity notification	15.4.0
2019-01	RP-82					- correction to ASN.1: addiming a missing change to "WriteReplaceWarningResponseIEs F1AP-PROTOCOL-IES ::= {"	15.4.1
2019-03	RP-83	RP-190555	0202	2	F	Indication that cells are only UL or DL on F1	15.5.0
2019-03	RP-83	RP-190554	0204	1	F	AMF intitiated UE Context Release failure cause	15.5.0
2019-03	RP-83	RP-190554	0220	1	F	Correction to reconfiguration with sync for gNB-DU	15.5.0
2019-03	RP-83	RP-190554	0225	1	F	Introduction of PH-InforSCG in DU to CU RRC Information	15.5.0
2019-03	RP-83	RP-190554	0226	1	F	CR to 38.473 on Measurement gap coordination	15.5.0
2019-03	RP-83	RP-190554	0228	1	F	CR for TS 38.473 for MR-DC coordination	15.5.0
2019-03	RP-83	RP-190554	0229	2	F	Condition for inclusion of the Dedicated SI Delivery Needed UE List IE	15.5.0
2019-03	RP-83	RP-190554	0230	1	F	Correction of the Transmission stop/restart indication	15.5.0
2019-03	RP-83	RP-190554	0231	-	F	Corrections on gNB-CU/gNB-DU Configuration Update	15.5.0
2019-03	RP-83	RP-190556	0236	2	F	Correction of QoS Flow Mapping Indication	15.5.0
2019-03	RP-83	RP-190554	0244	-	F	Release due to pre-emption	15.5.0
2019-03	RP-83	RP-190554	0245	2	F	CR on RRC container in UE context modification request message	15.5.0
2019-03	RP-83	RP-190554	0246	2	F	CR on UE context modification refuse	15.5.0
2019-03	RP-83	RP-190554	0247	-	F	Transaction ID in Error Indication procedure	15.5.0
2019-03	RP-83	RP-190554	0249	2	F	Cells to be deactivated over F1	15.5.0
2019-03	RP-83	RP-190554	0251	1	F	CR to 38.473 on SRB duplication and LCID	15.5.0
2019-03	RP-83	RP-190554	0258	-	F	CR to 38.473 on corrections for removal of PDCP duplication for SRB	15.5.0
2019-03	RP-83	RP-190554	0263	1	F	CR to 38.473 on transfering UEAssistanceInformation over F1	15.5.0
2019-03	RP-83	RP-190554	0265	-	F	Rapporteur updates	15.5.0
2019-03	RP-83	RP-190554	0266	1	F	Correction on gNB-DU Resource Coordination	15.5.0
2019-03	RP-83	RP-190554	0267	1	F	Endpoint IP address and port	15.5.0
2019-03	RP-83	RP-190554	0268	1	F	Correction to add paging origin IE	15.5.0
2019-03	RP-83	RP-190555	0269	2	F	Multiple SCTP associations over F1AP	15.5.0
2019-03	RP-83	RP-190554	0272	1	F	About Cells Failed to be Activated IE in gNB-CU Configuration Update Ack	15.5.0
2019-03	RP-83	RP-190556	0273	1	F	gNB-DU UE Aggregate Maximum Bit Rate Uplink correction	15.5.0
2019-03	RP-83	RP-190554	0276	1	F	RRC Reconfiguration failure	15.5.0
2019-03	RP-83	RP-190554	0278	1	F	Node behaviour at reception of DU to CU RRC Information	15.5.0
2019-03	RP-83	RP-190554	0281	-	F	Addition of Transaction ID to Initial UL RRC Message Transfer	15.5.0
2019-07	RP-84	RP-191397	0200	5	F	RAN sharing with multiple Cell ID broadcast	15.6.0
2019-07	RP-84	RP-191397	0270	5	F	Addition of Network Access Rate Reduction message	15.6.0
2019-07	RP-84	RP-191397	0271	3	F	RAN UE ID for F1	15.6.0
2019-07	RP-84	RP-191396	0283	2	F	MR-DC resource coordination in F1	15.6.0
2019-07	RP-84	RP-191396	0316	2	F	Full configuration indication from gNB-CU to gNB-DU.	15.6.0
2019-07	RP-84	RP-191396	0322	2	F	CR to 38.473 on clarification to RRC reconfigure complete indicator	15.6.0
2019-07	RP-84	RP-191394	0326	2	F	CR to 38.473 on deconfiguring CA based PDCP duplication for DRB	15.6.0
2019-07	RP-84	RP-191395	0330	3	F	CR to 38.473 on Removal of Multiple TNLAs	15.6.0
2019-07	RP-84	RP-191396	0348	-	F	Full configuration in UE Context Setup	15.6.0
2019-07	RP-84	RP-191396	0351	2	F	CR on PWS segmentation over F1	15.6.0
2019-07	RP-84	RP-191396	0352	1	F	CR on cell type over F1	15.6.0
2019-07	RP-84	RP-191396	0357	-	F	Rapporteur updates: Alignment and editorials	15.5.0
2019-07	RP-84	RP-191396	0358	-	F	Rapporteur update: Correction of Presence for DRB information	15.6.0
2019-07	RP-84	RP-191396	0359	-	F	Rapporteur updates: Correction of Presence for E-UTRA PRACH Configuration	15.6.0
2019-07	RP-84	RP-191396	0370	-	F	Full configuration IE included in the UE Context Modification Response.	15.6.0
2019-07	RP-84	RP-191396	0376		F	CR to 38.473 on clarification for UP TNL Information IE over F1	15.6.0
2019-07	RP-84	RP-191396	0377	2	F	Procedure description on optional IEs in CU to DU RRC information IE.	15.6.0
2019-09	RP-85	RP-192166	0343	3	F	CR on MR-DC low layer coordination with an MgNB-DU	15.7.0
2019-09	RP-85	RP-192166	0344	2	F	CR on MCG PHR format in MgNB-DU	15.7.0
2019-09	RP-85	RP-192166	0388		F	CR on DC Coordination for PDCCH Blind Detection	15.7.0
2019-09	RP-85	RP-192167	0393	1	F	Rapporteur update - clarification of semantics	15.7.0
2019-09	RP-85	RP-192166	0399	1	F	Clarification for TNLA removal	15.7.0
2019-12	RP-86	RP-192915	0318	5	F	Correction about gNB-CU System Information IE	15.8.0
2019-12	RP-86	RP-192915	0447	1	F	On CellGroupConfig handling	15.8.0

2019-12	RP-86	RP-192915	0458	1	F	Correction of S-NSSAI coding	15.8.0
2019-12	RP-86	RP-192915	0459	1	F	Removal of Requested P-MaxFR2	15.8.0
2019-12	RP-86	RP-192915	0479	2	F	Addition of Message Identifier and Serial Number to PWS Cancel Request	15.8.0
2019-12	RP-86	RP-192916	0482	2	F	Clarifications on SCell lists	15.8.0
2019-12	RP-86	RP-192916	0494	-	F	RRC Container in Modification Procedure	15.8.0
2019-12	RP-86	RP-192916	0508	0	F	CR to 38.473 on applicability of the IE Selected BandCombinationIndex and Selected FeatureSetEntryIndex	15.8.0
2019-12	RP-86	RP-192916	0509	1	F	CR to 38.473 on MeasGapSharingConfig and gNB-CU System Information	15.8.0
2019-12	RP-86	RP-192916	0510	1	F	CR to 38.473 on cause values over F1	15.8.0
2019-12	RP-86	RP-192916	0515	2	F	Clarification on Initial UL RRC Message Transfer procedure	15.8.0
2019-12	RP-86	RP-192913	0280	7	F	Trace function support for F1AP	16.0.0
2019-12	RP-86	RP-192908	0287	7	B	Support for CLI	16.0.0
2019-12	RP-86	RP-192913	0314	5	B	Introduction of Additional RRM Policy Index (ARPI)	16.0.0
2019-12	RP-86	RP-192908	0339	6	B	CR to F1-AP for RIM new message	16.0.0
2019-12	RP-86	RP-192915	0460		F	Removal of unused IEs	16.0.0
2019-12	RP-86	RP-192913	0463	1	C	Extending the MDBV Range	16.0.0
2019-12	RP-86	RP-192910	0514	3	B	CR for TS38.473 on supporting SN Resume during the RRCResume procedure	16.0.0
2019-12	RP-86	RP-192914	0518	2	F	Support for setting up IPSec a priori in F1	16.0.0
2020-03	RP-87-e	RP-200428	0522	1	A	Correction of PWS Failure Indication	16.1.0
2020-03	RP-87-e	RP-200428	0525	-	A	Correction of the presence of UL UP TNL Information to be setup List IE in tabular	16.1.0
2020-03	RP-87-e	RP-200425	0527	2	F	Corrections to CLI	16.1.0
2020-03	RP-87-e	RP-200425	0528	1	D	Rapporteur: Editorial updates	16.1.0
2020-03	RP-87-e	RP-200425	0530	2	B	E2E delay measurement for Qos monitoring for URLLC	16.1.0
2020-03	RP-87-e	RP-200428	0534	1	A	Correction relating to Initial UL RRC Message Transfer procedure CR 38.473	16.1.0
2020-07	RP-88-e	RP-201077	0285	17	B	BL CR to 38.473: Support for IAB	16.2.0
2020-07	RP-88-e	RP-201074	0432	12	B	Support of NR V2X over F1	16.2.0
2020-07	RP-88-e	RP-201082	0441	12	B	Addition of SON features	16.2.0
2020-07	RP-88-e	RP-201079	0477	8	B	Introduction of NR_IIOT support to TS 38.473	16.2.0
2020-07	RP-88-e	RP-201075	0481	10	B	Baseline CR for introducing Rel-16 NR mobility enhancement	16.2.0
2020-07	RP-88-e	RP-201082	0492	6	B	Addition of MDT features	16.2.0
2020-07	RP-88-e	RP-201080	0502	7	B	Introduction of NPN	16.2.0
2020-07	RP-88-e	RP-201076	0537	1	B	CR38.473 on TDD pattern for NR-DC power control cordination for sol1	16.2.0
2020-07	RP-88-e	RP-201085	0539	-	F	Rapporteur: Corrections after implementation	16.2.0
2020-07	RP-88-e	RP-201090	0543	2	A	Encoding PLMNs in served cell information NR	16.2.0
2020-07	RP-88-e	RP-201091	0545	1	A	Correction for usage of Cell Broadcast Cancelled List	16.2.0
2020-07	RP-88-e	RP-201091	0548	1	A	Correction on UE CONTEXT MODIFICATION REQUIRED message	16.2.0
2020-07	RP-88-e	RP-201085	0561	1	F	Correction on CLI	16.2.0
2020-07	RP-88-e	RP-201090	0567	-	A	Encoding PLMNs in served cell information IEs - semantics corrections	16.2.0
2020-07	RP-88-e	RP-201092	0570	1	A	Correction for UL UP TNL Information	16.2.0
2020-07	RP-88-e	RP-201092	0572	-	A	Correction on RRC Container in Initial UL RRC Messag Transfer	16.2.0
2020-07	RP-88-e	RP-201092	0576	1	A	Correction on RRC Connection Reconfiguration Complete Indicator	16.2.0
2020-07	RP-88-e	RP-201092	0581	2	F	Corrections of Inactive UE Context stored at gNB-DU	16.2.0
2020-07	RP-88-e	RP-201085	0600	2	F	Correction on RF parameters in NR cell information	16.2.0
2020-07	RP-88-e	RP-201090	0601	4	F	Correction of S-NSSAI range	16.2.0
2020-07	RP-88-e	RP-201092	0603	2	A	Correction for Handover Preparation Information	16.2.0
2020-07	RP-88-e	RP-201092	0607	1	A	CR on Concurrent Warning Message Indicator over F1 (Rel-16)	16.2.0
2020-07	RP-88-e	RP-201092	0615	-	A	Section renumbering for PWS cancel	16.2.0
2020-07	RP-88-e	RP-201092	0616	-	A	Correction on DL RRC MESSAGE TRANSFER	16.2.0
2020-07	RP-88-e	RP-201092	0618		A	Addition of abnormal conditions in PWS Cancel procedure	16.2.0
2020-09	RP-89-e	RP-201850	0495	10	B	Introduction of positioning support over F1AP	16.3.0
2020-09	RP-89-e	RP-201956	0557	2	A	Support of PSCell/SCell-only operation mode	16.3.0
2020-09	RP-89-e	RP-201956	0583	5	F	Cell Creation Rejection when max number of supported cells is exceeded at CU CR 38.473	16.3.0
2020-09	RP-89-e	RP-201956	0587	5	A	Measurement gap deactivation over F1AP CR 38.473	16.3.0
2020-09	RP-89-e	RP-201949	0619	2	F	Slot list length correction in TDD UL-DL Configuration	16.3.0
2020-09	RP-89-e	RP-201956	0625	1	F	Addition of abnormal conditions in Write-Replace Warning procedure	16.3.0
2020-09	RP-89-e	RP-201956	0628	2	A	Correction of PSCell/SCell-only mode	16.3.0
2020-09	RP-89-e	RP-201956	0634	1	A	Correction on UE Context Modification Procedure	16.3.0
2020-09	RP-89-e	RP-201956	0639	1	F	Rapporteur Corrections	16.3.0
2020-09	RP-89-e	RP-201949	0640	-	F	Correction of procedure ID	16.3.0
2020-09	RP-89-e	RP-201956	0642	-	A	Correction of PWS cancel	16.3.0
2020-09	RP-89-e	RP-201949	0643	1	F	Corrections on PC5 Link Aggregated Bit Rate	16.3.0
2020-09	RP-89-e	RP-201949	0660	-	F	Correction on the Maximum Number of CHO Preparations in F1AP	16.3.0

2020-09	RP-89-e	RP-201956	0663	1	F	Corrections to 38.473 on node name type	16.3.0
2020-09	RP-89-e	RP-201947	0664	1	F	Correction on IAB-DU configuration	16.3.0
2020-09	RP-89-e	RP-201982	0671		F	Correction on IAB-DU configuration	16.3.0
2020-09	RP-89-e					Correct wrong numbering of protocolIE-ID in clause 9.4.7	16.3.1
2020-12	RP-90-e	RP-202310	0645	2	F	Uniqueness of BH RLC channel ID	16.4.0
2020-12	RP-90-e	RP-202310	0658	3	F	Correction on V2X related information	16.4.0
2020-12	RP-90-e	RP-202310	0665	2	F	Correction on unsuccessful operations of IAB procedures	16.4.0
2020-12	RP-90-e	RP-202310	0666	1	F	Correction on the identification of IAB-donor-DU	16.4.0
2020-12	RP-90-e	RP-202310	0667	2	F	Correction on the Context Setup procedure for IAB node	16.4.0
2020-12	RP-90-e	RP-202310	0668	1	F	Correction on BAP address	16.4.0
2020-12	RP-90-e	RP-202310	0672	1	F	CR on F1-C transfer for Rel-16 IAB	16.4.0
2020-12	RP-90-e	RP-202311	0677	-	F	Correction of F1AP positioning procedures	16.4.0
2020-12	RP-90-e	RP-202311	0678	1	F	Corrections to tabular and asn.1 for NR positioning (F1AP)	16.4.0
2020-12	RP-90-e	RP-202310	0681	1	F	Correction of alternative QoS profile	16.4.0
2020-12	RP-90-e	RP-202313	0683	-	F	Removal of duplicated imports	16.4.0
2020-12	RP-90-e	RP-202312	0684	2	F	Corrections of UL and DL carrier list	16.4.0
2020-12	RP-90-e	RP-202311	0689	1	F	RRC alignement and various correction including ASN.1	16.4.0
2020-12	RP-90-e	RP-202311	0691	1	F	Correction of RLC Duplication Information over F1	16.4.0
2020-12	RP-90-e	RP-202288	0695	3	A	Correction on value range of UAC reduction Indication	16.4.0
2020-12	RP-90-e	RP-202311	0709	1	F	Coupling TRP ID and Cell ID in Measurement procedures	16.4.0
2021-03	RP-91-e	RP-210123	0431	7	B	Introduction of SFN Offset per cell over F1	16.5.0
2021-03	RP-91-e	RP-210240	0632	6	A	Correction on Overlapping Band Handling over F1	16.5.0
2021-03	RP-91-e	RP-210235	0676	2	F	Correction on PRACH coordination	16.5.0
2021-03	RP-91-e	RP-210239	0702	3	F	Cause value on F1 for insufficient UE capabilities CR 38.473	16.5.0
2021-03	RP-91-e	RP-210239	0711	1	F	Update on QoS monitoring control	16.5.0
2021-03	RP-91-e	RP-210233	0715	2	F	Stage-3 CR on transmission stop for Rel-16 DAPS handover	16.5.0
2021-03	RP-91-e	RP-210232	0720	1	F	Correction of NPN related Cell Information	16.5.0
2021-03	RP-91-e	RP-210231	0721	-	F	Correction on IAB configuration	16.5.0
2021-03	RP-91-e	RP-210231	0722	-	F	Correction on BAP address configuration for IAB-donor-DU	16.5.0
2021-03	RP-91-e	RP-210230	0725	1	F	Including SRS frequency information in Positioning Information Request	16.5.0
2021-03	RP-91-e	RP-210231	0728	2	F	CR to 38.473: Correction on IAB related definitions and unsuccessful establishment of a BH RLC channel	16.5.0
2021-03	RP-91-e	RP-210230	0736	-	F	Correction of the PCI IE presence in the ASN.1 for the SRS Configuration	16.5.0
2021-06	RP-92-e	RP-211334	0704	4	A	How to release SCG configuration between MN-CU and MN-DU CR 38.473	16.6.0
2021-06	RP-92-e	RP-211315	0712	2	F	Clarification on TAI Slice Support List	16.6.0
2021-06	RP-92-e	RP-211323	0740	2	F	Enabling CHO with SCG configuration	16.6.0
2021-06	RP-92-e	RP-211327	0743	-	F	Correction of Spatial Relation Information	16.6.0
2021-06	RP-92-e	RP-211317	0744	-	F	Correction on reference to RACH-Report	16.6.0
2021-06	RP-92-e	RP-211330	0753		F	Stage-3 CR on system information message over F1 (Rel-16)	16.6.0
2021-06	RP-92-e	RP-211333	0760	-	A	Correction on SRB ID	16.6.0
2021-06	RP-92-e	RP-211334	0762	3	A	gNB-DU UE Aggregate Maximum Bit Rate Uplink correction	16.6.0
2021-06	RP-92-e	RP-211322	0763	-	F	Miscellaneous corrections on IAB in TS 38.473	16.6.0
2021-06	RP-92-e	RP-211327	0765	1	F	Correction on SFN Initialisation Time	16.6.0
2021-06	RP-92-e	RP-211327	0766	-	F	Correction on relative cartesian coordinate	16.6.0
2021-06	RP-92-e	RP-211322	0770	1	F	Correction on BH RLC CH configured for BAP control PDU	16.6.0
2021-06	RP-92-e	RP-211322	0771	-	F	Correction on gNB-DU Resource Configuration	16.6.0
2021-06	RP-92-e	RP-211322	0772	1	F	Correction on UL BH information configuration for DRBs support CA based duplication	16.6.0
2021-06	RP-92-e	RP-211317	0776	1	F	Correction on MLB for TS 38.473	16.6.0
2021-09	RP-93-e	RP-211876	0790	1	F	Correction of served cell information for NPN	16.7.0
2021-09	RP-93-e	RP-211880	0792	1	F	Correction of wrong CR implementation for Stage-3 CR on transmission stop for Rel-16 DAPS handover	16.7.0
2021-09	RP-93-e	RP-211883	0796	1	F	Adding procedural text for System Frame Number and Slot Number	16.7.0
2021-09	RP-93-e	RP-211881	0800	-	A	Correction of the IE related to E-UTRA resource coordination in F1AP	16.7.0
2021-12	RP-94-e	RP-212864	0804	1	A	Correction on F1 Removal for RAN Sharing in Rel-16	16.8.0
2021-12	RP-94-e	RP-212864	0811	4	F	Incorrect Node Name IE in ASN.1	16.8.0
2021-12	RP-94-e	RP-213174	0822	3	F	Correction on PRS-only TRP	16.8.0
2021-12	RP-94-e	RP-212867	0827	1	F	Support of providing spatial relation per SRS resource from gNB-CU to gNB-DU	16.8.0
2022-03	RP-95-e	RP-220279	0778	4	F	Support of dynamic ACL during dual connectivity	16.9.0
2022-03	RP-95-e	RP-220276	0837	1	F	Correction on packet delay budget for IAB access link in TS 38.473	16.9.0
2022-03	RP-95-e	RP-220276	0838	-	F	CR to 38.473: Correction on IAB TNL Address Allocation procedure	16.9.0
2022-03	RP-95-e	RP-220242	0844	2	F	CR to TS38.473: Correction on PC5 QoS parameters for NR V2X	16.9.0
2022-03	RP-95-e	RP-220281	0847	1	F	Correction on positioning information configuration	16.9.0
2022-03	RP-95-e	RP-220281	0848	1	F	Correction on Measurement Periodicity	16.9.0
2022-03	RP-95-e	RP-220281	0849	1	F	Correction on PRS Beam Information	16.9.0

2022-03	RP-95-e	RP-220281	0850		F	CR for the correction on measurement gap configuration for position	16.9.0
2022-03	RP-95-e	RP-220278	0854	1	F	Correction of frequency information for DL only or UL only cell	16.9.0
2022-03	RP-95-e	RP-220276	0860		F	(Stage-3) Clarification on IAB Address Remove	16.9.0
2022-03	RP-95-e	RP-220221	0710	9	B	Addition of SON features enhancement	17.0.0
2022-03	RP-95-e	RP-220224	0716	6	B	Introduction of NR MBS	17.0.0
2022-03	RP-95-e	RP-220222	0737	13	B	CP-based Congestion Indication for IAB Networks	17.0.0
2022-03	RP-95-e	RP-220221	0738	7	B	BLCR to 38.473: Support of MDT enhancement	17.0.0
2022-03	RP-95-e	RP-220223	0751	7	B	Introduction of Enhanced IIoT support over F1	17.0.0
2022-03	RP-95-e	RP-220218	0777	10	B	SCG BL CR to TS 38.473	17.0.0
2022-03	RP-95-e	RP-220218	0795	5	B	BLCR to TS 38.473 for Conditional PScell Change/Addition	17.0.0
2022-03	RP-95-e	RP-220228	0803	6	B	Introduction of NR Positioning enhancements	17.0.0
2022-03	RP-95-e	RP-220230	0806	6	B	BL CR to F1AP on Rel-17 RedCap	17.0.0
2022-03	RP-95-e	RP-220236	0817	2	B	Addition of NR Timing Advance reporting for NR UL E-CID [NRTADV]	17.0.0
2022-03	RP-95-e	RP-220229	0826	7	B	Support of QoE information transfer	17.0.0
2022-03	RP-95-e	RP-220233	0833	3	B	CG-SDT BLCR to TS38.473	17.0.0
2022-03	RP-95-e	RP-220233	0834	3	B	Support of RACH-based SDT	17.0.0
2022-03	RP-95-e	RP-220231	0842	3	B	Introduction of SideLink Relay	17.0.0
2022-03	RP-95-e	RP-220219	0852	4	B	Introduction of MultiSIM support over F1	17.0.0
2022-03	RP-95-e	RP-220235	0855	4	B	Support for UE Power Saving Enhancements	17.0.0
2022-03	RP-95-e	RP-220232	0856	1	B	(BLCR to TS 38.473) Supporting network slicing enhancement	17.0.0
2022-03	RP-95-e	RP-220236	0858		D	Editorial corrections	17.0.0
2022-06	RP-96	RP-221143	0862	2	F	QoE Rel-17 Corrections	17.1.0
2022-06	RP-96	RP-221132	0863	-	F	Correction of PDC Measurement Periodicity values	17.1.0
2022-06	RP-96	RP-221141	0864	1	F	Correction of R17 SON features enhancement	17.1.0
2022-06	RP-96	RP-221134	0865	1	F	Corrections on NR MBS in F1AP	17.1.0
2022-06	RP-96	RP-221134	0866	-	F	NR MBS F1AP asn.1 correction	17.1.0
2022-06	RP-96	RP-221136	0868	2	F	Correction on CG based SDT	17.1.0
2022-06	RP-96	RP-221150	0879	1	A	F1AP CR for ACL remaining issues	17.1.0
2022-06	RP-96	RP-221145	0880	4	F	CR to 38.473 on Measurement Amount	17.1.0
2022-06	RP-96	RP-221139	0884	-	F	ASN.1 corrections on NR SL relay for 38.473	17.1.0
2022-06	RP-96	RP-221137	0887	2	F	Correction to MUSIM	17.1.0
2022-06	RP-96	RP-221139	0892	1	F	Corrections for SL_relay (F1AP)	17.1.0
2022-06	RP-96	RP-221126	0894	1	F	Correction on RedCap Broadcast Information for TS38.473	17.1.0
2022-06	RP-96	RP-221131	0896	1	F	F1AP ASN.1 review for NR Positioning Enhancements	17.1.0
2022-06	RP-96	RP-221134	0897		F	Remove the editor's notes	17.1.0
2022-06	RP-96	RP-221136	0900	1	F	Correction on SRB SDT indication	17.1.0
2022-06	RP-96	RP-221136	0901	1	F	Correction on SDT termination request in F1	17.1.0
2022-06	RP-96	RP-221141	0902	1	F	Correction on SON feature enhancements - F1AP	17.1.0
2022-06	RP-96	RP-221131	0905	3	F	Positioning corrections for F1AP	17.1.0
2022-06	RP-96	RP-221132	0908	2	F	Correction of PDC Measurement Initiation Failure message	17.1.0
2022-06	RP-96	RP-221805	0910	2	F	Correction for IAB inter-donor DU re-routing and resource multiplexing	17.1.0
2022-06	RP-96	RP-221149	0913	1	A	Correction on IAB-DU cell resource configuration	17.1.0
2022-06	RP-96	RP-221130	0916	2	F	CR to TS38.473: Correction on PC5 DRX parameters for NR V2X	17.1.0
2022-06	RP-96	RP-221139	0920		F	Corrections on Remote UE Local ID	17.1.0
2022-06	RP-96	RP-221141	0921	2	F	F1AP corrections for NR-U	17.1.0
2022-06	RP-96	RP-221152	0923	2	A	Correction for PRS Muting	17.1.0
2022-06	RP-96	RP-221132	0924	2	F	NR-IIoT F1AP correction	17.1.0
2022-06	RP-96	RP-221129	0927	2	F	Supporting network slice AS group	17.1.0
2022-06	RP-96	RP-221129	0928	1	F	Correction of the presence of UE-Slice-MBR	17.1.0
2022-06	RP-96	RP-221141	0929	-	F	Corrections to Load Balancing Enhancements	17.1.0
2022-06	RP-96	RP-221139	0930	2	F	Corrections for SL relay	17.1.0
2022-06	RP-96	RP-221145	0932	2	D	Editorial corrections	17.1.0
2022-06	RP-96	RP-221134	0938	1	F	Correction on MBS features	17.1.0
2022-06	RP-96	RP-221136	0939	1	F	Correction on SDT in F1AP	17.1.0
2022-06	RP-96	RP-221136	0940	1	F	Correction on Rel-17 SDT (F1AP)	17.1.0
2022-06	RP-96	RP-221141	0945	1	F	Correction on update management based MDT user consent	17.1.0
2022-06	RP-96	RP-221131	0948	2	F	Support of multiple measurement instances	17.1.0
2022-06	RP-96	RP-221145	0949	1	F	Supporting the disaster roaming information [MINT]	17.1.0
2022-06	RP-96	RP-221150	0951	2	A	SIB Issues Rel-17	17.1.0
2022-06	RP-96	RP-221150	0953	1	A	gNB-CU and gNB-DU Name in Configuration Update Procedures	17.1.0
2022-06	RP-96	RP-221137	0957	-	F	Clarification on the paging cause	17.1.0
2022-06	RP-96	RP-221143	0958	1	F	CR to 38.473 on ASN.1 corrections of QoE measurement	17.1.0
2022-06	RP-96	RP-221152	0964		A	ASN.1 correction for UL-AoA	17.1.0
2022-06	RP-96	RP-221141	0965	1	F	ASN.1 corrections	17.1.0
2022-06	RP-96	RP-221141	0968	1	F	CCO corrections	17.1.0
2022-06	RP-96	RP-221131	0969	-	F	Corrections to Measurement Pre-configuration Information Transfer	17.1.0
2022-06	RP-96	RP-221628	0971	-	F	Removal of UE Tx TEG Association from F1AP	17.1.0
2022-09	RP-97-e	RP-222201	0889	2	A	Correction on interFrequencyConfig-NoGap	17.2.0

2022-09	RP-97-e	RP-222189	0890	2	F	Correction to SDT for supporting delta signaling	17.2.0
2022-09	RP-97-e	RP-222188	0978	-	F	Correction on Broadcast and Unicast co-existence	17.2.0
2022-09	RP-97-e	RP-222183	0981	1	F	Miscellaneous Correction on IAB	17.2.0
2022-09	RP-97-e	RP-222188	0984	1	F	Further Corrections for NR MBS	17.2.0
2022-09	RP-97-e	RP-222188	0985	2	F	Corrections for the establishment of F1-U ptp retransmission tunnels	17.2.0
2022-09	RP-97-e	RP-222185	0986	3	B	CR for TS38.473 on Extending NR Operation to 71GHz	17.2.0
2022-09	RP-97-e	RP-222186	0988		F	Addition of SRS port index	17.2.0
2022-09	RP-97-e	RP-222190	0989	1	F	SL relay corrections	17.2.0
2022-09	RP-97-e	RP-222187	0994	1	F	Correction of UE Paging Capability	17.2.0
2022-09	RP-97-e	RP-222201	0997	1	F	Correction on measurement gap configuration over F1 in Rel-17	17.2.0
2022-09	RP-97-e	RP-222186	0998	2	F	Support of timing error margins for TEGs in F1AP	17.2.0
2022-09	RP-97-e	RP-222188	1000	1	F	Introduction of MBS specific cause values	17.2.0
2022-09	RP-97-e	RP-222188	1001	1	F	Correction on Multicast Group Paging	17.2.0
2022-09	RP-97-e	RP-222188	1002	1	F	Correction on MRB ID Change	17.2.0
2022-09	RP-97-e	RP-222603	1004	4	A	CR to 38.473 on E-CID measurement periodicity	17.2.0
2022-09	RP-97-e	RP-222088	1013	3	F	Correction on RedCap paging capability	17.2.0
2022-09	RP-97-e	RP-222186	1016	1	F	Rel-17 ePos correction for the missing support of SRS-PosRRC-InactiveConfig-r17 configuration	17.2.0
2022-09	RP-97-e	RP-222189	1017	1	F	Transferring CG-SDT configuration and SRS positioning Inactive configuration from DU to CU	17.2.0
2022-09	RP-97-e	RP-222088	1019	1	F	Correction of the maximum PTW length of IDLE eDRX	17.2.0
2022-09	RP-97-e	RP-222189	1021	1	F	Correction on Rel-17 SDT	17.2.0
2022-09	RP-97-e	RP-222638	1025	3	F	Introduction of uplink GapFR2 [NR_RF_FR2_req_enh2-Core]	17.2.0
2022-09	RP-97-e	RP-222186	1027	2	F	Correction to positioning gap configuration	17.2.0
2022-09	RP-97-e	RP-222191	1031	-	F	Correction to Report Caracteristics	17.2.0
2022-09	RP-97-e	RP-222186	1034	-	F	Correction on Measurement Time Occasion	17.2.0
2022-09	RP-97-e	RP-222191	1035	1	F	Correction on NR-U MLB	17.2.0
2022-12	RP-98-e	RP-222883	0975	4	A	R17CR for DAPS over F1 to TS38.473	17.3.0
2022-12	RP-98-e	RP-222884	1012	1	A	Correction of on-demand SI for connected UE	17.3.0
2022-12	RP-98-e	RP-222879	1039	1	F	Further correction to Report Caracteristics	17.3.0
2022-12	RP-98-e	RP-222882	1043	3	F	Provision of MBS Multicast F1-U references to UE Context in gNB-CU enabling retrieval of data forwarding progress information	17.3.0
2022-12	RP-98-e	RP-222886	1046	2	F	Correction of TRP TEG	17.3.0
2022-12	RP-98-e	RP-222884	1052	2	A	Correction on generation of gap type over F1 in Rel-17	17.3.0
2022-12	RP-98-e	RP-222886	1054	1	F	Correction of Timing Error Margin	17.3.0
2022-12	RP-98-e	RP-222887	1057	1	A	CR to 38.473 on SRS periodicity	17.3.0
2022-12	RP-98-e	RP-222886	1058	1	F	Correction of ASN.1 for UL RTOA Measurement	17.3.0
2022-12	RP-98-e	RP-222887	1060	3	A	Correction on positioning SI delivery over F1AP	17.3.0
2022-12	RP-98-e	RP-222881	1061	1	F	Correction on resource configuration for IAB	17.3.0
2022-12	RP-98-e	RP-222886	1072	-	F	Correction on presence of timing error margin for TRP TEGs	17.3.0
2022-12	RP-98-e	RP-222883	1075	1	A	CR for DAPS state transfer in case of split gNB deployment to Rel-17 38.473	17.3.0
2022-12	RP-98-e	RP-222888	1076	1	F	SL relay corrections	17.3.0
2022-12	RP-98-e	RP-222888	1077	4	F	Correction to 38.473 for SL relay (R17)	17.3.0
2022-12	RP-98-e	RP-222957	1081	5	A	Support of DC Location for two UL CCs in Split architecture	17.3.0
2022-12	RP-98-e	RP-222882	1083	1	F	Correction on NR MBS over F1AP	17.3.0
2022-12	RP-98-e	RP-222882	1084	1	F	Correction on MRB QoS Information	17.3.0
2022-12	RP-98-e	RP-222879	1091	2	F	Correction on Resource Status Reporting procedure over F1	17.3.0
2022-12	RP-98-e	RP-222885	1094	2	F	Support of DC Location for more carriers in Split architecture (The CR is not implemented. The CR is postponed to next plenary meeting due to the fact that it is impossible to modify a paragraph via 2 separate CRs without CR clash)	17.3.0
2022-12	RP-98-e	RP-222886	1100	1	F	Support of DC Location for more carriers in Split architecture	17.3.0
2023-03	RAN#99	RP-230591	1029	4	B	Introduction of two PHR mode [NR_feMIMO-Core]	17.4.0
2023-03	RAN#99	RP-230582	1090	4	F	Correction to conditional MCG configuration in CPAC	17.4.0
2023-03	RAN#99	RP-230589	1093	4	F	Correction to support NCD-SSB RedCap requirements in F1AP	17.4.0
2023-03	RAN#99	RP-230593	1094	3	F	Correction for supporting DC Location for more carriers in Split architecture	17.4.0
2023-03	RAN#99	RP-230585	1097	2	F	Correction of NR PRACH Configuration List for FR2-2	17.4.0
2023-03	RAN#99	RP-230581	1113	1	F	Correction on missing cause value for CG-SDT	17.4.0
2023-03	RAN#99	RP-230589	1114	2	F	Correction on the UE identity index for paging RedCap UE to TS38.473	17.4.0
2023-03	RAN#99	RP-230596	1116	-	A	Correction on IAB UP configuration update	17.4.0
2023-03	RAN#99	RP-230583	1117	1	F	Correction on NR MBS Broadcast aspects	17.4.0
2023-03	RAN#99	RP-230587	1118	1	F	Correction of gNB Rx-Tx Time Difference	17.4.0
2023-03	RAN#99	RP-230590	1120	1	F	Correction of RRC references for SLrelay	17.4.0
2023-03	RAN#99	RP-230593	1121	-	F	Correction of RRC references for DRX	17.4.0
2023-03	RAN#99	RP-230593	1122	-	F	Correction of RRC references	17.4.0
2023-03	RAN#99	RP-230591	1124	2	B	Missing transmission bandwidth configurations in F1AP	17.4.0
2023-03	RAN#99	RP-230583	1125	1	F	Correction of F1 Broadcast Setup	17.4.0
2023-03	RAN#99	RP-230588	1126	-	F	PRS configuration procedure correction	17.4.0

2023-03	RAN#99	RP-230581	1127	1	F	Correction to CellGroupConfig handling for SDT	17.4.0
2023-03	RAN#99	RP-230584	1131	2	F	Correction on NR-U Channel ID	17.4.0
2023-03	RAN#99	RP-230595	1137	2	A	ASN.1 Correction of M6 Configuration	17.4.0
2023-03	RAN#99	RP-230586	1138	1	F	Correction to TS 38.473 on Re-routing Enable Indicator	17.4.0
2023-04	RAN#99	-	-	-	-	Editorial Changes(font, style, line break)	17.4.1
2023-06	RAN#100	RP-231073	1119	2	F	Correction of Burst Arrival Time semantics description	17.5.0
2023-06	RAN#100	RP-231075	1135	3	A	Correction of SIType List	17.5.0
2023-06	RAN#100	RP-231075	1144	2	A	Corrections on TNL association addition, update and removal (F1AP)	17.5.0
2023-06	RAN#100	RP-231072	1145	2	F	Correction to TS 38.473 on RB Set Configuration	17.5.0
2023-06	RAN#100	RP-231079	1146	2	F	Introduction of the UE hashed ID to 38.473	17.5.0
2023-06	RAN#100	RP-231074	1147	2	F	Correction on Broadcast Partial Success	17.5.0
2023-06	RAN#100	RP-231081	1150	3	A	ASN.1 Correction of PRACH Configuration	17.5.0
2023-06	RAN#100	RP-231071	1158	2	F	F1AP Rel-17 correction for NR-U metrics	17.5.0
2023-06	RAN#100	RP-231080	1159	2	F	Correction on F1AP for L2 U2N Relay	17.5.0
2023-06	RAN#100	RP-231084	1162	2	F	Correction of Extended Packet Delay Budget	17.5.0
2023-06	RAN#100	RP-231075	1165	2	A	Correction on E-UTRA - NR Cell Resource Coordination	17.5.0
2023-06	RAN#100	RP-231074	1166	2	F	Transfer of MBSInterestIndication from CU to DU	17.5.0
2023-06	RAN#100	RP-231074	1170	1	F	Correction of MRB Setup	17.5.0
2023-06	RAN#100	RP-231079	1172	0	F	Correction of RedCap-specific initial DL BWP without CD-SSB for SDT	17.5.0
2023-06	RAN#100	RP-231077	1177	1	F	SRS Resource correction on Comb 8, Number of Symbols and Repetition Factor	17.5.0
2023-06	RAN#100	RP-231077	1179	1	F	Subcarrier Spacing correction	17.5.0
2023-09	RAN#101	RP-231896	1175	3	A	Correction on IAB bar configuration	17.6.0
2023-09	RAN#101	RP-231897	1191	1	F	Correction of Distribution procedure	17.6.0
2023-09	RAN#101	RP-231901	1193	1	F	Mapping of SRB1 for the remote UE	17.6.0
2023-09	RAN#101	RP-231899	1197	-	A	Correction to TS 38.473 on inter-node message for CU-DU split scenario	17.6.0
2023-09	RAN#101	RP-231896	1203	-	A	Configuration of BH information for DRBs support CA based duplication	17.6.0
2023-09	RAN#101	RP-231897	1205	1	F	Correction on condition of successful MBS Broadcast Context Setup	17.6.0
2023-09	RAN#101	RP-231900	1206	1	F	PRS CONFIGURATION REQUEST Correction	17.6.0
2023-09	RAN#101	RP-231900	1211	-	A	Correction of Positioning SIType List	17.6.0
2023-09	RAN#101	RP-231898	1215	1	F	Rel-17 Correction in the UE Context Modification procedure abnormal description for conditional mobility modification	17.6.0
2023-12	RAN#102	RP-233851	1220	3	F	Correction on SI delivery to RedCap UE	17.7.0
2023-12	RAN#102	RP-233851	1223	4	F	Support of preconfigured Measurement GAP	17.7.0
2023-12	RAN#102	RP-233848	1230	1	F	Clarification on gNB-DU Cell Resource Configuration for IAB	17.7.0
2023-12	RAN#102	RP-233849	1233	1	F	Correction of F1-U context Reference for PTM	17.7.0
2023-12	RAN#102	RP-233850	1237	-	F	Correction to F1AP for the misalignment on DL PRS	17.7.0
2023-12	RAN#102	RP-233850	1246	-	F	Correction on TRP Information Type Response Item IE of Positioning	17.7.0
2023-12	RAN#102	RP-233847	1247	1	F	Correction on NR-Mode-Info IE of SON	17.7.0
2023-12	RAN#102	RP-233818	1037	13	B	Additions for L1/L2 triggered mobility	18.0.0
2023-12	RAN#102	RP-233833	1070	11	B	Introduction of R18 QoE measurement enhancements	18.0.0
2023-12	RAN#102	RP-233832	1105	11	B	Addition of SON features enhancement	18.0.0
2023-12	RAN#102	RP-233814	1109	7	B	Support of Network-Controlled Repeater	18.0.0
2023-12	RAN#102	RP-233822	1123	10	B	Support for NR Sidelink Relay Enhancements	18.0.0
2023-12	RAN#102	RP-233817	1129	11	B	Introduction of Network Energy Saving	18.0.0
2023-12	RAN#102	RP-233819	1140	8	B	Introduction on MT-SDT	18.0.0
2023-12	RAN#102	RP-233845	1167	3	B	Support 1-symbol PRS [1symbol_PRS]	18.0.0
2023-12	RAN#102	RP-233838	1168	8	B	Introduction of 5G Timing Resiliency and URLLC enhancements	18.0.0
2023-12	RAN#102	RP-233816	1169	8	B	Introduction on NR Redcap Enhancement	18.0.0
2023-12	RAN#102	RP-233834	1176	10	B	(CR to 38.473): Support for mobile IAB	18.0.0
2023-12	RAN#102	RP-233829	1189	7	B	Introduction of NR MBS enhancements	18.0.0
2023-12	RAN#102	RP-233842	1194	2	B	Introduction of measurements without gap with interruption	18.0.0
2023-12	RAN#102	RP-233845	1213	3	B	Switching from SDT to RRC connected state [Large SDT Uplink Data]	18.0.0
2023-12	RAN#102	RP-233830	1219	5	B	Support for NR XR	18.0.0
2023-12	RAN#102	RP-233841	1221	2	B	Introduction of 3 MHz channel bandwidth	18.0.0
2023-12	RAN#102	RP-233839	1226	2	B	RAN impact on supporting Network Slice Service continuity scenario	18.0.0
2023-12	RAN#102	RP-233818	1227	2	B	On Subsequent CPAC	18.0.0
2023-12	RAN#102	RP-233845	1231	2	B	Introduction of RedCap UE MBS Broadcast reception [RedcapMBS]	18.0.0
2023-12	RAN#102	RP-233821	1232	1	B	Introduction of early capability restriction for Multi-SIM	18.0.0
2023-12	RAN#102	RP-233813	1248	1	B	Introduction of SL CA over F1 interface	18.0.0
2023-12	RAN#102	RP-233825	1250	0	B	A2X communication services support in F1AP	18.0.0
2024-03	RAN#103	RP-240620	1180	11	B	Support of NR Positioning Enhancements	18.1.0
2024-03	RAN#103	RP-240621	1252	2	F	Information about context handling for S-CPAC	18.1.0

2024-03	RAN#103	RP-240636	1267	2	F	Correction of SRB ID for QoE	18.1.0
2024-03	RAN#103	RP-240637	1269	3	F	Corrections for mobile IAB	18.1.0
2024-03	RAN#103	RP-240633	1275	2	F	Correction on MBS RAN sharing	18.1.0
2024-03	RAN#103	RP-240634	1278	1	F	Introduction of separate uplink and downlink PDU set QoS parameters	18.1.0
2024-03	RAN#103	RP-240621	1280	1	F	Correction on Reference configuration and RRC Complete configuration_Option 1	18.1.0
2024-03	RAN#103	RP-240630	1290	1	F	Corrections on multi-path relay	18.1.0
2024-03	RAN#103	RP-240639	1292	1	F	Correction of timing synchronisation status reporting procedure	18.1.0
2024-03	RAN#103	RP-240629	1293	2	F	Filter information for temporary capabilities restriction in Multi-SIM	18.1.0
2024-03	RAN#103	RP-240636	1300	3	F	Corrections on QoE enhancements	18.1.0
2024-03	RAN#103	RP-240633	1303	1	F	Correction to F1AP on Multicast reception in RRC_INACTIVE state	18.1.0
2024-03	RAN#103	RP-240626	1306	1	F	Correction of SSBs activation and deactivation for network energy saving	18.1.0
2024-03	RAN#103	RP-240637	1311	1	F	Correct the ASN.1 errors for mobile IAB	18.1.0
2024-03	RAN#103	RP-240630	1312	-	F	Correction on the Assigned Criticality for SL Relay related IE	18.1.0
2024-03	RAN#103	RP-240621	1314	1	F	Correction for S-CPAC in Access Success	18.1.0
2024-03	RAN#103	RP-240637	1315	1	F	Corrections on F1AP for mobile IAB	18.1.0
2024-03	RAN#103	RP-240635	1319	2	F	Correction on SON for NR-U	18.1.0
2024-03	RAN#103	RP-240635	1320	2	F	Correction on RACH Optimisation	18.1.0
2024-03	RAN#103	RP-240637	1323	-	F	Correction on mobile TRP location information	18.1.0
2024-03	RAN#103	RP-240634	1324	1	F	Correction on PSI based discard	18.1.0
2024-03	RAN#103	RP-240634	1325	-	F	Correction on the behavior description of ECN Marking or Congestion Information Reporting over F1AP	18.1.0
2024-03	RAN#103	RP-240645	1329	-	A	PRS Angle Item ASN Correction	18.1.0
2024-03	RAN#103	RP-240625	1332	-	F	Correction of NR Paging Long eDRX Information for RRC INACTIVE	18.1.0
2024-03	RAN#103	RP-240632	1334	1	F	Correction on the missing SIBs for MBS and ATG in TS 38.473	18.1.0
2024-03	RAN#103	RP-240621	1335	3	F	Essential corrections for LTM over F1	18.1.0
2024-03	RAN#103	RP-240630	1344	1	F	Correction on SLrelay	18.1.0
2024-03	RAN#103	RP-240623	1345	1	F	Corrections of SL CA	18.1.0
2024-03	RAN#103	RP-240617	1346	2	D	Rapporteur CR	18.1.0
2024-03	RAN#103	RP-240627	1348	2	F	Correction on SDT	18.1.0
2024-03	RAN#103	RP-240646	1355	1	A	Missing procedure text in F1AP	18.1.0
2024-03	RAN#103	RP-240645	1356	1	A	PRS bandwidth ASN Correction	18.1.0
2024-03	RAN#103	RP-240642	1359	1	A	F1 correction for NR-U	18.1.0
2024-03	RAN#103	RP-240642	1362	-	F	Clarification on MIMO PRB usage Information reporting over F1	18.1.0
2024-03	RAN#103	RP-240633	1364	-	F	ASN.1 correction to F1AP for MBS enh	18.1.0
2024-06	RAN#104	RP-241112	1339	3	B	Introduction of Measurement Quality and Time Stamp Information to E-CID [ECIDQualTimeStamp]	18.2.0
2024-06	RAN#104	RP-241106	1367	3	F	Correction for early sync and TCI state of LTM	18.2.0
2024-06	RAN#104	RP-241118	1369	2	A	Introduction of SL related information transmission over F1 interface	18.2.0
2024-06	RAN#104	RP-241102	1373	2	F	Correction on musim-CapabilityRestrictionIndication IE	18.2.0
2024-06	RAN#104	RP-241115	1377	1	A	Correction of Time Reference Information	18.2.0
2024-06	RAN#104	RP-241112	1378	1	B	Introduction of 2Rx XR support [2Rx_XR_Device]	18.2.0
2024-06	RAN#104	RP-241105	1379	2	F	Introduction of MDBV in the Alternative QoS	18.2.0
2024-06	RAN#104	RP-241106	1385	6	F	LTM corrections to F1AP	18.2.0
2024-06	RAN#104	RP-241106	1388	2	F	Correction on the LTM configuration ID and LTM configuration ID Mapping list	18.2.0
2024-06	RAN#104	RP-241106	1392	3	F	Further corrections on LTM over F1	18.2.0
2024-06	RAN#104	RP-241106	1394	4	F	Corrections for LTM for reconfiguration with sync	18.2.0
2024-06	RAN#104	RP-241105	1395	3	F	Correction on non-integer DRX cycle	18.2.0
2024-06	RAN#104	RP-241101	1400	3	F	Correction on measurement report for SRS Bandwidth Aggregation	18.2.0
2024-06	RAN#104	RP-241103	1401	3	F	Corrections on SL relay	18.2.0
2024-06	RAN#104	RP-241103	1405	3	F	Corrections on PC5 RLC channel configuration for U2U relay	18.2.0
2024-06	RAN#104	RP-241106	1409	2	F	LTM correction for UE-based TA information from CU to DU	18.2.0
2024-06	RAN#104	RP-241101	1410	3	F	Various corrections on Rel-18 Positioning	18.2.0
2024-06	RAN#104	RP-241104	1411	2	F	Correction of MBS NG-U Setup and non shared gNB-DU	18.2.0
2024-06	RAN#104	RP-241106	1416	2	F	Corrections for CFRA resource provision in LTM	18.2.0
2024-06	RAN#104	RP-241113	1417	1	D	Rapporteur Corrections	18.2.0
2024-06	RAN#104	RP-241113	1418	1	F	Correction on MBS Broadcast F1-U Path Failure	18.2.0
2024-06	RAN#104	RP-241101	1420	2	F	Correction to DL-PRS Aggregation	18.2.0
2024-06	RAN#104	RP-241106	1421	1	F	Correction on LTM	18.2.0
2024-06	RAN#104	RP-241106	1422	1	F	Correction on the Assigned Criticality of LTM Cells To Be Released List IE	18.2.0
2024-06	RAN#104	RP-241114	1424	1	A	UE Context Release Command ASN.1 correction	18.2.0
2024-06	RAN#104	RP-241101	1425	2	F	Correction on Measurement Reporting for BWA	18.2.0
2024-06	RAN#104	RP-241101	1430	2	F	Correction on UL-RSCP	18.2.0
2024-06	RAN#104	RP-241106	1435	1	F	Corrections for LTM CSI Report Config generation	18.2.0
2024-06	RAN#104	RP-241101	1437	-	F	Correction on SIB23 in TS38.473	18.2.0
2024-06	RAN#104	RP-241113	1439	1	F	Correction for CHO cancel in UE CONTEXT MODIFICATION REQUEST	18.2.0
2024-06	RAN#104	RP-241106	1440	-	F	Correction on TAG information transfer for LTM	18.2.0

2024-09	RAN#105	RP-241879	1415	3	B	Introduction of barring exemption for (e)RedCap and 2RX XR UEs [EM_Call_Exemption]	18.3.0
2024-09	RAN#105	RP-241869	1442	-	F	Triggering reporting of requested DL LBT information	18.3.0
2024-09	RAN#105	RP-241876	1444	2	F	Correction on new SIB17bis for TRS in Idle and Inactive	18.3.0
2024-09	RAN#105	RP-241880	1446	1	F	F1AP support for sub 1s location information reporting periodicity [Sub_1s_periodicity]	18.3.0
2024-09	RAN#105	RP-241871	1448	1	A	Correction on Multicast Group Paging	18.3.0
2024-09	RAN#105	RP-241873	1452	2	A	Correction on asymmetric UL and DL for TDD Carrier	18.3.0
2024-09	RAN#105	RP-241871	1460	1	F	Correction on configuration update for an inactive Multicast session	18.3.0
2024-09	RAN#105	RP-241874	1462	1	F	Positioning ASN.1 correction (AggregatedPRSResourceSetList, kminus values)	18.3.0
2024-09	RAN#105	RP-241871	1464	1	A	Changing presence of IE extensions for MBS related IEs	18.3.0
2024-09	RAN#105	RP-241874	1466	1	F	Positioning corrections to F1AP	18.3.0
2024-09	RAN#105	RP-241873	1471	1	F	Correction of CP Transport Layer Information and DL UP TNL Information	18.3.0
2024-09	RAN#105	RP-241871	1473	-	F	Add Cause IE in MULTICAST CONTEXT NOTIFICATION REFUSE message	18.3.0
2024-09	RAN#105	RP-241870	1476	1	A	Correction on SSB transmission periodicity for IAB	18.3.0
2024-09	RAN#105	RP-241872	1477	1	F	LTM supports feature of MIMO with 2TA	18.3.0
2024-09	RAN#105	RP-241872	1478	2	F	Correction on Early Sync Information from DU to CU	18.3.0
2024-09	RAN#105	RP-241878	1479	1	F	Correction on XR PSI based SDU Discard	18.3.0
2024-09	RAN#105	RP-241877	1481	1	F	Corrections on Uu Relay RLC channel setup	18.3.0
2024-09	RAN#105	RP-241876	1483	1	A	Corrections on PEI subgrouping	18.3.0
2024-09	RAN#105	RP-241872	1484	1	F	Correction on Complete Candidate Configuration	18.3.0
2024-12	RAN#106	RP-243102	1456	3	F	Clarification on SpCell ID replacement	18.4.0
2024-12	RAN#106	RP-243104	1457	2	F	Support of UE specific SRS reservation	18.4.0
2024-12	RAN#106	RP-243108	1467	4	F	Correction of TRP Geographical coordinates [PosLocalCoords]	18.4.0
2024-12	RAN#106	RP-243104	1489	1	A	Correction of Positioning SI for connected UE	18.4.0
2024-12	RAN#106	RP-243100	1491	2	A	Correction for the semantics description	18.4.0
2024-12	RAN#106	RP-243105	1495	1	F	Corrections on MP relay based PDCP duplication	18.4.0
2024-12	RAN#106	RP-243102	1496	2	F	L2 reset for intra-DU LTM	18.4.0
2024-12	RAN#106	RP-243104	1499	2	A	Positioning measurement correction related to "shall, if supported"	18.4.0
2024-12	RAN#106	RP-243100	1502	3	A	Correction on BAP mapping configuration for IAB	18.4.0
2024-12	RAN#106	RP-243104	1504	2	F	Correction to Requested DL PRS Transmission Characteristics	18.4.0
2024-12	RAN#106	RP-243107	1507	2	F	Correction on broadcast reception for eRedCap UE [RedcapMBS]	18.4.0
2024-12	RAN#106	RP-243099	1510	3	F	Option 2: Correction on DU collecting and reporting of DL LBT failure information	18.4.0
2024-12	RAN#106	RP-243104	1514	3	A	Correction on FR1 SRS Bandwidth in Requested SRS Transmission Characteristics	18.4.0
2024-12	RAN#106	RP-243104	1516	3	A	introduction of Missing bandwidths in FR1 SRS Bandwidth	18.4.0
2024-12	RAN#106	RP-243102	1520	1	F	Corrections for Early TA acquisition for PDCCH order in LTM	18.4.0
2024-12	RAN#106	RP-243104	1524	2	F	Correction on Positioning SRS Resource	18.4.0
2024-12	RAN#106	RP-243104	1528	1	F	Correction on Measurement Time Window	18.4.0
2025-03	RAN#107	RP-250139	1533	1	F	Clarification on UE Context Modification Failure for LTM	18.5.0
2025-03	RAN#107	RP-250141	1535	-	F	Correction on criticality of SRS Reservation Type	18.5.0
2025-03	RAN#107	RP-250138	1539	1	A	Correction on the support of MBS in QoS Flow Level QoS Parameters and QoS Flow Identifier	18.5.0
2025-03	RAN#107	RP-250141	1542	1	A	Correction on Positioning Assistance Information Control Procedure	18.5.0
2025-06	RAN#108	RP-251150	1536	5	F	Correction on the mobile IAB procedures	18.6.0
2025-06	RAN#108	RP-251154	1550	1	F	Corrections for multi-path relay	18.6.0
2025-06	RAN#108	RP-251144	1552	4	A	Correction to CPAC MCG configuration cancellation	18.6.0
2025-06	RAN#108	RP-251149	1555	2	F	Clarification on Carrier Bandwidth less than 5MHz	18.6.0
2025-06	RAN#108	RP-251152	1558	6	F	F1 support of L3 measurements-based LTM	18.6.0
2025-06	RAN#108	RP-251155	1561	2	F	Correction for MDBV in alternative QoS	18.6.0
2025-06	RAN#108	RP-251153	1569		F	Correction to SRS information	18.6.0
2025-06	RAN#108	RP-251154	1571	1	F	Corrections on PDCP duplication for inter-DU multi-path relay	18.6.0
2025-06	RAN#108	RP-251146	1574	1	F	NPN Assistance Information for RAN sharing with multiple cell-ID broadcast	18.6.0
2025-09	RAN#109	RP-252690	1586	-	F	Correction on Intra-CU LTM shared preamble index limit per DU	18.7.0
2025-09	RAN#109	RP-252690	1587	1	F	Essential corrections to TCI states for LTM	18.7.0
2025-09	RAN#109	RP-252674	1443	10	B	Introduction of low-power wake-up signal and receiver for NR	19.0.0
2025-09	RAN#109	RP-252679	1485	8	B	Support of XR enhancements	19.0.0
2025-09	RAN#109	RP-252681	1519	7	B	Addition of SON enhancements	19.0.0
2025-09	RAN#109	RP-252675	1531	8	B	Introduction of Network Energy Saving Enhancement	19.0.0
2025-09	RAN#109	RP-252682	1532	9	B	Support of enhancements on AI/ML for NG-RAN	19.0.0
2025-09	RAN#109	RP-252686	1543	3	B	E-CID measurement enhancement [ECID_enh1]	19.0.0
2025-09	RAN#109	RP-252677	1547	7	B	Support for Inter-CU LTM and intra-CU conditional LTM	19.0.0
2025-09	RAN#109	RP-252680	1548	6	B	Support of Multi-hop relay	19.0.0
2025-09	RAN#109	RP-252687	1564	3	B	Trace Depth for Vendor Specific Trace Record	19.0.0
2025-09	RAN#109	RP-252672	1566	4	B	Introduction of Evolution of NR duplex operation	19.0.0
2025-09	RAN#109	RP-252673	1575	2	B	Support of Sample-based measurement (case 3b)	19.0.0
2025-09	RAN#109	RP-252684	1580	1	B	Introduction of 7 MHz Channel Bandwidth	19.0.0
2025-12	RAN#110	RP-253357	1592	3	A	Correction to positioning activation and deactivation procedure	19.1.0
2025-12	RAN#110	RP-253353	1601	3	F	Miscellaneous corrections on AI/ML for NG-RAN	19.1.0

2025-12	RAN#110	RP-253360	1604	3	A	Correction (SSB) on the CU-DU Mobility Initiation Request message	19.1.0
2025-12	RAN#110	RP-253348	1605	2	F	Essential correction for inter-CU LTM on F1AP	19.1.0
2025-12	RAN#110	RP-253348	1606	3	F	Semi-Persistent CSI-RS activation with TCI state	19.1.0
2025-12	RAN#110	RP-253346	1609	3	F	Correction of OD-SIB1 procedure for network energy saving	19.1.0
2025-12	RAN#110	RP-253341	1615	1	D	Rapporteur Corrections	19.1.0
2025-12	RAN#110	RP-253348	1632	1	F	Correction related to LTM Security Information	19.1.0
2025-12	RAN#110	RP-253348	1633	3	F	Correction on CSI-RS Resource Set and CSI IM Resource Transfer for inter-CU LTM	19.1.0
2025-12	RAN#110	RP-253344	1636	2	F	Correction to F1AP on CLI Indication	19.1.0
2025-12	RAN#110	RP-253348	1638	2	F	Removal of Request for CSI-RS resource configuration for Early CSI acquisition	19.1.0
2025-12	RAN#110	RP-253344	1639	3	F	Corrections to SBFD operation	19.1.0
2025-12	RAN#110	RP-253351	1640	3	F	Corrections on Multi-hop relay	19.1.0
2025-12	RAN#110	RP-253352	1641	1	F	Correction for LTM failure without RLF-report	19.1.0
2025-12	RAN#110	RP-253346	1659	2	F	Corrections to F1 support of Network Energy Saving Enhancement	19.1.0
2025-12	RAN#110	RP-253352	1661	1	F	BFR in CSI-RS beam	19.1.0
2025-12	RAN#110	RP-253344	1667	1	F	Correction on SBFD RACH configuration for F1AP	19.1.0
2025-12	RAN#110	RP-253348	1670	1	F	Correction on Early UL Sync Configuration IE for inter-CU LTM	19.1.0
2025-12	RAN#110	RP-253348	1671	1	F	Correction on the LTM Indicator for C-LTM	19.1.0
2025-12	RAN#110	RP-253348	1672	-	F	R19 correction (CSI-RS) on the CU-DU Mobility Initiation Request message	19.1.0
2026-03	RAN#111	RP-260270	1645	2	F	Corrections of AI/ML-based CCO	19.2.0
2026-03	RAN#111	RP-260276	1650	2	A	Correction on prioritized alternative QoS profile	19.2.0
2026-03	RAN#111	RP-260271	1676	1	F	Correction on CSI Report Configuration for CSI Acquisition for inter-CU LTM	19.2.0
2026-03	RAN#111	RP-260270	1692	2	F	Abnormal conditions for F1AP gNB-CU Configuration Update	19.2.0
2026-03	RAN#111	RP-260270	1693	2	F	Correction for gNB-DU Configuration Update	19.2.0
2026-06	RAN#112	RP-261206	1597	4	F	Correction of Transmission Bandwidth Asymmetric	19.3.0
2026-06	RAN#112	RP-261205	1652	3	F	Correction to LTM MCG configuration management for inter-CU SCG LTM	19.3.0
2026-06	RAN#112	RP-261204	1679	4	F	Correction of paging capability for LP-WUS	19.3.0
2026-06	RAN#112	RP-261207	1680	3	F	Correction of Positioning System Information Delivery [OdPosSIB_Req]	19.3.0
2026-06	RAN#112	RP-261202	1687	2	F	Correction of SRB ID for network-side data collection	19.3.0
2026-06	RAN#112	RP-261205	1702	3	F	Support for modification of LTM configurations initiated by the candidate	19.3.0
2026-06	RAN#112	RP-261210	1708	2	A	Correction to BWP configuration for SRS frequency hopping	19.3.0
2026-06	RAN#112	RP-261206	1709	2	F	Correction on CCO Assistance Information over F1	19.3.0
2026-06	RAN#112	RP-261203	1712	3	F	Corrections on SBFD RACH	19.3.0
2026-06	RAN#112	RP-261205	1714	1	F	ASN.1 Issue in CSI-RS Coordination procedure	19.3.0
2026-06	RAN#112	RP-261205	1715	1	F	Clarification on the gNB-DU ID for inter-CU LTM	19.3.0
2026-06	RAN#112	RP-261202	1719	1	F	Correction for timing information and coverage modification cause in AI/ML-based CCO signalling	19.3.0
2026-06	RAN#112	RP-261206	1723	-	F	Correction of procedure texts for the broadcast information IEs	19.3.0

History

Version	Date	Status
V19.0.0	January 2026	Publication
V19.1.0	February 2026	Publication
V19.2.0	April 2026	Publication
V19.3.0	July 2026	Publication